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# Papers 1–4 Combined: All Answers by Section

All four answer books merged into one, section by section. Questions on the same topic from different papers are printed once as a single combined answer, with every fact from each version kept; the paper tags (P1–P4) show where each was asked. 386 topics in all, from 554 original answers. Each section has long answers first, then short answers, then its Rapid Revision Sheet.

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# Section 1 · Basic Principles

## Long answer questions (LAQ)

### LAQ 1 Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy

P1 **merged from 3**

*Asked as:* P1 LAQ "Iodinated and other contrast agents (use in urology) and their complications" · P1 SAQ "Non iodinated contrast materials" · P1 SAQ "Acetylcysteine and Contrast Induced Nephropathy (CIN)"

#### Introduction

- Iodine (atomic number 53, K-edge 33 keV) absorbs X-rays strongly. All iodinated agents are built on a **tri-iodinated benzene ring**.
- They are filtered by the glomerulus, not secreted or reabsorbed, so they concentrate in urine. Half-life is about 2 hours with normal function; biliary excretion rises in renal failure.
- Modern practice uses **non-ionic low-osmolar** (iohexol, about 600–850 mOsm/kg) or **iso-osmolar** (iodixanol, 290) agents. High-osmolar ionic agents caused far more reactions.
- Uses: CT (computed tomography) urography, IVU (intravenous urography), pyelography, cystourethrography, angiography.
- Non-iodinated agents are used when iodinated contrast is risky (allergy, renal impairment) or with MRI (magnetic resonance imaging) and ultrasound.

#### Classification of iodinated agents

| Class | Example | Osmolality (mOsm/kg) | Iodine : particles |
|---|---|---|---|
| High-osmolar ionic monomer | Diatrizoate | ~1500–2000 (5–8 × plasma) | 3 : 2 |
| Low-osmolar non-ionic monomer | Iohexol, iopamidol | ~600–850 | 3 : 1 |
| Low-osmolar ionic dimer | Ioxaglate | ~600 | 6 : 2 |
| Iso-osmolar non-ionic dimer | Iodixanol | ~290 | 6 : 1 |

- **Why it matters:** a higher ratio gives more opacity per particle. Fewer particles mean less fluid shift, pain and reactions. So non-ionic low- or iso-osmolar agents are standard intravascularly.

#### Types of contrast agent and their uses in urology

| Agent | Principle | Urological use |
|---|---|---|
| Iodinated: ionic high-osmolar (diatrizoate); non-ionic low-osmolar (iohexol); iso-osmolar dimer (iodixanol, 290 mOsm/kg) | Iodine absorbs X-rays; glomerular filtration | CT urography, angiography; retrograde pyelogram, urethrogram, cystogram |
| Gadolinium (GBCA, gadolinium-based contrast agent) | Shortens T1 relaxation; chelated | MRI of renal mass, prostate, MR urography |
| Microbubbles | Gas in shell, intravascular | CEUS (contrast-enhanced ultrasound): cysts; voiding urosonography |

| Agent | Principle | Urological use |
|---|---|---|
| Carbon dioxide | Negative contrast | Angiography in CKD (chronic kidney disease) |

## Complications of iodinated contrast

### A. Hypersensitivity (allergic-like) reactions

- Mostly non-IgE (immunoglobulin E) direct mast-cell release: idiosyncratic, not dose-related. Severe reactions: 0.22% with ionic vs 0.04% with non-ionic agents (Katayama, 1990).

| Type | Timing | Features |
|---|---|---|
| Immediate | Within about 1 hour (most within minutes) | Urticaria, angio-oedema, bronchospasm, hypotension; anaphylaxis if severe |
| Non-immediate | Hours to days | Maculopapular rash; T-cell mediated, milder |

- Risk factors: previous reaction (strongest), asthma, atopy, severe cardiac disease, drug allergy, female sex. **Shellfish ("iodine") allergy is not a cross-reaction.**
- **Physiological reactions:** warmth, metallic taste, nausea, vasovagal hypotension with bradycardia, arrhythmia, seizures.
- Retrograde studies are safe in allergy.

**B. Contrast-associated acute kidney injury** (see the separate CIN section below)

### C. Others

- **Metformin:** lactic acidosis only if AKI (acute kidney injury) develops. ACR (American College of Radiology): eGFR (estimated glomerular filtration rate) ≥30 and no AKI, continue. eGFR <30 or AKI, withhold 48 hours and restart when function (creatinine) is stable.
- **Thyroid:** iodine load can trigger hyperthyroidism/thyrotoxicosis (Jod-Basedow) in Graves' disease or nodular goitre. It blocks radio-iodine scans or therapy for several weeks.
- **Extravasation:** usually minor, but large power-injected volumes can rarely cause compartment syndrome or skin necrosis. Elevate the limb, apply compresses and observe. Get a surgical review if swelling increases, blisters form, or perfusion or sensation changes.
- **Also know:** iodide "mumps"; myasthenia flare; pregnancy (crosses placenta, not teratogenic); breastfeeding can continue; phaeochromocytoma safe with IV (intravenous) non-ionic agents; dialysis patients need no urgent dialysis.

## Prevention and treatment of contrast reactions

### Premedication for high-risk patients / prior reaction (ACR 2023)

| Regimen | Drugs |
|---|---|
| 1 | Prednisone 50 mg orally 13, 7 and 1 hour before + diphenhydramine 50 mg 1 hour before |
| 2 | Methylprednisolone 32 mg orally 12 and 2 hours before + diphenhydramine 50 mg 1 hour before |

- First consider ultrasound, MRI or non-contrast CT. Breakthrough reactions still occur, so keep resuscitation drugs ready. Retrograde pyelography is safe even after anaphylaxis, because little contrast is absorbed.

**Treating reactions** `Stop injection → airway, oxygen, call for help Urticaria → observe ± H1-antihistamine (may progress) Bronchospasm → salbutamol inhaler 2`

puffs (90 µg each), up to 3 times Anaphylaxis → intramuscular (IM) adrenaline 0.3–0.5 mg (1 mg/mL) every 5–15 min (repeat), max 1 mg; IV fluids Vasovagal → legs up, IV fluids, IV atropine Observe ≥30 min; later allergy testing; record the reaction

## Contrast-induced nephropathy (CIN) and contrast-associated AKI

### Definitions

- **CIN (contrast-induced nephropathy)** (older term): creatinine rise ≥25% or ≥0.5 mg/dL within 48–72 hours of iodinated contrast, with no other cause.
- Current terms (ACR):
- **CA-AKI (contrast-associated acute kidney injury):** AKI within 48 hours of contrast, whatever the cause.
- **CI-AKI (contrast-induced AKI):** the subset actually caused by contrast.
- **KDIGO (Kidney Disease: Improving Global Outcomes) AKI:** creatinine rise ≥0.3 mg/dL in 48 hours, ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for ≥6 hours.

### Pathophysiology

- **Renal (afferent) vasoconstriction** causes **medullary hypoxia** and ischaemic tubular injury (ischaemic ATN, acute tubular necrosis), plus **direct tubular toxicity**. Usually transient; dialysis rarely needed.

### Risk factors

- **Patient:** CKD (strongest), diabetes, age, dehydration, heart failure, anaemia, NSAIDs (non-steroidal anti-inflammatory drugs).
- **Procedure:** high contrast volume, repeated doses within 48–72 hours, intra-arterial route, high-osmolar agents.
- **eGFR thresholds:** real risk mainly at **eGFR <30 mL/min/1.73 m²** (IV contrast); ESUR (European Society of Urogenital Radiology) uses **<45** for intra-arterial first-pass exposure.
- ACR/NKF (National Kidney Foundation) 2020: older uncontrolled studies overstated the risk; it is rare but real.

### Prevention

- Consider ultrasound or non-contrast CT instead.
- Lowest dose of a **low- or iso-osmolar** agent; weight-based dosing; a low tube voltage (80 kVp, kilovolt peak) needs less contrast. Avoid repeat doses within 48 hours; stop NSAIDs.
- **IV isotonic saline** (e.g. 1 mL/kg/h before and after) for high-risk patients (eGFR <30 not on dialysis): the only proven measure.
- **PRESERVE (2018):** sodium bicarbonate and **NAC (N-acetylcysteine)** gave no benefit over saline.
- **AMACING (2017):** no prophylaxis was non-inferior to hydration at eGFR 30–59.
- Prophylactic dialysis (removing contrast by dialysis) does not reduce risk.
- **Metformin:** withhold for 48 hours only if eGFR <30 or AKI (see above).

## Acetylcysteine

### Introduction

- **NAC:** acetylated form of the amino acid cysteine; a **thiol (sulfhydryl, –SH) compound**. Given orally, intravenously or by nebuliser.

**Mechanisms of action**

- **Glutathione precursor:** supplies cysteine, the rate-limiting substrate for **glutathione** synthesis.
- **Antioxidant:** the free –SH group scavenges reactive oxygen species.
- **Mucolytic:** –SH breaks **disulfide bonds** in mucoproteins, lowering mucus viscosity.
- **Thiol–disulfide exchange:** in theory forms a soluble mixed disulfide with cystine.

**Clinical uses**

- **1. Paracetamol (acetaminophen) overdose: main indication.**
- Overdose saturates normal pathways; the toxic metabolite **NAPQI** (N-acetyl-p-benzoquinone imine) builds up once glutathione is exhausted, causing hepatic necrosis.
- NAC restores glutathione. It is most effective **within 8 hours**, but still worth giving later.
- **IV 21-hour regimen:** 150 mg/kg, then 50 mg/kg over 4 h, then 100 mg/kg over 16 h. Shorter two-bag regimens are now used in some countries.
- **2. Mucolytic:** COPD (chronic obstructive pulmonary disease), cystic fibrosis.

**Urological relevance**

- **Contrast-induced AKI:** once widely used (early small trials, KDIGO 2012 suggestion). **PRESERVE (2018):** NAC gave no benefit over saline. It is **no longer recommended**; isotonic saline hydration is the only proven measure.
- **Cystinuria:** thiol action is theoretical; standard thiols are **tiopronin** and D-penicillamine, not NAC.
- **Cyclophosphamide cystitis: mesna** (another thiol) is the agent used.
- **Mucus after enterocystoplasty or neobladder:** sometimes used orally or in irrigation; evidence weak.
- **Male infertility:** antioxidant for oxidative stress (high ROS (reactive oxygen species), sperm DNA fragmentation); limited evidence.
- **Ischaemia–reperfusion** (renal surgery, transplant): antioxidant protection studied; not established.

**Adverse effects**

- **Anaphylactoid reactions** with IV loading (flushing, rash, bronchospasm): slow the infusion, give antihistamine.
- Nausea, vomiting; unpleasant sulphur smell; bronchospasm when nebulised in asthma.
- Safe in pregnancy and renal impairment for paracetamol poisoning; benefit outweighs risk.

## Non-iodinated contrast agents

**1. Gadolinium-based contrast agents (GBCAs)**

- **Mechanism:** paramagnetic gadolinium **shortens T1 relaxation** of nearby water, giving bright enhancement. Free gadolinium is toxic, so it is **chelated** (linear or macrocyclic; macrocyclic is more stable). Usual dose 0.1 mmol/kg; renal excretion.
- **NSF (nephrogenic systemic fibrosis):** skin and organ fibrosis, joint contractures, no proven treatment. Risk: eGFR <30 mL/min/1.73 $m^2$, dialysis, AKI.

| ACR group | Agents | NSF risk |
|---|---|---|
| I (linear) | Gadodiamide, gadopentetate, gadoversetamide | Highest |
| **II** | Gadobenate, gadoteridol, gadoterate, gadobutrol | **Little or none, even in ESRD (end-stage renal disease)** |
| III | Gadoxetate (liver-specific agent) | Few NSF data; recent ACR editions have moved some agents between groups, so check the current manual |

- **Gadolinium deposition:** traces retained in brain (dentate nucleus) and bone, mainly with linear agents; no proven harm. Europe suspended most linear agents (2017).
- **Other:** allergic reactions rarer than with iodinated agents; not nephrotoxic at usual doses; avoid in pregnancy unless essential.
- **Uses:** renal mass, prostate mpMRI (multiparametric MRI), MR urography and angiography.

**2. Ultrasound contrast (microbubbles)**

- Gas core (perfluorocarbon or sulfur hexafluoride) in a lipid or albumin shell; strong harmonic echoes.
- Purely **intravascular**; gas exhaled by lungs. **No nephrotoxicity or radiation.** Rare cardiopulmonary reactions.
- **CEUS:** solid vs cystic renal mass (Bosniak), ablation monitoring, surveillance in CKD. Intravesical use: **voiding urosonography (ceVUS) for VUR (vesicoureteral reflux)**; ureteric patency after PCNL (percutaneous nephrolithotomy).

**3. Other agents**

- **$CO_2$ angiography:** negative contrast; not allergenic or nephrotoxic. Useful in CKD; **never above the diaphragm** (cerebral embolism).
- **Ferumoxytol** (iron oxide): MRI agent in renal failure. Lymphotropic iron nanoparticles help nodal staging (nodal metastases fail to darken).
- **Barium:** GI (gastrointestinal) studies only; contraindicated if perforation suspected.
- **Also know:** air as negative contrast (historical); radiotracers ($^{99m}$Tc, PET (positron emission tomography)) are not contrast agents; MRI safety (pacemakers, metal).

## Diagrams to draw

- Tri-iodinated benzene ring, monomer versus dimer, on an osmolality scale.
- Chelated gadolinium (macrocyclic vs linear ring).
- Contrast, then vasoconstriction plus tubular toxicity, then medullary hypoxia, then AKI.
- Paracetamol to NAPQI to glutathione conjugation, with NAC replenishing glutathione.

**Key points to remember:**

- Iodine : particle ratio 1.5 (ionic), 3 (non-ionic monomer), 6 (iodixanol, 290 mOsm/kg).
- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE 2018: bicarbonate and NAC useless; AMACING: no prophylaxis at eGFR 30–59.
- Metformin: withhold 48 h only if eGFR <30 or AKI.
- NAC: thiol; glutathione precursor, antioxidant, mucolytic; paracetamol antidote, best within 8 h; cystinuria uses tiopronin, not NAC.
- GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
- Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux. $CO_2$ angiography: never above diaphragm.

## LAQ 2 Laparoscopic access: Veress, Hasson, trocars, gases; open vs laparoscopic vs robotic prostatectomy

P1 · P4 **merged from 2**

*Asked as:* P1 LAQ "Laparoscopic access (Veress, Hasson cannula, extraperitoneal space), gases, and open vs laparoscopic vs robotic radical prostatectomy" · P4 SAQ "Different types of trocars in laparoscopy."

### Access in laparoscopy

**A. Transperitoneal: closed (Veress)**

- Veress needle: 14-gauge, spring-loaded blunt stylet. It is an entry needle, not a trocar. Check flow and tip recoil first. Two "pops": fascia, then peritoneum.
- **Sites:** umbilicus (thinnest wall; aim towards pelvis in non-obese); lower quadrants; **Palmer point** (left subcostal, midclavicular line) with suspected adhesions; avoid scars.
- **Confirmation tests:** aspiration–injection–aspiration of saline; **drop test**; advancement test (1 cm free). Then insufflate: opening pressure <8–10 mmHg, set limit 15 mmHg.
- First trocar inserted blind, or under vision with an **optical trocar** (0° lens inside a clear tip).

**B. Open (Hasson)**

- 2-cm infra-umbilical incision; fascia and peritoneum opened under vision; stay sutures; blunt cannula.
- Safer with adhesions or previous surgery and early in training; more gas leak.

**C. Retroperitoneal**

- 2-cm incision below the tip of the 12th rib; finger dissection behind Gerota's fascia, in front of psoas, then **balloon dilatation** (commercial or **Gaur** glove-on-catheter balloon), then a balloon-tip port to prevent gas leak and emphysema.

**D. Extraperitoneal (pelvic):** infra-umbilical, behind rectus into the Retzius space with a balloon (used for extraperitoneal prostatectomy).

**E. Others:** hand-assisted (6.5–7.5 cm port); single-port/LESS (laparoendoscopic single-site); direct trocar entry without pneumoperitoneum.

- **Access complications:** preperitoneal insufflation, vessel or bowel puncture, gas embolism, subcutaneous emphysema.

### Hasson cannula (Hassan's canula)

*Assumed to mean the Hasson cannula, used for open laparoscopic access.*

**Introduction**

- Described by **Harrith Hasson (1971)** for open ("direct vision") laparoscopy. The first port is placed under vision instead of a blind Veress needle and trocar.
- **Parts:** outer sleeve with a cone-shaped (olive) adapter, **blunt obturator**, struts to anchor fascial stay sutures, and a gas inlet with tap.

**Indications**

- **Previous abdominal surgery or expected adhesions** (main indication).
- Organomegaly, pregnancy, aortic aneurysm, children, thin patients.
- Retroperitoneoscopy and extraperitoneal access (open start, then balloon dilatation).

### Technique (transperitoneal)

1. 2-cm semicircular infra-umbilical incision.
2. Open fascia and then peritoneum separately, under vision, finger-sized.
3. Confirm entry by sight and finger sweep (free adhesions).
4. Place two 0 silk fascial stay sutures.
5. Insert the cannula with its **blunt obturator**.
6. Seat the cone in the defect; wrap stay sutures round the struts to seal.
7. Remove obturator; insufflate at high flow (to 12–15 mmHg).
8. At closure, tie the stay sutures to close the fascia. - **Retroperitoneal:** 2–2.5 cm incision below the 12th rib tip, finger dissection onto psoas, balloon dilatation, then port.

### Advantages over Veress

- No blind puncture: fewer major vessel and bowel injuries; no preperitoneal insufflation; no gas embolism from intravascular insufflation.
- Fast insufflation; easy fascial closure (less port-site hernia).

### Disadvantages

- Larger incision; **gas leak** around the port; slightly longer in obese patients.
- Early retroperitoneal use caused subcutaneous emphysema and hypercarbia, now solved by balloon-cuffed blunt ports.

### Complications

- Bowel injury if adherent under the wound (repair through the same incision); wound infection; port-site hernia; vascular injury is still possible.

### Evidence

- Reviews show no single entry method is clearly safest for major injury; open entry has fewer failed entries.

## Instrumentation for development of the extraperitoneal space

- Extraperitoneal laparoscopy works outside the peritoneum: the **flank retroperitoneum** (kidney, adrenal, ureter) or the **pelvic prevesical (Retzius) space** (extraperitoneal prostatectomy, lymphadenectomy).
- There is no natural cavity, so a working space must be **dissected before insufflation**: finger first, then balloon.

| Instrument | Description and use |
|---|---|
| Open (Hasson) access set | Scalpel, S-retractors, artery forceps |
| **Index finger** | Blunt first pocket; feels psoas behind, Gerota's fascia in front |
| **Commercial balloon dissector** | Transparent silicone balloon on a stiff hollow shaft; inflated with room air by bulb (maximum 800 mL, 40 pumps). Oblong for the retroperitoneum, round for the pelvis. Laparoscope inside confirms position |
| **Gaur balloon (1992)** | Size 7 glove (or glove fingers, condom) tied with 0 silk on an 8-Fr red rubber catheter; filled with saline (600–800 mL). Cheap, but no rigid shaft or view; latex can rupture into fragments |

| Instrument | Description and use |
|---|---|
| **Optical (visual) obturator trocar** with 0° scope | Enters under vision; yellow retroperitoneal fat confirms the plane; scope sweeps the space |
| **Blunt-tip balloon trocar** | Internal balloon plus external foam cuff gives an airtight seal; prevents gas leak, emphysema and hypercarbia |

### Technique: retroperitoneal

1. Full flank, table flexed; 1.5–2.5-cm incision below the 12th rib tip.
2. Split muscles; open the lumbodorsal fascia; finger dissection between psoas and Gerota's fascia.
3. Balloon inflated cephalad then caudad, in front of psoas, behind Gerota's fascia.
4. Blunt-tip balloon trocar; insufflate; secondary ports under vision.

### Technique: pelvic extraperitoneal

- Infraumbilical incision; open anterior rectus sheath, separate recti; finger tunnels behind the rectus to the pubis; pierce transversalis fascia; balloon midline, then each side.

### Pitfalls

- Balloon in muscle causes a flank hernia. Too anterior causes peritoneal tear or colon injury; too posterior causes psoas or quadratus bleeding.
- Peritoneal tear shrinks the space; convert if needed. Extraperitoneal $CO_2$ needs more ventilation.

## Trocars

### Definition

- A **trocar** is a **cannula (port) plus an obturator**; 3–20 mm diameter, 5–20 cm long (Chapter 18).
- Parts: obturator tip, cannula, **valve** (keeps gas in), insufflation side-arm, retention mechanism.

### Classification

| Basis | Types |
|---|---|
| **Obturator tip** | Bladed (cutting) with spring safety shield; **bladeless (blunt dilating)**; **optical (visual)**; blunt (Hasson) |
| **Material** | Reusable metal; disposable plastic |
| **Valve** | Flap or trumpet valve (needs reducer); **multiseal** (5–12 mm, no reducer); **valveless (AirSeal)** |
| **Retention** | Smooth (suture-anchored); threaded sleeve; expandable arms; **balloon (Blunt Tip)** |
| **Special** | Hasson open cannula; radially expanding; hand ports; single-site multichannel ports; robotic 8-mm metal ports |

### Main types explained

- **Bladed trocars:** cut through tissue; more epigastric injury and port-site hernia. Now of historical interest.
- **Bladeless (blunt dilating) trocars:** spread muscle fibres; have replaced bladed ones. About **fivefold less inferior epigastric injury**; trocar-site bleeding 3% vs 9%. Switching

cut port hernia from 1.83% to 0.19%. **Fascial closure only for midline ports >10 mm** or stretched sites (all ≥5 mm in children).

- **Optical trocars:** clear tip with a 0° scope inside; each layer is seen during entry.
- **Hasson cannula:** blunt obturator and cone, fixed with fascial stay sutures after open entry; safer with adhesions; leaks more gas.
- **Radially expanding trocar:** a sleeve over the Veress needle is dilated to 12 mm; less port-site bleeding and pain (Chapter 39).
- **Balloon (Blunt Tip) trocar:** internal balloon and external foam cuff seal the wall; prevents leak and **subcutaneous emphysema** in retroperitoneoscopy.
- **AirSeal valveless port:** stable pneumoperitoneum at 8–12 mmHg despite suction.
- **Single-site ports:** TriPort; da Vinci SP (single-port) 25–27-mm guide. **Hand ports:** GelPort.

**Safety points**

- **Never use a metal trocar with a plastic retaining ring**: capacitive coupling burns viscera out of view.
- Insert under vision; ports 8–10 cm apart.

## Gases used

| Gas | Advantages | Disadvantages |
|---|---|---|
| **$CO_2$ (carbon dioxide; standard)** | Non-combustible, cheap, **highly blood-soluble** (embolism less lethal), quickly absorbed | Absorbed, causing hypercarbia, acidosis, arrhythmia; dangerous in COPD (chronic obstructive pulmonary disease) |
| **Nitrous oxide** | Less peritoneal pain, fewer acid–base effects | **Supports combustion**: no electrosurgery |
| **Helium** | Inert, no hypercarbia; use in lung disease | Poorly soluble, so embolism risk; costly |
| **Argon** | Inert | Embolism risk |
| Air, oxygen | – | Embolism, combustion: abandoned |

- **Physiology:** keep pressure 10–15 mmHg (12 mmHg more physiological). Pressure reduces venous return, raises afterload, causes **oliguria** (renal vein and parenchymal compression) and splints the diaphragm.
- $CO_2$ absorption plateaus at 30–60 minutes; higher with extraperitoneal insufflation. Hypercarbia peaks after extubation.
- **Gas embolism:** sudden collapse, millwheel murmur, end-tidal $CO_2$ rise then fall. Treat: stop insufflation, **left lateral, head-down**, 100% oxygen, aspirate via central line.

## Comparison of open (RRP), laparoscopic (LRP) and robotic (RALP) radical prostatectomy

- **Background:** anatomical nerve-sparing open RRP (Walsh, 1982) was the standard. First LRP (Schuessler, 1997) was too slow; French centres standardised it in 2000, but laparoscopic suturing of the anastomosis remained a barrier. The robot (da Vinci, 2000) removed that barrier, and RALP is now dominant in the USA.

| Parameter | Open RRP | LRP | RALP |
|---|---|---|---|
| Vision | Direct, loupes | 2D, unstable | **3D, 10× magnified**, stable |

| Parameter | Open RRP | LRP | RALP |
|---|---|---|---|
| Operating time | Shortest | Longest early | Longer than open; falls with experience |
| Instruments | Hands, haptic feedback | Rigid, fulcrum effect | **Wristed** (7 degrees of freedom); no haptics |
| Learning curve | Moderate | **Steepest** (suturing) | Shorter |
| Blood loss | 300–1000 mL | Less (pneumoperitoneum tamponade) | Least; **transfusion 1–2%** |
| Hospital stay, catheter | Longer; catheter ~2 weeks | Shorter | Usually 1 day; catheter ~1 week |
| Pain, return to work | More | Less | Less; normal activity 3–4 weeks |
| Positive margins | Similar | Similar | Similar (~15%; ~9% if pT2) |
| Continence (1 year) | ~93% (expert) | Similar | Similar; possibly earlier |
| Potency | Surgeon-dependent | Similar | Similar |
| Bladder neck stricture | Up to 16% (older series) | Lower | **<2%** (mucosa-to-mucosa) |
| Unique risks | Wound infection, hernia | Conversion, $CO_2$ effects | Steep Trendelenburg (eye, nerve), robot failure (<0.5%) |
| Cost | Lowest | Moderate | **Highest** |

### Evidence

- **Yaxley 2016 (randomised, one open vs one robotic surgeon):** equal continence, potency and early margins at 3 months; less blood loss and shorter stay with robot.
- **Coughlin 2018:** no difference in urinary or sexual function at 24 months.
- **LAP-01 (robotic vs laparoscopic):** earlier continence with robot.
- **Concept:** the **surgeon, not the platform**, decides outcome; margins depend mainly on tumour (Gleason ≥7, stage) and surgeon volume. The robot's advantages are recovery and wider reach (obese patients, prior surgery).
- **Also know:** transperitoneal vs extraperitoneal approach (no consistent difference); Retzius-sparing RALP (earlier continence, more margins); single-port (SP) robot; new robotic platforms; telesurgery.

## Diagrams to draw

- Veress needle sites (umbilicus, Palmer point) and trocar layout for RALP.
- Hasson cannula with cone, stay sutures and blunt obturator in the abdominal wall.
- Flank section with balloon between psoas and Gerota's fascia.
- Trocar with obturator tip types (bladed, bladeless, optical) and valve.

**Key points to remember:**

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- Hasson 1971: open entry under vision; main indication previous surgery or adhesions; avoids blind Veress injuries; drawback is gas leak.
- Extraperitoneal space is created first: finger, then balloon. Gaur balloon: glove on red rubber catheter. Balloon in muscle causes hernia.
- Trocar = cannula + obturator. Bladeless: fewer epigastric injuries and hernias. Balloon trocar seals and prevents retroperitoneal emphysema.
- $CO_2$ standard (soluble, non-flammable); helium in COPD; $N_2O$ supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.

## LAQ 3 Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy

P1 · P4 **merged from 2**

*Asked as:* P1 LAQ "Pneumoperitoneum physiology, complications of retroperitoneoscopy and laparoscopic donor nephrectomy" · P4 SAQ "General complications related to laparoscopy."

*"d/t" is read as "due to" and "lap" as laparoscopy.*

### Introduction to pneumoperitoneum

- **Pneumoperitoneum** is insufflation of gas, usually $CO_2$ (carbon dioxide), into the peritoneal cavity at 10–15 mmHg to create a working space.
- Its effects have **two drivers**: **pressure** (compression of the cava, renal veins, kidneys and diaphragm) and **absorbed $CO_2$** (hypercarbia, acidosis, sympathetic drive). Patient position and volume status modify both.
- **$CO_2$** is standard: non-combustible, cheap, highly blood-soluble, so embolism is less lethal. Helium or argon in severe lung disease; $N_2O$ (nitrous oxide) supports combustion.

### Choice of insufflant

| Gas | Advantages | Disadvantages |
|---|---|---|
| **$CO_2$ (standard)** | Non-combustible; highly blood-soluble, so embolism is less lethal; cheap | Hypercarbia, acidosis, arrhythmia; COPD (chronic obstructive pulmonary disease) patients cannot clear it |
| Helium, argon | Inert; no hypercarbia | Poorly soluble, so gas embolism risk; costly. Helium used in lung disease or to salvage hypercarbia |
| $N_2O$ (nitrous oxide) | Less peritoneal irritation | Supports combustion; no electrosurgery |
| Room air, oxygen | — | Embolism, combustion; abandoned |

### Effects by pressure (supine, Campbell Table 18.1)

| Effect | 5 mmHg | 10 mmHg | 20 mmHg | 40 mmHg |
|---|---|---|---|---|
| Heart rate (HR) | ↑ | ↑ | ↑ | ↓ |
| MAP (mean arterial pressure), SVR (systemic vascular resistance) | ↑ | ↑ | ↑ | ↑ |
| Cardiac output | ↔/↓ | ↔/↑ | ↔/↓ | ↓ |
| Venous return, cardiac output (Summer 2019 source) | — | Variable | Variable | ↓ sharply |
| GFR (glomerular filtration rate), urine output | ↔ | ↓ | ↓↓ | ↓↓ |
| Arterial $PCO_2$ | ↔ | ↑ | ↑ | ↑ |
| Arterial pH | ↔ | ↔/↓ (another source: ↔) | ↓ | ↓ |

- Renal function is the first casualty (from 10 mmHg). Afterload rises at every pressure. Output collapses at 40 mmHg. Acidosis builds from 20 mmHg.

### Cardiovascular (haemodynamic) changes

- **Afterload rises** at every pressure: aortic and splanchnic compression plus catecholamine and **vasopressin** release raise MAP and SVR.
- **Preload depends on volume status.** In normovolaemia or hypovolaemia, pressure up to 20 mmHg compresses the IVC (inferior vena cava) and venous return falls. In hypervolaemia the cava resists and venous return may rise. At ≥40 mmHg venous return falls sharply in everyone and output collapses.
- **Arrhythmias:** hypercapnia causes tachycardia and ventricular ectopics; peritoneal stretch causes **vagal bradycardia** or asystole. A new arrhythmia may herald pneumothorax, hypoxia or gas embolism.
- **CVP (central venous pressure) reads falsely high** because intrathoracic pressure rises; do not guide fluids by it.
- Leg venous stasis: theoretical DVT (deep vein thrombosis) risk.

### Respiratory changes

- **Diaphragm splinted:** FRC (functional residual capacity) and compliance fall, peak airway pressure rises; atelectasis and V/Q (ventilation–perfusion) mismatch. Dead space is unchanged. PEEP (positive end-expiratory pressure) helps in lung disease.
- **Head-down** (steep Trendelenburg) adds a further fall in vital capacity, endobronchial tube slip and facial oedema.
- **Barotrauma** (pressure >15 mmHg for long): pneumothorax, pneumomediastinum.

### Acid–base changes ($CO_2$ absorption)

- $CO_2$ is absorbed through the peritoneum and, extraperitoneally, through fat and disrupted lymphatics. Absorption rises over 30–60 minutes, then plateaus.
- **Hypercarbia and respiratory acidosis** with rising end-tidal $CO_2$. Managed by raising minute ventilation and lowering pressure.
- **COPD** patients cannot blow it off: arterial gases if insufflation exceeds 1 hour, continued in recovery; consider helium. **Hypercarbia peaks after extubation**, as stored $CO_2$ is released.

### Renal changes

- **Oliguria** comes from **renal vein and parenchymal compression**, not ureteric compression or hormones. Cortical flow falls; medullary flow rises up to 20 mmHg, then falls. It reverses on desufflation.
- Keep pressure ≤10 mmHg (another source: ≤10–12 mmHg); avoid fluid overload. If needed give furosemide, mannitol 12.5–25 g or dopamine 2 µg/kg/min to keep urine >100 mL/hour, rather than fluid boluses (overload, oedema). Important in live donor nephrectomy.

### Other systems

- **Splanchnic:** hepatic and mesenteric flow fall (rare mesenteric thrombosis); less ileus than open surgery; no rise in reflux.
- **ICP (intracranial pressure) and IOP (intraocular pressure)** rise with hypercarbia and head-down tilt: caution in head injury, shunts and glaucoma.
- **Hormonal:** catecholamines, cortisol and **vasopressin** rise; renin–angiotensin activation adds to afterload.

- **Stress and immune response:** cortisol, catecholamines and prolactin rise as in any surgery, but hyperglycaemia, insulin resistance, catabolism and IL-6 (interleukin-6) rise less than after open surgery, which may explain faster recovery; oncological benefit unproven.
- **Also know:** hypothermia from cold dry gas; shoulder-tip pain from diaphragmatic $CO_2$ irritation.

### Effects of position and approach

| Position | Heart rate | MAP | Cardiac output | Other effects |
|---|---|---|---|---|
| Head-up | ↑ | ↓ | ↓ | Output ↓, hypotension |
| Head-down (Trendelenburg) | ↓ | ↑ | ↑ (but lungs worse) | Venous return and output ↑; lung volumes ↓, ICP and IOP ↑ |
| Lateral flank, kidney rest | — | — | — | Lung V/Q mismatch; rhabdomyolysis risk with kidney rest |

- **Summary:** head-down raises MAP and output but worsens lungs; head-up lowers output.
- **Retroperitoneal** insufflation causes smaller swings in output and venous pressures than transperitoneal; pelvic extraperitoneal insufflation raised arterial $CO_2$ more.

### Safe practice

- Preoperative cardiopulmonary assessment; correct arrhythmias; optimise volume.
- **Practical rule: work at 10–12 mmHg** (12 mmHg keeps stroke volume unchanged); 20 mmHg only briefly (first trocar, venous bleeding); never exceed 20 mmHg for long.
- Monitor end-tidal $CO_2$, saturation, airway pressure and urine output.
- **Recent advances:** low-pressure (8 mmHg) surgery with valveless insufflators (AirSeal); helium (or argon) for lung disease (poorly soluble, so embolism risk).

### General complications of laparoscopy

#### Incidence

- Historically about 4%, mortality 0.03–0.08%. Contemporary series 13–22% because procedures are complex. **Vascular, then adjacent-organ injuries are commonest.**
- Risk factors: complex procedures, operating time >4 h, creatinine >1.5 mg/dL, learning curve.

#### Complications by phase

| Phase | Complication | Key points |
|---|---|---|
| **Access (Veress, first trocar)** | Preperitoneal insufflation; vessel or bowel puncture; **major vascular injury (0.11–2%)** | Confirm Veress tests; open access with adhesions. Major vessel: leave trocar in, laparotomy |
| **Insufflation** | **Gas embolism**; subcutaneous emphysema; pneumothorax (1.6–4%); pneumomediastinum, pneumopericardium; hypercarbia | Embolism: see below |
| **Secondary trocars** | Epigastric bleeding; crossing swords, rollover | Transilluminate; bladeless trocars; insert under vision |
| **Dissection** | Thermal or mechanical bowel injury; vascular injury; bladder or ureter injury | Four electrosurgical mechanisms below |

| Phase | Complication | Key points |
|---|---|---|
| **Positioning** | Nerve injury (about 2.8%: brachial plexus, femoral, sciatic); **rhabdomyolysis** (about 1% retroperitoneal) | Padding, kidney rest <1 h, cases <5 h |
| **Exit** | Bowel entrapment, late port bleeding | Inspect each site at 5 mmHg |
| **Postoperative** | Port-site hernia (0.3–5.4%), wound infection, shoulder-tip pain, DVT | Close fascia of large ports |

**Physiological complications:** hypercarbia and acidosis; oliguria from renal vein compression; reduced venous return; raised intracranial pressure in head-down tilt (see above).

#### Gas embolism

- Gas enters an opened vein (renal vein, IVC or lumbar vein). Sudden collapse; end-tidal $CO_2$ rises abruptly, saturation falls, then end-tidal $CO_2$ falls; "mill-wheel" murmur.
- Treatment: stop insufflation, **left lateral, head down**, 100% oxygen, aspirate through a central line.

#### Electrosurgical (thermal) bowel injury: four mechanisms

1. **Inappropriate (direct) activation** out of view.
2. **Direct coupling** with another metal instrument.
3. **Capacitive coupling** (metal cannula in a plastic retainer).
4. **Insulation failure.** - Monopolar injury presents **late (up to 18 days)** with low-grade fever and pain; resect with margin.

#### Missed bowel injury (Bishoff)

- Not classic peritonitis: **pain at one trocar site, leucopenia, low fever, distension, diarrhoea** at 24–48 h.
- **CT (computed tomography) with oral contrast**; free gas is normal for up to 7–9 days.

### Complications of retroperitoneoscopy

#### Introduction

- **Retroperitoneoscopy** is laparoscopic or robotic surgery in the retroperitoneum without entering the peritoneum. It gives direct hilar access and avoids bowel, but the space must be **created**.
- **Why its complications differ:** small space with no natural cavity; space made by finger and **balloon dissection** (commercial or Gaur glove balloon) behind Gerota's fascia and in front of the psoas; fewer landmarks; dissection close to the lens; gas tracks along tissue planes; $CO_2$ is absorbed through fat and disrupted lymphatics, so hypercarbia appears early if the port leaks.
- Minimally invasive renal surgery: about 15% complications, mostly postoperative. Learning curve about 20 cases.

#### Classification (by phase of surgery)

| Phase | Complications |
|---|---|
| Access and space creation | Wrong plane, balloon rupture, peritoneal tear, muscle bleeding |
| Insufflation | Subcutaneous emphysema, hypercarbia, pneumothorax, gas embolism |
| Dissection | Vascular, visceral, nerve, thermal injury |

| Phase | Complications |
|---|---|
| Exit and recovery | Port bleeding, hernia, rhabdomyolysis, urinoma, lymphocele |

**1. Access and balloon-related**

- **Balloon in muscle** (wrong plane, intramuscular) gives a postoperative **flank hernia** or bulge.
- **Too anterior** gives peritoneal breach or colon injury; **too posterior** gives bleeding from quadratus lumborum or psoas.
- **Balloon rupture:** latex glove balloon breaks into many fragments (silicone leaves one) that must be retrieved.
- **Peritoneal tear:** $CO_2$ enters the peritoneum and collapses the small working space. Fix by venting with a Veress needle or an extra port, enlarging the tear, or converting to transperitoneal.
- Disorientation: keep the **psoas horizontal** as the landmark.

**2. Gas-related**

- **Subcutaneous emphysema** (commonest; commoner than transperitoneal): crepitus over flank, chest and neck; pneumoscrotum. Cause: leak around the port. Prevent with a **balloon-tip (blunt-tip) trocar**, tight incisions and 12 rather than 15 mmHg.
- **Hypercarbia**, especially with emphysema; a sealed port removes the difference from transperitoneal surgery.
- **Pneumothorax (1.6–4%, commoner retroperitoneally):** pleura is near the upper pole and 12th rib. Signs: neck and chest emphysema; tension gives hypotension, absent breath sounds and high airway pressure. Treat with needle decompression (16-G needle in the second or third intercostal space, midclavicular line), then chest drain. Pneumomediastinum and pneumopericardium (about 0.8%) also occur.
- **Gas embolism** (renal vein injury, IVC or lumbar vein): managed as above.

**3. Vascular injury**

- Commonest complication of minimally invasive urology; major injury <1 in 1000 (P4: 0.11–2% at access) but fatal in 8–17%. Upper-tract work carries about five times the pelvic vascular risk.
- Sites: renal artery and vein, **lumbar and gonadal veins**, adrenal vein, IVC, aorta. In the small posterior space the IVC or aorta can be mistaken for hilar vessels.
- **Management of venous bleeding:** raise pressure temporarily to 25 mmHg, suction to see the site, compress or tamponade with a sponge, haemostatic sealant, extra 5-mm port for suturing or a Satinsky clamp, hand-assist, add ports, suture or clip, or convert if uncontrolled. Arterial: clip or staple; aortic injury needs conversion. Never clip blindly.

**4. Visceral injury**

- **Peritoneum and colon**; duodenum (right); spleen, pancreas (left); liver; pleura; ureter; adrenal; usually by thermal spread.
- Thermal bowel injury follows the four mechanisms above; it may present up to 18 days later; diagnose by CT with oral contrast.

**5. Positioning and nerve injury**

- **Rhabdomyolysis** (about 1% retroperitoneal; 0.4% overall): men, BMI (body mass index) ≥25, kidney rest, full flexion, long cases. Brown urine, hip pain, CK (creatine kinase) >5,000

units. Prevent: limit kidney rest to <1 hour, cases <5 hours, avoid hypotension.

- **Nerves** (neuropraxia): brachial plexus (axillary roll), sciatic; in the field, **genitofemoral and lateral femoral cutaneous** on the psoas, ilioinguinal and iliohypogastric (thigh numbness, flank bulge).

**6. Postoperative**

- Bleeding or haematoma; port-site hernia or haematoma; wound infection.
- **Urinoma** (partial nephrectomy, pyeloplasty); **lymphocele** or chylous leak after hilar or node dissection.
- Delayed bowel injury; DVT.
- **Conversion** to transperitoneal or open surgery for bleeding, peritoneal tear or lost orientation.

**Retroperitoneal versus transperitoneal**

| Retroperitoneal more | Retroperitoneal less |
|---|---|
| Emphysema, pneumothorax, hypercarbia, disorientation | Bowel injury, ileus, adhesions; peritoneum spared |

**Prevention**

- Open (Hasson) access below the 12th rib tip; balloon wholly in the retroperitoneum; balloon-tip port; 12 mmHg; trocars under vision; respect the 12th rib and pleura; check haemostasis at low pressure.

## Laparoscopic donor nephrectomy

**Introduction**

- First described by **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first** at every stage.

**Donor evaluation** `Questionnaire → physical and psychosocial review, blood group, HLA (human leukocyte antigen) typing → GFR by at least two methods, one measured → metabolic, serology, cancer screening → CT angiography for vessels → cardiac review if needed`

- Exclusions: kidney disease, malignancy, active infection, cardiovascular disease, diabetes, significant obesity; bilateral stones generally exclude.

**Side selection**

- **Left preferred:** longer renal vein, simpler dissection. **Right** if the left has multiple arteries or complex veins, or the right kidney is smaller: **the donor keeps the better kidney.**
- **Anatomy matters:** arterial variation in 25–40%; early (prehilar) branching and accessory arteries must be seen on CT, as they may contraindicate the laparoscopic route.

**Techniques**

| Technique | Features |
|---|---|
| Pure transperitoneal | 3–4 ports; colon reflected; Pfannenstiel extraction |
| **Hand-assisted** (about two-thirds) | Hand port doubles as extraction site; tactile feedback, quick haemostasis |
| Retroperitoneoscopic | No bowel mobilisation; direct hilum; less room |

| Technique | Features |
|---|---|
| Robotic; LESS (laparoendoscopic single-site surgery) | Wristed instruments; single incision, less pain |

**Operative steps (left)**

1. Reflect colon; divide **gonadal, adrenal and lumbar veins**.
2. Preserve periureteric tissue and the **gonadal–ureter triangle** (ureteric blood supply).
3. Keep pressure ≤10 mmHg; hydration and mannitol or furosemide to keep urine >100 mL/hour (counters pneumoperitoneum oliguria).
4. Divide ureter, then **artery first, then vein**. Vein stapled lateral to the adrenal vein; Endo-TA (non-cutting) stapler gives extra vein length.
5. Extract quickly; cold flush. Warm ischaemia a few minutes. - **Hem-o-lok:** manufacturer contraindicates it on the donor renal artery after stump bleeds. - **Open donor nephrectomy** is kept for complex anatomy or previous surgery, more often on the right.

**Advantages versus open**

- Less pain, shorter stay, faster return to work, better cosmesis; **more people volunteer to donate**; equal graft function.

**Disadvantages and complications**

- Longer operation, learning curve, shorter vessels (right vein), slightly longer warm ischaemia.
- Complications about **16%** (mostly gastrointestinal); Clavien ≥IV 2.5%; **mortality 3 per 10,000**; conversion 0.26–3.34%, mainly for vascular injury.

**Long-term donor counselling**

- Function 60–80% of pre-donation; lifetime ESRD (end-stage renal disease) **90 per 10,000** versus 14 in matched non-donors (Muzaale, 2014); more gestational hypertension.
- **Also know:** kidney paired donation for ABO-incompatible or crossmatch-positive pairs; living-donor grafts survive longer than deceased-donor grafts.

## Diagrams to draw

- Abdomen under pressure with arrows to diaphragm (lungs), IVC (preload), aorta (afterload) and renal vein (oliguria), plus $CO_2$ absorption to blood.
- Abdominal wall showing capacitive coupling through a metal cannula in a plastic anchor.
- Flank cross-section showing balloon between psoas and Gerota's fascia, with peritoneum pushed medially and port positions for retroperitoneoscopy.
- Left renal hilum with gonadal, adrenal and lumbar veins, stapled artery and vein, and the gonadal–ureter triangle.

**Key points to remember:**

- Two drivers: pressure and absorbed $CO_2$; work at 10–12 mmHg.
- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.
- Vascular injury is commonest; leave the trocar in place.
- Gas embolism: left lateral, head down, oxygen, aspirate.
- Missed bowel injury: trocar-site pain, leucopenia.
- Retroperitoneoscopy: space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it.
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

## LAQ 4 Tissue engineering in urology: principles, status and recent advances

P1 · P3 · P4 **merged from 3**

*Asked as:* P1 LAQ "Principles of tissue engineering and its use in urology" · P3 SAQ "Describe principles of tissue engineering and what its status in managemet of urological diseases is." · P4 LAQ "Tissue engineering in urology, including recent advances"

### Definition and rationale

- **Tissue engineering** applies engineering and life sciences to build biological substitutes that restore, maintain or improve tissue function (Langer and Vacanti, 1993).
- **Regenerative medicine** is the wider term: cells, scaffolds, growth factors and gene therapy.
- **Why in urology:** bowel used for bladder and diversion causes mucus, stones, metabolic acidosis, infection and malignancy; grafts for long urethral defects are scarce. The aim is **to replace "wrong tissue" with "right tissue"**.

### Principles: the triad plus bioreactor and blood supply

``` Cells + Scaffold + Signals (growth factors, mechanical cycling in a bioreactor) -> construct -> implanted -> vascularised and remodelled by the host

Biopsy -> cells expanded in culture -> seeded on scaffold (+ signals) -> conditioned in bioreactor -> implanted -> host vascularises and remodels ```

### Cell sources

| Source | Features, advantages and limitations |
|---|---|
| **Autologous differentiated cells** (urothelial and smooth muscle cells from a bladder biopsy) | Standard; no rejection; expand well in culture; unusable if the bladder is malignant or diseased |
| **Adult stem cells:** bone marrow or adipose mesenchymal, muscle-derived, **urine-derived** | Easy, non-invasive harvest; act mainly by paracrine (growth factor) effects |
| **Embryonic stem cells** | Pluripotent; ethical issues, teratoma risk |
| **iPSCs** (induced pluripotent stem cells) | Adult cells reprogrammed; patient-specific, no ethical issue; tumour risk |

- Gene-edited and gene-modified cells are being explored. Cell sheet engineering builds layers without a scaffold.

### Scaffolds (biomaterials)

- **Requirements:** biocompatible; biodegradable at the rate new tissue forms; porous for cell entry, attachment and vessel ingrowth; mechanically suited (a compliant bladder, a non-collapsing tube).

| Type | Examples | Feature |
|---|---|---|
| **Natural polymers** | Collagen, alginate, silk fibroin | Good cell recognition; weak |
| **Acellular tissue matrices** | Bladder acellular matrix, SIS (small intestinal submucosa), decellularised dermis | Keep native architecture and stored growth factors |
```

| Type | Examples | Feature |
|---|---|---|
| **Synthetic polymers** | PGA (polyglycolic acid), PLA (polylactic acid), PLGA (their copolymer, poly(lactic-co-glycolic) acid) | Controlled shape and degradation; no bioactivity |
| **Newer** | Electrospun nanofibres, hydrogels, bio-ink; **"smart" scaffolds** releasing growth factors such as VEGF (vascular endothelial growth factor); silk fibroin | Mimic extracellular matrix |

- **Seeding:** urothelial cells on the luminal side and smooth muscle cells on the outer side, copying native layers.
- **Unseeded (acellular)** matrices rely on host cells migrating in, so they work only for short onlay defects (about 0.5 cm for tubes in animal work). Longer or tubular gaps need **seeded** scaffolds, or they fibrose and contract.

### Signals

- Growth factors (for example VEGF, for vessels) released from the scaffold, extracellular-matrix cues and mechanical stimuli direct cell growth and differentiation.

### Bioreactors

- Culture chambers that perfuse nutrients and oxygen through thick constructs and apply **physiological conditioning**, such as cyclic stretch and filling to train bladder smooth muscle, so the tissue is functional before implantation.
- Types: static culture, spinner flask, rotating wall, perfusion and mechanical-stretch bioreactors. They also allow monitoring and sterile, standardised production.

### Vascularisation and innervation (the main barrier)

- Oxygen diffuses only about **100–200 µm**, so thick constructs die centrally without vessels.
- Strategies: **omental wrap** (used by Atala), angiogenic factors, co-seeding endothelial cells, built-in (printed) vascular channels by bioprinting.
- Innervation and coordinated contraction cannot yet be recreated.

### Clinical applications and status in urology

**1. Bladder**

- **Atala et al. (2006):** first seeded biodegradable bladder construct (autologous urothelial and muscle cells on collagen–PGA, wrapped in omentum) for augmentation in **7 myelomeningocele patients**; compliance improved.
- **Industry (Tengion) multi-institution phase II trial** in spina bifida: **no clinically or statistically significant improvement in compliance or capacity** (Chapter 48).
- Chapter 48 conclusion: a bladder with specialised urothelium and contractile muscle has **not** been achieved; it remains experimental.
- Chapter 111: an engineered bladder gives **storage only**; innervation for voluntary emptying cannot yet be recreated.

**2. Urinary diversion: Neo-Urinary Conduit (Tengion, NCT01087687)** (Chapter 135)

- Seeded biodegradable conduit to avoid bowel. Small enrolment; conduits survived about 250 days on average; **half developed stenosis or stricture.**
- Lessons: stiff, non-collapsible material; no cells formed around the connective-tissue sleeve; **no blood supply**.

### 3. Urethra

- Acellular bladder matrix onlay for hypospadias and short strictures: success in short onlay repairs.
- **Raya-Rivera et al. (2011, Lancet):** tubular PGA–PLGA scaffolds seeded with autologous cells in **5 boys** with posterior urethral defects; functional urethras at up to 6 years.
- Tissue-engineered oral mucosa (MukoCell) for urethroplasty, avoiding donor-site morbidity.

### 4. Other organs

- **Sphincter, detrusor:** autologous muscle-derived cell injection for SUI (stress urinary incontinence), investigational; for underactive detrusor, open-label studies improved voiding efficiency (Chapter 111). Biodegradable scaffold slings (Chapter 48).
- **Vagina:** engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59).
- **Penis:** engineered corporal tissue and tunica grafts (stem cell-seeded SIS in Peyronie disease), animal or experimental stage. ED (erectile dysfunction) stem cell therapy is investigational (Chapter 72).
- **Kidney:** decellularised–recellularised scaffolds (animal); renal assist devices; kidney organoids.
- **Testis:** spermatogonial stem cell banking for prepubertal cancer patients (future, Chapter 4).

| Organ | Status |
|---|---|
| **Bladder** | **Atala (2006):** seeded collagen–PGA construct with omental wrap in **7 myelomeningocele patients**, compliance improved. Industry phase II trial: **no significant gain in compliance or capacity**. Still experimental (Chapter 48) |
| **Conduit** | Neo-Urinary Conduit trial (NCT01087687): survived about 250 days on average; **half developed stenosis or stricture**; stiff material, no blood supply (Chapter 135) |
| **Urethra** | Acellular matrix onlay for short strictures and hypospadias; **Raya-Rivera (2011):** seeded tubular PGA–PLGA urethras in **5 boys**, functional up to 6 years; tissue-engineered oral mucosa avoids donor-site morbidity |
| **Sphincter, detrusor** | Autologous muscle-derived cells for SUI (investigational); biodegradable scaffold slings; underactive detrusor: open-label improvement in voiding efficiency (Chapter 111) |
| **Vagina** | Engineered vaginas in MRKH syndrome (Raya-Rivera, 2014); off-the-shelf SIS grafts (Chapter 59) |
| **Penis** | Corporal tissue and tunica grafts: animal or experimental; stem cells for ED investigational |
| **Kidney** | Decellularised–recellularised scaffolds (animal), kidney organoids, renal assist devices: experimental |
| **Testis** | Spermatogonial stem cell banking in prepubertal cancer patients |

- **Overall status:** still experimental except short urethral onlays.

## Recent advances

### 3D bioprinting

- Layer-by-layer deposition of **"bio-ink"** (cells in hydrogel), with or without polymer, driven by patient CT (computed tomography) or MRI (magnetic resonance imaging) data.

- Advantages: exact patient-specific shape, several cell types placed precisely, built-in channels for vessels.
- Current urological use: 3D-printed (non-living) kidney and prostate models already aid planning and teaching; bioprinted urethra and bladder constructs are still at the animal stage.

**Organoids**

- 3D self-organising mini-organs grown from stem cells or tumour tissue.
- **Kidney organoids** from iPSCs: model development and disease, drug toxicity testing.
- **Tumour organoids** (bladder, prostate): personalised drug testing and biology (WNT/β-catenin work, Chapter 129).

**Whole-organ decellularisation**

- Kidneys are stripped of cells by detergent perfusion, leaving the matrix and vascular tree, then recellularised with the patient's cells; still at the animal stage. Renal assist devices are a parallel route.

**Other**

- Robotic bladder auto-transplantation and robotic retrieval of a **vascularised composite bladder allograft** (Chapter 135): a transplant, not engineered, route to "right tissue" that avoids bowel but needs immunosuppression.
- Smart (growth factor-releasing) scaffolds; urine-derived stem cells; gene-edited and gene-modified cells; cell sheet engineering.

## Challenges

- **Vascularisation** of thick constructs (diffusion limit), **innervation** and contractility.
- Fibrosis and contracture; scale-up, cost, culture time; regulatory approval.
- Stem cell tumour risk. Promising pilots followed by disappointing controlled trials.

## Future

- Vascularised bioprinted tissue, off-the-shelf cell-free smart scaffolds, combined gene and cell therapy.

## Diagram to draw

- Biopsy, then cell culture, then seeding on scaffold, then bioreactor, then implant with omental wrap.

> **Key points to remember:**
> - Triad: cells, scaffold, signals; plus bioreactor and blood supply.
> - Diffusion limit about 100–200 µm: vascularity is the main barrier.
> - Atala 2006 (7 patients) promising; phase II trial: no gain in compliance or capacity.
> - Neo-urinary conduit: half stenosed; no blood supply.
> - Raya-Rivera 2011: seeded tubular urethras in 5 boys.
> - Still experimental except short urethral onlays.
> - New: iPSCs, urine-derived cells, bioprinting, organoids.

## LAQ 5 Radionuclide (isotope) renal scans: MAG3 and DMSA

P1 · P2 **merged from 2**

*Asked as:* P1 LAQ "Radionuclide (isotope) scans in urology: principles, renal scans, MAG3 and applications" · P2 SAQ "Renal isotope scans and DMSA renal scan"

*"Uro" is read as urology. "MAG 3" is taken to mean the Tc-99m MAG3 (mercaptoacetyltriglycine) renal scan: principle, technique, interpretation and uses.*

### Principles

- Nuclear medicine shows **function, not structure**. A radiotracer, given in a tiny (subpharmacological) dose, follows a physiological or molecular process; the patient is the radiation source.
- **Gamma camera** (sodium iodide crystal, photomultipliers, collimator): photon, collimator (admits photons from one direction), sodium iodide crystal (light), photomultiplier tubes, image. Planar (about 20 mm resolution) or **SPECT (single-photon emission computed tomography)**, 3D, with CT (about 10 mm).
- **PET (positron emission tomography):** the positron annihilates with an electron, giving two 511 keV photons 180° apart, detected in coincidence. This rejects noise and allows true quantitation (SUV, standardised uptake value); sharper (about 5 mm).

| | Planar | SPECT | PET |
|---|---|---|---|
| Image | 2D | 3D, with CT (computed tomography) | 3D, with CT or MRI (magnetic resonance imaging) |
| Resolution | About 20 mm | About 10 mm | About 5 mm |
| Quantitation | Semiquantitative | Semiquantitative | SUV |
| Main use | Renal function | Problem-solving | Cancer |

- **$^{99m}$Tc (technetium-99m):** 6-hour half-life, single 140 keV photon: the workhorse. **$^{18}$F:** about 110 minutes, lowest positron energy, so the sharpest PET image. $^{68}$Ga comes from a generator; $^{11}$C (20 minutes) needs an on-site cyclotron.

### Applications in urology (overview)

| Area | Tracer | Use |
|---|---|---|
| **Kidney** | $^{99m}$Tc-DTPA (diethylenetriaminepentaacetic acid), -MAG3, -DMSA (dimercaptosuccinic acid) | GFR (glomerular filtration rate), split function, obstruction, scars, transplant |
| **Bone** | $^{99m}$Tc-MDP (methylene diphosphonate) | High-risk prostate cancer staging; low specificity |
| **Prostate** | **PSMA (prostate-specific membrane antigen) PET**; choline; fluciclovine | PSMA preferred (NCCN [National Comprehensive Cancer Network]) for staging and recurrence; specificity >90% |
| **Therapy** | $^{177}$Lu-PSMA-617; radium-223 | mCRPC (metastatic castration-resistant prostate cancer); VISION trial (2021) |

| Area | Tracer | Use |
|---|---|---|
| **FDG (fluorodeoxyglucose) PET** | Glucose analogue trapped in cells | **Residual seminoma >3 cm**; penile cancer with palpable nodes; distant urothelial disease. Poor in prostate and clear cell RCC (renal cell carcinoma) |
| **Renal mass** | Sestamibi SPECT; girentuximab PET | Oncocytoma "hot" on sestamibi; clear cell RCC "hot" on girentuximab |
| **Adrenal** | $^{123}$I-MIBG (metaiodobenzylguanidine), $^{68}$Ga-DOTATATE, FDG | Metastatic or extra-adrenal phaeochromocytoma; MIBG best in MEN-2 (multiple endocrine neoplasia type 2) |
| **Sentinel node** | $^{99m}$Tc-nanocolloid plus blue dye | Penile cancer, impalpable nodes |
| **Scrotum** | $^{99m}$Tc-pertechnetate | Torsion cold, epididymitis hot |
| **Reflux** | Direct radionuclide cystography | VUR (vesicoureteral reflux) follow-up: more sensitive, lower dose than VCUG (voiding cystourethrogram) |
| **Infection** | $^{111}$In-labelled leucocytes; $^{67}$Ga | Occult abscess; not in transplant |

## Renal tracers

- Each tracer images the part of the nephron that handles it. Plasma clearance is about 20% filtration and 80% tubular secretion.

| Feature | $^{99m}$Tc-DTPA | $^{99m}$Tc-MAG3 | $^{99m}$Tc-DMSA |
|---|---|---|---|
| Handling | Glomerular filtration only | Protein-bound; tubular secretion, high extraction | Filtered, then retained in proximal tubules (megalin, cubilin); tubular cortical binding |
| Study | Dynamic | Dynamic | **Static cortical**, at 2 hours (P2: 2–4 hours) |
| Main use | **GFR**; captopril study; renovascular | **Split function, obstruction (first choice)**, drainage; transplant | **Pyelonephritis and scars** |
| Renal failure | Poor images in renal failure | Good images in renal failure | Delay imaging |
| Dose | 5–10 mCi | 5–10 mCi | 5 mCi adult; 50 µCi/kg child |

## MAG3: principle

- **$^{99m}$Tc-MAG3:** protein-bound tracer cleared mainly by **proximal tubular secretion**. Extraction is nearly **60%** per pass, versus about 20% for DTPA.
- So it gives **good images even with poor GFR**, lower background and lower cost. **Agent of choice for split function and obstruction.**

## Dynamic and diuretic renography (MAG3): method and interpretation

- **Preparation:** good hydration; bladder catheter for outlet obstruction or neurogenic bladder, and in children; cap a nephrostomy. Dose 185–370 MBq (5–10 mCi) intravenous bolus in adults.

- **Perfusion phase** (first minute): frames every 1–2 seconds; the kidney appears within seconds of the aorta (renal curve follows aortic upstroke).
- **Function (uptake) phase:** regions of interest with background subtraction. **Split function (DRF, differential renal function) = area under each curve at 1–3 minutes**; **normal 45–55%** (45:55 to 55:45). Normal MAG3 peak at 3–5 minutes; bladder by 10–15 minutes.
- **Drainage phase:** a rising curve needs furosemide. Upright or post-void images test gravity drainage.
- **GFR:** DTPA camera method (Gates) or plasma-sample clearance.
- **Diuretic renography ("well-tempered" diuretic renogram):** dilatation is not obstruction. Furosemide 0.5 mg/kg (adult maximum 40 mg) in adults; 1 mg/kg in children, given when the system is full; separates a patulous system from obstruction. Timing varies (F+20 classic, F+15, F0, F−15), so the report must always state when it was given.

| T½ (half-clearance time) after furosemide | Meaning |
|---|---|
| <10 minutes | Not obstructed |
| 10–20 minutes | Indeterminate |
| >20 minutes | Obstructed |

- **Curve types (O'Reilly):** I normal; II rising (obstructed); IIIa dilated, not obstructed, washes out (falls) with diuretic; IIIb equivocal.

### MAG3 uses

- PUJ (pelviureteric junction) obstruction and megaureter: **surgery if DRF <40% with poor drainage or falling DRF**; follow-up after pyeloplasty.
- Transplant: acute rejection (perfusion falls) versus delayed graft function (perfusion kept); urine leak.
- **Also know:** indirect cystography for VUR; split function before nephrectomy.

### MAG3 pitfalls

- Poor function, dehydration, full bladder, wrong ROI (region of interest), mistimed diuretic, massive dilatation (capacious pelvis) and neonatal immaturity mimic obstruction. **Trend on serial scans** matters.

### DMSA renal scan

#### Principle

- **$^{99m}$Tc-DMSA** is filtered and then taken up by **proximal tubular cells** (megalin and cubilin receptors). It is retained in the cortex and excreted very slowly.
- So it gives a **static map of functioning cortex**, not drainage.

#### Technique

- Intravenous dose: **50 µCi/kg in children, 5 mCi (185 MBq) in adults**.
- Images at **2–4 hours** (P1: 2 hours; later in renal failure): posterior and oblique views, **pinhole collimator** or **SPECT**, which is slightly more sensitive.
- Radiation about 1 mSv.

#### What it measures

- **DRF:** from geometric-mean counts; normal 45–55%. Better than MAG3 for ectopic, horseshoe or duplex kidneys and very poor function.

- **Cortical defects:** photopenic areas.

**Indications**

- **Acute pyelonephritis** (gold standard), especially in infants with febrile UTI (urinary tract infection).
- **Renal scarring** in VUR and recurrent UTI; top-down approach (DMSA first, VCUG only if abnormal).
- Function of a duplex upper moiety, ectopic or horseshoe kidney; MCDK (multicystic dysplastic kidney: no uptake).
- Deciding whether to reconstruct or remove a poorly functioning kidney.

**Acute pyelonephritis versus scar**

| | Acute pyelonephritis | Scar |
|---|---|---|
| Timing | Within 1 week of fever (sensitivity falls by 1 month) | **≥6 months** after infection |
| Defect | Ill-defined, may be multiple | Sharp, with cortical thinning |
| Kidney size | Normal or swollen | Small or contracted (often small) |
| Repeat scan (at about 6 months) | Resolves (about 85%) | Persists |

**Goldraich scar grading**

| Type | Finding |
|---|---|
| 1 | No more than 2 scarred areas |
| 2 | More than 2 scars, normal tissue between |
| 3 | Generalised damage, small kidney |
| 4 | End-stage, shrunken kidney, little or no uptake |

**Limitations**

- Cannot tell **congenital reflux dysplasia** (boys, grade IV–V VUR, no UTI) from acquired scar.
- Gross hydronephrosis gives false defects; no information on drainage; poor uptake in renal failure and proximal tubular disorders.
- **Also know:** MAG3 is best for obstruction; NICE (National Institute for Health and Care Excellence): late DMSA 4–6 months after atypical or recurrent UTI.

## Other renal applications (special studies)

- **Captopril renography (DTPA):** the stenotic kidney keeps GFR by angiotensin II efferent constriction. Captopril removes this support, so uptake falls. Repeat off ACE (angiotensin-converting enzyme) inhibitors: improvement **≥10%** off ACE inhibitors means renovascular hypertension is likely. Largely replaced by CT angiography.
- **Transplant (MAG3):** acute rejection: perfusion falls and worsens; delayed graft function: perfusion kept, improves. Shows urine leak without iodinated contrast.
- **Radionuclide (nuclear) cystography:** direct (tracer instilled into the bladder) or indirect (after MAG3). More sensitive and lower radiation dose than VCUG, but little anatomy. So VCUG first; nuclear cystography for follow-up of reflux.

### Infection imaging

- **$^{111}$In-oxine leucocytes** (image at 24 hours): best for occult abdominal or genitourinary infection. $^{99m}$Tc-HMPAO (hexamethylpropyleneamine oxime) leucocytes give better images, but urinary excretion obscures the tract. Neither in a transplant (leucocytes home to the graft).
- **Also know:** $^{67}$Ga-citrate (non-specific, slow).

### Oncology

- **FDG ($^{18}$F-fluorodeoxyglucose):** a glucose analogue phosphorylated and trapped (Warburg effect). Urinary excretion limits it to distant disease.
- **Seminoma residual mass >3 cm** after chemotherapy (EAU [European Association of Urology]). Not for NSGCT (non-seminomatous germ cell tumour), because teratoma is not FDG-avid.
- Penile cancer with palpable nodes; equivocal bladder or upper-tract metastases.
- Poor in prostate cancer and clear cell RCC.
- **Prostate: PSMA PET** is preferred (NCCN) for high-risk staging and biochemical recurrence. Specificity >90%; nodal sensitivity only 30–40%. Calais trial (2019): detection 56% (PSMA) versus 26% (fluciclovine). Bone scan ($^{99m}$Tc-MDP) remains conventional.
- **Renal mass: sestamibi SPECT** "hot" in oncocytoma; **girentuximab** (anti-CAIX, carbonic anhydrase IX) PET "hot" in clear cell RCC (ZIRCON trial).
- **Also know:** $^{123}$I-MIBG and $^{68}$Ga-DOTATATE for phaeochromocytoma; $^{99m}$Tc-nanocolloid sentinel node biopsy in penile cancer; $^{99m}$Tc-pertechnetate scrotal scan (torsion cold, epididymitis hot) when ultrasound is equivocal.

### Therapy (theranostics)

- **$^{177}$Lu-PSMA-617:** beta emitter for mCRPC after ARPI (androgen receptor pathway inhibitor) and taxane: **VISION (2021)**. PSMA PET confirms the target first.
- **Radium-223:** alpha emitter for bone-only mCRPC.

### Advantages and limitations

- Quantitative function, low radiation, safe in contrast allergy and pregnancy (MAG3 fetal dose <5 mGy).
- Poor anatomy; PET is costly; results depend on technique.

### Diagrams to draw

- MAG3 renogram: perfusion spike, peak at 3–5 minutes, washout, 1–3 minute split window; renogram curve types I (normal), II (obstructed, rising), IIIa (dilated non-obstructed, falls after furosemide) and IIIb.
- Gamma camera chain: patient, collimator, crystal, photomultiplier, image.
- Posterior DMSA image: normal kidney and a scarred kidney with polar defects.

**Key points to remember:**

- $^{99m}$Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered = GFR; MAG3 secreted = obstruction and transplant; DMSA cortical = scars.
- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- Split function (DRF) from 1–3 minute counts; normal 45–55%; furosemide 0.5 mg/kg.
- T½ >20 minutes obstructed, <10 minutes not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.
- DMSA binds proximal tubules: a static cortical scan at 2–4 h; gold standard for acute pyelonephritis and scars.
- Scar only if the defect persists ≥6 months; pyelonephritis resolves.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with $^{99m}$Tc-nanocolloid.
- Sestamibi hot = oncocytoma; $^{177}$Lu-PSMA-617 (VISION).

# LAQ 6 Energy sources in urology and laparoscopic surgery, including bipolar energy

P2 · P3 · P4 **merged from 3**

*Asked as:* P3 LAQ "Therapeutic energy sources in urology (including laparoscopic energy devices)" · P4 LAQ "Basic energy modalities in urological surgery, including bipolar energy" · P2 SAQ "Energy sources in laparoscopic urology"

## Introduction and principles

- Therapeutic energy is absorbed by tissue or stone and released as **heat, mechanical stress, cavitation or a chemical reaction**. Wavelength or frequency sets penetration; absorption sets the effect.
- All haemostatic devices achieve haemostasis (seal) by **denaturing protein**, by heat (current, light) or mechanical vibration.
- Heat effects: about **60 °C** coagulates protein; **100 °C** vaporises cell water; higher temperatures carbonise.

## Classification

| Group | Modalities |
|---|---|
| Electrical | Monopolar and bipolar electrosurgery, vessel sealers, RFA (radiofrequency ablation), IRE (irreversible electroporation), EMDA (electromotive drug administration) |
| Electromagnetic | Lasers, MWA (microwave ablation), chemohyperthermia, PDT (photodynamic therapy), ionising radiation |
| Acoustic | Ultrasonic dissector, SWL (shock wave lithotripsy), intracorporeal ultrasonic and pneumatic lithotripsy, HIFU (high-intensity focused ultrasound), histotripsy, burst wave lithotripsy, Li-ESWT (low-intensity shock wave therapy) |
| Direct contact | Cryoablation, water vapour (Rezum), waterjet (Aquablation) |

## 1. Electrosurgery: physics

- Alternating current at about **500 kHz** (radiofrequency). Mains current (50 Hz) depolarises nerve and muscle; at 500 kHz ions only oscillate and produce heat, too fast to stimulate nerve or muscle.
- **Ohm's law** ($V = I \times R$): tissue is the resistance. **Current density** decides the effect: a small tip concentrates current and heats (burns); a large pad disperses it.
- **Monopolar:** generator, active electrode, patient, dispersive pad, back to generator. Partial pad contact causes burns.
- **Bipolar:** current passes only between two electrodes (jaws) on the same instrument; less power, less spread, no dispersive pad (the patient is not in the circuit).
- **Electrocautery** (heated wire) differs: no current enters the patient.

| Waveform | Character | Effect |
|---|---|---|
| **Cut** (50–80 W) | Continuous, low voltage | Cell water flash-boils; rapid boiling, clean incision, little char |
| **Coagulation** (30–50 W) | Interrupted (about 6% duty cycle), high voltage | Desiccation, deeper spread (up to 3 mm) |

| Waveform | Character | Effect |
|---|---|---|
| Fulguration (spray) | Higher voltage, arcs (sparks) across air | Non-contact surface haemostasis; surface charring of oozing beds |
| Blend | Cut with interruptions | Cutting plus haemostasis |

## Bipolar energy in detail

- Radiofrequency current (about 500 kHz). Less power and less lateral thermal spread (1–6 mm vs 5–7 mm monopolar). In endoscopy, **saline** irrigation can be used.

| | Monopolar | Bipolar |
|---|---|---|
| Circuit | Loop to patient to pad | Active and return poles on the instrument |
| Irrigant | Glycine 1.5%, sorbitol–mannitol | Normal saline |
| TUR (transurethral resection) syndrome | Yes | Theoretically eliminated |
| Capacitive coupling, pad burns | Yes | No |

- **Endoscopic uses:**
- **Bipolar TURP (transurethral resection of prostate):**
  - *Plasma system:* energy vaporises saline around the loop into plasma (orange glow of sodium); cuts at lower temperature while sealing vessels; platinum–iridium loop.
  - *TURis (transurethral resection in saline):* the sheath acts as return electrode; chapter notes report strictures and fistulae.
  - *True bipolar:* dual loops.
  - Outcomes: equal IPSS (International Prostate Symptom Score) and Qmax; less transfusion and clot retention; shorter catheter time and stay. NNT (number needed to treat) about 50 to prevent one TUR syndrome.
- **Bipolar TUVP (transurethral vaporisation):** PlasmaButton; AUA (American Urological Association) option for <30 and 30–80 cc.
- **Bipolar enucleation:** plasmakinetic energy can perform AEEP (anatomic endoscopic enucleation of prostate) satisfactorily; comparative data limited.
- **TURBT (transurethral resection of bladder tumour):** fewer obturator jerks and less bleeding in reviews; one RCT (randomised controlled trial) showed only less thermal artefact.
- **Laparoscopic and open uses:** bipolar forceps for pinpoint haemostasis; vessel sealers; prostatic pedicles, cystectomy.
- **Bipolar safety:** safer near pacemakers; touching tines complete the circuit without tissue. Obturator reflex reduced, not abolished.

## Electrosurgery in endoscopic urology: present status

- **TURP:** still the reference BPH (benign prostatic hyperplasia) operation. Monopolar TURP needs non-conducting **1.5% glycine**, risking hyponatraemia (TUR syndrome); **bipolar** in saline matches monopolar efficacy with better safety and avoids TUR syndrome.
- **TURBT:** standard; obturator reflex risk on lateral wall. En bloc resection is increasing.

- **Obturator reflex:** lateral-wall TURBT stimulates the nerve, causing adduction and perforation. Prevent with paralysis, obturator block, partial filling, short bursts, bipolar.
- **EMDA–MMC (mitomycin C):** iontophoresis raises drug uptake in NMIBC (non-muscle-invasive bladder cancer); evidence low quality, not standard.

### Laparoscopic energy devices

| Device | Principle | Lateral spread | Notes |
|---|---|---|---|
| **Monopolar** (hook, scissors) | Tip to patient to return pad | **5–7 mm** | Versatile; stray-current risk |
| **Bipolar** | Current only between jaws | 1–6 mm | No pad; no capacitive coupling |
| **Vessel sealer** (LigaSure, EnSeal, Caiman) | Bipolar current + jaw compression fuses collagen and elastin; impedance feedback stops output; built-in blade | 1–3 mm | Seals vessels **≤7 mm** in one application only |
| **Ultrasonic** (Harmonic) | Blade vibrating at **25–55 kHz** (P4: about 55.5 kHz); friction and cavitation denature protein; no current through the patient | **≈1 mm** | Safe with pacemakers; jaws **>200 °C**, slow to cool, so do not touch bowel |
| **Argon beam** | Monopolar current through argon, non-contact | Superficial | Liver, spleen, raw surfaces; **gas embolism** |

- Vessel sealers are routine in laparoscopic and robotic surgery.
- **Also know:** lasers ($CO_2$, KTP [potassium titanyl phosphate], holmium) now rarely used laparoscopically; laparoscopic RFA (target 60 °C) and cryoablation (below −40 °C, two freeze–thaw cycles) for small renal masses; robotic HotShears, bipolar Maryland, Vessel Sealer.

### Electrosurgical injury and safety

| Mechanism | Prevention |
|---|---|
| Direct (inadvertent) activation out of view | Activate only with tip in view |
| **Direct coupling**: active electrode touches another metal instrument | Whole metal tip visible |
| **Capacitive coupling**: current induced through intact insulation (metal cannula in plastic retainer) | Never mix metal and plastic ports (avoid hybrid trocars) |
| **Insulation failure** | Inspect insulation; **AEM (active electrode monitoring)** |
| **Thermal spread** to ureter, nerves or bowel | Bipolar near vital structures |
| Return pad burns; surgical fire (with oxygen) | Pad on muscle, away from implants; **REM (return electrode monitoring)** detects poor pad contact |

- **AEM** detects insulation failure and capacitive coupling.
- Unseen bowel injuries present late (days) as perforation or peritonitis.
- Use lowest effective power (cut 50–80 W, coagulation 30–50 W), tip in view; bipolar near vital structures and in pacemaker patients.

## 2. Ultrasonic energy

- **Ultrasonic dissector:** see the device table; little lateral spread, but the blade stays hot.
- **Intracorporeal ultrasonic lithotripter:** vibrating probe with suction of fragments.

## 3. Lasers

- **Principle:** pumped gain medium in a mirrored cavity gives stimulated emission and a **monochromatic, coherent, collimated** beam.
- **Effect** depends on wavelength (chromophore: water, haemoglobin) and pulse duration.
- **Mechanisms:** photothermal (long pulses), photomechanical (short pulses, shock waves), photochemical (PDT).
- **Penetration predicts behaviour:** deep (Nd:YAG) coagulates widely with delayed sloughing; shallow (thulium) vaporises precisely with char.

| Laser | nm | Penetration | Use and present status |
|---|---|---|---|
| **KTP/LBO** (potassium-titanyl-phosphate/lithium triborate; GreenLight) | 532 | 0.8 mm | Prostate photovaporisation (haemoglobin); can continue anticoagulation |
| Nd:YAG (neodymium:yttrium-aluminium-garnet) | 1064 | 10 mm | Deep coagulation (penile CIS [carcinoma in situ], warts); largely obsolete for BPH |
| **TFL** (thulium fibre laser) | 1940 | 0.08 mm | Stones: beats low-power holmium, equals high-power; less retropulsion; enucleation (ThuLEP, thulium laser enucleation of prostate) |
| Tm:YAG (thulium:YAG) | 2000 | 0.2 mm | Vaporesection, enucleation |
| **Ho:YAG** (holmium:YAG) | 2120 | 0.31 mm | Gold standard lithotripsy; **HoLEP** (holmium laser enucleation of prostate) size-independent; strictures, UTUC (upper tract urothelial carcinoma) ablation |

- Safety: eye protection for all; irrigant heating; fibre damage.

## 4. Acoustic energy and lithotripsy

- **SWL:** electrohydraulic, electromagnetic or piezoelectric sources. Shock waves break brittle stones by **spallation, shear and cavitation**. Slower rates (about 1 Hz) work better and injure less. **Status:** first line for renal stones ≤2 cm and favourable upper ureteric stones.
- **Intracorporeal:** pneumatic (ballistic), ultrasonic, combined, electrohydraulic, laser; pneumatic, ultrasonic and combined probes in PCNL (percutaneous nephrolithotomy); electrohydraulic now rare (ureteric injury).
- **HIFU:** focused beam (1500–10,000 W/cm$^2$) is absorbed and heats a rice-grain focus to necrosis; transrectal prostate. **Status:** focal or salvage prostate cancer (best for posterior tumours); investigational in guidelines.
- **Histotripsy** (cavitation liquefaction), **burst wave lithotripsy, ultrasonic propulsion:** investigational.
- **Li-ESWT:** for vasculogenic ED (erectile dysfunction) and Peyronie pain; evidence still limited.

## 5. Ablative energies (thermal ablation of tumours)

| Modality | Mechanism | Key points and present status |
|---|---|---|
| **Cryoablation** | Argon expansion (Joule–Thomson); **below −40 °C, two freeze–thaw cycles**; visible ice ball | Small renal mass ≤3–4 cm in frail patients; focal or salvage prostate |
| **RFA** | 375–900 kHz current; ionic agitation; target 60 °C; char at 150 °C | Small renal mass; oncological results similar to cryo; heat sink near vessels |
| **MWA** | 915 MHz or 2.45 GHz; dielectric hysteresis | Renal mass; hotter, larger zones, less heat sink |
| **HIFU** | Focused ultrasound absorbed, causing necrosis | See acoustic energy |
| **IRE** | High-voltage microsecond DC (direct current) pulses open permanent membrane pores; non-thermal | Spares vessels and ducts; focal prostate; good continence and potency; trials ongoing |

- **Cryo mechanism:** extracellular ice, then cell dehydration, then intracellular ice, then membrane rupture on thawing; ischaemia and apoptosis follow.
- **IRE mechanism:** electric pulses open pores in cell membranes that do not reseal, so cells die by apoptosis without heat; collagen scaffolds survive. It needs general anaesthesia with muscle paralysis and ECG (electrocardiogram)-synchronised pulses.
- **Guidelines:** AUA 2021: consider thermal ablation for **cT1a ≤3 cm**; biopsy before or at ablation. EAU (European Association of Urology): not routinely >3 cm (cryo >4 cm).

## 6. Light and chemical energy

- **PDT:** photosensitiser plus light gives singlet oxygen. **VTP (vascular-targeted photodynamic therapy, padeliporfin)** for low-risk prostate cancer: RCT (Azzouzi 2017) halved progression versus AS (active surveillance; 28% vs 58%); not widely adopted. **Photodynamic diagnosis** (blue light) aids TURBT.
- **Chemohyperthermia:** MMC with bladder wall heated to **42 °C** for NMIBC; used after BCG (bacillus Calmette-Guérin) failure or shortage; not standard. EMDA for mitomycin (see above).

## 7. Prostate (BPH): newer energies

- **Rezum:** steam (103 °C) releases latent heat by convection; no char, zone-confined; preserves ejaculation.
- **Aquablation:** robotic non-thermal saline waterjet; good for large glands; haemostasis needed.
- **Also know:** TUMT (transurethral microwave thermotherapy) and TUNA (transurethral needle ablation) largely abandoned.

## 8. Ionising radiation

- EBRT (external beam radiotherapy) and **brachytherapy** (LDR [low dose rate] or HDR [high dose rate]): standard for localised prostate cancer; trimodality bladder preservation; penile cancer (brachytherapy for small T1–T2 glans tumours).
- **Radioligand therapy** ($^{177}$Lu-PSMA [prostate-specific membrane antigen], VISION trial) and **radium-223** (ALSYMPCA) for mCRPC (metastatic castration-resistant prostate cancer).

### Diagrams to draw

- Monopolar vs bipolar circuit (monopolar loop-to-pad beside bipolar loop with local return), with cut and coagulation waveforms.
- Laser penetration depths compared, with cryoprobe ice ball.

**Key points to remember:**

- Electrosurgery at about 500 kHz; current density decides heating.
- Cut continuous low voltage; coagulation interrupted high voltage.
- Spread: monopolar 5–7 mm > bipolar 1–6 > sealer (LigaSure) 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz; ultrasonic blade hottest after use.
- Stray current: insulation failure, direct and capacitive coupling; avoid hybrid trocars; use AEM and REM.
- Bipolar: no pad, saline irrigation, no TUR syndrome; bipolar TURP equals monopolar efficacy with fewer complications.
- Holmium 2120 nm: gold standard for stones and HoLEP.
- Cryo below −40 °C, two cycles; RFA 60 °C; MWA less heat sink; thermal ablation for cT1a ≤3 cm.
- HIFU, IRE, VTP: focal prostate therapy, still evolving.

# LAQ 7 LASERs in urology: holmium, thulium, KTP and newer laser technologies

P4 **merged from 4**

*Asked as:* P4 LAQ "LASERs in urology: applications and newer generation laser technologies (new LASER for stone treatment)" · P4 SAQ "Holmium laser" · P4 SAQ "Thalium laser in urology" · P4 SAQ "KTP*** laser in urology"

*"Thalium" is taken to mean thulium (spelling error). "KTP***" is taken to mean the KTP (potassium titanyl phosphate) "GreenLight" laser; the asterisks are ignored.*

## Principle

- LASER (light amplification by stimulated emission of radiation): a "pumped" gain medium releases identical photons. A mirror resonator amplifies them, and a partial mirror lets the beam out.
- Light is **monochromatic, coherent and collimated**, so it passes down thin fibres.
- **Chromophore** decides the effect. Water absorbs holmium and thulium light. Haemoglobin absorbs KTP light.
- Absorption sets **penetration**. Shallow penetration vaporises and cuts. Deep penetration coagulates and later sloughs.
- Coagulation occurs at about 60 °C; vaporisation at ≥100 °C.
- **Pulse duration** decides the stone mechanism. Short pulses (<10 µs) act photomechanically (shock waves). Long pulses (>250 µs: Ho:YAG [holmium:yttrium-aluminium-garnet], TFL [thulium fibre laser]) act **photothermally**.

## Lasers used

| Laser | Wavelength (nm) | Chromophore | Penetration | Main uses |
|---|---|---|---|---|
| Ho:YAG | 2120 (Ch 19 quotes 2100; often quoted as 2100 nm) | Water | 0.3–0.4 mm (other sources: 0.4 mm; 0.31 mm) | Stones, HoLEP, incisions, tumours |
| TFL | 1940 | Water (peak) | <0.1 mm (other sources: about 0.2 mm; 0.08 mm; chapter values 0.08–0.2 mm) | Dusting, enucleation |
| Tm:YAG | ~2010 (other sources: 2013) | Water | 0.2 mm | Vaporesection, enucleation |
| KTP/LBO | 532 | Haemoglobin | 0.8 mm | PVP |
| Diode | 940–1470 | Water + haemoglobin | 0.5–5 mm | Prostate vaporisation |
| Nd:YAG | 1064 | Neither strongly | Up to 10 mm | Coagulation of tumours |
| $CO_2$ | 10,600 | Water | ~0.1 mm | Skin and penile lesions |

## Holmium:YAG laser

### Physics

- Solid-state, flashlamp-pumped, **pulsed** laser at **2120 nm** (often quoted as 2100 nm).

- Chromophore **water**; penetration about **0.4 mm**. It cuts and vaporises; heat spreads 2–3 mm and seals small vessels.
- Flexible quartz fibres; works in saline.

**Mechanism on stones**

- Long pulses (>250 µs) act **photothermally**, not by shock waves: long pulses heat the stone until it melts or decomposes.
- **Moses effect:** the start of each pulse makes a vapour channel; the rest of the pulse crosses it to the stone. So keep the fibre close and perpendicular.
- Breaks **all stone types**, including cystine and calcium oxalate monohydrate. It is the gold standard for stones and enucleation.

**Settings**

| Mode | Settings | Effect |
|---|---|---|
| Fragmenting | 0.6–1.0 J × 6–10 Hz | Pieces for basketing; more retropulsion |
| Dusting | 0.2–0.3 J × 40–80 Hz | Fine dust; "popcorning" in a calyx |
| HoLEP enucleation | 2.0 J × 40–50 Hz | Cutting along the capsule |

- **Retropulsion** rises with pulse energy (not frequency) and falls with longer pulses or pulse modulation.
- Heat tracks power: ≤10 W in the ureter; good outflow.

**Uses**

- **Stones:** URS (ureteroscopy), RIRS (retrograde intrarenal surgery), PCNL (percutaneous nephrolithotomy), bladder stones.
- **BPH (benign prostatic hyperplasia):** HoLEP (holmium laser enucleation of the prostate).
- **Incisions:** strictures, bladder neck contracture, endopyelotomy, ureterocele, calyceal diverticulum neck.
- **Tumours:** UTUC (upper tract urothelial carcinoma) ablation; en bloc bladder tumour resection.
- Drawback: steep HoLEP learning curve.

## Thulium lasers

**Types and physics**

| | Tm:YAG (thulium:yttrium-aluminium-garnet) | TFL (thulium fibre laser) |
|---|---|---|
| Wavelength | 2013 nm (P4 LAQ: ~2010 nm) | **1940 nm** (water absorption peak) |
| Source | Diode-pumped solid crystal | Diode-pumped thulium-doped silica fibre |
| Output | Continuous wave (now also pulsed) | Pulsed or continuous |
| Water absorption | High | About 4× holmium |

- Optical penetration is the shallowest of urological lasers: about 0.2 mm (chapter values for TFL: 0.08–0.2 mm), against 0.31–0.4 mm for Ho:YAG.
- Shallow absorption gives precise vaporisation and clean incision, but **carbonisation**.

**Thulium:YAG (2013 nm)**

- Originally continuous wave (0.2 mm penetration, carbonisation a drawback): ThuVARP (vaporesection), ThuVAP (vaporisation), ThuVEP (vapoenucleation), **ThuLEP** (laser enucleation, 2010; mostly blunt dissection; size-independent like HoLEP).
- **Pulsed Tm:YAG:** newer pulsed solid-state version of the continuous-wave thulium laser for photothermal lithotripsy and enucleation.

**TFL (1940 nm) for stones**

- **Diode-pumped silica fibre** laser near the **water absorption peak** (about **4× holmium absorption**).
- **Low peak power, long pulses, very high frequency** (up to about 2000 Hz [P4 LAQ: above 2000 Hz]; holmium about 120 Hz); pulse energy as low as 0.025 J. This gives photothermal fine "dusting" and **minimal retropulsion**.
- **Thin fibres** (50–150 µm; P4 LAQ: down to about 150 µm) give better irrigation and scope deflection. Small, quiet, compact unit; no dedicated high-voltage outlet.
- Settings: start 0.6 J × 6 Hz; dusting 0.2 J × 40 Hz; avoid >40 Hz (pulses miss the stone); settings ≤40 Hz are often enough. "Paint, do not drill."
- **RCTs (randomised controlled trials):** TFL beat **low-power (30 W) holmium** (stone-free rate, time); it **equalled 120 W pulse-modulated holmium** in stone-free rate, with less retropulsion.
- Concerns: efficacy on all stone types; thermal injury at high power.

**Thulium for prostate**

- Techniques: ThuVARP, ThuVAP, ThuVEP, ThuLEP, ThuFLEP (thulium fibre laser enucleation).
- Settings: 70–90 W enucleation, 30 W coagulation; morcellation as for HoLEP.
- Outcomes: ThuLEP equals HoLEP; size-independent; safe on anticoagulants (AUA [American Urological Association]). Retrograde ejaculation about four-fifths.

**Other thulium uses:** en bloc bladder tumour resection; UTUC ablation; incision of strictures.

## KTP/LBO (GreenLight) laser

**Physics**

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam passes through a **KTP or LBO (lithium triborate) crystal**. This doubles frequency and halves wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water. Optical penetration 0.8 mm; tissue effect 1–3 mm.
- Effect: **photoselective vaporisation** with a thin coagulation rim, so vaporisation is bloodless.

**Platforms**

| Generation | Power |
|---|---|
| KTP | 80 W |
| LBO HPS (High Performance System) | 120 W |
| LBO XPS | **180 W**; 750 µm MoXy fibre (metal tip, fluid cooling) |

- Side-firing fibre emits at 70°; 23–24 Fr cystoscope; saline irrigation.

**PVP (photoselective vaporisation of prostate) technique**

- Identify ureteric orifices; open the bladder neck first.
- Sweep 15–30° in 1–2 seconds at ≤2 mm working distance. Large bubbles mean efficient vaporisation.
- Pulsatile bleeding: coagulation mode (about 30 W). Energy >5 kJ/cc predicts complete vaporisation.

**Indications**

- AUA: prostates <30 cc and 30–80 cc; men at **higher bleeding risk** (anticoagulation continued). Not >80 cc (high retreatment).

**GOLIATH trial**

- RCT, 29 European centres, about 270 men, prostate ≤100 cc: **180 W PVP vs TURP** (transurethral resection of prostate).
- PVP **non-inferior** for IPSS (International Prostate Symptom Score), Qmax and complications at 6 months, 1 and 2 years.
- PVP: shorter catheter time and stay, fewer early reinterventions. TURP: faster. Retrograde ejaculation (about two-thirds) and 2-year reoperation equal.

**Complications**

- Dysuria and storage symptoms (highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum; capsular perforation 0.2–1%; retrograde ejaculation.
- No TUR (transurethral resection) syndrome; transfusion very rare.
- **No tissue** for histology.

**Other KTP uses:** laparoscopic non-contact cutting (largely supplanted); KTP enucleation possible.

## Applications by area

### 1. Stones

- **Ho:YAG** is the gold standard and breaks all stone types (see above). **TFL:** higher water absorption and low peak power give finer dust and less retropulsion.
- **Heat** depends on power, not laser type. Keep ≤10 W in the ureter; use good flow and an access sheath.

### 2. BPH

- **HoLEP:** endoscopic equivalent of open prostatectomy, **size-independent**. Enucleate at about 2 J × 40–50 Hz, then morcellate. The only endoscopic procedure superior to TURP. Less bleeding, shorter catheter time; safe on anticoagulants.
- **PVP (KTP/LBO 120–180 W):** haemoglobin absorption gives bloodless vaporisation, so anticoagulants can continue. **GOLIATH:** non-inferior to TURP. Avoid >80 cc (retreatment).
- **Thulium:** vaporisation, vaporesection, ThuLEP (equivalent to HoLEP). Diode vaporisation also used.
- Nd:YAG coagulation was abandoned (1 cm penetration, sloughing).
- AUA: HoLEP, ThuLEP or PVP for **high bleeding risk**.

### 3. Incisions

- Laser urethrotomy has no advantage over cold knife.

- Holmium incision of bladder neck contracture, ureteric stricture, endopyelotomy, ureterocele and calyceal diverticulum neck.

**4. Urothelial tumours**

- **NMIBC (non-muscle-invasive bladder cancer):** laser **en bloc resection** (Ho, Tm, KTP) gives detrusor in the specimen and no obturator jerk. Office ablation of small low-grade recurrences.
- **UTUC:** biopsy, then ablate by ureteroscopy:
- Ho:YAG: <0.5 mm, haemostatic.
- Nd:YAG: 5–6 mm coagulation, so stricture risk.
- Thulium: <0.2 mm.
- **PDT (photodynamic therapy):** a photosensitiser plus laser light makes singlet oxygen.

**5. Penile and genital lesions**

- PeIN (penile intraepithelial neoplasia) and small Ta/T1: **Nd:YAG preferred**, with base biopsies.
- $CO_2$ is too superficial for T1 (recurrence up to 50%). $CO_2$ or Nd:YAG for genital warts.

## Newer generation laser technologies (new lasers for stone treatment)

- **Aims:** less retropulsion, finer dust, faster work.
- **1. High-power holmium (60–120 W; other sources: 100–120 W machines) with pulse modulation:** high frequency allows **dusting** (0.2–0.3 J × 40–80 Hz) and "popcorning"; long-pulse modes reduce retropulsion.
- **2. Pulse modulation (Moses technology and similar):** the first part of a pulse boils a **vapour channel**; vapour absorbs less light, so the rest reaches the stone. Moses technology splits each pulse in two: the first opens the channel, the second strikes the stone. Less retropulsion; may shorten HoLEP.
- **3. TFL (1940 nm):** see above; ThuFLEP for prostate.
- **4. Pulsed Tm:YAG (2013 nm):** see above; it also does lithotripsy.
- **Others:** thin fibres; suction sheaths; AI (artificial intelligence) settings; 180 W GreenLight KTP/LBO (532 nm) vaporisation; diode lasers; burst wave lithotripsy (investigational).

### Holmium versus TFL

| | Ho:YAG | TFL |
|---|---|---|
| Wavelength | 2120 nm | 1940 nm |
| Water absorption | High | About 4× higher |
| Penetration | 0.31 mm | 0.08 mm |
| Peak power | High | Low |
| Retropulsion | More | Minimal |
| Fragments | Small | Smaller |
| Fibre | ≥200 µm | ≥150 µm |

### Principles of settings

- **Retropulsion rises with pulse energy**, not frequency; it falls with longer pulses.
- Keep the fibre close and perpendicular to the stone.
- TFL settings ≤40 Hz often enough.

### Safety

- Wavelength-specific eyewear for all, including the patient. Door signage, covered windows, trained staff and a laser safety officer.
- Check fibres; never fire inside the scope. Guard against fire and thermal (urothelial) injury.
- **Heat follows total power, not laser type:** ≤10 W in the ureter, ≤25 W in the kidney; use low power, pauses, high flow, cool irrigation and an **access sheath**; urothelial damage begins above about 43 °C.

### Diagrams to draw

- Water and haemoglobin absorption curves marking 532, 1064, 1940, 2013 and 2120 nm.
- Vapour bubble (Moses channel) between fibre tip and stone.
- Side-firing fibre at 70° vaporising lateral lobe.

> **Key points to remember:**
>
> - Chromophore sets penetration: TFL shallowest, Ho 0.4 mm, Nd:YAG deepest.
> - Holmium: 2120 nm; water chromophore; pulsed; photothermal; Moses effect: fibre close and perpendicular.
> - Holmium settings: fragment 0.6–1.0 J × 6–10 Hz; dust 0.2–0.3 J × 40–80 Hz.
> - HoLEP size-independent, superior to TURP.
> - TFL 1940 nm, Tm:YAG 2013 nm; water absorption about 4× holmium.
> - TFL: low peak power, fine dust, thin fibres, minimal retropulsion; settings ≤40 Hz often enough.
> - TFL beats low-power, equals high-power pulse-modulated holmium.
> - ThuLEP is size-independent and equal to HoLEP.
> - KTP 532 nm: frequency-doubled Nd:YAG; haemoglobin chromophore.
> - GOLIATH: 180 W PVP non-inferior to TURP; best for anticoagulated men; not >80 cc.
> - Nd:YAG for PeIN; eyewear for everyone.
> - Heat depends on power: ≤10 W in the ureter, ≤25 W in the kidney; access sheath; damage above 43 °C.

## LAQ 8 Describe IVU & its applications in Urology.

P1

**Type:** LAQ (25 marks)
**Source:** 2021-Summer Paper 1, Q2

### Definition and principle

- IVU (intravenous urography; also IVP, intravenous pyelography or excretory urography) is a series of timed abdominal radiographs after intravenous iodinated contrast.
- Contrast is **freely filtered by the glomerulus and not reabsorbed**. As water is reabsorbed it is concentrated in the tubules (**nephrogram**) and then outlines the calyces, pelvis, ureters and bladder (**pyelogram**).
- So IVU shows **anatomy plus a rough, qualitative idea of function and drainage** of each kidney.
- **Current status:** largely replaced by CTU (CT [computed tomography] urography). It is still used where CT is unavailable (common in India) or when dynamic excretion and drainage must be seen.

### Contrast media

| Type | Example | Feature |
|---|---|---|
| High-osmolar ionic | Diatrizoate | More reactions; obsolete for IV use |
| **Low-osmolar non-ionic** (standard) | Iohexol, iopamidol | Fewer reactions |
| Iso-osmolar non-ionic | Iodixanol | High-risk renal patients |

### Preparation

- Check creatinine or eGFR (estimated glomerular filtration rate), allergy, asthma, pregnancy, diabetes on metformin, myeloma.
- Laxative the night before (bowel gas hides the kidneys); short fast but **no dehydration**.
- Premedicate high-risk patients, e.g. prednisone 50 mg at 13, 7 and 1 h plus diphenhydramine 50 mg at 1 h.

### Technique: film sequence

| Film | Timing | Shows |
|---|---|---|
| **Scout (control) KUB** (kidney–ureter–bladder) | Before contrast | Calcification, bones, gas, technique |
| Contrast | Low-osmolar, about 1 mL/kg (usually 50–100 mL) IV | — |
| **Nephrogram / nephrotomogram** | About 1 min (60–90 s) | Renal size, outline, parenchyma |
| Pyelogram | 5 min | Calyces; then apply **ureteric compression** |
| Compression film | 10 min | Well-distended calyces and pelvis |
| **Release (full-length) film** | About 15 min | Whole ureters |

| Film | Timing | Shows |
|---|---|---|
| Bladder and **post-void** films | 20–30 min | Bladder outline, residue, lower ureters |
| Delayed films | Up to 24 h | Level of obstruction; **no excretion at 24 h = non-functioning kidney** |
| - **Also know:** films in expiration; prone film fills the ureters; oblique views; erect film for nephroptosis. | | |
| - Compression is avoided in suspected obstruction or colic, aortic aneurysm, recent abdominal surgery and trauma. | | |

## Contraindications

- Previous severe contrast reaction; significant renal impairment; pregnancy (relative); dehydration, myeloma.

## Nephrogram patterns

| Pattern | Causes |
|---|---|
| Absent | Renal artery occlusion, agenesis, avulsion |
| **Delayed, increasingly dense** | **Acute obstruction**, renal artery stenosis |
| Striated | Acute pyelonephritis, obstruction |
| **Rim (cortical rim) sign** | Renal infarction (capsular collaterals) |
| Bilateral persistent dense | Hypotension, ATN (acute tubular necrosis) |

## Applications in urology

| Condition | IVU features |
|---|---|
| **Calculi** | Confirms an opacity is in the tract; **radiolucent stone = filling defect**; site, obstruction, calyceal anatomy before SWL (shock wave lithotripsy) or PCNL (percutaneous nephrolithotomy) |
| **Acute obstruction** | Delayed dense nephrogram, delayed excretion, dilatation down to the block, **standing column** of contrast, forniceal rupture |
| **PUJ (pelviureteric junction) obstruction** | Ballooned pelvis, delayed drainage, non-dilated ureter; crossing-vessel indentation |
| **Duplex system** | Upper moiety obstructed or non-functioning: lower moiety displaced, "**drooping lily**" |
| **Ureterocele** | "**Cobra head**" (spring onion) at the bladder |
| **Horseshoe kidney** | Low, medially rotated; lower calyces point medially |
| **Retrocaval ureter** | "Reverse J" or **fish-hook** at L3 |
| **Medullary sponge kidney** | "**Paintbrush**" papillary striations |
| **Papillary necrosis** | **Ring sign**, "lobster claw" calyces |

| Condition | IVU features |
|---|---|
| **Genitourinary TB** (tuberculosis) | **Moth-eaten** calyx, **phantom calyx**, Kerr's kink, **pipestem** or beaded ureter, contracted bladder, autonephrectomy |
| **Urothelial tumour** | Irregular filling defect, non-visualised calyx |
| Retroperitoneal fibrosis | Medial deviation and tapering of ureters with hydronephrosis |
| Pelvic lipomatosis | Pear-shaped bladder |
| Schistosomiasis | Ureteric striations, calcified bladder |
| - **Trauma: one-shot intraoperative IVP:** 2 mL/kg IV push, single film at 10 min. It confirms a functioning contralateral kidney before exploration in the unstable patient. | |
| - **Also know:** nephroptosis (descent >5 cm or two vertebral bodies supine to erect); urinary fistulae (ureterovaginal); renal masses (distortion); post-operative assessment; diverticula. | |

## Complications

- **Contrast reactions:** immediate (urticaria to anaphylaxis) or delayed (rash). Severe reaction: stop contrast, **IM (intramuscular) adrenaline 0.3–0.5 mg** every 5–15 min (maximum 1 mg), oxygen, β2-agonist. Observe 30 min.
- **CA-AKI (contrast-associated acute kidney injury):** prevent with hydration, low or iso-osmolar contrast, lowest dose.
- Radiation; metformin-related lactic acidosis in renal impairment.

## Limitations and current role

- Two-dimensional; bowel gas; poor parenchymal detail; misses small masses; poor in renal failure.
- CTU is more sensitive for stones, masses and urothelial lesions, but gives a higher radiation dose than IVU. IVU remains useful in resource-limited settings.

## Recent advances

- Split-bolus CTU (fewer phases); dual-energy CT (virtual unenhanced images); MR urography without radiation.

**Diagram to draw:** normal IVU with minor and major calyces, pelvis, ureter and the three narrowings; beside it a "drooping lily".

> **Key points to remember:**
> - Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
> - Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
> - Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
> - Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

## LAQ 9 Robotics in urology (including robotic vs standard laparoscopy)

P4

**Asked as:**

- Role of robotics in urology — *Summer 2015 Paper IV Q1 (LAQ)*
- Discuss Robot and its role in urology. — *Winter 2017 Paper IV Q1 (LAQ)*
- Robots in urology — *Practice Set 4 Paper IV Q2 (LAQ)*
- Robotic vs std lap surgeries — *Practice Set 3 Paper IV Q3b (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2017-Winter Paper 4, Q1; 2015-Summer Paper 4, Q1) (robotics also answered in 2017-Winter Paper 4, Q1 and 2015-Summer Paper 4, Q1)

### Introduction

- Surgical robots are **master–slave telemanipulators**: the surgeon controls every movement; the robot acts on nothing by itself.
- Robotic surgery is **robot-assisted laparoscopy** with a **master–slave telemanipulator** (surgeon console, patient-side cart, vision cart). Access and physiology are the same; the interface differs.
- The robot did not change the operations. It removed the barrier of laparoscopic suturing, so RARP (robot-assisted radical prostatectomy) spread where laparoscopic prostatectomy had stalled. RARP now dominates in the USA.

### History

- **PUMA 560 (1985)** first surgical robot; **PROBOT** first urological robot (TURP (transurethral resection of prostate)); "Mona" first human trial (1997).
- **da Vinci** FDA (Food and Drug Administration) approval and first RARP (**Binder and Kramer**) in **2000**; robotic adrenalectomy 1999; first paediatric robotic procedure **2002**; robotic RPLND (retroperitoneal lymph node dissection) 2006.

### da Vinci system

1. **Surgeon console**: 3D (three-dimensional) view magnified about 10 times, motion scaling, tremor filtration.
2. **Patient-side cart**: **EndoWrist** instruments with **7 DOF (degrees of freedom)**; most have a 10-use limit; ports 8 mm, 8–10 cm apart to avoid arm collisions.
3. **Vision cart**: light source, camera electronics, energy generator.

| Generation | Added feature |
|---|---|
| Standard (2000) | Three arms |
| S (2006) / Si (2009) | High definition, fourth arm / dual console for training |
| Xi (2014) | 8-mm camera in any port, finer arms, side docking (docks from either side), no stirrups |
| SP (2018) | All instruments through one 2.5-cm cannula |
| da Vinci 5 (2024) | Force feedback |

- Patents expired in 2019 (after patent expiry), so other platforms appeared (e.g. **Hugo, Versius** and the low-cost Indian **SSI Mantra**).
- **Also know:** ports 8–10 cm apart to avoid arm collisions; Xi docks from any side; retroperitoneal cases dock over the head.

### Robotic vs standard laparoscopy: comparison

| Feature | Standard laparoscopy | Robotic |
|---|---|---|
| Vision | Mostly 2D; assistant holds camera | **3D**, 10× magnified, surgeon controls camera |
| Instruments | Rigid, **4 DOF (degrees of freedom)**; fulcrum effect reverses movement | **EndoWrist, 7 DOF**, 90° articulation; intuitive movement |
| Precision | Tremor transmitted | **Tremor filtration, motion scaling** |
| Ergonomics | Standing, awkward | Seated, less fatigue |
| Suturing | Hard; steep learning curve | Easy; shorter learning curve |
| Haptic feedback | Present (limited) | Absent (force feedback only in newest system) |
| Port placement | Flexible | Ports 8–10 cm apart; docking time |
| Bedside | Surgeon at patient | Surgeon remote; emergency undocking delays |
| Cost | Low, reusable | High capital; instruments limited to about 10 uses |
| Failure | — | Robot failure about 0.4% |

### Advantages and disadvantages

| Advantages | Disadvantages |
|---|---|
| 3D vision, wristed instruments, easy suturing | High capital and instrument cost; cost-effective only at high volume |
| Tremor filtration, seated ergonomics | Limited haptic feedback |
| Less blood loss and shorter stay than open; fewer conversions than laparoscopy | Docking time, bulk, surgeon away from patient |

### Applications (with evidence)

#### 1. RARP

- **Yaxley (Lancet 2016)**, RCT (randomised controlled trial): continence and potency **equal to open at 12 weeks**, with less blood loss and shorter stay; **Coughlin 2018**: equal at **24 months** on follow-up.
- Positive margins equal; the surgeon, not the platform, is the main variable. Bladder neck stricture under 2% vs up to 16% open.
- **Retzius-sparing** and **"hood"** techniques preserve continence structures and give earlier continence; ERAS (enhanced recovery after surgery) allows 23-hour or same-day discharge. Robotic surgery also widens candidacy (prior mesh, prior pelvic surgery) and allows salvage surgery.

- Versus laparoscopy: laparoscopic prostatectomy stalled on the vesicourethral anastomosis; the robot made it routine and dominant. A small RCT (randomised controlled trial; Porpiglia, 2013) favoured robotic continence and potency.

**2. RAPN (robot-assisted partial nephrectomy)**

- Versus laparoscopy (robotic vs laparoscopic gives): **fewer conversions**, **shorter warm ischaemia**, less (smaller) eGFR (estimated glomerular filtration rate) loss/fall and shorter stay; complications and margins similar. So more complex and hilar tumours are done minimally invasively.
- **Trifecta** = negative margin, no major complication within 30 days, warm ischaemia ≤25 minutes.
- ICG (indocyanine green) guides selective clamping.

**3. RARC (robot-assisted radical cystectomy)**

- **RAZOR (Lancet 2018)**: **non-inferior** 2-year PFS (progression-free survival) (72.3% vs 71.6%), with less blood loss and fewer transfusions (less transfusion).
- **iROC**: robotic cystectomy with **intracorporeal** diversion gave **more days alive and out of hospital at 90 days**, fewer VTE (venous thromboembolism) and wound complications. Continent diversion fell (11% in iROC).

**4. Other procedures**

| Procedure | Role of the robot / why the robot helps |
|---|---|
| Pyeloplasty, ureteric reimplantation | Easy intracorporeal suturing; also in children and infants (small working space: about 1 L vs 6 L in adults); similar success to laparoscopy; easier suturing |
| BMG (buccal mucosa graft) ureteroplasty | Long proximal or mid-ureteric strictures |
| RPLND (retroperitoneal lymph node dissection) | Template, nerve-sparing; short stay |
| Adrenalectomy, nephroureterectomy | Equivalent to laparoscopy; adrenalectomy suits the retroperitoneal approach |
| Simple prostatectomy | Large glands; less bleeding than open |
| Sacrocolpopexy, VVF (vesicovaginal fistula) / fistula repair | Deep pelvic suturing |
| **RAKT (robot-assisted kidney transplantation)** | Fewer wound complications in obese recipients |

- **Radical nephrectomy, adrenalectomy:** no outcome advantage over laparoscopy; robot only adds cost.
- Outcomes depend more on the **surgeon** than on the platform.

## Choosing robot vs laparoscopy

- Robot: complex excision plus reconstruction. Laparoscopy: extirpation, low-resource settings.

### Paediatric robotics

- Reconstruction (pyeloplasty, reimplantation, appendicovesicostomy) is where paediatric robotics has grown. Lower insufflation pressure; "burp" ports to enlarge the working space.

### Physiology and positioning

- **$CO_2$ pneumoperitoneum** (12–15 mm Hg, lower in children; another source: work at 10–12 mm Hg) causes hypercarbia, acidosis, reduced venous return and oliguria.
- **Steep Trendelenburg** for pelvic surgery increases venous return and cardiac output, and raises intracranial and intraocular pressure.
- The only strict contraindication to minimally invasive surgery is inability to tolerate pneumoperitoneum.

### Complications specific to robotics

| Problem | Details |
|---|---|
| **Steep Trendelenburg (~22–25°)** | Facial and laryngeal oedema, raised IOP (intraocular pressure), corneal abrasion |
| Nerve injury (up to ~6%) | Brachial plexus (avoid shoulder braces); femoral, peroneal |
| **Compartment syndrome, rhabdomyolysis** | Long lithotomy or flank position; fasciotomy if >30 mmHg with symptoms |
| Thermal bowel injury / bowel burns | Insulation failure, capacitive coupling; presents late |
| Robot malfunction | About 0.4% (another source: <1%); half found before the patient enters; rarely forces conversion |
| Port-site hernia | Close larger port sites; transverse incisions cut it from 5.4% to 0.4% after RARP |

- **Prevention**: secure inaccessible arms and lines before tilting, pad pressure points, limit tilt and lithotomy time, protect the eyes.

### Training and credentialing

- Box trainer, **VR (virtual reality)** simulator (simulators), animal and cadaver labs, then proctored cases (proctoring). **FRS (Fundamentals of Robotic Surgery)** curriculum improved skills in an RCT. Five-day mini-residencies also help. Dual console and simulators for training. Proficiency comes after about 40 cases, but outcomes keep improving (outcomes improve) with volume.

### Recent advances

- Single-port transvesical and extraperitoneal surgery; da Vinci SP (single port); force feedback (da Vinci 5); **AI (artificial intelligence)** video-based skill assessment; augmented reality 3D overlay; **telesurgery** (limited by latency; latency limits it).
- New platforms (Hugo RAS, Versius, Indian SSI Mantra) may cut costs.
- **Also know:** ERAS (enhanced recovery after surgery) with same-day RARP discharge; ICG (indocyanine green) fluorescence for selective clamping and ureteric perfusion.

**Diagram to draw:** console, vision cart and patient cart; RARP port placement; EndoWrist (7 DOF) beside a rigid instrument (4 DOF).

**Key points to remember:**

- da Vinci 2000; Xi 2014; SP 2018; patents expired 2019; 7 degrees of freedom.
- Yaxley and Coughlin: RARP equal function to open at 12 weeks and 24 months; RAZOR non-inferior; iROC more days out of hospital.
- RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Trendelenburg causes oedema and raised IOP; avoid shoulder braces.
- Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
- Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
- Robot made prostatectomy routine; no benefit for simple extirpation.
- Surgeon matters more than platform.

## LAQ 10 Flexible endourological instruments, digital and current ureteroscopes

P4

**Asked as:**

- Briefly outline flexible endourological instrument and its role in management of urological disease — *Practice Set 1 Paper IV Q1 (LAQ)*
- Digital flexible ureteroscopes — *Summer 2019 Paper IV Q3c (SAQ)*
- Current ureter scopes. — *Summer 2022 Paper IV Q3d (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)

(flexible ureteroscopy in detail: see Practice Set 1 Paper 2, Q3b; digital scopes also in 2019-Summer Paper 4, Q3c)

*Summer 2022 Q3d taken as "current ureteroscopes"; see also 2019-Summer Paper 4, Q3c.*

### Introduction

- A flexible endoscope has a **deflectable tip** and carries its image by a fibre bundle or a tip-mounted chip. It follows the natural curves of the urethra, ureter and collecting system.
- Physical basis: light travels along a curved glass fibre by **total internal reflection** (Tyndall).
- Marshall reported the first flexible URS (ureteroscopy) in 1964. It became therapeutic only when a **working channel** and **active tip deflection** were added in the early 1980s.
- **Fibreoptic flexible URS (ureteroscopy):** coherent glass-fibre bundle carries the image; fibres break → black spots; honeycomb (moiré) pattern.
- **Digital scopes** (first 2006): a sensor at the tip (**"chip-on-tip"**) converts light to an electrical signal, so the image does not degrade.

### Classification of current ureteroscopes

| Type | Key features |
|---|---|
| **Rigid rod-lens** | Larger calibre; largely historical |
| **Semirigid** | **Fibreoptic** image bundle; tip **≤7 Fr**; channel >3 Fr; often **two channels**; in-line or offset eyepiece |
| **Flexible fibreoptic** | Tip 4.5–7.5 Fr (smallest); image degrades as fibres break (black dots) |
| **Flexible digital** | "Chip-on-tip" CMOS (complementary metal-oxide semiconductor) or CCD (charge-coupled device) sensor plus LED (light-emitting diode); no camera head |
| **Single-use flexible** | Digital; LithoVue (2016) first; now several makers |
| **Ureteral resectoscope** | 11.5 Fr; largely replaced by laser |

### Semirigid ureteroscope

- Uses **coherent fibre bundles** for the image (random fibres carry only light).
- **Two channels** are preferred: irrigation through one while an instrument is in the other, or a laser through one to break a stone held in a basket.

- **Offset eyepiece** gives a straight channel for rigid probes (pneumatic, biopsy).
- **Rule:** semirigid below the iliac vessels, flexible above.

### Components

| Component | Key features |
|---|---|
| Optics | **Fibreoptic:** a *coherent* image bundle carries the picture; a *random* light bundle carries illumination; cladding improves reflection. **Digital:** CCD (charge-coupled device) or CMOS (complementary metal oxide semiconductor) chip at the tip with LED (light-emitting diode) light; no camera head or light cable |
| Deflection | Lever → control wires through movable metal rings → wire fixed at the tip (deflection mechanism: lever pulls wires). "Intuitive" = tip moves with the lever. The reticle (notch) marks the plane of deflection |
| Working channel | Single channel shared by irrigation and instruments, so instruments reduce flow and deflection |
| Shaft | Tapered tip; passive (secondary) deflection segment behind the tip helps lower-pole access |

- Flexible ureteroscope parts: optics, deflection mechanism (lever pulls wires), working channel.

### Sensor technology (digital)

| | CCD (charge-coupled device) | CMOS (complementary metal-oxide semiconductor) |
|---|---|---|
| Heat and power | More | Less (runs cooler) |
| Cost, speed | Higher, slower | Cheaper, faster |
| Status | Phased out | Current standard |

- **LED (light-emitting diode)** at the tip; no camera head or light cable → better ergonomics.

### Types

| Instrument | Specifications | Main use |
|---|---|---|
| Flexible cystoscope | 14–17 Fr; 37–40 cm; deflection 120–220°; channel 6.4–7.5 Fr; field about 120° | Office diagnosis and minor therapy |
| Flexible ureteroscope | Tip 4.5–7.5 Fr; deflection ≥270° both ways; channel **3.6 Fr**; dual-channel scope (2 × 3.3 Fr) exists | Kidney and upper ureter |
| Flexible nephroscope (or flexible cystoscope used through the tract) | Larger calibre; too wide for the UPJ (ureteropelvic junction) | Clearing calyces during PCNL (percutaneous nephrolithotomy) |

### Deflection and channel (flexible ureteroscope)

- Active deflection **≥270° both directions** (another source: **>270° both ways** in one plane); enhanced primary deflection eases lower-pole access / reaches the lower pole (ureteroinfundibular angle up to about 175°).
- Working channel **3.6 Fr**; dual-channel scopes exist.

### Fibreoptic vs digital (digital versus fibreoptic)

| Feature | Fibreoptic | Digital |
|---|---|---|
| Image | Honeycomb pattern; black dots as fibres break; degrades | Sharper, larger (magnified), stable image; no focusing or white balance |
| Size / Tip | Smallest tips, tapered | Larger (about 8.5 Fr), flatter; more access-sheath use |
| Outcome / Operative time | Reference; longer | Shorter operating time, equal (similar) stone-free rate (SFR) |
| Durability | Fragile; over-rotation damages fibres; repair about every 15 cases | **Not better** in later comparisons; not clearly better in trials |
| Processor | Platform-agnostic | Proprietary |

- Surprise finding: in vitro, illumination was better with the fibreoptic cystoscope.
- Digital: sharper image, **shorter operating time, same stone-free rate**. But larger tip, so **more access sheaths and sheath complications**.
- Larger tip → more access-sheath use. Keep a fibreoptic scope for narrow UPJ (pelviureteric junction) or infundibulum.

### Single-use scopes (single-use vs reusable)

- **LithoVue (2016):** first single-use digital ureteroscope; CMOS, LED, tip 7.7 Fr, shaft 9.5 Fr, 3.6 Fr channel, 270° each way. Single-use cystoscopes followed (2020).
- Rationale (reason): reusable ureteroscopes may need repair after only **3–5 cases**; reprocessing and repair add cost.
- Single-use advantages: maximal deflection each case, sterile, no reprocessing or repair; cost **lower below about 99 cases/year**.
- Best value in low-volume centres and for **large lower-pole stones**.
- Reusable: cheaper in high-volume centres; needs reprocessing (peracetic acid, glutaraldehyde, ethylene oxide).
- Cost advantage depends on case volume; environmental impact is similar to reusables. Most centres use a **hybrid fleet**.

### Accessories

- **Guidewires:** 0.035/0.038 inch PTFE (polytetrafluoroethylene)-coated safety wire; hydrophilic wires to pass impacted stones (exchange, as too slippery for safety); nitinol-core and hybrid wires.
- **Access sheath** (11–16 Fr): repeated passes, lower intrarenal pressure; small risk of ureteric injury.
- **Laser fibres** 200–365 µm; **nitinol tipless baskets** ≤3 Fr; three-prong graspers; 3 Fr cup biopsy forceps; 3 Fr balloon dilating to 12 Fr; fulguration electrodes; pressure irrigation pumps.
- Flexible EHL (electrohydraulic lithotripsy) probes work; flexible pneumatic probes restrict deflection.

## Role in urological disease

| Area | Role |
|---|---|
| **Lower tract** | Haematuria work-up; bladder cancer surveillance, office biopsy and fulguration; NBI (narrow-band imaging) and blue light; stent removal or exchange; bladder stones, foreign bodies, strictures; botulinum toxin injection; diversions through a catheterisable channel; difficult ureteric access via a suprapubic tract |
| **Stones** | RIRS (retrograde intrarenal surgery): renal stones <20 mm, lower pole <10 mm; >2 cm when PCNL is unsafe |
| **UTUC (upper tract urothelial carcinoma)** | Diagnostic URS and biopsy (predicts grade in about 9 of 10); laser ablation of low-grade tumours; prefer to percutaneous route (seeding) |
| **Obstruction** | Endopyelotomy without crossing vessels; endoureterotomy for strictures <1.5 cm |
| **Diagnosis** | Positive cytology, filling defects, lateralising haematuria (haemangioma fulgurated), recurrent one-sided infection; foreign body retrieval |
| **PCNL** | Antegrade or retrograde flexible endoscopy raises stone-free rate; ECIRS (endoscopic combined intrarenal surgery) |

## Care and complications

- **Scope damage:** deflection wear; laser fibre scraping and perforating the channel. Pass fibres with the scope straight; **fire laser only when the fibre tip is seen touching the stone** (fire only with the fibre tip seen on the stone); leave the bladder empty to avoid buckling.
- **Sterilisation:** sterilise with ethylene oxide, glutaraldehyde or peracetic acid.
- **Patient:** ureteric injury, stricture, sepsis from high pressure. Use saline irrigation and antibiotic prophylaxis for all upper tract procedures.

## Recent advances

- **Intrarenal pressure sensor** in the scope (LithoVue Elite): high pressure causes fever, SIRS (systemic inflammatory response syndrome), sepsis. Temperature sensing next.
- **AI (artificial intelligence):** stone recognition, laser setting adjustment, navigation.
- **Robotic flexible URS (ureteroscopy):** ergonomics, less radiation to surgeon.
- **Also know:** suction (access) sheaths, further miniaturisation.

**Diagram to draw:** cross-section of a flexible ureteroscope: chip, LED, working channel and deflection wires (scope tip cross-section showing CMOS chip, LEDs, 3.6 Fr channel and 270° deflection).

> **Key points to remember:**
> - Working channel plus active deflection made flexible scopes therapeutic.
> - fURS: 3.6 Fr channel, ≥270° (>270°) deflection; cystoscope 14–17 Fr.
> - Chip-on-tip CMOS digital: better image, larger tip, durability not better; shorter operation.
> - Single-use cheaper below about 99 cases/year; LithoVue first single-use (2016).
> - Semirigid: two channels; below the iliac vessels.
> - Pressure sensors and AI are the newest developments.
> - Golden rule: never fire unless the fibre tip is seen on the stone.

## LAQ 11 Natural orifice endoscopic surgery.

P4

**Type:** LAQ (25 marks)
**Source:** 2022-Summer Paper 4, Q1

*Taken as NOTES (natural orifice transluminal endoscopic surgery) and its relatives (hybrid NOTES, LESS [laparoendoscopic single-site surgery]).*

### Definition and concept

- **NOTES:** entry into the peritoneal or retroperitoneal cavity through a deliberate incision in a hollow viscus reached through a natural orifice (mouth, vagina, bladder, rectum). The aim is **"scarless" surgery**: no abdominal wall wound, so no wound pain, infection or hernia.
- It differs from **endoluminal** surgery (e.g. ureteroscopy), which stays inside the lumen.
- **Pure NOTES:** all instruments through the orifice. **Hybrid NOTES:** natural orifice plus one or more small abdominal ports. Hybrid is what is actually used in humans.
- Chapter 142 places NOTES at the end of a spectrum: transperitoneal → retroperitoneal → hand-assisted → LESS → SP (single-port) robotic → NOTES. **Each step to fewer incisions buys cosmesis and less pain at the cost of working space and triangulation.**

### History (milestones)

| Year | Event |
|---|---|
| 2002 | Transvaginal laparoscopic nephrectomy in pigs (Gettman, Cadeddu) |
| 2004 | Transgastric peritoneoscopy in pigs (Kalloo) |
| 2006 | NOSCAR (Natural Orifice Surgery Consortium for Assessment and Research) white paper listed barriers |
| 2008–2010 | Hybrid, then pure, transvaginal nephrectomy in humans |

### Routes of access

| Route | Features | Urological use |
|---|---|---|
| **Transvaginal** (posterior colpotomy) | Commonest; easy closure under vision; large specimens pass; women only | Nephrectomy, donor nephrectomy, specimen extraction |
| **Transgastric** | Upper abdomen reachable; gastrotomy closure and leak are the main risk | Experimental (peritoneoscopy, renal surgery in animals) |
| **Transvesical** | Sterile urine; bladder dome port; small calibre; closure difficult | Peritoneoscopy (animal, Lima), transvesical access to pelvis |
| **Transurethral** | Urologist's familiar route | Experimental transurethral prostatectomy in cadavers |
| **Transcolonic/transanal** | Faecal contamination risk | Rarely urological |

### Hybrid NOTES and natural orifice extraction

- Standard laparoscopy does the dissection; **the vagina is the access for a camera or instrument and the extraction route** (Chapter 142). This is the only NOTES form with

real clinical urological use.

- **NOSE (natural orifice specimen extraction):** kidney removed through the vagina after laparoscopic nephrectomy, avoiding the extraction incision, which is the largest and most painful wound.
- **NOTES-assisted donor nephrectomy:** laparoscopic or robotic donor nephrectomy with **transvaginal kidney extraction**. Reported benefits: less pain, faster recovery, no extraction scar; warm ischaemia acceptable.

### Steps of transvaginal (hybrid) nephrectomy

1. Lithotomy then modified flank; bowel and vaginal preparation; antibiotics.
2. Umbilical port (camera or instrument) plus one or two small ports.
3. **Posterior colpotomy** under laparoscopic vision; long port or flexible scope through the vagina.
4. Colon reflection, hilar dissection, ureter and vessels stapled.
5. Kidney in a bag, delivered through the colpotomy.
6. Colpotomy closed with absorbable sutures under vision.

### LESS: the close relative

- All ports through **one skin incision** (umbilicus or below the waistline) using several low-profile trocars or a multichannel port (TriPort; da Vinci SP 25–27-mm entry guide).
- Problems: **no triangulation**, instrument clashing, in-line view. Solutions: articulating or bent instruments, flexible-tip scopes, instruments of different lengths.
- Evidence (Chapter 142): randomised trial versus laparoscopy showed the same time, blood loss and complications, with less analgesia. Donor meta-analysis: lower pain scores. Oncological outcome unchanged.
- Drawback (Chapter 20): **more incisional and umbilical hernias** because of the larger single incision.
- **SP robot (FDA [Food and Drug Administration] approval 2018)** has revived single-site surgery.

### vNOTES (vaginal NOTES)

- A gynaecological development: a single-port device in the vagina gives access to the peritoneum.
- **HALON RCT (randomised controlled trial):** vNOTES hysterectomy was non-inferior to laparoscopic hysterectomy, with more same-day discharge and less pain.
- Urogynaecological uses: vNOTES apical prolapse repair (uterosacral suspension), adnexal surgery.

### Challenges (NOSCAR list)

- **Safe access** and **secure closure** of the viscerotomy (leak, fistula); peritoneal **infection**.
- **Spatial orientation**: retroflexed, off-axis view.
- **Platform**: flexible scopes lack stiffness, traction and triangulation.
- Control of **haemorrhage** and other complications.
- Training and cost; dyspareunia and fertility concerns after colpotomy; transvaginal route unavailable in men.

- Chapter 142: adoption is limited by **no clear therapeutic benefit**, safety concerns and very selective patients; **pure NOTES has very few urological cases.**

### Recent advances

- **Flexible robotic platforms** (Chapter 39): MASTER (NOTES, haptics), ViaCath.
- **Transvesical single-port** (LESS or da Vinci SP) surgery for diverticulectomy and foreign bodies (Chapter 124).
- Magnetic anchoring and guidance systems to restore triangulation.

**Also know:** LESSCAR consortium standardised the term LESS; "embryonic NOTES" (umbilicus as the natural scar); hidden-incision endoscopic surgery (HIdES) in children.

**Diagram to draw:** sagittal female pelvis showing posterior colpotomy with a transvaginal port and kidney extraction, plus umbilical port.

> **Key points to remember:**
> - NOTES = transluminal via natural orifice; hybrid transvaginal is the only routinely used urological form.
> - Transvaginal extraction removes the most painful wound.
> - LESS: one incision, lost triangulation, more hernias.
> - Barriers: closure, infection, orientation, platform; no proven benefit yet.

## LAQ 12 Laparoendoscopic single-site (single-port) surgery: LESS

P4

**Asked as:**

- Laproendoscopic single site surgery: LESS — *Winter 2023 Paper IV Q1 (LAQ)*
- Single port lap. Surgeries in urology — *Practice Set 2 Paper IV Q3c (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)
(also answered as a long answer in 2023-Winter Paper 4, Q1)

*Read as laparoendoscopic single-site surgery (LESS), i.e. single-port (single-site) laparoscopic and robotic surgery in urology. NOTES (natural orifice transluminal endoscopic surgery) is covered in 2022-Summer Paper 4, Q1.*

### Definition and concept

- **LESS:** laparoscopic or robotic surgery in which all instruments and the camera enter (pass) through **one skin incision**, usually hidden in the umbilicus (umbilical) or below the waistline.
- Aim: further reduce access trauma, giving **better cosmesis, less pain and shorter stay**. Chapter 142 places it on a spectrum: multiport → hand-assisted → LESS → SP (single-port) robot → NOTES. Each step to fewer incisions buys cosmesis at the price of working space and triangulation.
- First urological single-port nephrectomies were reported in 2007; the first single-port robotic urological surgery followed in 2009.

### Nomenclature (LESSCAR)

- Early names were many: SILS (single-incision laparoscopic surgery), SPA (single-port access), OPUS (one-port umbilical surgery), TUES (transumbilical endoscopic surgery), e-NOTES (embryonic NOTES).
- **LESSCAR (Laparoendoscopic Single-Site Surgery Consortium for Assessment and Research, 2008)** agreed the single umbrella term **LESS** and set standards for reporting, training and research.
- The term was agreed by LESSCAR (LESS Consortium for Assessment and Research, 2008).
- Sub-types (variants): transumbilical or extraumbilical; **pure LESS**; **"LESS + 1"** or needlescopic-assisted (an extra 2–5-mm port); **R-LESS** (robotic LESS); purpose-built **SP (single-port) robot**.

### Access: ports and devices

| Option | Description |
|---|---|
| Several low-profile trocars | Placed close together through one fascial incision (the future extraction incision) |
| **Multichannel ports** | TriPort 15 (two 5-mm + one 15-mm); TriPort+ (three 5-mm + one 10-mm); SILS port; GelPOINT (gel cap, trocars placed anywhere on it) |
| **da Vinci SP entry guide** | One 25–27-mm cannula carrying three articulating (wristed) 6-mm instruments + one flexible articulating camera |

- Position: modified flank (transperitoneal) or full flank (retroperitoneal SP, 2.5–3-cm incision near the 12th rib tip, balloon dilatation).

### The triangulation problem

- In standard laparoscopy ports are spread so instruments meet the target at an angle (the "diamond"). In LESS all shafts are **parallel and in-line** with the camera (instruments lie parallel to the camera).
- Results: **instrument clashing**, **"crossing swords"**, poor depth perception, poor retraction and hard suturing.
- Solutions:
- **Articulating instruments** (tip bends inside the abdomen) and **pre-bent or curved instruments**: they cross at the fascia, so the right hand works on the left side (**chiral manipulation**).
- **Flexible-tip or 5-mm low-profile scopes**, offset optics, extra-long instruments and instruments of **different lengths** to avoid handle clash.
- A 2-mm needlescopic instrument or extra 5-mm port (an extra needlescopic port) without hesitation.
- **Robotic platforms** restore wristed movement.

### Robotic single-site surgery

| Platform | Features |
|---|---|
| da Vinci Single-Site (Si/Xi) | Curved crossing cannulas; semi-rigid **non-wristed** instruments; software swaps hand control |
| **da Vinci SP (FDA [Food and Drug Administration] approval 2018)** | Purpose-built; one arm, no external collisions; instruments with elbow and wrist; 3D articulating camera; on-screen navigator shows instrument positions |

- SP needs **10–25 cm between (from) cannula and (to) target** to deploy the joints; a GelPort lets instruments "float" outside the skin. Assistance through an **"SP + 1" port** or ROSI (remote-operated suction-irrigation). Limited by cost and availability.

### Indications

- **Kidney:** simple and **donor nephrectomy**, radical nephrectomy, partial nephrectomy, **pyeloplasty**, nephroureterectomy, cyst decortication, cryoablation.
- **Adrenalectomy**; varicocelectomy; orchidopexy and pyeloplasty in children (paediatric).
- **Pelvis:** SP radical prostatectomy (transperitoneal, extraperitoneal); **SP transvesical simple prostatectomy** and bladder diverticulectomy.
- **Best case:** thin patient (patients), small or benign lesion (lesions), experienced surgeon.
- **Relative contraindications (avoid in):** obesity, large or hilar tumours, complex vascular anatomy, dense adhesions, unfit for pneumoperitoneum.

### Outcomes versus conventional laparoscopy (Chapter 142)

- **Randomised trial (RCT, randomised controlled trial):** same operative time, blood loss, complications and stay; **less analgesia**.
- **Donor nephrectomy meta-analysis:** same stay, recovery, transfusion and conversion (otherwise equal); **lower pain scores**.
- **Renal mass:** same perioperative outcome and survival.
- **SP robotic partial nephrectomy vs multiport:** shorter stay, less opioid; same margins and complications; **longer warm ischaemia** in one review.

- An **extra port is often needed**; conversion is mostly to laparoscopy, rarely open.
- **Net effect:** gains are confined (limited) to **pain, cosmesis and stay**; oncological and functional outcomes are unchanged, but the operation is technically harder and longer.

### Complications

- All laparoscopic complications (usual laparoscopic injuries: access injury, vascular and bowel injury, $CO_2$ effects).
- **Incisional and umbilical hernia** is higher (larger fascial defect): about 5% with da Vinci Xi single-site, lower with SP.
- Wound infection; conversion; longer operation; surgeon ergonomic strain. Prevent by careful fascial closure and case selection.

### Recent advances

- **da Vinci SP** with same-day discharge pathways; **SP retroperitoneal** renal surgery.
- **Transvesical SP** prostatectomy; hidden incision endoscopic surgery (HIdES) in children.
- New single-port platforms; magnetic anchoring and guidance systems / instruments (restore triangulation).

**Diagram to draw:** umbilical multichannel port with articulating (crossed) instruments crossing at the fascia (chiral) versus a triangulated multiport diamond (layout).

> **Key points to remember:**
> - LESS = one incision; LESSCAR (2008) standardised the term.
> - Core problem: lost triangulation; solved by articulating or curved instruments and robots.
> - Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
> - da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

## Short answer questions (SAQ)

## SAQ 1 Pioneers of urology: Novick, Chaussy, Hopkins, Huggins, Calne, Gaur and Asopa

P1 **merged from 3**

*Asked as:* P1 SAQ "Pioneers of urology: Hopkins, Huggins, Calne, Gaur and Asopa" · P1 SAQ "Andrew Novick." · P1 SAQ "Christian Chaussey"

*Names read as Prof. Harold Hopkins ("Herald"), Dr Charles Huggins, Prof. Sir Roy Calne ("Caine"), Dr D. D. Gaur ("Gaue"), Hari Asopa, Andrew C. Novick and Christian Chaussy ("Chaussey"). The question is taken to ask for each person's contributions to urology. Novick is answered mainly through nephron-sparing surgery, and Chaussy through the history and principles of SWL.*

### Prof. Harold Hopkins (British physicist)

- **Fibreoptics (1954, with Kapany):** coherent bundles of glass fibres transmit an image by total internal reflection (Tyndall's principle). This made flexible endoscopes possible.
- **Rod-lens system (1966):** the old scope used thin glass lenses separated by long air spaces. Hopkins replaced them with **long glass rods separated by short air gaps**.

Result: more light transmission, a brighter and wider image, and a **smaller calibre**. Developed commercially with Karl Storz.
- It underpins all modern rigid endoscopes and enabled the first rigid ureteroscope (1979).

### Dr Charles B. Huggins (1901–1997)

- Canadian-born American urologist, University of Chicago.
- **1941, with Clarence Hodges:** castration (or oestrogen) relieved far-advanced disease and caused **regression of metastatic prostate cancer**, with falling serum acid phosphatase, pain relief and weight gain; injected androgens made it worse. This proved that **prostate cancer is hormone (androgen) dependent**.
- Used serum **acid phosphatase** to monitor response: the first tumour marker.
- **1945:** bilateral adrenalectomy for castration-resistant disease (relapse after castration, suspecting adrenal androgens); modest, brief responses; morbidity high.
- **Nobel Prize in Physiology or Medicine, 1966** (shared with Peyton Rous) for hormonal treatment of prostate cancer: the basis of hormonal cancer therapy.
- **Legacy:** basis of all ADT (androgen deprivation therapy): orchidectomy, oestrogens, LHRH (luteinising hormone-releasing hormone) agonists and antagonists, and later abiraterone (blocking adrenal androgens).
- **Also know:** canine prostate physiology; adrenalectomy for breast cancer.

### Prof. Sir Roy Calne (Cambridge transplant surgeon)

- Showed in dogs that **6-mercaptopurine and then azathioprine** prolong renal allograft survival; azathioprine with steroids became standard immunosuppression.
- Performed the **first liver transplant in the UK (1968)**.
- **First clinical use of ciclosporin** in kidney transplantation (late 1970s), the calcineurin-inhibitor era; acute rejection fell markedly.

### Dr D. D. Gaur (Mumbai, India)

- **1992: balloon dissection of the retroperitoneum.** A size 7 surgeon's glove tied on an 8-Fr red rubber catheter, inflated inside the retroperitoneum, creates a working space.
- This made **retroperitoneoscopy** practical and cheap; used for ureterolithotomy, nephrectomy and other renal surgery.

### Hari S. Asopa (Indian urologist)

**1. Asopa dorsal inlay urethroplasty (2001)**

- Single-stage graft urethroplasty for anterior urethral stricture by a **ventral sagittal urethrotomy** approach.
- **Steps:** expose the urethra ventrally without circumferential mobilisation; open the urethra ventrally through the stricture; incise the **dorsal urethral plate** in the midline down to the corpora; **inlay** a free graft (buccal mucosa or penile/preputial skin) and quilt it to the tunica albuginea; close the urethra over a catheter.
- **Rationale:** the graft lies on the well-vascularised corporal bed; no mobilisation preserves spongiosal blood supply; suits bulbar and penile strictures.
- Ranks with **Barbagli dorsal onlay** and **Kulkarni one-sided** repair among dorsal graft techniques.

**2. Asopa hypospadias repair (1970s)**

- One-stage repair using a **vascularised inner preputial island flap** still attached to the outer preputial skin, transferred ventrally to form the neourethra; a forerunner of the Duckett transverse preputial island flap.

## Andrew C. Novick

### Who he was

- A **Cleveland Clinic** urologist and long-time chairman of its urology department. He was a world authority on renal surgery, an editor of *Campbell-Walsh Urology* (9th edition), and died in 2008.

### Contributions

1. **Nephron-sparing surgery (NSS) / partial nephrectomy (PN):** he established its technique, indications and long-term cancer control. His group showed tumours ≤4 cm did as well with NSS as with radical nephrectomy, supporting the T1a/T1b split.
2. **Renovascular surgery:** aortorenal, splenorenal and hepatorenal bypass; **bench (ex vivo) surgery with autotransplantation** for complex lesions.
3. **Renal transplantation** surgery.

### Indications for PN

| Category | Situations |
|---|---|
| **Imperative** | Bilateral RCC (renal cell carcinoma); solitary functioning kidney |
| **Relative** | Contralateral kidney threatened (stones, stenosis, diabetes, hypertension); hereditary RCC |
| **Elective** | cT1a (≤4 cm) with a normal other kidney; now larger selected tumours |

- **Relative contraindications:** <20% of nephron mass left; pedicle encasement; venous thrombus.

### Principles of NSS

1. Three-dimensional CT (computed tomography) maps tumour, vessels and collecting system.
2. **Early hilar control.** Keep warm ischaemia under about 25–30 minutes; if longer, use **ice-slush hypothermia** and mannitol.
3. Excise with a **negative margin** (width immaterial); intraoperative ultrasound for endophytic tumours.
4. **Watertight collecting-system closure**, vessel ligation, capsular renorrhaphy.
5. Drain for urine leak. - Function afterwards depends on parenchymal **quality and quantity preserved**; ischaemia matters less. At least 20–30% of one kidney is needed to avoid dialysis.

### Legacy

- PN is now the AUA (American Urological Association) and EAU (European Association of Urology) standard for cT1a tumours.

## Christian Chaussy (Munich): extracorporeal shock wave lithotripsy

### Contribution and history

- Engineers at **Dornier** found by chance that shock waves from raindrop and micrometeorite research affected tissue.
- Chaussy, with Brendel and Schmiedt in **Munich**, developed the technique in animals and performed the **first human SWL (shock wave lithotripsy) on 7 February 1980 on the**

**Dornier HM1**. They reported it in the *Lancet* in 1980.

- **HM3** (water bath, electrohydraulic spark, biplanar fluoroscopy): first installations Stuttgart 1983 and Indianapolis 1984. It remains the reference for efficacy.
- SWL made most open stone surgery obsolete.

**Principles**

- A shock wave is a short acoustic pulse: **peak positive 30–100 MPa, then a negative tail of about −10 MPa**.
- Generated at **F1**, focused onto the stone at **F2** through water or gel coupling (water and tissue have similar acoustic impedance); targeting by fluoroscopy or ultrasound.

| Energy source | Feature |
|---|---|
| Electrohydraulic (HM1, HM3) | Underwater spark, ellipsoid reflector; most effective; electrode wear |
| Piezoelectric | Ceramic elements; long life, little pain; small focal zone, low energy |
| **Electromagnetic** (modern) | Coil and membrane with lens; consistent; dominant today |

**Mechanisms of fragmentation**

- **Spallation:** reflected wave at the back surface creates tension.
- **Shear and tear** inside the stone.
- **Quasistatic squeezing** and **dynamic squeezing** (strongest force in initial breakage).
- **Cavitation:** collapsing bubbles erode the surface; excess bubbles shield the stone and injure tissue.

**Practical points**

- Better results at **60–90 shocks/min**, with power **ramping** and air-free coupling.
- Poor results: stone >1000 HU (Hounsfield units), skin-to-stone distance >10 cm, lower pole >1 cm.
- **Contraindications:** pregnancy, uncorrected coagulopathy, untreated UTI (urinary tract infection), distal obstruction, aortic aneurysm >4 cm.
- **Also know:** burst wave lithotripsy and ultrasonic propulsion are newer descendants.

## Diagrams to draw

- Conventional lens scope versus Hopkins rod-lens scope (glass rods with short air gaps).
- Dorsal inlay: urethra opened ventrally, dorsal plate incised, graft on corpora (cross-section).
- PN steps: clamp, ice slush, excision, closure, renorrhaphy.
- Ellipsoid reflector with spark at F1 and stone at F2.

**Key points to remember:**

- Hopkins: rod-lens system 1966; fibreoptics 1954.
- Huggins and Hodges 1941: androgen dependence; castration regresses prostate cancer; Nobel 1966; 1945 adrenalectomy for castration-resistant disease.
- Calne: azathioprine, ciclosporin, first UK liver transplant 1968.
- Gaur 1992: glove-balloon retroperitoneoscopy.
- Asopa: ventral sagittal urethrotomy, dorsal plate incised, free graft inlaid (2001).
- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery; PN indications imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer; negative margin is enough; keep ≥20–30% of one kidney.
- Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1; HM3 (1983) is still the efficacy benchmark.
- Spallation, squeezing, shear, cavitation; electromagnetic sources today.

## SAQ 2 Bacteriuria in pregnancy and antibiotic policy

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "Bacteriuria in pregnancy." · P3 SAQ "Baceriuria and use of antibiotic policy in pregnancy"

*Read as: bacteriuria in pregnancy and the antibiotic policy for it.*

### Definition

- **ASB (asymptomatic bacteriuria):** ≥$10^5$ CFU (colony-forming units)/mL of one uropathogen in a midstream sample without urinary symptoms (≥$10^2$ CFU/mL on a catheter sample).

### Epidemiology

- Prevalence is **not raised** by pregnancy (about 2–10%; P3: about 2–7%), but ASB **persists and ascends**. Higher with low socioeconomic status, multiparity, sickle cell trait. Organisms as in non-pregnant women (mainly *E. coli*).

### Why pregnancy matters (pathophysiology)

- **Progesterone** relaxes ureteric smooth muscle; the **uterus compresses the ureters at the pelvic brim** (right more). Dilatation from about 15 weeks; normal by 6 weeks postpartum.
- Result: **stasis and hydroureteronephrosis**; bacteriuria persists and ascends.
- GFR (glomerular filtration rate) rises 30–50%: creatinine and drug levels fall.
- **Untreated ASB leads to pyelonephritis in 15–45%** (historical), mostly in the third trimester (60–75% of cases). Treatment cuts this to **1–4%**.
- Pyelonephritis causes **preterm birth, low birth weight** and maternal sepsis. **Cochrane 2019:** treating ASB reduces pyelonephritis, preterm delivery and low birth weight.

### Screening

- **Urine culture at 12–16 weeks** or the first visit (USPSTF [US Preventive Services Task Force], IDSA [Infectious Diseases Society of America]). Dipstick misses many cases.

- Treat all ASB found in pregnancy.

### Antibiotic policy

| Safe | Restricted or avoided |
|---|---|
| **β-lactams** (penicillins such as amoxicillin; cephalosporins such as cephalexin): throughout | **Nitrofurantoin:** stop near term (from 35 weeks): neonatal haemolysis (G6PD [glucose-6-phosphate dehydrogenase] deficiency); not for pyelonephritis |
| Fosfomycin: lower tract (cystitis) only | **TMP-SMX (trimethoprim-sulfamethoxazole):** avoid 1st trimester (folate, neural tube); caution near term (kernicterus) |
| Gentamicin: pyelonephritis | **Fluoroquinolones** (cartilage), **tetracyclines** (teeth, bone), chloramphenicol: avoid |

- ACOG (American College of Obstetricians and Gynecologists) 2017: nitrofurantoin and sulfonamides first-line in 2nd–3rd trimester; 1st trimester only if no alternative.
- **ASB and cystitis:** culture-directed, **4–7 days** (cures more than a single dose); **repeat culture** after treatment.

### Pyelonephritis

- **Admit**; IV (intravenous) β-lactam (ampicillin plus gentamicin, ceftriaxone or aztreonam) until afebrile, then oral (cephalexin) to complete ≥14 days (P3: 10–14 days).
- No improvement by **72 h: renal ultrasound** (stone, obstruction, abscess).

### Prophylaxis

- After one pyelonephritis or two cystitis episodes: nightly nitrofurantoin or cephalexin until delivery, or close culture surveillance (Cochrane 2015: close surveillance equally effective).
- **Also know:** urinary diversion or augmentation, SCI (spinal cord injury): most treat ASB in pregnancy and consider prophylaxis.

### Diagram to draw

- Right hydroureter compressed at the pelvic brim.

> **Key points to remember:**
> - Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
> - Screen by culture at 12–16 weeks; treat all ASB for 4–7 days.
> - β-lactams safe throughout; no fluoroquinolones or tetracyclines.
> - Nitrofurantoin not near term (after 35 weeks); TMP not in 1st trimester.

## SAQ 3 Evidence-based medicine and critical appraisal

P1 · P4 **merged from 2**

*Asked as:* P1 SAQ "Evidence-based medicine and critical appraisal of research papers" · P4 SAQ "Describe evidence based medicine."

*"quaLliy" is taken to mean quality, i.e. critical appraisal.*

### Definition

- **EBM (evidence-based medicine):** "the conscientious, explicit and judicious use of current best evidence in making decisions about the care of individual patients" (**Sackett, 1996**).
- It integrates best research evidence, **clinical expertise** and **patient values**.

- **Critical appraisal:** systematic assessment of a paper's **validity, results and applicability**; the "Appraise" step of EBM.

### Steps ("5 As")

1. **Ask** a focused question in **PICO** form (Population, Intervention, Comparison, Outcome).
2. **Acquire**: PubMed, Cochrane Library, guidelines.
3. **Appraise**: validity (randomisation, blinding), effect size (ARR [absolute risk reduction], **NNT [number needed to treat] = 1/ARR**), precision (95% CI [confidence interval]), applicability.
4. **Apply** with expertise and patient preference.
5. **Assess** (audit) the outcome.

### Critical appraisal: three core questions

#### 1. Is it valid? (internal validity)

- **PICO question**; suitable design.
- **RCTs (randomised controlled trials):** true randomisation, **allocation concealment**, comparable groups at baseline, **blinding**, complete follow-up (>80%), **ITT (intention-to-treat) analysis**.
- **Bias** checked:

| Bias | Meaning |
|---|---|
| Selection | Groups differ at entry |
| Confounding | Third factor (comorbidity, frailty) linked to exposure and outcome; needs risk adjustment |
| Performance, detection, attrition | Unequal care, outcome assessment or dropouts |
| Lead-time, length-time; stage migration (Will Rogers) | Screening or better tests make survival look longer |
| Publication | Negative studies unpublished |

- **Pitfalls in appraisal:** selection, confounding, lead-time, length-time, stage migration (Will Rogers) and attribution bias.

#### 2. What are the results?

- Effect: RR (relative risk), **ARR, NNT = 1/ARR**.
- Precision: **95% CI**; p value; **sample size and power** (usually 80%; α 0.05).
- **Surrogate end points** must meet the **Prentice criteria** (prostate cancer: only metastasis-free survival does); validated outcome tools (validated PROs [patient-reported outcomes]); Clavien–Dindo grading.

#### 3. Does it apply? (external validity)

- Similar patients and setting; benefit outweighs harm.

### Levels of evidence (Oxford CEBM [Centre for Evidence-Based Medicine], as adapted by EAU [European Association of Urology])

| Level | Type of evidence / design |
|---|---|
| **1a** | Systematic review or meta-analysis of RCTs |
| **1b** | At least one RCT |

| Level | Type of evidence / design |
|---|---|
| 2a | Well-designed controlled study without randomisation |
| 2b | Quasi-experimental (cohort) study |
| 3 | Non-experimental: case-control, correlation, case reports, case series |
| 4 | Expert committee reports or opinion |

- **Grades:** A (consistent level 1), B (level 2–3), C (level 4 or consensus); D (some systems) = no recommendation possible.
- **GRADE** (Grading of Recommendations Assessment, Development and Evaluation): certainty high to very low; RCTs start high, observational studies low; **downgrade** for bias, inconsistency, indirectness, imprecision, publication bias; **upgrade** for large effect or dose–response. Recommendations **strong or weak** (now used by EAU).
- AUA (American Urological Association): evidence strength A/B/C; strong, moderate or conditional recommendations.

### Reporting checklists and quality tools

| Design | Checklist |
|---|---|
| RCT | **CONSORT** (Consolidated Standards of Reporting Trials) |
| Systematic review / meta-analysis | **PRISMA** (Preferred Reporting Items for Systematic reviews and Meta-Analyses) |
| Observational | **STROBE** (Strengthening the Reporting of Observational studies in Epidemiology) |
| Diagnostic accuracy | **STARD** (Standards for Reporting Diagnostic accuracy studies) |
| Surgical innovation | **IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study) |

- **Also know:** Cochrane RoB 2 (risk-of-bias) tool; trial registration; conflicts of interest; Clavien–Dindo; validated PROs; Donabedian model.

### Limitations of EBM

- Few surgical RCTs; trial patients differ from real ones; publication bias; "cookbook medicine" if patient values are ignored.

### Diagram to draw

- Evidence pyramid and the three overlapping circles of EBM.

> **Key points to remember:**
> - Sackett 1996: best evidence + clinical expertise + patient values.
> - 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
> - Level 1a = systematic review of RCTs; GRADE: strong or weak.
> - NNT = 1/ARR; beware lead-time, length-time and confounding bias.
> - Three questions: valid, results, applicable.
> - Randomisation, concealment, blinding, ITT, follow-up >80%.
> - CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

## SAQ 4 Microscopic / asymptomatic haematuria and urinary biomarkers

P2 · P3 **merged from 2**

*Asked as:* P3 SAQ "Asymptomatic haematuria and urinary biomarkers" (Asymptomatic Hematuria, Summer 2016 Paper III Q3d; Urinary biomarkers in diagnostic evaluation of hematuria, Summer 2021 Paper III Q3f) · P2 SAQ "Microscopic Hematuria" (Practice Set 1 Paper 2 Q3e; also 2016 Summer Paper 3 Q3d)

### Definitions and epidemiology

- **MH (microscopic haematuria)**: **3 or more RBCs/HPF (red blood cells per high-power field)** on microscopy of one properly collected sample (AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2020).
- **A dipstick alone is not enough**: myoglobin, semen and bacterial peroxidases give false positives.
- **GH (gross haematuria)**: visible; exclude pigmenturia and menstrual blood.
- Prevalence about 6.5%. Malignancy in about 3% of MH (over 85% of these are bladder cancer) and in 14–26% of GH (DETECT I, IDENTIFY).
- Haematuria is evaluated by **cystoscopy and upper tract imaging**; urine tests are non-invasive **adjuncts** for bladder cancer.

### Causes

- **Neoplasm**: bladder (over 85% of cancers found), UTUC (upper tract urothelial carcinoma) and RCC (renal cell carcinoma) about 1 in 1000 each, prostate.
- **Benign**: BPE (benign prostatic enlargement), stones, UTI (urinary tract infection), stricture, TB (tuberculosis), schistosomiasis.
- **Medical renal** (IgA nephropathy): dysmorphic RBCs, casts, proteinuria.
- **Others**: exercise, endometriosis, sickle cell.
- Anticoagulants do not explain haematuria; evaluate as usual.

### Initial evaluation

- History (smoking, occupation, cyclophosphamide, pelvic radiation, Lynch syndrome), examination, **blood pressure, creatinine**, urine culture.
- Suspected UTI: culture, treat, then **repeat urinalysis**; persistent MH means full evaluation.
- Glomerular or renal picture (dysmorphic RBCs, casts or proteinuria suggest a glomerular source): nephrology **and** urological evaluation.

### AUA/SUFU 2020 risk stratification

| | Low (all) | Intermediate (any) | High (any) |
|---|---|---|---|
| Age | Women under 50, men under 40 | Women 50–59, men 40–59 | 60 or over |
| Smoking (pack-years) | Never or under 10 | 10–30 | Over 30 |
| RBCs/HPF | 3–10 | 11–25 | Over 25 |

| | Low (all) | Intermediate (any) | High (any) |
|---|---|---|---|
| Gross haematuria history | No | No | **Yes** |
| Cancer risk | 0–0.4% | 1–2.5% | About 6% |
| **Work-up** | Shared decision: repeat urinalysis in 6 months **or** cystoscopy + RUS (renal ultrasound) | Cystoscopy + RUS | Cystoscopy + **CT (computed tomography) urography** |

- Additional risk factors (irritative symptoms, radiation, cyclophosphamide, occupational exposure) move a patient from low to intermediate.
- CT contraindicated: MR (magnetic resonance) urography, or retrograde pyelograms.
- Persistent MH after a negative ultrasound: CT or retrograde imaging.
- Hereditary RCC risk (VHL (von Hippel–Lindau)): upper tract imaging in any stratum.
- White-light cystoscopy. Cytology and urine markers are **not** used initially; cytology if CIS (carcinoma in situ) is suspected.
- **Gross haematuria**: cystoscopy, CT urography and cytology at any age.

### Urine cytology

- **Sensitivity 31–62%, specificity 94–100%**: good for **high-grade tumours and CIS**, poor for low grade.
- **Paris System:** negative for HGUC (high-grade urothelial carcinoma), atypical, suspicious, HGUC (graded by nuclear-to-cytoplasmic ratio and nuclear features).
- Positive cytology with normal cystoscopy: random and prostatic urethral biopsies, upper tract washings.

### Urinary biomarkers (sensitivity/specificity, Chou 2015 meta-analysis)

| Marker | Principle | Sens / spec | Note |
|---|---|---|---|
| **NMP22 (nuclear matrix protein 22)** | Nuclear protein; lab or point-of-care | 0.69/0.77 | False positives: inflammation, stones |
| **BTA (bladder tumour antigen)** stat / TRAK | Complement factor H–related protein; dipstick or ELISA (enzyme-linked immunosorbent assay) | 0.64/0.77 | **False positive with blood** (factor H in serum) |
| **UroVysion FISH (fluorescence in situ hybridisation)** | Gain of chromosomes 3, 7, 17; loss of 9p21 | 0.63/0.87 | After atypical cytology |
| **ImmunoCyt** | Antibodies to CEA (carcinoembryonic antigen) + two mucins | 0.78/0.78 | Highest sensitivity; operator dependent |
| **Cxbladder** | RT-PCR (reverse transcription polymerase chain reaction), 5 mRNAs | about 0.82/0.85 | Needs validation |

- **Also know:** Xpert Bladder (mRNA); AssureMDx (FGFR3, TERT, HRAS mutations + methylation).
- **Pattern:** more sensitive and less specific than cytology; they miss small low-grade tumours.

### Role of markers in haematuria (guidelines)

- **AUA/SUFU 2020: no cytology or markers in the initial evaluation of microhaematuria**; cytology only after a negative work-up if CIS is suspected.
- **Gross haematuria:** cytology with cystoscopy and CT urography. A negative marker still leaves **≥10% cancer probability**, so cystoscopy is still needed.
- **No marker replaces cystoscopy** (AUA, EAU (European Association of Urology)); a negative test may one day help triage low-risk patients.
- **Recent advances:** urinary tumour DNA; methylation assays (Bladder EpiCheck).

### Follow-up

- Negative evaluation: consider repeat urinalysis within 12 months; release if negative.
- Re-evaluate for gross haematuria, rising MH or new symptoms.

**Diagram to draw:** AUA/SUFU flowchart from MH to risk-based work-up (haematuria pathway), with markers shown as adjuncts.

> **Key points to remember:**
> - MH: 3 or more RBCs/HPF on microscopy of one sample; dipstick not enough.
> - Low: optional (repeat urinalysis or cystoscopy + ultrasound); intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
> - Any gross haematuria = high risk.
> - Cytology: specific for high grade/CIS, insensitive for low grade.
> - BTA is falsely positive with blood.
> - No markers in initial microhaematuria work-up; none replaces cystoscopy.

## SAQ 5 Hemostasis in laparoscopic surgery

P3 · P4 **merged from 2**

*Asked as:* P3 SAQ "Hemostasis in laparoscopic surgery." · P4 SAQ "Hemostasis in lap surgery"

### Principles

- Devices seal by **denaturing protein**. Plan from imaging.
- Keep a **haemorrhage tray** ready: laparoscopic Satinsky, 10-mm suction, vascular suture with Lapra-Ty, haemostatic agent.

### Effect of pneumoperitoneum

- $CO_2$ at 12–15 mmHg **tamponades venous bleeding**, hiding it.
- Raise the pressure briefly for venous bleeding: 25 mmHg (P4: 20–25 mmHg).
- Before exit, **lower the pressure and inspect** the operative site and each port to unmask hidden bleeders and port-site bleeding: about 5 mmHg (P4: 5–10 mmHg).

### Energy devices

| Device | Lateral thermal spread | Note |
|---|---|---|
| Monopolar | 5–7 mm | Capacitive coupling, insulation failure, direct coupling |
| Bipolar | 1–6 mm | Current only between the jaws |

| Device | Lateral thermal spread | Note |
|---|---|---|
| Vessel sealer (LigaSure) | 1–3 mm | Seals vessels **≤7 mm**; one application only |
| Ultrasonic (Harmonic) | ≈1 mm | No current; blade >200 °C, cools slowly |
| Argon beam | Superficial | Raw surfaces; gas embolism risk |

### Clips

- **Titanium clips:** no diathermy near them (they slip).
- **Hem-o-lok** (locking polymer): circumferential dissection, see the tip beyond the vessel, feel the snap, at least two clips on the patient side with a 2-mm cuff.
- **FDA (Food and Drug Administration) warning:** after donor renal artery stump haemorrhages, the manufacturer **contraindicated Hem-o-lok on the living-donor renal artery**. A later series found it safe with correct technique, but use a stapler there.
- **Lapra-Ty** absorbable clips act as knots (renorrhaphy).

### Staplers

- Endo GIA through a 12-mm port; **grey (2.0 mm) or white (2.5 mm)** loads for renal vessels; blue or green for thicker tissue.
- **Never fire across clips** (misfire). Reinforce a bleeding staple line with clips.

### Sutures and haemostatic agents

- Intracorporeal or barbed sutures.
- Adjuncts: fibrin sealant, gelatin–thrombin matrix, oxidised cellulose.

### Managing bleeding

- **Venous injury** (algorithm): 1. Raise pneumoperitoneum pressure (25 mmHg; P4: 20–25 mmHg). 2. Suction to identify the bleeding point. 3. Pledget tamponade. 4. Sealant. 5. Extra 5-mm port for suturing or Satinsky. 6. Hand-assist. 7. Convert.
- Keep the patient well filled (gas embolism).
- **Arterial:** minor bleeding, tamponade; renal artery, staple; aorta, convert.
- **Port-site bleeding:** cantilever the trocar to find it; diathermy or transfascial suture. Blunt trocars reduce it.

**Diagram to draw:** Hem-o-lok placement with two patient-side clips and a 2-mm cuff.

> **Key points to remember:**
> - Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
> - LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
> - Hem-o-lok contraindicated on the donor renal artery.
> - Never staple across clips; grey or white loads for vessels.

## SAQ 6 Newer generation CT scan

P1

**Type:** SAQ (10 marks)
**Source:** 2015-Summer Paper 1, Q3a

## Introduction

- Computed tomography (CT) is the reference imaging for stones, masses and trauma. Attenuation is expressed in Hounsfield units (HU): water 0, air −1000.

## Evolution

| Generation | Feature |
|---|---|
| Helical (~1989) | Continuous table movement, volumetric data |
| Multidetector (1998 → 320-row) | Sub-millimetre slices, 3D reconstruction |
| Dual-source/dual-energy (2005) | Two energies: material analysis |
| Photon-counting (US approval 2021) | Energy-sorts each photon: less noise, dose and metal artefact |

## Dual-energy CT (DECT)

- Two energies (e.g. 80 and 140 kVp) compared pixel by pixel: **uric acid vs non-uric-acid stones**, **virtual unenhanced** images (one phase fewer) and **iodine maps** (hyperdense cyst vs enhancing mass).

## Low-dose CT

| Protocol | Dose |
|---|---|
| Conventional abdomen/pelvis | 10–20 mSv |
| Low-dose | <3–4 mSv |
| Ultra-low-dose | 1–2 mSv |
| - Stone sensitivity stays ~95%; use standard dose if body mass index >30. EAU (European Association of Urology): ultrasound first; AUA (American Urological Association): CT first. | |

## Reconstruction

- Filtered back projection is noisy; iterative reconstruction halves dose; **deep learning reconstruction** gives good images at ~1 mSv.

## Contrast phases and protocols

| Phase | Timing | Use |
|---|---|---|
| Arterial | 20–30 s | Renal arteries, donor anatomy, active bleeding |
| Corticomedullary | 30–70 s | Hypervascular (clear cell) tumours, renal vein |
| Nephrographic | 90–120 s | Most sensitive for renal masses |
| Excretory | 7–10 min | Collecting system, ureter, urine leak |

| Phase | Timing | Use |
|---|---|---|
| - **CT urography:** non-contrast, nephrographic and excretory phases (~20–30 mSv). **Split-bolus** merges phases, lowering dose but may miss subtle urothelial lesions. | | |
| - **Renal mass:** enhancement ≥20 HU = solid, 10–20 equivocal, <10 cyst; Bosniak 2019 for cysts. | | |
| - **Adrenal:** <10 HU unenhanced = lipid-rich adenoma; otherwise 15-minute washout (absolute >60% or relative >40% = adenoma). | | |
| - **Also know:** stone HU (uric acid 200–400, calcium oxalate 600–1200; indinavir invisible); bone window for fragments; partial volume causes pseudoenhancement of small cysts; ALARA (as low as reasonably achievable) dosing, since radiation risk has no threshold. | | |

**Diagram to draw:** Timeline: axial → helical → multidetector → dual-source → photon-counting.

> **Key points to remember:**
> - Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
> - DECT: uric acid stones, virtual unenhanced, iodine maps.
> - Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

## SAQ 7 History of Laparoscopy

P1

**Type:** SAQ (10 marks)
**Source:** 2015-Summer Paper 1, Q3b

### Introduction

- Laparoscopy is endoscopic viewing of the peritoneal cavity through a pneumoperitoneum; it grew from optics to robotic surgery.

### Optics and access

| Year | Pioneer | Milestone |
|---|---|---|
| **1806** | **Bozzini** | **Lichtleiter**, candle-lit light conductor |
| 1877 | Nitze | Distal electric light and lenses |
| **1901** | **Kelling** | Insufflated abdomen of a **dog** |
| **1910** | **Jacobaeus** | First human series; "laparoscopy" |
| **1938** | **Veress** | Spring-loaded needle |
| **1966** | **Hopkins** | **Rod-lens system** |
| 1970s–80s | **Semm** | Automatic insufflator; operative laparoscopy |
| 1970s | Hasson | Open access |
| 1987 | Mouret | Video cholecystectomy |

| Year | Pioneer | Milestone |
|---|---|---|
| - **Also know:** Desormeaux 1853, first therapeutic endoscopy; fibreoptics (Tyndall's principle) for flexible scopes; $CO_2$ became the standard insufflant. | | |

### Urological milestones

| Year | Pioneer | Milestone |
|---|---|---|
| **1976** | **Cortesi** | Impalpable testis, first urological use |
| **1990–91** | **Clayman** | **First laparoscopic nephrectomy**; nephroureterectomy 1991 |
| **1992** | **Gaur (India)** | **Balloon retroperitoneal access** |
| **1995** | **Ratner** | Live-donor nephrectomy |
| 1997 | Schuessler | Laparoscopic radical prostatectomy (slow, not adopted) |
| 2000 | Guillonneau & Vallancien | Standardised prostatectomy |
| **2000** | **da Vinci**; **Binder** | Robot cleared in US; first robotic prostatectomy |
| 2000s | **Menon** | Vattikuti Institute Prostatectomy |
| 2007 | Raman; Rane | Single-port nephrectomy |

### Why the robot spread

- Laparoscopic prostatectomy stalled because intracorporeal suturing was hard. The robot gave wristed instruments and a 3D magnified view.
- da Vinci generations: S 2006 (fourth arm), Si 2009 (dual console), Xi 2014 (flexible docking), SP 2018 (single port).

### Indian contribution and recent advances

- Gaur's balloon made retroperitoneoscopy practical; indigenous robot SSI Mantra with telesurgery.
- New robots (Versius, Hugo, Hinotori), telesurgery, haptic feedback.

**Diagram to draw:** Timeline 1806 → 1901 → 1910 → 1938 → 1990 → 2000.

> **Key points to remember:**
> - Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
> - Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
> - da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

## SAQ 8 Foley's (urethral) catheters, guidewires and DJ stents with their complications

P1

**Asked as:**

- Foley's catheter. — *Summer 2016 Paper I Q3e (SAQ)*
- Urethral catheters, guidewires and DJ stents — *Practice Set 18 SAQ Q3f (SAQ)* (also answered in part in 2016-Summer Paper 1, Q3e and 2015-Summer Paper 3, Q1)

- Complications if DJS — *Summer 2021 Paper I Q3c (SAQ) — Taken as complications of a DJ (double-J) ureteric stent.*

**Type:** SAQ (10 marks)

## Foley's / urethral catheters

### History

- **Frederic Foley**, a Boston urologist, developed the modern balloon catheter (Campbell: 1929; published in the 1930s) for **haemostasis after prostatectomy**. Reybard (1855) had made the first self-retaining one.

### Design

- **Size: Fr (French) = 3 × diameter in mm.** Adults (adult) usually 14–16 Fr; clot retention 20–24 Fr three-way. Newborn 6 Fr.
- Normal calibre: (male) meatus 24 Fr (narrowest point), bladder neck 28 Fr, prostatic urethra 32 Fr, female urethra 22 Fr.
- **Balloon:** usually 10 mL; larger (30 mL) for traction. Use (inflate balloon with) **sterile water**, only after urine flows at the hub; volume is printed on the port.
- **Lumens:** one (intermittent, specimen); two-way (drainage + balloon) is routine; three-way adds irrigation for haematuria (stiff wall resists collapse). Each lumen reduces (shrinks) drainage size.
- **Tips:** straight; **coudé** (curved, for prostatic enlargement, tip up); **Councill** (open tip, over a wire).
- **Materials:** latex (cheap, allergy); silicone (long term, less urethritis and encrustation; better urethral tolerance); PVC (polyvinyl chloride); coatings (silver, antibiotic) reduce bacteriuria but not clinical infection.
- **Special:** Malecot (winged, no balloon); Lapides diagnostic (rings 1 cm apart); urodynamic.

### Indications and contraindications

- **Indications:** retention, clot retention, output monitoring, long surgery, immobility, after urological surgery.
- **Absolute contraindication:** suspected urethral injury (blood at meatus, pelvic fracture). Do an RUG (retrograde urethrogram) first (**suspected urethral injury → RUG first**).
- **Relative:** AUS (artificial urinary sphincter): deactivate, then 12–14 Fr; stricture.

### Technique

- Aseptic; 10–15 mL lignocaine gel. In men, advance **to the hub** and see urine before inflating. Replace the foreskin.
- **Difficult:** 18 Fr coudé → guidewire → cystoscopy, Councill over wire → suprapubic catheter.

### Complications

- **CAUTI (catheter-associated urinary tract infection):** 75% of hospital UTIs; bacteriuria 5–10% per catheter-day; **biofilm** forms within minutes.
- Trauma (about 0.3% of inpatients): false passage, balloon inflated in the urethra. Also stricture, traumatic hypospadias, encrustation, paraphimosis, non-deflating balloon (puncture with a wire).

### CAUTI prevention

- Review daily and remove early; closed dependent system; clean meatus with water; culture only if symptomatic. Treat only symptomatic infection and **change the catheter**.

## Guidewires

| Feature | Details |
|---|---|
| Uses | Access, safety wire, straightening ureter, guiding dilation and stents |
| Construction | Stainless-steel core with outer wrap; **nitinol** core is kink-resistant |
| Coating | PTFE (polytetrafluoroethylene): standard safety wire. **Hydrophilic** "glide": passes impacted stones, but too slippery to be a safety wire. Hybrid: hydrophilic tip, PTFE shaft |
| Size | 0.018–0.038 inch (commonly 0.035/0.038); 80–260 cm, 145 cm most useful; floppy tip 1–3 cm |
| Tip and stiffness | Straight, angled, J; standard to super-stiff (straightens tortuous ureter) |

## DJ (double-J) stents: introduction

- The **double-J (double-pigtail)** stent (Finney 1978) is a self-retaining polymer tube with coils in the pelvis and bladder; adult usually 6 Fr, length by height. Consent should cover its symptoms and complications.
- **Materials:** polyurethane, silicone (hydrocoated silicone less painful); **metallic (Resonance)** for extrinsic malignant compression.
- **Indications:** obstruction (infected obstructed kidney, malignancy, pregnancy), after URS (ureteroscopy) with injury, pyeloplasty, transplant, leak.
- **Placement:** retrograde pyelogram → wire → stent over wire under fluoroscopy or vision.
- **Problems:** stent symptoms (bladder irritation and reflux) eased by α-blockers and antimuscarinics; migration; encrustation (forgotten stent); failure with extrinsic compression.

### Classification of DJ stent complications

| Timing | Complications |
|---|---|
| **At insertion** | **Misplacement**: proximal end below the obstruction; **submucosal** passage or perforation (suspect when force is needed) |
| **Early** | **Stent-related symptoms**; haematuria; infection and urosepsis; **vesicoureteric reflux** up the stent (flank pain on voiding) |
| **Late** | **Migration** (proximal or distal; short stent); **encrustation**; **stent failure** (obstruction); biofilm and colonisation; **fracture**; **forgotten stent**; rarely **ureteroarterial fistula** |

### Stent-related symptoms

- About **4 in 5 patients**: urgency, frequency, dysuria, haematuria, flank and groin pain; sexual and work problems.
- **Mechanism:** the **intravesical coil irritates the trigone**, and **reflux up the stent** distends the kidney on voiding.
- Measured by **USSQ** (Ureteral Stent Symptom Questionnaire).
- **Reduced by: α-blockers and antimuscarinics**; intraureteral placement; hydrocoated silicone; shortest dwell. Diameter does not matter.

**Encrustation and forgotten stent**

- Minerals and biofilm deposit on the stent. Risk rises with **dwell time**, infection, stone-forming urine and pregnancy. Leads to obstruction, sepsis, fracture, **ureteric avulsion on removal**, renal loss.
- **FECal grading (Forgotten, Encrusted, Calcified):** I minimal linear; II circular; III partial on one or both coils; IV complete on one coil; V both coils.
- **Management:** non-contrast CT (computed tomography); never pull by force; clear **bladder coil** (cystolithotripsy), then **ureter** (ureteroscopy), then **renal coil** (flexible ureteroscopy or PCNL [percutaneous nephrolithotomy]).

**Stent failure**

- Extrinsic compression (malignancy, radiation), debris, encrustation. Obstructive symptoms → image; change stent, **PCN** (percutaneous nephrostomy) or **metallic stent**.

**Prevention**

- Stent only when indicated; correct length; **stent register**; long-term stents changed every 3–6 months (local practice); **pregnancy: about every 4 weeks**.

**Diagram to draw:** cross-section of a three-way Foley; DJ stent in situ with bladder coil on the trigone and arrows showing reflux.

> **Key points to remember:**
>
> - Foley balloon catheter: designed for post-prostatectomy haemostasis.
> - Fr = 3 × diameter (mm); inflate with sterile water; coudé for prostatic enlargement, Councill over a wire.
> - Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
> - CAUTI 5–10% per day; remove early; change the catheter when treating.
> - Hydrophilic wire to pass, PTFE wire to stay (safety).
> - DJ symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
> - Long-dwelling stents: monitor for encrustation; encrustation rises with dwell time; FECal I–V.
> - Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

## SAQ 9 Proteinuria

P1

**Asked as:**

- Proteinuria — *Summer 2017 Paper I Q3f (SAQ)*
- Proteinuria — *Summer 2021 Paper I Q3d (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2017-Summer Paper 1, Q3f)

### Definition

- Normal protein excretion is **<150 mg/day** (another source: **80–150 mg/day**) (albumin <30 mg/day). The filter (glomerulus) blocks (holds back) albumin and the proximal tubule reabsorbs filtered small proteins.

- Normal mix: about 30% albumin, 30% globulins, 40% tissue proteins (Tamm–Horsfall).
- **Nephrotic range:** >3.5 g/day, with hypoalbuminaemia, oedema and hyperlipidaemia.

### Detection and quantification (KDIGO, Kidney Disease: Improving Global Outcomes, categories by ACR)

| Category | ACR (albumin-to-creatinine ratio) | PCR (protein-to-creatinine ratio) | 24-h protein |
|---|---|---|---|
| A1 normal–mild | <30 mg/g (<3 mg/mmol) | <150 mg/g (<15 mg/mmol) | <150 mg |
| A2 moderate | 30–300 mg/g | 150–500 mg/g | 150–500 mg |
| A3 severe | >300 mg/g (>30 mg/mmol) | >500 mg/g | >500 mg |
| - **Spot early-morning ACR is first line**; PCR if non-albumin protein suspected; 24-h urine with creatinine check. Alternatives: PCR, 24-h urine. | | | |
| - **Dipstick** (tetrabromophenol blue) detects albumin only (mainly albumin), from 20–30 mg/dL; misses dilute urine and **light chains**. **Sulfosalicylic acid** detects (precipitates) all proteins; dipstick negative but sulfosalicylic acid positive → suspect myeloma. | | | |

### Types

| Type | Mechanism | Amount | Example |
|---|---|---|---|
| **Glomerular** (commonest) | Leaky filter | Often >2 g; albumin ≈70% (about 70%) | Diabetes, glomerulonephritis, FSGS (focal segmental glomerulosclerosis), amyloid |
| **Tubular** | Failed reabsorption of small proteins | <2 g (seldom >2–3 g); mainly globulins | Fanconi syndrome, heavy metals, drugs |
| **Overflow** | Excess plasma protein | Variable (0.3–2 g) | **Myeloma** (Bence Jones), myoglobin, haemoglobin |

| By timing | Features |
|---|---|
| **Transient** | Fever, exercise, heart failure; commonest in children |
| **Orthostatic** | Young men, upright only, <1 g; benign (normal first-morning sample). Test with a split (supine overnight vs upright day) collection |
| **Persistent** | Mostly glomerular; needs full work-up |

### Evaluation

`Positive dipstick → repeat; exclude UTI (urinary tract infection), fever, exercise → persistent → quantify (ACR, PCR or 24-h protein) → creatinine/eGFR (estimated glomerular filtration rate), sediment (dysmorphic red cells / RBCs, casts), ultrasound → glomerular pattern (>2 g, albumin) → nephrology ± renal`

`biopsy; C3, C4, ANA if systemic → globulins → electrophoresis, immunofixation (myeloma)`

- UTI: urinary tract infection; eGFR: estimated glomerular filtration rate; RBC: red blood cell; ANA: antinuclear antibody.

### Urological relevance

- **Haematuria work-up:** proteinuria, casts or dysmorphic cells suggest (suggests) medical renal disease; urological bleeding rarely raises protein much (adds little protein).
- **Reflux nephropathy:** proteinuria marks progression to renal failure.
- **After nephrectomy or partial nephrectomy:** >50% loss of mass causes hyperfiltration, proteinuria and FSGS. AUA (American Urological Association) 2021 counts proteinuria as a reason to prefer (favour) nephron sparing and refer to nephrology.
- **Living donors:** significant proteinuria excludes donation. **Also know:** mTOR (mammalian target of rapamycin) inhibitors cause it.

### Management

- Treat the cause; blood pressure control; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)**; SGLT2 (sodium–glucose cotransporter-2) inhibitors.

**Diagram to draw:** glomerulus and proximal tubule marking glomerular, tubular and overflow sites.

> **Key points to remember:**
> - Normal <150 mg/day; nephrotic >3.5 g/day.
> - ACR A2 30–300, A3 >300 mg/g.
> - Glomerular, tubular, overflow; orthostatic proteinuria is benign.
> - Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

## SAQ 10 Health-related quality of life (HRQOL) instruments in urology, including after urinary diversion

P1

**Asked as:**

- Health related quality of life instruments (HRQOL) in urology. — *Summer 2018 Paper I Q3c (SAQ)*
- Assessment of quality of life after urinary diversion — *Practice Set 3 Paper I Q3c (SAQ)*

**Type:** SAQ (10 marks)

### Definition

- **HRQOL (HRQoL, health-related quality of life):** the parts of life affected by health: physical, emotional and social function and the patient's bother; the patient's view of how disease and treatment affect physical, emotional and social function, plus **bother**. It differs from a **symptom score**, which only counts symptoms.
- Must be **patient-reported**; doctors underestimate dysfunction (urinary, sexual and bowel dysfunction).

## Types of instruments

| Type | Measures | Examples | Limitation |
|---|---|---|---|
| **Generic** | Domains common to all diseases; allow comparison across diseases | **SF-36** (Short Form-36), SF-12, **EQ-5D** (EuroQol five-dimension; 5 items + VAS (visual analogue scale); gives utilities for QALYs (quality-adjusted life-years)), Sickness Impact Profile | Insensitive to diversion-specific issues |
| **Cancer-specific** | Cancer effects | **FACT-G** (Functional Assessment of Cancer Therapy–General), **EORTC QLQ-C30** (European Organisation for Research and Treatment of Cancer core questionnaire) | General cancer domains |
| **Disease-specific** | Domains of one disease; more responsive | See below | |
| **Bladder-cancer-specific** | | **FACT-BI** (40 items), **EORTC QLQ-BLM30** (muscle-invasive module, 30 items), **BCI (Bladder Cancer Index**, 34 items: urinary, bowel, sexual domains), FACT-VCI (Vanderbilt Cystectomy Index) | Weight symptoms of one diversion, so biased in cross-diversion comparisons |
| **Neobladder-specific** | | IONB-PRO (23 items) | Not usable across diversions |

## Urology-specific instruments

- **Prostate cancer: EPIC (Expanded Prostate Cancer Index Composite)**, EPIC-26, UCLA (University of California, Los Angeles) Prostate Cancer Index, FACT-P.
- **Bladder cancer:** FACT-BI, **Bladder Cancer Index**, EORTC QLQ-BLM30 and NMIBC-24, FACT-Vanderbilt Cystectomy Index.
- **Kidney cancer:** FKSI (FACT–Kidney Symptom Index).
- **LUTS (lower urinary tract symptoms): IPSS (International Prostate Symptom Score)** 7 items + 1 quality-of-life item; BPH Impact Index.
- **Incontinence and OAB (overactive bladder): ICIQ** (International Consultation on Incontinence Questionnaire), **OAB-q**, King's Health Questionnaire, I-QOL.
- **Sexual function: IIEF** (International Index of Erectile Function; 15 items), SHIM (5 items; no bother, so not true HRQOL), FSFI (Female Sexual Function Index).
- **Others:** NIH-CPSI (chronic prostatitis), O'Leary–Sant (interstitial cystitis), Wisconsin Stone QOL, USSQ (ureteric stent).
- **PROMIS (Patient-Reported Outcomes Measurement Information System):** free item banks, **computer-adaptive testing**, common T-score.

## Psychometric properties (choosing a tool)

- A good tool is **reliable** (Cronbach α >0.7), **valid** in that population and **responsive** to change.

| Property | Meaning | Threshold |
|---|---|---|
| Internal consistency | Items agree | Cronbach α >0.7 (>0.9 = redundant) |

| Property | Meaning | Threshold |
|---|---|---|
| Test–retest reliability | Stable over time | r >0.7 |
| Content (face) validity | Right items present | Expert judgement |
| Criterion validity | Agrees with gold standard | r ≥0.7 |
| Construct validity | Convergent and divergent | Correlations |
| **Responsiveness** | Detects real change | **MCID (minimal clinically important difference)**: AUA (American Urological Association) Symptom Index 3.1 points; otherwise ≥½ SD |

- Validity is population-specific; translations need revalidation.
- Development: define domains → generate items → pilot → psychometric testing (Likert scales).

## Quality of life after urinary diversion

### Domains to assess

- Urinary: continence, catheterisation, stoma problems.
- **Body image**, sexual function, bowel function, social and emotional function, global HRQoL.

### Evidence: ileal conduit versus neobladder

- **No diversion proven superior.** Studies are mostly cross-sectional and retrospective, without baseline data or randomisation.
- Verdict depends on instrument: generic tools mostly show **no difference** (largest group of studies); EORTC tools tend to **favour neobladder**; the urinary-focused BCI **penalises neobladder** for leakage.
- **Consistent signal:** body image favours neobladder; urinary function favours conduit.
- Large prospective cohort: global HRQoL exceeded baseline in all diversions by 6 months.
- Neobladder: daytime continence improves over 6–12 months; **night-time incontinence** (up to 50%) is the main bother.
- **Also know:** informed patients have less decision regret; most adapt to their diversion; PROMIS (Patient-Reported Outcomes Measurement Information System).

### Practical approach

- Baseline and serial validated questionnaires (3, 6, 12 months), a disease-specific plus a generic tool, and realistic counselling.

**Diagram to draw:** Pyramid: symptoms → function → bother → overall HRQOL. Bar sketch: body image (neobladder better) versus urinary function (conduit better).

**Key points to remember:**

- Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF) plus bladder-specific (FACT-BI, BLM30, BCI) tools.
- SHIM lacks bother; I-QOL lacks symptoms.
- Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

## SAQ 11 PET scans in urology.

P1

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 1, Q3e

### Principle

- **PET (positron emission tomography):** tracer emits a positron → annihilation with an electron → **two 511-keV photons at 180°** → coincidence detection. Fused with CT (computed tomography) or MRI.
- Quantitative (**SUV, standardised uptake value**); costly. Half-lives: $^{18}$F 110 min, $^{68}$Ga 68 min, $^{11}$C 20 min (on-site cyclotron).

### Tracers

| Tracer | Mechanism | Main use |
|---|---|---|
| **$^{18}$F-FDG (fluorodeoxyglucose)** | Glucose analogue trapped after phosphorylation (Warburg effect) | Bladder, UTUC (upper tract urothelial carcinoma), penile, seminoma |
| **PSMA (prostate-specific membrane antigen)** ligands ($^{68}$Ga-PSMA-11, $^{18}$F-DCFPyL, $^{18}$F-PSMA-1007, rhPSMA-7.3) | Bind cell-surface PSMA | Prostate staging and recurrence |
| $^{11}$C/$^{18}$F-choline | Cell membrane synthesis | Prostate recurrence (older) |
| $^{18}$F-fluciclovine | Amino acid transport | Prostate recurrence |
| $^{18}$F-NaF (sodium fluoride) | Bone turnover | Bone metastases |

### Uses by organ

#### Prostate cancer

- FDG is poor (low glucose metabolism). PSMA has specificity >90%.
- **Primary staging (high risk): proPSMA (2020)**: PSMA PET/CT more accurate than CT plus bone scan (92% vs 65%). Nodal sensitivity only 30–40%, so a negative scan does not exclude micrometastases.
- **BCR (biochemical recurrence):** detection rises with PSA (prostate-specific antigen); about half positive below 0.5 ng/mL. PSMA beats fluciclovine (56% vs 26%). CONDOR: management changed in 63.9%.

- **Theranostics:** PSMA-positive mCRPC (metastatic castration-resistant prostate cancer) → **$^{177}$Lu-PSMA-617 (VISION 2021)**.
- NCCN (National Comprehensive Cancer Network): PSMA PET preferred; outcome benefit still unproven.

**Testis**

- **Seminoma residual mass >3 cm:** FDG-PET **≥6 weeks after chemotherapy**. Negative scan reliable (observe); positive → consider resection.
- **Not for NSGCT (non-seminomatous germ cell tumour):** teratoma is not FDG-avid.

**Bladder and UTUC**

- FDG for nodal and distant staging in MIBC (muscle-invasive bladder cancer): modest sensitivity, high specificity; urinary excretion hides the primary. Use when CT is equivocal.

**Kidney**

- FDG not routine (clear cell RCC (renal cell carcinoma) poorly avid). **$^{89}$Zr-girentuximab** (anti-CAIX, carbonic anhydrase IX) identifies clear cell RCC (ZIRCON).

**Penis**

- FDG confirms metastases in **palpable inguinal nodes**; poor for impalpable nodes.
- **Also know:** EMPIRE-1 (fluciclovine-guided salvage radiotherapy); false positives from inflammation; neuroendocrine cancer loses PSMA.

**Diagram to draw:** Positron annihilation with two 511-keV photons hitting a detector ring.

> **Key points to remember:**
> - proPSMA: PSMA PET 92% vs 65% accuracy.
> - Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
> - FDG poor in prostate and clear cell RCC.
> - VISION: $^{177}$Lu-PSMA-617 in mCRPC.

## SAQ 12 Recist Criteria.

P1

**Type:** SAQ (10 marks)
**Source:** 2024-Winter Paper 1, Q3a

### Definition and evolution

- **RECIST (Response Evaluation Criteria In Solid Tumours):** standard imaging rules for tumour response and progression.
- **WHO (World Health Organization) 1981** (bidimensional) → **RECIST 2000** (unidimensional) → **RECIST 1.1 (2009)**, now standard. Used in trials for metastatic RCC (renal cell carcinoma), urothelial and prostate cancer.

### Lesions at baseline (RECIST 1.1)

| Category | Definition |
|---|---|
| Measurable lesion | Longest diameter **≥10 mm on CT (computed tomography)**, ≥20 mm on chest X-ray, ≥10 mm with callipers |

| Category | Definition |
|---|---|
| Measurable node | **≥15 mm short axis**; 10–15 mm pathological but non-measurable; <10 mm normal |
| Non-measurable | Small lesions, blastic bone metastases, effusions, ascites |
| Target lesions | Up to **5 (maximum 2 per organ)**; largest, reproducible; **sum of diameters** recorded |
| Non-target | All other lesions, recorded as present or absent |

### Response categories

| Response | Target lesions |
|---|---|
| **CR** (complete response) | All disappear; nodes <10 mm |
| **PR** (partial response) | **≥30% fall** in sum versus **baseline** |
| **PD** (progressive disease) | **≥20% rise** versus **nadir** and **≥5 mm absolute**, or any **new lesion** |
| **SD** (stable disease) | Neither PR nor PD |

- **Overall response** combines target, non-target and new lesions; confirm response at ≥4 weeks in single-arm trials.
- **End points:** ORR (objective response rate = CR + PR), PFS (progression-free survival), duration of response.

### Limitations

- Size only, ignoring necrosis (TKIs (tyrosine kinase inhibitors)); bone poorly assessed; observer variability.

### Modifications

| Criteria | Use |
|---|---|
| **iRECIST (2017)** | Immunotherapy: first progression is "unconfirmed", confirmed only on the next scan, allowing for **pseudoprogression** |
| **Choi** | Density fall counts as response |
| **PCWG3 (Prostate Cancer Working Group 3)** | Bone: 12-week scan is the new baseline (flare), then **≥2 new lesions** = progression; PSA (prostate-specific antigen) progression ≥25% and ≥2 ng/mL above nadir; soft tissue by RECIST |
| PERCIST | PET (positron emission tomography) metabolic response |

**Diagram to draw:** number line of change in sum: −30% (PR), +20% (PD).

> **Key points to remember:**
> - RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
> - PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
> - iRECIST for immunotherapy; PCWG3 for prostate.

## SAQ 13 Outline algorithm for difficult catheter placement.

P1

**Type:** SAQ (10 marks)
**Source:** 2024-Winter Paper 1, Q3b

## Introduction

- **Normal calibre:** meatus 24 Fr (narrowest), bladder neck 28 Fr, prostatic urethra 32 Fr.
- **Causes:** BPH (benign prostatic hyperplasia, commonest), urethral stricture, false passage, meatal stenosis, bladder neck contracture, urethral trauma, abnormal anatomy.
- **Golden rule:** escalate **blind → wire → vision → suprapubic**; never repeat forceful blind passage.

## Before starting

- History (surgery, radiotherapy, AUS (artificial urinary sphincter), reconstruction), DRE (digital rectal examination).
- **Absolute contraindication:** suspected urethral injury (pelvic fracture, blood at meatus, high-riding prostate) → **RUG (retrograde urethrogram)**; pelvic fracture urethral injury → suprapubic tube.
- **Technique:** 10–15 mL lubricant (lidocaine gel), penis stretched, steady pressure; inflate only after urine flows at the hub.

## Algorithm (Campbell Fig. 15.6)

```
First attempt fails ├─ Prostatic obstruction suspected │ → 18-Fr coudé, tip (marker) pointing up │ → fails → flexible cystoscopy └─ Not suspected (stricture likely) → smaller catheter → fails → hydrophilic glidewire → fails → flexible cystoscopy At cystoscopy / with wire in bladder: ├─ Stricture → wire across → 5-Fr catheter, aspirate urine to confirm │ → super-stiff wire → serial dilatation (1 Fr above catheter size) │ → Councill (open-tip) catheter over wire └─ False passage → pass under vision (scope anterior) → wire → Councill over wire Wire cannot be passed / dense stricture / urethral disruption → suprapubic catheter (ultrasound-guided percutaneous, or open)
```

- **Why 18-Fr coudé:** stiffer, larger tip area, curved tip follows the bulbar-prostatic angle; 14–16 Fr usually buckle.
- **Blood on the tip** suggests false passage: stop blind attempts.

## Special situations

| Situation | Approach |
|---|---|
| AUS | Deactivate; smallest catheter 12–14 Fr; remove early |
| Obese or female | Exposure: Trendelenburg, retract pannus, two fingers in vagina to guide |
| Buried penis | Infrapubic pressure; flexible cystoscopy |
| Neobladder, continent pouch | Never force; scope + wire; suprapubic only with imaging guidance |
| PUV (posterior urethral valves) neonate | 6–8 Fr, or 5-Fr open-ended over glidewire; else suprapubic or vesicostomy |

**Also know:** filiforms, balloon dilators; complications: false passage, haematuria, stricture.

**Diagram to draw:** coudé catheter following the bulbar-prostatic curve with tip up.

**Key points to remember:**

- Blind → wire → vision → suprapubic.
- BPH: 18-Fr coudé, tip up.
- Councill catheter over wire after dilatation.
- Suspected urethral injury: RUG first, no blind catheter.

## SAQ 14 Indications for outpatient cystourethroscopy.

P1

**Type:** SAQ (10 marks)
**Source:** 2024-Winter Paper 1, Q3e

### Introduction

- **Flexible cystourethroscopy** under local anaesthesia is the **commonest office procedure** in urology; no lithotomy needed.
- **Prerequisites:** consent; treat any UTI (urinary tract infection) first; lidocaine gel with **≥5 minutes dwell** (ideally 10–20); no antibiotic prophylaxis unless risk factors (age, diabetes, immunosuppression, catheters, prior UTI).

### Diagnostic indications

| Indication | Comment |
|---|---|
| **Haematuria** | All **gross haematuria** (with CT (computed tomography) urography, cytology). Microhaematuria by AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) risk: intermediate and high risk; shared decision in low risk |
| **Bladder cancer surveillance** | NMIBC (non-muscle-invasive bladder cancer): first check at 3 months; low risk then yearly; high risk 3–4-monthly for 2 years; also upper-tract follow-up of UTUC (upper tract urothelial carcinoma) |
| Recurrent UTI | Look for stone, diverticulum, stricture, fistula, foreign body |
| LUTS (lower urinary tract symptoms) | Not routine; before invasive BPH (benign prostatic hyperplasia) treatment (median lobe before urethral lift, bladder-neck contracture, stricture) |
| Pain syndromes | IC/BPS (interstitial cystitis/bladder pain syndrome): **Hunner lesion**; chronic pelvic pain with bladder pain or low flow |
| Women | Suspected urethral diverticulum, vesicovaginal fistula, mesh or suture erosion after sling, recurrent UTI after surgery |
| Others | Urethral stricture assessment and follow-up; abnormal urine cytology; radiation cystitis; foreign body |

### Therapeutic indications

- **Removal or exchange of ureteric stents** (exchange with fluoroscopy).
- **Biopsy and fulguration** of small (≤1 cm) low-grade recurrences; office laser ablation.
- **Intradetrusor botulinum toxin** injection (OAB (overactive bladder), neurogenic detrusor overactivity).
- Urethral bulking agents; stricture dilatation; small stones or foreign bodies; blue-light cystoscopy where available.

### When not indicated or done in theatre

- Not needed for the index stress incontinence patient, initial OAB or uncomplicated LUTS.
- **No surveillance cystoscopy** in chronic indwelling catheters (AUA/SUFU 2021).
- Theatre if TURBT (transurethral resection of bladder tumour) needed, poor tolerance, children, or complex diversions.

**Diagram to draw:** haematuria risk-group flowchart.

> **Key points to remember:**
> - Haematuria and NMIBC surveillance are the two commonest indications.
> - Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
> - Therapeutic: stent removal, botulinum toxin, fulguration of small recurrences.

## SAQ 15 MR urography

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 1 Paper 1, Q3a

### Definition

- MRU (magnetic resonance urography): magnetic resonance imaging of kidneys, ureters and bladder giving anatomy and function **without ionising radiation**.

### Techniques

| | Static MRU | Excretory MRU |
|---|---|---|
| Sequence | Heavily T2-weighted; urine is bright | Gadolinium-enhanced T1 |
| Contrast | None | GBCA (gadolinium-based contrast agent) |
| Depends on function | No | Yes; low-dose furosemide improves it |
| Best for | Dilated or obstructed system | Non-dilated system, excretion |

- **Functional MRU (children):** hydration, bladder catheter, furosemide 1 mg/kg (maximum 20 mg), T2 images, then dynamic T1 after slow contrast injection. Patlak analysis gives differential function, GFR (glomerular filtration rate) and drainage.

### Appearances

- Stones and calcification give **signal voids**; tumour, clot, gas and sloughed papillae are low-signal defects in bright urine.

### Indications

- CT (computed tomography) urography contraindicated: contrast allergy, renal insufficiency, **pregnancy** (separates physiological from calculous dilatation; small stones missed).
- **Children:** complex anomalies, ectopic ureter in the incontinent girl, duplex systems, megaureter, crossing vessels in UPJ (ureteropelvic junction) obstruction, ectopic kidneys.
- Hydronephrosis: alternative to MAG3 (mercaptoacetyltriglycine) renography; equivalence not yet proven.
- Upper tract tumours and renal masses.

### Advantages and limitations

| Advantages | Limitations |
|---|---|
| No radiation; best anatomical detail | Cost, availability, long study (up to 1 hour) |
| Anatomy, function and drainage in one test | Sedation often needed under 6 years |
| As accurate as CTU (CT urography) for obstruction | Poor for stones; complex post-processing |
| Safe in iodinated contrast allergy | Pacemakers, metal implants, claustrophobia |

### Gadolinium safety

- **NSF (nephrogenic systemic fibrosis)** was the concern at GFR <30 mL/min/1.73 m². Group II agents (gadobenate, gadoteridol, gadoterate, gadobutrol) carry little or no risk even in end-stage disease.
- **Pregnancy:** gadolinium only if it clearly helps.
- **EAU (European Association of Urology):** CTU is preferred to MRU for upper tract urothelial cancer.

**Diagram to draw:** coronal static T2 MRU of a hydronephrotic kidney with a dark stone in the ureter.

> **Key points to remember:**
> - Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
> - Stones are signal voids; CTU preferred for urothelial cancer.
> - Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
> - NSF risk minimal with group II agents.

## SAQ 16 Urine cytology

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 1 Paper 1, Q3b

### Definition and principle

- Microscopy of shed urothelial cells (Papanicolaou, 1945).
- **Principle:** high-grade cells are poorly cohesive, shed readily and look atypical; low-grade cells look almost normal. So cytology detects **high-grade tumours and CIS (carcinoma in situ)** but misses low grade.

### Specimens

- Voided urine (not the first morning sample, whose cells have degenerated), at least 30 mL; bladder washing; upper tract washings.

### Performance

- Sensitivity 31–62%, specificity 94–100%; strongly grade-dependent.
- **False positives:** instrumentation, stones, infection, radiotherapy, intravesical BCG (bacille Calmette–Guérin), polyomavirus ("decoy" cells), ileal conduit.
- **False negatives:** low grade, low cellularity, degeneration.

### The Paris System

- Published 2016; **2nd edition 2022**. Aim: detect **HGUC (high-grade urothelial carcinoma)**. Key criterion: **N:C (nuclear-to-cytoplasmic) ratio**, plus hyperchromasia, coarse chromatin and irregular nuclear membrane.

| Category | Criteria |
|---|---|
| Non-diagnostic | Inadequate |
| Negative for HGUC | N:C <0.5, or atypia explained by a benign cause |
| AUC (atypical urothelial cells) | N:C ≥0.5 and one feature |
| Suspicious for HGUC | N:C ≥0.7, at least two features, few cells (<5–10) |
| HGUC | N:C ≥0.7, at least two features, many cells |
| LGUN (low-grade urothelial neoplasm) | Fibrovascular cores; rarely diagnosable |
| Other malignancy | Squamous, adenocarcinoma, secondary |

- It reduces equivocal "atypia" reports; cancer risk rises with each category.

### Clinical uses

- **Not** for screening or initial microhaematuria evaluation (AUA/SUFU [American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2020).
- Gross haematuria; NMIBC (non-muscle-invasive bladder cancer) surveillance (mandatory in high risk); suspected CIS; upper tract tumours.
- **Positive cytology, normal cystoscopy:** image upper tracts, selective washings ± ureteroscopy, random bladder and prostatic urethral biopsies.
- **Also know:** FISH (fluorescence in situ hybridisation; chromosomes 3, 7, 17, 9p21), NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), CxBladder: more sensitive, less specific; none replaces cystoscopy.

**Diagram to draw:** normal urothelial cell beside an HGUC cell with high N:C ratio.

> **Key points to remember:**
> - High specificity, low sensitivity; good for high grade and CIS.
> - Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
> - Positive cytology, normal cystoscopy: upper tracts, random biopsies.
> - Not for screening or initial microhaematuria.

## SAQ 17 Patient positioning in lap urology

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 1, Q3a (pneumoperitoneum physiology also answered in 2019-Summer Paper 1, Q2)

### Principles

- Positioning uses **gravity** to move bowel away and gives port or robot access while keeping ventilation safe.

- Injury comes from **stretch, compression or ischaemia**, so position and duration act together; nerve injury occurs in about 2.8%. Shared team responsibility (ASA [American Society of Anesthesiologists] advisory, 2000).
- Secure IV (intravenous) access, oximetry and BP (blood pressure) cuff before final positioning: tucked arms become inaccessible.

### Positions

| Position | Used for | Set-up | Hazards |
|---|---|---|---|
| **Steep Trendelenburg**, low lithotomy or split legs | RARP (robot-assisted radical prostatectomy), cystectomy | Arms tucked and padded; chest strap over egg-crate; **no shoulder braces**; tilt about 22–30° | Brachial plexus; femoral nerve (excess hip flexion); facial and laryngeal oedema; corneal abrasion; tube into right bronchus |
| **Modified flank** (30–45°) | Transperitoneal nephrectomy, pyeloplasty | Back on gel rolls; upper arm across chest; wide tape; tilt to test | Pressure points, slippage |
| **Full lateral (90°)**, table flexed | Retroperitoneoscopy, adrenalectomy | **Axillary roll** 3–4 fingerbreadths below axilla; lower leg flexed 45°, upper straight on pillows | **Rhabdomyolysis**; sciatic stretch |
| **Prone** | Posterior adrenalectomy | Pad torso, elbows, hips | Airway, face and eyes |

### Rhabdomyolysis

- About 0.4% (1% retroperitoneal). Risk: men, BMI (body mass index) ≥25, **kidney rest**, full flexion, long cases. Downside hip pain, brown urine, CK (creatine kinase) >5,000 units. Prevent: limit kidney rest (Campbell gives 20–30 minutes in one place, under 1 hour in another), avoid hypotension, keep cases under 5 hours.

### Prevention and physiology

- Abduction ≤90°; pad ulnar groove and fibular head; recheck after every table move.
- **Head-down** raises cardiac output but splints the diaphragm; restrict fluids.
- Nerve injury on day 1: neurology review, conduction studies, physiotherapy.
- **Also know:** forearm compartment syndrome from infiltrated IV lines; Xi side-docking avoids stirrups.

**Diagram to draw:** full flank position with axillary roll and table break.

> **Key points to remember:**
> - Nerve injury about 2.8%; brachial plexus is commonest at risk.
> - Never shoulder braces in steep Trendelenburg.
> - Axillary roll below, not in, the axilla.
> - Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

## SAQ 18 Clinical trial methodology

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 1, Q3a

## Definition

- A **clinical trial** is a prospective study that assigns people to an intervention to test its effect on health outcomes. The **RCT (randomised controlled trial)** is the gold standard because randomisation balances known and unknown confounders.

## Phases

| Phase | Purpose | Subjects |
|---|---|---|
| 0 | Microdose pharmacokinetics | Very few |
| I | Safety, dose, toxicity | 20–80, often volunteers |
| II | Activity, efficacy signal | Patients, about 100–300 |
| III | Comparative efficacy versus standard | Hundreds to thousands; basis of approval |
| IV | Post-marketing surveillance | General population |

## Design

- **Parallel**, **crossover** (own control; stable chronic disease), **factorial** (two questions at once), **cluster** or **adaptive**.
- Hypothesis: **superiority**, **non-inferiority** (pre-set margin) or equivalence.
- Surgical innovation follows the **IDEAL** framework (Idea, Development, Exploration, Assessment, Long-term study).

## Randomisation and allocation

- Simple, **block** (equal numbers), **stratified** (by prognostic factors) or **minimisation**.
- **Allocation concealment** (central or web-based) stops selection bias.

## Blinding

- Single (patient), double (plus assessor) or triple (plus analyst). Surgery: sham procedures or blinded assessors.

## Endpoints

- One **primary endpoint** sets sample size; secondary endpoints are exploratory.
- **Hard** (overall survival) versus **surrogate** endpoints, valid only if they meet the **Prentice criteria**; in prostate cancer only MFS (metastasis-free survival) does.
- **PROs (patient-reported outcomes)** with validated tools; complications graded by **Clavien–Dindo** or CTCAE (Common Terminology Criteria for Adverse Events).

## Sample size and analysis

- Set α (type I error, usually 0.05), power (80–90%) and expected effect.
- **ITT (intention-to-treat)** preserves randomisation; per-protocol supports non-inferiority claims.
- Kaplan–Meier curves for time-to-event data.

## Ethics and reporting

- Declaration of Helsinki, ICH-GCP (International Council for Harmonisation–Good Clinical Practice), informed consent, ethics committee, DSMB (data safety monitoring board).

- **Registration** before enrolment (CTRI [Clinical Trials Registry–India]).
- **CONSORT** (Consolidated Standards of Reporting Trials): checklist plus **participant flow diagram** (25 items in the 2010 version; updated 2025).
- **Also know:** lead-time bias; Will Rogers phenomenon.

**Diagram to draw:** CONSORT flow diagram: enrolment → randomisation → allocation → follow-up → analysis.

> **Key points to remember:**
> - Phase III RCT with concealed allocation and blinding is the gold standard.
> - Primary endpoint drives sample size; surrogates need the Prentice criteria.
> - ITT analysis; CONSORT flow diagram for reporting.

## SAQ 19 Doppler in urology

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 1, Q3b

### Principle

- **Doppler shift:** sound reflected from moving red cells changes frequency in proportion to velocity and the cosine of the insonation angle: $\Delta f = 2f_0 v \cos\theta / c$. Keep the angle **below 60°**.
- **RI (resistive index) = (PSV − EDV) / PSV** (peak systolic and end-diastolic velocity); reflects downstream resistance.

### Modes

| Mode | Shows | Use |
|---|---|---|
| **Colour** | Velocity and direction (red towards, blue away by convention) | Flow in kidney, testis, ureteric jets |
| **Spectral (duplex)** | Waveform at one point; velocities, RI | Penile and renal arteries |
| **Power** | Amplitude only; 3–5 times more sensitive, less angle-dependent | Low flow: torsion, prostate lesions |

### Applications

| Organ | Findings |
|---|---|
| **Scrotum** | **Torsion:** absent intratesticular flow (compare with the other side); epididymitis: increased flow; **varicocele:** venous reflux on Valsalva; intratesticular RI >0.6 suggests poor spermatogenesis |
| **Penis (after intracavernosal injection)** | **PSV <25 cm/s = arterial ED (erectile dysfunction)**, 25–35 equivocal, >35 normal; **EDV >5 cm/s and RI <0.75 = veno-occlusive dysfunction**; Peyronie plaque assessment |
| **Priapism** | Ischaemic: absent cavernosal flow; non-ischaemic: high flow (PSV >50 cm/s), locates the fistula |

| Organ | Findings |
|---|---|
| **Kidney** | Normal RI 0.47–0.70; **raised RI in acute obstruction** (useful in pregnancy, where change in RI is more accurate); renal artery stenosis: high velocity and parvus-tardus waveform; arteriovenous fistula after biopsy or PCNL (percutaneous nephrolithotomy) |
| **Transplant** | Arterial thrombosis: absent signal; **venous thrombosis: reversed diastolic flow**; stenosis: high velocity; rejection: loss of diastolic flow, high RI |
| **Bladder** | Ureteric jets (absent on the obstructed side) |
| **Tumours** | Vascularity of renal and testicular masses; tumour thrombus; hypervascular prostate cancer foci |

- **Also know:** intraoperative micro-Doppler in microsurgical varicocelectomy; contrast-enhanced ultrasound and elastography in multiparametric ultrasound.

### Limitations

- Operator- and angle-dependent; poor in obesity and deep vessels; early torsion may show collateral flow, so **Doppler cannot exclude torsion**.

**Diagram to draw:** spectral waveform marking PSV and EDV with the RI formula.

> **Key points to remember:**
> - RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
> - Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
> - Absent intratesticular flow = torsion, but explore on clinical grounds.

## SAQ 20 Flow cytometry

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 1, Q3d

### Definition

- **Flow cytometry** measures physical and fluorescent properties of **single cells in suspension** flowing one at a time through a laser beam, thousands per second.

### Principle

1. **Sample:** fresh cells, urine, bladder washings, or nuclei from paraffin blocks.
2. **Staining:** fluorochrome-labelled antibodies to CD (cluster of differentiation) antigens, or **DNA-binding dyes** (propidium iodide, DAPI [4′,6-diamidino-2-phenylindole]).
3. **Hydrodynamic focusing** lines the cells up in single file.
4. **Laser interrogation:** - **Forward scatter** = cell size. - **Side scatter** = internal complexity (granularity). - **Fluorescence** = amount of bound antibody or DNA.
5. Photomultiplier detectors → histograms and dot plots; **gating** selects populations. - **FACS (fluorescence-activated cell sorting)** physically separates selected cells.

### DNA content analysis

- **DNA index** = tumour G0/G1 peak ÷ normal diploid peak. DI 1.0 = diploid; about 2.0 = tetraploid; any other = **aneuploid**.

- **S-phase fraction** estimates proliferation.
- Aneuploidy and high S-phase generally mean higher grade and worse prognosis.

### Applications in urology

| Area | Use |
|---|---|
| **Bladder cancer** | Ploidy of bladder washings: aneuploidy with high grade, CIS (carcinoma in situ), recurrence; replaced by **UroVysion FISH (fluorescence in situ hybridisation)** |
| **Upper-tract urothelial cancer** | Aneuploidy → poor long-term survival |
| **Prostate cancer** | Aneuploid tumours progress more after treatment than diploid |
| **Neuroblastoma** | **Hyperdiploid favourable**; diploid or tetraploid adverse; part of COG (Children's Oncology Group) risk |
| **Urinalysis** | Automated urine analysers count RBC (red blood cells), WBC (white blood cells) and bacteria |
| **Transplantation** | **Flow cytometric crossmatch**: more sensitive than CDC (complement-dependent cytotoxicity) crossmatch for donor-specific antibody; Luminex single-antigen beads; CD3 count to guide antithymocyte globulin |
| **Male infertility** | Sperm DNA fragmentation (SCSA [sperm chromatin structure assay]); haploid cells in testicular aspirates |
| **Lymphoma** | Immunophenotyping of testicular or renal lymphoma |

### Advantages and limitations

- **Advantages:** rapid, objective, quantitative, many parameters per cell.
- **Limitations:** architecture lost; near-diploid aneuploidy missed; normal cells dilute tumour peaks; ploidy now adds little independent value.

**Diagram to draw:** DNA histogram with diploid G0/G1 peak, S-phase and aneuploid peak.

> **Key points to remember:**
> - Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
> - DNA index 1.0 diploid; aneuploidy = worse prognosis.
> - Flow crossmatch is more sensitive than CDC crossmatch.

## SAQ 21 VCUG

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 1, Q3c

### Definition

- **VCUG (voiding cystourethrogram; MCU, micturating cystourethrogram):** fluoroscopic study of bladder filling and voiding after catheter instillation of contrast. It shows the bladder, **posterior urethra** and UVJ (ureterovesical junction) competence.
- **Gold standard for VUR (vesicoureteral reflux)** and PUV (posterior urethral valves).

### Indications

- **Children:** febrile UTI (urinary tract infection) with abnormal ultrasound or recurrent infection (EAU [European Association of Urology]: also age <12 months); bilateral hydroureteronephrosis (PUV); dilated ureter; ureterocele; neurogenic bladder.
- **Adults:** stricture (with RGU [retrograde urethrogram]), diverticula, fistula, post-operative leak, voiding dysfunction.

### Technique

- **Sterile urine** first. After UTI, wait until urine is sterile and the child is improving (about a week); the old 4–6-week delay is unnecessary.
- Scout film → catheter: **5 Fr feeding tube (infants), 6–8 Fr (children), 14–16 Fr (adults)** → confirm tip position.
- **Gravity fill** (≤70 cm $H_2O$) to expected capacity: <2 years weight (kg) × 7; 2–14 years (age × 30) + 30 mL; >14 years 300–500 mL. (Koff formula: (age + 2) × 30 mL.)
- **Oblique views** (UVJ, prostatic urethra) → remove catheter → **void under fluoroscopy** → **post-void film**.
- **Cyclic VCUG** (2–3 fill–void cycles) detects 12–20% more VUR.

### Findings

- **VUR grade (International Reflux Study, 1981):**

| Grade | Appearance |
|---|---|
| I | Ureter only, not dilated |
| II | Pelvis and calyces, no dilatation |
| III | Mild–moderate dilatation, minimal blunting |
| IV | Moderate dilatation and tortuosity |
| V | Gross dilatation; papillary impressions lost |

- Filling-phase reflux has a poorer resolution prognosis than voiding-only reflux.
- **PUV:** dilated, elongated posterior urethra, prominent bladder neck, trabeculation.
- Ureterocele, diverticula, residual urine; post-void pelvic retention suggests PUJ (pelviureteric junction) obstruction.

### Limitations and complications

- Invasive, radiation, UTI, catheter trauma; snapshot (occult VUR); interobserver variability; autonomic dysreflexia (injury above T6).

### Alternatives

- **Direct radionuclide cystography:** more sensitive, lower dose, poor anatomy.
- **CeVUS (contrast-enhanced voiding urosonography):** no radiation, accuracy close to VCUG.
- **PIC (positional instillation of contrast) cystography:** occult VUR with negative VCUG.

**Diagram to draw:** VUR grades I–V.

**Key points to remember:**

- Gold standard for VUR and PUV; sterile urine first.
- Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
- Grades I–II non-dilating, III–V dilating.

## SAQ 22 Interventional radiology in urology and therapeutic angioembolisation

P1

**Asked as:**

- Therapeutic angioembolization. — *Practice Set 7 Paper I Q3e (SAQ)*
- Interventional urological treatment in urology — *Practice Set 8 Paper I Q3e (SAQ)*

**Type:** SAQ (10 marks)

*Assumed meaning: interventional radiology (image-guided percutaneous and endovascular) treatment in urology.*

### Principles

- Guidance by ultrasound, fluoroscopy or CT (computed tomography), usually under local anaesthesia. **Seldinger technique:** needle, guidewire, then devices over the wire.
- **Beforehand:** correct coagulopathy, manage anticoagulants, urine culture, antibiotic prophylaxis.
- **Requirements:** experienced team, resuscitation in the suite; unstable non-responders need surgery.

### Non-vascular procedures

| Procedure | Indications and points |
|---|---|
| **PCN (percutaneous nephrostomy)** | Obstructed infected kidney (as effective as retrograde stent), malignant ureteric obstruction, urinary leak or fistula, access for PCNL (percutaneous nephrolithotomy). Puncture a **posterior calyx** through the avascular Brödel plane. Transfusion 0.5–4% |
| **Antegrade stenting** | When retrograde stenting fails: malignant obstruction, ileal conduit, transplant ureter |
| Antegrade pyelography, Whitaker test | Define strictures; pressure–flow test for obstruction |
| **Percutaneous drainage** | Renal or perinephric abscess >3 cm, urinoma, lymphocele |
| Image-guided biopsy | Renal mass biopsy |

### Vascular (endovascular) procedures: therapeutic angioembolisation

#### Definition

- Deliberate **endovascular occlusion** of a target vessel, usually by transfemoral selective catheterisation under fluoroscopy, to stop bleeding, devascularise a tumour or close an abnormal vessel.
- **Superselective** embolisation of a segmental branch preserves most parenchyma.

### Embolic agents

| Type | Examples | Use |
|---|---|---|
| Temporary | Autologous clot; gelatin sponge (2–6 weeks) | Priapism, prostate, bladder bleeding |
| Permanent particles | PVA (polyvinyl alcohol), microspheres | Tumours, PAE (prostate artery embolisation) |
| Mechanical | Microcoils, vascular plugs | Pseudoaneurysm, AVF (arteriovenous fistula), varicocele |
| Liquid | NBCA (N-butyl cyanoacrylate) glue, absolute ethanol | Tumour ablation, recurrent priapism |

### Urological indications

| Procedure | Indications |
|---|---|
| **Renal angioembolisation** | Grade IV–V renal trauma; **pseudoaneurysm or arteriovenous fistula** after PCNL or partial nephrectomy (delayed bleed); bleeding AML (angiomyolipoma, first line); palliative embolisation of RCC (renal cell carcinoma) |
| **Internal iliac (anterior division) embolisation** | Intractable bladder haemorrhage; initial control about 90%. Spare the posterior division (gluteal ischaemia) |
| **PAE (prostate artery embolisation)** | LUTS (lower urinary tract symptoms) from BPH (benign prostatic hyperplasia); EAU (European Association of Urology) for men accepting lesser outcomes; AUA (American Urological Association) 2023 may offer |
| **Varicocele embolisation** | Coils or liquid sclerosants via femoral or jugular vein |
| **High-flow priapism** | Superselective cavernosal artery embolisation; resolution ≥75%; recurrence 30–40% |
| **PTRA (percutaneous transluminal renal angioplasty)** | Fibromuscular dysplasia; transplant renal artery stenosis |

- **Renal trauma:** active contrast extravasation, pseudoaneurysm, AVF; success high in grade IV, lower in grade V. Failure predictors: instability, grade V, urinary extravasation. Repeat embolisation often succeeds.
- **Iatrogenic bleeding** after PCNL (percutaneous nephrolithotomy) or partial nephrectomy: **treatment of choice**; superselective coils nearly 100% technically successful.
- **AML (angiomyolipoma): first line for acute haemorrhage**; prophylactic for >6 cm, aneurysm >5 mm, growth or pregnancy plans. In TSC (tuberous sclerosis complex), embolise then give steroids.
- **RCC:** palliation of haematuria or pain in unfit patients; preoperative use no longer routine.
- **Intractable bladder haemorrhage** (radiation cystitis, cancer): bilateral **anterior division of internal iliac**; avoid the posterior division (gluteal pain).
- **Prostate:** PAE for BPH (benign prostatic hyperplasia) and refractory prostatic haematuria.
- **High-flow priapism:** superselective cavernosal artery embolisation; resolution ≥75%; recurrence → repeat with a non-resorbable agent.
- **Also know:** varicocele (venous coils); lymphatic embolisation for chylous ascites; ADPKD (polycystic kidney) pain.

### Ablation

- **Cryoablation or RFA (radiofrequency ablation)**, percutaneous, for cT1a renal masses <3 cm; biopsy first. Higher local recurrence than partial nephrectomy, salvaged by repeat ablation.

### Complications

- Contrast nephropathy, access-site haematoma, coil migration.
- **Post-embolisation syndrome:** fever, flank pain, nausea, leucocytosis; up to 72 hours; self-limiting.
- **Non-target embolisation:** bowel, bladder, rectum, gluteal muscle, penile skin.
- Renal infarction or abscess; ED (erectile dysfunction) after priapism embolisation.

**Diagram to draw:** prone PCN through a posterior calyx along the Brödel line; renal angiogram showing segmental pseudoaneurysm before and after coil occlusion.

> **Key points to remember:**
> - PCN equals retrograde stent for decompressing an obstructed infected kidney.
> - Superselective coils: treatment of choice for pseudoaneurysm and AVF.
> - Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula; prophylactic >6 cm or aneurysm >5 mm.
> - Bladder bleeding: internal iliac anterior division; spare posterior division.
> - Post-embolisation syndrome is common and self-limiting.

## SAQ 23 Tumour markers in urology

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 18 SAQ Sheet, Q3d

### Definition and uses

- A substance made by a tumour, measured in blood, urine or tissue.
- Uses: screening, diagnosis, staging, prognosis, response, and detecting relapse. Ideal: specific, sensitive, proportional to bulk, short half-life.

### Prostate

- **PSA (prostate-specific antigen):** a kallikrein serine protease (hK3) from luminal cells; **organ-specific, not cancer-specific** (BPH (benign prostatic hyperplasia), prostatitis, instrumentation raise it). Best marker for post-treatment follow-up.
- **Refinements:** density (PSA ÷ volume; cut-offs 0.08–0.15); %free PSA (lower in cancer; for PSA 4–10 ng/mL); velocity (little value); **PHI (Prostate Health Index) = (p2PSA ÷ free PSA) × √total PSA**; 4Kscore.
- **Urine:** PCA3 (prostate cancer antigen 3) after DRE (digital rectal examination); score = PCA3 mRNA ÷ KLK3 mRNA × 1000.
- **Also know:** tissue genomic classifiers.

### Testis (germ cell tumours)

| Marker | Half-life | Source | Never raised in |
|---|---|---|---|
| AFP (α-fetoprotein) | 5–7 days | Embryonal, yolk sac | Pure seminoma, choriocarcinoma |
| β-hCG (human chorionic gonadotrophin) | 24–36 hours | Choriocarcinoma, embryonal; seminoma in ~15% | Yolk sac tumour |
| LDH (lactate dehydrogenase) | 24 hours | Non-specific; reflects bulk | — |

- Measure before and after orchidectomy, on chemotherapy, in follow-up. Post-orchidectomy values give the **S stage** (S1: LDH <1.5 × normal, hCG <5000 mIU/mL, AFP <1000 ng/mL) and the **IGCCCG (International Germ Cell Cancer Collaborative Group)** risk group.
- Raised AFP with "pure seminoma" histology → treat as NSGCT (non-seminomatous germ cell tumour). False positives: liver disease, marijuana, LH (luteinising hormone) cross-reaction.
- **miR-371a-3p:** sensitive new marker; not raised by teratoma.

### Bladder and upper tract

- **Urine cytology:** sensitive and specific for high-grade cancer, poor for low grade; for surveillance, not screening.
- NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), ImmunoCyt, UroVysion FISH (fluorescence in situ hybridisation): selected use only.

### Others

- **Kidney and penis:** no validated serum marker.
- **Adrenal:** metanephrines, cortisol, aldosterone–renin ratio. **Neuroblastoma:** urinary VMA (vanillylmandelic acid)/HVA (homovanillic acid) raised in 90–95%.

**Diagram to draw:** marker decline curve after orchidectomy using half-lives.

> **Key points to remember:**
> - PSA: organ-specific, not cancer-specific.
> - AFP never in pure seminoma; half-lives AFP 5–7 days, hCG 24–36 hours.
> - S stage uses post-orchidectomy values.
> - Cytology: good for high grade only.

## SAQ 24 Haemostatic agents in urology

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 18 SAQ Sheet, Q3e

### Introduction

- Topical agents are **adjuncts** to surgical haemostasis (suture, clip, energy), used for diffuse ooze. No randomised study shows clear benefit of any single agent in partial nephrectomy.

## Classification

| Class | Examples | Mechanism / note |
|---|---|---|
| **Passive (mechanical)** | Gelatin sponge (Gelfoam), **oxidised regenerated cellulose (Surgicel)**, microfibrillar collagen, polysaccharide spheres | Matrix for platelets; needs intact clotting. Cellulose is acidic (antibacterial) |
| **Active** | Topical thrombin | Converts the patient's fibrinogen to fibrin |
| **Flowable** | Gelatin matrix + thrombin (**Floseal**) | Fills irregular beds, swells |
| **Fibrin sealants** | Fibrinogen + thrombin (Tisseel, Evicel) | Final clotting step; works despite coagulopathy; also seals leaks |
| **Synthetic sealants and glues** | PEG (polyethylene glycol) hydrogel (Coseal), cyanoacrylate, albumin–glutaraldehyde (BioGlue) | Seal; adhesive |
| **Systemic** | Tranexamic acid, aminocaproic acid, desmopressin | Antifibrinolytic; desmopressin for uraemic platelets |

## Urological uses

- **Partial nephrectomy:** renorrhaphy ± sealant or Surgicel bolster; sutureless repair in selected cases.
- **Laparoscopic venous injury:** pressure → sealant → suture.
- PCNL (percutaneous nephrolithotomy) tract sealing in tubeless cases; adrenal and spleen/liver ooze; prostatectomy and lymphadenectomy beds.
- **Fibrin glue:** ureteroscopic fulguration for upper-tract bleeding; fistula plugs.
- **Tranexamic acid:** lowers transfusion at radical cystectomy.

## Intravesical agents (haemorrhagic cystitis)

| Agent | Note |
|---|---|
| Alum 1% (50 g in 5 L) | First line; avoid in renal failure (aluminium toxicity) |
| Silver nitrate 0.5–1% | Chemical coagulation |
| Aminocaproic acid 200 mg/L irrigation | Thromboembolism and clot retention |
| Formalin 1–4% | Most effective; under anaesthesia; **cystogram first** to exclude reflux |

## Complications

- Swelling compresses nerves, ureter or spinal cord (gelatin, cellulose): use the least amount.
- Mimics abscess on imaging; foreign-body reaction; infection.
- Flowables entering open vessels → embolism or thrombosis.
- Bovine thrombin → antibodies and coagulopathy; human products → theoretical viral risk.
- **Also know:** energy adjuncts (bipolar, argon beam, vessel sealers); hyperbaric oxygen in radiation cystitis.

**Diagram to draw:** clotting cascade showing where passive agents, thrombin and fibrin sealant act.

> **Key points to remember:**
>
> - Agents are adjuncts, not substitutes for sutures.
> - Fibrin sealant works despite coagulopathy; passive agents need normal clotting.
> - Surgicel: acidic, may mimic abscess.
> - Alum first line; formalin last, after cystogram.

## SAQ 25 Sterilisation of urological instruments

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 19 SAQ Sheet, Q3a

### Definitions

- **Cleaning:** removes debris that shields organisms; always first.
- **Disinfection:** kills most microbes but not all spores. HLD (high-level disinfection) kills all except large numbers of spores.
- **Sterilisation:** kills all life, including spores.

### Spaulding classification

| Category | Contact | Requirement | Urological examples |
|---|---|---|---|
| Critical | Sterile tissue, vessels | Sterilisation | Resectoscope, ureteroscope, nephroscope, laparoscopic and robotic instruments |
| Semi-critical | Mucosa | HLD at least (sterilise if possible) | Flexible cystoscope, TRUS (transrectal ultrasound) probe |
| Non-critical | Intact skin | Low-level disinfection | Abdominal ultrasound probe |

### Reprocessing steps

1. Bedside pre-clean; leak test for flexible scopes.
2. Manual cleaning with enzymatic detergent; brush and flush channels.
3. Rinse → disinfect or sterilise → rinse → dry channels (alcohol, air) → store hanging; record tracking.

### Methods

| Method | Conditions | Use and drawbacks |
|---|---|---|
| **Steam autoclave** | 121 °C for 15 min (15 psi) or 134 °C for 3–4 min | Cheapest, most reliable; metal instruments, autoclavable optics |
| Dry heat | 160 °C for 2 hours | Glass; rarely used |
| **Ethylene oxide** | Low temperature | Heat-sensitive scopes, cables; toxic, carcinogenic, long aeration |
| **Hydrogen peroxide gas plasma** | Low temperature, short cycle | No toxic residue; not for cellulose; long narrow lumens limited |
| **Peracetic acid (Steris)** | Automated liquid, low temperature | Sporicidal; ureteroscopes |

| Method | Conditions | Use and drawbacks |
|---|---|---|
| **Glutaraldehyde 2%** | HLD in 20 min; sterilant after about 10 hours | Cheap; irritant, sensitiser; must rinse well |
| Gamma irradiation | Industrial | Single-use catheters, stents, sutures |

### Monitoring

- **Physical:** time, temperature, pressure records.
- **Chemical indicators:** autoclave tape; Bowie–Dick test (air removal).
- **Biological indicators:** *Geobacillus stearothermophilus* spores (steam, plasma); *Bacillus atrophaeus* (ethylene oxide, dry heat).

### Urology-specific points

- A laser-perforated ureteroscope channel lets fluid in during sterilisation, destroying the scope.
- Poor channel cleaning causes outbreaks (Pseudomonas, mycobacteria).
- **Prions** resist routine methods: single-use or special protocols.
- **Single-use scopes** avoid reprocessing (about $107 per case for reusable ureteroscopes).
- **Also know:** laser fibre re-sterilisation; limited-life robotic instruments.

**Diagram to draw:** reprocessing cycle flowchart.

> **Key points to remember:**
> - Spaulding: critical sterilise, semi-critical HLD, non-critical low-level.
> - Autoclave 121 °C 15 min or 134 °C 3–4 min.
> - Ureteroscopes: ethylene oxide, glutaraldehyde or peracetic acid.
> - Biological indicator: *G. stearothermophilus*.

## SAQ 26 Ethics and informed consent in urology

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 19 SAQ Sheet, Q3e

### Four principles (Beauchamp and Childress, 1979)

| Principle | Meaning |
|---|---|
| **Autonomy** | Patient's deliberated self-rule; basis of consent and confidentiality |
| **Beneficence** | Act for the patient's benefit, judged by the patient's values |
| **Non-maleficence** | Avoid harm (*primum non nocere*); requires training and honest risk data |
| **Justice** | Fair distribution of resources, benefits and risks |

| Principle | Meaning |
|---|---|
| - Hippocratic Oath gave beneficence, non-maleficence, confidentiality; autonomy and justice are modern additions (autonomy is bounded by justice). | |
| - **Jonsen's four-box method:** medical indications; patient preferences; quality of life; contextual features (family, cost, law). | |

### Informed consent: legal evolution

- From battery to negligence. **Schloendorff (1914):** right to bodily integrity. **Salgo (1957):** term "informed consent". **Canterbury v Spence (1972): reasonable-patient** disclosure standard.
- India: **Samira Kohli (2008)**: consent for one procedure does not cover another, except in emergency.

### Elements

1. Diagnosis, proposed treatment and alternatives, including no treatment.
2. Risks, benefits and uncertainty compared with alternatives.
3. **Check understanding** (most often missed).
4. Voluntary decision by a patient with capacity; documented. - Material risks: no agreed cut-off (1% proposed); disclose rarer ones by severity; lay language. - Urology examples: incontinence and erectile dysfunction after prostatectomy; retrograde ejaculation after TURP (transurethral resection of the prostate); stent symptoms; conversion to open surgery.

### Exceptions

- **Emergency** in an incapable patient (presumed consent); a conscious patient may still refuse.
- **No capacity:** surrogate using substituted judgement, then best interests.
- **Minors (<18 years):** parents consent; child gives assent.

### Improving consent

- Decision aids, repeat-back, simple written material.
- **Shared decision making:** two-way exchange of options and patient values; vital when several options are reasonable (e.g. localised prostate cancer).

### Special areas

- **End of life:** early goals-of-care talks, advance directives, DNR (do-not-resuscitate) orders.
- **Research:** written consent, ethics committee approval, extra protection for vulnerable groups.
- **Also know:** confidentiality; conflicts of interest; futility policies.

**Diagram to draw:** Jonsen four-box grid.

> **Key points to remember:**
>
> - Four principles: autonomy, beneficence, non-maleficence, justice.
> - Canterbury v Spence: reasonable-patient standard; Samira Kohli in India.
> - Commonest gap: not checking understanding.
> - Exceptions: emergency, incapacity (surrogate), minors.

## SAQ 27 Operating room set-up

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 19 SAQ Sheet, Q3f (see also 2021-Summer Paper 4, Q3a for operating room safety)

### Physical design

- **Zones:** protective (reception, changing) → clean (pre-op, recovery) → sterile (theatre, scrub) → disposal; one-way flow.
- **Ventilation:** positive pressure, about 20 air changes an hour, filtered air; laminar flow for implants.
- **Environment:** temperature about 20–24 °C, relative humidity about 20–60%. Patient warming: even 1–2 °C hypothermia raises SSI (surgical site infection) and blood loss (**Kurz 1996**: 3-fold wound infection).
- Power backup, piped gases, anaesthetic scavenging, radiolucent table.

### Equipment layout by procedure

| Setting | Arrangement |
|---|---|
| **Endourology** (TURP (transurethral resection of the prostate), URS (ureteroscopy), PCNL (percutaneous nephrolithotomy)) | Lithotomy or prone; tower and monitor directly ahead; C-arm opposite surgeon; wires, baskets, dilators, spare scopes ready |
| **Laparoscopic, upper abdomen** | Surgeon and assistant face the abdomen; monitor opposite; scrub nurse near the foot |
| **Laparoscopic, pelvic** | 30° Trendelenburg, split legs; surgeon on preferred side, assistant opposite |
| **Robotic** | Patient cart, surgeon console, vision cart; Si docks between legs, Xi from the side |

- **Checklist before laparoscopic incision:** irrigation–suction, electrosurgical unit, full $CO_2$ cylinder plus spare, white-balanced camera, insufflator and Veress needle checked, **open tray ready**.
- Cables fixed; camera cable apart from diathermy; insufflator at eye level.

### Positioning safety

- Pad bony prominences; axillary roll (lateral); avoid prolonged kidney rest and full flexion (rhabdomyolysis); no shoulder braces in steep Trendelenburg (brachial plexus).

### Ergonomics and staff protection

- Monitor straight ahead, top at eye level; elbows 90–120°; table at the surgeon's pubic symphysis (laparoscopy).
- **Radiation:** ALARA (as low as reasonably achievable): time, distance, shielding; lead apron, thyroid shield, glasses; dosimeter.
- Smoke evacuation; sharps safety; vaccination.

### Safety systems

- **WHO (World Health Organization) Surgical Safety Checklist** (sign in, time out, sign out); counts; fire-risk assessment (oxidiser, fuel, ignition).

**Diagram to draw:** theatre plan: table, tower, C-arm, anaesthetist, surgeon, scrub nurse.

> **Key points to remember:**
> - Zones with one-way flow; positive pressure, about 20 air changes an hour.
> - Monitor directly ahead at eye level; open tray ready for laparoscopy.
> - Hypothermia triples wound infection (Kurz).
> - ALARA: time, distance, shielding.

## SAQ 28 Amyloidosis in urology

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 21 SAQ Sheet, Q3c

### Definition

- Extracellular deposits of misfolded protein in **β-pleated sheet fibrils**. **Congo red stain gives apple-green birefringence** under polarised light.

### Classification

| Type | Protein | Cause |
|---|---|---|
| **AL** (amyloid light-chain; primary) | Immunoglobulin light chain | Plasma cell dyscrasia, myeloma |
| **AA** (amyloid A; secondary) | Serum amyloid A | Chronic inflammation: TB (tuberculosis), rheumatoid arthritis, osteomyelitis, pressure sores, hidradenitis |
| **ATTR** (hereditary transthyretin) | Transthyretin | Autosomal dominant neuropathy |
| Dialysis-related | β2-microglobulin | Long-term dialysis |
| **Localised** | Usually AL | Deposits in one organ only |

### Urological sites

- **Kidney (systemic):** nephrotic proteinuria, CKD (chronic kidney disease), enlarged kidneys, renal vein thrombosis, proximal tubular defects.
- **Bladder (usually localised): painless gross haematuria**; yellowish plaques or nodules that **mimic carcinoma**. Secondary bladder amyloid can bleed severely; Campbell lists amyloidosis among systemic causes of haemorrhagic cystitis.
- **Ureter and renal pelvis:** stricture or filling defect, often calcified; mimics urothelial cancer; obstruction.
- **Urethra:** rare; mass, obstructive voiding, haematuria.
- **Seminal vesicles:** senile deposits in older men; can mimic cancer invasion on MRI (magnetic resonance imaging).
- **Also know:** ATTR gives neurogenic bladder (underactive detrusor, LUTS [lower urinary tract symptoms] >80%) and erectile dysfunction; adrenal insufficiency; papillary necrosis.

### Evaluation

- **Biopsy is diagnostic** (avoids unnecessary nephroureterectomy or cystectomy).

- Exclude systemic disease: serum and urine protein electrophoresis, immunofixation, free light chains, echocardiogram, renal function.

### Management

- **Localised bladder:** TUR (transurethral resection) or fulguration; intravesical DMSO (dimethyl sulfoxide) reported; recurrence is common, so follow-up cystoscopy is needed.
- **Ureter:** ureteroscopic resection or segmental excision; stent for obstruction.
- **Urethra:** observe if mild; dilatation or urethroplasty if obstructive.
- **Refractory bleeding:** irrigation, embolisation, cystectomy.
- **Systemic:** AL: chemotherapy (daratumumab with bortezomib regimen, ANDROMEDA trial) or stem cell transplant; AA: treat inflammation (colchicine in familial Mediterranean fever); ATTR: tafamidis, patisiran. AA amyloid can recur in a renal transplant.

**Diagram to draw:** Congo red stained fibrils with apple-green birefringence.

> **Key points to remember:**
> - Congo red, apple-green birefringence.
> - Localised bladder amyloid mimics cancer; biopsy then TUR.
> - Always exclude systemic AL disease.

## SAQ 29 Collin's knife.

P3

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 3, Q3b

*Taken to mean the **Collings knife**, the knife-shaped cutting electrode of the resectoscope (the reference chapters spell it both "Collins" and "Collings").*

### Description

- A **hot (electrosurgical) knife electrode** fitted to the **resectoscope working element** in place of the loop.
- Thin **straight or angled (L- or hook-shaped) blade**; cuts a precise line with **cutting current** under irrigation and direct vision.
- **Monopolar** versions need **non-conducting irrigant** (glycine, sorbitol or mannitol); bipolar versions allow normal saline.
- Adult and **infant (9.5 Fr) resectoscope** sizes.
- **Principle:** it **incises rather than resects**; a deep, controlled cut releases a fibrous ring with little tissue loss.

### Uses (from the chapters)

| Use | Technique point |
|---|---|
| **TUIP** (transurethral incision of prostate) | Gland under 30 mL, no median lobe; incise at 5 and 7 o'clock from below the ureteric orifice to just proximal to the verumontanum, down to the capsule |
| **Bladder neck contracture** (after TURP (transurethral resection of prostate) or simple prostatectomy) | Incise until the scar springs open, to about 22 Fr; mitomycin C for recurrence |

| Use | Technique point |
|---|---|
| **Ureterocele incision** (older children) | Low transverse full-thickness incision near the bladder floor to reduce reflux |
| **PUV (posterior urethral valve) ablation** | Infant resectoscope only if urethral calibre is normal; incise at 5, 7 (± 12) o'clock; avoid hot-loop resection (stricture) |
| **Bladder diverticulum neck** | Incise the narrow neck so the diverticulum drains |
| **PCNL (percutaneous nephrolithotomy) tract** | Incise scarred fascia when rigid dilators fail |

### Alternatives

- **Cold knife (Sachse)** urethrotome: preferred for urethral stricture; hot knife unnecessary there.
- **Holmium or thulium laser**, Bugbee electrode, hook electrode, **Mohan's valvotome** (for valves).

### Precautions and complications

- Stay away from the **ureteric orifices** and external sphincter; control depth (perforation, extravasation).
- TUR (transurethral resection) syndrome with monopolar irrigant if veins open.
- **Retrograde ejaculation** after bladder neck incision; stricture if the urethra is too small for the sheath.

**Diagram to draw:** resectoscope tip with L-shaped knife electrode, and the 5 and 7 o'clock TUIP incisions.

> **Key points to remember:**
> - Knife-shaped resectoscope electrode for incision, not resection.
> - Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
> - Monopolar needs glycine or sorbitol; bipolar uses saline.

## SAQ 30 APOGEE

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 7 Paper 3, Q3c

*Assumed meaning: the Apogee transvaginal mesh kit (American Medical Systems) for posterior vaginal wall and vault prolapse. It is not described in the chapter notes; facts below are general.*

### Definition

- A **prepackaged polypropylene mesh kit** placed transvaginally to support the **posterior compartment and vaginal apex** (DeLancey levels I and II). Anterior companion: **Perigee** (transobturator).
- Mesh: **type I macroporous monofilament polypropylene** (Amid classification), the only type that integrates into tissue.

### Indications (historical)

- Symptomatic posterior and vault prolapse, mostly **recurrent** or with poor native tissue, avoiding abdominal surgery.

### Technique (principle)

1. Midline posterior vaginal incision; dissect the rectovaginal space to the ischial spines.
2. Two small perianal incisions lateral and posterior to the anus.
3. Curved trocars pass through the **ischiorectal fossa**, piercing the iliococcygeus near the ischial spine, and deliver the mesh arms.
4. Mesh body fixed to the apex and laid over the rectovaginal septum **tension-free**; epithelium closed without trimming.

### Complications

- **Mesh exposure** (IUGA/ICS [International Urogynecological Association/International Continence Society] term; "erosion" deprecated), extrusion, **perforation** into rectum or bladder.
- **Dyspareunia**, buttock and pelvic pain, mesh contraction, defecatory dysfunction, ischiorectal haematoma or abscess, rectal injury.
- **Management:** small exposure → vaginal oestrogen, then partial excision; pain or perforation → excision of mesh and arms.

### Regulatory status

- **FDA (US Food and Drug Administration) 2011:** serious complications of transvaginal POP (pelvic organ prolapse) mesh "not rare", with no clear benefit over native repair. Reclassified high-risk in 2016.
- **2019:** FDA ordered remaining manufacturers to **stop selling transvaginal POP mesh**; Apogee-type kits are off the market. Synthetic mesh improves anatomy but complicates more (Cochrane 2016).
- This does **not** apply to mid-urethral slings or **sacrocolpopexy mesh**.

### Current alternatives

- Native posterior colporrhaphy or site-specific repair; uterosacral suspension or **sacrospinous fixation**; **sacrocolpopexy** (abdominal, laparoscopic, robotic) for apical prolapse; colpocleisis in the frail.

**Diagram to draw:** posterior mesh with two arms through the ischiorectal fossae.

> **Key points to remember:**
> - Apogee: transvaginal posterior/apical polypropylene mesh kit.
> - Exposure, dyspareunia and pain drove withdrawal.
> - FDA 2019 stopped transvaginal POP mesh sales.

## SAQ 31 Urethral stents and metallic stents in urology

P4

**Asked as:**

- Urethral stents — *Summer 2015 Paper IV Q3d (SAQ)*
- Metallic stents in urology — *Practice Set 3 Paper IV Q3c (SAQ)*

- Urethral stents. — *Practice Set 4 Paper IV Q3f (earlier version) (SAQ)*

**Type:** SAQ (10 marks)

(urethral stents also answered in 2015-Summer Paper 4, Q3d; ureteric stents in 2015-Summer Paper 3, Q1)

## Definition and introduction

- A urethral stent is an endoscopically placed tube that keeps the **prostatic or bulbar urethra** open, mainly in men unfit for surgery.
- Polymer stents (polyurethane, silicone) fail early under **extrinsic compression** and need frequent changes. Metal stents **resist compression** and stay longer, but suffer **encrustation, tissue ingrowth and migration**.

## Classification (urethral stents)

| Type | Examples | Features |
|---|---|---|
| **Temporary** | **Memokath** (nitinol), **Allium** | Not epithelialised (designed to avoid epithelial incorporation); removable; left 6–12 months |
| **Permanent, epithelialising** | **UroLume** | Urothelium grows over it; **withdrawn** |
| Biodegradable | Experimental | Dissolve |

## Classification of metallic stents (by site)

| Site | Type | Example | Features |
|---|---|---|---|
| Ureter | Full-length coiled metal, double pigtail | **Resonance** (2006) | Tightly coiled alloy; changed every 12 months |
| Ureter | Segmental self-expanding mesh, bare or covered | Wallstent-type; covered nitinol (Allium) | Placed across malignant stricture; bare mesh allows ingrowth |
| Ureter | Thermo-expandable nitinol coil | Memokath 051 | Removable |
| Prostate | Temporary thermo-expandable nitinol coil | **Memokath** | Not epithelialised; removable |
| Prostate, bulbar urethra | Covered self-expanding, removable | Allium | Left 6–12 months |
| Bulbar urethra | Permanent epithelialising mesh | **UroLume** | Withdrawn |

## Ureteric metal stents

- **Indications:** malignant extrinsic obstruction; retroperitoneal fibrosis (first Resonance use); benign stricture in a patient unfit for repair; ureteroenteric stricture with limited life expectancy; failed tandem polymer stents.
- **Outcomes:** full-length metal stent: up to 3 years of drainage, pooled success about 72.5%, migration about 1%, but about one-third failed in one malignant series. Ureteroenteric strictures: patency >80% at 6–22 months.
- **Problems:** encrustation, difficult exchange, ingrowth, migration.

## Urethral and prostatic stents

### Indications

1. BPH (benign prostatic hyperplasia) retention in men unfit for TURP (transurethral resection of prostate) / unfit for surgery.
2. Recurrent short bulbar stricture in men unfit for urethroplasty.
3. **DSD (detrusor–sphincter dyssynergia)**: formerly (once) an alternative to sphincterotomy; now **abandoned** for complications.

### Technique

- Local or regional anaesthesia with endoscopic or fluoroscopic guidance. Memokath expands when warmed (with warm water) and softens when cooled (with cold water) for removal. A bulbar stent must **not cross the external sphincter**, or incontinence follows.

### Complications

- **Migration** (temporary stents), **encrustation**, **hyperplastic ingrowth** with restenosis (UroLume), incontinence, pain and painful erections.

### Results (outcomes)

- Prostatic stents free most unfit men from catheters short term, but many are later removed. Memokath variable. Removable bulbar stents succeed only in small series. UroLume worked early (good early results, then ingrowth) but failed long term from **hyperplastic ingrowth** and restenosis, so withdrawn, and many men still carry one.
- No stent removes **spongiofibrosis**, so none cures a stricture.

## Guidelines

- **AUA (American Urological Association)**: **no (urethral) stent if urethroplasty is considered** (planned); stents make reconstruction harder.

## Recent advances

- **Optilume** paclitaxel balloon (FDA (Food and Drug Administration) approval 2023 for anterior stricture): **ROBUST III**, after failed endoscopy, about 75% vs 27% stricture-free. Optilume BPH (EVEREST-1).
- **iTind** (temporary nitinol prostatic device): nitinol device left 5–7 days; AUA 2023 option for 25–75 mL glands without median lobe.
- **Also know:** biodegradable and drug-eluting stents (experimental).

**Also know:** a soft silicone "urethral stent" (splint) after hypospadias repair or urethroplasty is a different device; in complicated hypospadias repairs it reduces meatal stenosis and fistula.

**Diagram to draw:** sagittal urethra with prostatic stent and bulbar stent distal to sphincter; prostatic Memokath ending above the external sphincter.

**Key points to remember:**

- Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
- Metal resists extrinsic compression; weakness is ingrowth and encrustation.
- Resonance: coiled, full-length, changed every 12 months.
- Memokath: thermo-expandable nitinol, removable.
- Migration, encrustation, ingrowth; never cross the sphincter.
- AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

## SAQ 32 Operating room safety : the role of surgeon.

P4

**Type:** SAQ (10 marks)
**Source:** 2021-Summer Paper 4, Q3a

### Introduction

- About half of hospital adverse events involve surgery; half of these are preventable. The surgeon is **team leader** and ultimately responsible.
- **Swiss cheese model (Reason):** harm occurs when holes in several barriers line up, so safety needs layered defences.

### WHO (World Health Organization) Surgical Safety Checklist (2008, 19 items)

| Phase | Key checks |
|---|---|
| **Sign in** (before induction) | Identity, site, procedure, consent; site marked; allergy; airway; blood loss >500 mL |
| **Time out** (before incision) | Team introductions; confirm patient, site, procedure; surgeon states critical steps, duration, blood loss; antibiotic within 60 min; imaging displayed |
| **Sign out** (before leaving theatre) | Procedure recorded; counts correct; specimen labelled; equipment problems; recovery concerns |

- Pilot in 8 hospitals: deaths fell 47%, major complications 36%.

### Surgeon's responsibilities

1. **Universal Protocol** (Joint Commission, 2004): verification, **site marking** by the operating surgeon, patient awake, correct side only, time-out. Wrong-site surgery is a never event.
2. **Lead the time-out:** whole team engaged; repeat when procedure or surgeon changes.
3. **Retained surgical items** (sponges commonest): a count discrepancy raises risk about 100-fold; do not close until resolved; X-ray or RF (radiofrequency) tags.
4. **Fire safety:** the surgeon owns the **ignition source** (diathermy, light cables, laser). Let alcohol prep dry; $FiO_2$ (inspired oxygen fraction) ≤30% near the airway.
5. **Sharps:** double gloves, blunt needles, neutral zone (no hand-to-hand passing).
6. **Distractions:** "sterile cockpit" during critical steps (e.g. hilar clamping).
7. **Communication** (failure underlies up to 70% of events): Use **TeamSTEPPS** (Team Strategies and Tools to Enhance Performance and Patient Safety) tools: brief, debrief, **SBAR** (situation, background, assessment, recommendation), closed-loop check-back, **CUS** ("concerned, uncomfortable, safety issue"), two-challenge rule. Flatten hierarchy so juniors speak up.

8. **Self-monitoring: I'M SAFE** (illness, medication, stress, alcohol, fatigue, eating).

- **Also know:** positioning and padding; VTE (venous thromboembolism) prophylaxis; simulation.

**Diagram to draw:** Swiss cheese model with aligned holes.

> **Key points to remember:**
> - WHO checklist: sign in, time out, sign out.
> - Universal Protocol: verify, mark, time-out.
> - Count discrepancy: ~100-fold risk of a retained item.
> - Fire triangle: surgeon owns ignition.

## SAQ 33 Urological applications in nanotechnology.

P4

**Type:** SAQ (10 marks)
**Source:** 2022-Summer Paper 4, Q3a

*Taken as "applications of nanotechnology in urology".*

### Definition

- **Nanotechnology:** design and use of materials of about **1–100 nm**. At this scale, surface area and optical, magnetic and thermal properties change.
- **Why useful:** particles pass through leaky tumour vessels (**enhanced permeability and retention**), carry drugs or tracers, and can be targeted with ligands such as PSMA (prostate-specific membrane antigen) antibodies.

### Applications

| Area | Application |
|---|---|
| **Imaging and staging** | **USPIO (ultrasmall superparamagnetic iron oxide) lymphotropic MRI (magnetic resonance imaging):** normal nodes take up particles and lose T2 signal; metastatic nodes do not (Chapter 6; investigational in penile cancer, Chapter 84). Gold-nanoparticle CT (computed tomography) contrast, quantum dots |
| **Sentinel node** | Technetium-99m **nanocolloid** for dynamic sentinel node biopsy in penile cancer (Chapter 84) |
| **Diagnosis** | Nanowire and nanoparticle biosensors detecting PSA (prostate-specific antigen) and urinary markers at very low levels |
| **Drug delivery** | Nanoparticle (albumin-bound) paclitaxel in urothelial and prostate cancer (Chapter 164: tolerated, no pathological complete response); **NanoPac** intratumoral paclitaxel (Chapter 163); liposomal drugs; mucoadhesive intravesical carriers; siRNA (small interfering RNA) delivery |
| **Focal ablation** | **Gold nanoshells (AuroLase):** collect in tumour vessels and are heated by near-infrared laser; about three-quarters success on 12-month biopsy (Chapter 163). Magnetic nanoparticle hyperthermia |
| **Irreversible electroporation** | Electric pulses create **nanoscale membrane pores** ("NanoKnife"); spares matrix, vessels and ducts; investigational (Chapter 143) |
| **Devices** | Silver and other nanocoatings on catheters and stents against biofilm and encrustation |

| Area | Application |
|---|---|
| **Tissue engineering** | Electrospun nanofibre scaffolds that mimic extracellular matrix |
| **Stone disease** | "Calcifying nanoparticles" (nanobacteria) linked to Randall plaque, not proven causal (Chapter 95); nano-CT of stones and tissue |

### Limitations

- Toxicity and long-term tissue accumulation; uptake by liver and spleen; manufacturing cost; regulatory hurdles. Most uses remain experimental.

**Diagram to draw:** nanoparticle with drug core, polymer shell and surface PSMA ligand entering a leaky tumour vessel.

> **Key points to remember:**
> - 1–100 nm; enhanced permeability and retention allows tumour targeting.
> - USPIO MRI: benign nodes lose signal; metastatic nodes do not.
> - AuroLase gold nanoshells: photothermal focal therapy.

## SAQ 34 Morcellation

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 1 Paper 4, Q3d

### Definition

- Cutting a large specimen into small pieces so it leaves through a port, small incision or endoscope.
- **Manual:** ring forceps or Kelly clamp inside a bag. **Power (electromechanical):** rotating cylindrical blade (laparoscopy) or endoscopic tissue morcellator (prostate).

### 1. Endoscopic morcellation after enucleation

- Made **HoLEP (holmium laser enucleation of prostate)** practical; also ThuLEP (thulium laser enucleation).
- **Steps:** 1. Achieve good haemostasis first. 2. Pass a long nephroscope with a 5 mm channel. 3. Run two inflow lines fully open to keep the bladder distended. 4. Engage lobes with suction; cut only in mid-bladder, away from the wall. 5. Collect fragments in a trap for histology.
- Devices: Piranha (rotating blades), VersaCut (oscillating).
- **Complications:** bladder mucosal injury (stop, release suction, disengage; mostly catheter only); intraperitoneal perforation needs laparotomy; **retained adenoma** (from malfunction or poor vision) needs repeat morcellation.

### 2. Laparoscopic morcellation (kidney, spleen)

- **LapSac** (nylon with polyurethane lining) is impermeable to tumour cells and bacteria.
- **Rules:** purpose-built bag; drapes; change gloves and instruments after; no deep forceps passes; keep pneumoperitoneum and watch the bag.

| | Intact extraction | Morcellation |
|---|---|---|
| Incision | Larger | Small, better cosmesis |
| Pain, time, stay | Similar | Similar |
| Pathology | Full staging | Type and grade kept; size, capsule, vein invasion, and depth in UTUC (upper tract urothelial carcinoma) lost |
| Seeding | — | Rare port-site recurrence |

- Splenectomy morcellation risks **splenosis** if the bag ruptures.

### 3. FDA 2014 warning (gynaecology)

- In 2014 the FDA (Food and Drug Administration) warned against laparoscopic power morcellation in hysterectomy or myomectomy for fibroids, because unsuspected uterine sarcoma (about 1 in 350) was spread and upstaged.
- A boxed warning followed; contained (in-bag) morcellation is advised.
- Lesson for urology: avoid uncontained power morcellation of possible malignancy.

### Complications summary

- Bag rupture, seeding, lost staging, visceral or vascular injury, retained tissue.

**Diagram to draw:** enucleated lobes in a distended bladder with morcellator suction.

> **Key points to remember:**
> - Morcellator made HoLEP practical; distend bladder, haemostasis first.
> - Morcellation loses stage information; histology still obtained.
> - FDA 2014: uncontained power morcellation spreads occult sarcoma.

## SAQ 35 Stem cell therapy

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 1 Paper 4, Q3f (tissue engineering also in 2023-Winter Paper 1, Q1)

### Definition

- Stem cells **self-renew** and **differentiate** into specialised cells. Therapy uses them to repair or replace damaged tissue.

### Types

| Type | Source | Comment |
|---|---|---|
| Embryonic | Blastocyst | Pluripotent; ethics, teratoma |
| Adult mesenchymal | Bone marrow, adipose (ADSC, adipose-derived stem cells), umbilical cord | Easy harvest; mainly paracrine |
| Muscle-derived (MDSC) | Skeletal muscle biopsy | Used for sphincter and detrusor |
| iPSCs (induced pluripotent stem cells) | Reprogrammed adult cells | Patient-specific; tumour risk |

| Type | Source | Comment |
|---|---|---|
| Urine-derived, spermatogonial | Urine; testis | Research |

### Mechanisms

- Differentiation into muscle, nerve or endothelium.
- **Paracrine action:** growth factors, anti-inflammatory, anti-fibrotic and neurotrophic signals.
- Homing to injured tissue; seeding of scaffolds.

### Applications

- **SUI (stress urinary incontinence):**
- Autologous muscle-derived cells: thigh muscle biopsy → culture → transurethral or periurethral injection into the sphincter.
- Aim: regenerate the striated sphincter, not just bulk it.
- Early studies improved; placebo-controlled trials have not shown clear superiority, so it remains investigational.
- After childbirth injury, stem cell homing chemokines (CXCL12, CCL7) rise.
- **ED (erectile dysfunction):** intracavernosal bone-marrow, adipose or cord cells; animal and small uncontrolled human series only. SMSNA (Sexual Medicine Society of North America): experimental, IRB (institutional review board) protocols only.
- **Underactive detrusor:** muscle-derived cells injected into the bladder wall; two open-label studies improved voiding; research.
- **Tissue engineering:** cell-seeded scaffolds for bladder (Atala, 2006) and urethra (Raya-Rivera, 2011); adipose stem cell-seeded grafts for Peyronie disease (experimental).
- **Fertility:** spermatogonial stem cell banking in prepubertal cancer patients.
- **Kidney:** iPSC kidney organoids (research).

### Challenges

- Poor cell survival and retention; tumour formation (embryonic, iPSC); standardisation, cost and regulation.
- Unregulated "stem cell clinics" offer unproven treatments outside trials, a growing ethical concern.
- **Also know:** cancer stem cells (a different concept) may drive tumour recurrence and resistance.

**Diagram to draw:** muscle biopsy → culture → periurethral injection into sphincter.

> **Key points to remember:**
> - Adult stem cells act mainly by paracrine signals.
> - SUI muscle-derived cells and ED stem cells are investigational.
> - Embryonic and iPSCs carry tumour risk.

## SAQ 36 Advances in lap surgeries

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 4, Q3a (related: LESS in 2023-Winter Paper 4, Q1; energy sources

in 2015-Summer Paper 2, Q3d)

### Introduction

- Since the first laparoscopic nephrectomy (Clayman, 1991), advances aim at **less access trauma, better vision, safer haemostasis, gentler physiology and faster training**.

### 1. Access and insufflation

- **Non-bladed dilating and optical trocars**: port-site hernia fell from 1.83% to 0.19%, wall bleeding from 0.83% to 0.16%.
- **Balloon dilators** for retroperitoneal and extraperitoneal space.
- **AirSeal** valveless system: stable pneumoperitoneum at **8–12 mmHg** despite suction, smoke evacuation, less $CO_2$ absorption.
- Lower pressure (12 rather than 15 mmHg): less pain and oliguria.

### 2. Fewer and smaller incisions

| Technique | Gain | Limitation |
|---|---|---|
| **LESS** (laparoendoscopic single-site surgery) | Cosmesis, less analgesia | No triangulation, clashing |
| **NOTES** (natural orifice transluminal endoscopic surgery) | No abdominal scar (hybrid transvaginal extraction) | No proven benefit |
| Mini-laparoscopy (2–3 mm instruments) | Less pain, cosmesis | Instrument flexibility |
| Hand-assisted | Tactile feedback, fast bleeding control | Larger wound |

### 3. Vision

- Chip-on-tip HD (high-definition) and **3D** laparoscopes; deflectable tips.
- **ICG (indocyanine green) near-infrared fluorescence**: vessel mapping for selective clamping in partial nephrectomy, ureteric identification, sentinel nodes.
- Laparoscopic ultrasound; 3D-printed planning models.

### 4. Haemostasis and suturing

- **Advanced bipolar sealers** (LigaSure: vessels ≤7 mm, 1–3 mm spread) and **ultrasonic shears** (about 1 mm spread, but very hot jaws).
- **Hem-o-lok** clips, **barbed sutures** (shorter anastomosis), Lapra-Ty clips, powered articulating staplers.

### 5. Perioperative care

- **ERAS (enhanced recovery after surgery)**: stay about 30% shorter, complications halved; no routine bowel preparation.

### 6. Training

- Box trainers, VR (virtual reality) simulators, EAU **E-BLUS** (European Basic Laparoscopic Urological Skills) certification, 5-day mini-residencies.

### 7. Robot-assisted laparoscopy

- Wristed 3D platforms (da Vinci Xi, SP 2018; Hugo, Versius, SSI Mantra) solved laparoscopic suturing (**see Q3b**).
- **Also know:** augmented reality, artificial intelligence, telesurgery.

**Diagram to draw:** LESS multichannel port in the umbilicus with articulating instruments.

> **Key points to remember:**
> - Blunt trocars cut port-site hernia to 0.19%.
> - AirSeal allows 8–12 mmHg working pressure.
> - ICG fluorescence guides selective clamping.
> - LESS: cosmesis at the cost of triangulation.

## SAQ 37 Artificial intelligence in urology

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 4, Q3d

### Definitions

- **AI (artificial intelligence):** computer systems doing tasks that need human intelligence: recognising patterns, predicting, deciding.
- **ML (machine learning):** algorithms learn from data rather than fixed rules. **Supervised** (labelled data, e.g. tumour or not), **unsupervised** (finds clusters), reinforcement.
- **Deep learning:** multilayer **ANNs (artificial neural networks)**; **CNNs (convolutional neural networks)** are the commonest for images.
- **NLP (natural language processing)** and **LLMs (large language models)** handle text; **radiomics** extracts quantitative image features.

### Development steps

`Data collection and labelling → training → internal validation → external validation → regulatory approval → use and monitoring`

### Applications

| Area | Examples |
|---|---|
| Imaging | CNNs detect stones on CT (computed tomography) with very high sensitivity; ANNs detect renal tumours; prostate MRI (magnetic resonance imaging) lesion detection; deep-learning reconstruction allows ultra-low-dose CT (about 1 mSv) |
| Pathology | Automated cancer detection and Gleason grading on digital slides; Paige Prostate was the first FDA (Food and Drug Administration)-authorised AI pathology tool (2021) |
| Endoscopy | Real-time bladder-tumour detection at cystoscopy; future ureteroscopes setting laser energy, irrigation, pressure and temperature |
| Stones | Predict composition and stone-free outcome |
| Prognosis | Radiomics, pathomics and genomics combined for RCC (renal cell carcinoma) prognosis; AI on biopsy slides predicting benefit of ADT (androgen deprivation therapy) with radiotherapy |
| Robotic surgery | Video and kinematic data for skill assessment, outcome prediction, real-time guidance; future semi-autonomy |
| Paediatrics | Deep learning on ultrasound predicts CKD (chronic kidney disease) progression in valves; VUR (vesicoureteral reflux) infection-risk models |
| Others | Urodynamics interpretation; LLM chatbots for patient information |

### Advantages

- Fast big-data processing; consistency; decision support.

### Limitations and ethics

- Needs large, high-quality labelled data; **bias** and poor generalisability (overfitting); "black box" decisions; few prospective or external validations.
- Privacy and data sharing; medico-legal liability; LLM "hallucinations"; regulation.
- AI assists, not replaces, the urologist or radiologist.

**Diagram to draw:** neural network: input layer (image pixels), hidden layers, output (stone or no stone).

> **Key points to remember:**
> - Deep learning is part of ML, which is part of AI; CNNs for images.
> - CNNs: very high sensitivity for CT stones.
> - Paige Prostate: first FDA-authorised AI pathology tool (2021).
> - Limits: data, bias, black box, validation, liability.

## SAQ 38 3D laparoscopy

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 4, Q3b

### Introduction

- Standard laparoscopy gives a **two-dimensional (2D)** image. Depth is judged from monocular cues (shadow, size, motion parallax, touch), which slows suturing.
- **Three-dimensional (3D) laparoscopy** restores **stereoscopic depth perception**, as in robotic surgery (Chapter 18: two-lens binocular systems give depth perception).

### Principle

- Two lenses or two chip-on-tip sensors, a few millimetres apart, capture two images from slightly different angles, like the two eyes.
- The monitor shows both images; **polarised glasses** send one to each eye. The brain fuses them, and the **binocular disparity** is perceived as depth.
- **Also know:** active-shutter glasses; head-mounted displays; glasses-free (autostereoscopic) screens; 3D scopes at 0° and 30°, and with a deflectable tip; 3D 4K systems.

### Advantages

- Better depth judgement and hand–eye coordination.
- **Shorter time for complex tasks**, especially intracorporeal suturing and knotting, with fewer errors.
- **Shorter learning curve**; novices gain more than experts.
- Gives robotic-type vision at much lower cost.

### Disadvantages

- Visual strain, headache, dizziness or nausea in some users; glasses dim the image.
- Every team member needs glasses and a correct viewing distance and angle.

- Higher cost than 2D; 30° rotation needs care.
- No wristed instruments, tremor filtering or motion scaling: these come only with the robot.

### Urological uses

- Reconstructive steps: **pyeloplasty**, vesicourethral anastomosis in laparoscopic radical prostatectomy, **partial nephrectomy** renorrhaphy, ureteric reimplantation, donor nephrectomy.

### Evidence

- Randomised trials and meta-analyses in laboratory tasks and laparoscopic surgery show shorter operating and suturing times with 3D. Differences in complications and oncological outcomes are not established.

| Feature | 2D laparoscopy | 3D laparoscopy | Robot |
|---|---|---|---|
| Depth perception | No | Yes | Yes (magnified) |
| Wristed instruments | No | No | Yes |
| Cost | Low | Moderate | High |

**Diagram to draw:** two lenses at the scope tip, two images, polarised glasses, fused 3D image in the brain.

> **Key points to remember:**
> - Two lenses, two images, polarised glasses: binocular disparity gives depth.
> - Main gain: faster, more accurate suturing; shorter learning curve.
> - Drawbacks: eye strain, headache, cost, glasses for all.
> - Robotic vision at low cost, but without wristed instruments.

## SAQ 39 Telemedicine in urology

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 4, Q3c

### Definition

- **Telemedicine** is delivery of health care at a distance using information and communication technology (video, phone, messaging, data transfer).
- **COVID-19 (coronavirus disease 2019)** distancing accelerated its adoption. Chapter 2: urologists' telehealth use rose sharply in 2019–2020.

### Types

| Type | Method | Urological example |
|---|---|---|
| Synchronous | Real-time video or phone consultation | Stone, PSA (prostate-specific antigen) and post-operative follow-up |
| Asynchronous (store-and-forward) | Reports, images, bladder diaries sent for later review | Scan reports, uroflow traces |
| Remote monitoring | Apps and home devices | Bladder diary apps, home uroflowmetry |

| Type | Method | Urological example |
|---|---|---|
| Provider-to-provider | E-consults, telementoring, tele-tumour boards | Haematuria triage (Chapter 8) |
| Telesurgery | Remote robotic surgery | Experimental; limited by latency |

### Uses in urology

- **History-driven care:** LUTS (lower urinary tract symptoms) review, results, drug titration.
- ED (erectile dysfunction) evaluation by selected telemedicine (Chapter 72); men's health in large, sparse regions (Chapter 68).
- After mid-urethral sling, telemedicine follow-up is safe, and app-based follow-up gave higher satisfaction than office visits (Chapter 116).
- Internet-based pelvic floor training equalled a postal programme in an RCT (randomised controlled trial) (Chapter 114).

### Benefits (Chapter 2)

- Access for rural and underserved patients; less travel; shorter waits; relatives join remotely; sustained satisfaction.

### Limitations (Chapter 2)

- **No examination or in-office tests**: vital signs, cystoscopy, PVR (post-void residual).
- Needs connectivity and a secure, simple platform; hard for elderly or non-technical patients.
- No assistant to pre-gather results; more no-shows.
- **Medico-legal:** US licensure is state-based, and pandemic payment waivers have ended.

### Indian framework

- **Telemedicine Practice Guidelines (2020)**: only a registered medical practitioner may consult; identity of patient and doctor confirmed; consent and records kept; narcotic and Schedule X drugs may not be prescribed.
- Government platform **eSanjeevani**; patient data covered by the Digital Personal Data Protection Act 2023.

**Diagram to draw:** patient (home or district hospital) linked by video and data to the urologist, with remote monitoring feeding in.

> **Key points to remember:**
> - Telemedicine trades examination for access.
> - Best for follow-up, results and history-driven visits.
> - Limits: no examination, connectivity, elderly, medico-legal.
> - India: Telemedicine Practice Guidelines 2020; eSanjeevani.

## SAQ 40 Simulators in urology

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 4, Q3d

### Rationale

- Fewer training cases and complex technology mean skills must be learnt **outside the patient**.
- Chapter 1: simulation is used to **acquire and maintain skills and to warm up** before operating.

### Types

| Type | Examples | Strengths and limits |
|---|---|---|
| **Bench / box trainer** | Peg transfer, cutting, suturing; chicken oesophagus anastomosis; 3D (three-dimensional)-printed perfused hydrogel kidney | Cheap, proven skill gains; limited realism |
| **VR (virtual reality)** | LapSim, LapMentor, RobotiX Mentor, SimNow (robotic prostatectomy); endourology trainers for cystoscopy, ureteroscopy, PCNL (percutaneous nephrolithotomy), TURP (transurethral resection of the prostate) | Records metrics (precision, economy of motion), builds curricula, sets expert benchmarks; costly; poor tissue feel |
| **Live animal** | Pig | Real bleeding; ethics, cost |
| **Cadaver** | Fresh-frozen | Real human anatomy; scarce |
| **Team (non-technical)** | High-fidelity theatre crises | Communication and teamwork |

### Validity (needed before a simulator is trusted)

- **Face:** looks real. **Content:** teaches the right skills. **Construct:** experts score better than novices. **Concurrent:** agrees with an established test. **Predictive:** simulator score forecasts operating performance.

### Evidence (Chapter 18, Chapter 12)

- VR training **beats no training or box training** for novices and improves live performance and operating time.
- Robotic simulator scores correlate with live RARP (robot-assisted radical prostatectomy) performance.
- **FRS (Fundamentals of Robotic Surgery):** a multi-institutional RCT (randomised controlled trial) showed better speed and accuracy.
- Ureteroscopy team simulation taught non-technical skills better than lectures (randomised trial).

### Structured curricula and assessment

- **EAU (European Association of Urology) E-BLUS** (European Basic Laparoscopic Urological Skills): online theory, then dry-model tasks, then examination.
- **Proficiency-based progression:** move on only after reaching expert benchmarks.
- Scoring: **GEARS (Global Evaluative Assessment of Robotic Skills)** and **OSATS (Objective Structured Assessment of Technical Skills)**.

### Limitations

- Cost, poor tissue feel, limited proof of better patient outcomes.

**Diagram to draw:** pyramid: box trainer at the base, then VR, then animal or cadaver, then supervised surgery at the top.

> **Key points to remember:**
>
> - Bench, VR, animal, cadaver and team simulation.
> - Validity: face, content, construct, concurrent, predictive.
> - VR beats box or no training for novices.
> - E-BLUS and FRS; proficiency-based progression; GEARS.

## SAQ 41 Virtual and augmented reality in urology

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 4, Q3e

### Definitions

- **Virtual reality (VR):** the user is immersed in a fully computer-generated environment, usually through a head-mounted display.
- **Augmented reality (AR):** digital images are laid over the real view, for example a 3D (three-dimensional) tumour model on the robotic camera image.
- **Mixed reality:** virtual holograms are anchored in the room and can be handled.

### Workflow

`CT (computed tomography) / MRI (magnetic resonance imaging) -> segmentation of organ, tumour, vessels, collecting system -> 3D virtual model -> viewed in VR, 3D-printed, or overlaid (AR) -> registration to the live surgical view`

### Applications

| Use | Examples |
|---|---|
| **Pre-operative planning** | Partial nephrectomy: tumour depth, arterial branches, selective clamping. **PCNL (percutaneous nephrolithotomy)**: calyx and tract choice (Chapter 98: immersive VR and 3D printing show early promise). Prostatectomy: MRI lesion model to plan nerve-sparing |
| **Intra-operative navigation** | AR overlay in the robot console to locate an endophytic tumour, hilar vessels, or a prostate lesion near the capsule |
| **Education** | VR anatomy, 360° operation videos; simulation, see Q3d |
| **Patient counselling** | Patients understand their anatomy and planned operation; better-informed consent |
| **Paediatrics** | VR distraction reduces anxiety before surgery and procedures such as catheterisation and VCUG (voiding cystourethrogram) (Chapters 35 and 38) |

### Limitations

- **Registration error:** organs move and deform with breathing, pneumoperitoneum and dissection, so a rigid overlay drifts; manual alignment is still common.
- Segmentation time and cost.
- Chapter 161: promising but **resource-intensive and hard to validate**.
- Chapter 98: 3D reformatting added nothing over orthogonal CT measurements for predicting stone-free rate.
- Little outcome evidence; cybersickness with headsets.

### Future

- **AI (artificial intelligence)** for automatic segmentation and real-time tracking.
- **Digital surgery:** reconstruction of the patient's own anatomy allows rehearsal before the operation (Chapter 147).

**Diagram to draw:** CT kidney, segmented 3D model, and overlay of the model on the robotic view of a renal tumour.

> **Key points to remember:**
> - VR immerses; AR overlays on reality.
> - Main uses: planning partial nephrectomy, PCNL and prostatectomy; navigation; teaching; counselling.
> - Biggest problem: registration on moving, deforming organs.
> - Evidence for outcomes still limited.

## SAQ 42 3D printing in urology

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 4, Q3a

### Definition

- **3D (three-dimensional) printing** is additive manufacturing: a solid object is built layer by layer from a digital model.
- In urology it turns a patient's CT (computed tomography) or MRI (magnetic resonance imaging) into a physical replica of kidney, tumour, vessels or prostate.

### Workflow

`Thin-slice CT/MRI (DICOM files) → segmentation of kidney, tumour, vessels, calyces → STL (standard tessellation language) mesh → printer → post-processing`

- Takes hours to days; segmentation needs expertise.

### Printing technologies

| Method | Principle | Use |
|---|---|---|
| FDM (fused deposition modelling) | Melted plastic filament extruded | Cheap, rigid models |
| SLA (stereolithography) | Laser cures liquid resin | Fine detail, transparent models |
| Material jetting (PolyJet) | Jetted photopolymers, multi-colour | Tumour, vessels and calyces in different colours |
| SLS (selective laser sintering) | Laser fuses powder | Durable parts, devices |
| **Bioprinting** | Living cells in hydrogel "bio-ink" | Tissue engineering (see Practice Set 6 Paper 4, Q2) |

### Applications

1. **Partial nephrectomy planning:** complex hilar or endophytic tumours. The model shows the tumour against vessels and collecting system, so the surgeon plans the approach and rehearses steps. Promising.

2. **PCNL (percutaneous nephrolithotomy) planning:** staghorn stones, anomalous kidneys; choice of calyx. Early promise; 3D reformatting did not predict stone-free rate better.
3. **Prostate:** models with MRI lesions for margin and nerve-sparing planning and focal therapy.
4. **Patient education and consent:** patients understand their anatomy and the operation better.
5. **Training and simulation:** silicone or hydrogel phantoms for PCNL puncture, partial nephrectomy, pyeloplasty, anastomosis.
6. **Patient-specific devices:** prototype stents, catheters and surgical guides.

### Limitations

- Cost, printing time and segmentation errors.
- Materials do not reproduce tissue feel or bleeding.
- Evidence mostly small series; no proven outcome benefit.

### Future

- Virtual and augmented reality models on the robotic console, point-of-care printing, drug-eluting printed stents.

**Diagram to draw:** CT → segmentation → STL → printer → colour kidney model with tumour, artery and calyces.

> **Key points to remember:**
> - Additive manufacturing from CT/MRI via segmentation and an STL file.
> - Main uses: partial nephrectomy and PCNL planning, consent, simulation.
> - Limits: cost, time, unrealistic tissue, weak outcome evidence.

# Rapid Revision Sheet · Section 1

### Contrast agents in urology: iodinated and non-iodinated agents, complications and contrast-induced nephropathy (LAQ)

- Iodine : particle ratio 1.5 (ionic), 3 (non-ionic monomer), 6 (iodixanol, 290 mOsm/kg).
- Non-ionic agents: severe reactions 0.04% vs 0.22% ionic.
- Anaphylaxis: IM adrenaline 0.3–0.5 mg, max 1 mg; shellfish allergy is not a risk.
- Premedication: prednisone 50 mg at 13-7-1 h or methylprednisolone 32 mg at 12-2 h, plus diphenhydramine.
- CIN = rise ≥25% or 0.5 mg/dL in 48–72 h; now CA-AKI vs CI-AKI.
- Main CA-AKI risk: eGFR <30 (IV); <45 intra-arterial.
- Isotonic saline only proven; PRESERVE 2018: bicarbonate and NAC useless.
- Metformin: withhold 48 h only if eGFR <30 or AKI.
- NAC: thiol; glutathione precursor, antioxidant, mucolytic.
- NAC is the antidote for paracetamol overdose; best within 8 h.
- Cystinuria uses tiopronin, not NAC.
- GBCAs shorten T1; NSF risk eGFR <30; group II agents almost no risk.
- Microbubbles: intravascular, no nephrotoxicity; ceVUS for reflux.
- $CO_2$ angiography: never above diaphragm.

### Laparoscopic access: Veress, Hasson, trocars, gases; open vs laparoscopic vs robotic prostatectomy (LAQ)

- Veress tests: aspiration, drop, advancement; opening pressure <10 mmHg.
- Hasson 1971: open entry under vision.
- Main indication for Hasson: previous surgery or adhesions.
- Hasson avoids blind Veress injuries; drawback is gas leak.
- Extraperitoneal space is created first: finger, then balloon.
- Gaur balloon: glove on red rubber catheter.
- Balloon in muscle causes hernia.
- Trocar = cannula + obturator.
- Bladeless trocars: fewer epigastric injuries and hernias.
- Balloon trocar seals and prevents retroperitoneal emphysema.
- $CO_2$ standard (soluble, non-flammable); helium in COPD; $N_2O$ supports combustion.
- Gas embolism: left lateral head-down, oxygen, aspirate.
- Yaxley 2016: equal function and margins; robot less blood loss and stay.

### Pneumoperitoneum physiology and complications of laparoscopy and retroperitoneoscopy (LAQ)

- Two drivers: pressure and absorbed $CO_2$; work at 10–12 mmHg.
- Afterload rises at all pressures; output collapses at 40 mmHg.
- Oliguria from renal vein compression; reversible.
- Hypercarbia peaks after extubation; COPD needs arterial gases.
- Vascular injury is commonest; leave the trocar in place.
- Gas embolism: left lateral, head down, oxygen, aspirate.

- Missed bowel injury: trocar-site pain, leucopenia.
- Retroperitoneoscopy: space is created, not natural: wrong plane gives hernia, peritoneal tear or bleeding.
- Emphysema commonest; balloon-tip port and 12 mmHg prevent it (prevents leak).
- Pneumothorax 1.6–4%, commoner retroperitoneally.
- Rhabdomyolysis about 1%: limit kidney rest to <1 hour.
- Ratner 1995; left kidney preferred; hand-assisted in about two-thirds.
- Mortality 3 per 10,000; conversion 0.26–3.34%.
- Donor ESRD 90 versus 14 per 10,000 (matched non-donors).

**Tissue engineering in urology: principles, status and recent advances (LAQ)**

- Triad: cells, scaffold, signals; plus bioreactor and blood supply.
- Diffusion limit about 100–200 µm: vascularity is the main barrier.
- Atala 2006 (7 patients) promising; phase II trial: no gain in compliance or capacity.
- Neo-urinary conduit (conduit trial): half stenosed; no blood supply.
- Raya-Rivera 2011: seeded tubular urethras in 5 boys.
- Still experimental except short urethral onlays.
- New: iPSCs, urine-derived cells, bioprinting, organoids.

**Radionuclide (isotope) renal scans: MAG3 and DMSA (LAQ)**

- $^{99m}$Tc: 6 hours, 140 keV; PET: two 511 keV photons.
- DTPA filtered = GFR; MAG3 secreted = obstruction and transplant; DMSA cortical = scars.
- MAG3: tubular secretion, about 60% extraction; best for split function and drainage.
- Split function (DRF) from 1–3 minute counts; normal 45–55%; furosemide 0.5 mg/kg.
- T½ >20 min obstructed, <10 min not.
- Hydration, catheter, correct diuretic timing: the well-tempered renogram.
- DMSA binds proximal tubules: a static cortical scan at 2–4 h.
- DMSA is the gold standard for acute pyelonephritis and scars.
- Scar only if the defect persists ≥6 months; pyelonephritis resolves.
- Captopril renogram: ≥10% improvement off ACE inhibitors.
- PSMA PET preferred; FDG for residual seminoma >3 cm.
- MIBG best in MEN-2; sentinel node with $^{99m}$Tc-nanocolloid.
- Sestamibi hot = oncocytoma; $^{177}$Lu-PSMA-617 (VISION).

**Energy sources in urology and laparoscopic surgery, including bipolar energy (LAQ)**

- Electrosurgery at about 500 kHz; current density decides heating.
- Cut continuous low voltage; coagulation interrupted high voltage.
- Spread: monopolar 5–7 mm > bipolar 1–6 > sealer (LigaSure) 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; Harmonic vibrates at 25–55 kHz; ultrasonic blade hottest after use.
- Stray current: insulation failure, direct and capacitive coupling; avoid hybrid trocars; use AEM and REM.
- Bipolar: no pad, saline irrigation, no TUR syndrome.
- Bipolar TURP equals monopolar efficacy with fewer complications.
- Holmium 2120 nm: gold standard for stones and HoLEP.
- Cryo below −40 °C, two cycles; RFA 60 °C; MWA less heat sink.

- Thermal ablation for cT1a ≤3 cm.
- HIFU, IRE, VTP: focal prostate therapy, still evolving.

**LASERs in urology: holmium, thulium, KTP and newer laser technologies (LAQ)**

- Chromophore sets penetration: TFL shallowest, Ho 0.4 mm, Nd:YAG deepest.
- Holmium: 2120 nm; water chromophore; pulsed; 0.4 mm penetration.
- Holmium lithotripsy is photothermal; Moses effect: fibre close and perpendicular.
- Moses: two-part pulse through a vapour channel.
- Holmium: fragment 0.6–1.0 J × 6–10 Hz; dust 0.2–0.3 J × 40–80 Hz.
- HoLEP size-independent, superior to TURP; PVP for anticoagulated men (GOLIATH).
- TFL 1940 nm, Tm:YAG 2013 nm; water absorption about 4× holmium.
- TFL: water peak, low peak power, fine dust, thin fibres, minimal retropulsion; settings ≤40 Hz often enough.
- TFL beats low-power, equals high-power pulse-modulated holmium.
- ThuLEP is size-independent and equal to HoLEP.
- KTP 532 nm: frequency-doubled Nd:YAG; haemoglobin chromophore.
- GOLIATH: 180 W PVP non-inferior to TURP; best for anticoagulated men; not >80 cc.
- Nd:YAG for PeIN; eyewear for everyone.
- Heat depends on power: ≤10 W in the ureter, ≤25 W in the kidney; access sheath; damage above 43 °C.

**Describe IVU & its applications in Urology. (LAQ)**

- Contrast is filtered, not reabsorbed: nephrogram then pyelogram.
- Delayed, increasingly dense nephrogram = acute obstruction; no excretion at 24 h = non-functioning.
- Classic signs: drooping lily, cobra head, fish-hook, paintbrush, ring sign, phantom calyx.
- Severe reaction: IM adrenaline 0.3–0.5 mg; CTU has replaced IVU.

**Robotics in urology (including robotic vs standard laparoscopy) (LAQ)**

- da Vinci 2000; Xi 2014; SP 2018; patents expired 2019; 7 degrees of freedom.
- Yaxley and Coughlin: RARP equal function to open at 12 weeks and 24 months; RAZOR non-inferior; iROC more days out of hospital.
- RAPN trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Trendelenburg causes oedema and raised IOP; avoid shoulder braces.
- Robot: 3D, 7 DOF, tremor filtration, but no haptics and high cost.
- Robotic partial nephrectomy: shorter warm ischaemia, fewer conversions.
- Robot made prostatectomy routine; no benefit for simple extirpation.
- Surgeon matters more than platform.

**Flexible endourological instruments, digital and current ureteroscopes (LAQ)**

- Working channel plus active deflection made flexible scopes therapeutic.
- fURS: 3.6 Fr channel, ≥270° (>270°) deflection; cystoscope 14–17 Fr.
- Chip-on-tip CMOS digital: better image, larger tip, durability not better; shorter operation.
- Single-use cheaper below about 99 cases/year; LithoVue first single-use (2016).
- Semirigid: two channels; below the iliac vessels.
- Pressure sensors and AI are the newest developments.

- Golden rule: never fire unless the fibre tip is seen on the stone.

**Natural orifice endoscopic surgery. (LAQ)**

- NOTES = transluminal via natural orifice; hybrid transvaginal is the only routinely used urological form.
- Transvaginal extraction removes the most painful wound.
- LESS: one incision, lost triangulation, more hernias.
- Barriers: closure, infection, orientation, platform; no proven benefit yet.

**Laparoendoscopic single-site (single-port) surgery: LESS (LAQ)**

- LESS = one incision; LESSCAR (2008) standardised the term.
- Core problem: lost triangulation; solved by articulating or curved instruments and robots.
- Same outcomes as laparoscopy with less pain and better cosmesis; more hernias.
- da Vinci SP (FDA 2018): one 25–27-mm cannula, wristed instruments.

**Pioneers of urology: Novick, Chaussy, Hopkins, Huggins, Calne, Gaur and Asopa (SAQ)**

- Hopkins: rod-lens system 1966; fibreoptics 1954.
- Huggins and Hodges 1941: androgen dependence; castration regresses prostate cancer; Nobel 1966.
- Huggins 1945: adrenalectomy for castration-resistant disease.
- Calne: azathioprine, ciclosporin, first UK liver transplant 1968.
- Gaur 1992: glove-balloon retroperitoneoscopy.
- Asopa: ventral sagittal urethrotomy, dorsal plate incised, free graft inlaid (2001).
- Novick (Cleveland Clinic): nephron-sparing, renovascular and bench surgery.
- PN indications: imperative, relative, elective (cT1a ≤4 cm).
- Warm ischaemia under about 25–30 minutes; ice slush if longer.
- Negative margin is enough; keep ≥20–30% of one kidney.
- Chaussy, Munich: first human SWL 7 February 1980, Dornier HM1.
- HM3 (1983) is still the efficacy benchmark.
- Spallation, squeezing, shear, cavitation; electromagnetic sources today.

**Bacteriuria in pregnancy and antibiotic policy (SAQ)**

- Prevalence unchanged; risk of pyelonephritis raised (15–45% untreated).
- Screen by culture at 12–16 weeks; treat all ASB for 4–7 days.
- β-lactams safe throughout; no fluoroquinolones or tetracyclines.
- Nitrofurantoin not near term (after 35 weeks); TMP not in 1st trimester.

**Evidence-based medicine and critical appraisal (SAQ)**

- Sackett 1996: best evidence + clinical expertise + patient values.
- 5 As: Ask (PICO), Acquire, Appraise, Apply, Assess.
- Level 1a = systematic review of RCTs; GRADE: strong or weak.
- NNT = 1/ARR; beware lead-time, length-time and confounding bias.
- Three questions: valid, results, applicable.
- Randomisation, concealment, blinding, ITT, follow-up >80%.
- CONSORT (RCT), PRISMA (reviews), STROBE (observational), STARD (diagnostic).

**Microscopic / asymptomatic haematuria and urinary biomarkers (SAQ)**

- MH: 3 or more RBCs/HPF on microscopy of one sample; dipstick not enough.
- Low: optional (repeat urinalysis or cystoscopy + ultrasound); intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
- Any gross haematuria = high risk.
- Cytology: specific for high grade/CIS, insensitive for low grade.
- BTA is falsely positive with blood.
- No markers in initial microhaematuria work-up; none replaces cystoscopy.

**Hemostasis in laparoscopic surgery (SAQ)**

- Pneumoperitoneum hides venous bleeding: inspect at low pressure before exit.
- LigaSure seals ≤7 mm; spread monopolar 5–7 mm, ultrasonic ≈1 mm.
- Hem-o-lok contraindicated on the donor renal artery.
- Never staple across clips; grey or white loads for vessels.

**Newer generation CT scan (SAQ)**

- Doses: conventional 10–20, low-dose <3–4, ultra-low 1–2 mSv.
- DECT: uric acid stones, virtual unenhanced, iodine maps.
- Enhancement ≥20 HU = solid; deep learning enables ~1 mSv CT.

**History of Laparoscopy (SAQ)**

- Bozzini 1806; Kelling 1901 (dog); Jacobaeus 1910 (human); Veress 1938; Hopkins rod-lens.
- Urology: Cortesi 1976; Clayman 1990; Gaur 1992; Ratner 1995.
- da Vinci 2000; Binder first robotic prostatectomy; robot solved suturing barrier.

**Foley's (urethral) catheters, guidewires and DJ stents with their complications (SAQ)**

- Foley balloon catheter: designed for post-prostatectomy haemostasis.
- Fr = 3 × diameter (mm); inflate with sterile water; coudé for prostatic enlargement, Councill over a wire.
- Suspected urethral injury → RUG first; AUS → deactivate, 12–14 Fr.
- CAUTI 5–10% per day; remove early; change the catheter when treating.
- Hydrophilic wire to pass, PTFE wire to stay (safety).
- DJ symptoms in about 80%: trigonal irritation and reflux; α-blockers and antimuscarinics help.
- Long-dwelling stents: monitor for encrustation; encrustation rises with dwell time; FECal I–V.
- Encrusted stent: bladder, then ureter, then kidney; keep a stent register.

**Proteinuria (SAQ)**

- Normal <150 mg/day; nephrotic >3.5 g/day.
- ACR A2 30–300, A3 >300 mg/g.
- Glomerular, tubular, overflow; orthostatic proteinuria is benign.
- Dipstick misses light chains; use sulfosalicylic acid or immunofixation.

**Health-related quality of life (HRQOL) instruments in urology, including after urinary diversion (SAQ)**

- Generic (SF-36, EQ-5D) vs disease-specific (EPIC, IPSS, ICIQ, IIEF) plus bladder-specific (FACT-BI, BLM30, BCI) tools.

- SHIM lacks bother; I-QOL lacks symptoms.
- Reliable, valid, responsive; Cronbach α >0.7; IPSS MCID ~3 points.
- No diversion proven superior; studies flawed.
- Neobladder better body image; conduit better urinary function.

**PET scans in urology. (SAQ)**

- proPSMA: PSMA PET 92% vs 65% accuracy.
- Seminoma residual >3 cm: FDG-PET at ≥6 weeks; not NSGCT.
- FDG poor in prostate and clear cell RCC.
- VISION: $^{177}$Lu-PSMA-617 in mCRPC.

**Recist Criteria. (SAQ)**

- RECIST 1.1: ≥10 mm lesion, node ≥15 mm short axis; 5 targets, 2 per organ.
- PR ≥30% fall from baseline; PD ≥20% and ≥5 mm rise from nadir, or new lesion.
- iRECIST for immunotherapy; PCWG3 for prostate.

**Outline algorithm for difficult catheter placement. (SAQ)**

- Blind → wire → vision → suprapubic.
- BPH: 18-Fr coudé, tip up.
- Councill catheter over wire after dilatation.
- Suspected urethral injury: RUG first, no blind catheter.

**Indications for outpatient cystourethroscopy. (SAQ)**

- Haematuria and NMIBC surveillance are the two commonest indications.
- Flexible scope, lidocaine dwell ≥5 minutes, no routine prophylaxis.
- Therapeutic: stent removal, botulinum toxin, fulguration of small recurrences.

**MR urography (SAQ)**

- Static T2 (no contrast, function-independent) versus excretory gadolinium T1.
- Stones are signal voids; CTU preferred for urothelial cancer.
- Functional MRU: furosemide 1 mg/kg, Patlak analysis for split function.
- NSF risk minimal with group II agents.

**Urine cytology (SAQ)**

- High specificity, low sensitivity; good for high grade and CIS.
- Paris System: N:C ≥0.7 for suspicious and HGUC; AUC N:C ≥0.5.
- Positive cytology, normal cystoscopy: upper tracts, random biopsies.
- Not for screening or initial microhaematuria.

**Patient positioning in lap urology (SAQ)**

- Nerve injury about 2.8%; brachial plexus is commonest at risk.
- Never shoulder braces in steep Trendelenburg.
- Axillary roll below, not in, the axilla.
- Rhabdomyolysis: men, BMI ≥25, kidney rest; CK >5,000.

**Clinical trial methodology (SAQ)**

- Phase III RCT with concealed allocation and blinding is the gold standard.
- Primary endpoint drives sample size; surrogates need the Prentice criteria.

- ITT analysis; CONSORT flow diagram for reporting.

**Doppler in urology (SAQ)**

- RI = (PSV − EDV)/PSV; normal renal 0.47–0.70.
- Penile PSV <25 cm/s arterial; EDV >5 cm/s veno-occlusive.
- Absent intratesticular flow = torsion, but explore on clinical grounds.

**Flow cytometry (SAQ)**

- Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
- DNA index 1.0 diploid; aneuploidy = worse prognosis.
- Flow crossmatch is more sensitive than CDC crossmatch.

**VCUG (SAQ)**

- Gold standard for VUR and PUV; sterile urine first.
- Capacity (age × 30) + 30 mL (2–14 years); cyclic study detects 12–20% more.
- Grades I–II non-dilating, III–V dilating.

**Interventional radiology in urology and therapeutic angioembolisation (SAQ)**

- PCN equals retrograde stent for decompressing an obstructed infected kidney.
- Superselective coils: treatment of choice for pseudoaneurysm and AVF.
- Embolisation is first line for bleeding AML and post-PCNL pseudoaneurysm or arteriovenous fistula; prophylactic >6 cm or aneurysm >5 mm.
- Bladder bleeding: internal iliac anterior division; spare posterior division.
- Post-embolisation syndrome is common and self-limiting.

**Tumour markers in urology (SAQ)**

- PSA: organ-specific, not cancer-specific.
- AFP never in pure seminoma; half-lives AFP 5–7 days, hCG 24–36 hours.
- S stage uses post-orchidectomy values.
- Cytology: good for high grade only.

**Haemostatic agents in urology (SAQ)**

- Agents are adjuncts, not substitutes for sutures.
- Fibrin sealant works despite coagulopathy; passive agents need normal clotting.
- Surgicel: acidic, may mimic abscess.
- Alum first line; formalin last, after cystogram.

**Sterilisation of urological instruments (SAQ)**

- Spaulding: critical sterilise, semi-critical HLD, non-critical low-level.
- Autoclave 121 °C 15 min or 134 °C 3–4 min.
- Ureteroscopes: ethylene oxide, glutaraldehyde or peracetic acid.
- Biological indicator: *G. stearothermophilus*.

**Ethics and informed consent in urology (SAQ)**

- Four principles: autonomy, beneficence, non-maleficence, justice.
- Canterbury v Spence: reasonable-patient standard; Samira Kohli in India.
- Commonest gap: not checking understanding.
- Exceptions: emergency, incapacity (surrogate), minors.

**Operating room set-up (SAQ)**

- Zones with one-way flow; positive pressure, about 20 air changes an hour.
- Monitor directly ahead at eye level; open tray ready for laparoscopy.
- Hypothermia triples wound infection (Kurz).
- ALARA: time, distance, shielding.

**Amyloidosis in urology (SAQ)**

- Congo red, apple-green birefringence.
- Localised bladder amyloid mimics cancer; biopsy then TUR.
- Always exclude systemic AL disease. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Collin's knife. (SAQ)**

- Knife-shaped resectoscope electrode for incision, not resection.
- Uses: TUIP, bladder neck contracture, ureterocele, PUV, diverticulum neck.
- Monopolar needs glycine or sorbitol; bipolar uses saline.

**APOGEE (SAQ)**

- Apogee: transvaginal posterior/apical polypropylene mesh kit.
- Exposure, dyspareunia and pain drove withdrawal.
- FDA 2019 stopped transvaginal POP mesh sales.

**Urethral stents and metallic stents in urology (SAQ)**

- Temporary (Memokath, Allium) vs permanent (UroLume, withdrawn).
- Metal resists extrinsic compression; weakness is ingrowth and encrustation.
- Resonance: coiled, full-length, changed every 12 months.
- Memokath: thermo-expandable nitinol, removable.
- Migration, encrustation, ingrowth; never cross the sphincter.
- AUA: no stent if urethroplasty planned; Optilume and iTind are newer.

**Operating room safety : the role of surgeon. (SAQ)**

- WHO checklist: sign in, time out, sign out.
- Universal Protocol: verify, mark, time-out.
- Count discrepancy: ~100-fold risk of a retained item.
- Fire triangle: surgeon owns ignition.

**Urological applications in nanotechnology. (SAQ)**

- 1–100 nm; enhanced permeability and retention allows tumour targeting.
- USPIO MRI: benign nodes lose signal; metastatic nodes do not.
- AuroLase gold nanoshells: photothermal focal therapy.

**Morcellation (SAQ)**

- Morcellator made HoLEP practical; distend bladder, haemostasis first.
- Morcellation loses stage information; histology still obtained.
- FDA 2014: uncontained power morcellation spreads occult sarcoma.

**Stem cell therapy (SAQ)**

- Adult stem cells act mainly by paracrine signals.

- SUI muscle-derived cells and ED stem cells are investigational.
- Embryonic and iPSCs carry tumour risk.

**Advances in lap surgeries (SAQ)**

- Blunt trocars cut port-site hernia to 0.19%.
- AirSeal allows 8–12 mmHg working pressure.
- ICG fluorescence guides selective clamping.
- LESS: cosmesis at the cost of triangulation.

**Artificial intelligence in urology (SAQ)**

- Deep learning is part of ML, which is part of AI; CNNs for images.
- CNNs: very high sensitivity for CT stones.
- Paige Prostate: first FDA-authorised AI pathology tool (2021).
- Limits: data, bias, black box, validation, liability.

**3D laparoscopy (SAQ)**

- Two lenses, two images, polarised glasses: binocular disparity gives depth.
- Main gain: faster, more accurate suturing; shorter learning curve.
- Drawbacks: eye strain, headache, cost, glasses for all.
- Robotic vision at low cost, but without wristed instruments.

**Telemedicine in urology (SAQ)**

- Telemedicine trades examination for access.
- Best for follow-up, results and history-driven visits.
- Limits: no examination, connectivity, elderly, medico-legal.
- India: Telemedicine Practice Guidelines 2020; eSanjeevani.

**Simulators in urology (SAQ)**

- Bench, VR, animal, cadaver and team simulation.
- Validity: face, content, construct, concurrent, predictive.
- VR beats box or no training for novices.
- E-BLUS and FRS; proficiency-based progression; GEARS.

**Virtual and augmented reality in urology (SAQ)**

- VR immerses; AR overlays on reality.
- Main uses: planning partial nephrectomy, PCNL and prostatectomy; navigation; teaching; counselling.
- Biggest problem: registration on moving, deforming organs.
- Evidence for outcomes still limited.

**3D printing in urology (SAQ)**

- Additive manufacturing from CT/MRI via segmentation and an STL file.
- Main uses: partial nephrectomy and PCNL planning, consent, simulation.
- Limits: cost, time, unrealistic tissue, weak outcome evidence. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 2 · Oncology, DSD, Gender & Ageing

## Long answer questions (LAQ)

### LAQ 1 Immunology, immunotherapy and cancer vaccines in urological oncology (including Sipuleucel-T)

P2 · P4 **merged from 2**

*Asked as:* P2 LAQ "Describe use and principles of immunology and Immunotherapy in Urological Oncology" · P4 LAQ "Rationale of using cancer vaccines in Urology. Explain their mechanism of action. Enumerate various vaccines for renal, prostate and bladder malignancies . Write a note on Sipuleucel-T."

#### Introduction

- Immunotherapy works by either **supplying a danger signal** that starts a Th1 response (BCG (bacille Calmette–Guérin), vaccine adjuvant) or **removing a brake** on an existing response (PD-1 (programmed death-1), CTLA-4 (cytotoxic T-lymphocyte antigen-4)).
- A **cancer vaccine** is a therapeutic agent that trains the patient's own T cells to recognise **tumour antigens**. It treats existing cancer; it does not prevent it.
- **BCG**, a tuberculosis vaccine, was among the first immune treatments for any cancer and is the original urological immunotherapy. It is non-specific; cancer vaccines aim for a **specific** anticancer response.

#### Principles of tumour immunology

| Arm | Cells and role |
|---|---|
| **Innate** (rapid, pattern-based) | PAMPs (pathogen-associated molecular patterns) bind **TLRs (Toll-like receptors)**; macrophages, neutrophils; urothelium as sentinel (IFN (interferon), IL-8 (interleukin-8)) |
| **Bridge** | **Dendritic cell (DC)**: takes up antigen, matures on danger signal, migrates to lymph node |
| **Adaptive** (specific, memory) | **CD8 killer** (sees class I MHC (major histocompatibility complex), ~9 amino acids); **CD4 helper** (class II); B cells |
| **Brake** | **Treg** (regulatory T cell, FoxP3+ CD4); MDSCs (myeloid-derived suppressor cells); M2 macrophages |

- **Two signals**: TCR (T-cell receptor) meets MHC–peptide (signal 1) plus **CD28–B7** costimulation (signal 2). Without signal 2 the T cell becomes tolerant.
- **Th1 vs Th2**: IL-12 gives Th1 (IL-2, IFN-γ, TNF-α (tumour necrosis factor-α)), which is antitumour. IL-4 gives Th2 (antibody, chronic inflammation).
- **Checkpoints**: CTLA-4 (on Tregs) competes for B7; PD-1 on CD8 cells binds **PD-L1** (programmed death ligand-1); LAG-3 binds class II.
- **Adaptive immune resistance**: CD8 cells secrete IFN-γ, which induces PD-L1 on the tumour, which switches the same T cells off. So PD-L1 marks a suppressed ongoing response.

### Immunoediting (three Es) and the rationale for immunotherapy in urology

- **Elimination** (small tumours destroyed), then **Equilibrium** (growth held), then **Escape** (clinical cancer).
- Escape is by loss of antigen or MHC, Treg expansion and checkpoint ligands. Vaccines try to restart the response.
- **Bladder:** inflammation-driven (schistosomiasis); CD8 infiltration is good; BCG success; high mutation load gives many neoantigens.
- **Kidney: CD8 paradox** (more CD8, worse outcome); rare spontaneous regression of metastases after nephrectomy; responses to IL-2 and interferon.
- **Prostate:** Treg-dominated and PD-L1 low, so checkpoint inhibitors fail; slow growth gives time to build a vaccine response; **organ-specific antigens** (PSA (prostate-specific antigen), PAP (prostatic acid phosphatase)).
- Vaccines have low toxicity and offer possible lasting memory.

### BCG in NMIBC (non-muscle-invasive bladder cancer)

- Intravesical live attenuated mycobacteria.
- Mechanism (sequence): 1. Bacilli attach via **fibronectin**. 2. TLR2/4 activation → IL-8 and TNF-α. 3. Neutrophil wave → DC priming. 4. **Th1 CD4** response → memory.
- Th1 predicts success, Th2 failure. About half stay in remission for years.
- Used for intermediate/high-risk NMIBC and CIS (carcinoma in situ) with induction plus maintenance; standard for high-risk NMIBC.

### Cytokines and gene/viral therapy

- **IFN-α** and **high-dose IL-2**: durable complete responses in a few mRCC (metastatic renal cell carcinoma) patients; capillary leak. Intravesical IFN-α: mixed results.
- **Nadofaragene firadenovec** (adenovirus carrying the IFN-α2b gene + Syn3): BCG-unresponsive CIS, CR (complete response) 53.4%; FDA (Food and Drug Administration) approved December 2022.
- **Cretostimogene**: oncolytic adenovirus with GM-CSF (granulocyte-macrophage colony-stimulating factor) gene.
- **N-803** (IL-15 superagonist) with BCG: FDA approved 2024 for BCG-unresponsive CIS.

### Cancer vaccines: mechanism of action

- Two components: **antigen** (overexpressed in tumour) + **adjuvant** (a danger signal, often GM-CSF, which activates DCs).
- Sequence: `Intradermal antigen + adjuvant → immature DC takes up antigen; adjuvant matures it → DC migrates to lymph node → peptide on MHC I/II = signal 1; B7 costimulation = signal 2 → CD4 Th1 help (IL-2, interferon-γ) + CD8 cytotoxic T cells → T cells traffic to tumour → apoptosis of antigen-bearing cells → memory T cells`
- Without signal 2 the T cell becomes tolerant, hence the adjuvant.
- **Prime–boost:** the first dose primes, later doses boost.
- **Delayed effect:** survival may improve without tumour shrinkage or PSA fall.
- **Resistance:** PD-L1 on tumour, Tregs, MDSCs, MHC loss. This is the basis for combining vaccines with checkpoint inhibitors.

## Vaccine platforms

| Platform | Example |
|---|---|
| Whole-cell (autologous or allogeneic) | Reniale (RCC (renal cell carcinoma)); GVAX (prostate) |
| Peptide/protein | IMA901 |
| Dendritic-cell (ex vivo loaded) | Sipuleucel-T; AGS-003 (rocapuldencel-T) |
| Viral vector (poxvirus) | PROSTVAC-VF; TroVax (MVA-5T4) |
| Heat-shock protein–peptide | Vitespen (HSPPC-96) |
| DNA/mRNA, personalised neoantigen | Phase I–II trials |

## Vaccines by organ

### Renal cell carcinoma

- **IMA901:** 10 tumour-associated peptides + GM-CSF, with low-dose cyclophosphamide to deplete Tregs; HLA-A*02+ (human leucocyte antigen) metastatic RCC with sunitinib. **No OS (overall survival) benefit.**
- **AGS-003 (rocapuldencel-T):** autologous DCs loaded with tumour RNA, with sunitinib. **Terminated for lack of efficacy.**
- **TroVax** with standard first-line therapy: no OS benefit. **Vitespen** (adjuvant): no overall benefit.
- **Reniale** (adjuvant autologous tumour-cell vaccine): claimed benefit, not adopted.
- Checkpoint inhibitors, not vaccines, are today's RCC immunotherapy.

### Prostate cancer

- **Sipuleucel-T:** the **only approved** vaccine (see note below).
- **PROSTVAC-VF:** vaccinia prime, fowlpox boost, encoding PSA + TRICOM (three costimulatory molecules). Phase III **closed for futility**.
- **GVAX:** allogeneic whole-cell lines secreting GM-CSF. Phase III trials **failed**.
- Also: DCVAC/PCa (phase III negative); DNA vaccines against PAP (trials).

### Bladder cancer

- **BCG** (above); nadofaragene firadenovec, cretostimogene and N-803 (above).
- Peptide and neoantigen vaccines: experimental only.

## Why most vaccines failed

- Tolerance to self-antigens; immunosuppressive tumour environment; dysfunctional patient DCs; bulky late disease; progression endpoints miss delayed effects.
- **Pattern:** IMA901, AGS-003, PROSTVAC and GVAX all failed; **only sipuleucel-T succeeded in phase III.**

## Note on Sipuleucel-T

- **Nature:** autologous cellular immunotherapy: the patient's **CD54+ APCs (antigen-presenting cells)** activated with **PA2024**, a fusion protein of **PAP + GM-CSF**.
- **Manufacture and dosing:** 1. Leukapheresis collects PBMCs (peripheral blood mononuclear cells). 2. Cells incubated about 36 h with PA2024 at a central facility. 3. Infused intravenously about 3 days after collection. 4. **Three doses at 2-week intervals** (one-month course).
- **Mechanism:** primes PAP-specific CD4 and CD8 T cells; later infusions boost.

- **IMPACT trial (Kantoff, NEJM 2010):** 512 men, 2:1 versus placebo, asymptomatic or minimally symptomatic mCRPC (metastatic castration-resistant prostate cancer).
- **OS 25.8 vs 21.7 months (HR (hazard ratio) 0.78)**, a gain of 4.1 months, despite placebo crossover.
- **No change in time to progression or PSA.**
- **FDA approval April 2010:** first therapeutic vaccine for a solid tumour.
- **Indication:** asymptomatic or minimally symptomatic mCRPC, good performance status, **no visceral metastases, no opioid need**; usually before chemotherapy.
- **Toxicity:** mild infusion reactions (fever, chills, fatigue); premedicate with paracetamol and antihistamine.
- **Limitations:** cost, logistics, no response marker; not available in India.

### Checkpoint inhibitors

| Site | Use (trial) |
|---|---|
| **BCG-unresponsive CIS** | Pembrolizumab if cystectomy declined/unfit (KEYNOTE-057: CR 41%) |
| **Adjuvant MIBC (muscle-invasive bladder cancer) and UTUC** (upper tract urothelial carcinoma) | Nivolumab after radical surgery (**CheckMate 274**: DFS (disease-free survival) 20.8 vs 10.8 months; included UTUC, benefit uncertain in that small subgroup). Atezolizumab (IMvigor010) negative |
| **Metastatic UC (urothelial carcinoma)** | Avelumab **maintenance** after platinum (JAVELIN Bladder 100, OS gain); pembrolizumab second line (**KEYNOTE-045**: OS 10.3 vs 7.4 months); **enfortumab vedotin (EV) + pembrolizumab** |
| **RCC** | Nivolumab + ipilimumab (CheckMate 214); IO (immuno-oncology) + TKI (tyrosine kinase inhibitor) combinations; adjuvant pembrolizumab (KEYNOTE-564, DFS HR 0.68) |
| **Prostate** | Monotherapy poor; pembrolizumab only for **MSI-H/dMMR** (microsatellite instability-high/mismatch repair-deficient) tumours |

- VEGF (vascular endothelial growth factor) suppresses DCs and T-cell entry, so TKIs synergise with IO.
- **Biomarkers**: PD-L1 CPS (combined positive score) only for first-line single-agent pembrolizumab in cisplatin-ineligible UC; TMB (tumour mutational burden), MSI.

### Adoptive cell therapy (brief)

- **CAR-T (chimeric antigen receptor T cells)**: T cells engineered against a surface antigen (e.g. PSMA (prostate-specific membrane antigen)). Investigational in GU (genitourinary) cancers; limited by poor trafficking, hostile microenvironment and cytokine release syndrome.

### Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

| Grade | Action |
|---|---|
| 1 | Continue, monitor |
| 2 | Hold; prednisolone 0.5–1 mg/kg |
| 3–4 | Stop; methylprednisolone 1–2 mg/kg; infliximab or mycophenolate if refractory |

- Endocrinopathies need hormone replacement.
- BCG toxicity: cystitis, BCG sepsis.

### Recent advances

- **EV-302**: enfortumab vedotin + pembrolizumab first line in metastatic UC, OS 31.5 vs 16.1 months versus platinum; now the preferred first line.
- **NIAGARA**: perioperative durvalumab with neoadjuvant cisplatin chemotherapy improved event-free and overall survival in MIBC.
- **Also know:** CheckMate 901 (nivolumab + gemcitabine–cisplatin); ADT (androgen deprivation) + IO trials; anti-CD38 and LAG-3 targets; DNA/mRNA personalised neoantigen vaccines (phase I–II).

**Diagrams to draw:**

- T cell–tumour synapse with TCR–MHC, CD28–B7, CTLA-4 and PD-1/PD-L1 blocked by antibodies.
- Sipuleucel-T flow: leukapheresis, PA2024 loading, reinfusion, T-cell priming.

> **Key points to remember:**
> - Three Es: elimination, equilibrium, escape; Th1 is antitumour.
> - BCG is the original urological immunotherapy: TLR2/4 → neutrophils → Th1 memory.
> - Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
> - IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
> - Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months (+4.1 months); no PSA or progression effect.
> - CheckMate 274 adjuvant nivolumab; EV-302 EV + pembrolizumab first line.

## LAQ 2 DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia

P3

**Asked as:**

- Management of DSD — *Summer 2019 Paper III Q3f (SAQ)*
- Evaluation and treatment of newborn with ambiguous genitalia — *Practice Set 4 Paper III Q2 (LAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)

(DSD management also answered as an SAQ in 2019-Summer Paper 3, Q3f)

*DSD read as: disorders/differences of sex development. Detrusor–sphincter dyssynergia is covered in 2016-Summer Paper 1, Q3a.*

### Introduction and definition

- Atypical ("ambiguous") genitalia point to a **DSD (difference/disorder of sex development)**: congenital conditions with atypical chromosomal, gonadal or anatomical sex (Chicago consensus 2006).
- It is a **medical emergency only for salt-wasting CAH (congenital adrenal hyperplasia)**; sex designation is not a "social emergency". Evaluate within 24–48 hours (neonatal evaluation, first 24–48 h).

### Classification (by karyotype)

| Group | Main causes (examples) |
|---|---|
| **46,XX DSD** | **CAH** (commonest; 21-hydroxylase 95%, 11β-hydroxylase about 5%, 3β-HSD (hydroxysteroid dehydrogenase) <1%); maternal androgens or aromatase deficiency; ovotesticular; testicular (SRY (sex-determining region Y) positive, SRY+) |
| **46,XY DSD** | Partial/complete gonadal dysgenesis; androgen synthesis defects (17β-HSD3, 5α-reductase); androgen insensitivity (CAIS/PAIS, complete/partial; PAIS = partial androgen insensitivity); LH (luteinising hormone) receptor defect |
| **Sex chromosome DSD** | 45,X/46,XY MGD (mixed gonadal dysgenesis); 45,X Turner; 47,XXY Klinefelter; chimerism |

### History

- Consanguinity, **unexplained infant deaths** (CAH), infertility or amenorrhoea in relatives; maternal virilisation, progestogens or androgens; prenatal cfDNA (cell-free DNA) karyotype versus ultrasound discordance.

### Examination (gender-neutral terms)

- Stretched phallic length, glans width, meatus, number of perineal openings, labioscrotal fusion and rugosity, pigmentation, anogenital distance. **Prader grade (stage) I–V** for virilisation.
- Dehydration, hypotension, hyperpigmentation (CAH); dysmorphism.
- **Gonads are the key sign.** Ovaries never descend, so **a palpable gonad means testicular tissue**. Bilateral impalpable gonads with atypical genitalia = CAH until proven

otherwise (impalpable gonads suggest CAH). Asymmetry suggests MGD or ovotesticular DSD.
- Impalpable testis with hypospadias, or proximal hypospadias, also needs DSD work-up (Kaefer: DSD in about 30%, 50% if the testis is impalpable).

### Investigations (tests)

- **Karyotype** at once (preliminary 2–3 days); **FISH (fluorescence in situ hybridisation) for X/Y/SRY** within hours.
- **CAH screen: 17-OHP (17-hydroxyprogesterone) at about 36 hours** (after 36 h; earlier values are falsely high); electrolytes and glucose daily. **Emergency:** exclude **salt-wasting CAH** (low sodium, high potassium, shock); **salt-wasting appears around day 5–7** (usually after day 5–7), so normal early electrolytes do not exclude CAH.
- **At about 1 week:**

| Test | Meaning |
|---|---|
| AMH (anti-Müllerian hormone) | Present = functioning Sertoli cells |
| Testosterone, LH, FSH (follicle-stimulating hormone) | Testicular function (mini-puberty) |
| Androstenedione, DHEA-S (dehydroepiandrosterone sulfate) | Adrenal enzyme block |
| Testosterone:DHT (dihydrotestosterone) >20:1 | 5α-reductase deficiency |
| hCG (human chorionic gonadotropin) stimulation | No rise = no Leydig tissue or LH receptor defect |

- **Pelvic ultrasound:** uterus (well seen neonatally), gonads, kidneys, cerebriform adrenals (CAH); then genitography, endoscopy, laparoscopy ± biopsy, genetic panels.
- **Genitography / endoscopy:** urogenital sinus, confluence level relative to bladder neck, cervix.
- If unclear: MRI (magnetic resonance imaging), **laparoscopy ± deep longitudinal gonadal biopsy**, gene panels (genetic panels), microarray, exome. **No gonadectomy during diagnosis (while diagnosing).**

### Diagnostic algorithm

```
Palpable gonads: 0 -> CAH likely; 1 -> MGD, PAIS, ovotesticular; 2 -> 46,XY DSD (PAIS, 5α-reductase, synthesis defect) 46,XX + uterus + high 17-OHP -> CAH 46,XY, no uterus -> androgen synthesis/action defect 46,XY or 45,X/46,XY + uterus -> gonadal dysgenesis / MGD
```

- A precise diagnosis is reached in nearly all 46,XX but under 50% of 46,XY DSD.

### Treatment

#### 1. Emergency and medical management (CAH)

- Salt-wasting crisis: IV (intravenous) saline, dextrose, stress-dose hydrocortisone.
- Maintenance (Endocrine Society 2018): hydrocortisone 10–15 mg/m²/day in divided doses, fludrocortisone, salt supplements in infancy; stress dosing during illness and surgery.
- **Puberty:** oestrogen or testosterone induction if gonads fail.

#### 2. Team and family

- **MDT (multidisciplinary team):** urologist, endocrinologist, geneticist, psychologist, neonatologist, ethicist; gender-neutral language. Congratulate parents; say "your baby", never "it"; open disclosure, peer support.

**3. Sex of rearing (gender assignment)**

- Weighs diagnosis, prenatal androgen exposure, likely gender identity, fertility, sexual function, family wishes and culture.
- Usually female: 46,XX CAH, CAIS (complete androgen insensitivity), complete gonadal dysgenesis. Trend male: PAIS, 5α-reductase and 17β-HSD3 deficiency.
- Micropenis (stretched length under 2 cm) alone is no reason for female rearing.

**4. Surgery: timing and consent**

- Irreversible surgery is ideally deferred until the child can participate (consent); parents decide with the team after balanced counselling.
- **Feminising** surgery (clitoroplasty, vaginoplasty, labioplasty) is irreversible, so deferral until the child can consent is favoured unless obstructed; severe CAH is debated.
- Clitoroplasty (ventral, nerve-sparing, glans preserved; Prader III–V); vaginoplasty by confluence (flap for low, partial/total urogenital mobilisation, pull-through for high); labioplasty. The vagina's position relative to the bladder neck, not common-channel length, decides the vaginoplasty. Stress steroids in CAH.
- **Masculinising:** proximal hypospadias repair, orchidopexy, often done in infancy (complication rates high).

**5. Gonads (gonadal tumour risk)**

| Risk | Condition | Plan / management |
|---|---|---|
| High | Dysgenesis with Y, intra-abdominal (15–35%); PAIS non-scrotal (50%) | Gonadectomy |
| Intermediate | Turner with Y (12%) | Gonadectomy |
| Low | CAIS (2%); ovotesticular (3%) | Keep CAIS gonads until after puberty; ovotesticular: remove discordant tissue |

## Follow-up and recent advances

- Pubertal induction, psychology, fertility preservation (experimental gonadal tissue cryopreservation), transition to adult care.
- Prenatal cfDNA diagnosis; rapid exome/genome sequencing. Prenatal dexamethasone for CAH is experimental.

**Diagram to draw:** Prader stages I–V and the urogenital sinus with confluence level; karyotype-based DSD algorithm.

> **Key points to remember:**
> - Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
> - 17-OHP after 36 h; salt crisis around day 5–7.
> - Karyotype + uterus + gonads triad guides diagnosis; no gonadectomy while diagnosing.
> - Defer irreversible surgery; CAIS gonads kept until after puberty (till puberty).

## LAQ 3 Discuss in brief, present concepts in various GU cancers

P4

**Type:** LAQ (25 marks)
**Source:** Practice Set 4 Paper 4, Q1

*Read as: the current key concepts in diagnosis and management of each GU (genitourinary) cancer: kidney, upper tract, bladder, prostate, testis and penis.*

### Common themes

- **De-escalation:** surveillance and organ sparing for low-risk disease.
- **Precision:** molecular classification, germline testing and better imaging (mpMRI (multiparametric MRI, magnetic resonance imaging), PSMA (prostate-specific membrane antigen) PET (positron emission tomography)).
- **Perioperative systemic therapy:** immune checkpoint inhibitors (ICIs), ADCs (antibody–drug conjugates) and targeted drugs.
- **Robotic surgery** as the default approach.

### 1. Kidney: RCC (renal cell carcinoma)

- **WHO (World Health Organization) 2022:** new molecularly defined types (FH (fumarate hydratase)-deficient, SDH (succinate dehydrogenase)-deficient, TFE3-rearranged RCC).
- **Mechanism:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (vascular endothelial growth factor). This explains why VEGF TKIs (tyrosine kinase inhibitors) and the HIF-2α inhibitor **belzutifan** work.
- **Small renal mass:** active surveillance in the elderly or unfit; renal mass biopsy; **partial nephrectomy** for cT1 to avoid CKD (chronic kidney disease); thermal ablation for small tumours.
- **Adjuvant pembrolizumab** after nephrectomy for high-risk RCC (**KEYNOTE-564**: better disease-free and overall survival).
- **Metastatic:** IMDC (International Metastatic RCC Database Consortium) risk groups. First line is an **ICI combination**: nivolumab plus ipilimumab, or an ICI plus TKI (pembrolizumab–axitinib, nivolumab–cabozantinib, pembrolizumab–lenvatinib). **CARMENA (2018):** sunitinib alone was non-inferior to cytoreductive nephrectomy plus sunitinib in intermediate/poor risk, so cytoreduction is now selective. SBRT (stereotactic body radiotherapy) for oligometastases.

### 2. Upper tract urothelial carcinoma (UTUC)

- **Risk stratification (EAU (European Association of Urology)):** low risk (unifocal, <2 cm, low-grade, non-invasive on CT (computed tomography) urography) goes to **kidney-sparing** ureteroscopic ablation or segmental ureterectomy; UGN-101 (mitomycin gel) is an option. High risk goes to **RNU (radical nephroureterectomy) with bladder cuff** and template lymph node dissection.
- **Single postoperative intravesical mitomycin** after RNU cuts bladder recurrence (**ODMIT-C**: 28% to 16% at 1 year).
- **POUT:** adjuvant platinum chemotherapy improved disease-free survival in pT2–T4 or N+ disease.
- **Also know:** screen for **Lynch syndrome**; aristolochic acid (Balkan) nephropathy.

## 3. Bladder

- **NMIBC (non-muscle-invasive bladder cancer):** EAU risk groups; en bloc TURBT (transurethral resection of bladder tumour); blue-light or narrow-band cystoscopy; re-TURBT for T1; BCG (bacillus Calmette–Guérin) induction plus maintenance. **BCG-unresponsive** disease: radical cystectomy is standard; bladder-sparing options are pembrolizumab, nadofaragene firadenovec (gene therapy) and N-803 with BCG.
- **MIBC (muscle-invasive):** VI-RADS (Vesical Imaging Reporting and Data System) on MRI; **cisplatin-based neoadjuvant chemotherapy** then cystectomy; **trimodality** bladder preservation in selected patients; adjuvant **nivolumab** (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months); perioperative durvalumab with chemotherapy (**NIAGARA, 2024**) improved event-free and overall survival.
- **Metastatic: enfortumab vedotin plus pembrolizumab** (**EV-302**) replaced platinum first line; avelumab maintenance; erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

## 4. Prostate

- **Diagnosis: MRI before biopsy** (**PRECISION**: more significant and fewer insignificant cancers); **transperineal biopsy** (less sepsis); **PSMA PET** staging (**proPSMA**: more accurate than CT plus bone scan); ISUP (International Society of Urological Pathology) grade groups; germline testing (BRCA2).
- **Localised: active surveillance** for low risk and selected favourable intermediate risk (**ProtecT, 15 years**: cancer mortality low and similar with surveillance, surgery or radiotherapy); robotic prostatectomy; **hypofractionated** radiotherapy; focal therapy in selected men.
- **Advanced:** mHSPC (metastatic hormone-sensitive) gets ADT (androgen deprivation therapy) plus an **ARPI** (androgen receptor pathway inhibitor); **triplet** with docetaxel (**PEACE-1, ARASENS**); prostate radiotherapy for low volume. EMBARK: enzalutamide for high-risk biochemical recurrence. mCRPC (metastatic castration-resistant): **PARP (poly-ADP ribose polymerase) inhibitors** for HRR (homologous recombination repair) mutations; **$^{177}$Lu-PSMA-617** (**VISION**: overall survival HR (hazard ratio) 0.62).

## 5. Testis

- **Stage I: surveillance** preferred for most seminomas and NSGCTs (non-seminomatous germ cell tumours); alternatives are one cycle of carboplatin (seminoma) or one BEP (bleomycin, etoposide, cisplatin) for high-risk NSGCT.
- **miR-371a-3p:** new serum marker for seminoma and NSGCT.
- **Low-volume stage II seminoma: primary RPLND (retroperitoneal lymph node dissection)** (SEMS, PRIMETEST) spares chemotherapy toxicity; robotic, nerve-sparing templates preserve ejaculation.
- **Also know:** IGCCCG (International Germ Cell Cancer Collaborative Group) 2021 update; survivorship care.

## 6. Penis

- **WHO 2022: HPV (human papillomavirus)-associated** vs **HPV-independent** squamous cell carcinoma; p16 staining; HPV vaccination for prevention.
- **Organ sparing:** topical treatment or laser for PeIN (penile intraepithelial neoplasia), glans resurfacing, wide excision, glansectomy; a negative margin of a few millimetres is enough.

- **Nodes:** for cN0 pT1b or higher, **DSNB (dynamic sentinel node biopsy)** or modified inguinal dissection. Bulky nodes: neoadjuvant TIP (paclitaxel, ifosfamide, cisplatin), then consolidative dissection.

**Diagram to draw:** six-row table: cancer, key diagnostic advance, localised treatment, advanced treatment.

> **Key points to remember:**
> - RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
> - UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
> - Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
> - Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

## Short answer questions (SAQ)

### SAQ 1 46,XX male

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "46-XX Male" · P2 SAQ "46 XX Male"

*Assumed meaning: 46,XX testicular DSD (difference of sex development), "XX male" or de la Chapelle syndrome.*

#### Definition and classification

- **Testes and male genitalia in a person with a 46,XX karyotype.** Old term "XX sex reversal"; renamed **46,XX testicular DSD** (Chicago consensus 2006).
- First described by **de la Chapelle (1964)**. About **1 in 20,000** males.
- In the DSD classification it is a 46,XX DSD of gonadal development, together with ovotesticular DSD; **CAH (congenital adrenal hyperplasia)** is the androgen-excess type.

#### Genetics

| Type | Mechanism | Phenotype |
|---|---|---|
| **SRY-positive** (about 80–90%) | **SRY** (sex-determining region Y) translocated from Yp to the X chromosome during paternal meiosis | Usually normal male genitalia |
| SRY-negative | Testis genes switched on without SRY: **SOX9** duplication or upregulation, SOX3, loss of RSPO1 or WNT4 | More hypospadias, undescended testes, ovotestis |

- SRY activates SOX9, so Sertoli cells form and secrete **AMH (anti-Müllerian hormone)**: no uterus. Leydig cell testosterone virilises.
- The **AZF (azoospermia factor) region of Yq is absent**, so spermatogenesis always fails.

#### Clinical features

- Normal male identity; often found at **infertility work-up** for azoospermia, or prenatally (cell-free DNA 46,XX but male ultrasound).
- Small firm testes, **azoospermia**, hypergonadotropic hypogonadism, gynaecomastia in some; progressive testicular regression.
- Height normal or **shorter than average** (no Y growth genes), unlike Klinefelter.

| Feature | 46,XX male | Klinefelter (47,XXY) |
|---|---|---|
| Frequency | 1 in 20,000 | 1 in 500–1000 |
| Height | Normal or short | Often tall |
| Genital anomalies | More common | Rare |
| Sperm retrieval | **Never (no AZF)** | Possible by micro-TESE (microdissection testicular sperm extraction) |

### Evaluation

- Semen analysis (azoospermia); FSH (follicle-stimulating hormone) and LH (luteinising hormone) high, testosterone low or low-normal.
- **Karyotype** 46,XX; **FISH (fluorescence in situ hybridisation) for SRY** confirms; DSD gene panel if SRY-negative; scrotal ultrasound.
- In an infant with atypical genitalia, exclude CAH (17-hydroxyprogesterone) and ovotesticular DSD.

### Management

- **Testosterone replacement** when hypogonadal (libido, bone density, muscle).
- **Infertility:** testicular sperm extraction is futile; counsel for **donor sperm or adoption**.
- Repair hypospadias or undescended testis; reduction mammoplasty for persistent gynaecomastia.
- Genetic counselling, psychological support and bone density monitoring.

**Diagram to draw:** X and Y chromosomes showing Xp–Yp exchange carrying SRY onto X.

> **Key points to remember:**
> - 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
> - Azoospermia always (AZF absent): no sperm retrieval.
> - Shorter than Klinefelter; FISH for SRY confirms.

## SAQ 2 Female circumcision

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "Female circumcision" · P2 SAQ "Female Circumcision"

*Taken as female genital mutilation/cutting (Chapter 58 calls it "female circumcision (infibulation)"); the answer follows WHO (World Health Organization) definitions.*

### Definition and epidemiology

- **FGM (female genital mutilation):** all procedures that partly or totally remove the external female genitalia, or otherwise injure them, for non-medical reasons (WHO).
- About **200 million** living women and girls are affected; about 3 million African girls a year. Mainly Africa, the Middle East and Asia (Indonesia, Malaysia); practised in some Indian communities ("khatna").
- Usually done in childhood, at 4–10 years, without anaesthesia, as a rite of passage; it predates Islam. Pharaonic type = infibulation.

### WHO classification

| Type | Description |
|---|---|
| I (clitoridectomy) | Partial or total removal of the clitoral glans and/or prepuce |
| II (excision) | Removal of the clitoral glans and labia minora, with or without the labia majora |
| III (infibulation) | Narrowing of the vaginal opening (introitus) by cutting and joining the labia into a seal, with or without clitoral excision |
| IV (other) | Pricking, piercing, incising, scraping, cauterisation |

- The older Toubia scheme numbers differ: infibulation is Type III in WHO.

### Complications

- **Immediate:** pain, haemorrhage, shock, infection (tetanus), **acute urinary retention**, urethral or rectal injury, death.
- **Urological:** slow stream, dysuria, **recurrent UTI (urinary tract infection)**, meatal obstruction under the scar, urethral stenosis, incontinence, vaginal stone from urine stasis, **vesicovaginal fistula** after obstructed labour.
- **Gynaecological and sexual:** retained menses, dysmenorrhoea, inclusion cysts, keloid, dyspareunia, apareunia, infertility.
- **Obstetric:** obstructed labour, perineal tears, postpartum haemorrhage, neonatal death.
- **Psychological:** anxiety, depression, PTSD (post-traumatic stress disorder).

### Evaluation

- Sensitive, culturally respectful history; examination to classify the type; drawings or photographs for consent.
- Flow rate and residual urine; cystoscopy if fistula is suspected.

### Management

- **Deinfibulation (type III):** anterior **midline division of the scar** under local or regional anaesthesia to expose the urethral meatus and introitus; raw edges oversewn (subcuticular closure of each edge). High success.
- **Timing:** ideally before pregnancy, or antenatally, or in labour. Relieves retention, UTI and dyspareunia.
- **Never reinfibulate** (illegal in the UK).
- Excise cysts, treat stenosis, repair fistula; counselling.
- Clitoral reconstruction: less predictable, limited evidence and follow-up.

### Legal and ethical aspects

- WHO and the UN (United Nations) condemn FGM as a human rights violation; WHO opposes its performance by health workers.
- Illegal in many countries (e.g. UK FGM Act 2003, with mandatory reporting in girls under 18). India has no specific law.

**Diagram to draw:** vulva showing types I–III and the deinfibulation incision line.

**Key points to remember:**

- WHO types I–IV: I clitoridectomy, II excision, III infibulation, IV others.
- Urological: retention, recurrent UTI, stenosis, stones, fistula.
- Midline deinfibulation for type III; never reinfibulate.

## SAQ 3 Familial / hereditary urological malignancies

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "Familial malignancies in urology" (Familial malignancies in Urology, Practice Set 2 Paper I Q3b; Familial malignancy in urology, Practice Set 8 Paper I Q3f) · P2 SAQ "Hereditary urological malignancies."

### Principles

- **Knudson two-hit hypothesis:** every cell carries one inherited mutant tumour-suppressor allele, so one more hit causes cancer: tumours are **early, bilateral and multifocal**.
- **4–6% of RCC (renal cell carcinoma)** is familial. All familial RCC syndromes are **autosomal dominant**; only HPRC (hereditary papillary renal carcinoma) involves an activated oncogene.

### Hereditary kidney cancer syndromes

| Syndrome | Gene (chromosome) | Kidney tumour | Other features |
|---|---|---|---|
| **VHL** (von Hippel–Lindau) | VHL (3p25) | Clear cell, bilateral, multifocal (~50%) | Retinal angioma, CNS (central nervous system) haemangioblastoma, phaeochromocytoma, pancreatic cysts |
| **HPRC** | MET (7q31) oncogene | Multiple bilateral papillary (type 1, type 1-like) | None |
| **HLRCC** (hereditary leiomyomatosis and RCC) | FH (fumarate hydratase, 1q42–43) | **Aggressive**, often solitary papillary; collecting duct | Skin and uterine leiomyomas |
| **BHD** (Birt–Hogg–Dubé) | FLCN (folliculin, 17p11.2) | Chromophobe, hybrid oncocytic, oncocytoma | Fibrofolliculomas, lung cysts, pneumothorax |
| **TSC** (tuberous sclerosis complex) | TSC1 (9q34), TSC2 (16p13.3) | Multiple AMLs (angiomyolipomas); RCC 2–3% (P2: 1–3%) | Angiofibromas, cortical tubers |

- **VHL type 1** (deletions, truncations): no phaeochromocytoma; **type 2** (missense): phaeochromocytoma.
- **Also know:** SDH (succinate dehydrogenase)-deficient RCC (paraganglioma); BAP1 (uveal melanoma, mesothelioma); Cowden (PTEN; breast, thyroid).

### Management of familial kidney tumours

- **3 cm rule:** watch tumours <3 cm; nephron-sparing surgery at ≥3 cm (VHL, HPRC, BHD) to save function for repeat operations.

- **Exception: HLRCC**: excise promptly, with wide margins.
- TSC AML: everolimus at 3–5 cm (P1; another source: if >3 cm); embolisation or excision (partial nephrectomy) if ≥5 cm.
- **Belzutifan** (HIF-2α (hypoxia-inducible factor 2α) inhibitor), FDA (Food and Drug Administration)-approved 2021 for VHL tumours.

### Other urological cancers

- **Prostate: BRCA2**, BRCA1, ATM, CHEK2, PALB2, HOXB13, mismatch repair genes. Father and brother affected: 9-fold risk. BRCA2 gives aggressive prostate cancer. Earlier screening, cautious active surveillance, **PARP (poly ADP-ribose polymerase) inhibitors**.
- **UTUC (upper tract urothelial carcinoma): Lynch syndrome** (MLH1, MSH2, MSH6, PMS2): about 14-fold risk, about 75-fold with MSH2. Suspect if age <60 or a history of Lynch cancer; screen with MMR (mismatch repair) immunohistochemistry, then germline testing.
- **Testis:** affected brother RR (relative risk) 8–12, father 2–4.
- **Phaeochromocytoma:** about one-third hereditary: MEN-2 (multiple endocrine neoplasia type 2; RET), VHL, NF1 (neurofibromatosis type 1), SDHB (highest malignancy).
- **Also know:** Wilms tumour (WT1 gene: WAGR (Wilms, aniridia, genitourinary anomalies, retardation) and Denys–Drash; Beckwith–Wiedemann).

### Approach

- Young age, multifocality or family history: genetic counselling, germline testing, screening of relatives.

**Diagram to draw:** VHL loss leading to HIF accumulation, VEGF (vascular endothelial growth factor) production and angiogenesis.

> **Key points to remember:**
> - All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
> - VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
> - Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

## SAQ 4 Penile inversion vaginoplasty

P2 · P4 **merged from 2**

*Asked as:* P2 SAQ "Penile inversion vaginoplasty technique." · P4 SAQ "Penile Inversion Vaginoplasty Technique"

### Introduction

- The commonest gender-affirming genital operation for transgender women: a functional, aesthetic, sensate vulva and vagina.
- **Principle:** homologous penile and scrotal tissues transferred on their own blood supply.

### Preparation

- **WPATH (World Professional Association for Transgender Health) SOC8 (Standards of Care, version 8):** one assessment, **6 months of hormones** before gonadectomy.
- **Permanent hair removal** of the scrotal graft area (prevents intravaginal hair).
- BMI (body mass index) usually ≤35 kg/m²; stop smoking; sperm banking (fertility preservation); continue oestrogen.

- **Choose depth:** full-depth vs **vulvoplasty (zero-depth)**: a vulva without a canal, chosen for high surgical risk, prior prostatectomy, or no wish for penetrative intercourse.

### Tissue transfer (homologous tissues)

| Male structure | Becomes |
|---|---|
| Glans on dorsal neurovascular bundle | **Clitoris** |
| Preputial mucosal collar | Clitoral hood, inner labia minora |
| Dorsal bulbar urethral plate (spatulated urethra) | Vestibule |
| Penile shaft skin (inverted) | **Neovaginal lining** |
| Scrotal skin (full-thickness graft) | Extra (proximal) canal lining |
| Lateral scrotal skin + cord fat | Labia majora |

### Operative steps

`Lithotomy; tranexamic acid 1 g; catheter → harvest, defat and tubularise scrotal skin graft → bilateral orchidectomy (leave cord fat for labial bulk) → deglove penis between dartos and Buck's fascia (keep dartos with skin = blood supply) → separate dorsal neurovascular bundle + glans; excise ALL corporal erectile tissue → clitoroplasty; fix at level of adductor longus tendons → shorten and spatulate urethra; excise bulbospongiosus → canal dissection between prostate and rectum: central tendon, behind prostate, preserve Denonvilliers fascia, repeated rectal checks → invert penile skin tube + graft into canal; labiaplasty → pack canal; negative-pressure dressing`

- **Why excise all erectile tissue:** residual tissue engorges and obstructs (narrows) the canal.
- Depth depends on canal dissection, not penile length.
- **Robotic peritoneal flaps** add depth with less rectal injury; **intestinal vaginoplasty** if skin is inadequate or for revision.

### Postoperative care

- Pack and catheter out on about day 5.
- **Lifelong dilation**: several times daily at first, e.g. 4 times daily for 6 weeks, tapering to daily after 1 year; pelvic floor physiotherapy; continue oestrogen.

### Complications and outcomes

- **Wound dehiscence** (commonest, 12–33%); granulation tissue.
- **Rectovaginal fistula** (0.8% primary, 6.3% revision): the worst complication. Urethrovaginal fistula (0.8–3.9%).
- Urethral stricture, meatal stenosis or malposition (spraying).
- **Vaginal stenosis** (poor dilation); prolapse 4–10%.
- Most achieve orgasm (median about 6 months).

**Diagram to draw:** sagittal view of the lined neovaginal canal between prostate and rectum.

> **Key points to remember:**
> - Glans → clitoris; penile skin → canal; scrotal skin → graft.
> - Excise all erectile tissue; keep Denonvilliers fascia intact.
> - Lifelong dilation prevents stenosis; rectovaginal fistula is the worst complication.

## SAQ 5 Prepucioplasty

P3 · P4 **merged from 2**

*Asked as:* P3 SAQ "Prepucioplasty" · P4 SAQ "Prepucioplasty."

*Read as: preputioplasty, a foreskin-preserving operation for phimosis.*

### Definition

- An operation that widens the tight preputial ring but keeps the foreskin. It is an alternative to circumcision.

### Background

- **Physiological phimosis** at birth is due to glans–preputial adhesions. Smegma and erections separate them. About 90% retract by 3 years; fewer than 1% are phimotic at 17 years.
- **Secondary phimosis** follows scarring (forced retraction, balanoposthitis) or BXO (balanitis xerotica obliterans).

### Indications

- Persistent symptomatic phimosis: recurrent balanoposthitis, UTI (urinary tract infection), paraphimosis, painful erection.
- Failed **topical steroid** (betamethasone 0.05% twice daily for about 4 weeks with gentle retraction). About 1 in 6 fail.
- Wish to keep the foreskin (cultural reasons, cosmesis).

### Contraindications

- **BXO** with scarring: circumcision is preferred. Triradiate incision with triamcinolone has been reported in selected boys.
- Physiological non-retractility needs only reassurance.
- Treat active infection first.

### Techniques

| Technique | Method |
|---|---|
| Dorsal slit with transverse closure | Longitudinal incision through the ring, closed transversely (Heineke–Mikulicz principle) |
| Triple incision | Three longitudinal cuts over the ring, each closed transversely |
| Y–V plasty | V-flap of outer skin advanced into a Y-cut |

- **Principle:** cutting the ring lengthwise and closing it crosswise widens its circumference.
- General anaesthesia or penile block; incise only the band; confirm full retraction; absorbable sutures.

### Postoperative care

- **Daily retraction from the first week**, continued for weeks, prevents re-stenosis. A topical steroid may help.

### Outcomes and complications

- Most achieve a retractile foreskin. It is quicker and less painful than circumcision.
- Complications: re-phimosis (needs circumcision), bleeding, infection, poor cosmesis, paraphimosis.
- **Also know:** preputial reconstruction during hypospadias repair is also called preputioplasty; dehiscence is its risk.

**Diagram to draw:** dorsal incision through the ring, closed transversely.

> **Key points to remember:**
> - Steroid first; about 1 in 6 fail.
> - Cut lengthwise, close crosswise; daily retraction after.
> - BXO: circumcision preferred.

## SAQ 6 Oncogenes

P1

**Asked as:**

- Oncogenes — *Winter 2017 Paper I Q3f (SAQ)*
- Oncogene — *Practice Set 6 Paper I Q3f (SAQ)* (also answered in 2017-Winter Paper 1, Q3f)

**Type:** SAQ (10 marks)

### Definition

- **Proto-oncogene:** a normal gene that promotes cell growth and division (an "accelerator").
- **Oncogene:** its mutated or overactive form that drives (driving) uncontrolled proliferation. The change is **gain of function** and **dominant**: one altered allele is enough.
- The first human oncogene, **HRAS**, was isolated from (found in) the **T24 bladder cancer** cell line.

### Mechanisms of activation

| Mechanism | Effect | Example |
|---|---|---|
| **Point (activating) mutation** | Constitutively active protein (protein permanently "on") | RAS (bladder); FGFR3 (fibroblast growth factor receptor 3) in low-grade papillary bladder cancer; MET germline (germline MET) in hereditary papillary RCC (renal cell carcinoma) |
| **Gene amplification** | Increased copy number (more copies) → overexpression | MYC (prostate, bladder); HER2 (human epidermal growth factor receptor 2); AR (androgen receptor) in castration-resistant prostate cancer |
| **Chromosomal translocation / fusion** | Gene placed under a strong promoter, or fusion protein | **TMPRSS2–ERG** (androgen-driven ERG in about half of prostate cancers); BCR–ABL; TFE3 fusions (translocation RCC) |

| Mechanism | Effect | Example |
|---|---|---|
| - **Also know:** promoter hypomethylation can switch oncogenes on. | | |

## Functional classes

| Class | Examples |
|---|---|
| Growth factors | PDGF (platelet-derived growth factor), sis |
| **Receptor tyrosine kinases** | EGFR (epidermal growth factor receptor), HER2, FGFR3, MET, KIT (seminoma) |
| **Signal transducers** | **RAS** (binds GTP [guanosine triphosphate]; GTP-binding; mutation locks it "on"), BRAF, PI3K |
| **Transcription factors** | **MYC**, ERG, ETS family |
| Cell cycle / apoptosis regulators | Cyclin D1, BCL-2 |

## Oncogene versus tumour suppressor gene

| | Oncogene | Tumour suppressor |
|---|---|---|
| Normal role | Accelerator | Brake, DNA repair |
| Change | Gain of function | Loss of function |
| Alleles needed | One | Usually both (**Knudson two-hit**: retinoblastoma) |
| GU (genitourinary) examples | MYC, MET, FGFR3, ERG | RB1, TP53, VHL, PTEN |

## Urological significance

- **Bladder:** FGFR3 mutations in up to 80% of low-grade Ta tumours (10–20% of MIBC (muscle-invasive bladder cancer)); FGFR3 and TERT promoter mutations in >60% of low-grade papillary tumours; RAS (HRAS) mutually exclusive with FGFR3; HER2 in 9–12% of MIBC; MYC overexpressed in over half.
- **Kidney:** MET activation in papillary RCC.
- **Prostate:** TMPRSS2–ERG fusion, MYC (8q gain), AR amplification after (under) hormone therapy.
- **Testis:** KIT and KRAS mutations (seminoma).

## Targeted therapy ("oncogene addiction")

- **Erdafitinib** (pan-FGFR inhibitor) for FGFR-altered (FGFR3/2-altered) advanced (metastatic) urothelial cancer; MET-active TKIs (tyrosine kinase inhibitors) in papillary RCC; AR-targeted drugs in prostate cancer; HER2-directed drugs (trastuzumab deruxtecan received a tumour-agnostic FDA approval in 2024 for HER2-positive solid tumours; disitamab vedotin is under study in urothelial cancer).

**Diagram to draw:** growth factor → receptor → RAS → RAF → MEK → ERK → MYC pathway.

> **Key points to remember:**
>
> - Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
> - HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
> - TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

## SAQ 7 Premalignant conditions in urology (including premalignant lesions of penis)

P1

**Asked as:**

- Premalignant conditions in urology — *Summer 2021 Paper I Q3a (SAQ)*
- Premalignant lesion of penis — *Practice Set 6 Paper I Q3a (SAQ)*

**Type:** SAQ (10 marks)

(premalignant conditions also answered in 2021-Summer Paper 1, Q3a)

### Definition

- Altered tissue with a raised risk of invasive cancer, often **intraepithelial neoplasia with an intact basement membrane**. Treating it prevents cancer.
- Penis: a lesion with raised risk of invasive SCC (squamous cell carcinoma). The key entity is **PeIN (penile intraepithelial neoplasia)**: dysplastic squamous epithelium with an **intact basement membrane**.

### Penis

#### Classification

**1. PeIN (HPV-associated or HPV-independent)**

- **HPV (human papillomavirus)-associated** (commonest; basaloid and warty) or differentiated (HPV-independent).
- **Mechanism (HPV):** viral E6 inactivates p53, E7 inactivates Rb (retinoblastoma protein); **p16** is the marker.
- **HPV-independent (differentiated):** arises in lichen sclerosus.
- **WHO (World Health Organization) 2022:** all PeIN is high grade; grades I–III abandoned.

| Clinical entity / lesion | Site, appearance / key points | Risk of invasion |
|---|---|---|
| **Erythroplasia of Queyrat** | Glans or prepuce; red velvety (glans) plaque | **Progresses in 10–33%** |
| **Bowen disease** | Shaft; scaly red (shaft) plaque | About 5% progress |
| Bowenoid papulosis | Shaft papules, young men; HPV 16 | Does not progress; may regress |

**2. Other lesions**

- **Lichen sclerosus (BXO [balanitis xerotica obliterans]):** SCC (squamous cell carcinoma) in **2.3–9%** (HPV-independent type).
- **Buschke–Löwenstein tumour** (giant condyloma; verrucous carcinoma): HPV 6/11; locally invasive, locally destructive, rarely metastasises; may harbour SCC.

- **Also know:** cutaneous horn; PEKMB (pseudoepitheliomatous keratotic and micaceous balanitis, controversial); condyloma acuminatum; (chronic) phimosis; genital lichen planus; Kaposi sarcoma (HHV-8 [human herpesvirus 8], check HIV [human immunodeficiency virus]).

**Evaluation**

- Biopsy first: **multiple biopsies of adequate depth** to exclude invasion; p16 staining. Examine inguinal nodes; partner screening for cervical neoplasia.

**Treatment (organ-preserving)**

| Situation | Treatment |
|---|---|
| Prepuce (preputial lesions) | Circumcision or excision with clear margin |
| Glans | **Topical 5-FU (5-fluorouracil) or 5% imiquimod**; **laser** (Nd:YAG [neodymium:yttrium-aluminium-garnet] preferred; $CO_2$ too superficial); **glans resurfacing** with skin graft |
| Shaft | Excision; Mohs surgery |
| Lichen sclerosus | Clobetasol 0.05% for 3 months then maintenance; circumcision; lifelong follow-up |
| Buschke–Löwenstein | Wide excision; avoid radiotherapy (anaplastic change) |

- Long-term follow-up (recurrence common after organ-preserving treatment). **Prevention:** neonatal circumcision, HPV vaccination, avoiding smoking.

### Bladder

- **CIS** (carcinoma in situ): flat high-grade lesion, umbrella cells lost; untreated, **40–83%** invade muscle. Treat with BCG (bacillus Calmette–Guérin).
- **Urothelial dysplasia:** 14–19% progress; 60% to CIS with prior cancer.
- **Cystitis glandularis** (intestinal type): low risk of adenocarcinoma; resect and survey.
- **Leukoplakia** (keratinising squamous metaplasia): traditionally premalignant; Campbell now says **not** (no TP53 changes). Schistosomiasis and chronic catheters predispose to SCC.

### Prostate

- **HGPIN** (high-grade prostatic intraepithelial neoplasia): precursor; isolated HGPIN no longer needs repeat biopsy. IDC-P (intraductal carcinoma of the prostate) usually means invasive cancer nearby.

### Testis

- **GCNIS** (germ cell neoplasia in situ): precursor of all postpubertal GCTs (germ cell tumours) except spermatocytic tumour; **50% invasive within 5 years**. Treat by orchidectomy or **radiotherapy ≥20 Gy**.
- **Gonadoblastoma** in dysgenetic gonads with Y material; cryptorchidism.

### Kidney and others

- **Also know:** papillary adenoma; acquired cystic kidney disease on dialysis; urethral diverticulum; bowel in augmentation.

**Diagram to draw:** normal urothelium vs dysplasia vs CIS (umbrella cells lost); penile epithelium with full-thickness atypia above an intact basement membrane.

**Key points to remember:**

- PeIN: intact basement membrane; WHO 2022 all high grade.
- Queyrat 10–33% vs Bowen (about) 5% progression; bowenoid papulosis does not progress.
- Lichen sclerosus: SCC 2.3–9%.
- Untreated bladder CIS: 40–83% invade.
- GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

## SAQ 8 Secondary malignancies in urology

P1

**Asked as:**

- Secondary malignancies in Urology. — *Practice Set 3 Paper I Q3f (SAQ)*
- Secondary malignancies in Uro — *Practice Set 9 Paper I Q3c (SAQ)* (also answered in Practice Set 3 Paper 1, Q3f)

**Type:** SAQ (10 marks)

*Interpreted (read) as second cancers caused by urological or oncological treatment (radiotherapy, chemotherapy, urinary diversion, immunosuppression).*

### Definition

- A **second (treatment-related) malignancy** is a new, histologically distinct cancer caused by treatment of a first disease. It appears after a latency of years, often **in the radiation field**.

### Causes and risks

| Cause | Second cancer | Key facts |
|---|---|---|
| **Pelvic radiotherapy** | **Bladder cancer**, rectal cancer, sarcoma (in-field) | Bladder RR (relative risk) 2–4 after cervical EBRT (external beam radiotherapy), 1.5–2 after prostate EBRT, smaller after brachytherapy, little with IMRT (intensity-modulated radiotherapy). Latency 15–30 years |
| **Cyclophosphamide** | **Bladder cancer**, UTUC (upper tract urothelial carcinoma) | Only chemotherapy proven to cause bladder cancer; metabolites phosphoramide mustard and acrolein. About 4.5-fold, dose-dependent (>20 g cumulative) |
| **Urinary diversion / augmentation** | Adenocarcinoma at the uro-intestinal junction | Ureterosigmoidostomy 6–29% after 10–20 years (registry 2.6%); cystoplasty 1.6%; ileal conduit 0.02%. Risk tracks colonic mucosa exposed to urine and faeces |
| **Transplant immunosuppression** | Skin cancer, PTLD (post-transplant lymphoproliferative disorder), RCC (renal cell carcinoma), bladder cancer | RCC mostly in native kidneys; bladder cancer about 3-fold and aggressive |

| Cause | Second cancer | Key facts |
|---|---|---|
| **Testicular GCT (germ cell tumour) treatment** | Solid tumours, leukaemia | 60–100% excess after cisplatin or radiotherapy, 200% (highest) after both; etoposide-related leukaemia |
| **Paediatric tumours** | In-field solid tumours, leukaemia | Wilms tumour 1% at 10 years |

### Mechanisms

- Radiation and alkylating (drugs) or topoisomerase-inhibitor chemotherapy: mutagenic DNA damage in normal tissue.
- Immunosuppression: loss of (lost) immune surveillance and oncogenic viruses (EBV [Epstein–Barr virus], HPV [human papillomavirus]).
- Diversion: urine–faeces contact at the anastomosis.

### Prevention and surveillance

- IMRT, smaller fields; **mesna** with cyclophosphamide (reduces cystitis).
- **Haematuria after pelvic radiotherapy or cyclophosphamide: always cystoscopy.**
- Endoscopy from 5 years after ureterosigmoidostomy; augmentation surveillance from about 10 years; excise a defunctioned ureterosigmoid anastomosis.
- Testis cancer: fewer treatment modalities; low-dose CT (computed tomography) in surveillance.
- **Also know:** pioglitazone, aristolochic acid and phenacetin as drug-related urothelial carcinogens.

**Diagram to draw:** ureterosigmoid anastomosis showing (with) adenocarcinoma at the junction.

> **Key points to remember:**
> - Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
> - Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
> - Ureterosigmoidostomy cancer 6–29% after 10–20 years.
> - Transplant recipients: bladder cancer about 3-fold.

## SAQ 9 Genetic counselling

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 7 Paper 1, Q3d

*Assumed meaning: genetic counselling in urological practice.*

### Definition

- A communication process that helps patients and families understand the **medical, psychological and family implications** of a genetic condition, so they can make informed choices. It is **non-directive, voluntary and confidential**.
- **Germline testing** (blood) finds inherited variants with family implications; **somatic testing** (tumour) guides treatment.

### Steps

1. **Three-generation pedigree**.
2. **Risk assessment:** inheritance and recurrence risk (autosomal dominant 50%; autosomal recessive 25%).
3. **Pre-test counselling:** which test; possible results (pathogenic, **VUS [variant of uncertain significance]**, negative); insurance and psychological harms; consent.
4. **Post-test counselling:** interpret (do not act on a VUS); **cascade testing of relatives**; surveillance; targeted therapy.
5. **Reproductive options:** prenatal diagnosis, PGT (preimplantation genetic testing), donor gametes; psychosocial support.

### Urological applications

| Condition | Genes | Key point |
|---|---|---|
| **Prostate cancer** | BRCA2 (> BRCA1), ATM, CHEK2, PALB2, mismatch repair (Lynch), HOXB13 | NCCN (National Comprehensive Cancer Network): test metastatic, high-risk, strong family history; BRCA2: aggressive disease, PARP (poly-ADP-ribose polymerase) inhibitors, earlier screening |
| **Hereditary RCC (renal cell carcinoma)** | VHL, MET, FH, FLCN, TSC1/2, SDH, BAP1 | Counsel RCC at ≤46 years, bilateral or multifocal, family history |
| **UTUC (upper tract urothelial carcinoma)** | Lynch (MSH2 highest) | Screen if <60 years or Lynch-spectrum cancer history |
| **ADPKD (autosomal dominant polycystic kidney disease)** | PKD1 (chromosome 16), PKD2 (chromosome 4) | Testing selects living related donors |
| **Cystinuria** | SLC3A1, SLC7A9 (recessive) | Screen siblings |
| **Male infertility** | Karyotype, AZF (azoospermia factor) deletion, CFTR (cystic fibrosis transmembrane conductance regulator) | AZF deletions pass to sons by ICSI (intracytoplasmic sperm injection); absent vasa: test partner for CFTR |
| **DSD (differences of sex development)** | e.g. CAH (congenital adrenal hyperplasia, recessive), AIS (androgen insensitivity, X-linked) | Recurrence and gonadal tumour risk |

### Ethics and law

- Confidentiality; right not to know.
- **India:** PCPNDT Act (Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994) bans disclosure of fetal sex.

**Diagram to draw:** three-generation pedigree with standard symbols.

**Key points to remember:**

- Non-directive; pre-test and post-test counselling; cascade testing.
- Prostate: BRCA2 most important; test all metastatic men.
- RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

## SAQ 10 Congenital Penile curvature..

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 10 Paper 1, Q3d

### Definition and principle

- Curvature of the erect penis present from birth, with an **orthotopic meatus**.
- Rigidity needs **symmetric compliance** of the tunica albuginea; curvature means one aspect is less compliant.

### Embryology

- DHT (dihydrotestosterone) drives fusion of the urethral folds and maturation of spongiosum, Buck's and dartos fascia. The urethra may form normally while the mesenchyme stays deficient → dysgenetic, inelastic ventral tissue.
- Normal fetal ventral curvature resolves by 20 weeks.

### Devine and Horton classification (1973)

| Type | Abnormal tissue |
|---|---|
| I | Corpus spongiosum deficient (chordee without hypospadias) |
| II | Dysgenetic Buck's and dartos fascia; normal spongiosum |
| III | Dartos only |
| IV | Normal layers; **asymmetric corporal compliance** (true congenital curvature) |
| V | Congenitally short urethra (very rare; doubted) |

### Two clinical groups

| Feature | Chordee without hypospadias (I–III) | Congenital curvature (IV) |
|---|---|---|
| Direction | Ventral ± torsion | Ventral, lateral (often left), rarely dorsal |
| Erection size | In proportion to flaccid size | **Noticeably large** |
| Presentation | Childhood | 18–30 years, noticed at puberty |
| - **Versus acquired** (Peyronie disease, fracture): congenital is painless, no plaque, no shortening. | | |

### Evaluation

- Home photographs of erection; **artificial erection** (Gittes–McLaughlin saline test or prostaglandin) in theatre.

- Grade: mild <30°, moderate 30–45°, severe ≥45°.

### Management

- Surgery if curvature hinders intercourse or distresses. 1. Deglove; excise ventral dysgenetic tissue; mobilise spongiosum from glans to perineum. 2. Residual curve: **midline ventral septotomy** under artificial erection. 3. Still curved: **dorsal plication** contralateral to the curve: Nesbit (excise ellipses, close transversely), Yachia (Heineke–Mikulicz), 16-dot or TAP (tunica albuginea plication). **12 o'clock is nerve-free**, so midline plication is safe. 4. Severe: ventral **corporotomy with graft** (dermis, tunica vaginalis).
- **Rule:** if the meatus reaches the glans tip, do not divide the urethra; the corpora usually limit straightening.
- **Results:** single operation in almost all; recurrence 2–15%, mostly after plication.

**Diagram to draw:** penile cross-section: nerves at 11 and 1 o'clock, plication at 12.

> **Key points to remember:**
> - Devine–Horton I–III = chordee without hypospadias; IV = corporal asymmetry.
> - Congenital curvature: young man, large erection, orthotopic meatus.
> - Dorsal 12 o'clock plication; never divide a urethra that reaches the glans tip.

## SAQ 11 Circumcision undone

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 1, Q3d

*"Circumcision undone" is read as an incomplete or complicated circumcision needing revision. Foreskin restoration is mentioned briefly at the end.*

### Introduction

- Circumcision complications: **0.2–5%**, rising **10- to 20-fold after infancy**.
- Most late problems concern **skin quantity** (too little or too much) or the **exposed meatus**.

### Inadequate ("undone") circumcision

- **Too little skin removed** or asymmetrical excision: redundant prepuce covering the glans.
- Leads to **secondary (cicatricial) phimosis**, smegma retention, balanitis, adhesions and cosmetic distress.
- A large hydrocele or hernia pushes skin forward and raises this risk.

### Other complications

| Complication | Management |
|---|---|
| **Bleeding** (commonest, about 0.1%), usually frenular | Pressure, cautery, silver nitrate or suture |
| Glanular adhesions; **skin bridges** (can cause chordee, torsion) | Office lysis or excision; steroids do not help |
| **Excess skin removal**, degloving | Heals by secondary intention with ointment; severe: grafts |

| Complication | Management |
|---|---|
| **Trapped penis** (cicatrix over glans) | Betamethasone 0.05% and retraction; else relaxing incision, flaps, Z-plasty |
| **Meatal stenosis** (almost only after circumcision) | Meatotomy or meatoplasty |
| Subcoronal fistula; glans amputation; necrosis (cautery on clamp) | Hypospadias-type repair; reattach glans within 8 hours |

### Evaluation

- Original procedure (age, device); examine skin quantity, meatus, curvature; exclude hypospadias, buried or webbed penis and BXO (balanitis xerotica obliterans).

### Circumcision revision

- **Indications:** redundant or asymmetrical skin, phimosis, adhesions, cosmesis.
- **Needs general anaesthesia** in children, unlike neonatal circumcision.
- **Technique (sleeve):** mark inner and outer incisions (inner 5–10 mm from the corona), excise the excess collar, haemostasis, interrupted absorbable sutures.
- Lyse adhesions and skin bridges; correct torsion or chordee; meatotomy if needed; histology if BXO suspected.

### Prevention

- Separate prepuce fully and inspect the meatus before using a device (Gomco, Mogen, Plastibell); defer with hypospadias, curvature, buried or webbed penis; trained operator.

### Also know: foreskin restoration

- Traction tissue expansion or skin grafting, for men wanting a foreskin; results variable.

**Diagram to draw:** sleeve revision showing two circumferential incisions.

> **Key points to remember:**
> - Complications 0.2–5%; bleeding commonest.
> - Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
> - Meatal stenosis almost only in circumcised boys.
> - Revision needs general anaesthesia.

## SAQ 12 Noonan syndrome

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 22 SAQ Sheet, Q3a

### Definition and genetics

- Noonan syndrome is an **autosomal dominant RASopathy** (disorder of the RAS–MAPK [mitogen-activated protein kinase] signalling pathway) with variable expression; many cases are de novo. Incidence about 1 in 1,000–2,500.
- **Karyotype is normal** (46,XY or 46,XX). The phenotype resembles Turner syndrome, hence the old name "male Turner".

- **Mechanism:** gain-of-function variants increase RAS–MAPK signalling, which disturbs growth and organ development.

| Gene | Approximate share | Note |
|---|---|---|
| **PTPN11** (encodes SHP2 phosphatase) | About 50% | Pulmonary stenosis common |
| SOS1 | About 10% | Milder growth and learning effects |
| RAF1 | About 5–10% | Hypertrophic cardiomyopathy |
| RIT1, KRAS, others | Smaller | KRAS: more learning difficulty |

- Related RASopathies: Costello, cardiofaciocutaneous, NF-1 (neurofibromatosis type 1).

### Clinical features

- Short stature; webbed neck; hypertelorism, ptosis, low-set posteriorly rotated ears; pectus deformity.
- **Heart:** pulmonary valve stenosis (commonest), HCM (hypertrophic cardiomyopathy), septal defects.
- **Bleeding tendency:** factor XI deficiency, platelet dysfunction.
- Lymphatic dysplasia: genital (penoscrotal) lymphoedema; mild learning difficulty.

### Urological aspects

- **Cryptorchidism:** the main urological feature, in most affected boys (often bilateral).
- **Testicular dysfunction:** primary Sertoli cell dysfunction (low inhibin B, high FSH [follicle-stimulating hormone]) with poor spermatogenesis, even after orchidopexy. Leydig function is usually adequate; puberty is often delayed. Female fertility is usually normal.
- **Renal anomalies (about 10%):** mostly mild renal pelvic dilatation; also duplex system, solitary kidney, hypoplasia.
- Small excess of childhood cancers (JMML [juvenile myelomonocytic leukaemia], embryonal rhabdomyosarcoma).

### Evaluation

- Clinical diagnosis (van der Burgt criteria), confirmed by gene panel; karyotype excludes Turner in girls.
- Before surgery: **echocardiography, coagulation screen and factor XI level**, anaesthetic review.
- Renal and bladder ultrasound at diagnosis.

### Management

- **Orchidopexy by 18 months** (EAU [European Association of Urology] and AUA [American Urological Association]); laparoscopic for non-palpable testes.
- Follow-up: pubertal hormones, semen analysis, fertility counselling.
- **Also know:** growth hormone for short stature; multidisciplinary cardiac and haematology care.

**Diagram to draw:** RAS–MAPK cascade with SHP2 (PTPN11) acting upstream of RAS.

> **Key points to remember:**
> - Autosomal dominant, normal karyotype; PTPN11 in about half.
> - Cryptorchidism plus Sertoli dysfunction: male subfertility.
> - Check heart and clotting (factor XI) before orchidopexy.

## SAQ 13 Mixed gonadal dysgenesis.

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 3, Q3e (DSD management in general in 2019-Summer Paper 3, Q3f)

### Definition

- MGD (mixed gonadal dysgenesis): a **dysgenetic testis on one side and a streak gonad on the other**, with persistent Müllerian structures and variable virilisation.
- A sex chromosome DSD (difference of sex development), usually **45,X/46,XY mosaicism** (about 1 in 5,000 births). **Second commonest cause of neonatal atypical genitalia after CAH (congenital adrenal hyperplasia).**

### Pathogenesis

- The Y-bearing line forms a testis; the 45,X line leaves a streak.
- AMH (anti-Müllerian hormone) and testosterone act **locally**, so anatomy is asymmetric: Wolffian duct on the **testis side**; tube and hemiuterus on the **streak side**.
- 45,X/46,XY spectrum: female (Turner + Y), atypical (MGD) or male with normal testes (about 85%).

### Clinical features

- **Asymmetric genitalia:** phallus with chordee, perineal hypospadias, urogenital sinus; one testis in a hemiscrotum, the other side impalpable. Hernia may contain a uterus or tube.
- **Turner features:** short stature, coarctation, horseshoe kidney.

### Evaluation (MDT, multidisciplinary team)

- **Karyotype** (many cells, for mosaicism); **17-OHP (17-hydroxyprogesterone)** to exclude CAH; LH (luteinising hormone), FSH (follicle-stimulating hormone), testosterone, AMH.
- **Pelvic ultrasound** (uterus, asymmetric gonads); genitography, endoscopy; **laparoscopy with gonadal biopsy**. No gonadectomy during diagnosis.
- Echocardiography, renal ultrasound (Turner screening).

| | MGD | Ovotesticular DSD |
|---|---|---|
| Karyotype | 45,X/46,XY | Mostly 46,XX |
| Gonads | Dysgenetic testis + streak | Ovary + testis, or ovotestis |
| Tumour risk | 15–35% | About 3% |

### Management

- **Sex of rearing:** individualised; more androgen exposure predicts male identity; dysphoria is commoner when raised female. Avoid irreversible infant surgery.

- **Gonads:** germ cell neoplasia 15–35% (gonadoblastoma, then dysgerminoma or seminoma) in streak and testis alike. **Remove streaks** (biopsy first if tissue uncertain).
- Raised male: keep a scrotal testis under surveillance; bring down an intra-abdominal one if possible.
- Raised female: remove or relocate the testis.
- **Surgery:** male: staged hypospadias repair, orchidopexy; female: feminising surgery deferred.
- **Hormones:** pubertal sex steroids; growth hormone. Infertility usual.

**Diagram to draw:** asymmetric anatomy: testis with vas; streak with tube and hemiuterus.

> **Key points to remember:**
> - 45,X/46,XY; testis + streak + Müllerian remnants; asymmetric.
> - Second commonest cause of atypical genitalia after CAH.
> - Tumour risk 15–35%: remove streaks.
> - Screen for Turner features.

## SAQ 14 Telomerase

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 3, Q3b

### Telomeres

- **Telomeres** are repetitive sequences (TTAGGG repeats) at chromosome ends. They protect ends from degradation and fusion.
- DNA polymerase cannot copy the very end of the lagging strand (end-replication problem). So about **100 base pairs are lost per division**: the "mitotic clock".
- Critical shortening triggers **replicative senescence** (Hayflick limit).

### Telomerase: structure and function

- A specialised **reverse transcriptase** that adds telomeric repeats.
- Two core parts: **hTERT** (human telomerase reverse transcriptase, catalytic protein) and **hTR** (telomerase RNA, the template).
- **Normal expression:** only rapidly dividing tissues: germ cells (testis), intestinal epithelium, endometrium, activated lymphocytes. Most somatic cells lack it.

### Role in cancer

```
Proliferation -> moderate shortening -> senescence/apoptosis (barrier 1) -> checkpoint lost (p53) -> severe shortening -> chromosomal instability (barrier 2) -> telomerase activation -> immortal cancer cell
```

- **85–90% of human tumours** express high telomerase, giving unlimited replication.
- **ALT (alternative lengthening of telomeres):** telomerase-independent recombination; linked to ATRX or DAXX loss.
- **TERT promoter mutations** switch on hTERT.

### Urological relevance

| Organ | Finding |
|---|---|
| **Bladder** | TERT promoter mutation is the commonest mutation (over 60% even in PUNLMP, papillary urothelial neoplasm of low malignant potential). Urine markers: telomerase activity by **TRAP (telomeric repeat amplification protocol)**, a PCR (polymerase chain reaction)-based assay; hTERT messenger RNA; AssureMDx (FGFR3, TERT, HRAS mutations plus methylation). Inflammation gives false positives; not routine |
| **Upper tract** | hTR positive in almost all UTUC (upper tract urothelial carcinoma); rises with stage |
| **Prostate** | Short telomeres in cancer and premalignant lesions; short tumour telomeres predict recurrence; variable stromal telomeres predict cancer death |

- **Also know:** senescent cells secrete SASP (senescence-associated secretory phenotype) cytokines in ageing and BPH (benign prostatic hyperplasia).

### Therapy

- **Imetelstat** (telomerase inhibitor) is used in myelodysplastic syndrome.
- hTERT-based vaccines (UV1, GX301, INO-9012) are in early prostate, renal and bladder trials.

**Diagram to draw:** telomere-to-cancer flowchart with two barriers.

> **Key points to remember:**
> - Telomerase = hTERT (catalytic) + hTR (RNA template).
> - About 100 base pairs lost per division; 85–90% of tumours express telomerase.
> - TERT promoter mutation is the commonest mutation in bladder cancer.
> - TRAP assay detects urinary telomerase; inflammation gives false positives.

## SAQ 15 Enlist the steps in female to male gender reassignment surgery.

P4

**Type:** SAQ (10 marks)
**Source:** 2024-Winter Paper 4, Q3b

### Preconditions

- Gender dysphoria diagnosed per **WPATH (World Professional Association for Transgender Health)** by an MDT (multidisciplinary team).
- **Testosterone:** chapter cites **at least 12 months** before genital surgery (clitoral growth plateaus at about 2 years). WPATH SOC8 (Standards of Care 8, 2022) relaxed this to about 6 months.
- **Urological screening:** uroflowmetry, PVR (post-void residual), renal ultrasound; treat LUTD (lower urinary tract dysfunction) first.

### Stage 1: removal of female structures

1. **Subcutaneous mastectomy** (usually first).
2. **Hysterectomy + bilateral salpingo-oophorectomy**, preferably laparoscopic; remove whole cervix ± upper third of vagina.

### Stage 2: genitoperineal transformation

3. **Vaginectomy** (needed if the urethra is lengthened): - Hydrodissection; circumferential introital incision. - Rectovaginal plane bluntly; urethrovaginal plane sharply, **no diathermy near the sphincter**. - Paravaginal haemostasis; levators closed in midline.
4. **Urethral lengthening, fixed (perineal) part:** vestibular mucosa and labia minora tubularised over 18 Fr; covered with bulbospongiosus.
5. **Phallus construction**, either: - **Metoidioplasty:** release clitoral chordee and urethral plate; lengthen urethra to the glans (preputial or labial flap ± buccal graft). Gives a 3–8 cm phallus with erection and sensation. - **Phalloplasty: RFFF (radial forearm free flap)** or ALT (anterolateral thigh) flap with **tube-in-tube urethra**; microvascular anastomosis; **clitoral nerve coapted** to a flap nerve.
6. **Urethral anastomosis** of phallic to fixed part; neomeatus at the tip.
7. **Scrotoplasty:** labia majora flaps rotated 180° in front (Ghent type). Monsplasty and perineal closure.
8. **SPC (suprapubic catheter)** + urethral catheter; urethral catheter out at 14–21 days.

### Stage 3: prostheses

9. **Testicular prostheses** (17 mL) at 6 months or more.
10. **Erectile prosthesis** after 1 year, once protective sensation returns; single cylinder fixed to pubic bone (dedicated devices, e.g. ZSI 475 FTM [female-to-male]).

### Complications

- **Fistula** (phallic base), **stricture** (fixed–phallic anastomosis), hair growth, dribbling, prosthesis infection or erosion, vaginectomy bleeding or rectal/bladder injury.

**Diagram to draw:** three parts of the neourethra (native, fixed, phallic).

> **Key points to remember:**
> - Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
> - No urethral lengthening without vaginectomy (fistula risk).
> - Commonest stricture: fixed–phallic anastomosis.

## SAQ 16 Antioxidants in urological malignancy

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 1 Paper 4, Q3a (prevention trials also in 2017-Summer Paper 4, Q3a)

### Rationale

- ROS (reactive oxygen species) cause **oxidative DNA damage**; polyunsaturated fat and excess iron increase it. Antioxidants scavenge ROS, so diet studies suggested protection.
- Randomised trials of single high-dose supplements have been **null or harmful**.

### Prostate cancer

| Agent | Evidence | Result |
|---|---|---|
| **SELECT** (Selenium and Vitamin E Cancer Prevention Trial) | 35,533 men; selenium 200 µg, vitamin E 400 IU, both, or placebo | Stopped early (median 5.46 years) for futility. No benefit. Vitamin E alone trended to more cancer (P = 0.06); extended follow-up showed a significant **increase (about 17%)** |
| ATBC (Alpha-Tocopherol, Beta-Carotene) | Low-dose vitamin E in heavy smokers | Secondary analysis: prostate cancer ↓32%, deaths ↓41%; beta-carotene ↑ lung cancer |
| Selenised yeast | Skin-cancer trial | Secondary fall in prostate cancer prompted SELECT |
| Lycopene (cooked tomato) | Meta-analysis: modestly lower risk, mainly advanced cancer | WCRF (World Cancer Research Fund) downgraded "probable" (2007) to "no conclusion" (2018); supplement trials mostly null |
| Green tea catechins | HGPIN (high-grade prostatic intraepithelial neoplasia)/ASAP trials | Large fall in progression to cancer (meta-analysis); dose impractical |
| Vitamin C, high-dose antioxidants | High doses act as pro-oxidants | Supplements above median → more advanced cancer |

- **Also know:** vitamin D (VITAL, no effect); coffee (modestly lower advanced risk); pomegranate, soy and curcumin unproven.

### Bladder cancer

- Supplemental beta-carotene, vitamins C and E and folate do not lower risk; SELECT showed no effect.
- Fruit and vegetable benefit disappears after adjusting for smoking. Smoking cessation is the real prevention.

### Kidney cancer

- No antioxidant trial has shown benefit.

### Why the trials failed

- Single nutrient at high dose; wrong chemical form; non-smoking populations; PSA-era indolent cancers; U-shaped dose effects.

### Current position

- Antioxidant supplements are **not recommended** for cancer prevention. Advise a balanced diet, no smoking and normal weight.

**Diagram to draw:** ROS → DNA damage → cancer, with antioxidants blocking the first arrow.

> **Key points to remember:**
> - SELECT: no benefit; vitamin E alone increased prostate cancer.
> - ATBC benefit limited to smokers on low dose.
> - Lycopene downgraded to "no conclusion" (2018).

## SAQ 17 Gene therapy in urology

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 4, Q3f (intravesical gene therapy also answered in 2022-Summer Paper 2, Q3a)

## Definition

- Delivery of nucleic acid (DNA or RNA) into cells to treat disease: add, replace or silence a gene.

## Vectors

| Vector | Features |
|---|---|
| **Adenovirus** | Efficient, non-integrating, transient expression; replication-incompetent or oncolytic |
| Retrovirus, lentivirus | Integrate: long expression, risk of insertional mutagenesis |
| AAV (adeno-associated virus) | Low immunogenicity, long expression |
| Non-viral | Naked plasmid, liposomes, nanoparticles: safer, less efficient |
| Gene silencing | siRNA (small interfering RNA), antisense; CRISPR (clustered regularly interspaced short palindromic repeats) editing experimental |

## Strategies in urological cancer

1. **Immunogene (cytokine) therapy: nadofaragene firadenovec**: replication-incompetent adenovirus type 5 carrying the **interferon-α2b gene** with **Syn3** surfactant, which crosses the bladder's protective barrier. The urothelium keeps secreting interferon, overcoming the 1–2-hour dwell of protein interferon. 75 mL every 3 months; CIS (carcinoma in situ) complete response **53% at 3 months, 24% at 12 months**. **FDA (Food and Drug Administration) approved December 2022** for BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer).
2. **Oncolytic virus: cretostimogene (CG0070)**: replication-competent adenovirus; E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity.
3. **Suicide gene:** HSV-tk (herpes simplex virus thymidine kinase) turns ganciclovir or valacyclovir into a toxin; prostate cancer with radiotherapy (trials).
4. **Toxin gene under a tumour promoter:** BC-819 (diphtheria toxin A, H19 promoter).
5. **Gene-modified cells:** CAR-T (chimeric antigen receptor T cells) against PSMA (prostate-specific membrane antigen): early trials, toxic. Viral-vector vaccine PROSTVAC failed.

## Benign conditions

- **OAB (overactive bladder) and ED (erectile dysfunction):** URO-902 (hMaxi-K) plasmid for the BK potassium channel relaxes smooth muscle; phase 1 only.
- **Primary hyperoxaluria type 1:** lumasiran (RNA interference against glycolate oxidase) lowers oxalate.
- Underactive detrusor: experimental.

## Challenges

- Delivery barriers, vector immunity, short expression, cost, biosafety.

**Diagram to draw:** adenovirus → urothelial cell → interferon secretion → tumour apoptosis.

**Key points to remember:**

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: oncolytic adenovirus + GM-CSF.
- Lumasiran: RNA interference for primary hyperoxaluria type 1.

## SAQ 18 Molecular targeted therapy in urological cancers

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 4, Q3c

### Concept

- **Targeted therapy** blocks a molecule the cancer depends on (receptor, kinase, transcription factor, DNA-repair enzyme, surface antigen), not all dividing cells.
- Classes: small-molecule TKIs (tyrosine kinase inhibitors, "-nib"), monoclonal antibodies ("-mab"), mTOR (mammalian target of rapamycin) inhibitors, HIF-2α (hypoxia-inducible factor-2α) inhibitor, PARP (poly-ADP-ribose polymerase) inhibitors, ADCs (antibody–drug conjugates), radioligands.

### RCC (renal cell carcinoma)

- **Basis:** VHL (von Hippel–Lindau) loss → HIF accumulates even in normoxia → constant VEGF (vascular endothelial growth factor), PDGF (platelet-derived growth factor) and other targets.

| Target | Drugs | Key evidence |
|---|---|---|
| VEGF receptors | Sunitinib, pazopanib, axitinib, cabozantinib (also MET, AXL), lenvatinib | COMPARZ: pazopanib noninferior to sunitinib; METEOR: cabozantinib beat everolimus |
| VEGF ligand | Bevacizumab | Modest PFS (progression-free survival) gain |
| mTOR | Temsirolimus (IV, poor risk), everolimus (oral) | Temsirolimus improved OS (overall survival) over interferon |
| HIF-2α | **Belzutifan** | VHL disease tumours; LITESPARK-005 beat everolimus after immunotherapy and TKI |

- First line now: TKI **plus** checkpoint inhibitor. Everolimus also treats tuberous sclerosis angiomyolipoma.

### Urothelial carcinoma

- **FGFR3 (fibroblast growth factor receptor 3)** alterations, typical of luminal tumours: **erdafitinib** (oral) after prior therapy.
- **ADCs: enfortumab vedotin** (Nectin-4) improved OS after platinum and immunotherapy; sacituzumab govitecan (Trop-2); HER2 (human epidermal growth factor receptor 2)-directed ADCs emerging.
- **EV-302 (2024):** enfortumab vedotin plus pembrolizumab, first line, roughly doubled OS versus platinum chemotherapy and is now standard (chapter notes predate it).

### Prostate cancer (mCRPC, metastatic castration-resistant)

- **PARP inhibitors** exploit **synthetic lethality** in HRR (homologous recombination repair)-deficient tumours, especially BRCA2. **PROfound:** olaparib beat a second hormonal agent (imaging PFS 7.4 vs 3.6 months in BRCA/ATM).
- **PSMA (prostate-specific membrane antigen) radioligand: 177Lu-PSMA-617; VISION**: OS 15.3 vs 11.3 months.

### Toxicity

- VEGF TKIs: hypertension, hand–foot syndrome, hypothyroidism. mTOR: mucositis, hyperglycaemia. Erdafitinib: hyperphosphataemia. Enfortumab: neuropathy, rash. PARP inhibitors: anaemia.

### Limitations

- Resistance develops; cures rare; biomarker testing needed.

**Diagram to draw:** VHL–HIF–VEGF pathway with drug sites (TKI at receptor, bevacizumab at VEGF, mTOR and HIF-2α inhibitors).

> **Key points to remember:**
>
> - RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
> - Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
> - Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION).

# Rapid Revision Sheet · Section 2

### Immunology, immunotherapy and cancer vaccines in urological oncology (including Sipuleucel-T) (LAQ)

- Three Es: elimination, equilibrium, escape; Th1 is antitumour.
- BCG is the original urological immunotherapy: TLR2/4 → neutrophils → Th1 memory.
- Vaccine = antigen + adjuvant; DC gives signal 1 (MHC–peptide) and signal 2 (costimulation).
- IMA901, AGS-003, PROSTVAC, GVAX all failed in phase III.
- Sipuleucel-T: PAP–GM-CSF; IMPACT OS 25.8 vs 21.7 months (+4.1 months); no PSA or progression effect.
- CheckMate 274 adjuvant nivolumab; EV-302 EV + pembrolizumab first line.

### DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia (LAQ)

- Palpable gonad = testicular tissue; impalpable gonads = CAH until proven otherwise.
- 17-OHP after 36 h; salt crisis around day 5–7.
- Karyotype + uterus + gonads triad guides diagnosis; no gonadectomy while diagnosing.
- Defer irreversible surgery; CAIS gonads kept until after puberty (till puberty).

### Discuss in brief, present concepts in various GU cancers (LAQ)

- RCC: adjuvant pembrolizumab (KEYNOTE-564); first-line ICI combinations; CARMENA.
- UTUC: single post-RNU instillation; POUT adjuvant chemotherapy.
- Bladder: EV-302 enfortumab–pembrolizumab; CheckMate 274; NIAGARA.
- Prostate: MRI first, PSMA PET, ARPI triplets, Lu-PSMA (VISION); testis: miR-371, RPLND for stage II seminoma.

### 46,XX male (SAQ)

- 46,XX testicular DSD: SRY on X in about 80–90%; 1 in 20,000.
- Azoospermia always (AZF absent): no sperm retrieval.
- Shorter than Klinefelter; FISH for SRY confirms.

### Female circumcision (SAQ)

- WHO types I–IV: I clitoridectomy, II excision, III infibulation, IV others.
- Urological: retention, recurrent UTI, stenosis, stones, fistula.
- Midline deinfibulation for type III; never reinfibulate.

### Familial / hereditary urological malignancies (SAQ)

- All familial RCC is autosomal dominant; 3 cm rule except HLRCC.
- VHL 3p25, MET 7q31, FH 1q42, FLCN 17p11.
- Lynch (MSH2) → UTUC; BRCA2 → aggressive prostate cancer.

### Penile inversion vaginoplasty (SAQ)

- Glans → clitoris; penile skin → canal; scrotal skin → graft.
- Excise all erectile tissue; keep Denonvilliers fascia intact.
- Lifelong dilation prevents stenosis; rectovaginal fistula is the worst complication.

### Prepucioplasty (SAQ)

- Steroid first; about 1 in 6 fail.
- Cut lengthwise, close crosswise; daily retraction after.

- BXO: circumcision preferred.

**Oncogenes (SAQ)**

- Oncogenes: gain of function, one allele; activated by mutation, amplification, translocation.
- HRAS first found in T24 bladder cancer; FGFR3 in up to 80% low-grade Ta.
- TMPRSS2–ERG in about half of prostate cancers; MET in papillary RCC.

**Premalignant conditions in urology (including premalignant lesions of penis) (SAQ)**

- PeIN: intact basement membrane; WHO 2022 all high grade.
- Queyrat 10–33% vs Bowen (about) 5% progression; bowenoid papulosis does not progress.
- Lichen sclerosus: SCC 2.3–9%.
- Untreated bladder CIS: 40–83% invade.
- GCNIS: 50% invasive in 5 years; radiotherapy ≥20 Gy.

**Secondary malignancies in urology (SAQ)**

- Pelvic EBRT raises bladder cancer 1.5–4-fold; IMRT little.
- Cyclophosphamide: only chemotherapy proven to cause bladder cancer.
- Ureterosigmoidostomy cancer 6–29% after 10–20 years.
- Transplant recipients: bladder cancer about 3-fold.

**Genetic counselling (SAQ)**

- Non-directive; pre-test and post-test counselling; cascade testing.
- Prostate: BRCA2 most important; test all metastatic men.
- RCC ≤46 years, ADPKD, cystinuria, AZF/CFTR, DSD are classic urological indications.

**Congenital Penile curvature.. (SAQ)**

- Devine–Horton I–III = chordee without hypospadias; IV = corporal asymmetry.
- Congenital curvature: young man, large erection, orthotopic meatus.
- Dorsal 12 o'clock plication; never divide a urethra that reaches the glans tip.

**Circumcision undone (SAQ)**

- Complications 0.2–5%; bleeding commonest.
- Too little skin: redundancy, phimosis, adhesions; too much: trapped penis, chordee.
- Meatal stenosis almost only in circumcised boys.
- Revision needs general anaesthesia.

**Noonan syndrome (SAQ)**

- Autosomal dominant, normal karyotype; PTPN11 in about half.
- Cryptorchidism plus Sertoli dysfunction: male subfertility.
- Check heart and clotting (factor XI) before orchidopexy. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Mixed gonadal dysgenesis. (SAQ)**

- 45,X/46,XY; testis + streak + Müllerian remnants; asymmetric.
- Second commonest cause of atypical genitalia after CAH.
- Tumour risk 15–35%: remove streaks.
- Screen for Turner features.

**Telomerase (SAQ)**

- Telomerase = hTERT (catalytic) + hTR (RNA template).
- About 100 base pairs lost per division; 85–90% of tumours express telomerase.
- TERT promoter mutation is the commonest mutation in bladder cancer.
- TRAP assay detects urinary telomerase; inflammation gives false positives. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Enlist the steps in female to male gender reassignment surgery. (SAQ)**

- Mastectomy → hysterectomy/oophorectomy → vaginectomy + urethroplasty + phallus + scrotum → prostheses.
- No urethral lengthening without vaginectomy (fistula risk).
- Commonest stricture: fixed–phallic anastomosis.

**Antioxidants in urological malignancy (SAQ)**

- SELECT: no benefit; vitamin E alone increased prostate cancer.
- ATBC benefit limited to smokers on low dose.
- Lycopene downgraded to "no conclusion" (2018).

**Gene therapy in urology (SAQ)**

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3; FDA 2022.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: oncolytic adenovirus + GM-CSF.
- Lumasiran: RNA interference for primary hyperoxaluria type 1.

**Molecular targeted therapy in urological cancers (SAQ)**

- RCC: VEGF TKIs, mTOR inhibitors, belzutifan (HIF-2α).
- Bladder: erdafitinib (FGFR3), enfortumab vedotin (Nectin-4).
- Prostate: PARP inhibitors (BRCA2), 177Lu-PSMA (VISION). *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 3 · Infections & Inflammation

## Long answer questions (LAQ)

### LAQ 1 Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder

P2 · P3 **merged from 2**

*Asked as:* P3 LAQ "Renal and genitourinary TB: etiopathogenesis, presentation, (laboratory and newer) diagnosis and management" (Describe etiopathogenisis, clinical presentation, diagnosis and management of Renal TB?, Summer 2022 Paper III Q2 LAQ; Laboratory Diagnosis of Genitourinary Tuberculosis, Winter 2023 Paper III Q3f SAQ; Describe newer diagnostic modalities for Gu Kochs. Briefly outline role of all present diagnostic tests and give diagnostic algorithm, Practice Set 6 Paper III Q3d SAQ) · P2 LAQ "Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30-year male."

*Assumed meaning: "Gu Kochs" = genitourinary tuberculosis.*

#### Introduction

- Caused by the MTBC (*Mycobacterium tuberculosis* complex); *M. bovis* (and BCG (bacille Calmette–Guérin)) is innately resistant to pyrazinamide.
- UG (urogenital) TB (tuberculosis) is 15–20% of pulmonary TB in developing countries; the second commonest extrapulmonary site after lymph nodes. **Kidney is involved in 80%.**
- Untreated, it destroys the kidney and contracts the bladder.

#### Etiopathogenesis

- **Primary infection:** inhaled bacilli spread to hilar nodes, then by lymph and blood to the well-perfused kidney and epididymis, before immunity develops (up to 12 weeks).
- **Granuloma:** macrophages, epithelioid cells and Langhans giant cells in a lymphocyte cuff, maintained by IFN-γ (interferon-γ). In 90–95% it contains the infection and it stays latent; viable bacilli stay dormant.
- **Reactivation** (lifetime 5–15%), even decades later: old age, diabetes, renal failure, HIV (human immunodeficiency virus), malnutrition, steroids or other immunosuppression. **Caseation** then releases bacilli into urine.

| Route | Lesion |
|---|---|
| **Haematogenous** (principal) | Parenchymal disease: kidney, epididymis, prostate |
| **Descending urinary** | Luminal disease: ureteric stricture, cystitis, urethra |
| **Ascending** (intravesical BCG, about 1%) | Cystitis, prostatitis, epididymo-orchitis |
| **Contiguous** | From psoas, bowel |

- **Renal sequence:** 1. Bacilli lodge in **periglomerular capillaries**. 2. Cortical granulomas coalesce. 3. **Caseation**; cavities open into calyces. 4. Papillary necrosis, abscess, calcification. 5. Bacilluria seeds ureter and bladder.
- **Autonephrectomy:** caseo-cavernous type, or fibrotic calcified "putty" kidney.

| Site | Frequency | Pathology |
|---|---|---|
| Kidney | 80% | Periglomerular granulomas, then caseating cavities, papillary necrosis, calcification; **autonephrectomy** ("putty kidney") |
| Ureter | — | Mural granulomas; stricture, commonest at the **VUJ (vesicoureteric junction)**; "beaded corkscrew" or "pipestem" ureter |
| Bladder | — | Cystitis starts around the ureteric orifices ("golf-hole" orifice); ulcers; after about a year of fibrosis it contracts (<100 mL); **thimble bladder <20 mL** |
| Epididymis | Commonest genital site | Starts in globus minor; beaded vas, posterior scrotal sinus |
| Prostate, seminal vesicles | 22–49% | Calcification, abscess; infertility |

- **Key concept:** healing occurs by fibrosis, and fibrosis obstructs. Infundibular and ureteric strictures and bladder contraction can **worsen during treatment**.
- **Also know:** granulomatous interstitial nephritis; miliary renal TB; amyloidosis; squamous metaplasia of the pelvis (squamous cell carcinoma risk); ESRD (end-stage renal disease) in about 7%.

## Clinical presentation

- Renal TB is silent until it reaches the bladder; asymptomatic in about 8%.
- **Storage LUTS (lower urinary tract symptoms)** and dysuria (commonest); haematuria and flank pain (about one-third); colic <10% (sloughed papillae, clot, debris).
- Constitutional symptoms <20%. Recurrent "UTI (urinary tract infection)" not responding to antibiotics.
- Signs: renal angle tenderness, flank sinus, beaded vas, epididymal nodule; hypertension; renal failure if bilateral.

## Diagnosis: principles and specimen

- GUTB (genitourinary tuberculosis) is paucibacillary with intermittent shedding, so **multiple samples** are needed.
- **Hierarchy of proof:** positive culture or DNA, then caseating granulomas on histology, then clinical picture with response to treatment.
- **Specimen:** early-morning first-void urine (EMU), **3–5 samples on consecutive days**, processed at once (acidic urine kills bacilli). Avoid prior fluoroquinolones.
- Also tissue, pus, FNAC (fine-needle aspiration cytology) or semen.

## Laboratory tests

| Test | Method | Comment / role |
|---|---|---|
| **Urine routine** | Microscopy | **Sterile (acid) pyuria** with or without microscopic haematuria in >90% |
| **ZN (Ziehl–Neelsen) smear** | Carbol fuchsin, acid-alcohol, methylene blue: red bacilli. Auramine fluorescence is more sensitive | Quick; sensitivity under 50%; saprophytic mycobacteria give false positives |

| Test | Method | Comment / role |
|---|---|---|
| **Urine AFB (acid-fast bacilli) culture: gold standard** | Decontaminate and concentrate. **LJ (Löwenstein–Jensen)** solid medium: 4–6 weeks. **MGIT (mycobacteria growth indicator tube)** liquid: about 10 days; use a liquid system as well | 80–90% sensitivity with ideal technique; diagnosis; gives DST (drug-susceptibility testing); **the only test that monitors treatment** |
| **NAAT (nucleic acid amplification test): GeneXpert MTB/RIF** | Automated cartridge PCR (polymerase chain reaction); about 2 hours (result in 1–2 days) | Rapid diagnosis; detects MTB and **rifampicin resistance** (MDR (multidrug-resistant) TB surrogate). **Xpert Ultra** is more sensitive. Stays positive after cure, so cannot monitor treatment |
| TrueNat | Indian chip-based NAAT | NTEP (National TB Elimination Programme) |
| **LPA (line probe assay)** | DNA strip hybridisation | Rifampicin, isoniazid or second-line drug resistance |
| **LF-LAM (lateral-flow lipoarabinomannan)** | Urine antigen strip, under 30 minutes; point-of-care | HIV with symptoms, CD4 <200 or advanced disease (WHO (World Health Organization) 2019) |
| **Histopathology** | Biopsy (bladder, epididymis) | Caseating granulomas; NAAT on tissue raises yield |
| **TST (tuberculin skin test)** | 5 TU (tuberculin units) PPD intradermal; read induration at 48–72 h | Proves infection only, not active disease; BCG false positive |
| **IGRA (interferon-γ release assay)** | ESAT-6/CFP-10 blood test (QuantiFERON, T-SPOT) | Unaffected by BCG; cannot separate latent and active; shows infection, not active disease |

## Newer diagnostic modalities

- **Liquid culture (MGIT):** about 10 days versus 4–6 weeks on LJ.
- **NAAT:** GeneXpert MTB/RIF (Xpert Ultra more sensitive), about 2 hours, detects rifampicin resistance; TrueNat; **LPA** for isoniazid resistance.
- **Urine LF-LAM:** strip, under 30 minutes; WHO 2019.
- **IGRA:** unaffected by BCG.
- **Imaging:** CT (computed tomography) urography; **MR (magnetic resonance) urography with DWI (diffusion-weighted imaging)** grades fibrosis.

## Supportive tests

- ESR, blood count, renal function, HIV, blood sugar.
- Sputum and **chest radiograph in all** (mandatory; lung TB coexists in up to one-fifth).

## Imaging and endoscopy

| Imaging | Findings / role |
|---|---|
| KUB (kidney–ureter–bladder) film | Calcification (24–44%); lobar "putty" kidney |

| Imaging | Findings / role |
|---|---|
| IVU (intravenous urography) / CT urography | **Moth-eaten calyx**, **phantom calyx** (infundibular stenosis), cavities, hydrocalycosis, **Kerr's kink**, pipestem or beaded corkscrew ureter, ureteric strictures, calcification, small bladder; IVU replaced by CT |
| Ultrasound | Children, pregnancy, scrotum; follows hydronephrosis during treatment |
| MRU (magnetic resonance urography) | Children, pregnancy |
| Cystoscopy and biopsy | **"Golf-hole" ureteric orifice**, ulcers; biopsy if culture negative (yield 19–52%) |
| Chest radiograph | Mandatory (lung TB in up to one-fifth) |

### Diagnostic algorithm

```
Suspicion (sterile pyuria, storage symptoms, epididymal nodule) -> 3-5 EMU: ZN + culture (LJ + MGIT) + Xpert -> chest X-ray; CT urography Positive -> treat; DST guides regimen Negative -> cystoscopy/biopsy, aspiration + NAAT on tissue Still negative -> empirical trial if clinically compatible
```

### Medical management (first step in all)

- **Multidrug therapy** because bacilli exist in different metabolic states and single drugs select resistance.
- Baseline liver and renal function, HIV, hepatitis, diabetes; DOT (directly observed therapy).
- **2HRZE/4HR:** isoniazid, rifampicin, pyrazinamide, ethambutol for 2 months, then isoniazid and rifampicin for 4 months, with **pyridoxine** against isoniazid neuropathy. India's NTEP uses 2HRZE/4HRE.
- Complicated disease: some treat 12–24 months, or ≥6 months after surgery (P2: some extend to 9–12 months). The new 4-month rifapentine–moxifloxacin regimen is not used for UG TB.
- Toxicity: hepatitis (all except ethambutol); optic neuritis (ethambutol).
- **Monitor with ultrasound** for worsening obstruction; stent or PCN (percutaneous nephrostomy) early. Steroids for strictures: anecdotal only.
- MDR-TB: second-line drugs (bedaquiline, pretomanid, linezolid).

### Surgical management

- About half need surgery, **after at least 4–6 weeks of ATT (antituberculous therapy)**, when inflammation and bacillary load have settled.

| Problem | Procedure |
|---|---|
| Obstruction | Early **DJ (double-J) stent** (succeeds in 41%) or **PCN** |
| Localised polar lesion not responding | Partial nephrectomy |
| Non-functioning kidney + persistent disease or hypertension | **Nephrectomy** (laparoscopic preferred; ligate artery and vein separately) |
| PUJ (pelviureteric junction) stricture | Pyeloplasty; ureterocalicostomy |
| Lower ureteric stricture | Reimplantation, psoas hitch, Boari flap; ileal ureter if long |
| Contracted bladder | Augmentation; substitution if <20 mL |

- A stricture not improving after 6 weeks of drugs needs open repair.
- Contested: nephrectomy of a non-functioning kidney without active disease.

### Thimble bladder in a 30-year-old man: assessment

| Assess | How | Why |
|---|---|---|
| Disease activity | Repeat urine AFB culture; bladder biopsy | Do not reconstruct with active TB |
| Capacity and compliance | Cystogram, cystoscopy under anaesthesia, urodynamics | Size and type of reconstruction |
| Upper tracts | Ultrasound, CT urography, MCU (micturating cystourethrogram) | VUJ stricture or reflux needs reimplantation |
| Renal function | Creatinine, GFR (glomerular filtration rate), split function | Bowel reservoir needs creatinine <2 mg/dL, GFR >35–40 mL/min |
| Urethra, bladder neck, sphincter | Urethrogram, endoscopy | Orthotopic reservoir needs a patent, continent outlet |
| Patient | Dexterity, motivation, fertility wishes | May need CIC (clean intermittent catheterisation) |

### Thimble bladder: surgical options

| Bladder state | Operation |
|---|---|
| Half the bladder diseased, capacity <100 mL | **Augmentation ileocystoplasty** ("clam" or onlay of detubularised ileum) |
| Severely contracted | Augmentation with **ileocaecum** or sigmoid |
| **Thimble bladder (<20 mL)** | **Substitution cystoplasty:** supratrigonal excision of the diseased bladder, keeping trigone and bladder neck, and an orthotopic detubularised ileal or ileocaecal reservoir |

- **Principles:** detubularise and fold the bowel into a spherical, low-pressure reservoir (Laplace's law; interrupts peristalsis). Anastomose widely to the trigone.
- **Ureters:** if the VUJ is strictured or refluxing, excise to healthy ureter and **reimplant into the reservoir**: an antireflux tunnel under a colonic taenia, or an isoperistaltic afferent ileal limb. A Boari flap or psoas hitch is not possible in a contracted bladder.
- **Avoid orthotopic reconstruction** with poor renal function, a strictured or incompetent urethra, inability to perform CIC, active TB, or short bowel or inflammatory bowel disease. Then use a continent catheterisable pouch (Mitrofanoff) or an **ileal conduit**.
- **Also know:** laparoscopic and robotic augmentation; transurethral incision of bladder neck contracture.

### Complications of bladder reconstruction

- **Mucus** and catheter blockage; **stones**; bacteriuria and UTI; **hyperchloraemic metabolic acidosis**; vitamin B12 deficiency if ileum is lost; **perforation** (needs regular emptying); incomplete emptying needing CIC; nocturnal incontinence; late tumour in the reservoir.

### Follow-up

- Relapse 6.3–22% (mean about 5 years), so follow up with urine AFB culture and imaging for up to 10 years.
- After reconstruction: also creatinine, bicarbonate, B12 and ultrasound.

**Diagrams to draw:**

- Kidney showing caseous cavity, phantom calyx and VUJ stricture.
- Stepwise diagnostic flowchart (algorithm above) from sterile pyuria to culture and DST.
- Supratrigonal excision of a thimble bladder with an ileocaecal substitute and ureteric reimplantation.

**Key points to remember:**

- Kidney 80%; periglomerular seeding; bladder disease descends from the kidney; heals by fibrosis, so obstruction worsens on treatment.
- Sterile pyuria; 3–5 early-morning urine AFB cultures; culture is the gold standard and the only test that monitors treatment.
- GeneXpert (Xpert) detects rifampicin resistance in 2 hours; cannot monitor treatment.
- LF-LAM for HIV with CD4 under 200.
- TST and IGRA show infection, not active disease.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Thimble bladder <20 mL: augment if capacity <100 mL; substitute if <20 mL; needs creatinine <2 mg/dL and a good urethra.
- Relapse up to 22%; follow up 10 years.

## LAQ 2 Interstitial cystitis / bladder pain syndrome

P2 · P4 **merged from 2**

*Asked as:* P4 LAQ "Interstitial cystitis / painful bladder syndrome: recent trends in evaluation, diagnosis and management" (Discuss the recent trends in the evaluation. Diagnosis and management of interstitial cystitis, Winter 2024 Paper IV Q1 LAQ; Painful bladder syndrome management, Summer 2021 Paper IV Q3d SAQ) · P2 SAQ "Interstitial cystitis (Bladder pain syndrome)."

*Read as: recent trends in the evaluation, diagnosis and management of IC/BPS (interstitial cystitis/bladder pain syndrome).*

### Definitions: how they have changed

- **NIDDK (National Institute of Diabetes and Digestive and Kidney Diseases, 1987):** research criteria; missed about 60% of patients, so the definition was broadened.
- **AUA (American Urological Association), current:** an unpleasant sensation (pain, pressure, discomfort) perceived to be related to the bladder, with LUTS (lower urinary tract symptoms) for **more than 6 weeks**, without infection or another identifiable cause.
- **ESSIC (European Society for the Study of Interstitial Cystitis, 2008):** chronic (**>6 months**) bladder-related pain with at least one other urinary symptom (persistent urge or frequency), after confusable diseases are excluded.
- **ICD-11 (International Classification of Diseases, 11th revision):** "chronic primary bladder pain syndrome".
- **Trend: splitting the syndrome.** "IC" is reserved for **Hunner lesion (HL) disease**, a true inflammatory disease; BPS without HL is a symptom complex. Both sit within UCPPS (urological chronic pelvic pain syndrome).

### Phenotypes

- **Hunner lesion disease ("IC"):** about 20%, mainly over 50; true inflammation (dense lymphoplasmacytic infiltrate, mast cells).
- **Non-Hunner BPS (bladder pain syndrome):** bladder often histologically normal, or shows glomerulations on cystoscopy.
- Also pelvic-floor-dominant and widespread (centralised) pain phenotypes.

### Pathophysiology (current view)

- Multifactorial: leaky GAG (glycosaminoglycan) layer; mast-cell activation; neurogenic inflammation; **central sensitisation**; stress-axis dysfunction; small-fibre neuropathy. No infection proven.
- **Associated conditions:** 75–80% have pain beyond the pelvis; IBS (irritable bowel syndrome, >30%), fibromyalgia, vulvodynia, endometriosis; tender pelvic floor in about 80%.

### Recent trends in evaluation (AUA 2022), before treatment

- **Diagnosis of exclusion plus localisation:** tests aim to find the pain generator.
- **History:** pain worse on filling, relieved by voiding; "painful urgency" without urge incontinence, unlike OAB (overactive bladder). **Ask about ketamine.**
- **Examination: six-site pelvic-floor palpation** (anterior and posterior levator ani, obturator internus, each side); vulval swab test; DRE (digital rectal examination) in men.

- **Basic tests:** PVR (post-void residual), urinalysis, culture; cytology if risk factors; frequency–volume (voiding) chart.
- **Scores:** O'Leary–Sant ICSI/ICPI (Interstitial Cystitis Symptom/Problem Index), GUPI (Genitourinary Pain Index), pain VAS (visual analogue scale); **score pain and urinary symptoms separately.**
- **Body map:** widespread pain = 3 or more of 12 regions besides the pelvis.
- **Office cystoscopy if over 50 or HL suspected** (AUA 2022 change): HL is a stellate red area with radiating vessels, found only by looking.
- **Local anaesthetic challenge** (lidocaine with or without bupivacaine) localises pain to the bladder and selects intravesical therapy.
- **Dropped:** potassium sensitivity test (neither sensitive nor specific; obsolete); routine hydrodistension (glomerulations are non-specific); routine urodynamics (optional only).
- **Biopsy** only to exclude CIS (carcinoma in situ).

### ESSIC classification

Cystoscopy with hydrodistension plus biopsy, e.g. "3C":

| Cystoscopy | Biopsy |
|---|---|
| X not done; 1 normal; 2 glomerulations; 3 Hunner lesion | X not done; A normal; B inconclusive; C positive (inflammation, mastocytosis, fibrosis) |

### Differential diagnosis (exclude)

- UTI (urinary tract infection) including tuberculosis, CIS, stones, radiation or cyclophosphamide cystitis, **ketamine cystitis**, OAB, diverticulum, endometriosis.

### Phenotype-based diagnosis and treatment (key trend)

| Phenotype | Identified by | Treatment |
|---|---|---|
| Hunner lesion | Cystoscopy | Fulguration and/or triamcinolone; ciclosporin A if refractory |
| Anaesthetic-responsive bladder pain | Local anaesthetic challenge | Intravesical therapy |
| Pelvic-floor tenderness | Pelvic-floor examination | Myofascial physiotherapy |
| Widespread (centralised) pain | Body map, associated pain conditions | Systemic: amitriptyline, gabapentinoids, CBT (cognitive behavioural therapy), pain team |

- **MAPP (Multi-Disciplinary Approach to the Study of Chronic Pelvic Pain) reanalysis, 2023:** hydroxyzine/PPS (pentosan polysulfate) helped localised pain; amitriptyline helped widespread pain.

### Management principles (AUA 2022)

- **Old AUA (2011/2014): six tiers**, from self-care to surgery.
- **AUA 2022 dropped the tiers:** grouped options, phenotype-guided, shared decisions; multimodal care allowed early; major surgery stays last.
- Stop treatments that fail; monitor with symptom scores.

## Treatment options

| Group | Options |
|---|---|
| Behavioural | Education; avoid trigger foods (coffee, citrus, spicy food, alcohol); stress management; flare plans |
| Physical | **Myofascial pelvic-floor physiotherapy** if tender (Standard); **avoid Kegel exercises** |
| Oral | **Amitriptyline** 10 mg at night, rising to about 50 mg; hydroxyzine; cimetidine; **PPS 100 mg three times daily** (**pigmentary maculopathy**: baseline and periodic retinal checks) |
| Intravesical | **DMSO (dimethyl sulfoxide) 50%, 50 mL, 15–20 min dwell**; heparin; alkalinised lidocaine (flares); also hyaluronic acid, chondroitin |
| Procedures | **HL: fulguration and/or triamcinolone injection** (75–90% improve); short (1–3 min) low-pressure (60–80 $cmH_2O$) hydrodistension; **onabotulinumtoxinA 100 U into the trigone** (must accept self-catheterisation; risk of retention); SNM (sacral neuromodulation) |
| Refractory HL | **Oral ciclosporin A** (watch kidney function and blood pressure) |
| Last resort (<10%) | Cystoplasty (augmentation) or diversion with or without cystectomy; pain may persist |

- **Do not offer:** long-term antibiotics, intravesical BCG (bacille Calmette–Guérin), long-term oral (systemic) steroids.
- Pain management: gabapentinoids; opioids only via pain specialists.
- Most hydrodistension responses last under 2 months.
- Treat associated pain conditions (IBS, fibromyalgia) with other specialists.

## Recent advances

- Urinary biomarkers and functional MRI (magnetic resonance imaging) for phenotyping (research); phenotype-stratified trials.
- **Also know:** tanezumab (anti-NGF (nerve growth factor)) not adopted; liposomal intravesical botulinum toxin no better than placebo; pudendal neuromodulation.

**Diagram to draw:** cystoscopic stellate Hunner lesion with radiating vessels, and the four-phenotype treatment chart.

> **Key points to remember:**
>
> - AUA definition: bladder pain with LUTS >6 weeks, no other cause; ESSIC >6 months.
> - AUA 2022: no tiers; phenotype-based; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test (obsolete).
> - HL: fulguration or triamcinolone; ciclosporin if refractory.
> - Amitriptyline, PPS, DMSO; PPS causes maculopathy; antibiotics and BCG not offered.

## LAQ 3 Genitourinary infections, common organisms, bacteraemia and urosepsis: pathophysiology and management

P1

**Asked as:**

- Discuss general principles of genitourinary infections, characteristics of common organisms and management of bacteremia and septic shock. — *Summer 2016 Paper I Q2 (LAQ)*
- Describe Pathophysiology of urosepsis and its management. — *Summer 2017 Paper I Q2 (LAQ)*
- Describe pzthophysiology of urosepsis and its management. — *Practice Set 11 Paper I Q1 (LAQ)* (also answered in 2017-Summer Paper 1, Q2) — *"Pzthophysiology" is read as pathophysiology.*

**Type:** LAQ (25 marks)

### Definitions

- **UTI (urinary tract infection):** urothelial inflammation from bacterial invasion, usually with bacteriuria and pyuria. **Sterile pyuria** needs evaluation (tuberculosis, stone, tumour).
- **Uncomplicated:** healthy non-pregnant woman, normal tract. **Complicated:** men, pregnancy, obstruction, stones, catheter, neurogenic bladder, diabetes, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months. Unresolved = never cleared; reinfection = new organism; **bacterial persistence** = same organism from a focus (stone, prostate).
- **Prophylaxis** assumes a sterile tract; **suppression** controls a focus that cannot be eradicated.
- **Urosepsis:** sepsis whose source is the urogenital tract (the urinary tract or male genital organs). It is usually Gram-negative and usually follows obstruction or instrumentation.

### Pathogenesis of UTI

- **Routes:** ascending from bowel flora (commonest); haematogenous (*S. aureus*, *Candida*); lymphatic (rare).
- **Virulence (uropathogenic *E. coli*): type 1 pili** (mannose-sensitive) bind bladder uroplakins and cause cystitis. **P pili** (mannose-resistant) bind renal glycolipids and are found in about 90% of pyelonephritis strains. Haemolysin damages tissue. Bacteria invade umbrella cells and form intracellular communities; quiescent reservoirs cause recurrence.
- **Principle:** a strong host is infected only by virulent strains; a compromised host (obstruction, catheter) by almost any organism.
- **Host defences:** voiding washout, lactobacilli, urine osmolality and pH, uromodulin (Tamm–Horsfall protein), urothelial exfoliation.
- **Risk factors: obstruction** (stasis impairs defences and turns UTI life-threatening), VUR (vesicoureteral reflux; with infection it scars the growing kidney), stones, catheters, diabetes, pregnancy, spermicides.
- **Diagnosis:** dipstick (nitrite, leucocyte esterase) rules out better than it rules in. Culture is the gold standard: $\geq 10^5$ CFU (colony-forming units)/mL (Kass); $\geq 10^2$ in a symptomatic woman or a catheter specimen. Image (ultrasound, CT (computed tomography)) in complicated infection or fever beyond 72 hours.

## Characteristics of common organisms

| Organism | Key features |
|---|---|
| ***E. coli*** | ~85% of community and ~50% of hospital UTI; commonest cause of urosepsis; ST131 clone is ESBL (extended-spectrum β-lactamase) and quinolone resistant |
| ***Proteus*** | Urease splits urea into ammonia: alkaline urine (pH >7), **struvite stones**, persistence in stones |
| ***Klebsiella*** | Capsule; urease; ESBL and KPC (*K. pneumoniae* carbapenemase) producers |
| ***Pseudomonas*** | Nosocomial, catheter biofilm; needs ceftazidime, piperacillin, aminoglycoside, fluoroquinolone or meropenem |
| ***Enterococcus*** | **Always resistant to cephalosporins**; amoxicillin, nitrofurantoin; VRE (vancomycin-resistant) |
| ***S. saprophyticus*** | ~10% of cystitis in young sexually active women |
| ***S. aureus*** | Haematogenous renal abscess; in urine, look for bacteraemia |
| **Anaerobes, *Candida*** | Abscesses, fistulae; *Candida* with catheters and diabetes (*Candida* in catheterised diabetics) |

- **Urosepsis organisms:** *E. coli* (commonest), *Proteus*, *Klebsiella*, *Pseudomonas*, *Enterococcus*; ESBL (extended-spectrum β-lactamase) producers rising; *Candida* in catheterised diabetics.
- **Also know:** emerging *Aerococcus urinae*; *Ureaplasma* and *Mycoplasma* with negative routine cultures; *Chlamydia* and tuberculosis do not grow on routine culture (sterile pyuria).
- **Antibiotic principles (see 2015-Summer Paper 1, Q3f):** urine levels matter in cystitis, tissue levels in pyelonephritis and bacteraemia. Use a narrow spectrum and short course; adjust for renal function; **drain obstruction**.

## Bacteraemia, sepsis, septic shock and urosepsis

### Sepsis-3 (2016)

| Term (Sepsis-3, 2016) | Definition |
|---|---|
| **Sepsis** | Life-threatening organ dysfunction from a dysregulated host response to infection: acute rise in **SOFA (Sequential Organ Failure Assessment) score ≥2** |
| **qSOFA (quick SOFA)** | Respiratory rate ≥22/min, altered mentation, systolic blood pressure ≤100 mmHg; ≥2 = high risk (of death) (prompt only) |
| **Septic shock** | Vasopressor needed for MAP (mean arterial pressure) ≥65 mmHg **and** lactate >2 mmol/L despite (adequate) fluids; mortality >40% |
| - SIRS (systemic inflammatory response syndrome) criteria were dropped from the definition as non-specific. | |

### Causes and risk factors (urological triggers)

- **Urological triggers:** infected obstructed kidney, PCNL (percutaneous nephrolithotomy), high-pressure ureteroscopy, prostate biopsy, TURP (transurethral resection of the prostate), infected stones, emphysematous pyelonephritis, abscess.

- **Obstruction with infection:** stone, PUJ (pelviureteric junction) obstruction, tumour, pregnancy, giving infected hydronephrosis or pyonephrosis. Pyonephrosis, emphysematous pyelonephritis, renal, perinephric and prostatic abscess.
- **Parenchymal infection:** emphysematous pyelonephritis, renal or perinephric abscess, prostatic abscess, Fournier gangrene.
- **Instrumentation:** PCNL (percutaneous nephrolithotomy), ureteroscopy with high irrigation pressure, TURP (transurethral resection of prostate), prostate biopsy (sepsis 0.3–3.1%), catheters.
- **Host:** diabetes, immunosuppression, old age, neurogenic bladder.

**Pathophysiology**

- **Pathophysiology (summary):** endotoxin binds TLR4 (Toll-like receptor 4). Cytokines (TNF-α (tumour necrosis factor-α), interleukins IL-1 and IL-6) and nitric oxide cause vasodilatation, capillary leak, myocardial depression and DIC (disseminated intravascular coagulation), then organ failure.

**1. Entry of bacteria (and endotoxin) into blood**

- Obstruction causes stasis, impairs washout and raises pelvic pressure. Bacteria and endotoxin then enter the blood (circulation) by **pyelovenous and pyelolymphatic backflow**.
- High irrigation pressure at endoscopy does the same, and so does fragmenting infected stones, which release endotoxin (fragmenting infected stones releases endotoxin). Stone culture predicts postoperative sepsis better than bladder urine culture.
- An obstructed infected kidney is (behaves like) an **abscess with a collecting system**. Antibiotics barely reach it, and bladder urine may be sterile.

**2. Host response** `Endotoxin (LPS, lipopolysaccharide) / (or) Gram-positive wall products → TLR4 (Toll-like receptor 4) on macrophages → NF-κB (nuclear factor κB) → cytokines TNF-α (tumour necrosis factor α), interleukins IL-1, IL-6, IL-8 → complement + coagulation activation + endothelial injury (activation)`

- **Vasodilatation:** inducible nitric oxide causes (lowers) low systemic vascular resistance ("warm" shock).
- **Capillary leak:** (endothelial) glycocalyx damage leads to hypovolaemia and tissue oedema.
- **Coagulopathy:** tissue factor release causes microthrombi, platelet consumption and DIC (disseminated intravascular coagulation).
- **Myocardial depression** and **mitochondrial dysfunction (failure):** cells cannot use oxygen, so lactate rises.
- **Immunoparalysis:** a later anti-inflammatory phase predisposes to (invites) secondary infection.

**3. Organ dysfunction:** AKI (acute kidney injury), ARDS (acute respiratory distress syndrome), encephalopathy, liver dysfunction (failure), DIC, multi-organ failure.

**Clinical features**

- Fever or hypothermia, rigors, flank pain, tachycardia, tachypnoea, confusion, oliguria, hypotension. Often starts (begins) within hours of an endourological procedure (endourology).

**Investigations**

- **Lactate**, blood cultures ×2 (two blood cultures), urine culture (and pelvic urine at drainage), blood count, creatinine, bilirubin, platelets, coagulation, arterial (blood) gas, glucose. Procalcitonin helps decide when to stop antibiotics.
- **Imaging:** bedside ultrasound (hydronephrosis, debris); **CT (computed tomography)** for stone, gas or abscess.

## Management (Surviving Sepsis Campaign: hour-1 bundle 2018, guidelines 2021)

**A. Resuscitation and antibiotics (SSC, Surviving Sepsis Campaign 2021)** Recognise (SSC: do not use qSOFA alone for screening) → Lactate → blood cultures ×2 + urine culture → IV (intravenous) broad-spectrum antibiotics within 1 hour (shock or probable sepsis; within 3 h if sepsis only possible) → 30 mL/kg (balanced) crystalloid if hypotensive or lactate ≥4 mmol/L (within 3 h if hypotensive or hypoperfused) → noradrenaline first line for MAP ≥65 (target MAP 65); add vasopressin → SOURCE CONTROL: stent or nephrostomy (equally effective), drain abscess, change catheter; stone surgery later → refractory shock (persistent vasopressor need): hydrocortisone 200 mg/day; ICU (intensive care unit) → re-measure (repeat) lactate, capillary refill → de-escalate by culture at 48–72 h

- **Empirical antibiotics:** by local resistance: third-generation cephalosporin or piperacillin–tazobactam ± aminoglycoside; carbapenem if ESBL risk. Adjust for renal function.

**B. Source control: the key urological step**

- **Drain the obstructed infected kidney urgently:** retrograde **double-J stent** or **percutaneous nephrostomy (PCN)**. Randomised trials show equal effect on sepsis (equally effective in a randomised trial (Pearle, 1998)).
- Stent: suits coagulopathy; avoid high-pressure contrast injection.
- PCN: better for (suits) large or impacted stones, pyonephrosis; allows (gives) pelvic urine (for) culture.
- **No definitive stone surgery during sepsis.** Treat the stone after (once) the infection clears.
- Other sources: drain renal or perinephric abscess; emphysematous pyelonephritis (percutaneous drainage, nephrectomy if diffuse gas without fluid); prostatic abscess drainage; change or remove (the) catheter; debride Fournier gangrene.

**C. Supportive care (Support)**

- Oxygen and (or) ventilation, glucose control (insulin if glucose ≥180 mg/dL), transfusion (transfuse) below haemoglobin 7 g/dL, thromboprophylaxis, stress-ulcer prophylaxis, renal replacement if needed. Watch for AKI (acute kidney injury), ARDS (acute respiratory distress syndrome) and DIC.

**D. De-escalation and duration**

- Narrow antibiotics by (to) culture at 48–72 h. Usual course (usually) 7–10 days (another source: 7–14 days); longer with an undrained focus or *S. aureus* bacteraemia; shorter courses suffice after good source control. Step down (switch) to oral therapy when (once afebrile and) stable.

### Special situations

- **Emphysematous pyelonephritis** (diabetics): fluids, glucose control, antibiotics, drain; nephrectomy for diffuse gas without fluid.
- **Pregnancy:** stent or nephrostomy; avoid fluoroquinolones.
- **Fungal urosepsis:** remove catheter, antifungal, drain fungal balls.
- **Prognosis:** urosepsis does better than sepsis from other sites if the source is drained early; delay in drainage (delayed drainage) is the main avoidable cause of death.

### Prevention

- Sterile urine before elective endourology; antibiotic prophylaxis; low irrigation pressure and access sheath; short operating time with infected stones; stage the procedure if pus is found.

**Diagram to draw:** pathway from *E. coli* adherence to bacteraemia and cytokine-driven shock; obstructed kidney leading to backflow, cytokine cascade and shock; obstructed kidney → pyelovenous backflow → cytokine cascade → shock and organ failure.

> **Key points to remember:**
>
> - *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
> - Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
> - Obstruction plus infection is the emergency; bladder urine may be sterile.
> - Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
> - Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.
> - Stent and nephrostomy equally effective (stent = nephrostomy for drainage); delay definitive stone surgery.

## LAQ 4 Describe clinical implication of AIDS on the Genito-Urinary Tract?

P1

**Type:** LAQ (25 marks)
**Source:** 2020-Summer Paper 1, Q1

### Introduction

- AIDS (acquired immunodeficiency syndrome) is the late stage of HIV (human immunodeficiency virus) infection. HIV is a retrovirus that destroys CD4 T lymphocytes. It spreads by blood, semen, vaginal fluid and breast milk.

| Stage | Features |
|---|---|
| 1. Acute | 2–4 weeks after infection; flu-like illness; high viral load; antibody tests may be negative |
| 2. Clinical latency | Years; usually no symptoms |
| 3. AIDS | **CD4 <200 cells/mm³** or an AIDS-defining condition |

- **Concept:** genitourinary (GU) disease follows the falling CD4 count. Organisms become opportunistic, infections become chronic and the GU tract becomes a reservoir. With HAART (highly active antiretroviral therapy), the burden shifts to drug toxicity and diseases of ageing.

### Diagnosis (urologist's role)

- Test anyone with, or at risk of, an STD (sexually transmitted disease), and every TB (tuberculosis) patient.
- Antigen–antibody immunoassay → antibody differentiation assay; if discordant, HIV RNA (acute infection).

### Manifestations by site

| Site | Manifestations |
|---|---|
| **Genital skin and STDs** | Ulcerative STDs (herpes, syphilis, chancroid) increase HIV acquisition and transmission; HSV-2 (herpes simplex virus 2) in 50–90%. Warts (HPV (human papillomavirus) 6, 11) in about 20%, may be giant; molluscum 5–18%. Perineal cellulitis (cover MRSA (methicillin-resistant *Staphylococcus aureus*)); Fournier gangrene commoner |
| **UTI (urinary tract infection)** | About fivefold commoner. Bacteriuria about 30% at CD4 <200, 11% at 200–500, none above 500. *E. coli* and other gram-negatives; *Pseudomonas* up to 30%. *Salmonella* UTI needs lifelong prophylaxis. Culture before treatment |
| **Opportunistic UTI** | No response, or CD4 <500: look for fungi (*Candida*, *Cryptococcus*, *Histoplasma*), TB, CMV (cytomegalovirus), adenovirus |
| **Kidney** | Opportunistic pyelonephritis (*Pneumocystis*, *Candida*, mycobacteria). AKI (acute kidney injury) from sepsis, dehydration, nephrotoxins (tenofovir, indinavir, amphotericin). Obstruction by stones, lymphoma, fungal balls |

| Site | Manifestations |
|---|---|
| **GU TB** | HIV raises active TB 15–20-fold. Disease is disseminated and bilateral, with **less fibrosis, fewer strictures and fewer contracted bladders**, because fibrosis needs an intact immune response. Mortality is higher. Same 6-month regimen; rifampicin lowers protease inhibitor levels |
| **Prostate** | Bacterial, TB and fungal prostatitis; prostatic abscess and urosepsis; relapse up to 70%. Cryptococcal prostatitis is a reservoir for relapsing meningitis, so give long-term antifungals |
| **Testis and epididymis** | Recurrent, chronic epididymo-orchitis (including TB). Testis harbours opportunists. Testicular atrophy and hypogonadism |
| **Voiding** | LUTS (lower urinary tract symptoms) worsen with disease stage. Retention may herald seroconversion. Neurogenic causes: HIV encephalopathy, toxoplasmosis, **CMV polyradiculopathy** (bladder and bowel dysfunction with sciatic pain: lumbar puncture, treat early), neuropathy |
| **Sexual function** | ED (erectile dysfunction) in about half of men under 50; protease inhibitors worsen it. **Protease inhibitors raise sildenafil levels 3–10-fold: start low.** Low libido, ejaculatory disorders |

### HIVAN (HIV-associated nephropathy)

- Nephrotic-range proteinuria and rising creatinine, often without oedema or hypertension. Typically CD4 <200 and viral load >1000 copies/mL; about 12 times commoner in Black patients.
- Biopsy: **collapsing FSGS (focal segmental glomerulosclerosis)** with tubular microcysts.
- Treatment: HAART plus ACE (angiotensin-converting enzyme) inhibitor. Without it, dialysis follows within about 10 months.

### Stones

- **Indinavir** stones in up to 22% of treated patients. Indinavir precipitates in alkaline urine. Stones are radiolucent, and pure stones may be invisible even on CT (computed tomography).
- Stop the drug and hydrate: succeeds in up to 70%. Newer protease inhibitors rarely cause stones.

### Malignancy

| AIDS-defining | Non-AIDS-defining |
|---|---|
| **Kaposi sarcoma** (HHV-8 (human herpesvirus 8)), non-Hodgkin lymphoma, cervical cancer | Penile, anal, testicular, prostate cancers; Hodgkin lymphoma |
| Falls with HAART | Now about 70% of cancers in treated HIV |

- **Kaposi sarcoma:** genital in about 3% of AIDS; glans lesions can block the meatus. Treat with ART (antiretroviral therapy) first. Local excision, laser or intralesional vinblastine; radiotherapy 15–30 Gy; systemic chemotherapy if disseminated.
- **Penile cancer:** HIV raises HPV infection; about fourfold risk of death.

### Implications for urological practice

- **Universal precautions** for every patient: double gloves, no hand-to-hand passing of sharps.

- Occupational risk: percutaneous about 0.3%, mucous membrane about 0.09%.
- **PEP (post-exposure prophylaxis)** within 72 hours, for 28 days: tenofovir + emtricitabine plus raltegravir or dolutegravir. HIV test at 6 weeks, 3 and 6 months.
- CD4 <200: TMP-SMX (trimethoprim-sulfamethoxazole) *Pneumocystis* prophylaxis also lowers UTI.
- **Prevention:** male circumcision cuts heterosexual acquisition by 50–60% (three African RCTs (randomised controlled trials)); consistent condom use cuts risk about 80%; PrEP (pre-exposure prophylaxis).

**Also know:** IRIS (immune reconstitution inflammatory syndrome) on starting ART; serodiscordant couples: suppressed viral load with sperm washing and assisted reproduction; transplantation is possible with controlled HIV.

**Diagram to draw:** male GU tract outline labelled with the lesion at each site.

> **Key points to remember:**
> - AIDS = CD4 <200; UTI risk rises as CD4 falls.
> - HIV-TB: disseminated, less fibrosis, fewer strictures.
> - Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
> - PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

## LAQ 5 Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria.

P3

**Type:** LAQ (25 marks)
**Source:** 2024-Winter Paper 3, Q2 (also answered in 2018-Summer Paper 1, Q1, with lymphatic anatomy)

### Definition

- **Chyluria** is the passage of **chyle** (intestinal lymph rich in chylomicrons and protein) in the urine, making it milky.
- It needs an abnormal **lymphourinary fistula**, usually at a renal calyceal fornix (pyelolymphatic fistula).

### Relevant lymphatic anatomy

- Intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction.
- Lumbar (para-aortic) trunks from the kidneys join the cisterna. Lymph flows cephalad and right to left.
- Valves normally stop backflow from the cisterna into renal hilar lymphatics.

### Aetiology

| Parasitic (commonest; India, China) | Non-parasitic |
|---|---|
| **Wuchereria bancrofti** (lymphatic filariasis; Culex mosquito in urban India) | Congenital: lymphangiectasia, renal lymphangioma |
| Rarely other parasites | Trauma or surgery (partial nephrectomy, RPLND [retroperitoneal lymph node dissection]) |
| | Retroperitoneal tumour, fibrosis, tuberculosis, pregnancy |
| - Genital and urinary filariasis is almost always *W. bancrofti*; Brugia species spare the genitalia. | |

### Pathogenesis

- **Life cycle:** a mosquito deposits infective larvae, which mature in central lymphatics over 6–9 months. Adults live 5–7 years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.
- **Obstructive theory:** adult worms and the host reaction cause **lymphangitis and fibrosis** of lymphatics between the intestinal trunks and thoracic duct.
- Lymphatics below the block **dilate**, their **valves fail**, and chyle **refluxes** into renal hilar and peripelvic lymphatics.
- Rising pressure ruptures the thin-walled lymphatics into a calyceal fornix, creating a **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- Chyluria appears **earlier than elephantiasis**, is usually **intermittent** and may remit spontaneously.

### Clinical features

- **Milky urine**, worse after fatty meals and on standing; may clear after fasting.
- **Chylous clots:** colic, retention.
- **Haematochyluria** (blood-stained chylous urine).
- **Systemic effects** of fat and protein loss: weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.
- Endemic residence, filarial fever, hydrocele or lymphoedema.
- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

### Evaluation

**1. Confirm chyle**

- **Ether (chloroform) test:** milky urine shaken with ether clears, because the fat dissolves.
- **Urine triglycerides** positive; Sudan III staining shows fat globules; urine protein raised.
- Differentiate from **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

**2. Filarial tests**

- Night (midnight) blood smear: often negative by this stage.
- **Circulating filarial antigen** (card test), eosinophilia, PCR (polymerase chain reaction); ultrasound "filarial dance sign" in scrotal lymphatics.

**3. Lateralise and localise the fistula**

- **Cystoscopy after a fatty meal:** milky efflux from one ureteric orifice; bilateral in some.
- **Retrograde pyelography:** pyelolymphatic backflow.
- **Lymphoscintigraphy:** non-invasive; tracer seen in the kidney and bladder. MR lymphangiography or CT; pedal lymphangiography is the historical gold standard.

**4. Baseline:** albumin, haemoglobin, renal function.

### Management

`Mild -> diet + antifilarial drugs Moderate/persist -> add endoscopic sclerotherapy (repeatable) Severe or failed -> renal hilar lymphatic disconnection` **1. Conservative**

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids. MCTs are absorbed directly into portal blood, not lacteals, so less chyle forms.
- **DEC (diethylcarbamazine)** 2 mg/kg three times daily, traditionally for 12 days (a 1-day course is as effective). It kills microfilariae but only modestly kills adults.
- **Doxycycline** (anti-*Wolbachia*, 200 mg daily for 4–8 weeks); albendazole; ivermectin.

**2. Endoscopic sclerotherapy (renal pelvic instillation)**

- Ureteric catheter on the affected side, head-down position. Instil **silver nitrate (0.5–1%)** or **povidone-iodine**, repeated over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- Good initial response; recurrence common.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

**3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)**

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:** 1. Full renal mobilisation (nephrolysis). 2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every hilar lymphatic. 3. Strip the upper ureter; remove perinephric fat. 4. Nephropexy; optional **omental wrap** around the hilum. 5. On-table cystoscopy confirms clear efflux.
- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.
- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open (shorter operation and stay, less blood loss, fewer complications).
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy obsolete.

**4. Prevention**

- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

**Diagram to draw:** retroperitoneal lymphatics with obstructed thoracic duct, dilated hilar lymphatics and pyelolymphatic fistula at a calyceal fornix.

> **Key points to remember:**
> - Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
> - Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
> - Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
> - Chyluria precedes elephantiasis and is often intermittent.

## Short answer questions (SAQ)

### SAQ 1 Chronic (non-bacterial) prostatitis: diagnosis, classification and management

P1 · P4 **merged from 2**

*Asked as:* P1 SAQ "Chronic Prostatitis : diagnosis, classification and management." · P4 SAQ "Chronic non bacterial prostatitis."

#### Definition and classification: NIH (National Institutes of Health), 1999

| Category | Description |
|---|---|
| I | Acute bacterial prostatitis |
| II | **Chronic bacterial prostatitis (CBP)**: recurrent UTI (urinary tract infection) with the **same organism**; well between episodes |
| **IIIA** | **CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)**, **inflammatory**: leucocytes in EPS (expressed prostatic secretion), VB3 (post-massage urine) or semen |
| **IIIB** | CP/CPPS, **non-inflammatory** |
| IV | Asymptomatic inflammatory prostatitis (incidental) |

- Chronic non-bacterial prostatitis is now called **CP/CPPS**, NIH category III: pelvic pain for **≥3 of the last 6 months** without uropathogenic infection. It accounts for 90–95% of prostatitis.
- The IIIA vs IIIB split has no treatment value.

### Aetiology

- Interplay of past infection, neurogenic inflammation, **pelvic floor muscle dysfunction**, autoimmunity, psychological stress and **central sensitisation**.
- The prostate may not be the pain source.

### Clinical features

- Perineal (commonest), testicular, suprapubic or penile pain; ejaculatory pain; LUTS (lower urinary tract symptoms); sexual dysfunction.

### Evaluation (diagnosis of exclusion)

- **NIH-CPSI (Chronic Prostatitis Symptom Index):** pain, urinary and quality-of-life domains (score 0–43).
- **DRE (digital rectal examination):** prostate tender in fewer than half; palpate the pelvic floor for **trigger points** (tender in about half).
- **Localisation tests:**
- **Meares–Stamey 4-glass:** VB1 (urethra), VB2 (bladder), EPS, VB3 (prostate).
- **2-glass** (pre- and post-massage urine): matches the 4-glass test in >95%.
- Urinalysis and culture, PVR (post-void residual), flow rate.
- Cytology if irritative symptoms or haematuria.
- Selected patients: cystoscopy (if low flow or bladder pain), urodynamics.
- **UPOINT phenotyping: U**rinary, **P**sychosocial, **O**rgan-specific, **I**nfection, **N**eurological/systemic, **T**enderness of muscles.

### Management of category II (chronic bacterial prostatitis)

- Fluoroquinolone (lipid-soluble zwitterion, enters prostate) for **4–6 weeks** (EAU (European Association of Urology)).
- Chlamydia: azithromycin. Resistant organisms: TMP-SMX (trimethoprim-sulfamethoxazole) for 3 months. Refractory: suppression or TURP (transurethral resection of prostate).

### Management of category III (CP/CPPS): multimodal, UPOINT-directed

Monotherapy trials fail.

| Domain | Therapy |
|---|---|
| Urinary | **α-blockers** in recent-onset (<1 year), α-blocker-naive men; antimuscarinics |
| Psychosocial | CBT (cognitive behavioural therapy), stress reduction |
| Organ (prostate) | **Quercetin**, **pollen extract** (small RCTs, randomised controlled trials); 5-ARI (5α-reductase inhibitor) if BPH (benign prostatic hyperplasia) |
| Infection | Quinolone or tetracycline for ≥6 weeks only if treatment-naive and <1 year (EAU), or positive VB3; no repeated courses |
| Neurological | Pregabalin, amitriptyline, duloxetine |
| Tenderness | **Pelvic floor physiotherapy, myofascial trigger-point release**, biofeedback |

- **Others: acupuncture** (beat sham in a large RCT: 61.5% vs 38.3% response), aerobic exercise, PTNS (percutaneous tibial nerve stimulation), shockwave therapy; NSAIDs (non-steroidal anti-inflammatory drugs) only in combination.
- **Avoid (not recommended):** TURP or radical prostatectomy for pain.

**Diagram to draw:** UPOINT snowflake with a therapy for each domain.

> **Key points to remember:**
> - NIH I–IV; III = pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
> - CBP: same organism, fluoroquinolone 4–6 weeks.
> - CP/CPPS: NIH-CPSI to score, UPOINT to phenotype and treat.
> - Multimodal therapy; α-blocker plus antibiotic only early.

## SAQ 2 Emphysematous pyelonephritis: prognostic factors and conservative management

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "EPN. Prognostic factors & conserv. mgt." · P3 SAQ "Prognostic indicators in emphysematous pyelonephritis."

*Assumed meaning: EPN = emphysematous pyelonephritis.*

### Definition and pathogenesis

- EPN is an acute **necrotising** infection of the renal parenchyma and perirenal tissue by **gas-forming organisms**. It is a urological emergency.
- Organisms: *E. coli* (commonest), *Klebsiella*, *Proteus*.
- Most patients are **diabetic**; obstruction in about a quarter.
- **Mechanism (proposed):** high tissue glucose fermented to $CO_2$ in poorly perfused tissue.
- Reported mortality 19–43%.
- **CT (computed tomography)** is the imaging of choice.

### Imaging-based prognostic classifications

#### 1. Huang–Tseng CT classification (2000)

| Class | Gas location |
|---|---|
| 1 | Collecting system only (emphysematous pyelitis) |
| 2 | Renal parenchyma only, no extrarenal extension |
| 3A | Extension into the **perinephric** space |
| 3B | Extension into the **pararenal** space |
| 4 | **Bilateral** EPN or EPN in a **solitary kidney** |

#### 2. Wan classification (CT pattern)

| Type | CT | Mortality |
|---|---|---|
| I | Streaky or mottled gas, **no fluid**; rapid parenchymal destruction | 50–60% |
| II | Renal or perirenal **fluid**, bubbly or loculated gas, or gas in collecting system | <20% |

- **Why:** fluid means a walled-off (contained) collection that can be drained. Streaky gas without fluid means necrosis faster than the host can contain it.

### Clinical and laboratory prognostic factors

- **Huang–Tseng risk factors: thrombocytopenia**, **AKI (acute kidney injury)/raised creatinine**, **altered sensorium**, **shock**.
- **Other adverse factors:** hypoalbuminaemia, bacteraemia, need for haemodialysis, polymicrobial infection.

### Conservative management (now standard)

**1. Resuscitation:** fluids, inotropes, ICU (intensive care unit) if shocked; **tight glucose control**; correct electrolytes.

**2. Antibiotics:** broad-spectrum IV (intravenous), adjusted to culture; usually several weeks in total.

**3. Drainage**

- **Relieve obstruction:** DJ (double-J) stent or PCN (percutaneous nephrostomy).
- **PCD (percutaneous catheter drainage)** of gas and fluid collections under CT or ultrasound; multiple drains if needed; repeat CT to check.

### How prognosis guides management (algorithm)

```
All: resuscitation + glucose control + IV antibiotics + relieve obstruction (stent/PCN) Class 1–2 -> antibiotics ± PCD (P3: antibiotics + PCD) Class 3–4, <2 risk factors -> antibiotics + PCD (usually succeeds) Class 3–4, >=2 risk factors, or PCD failure -> nephrectomy (emergency or delayed)
```

- Routine emergency nephrectomy carried 40–50% mortality, so drainage-first care replaced it.
- Delayed nephrectomy for a kidney that is non-functioning on renal scan.
- **Emphysematous pyelitis** (class 1, gas in the collecting system only) is often non-diabetic and often responds to antibiotics alone.

**Diagram to draw:** CT sketches of Huang–Tseng classes 1, 2, 3A/3B and 4.

> **Key points to remember:**
> - Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
> - Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
> - Wan type I (no fluid) 50–60% mortality; type II <20%.
> - Drainage first; nephrectomy for extensive disease with ≥2 risk factors or drainage failure.

## SAQ 3 Principles of antimicrobial therapy in UTI

P1

**Type:** SAQ (10 marks)
**Source:** 2015-Summer Paper 1, Q3f

### Definitions

- **Uncomplicated** urinary tract infection (UTI): healthy non-pregnant women, normal tract. **Complicated:** men, pregnancy, abnormal tract, catheter, immunosuppression.
- **Recurrent:** ≥2 in 6 months or ≥3 in 12 months.

## Principles

- **Site decides the level:** urine in cystitis, tissue elsewhere; nitrofurantoin and fosfomycin lack tissue levels: **cystitis only**.

| Killing pattern | Drugs | Dosing goal |
|---|---|---|
| Time-dependent (time above MIC [minimum inhibitory concentration]) | β-lactams, carbapenems | Frequent doses or extended infusion |
| Concentration-dependent (peak:MIC) | Aminoglycosides, fluoroquinolones (FQs) | High once-daily dose |

- **Resistance thresholds:** trimethoprim–sulfamethoxazole (TMP-SMX) if local resistance ≤20%; oral FQ for pyelonephritis if ≤10%.
- **Collateral damage:** FQs and cephalosporins select resistance and *C. difficile*; avoid them in cystitis.
- Culture all but simple cystitis; **drain** obstructed infected kidneys; shortest course.

## Empirical choices

| Scenario | Drug | Duration |
|---|---|---|
| Cystitis (women) | Nitrofurantoin 100 mg twice daily; fosfomycin 3 g; pivmecillinam 400 mg; TMP-SMX 160/800 mg twice daily | 5 d; single dose; 3–5 d; 3 d |
| Cystitis (men) | TMP-SMX or per culture | 7 d (= 14 d, Drekonja 2021) |
| Pyelonephritis (oral) | Ciprofloxacin 500 mg twice daily; levofloxacin 750 mg daily | 7 d; 5 d |
| Pyelonephritis (intravenous) | Ceftriaxone 1 g daily; gentamicin 5–7 mg/kg once daily | 10–14 d; oral when afebrile |
| Complicated / catheter UTI | Aminoglycoside or 3rd-generation cephalosporin per culture; change catheter | 7–14 d |
| Urosepsis | Broad-spectrum **within 1 h**, source control | – |

## Asymptomatic bacteriuria

- Treat **only in pregnancy and before mucosa-breaching urological procedures**. Not in elderly, diabetic, catheterised or spinal cord injury patients.
- **Pregnancy:** avoid FQs and TMP-SMX; cephalexin is suitable.

## Prophylaxis

- Recurrent: vaginal oestrogen, **methenamine hippurate** (non-inferior to antibiotics, ALTAR 2022), cranberry, post-coital antibiotics.
- Surgical: single dose within 1 hour; ≤24 h for percutaneous nephrolithotomy and prostheses.
- **Also know:** ESBL (extended-spectrum β-lactamase) producers; nitrofurantoin avoided if creatinine clearance <30 mL/min; gepotidacin (US 2025).

**Diagram to draw:** Concentration–time curve showing peak:MIC and time above MIC.

**Key points to remember:**

- Recurrent = ≥2 in 6 months or ≥3 in 12 months.
- Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
- Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).

## SAQ 4 Renal and perinephric abscess

P1

**Asked as:**

- Perinephric abscess — *Summer 2017 Paper I Q3c (SAQ)*
- Treatment of renal abscess — *Winter 2017 Paper I Q3b (SAQ)*

**Type:** SAQ (10 marks)

### Definition and types

- **Renal (cortical/corticomedullary) abscess:** pus confined to the renal parenchyma.
- **Cortical (renal carbuncle):** haematogenous *Staphylococcus aureus* from a skin source; formerly about 80% of cases.
- **Corticomedullary (now commonest):** ascending Gram-negative infection (*E. coli*, *Klebsiella*, *Proteus*) in a damaged or obstructed kidney (stones, reflux, diabetes).
- **Perinephric abscess:** pus **outside the renal capsule but within Gerota fascia**. Rupture through Gerota fascia gives a **paranephric** abscess (paranephric beyond it).

### Aetiology (perinephric abscess)

- Rupture of a renal cortical abscess or pyonephrosis (commonest); extravasation of infected urine from obstruction (stone).
- **Haematogenous** spread in about a third (skin source, *S. aureus*).
- Infected perirenal haematoma or urinoma (trauma, surgery).
- Spread from bowel (Crohn disease, perforation), pancreas, spine.
- Risk factors: **diabetes (about a third)**, stones, obstruction, polycystic kidneys on dialysis.
- Organisms: *E. coli*, *Proteus*, *S. aureus*; anaerobes with bowel sources.

### Clinical features

- Fever, flank pain, leucocytosis.
- Perinephric abscess is **insidious:** symptoms >5 days in most; more than a third afebrile.
- Flank pain, tenderness, flank mass (about half); psoas sign (flexed, externally rotated hip).
- **Suspect it when "pyelonephritis" persists:** pyelonephritis defervesces within 4–5 days; perinephric abscess does not.

| | Pyelonephritis | Perinephric abscess |
|---|---|---|
| Symptoms before admission | ≤5 days | >5 days |
| Fever on antibiotics | Never >4 days | ≥5 days (median 7) |

### Investigations / diagnosis (needed to plan treatment)

- Leucocytosis, raised creatinine, pyuria; urine culture identifies the organism in only about a third; urine may be sterile (haematogenous or non-communicating abscess). Blood cultures positive in a quarter to a third.

- **CT (computed tomography): investigation of choice.** Renal abscess: round low-attenuation lesion with enhancing "ring sign". It shows the collection, its spread (psoas, flank), the cause (stone, obstruction) and plans drainage.
- Ultrasound: quick; guides drainage. Plain X-ray: loss of psoas shadow, scoliosis.
- AFBN (acute focal bacterial nephritis) is wedge-shaped without a wall and is treated medically.

## Treatment

### 1. Antibiotics immediately (after blood and urine cultures)

| Presumed source | Empirical therapy |
|---|---|
| Haematogenous (Gram-positive) | Penicillinase-resistant penicillin; vancomycin if allergic or MRSA (methicillin-resistant *S. aureus*) |
| Ascending (Gram-negative) | IV (intravenous) third-generation cephalosporin, antipseudomonal penicillin or aminoglycoside |
| Perinephric abscess | Aminoglycoside plus antistaphylococcal penicillin; cephalosporin or vancomycin if allergic. Adjust to culture |
| - Switch to oral once afebrile; continue until clinical and imaging resolution. | |

### 2. Size-based drainage (renal abscess)

| Size | Management |
|---|---|
| **≤3 cm** | Antibiotics alone (stable patient) |
| **3–5 cm** | Antibiotics alone or drainage; no consistent pathway |
| **>5 cm** | **Percutaneous drainage** (CT/ultrasound-guided) first line; may need several drains |
| Any size: immunocompromised or not responding | Percutaneous drainage |

- **Perinephric abscess (if present) — lower threshold:** small (<3 cm) in an immunocompetent patient may resolve with antibiotics; drain if >3 cm. Larger or non-responding: **CT- or ultrasound-guided percutaneous drainage** (first line).
- Aspiration also distinguishes abscess from a cystic tumour; culture the pus.
- Serial ultrasound or CT until resolution. Worsening suggests perinephric spread, wrong diagnosis or resistant organism.

### 3. Treat the cause

- Relieve obstruction (stent or nephrostomy); treat stones once infection settles; control diabetes.

### 4. Surgery

- Open drainage for loculated or failed (percutaneous drainage) cases; **nephrectomy** for a non-functioning, destroyed or pyonephrotic kidney, at the same time or staged.

## Outcome

- Perinephric abscess: mortality fell from 40–50% to about 12% with CT; delay in diagnosis causes most deaths.

**Diagram to draw:** coronal kidney showing renal, perinephric and paranephric spaces / abscess sites.

> **Key points to remember:**
> - Now mostly ascending Gram-negative abscess in a damaged kidney.
> - Perinephric: within Gerota fascia; diabetes in a third.
> - Fever beyond 4–5 days of antibiotics: get a CT.
> - CT is the investigation of choice; follow until resolution.
> - ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
> - Percutaneous drainage first line; nephrectomy if non-functioning.

## SAQ 5 Ciprofloxacin

P1

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 1, Q3d

### Class and mechanism

- Second-generation **fluoroquinolone**, derived from nalidixic acid.
- Inhibits **DNA gyrase** (topoisomerase II, main target in Gram-negatives) and **topoisomerase IV** (Gram-positives). This blocks DNA supercoiling and replication: **bactericidal**.

### Spectrum

- Enterobacteriaceae, ***Pseudomonas aeruginosa***, *Haemophilus*, *S. saprophyticus*, *S. aureus* (methicillin-sensitive); *Legionella*, some mycobacteria.
- Weak against streptococci and enterococci; **no anaerobic cover**, so bowel and vaginal flora are partly spared.

### Pharmacokinetics

- Oral bioavailability about 70%; IV (intravenous) and oral forms; half-life about 4 h.
- **Concentration-dependent killing:** AUC/MIC (area under curve to minimum inhibitory concentration) predicts effect; post-antibiotic effect.
- High urine and tissue levels; **good prostatic penetration**.
- Renal excretion: reduce dose if creatinine clearance <30 mL/min.

### Doses

- Cystitis 250 mg twice daily × 3 days; pyelonephritis 500 mg twice daily × 7 days (or 1 g extended-release daily); IV 400 mg 12-hourly.
- Chronic bacterial prostatitis: 4–6 weeks.

### Indications and restrictions

- Pyelonephritis and complicated UTI (urinary tract infection) **only where local resistance ≤10%**; prostatitis; prophylaxis for transrectal biopsy (falling, because of resistance).
- **Not first line for uncomplicated cystitis** (FDA (US Food and Drug Administration) 2016; European restrictions 2019): use only when no alternative.

### Resistance

- *gyrA*/*parC* mutations, efflux pumps, reduced porins, plasmid *qnr* genes. Linked to ESBL (extended-spectrum β-lactamase) *E. coli* (ST131). Main cause of post-biopsy sepsis.

### Adverse effects

- GI (gastrointestinal) upset (commonest), photosensitivity, *C. difficile*.
- **FDA boxed warnings:** tendinopathy and **Achilles rupture** (2008; risk higher with age, steroids, renal failure); peripheral neuropathy (2013); disabling effects (2016); psychiatric effects and dysglycaemia (2018); aortic aneurysm and dissection (2018).
- CNS (central nervous system): seizures, confusion; QT prolongation.

### Interactions

- Antacids, iron, zinc, calcium block absorption (separate by ≥2 h). Raises theophylline and tizanidine levels; potentiates warfarin.

### Pregnancy and children

- Avoid: **cartilage damage** in immature animals. Also avoid in breastfeeding.

> **Key points to remember:**
> - DNA gyrase and topoisomerase IV; concentration-dependent.
> - Use for pyelonephritis only if local resistance ≤10%.
> - Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

## SAQ 6 Catheter Associated Urinary Tract Infections (CAUTI)

P1

**Type:** SAQ (10 marks)
**Source:** 2017-Winter Paper 1, Q3c

### Definition

- **CAUTI (catheter-associated urinary tract infection):** symptomatic UTI (urinary tract infection) with a catheter in place >2 days (or removed within 48 hours). Distinguish from **asymptomatic bacteriuria**, which needs no treatment.

### Epidemiology and risk factors

- Commonest hospital-acquired infection (about 75–80% of hospital UTIs); commonest cause of secondary bacteraemia.
- Bacteriuria develops at about **5–10% per catheter-day**; 100% with long-term catheters. Intermittent catheterisation: 1–3% per catheterisation.
- **Risk factors:** duration (most important), female sex, older age, diabetes, breaks in the closed system, no systemic antibiotics.

### Pathogenesis

- Entry at insertion, then **extraluminal** (along the outer surface; commoner in women) or **intraluminal** (from the drainage bag).
- **Biofilm:** bacteria in a matrix on the catheter surface, formed within minutes; it shields them from antibiotics. Urease-producers (*Proteus*) cause encrustation.
- Organisms: *E. coli*, *Pseudomonas*, *Proteus*, *Enterococcus*; long-term catheters polymicrobial.

### Diagnosis

- Fever, rigors, flank or suprapubic pain, or delirium with no other source. Pyuria does not distinguish infection.
- Culture from a **freshly replaced catheter**, not the bag. **IDSA (Infectious Diseases Society of America): ≥$10^3$ CFU/mL** (colony-forming units) with symptoms; surveillance (CDC (Centers for Disease Control and Prevention)) definition uses ≥$10^5$.

### Prevention (bundle)

- **Avoid unnecessary catheters;** review daily, remove early. Aseptic insertion.
- **Closed, dependent drainage**; bag below bladder; no routine irrigation.
- Meatal cleaning with water; no prophylactic antibiotics (delay bacteriuria only 3–4 days, select resistance).
- Alternatives: **clean intermittent catheterisation**, condom or suprapubic catheter.
- Silver-alloy catheters reduce bacteriuria in short-term use; no proven fall in symptomatic CAUTI.
- Do not treat asymptomatic bacteriuria (except pregnancy, before urological surgery).

### Treatment

- **Replace or remove the catheter** before taking the culture and starting antibiotics, if in place >2 weeks (biofilm source).
- Empirical, then culture-directed antibiotics: **7 days** if prompt response; **10–14 days** if delayed.

**Diagram to draw:** catheter in bladder showing extraluminal and intraluminal routes and biofilm.

> **Key points to remember:**
> - Bacteriuria 5–10% per catheter-day; duration is the key risk.
> - Diagnosis: symptoms + ≥$10^3$ CFU/mL from a fresh catheter.
> - Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

## SAQ 7 Complicated urinary tract infections.

P1

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 1, Q3f

### Definition

- **Complicated UTI (urinary tract infection):** infection in a **structurally or functionally abnormal tract** or a **compromised host**, or with a resistant organism; more likely to fail or recur.
- **FDA (US Food and Drug Administration) 2018:** pyuria + pathogen in urine or blood + fever, chills, flank pain, with an abnormality or catheter.
- **Uncomplicated:** healthy non-pregnant woman, normal tract.

### Complicating factors

| Category | Examples |
|---|---|
| Obstruction | BOO (bladder outlet obstruction), PUJ (pelviureteric junction) obstruction, stricture, stone, tumour |
| Voiding dysfunction | Neurogenic bladder, high PVR (post-void residual), VUR (vesicoureteral reflux) |
| Foreign body / iatrogenic | Catheter, stent, nephrostomy, recent instrumentation, hospital-acquired |
| Anatomical | Diverticulum, fistula, cystocele |
| Host | **Male sex**, pregnancy, diabetes, immunosuppression, renal failure, transplant, elderly |
| Other | Symptoms >7 days; recent antibiotics |

- Each causes stasis, biofilm or weak defences.

### Organisms

- ***E. coli*** under half; more *Klebsiella*, *Proteus*, *Pseudomonas*, *Enterococcus*, *Candida*. Often **MDR (multidrug-resistant)** or ESBL (extended-spectrum β-lactamase) producers.

### Evaluation

- **Urine culture mandatory before antibiotics**; blood cultures if febrile.
- Renal function, glucose, lactate.
- **Imaging:** ultrasound first; **CT (computed tomography)** if no response in 72 hours, stones, abscess or emphysematous infection suspected.

### Treatment

- **Empirical IV (intravenous) therapy** by local resistance (EAU (European Association of Urology)): amoxicillin + aminoglycoside, 2nd-generation cephalosporin + aminoglycoside, or IV 3rd-generation cephalosporin. Avoid fluoroquinolones if used in the last 6 months or resistance >10%. Carbapenem or newer agents for ESBL (ceftolozane–tazobactam, ceftazidime–avibactam).
- **Source control is essential:** drain obstruction (**stent or nephrostomy, equally effective**), change catheter, drain abscess, treat stones later.
- Switch to oral when afebrile and stable; de-escalate by culture.
- **Duration:** usually 7–14 days (14 days in men if prostatitis not excluded); 7 days may suffice once afebrile ≥48 hours.
- **Sepsis:** antibiotics within 1 hour, fluids, vasopressors, ICU (intensive care unit).
- Correct the underlying cause, or infection recurs.

### Special forms

- **Also know:** emphysematous pyelonephritis and cystitis (diabetics); pyonephrosis; renal and perinephric abscess; XGP (xanthogranulomatous pyelonephritis); CAUTI (catheter-associated UTI); funguria; unresolved bacteriuria (resistance, staghorn "critical mass").

**Diagram to draw:** Flowchart: suspect cUTI → culture → imaging → empirical IV → drain obstruction → oral step-down.

> **Key points to remember:**
> - Complicated = abnormal tract or compromised host; male sex alone qualifies.
> - Culture mandatory; often MDR.
> - Drain obstruction: stent = nephrostomy.
> - 7–14 days; oral step-down when afebrile.

## SAQ 8 ESBL producing enterobacterial UTI

P1

**Asked as:**

- ESBL producing enterobacterial UTI — *Summer 2019 Paper I Q3f (SAQ)*
- ESBL producing enterobacterial UTI. — *Practice Set 5 Paper I Q3c (SAQ)* (also answered in 2019-Summer Paper 1, Q3f)

**Type:** SAQ (10 marks)

### Definition

- **ESBLs (extended-spectrum β-lactamases): plasmid-encoded** enzymes that hydrolyse penicillins, **third-generation cephalosporins** and aztreonam. They are **inhibited by clavulanate** and spare **carbapenems** and cephamycins.
- Mainly *E. coli* and *Klebsiella* (Enterobacterales); first reported 1983.

### Mechanism and spread

- Older types **TEM** and **SHV**; now **CTX-M** dominates worldwide. Spread through the community is linked to *E. coli* **ST131**, whose hallmark is fluoroquinolone resistance (fluoroquinolone resistance is its hallmark).
- Plasmids carry co-resistance genes: **fluoroquinolones, aminoglycosides, TMP-SMX (trimethoprim–sulfamethoxazole)**. So oral options are few.
- India has one of the highest prevalence rates.

### Risk factors

- Prior antibiotics (especially cephalosporins, quinolones); recent hospital stay; **indwelling catheter**, urological instrumentation; recurrent UTI (urinary tract infection); diabetes; nursing-home residence; travel to high-prevalence areas; transrectal prostate biopsy.

### Detection

- Screening: reduced susceptibility to ceftriaxone, cefotaxime or ceftazidime.
- **Phenotypic confirmation:** clavulanate synergy (**combination disc** or double-disc synergy test); automated systems.
- Genotypic PCR (polymerase chain reaction) for CTX-M.

### Treatment

| Infection | Options |
|---|---|
| **Cystitis** | **Nitrofurantoin** 5 days, **fosfomycin** 3 g single dose (E. coli), **pivmecillinam**; TMP-SMX or quinolone only if susceptible; single-dose aminoglycoside |
| **Pyelonephritis, complicated UTI** | Quinolone or TMP-SMX if susceptible; else (otherwise) **aminoglycoside** |

| Infection | Options |
|---|---|
| **Urosepsis, bacteraemia** | **Carbapenem** (meropenem, ertapenem), then oral step-down by culture |

- **MERINO trial (2018):** in ceftriaxone-resistant *E. coli*/*Klebsiella* bacteraemia, **piperacillin–tazobactam** had higher 30-day mortality than meropenem (12.3% vs 3.7%). Avoid pip-taz for serious ESBL infection.
- **Ceftolozane–tazobactam, ceftazidime–avibactam:** active but **reserved** for carbapenem-resistant organisms (stewardship).
- **Source control:** remove or change catheters; drain obstruction.
- Treat **asymptomatic bacteriuria** only before urological procedures or in pregnancy.
- **Febrile prostatitis after transrectal biopsy:** carbapenem is favoured.

### Prevention

- **Antibiotic stewardship:** avoid broad-spectrum cephalosporins and quinolones; narrowest spectrum; shortest effective course; culture before treatment.
- Hand hygiene, contact precautions, catheter care.
- Pre-biopsy rectal swab culture for targeted prophylaxis; transperineal biopsy.

**Also know:** AmpC (not clavulanate-inhibited) and carbapenemases **KPC (*Klebsiella pneumoniae* carbapenemase)**, NDM (New Delhi metallo-β-lactamase) are wider resistance.

**Diagram to draw:** treatment flowchart from culture result to oral or IV (intravenous) choice.

> **Key points to remember:**
> - Plasmid CTX-M; resists 3rd-gen (third-generation) cephalosporins; inhibited by clavulanate (clavulanate inhibits).
> - Oral: nitrofurantoin, fosfomycin, pivmecillinam.
> - Serious infection: carbapenem; MERINO: pip-taz (piperacillin–tazobactam) inferior.
> - Stewardship prevents spread.

## SAQ 9 Lichen sclerosus (BXO, balanitis xerotica obliterans)

P1

**Asked as:**

- Lichen sclerosis. — *Summer 2020 Paper I Q3c (SAQ)*
- BXO — *Practice Set 10 Paper I Q3b (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2020-Summer Paper 1, Q3c)

*Taken as lichen sclerosus (LS), formerly balanitis xerotica obliterans (BXO). BXO (balanitis xerotica obliterans) is taken as genital lichen sclerosus (LS) in the male.*

### Definition

- Chronic, progressive, **scarring inflammatory disease of genital skin**; possibly premalignant. BXO is its late stage in men.

### Epidemiology

- 6–10 times commoner in females (prepubertal and menopausal).

- Men: peak 30–50 years in one chapter, older men in another; also children (boys with phimosis). Commonly found at circumcision for phimosis after infancy. 15–20% have extragenital disease.

### Aetiology

- Cause unknown. Autoimmune links (thyroid disease, vitiligo, alopecia); familial clustering.
- **Chronic occlusion by urine** under the foreskin (dribbling, obesity) definitely increases risk.
- Infection and trauma are proposed but unproven.

### Clinical features

- White, atrophic, sclerotic plaques with loss of architecture; purpura, fissures; itch, pain, painful erection, poor stream, spraying.
- Sequence: prepuce → **phimosis** → glans → **meatal stenosis** (sentinel site) → fossa navicularis and penile urethral stricture.
- **Spares the proximal (bulbar and posterior) urethra.**
- Buried penis from loss of shaft skin.
- **Mechanism of stricture:** voiding against meatal stenosis forces urine into the glands of Littre → inflammation → spongiofibrosis.

### Evaluation

- Clinical diagnosis. **Biopsy** if doubtful or suspicious: hyperkeratosis, epidermal atrophy, basal vacuolation, dermal oedema with homogenised collagen, band-like lymphocytic infiltrate (lymphocytes).
- Uroflowmetry; **RUG (retrograde urethrogram) before meatal surgery**, because a proximal stricture often coexists (may coexist).
- **Differential:** lichen planus, Zoon balanitis, penile intraepithelial neoplasia.

### Management

| Situation | Treatment |
|---|---|
| Skin disease | **Clobetasol propionate 0.05%** daily for 1–3 months (another source: for about 3 months), then taper to twice-weekly maintenance; calcineurin inhibitors second line; retinoids for refractory disease |
| Phimosis | **Circumcision**, usually curative |
| Meatal stenosis | Dilatation with clobetasol; extended meatotomy; BMG (buccal mucosal graft) meatoplasty |
| Anterior stricture | **Oral mucosal graft, never genital skin** (LS recurs in skin); staged or single stage |
| Panurethral or unfit | **Perineal urethrostomy** (easy because the proximal urethra is spared); self-dilatation |

- Recurrence 40–50% after medical treatment (recurrence is common); restricture higher than in other aetiologies (restricture rates exceed other aetiologies); treatment follows the patient's goals.

### Malignancy

- Penile SCC (squamous cell carcinoma) in **2.3–9%** on long follow-up; HPV (human papillomavirus)-independent type.
- **Lifelong follow-up**; biopsy any non-healing ulcer or nodule.

**Diagram to draw:** penis with LS at (showing) prepuce, glans, meatus and fossa involved, proximal urethra spared.

> **Key points to remember:**
> - Genital skin disease; meatus is the sentinel site.
> - Spares proximal urethra; perineal urethrostomy easy.
> - Clobetasol first; oral mucosa, never skin, for repair.
> - SCC in 2.3–9%: lifelong follow-up.

## SAQ 10 Drug Fosfomycin

P1

**Type:** SAQ (10 marks)
**Source:** 2022-Summer Paper 1, Q3c

### Introduction

- Bactericidal **phosphonic acid derivative** antibiotic, chemically unrelated to other classes. Oral form: **fosfomycin trometamol**; intravenous form: fosfomycin disodium.

### Mechanism of action

- Enters bacteria through glycerol-3-phosphate and glucose-6-phosphate transporters.
- Irreversibly inhibits **MurA** (enolpyruvyl transferase), blocking the **first step of peptidoglycan (cell wall) synthesis**. β-lactams and vancomycin act at later steps, so **cross-resistance is limited**.

### Pharmacokinetics

- Oral bioavailability is modest, but the drug is excreted **unchanged in urine**. One 3 g dose keeps high urinary concentrations for about 1–2 days, which permits single-dose therapy. Low tissue levels.
- Concentrations fall in renal failure; avoid at very low creatinine clearance.

### Spectrum

- **Escherichia coli**, including **ESBL (extended-spectrum β-lactamase)** producers; most Enterobacteriaceae (Klebsiella variable); *Enterococcus* including **VRE (vancomycin-resistant enterococcus)**; staphylococci.
- **Not reliable:** *Pseudomonas aeruginosa*, *Acinetobacter*.

### Uses in urology

| Indication | Dose |
|---|---|
| Uncomplicated cystitis in women (first-line, EAU (European Association of Urology) and IDSA (Infectious Diseases Society of America)) | 3 g oral sachet, single dose |

| Indication | Dose |
|---|---|
| Cystitis in pregnancy, asymptomatic bacteriuria (safe, category B) | 3 g single dose |
| MDR (multidrug-resistant) lower UTI (urinary tract infection) | Oral option avoiding IV (intravenous) therapy; repeated doses used off-label |
| Prophylaxis for transrectal prostate biopsy | As effective as fluoroquinolone |
| Recurrent UTI prophylaxis | 3 g every 10 days (EAU) |

- **Not for pyelonephritis** or urosepsis orally (poor tissue levels). Emerging data suggest IV fosfomycin may help in bacteraemic MDR E. coli UTI.
- Not needed as prophylaxis for flexible cystoscopy.

### Adverse effects

- Diarrhoea, GI (gastrointestinal) upset, headache, vaginitis; rare hypersensitivity. IV form: **sodium load**, hypokalaemia.

### Resistance

- Transporter mutations, MurA changes, plasmid **fosA** enzymes. Resistance develops quickly in vitro, so systemic infections use combinations.

### Evidence

- Huttner (2018) RCT (randomised controlled trial): nitrofurantoin 5 days gave higher clinical resolution than single-dose fosfomycin (70% vs 58%).

**Diagram to draw:** peptidoglycan synthesis showing fosfomycin (MurA), β-lactams and vancomycin sites.

> **Key points to remember:**
> - Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
> - 3 g single dose for cystitis; active against ESBL E. coli and VRE.
> - Not for pyelonephritis; poor Pseudomonas cover.

## SAQ 11 Lymphogranuloma venereum

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 1, Q3c

### Definition

- **LGV (lymphogranuloma venereum)** is an invasive, lymphotropic STD (sexually transmitted disease) caused by ***Chlamydia trachomatis*** **serovars L1, L2 and L3**.
- Of 15 serovars, A–C cause trachoma and D–K anogenital infection. The L serovars invade lymphatics, causing lymphadenitis and scarring.

### Epidemiology

- Endemic in the tropics, including India; increasing as **proctitis in MSM (men who have sex with men)**. Not more frequent or virulent in HIV (human immunodeficiency virus).

### Clinical stages

| Stage | Timing | Features |
|---|---|---|
| **Primary** | 3–30 days | Small painless papule or ulcer, often unnoticed. Rectal inoculation → **proctitis** (bleeding, pain, tenesmus, discharge), now the commonest presentation |
| **Secondary (inguinal syndrome)** | 2–6 weeks later | Fever, malaise; **painful, matted, unilateral inguinal or femoral buboes** that may form fistulae. **Groove sign**: nodes above and below the inguinal ligament |
| **Tertiary** | Untreated | Fibrosis and lymphatic block: **genital elephantiasis** (esthiomene in women), perirectal abscess, fistulae, **rectal stricture** |

- Inguinal disease is commoner in men: the cervix and vagina drain to pelvic nodes.

### Differential diagnosis of genital ulcer

- Syphilis (painless ulcer and nodes); herpes (painful vesicles); chancroid (painful ulcer, suppurative nodes); donovanosis (beefy ulcer, no nodes); **LGV: transient painless lesion, then painful matted nodes**.

### Investigations

- **NAAT (nucleic acid amplification test)** of ulcer, rectal or node aspirate; LGV-specific genotyping confirms where available. Rectal chlamydia NAAT in all proctocolitis.
- Serology not routine. Screen for HIV, syphilis, gonorrhoea; offer PrEP (pre-exposure prophylaxis).

### Treatment

- **Doxycycline 100 mg twice daily for 21 days** (first line).
- Alternatives: erythromycin 500 mg four times daily for 21 days, or azithromycin 1 g weekly for 3 weeks (pregnancy; consider test of cure by NAAT).
- Fluctuant buboes: aspirate through healthy skin to prevent sinus formation.
- **Late disease:** stricture dilatation or resection, fistula repair, reduction of elephantiasis.

**Diagram to draw:** groin with buboes above and below the inguinal ligament (groove sign).

> **Key points to remember:**
> - Serovars L1–L3; invasive and lymphotropic.
> - Proctitis is now the commonest presentation.
> - Groove sign; esthiomene; rectal stricture.
> - Doxycycline 100 mg twice daily for 21 days.

## SAQ 12 Xanthogranulomatous pyelonephritis.

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 1, Q3a

### Definition

- XGP (xanthogranulomatous pyelonephritis) is a rare, severe, **chronic** infection causing diffuse destruction of an enlarged, non-functioning kidney, with **lipid-laden foamy**

**macrophages** (xanthoma cells).

- Up to a fifth of nephrectomies for pyelonephritis. Peak age 50–70 years; women; diabetes is a risk factor.

### Pathogenesis

- **Triad: stone, obstruction, infection.** Stones in up to 80% (half staghorn).
- Low-virulence organisms (**Proteus** commonest, *E. coli*) in an obstructed, ischaemic kidney with an inadequate acute inflammatory response.

### Pathology

- Diffuse (>80%) or segmental (focal).
- Gross: yellow nodules, pus-filled dilated calyces, a **contracted pelvis** encased by fibrosis, frequent **perinephric extension**.
- Microscopy: sheets of foamy histiocytes with lymphocytes, plasma cells and giant cells.
- **Malek staging:** I confined to kidney; II into perinephric fat; III into paranephric tissue or retroperitoneum.

### Clinical features

- Flank pain, fever (about two-thirds), flank mass (about 60%), weight loss; persistent bacteriuria.
- Labs: anaemia, raised inflammatory markers, pyuria; liver dysfunction in up to half. One-third have **sterile urine** (prior antibiotics).

### Imaging

- **CT (computed tomography), most useful:** enlarged kidney, little or no function, central calculus in a contracted pelvis; low-density cavities with rim enhancement (**"bear paw" sign**); extrarenal spread.
- Ultrasound: enlarged kidney with hypoechoic areas. DMSA (dimercaptosuccinic acid) scan quantifies lost function.
- **Differentials:** RCC (renal cell carcinoma), pyonephrosis, malakoplakia, lymphoma, tuberculosis.

### Management

- Antibiotics to stabilise; drain if septic.
- **Nephrectomy** (with perinephric fat) for diffuse disease; partial nephrectomy for focal disease.
- Surgery is difficult: dense adhesions to psoas, great vessels, bowel and diaphragm. Remove the whole inflammatory mass, because simple drainage leads to **nephrocutaneous fistula**.
- Laparoscopic nephrectomy is feasible; conversion rates are high.
- Frozen section can mimic clear cell carcinoma, and XGP can **coexist with RCC** or urothelial carcinoma. If cancer is not excluded, do nephrectomy.

**Diagram to draw:** CT sketch of an enlarged kidney with central staghorn and "bear paw" cavities.

> **Key points to remember:**
> - Stone, obstruction, Proteus infection → foamy macrophages.
> - CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
> - Mimics RCC; treat by complete nephrectomy, not drainage.

## SAQ 13 Papillary necrosis.

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 7 Paper 1, Q3a

### Definition

- **RPN (renal papillary necrosis):** ischaemic necrosis of the renal papillae (inner medulla), which may slough into the collecting system.

### Pathophysiology

- The papilla is vulnerable: it depends on the **vasa recta**, has **low oxygen tension** and a **hypertonic** environment. Almost every cause reduces medullary blood flow or oxygenation.
- **Sickle cell:** hypoxia and hypertonicity make cells sickle and block the vasa recta, causing microinfarcts.
- **Analgesics:** NSAIDs (non-steroidal anti-inflammatory drugs) block vasodilator prostaglandins; metabolites concentrate in the medulla.
- **Diabetes:** microangiopathy plus infection.

### Causes ("POSTCARDS")

| Letter | Cause |
|---|---|
| P | Pyelonephritis |
| O | Obstruction |
| S | Sickle cell disease or trait |
| T | Tuberculosis |
| C | Cirrhosis |
| A | Analgesic abuse |
| R | Renal transplant rejection; renal vein thrombosis |
| D | **Diabetes mellitus** |
| S | Systemic vasculitis |
| - **Also know:** infant dehydration and hypoxia; shock; renal candidiasis. | |

### Clinical features

- **Acute:** fever, flank pain, haematuria, **ureteric colic from a sloughed papilla**, sepsis; bilateral cases may cause AKI (acute kidney injury).
- **Chronic or incidental:** polyuria (poor concentration), sterile pyuria, recurrent UTI (urinary tract infection), CKD (chronic kidney disease). Retained papillae may **calcify** and harbour infection, including fungi.

### Investigations

- Urine microscopy and culture; **strain urine** and send tissue for histology; creatinine, glucose, haemoglobin electrophoresis.
- **CT urography or IVU (intravenous urography):** early contrast tracking into the papilla ("**ball-on-tee**", "**lobster claw**"); "**ring sign**" (contrast around a detached papilla); clubbed calyces; filling defects; triangular ring-shaped calcification.
- Retrograde pyelography when renal function is poor.
- **Differential diagnosis:** tuberculosis, medullary sponge kidney, calyceal diverticulum, upper tract urothelial carcinoma.

### Management

- **Treat the cause:** stop analgesics, control diabetes, hydrate, manage sickle crisis.
- **Culture-guided antibiotics.**
- **Obstruction by a sloughed papilla:** urgent drainage (stent or nephrostomy), then ureteroscopic extraction.
- Remove calcified retained papillae that cause persistent infection.
- **Follow-up:** renal function; analgesic abusers risk urothelial cancer.

**Diagram to draw:** calyx showing ring sign and lobster-claw appearance.

> **Key points to remember:**
> - Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
> - POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
> - Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

## SAQ 14 Cephalsporins

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 1, Q3c

*Assumed meaning: cephalosporins (generations, spectrum, urological use).*

### Introduction and mechanism

- **β-lactam** antibiotics. They bind **PBPs (penicillin-binding proteins)** and block peptidoglycan cross-linking, so the cell wall fails: **bactericidal**, time-dependent killing.
- Mostly renal excretion (adjust dose in renal failure); **ceftriaxone** also has biliary excretion (no adjustment).
- Resistance: β-lactamases (**ESBL [extended-spectrum β-lactamase]**, AmpC), altered PBPs, reduced porin entry.

### Generations

| Generation | Examples | Spectrum | Urological use |
|---|---|---|---|
| **First** | Cefazolin (intravenous), cephalexin, cefadroxil (oral) | Gram-positive (not MRSA [methicillin-resistant *Staphylococcus aureus*]); *E. coli*, *Klebsiella*, *P. mirabilis* | **Surgical prophylaxis (cefazolin)**; cystitis in pregnancy; paediatric prophylaxis |

| Generation | Examples | Spectrum | Urological use |
|---|---|---|---|
| **Second** | Cefuroxime, cefaclor; cefoxitin, cefotetan | Wider Gram-negative; cefoxitin and cefotetan add **anaerobes** | Prophylaxis for vaginal and bowel surgery; SWL (shock-wave lithotripsy) |
| **Third** | Ceftriaxone, cefotaxime, cefixime, cefpodoxime; **ceftazidime** | Most Enterobacterales; **only ceftazidime covers *Pseudomonas*** | Pyelonephritis, urosepsis; **gonorrhoea (ceftriaxone 500 mg IM [intramuscular])**; prostate biopsy prophylaxis |
| **Fourth** | Cefepime | Gram-negatives incl. *Pseudomonas*, stable to AmpC; Gram-positive | Complicated, nosocomial UTI (urinary tract infection) |
| **Fifth** | Ceftaroline | MRSA + Gram-negatives | Limited urological role |

- **Newer combinations** for complicated UTI with multidrug resistance: ceftazidime–avibactam, ceftolozane–tazobactam, cefiderocol.

### Gaps in spectrum

- **Enterococci** (intrinsically resistant; never treat with cephalosporins), *Listeria*, MRSA (except ceftaroline), atypical organisms.
- ESBL and AmpC producers hydrolyse third-generation agents.

### Adverse effects

- Hypersensitivity, less than penicillins. Cross-reactivity with third-generation agents is <1%; avoid after immediate (IgE [immunoglobulin E]-mediated) penicillin reaction.
- ***C. difficile* colitis**; "collateral damage" (selection of ESBL organisms).
- Positive Coombs test; bleeding with cefotetan and cefamandole.
- Ceftriaxone: biliary sludge; avoid in jaundiced neonates and with intravenous calcium.

### Paediatric use

- Febrile UTI in neonates: **ampicillin + third-generation cephalosporin** (covers enterococcus). Oral cefixime 8 mg/kg/day. A first-generation agent is a good narrow-spectrum choice for many outpatient UTIs.

**Diagram to draw:** β-lactam ring binding a PBP and blocking peptidoglycan cross-links.

> **Key points to remember:**
> - Cefazolin = standard surgical prophylaxis.
> - Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
> - No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

## SAQ 15 Fungal UTI

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 18 SAQ Sheet, Q3a

### Introduction and organisms

- Fungal UTI (urinary tract infection) is rising with antibiotics, catheters and immunosuppression. **Candida** is commonest: *C. albicans*, then *C. glabrata* (often

fluconazole-resistant); *C. krusei* is intrinsically resistant.
- Others in AIDS (acquired immunodeficiency syndrome) or transplant patients: Aspergillus, Cryptococcus, Mucorales, Histoplasma.

### Risk factors and routes

- Antibiotics, indwelling catheters and stents, diabetes, steroids, neutropenia, prematurity, parenteral nutrition, obstruction.
- **Ascending** (catheter) causes most candiduria. **Haematogenous** spread from candidaemia: in disseminated candidiasis the kidney is the organ most often involved.

### Clinical spectrum

- Asymptomatic candiduria (most) → cystitis → pyelonephritis → **fungus ball (bezoar)** causing obstruction → renal abscess or candidaemia.

### Diagnosis

- Microscopy: budding yeasts and pseudohyphae (may mimic red cells). Repeat culture: contamination is common.
- Children: treat when repeated cultures exceed **10,000–15,000 CFU (colony-forming units)/mL**.
- Persistent funguria: ultrasound or CT (computed tomography) for bezoar; blood cultures if high risk.

### Management

| Situation | Treatment |
|---|---|
| Asymptomatic | Remove or change catheter, stop antibiotics; no antifungal |
| Treat asymptomatic only if | Neutropenia, very-low-birth-weight infant, before a urological procedure |
| Cystitis | Fluconazole 200 mg (3 mg/kg) daily for 2 weeks |
| Pyelonephritis | Fluconazole 200–400 mg (3–6 mg/kg) daily for 2 weeks |
| Fluconazole-resistant (*C. glabrata*, *C. krusei*) | Amphotericin B deoxycholate 0.3–0.6 mg/kg/day for 1–7 days ± flucytosine |
| Resistant cystitis | Amphotericin B bladder irrigation 50 mg/L |
| Fungus ball | PCN (percutaneous nephrostomy) drainage + systemic antifungal + local amphotericin irrigation; endoscopic or open removal if persistent |

- Fluconazole concentrates in urine; **echinocandins and lipid amphotericin do not.**
- **Before procedures:** IDSA (Infectious Diseases Society of America) 2016 treats all candiduria (fluconazole 400 mg daily for several days before and after). AUA (American Urological Association) Best Practice Statement 2019 gives single-dose antifungal prophylaxis only for urothelium-breaching surgery. The two disagree.
- **Also know:** alkalinisation for bezoar; fluconazole caution under 6 months.

**Diagram to draw:** budding yeast with pseudohyphae; fungus ball in renal pelvis with nephrostomy.

**Key points to remember:**

- Candida commonest; *C. glabrata* often fluconazole-resistant.
- Asymptomatic candiduria: remove catheter, no antifungal (except neutropenia, neonate, procedure).
- Fluconazole 2 weeks; echinocandins fail in urine.
- Fungus ball: drain, antifungal, remove.

## SAQ 16 Antimicrobial resistance

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 18 SAQ Sheet, Q3b (also answered in part in 2015-Summer Paper 1, Q3f)

### Definition

- AMR (antimicrobial resistance): a microbe grows despite drug levels achievable in the patient. Reported as S (sensitive), I (intermediate), R (resistant) with the MIC (minimum inhibitory concentration).
- One of the most serious global health threats.

### Types

| Type | Mechanism | Relevance |
|---|---|---|
| Intrinsic | Species lacks the target | Predictable (enterococci and cephalosporins) |
| Acquired chromosomal | Mutants selected when urine level is close to MIC | Underdosing, poor compliance |
| Plasmid (extrachromosomal) | Horizontal transfer in bowel flora | **Commonest**; multidrug resistance after antibiotics |

### Mechanisms

- **Enzymes:** β-lactamases; aminoglycoside-modifying enzymes.
- **Altered target:** penicillin-binding protein; DNA gyrase mutation (quinolones).
- **Reduced entry** (porin loss) and **efflux pumps**.
- **Bypass:** enterococci scavenge folate (escape TMP-SMX (trimethoprim–sulfamethoxazole)).

### Key resistant organisms

- **ESBL (extended-spectrum β-lactamase)** (plasmid, 1983): resists 3rd-generation cephalosporins; often co-resistant to fluoroquinolones and aminoglycosides. *E. coli* ST131 carries it.
- **AmpC; KPC (*Klebsiella pneumoniae* carbapenemase, 2001); NDM (New Delhi metallo-β-lactamase)**: carbapenem resistance.
- MRSA (methicillin-resistant *S. aureus*), VRE (vancomycin-resistant enterococci).

### Drivers in urology

- Overuse of fluoroquinolones and cephalosporins; treating ASB (asymptomatic bacteriuria); long prophylaxis; catheters, stents, stones (biofilm).
- Transrectal biopsy: fluoroquinolone-resistant rectal *E. coli* causes sepsis.

### Prevention: stewardship

1. Culture before treatment; follow the **local antibiogram**.
2. TMP-SMX only if local resistance ≤20%; oral fluoroquinolone for pyelonephritis only if ≤10%.
3. Prefer narrow, urine-confined drugs for cystitis (nitrofurantoin, fosfomycin, pivmecillinam: low resistance).
4. Shortest course; de-escalate to culture.
5. Do not treat ASB, except in pregnancy and before mucosa-breaching procedures.
6. Single-dose surgical prophylaxis; transperineal biopsy or rectal-swab-targeted prophylaxis.
7. Infection control and hand hygiene. - **Also know:** FDA (Food and Drug Administration) 2016 fluoroquinolone warning; *C. difficile*; ESBL: carbapenem, or fosfomycin/nitrofurantoin for cystitis.

### Recent advances

- New β-lactam/β-lactamase inhibitor combinations (ceftazidime–avibactam, meropenem–vaborbactam); cefiderocol; gepotidacin.

**Diagram to draw:** Gram-negative cell showing β-lactamase, porin loss, efflux pump, altered target.

> **Key points to remember:**
> - Plasmid-mediated resistance is commonest.
> - ESBL: plasmid, co-resistant; KPC/NDM: carbapenem resistance.
> - TMP-SMX if resistance ≤20%; fluoroquinolone for pyelonephritis if ≤10%.
> - Stewardship: culture, narrow, short, no ASB treatment.

## SAQ 17 Biofilm

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 18 SAQ Sheet, Q3c (also answered in part in 2015-Summer Paper 3, Q3d)

### Definition

- A **structured microbial community in a self-made polymeric matrix**, adherent to a living or inert surface. The matrix contains exopolysaccharide, extracellular DNA and proteins.

### Stages of formation

1. **Conditioning film:** host proteins coat the device within minutes.
2. **Attachment:** reversible, then irreversible (pili, adhesins, flagella).
3. **Matrix production and microcolonies.**
4. **Maturation:** 3-D (three-dimensional) structure with water channels; cell-to-cell signalling (quorum sensing).
5. **Dispersion:** free bacteria seed new infection.

### Why antibiotics and immunity fail

- Poor drug penetration through the matrix.
- Slow-growing, dormant cells resist drugs that need active growth.

- Altered genes, envelope and targets.
- Survive **1000–1500 times** the usual killing concentration. Neutrophils are excluded.

### Urological examples

| Site | Point |
|---|---|
| Urethral catheter | Forms within minutes; **the catheter, not the bladder, is infected**. Proteus urease makes crystalline biofilm → encrustation and blockage |
| Ureteric stent, nephrostomy | Stent colonised more often than urine shows; nephrostomy changed every 3 months |
| Penile prosthesis, AUS (artificial urinary sphincter) | Revision of an "uninfected" device: about 10% infection |
| Infection stones | Bacteria inside struvite; antibiotics cannot sterilise |
| Chronic bacterial prostatitis | Biofilm commoner in prostatitis *E. coli*; needs longer, penetrating therapy |
| Recurrent cystitis | **Intracellular bacterial communities**: biofilm-like pods inside umbrella cells |

- Common organisms: Pseudomonas, Proteus, *E. coli*, Enterococcus, coagulase-negative staphylococci, Candida.

### Prevention and management

- Avoid devices; shortest dwell; aseptic insertion; closed dependent drainage.
- **Change the catheter** when treating symptomatic CAUTI (catheter-associated UTI (urinary tract infection)); remove infected stent or implant.
- Silver-alloy catheters reduce bacteriuria, not clinical UTI.
- Implants: antibiotic-coated devices, no-touch technique, **antiseptic washout** at revision (infection about 10% → 3%).
- Stones: complete clearance. Chemodissolution for catheter encrustation. Macrolides penetrate biofilm (prostatitis).
- **Also know (research):** quorum-sensing inhibitors, bacteriophages, new coatings.

**Diagram to draw:** five-stage biofilm cycle on a catheter surface.

> **Key points to remember:**
> - Biofilm: microbes in self-made matrix on a surface.
> - Survives 1000–1500 times normal antibiotic levels.
> - Treat CAUTI by changing the catheter.
> - Revision washout cuts prosthesis infection from about 10% to 3%.

## SAQ 18 Renal mucormycosis

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 19 SAQ Sheet, Q3c

### Introduction

- Mucormycosis: an **angioinvasive** fungal infection by Mucorales (*Rhizopus* commonest; *Mucor*, *Lichtheimia*).

- Kidney involvement occurs in **disseminated** disease, or as **isolated renal mucormycosis**, reported particularly from India, sometimes in apparently immunocompetent young adults. Cases rose during the 2021 COVID-19 (coronavirus disease 2019) wave.

### Risk factors

- Uncontrolled diabetes, especially **ketoacidosis**; corticosteroids; neutropenia and haematological cancer; transplantation; iron overload or deferoxamine; intravenous drug use; AIDS (acquired immunodeficiency syndrome).

### Pathogenesis

- Spores inhaled or ingested → **haematogenous** seeding of the kidney.
- Hyphae invade arteries and veins → **thrombosis → infarction and haemorrhagic necrosis**.
- Acidosis and free iron favour growth (hence the ketoacidosis risk).

### Clinical features

- Fever, flank pain, tender flank mass, haematuria, pyuria with sterile bacterial cultures.
- **Bilateral** disease → AKI (acute kidney injury) or anuria.
- Suspect when "pyelonephritis" fails antibiotics in a high-risk patient.

### Investigations

- **CECT (contrast-enhanced computed tomography):** enlarged kidney(s), wedge-shaped or diffuse **non-enhancing** areas (infarcts), perinephric stranding or collection, renal vessel thrombosis.
- **Tissue diagnosis** (biopsy or nephrectomy specimen): **broad, ribbon-like, aseptate or pauciseptate hyphae branching at right angles**, invading vessels.
- KOH (potassium hydroxide) mount, calcofluor stain, fungal culture (often negative), PCR (polymerase chain reaction).
- Differential: *Aspergillus* has narrow septate hyphae with acute-angle branching.
- Image sinuses, lungs, brain.

### Management (three pillars)

1. **Reverse predisposition:** control glucose, correct ketoacidosis, taper steroids and immunosuppression, stop deferoxamine.
2. **Early surgical debridement:** nephrectomy for a unilateral necrotic kidney; drain collections.
3. **Antifungal: liposomal amphotericin B 5 mg/kg/day** first line (ECMM/MSG ERC (European Confederation of Medical Mycology/Mycoses Study Group Education and Research Consortium) guideline, 2019). Step-down or salvage: **posaconazole or isavuconazole**. - **Fluconazole, voriconazole and echinocandins are inactive.** - Bilateral: antifungal plus dialysis.

### Prognosis

- Mortality is high, especially with bilateral or disseminated disease and delayed diagnosis.

**Diagram to draw:** right-angle branching aseptate hypha invading a renal vessel.

> **Key points to remember:**
> - Angioinvasion → infarction; diabetic ketoacidosis is the key risk.
> - CECT: non-enhancing enlarged kidney; histology: broad aseptate right-angle hyphae.
> - Liposomal amphotericin B 5 mg/kg/day plus nephrectomy and glucose control.
> - Voriconazole and echinocandins do not work.

## SAQ 19 Hydatid disease of kidney

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 19 SAQ Sheet, Q3d

### Introduction

- Cystic echinococcosis (CE) by *Echinococcus granulosus*, a cestode. Dog: definitive host; sheep and humans: intermediate.
- Ingested eggs → liver (commonest), lung, then **kidney (third site, about 2–3%)** via the systemic circulation. Usually single, unilateral, cortical.

### Pathology

- Wall: **pericyst** (host fibrous layer), **laminated membrane** (ectocyst), **germinal layer** (endocyst) producing brood capsules, scolices and daughter cysts ("hydatid sand").
- **Closed** versus **open** cyst (ruptured into the collecting system → **hydatiduria**).

### Clinical features

- Grows 1–2 cm a year; long asymptomatic; flank mass, dull pain.
- **Hydatiduria** ("grape-skin" membranes in urine; pathognomonic) with colic, in a minority.
- Infection; rupture (anaphylaxis). Renal function usually preserved.

### Investigations

- **Serology** (ELISA (enzyme-linked immunosorbent assay)) is adjunctive only (sensitivity 60–90%); eosinophilia; Casoni test obsolete.
- **Ultrasound and CT (computed tomography):** thick-walled cyst, wall calcification, daughter cysts, detached membranes ("water-lily sign"). Stage guides treatment.

| WHO (World Health Organization) stage | Appearance | Activity |
|---|---|---|
| CE1 | Unilocular, double-line wall | Active |
| CE2 | Multivesicular, daughter cysts | Active |
| CE3a / CE3b | Detached membranes / daughter cysts in solid matrix | Transitional |
| CE4 / CE5 | Solid "ball of wool" / calcified wall | Inactive |

- Retrograde pyelogram shows communication; image liver and lungs. **Never aspirate blindly** (anaphylaxis, seeding).

### Management

- **Albendazole 400 mg twice daily**, 1–6 months: alone for small or inoperable cysts; around surgery to reduce seeding (± praziquantel).

- **Surgery is the mainstay** (open, laparoscopic or robotic), nephron-sparing when possible: 1. Pack the field with scolicide-soaked swabs (hypertonic saline, povidone-iodine). 2. Inject scolicide only if **not** communicating (chemical injury to the collecting system). 3. **Pericystectomy, partial nephrectomy or deroofing**; close communications; omentoplasty or drain. 4. **Nephrectomy** for a non-functioning kidney.
- **PAIR (puncture, aspiration, injection, re-aspiration):** chapter says liver only, not kidney; some series report percutaneous renal treatment.
- Calcified inactive cysts (CE4, CE5) may be observed.

**Diagram to draw:** layers of a hydatid cyst with daughter cysts and a communication into a calyx.

> **Key points to remember:**
> - Kidney: third site (about 2–3%).
> - Hydatiduria (grape-skin membranes) is pathognomonic of a communicating cyst.
> - Albendazole 400 mg twice daily around surgery; no scolicide if communicating.

## SAQ 20 Malakoplakia

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 2, Q3e

### Definition

- "Soft plaque": a rare chronic granulomatous inflammation caused by **defective macrophage killing of bacteria**, mostly *E. coli*. Described by Michaelis and Gutmann (1902) and von Hansemann (1903).
- Commonest in the **bladder**; also ureter, kidney, prostate, testis, GI (gastrointestinal) tract, skin, lung.

### Pathogenesis

- Macrophages ingest bacteria but cannot digest them (lysosomal defect, low intracellular cGMP [cyclic guanosine monophosphate]).
- Bacterial remnants become mineralised with calcium phosphate, forming laminated **Michaelis–Gutmann bodies**.
- Hosts are often immunocompromised, diabetic, chronically ill or **renal transplant recipients**.

### Pathology

- Gross: soft yellow-brown plaques or nodules; renal tumour-like masses.
- Micro: sheets of large foamy histiocytes (**von Hansemann cells**) with **Michaelis–Gutmann bodies** (pathognomonic; "owl's eye" targetoid; stain with von Kossa, PAS [periodic acid–Schiff] and Prussian blue).
- MG bodies may be absent early; **α1-antitrypsin** immunostaining helps.

### Clinical features

- Mostly >50 years; urinary tract **female:male 4:1**.

- Bladder: irritative LUTS (lower urinary tract symptoms), haematuria, recurrent *E. coli* UTI (urinary tract infection); plaques or masses mimicking cancer; distal ureteric stricture.
- Kidney: masses, fever, flank pain; renal vein or IVC thrombosis. Testis: epididymo-orchitis. Prostate: mimics carcinoma.

### Investigations

- Urine culture (*E. coli*). Cystoscopy and **biopsy is diagnostic**.
- Imaging: enlarged kidneys with filling defects, **no stones, calcification or hydronephrosis** (unlike XGP).

| | **XGP (xanthogranulomatous pyelonephritis)** | **Malakoplakia** |
|---|---|---|
| Cell | Lipid-laden foamy macrophages | Von Hansemann cells with MG bodies |
| Stone/obstruction | Usually (staghorn) | Absent |
| Laterality | Unilateral | Uni- or multifocal, bilateral |
| Treatment | Nephrectomy | Antibiotics; nephrectomy if unilateral, symptomatic |

### Management

- **Long-term antibiotics active inside macrophages:** fluoroquinolones (concentrate in macrophages), trimethoprim–sulfamethoxazole, rifampicin, doxycycline.
- **Bethanechol + ascorbic acid** raise intracellular cGMP and correct the degradative defect.
- Reduce immunosuppression where possible.
- Surgery if progression despite medical therapy: TUR (transurethral resection) of bladder lesions; **nephrectomy** for symptomatic unilateral renal disease.

### Prognosis

- Bilateral or transplant-kidney disease: death often within 6 months; overall mortality can exceed 50%. Unilateral: long survival after nephrectomy.

**Diagram to draw:** histiocyte with Michaelis–Gutmann bodies.

> **Key points to remember:**
> - Defective macrophage digestion of *E. coli*.
> - Michaelis–Gutmann bodies are pathognomonic.
> - Fluoroquinolones + bethanechol/vitamin C; nephrectomy for unilateral renal disease.

## SAQ 21 Nitrofurantoin.

P4

**Type:** SAQ (10 marks)
**Source:** 2020-Summer Paper 4, Q3e

### Introduction

- A synthetic **nitrofuran**; **first-line for acute uncomplicated cystitis** (EAU (European Association of Urology)) because it concentrates in urine and spares bowel and vaginal flora.

## Mechanism

- Reduced by bacterial flavoproteins (nitroreductases) to **reactive intermediates** that damage ribosomal proteins, DNA and several enzyme systems.
- **Bacteriostatic**, bactericidal at high urinary concentration.
- Multiple targets keep **acquired resistance low**.

## Pharmacokinetics

- Well absorbed (better with food); **rapidly excreted in urine**. Serum and tissue levels are sub-therapeutic.
- Hence **useless for pyelonephritis, prostatitis, febrile or bloodstream infection**.
- Macrocrystals: slower absorption, less nausea.

## Spectrum

| Susceptible | Resistant |
|---|---|
| ***E. coli*** (most), ***Enterococcus faecalis***, *Staphylococcus saprophyticus*, *S. aureus* (not MRSA (methicillin-resistant *S. aureus*)), many *Klebsiella*, *Citrobacter* | ***Proteus*** (intrinsic), *Pseudomonas*, *Serratia*, *Providencia*, *Morganella*, *Acinetobacter* |

- Active against many ESBL (extended-spectrum β-lactamase)-producing *E. coli*.

## Uses and doses

| Indication | Regimen |
|---|---|
| Uncomplicated cystitis (women) | **100 mg twice daily × 5 days** (modified release) |
| Recurrent UTI (urinary tract infection) prophylaxis, adults | 50–100 mg at night, or post-coital |
| Children: treatment / prophylaxis | 3–5 mg/kg/day in 2 doses / 1–2 mg/kg at night (VUR (vesicoureteral reflux), after 3 months of age) |
| Pregnancy | Second and third trimesters; first trimester only if no alternative; **avoid near term** |

## Adverse effects

- **GI (gastrointestinal) upset** (commonest).
- **Pulmonary**: acute hypersensitivity (fever, cough, dyspnoea); **chronic interstitial fibrosis** with long-term use.
- **Peripheral neuropathy** (worse with renal impairment, diabetes, anaemia, B-vitamin deficiency).
- **Haemolysis in G6PD (glucose-6-phosphate dehydrogenase) deficiency**, including the neonate (hence avoid at term and <3 months of age).
- Hepatotoxicity (cholestatic or chronic hepatitis), rare.

## Precautions

- **Renal impairment**: inadequate urine levels and more toxicity. Historically avoided if CrCl (creatinine clearance) <60; Beers Criteria (2015) allow short courses down to **CrCl 30 mL/min**; avoid long-term use in the elderly.
- Interactions: probenecid (blocks excretion), magnesium antacids, quinolones (antagonism).

**Diagram to draw:** spectrum table (susceptible versus resistant).

**Key points to remember:**

- Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
- 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
- Proteus intrinsically resistant; resistance in E. coli low.
- Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015).

# Rapid Revision Sheet · Section 3

### Genitourinary tuberculosis: pathogenesis, diagnosis, management and thimble bladder (LAQ)

- Kidney 80%; periglomerular seeding; bladder disease descends from the kidney; heals by fibrosis, so obstruction worsens on treatment.
- Sterile pyuria; 3–5 early-morning urine AFB cultures; culture is the gold standard and the only test that monitors treatment.
- GeneXpert (Xpert) detects rifampicin resistance in 2 hours; cannot monitor treatment.
- LF-LAM for HIV with CD4 under 200.
- TST and IGRA show infection, not active disease.
- 2HRZE/4HR with pyridoxine; surgery after 4–6 weeks of ATT.
- Thimble bladder <20 mL: augment if capacity <100 mL; substitute if <20 mL; needs creatinine <2 mg/dL and a good urethra.
- Relapse up to 22%; follow up 10 years.

### Interstitial cystitis / bladder pain syndrome (LAQ)

- AUA definition: bladder pain with LUTS >6 weeks, no other cause; ESSIC >6 months.
- AUA 2022: no tiers; phenotype-based; cystoscopy if over 50 or HL suspected; anaesthetic challenge, not potassium test (obsolete).
- HL: fulguration or triamcinolone; ciclosporin if refractory.
- Amitriptyline, PPS, DMSO; PPS causes maculopathy; antibiotics and BCG not offered.

### Genitourinary infections, common organisms, bacteraemia and urosepsis: pathophysiology and management (LAQ)

- *E. coli* ~85% community UTI; P pili cause pyelonephritis; *Proteus* causes struvite; *Enterococcus* resists cephalosporins.
- Sepsis = SOFA rise ≥2; septic shock = vasopressor for MAP ≥65 plus lactate >2 mmol/L.
- Obstruction plus infection is the emergency; bladder urine may be sterile.
- Hour-1: lactate, cultures, antibiotics, 30 mL/kg, noradrenaline (SSC 2021).
- Infected obstructed kidney: drain urgently; no definitive stone surgery in sepsis.
- Stent and nephrostomy equally effective (stent = nephrostomy for drainage); delay definitive stone surgery.

### Describe clinical implication of AIDS on the Genito-Urinary Tract? (LAQ)

- AIDS = CD4 <200; UTI risk rises as CD4 falls.
- HIV-TB: disseminated, less fibrosis, fewer strictures.
- Indinavir stones: radiolucent, may be CT-invisible; stop drug and hydrate.
- PEP within 72 hours for 28 days; circumcision cuts acquisition 50–60%.

### Discuss the Aetiopathogenesis, Clinical feature and management of Chyluria. (LAQ)

- Chyluria = pyelolymphatic fistula, usually from *W. bancrofti* obstruction.
- Ether test and urine triglycerides confirm; cystoscopy after a fatty meal lateralises.
- Diet with MCT and DEC → silver nitrate sclerotherapy → laparoscopic hilar lymphatic disconnection.
- Chyluria precedes elephantiasis and is often intermittent.

**Chronic (non-bacterial) prostatitis: diagnosis, classification and management (SAQ)**

- NIH I–IV; III = pain ≥3 of 6 months, no infection; IIIA inflammatory, IIIB not.
- CBP: same organism, fluoroquinolone 4–6 weeks.
- CP/CPPS: NIH-CPSI to score, UPOINT to phenotype and treat.
- Multimodal therapy; α-blocker plus antibiotic only early.

**Emphysematous pyelonephritis: prognostic factors and conservative management (SAQ)**

- Huang–Tseng: 1 pelvis, 2 parenchyma, 3A perinephric, 3B pararenal, 4 bilateral or solitary.
- Four risk factors: thrombocytopenia, AKI, altered sensorium, shock.
- Wan type I (no fluid) 50–60% mortality; type II <20%.
- Drainage first; nephrectomy for extensive disease with ≥2 risk factors or drainage failure.

**Principles of antimicrobial therapy in UTI (SAQ)**

- Recurrent = ≥2 in 6 months or ≥3 in 12 months.
- Cystitis: nitrofurantoin 5 d, fosfomycin single, TMP-SMX 3 d (resistance ≤20%).
- Pyelonephritis: ciprofloxacin 7 d or levofloxacin 5 d (FQ resistance ≤10%).

**Renal and perinephric abscess (SAQ)**

- Now mostly ascending Gram-negative abscess in a damaged kidney.
- Perinephric: within Gerota fascia; diabetes in a third.
- Fever beyond 4–5 days of antibiotics: get a CT.
- CT is the investigation of choice; follow until resolution.
- ≤3 cm antibiotics; 3–5 cm either; >5 cm percutaneous drainage.
- Percutaneous drainage first line; nephrectomy if non-functioning.

**Ciprofloxacin (SAQ)**

- DNA gyrase and topoisomerase IV; concentration-dependent.
- Use for pyelonephritis only if local resistance ≤10%.
- Tendon rupture, neuropathy, aortic aneurysm; avoid in pregnancy and children.

**Catheter Associated Urinary Tract Infections (CAUTI) (SAQ)**

- Bacteriuria 5–10% per catheter-day; duration is the key risk.
- Diagnosis: symptoms + ≥$10^3$ CFU/mL from a fresh catheter.
- Remove catheter early; closed drainage; treat only symptomatic CAUTI, 7 or 10–14 days.

**Complicated urinary tract infections. (SAQ)**

- Complicated = abnormal tract or compromised host; male sex alone qualifies.
- Culture mandatory; often MDR.
- Drain obstruction: stent = nephrostomy.
- 7–14 days; oral step-down when afebrile.

**ESBL producing enterobacterial UTI (SAQ)**

- Plasmid CTX-M; resists 3rd-gen (third-generation) cephalosporins; inhibited by clavulanate (clavulanate inhibits).
- Oral: nitrofurantoin, fosfomycin, pivmecillinam.
- Serious infection: carbapenem; MERINO: pip-taz (piperacillin–tazobactam) inferior.
- Stewardship prevents spread.

**Lichen sclerosus (BXO, balanitis xerotica obliterans) (SAQ)**

- Genital skin disease; meatus is the sentinel site.
- Spares proximal urethra; perineal urethrostomy easy.
- Clobetasol first; oral mucosa, never skin, for repair.
- SCC in 2.3–9%: lifelong follow-up.

**Drug Fosfomycin (SAQ)**

- Inhibits MurA: first step of cell wall synthesis; no cross-resistance.
- 3 g single dose for cystitis; active against ESBL E. coli and VRE.
- Not for pyelonephritis; poor Pseudomonas cover.

**Lymphogranuloma venereum (SAQ)**

- Serovars L1–L3; invasive and lymphotropic.
- Proctitis is now the commonest presentation.
- Groove sign; esthiomene; rectal stricture.
- Doxycycline 100 mg twice daily for 21 days.

**Xanthogranulomatous pyelonephritis. (SAQ)**

- Stone, obstruction, Proteus infection → foamy macrophages.
- CT: non-functioning enlarged kidney, contracted pelvis, "bear paw" sign.
- Mimics RCC; treat by complete nephrectomy, not drainage.

**Papillary necrosis. (SAQ)**

- Vasa recta, low oxygen and hypertonicity make the papilla vulnerable.
- POSTCARDS; diabetes, analgesics, sickle cell, obstruction with infection.
- Ring sign; sloughed papilla causes colic and obstruction: drain urgently.

**Cephalsporins (SAQ)**

- Cefazolin = standard surgical prophylaxis.
- Ceftriaxone misses *Pseudomonas*; ceftazidime and cefepime cover it.
- No cephalosporin covers enterococci; ESBL hydrolyses third-generation agents.

**Fungal UTI (SAQ)**

- Candida commonest; *C. glabrata* often fluconazole-resistant.
- Asymptomatic candiduria: remove catheter, no antifungal (except neutropenia, neonate, procedure).
- Fluconazole 2 weeks; echinocandins fail in urine.
- Fungus ball: drain, antifungal, remove.

**Antimicrobial resistance (SAQ)**

- Plasmid-mediated resistance is commonest.
- ESBL: plasmid, co-resistant; KPC/NDM: carbapenem resistance.
- TMP-SMX if resistance ≤20%; fluoroquinolone for pyelonephritis if ≤10%.
- Stewardship: culture, narrow, short, no ASB treatment.

**Biofilm (SAQ)**

- Biofilm: microbes in self-made matrix on a surface.
- Survives 1000–1500 times normal antibiotic levels.

- Treat CAUTI by changing the catheter.
- Revision washout cuts prosthesis infection from about 10% to 3%.

**Renal mucormycosis (SAQ)**

- Angioinvasion → infarction; diabetic ketoacidosis is the key risk.
- CECT: non-enhancing enlarged kidney; histology: broad aseptate right-angle hyphae.
- Liposomal amphotericin B 5 mg/kg/day plus nephrectomy and glucose control.
- Voriconazole and echinocandins do not work.

**Hydatid disease of kidney (SAQ)**

- Kidney: third site (about 2–3%).
- Hydatiduria (grape-skin membranes) is pathognomonic of a communicating cyst.
- Albendazole 400 mg twice daily around surgery; no scolicide if communicating. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Malakoplakia (SAQ)**

- Defective macrophage digestion of *E. coli*.
- Michaelis–Gutmann bodies are pathognomonic.
- Fluoroquinolones + bethanechol/vitamin C; nephrectomy for unilateral renal disease. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Nitrofurantoin. (SAQ)**

- Urine-only drug: cystitis yes, pyelonephritis or prostatitis no.
- 100 mg twice daily × 5 days; prophylaxis 50–100 mg at night (children 1–2 mg/kg).
- Proteus intrinsically resistant; resistance in E. coli low.
- Toxicity: pulmonary fibrosis, neuropathy, G6PD haemolysis; CrCl ≥30 (Beers 2015). *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 4 · Paediatric Urology 1

## Long answer questions (LAQ)

### LAQ 1 Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING

P1 · P3 · P4 **merged from 6**

*Asked as:* P1 LAQ "Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management" (Describe embryological development of VU Junction and outline management of primary VUR?, Summer 2020 Paper I Q2; B/l congenital VUR - Pathophysiology and natural history, Practice Set 1 Paper I Q3f SAQ) · P3 LAQ "Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome" (Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR, Summer 2015 Paper III Q2; Discuss mgt. of b/l VUR, Practice Set 2 Paper III Q2; Discuss treatment of b/l VUR. Describe surgical techniques and criteria for their selection, Practice Set 2 Paper III Q3d SAQ; Discuss the management of B/l VUR and discuss its long term outcome, Practice Set 6 Paper III Q1) · P4 LAQ "VUR: etiopathogenesis, recent trends in management and STING technique" (Describe etiopathogenesis of VUR and recent trends in management, Practice Set 2 Paper IV Q2; STING technique for VUR, Summer 2020 Paper IV Q3c and Practice Set 5 Paper IV Q3e SAQ) · P3 LAQ "Development and clinical anatomy of Vesico-Ureteric junction and management of grade 3 VUR in adult?" · P3 SAQ "Endoscopic correction of VUR (STING procedure): indications and complications" (Endoscopic correction of VUR : indications and complications, Summer 2019 Paper III Q3e; Sting procedure, Winter 2024 Paper III Q3f) · P3 SAQ "Paquin and Gil-Vernet techniques of ureteric reimplantation"

*"B/l congenital VUR" is read as bilateral primary (congenital) VUR (vesicoureteral reflux), often detected antenatally. STING is taken as subureteric Teflon injection, the endoscopic treatment of VUR.*

#### Definition and epidemiology

- **VUR**: retrograde flow of urine from the bladder into the upper tract.
- **Primary**: congenitally deficient UVJ (ureterovesical junction) flap valve with a short intravesical tunnel and a normal bladder. **Secondary**: high bladder pressure (PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction)).
- Found in about **30% of children with UTI (urinary tract infection)**, about 30% of siblings and about 17% of children imaged for hydronephrosis. The younger the child, the likelier VUR.
- Infant boys predominate (higher voiding pressures, more UTI if uncircumcised); overall girls are affected twice as often. Less common in Black children.
- **Bilateral VUR is a high-risk feature:** it resolves less and scars more.
- **Infection, not sterile reflux, injures the kidney.** Infected urine plus intrarenal reflux causes scars, hypertension and CKD (chronic kidney disease); some high-grade "scars" are congenital dysplasia. So management targets infection risk, not the reflux itself.

## Embryology of the VUJ (vesicoureteric junction)

1. **Ureteric bud (weeks 4–6)** arises from the caudal mesonephric (Wolffian) duct: ureter and duct form a "Y", and the common nephric (excretory) duct is the stem.
2. **Reciprocal induction:** the bud enters the metanephric blastema, branches into the collecting system and induces nephrons.
3. **Weeks 5–6:** the urorectal septum divides the cloaca into urogenital sinus (UGS) and anorectal canal.
4. **Absorption:** the common nephric duct is taken into the UGS (current view: removed by **apoptosis**), so ureter and duct open separately.
5. **Rotation (Mackie and Stephens):** the ureteric orifice moves **cranially and laterally** to the bladder base; the duct stays caudal as the ejaculatory duct, ending in the prostatic urethra (Gartner duct in females).
6. **Trigone:** classically ureteral; lineage tracing shows it is mostly bladder muscle. The **Chwalla membrane** over the orifice ruptures; its persistence explains ureterocele.
7. **Maturation:** ureteric muscle joins the detrusor as **Waldeyer's sheath**; the tunnel lengthens with growth, so VUR resolves with age.

| Bud origin | Orifice | Result |
|---|---|---|
| Normal | Trigone corner | Tunnel length to diameter about **5:1** (Paquin); refluxing junctions about 1.4:1 |
| Too low (early budding, early contact with UGS) | High, **lateral** | Short tunnel, so **primary VUR**; abnormal bud–metanephros contact gives **dysplasia** |
| Too high (late budding, late contact) | Caudal (distal), **medial** | **Ectopic ureter** (bladder neck, urethra, Wolffian structures), often obstructed |
| Two buds | Duplex | **Weigert–Meyer rule:** upper pole caudal-medial (ectopia, ureterocele); lower pole lateral (reflux) |

- **Concept:** bud position fixes both orifice and kidney, so this single embryological fault (lateral orifice, short tunnel, dysplasia) explains why high-grade VUR and dysplasia coexist.

## Clinical anatomy of the VUJ

- **Course:** juxtavesical ureter in **Waldeyer's sheath**, then an oblique intramural segment through the detrusor hiatus, then the submucosal tunnel, then the orifice (caudal and medial to the entry).
- **Intramural length 1.2–2.5 cm in adults** (0.5–0.8 cm in neonates). The **narrowest part** of the ureter (about 3–4 mm): the commonest stone impaction site.
- **Trigone:** superficial trigone (ureteric longitudinal muscle, running to the verumontanum); deep trigone (Waldeyer's sheath plus detrusor, to the bladder neck). Its base is the interureteric ridge. Waldeyer's sheath anchors the intramural ureter to the deep trigone.
- **Orifice (Lyon):** position A (medial trigone) to C (lateral); shape cone (normal), stadium, horseshoe, **golf-hole**. The more lateral and open, the more reflux.
- **Relations:** the vas (male) or uterine artery (female) crosses the ureter. **Pelvic plexus nerves lie dorsomedial to the VUJ**; wide extravesical dissection causes voiding dysfunction.
- **Clinical:** Hutch diverticulum at the hiatus; "fish-hook" ureter with BPH (benign prostatic hyperplasia).

### Normal antireflux mechanism

- Balance of three factors: **ureteric peristalsis**, **UVJ anatomy** and **bladder dynamics** (low bladder pressure).
- **Passive flap valve:** the submucosal (intramural) ureter is compressed against firm detrusor backing as the bladder fills. This needs adequate tunnel length and fixation.
- **Paquin:** tunnel length to diameter about **5:1** normally versus 1.4:1 in refluxing units. Orifice shape also matters.
- **Active:** ureteric peristalsis opens the tunnel for each bolus, then it closes.
- **Bladder pressure:** high storage or voiding pressure overwhelms a normal valve.

### Etiopathogenesis

**1. Embryology (Mackie and Stephens):** early budding gives a high, lateral orifice with a short tunnel (VUR) and dysplasia; late budding gives a distal, medial, ectopic and often obstructed ureter (see above).

**2. Primary versus secondary**

| Primary | Secondary |
|---|---|
| Congenitally deficient UVJ: short tunnel, weak detrusor backing, lateral orifice | Normal UVJ overwhelmed by pressure: PUV, neurogenic bladder (VUR in up to 30% of spina bifida), dysfunctional voiding; or UVJ anomaly (ureterocele, duplication, paraureteral diverticulum) |

- They overlap: "primary" VUR often has a treatable secondary component.

**3. Bladder and bowel dysfunction (BBD):** high voiding pressures and incomplete emptying distort the UVJ. Detrusor overactivity is the commonest urodynamic finding. Infant boys have high voiding pressures that settle with maturation.

**4. Genetics:** sibling prevalence about 30%, highest under 3 years; identical twins almost fully concordant; polygenic.

**5. Infection:** cystitis lowers the reflux threshold; *E. coli* endotoxin causes ureteric atony.

### Pathogenesis of renal damage

| | Reflux-associated dysplasia | Acquired scar |
|---|---|---|
| Cause | Abnormal bud (congenital) | Infected urine plus VUR |
| Typical patient | Boy, grade IV–V, antenatal hydronephrosis | Girl, recurrent febrile UTI, BBD |
| Preventable | No | Yes |

- **Intrarenal reflux** through flat compound papillae at the poles needs only 2 mmHg in neonates, so scars are polar.
- **Sterile reflux rarely scars** unless pressures are high (Ransley and Risdon). Most scars follow early infections ("big bang", now debated).
- Sequelae: hypertension in **34–38%** of scarred children; CKD; poor growth; pre-eclampsia.
- **Also know:** massive bilateral VUR gives the megacystis–megaureter association.

### International grading (International Reflux Study, 1981; VCUG (voiding cystourethrogram))

| Grade | Findings |
|---|---|
| I | Non-dilated ureter only |

| Grade | Findings |
|---|---|
| II | Ureter, pelvis and calyces reached; no dilatation, sharp fornices |
| III | Mild–moderate dilatation of ureter, pelvis and calyces; minimal forniceal blunting |
| IV | Moderate dilatation and tortuosity; fornices obliterated but **papillary impressions kept** |
| V | Gross dilatation and tortuosity; **papillary impressions lost**; intrarenal reflux |

- **Also know:** grading varies between observers; UDR (ureteral diameter ratio) predicts resolution better.

### Evaluation (children)

- History of febrile UTIs and BBD; urine culture; BP (blood pressure), growth, spine, urine protein, creatinine.
- **RBUS (renal and bladder ultrasound)**: a normal scan does not exclude VUR.
- **VCUG** (gold standard): grade and urethra (excludes PUV). **Nuclear cystogram** (lower dose) for follow-up; **ceVUS (contrast-enhanced voiding urosonography)** avoids radiation.
- **DMSA (dimercaptosuccinic acid) scan**: scars and split function; a defect persisting 6 months is a scar.
- **UDR** predicts resolution and breakthrough UTI better than grade.

### Natural history

- Resolution comes from **elongation of the intravesical ureter** with growth.
- Resolution falls with grade and age: grade II 63–85%, grade III about 50%, **grades IV–V and bilateral grade III rarely** (International Reflux Study). Younger age resolves better; infants with grade IV–V may still resolve fairly well.
- **Bilaterality and older age delay resolution:** persistence is highest in bilateral grade IV (AUA (American Urological Association) 1997: about 10% resolve in 5 years), and in bilateral grade III at 6–10 years.
- **Consequences of bilateral damage:** adult hypertension in 34–38% of scarred children; CKD, with reflux nephropathy a leading paediatric transplant diagnosis; poor growth; pre-eclampsia (worse with bilateral scars). Correction cannot reverse dysplasia.

### Risk stratification (Hidas)

| Low risk | High risk |
|---|---|
| Male, >5 years | Female, <1 year (infant) |
| Unilateral, grade I–II | **Bilateral**, grade III–V |
| No BBD, no scars | BBD, scars, multiple febrile UTIs |

- Two-year breakthrough UTI on CAP (continuous antibiotic prophylaxis): 8.6% low, 26% intermediate, 62.5% high risk.

### Recent trends in management

1. **Treat infection risk, not reflux:** risk stratification by sex, age, grade, laterality, BBD, scars and presentation (Hidas model).
2. **Fewer VCUGs:** ultrasound first (AAP (American Academy of Pediatrics) 2011, NICE (National Institute for Health and Care Excellence)); or "top-down" DMSA first. Contrast-enhanced voiding urosonography avoids radiation.

3. **Objective measures:** UDR predicts resolution and breakthrough UTI better than grade; machine-learning models.
4. **Selective CAP:** RIVUR halved recurrent UTI but did not reduce scarring; NNT (number needed to treat) 5 in high-risk and 18 in low-risk children. The Swedish Reflux Trial showed benefit in girls with dilating VUR.
5. **BBD first:** timed voiding, constipation treatment, anticholinergics. Topical steroid for phimosis in boys.
6. **Endoscopic injection:** Dx/HA (dextranomer/hyaluronic acid); HIT and double HIT (hydrodistension implantation technique) give about 90% resolution. Vantris works better but can obstruct late.
7. **RALUR-EV (robot-assisted extravesical reimplantation):** less pain and shorter stay; results approaching the open benchmark of **95.9%**.
8. **Less follow-up imaging:** VCUG may be omitted after surgery for low-grade VUR if ultrasound is normal.

### Medical management

- **Aim:** prevent febrile UTI and scars while VUR resolves.
- **Treat BBD first:** timed voiding, hydration, constipation treatment, anticholinergics if needed. BBD raises breakthrough UTI and lowers resolution and endoscopic success.
- **CAP:** one low dose at night, a quarter of the treatment dose.

| Agent | Age | Dose | Note |
|---|---|---|---|
| Trimethoprim or amoxicillin | <2 months (young infants) | — | Avoid sulfonamide (neonatal jaundice) |
| TMP-SMX (trimethoprim-sulfamethoxazole) | >2 months | 2 mg/kg TMP | Avoid under 2 months (kernicterus); used in RIVUR |
| Nitrofurantoin | >2 months | 1–2 mg/kg | Avoid in G6PD (glucose-6-phosphate dehydrogenase) deficiency; rare lung, liver toxicity |

- **RIVUR (Randomized Intervention for Children with Vesicoureteral Reflux, 2014)**, 607 children: recurrent UTI fell from **23.6% to 12.9%** on TMP-SMX; **scarring unchanged**; most benefit with BBD and in high risk.
- **Swedish Reflux Trial** (grade III–IV, age 1–2 years): febrile UTI in girls 19% on CAP, 23% after endoscopy, 57% on surveillance; **no difference in boys**.
- **IRSC (International Reflux Study in Children) and Birmingham trials:** surgery equalled medical care for (new) scarring. **PRIVENT:** 6% absolute UTI reduction.
- **Circumcision** or topical steroid for phimosis: about 4 boys with VUR circumcised prevent one UTI (NNT about 4).
- After stopping CAP, 8–10% have a febrile UTI.
- Low-risk children: observation without CAP.
- **Guidelines:** AUA 2010/2017 recommends CAP for VUR with BBD and treating BBD first; EAU/ESPU (European Association of Urology/European Society for Paediatric Urology) give high-risk children CAP and correct early after breakthrough infection.

### Indications for correction

- Breakthrough febrile UTI on CAP despite good bladder and bowel habits.

- New scars or falling function.
- Persistent high-grade (IV–V) bilateral VUR (girls traditionally corrected before puberty; benefit unproven).
- Non-adherence or intolerance to CAP; parental choice.
- **Controversy:** asymptomatic bilateral high-grade VUR in an older boy with normal kidneys can be observed.

### Case: 8-year-old boy with bilateral grade IV VUR

- Exclude PUV, neurogenic bladder and BBD; DMSA, creatinine, BP.
- This rarely resolves at 8 years, but older boys have low UTI risk, so treating an **asymptomatic** boy is **controversial**.
- **No UTI, no scars:** observation or CAP with BBD therapy. **Breakthrough UTI, scars or falling function:** correct surgically.
- **Choice here: bilateral Cohen.** `Exclude PUV/neurogenic bladder, treat BBD -> febrile UTI, scars or renal loss? Yes -> bilateral Cohen (or Dx/HA) No -> observe or CAP; correct if UTI/scar`

### Endoscopic correction: STING procedure

#### Definition and principle

- **STING (subureteric Teflon injection):** endoscopic injection of a **bulking agent** beneath the ureteric orifice. The mound supports and lengthens the intramural tunnel, restoring the **flap valve**.
- First done with Teflon by Matouschek (1981); popularised by **O'Donnell and Puri, 1986**. **Dx/HA (Deflux)** is now standard.

#### Bulking agents

| Agent | Particle / property | Problem |
|---|---|---|
| Teflon (PTFE (polytetrafluoroethylene) paste) | 90% <40 µm, permanent | **Migration**; never FDA (US Food and Drug Administration) approved |
| **Dx/HA (Deflux)** | Microspheres **80–250 µm**; gel resorbed | Volume loss about 23%; late recurrence; FDA approved 2001; standard agent |
| Polyacrylate (Vantris) | 300 µm, permanent | Works better but late ureteric obstruction |

- **Also know:** Macroplastique.

#### Indications

- Persistent primary VUR (grades II–IV) with **breakthrough febrile UTI** on prophylaxis, **new scars**, poor adherence or parental preference.
- Older girls; residual VUR after reimplantation; duplex or secondary VUR (lower success).
- **Relative contraindications:** grade V, **UVJ obstruction**, untreated BBD, large diverticulum.

#### Technique

1. Antibiotic at induction; offset-lens cystoscope; rigid 3.7–5 Fr needle; bladder partly filled.
2. **STING:** needle enters **2–3 mm below the orifice at 6 o'clock**, bevel up, advanced **4–5 mm** submucosally.
3. Inject slowly; a mound should appear within **0.1–0.2 mL**, else reposition.

4. Aim for a **"volcano" mound** with a slit-like orifice. Do not cauterise bleeding (the implant extrudes). Day case.

| Technique | Needle entry | Result |
|---|---|---|
| STING | 2–3 mm below the orifice, at 6 o'clock | Baseline |
| **HIT** (hydrodistension implantation technique) | Inside the hydrodistended orifice | Better than STING, all grades |
| **Double HIT** | Two intraureteric sites | About 90% resolution |

**Results**

- Low-grade VUR about 90% after one injection.
- **Elder 2006 meta-analysis** (one injection): grade I–II about 79%, III 72%, **IV 63%**, V 51%; repeat injections raise success.
- **Swedish Reflux Trial:** 20% recurrence at 2 years after initial success.
- Less effective in adults.

**Complications**

- **Failure or recurrence** (commonest): repeat injection once or twice, then reimplant (reimplantation is not harder after Dx/HA).
- **New contralateral VUR** after unilateral injection.
- **Ureteric obstruction:** early or **late-onset**, so long-term ultrasound.
- Febrile UTI despite a negative VCUG.
- **Calcified mound** (over half) mimics a ureteric stone. Teflon migration.

**Follow-up**

- Antibiotics 3 months; ultrasound at 2–6 weeks; ultrasound with or without VCUG at 3 months.

## Ureteric reimplantation: principles and techniques

- **Principles:** exclude secondary VUR; gentle handling; tension-free; straight course; tunnel **5:1** (Paquin's rule).
- Open success **95.9%** overall (Elder 1997; AUA meta-analysis benchmark).

| Technique | Method / tunnel | Strength / weakness |
|---|---|---|
| **Cohen** (intravesical) | Cross-trigonal | Commonest; success over 98%; one cystotomy for both sides; later ureteroscopy or retrograde access hard |
| Politano–Leadbetter (intravesical) | New superomedial hiatus above the old one | Long tunnel; blind hiatus risks bowel injury |
| Glenn–Anderson (intravesical) | Orifice advanced towards the bladder neck | Least kinking; short tunnel |
| **Lich–Gregoir** (extravesical) | Detrusorotomy, ureter laid in | No haematuria or spasm; no cystotomy; **retention up to 10% after bilateral repair** (pelvic plexus injury); stay close to the ureter |

| Technique | Method / tunnel | Strength / weakness |
|---|---|---|
| **RALUR / RALUR-EV** (robot-assisted laparoscopic ureteral reimplantation, extravesical) | Y-shaped detrusorotomy | Less pain, shorter stay; results approaching open |
| **Paquin** | Combined extra- and intravesical, suprahiatal | Long tunnel; for dilated ureters |
| **Gil-Vernet** | Intravesical trigonoplasty, no tunnel | Low-grade bilateral VUR |

### Paquin technique

- **Combined extravesical–intravesical** reimplant described by Paquin (1959), with the 5:1 rule.
- Steps: 1. Ureter mobilised **extravesically** and divided at the bladder. 2. Bladder opened; a **new hiatus** created higher on the posterior wall (**suprahiatal**). 3. Long **submucosal tunnel** from the new hiatus down to the original orifice site, length about 5 × diameter. 4. Ureter drawn through, spatulated, sutured to mucosa and anchored to detrusor. 5. Old hiatus closed.
- **Advantages:** long tunnel under direct vision; good for **dilated ureters and megaureter** (can combine with tapering).
- **Disadvantages:** kinking or obstruction at the new hiatus.

### Gil-Vernet trigonoplasty

- **Intravesical, with no tunnelling and no ureteric mobilisation.**
- Steps: 1. Bladder opened. 2. **Transverse mucosal incision** across the trigone between the two orifices. 3. Intramural ureters with their muscle sheath **advanced medially** by one non-absorbable or absorbable stitch across the midline. 4. Mucosa closed.
- **Effect:** lengthens and medialises the intravesical ureters; corrects both sides at once.
- **Indications:** low to moderate grade (I–III), usually bilateral VUR, with non-dilated ureters.
- **Advantages:** quick, simple, low obstruction risk; easy later retrograde access.
- **Disadvantages:** unsuitable for high grade, dilated ureters or paraureteral diverticulum.

| Feature | Paquin | Gil-Vernet |
|---|---|---|
| Approach | Extra- + intravesical | Intravesical |
| Ureter divided | Yes | No |
| Tunnel | New long submucosal | None (advancement) |
| Best for | High grade, dilated | Low grade, bilateral |

## Criteria for selecting the technique

| Situation | Preferred option |
|---|---|
| Grades II–IV, parents want least invasive | Dx/HA injection (lower success at higher grade) |
| **Bilateral VUR, small or trabeculated bladder** | **Cohen** (both sides through one cystotomy) |
| Grade V, dilated ureter | Intravesical reimplant with or without tapering (Politano–Leadbetter or Cohen) |

| Situation | Preferred option |
|---|---|
| Unilateral, older child | Lich–Gregoir or RALUR-EV |
| Bilateral extravesical | Nerve-sparing dissection; warn about retention |
| Duplex system | Common sheath reimplant |
| Failed injection | Reimplant (not harder after Dx/HA) |
| Neurogenic or PUV bladder (secondary VUR) | Treat the bladder first (anticholinergics, CIC (clean intermittent catheterisation)) |

### Complications and follow-up after reimplantation

- Ultrasound at 6–12 weeks.
- Complications: transient obstruction; persistent reflux (usually resolves by 1 year); contralateral VUR (about 1 in 10, usually resolves); leak; retention after bilateral extravesical repair.
- Obstruction early (oedema): stent or nephrostomy; late (ischaemia, kink): redo.
- Lifelong BP and renal checks if scarred; scarred women risk pre-eclampsia.

### Long-term outcome

| Outcome | Facts |
|---|---|
| **Renal scarring** | About 15% get a febrile UTI within 2 years; about 15% of those scar. Risk highest under 1 year. Much "scarring" in boys with high grade is **congenital dysplasia**, not preventable by any treatment |
| **Hypertension** | Reflux nephropathy may be the commonest cause of childhood hypertension after essential hypertension (renin–angiotensin). **34–38%** of scarred children are hypertensive as adults. Correcting VUR does not lower BP; nephrectomy of a small scarred kidney helps selected cases |
| **CKD and ESKD (end-stage kidney disease)** | Now rare: RRT (renal replacement therapy) from reflux nephropathy fell from 1.9 to 0.6 per million (USRDS (United States Renal Data System), 1996–2014). Bilateral scars cause hyperfiltration, proteinuria, concentrating defect, tubular acidosis |
| **Pregnancy** | VUR in late pregnancy: pyelonephritis about one-third versus under 5%. Scars: about **3× hypertension, 8× pre-eclampsia**; bilateral worse. Impaired function: fetal loss. Reimplanted women still get UTIs |
| Growth; offspring | Catch-up growth after infection control; about 43% of screened offspring have VUR |

- **Concept:** in adult life **the scar, not the reflux, carries the risk**, so correction does not remove it. Scarred patients need lifelong BP, proteinuria and creatinine checks.

### Grade 3 VUR in an adult

- **Grade 3:** mild–moderate dilatation of ureter, pelvis and calyces, minimal forniceal blunting.
- **Adults rarely lose function from reflux alone**, unless high bladder pressure, stasis and UTI coexist. The **scar**, not the reflux, carries the risk. Spontaneous resolution is unlikely.
- **Causes:** persistent primary VUR; **secondary**: BOO (bladder outlet obstruction), neurogenic bladder, resection near the orifice, ureterocele incision, paraureteral diverticulum, transplant ureter, TB (tuberculosis).

- **Presentation:** recurrent febrile UTI, pyelonephritis (especially in pregnancy), loin pain on voiding, hypertension, proteinuria, CKD.
- **Evaluation:** VCUG (grade, bladder, diverticulum, outlet); ultrasound and **DMSA** for scars and split function; MAG3 (mercaptoacetyltriglycine) renogram if obstruction is suspected; **UDS (urodynamics)**: compliance, voiding pressure, residual; cystoscopy; blood pressure, creatinine, urine protein.
- **Treat secondary causes first** (relieve BOO; CIC and antimuscarinic for neurogenic bladder), then reassess.

| Situation (primary grade 3) | Management |
|---|---|
| Incidental; no UTI, no scars | Observe: timed voiding, bowel care, prompt UTI treatment |
| Recurrent febrile UTI, new scars, falling function | **Correct the reflux** |
| Woman planning pregnancy with scars or prior pyelonephritis | Consider correction before pregnancy |
| Poorly functioning scarred kidney with infection or hypertension | Nephrectomy |

- Low-dose prophylaxis selectively; adult evidence is limited.
- **Correction:** endoscopic Dx/HA injection (day case, repeatable, but less effective in adults); or **ureteroneocystostomy** (open, laparoscopic or robotic). Lich–Gregoir/RALUR-EV suits adults (no cystotomy; stay close to the ureter to protect nerves); Politano–Leadbetter (long tunnel); Cohen (over 98%; later retrograde access hard).
- **Psoas hitch** if length is needed; tapering seldom needed in grade 3.
- Follow-up: ultrasound at 6–12 weeks; lifelong blood pressure and renal checks if scarred.

### Overall management algorithm

```
Treat BBD -> low risk: observe | high risk (bilateral): CAP Breakthrough UTI / new scar / persistent IV-V -> Dx/HA injection or reimplant (Cohen) Scars -> lifelong BP and proteinuria checks
```

**Diagrams to draw:**

- "Y" of ureteric bud and mesonephric duct with orifice migration (early-budding Y diagram).
- Long oblique tunnel versus short lateral refluxing tunnel; oblique intramural ureter in Waldeyer's sheath with submucosal tunnel and detrusor backing.
- Grades I–V.
- Cohen cross-trigonal reimplant; Gil-Vernet transverse trigonal incision and medial advancement stitch.
- Needle sites: STING below the orifice, HIT and double HIT inside the tunnel, with volcano mound.

**Key points to remember:**

- Low (early) bud → high lateral orifice → short tunnel → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 normal versus 1.4:1 refluxing (Paquin); adult intramural ureter 1.2–2.5 cm.
- Dysplasia (boys, congenital) versus scar (girls, infection); sterile reflux rarely scars.
- Grade V loses papillary impressions; grade IV–V and bilateral grade III rarely resolve; bilateral grade IV persists most; bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Treat BBD and give CAP first; correct for breakthrough febrile UTI, new scars or persistent grade IV–V.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- STING: O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%; Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction; calcified mound mimics a stone.
- Open reimplantation 95.9%; Cohen cross-trigonal (over 98%) suits bilateral VUR through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- Paquin: tunnel 5:1; combined, suprahiatal reimplant. Gil-Vernet: trigonoplasty, medial advancement, no tunnel; for low-grade bilateral VUR.
- 34–38% of scarred children become hypertensive; correction does not lower BP; scars bring 8× pre-eclampsia risk.
- In adults the scar, not the reflux, carries the risk; exclude secondary causes first; correct for recurrent pyelonephritis, new scars or pregnancy plans.

## LAQ 2 Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods

P1 · P2 · P3 **merged from 4**

*Asked as:* P1 LAQ "Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis" (Write a note on the embryological descent of the gonads, explaining the various phases of descent. Outline the evaluation and management of bilateral undescended testis in a 3 year old child, Winter 2024 Paper I Q2; Describe developmental embryology of testes and its physiological descent. Mention various abnormalities associated with testicular descent, Practice Set 14 Paper I Q1) · P3 SAQ "Maldescended (undescended) testes and treatment algorithm" (Treatment algorithm of undescended testis, Summer 2018 Paper III Q3a; Maldescended testes, Practice Set 5 Paper III Q3b) · P2 SAQ "Undescended testis" · P1 SAQ "Methods of orchidopexy"

*Maldescent is taken as any testis not in a stable scrotal position: undescended, ectopic, retractile and acquired.*

### Development of the testis

- **Indifferent gonad:** the gonadal (germinal) ridge forms at **4 weeks** on the medial urogenital ridge. It has four components: coelomic epithelium, underlying mesenchyme, adjacent mesonephros and **PGCs (primordial germ cells)**.
- PGCs arise in the **yolk sac wall in week 3** and migrate along the dorsal mesentery of the hindgut to the ridge (about 6 weeks).
- **SRY (sex-determining region Y)** on the short arm of Y, partly activated by SF1 (steroidogenic factor 1), switches on **SOX9** in pre-Sertoli cells. Sertoli cells then appear (about 40 days). Without SRY no Sertoli cells form and the gonad becomes an ovary.
- Sertoli cells enclose germ cells as **testicular cords (about 6 weeks)**, which canalise into seminiferous tubules.
- **Rete testis** forms with the cords and joins **5–12 mesonephric tubules**, which become efferent ductules. Full continuity (tubules, rete, efferent ductules, epididymis) is established by the **third month**.
- The testis turns round and a thick **tunica albuginea** forms. Testes become spherical after 7–8 weeks; ovaries stay oblong.

| Hormone | Cell | Action | Male remnants |
|---|---|---|---|
| **AMH (anti-Müllerian hormone)** | Sertoli | Müllerian regression, weeks 8–10 | Appendix testis, prostatic utricle |
| **Testosterone** | Leydig (from 8 weeks; driven by placental hCG (human chorionic gonadotrophin), later fetal LH (luteinising hormone)) | Wolffian duct to epididymis, vas, seminal vesicle (weeks 8–12) | Appendix epididymis, paradidymis |
| **DHT (dihydrotestosterone)** | Formed locally by 5α-reductase type 2 | External genitalia, prostate, scrotum (fused by 10–12 weeks) | – |

| Hormone | Cell | Action | Male remnants |
|---|---|---|---|
| **INSL3 (insulin-like peptide 3)** | Leydig | Gubernacular swelling (via RXFP2 receptor) | – |

### Gonadal descent: starting point

- At **7–8 weeks** the testis lies near the kidney (upper lumbar), held by the **cranial suspensory ligament (CSL)** above and the **gubernaculum** below. The gubernaculum appears at 7 weeks, attaches to the **epididymis** and runs to the future inguinal canal.
- **Female:** the CSL persists; the gubernaculum becomes the ovarian and round ligaments; the ovary descends only into the pelvis.

### Phases of testicular descent (classic two-phase model)

| | Transabdominal phase | Inguinoscrotal phase |
|---|---|---|
| Timing | Weeks 10–15 | From the 7th month; inguinal passage rarely before 22 weeks; most testes scrotal after 27 weeks |
| Events | **CSL regresses**; **gubernaculum swells** (outgrowth); its cord shortens, holding the testis at the internal ring while the kidney and abdomen grow away (the kidney moves cranially) | Gubernaculum migrates **3–5 cm** to the scrotum; **processus vaginalis** forms within it; the bulb then regresses; the processus closes |
| Control | CSL regression by **androgen**; swelling by **INSL3** from Leydig cells, via its receptor RXFP2 (occurs even in androgen resistance) | **Androgen**, mainly indirectly via the **genitofemoral nerve (GFN)** releasing **CGRP (calcitonin gene-related peptide)**; intra-abdominal pressure |
| Failure gives | Intra-abdominal testis | Canalicular or ectopic testis |

- **Modern view:** the "transabdominal descent" is largely relative, not true migration. Barteczko and Jacob describe **five phases** (5 weeks to 28 weeks). The gubernaculum **swells to make room** (outgrowth), then **shrinks to let the testis pass** (regression). Testosterone (peak 14–16 weeks) and INSL3 (15–17 weeks) act together on gubernacular and cremaster growth (myogenesis).
- Spontaneous postnatal descent occurs mainly in the first 3–6 months.
- **Also know:** AMH defects (persistent Müllerian duct syndrome) cause cryptorchidism; Insl3-knockout mice are cryptorchid; the scrotum forms by 10–12 weeks under DHT.

| Failed step | Result |
|---|---|
| CSL retained | Intra-abdominal testis |
| Abnormal or ectopic gubernacular attachment | Canalicular testis; ectopic (superficial inguinal pouch, perineal, femoral, prepenile) |
| Processus not closed | Hernia, hydrocele (patent in 87% of unilateral cases) |
| Fibrous remnant of processus / processus not fully involuted | Acquired (ascending) testis; acquired cryptorchidism |

## Abnormalities of descent: definitions and classification

| Type | Description |
|---|---|
| **Undescended (cryptorchid)** | Arrested on the normal path: abdominal, "peeping" at the ring, canalicular, high scrotal/prescrotal |
| **Ectopic** | Off the normal path: **superficial inguinal pouch** (commonest), perineal, femoral, prepubic, penile, contralateral scrotum (transverse, crossed) |
| **Retractile** | Scrotal testis pulled up by the cremaster; stays down after sustained traction; review yearly (risk of ascent); some become acquired |
| **Acquired (ascending)** | Documented scrotal earlier, ascends later; diagnosed at 8–11 years; fibrous processus remnant |
| Recurrent, secondary, iatrogenic | Re-ascent after spontaneous descent; after hernia repair or orchidopexy |
| **Vanishing** vs agenesis | Lost after formation, in utero or perinatally (torsion; monorchia, anorchia) vs never formed |

- Clinically: **palpable** (canalicular, high scrotal, ectopic) or **non-palpable** (intra-abdominal, "peeping", vanished/absent).

## Epidemiology and aetiology

- 1–4% of term boys (P2: 2–4% at birth, 1–2% at 3 months); up to 45% of preterm boys. Most descend by 6 months.
- 75–80% palpable; 60–70% unilateral.
- Descent needs Leydig hormones: **INSL3** (transabdominal phase) and **androgen** (inguinoscrotal phase), with the genitofemoral nerve/CGRP guiding the gubernaculum.
- Risk factors: prematurity, low birth weight, syndromes (prune belly, cerebral palsy).

## Associated anomalies and sequelae

- **Testicular: TTE (transverse testicular ectopia)**, polyorchidism, splenogonadal fusion; epididymal non-fusion.
- **PMDS (persistent Müllerian duct syndrome):** mutation of AMH or its receptor (AMHR2); bilateral cryptorchidism or TTE.
- **Genital and DSD (differences of sex development):** hypospadias (12–24%); bilateral non-palpable testes need karyotype and 17-OHP (17-hydroxyprogesterone) to exclude **CAH (congenital adrenal hyperplasia)**.
- **Syndromes:** prune belly (all), Klinefelter, androgen insensitivity, hypogonadotrophic hypogonadism, cerebral palsy, gastroschisis, omphalocele, myelomeningocele, PUV (posterior urethral valves).
- **Sequelae:** subfertility (worse if bilateral), **TGCT (testicular germ cell tumour) 2–5-fold**, torsion, hernia, trauma.

## Evaluation

- **History:** prematurity, testes once scrotal (acquired), hernia, family history, salt-wasting illness.
- **Examination:** warm room; supine and frog-leg, then cross-legged or squatting. Sweep ("milk") from the anterior superior iliac spine down the canal to the scrotum to find the lowest position; check the superficial inguinal pouch.

- Sustained traction fatigues the cremaster: a **retractile** testis stays in the scrotum after gentle traction.
- Note penis size, meatus (hypospadias), scrotal development, syndromic features. Contralateral hypertrophy suggests a vanished testis.
- **No ultrasound or MRI (magnetic resonance imaging)**: inaccurate (ultrasound sensitivity about 45% for non-palpable testes); they never change the operation or remove the need for surgery, and delay surgery.

### Bilateral undescended testis at 3 years

- This is **late presentation**; surgery is advised from 6 months, preferably by 12 months and by 18 months at the latest.
- **Bilateral palpable:** no investigations needed; plan orchidopexy, both sides usually in one sitting.
- **Bilateral non-palpable (or with hypospadias, small phallus or micropenis): treat as DSD until proven otherwise**, with urgent work-up:
- **Karyotype**; electrolytes and **17-OHP** to exclude **CAH** in a virilised 46,XX child.
- **Hormones:** LH, FSH (follicle-stimulating hormone), testosterone, **AMH and inhibin B**. Measurable AMH or inhibin B confirms testicular tissue; high gonadotrophins with absent AMH suggest **anorchia**. The **hCG stimulation test** (testosterone rise) is now less used.
- Pelvic ultrasound for Müllerian structures.
- **Both testes very high or of doubtful viability:** stage the sides, the second at 6 months, to avoid bilateral atrophy.
- **Biopsy** only if DSD or dysgenesis is suspected. **Bilateral anorchia:** testosterone at puberty, prostheses later.
- Adjuvant LHRH (buserelin) in bilateral cases is experimental.

### Management principles (AUA (American Urological Association) 2014; EAU (European Association of Urology) similar)

- **Observe to 6 months** (corrected age) for spontaneous descent; refer if still undescended.
- **Orchidopexy from 6 months, preferably by 12 months, and by 18 months at the latest.** Reason: germ-cell and Leydig-cell loss starts in infancy; early surgery restores testicular growth and lowers cancer risk. **Kollin RCT (randomised controlled trial):** surgery at 9 months gave better testicular growth than at 3 years.
- **No hormonal therapy** (hCG or LHRH (luteinising hormone-releasing hormone)) to induce descent: descent in about 20% only; hCG may increase germ-cell apoptosis.
- **Goals of surgery:** tension-free **dependent scrotal** testis with intact blood supply; ligate the patent processus.

### Treatment algorithm

```
Boy with empty scrotum ├─ Bilateral non-palpable ± abnormal phallus → DSD work-up (karyotype, hormones, electrolytes) └─ Otherwise → re-examine at 6 months corrected age Still undescended → refer → examination under anaesthesia (EUA) ├─ Retractile → yearly review (risk of ascent) ├─ Palpable (undescended or ectopic) → inguinal or scrotal (Bianchi) orchidopexy └─ Non-palpable → diagnostic laparoscopy ├─ Testis seen → laparoscopic orchidopexy │ vessels short (testis not within 2–4 cm of ring) → Fowler–Stephens │ (clip testicular
```

vessels; testis lives on vasal collaterals; │ one-stage, or two-stage with second stage at 6 months) or Shehata traction ├─ Vessels + vas enter ring → inguinal or scrotal exploration (excise nubbin) └─ Blind-ending vessels → stop (vanished testis)

- **Scrotal nubbin with contralateral hypertrophy** may start with scrotal exploration.
- **Postpubertal intra-abdominal testis with a normal contralateral testis: orchidectomy** (poor function, cancer risk); offer a prosthesis.

### Methods of orchidopexy: palpable testis

- **Inguinal orchidopexy** (steps): 1. Groin incision. 2. Open external oblique (spare the ilioinguinal nerve). 3. Free the cord; divide the gubernaculum. 4. Separate and **ligate the sac (patent processus) at the internal ring**. 5. Retroperitoneal mobilisation; divide lateral bands. 6. **Prentiss manoeuvre** if needed (medial transposition under the inferior epigastric vessels) for extra length. 7. **Subdartos pouch.** Transalbugineal sutures are unnecessary.
- **Trans-scrotal (Bianchi) orchidopexy** for low testes: high scrotal incision; sac ligated through it; inguinal conversion 3.5–4.4%; atrophy 0.3–0.6%; less pain.

### Methods of orchidopexy: non-palpable testis

Examination under anaesthesia, then laparoscopy.

| Method | Principle | Success |
|---|---|---|
| **Primary laparoscopic** | Mobilise vessels, preserve vasal collaterals, new hiatus medial to epigastric vessels | >90% (97%); primary orchidopexy, when possible, is most successful |
| **Open transabdominal (Jones and Bagley)** | Preperitoneal high mobilisation | 95% |
| **Fowler–Stephens** (one- or two-stage) | Divide testicular vessels high; testis survives on **vasal and gubernacular collaterals**; second stage after 6 months | 60–97% (one-stage 63%, two-stage 68%) |
| **Shehata traction** | Fix testis near the opposite anterior superior iliac spine; vessels lengthen; orchidopexy at 12 weeks | 84–85%; atrophy 10% |
| **Microvascular autotransplantation** | Testicular vessels to inferior epigastric vessels | 96% |
| **Orchidectomy** | Postpubertal, dysgenetic or atrophic testis | — |

- Fowler–Stephens is for testes **not within 2–4 cm of the internal ring**.
- **Also know:** staged (two-stage) inguinal orchidopexy; Dessanti PTFE (polytetrafluoroethylene) sleeve traction; gubernaculum-sparing technique; redo approaches (Redman lateral, Cartwright fascial patch).

### Complications of orchidopexy

- **Atrophy** (highest after Fowler–Stephens), retraction or re-ascent, vas or vessel injury, haematoma, infection; torsion very rare after subdartos fixation.

### Follow-up and prognosis

- Check size and position at 6 months. Offer a prosthesis for an atrophic testis.

- **Fertility:** paternity about 80% (four-fifths) after unilateral and one-third after bilateral; worse when bilateral (oligospermia 42%, azoospermia 31% in one large review).
- **Cancer:** 2–5-fold TGCT risk (P1: about 3-fold), higher with late surgery; orchidopexy after 10–11 years carries relative risk 5.8 (Walsh).
- Counsel on fertility and cancer risk; teach **self-examination** after puberty.

**Diagrams to draw:**

- Sagittal fetus showing CSL, gubernaculum, processus vaginalis and both phases of descent.
- Line of descent with ectopic sites.
- Laparoscopic findings at the internal ring: testis, vessels entering the ring, blind-ending vessels.
- Fowler–Stephens collateral supply: testicular artery clipped, deferential and cremasteric arteries intact.

> **Key points to remember:**
>
> - SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
> - Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
> - Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP; most testes scrotal after 27 weeks.
> - Ectopic = off the path; commonest site superficial inguinal pouch.
> - Operate from 6 months, preferably by 12 months, and by 18 months at the latest.
> - No imaging, no hormones.
> - Non-palpable: EUA then diagnostic laparoscopy; laparoscopy decides.
> - Bilateral non-palpable: DSD work-up first (karyotype, 17-OHP, AMH) to exclude CAH/DSD.
> - Subdartos pouch; Prentiss manoeuvre gives extra length.
> - Fowler–Stephens relies on vasal collaterals.

## LAQ 3 Retrocaval ureter and IVC anomalous development

P1 · P3 **merged from 2**

*Asked as:* P1 LAQ "IVC anomalous development and retrocaval ureter" · P3 SAQ "Retrocaval ureter"

### Introduction

- The **IVC (inferior vena cava)** forms between the 6th and 8th weeks from three pairs of fetal veins by anastomosis, regression and persistence. The final IVC is **right-sided**, so long left veins must cross the aorta to reach it.
- Most anomalies are silent CT (computed tomography) findings. They **bleed at surgery, mimic lymph nodes, trap the ureter and alter IVC filter placement**.

### Embryology of the IVC

#### Three paired venous systems

- Fetal veins (posterior cardinal, supracardinal, subcardinal) form a venous collar around each ureter and ascending kidney: posterior cardinal and supracardinal veins dorsally, subcardinal veins ventrally.

| Vein | Position | Adult derivatives |
|---|---|---|
| **Posterior cardinal** (first) | Dorsolateral | Iliac confluence; mostly regresses |
| **Subcardinal** | Ventromedial | Right: **suprarenal IVC**; intersubcardinal anastomosis: **left renal vein**; left: adrenal and gonadal veins |
| **Supracardinal** (last) | Dorsomedial | Right: **infrarenal IVC**; thoracic parts: **azygos and hemiazygos** |

#### Segments of the definitive IVC (top to bottom)

1. **Hepatic**: right vitelline vein.
2. **Suprarenal**: right subcardinal vein (via the subcardinal–hepatic anastomosis).
3. **Renal**: subcardinal–supracardinal anastomosis.
4. **Infrarenal**: right supracardinal vein.
5. **Iliac confluence**: posterior cardinal veins.

#### Renal collar

- A venous ring encircles the aorta at renal level. The **ventral (pre-aortic) limb** persists as the left renal vein; the **dorsal (retro-aortic) limb** regresses.
- Normally the right **supracardinal** vein, which lies **dorsal** to the ureter, forms the infrarenal IVC, so the normal ureter lies in front of the IVC.

```
Left-sided channels regress -> right-sided persist -> single right IVC + pre-aortic left renal vein
```

**Diagram to draw:** three stages: paired cardinal veins, anastomoses with renal collar, definitive IVC.

## Anomalies of IVC development

| Anomaly | Embryological error | Key features |
|---|---|---|
| **Retrocaval ureter** ("preureteral vena cava") | Infrarenal IVC forms from a vein **ventral** to the ureter | Right only; 1 in 1500 autopsies; male 3:1; presents 3rd–4th decade. **Type I** low loop, fishhook, obstructs; **type II** high loop, sickle, mild |
| **Circumaortic left renal vein** | Dorsal limb persists with ventral | Commonest (2–16%); posterior limb runs lower into lumbar IVC |
| **Retroaortic left renal vein** | Ventral limb regresses, dorsal persists | Less common; may be compressed ("posterior nutcracker") |
| **Left IVC** | Left supracardinal persists, right regresses | Crosses to the right at renal level, in front of the aorta |
| **Duplicated IVC** | Both supracardinals persist | Left IVC joins left renal vein; double right cava can trap the ureter |
| **Azygos continuation** | Subcardinal–hepatic anastomosis fails | Large azygos; hepatic veins drain directly to right atrium; polysplenia, heterotaxy |
| **Absent infrarenal IVC** | Supracardinal and posterior cardinal failure | Deep vein thrombosis in the young |

## Retrocaval ureter: definition

- The **right ureter** passes **behind the IVC**, then crosses in front of it from medial to lateral to resume a normal course.
- Better termed "**preureteral vena cava**": the anomaly is venous; the ureter is trapped.
- In P3 the question was also asked as "Retrocecal ureter"; this is taken as a typo for retrocaval (circumcaval) ureter.

## Retrocaval ureter: embryology

- If a vein **ventral** to the ureter persists as the infrarenal IVC, the ureter is trapped behind it.
- Campbell names a persistent **lumbar subcardinal** vein (the lumbar part of the ventral subcardinal vein; Chapter 52) or a **persistent posterior cardinal** vein (Chapter 93, and most texts); both mean that a ventral vein forms the IVC.
- Variant: double right IVC, with the ureter between the limbs.

## Retrocaval ureter: incidence and presentation

- About **1 in 1500** autopsies; **male 3:1**; right side only.
- Presents in the **third–fourth decade** with right flank pain, UTI (urinary tract infection), stones or incidental hydronephrosis.
- Associations: horseshoe kidney, left renal anomalies; bilateral cases occur in situs inversus.

## Retrocaval ureter: classification (Bateson and Atkinson)

| | Type I (low loop), commoner | Type II (high loop) |
|---|---|---|
| Crossing | Lower, at L3–L4 | Higher, at or near renal pelvis level |
| Shape | **J or "fishhook" / reverse-J**, S-curve | Smooth horizontal **sickle** curve |
| Obstruction | Yes, moderate–severe, at the iliopsoas edge | Mild or none (pelvis also crosses) |

### Retrocaval ureter: investigations

- **CT urography (computed tomography urography): investigation of choice**, showing ureter and cava together; MRI (magnetic resonance imaging) is the alternative.
- IVU (intravenous urography): fishhook; ureter not seen beyond the J hook.
- Retrograde pyelography: **S-curve** or fishhook at L3–L4.
- **Diuretic renogram:** grades the degree of obstruction and gives split function.
- Pitfall: mistaken for **PUJ (pelviureteric junction) obstruction**, so pyeloplasty fails.

### Retrocaval ureter: management

- **Non-obstructed (usually type II): observe.**
- **Obstructed: dismembered ureteroureterostomy** (or ureteropyelostomy): divide the ureter, transpose it **anterior to the IVC**, and do a spatulated, tension-free anastomosis over a **double-J stent**. Excise or bypass the **aperistaltic retrocaval segment**; respect and preserve the ureteric blood supply.
- **Laparoscopic or robotic** repair (transperitoneal or retroperitoneal) is now preferred: equal success and faster recovery.
- Nephrectomy if the kidney is non-functioning.
- A pyeloplasty that misses the diagnosis fails.
- **Also know:** retroiliac ureter (L5–S1, with vasal anomalies); mimics of extrinsic ureteric crossing: spermatic / gonadal or lumbar vein, psoas (iliopsoas) tendon.

**Diagram to draw:** right ureter hooking behind the IVC with a fishhook pelvis.

### Clinical implications of IVC anomalies

**1. Diagnosis**

- Venous-phase CT or MRI. A left or duplicated IVC mimics **para-aortic lymphadenopathy**; a large azygos mimics a mediastinal mass, risking needless biopsy.
- Retrocaval ureter: S-shaped or fishhook ureter at L3–L4 on retrograde pyelography; CT urography confirms; diuretic renography grades obstruction.

**2. RPLND (retroperitoneal lymph node dissection)**

- Preoperative CT must look for circumaortic or retroaortic veins and retrocaval ureter. A left or duplicated IVC lies in the left template; a retroaortic limb tears when lumbar vessels are divided.

**3. Radical nephrectomy**

- The hidden posterior limb of a circumaortic vein bleeds when the artery is approached from behind: ligate both limbs. Caval thrombus in a duplicated IVC needs rerouting of the left IVC.

**4. Living donor nephrectomy**

- The left kidney is preferred for its long vein. A circumaortic or retroaortic vein shortens it and adds bleeding risk; CT angiography is mandatory, and the right kidney may be chosen.

**5. Retrocaval ureter repair**

- As above: observe non-obstructed type II; for obstruction, divide, transpose in front of the IVC and re-anastomose over a stent, excising the aperistaltic segment.

**6. Interventional and vascular**

- **IVC filter:** with duplication, emboli bypass a single filter, so filter both cavae or place the filter suprarenally.
- Aortic surgery risks retroaortic vein injury. A retroaortic vein can cause haematuria or left varicocele.
- **Also know:** variable left adrenal vein entry (relevant to adrenal vein sampling); right gonadal vein drains into the right renal vein in up to 10%.

> **Key points to remember:**
> - Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
> - Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
> - Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
> - Type I fishhook obstructs; type II sickle often does not.
> - CT urography; anterior transposition and ureteroureterostomy.
> - Duplicated IVC: filter both cavae; do not mistake it for nodes.

## LAQ 4 Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula

P1 · P4 **merged from 2**

*Asked as:* P4 LAQ "Hypospadias: present concepts in management, and hypospadias fistula" · P1 SAQ "Hypospadias repairs – single-stage versus staged techniques"

### Introduction

- Read "present concepts" as the present (current) concepts in the management of hypospadias.
- Hypospadias is an arrest of the androgen-dependent "closing zipper" of urethral fold fusion (weeks 8–16). Triad: ventral ectopic meatus, ventral curvature, dorsal hooded prepuce.
- 1 in 150–300 male births; distal 70–85%, proximal 10–25%.
- **Goals:** a straight erection, a urethra to the tip (standing voiding, semen delivery) and acceptable appearance, with the fewest operations and a result lasting into adulthood.
- The choice of repair depends on **curvature after degloving**, plate quality and glans size, not on meatal position alone.

### Aetiology and associations

- **Aetiology:** causes converge on deficient androgen action (placental insufficiency, assisted reproduction, genes, endocrine disruptors).
- Associations: hernia or hydrocele 9–16%, cryptorchidism about 7% (10% in proximal), enlarged utricle.

### Evaluation

- **DSD (disorder of sex development) work-up** if hypospadias plus undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).
- No routine VCUG (voiding cystourethrogram) or renal ultrasound.

### Classification: severity is judged in theatre, not by the meatus

- A "distal" meatus may hide absent spongiosum and severe curvature, so classify **after degloving**.

| GMS (glans–meatus–shaft) score | Glans | Meatus | Shaft curvature |
|---|---|---|---|
| 1 | Good size, deep groove | Glanular | None |
| 2 | Adequate, grooved | Coronal | <30° |
| 3 | Small; narrow or flat plate | Mid or distal shaft | 30–60° |
| 4 | Very small; flat plate | Proximal shaft, penoscrotal | >60° |

- Total 3–12; a higher score predicts more complications.
- **POST (plate objective scoring tool):** a low ratio of neomeatus length to glans fusion length means a poor plate, which favours an inlay graft over plain TIP (tubularised incised

plate).

### Timing and preparation

- Repair at **6–18 months** (AAP (American Academy of Pediatrics): 6–12 months), before genital awareness at about 18 months. One short anaesthetic appears low-risk.
- A straight penis that voids standing may gain little from surgery: counsel honestly.
- **Preoperative androgen** if glans width <14–15 mm: testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Evidence mixed; two RCTs (randomised controlled trials) showed fewer complications.

### Curvature decides the operation

```
Deglove + excise dysplastic dartos + artificial erection ├─ straight → urethroplasty (single-stage) ├─ residual <30° → dorsal midline plication (12 o'clock nerve-free zone), single-stage └─ >30° → divide plate → still >30° → ventral corporotomy ± graft → staged repair
```

- Recurrence 28% after plication versus 9% after grafting. Under-correction causes repeated surgery, so correct fully first time.

### Choice of technique

| Situation | Current choice |
|---|---|
| Glanular meatus, no curvature | Advancement: MAGPI (meatal advancement and glanuloplasty), MIV (M inverted V) glansplasty; selectively |
| Distal or midshaft, good plate | **TIP (Snodgrass)**: >90% of distal, about 80% of midshaft repairs; Thiersch–Duplay; GAP (glans approximation procedure) for a deep groove |
| Narrow or shallow plate | TIP with **dorsal inlay graft**; onlay island flap; Mathieu (SLAM modification) |
| Proximal, curvature >30° | **Two-stage**: STAG (staged tubularised autograft) or Bracka graft; Byars flaps; distal-first (Ulaanbaatar) |
| Proximal, curvature corrected, good tissue | One-stage onlay or TPIF (transverse preputial island flap); proximal TIP abandoned |

### Single-stage techniques

| Group | Examples | Indication |
|---|---|---|
| Advancement | **MAGPI** (Duckett 1981); MIV | Glanular meatus, no curvature |
| Tubularisation | **TIP** (Snodgrass 1994); Thiersch–Duplay; GAP | Distal–midshaft; TIP in >90% of distal |
| Flap | Mathieu (flip-flap); **OIF** (onlay island flap) | Thin plate |
| Proximal one-stage | TPIF (Duckett tube); proximal OIF | Curvature corrected; good tissue |
| Graft | Dorsal buccal inlay | Narrow plate |

- Pros: one anaesthetic.
- Proximal TIP is abandoned: devascularised plate, stricture, recurrent curvature.

### Staged techniques

- Now the approach of choice for most proximal hypospadias.
- **Rationale (why stage):** a urethroplasty laid over a fresh corporal graft fails more often; stage 1 builds a supple, vascular bed. Stage 2 after **≥6 months**.

| Type | Stage 1 | Stage 2 (≥6 months) |
|---|---|---|
| **Bracka / STAG** (staged tubularised autograft) | Correct curvature, divide plate, quilt inner preputial or buccal graft | Tubularise, glansplasty, tunica vaginalis cover |
| **Byars flaps** | Split dorsal prepuce, transpose ventrally | Tubularise |
| Distal-first (Ulaanbaatar) | Curvature and distal urethra | Join proximal urethra |

- Repeat the artificial erection before stage 2.

### Single-stage versus staged: comparison of outcomes

| Measure | Single-stage | Staged |
|---|---|---|
| Distal success | 83–95% | – |
| Proximal complications | About 60% (long TIP, mixed) | STAG about 23% (lowest); Byars 43–81% |
| Recurrent curvature | Higher after plication | 0–10% |
| Penile length, perception | Lower | Better |
| Extra procedures | Fewer planned | 25–75% need more |

### Operative principles: protect blood supply

- Magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), rapidly absorbed fine sutures, no hair-bearing skin, haemostasis.
- Postoperative: stent 3–21 days by complexity, compressive dressing, oxybutynin; antibiotic prophylaxis debated.

### Complications and outcomes (standardised)

- Definitions: glans dehiscence (fusion <2 mm); meatal stenosis (symptoms plus <8 Fr before puberty, <12 Fr after); recurrent curvature >30°.
- TIP meta-analysis (Pfistermuller 2015): complications 10.6%, fistula 5.7%. Proximal complications 23–68%; STAG about 23% (lowest); Byars 43–81%.
- Fistula is the commonest complication. A long, rigid, spongiosum-free neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum.
- **Also know:** BXO (balanitis xerotica obliterans): excise and stage with buccal graft; redo repairs use buccal (lip, inside glans) mucosa, Cecil scrotal parasitisation, tissue expansion.

### Hypospadias fistula: definition and incidence

- A UCF (urethrocutaneous fistula) is a urethral leak anywhere below the meatus: the **commonest complication** of hypospadias repair, just under 10% at short-term follow-up.
- After TIP: 5.7% (**Pfistermuller 2015**); after proximal repair 3–45%. Usually at the original meatus or coronal margin.

### Hypospadias fistula: aetiology

- **Distal obstruction** (meatal stenosis, stricture): raises voiding pressure and turbulence; the **key correctable cause**.
- **Ischaemia**, oedema, infection, haematoma, tension.
- **Technical:** overlapping suture lines, poorly inverted epithelium, slowly absorbed sutures, **no barrier layer**.
- **Patient:** proximal meatus, glans <14 mm, redo surgery. A long, rigid neourethra raises resistance (Poiseuille's law).

### Hypospadias fistula: prevention

- **Vascularised barrier** (dartos; **tunica vaginalis** in proximal or redo), offset suture lines, stent in complex repairs.

### Hypospadias fistula: evaluation

- Split stream / multiple streams, usually within 2 years.
- Under anaesthesia: **bougie calibration** for distal narrowing, **urethroscopy** if irregular, and a **stress test** (compress proximally and distally, inject saline or methylene blue) for hidden tracts.

### Hypospadias fistula: management

- Rarely closes spontaneously. **Wait 6–12 months**; **correct distal obstruction at the same sitting**.

| Fistula | Repair |
|---|---|
| Small, shaft | Excise the epithelialised tract, inverting subepithelial closure with fine absorbable suture, multiple offset layers, **dartos barrier** |
| Large or thin tissue | Trap-door or island skin flap plus second layer |
| Coronal margin (glans dehiscence) | Implies distal narrowing or poor tissue: **redo urethroplasty and glansplasty** |
| With diverticulum | Excise, or reuse the redundant wall as a flap |

- Recurrence about **20%**; reduce with interposed vascularised layers (dartos, tunica vaginalis, Cecil modification when tissue is scarce).
- Catheter: none for simple repairs, 7–14 days for complex.
- **Also know:** preputial fistula does not involve the urethra; congenital urethral fistula is closed the same way; follow through puberty.

```
UCF → wait 6–12 months → calibrate, urethroscopy, stress test ├─ distal obstruction → correct at same sitting ├─ small → excise + layered closure + barrier flap └─ large/coronal → flap repair or redo urethroplasty
```

### Follow-up and patient-reported outcomes

- About three-quarters of complications appear by 3 years; stricture and curvature appear late. Follow proximal cases **through puberty**.
- Tools: HOSE (Hypospadias Objective Scoring Evaluation, surgeon), PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation, photo panels). Decision aids reduce parental regret.

### Recent advances

- Registries, AI (artificial intelligence) video review, genomic risk profiling, a quality-of-life measure, engineered tissue.

**Diagrams to draw:** curvature algorithm; STAG stage 1 graft and stage 2 tubularisation; fistula repair showing neourethra, dartos barrier and skin with offset suture lines.

**Key points to remember:**

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages: curvature >30° after degloving means stage.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal; STAG has the lowest proximal complications.
- Stage 2 after ≥6 months.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.
- Fistula: commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

## LAQ 5 Wilms tumour, including bilateral Wilms tumour

P2 · P3 **merged from 2**

*Asked as:* P2 LAQ "Wilms tumour: classification, staging and management" · P3 SAQ "Bilateral Wilms tumour: treatment strategies"

### Introduction

- Read as "Wilms tumour" (nephroblastoma); "Wilm's bladder" in one P2 wording is taken as a typo.
- WT (Wilms tumour) is an embryonal tumour of metanephric blastema. It forms 95% of kidney cancers under 15 years; median age 3.5 years; over 80% present before 5 years.

### Genetics, syndromes and precursors

- **Genetics:** mostly sporadic; familial 1–2%. Genes: **WT1 (11p13)**, IGF2/11p15 (commonest change, about 70%), WTX, CTNNB1; **TP53 in anaplasia (in most anaplastic tumours)**.

| Syndrome | WT risk |
|---|---|
| **WAGR** (Wilms, aniridia, genitourinary anomalies, intellectual disability / retardation; WT1 deletion) | 50% |
| **Denys–Drash** (46,XY DSD [disorder of sex development], mesangial sclerosis; WT1 point mutation) | 50% |
| **BWS (Beckwith–Wiedemann syndrome)**, hemihypertrophy (11p15) | 5–10% |

- **Screening:** ultrasound every 3–4 months until 7 years when risk is >1%.
- **Nephrogenic rests** (precursors) occur in over one-third of WT kidneys and over 90% of bilateral cases.

### Classification: histology

| Group | Features | Significance |
|---|---|---|
| **Favourable histology** | **Triphasic:** blastema, epithelium (tubules), stroma; pseudocapsule | Survival >90% |
| **Anaplasia** (unfavourable) | Nuclei ≥3× neighbours (larger), hyperchromasia, abnormal mitoses; **focal** or **diffuse** (diffuse worse than focal) | Chemoresistant; poor even at stage I |

- **SIOP (International Society of Paediatric Oncology) post-chemotherapy risk groups:**

| Risk | Histology |
|---|---|
| Low | Completely necrotic |
| Intermediate | All others (regressive, mixed, epithelial, stromal, focal anaplasia) |
| High | **Blastemal-predominant**; diffuse anaplasia |

- Residual blastema predicts relapse; stromal and epithelial tumours shrink little but are cured by excision.

### Staging

#### COG (Children's Oncology Group): surgical-pathological, before chemotherapy

| Stage | Description |
|---|---|
| I | Confined to kidney, completely resected, capsule intact, no renal sinus vessel invasion |
| II | Beyond kidney (renal sinus vessels, perirenal fat, vein thrombus) but completely resected |
| III | Residual abdominal disease: **positive nodes, any spill, prior biopsy**, positive margin, peritoneal implants |
| IV | Haematogenous metastases (lung commonest, liver, bone, brain) |
| V | Bilateral at diagnosis (each side staged separately) |

- **SIOP staging** uses the same five stages but is applied **after preoperative chemotherapy**. Necrotic, chemotherapy-induced change in the sinus or perirenal fat does not upstage.
- **Biology:** combined **1p and 16q LOH (loss of heterozygosity)** predicts relapse; 1q gain is the next stratifier.

### Clinical features

- Abdominal mass; haematuria about 20%; hypertension about 25% (renin).
- Look for aniridia, hemihypertrophy and varicocele. Varicocele or hepatomegaly suggests IVC (inferior vena cava) thrombus.
- Acquired von Willebrand disease in up to 8%.

### Investigations

- Ultrasound with Doppler (IVC), then contrast CT (computed tomography) or MRI (magnetic resonance imaging) abdomen, then **chest CT** (lung is the commonest metastatic site; do it before anaesthesia).
- Blood count, renal and liver function, coagulation; syndrome review.

### Management: two philosophies

| | COG (North America) | SIOP (Europe) |
|---|---|---|
| Sequence | **Upfront transperitoneal radical nephrectomy**, then chemotherapy | **Preoperative VCR + AMD** (4 weeks; add DOX for 6 weeks if metastatic), no biopsy, then nephrectomy |
| Advantage | Untreated histology and biology for risk assignment | Tumour shrinks; spill almost zero |
| Risk basis | Stage + histology + LOH | Post-chemotherapy histology + stage |

(VCR = vincristine; AMD = actinomycin D (dactinomycin); DOX = doxorubicin; RT = radiotherapy; NSS = nephron-sparing surgery.)

### Surgical principles (radical nephrectomy)

- Transperitoneal approach; palpate renal vein and IVC before ligation; spare a tumour-free adrenal.
- **Avoid spill** (it makes the tumour stage III).
- **Sample at least 5 lymph nodes** (hilar, paracaval, para-aortic, interaortocaval); this is essential, formal dissection is not. Omitted sampling is the commonest protocol violation.
- IVC thrombus at or above the hepatic veins: preoperative chemotherapy to avoid bypass.

### Postoperative therapy (COG, favourable histology)

| Group | Treatment |
|---|---|
| Very low risk (<2 years, <550 g, stage I) | Nephrectomy alone (AREN0532) |
| Stage I–II | VCR + AMD |
| Stage III | VCR + AMD + DOX + flank RT |
| Stage IV | Three drugs + whole-lung RT; lung RT omitted if lung nodules clear by week 6 (AREN0533) |
| Anaplasia or 1p/16q LOH | Intensified regimens (regimen M adds cyclophosphamide + etoposide) |

### Bilateral WT (stage V): principles

- **Synchronous bilateral WT** is **5–7%**: **COG stage V**. Each kidney is staged separately.
- Linked to WAGR, Denys–Drash (about 50% risk) and BWS; **nephrogenic rests in over 90%**.
- Aim: **cure while preserving renal tissue (nephrons)**. Renal failure is about 15% at 15 years; ESRD (end-stage renal disease) mainly follows bilateral nephrectomy.
- **No upfront biopsy and no upfront nephrectomy.**

### Bilateral WT: COG approach (AREN0534)

```
VCR + AMD + DOX × 6 weeks (no biopsy) → imaging at week 6 ├─ resectable / NSS feasible → bilateral NSS └─ not → 6 more weeks chemotherapy → surgery by week 12 poor response → bilateral OPEN biopsy → further therapy by each kidney's stage and histology ± flank RT
```

- **Why stop at 12 weeks:** in NWTS-4 (National Wilms Tumor Study 4), prolonged chemotherapy in poor responders delayed surgery and some tumours became anaplastic. A non-shrinking tumour is often differentiated and is cured by excision.
- **Why an open, bilateral biopsy:** anaplasia is discordant between kidneys in about 4 of 5.

### Bilateral WT: surgery

- **Bilateral NSS** (partial nephrectomy or enucleation); frozen section; lymph node sampling; avoid spill.
- Compressed normal parenchyma is often underestimated, so try NSS in every case. Expert centres achieve bilateral NSS in over 90%.
- Enucleation: positive margins 23–36%, good outcome with flank radiotherapy.
- Unresectable both sides: bilateral nephrectomy and dialysis are rare and only for refractory tumour; transplant after **1–2 years** tumour-free.

### Bilateral WT: adjuvant therapy and SIOP approach

- Adjuvant therapy is guided by the worst histology and margins: anaplasia or positive margin means intensified chemotherapy and flank radiotherapy.
- **SIOP:** same principle (no biopsy, NSS by week 12) but **two drugs, VCR + AMD**, if non-metastatic.

### Unilateral nephron-sparing surgery

- For syndromic children and solitary kidney; SIOP UMBRELLA criteria are volume <300 mL, a substantial remnant and negative nodes.

### Prognosis and outcomes

- Favourable histology: survival >90%.
- **AREN0534 (bilateral):** surgery by week 12 in 84%; bilateral NSS 39%; **4-year EFS (event-free survival) 82.1%, OS (overall survival) 94.9%**, versus 56% and 80.8% on NWTS-5.
- Salvage survival after relapse: 70–80% after VCR/AMD only; about 10% for anaplastic or blastemal relapse.

### Late effects and follow-up

- **Late effects:** cardiotoxicity (DOX), second cancers (RT), infertility, renal failure (bilateral).
- **Follow-up (bilateral):** lifelong blood pressure, proteinuria, creatinine (renal function) and imaging, as late relapse occurs.

**Diagrams to draw:** COG staging diagram showing capsule, renal sinus, nodes and spill; bilateral tumours with planned partial nephrectomies; AREN0534 timeline (weeks 0, 6, 12).

> **Key points to remember:**
>
> - WT1 at 11p13 (WAGR, Denys–Drash); BWS at 11p15.
> - Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
> - Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
> - COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
> - Bilateral: stage V, 5–7%; no upfront nephrectomy or biopsy; three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12, NSS; SIOP uses two drugs.
> - AREN0534: 4-year OS about 95%; bilateral NSS to avoid ESRD.

## LAQ 6 Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair

P2 · P3 **merged from 2**

*Asked as:* P3 LAQ "Embryology of the penis and management of proximal hypospadias (including Koyanagi repair)" · P2 SAQ "Perineo-scrotal hypospadias"

### Embryology of the penis

#### Indifferent stage (up to 8–9 weeks)

- Cloacal folds form the **genital tubercle**; after the cloaca divides (7 weeks), **urethral folds** flank the urogenital membrane, with **labioscrotal swellings** laterally. Both sexes look identical until about 9 weeks.

#### Hormonal control

- SRY (sex-determining region Y) drives the testis, whose **Leydig cells (about 8 weeks)** secrete testosterone.
- **Type 2 5α-reductase** in the tubercle converts it to **DHT (dihydrotestosterone)**, which acts on androgen receptors. **DHT at 8–12 weeks** is the key mediator; the critical window is weeks 8–16. Without androgen action the default is female.
- First sign: longer anogenital distance, then tubercle elongation (sonic hedgehog-driven).

| Primordium (germ layer) | Male | Female |
|---|---|---|
| Genital tubercle (mesoderm: corpora; ectoderm: skin, prepuce) | Glans, corpora | Clitoris |
| Urethral folds | Fuse to form penile urethra, median raphe | Labia minora |
| Labioscrotal swellings | Scrotum | Labia majora |
| Urethral plate (endoderm) | Whole urethra, including glanular | Vestibule |

#### Urethral formation: the "double zipper"

1. **Opening zipper** (both sexes): the urethral plate canalises proximal to distal into a urethral groove.
2. **Closing zipper** (androgen-dependent, male only): the folds fuse in the ventral midline, forming the tubular urethra and raphe. - The **glanular urethra** forms by direct canalisation of the endodermal plate; Glenister's ectodermal ingrowth theory (1954) is outdated. - Timetable: coronal sulcus 11–12 weeks; shaft fused by 16 weeks; complete by about 18 weeks; fetal ventral curvature gone by 20 weeks. - **Prepuce** fuses dorsally first; ventral fusion awaits urethral closure, hence the **dorsal hood** in hypospadias. - Dorsal nerves run at 11 and 1 o'clock, leaving a **12 o'clock nerve-free zone** (safe plication site). - **Hypospadias = arrest of the closing zipper**: the meatus stops where closure stopped, and unfused dysplastic ventral tissue tethers the shaft into curvature.

**Diagram to draw:** double-zipper urethral formation.

### Definition of proximal (perineoscrotal) hypospadias

- Meatus on the proximal shaft, at the penoscrotal junction, scrotum or perineum (10–25% of hypospadias).

- Perineoscrotal hypospadias is a severe form, with severe ventral curvature, a short penis, bifid scrotum and often **penoscrotal transposition**.
- Cause: failed androgen-dependent urethral fold fusion (8–16 weeks).

### Evaluation

- Severity is judged **after degloving**: meatus, plate and curvature (GMS (glans–meatus–shaft) score).
- Examine stretched penile length, glans width, penoscrotal transposition, bifid scrotum and **both testes**.
- **DSD (disorder of sex development) work-up** if hypospadias plus an undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).
- DSD risk is 64% in proximal versus 7% in distal cases. A 46,XY karyotype can still hide 5α-reductase deficiency or androgen insensitivity.
- Prostatic utricle is commoner; cystoscopy in severe cases. No routine VCUG (voiding cystourethrogram) or renal ultrasound.

### Preparation

- **Timing:** 6–18 months (AAP (American Academy of Pediatrics): 6–12 months).
- **Preoperative androgen** if glans width <14–15 mm: e.g. testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Side effects are transient; evidence is mixed, but two RCTs (randomised controlled trials) showed fewer complications.

### Curvature correction (decides one- versus two-stage)

```
Deglove + excise dysplastic dartos + artificial erection ├─ straight → urethroplasty ├─ residual <30° → dorsal midline plication (12 o'clock) └─ >30° → divide urethral plate └─ still >30° → ventral corporotomy ± graft → staged repair
```

| | Dorsal plication (Nesbit, Baskin) | Ventral lengthening |
|---|---|---|
| Method | Shortens the convex dorsal side | 1–3 transverse corporotomies, or one full-thickness 9-to-3 o'clock corporotomy with dermis, tunica vaginalis or SIS (small intestinal submucosa) graft |
| Use | Mild (≤15–30°) | Moderate–severe |
| Recurrence | 28% | 9% |

**Diagram to draw:** curvature algorithm.

### Urethroplasty options

| Approach | Technique | Comment |
|---|---|---|
| One-stage **proximal TIP (tubularised incised plate)** | Plate kept, incised and tubularised | Recurrent curvature, stricture: largely abandoned for severe cases |
| One-stage **onlay island flap** | Inner preputial flap on preserved plate | Diverticulum, fistula; higher complications |

| Approach | Technique | Comment |
|---|---|---|
| One-stage **TPIF (transverse preputial island flap, Duckett tube)** | Tube after plate division | High complications |
| One-stage **Koyanagi** | Meatus-based parameatal foreskin flap (see below) | No proximal circular anastomosis |
| **Two-stage Bracka / STAG (staged tubularised autograft)** | Stage 1: correct curvature, divide plate, quilt inner-prepuce or **buccal graft**. Stage 2 (≥6 months): tubularise, glansplasty, barrier layer | **Preferred for most proximal cases**; STAG lowest complications (about 23%) |
| **Two-stage Byars** | Dorsal preputial flaps moved ventrally, tubularised later | Supple flap bed; complications 43–81% |

- **Why stage?** A urethroplasty laid over a fresh corporal graft fails more often; staging gives a supple, vascular bed.
- Principles: magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), fine absorbable sutures, no hair-bearing skin. Stent, compressive dressing, oxybutynin; **scrotoplasty** for transposition.

**Diagram to draw:** two-stage graft repair: stage 1 graft bed, stage 2 tubularisation.

### Koyanagi repair: definition and principle

- **One-stage repair of proximal hypospadias** with a **meatus-based parameatal foreskin flap** (Koyanagi). Skin strips round the meatus continue into the inner prepuce on both sides; they are swung ventrally and tubularised.
- **Principle:** the neourethra stays attached to the native meatus, so there is **no circular proximal anastomosis**, and the urethral plate can be divided to correct severe chordee.

### Koyanagi repair: indications and contraindications

- **Indications:** proximal hypospadias (penoscrotal, scrotal, perineal) with **severe curvature** needing plate transection, and an **ample, unoperated prepuce**.
- **Contraindications:** prior circumcision or failed repair; small glans (give preoperative testosterone); curvature needing a ventral corporal graft (stage instead).

### Koyanagi repair: procedure

`Stay suture; artificial erection → incise round the meatus, along both sides of the plate to the corona and round the inner prepuce (meatus-based "wings") → deglove; divide the plate; excise dysplastic dartos → residual curvature <30° → dorsal midline plication → split the dorsal prepuce; bring both wings ventrally on dartos pedicles → join inner edges, then outer edges, over a catheter → tube → glanuloplasty, meatus at the tip → dartos barrier layer; skin cover ± scrotoplasty → stent 7–10 days; compressive dressing`

- Modified and "neo-modified" versions add flap fixation and barrier layers to reduce fistula and diverticulum.

**Diagram to draw (Koyanagi):** meatus-based foreskin wings split dorsally and joined ventrally into a tube.

### Koyanagi repair: complications

| Complication | Cause |
|---|---|
| **Fistula** (commonest) | Long suture line; repair ≥6 months later |
| Meatal stenosis, stricture | Distal flap ischaemia |
| **Diverticulum** | No spongiosum; flap not fixed to the shaft |
| Glans dehiscence | Tension |
| Residual or recurrent curvature | Plication alone in severe cases |
| Flap necrosis, infection, poor cosmesis | Pedicle injury |

- Because proximal repairs overall have **23–68%** complications, many now prefer **two-stage repair** (STAG, Bracka).

### Complications and outcomes of proximal repair

- Complications **23–68%**; 25–75% need further surgery after planned staging.
- **Fistula** is the commonest (3–45%; repair ≥6 months later; exclude distal obstruction).
- Others: glans dehiscence, meatal stenosis (<8 Fr before puberty), stricture, **diverticulum** (no spongiosum), **recurrent curvature >30°**, BXO (balanitis xerotica obliterans), hair-bearing urethra.
- A long, rigid neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum.

### Follow-up

- Follow through puberty; patient-reported outcomes (PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation) score).

> **Key points to remember:**
> - DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
> - Hypospadias (including perineoscrotal) plus undescended testis: DSD work-up.
> - <30° plication; >30° divide plate ± ventral graft and stage.
> - STAG/Bracka staged repair preferred; fistula commonest complication.
> - Koyanagi: one stage; meatus-based parameatal foreskin flap; plate divided; no proximal circular anastomosis.
> - Koyanagi: fistula commonest; diverticulum, meatal stenosis.

## LAQ 7 Urological evaluation and management of pediatric kidney transplant patient.

P1

**Type:** LAQ (25 marks)
**Source:** 2022-Summer Paper 1, Q2

### Introduction

- Children needing KRT (kidney replacement therapy) often have CAKUT (congenital anomalies of the kidney and urinary tract). The urologist must ensure that the bladder that destroyed the native kidneys does not destroy the graft.
- **Five goals:** unobstructed drainage; a capacious, low-pressure reservoir with social continence; volitional emptying; no infection; the fewest operations.

### Causes of ESKD (end-stage kidney disease) in children

- Commonest age at transplant 13–17 years.
- **Younger:** dysplasia, hypoplasia, aplasia and obstructive uropathy, mainly PUV (posterior urethral valves). **Older:** FSGS (focal segmental glomerulosclerosis) and neuropathic bladder.

### Pretransplant evaluation

- **Every child:** urological history, renal ultrasound, uroflow with PVR (post-void residual) if toilet-trained.
- **VCUG (voiding cystourethrogram)** only with neuropathic bladder, PUV, VUR (vesicoureteral reflux), previous lower-tract surgery, recurrent febrile UTI (urinary tract infection) or unexplained hydronephrosis.
- **Urodynamics** for neuropathic bladder, PUV or large ureterocele, refractory voiding dysfunction, high-grade hydronephrosis or recurrent UTI. They measure capacity, compliance, detrusor activity, emptying and sphincter function, and are repeated.

```
History positive → ultrasound + VCUG → urodynamics Poor capacity/compliance → anticholinergic, CIC, cycling → still poor → augment Poor emptying → CIC ± catheterisable stoma Adequate → clear for transplant History negative → ultrasound + PVR → normal → clear
```

- An anuric child with no urological history needs no further tests.

### Pretransplant management

**1. Bladder optimisation (drugs → CIC → cycling → augmentation)**

- Commonest problem: low capacity, detrusor overactivity, poor compliance. High pressure harms the kidney even without reflux.
- **Target:** age-expected capacity at detrusor pressure **<40 cm $H_2O$**.
- Antimuscarinics (oxybutynin titrated; solifenacin), mirabegron, intradetrusor botulinum toxin; α-blockers aid emptying. Better storage may worsen emptying, so add **CIC (clean intermittent catheterisation)**. Prove volitional emptying by repeated low PVRs.
- A family's adherence to CIC predicts adherence to immunosuppression.

**2. Valve bladder versus neuropathic bladder**

- PUV bladder moves from overactivity to **myogenic failure**: postpone augmentation. Neuropathic bladder may deteriorate under high post-transplant urine output: reconstruct

earlier.

3. **Defunctionalised bladder** (anuria or diversion)

- **Bladder cycling:** catheter instillation of increasing saline volumes, with drugs, until age-expected capacity; confirm by urodynamics. **Never use the graft to test the bladder.**

4. **Augmentation**

- Indication: cannot reach **>75% expected capacity at <40 cm $H_2O$** with CIC every 3–4 hours despite maximal therapy. It does not clearly raise transplant complications.

| Segment | Point |
|---|---|
| Dilated ureter (ureterocystoplasty) | Explore first: urothelium, no mucus or acidosis |
| Ileum, sigmoid | Most used; mucus, stones, infection, malignancy; lifelong follow-up |
| Stomach | Haematuria–dysuria syndrome in anuric patients; rarely used |

- Continent stoma (Mitrofanoff) if the urethra is sensate or difficult; site it medial to the planned incision. Allow **6–12 weeks** healing before transplant.

5. **Infection, nephrectomy and dialysis**

- Treat stasis (double voiding, CIC); distinguish colonisation from UTI; consider circumcision in boys with valves.
- **Native nephrectomy** (default is to keep): uncontrolled hypertension, nephrotic syndrome, massive polyuria, recurrent pyelonephritis, large stones, malignancy risk, very large kidneys, massive VUR. Retroperitoneal route keeps PD (peritoneal dialysis).
- Keep a normal native ureter for later ureteroureterostomy. Intraperitoneal surgery on PD needs temporary haemodialysis.

## At transplantation

- Retroperitoneal Gibson incision; children **<15 kg** may need aorta–caval anastomosis, sometimes transperitoneal.
- **Ureter:** extravesical non-refluxing ureteroneocystostomy (3–4 cm detrusorotomy) preferred; Barry tunnel; ureteroureterostomy to a normal native ureter.
- **Stent:** selective; double-J removed at 4–6 weeks, or early removal at 5–7 days.

## Post-transplant complications

| Complication | Rate | Management |
|---|---|---|
| Ureteric obstruction | 6% | Usually distal ischaemic stenosis; >half within 100 days. Stent or nephrostomy; balloon dilatation; open repair (native ureter, bladder flap) |
| Urine leak | 3% | Early; drain creatinine; catheter, nephrostomy, stent, then reimplantation |
| Graft VUR | 2% (significant) | Febrile UTI → reimplant; asymptomatic → observe; injection works in one-third |
| Stones | 2% | Treat if >1 cm and growing, obstructing or infected |

- **Rising creatinine with hydronephrosis:** catheter drainage 1–2 weeks (bladder cause?) → biopsy if rejection is suspected → stent or nephrostomy. Obstruction and rejection can coexist.

- Also: lymphocele, **PTLD (post-transplant lymphoproliferative disorder)**, graft thrombosis (a leading cause of early graft loss), late infection reflecting bladder dysfunction.

### Follow-up

- Lifelong ultrasound, PVR and bladder surveillance; recurrent UTIs lower graft GFR (glomerular filtration rate).

**Diagram to draw:** extravesical ureteroneocystostomy with detrusor flaps closed over the ureter.

> **Key points to remember:**
> - Bladder target: age-expected capacity at <40 cm $H_2O$; augment if <75% despite CIC.
> - PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.
> - Never test a bladder with a graft; cycle the defunctionalised bladder.
> - Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

## LAQ 8 Embryology and management of ectopic ureter.

P3

**Type:** LAQ (25 marks)
**Source:** 2018-Summer Paper 3, Q1

### Definition

- An **ectopic ureter** is any ureter, single or duplex, that does **not open on the trigone**. True ectopia means an orifice **at or distal to the bladder neck**.
- Commoner in girls. In girls most ectopic ureters drain the **upper pole of a duplex kidney**; in boys most are **single-system**.

### Embryology

- **Week 4–6:** the **ureteric bud** arises from the **mesonephric (wolffian) duct** and induces the metanephric blastema to form the kidney.
- The duct segment distal to the bud is absorbed into the **urogenital sinus**; the ureteric orifice then moves cranially and laterally, the wolffian orifice caudally to the verumontanum.
- **Mackie–Stephens bud theory:**

| Bud position on wolffian duct | Absorption | Orifice | Result |
|---|---|---|---|
| Too caudal | Early | High and lateral, short tunnel | VUR (vesicoureteral reflux) |
| Too cranial | Late or never | Low and medial (bladder neck, urethra) or stays on wolffian duct | **Ectopic ureter**, often obstructed |

- A bud from the wrong place meets poor blastema, so **renal dysplasia increases with the degree of ectopia**.
- **Duplex system:** the upper-pole bud is more cranial, absorbed later, and ends **lower and more medial** (**Weigert–Meyer law**): upper pole obstructs; lower pole refluxes.

### Sites of ectopia (the key sex difference)

| | Female | Male |
|---|---|---|
| Sites (commonest first) | Urethra, vestibule, vagina; rarely cervix, uterus, rectum (via **Gartner duct**) | Posterior urethra, seminal vesicle; also vas, ejaculatory duct |
| Relation to sphincter | Can open **below** the external sphincter | Always **above** the external sphincter |
| Continence | **Continuous dribbling with normal voiding** | **Never incontinent** |
| Typical presentation | Constant dampness, perineal pus, UTI (urinary tract infection) | **Epididymo-orchitis**, UTI, flank pain, hydronephrosis |

- **Zinner syndrome:** ectopic ureter into the seminal vesicle with ipsilateral renal agenesis or dysplasia and a seminal vesicle cyst.

### Clinical features

- **Prenatal hydronephrosis**: most are detected this way.

- **Girl:** wet day and night despite normal voids; she is never dry for 30–60 minutes. Often misdiagnosed as voiding dysfunction.
- **Infection:** low-grade fever with spikes; urine culture may be negative because the infected moiety does not drain into the bladder.
- **Boy:** recurrent epididymitis in a young boy demands an upper-tract ultrasound.
- **Also know:** pain then perineal discharge; bilateral single-system ectopia with small bladder and wide bladder neck.

### Evaluation

| Test | What it shows |
|---|---|
| **Ultrasound** (first) | Dilated upper pole and ureter behind the bladder; parenchyma. Bladder view separates it from ureterocele |
| **VCUG** (voiding cystourethrogram) | Lower-pole or contralateral VUR ("drooping lily"), reflux into the ectopic ureter |
| **DMSA** (dimercaptosuccinic acid) | Gold standard for moiety function ("some" versus "none"); MAG3 (mercaptoacetyltriglycine) if drainage needed |
| **MR (magnetic resonance) urography** | Non-dilated upper pole, small dysplastic kidney, complex anatomy |
| **Examination / cystovaginoscopy** | Perineal orifice between urethra and vagina, cannulated for contrast |

- **Also know:** clear urine leaking despite methylene blue in the bladder suggests an ectopic ureter; CT for a small unseen kidney.

### Management

**Goals:** preserve renal function; remove infection, obstruction and reflux; achieve continence; with least morbidity. Start antibiotic prophylaxis in the neonate with significant dilation.

`Neonate with sepsis or massive dilation → end cutaneous ureterostomy (or nephrostomy); DMSA later DUPLEX system Upper pole non-functioning → upper-pole heminephroureterectomy Upper pole worth keeping → ureteroureterostomy (UU) / pyeloureterostomy + lower-pole VUR → common sheath reimplant (or lower-pole reimplant + distal UU) SINGLE system Functioning → ureteral reimplant (± tapering) Non-functioning → nephroureterectomy (laparoscopic) Bilateral → bladder neck reconstruction ± augmentation`

- **Upper-pole heminephroureterectomy** (open, laparoscopic or robotic): upper-pole ureter as a handle, passed behind the hilum; ligate polar vessels; excise the pole; close over fat. Dissect on the ureter wall to protect the **lower-pole blood supply**. The stump is rarely a problem.
- **Ureteroureterostomy (UU):** end-to-side upper into lower pole ureter, distal (inguinal incision) or proximal (preferred with big size mismatch). **Stent the recipient lower-pole ureter** cystoscopically for identification. Now used even for non-functioning poles; feared hypertension has not been shown.
- **Common sheath reimplantation:** do not separate the two ureters (they share blood supply).

### Complications

- Loss of lower-pole function (most serious); urinoma (up to 20% after minimally invasive partial nephrectomy, mostly harmless); stump infection; new reflux; persistent incontinence.

### Recent advances

- Robotic UU and heminephrectomy; single-site surgery; non-dismembered side-to-side ureterovesicostomy in infants.

**Diagram to draw:** duplex kidney showing Weigert–Meyer orifices, and female versus male ectopic sites relative to the external sphincter.

> **Key points to remember:**
> - Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
> - Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
> - DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
> - Protect the lower-pole blood supply; stent the recipient ureter for UU.

## Short answer questions (SAQ)

## SAQ 1 TIP (Snodgrass) repair for hypospadias

P1 · P3 **merged from 2**

*Asked as:* P3 SAQ "Snodgrass (TIP) repair for hypospadias, including extended indications" · P1 SAQ "TIP repair in hypospadias"

### Introduction

- "Shodgrass" in one P3 wording is taken as the Snodgrass TIP repair, and "distal hypos." as distal hypospadias.
- **TIP (tubularised incised plate) urethroplasty** was described by Snodgrass in 1994, built on Rich's hinging incision (1989).
- Designed for distal hypospadias, it is now used for **over 90% of distal and about 80% of midshaft repairs**.
- Distal hypospadias forms 70–85% of cases.

### Principle

- A **deep midline relaxing incision** of the urethral plate widens a narrow plate so that it can be tubularised **without tension**.
- The incised raw area **heals by re-epithelialisation** without a graft, giving a **vertical slit-like meatus**.

### Indications, prerequisites and assessment

- Distal and midshaft hypospadias, and redo cases with a healthy plate, when curvature is mild (**<30°** after degloving) and corrected by degloving or dorsal plication **without dividing the plate**.
- Supple, well-vascularised plate of adequate width and depth, and an adequate glans.
- **Glans width <14 mm** raises complications (glans width ≥14 mm is needed): give preoperative testosterone to enlarge it.

- **POST (plate objective scoring tool):** a low ratio (AB/BC ratio) of neomeatus length to glans fusion length means a poor plate, which favours a dorsal inlay graft.
- Timing: 6–12 months of age.
- Not for severe curvature needing plate division; proximal TIP is largely abandoned.

### Extended indications

| Indication | Rationale | Caveat |
|---|---|---|
| **Midshaft hypospadias** | Same principle | Good results |
| **Proximal hypospadias with mild curvature**, good plate | Single stage; no flaps | More fistula, stricture, recurrent curvature; Campbell-Walsh authors now prefer staging |
| **Redo or failed repairs** with preserved plate | Plate often survives | Worse than primary; inlay graft if scarred |
| **MIP (megameatus intact prepuce)** | Wide, deep plate | GAP (glans approximation procedure) alternative |
| **Narrow or shallow plate**: TIP + **dorsal inlay graft** ("Snodgraft") | Preputial or buccal graft fills the incision, so it cannot contract | Less meatal stenosis |
| **Adults, circumcised** boys | No preputial tissue needed | Success similar |

- **Also know:** in redo cases with poor blood supply a dorsal buccal inlay beats onlay or tube; incise-and-graft is also used in adult urethral stricture (Asopa dorsal inlay) and female urethroplasty.

### Contraindications

- Curvature >30° needing plate division; absent or scarred plate; penoscrotal transposition. Use staged repair instead (Bracka, STAG [staged tubularised autograft], Byars).

### Operative steps

1. Glans stay (traction) suture; tourniquet.
2. U-incision around the plate and meatus; circumscribe and deglove; artificial erection.
3. Separate (mobilise) the glans wings deeply from the plate.
4. **Midline incision of the plate** from inside the meatus to the end of the plate, down to the corpora.
5. Tubularise over a **6–8 Fr stent** with subepithelial 6-0/7-0 absorbable sutures, from distal to proximal, placing the first stitch about 3 mm (2–3 mm) short of the plate end to leave an oval meatus.
6. **Dartos flap** (dorsal preputial) over the neourethra as a barrier layer.
7. Two-layer glansplasty; mucosal collar; skin closure with Byars flaps, or preputioplasty.
8. Stent 5–7 days; compressive dressing; oxybutynin for bladder spasm.

### Complications and outcomes

- Pfistermuller meta-analysis (2015, 4,675 boys): complications 10.6%, reoperation 4.5%; **fistula 5.7%**, **meatal stenosis 3.6%**, stricture 1.3%.
- **Meatal stenosis is commoner than after Mathieu**; glans dehiscence also occurs.
- Distal repair success 83–95%.

- **Proximal and redo cases do significantly worse**; a dartos barrier layer improves results and reduces fistula.

### Modifications

- **Dorsal inlay graft (Snodgraft)**: inner preputial skin graft in the incised bed for a shallow plate.
- Lateral plate incisions to 12 mm width.
- **Also know:** MAGPI (meatal advancement and glanuloplasty incorporated), Mathieu, Thiersch–Duplay, onlay flap.

**Diagram to draw:** cross-sections showing plate incision, inlay graft, tubularisation and dartos cover.

> **Key points to remember:**
> - Snodgrass 1994; incise and tubularise; heals by epithelialisation.
> - Slit-like meatus; dartos barrier reduces fistula.
> - Glans <14 mm and poor plate raise complications.
> - Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
> - Extended to midshaft, selected proximal, redo, MIP and adults.
> - Key condition: curvature <30° without plate division.
> - Poor plate: dorsal inlay graft ("Snodgraft").
> - Proximal and redo TIP fail more often.

## SAQ 2 Glandular hypospadias

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 11 Paper 1, Q3b

*"Glandular" is read as glanular hypospadias: meatus on the glans.*

### Definition

- The mildest **distal (anterior)** hypospadias: the meatus opens on the ventral glans, proximal to the normal tip.
- Hypospadias occurs in 1 in 150–300 male births; distal forms are 70–85%.
- Usual features: **dorsal hooded foreskin**, ventral glans tilt, little or no curvature. **MIP (megameatus intact prepuce)** variant (up to 5%): wide meatus with normal foreskin.

### Embryology

- Urethral folds close from proximal to distal during weeks 8–16 under **DHT (dihydrotestosterone)**.
- Glanular urethra: ectodermal ingrowth (classic) or endodermal plate ("two-zipper", current view). Arrest at the last step gives a glanular meatus.

### Evaluation

- Examine meatus, plate, glans size, curvature, torsion; retract the foreskin before any **newborn circumcision**.
- **GMS (glans–meatus–shaft) score** (each 1–4, total 3–12) predicts complications.

- No workup needed for isolated distal hypospadias; DSD (difference of sexual development) workup if a testis is undescended.

### Indications and timing

- Deflected or spraying stream, meatal stenosis, curvature, cosmesis. A straight penis that voids standing may need no surgery; counsel parents.
- **Age 6–18 months** (AAP, American Academy of Pediatrics, 6–12 months).

### Operations

| Procedure | Suitable for | Key point |
|---|---|---|
| Urethromeatoplasty | Stenotic meatus, blind pit | Heineke–Mikulicz: vertical incision, horizontal closure |
| **MAGPI** (meatal advancement and glanuloplasty, Duckett 1981) | Glanular meatus, mobile skin | Advance meatus, two-layer glanuloplasty; retracts if used more proximally |
| MIV (M inverted V) glansplasty | Glanular or coronal | Avoids MAGPI retraction |
| **GAP** (glans approximation) | Wide deep groove, fish-mouth meatus (MIP) | Tubularise and close glans wings |
| **TIP** (tubularised incised plate, Snodgrass 1994) | Narrow plate | Most used; slit-like meatus |
| Mathieu flip-flap; Thiersch–Duplay | Selected | Stenosis, fistula |

- Preputioplasty or circumcision; dartos barrier layer.

### Complications

- Meatal stenosis or retraction, fistula, **glans dehiscence**, residual curvature, poor cosmesis. Distal repairs succeed in about 80%.

**Diagram to draw:** MAGPI steps (meatal advancement, glans wings closed beneath).

> **Key points to remember:**
> - Glanular = most distal hypospadias; dorsal hood, minimal curvature.
> - MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
> - Repair at 6–18 months; may be left alone if function is normal.

## SAQ 3 Vanishing testes

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 1, Q3b

### Definition

- A testis that **developed normally and was then lost**, usually from **torsion or a vascular accident in fetal life**, after genital masculinisation but before scrotal fixation.
- Unilateral loss gives **monorchia**; bilateral loss gives **anorchia** (testicular regression syndrome), which is rare.
- Excised "nubbins" contain haemosiderin.

### Incidence

- **15–40% of nonpalpable testes** at surgery.

### Clinical features

- **Contralateral compensatory hypertrophy:** volume >2 mL (sensitivity 72%, specificity 100%) or length about 19–20 mm suggests a solitary testis.
- No palpable intrascrotal "appendage" tissue; sometimes a scrotal nubbin.
- These aid counselling but **do not replace exploration**.
- **Anorchia:** micropenis in nearly half.

### Evaluation

- **Imaging** (ultrasound, MRI [magnetic resonance imaging]) is not recommended: it does not change the operation.
- **Bilateral nonpalpable:** exclude DSD (difference of sex development) and CAH (congenital adrenal hyperplasia) first.
- Anorchia: high FSH (follicle-stimulating hormone) and LH (luteinising hormone); low AMH (anti-Müllerian hormone) and inhibin B; no rise after hCG (human chorionic gonadotrophin). Most reliable during mini-puberty.

### Management: diagnostic laparoscopy (after examination under anaesthesia)

| Laparoscopic finding | Action |
|---|---|
| **Blind-ending vessels and vas** above a closed ring | Diagnosis confirmed; stop |
| Vessels and vas entering the internal ring | Inguinal or scrotal exploration; excise the nubbin |
| Testis seen | Orchidopexy (single stage or Fowler–Stephens) |
| Neither vas nor vessels | Search up to the kidney (agenesis very rare) |

- **Excise the remnant:** viable germ cells or tubules in 5–15%.
- **Scrotal-first exploration** is reasonable with a palpable nubbin and contralateral hypertrophy (length ≥1.8 cm); laparoscopy is then needed in about a quarter.
- **Contralateral fixation:** debated; not required by the AUA (American Urological Association) algorithm.
- **Prosthesis:** offer at least 6 months after scrotal surgery or after puberty, inguinal approach.
- **Anorchia:** testosterone from puberty. **Also know:** agenesis shows ipsilateral Müllerian persistence.

**Diagram to draw:** laparoscopic view of blind-ending vessels at a closed ring.

> **Key points to remember:**
> - Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
> - Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
> - Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
> - Anorchia: high FSH/LH, low AMH and inhibin B.

## SAQ 4 Pelvicalyceal anomalies other than PUJ obstruction

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 1, Q3d

## Introduction

- Congenital pelvicalyceal variants occur in about 4% of people.
- **Key question:** is there obstruction? Dilatation without obstruction needs no treatment; stasis causes stones and infection.

## Anomalies

| Anomaly | Features | Management |
|---|---|---|
| **Calyceal diverticulum** | Transitional-lined intrarenal cavity joined to a calyx (Type I, commoner, small) or pelvis (Type II, larger, symptomatic) by a narrow neck; 4.5 per 1000; congenital (persistent 3rd–4th generation ureteric branches) or acquired. **Stones in almost 40%**; milk of calcium | Asymptomatic: observe. Pain, infection, stones: ureteroscopic neck incision, **percutaneous ablation with fulguration**, laparoscopic marsupialisation; SWL (shock wave lithotripsy) ineffective |
| **Hydrocalycosis** | Dilated major calyx connected to the pelvis; infundibular obstruction by vessel, stenosis, scarring or achalasia | Symptomatic only: infundibulotomy or dilatation |
| **Megacalycosis** | **Non-obstructive**; malformed papillae; **12–20 calyces**; pelvis normal; male 6:1; normal diuretic renogram | **None**; may coexist with segmental megaureter |
| **Infundibular / infundibulopelvic stenosis** | Narrow infundibula ± small pelvis; usually bilateral, with VUR (vesicoureteric reflux), dysplasia; over a third reach renal failure (hyperfiltration) | Monitor creatinine and eGFR (estimated glomerular filtration rate); ACE (angiotensin-converting enzyme) inhibitors; surgery only if obstruction harms function: endoscopic, calicocalicostomy, ureterocalicostomy |
| **Extrarenal pelvis** | Large pelvis outside the sinus; mimics dilatation | None |
| **Bifid pelvis** | About 10%; normal variant | None |
| **Unipapillary kidney** | Single papilla and calyx; rare; usually hypoplastic kidney | None |

- **Also know:** acquired infundibular stenosis (tuberculosis, post-PCNL (percutaneous nephrolithotomy)); Fraley syndrome (crossing artery on upper infundibulum).

## Evaluation

- **Ultrasound:** cyst-like space; milk of calcium layers and shifts with position.
- **CT (computed tomography) urography or MRU (magnetic resonance urography): delayed contrast pooling** in a diverticulum.
- **Diuretic renogram (MAG3 (mercaptoacetyltriglycine)):** separates obstruction (hydrocalycosis) from non-obstruction (megacalycosis, extrarenal pelvis).
- **Retrograde pyelography:** maps the neck.

### Differential diagnosis

- Obstructive hydronephrosis, papillary necrosis, pyelonephritic clubbing, tuberculosis, renal cyst.
- **Stones:** retrograde first (find, dilate or laser-incise the ostium); percutaneous for lower-pole diverticula.

**Diagram to draw:** kidney showing diverticulum with narrow neck, hydrocalyx and megacalyces.

> **Key points to remember:**
> - Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
> - Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
> - Infundibulopelvic stenosis: bilateral, renal failure risk.

## SAQ 5 Retractile testis?

P2

**Type:** SAQ (10 marks)
**Source:** 2020-Summer Paper 2, Q3b

### Definition

- A **descended testis** that moves up out of the scrotum because of an active **cremasteric reflex**. It can be brought to the bottom of the scrotum by hand and **stays there**, at least for a while, after release.
- It is not undescended, but may be a mild form of the same disease and can later become an acquired undescended testis.

### Epidemiology

- Common: suprascrotal in up to one-third of 4-year-olds; 11–15% of boys up to 11 years. Nearly all are scrotal after 12 years.

### Examination (diagnosis is clinical)

- Warm room and hands, relaxed child, soap or gel; supine frog-leg (thigh abduction inhibits the cremasteric reflex), cross-legged or squatting.
- One hand sweeps from the internal ring down the canal; the other pulls the testis to the scrotal bottom. **Sustained gentle traction tires the cremaster.**
- Note size, mobility, contralateral testis and hernia.

| Testis | Finding | Action |
|---|---|---|
| **Retractile** | Reaches scrotal bottom and stays | Observe yearly |
| Gliding (unstable high scrotal) | Pulled down but springs back | Treat as undescended |
| Undescended (palpable) | Cannot reach mid-scrotum | Orchidopexy by 18 months |
| Ascending (acquired) | Previously scrotal, now not | Orchidopexy |

### Natural history

- Ascent: 2.6% in a prospective birth cohort; 7–32% in specialist series.
- Operated retractile testes show small volume, patent processus vaginalis and epididymal anomalies, suggesting a continuum with cryptorchidism.

### Management

- No ultrasound (adds nothing); no hormones (hCG [human chorionic gonadotrophin] neither separates nor treats).
- Reassure parents; **examine yearly** (AUA [American Urological Association] 2014) until puberty, because some testes ascend.
- **Orchidopexy** if it ascends, cannot be brought down, rests mostly high or becomes smaller. If unsure, examine under anaesthesia.

**Diagram to draw:** inguinoscrotal sketch: testis drawn by traction to the scrotal bottom.

> **Key points to remember:**
> - Retractile = comes down and stays down.
> - Gliding testis springs back: treat as undescended.
> - Yearly examination until puberty; operate if it ascends.

## SAQ 6 Retroileal Ureter

P2

**Type:** SAQ (10 marks)
**Source:** 2025-Winter Paper 2, Q3b

*"Retroileal" is taken as **retroiliac ureter**, the nearest real entity; retrocaval ureter is compared below (full answer: 2017-Winter Paper 3, Q3d).*

### Definition

- The ureter passes **behind the common iliac artery** and is compressed there.
- Better called "**preureteral iliac artery**": the anomaly is presumed vascular; the ureter is trapped.

### Embryology

- Normally the **ventral root of the umbilical artery** is replaced by a more **dorsal** branch, which lies behind the ureter.
- If the **ventral root persists** and the dorsal root fails, the ureter is trapped behind the artery.

### Features

- **Very rare**; either side; **bilateral** cases reported.
- Obstruction at **L5–S1**.
- **Associated anomalies are common**, especially **vasal and mesonephric duct ectopia**: ureter and vas draining together into the posterior urethra; vasa ending in the ureters; ectopic vaginal ureter.

### Presentation and diagnosis

- Flank pain, UTI (urinary tract infection) or hydroureteronephrosis down to the iliac crossing.
- **CT urography (computed tomography urography) with arterial phase** or MRI (magnetic resonance imaging) shows ureter and artery together; **diuretic renogram** for function and drainage.
- Caution: a vessel indenting an already dilated ureter may not be the cause.

### Management

- Non-obstructed: observe.
- Obstructed: **divide the ureter, re-route it in front of the artery** and do a spatulated ureteroureterostomy (or reimplantation) over a double-J stent; open, laparoscopic or robotic.

### Comparison with retrocaval ureter

| | **Retroiliac** | **Retrocaval (circumcaval)** |
|---|---|---|
| Vessel | Common iliac artery | IVC (inferior vena cava) |
| Embryology | Persistent ventral umbilical artery root | Persistent subcardinal vein |
| Side | Either, may be bilateral | Right |
| Level | L5–S1 | L3–L4, "fishhook" |
| Associations | Vasal/mesonephric ectopia | Horseshoe kidney, left renal anomalies |
| - **Also know:** distal ureter obstruction by uterine, umbilical, obturator or hypogastric vessels. | | |

**Diagram to draw:** ureter behind the common iliac artery at L5–S1, dilated above.

> **Key points to remember:**
> - Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
> - Obstruction at L5–S1; either side or bilateral.
> - Look for vasal and ureteric ectopia.
> - Retrocaval: right, L3–L4, fishhook, 1 in 1500.

## SAQ 7 Medullary sponge kidney

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 9 Paper 2, Q3e

### Definition

- **MSK (medullary sponge kidney; precalyceal canalicular ectasia):** dilated (ectatic) distal **collecting ducts** with small cysts and diverticula **confined to the medullary pyramids**. Nephrons are initially normal.
- Usually **non-heritable** (sporadic); usually bilateral.

### Pathogenesis of stones and nephrocalcinosis

- **Stasis and infection** in the ectatic ducts favour crystal deposition.
- **Tubular defects:**
- **Hypercalciuria** in one-third to one-half (mostly absorptive; sometimes renal leak).
- **Incomplete distal RTA (renal tubular acidosis)** in 30–40%, shown on acid loading. This causes **hypocitraturia** and alkaline urine.
- Impaired urinary concentration.

- Mutations in $H^+$-ATPase genes (ATP6V1B1, ATP6V0A4) support an MSK–distal RTA link.
- Result: calcium stones within dilated ducts and medullary **nephrocalcinosis**.

### Clinical features

- Often silent for life; presents **after 20 years**.
- **Renal colic 50–60%**, UTI (urinary tract infection) 20–33%, gross haematuria 10–18%.
- Found in **2.6–21% of stone formers**, more often women.

### Diagnosis

- **Urographic triad:** 1. Enlarged kidneys, with or without papillary calcification. 2. Elongated papillary tubules or cavities filling with contrast: **"bristles on a brush"** (paintbrush); when calcified, a **"bouquet of flowers"**. 3. Papillary blush and persistent medullary opacification.
- CT urography now replaces IVU (intravenous urography); plain CT shows medullary nephrocalcinosis.
- **Metabolic work-up:** 24-hour urine (calcium, citrate, pH), serum bicarbonate and potassium.
- **Differentials:** papillary necrosis, other medullary nephrocalcinosis (distal RTA, hyperparathyroidism), calyceal diverticulum.

### Management

| Agent | Target | Note |
|---|---|---|
| **Potassium citrate** | Hypocitraturia, hypercalciuria | **First choice**; reduces calciuria and recurrence |
| Thiazide | Hypercalciuria | Useful even without hypercalciuria |
| Inorganic phosphate | Hypercalciuria | If thiazide unsuitable; avoid with urease infection |
| Fluids, low-sodium diet | Volume, calciuria | All patients |
| - Treat infection; stones by SWL (shock-wave lithotripsy) or ureteroscopy; stones inside ducts may be unreachable. | | |
| - **Prognosis:** usually benign; renal failure is rare. | | |

**Diagram to draw:** pyramid with ectatic ducts filled with contrast ("brush") and small calculi.

> **Key points to remember:**
> - Ectatic collecting ducts; sporadic; "bristles on a brush".
> - Hypercalciuria, incomplete distal RTA, hypocitraturia.
> - Potassium citrate is first-line prevention.

## SAQ 8 Megameatus intact prepuce

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 15 Paper 2, Q3f

### Definition

- **MIP (megameatus intact prepuce)** is a hypospadias variant (up to 5%) with a **wide, "fish-mouth" glanular or coronal meatus**, a **completely formed prepuce** and **no significant curvature**. Described by Duckett and Keating (1989).

### Embryology

- Normally the prepuce closes ventrally only after the urethral folds fuse (about 20 weeks), so 95% of hypospadias have a hooded, incomplete foreskin.
- In MIP the foreskin is complete, showing that **preputial and urethral development can dissociate**.

### Clinical features

- Hidden under a normal foreskin; often found on **retraction before circumcision** or during circumcision.
- Wide, deep glans groove with a large, patulous meatus; well-developed spongiosum.
- Symptoms: spraying or deflected stream; cosmetic concern.
- **Advice:** retract the foreskin before every newborn circumcision. If found, the circumcision may be completed; prior circumcision does not harm later repair.

### Evaluation

- Examine meatal site and size, glans and plate, curvature (artificial erection at surgery). GMS (glans–meatus–shaft) score.
- No routine upper-tract imaging for isolated distal hypospadias.

### Management

- **Indication:** cosmesis and a straight stream; repair is optional. Usual timing 6–18 months.
- **Pyramid procedure (Duckett and Keating):** 1. Glans traction suture; retract prepuce. 2. Incise around the megameatus in a **pyramid shape**, apex on the ventral shaft. 3. Dissect the wide distal urethra free of the glans wings. 4. **Excise a ventral wedge** of redundant urethra to taper it. 5. Tubularise over a catheter; cover with dartos. 6. Approximate glans wings in two layers (glansplasty); preserve or reconstruct the prepuce.
- **Alternatives:** GAP (glans approximation procedure; pyramid = GAP with intact prepuce), TIP (tubularised incised plate) urethroplasty, Thiersch–Duplay tubularisation, Mathieu.

### Complications

- Fistula, meatal stenosis or regression, glans dehiscence; generally low after distal repair.

**Diagram to draw:** fish-mouth glanular meatus with intact prepuce, and the pyramid incision.

> **Key points to remember:**
> - Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
> - Often found at circumcision; retract foreskin first.
> - Pyramid repair: dissect, taper and tubularise; glansplasty.

## SAQ 9 Techniques of chordee correction

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 17 Paper 2, Q3b (related: Practice Set 10 Paper 1, Q3d)

### Definition and causes

- Chordee: **ventral penile curvature**, with or without hypospadias. It, not meatal position, decides one-stage versus staged repair.
- Causes: tethered ventral skin, fibrous dysplastic dartos and Buck's fascia, short urethral plate, **corporal disproportion**.
- Devine–Horton (1973) graded the dysplastic layers: I spongiosum, II Buck's and dartos, III dartos only, IV corporal disproportion.

### Assessment

- Parents' photographs; **artificial erection** in theatre: Gittes–McLaughlin (tourniquet at the base plus saline), corporal compression, or prostaglandin. A tourniquet can hide basal curvature.
- Grade: mild <30°, moderate 30–45°, severe ≥45° (goniometer; eye underestimates).

### Stepwise correction

```
Deglove to penoscrotal junction + excise dysplastic dartos → erection test ├─ Straight → urethroplasty ├─ Residual <30° → dorsal plication └─ >30° → divide urethral plate → still >30° → ventral lengthening (stage)
```

| | **Dorsal shortening (plication)** | **Ventral lengthening (corporotomy ± graft)** |
|---|---|---|
| Methods | **Nesbit:** excise dorsal tunical ellipses, close transversely. **Baskin:** single midline plication in the **12 o'clock nerve-free zone**. Heineke–Mikulicz dorsal incision closed transversely; 16-dot plication | 1–3 transverse **full-thickness corporotomies** (STAG, staged tubularised autograft); single **9-to-3 o'clock corporotomy** with graft: **dermis, tunica vaginalis**, SIS (small intestinal submucosa); dermal graft 20% larger than defect |
| Use | ≤15–30° | Moderate–severe after plate division |
| Recurrence | 28% (one series) | 9% |
| Risk | Shortening; dorsal nerve injury off midline | Longer, staged; theoretical erectile dysfunction |

- **Plate division** (transection) releases a short plate; not dividing it is an independent risk factor for recurrence. Repair is then staged.
- **Also know:** "fairy cuts" (superficial tunical incisions); lateral or dorsal curves need multipoint plication; correct **torsion** by dartos flap rotation.
- **Rule:** correct fully the first time and repeat the erection test before urethroplasty. Curvature may recur at puberty.

**Diagram to draw:** penile cross-section with dorsal nerves at 11 and 1 o'clock and midline plication at 12.

> **Key points to remember:**
> - Artificial erection after degloving decides the technique.
> - <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
> - Plication recurrence 28% vs grafting 9%.

## SAQ 10 Congenital ureteral anomalies

P3

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 3, Q3f

### Embryology basis

- The **ureteric bud** arises from the **mesonephric (wolffian) duct** at about week 5 and induces the metanephric blastema.
- Anomalies arise from **abnormal budding**, **failed distal canalisation**, **abnormal muscle maturation** or **persistent embryonic vessels**.

### Classification

| Category | Anomaly | Key points |
|---|---|---|
| **Number** | **Duplication**: incomplete (bifid, Y-ureter) or complete | About 1 in 125; unilateral 6× bilateral. Lower pole: VUR (vesicoureteric reflux), PUJ (pelviureteric junction) obstruction. Upper pole: ectopia, ureterocele |
| **Termination (orifice)** | **Ectopic ureter** | Opens at or below the bladder neck. **Females**: bladder neck to vagina, uterus; **continuous dribbling with normal voiding**. **Males**: always above external sphincter (posterior urethra, seminal vesicle, vas); **never incontinent**; epididymitis. Zinner syndrome |
| **Structure** | **Ureterocele** | Cystic dilation of distal submucosal ureter (Chwalle membrane). Intravesical vs ectopic (Glassberg); cecoureterocele (Stephens) |
| | **PUJ obstruction** | Intrinsic aperistaltic segment (infants) or crossing lower-pole vessel (older children) |
| | **Megaureter** (≥7 mm) | Obstructed, refluxing, non-refluxing non-obstructed, refluxing-obstructed; primary obstructive = adynamic distal segment |
| | **Primary VUR** | Short submucosal tunnel, lateral orifice |
| **Position** | **Preureteral (retrocaval) vena cava** | Persistent subcardinal vein traps the right ureter; fishhook (S) shape at L3–L4; type I obstructs, type II does not |
| | Retroiliac ureter; herniation | Rare |

### Weigert–Meyer law

- In complete duplication the **upper-pole ureter opens lower and more medial** (ectopic, obstructs, ureterocele); the **lower-pole ureter opens higher and lateral** (refluxes).
- Reason: the upper-pole bud arises more cephalad on the wolffian duct, is incorporated later and ends more caudally.
- **Also know:** blind-ending ureter, inverted-Y ureter, triplication, congenital stricture, ureteral valves, fibroepithelial polyp.

### Evaluation

- Ultrasound, VCUG (voiding cystourethrogram), **DMSA (dimercaptosuccinic acid)** scan, MR (magnetic resonance) urography.

### Management principles

- **Ureterocele:** endoscopic incision, upper-pole heminephrectomy or ureteroureterostomy.
- **Ectopic ureter:** heminephroureterectomy if non-functioning; otherwise ureteroureterostomy or reimplantation.
- **PUJ obstruction:** pyeloplasty. **Megaureter:** observe most; taper and reimplant (5:1) if function falls.
- **Retrocaval ureter:** divide, bring anterior to the cava, excise the aperistaltic segment, reanastomose.

**Diagram to draw:** complete duplication showing Weigert–Meyer orifice positions.

> **Key points to remember:**
> - Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
> - Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
> - Retrocaval ureter is a venous anomaly; always right-sided.

## SAQ 11 Autosomal recessive polycystic kidney disease.

P3

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 3, Q3c

### Definition and genetics

- **ARPKD (autosomal recessive polycystic kidney disease)**: bilateral, symmetrical renal enlargement from **dilated collecting ducts**, **always with congenital hepatic fibrosis**.
- Gene **PKHD1 on chromosome 6p**; protein **fibrocystin (polyductin)** on the **primary cilium** of tubular and bile-duct cells, so it is a **ciliopathy**.
- Incidence about 1 in 10,000–50,000 live births. Siblings have a 25% risk.
- "Infantile" is a misnomer: a **spectrum**; earlier onset means more severe disease.

### Pathology

- Radially arranged dilated collecting ducts (**microcysts**).
- **Liver:** periportal fibrosis, biliary ectasia; **Caroli syndrome** if intrahepatic ducts dilate.

### Clinical features

| Age | Features |
|---|---|
| **Fetus** | Large echogenic kidneys, **oligohydramnios** → Potter facies, limb deformity, **pulmonary hypoplasia** |
| **Neonate** | Huge, smooth, hard flank masses; respiratory distress; rising creatinine; **30–50% die early** |
| **Child** | Hypertension, CKD (chronic kidney disease), growth failure, anaemia, osteodystrophy |
| **Older child** | **Portal hypertension**: varices, splenomegaly, cholangitis; liver function stays normal |

- No association with renal tumours.

### Investigations

- **Ultrasound:** bilateral very large, **homogeneously hyperechoic** kidneys (brighter than liver) from countless microcyst interfaces; discrete cysts appear with age.
- **Parents' ultrasound** normal (excludes ADPKD); **PKHD1 genetic testing**; liver ultrasound.
- **Differential:** neonatal ADPKD (autosomal dominant PKD), bilateral hydronephrosis, MCDK (multicystic dysplastic kidney), glomerulocystic disease, bilateral tumours, renal vein thrombosis.

### Management (no cure)

- **Neonatal:** ventilation, fluid and electrolyte care, antihypertensives (ACE (angiotensin-converting enzyme) inhibitors).
- **CKD care:** nutrition, growth hormone, anaemia and bone treatment; dialysis and **kidney transplantation**.
- **Nephrectomy** (breathing, feeding, blood pressure, room for peritoneal dialysis): cautiously; **very early bilateral nephrectomy** (first 3 months) had the worst neurological outcomes (ARegPKD registry).
- **Liver:** variceal sclerotherapy, portosystemic shunt; **combined liver–kidney transplant** for severe liver disease.
- **Genetic counselling** and prenatal diagnosis.

### ARPKD versus ADPKD

| | ARPKD | ADPKD |
|---|---|---|
| Gene | PKHD1 (chromosome 6) | PKD1 (16), PKD2 (4) |
| Age | Perinatal | 30–50 years |
| Cysts from | Collecting duct | Whole nephron |
| Ultrasound | Symmetric, homogeneous, echogenic | Large discrete cysts, asymmetric |
| Liver | Fibrosis (always) | Cysts |
| Other | None | Berry aneurysm, mitral prolapse |

**Diagram to draw:** cut kidney with radial dilated collecting ducts.

> **Key points to remember:**
> - PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
> - Huge homogeneous echogenic kidneys with oligohydramnios.
> - Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

## SAQ 12 Ureterocele

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 3, Q3c (ectopic ureter also answered in 2018-Summer Paper 3, Q1)

### Definition

- Cystic dilatation of the distal ureter, in the bladder or spanning bladder neck and urethra; **never wholly urethral**, never joined to Wolffian structures.

- Commoner in girls (about 4:1); about 80% drain the **upper pole of a duplex kidney** (Weigert–Meyer rule: upper pole orifice caudal-medial); about 10% bilateral.
- **Cause:** persistent **Chwalla membrane**; upper pole often dysplastic.

### Classification

| System | Basis | Types |
|---|---|---|
| **Glassberg (most useful)** | Position | **Intravesical** vs **ectopic (extravesical)**: part at bladder neck or urethra |
| Stephens | Orifice morphology | Stenotic, sphincteric, sphincterostenotic, **cecoureterocele** (cavity extends under urethra), blind, non-obstructed |
| Churchill | Function | Upper pole only / whole kidney / contralateral at risk |

### Clinical features

- Antenatal hydronephrosis (most); febrile UTI (urinary tract infection) or urosepsis; adult single-system type with stone.
- **Prolapse:** smooth interlabial mass in girls; BOO (bladder outlet obstruction).
- **No incontinence when untreated** (unlike female ectopic ureter).

### Evaluation

- **Ultrasound:** thin-walled cyst in bladder, dilated upper pole and ureter.
- **VCUG (voiding cystourethrogram):** lower pole reflux ("drooping lily"), eversion (weak floor), prolapse.
- **DMSA (dimercaptosuccinic acid):** upper pole function; cystoscopy before incision.

### Management

- **Goals:** preserve function; remove infection, obstruction, reflux; keep continence.
- **CAP (continuous antibiotic prophylaxis)** from birth if hydronephrotic.
- **Observation:** small, non-obstructing, low-grade reflux, non-functioning upper pole.

| Option | Indication / result |
|---|---|
| **TUI (transurethral incision)** | Sepsis, neonates, intravesical type. Low transverse full-thickness incision; relieves obstruction in 78–97%; new reflux 0–50%. Intravesical without reflux: 70–80% cured; ectopic: secondary surgery up to 70% |
| Upper pole heminephrectomy | Non-functioning upper pole, no reflux; secondary surgery 40–50% |
| Ureteroureterostomy / pyeloureterostomy | Functioning upper pole |
| Excision + common sheath reimplant ± bladder neck repair | Persistent reflux, ectopic or cecoureterocele; persisting reflux 5–10% |
| Total reconstruction | Older child, non-functioning upper pole, lower pole reflux |

- **Also know:** "watering-can" laser punctures (less reflux); pseudoureterocele; ureterocele disproportion.

**Diagram to draw:** duplex kidney with upper pole ureterocele and lower pole reflux.

> **Key points to remember:**
> - Upper pole of duplex; never wholly urethral; Chwalla membrane.
> - Intravesical vs ectopic predicts TUI success.
> - TUI relieves obstruction in 78–97%; new reflux 0–50%.

## SAQ 13 Primary obst. megaureter.

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 3, Q3f

*Assumed meaning: primary obstructive megaureter (POM).*

### Definition

- **Megaureter:** ureter diameter **≥7 mm** (normal up to 16 years is 5.0–6.5 mm). It is a descriptive term.
- **Classification:** primary (intrinsic) or secondary (neurogenic bladder, outlet obstruction); and **obstructed, refluxing, non-refluxing non-obstructed, or refluxing and obstructed**.
- **POM:** an **adynamic juxtavesical segment** at the UVJ (ureterovesical junction) blocks peristalsis and causes functional obstruction above.

### Pathogenesis

- Ureteric muscle matures cranially to caudally; the juxtavesical segment matures last, even after birth. This explains spontaneous resolution.
- Excess collagen, thick muscle sleeve, abnormal interstitial cells of Cajal.

### Clinical features

- 10–23% of antenatal upper-tract dilatation. Boys, **left side**; **bilateral 25%**; contralateral dysplasia or obstruction 10–15%.
- Mostly asymptomatic; or UTI (urinary tract infection), pain, haematuria, stones.

### Evaluation

- **Ultrasound:** hydroureteronephrosis.
- **VCUG (voiding cystourethrogram):** excludes reflux and PUV (posterior urethral valves).
- **MAG3 (mercaptoacetyltriglycine) renogram:** split function; washout T½ unreliable (false positive up to 44%).
- MRU (magnetic resonance urography) for anatomy.

### Management

#### Conservative (most)

- CAP (continuous antibiotic prophylaxis) in infants; serial ultrasound and renograms; most resolve.
- Diameter <10 mm: virtually never needs surgery; ≥10 mm: a substantial minority do.
- Follow to early adulthood.

#### Indications for surgery

- Recurrent febrile UTI; progressive dilatation; **DRF (differential renal function) <40%**; **fall in DRF ≥5%**.

**Infant (small bladder, huge ureter): temporise**

- Cutaneous ureterostomy; refluxing reimplant; nephrostomy; **endoscopic balloon dilatation with double-J** for 2–6 months (70–80% success; complications up to a third).

**Definitive repair (child)**

- Excise stenotic segment → straighten → **taper** → **non-refluxing reimplant at 5:1** (Cohen or Politano–Leadbetter; extravesical for huge ureters). Stent afterwards.

| Tailoring | Method | Use |
|---|---|---|
| Starr plication | Imbricate lateral wall | Moderate (<1.75 cm) |
| Kalicinski folding | Exclude and fold lumen | Moderate |
| **Hendren excision** | Excise lateral wall | Massive; risk of ischaemia |

- Success over 90%.
- **Complications:** obstruction (stent first), VUR (vesicoureteral reflux; observe or inject), persistent dilatation.

**Diagram to draw:** dilated ureter above narrow adynamic segment; tapered reimplant.

> **Key points to remember:**
> - Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
> - Surgery: UTIs, progression, DRF <40% or fall ≥5%.
> - Excise, taper, reimplant 5:1; success >90%.

# Rapid Revision Sheet · Section 4

**Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING (LAQ)**

- Low (early) bud → high lateral orifice → short tunnel → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 normal versus 1.4:1 refluxing (Paquin); adult intramural ureter 1.2–2.5 cm.
- Dysplasia (boys, congenital) versus scar (girls, infection); sterile reflux rarely scars.
- Grade V loses papillary impressions; grade IV–V and bilateral grade III rarely resolve; bilateral grade IV persists most; bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Treat BBD and give CAP first; correct for breakthrough febrile UTI, new scars or persistent grade IV–V.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- STING: O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%; Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction; calcified mound mimics a stone.
- Open reimplantation 95.9%; Cohen cross-trigonal (over 98%) suits bilateral VUR through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- Paquin: tunnel 5:1; combined, suprahiatal reimplant. Gil-Vernet: trigonoplasty, medial advancement, no tunnel; for low-grade bilateral VUR.
- 34–38% of scarred children become hypertensive; correction does not lower BP; scars bring 8× pre-eclampsia risk.
- In adults the scar, not the reflux, carries the risk; exclude secondary causes first; correct for recurrent pyelonephritis, new scars or pregnancy plans.

**Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods (LAQ)**

- SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
- Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
- Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP; most testes scrotal after 27 weeks.
- Ectopic = off the path; commonest site superficial inguinal pouch.
- Operate from 6 months, preferably by 12 months, and by 18 months at the latest.
- No imaging, no hormones.
- Non-palpable: EUA then diagnostic laparoscopy; laparoscopy decides.
- Bilateral non-palpable: DSD work-up first (karyotype, 17-OHP, AMH) to exclude CAH/DSD.
- Subdartos pouch; Prentiss manoeuvre gives extra length.
- Fowler–Stephens relies on vasal collaterals.

**Retrocaval ureter and IVC anomalous development (LAQ)**

- Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
- Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
- Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
- Type I fishhook obstructs; type II sickle often does not.
- CT urography; repair by anterior transposition and ureteroureterostomy.
- Duplicated IVC: filter both cavae; do not mistake it for nodes.

**Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula (LAQ)**

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages.
- Curvature >30° after degloving: stage.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal; STAG has the lowest proximal complications.
- Stage 2 after ≥6 months.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.
- Fistula: commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

**Wilms tumour, including bilateral Wilms tumour (LAQ)**

- WT1 at 11p13 (WAGR, Denys–Drash); BWS at 11p15.
- Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
- Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
- COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
- Bilateral: stage V, 5–7%; no upfront nephrectomy or biopsy.
- Bilateral: COG three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12, NSS; SIOP two drugs.
- AREN0534: 4-year OS about 95%.
- Bilateral NSS to avoid ESRD.

**Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair (LAQ)**

- DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
- Hypospadias (including perineoscrotal) plus undescended testis: DSD work-up.
- <30° plication; >30° divide plate ± ventral graft and stage.
- STAG/Bracka staged repair preferred; fistula commonest complication.
- Koyanagi: one stage; meatus-based parameatal foreskin flap.
- Koyanagi: plate divided; no proximal circular anastomosis.
- Koyanagi: fistula commonest; diverticulum, meatal stenosis.

**Urological evaluation and management of pediatric kidney transplant patient. (LAQ)**

- Bladder target: age-expected capacity at <40 cm $H_2O$; augment if <75% despite CIC.
- PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.

- Never test a bladder with a graft; cycle the defunctionalised bladder.
- Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

**Embryology and management of ectopic ureter. (LAQ)**

- Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
- Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
- DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
- Protect the lower-pole blood supply; stent the recipient ureter for UU.

**TIP (Snodgrass) repair for hypospadias (SAQ)**

- Snodgrass 1994; incise and tubularise; heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Glans <14 mm and poor plate raise complications.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
- Extended to midshaft, selected proximal, redo, MIP and adults.
- Key condition: curvature <30° without plate division.
- Poor plate: dorsal inlay graft ("Snodgraft").
- Proximal and redo TIP fail more often.

**Glandular hypospadias (SAQ)**

- Glanular = most distal hypospadias; dorsal hood, minimal curvature.
- MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
- Repair at 6–18 months; may be left alone if function is normal.

**Vanishing testes (SAQ)**

- Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
- Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
- Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
- Anorchia: high FSH/LH, low AMH and inhibin B.

**Pelvicalyceal anomalies other than PUJ obstruction (SAQ)**

- Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
- Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
- Infundibulopelvic stenosis: bilateral, renal failure risk.

**Retractile testis? (SAQ)**

- Retractile = comes down and stays down.
- Gliding testis springs back: treat as undescended.
- Yearly examination until puberty; operate if it ascends.

**Retroileal Ureter (SAQ)**

- Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
- Obstruction at L5–S1; either side or bilateral.
- Look for vasal and ureteric ectopia.
- Retrocaval: right, L3–L4, fishhook, 1 in 1500.

**Medullary sponge kidney (SAQ)**

- Ectatic collecting ducts; sporadic; "bristles on a brush".

- Hypercalciuria, incomplete distal RTA, hypocitraturia.
- Potassium citrate is first-line prevention.

**Megameatus intact prepuce (SAQ)**

- Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
- Often found at circumcision; retract foreskin first.
- Pyramid repair: dissect, taper and tubularise; glansplasty.

**Techniques of chordee correction (SAQ)**

- Artificial erection after degloving decides the technique.
- <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
- Plication recurrence 28% vs grafting 9%.

**Congenital ureteral anomalies (SAQ)**

- Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
- Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
- Retrocaval ureter is a venous anomaly; always right-sided.

**Autosomal recessive polycystic kidney disease. (SAQ)**

- PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
- Huge homogeneous echogenic kidneys with oligohydramnios.
- Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

**Ureterocele (SAQ)**

- Upper pole of duplex; never wholly urethral; Chwalla membrane.
- Intravesical vs ectopic predicts TUI success.
- TUI relieves obstruction in 78–97%; new reflux 0–50%.

**Primary obst. megaureter. (SAQ)**

- Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
- Surgery: UTIs, progression, DRF <40% or fall ≥5%.
- Excise, taper, reimplant 5:1; success >90%.

# Section 5 · Paediatric Urology 2

## Long answer questions (LAQ)

### LAQ 1 Bladder embryology and exstrophy-epispadias complex

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 LAQ "Bladder and ureter development, exstrophy–epispadias complex and its management" · P1 LAQ "Describe embryology of bladder. What are the proposed theories for extrophy epispadias complex? Describe skeletal abnormalities." · P2 SAQ "Embryogenesis of urinary bladder"

#### Introduction

- Assumed meanings: "ectopia vesicae" means classic bladder exstrophy (CBE); "single stage repair of exstrophy of UB" means single-stage (complete primary) repair of exstrophy of the urinary bladder, mainly CPRE (complete primary repair of exstrophy, Mitchell).
- **EEC (exstrophy–epispadias complex)** is one spectrum: glanular epispadias, then **CBE (classic bladder exstrophy)** (over 50%), then cloacal exstrophy.
- CBE incidence about 1 in 46,000; male:female 2.3:1. Recurrence in offspring of an affected parent: 1 in 70.

#### Development of the ureter

- **Weeks 4–6:** the **ureteric bud** arises from the distal **Wolffian (mesonephric) duct**. GDNF (glial cell line-derived neurotrophic factor) from the metanephric mesenchyme induces a single bud (extra buds cause duplication).
- **Reciprocal induction:** the bud invades the metanephric blastema. The bud stalk forms the ureter; its branches form the pelvis, calyces and collecting ducts; the blastema forms the nephrons.
- **Wall:** urothelium, lamina propria, smooth muscle and serosa, the same plan as the bladder.
- **Chwalla membrane** at the future orifice normally ruptures; failure is the classic explanation of ureterocele.

#### Development of the bladder: division of the cloaca

- The **cloacal membrane** (bilaminar ectoderm–endoderm) occupies the infraumbilical wall. **Mesenchyme grows in** between its layers and forms the lower abdominal muscles and pelvic bones.
- **Cloacal division (weeks 5–6; P2: weeks 4–6):** the **urorectal septum** (Tourneux and Rathke folds) grows caudally and divides the endodermal cloaca into a dorsal **anorectal canal** (rectum, posterior) and a ventral **UGS (urogenital sinus)** (bladder, anterior). It meets the cloacal membrane, which perforates to form the urogenital and anal openings.
- Paired genital tubercles fuse **cephalad** to the membrane.
- **Wolffian (mesonephric) duct entry** is the landmark that divides the UGS:
- **Cranial vesicourethral canal** (continuous with the allantois) forms the **bladder and pelvic urethra**.
- **Caudal (definitive) UGS** forms the phallic urethra in males, and the vestibule and distal vagina in females.

### Development of the bladder: wall layers

- **Urothelium:** UGS endoderm.
- **Lamina propria, detrusor, adventitia / serosa:** surrounding splanchnic mesenchyme, by epithelial–mesenchymal signalling (epithelium and mesenchyme induce each other).

### Trigone and ureterovesical junction (weeks 5–6)

- The **common excretory duct** (Wolffian duct distal to the bud) dilates and is absorbed into the UGS; **apoptosis** separates the ureter from the Wolffian duct.
- The arms rotate (**Mackie–Stephens**): the ureteric orifice moves **cranially and laterally**; the Wolffian duct ends caudally as the ejaculatory duct at the verumontanum.
- **Trigone muscle:** classic view, ureteric extension; current view (mouse lineage tracing), mostly bladder smooth muscle.

| Bud origin | Orifice | Result |
|---|---|---|
| Too low (early contact) | High, lateral, short tunnel | VUR (vesicoureteral reflux) |
| Too high (late contact) | Caudal, medial | Ectopic ureter, often obstructed |
| Either | Abnormal bud–blastema contact | Dysplasia (VUR and dysplasia coexist) |

- **Weigert–Meyer:** in duplex systems the upper-pole orifice is caudal and medial.

### Allantois, urachus and fetal bladder function

- **Months 4–5:** the bladder descends into the pelvis; the allantoic duct elongates and obliterates as the **urachus** (median umbilical ligament).
- Urine production starts at about **week 10**. The fetal bladder is seen in about half at 10 weeks and almost all by 13 weeks; it empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

### Bladder anomalies explained by embryology

| Anomaly | Basis |
|---|---|
| **Exstrophy** | Cloacal membrane not reinforced by infraumbilical mesoderm |
| Urachal (patent urachus, sinus, cyst, diverticulum) | Failed obliteration at different levels |
| Agenesis, duplication | UGS or trigone failure; septation |
| Congenital (Hutch) diverticulum | Detrusor deficiency at the ureteric hiatus |
| VUR, ectopic ureter | Abnormal bud position |

### Developmental basis of EEC: core defect

- The **cloacal membrane is not reinforced by mesodermal ingrowth**. This mesoderm normally forms the lower abdominal muscles and pelvic bones, so the thin membrane ruptures (**Marshall–Muecke** overdevelopment theory).
- **Timing and extent of rupture decide the variant:** before septum descent gives cloacal exstrophy; after gives CBE; limited rupture gives epispadias.

## Developmental basis of EEC: theories

| Theory | Proposed defect | Result |
|---|---|---|
| **Cloacal membrane overdevelopment (Marshall–Muecke)** | Abnormally large membrane in week 4 blocks medial mesenchymal migration | Rupture before septum descent: cloacal exstrophy; after: bladder exstrophy; limited: epispadias |
| Lateral body-wall fold failure | Folds fail to meet in midline | Abdomen and pelvis: cloacal; pelvis alone: classic |
| Caudal genital hillocks | Genital tubercles fuse **below** rather than above the membrane | Dorsal urethral defect (epispadias) |
| Abnormal body-stalk insertion | Caudal insertion prevents mesenchyme interposition | Superficial unstable membrane disintegrates |
| **Bony pelvis first** | Pelvic ring primordium fails to rotate | Pelvic structures cannot meet, so the bladder herniates |

- All theories converge on **absent infraumbilical midline mesoderm**.

## Aetiology of EEC

- Multifactorial (inconsistent twin concordance). **22q11.2 duplication** is the commonest variant; **ISL1** susceptibility gene; p63 under-expressed in exstrophic urothelium.
- **IVF (in vitro fertilisation)** raises risk about 7.5-fold and early high-dose progesterone about tenfold; maternal smoking is linked to cloacal exstrophy; folate to the mildest form.

## Resulting anatomy in classic exstrophy

- Exposed (open) bladder plate; epispadias; low umbilicus; anterior anus; wide levator hiatus; inguinal hernias.
- **Male genitalia:** anterior corpora about 50% shorter; short, upturned penis with dorsal chordee.
- **Female:** bifid clitoris, short vagina, anterior stenotic introitus; later prolapse.
- **Bladder:** immature, not neuropathic (raised type III collagen falls after closure). Ureters enter with no obliquity, so **VUR (vesicoureteric reflux) occurs in 100%** after closure.

## Skeletal abnormalities: classic bladder exstrophy

### 3D (three-dimensional) CT (computed tomography) studies

- **Rotational:** posterior pelvis externally rotated (mean 12° each side); anterior segment externally rotated (mean 18° each side); sacroiliac joint rotated; acetabular and femoral retroversion.
- **Dimensional: pubic diastasis** (mean 4.8 cm); **anterior pubic segment shortened by 30%**; intertriradiate distance increased by about 31%; posterior segment length normal; larger sacrum.
- Bone histology is normal, so growth potential is normal.
- **Clinical effects:** short pendular penis (anterior segment short and splayed); wide hips, waddling gait and out-toeing in early childhood. These cause little disability and improve with growth.
- **Pelvic floor:** levator ani displaced posteriorly and flattened; anterior anus; later pelvic organ prolapse.

- **Spine:** minor variants (spina bifida occulta, lumbarisation, sacralisation) fairly common; scoliosis and dysraphism uncommon.

### Skeletal abnormalities: cloacal exstrophy and isolated epispadias

- **Cloacal exstrophy:** widest, **asymmetric** diastasis (about twice CBE), sacroiliac malformation, hip dislocation.
- **Spinal anomalies in 85–100%** (lumbar myelodysplasia commonest; lipomeningocele); MRI (magnetic resonance imaging) for all.
- Limb defects: **clubfoot** (commonest), limb shortening, tibial and fibular deformity.
- **Isolated epispadias:** male symphysis widened (less than CBE); female symphysis usually closed or narrow.

### Management: prenatal and at birth

- **Prenatal:** prenatal diagnosis in only about 25%. Ultrasound shows no bladder filling, low umbilicus, wide pubic rami, small genitalia and an enlarging lower abdominal mass. Counsel; deliver at an exstrophy centre.
- **At birth:** tie the cord with silk (no clamp); cover the plate with plastic wrap; saline irrigation at nappy change. Renal ultrasound; multidisciplinary team.
- **Assess:** template size and polyps (**≥3 × 3 cm** suits newborn closure), diastasis and pelvic malleability under anaesthesia, urethral plate and penile length, associated anomalies.
- Small template, penoscrotal duplication, hypoplastic bladder: **delayed closure at 6–9 months** (delay closure rather than do a bad early closure).

### Pelvic osteotomy and immobilisation

- **Purpose:** closing the pelvic ring places the bladder neck deep in the pelvis, reduces tension and helps continence.
- **Indications:** closure **after 72 hours**, diastasis **>4 cm**, stiff (poorly malleable) pelvis, failed closure, or cloacal exstrophy (always).
- **Combined bilateral anterior innominate + vertical iliac osteotomy** (most used) with external fixator and Buck traction for 4–6 weeks.
- Under 72 hours with malleable pubes: closure without osteotomy, with modified Bryant traction for 4 weeks.
- Spica casting fails more often (failure with external fixation plus traction 7.5% versus spica 47%).

### Modern staged repair of exstrophy (MSRE)

- Aim of closure: convert exstrophy into incontinent epispadias, with enough outlet resistance for bladder growth but not enough to harm the kidneys.

| Stage | Age | Procedure | Purpose |
|---|---|---|---|
| 1 | Newborn | Bladder, posterior urethra and abdominal wall closure; ± osteotomy | Converts to complete epispadias; outlet resistance stimulates bladder growth |
| 2 | 6–12 months | Epispadias repair (**modified Cantwell-Ransley**), after testosterone | Raises capacity; straightens penis |

| Stage | Age | Procedure | Purpose |
|---|---|---|---|
| 3 | 5–7 years (another source: 4–7 years) | BNR (bladder neck reconstruction): modified **Young-Dees-Leadbetter (YDL)** + bilateral **Cohen** reimplantation | Continence; needs capacity **≥100 mL** and a motivated child |

- **Closure keys: radical division of the urogenital diaphragm fibres** so the vesicourethral unit sits deep in the pelvis (missing this is the commonest error); ureteric stents; suprapubic tube 4 weeks; pubic nylon mattress suture.
- **Alternatives:** CPRE (Mitchell) with penile disassembly; Kelly radical soft-tissue mobilisation; Erlangen total repair (single stage); Mainz ureterosigmoidostomy. Most still need an outlet procedure.

## Single-stage repair (CPRE and others)

- Single-stage repairs aim for fewer operations and continence without bladder neck repair.

| Repair | Features |
|---|---|
| **CPRE (Mitchell)** | Newborn closure + penile disassembly + bladder neck tailoring |
| Erlangen ("total") | All in one at about 8 weeks, no osteotomy |
| Combined closure + epispadias repair | Delayed or failed closures, boys >6 months |

- **Selection:** template ≥3 × 3 cm, elastic, few polyps; experienced team. Not for penoscrotal duplication, hypoplastic bladder or bilateral hydronephrosis.
- **CPRE technique:** 1. Combined general and epidural anaesthesia; **osteotomy if >72 hours** (3 days) old, diastasis >4 cm or stiff pelvis. 2. Incise around the plate; free the bladder; **divide the urogenital diaphragm fibres** to the levator hiatus so the unit sits deep. 3. **Penile disassembly:** right and left hemicorporoglanular bodies plus a urethral wedge on spongiosum (now often left attached to the glans). 4. Urethra placed ventrally between the corpora; corpora joined; meatus usually **hypospadiac**. 5. Bladder and neck closed and tapered; ± bilateral ureteric reimplantation (BUR-CPRE). 6. Pubic No. 2 nylon mattress suture; suprapubic catheter 4 weeks, ureteric stents. 7. Immobilise: external fixation plus traction (failure 7.5% versus spica 47%).
- **Complications:** same frequency as MSRE but **more serious**: dehiscence, prolapse, vesicocutaneous fistula, **glans or corporal loss** (ischaemia), early reimplantation for VUR or hydronephrosis; retention in girls.

## Follow-up, failures and salvage

- **Follow-up:** prophylaxis; ultrasound; yearly cystogram and cystoscopy under anaesthesia.
- Dehiscence or prolapse: wait 4–6 months, then reclose with osteotomy.
- Failed BNR (no 3-hour dry interval by 2 years): bulking, redo YDL, or **augmentation + bladder neck closure + Mitrofanoff** (continent catheterisable stoma).

## Cloacal exstrophy (OEIS) management

- OEIS: omphalocele, exstrophy, imperforate anus, spinal defects.
- Stabilise; neurosurgery (spinal defects 85–100%); separate bowel from hemibladders; colostomy; osteotomy always; gender congruent with karyotype; continence mostly by augmentation with CIC (clean intermittent catheterisation).

### Outcomes and long-term issues

- Daytime dryness after MSRE about **70–74%** (another source: 70–75%), versus about 56% after CPRE with YDL; most still need bladder neck surgery after CPRE.
- **Bladder capacity at neck repair is the main predictor**; each failed closure lowers it. **Best continence predictors: template size at birth and a successful first closure.**
- **Also know:** variants (superior vesical fissure, duplicate exstrophy, pseudoexstrophy); adenocarcinoma risk (cytology, cystoscopy; colonoscopy after ureterosigmoidostomy); reduced male fertility and uterine prolapse in adults; female epispadias; psychosocial support.

**Diagrams to draw:** cloaca divided by the urorectal septum (vesicourethral canal, UGS, allantois) with failed mesodermal ingrowth of the cloacal membrane; Wolffian duct with ureteric bud and its absorption into the UGS; exstrophy pelvis with diastasis; MSRE timeline; penile disassembly into two hemicorpora and urethral wedge.

> **Key points to remember:**
>
> - Bud from Wolffian duct; position decides VUR or ectopia.
> - Urorectal septum weeks 5–6 (P2: 4–6); mesonephric duct entry is the landmark; vesicourethral canal forms the bladder; urachus from allantois.
> - Trigone: common excretory duct absorption; mostly bladder muscle (current view).
> - Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.
> - Marshall–Muecke: overdeveloped cloacal membrane, failed mesodermal reinforcement; rupture timing decides variant.
> - CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis about 4.8 cm; VUR in 100%.
> - Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.
> - MSRE: closure at birth, epispadias repair 6–12 months, YDL / BNR at 5–7 years at ≥100 mL.
> - Osteotomy if closure after 72 h or diastasis >4 cm.
> - CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
> - Capacity ≥100 mL, template size and a successful first closure predict continence.

## LAQ 2 Prune belly syndrome

P1 · P3 **merged from 2**

*Asked as:* P1 LAQ "Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management." · P3 SAQ "Prune belly syndrome"

### Definition

- **PBS (prune belly syndrome)**, also called Eagle–Barrett or triad syndrome: **deficient abdominal wall muscles**, **bilateral intra-abdominal testes** and a **dilated, poorly contracting urinary tract**.
- 1 in 29,000–44,000 live births; **95% male**.
- **Survival depends on renal dysplasia and lung hypoplasia, not on the abdominal wall.**

### Antenatal diagnosis

- **Ultrasound** may suggest PBS from 11–14 weeks, but the classic signs are not consistent even at 30 weeks.
- **Findings:** huge bladder filling the abdomen; bilateral hydroureteronephrosis; urachal diverticulum; dilated posterior urethra; echogenic kidneys with cortical cysts (dysplasia); irregular abdominal circumference; fetal ascites; oligohydramnios when severe.
- **Differential diagnosis:** PUV (posterior urethral valves), urethral atresia, megacystis–megaureter.
- Fetal dilatation does **not** predict postnatal renal function.
- **Prenatal management:** intervention has no proven renal benefit. Vesicoamniotic shunting is justified only for **urethral atresia with progressive oligohydramnios**, or to decompress the abdomen and avoid dystocia. Termination is hard to justify except with early, severe oligohydramnios. Plan delivery in a centre with neonatal intensive care.

**Diagram to draw:** fetal ultrasound sketch: huge bladder, dilated posterior urethra, bilateral hydroureteronephrosis.

### Pathogenesis (theories, none proven)

| Theory | Mechanism |
|---|---|
| Early (fetal) urethral obstruction | Back-pressure dilates the tract and stretches the wall; but most have no obstruction after birth |
| Lateral plate mesoderm defect | This mesoderm forms ureters, bladder, prostate, urethra and gubernaculum, so one defect explains the triad |
| Intrinsic urinary tract (ureteric) defect | Primary ureteric dilatation and fetal ascites |
| Yolk sac defect | Primary defect of the yolk sac |

- **Genetics:** mostly sporadic, normal karyotype; smooth-muscle genes **MYOCD (myocardin)** and **FLNA (filamin A)** in multiplex families.

### Pathophysiology and features by organ

- **Kidneys:** dysplasia in about 50% (Potter type II from mesenchymal defect; type IV cystic from obstruction).

- **Ureters:** long and tortuous, worse distally. Collagen replaces smooth muscle, so walls do not coapt and peristalsis fails: **a failed pump, not a blocked pipe**. The result is stasis and infection. **VUR (vesicoureteric reflux) in about 75%.** The proximal ureter has better muscle.
- **Bladder:** large, smooth, collagen-rich; urachal pseudodiverticulum (urachus patent in 25–30%); splayed trigone. Normal compliance but poor contractility, so voiding is "unbalanced", with residual urine.
- **Prostate and urethra:** prostatic hypoplasia dilates the posterior urethra; distal obstruction in about 20%; urethral atresia; **megalourethra** (scaphoid: spongiosum deficient; fusiform: all erectile bodies, more severe).
- **Testes:** intra-abdominal; azoospermia usual; paternity only by sperm retrieval and ICSI (intracytoplasmic sperm injection).
- **Extra-genitourinary (about 75%):** pulmonary hypoplasia; cardiac about 10%; gastrointestinal at least 30% (malrotation); orthopaedic up to 65% (talipes, hip dysplasia).

### Classification (Woodard)

| Category | Features | Outlook | Management |
|---|---|---|---|
| I | Oligohydramnios, **pulmonary hypoplasia**, dysplasia, urethral atresia | Perinatal death | Supportive; bladder drainage only |
| II | Full triad, moderate or unilateral dysplasia / moderate uropathy, **no pulmonary hypoplasia** | Risk of CKD (chronic kidney disease) | Individualised (controversial, debated group) |
| III | Mild or incomplete; **stable function** | Good; majority | Surveillance |

- **Also know:** RUBACE score (Renal, Ureter, Bladder, Abdominal wall, Cryptorchidism, Extra-genitourinary) grades morbidity / severity in survivors; "pseudo-prune" presents late with renal failure.

### Postnatal management: neonatal

Team: neonatology, nephrology, urology, cardiology.

1. Heart and lungs first: **chest radiograph** for pneumothorax and hypoplasia.
2. Creatinine, BUN (blood urea nitrogen), electrolytes; follow the **trend**, because the first value reflects the mother.
3. Renal and bladder ultrasound once stable.
4. Antibiotic prophylaxis; **circumcision**; strict asepsis.
5. **VCUG (voiding cystourethrogram)** only with renal insufficiency or suspected outlet obstruction.
6. **DMSA (dimercaptosuccinic acid) at 4–6 weeks** for split function; MAG3 (mercaptoacetyltriglycine) for drainage.
7. Outlet obstruction: catheter or suprapubic tube.

**Diagram to draw:** VCUG showing large bladder and dilated posterior urethra.

### Postnatal management: urinary tract

- **Conservative** for category III: surveillance, treat infection.
- **Surgery for:** recurrent upper tract infection, sepsis, progressive dilatation or rising creatinine / renal decline, high-grade VUR, true obstruction.

- **Temporary diversion:** wide **cutaneous vesicostomy (Blocksom)**. Avoid high ureterostomy, which wastes the good proximal ureter.
- **Reconstruction after 3–6 months** (lung maturity): excise the distal ureter, taper and reimplant the healthier proximal ureter; reduction cystoplasty gives little lasting benefit (capacity recurs). **CIC (clean intermittent catheterisation)** or Mitrofanoff for poor emptying.

### Orchidopexy

- **Bilateral transabdominal at about 6 months**, often with other surgery; later, **staged Fowler–Stephens** (better than single-stage) or laparoscopic.

### Abdominoplasty

- Aims: cosmesis; better cough, voiding / emptying and defecation.

| Technique | Key feature |
|---|---|
| Randolph | Transverse; excises the worst lower wall |
| Ehrlich | Vertical; umbilicus kept; vest-over-pants overlap |
| Monfort | Vertical; umbilicus kept; central fascial plate with lateral flaps overlapped; transperitoneal access, good exposure for combined surgery |

- **Also know:** Furness (extraperitoneal plication); Dénes; corset.

### Prognosis and follow-up

- About 20% die perinatally; **up to 30% develop chronic renal failure**.
- **Nadir creatinine <0.7 mg/dL at 1 year** predicts stable function.
- Lifelong timed voiding, cultures and imaging; transplant when needed.

> **Key points to remember:**
> - Triad; 95% male; dysplasia and lungs decide survival.
> - Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
> - Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
> - Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

## LAQ 3 Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency

P1 · P3 **merged from 2**

*Asked as:* P1 LAQ "Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children" · P3 SAQ "DD of frequency and urgency in 3 yr male child and treatment"

### Introduction

- The LUT (lower urinary tract) has two opposite functions: **storage** at low pressure with a closed outlet, and **emptying** by a coordinated detrusor contraction with a relaxed outlet. Bladder and sphincter act reciprocally.

### Neurophysiology: peripheral innervation

| | Parasympathetic | Sympathetic | Somatic |
|---|---|---|---|
| Origin | **S2–S4** (sacral parasympathetic nucleus) | Thoracolumbar (T10–L2) | **Onuf nucleus** (S2–S4 ventral horn) |
| Nerve | Pelvic | Hypogastric | Pudendal |
| Bladder | Contracts (ACh [acetylcholine] on **M3**; ATP) | Relaxes (noradrenaline on **β3**) | – |
| Outlet | Relaxes urethra (**NO**, nitric oxide) | Contracts bladder neck (**α1**) | Contracts EUS (external urethral sphincter; nicotinic) |
| Role | Voiding | Storage | Voluntary closure |

- All three nerves carry afferents. **Aδ fibres** (tension receptors in muscle) signal normal fullness. **Unmyelinated C fibres** are "silent" normally; after inflammation, SCI (spinal cord injury) or obstruction they are recruited and drive overactivity (the basis of capsaicin and resiniferatoxin).
- Detrusor has more M2 receptors, but **M3 mediates contraction**.

### Neurophysiology: reflex and central control

- **Storage (spinal reflexes):** low afferent activity triggers sympathetic outflow (detrusor inhibition, bladder-neck closure) and the **guarding reflex** (pudendal; sphincter activity rises with filling).
- **Voiding (spinobulbospinal reflex):** high afferent activity reaches the **PAG (periaqueductal grey)** and then the **PMC (pontine micturition centre, Barrington nucleus)**; the sphincter relaxes first, then the detrusor contracts. The PMC is an on–off switch.
- **Forebrain decides:** the insula senses fullness; the **medial prefrontal cortex** tonically holds the switch "off" until voiding is socially appropriate.
- Lesion patterns: above the pons, overactivity with a coordinated sphincter; between pons and sacral cord, overactivity with **DSD (detrusor–sphincter dyssynergia)**; sacral or peripheral, underactive bladder.
- **Central transmitters:** glutamate excitatory; GABA (γ-aminobutyric acid) and glycine inhibitory; serotonin inhibits voiding; dopamine tonically inhibits the PMC (lost in Parkinson disease).

`Filling → Aδ afferents → sacral cord → PAG low traffic: spinal storage reflexes (sympathetic + guarding) high traffic + cortex "permits": PMC → EUS relaxes → detrusor contracts`

**Diagram to draw:** bladder with pelvic, hypogastric and pudendal nerves, plus brainstem loop (PAG–PMC).

### Maturation in children

- Bladder visible on fetal ultrasound by 14 weeks. Infants void reflexly but **wake before voiding** (cortical arousal gates voiding).
- Infant voiding is often interrupted (transient dyscoordination), with detrusor pressure often **>100 cmH2O**; frequent small voids largely disappear by **2 years**.
- Capacity rises: **EBC (expected bladder capacity) = (age + 2) × 30 mL** (infants: weight in kg × 7 mL).
- Toilet training: bowel control, then daytime bladder, then night-time bladder. Persisting dyscoordination after toddlerhood is dysfunction, not immaturity.

### Definitions: frequency and urgency

- **Frequency:** ≥8 voids a day (ICCS, International Children's Continence Society). **Urgency:** sudden, compelling need to void.
- At 3 years toilet training may still be ongoing, so decide first whether the pattern is abnormal.

### Types of voiding dysfunction: non-neurogenic (functional) LUT dysfunction (ICCS)

| Type | Key feature |
|---|---|
| **OAB (overactive bladder)** | Urgency ± frequency (≥8 voids a day), urgency incontinence; tower flow |
| **Voiding postponement** | Habitual holding; holding manoeuvres; infrequent voids (≤3 a day) |
| **Dysfunctional voiding** | Habitual sphincter/pelvic-floor contraction during voiding in a neurologically normal child; **staccato** flow |
| **Underactive bladder** | Needs abdominal straining to void; interrupted flow |
| Giggle incontinence | Complete emptying with laughter |
| Vaginal reflux | Small leak just after voiding in girls |
| **Enuresis** | Wetting in sleep, ≥5 years; monosymptomatic or non-monosymptomatic |

- **BBD (bladder and bowel dysfunction):** LUT dysfunction plus constipation; rectal loading worsens symptoms and UTI (urinary tract infection).
- **Hinman (non-neurogenic neurogenic bladder):** the severe end of dysfunctional voiding, with trabeculated bladder, VUR (vesicoureteric reflux) and renal damage.

### Types of voiding dysfunction: neurogenic LUT dysfunction

- Causes: **myelomeningocele** (commonest), occult dysraphism / **tethered cord**, sacral agenesis, anorectal malformation, cerebral palsy, SCI, transverse myelitis, pelvic tumour surgery (sacrococcygeal teratoma).

- Pattern = detrusor (overactive or underactive) × outlet (competent, incompetent or dyssynergic). Bony level does not predict bladder function.
- **Hostile bladder: DLPP (detrusor leak point pressure) ≥40 cmH2O** (McGuire 1981), poor compliance, or NDO (neurogenic detrusor overactivity) with DSD.

### Types of voiding dysfunction: anatomical

- **PUV (posterior urethral valves)** "valve bladder": overactivity, then poor compliance, then myogenic failure; ectopic ureter.

### Differential diagnosis of frequency and urgency (3-year-old boy)

| Group | Causes and clues |
|---|---|
| **Infection** | **UTI**: fever, dysuria; balanoposthitis; viral haemorrhagic cystitis |
| **Functional** | **BBD**/OAB; voiding postponement; **extraordinary daytime urinary frequency** (sudden very frequent small voids, no nocturia or wetting, often after stress; self-limiting) |
| **Bowel** | **Constipation**: rectal mass presses on bladder; commonest associated cause |
| **Obstruction** | **PUV**, mild late form; **meatal stenosis** (circumcised boys; thin deflected stream); severe phimosis |
| **Neurogenic** | Occult spinal dysraphism, tethered cord (sacral dimple, hair tuft, gait or foot abnormality) |
| **Bladder irritants** | Stone, foreign body, hypercalciuria, pinworms, rarely **rhabdomyosarcoma** |
| **Polyuria** | Diabetes mellitus, diabetes insipidus, chronic kidney disease |
| **Behavioural** | Anxiety, family stress, developmental disorders |

### Evaluation

- **History:** onset, nocturia, wetting, stream, fever, **bowel habit (Bristol stool chart)**, thirst, stressors.
- **Examination:** abdomen, meatus and foreskin, **lumbosacral spine**, lower-limb neurology.
- **Urinalysis and culture**; glucose; specific gravity >1.005 makes diabetes insipidus unlikely.
- **Bladder and bowel diary.**
- **Ultrasound** of kidneys and bladder: wall thickness, residual, dilation, stones.
- **Uroflowmetry with PVR (post-void residual) and patch EMG (electromyography)** if the child cooperates: bell (normal), tower (OAB), staccato (dysfunctional voiding), interrupted (underactive/DSD), plateau (obstruction).
- **Selective:** VCUG (voiding cystourethrogram) for suspected PUV or febrile UTI with abnormal ultrasound; spinal MRI (magnetic resonance imaging) for neurological signs; urine calcium/creatinine ratio.
- Invasive (video)urodynamics only for neurogenic, anatomical or refractory cases.

**Diagram to draw:** flowchart: urinalysis, then ultrasound with residual, then targeted tests.

### Management

- **Treat constipation first:** disimpaction, then maintenance **polyethylene glycol**, diet and toilet routine.

- **Standard urotherapy** for 4 weeks: timed voiding every 2–3 hours, relaxed posture, adequate fluids, avoid caffeine.
- **UTI:** antibiotics per culture.
- **OAB persisting after urotherapy: oxybutynin 0.3–0.6 mg/kg/day** in 2–3 doses (maximum 15 mg/day; watch for constipation); mirabegron; botulinum toxin; parasacral TENS (transcutaneous electrical nerve stimulation).
- **Dysfunctional voiding: biofeedback**; α-blocker; sphincter botulinum toxin if refractory.
- **Extraordinary daytime frequency:** reassurance; resolves in weeks to months.
- **Enuresis:** alarm or desmopressin.
- **Meatal stenosis:** meatotomy. **PUV:** endoscopic valve ablation. **Tethered cord:** neurosurgical release.
- **Neurogenic: CIC (clean intermittent catheterisation; Lapides 1971) + antimuscarinic**, then intradetrusor botulinum toxin, then augmentation.

> **Key points to remember:**
> - S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
> - PMC = switch; medial prefrontal cortex holds it off.
> - EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
> - DLPP ≥40 cmH2O = hostile.
> - Always ask about bowels; treat constipation first.
> - Frequency/urgency: rule out UTI, PUV, meatal stenosis and occult dysraphism.
> - Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

## LAQ 4 Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies

P1 · P3 **merged from 2**

*Asked as:* P3 LAQ "Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO)" · P1 SAQ "Anomalies detected on Prenatal scan between 16th and 20th weeks of gestation"

### Introduction and terminology

- Assumed meaning: HN (hydronephrosis) and ANH (antenatal hydronephrosis) are now called **UTD (urinary tract dilation)**.
- Major congenital anomalies occur in about **2–3% of births**. Urinary tract anomalies are among the commonest detected prenatally (commonly quoted as 20–30% of all prenatal anomalies).
- **UTD is the second commonest prenatal anomaly after cardiac defects, in 1–3% of pregnancies.**
- Mostly UTD on the 20-week scan, usually mild and transient: mild 57–88%, moderate 10–30%, severe 1.5–13% of UTD. The serious group is **LUTO (lower urinary tract obstruction)**: PUV (posterior urethral valves), urethral atresia, PBS (prune-belly syndrome).

### The prenatal anomaly scan (16–20 weeks)

- The **anomaly scan is done at 18–20 weeks** (ACOG (American College of Obstetricians and Gynecologists)); by 20 weeks most urological anomalies are visible.
- Screening US (ultrasound) at 18–20 weeks detects dilation well (sensitivity about 90%) but predicts the diagnosis poorly.
- **Why this window:** fetal urine production begins at about 14 weeks, and after about 16 weeks amniotic fluid is mainly fetal urine, so **oligohydramnios reflects renal failure or outlet obstruction** and causes **pulmonary hypoplasia**. The bladder should be seen filling and emptying.
- Survey kidneys, collecting system, ureters, bladder, genitalia, amniotic fluid, abdominal wall and spine. AFI (amniotic fluid index): normal 8–18 cm; oligohydramnios <5–6 cm.

### Anomalies detected and their signs

| Anomaly | Signs |
|---|---|
| **PUV (posterior urethral valves)** | Male; thick bladder; **keyhole sign**; bilateral hydroureteronephrosis; oligohydramnios |
| **Bilateral renal agenesis** | **Anhydramnios after 16 weeks**, no bladder, no renal arteries: lethal Potter sequence |
| **MCDK (multicystic dysplastic kidney)** | Non-communicating cysts, no central pelvis |
| **ARPKD (autosomal recessive polycystic kidney disease)** | Large, homogeneously **bright** kidneys; oligohydramnios |
| **Bladder exstrophy** | **Bladder never seen**, low cord insertion, lower abdominal mass, wide pubis |
| **Prune belly** | Thin-walled large bladder, empty scrotum |
| **Myelomeningocele** | Spinal defect, lemon and banana signs |

- **Also know:** duplex kidney with ureterocele (intravesical cyst); cloacal exstrophy (adds omphalocele, spinal defects); hydrocolpos; ambiguous genitalia; renal ectopia.

**Diagram to draw:** fetal keyhole sign.

### Causes of antenatal UTD (SFU [Society for Fetal Urology] consensus)

| Cause | Incidence |
|---|---|
| Transient/physiological dilation | 50–70% |
| UPJ (ureteropelvic junction; PUJ, pelviureteric junction) obstruction pattern | 10–30% |
| VUR (vesicoureteral reflux) | 10–40% |
| UVJ (ureterovesical junction) obstruction/megaureter | 5–15% |
| MCDK | 2–5% |
| Duplex system, ureterocele, ectopic ureter | 5–7% |
| LUTO (PUV, prune-belly) | 0.1–0.5% |

- The chance of postnatal pathology rises with grade: about 12% (mild) to 88% (severe). UPJ obstruction and PUV track grade; **VUR is not predicted by grade**.

### Grading systems

**APD (anteroposterior renal pelvic diameter, APRPD):** transverse image at the hilum. APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.

| Grade | Second trimester | Third trimester |
|---|---|---|
| Mild | 4 to <7 mm | 4 to <9 mm |
| Moderate | 7–10 mm | 9–15 mm |
| Severe | >10 mm | >15 mm |

**SFU (postnatal only):** 1 pelvis barely split; 2 pelvis and major calyces; 3 minor calyces uniformly dilated; 4 plus parenchymal thinning.

**UTD consensus 2014 (risk stratification):** six parameters: APD, central/peripheral calyces, parenchymal thickness, parenchymal appearance, ureter, bladder (plus amniotic fluid prenatally).

| Prenatal parameter | A1 low risk | A2–3 increased risk (any one) |
|---|---|---|
| APD 16–27 weeks | 4 to <7 mm | ≥7 mm |
| APD ≥28 weeks | 7 to <10 mm | ≥10 mm |
| Calyces | Central or none | Peripheral |
| Parenchyma, ureter, bladder | All normal | Abnormal |
| Amniotic fluid | Normal | Unexplained oligohydramnios |

- **A1:** one repeat US after 32 weeks. **A2–3:** US every 4 weeks to delivery. Both: postnatal US after 48 hours.

### Prenatal algorithm (unilateral UTD, male fetus)

```
Anomaly scan: unilateral UTD → Detailed scan: contralateral kidney, ureter, bladder wall and emptying, posterior urethra (keyhole), amniotic fluid, other anomalies ├─ Bilateral / thick bladder / keyhole / oligohydramnios ─► LUTO
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pathway │ (fetal medicine; karyotype; Ruano staging; possible shunt) ├─ A1 ─► repeat US after 32 weeks └─ A2–3 ─► US every 4 weeks to delivery → Normal delivery at local centre; no fetal intervention for unilateral UTD → Postnatal US after 48 hours (earlier if solitary kidney or suspected LUTO)

- Repeat scans because the lesion may progress and a second anomaly is more likely. **Fetal MRI** (magnetic resonance imaging) only for complex anatomy.
- Counsel the parents: most cases resolve; prognosis with a normal opposite kidney is excellent.

### Antenatal management of fetal uropathy (suspected LUTO): evaluation

- Male fetus, thick-walled "keyhole" bladder, bilateral hydroureteronephrosis, oligohydramnios: LUTO until proved otherwise.
- Anomaly scan, **fetal echocardiogram, karyotype** (one-fifth of megacystis is aneuploid); fetal MRI if unclear.
- **Serial fetal urine** (3 samples, 48–72 hours apart). Favourable: Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L, β2-microglobulin <6 mg/L; not validated.
- **Ruano staging:** I mild, normal fluid: observe. **II severe, function preserved** (oligohydramnios, favourable urine): intervene. III cysts, dysplasia, unfavourable urine: intervention for the lungs only.

### Antenatal management of fetal uropathy: options

| Option | Notes |
|---|---|
| Observation, serial scans | Most cases |
| **VAS (vesicoamniotic shunt)** | Dislodgement, blockage, preterm labour |
| **Fetal cystoscopy + laser valve ablation** | Diagnoses and treats; fistula up to 10% |
| Serial amnioinfusion | Lungs in anhydramnios; under trial |
| Early delivery / termination | Mature lungs, falling fluid / poor prognosis |

- Shunt or fetal cystoscopy is done at 18–24 weeks in severe LUTO.
- **PLUTO (2013):** shunt versus conservative (31 fetuses): better perinatal survival, **no renal benefit**.

### Antenatal management of fetal uropathy: counselling and delivery

- Multidisciplinary team: fetal medicine, paediatric urology, nephrology, neonatology.
- Intervention improves survival, not renal function; complications **over 40%**.
- Deliver where the bladder can be decompressed at once; prophylaxis, then VCUG (voiding cystourethrogram).

### Postnatal algorithm

- **First US after 48 hours**, because neonatal dehydration under-reads dilation. Postnatal normal: APD <10 mm, no calyceal or ureteral dilation.

| | P1 (SFU 1–2) | P2 (SFU 3) | P3 (SFU 4) |
|---|---|---|---|
| Findings | APD 10–15 mm, central calyces | APD ≥15 mm, peripheral calyces or dilated ureter | Abnormal parenchyma or bladder |
| Repeat US | 3–6 months | Individualised | 1 month |

| | P1 (SFU 1–2) | P2 (SFU 3) | P3 (SFU 4) |
|---|---|---|---|
| Antibiotic prophylaxis | No | Discretion; **yes if ureter >7 mm** | Yes |
| VCUG | No | Discretion | Yes |
| MAG3 (mercaptoacetyltriglycine) renogram | No | Discretion | If persistent |

- **Male-specific points:** VCUG confirms a normal posterior urethra. **Circumcision** protects (NNT [number needed to treat] 10). UTI (urinary tract infection) risk: female sex, uncircumcised boys, high grade, dilated ureter.
- Prophylaxis: amoxicillin or a first-generation cephalosporin from birth; nitrofurantoin and trimethoprim only after 8 weeks.
- MAG3: DRF (differential renal function) <40%, T½ >20 minutes, or delayed retention suggest obstruction. MRU (magnetic resonance urography) for anatomy; DMSA (dimercaptosuccinic acid) for MCDK or duplex function.
- **Surgery indications (UPJ):** rising APD, DRF <40% with poor drainage, falling DRF, symptoms (UTI, stone, pain). Megaureter surgery: DRF <40%, febrile UTI; defer reimplant beyond 1 year.
- Outcome: SFU 1 resolves in nearly all by 2.5 years; SFU 2 about 95% by 4 years.

### Bilateral dilation and PUV: how they differ

- Bilateral hydroureteronephrosis, thick bladder, keyhole sign and oligohydramnios mean LUTO until proved otherwise.
- At birth: 5 Fr feeding tube, prophylaxis, creatinine, early VCUG; prenatal intervention improves survival but not renal function (PLUTO).

**Diagram to draw:** UTD A1/A2–3 prenatal and P1–P3 postnatal flowchart; keyhole bladder with bilateral hydroureteronephrosis and shunt.

> **Key points to remember:**
> - UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
> - Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
> - ANH = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
> - UTD A2–3 = APRPD ≥7 mm (16–27 weeks) or any abnormal feature.
> - Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
> - First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.
> - Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
> - Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
> - ARPKD large bright kidneys; MCDK non-communicating cysts.
> - Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
> - PLUTO: survival up, renal function unchanged.

## LAQ 5 Embryological development of urinary bladder and management of Hutch Diverticulum?

P3

**Type:** LAQ (25 marks)
**Source:** 2021-Summer Paper 3, Q1 (bladder diverticulum also answered in 2016-Summer Paper 2, Q3c)

### A. Embryology of the bladder

#### 1. Division of the cloaca (weeks 4–6)

- The **urorectal septum** (Tourneux and Rathke folds) divides the endodermal cloaca into a dorsal **anorectal canal** and a ventral **UGS (urogenital sinus)**. The cloacal membrane perforates to form the urogenital and anal openings.
- The **Wolffian (mesonephric) duct entry** is the landmark:
- **Cranial vesicourethral canal** (continuous with the allantois) → bladder and pelvic urethra.
- **Caudal (definitive) UGS** → phallic urethra in males; distal vagina and vestibule in females. The whole female urethra comes from the pelvic part.

#### 2. Wall layers

- **Urothelium:** endoderm of the UGS.
- **Lamina propria, detrusor and adventitia:** surrounding splanchnic mesenchyme. Epithelium and mesenchyme induce each other (epithelial–mesenchymal signalling).

#### 3. Trigone (weeks 5–6)

- The **common excretory ducts** (mesonephric duct distal to the ureteric bud) dilate and are absorbed into the UGS. **Apoptosis** separates the ureters from the mesonephric ducts.
- Ureteric orifices migrate **cranially and laterally**; the Wolffian ducts end caudally as ejaculatory ducts at the verumontanum.
- Classic view: trigone muscle is ureteric (Wolffian) tissue. Current view (mouse lineage tracing): **mostly bladder smooth muscle**, minor ureteric contribution.
- **Bud position (Weigert–Meyer):** bud too low → lateral orifice, short tunnel → VUR (vesicoureteral reflux); too high → caudal, medial, ectopic orifice.

#### 4. Allantois and urachus

- **Months 4–5:** the bladder descends into the pelvis. The allantoic duct elongates and obliterates into the **urachus** (median umbilical ligament).

#### 5. Function

- Urine production starts at **week 10**. Fetal bladder is seen from 10 weeks, almost always by 13 weeks, and empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

### B. Anomalies explained by embryology

| Anomaly | Embryological basis |
|---|---|
| Agenesis | Cranial UGS atrophy, or trigone not formed; survival only with ectopic ureters |
| Exstrophy | Cloacal membrane not reinforced by mesoderm |

| Anomaly | Embryological basis |
|---|---|
| Duplication | Complete or incomplete; sagittal or coronal |
| Urachal (patent urachus, sinus, cyst, vesicourachal diverticulum) | Urachus not obliterated at different levels |
| Congenital diverticulum | Focal detrusor deficiency, often at the ureteric hiatus |

### C. Hutch diverticulum: definition

- **Paraureteral diverticulum**: mucosa herniates through the **ureteric hiatus**, between the intravesical ureter and the roof of the hiatus, superolateral to the orifice, sparing the trigone. **Hutch** divided them into primary and secondary.

| | Primary (congenital) | Secondary (acquired) |
|---|---|---|
| Cause | Congenitally deficient detrusor at the hiatus | High pressure: PUV (posterior urethral valves), BOO (bladder outlet obstruction), neurogenic bladder (described in paraplegics), dysfunctional voiding, prior surgery |
| Bladder | Smooth; single, often intermittent | Trabeculated; one of many |

### D. Pathophysiology

- No muscle backing → **VUR**; no muscle in the sac → cannot empty → stasis, UTI (urinary tract infection), stones; tumour in adults.
- An enlarging sac damages **Waldeyer's sheath** and pulls the ureter out → UVJ (ureterovesical junction) dysfunction, reflux, later **obstruction** and renal damage.
- Large sacs compress the bladder neck → retention (vicious circle).

### E. Evaluation

- **US (ultrasound):** sac, residual urine, hydronephrosis.
- **VCUG (voiding cystourethrogram), gold standard:** oblique and post-void films show size, VUR and whether the ureter enters the sac. Repeat if suspicion persists (sac is dynamic).
- **DMSA (dimercaptosuccinic acid):** scars; **MAG3 (mercaptoacetyltriglycine) renogram** if obstruction.
- **Urodynamics** if secondary; cystoscopy (orifice, neck); in adults enter the sac, cytology.

### F. Management

**1. Treat the cause first (secondary):** valve ablation, BOO surgery, CIC (clean intermittent catheterisation) and antimuscarinics. Remodelling may make excision unnecessary.

**2. Primary diverticulum:**

| Situation | Management |
|---|---|
| Small, asymptomatic, no VUR | Observe |
| VUR, ureter beside the sac | As primary VUR (resolves like VUR of the same grade); many excise if it persists. Dx/HA (dextranomer/hyaluronic acid) injection possible |
| Ureter enters the sac | VUR will not resolve → surgery |

| Situation | Management |
|---|---|
| Breakthrough febrile UTI, new scars, large sac with residual, stones, obstruction, retention | **Diverticulectomy + ipsilateral reimplantation** |

**3. Operative steps (open, laparoscopic or robotic):** `Cystoscopy ± ureteric catheter → cystotomy → evert sac or dissect it from its pseudocapsule (0.5 cm mucosal cuff round the neck) → close detrusor defect in two layers (firm backing) → reimplant ureter: Cohen cross-trigonal or Politano–Leadbetter (tunnel 5:1) → catheter ± suprapubic tube, drain`

- Extravesical or combined approaches suit selected cases. In adults, endoscopic outlet surgery is done first.

## G. Complications and follow-up

- Ureteric injury, leak, persistent VUR, obstruction, recurrence (syndromic patients).
- US at 6–12 weeks; VCUG for febrile UTI; urodynamics if voiding is abnormal.

**Also know:** posterolateral multiple diverticula (Ehlers–Danlos, Williams, Menkes); hypoplasia; megacystis.

**Diagram to draw:** cloacal division by the urorectal septum; Hutch sac beside the ureteric orifice.

> **Key points to remember:**
> - Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
> - Hutch: herniation through the ureteric hiatus → VUR.
> - Ureter entering the sac: VUR will not resolve.
> - Secondary: treat the outlet first; else diverticulectomy + reimplantation.

# LAQ 6 Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias)

P3

**Asked as:**

- Discuss development of urethra. Classify congenital anomalies and discuss management of epispadias — *Practice Set 6 Paper III Q2 (LAQ)*
- Isolated penopubic Epispadias. — *Summer 2022 Paper III Q3f (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)
(isolated penopubic epispadias also answered in 2022-Summer Paper 3, Q3f)

## Development of the urethra

### 1. Division of the cloaca (weeks 5–6)

- The urorectal septum divides the cloaca into the dorsal anorectal canal and the ventral **UGS (urogenital sinus)**.
- Wolffian duct entry is the landmark. Above it, the **vesicourethral canal** forms the bladder and pelvic urethra. Below it, the **definitive UGS** forms the phallic urethra in males, and the urethra and vestibule in females.
- UGS also forms the prostate (buds at 10 weeks) and Cowper and Littre glands; the **utricle** is Müllerian.
- The epithelium of the **entire male urethra is endodermal** (UGS origin).

### 2. Penile urethra: the "double zipper" (androgen-dependent)

- Indifferent genital tubercle until 8–9 weeks. Testosterone (Leydig cells, 8 weeks) is converted to DHT (dihydrotestosterone) by **type 2 5α-reductase**.
- The **urethral plate** (endoderm) forms in the tubercle.
- **Opening zipper:** the plate canalises proximal to distal into a urethral groove bounded by the urethral folds.
- **Closing zipper:** the folds fuse in the ventral midline, forming the tubular urethra and median raphe.
- **Glanular urethra:** formed by direct canalisation of the plate inside the glans (current endodermal view; Glenister's ectodermal ingrowth theory is outdated).
- **Timetable:** coronal sulcus 11–12 weeks; fusion complete 16 weeks; urethra complete 18 weeks; fetal curvature gone 20 weeks.
- The prepuce forms with the glanular urethra, so hypospadias has a hooded foreskin.
- **Female:** the vestibular plate canalises but the folds do not fuse, giving an open vestibule and labia minora.

## Classification of congenital urethral anomalies

| Mechanism | Anomaly |
|---|---|
| Failed ventral fold fusion | **Hypospadias** (1 in 250 males; commonest at shaft–glans junction); congenital urethral fistula |
| Dorsal defect | **Epispadias** (exstrophy–epispadias complex) |

| Mechanism | Anomaly |
|---|---|
| Obstruction | **PUV (posterior urethral valves)**, Young type 1 (95%) and type 3; **anterior urethral valves** with diverticulum; urethral atresia or hypoplasia; meatal stenosis |
| Duplication | **Urethral duplication**: sagittal; ventral channel is functional; Effmann type I blind, II complete, III with caudal duplication |
| Dilatation | **Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); prune-belly association |
| Cystic or remnant | Cowper's syringocele; enlarged utricle (proximal hypospadias); parameatal cyst |

## Epispadias (including isolated penopubic epispadias): definition, embryology, classification and clinical features

- Part of the **exstrophy–epispadias complex**: a **dorsal urethral defect** (defect of the dorsal urethral wall): the roof is replaced by a broad (dorsal) mucosal strip (urethral plate). Cause (embryology): absent mesodermal reinforcement of the cloacal membrane; the genital tubercles fuse **caudal** to it, giving a dorsal urethral defect.
- **Male (1 in 117,000; incidence of isolated male epispadias about 1 in 117,000), classification by meatus:** glanular, penile, **penopubic** (subsymphyseal) (about 70% of male cases are complete (penopubic), with incontinence). **Female (1 in 484,000), Davis:** vesicular (patulous), subsymphyseal, retrosymphyseal (incontinent).
- **Features (clinical):** penopubic meatus; outlet may admit a finger → **total incontinence**. **Short, broad, upturned penis with dorsal chordee**, like exstrophy; pubic diastasis (less than exstrophy); small bladder from no outlet resistance; **VUR (vesicoureteral/vesicoureteric reflux) 30–40% (male), 30–75% (female)**. Female (Davis retrosymphyseal type): bifid clitoris, depressed mons, incontinence.

## Evaluation

- (Renal) ultrasound; VCUG (voiding cystourethrogram); cystoscopy and capacity under anaesthesia.
- **Key decision:** epispadias or exstrophy pathway. Treat as exstrophy (osteotomy and closure) if pubic diastasis is over 3 cm, the intrasymphyseal bar is poor, sphincter is absent or capacity is tiny (tiny bladder).

## Management of epispadias

- **Aims:** continence, safe upper tracts, a straight functional penis of good length.
- **Preoperative testosterone** 2 mg/kg intramuscularly (intramuscular), 5 and 2 weeks before urethroplasty.

### Stage 1: penile reconstruction and urethroplasty (6–12 months)

- Release chordee: divide suspensory ligaments and scar; free corpora from the inferior pubic rami.
- **Modified Cantwell–Ransley:** outline an 18-mm plate to the tip; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; free the plate from the corpora (distal 1 cm left on the glans); tubularise over 8 Fr; transpose it (the tubularised plate is moved) **ventral to the corpora**, which are rotated medially ± cavernocavernostomy; close glans wings; Z-plasty at the base; stent 10–12 days.
- **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias.

- Outlet resistance rises, so capacity grows.

**Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)**

- **YDL (Young–Dees–Leadbetter)** bladder neck repair + Marshall–Marchetti–Krantz suspension + **ureteric reimplantation**. Osteotomy usually unnecessary.
- **Good capacity:** single-stage repair possible.

**Female epispadias:** at 6–12 months excise the urethral roof, close over 10 Fr, unite clitoral halves and labia minora, monsplasty; YDL at 4–5 years.

### Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula up to (about) one-fifth (at the penile base); long-term fistula and stricture about 5% (stricture rare); erections and paternity (usually) preserved.
- Better than exstrophy (bladder never exposed or scarred). Failures (failure): bulking injection (injectables), augmentation, continent stoma.

**Diagram to draw:** double zipper (canalisation then fusion); penopubic epispadias (dorsal penopubic meatus and plate).

> **Key points to remember:**
> - Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
> - About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
> - Diastasis over 3 cm (>3 cm) → exstrophy pathway.
> - Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
> - Continence after YDL: males 47–82%, females 70–87%.

## Short answer questions (SAQ)

## SAQ 1 Posterior urethral valves: embryology, management and valve bladder

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 SAQ "Embryology and Management of P.U. valve patient." · P1 SAQ "Embryology of Posterior Urethral Valves." · P2 SAQ "Valve bladder: management"

### Introduction

- PUV (posterior urethral valves): the commonest cause of bladder outlet obstruction in boys (1.4–2.2 per 10,000 births).
- Morgagni described them (1769); Hugh Hampton Young named and classified them (1919).

### Normal development

- **Weeks 5–6:** the **urorectal septum** divides the cloaca into the anorectal canal (rectum) and the **urogenital sinus**.
- The **Wolffian (mesonephric) ducts** join the cloaca just before division. Their orifices normally migrate to the **verumontanum**, leaving small folds (plicae colliculi).

- The **urogenital membrane** normally disappears completely.
- Fetal urine starts at 10–14 weeks; after 16 weeks amniotic fluid is mainly fetal urine. So obstruction in this period damages bladder, kidneys and lungs.

### Embryology of the valves: Young types and their origin

| Type | Anatomy | Embryology |
|---|---|---|
| **I (about 95%)** | Leaflets from the verumontanum fuse anteriorly just proximal to the external sphincter | Abnormal anterior / anterolateral insertion of the Wolffian duct ends into the cloacal wall: hypertrophied inferior urethral crest |
| **II** | Folds from verumontanum to bladder neck | Not real (Stephens 1955): secondary bladder-neck hypertrophy |
| **III (5–10%)** | Annular (ring) diaphragm with a central hole | **Persistent urogenital membrane** |

### Other theories and genetics

- Hypertrophied mucosal folds (oldest); cloacal remnants; Wolffian–Müllerian maldevelopment.
- **COPUM (congenital obstructing posterior urethral membrane, Dewan):** one oblique membrane, torn by the first catheter into apparent type I leaflets. Plausible but unproven.
- **Cobb's collar (Moormann's ring):** a congenital ring-like narrowing of the proximal bulbar urethra, just distal to the external sphincter; some authors group it with type III valves.
- **Genetics:** mostly sporadic; chromosome 11 duplications; copy number variants (worse renal outcome); TBX5, PTK7.

**Diagram to draw:** posterior urethra with type I leaflets versus a type III membrane, with keyhole VCUG.

### Consequences of obstruction

- Thick bladder, dilated posterior urethra (**keyhole sign**), hydroureteronephrosis, **renal dysplasia**, oligohydramnios, pulmonary hypoplasia.
- Pop-offs (VUR (vesicoureteral reflux), urinoma, ascites) relieve pressure.
- Dysplasia is fixed in utero; ablation cannot reverse it.

### Antenatal management

- Bilateral hydroureteronephrosis, thick bladder, **keyhole sign**, oligohydramnios.
- **Vesicoamniotic shunt** helps survival (lungs), not kidneys (PLUTO trial, 2013).

### Neonatal management

1. **Drain the bladder:** 5–7 Fr feeding tube (coudé if it curls); minimal balloon inflation.
2. **Stabilise:** ventilation; correct **post-obstructive diuresis**, hyponatraemia, acidosis, hyperkalaemia; antibiotics.
3. **Confirm:** VCUG (voiding cystourethrogram), renal ultrasound, MAG3 (mercaptoacetyltriglycine) at 2–3 months.
4. **Primary endoscopic valve ablation** (preferred): laser, Bugbee or hook at 5 and 7 o'clock ± 12; VCUG within 1 month.
5. **Vesicostomy** if the urethra is too small or drainage fails. **Upper-tract diversion** rarely: no proven renal benefit.

6. **Circumcision:** cuts febrile UTI (urinary tract infection) sharply. - **VUR** (50–80%) is secondary; treat the bladder, not the ureter.

### Prognosis

| Nadir creatinine (first year) | ESRD (end-stage renal disease) risk |
|---|---|
| < 0.4 mg/dL | Nil |
| 0.4–0.69 | 2% |
| 0.7–0.99 | 27% |
| > 1.0 | Up to 100% |

- Lifetime ESRD risk 20–50%.

### Valve bladder: definition

- **Valve bladder syndrome** (Mitchell, 1982): upper tracts and renal function worsen after successful PUV ablation, without residual obstruction, because of bladder dysfunction.

### Valve bladder: pathophysiology (vicious cycle)

``` Detrusor hypertrophy, collagen, low compliance → poor sensation, rising residual

- polyuria (nephrogenic diabetes insipidus) → bladder never empty → pressure transmitted to dilated upper tracts → they refill the bladder → renal damage ```

- Three processes: **polyuria, poor sensation, residual urine**; together they keep the upper tracts dilated.
- Age pattern: hyperreflexia in infancy, better compliance in childhood, then a large atonic bladder (myogenic failure) after puberty.

**Diagram to draw:** vicious-cycle diagram of the valve bladder.

### Valve bladder: clinical features and evaluation

- Late continence (about 5.5 years), incontinence, UTI, worsening hydronephrosis, rising creatinine. **New incontinence after being dry is a red flag.**
- Uroflowmetry and PVR (post-void residual) at each visit; ultrasound; creatinine; voiding diary.
- **Videourodynamics** when the picture changes and before transplantation.

### Valve bladder: management (escalating ladder)

1. **Behavioural:** adequate fluids, timed and double voiding, biofeedback.
2. **Drugs:** anticholinergic (oxybutynin 0.1 mg/kg twice daily) for a small high-pressure bladder; α-blocker for high residual. Stop oxybutynin if residual rises.
3. **CIC (clean intermittent catheterisation)** if residual rises despite drugs.
4. **Overnight bladder drainage** (catheter for 7–10 hours) for worsening hydronephrosis, UTI or renal function; it gives bladder and kidneys a daily rest.
5. **Mitrofanoff appendicovesicostomy** if urethral CIC is difficult (sensate urethra, high bladder neck).
6. **Augmentation** (rare) for a small, high-pressure bladder despite the above: **ureterocystoplasty** with the dilated ureter is preferred, avoiding bowel (mucus, acidosis, stones). - Reflux is treated through the bladder; reimplantation is rarely needed.
```

### Long-term care and transplantation

- **CKD (chronic kidney disease)** care.
- ESRD in 20–50%. Before **transplantation**, optimise the bladder (videourodynamics, CIC or overnight drainage); graft survival is then similar.

> **Key points to remember:**
> - Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
> - Urorectal septum divides the cloaca at weeks 5–6.
> - COPUM: one membrane torn into leaflets by catheterisation.
> - Dysplasia is fixed in utero; ablation cannot reverse it.
> - Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
> - Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
> - Valve bladder: polyuria, poor sensation, residual urine; overnight drainage and CIC break the cycle.
> - Ureteric augmentation rather than bowel; optimise before transplant.

## SAQ 2 Ultrasonography in neonates

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "USG in neonates" · P3 SAQ "HSG in neonates"

### Introduction

- Assumed meaning: ultrasonography (USG) of the neonatal urinary tract, including antenatal hydronephrosis follow-up. "HSG" (hysterosalpingography) in the P3 wording is not done in neonates, so it is taken as USG.
- First-line imaging, used for antenatal **UTD (urinary tract dilation)** (1–3% of pregnancies), febrile UTI (urinary tract infection), mass and poor stream.

### Why ultrasound first

- No radiation or sedation; repeatable. Shows **anatomy, not function**.

### Timing and technique

- **First scan after 48 hours**: earlier, physiological (relative) dehydration under-reads dilation.
- **Scan early if outlet obstruction (PUV, posterior urethral valves) is suspected or the kidney is solitary**; catheterise in suspected obstruction.
- Higher-frequency probes than in adults; kidneys supine and prone; length against age nomograms; bladder full and after voiding; Doppler.
- Measure **APD (anteroposterior pelvic diameter)** on a transverse mid-kidney view; assess calyces, parenchyma, ureters and bladder wall (thick if over 3 mm when full).

### Normal neonatal kidney (pitfalls)

- Cortex **iso- or hyperechoic** to liver.
- **Prominent hypoechoic pyramids** are normal and mimic dilated calyces or cysts (not hydronephrosis).
- Normal postnatal **APD <10 mm**, with no calyceal or ureteric dilation.

### Grading

- **SFU (Society for Fetal Urology) grades:** 0 none; 1 pelvis only; 2 major calyces; 3 major and minor calyces; 4 grade 3 plus parenchymal thinning.
- **UTD 2014, prenatal:** A1 (APD 4 to <7 mm before 28 weeks, 7 to <10 mm after) versus A2–3 (larger, or any other abnormality).

### Antenatal hydronephrosis follow-up (UTD 2014, postnatal)

| UTD risk | Findings | Action |
|---|---|---|
| **P1** (SFU 1–2) | APD 10 to <15 mm, central calyces | Repeat US at 3–6 months; no VCUG (voiding cystourethrogram) or prophylaxis |
| **P2** (SFU 3) | APD ≥15 mm, peripheral calyces or dilated ureter | Repeat US; VCUG and prophylaxis at discretion / optional (give prophylaxis if ureter >7 mm) |
| **P3** (SFU 4) | Abnormal parenchyma or bladder | Repeat US at 1 month; VCUG, prophylaxis, MAG3 (mercaptoacetyltriglycine) scan |

**Diagram to draw:** SFU grades 0–4 as kidney outlines.

### Specific neonatal findings

| Condition | Ultrasound finding |
|---|---|
| PUJO (pelviureteric junction obstruction) | Dilated pelvis, normal ureter; pyramids under 3 mm predict pyeloplasty |
| VUR (vesicoureteral reflux) | Normal first scan in up to 25% |
| **PUV** | Thick bladder, dilated posterior urethra (**keyhole sign**), bilateral hydroureteronephrosis; VCUG before discharge |
| Duplex anomalies | Ureterocele (thin-walled intravesical cyst), ectopic ureter, megaureter |
| MCDK (multicystic dysplastic kidney) | Non-communicating cysts of varying size, no central pelvis, scant parenchyma |
| ARPKD (autosomal recessive polycystic kidney disease) | Bilateral large, bright (echogenic) kidneys |
| Renal vein thrombosis | Large echogenic kidney, no venous flow; haematuria, mass, thrombocytopenia, hypertension |
| Adrenal haemorrhage | Avascular, evolving suprarenal mass (neuroblastoma is vascular) |
| Other masses | Mesoblastic nephroma |
| Spine | Occult dysraphism before arch ossification (up to 6 months) |
| DSD (differences of sex development) | Müllerian structures, gonads |

- **Also know:** hydrometrocolpos, scrotal masses, twinkling artefact (stones).

### Limitations

- Normal US does not exclude VUR. Function needs MAG3 renography (after renal maturation) or DMSA (dimercaptosuccinic acid).

### Recent advances

- ceVUS (contrast-enhanced voiding urosonography).

> **Key points to remember:**
> - First postnatal scan after 48 h, except suspected PUV; normal APD <10 mm.
> - Echogenic cortex and prominent pyramids are normal in neonates.
> - UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
> - Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

## SAQ 3 Nocturnal enuresis

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "Nocturnal enuresis" · P3 SAQ "Enuresis"

### Definition (ICCS [International Children's Continence Society])

- Intermittent urinary incontinence **during sleep** in a child **aged 5 years or more**.
- **Primary:** never dry. **Secondary:** wetting after at least **6 months** dry (look for stressors, diabetes, UTI [urinary tract infection], neurological disease).
- **MNE (monosymptomatic):** no daytime LUTS (lower urinary tract symptoms). **NMNE (non-monosymptomatic):** with daytime LUTS.

### Epidemiology

- About **5–10% at 7 years**; falls with age, with spontaneous resolution about 15% a year; about 2% of adults; familial.

### Pathophysiology: three-way mismatch

1. **Nocturnal polyuria** (poor or reduced night-time ADH [antidiuretic hormone] rise).
2. **Small (reduced) nocturnal bladder capacity** or DO (detrusor overactivity).
3. **Failure to arouse** from sleep. - **Why sleep apnoea matters:** obstructed breathing makes arousal difficult and releases atrial natriuretic peptide, which raises urine output. - Comorbidities: constipation, sleep-disordered breathing, ADHD (attention deficit hyperactivity disorder).

**Diagram to draw:** triad: nocturnal polyuria, small bladder capacity, arousal failure.

### Evaluation

- **History:** day symptoms, bowels, snoring, fluids, family history.
- **Examination:** spine, genitalia, abdomen for faecal loading; urinalysis.
- **Bladder diary:** daytime voided volumes against EBC (expected bladder capacity) = (age + 2) × 30 mL in the chapter (ICCS uses [age + 1] × 30 mL); night urine output by weighing nappies.
- Uroflowmetry, PVR (post-void residual) and ultrasound only for NMNE or before adding anticholinergics. No invasive tests in MNE.

### Management: general

- **Treat daytime LUTS and constipation first** (in NMNE).
- Education removes blame; standard urotherapy. Waking, fluid restriction alone and punishment do not work.

### Management: first line

| | Alarm | Desmopressin |
|---|---|---|
| Target | Arousal (conditioning) | Night urine output |
| Use | Nightly for months; parent wakes child; continue to 14 dry nights | 0.2–0.4 mg tablet or 120–240 µg melt, 1 hour before bed |
| Response | 50–70%; durable | Rule of thirds; immediate but relapses off the drug |
| Caution | Needs motivated family | Restrict / limit evening fluids (hyponatraemia) |

- Overlearning (extra evening fluid once dry) reduces alarm relapse.

### Management: failure and resistant cases

- **Failure:** switch or combine.
- **Second line:** add an anticholinergic (oxybutynin) after checking PVR and bowels.
- **Resistant:** imipramine (cardiotoxic; ECG [electrocardiogram]), off-label β3-agonists; treat sleep apnoea (adenotonsillectomy).

### Adult enuresis

- New enuresis in an adult may signal high-pressure chronic retention: catheterise and treat the obstruction.

> **Key points to remember:**
> - ≥5 years; secondary = after 6 months dry.
> - Alarm (durable) and desmopressin (rule of thirds) are first line.
> - Treat daytime symptoms and constipation first.

## SAQ 4 Paediatric neurogenic bladder (myelodysplasia): assessment, management and bladder myoplasty

P1 · P3 **merged from 2**

*Asked as:* P3 SAQ "Neurogenic bladder: prenatal/neonatal assessment and bladder myoplasty" · P1 SAQ "Neurogenic bladder in myelodysplasia"

### Introduction

- Myelodysplasia (NTD [neural tube defect], mainly **myelomeningocele**, an open NTD) is the commonest cause of neurogenic LUT (lower urinary tract) dysfunction in children; other causes are occult dysraphism, sacral agenesis and anorectal malformation.
- Lesions: lumbosacral 47%, lumbar 26%, sacral 20%.
- **Bony or motor level does not predict bladder function**, and the lesion can progress (re-tethering), so every newborn needs urodynamic testing.
- About 10% of deaths in SB (spina bifida) are urological. Folic acid 400 µg daily before conception prevents 50–70% of NTDs.

### Patterns of dysfunction

- Detrusor: overactive (NDO, neurogenic detrusor overactivity) or underactive. Outlet: competent, incompetent or dyssynergic.

- Newborns: most have detrusor contractions; over a third are acontractile. Needle EMG (electromyography): intact sacral arc about 40%, partial denervation 25%, complete 35%.

### Prenatal assessment

- **Maternal serum AFP (alpha-fetoprotein)** is raised in open defects; amniotic AFP and acetylcholinesterase confirm.
- **Ultrasound:** spinal defect; "lemon" and "banana" signs of Chiari II; ventriculomegaly; hydronephrosis, thick bladder. Fetal MRI (magnetic resonance imaging) for level.
- Multidisciplinary counselling at a fetal centre.
- **MOMS (Management of Myelomeningocele Study) 2011:** closure **before 26 weeks** reduced shunt need (about 40% versus 82%) and improved motor and cognitive scores, but caused prematurity (34 versus 37 weeks) and uterine thinning. **No proven bladder benefit**: CIC (clean intermittent catheterisation) and reconstruction rates unchanged.

### Neonatal assessment (baseline; EAU [European Association of Urology] 2022)

| Test | Timing / purpose |
|---|---|
| Examination | Spine, legs, anal tone, perineal sensation, bladder |
| **Renal and bladder ultrasound** | Before discharge, then 6-monthly for 2 years (serially); hydronephrosis, wall thickness |
| Residual urine | Start CIC if not emptying |
| Creatinine | Within 3 months (cystatin C later; better with low muscle mass) |
| **VCUG (voiding cystourethrogram) or video-UDS** | Reflux; dilating VUR (vesicoureteric reflux) in about 7% |
| **UDS (urodynamics)** with EMG | Within 3 months, once stable after closure |
| DMSA (dimercaptosuccinic acid) | Baseline scarring |

### Urodynamic risk (McGuire 1981; UMPIRE [Urologic and Renal Protocol for the Newborn and Young Child with Spina Bifida])

| Risk | DLPP (detrusor leak point pressure) or end-filling pressure |
|---|---|
| High (hostile) | 40 cm $H_2O$ or more |
| Intermediate | 25–39 cm $H_2O$, NDO or poor compliance |
| Low (safe) | Under 25 cm $H_2O$ |

- Poor compliance and DSD (detrusor–sphincter dyssynergia) also predict upper-tract damage.
- Repeat studies: the lesion can change (re-tethering).

**Diagram to draw:** neonatal UDS trace with DLPP over 40 cm $H_2O$.

### Management: kidneys first, then continence

- **Early CIC (Lapides 1971) plus oxybutynin** for high-risk (hostile) bladders: hydronephrosis 0–15% versus 18–80% expectant; ESRD (end-stage renal disease) falls from about 30% to under 2% (P1: 0–1.6%).
- Other drugs: solifenacin, mirabegron; **botulinum toxin A** about 10 U/kg (maximum 360 U), continence 65–87%.

- Surgery: vesicostomy (bridge); **augmentation cystoplasty**; Mitrofanoff channel; bladder-neck sling, reconstruction or artificial sphincter.
- VUR: CIC and antimuscarinic first; avoid Credé; reimplant during augmentation if grade ≥III.
- **Also know:** latex allergy; bowel programme; sexual dysfunction (about 75% of men have ED [erectile dysfunction]; sildenafil helps); precocious puberty.

### Bladder myoplasty: definition

- The P3 wording "bladder myeloplsty" is assumed to mean "bladder myoplasty" (a muscle procedure on the bladder wall).
- **Detrusor myoplasty:** wrapping a **vascularised, innervated skeletal muscle flap** round the bladder so that voluntary contraction of that muscle empties an **acontractile bladder**.

### Bladder myoplasty: indications

- **Acontractile or underactive detrusor** with catheter dependence: lower motor neuron lesions (spina bifida, cauda equina injury), myogenic failure.
- Needs good bladder capacity and compliance, and a competent but not obstructing outlet.

### Latissimus dorsi detrusor myoplasty (LDDM; Stenzl)

- **Free latissimus dorsi flap:** artery and vein joined to the **inferior epigastric vessels**; motor nerve joined to an **intercostal nerve**.
- Muscle wrapped **spirally over more than three-quarters of the bladder** and anchored to pelvic floor fascia and ligaments.
- The patient voids by contracting the transferred muscle (abdominal effort).
- **Results:** most voided with low residual; bladder contractility index rose to the strong range.
- **Complications** in about one-third: thromboembolism, abscess, wound infection; none long term.
- **Rectus abdominis detrusor myoplasty:** simpler, less morbid; small spinal cord injury series with spontaneous voiding.
- **Status: investigational.** Standard care stays **clean intermittent catheterisation**.

**Diagram to draw:** latissimus flap spiralled round the bladder with vascular and nerve anastomoses.

### Other "myoplasty": detrusor myotomy/myectomy (autoaugmentation)

- Purpose is the opposite: **increase capacity and compliance** in a small, poorly compliant or overactive bladder.
- **Technique (Cartwright and Snow, 1989):** strip and excise the detrusor over the dome, leave the mucosa intact, and anchor the detrusor edges (psoas hitch) so the mucosa bulges as a **wide-mouthed diverticulum**.
- **Pros:** urothelial lining, no mucus, no metabolic change, no bowel.
- **Cons:** seldom adds significant capacity; compliance gain may not last; **not recommended for adult detrusor overactivity**.
- **Also know:** seromuscular enterocystoplasty covers the bulge with demucosalised sigmoid.

### Alternatives

- **Also know:** sacral neuromodulation, sacral anterior root stimulation, reduction cystoplasty (abandoned), muscle-derived stem cells (research).

**Key points to remember:**

- Level does not predict bladder: UDS within 3 months.
- DLPP 40 cm $H_2O$ or more = hostile (McGuire 1981); early CIC + antimuscarinic protects kidneys.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

## SAQ 5 Anterior urethral valve

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 17 Paper 2, Q3c

### Definition and epidemiology

- AUV (anterior urethral valve): a congenital obstructing fold or cusp in the anterior urethra (bulbar, penoscrotal or penile). It is the **commonest congenital obstructive lesion of the anterior urethra** but **25–30 times less common than PUV (posterior urethral valves)**.
- Often linked to an **anterior urethral diverticulum**: the distal lip of the diverticulum fills and acts as a valve, or a semilunar fold hangs from the roof.

### Embryology (unclear)

- Hypoplastic corpus spongiosum over the segment suggests a place on the **hypospadias spectrum**, or faulty union of urethral mucosa and fossa navicularis epithelium; a ruptured dilated Cowper (bulbourethral) gland duct is also proposed.

### Clinical features (vary with severity and age)

- Antenatal hydronephrosis, LUTO (lower urinary tract obstruction) in severe cases.
- **Ventral penile bulge** during voiding, with urine expressed by compressing the shaft; post-void dribbling, weak stream, incontinence.
- Palpable bladder, UTI (urinary tract infection), renal insufficiency.

### Evaluation

- **VCUG (voiding cystourethrogram):** dilated anterior urethra proximal to the valve, diverticulum, trabeculated bladder, bladder diverticula, often VUR (vesicoureteric reflux).
- Ultrasound, creatinine; urethroscopy confirms.
- **Differential:** Cowper syringocele, megalourethra, acquired diverticulum, stricture.

### Management

- **Premature or small infant:** vesicostomy until the urethra takes a cystoscope.
- **Most cases: endoscopic valve ablation** (Bugbee electrode or laser), usually definitive.
- **Large diverticulum:** open excision and urethroplasty over a catheter.
- Then manage VUR, infection and bladder dysfunction.

### AUV versus PUV

| Feature | AUV | PUV |
|---|---|---|
| Frequency | 25–30 times rarer | Commonest infant outlet obstruction |
| Site | Bulbar, penoscrotal, penile | From the verumontanum |
| Sign | Penile bulge | Weak stream, palpable bladder |
| Bladder dysfunction | Up to 80% | Near universal |
| Renal outcome | 78% normal function | ESRD (end-stage renal disease) 20–50% |

### Prognosis

- Depends on pre-treatment creatinine and GFR (glomerular filtration rate). Azotaemia, VUR and UTI together raise the risk of poor renal outcome 25-fold.

**Diagram to draw:** VCUG sketch: dilated bulbar urethra with diverticulum ending at a valve lip.

> **Key points to remember:**
> - 25–30 times rarer than PUV; diverticulum common.
> - Penile bulge on voiding; diagnosis by VCUG.
> - Endoscopic ablation usually definitive; 78% keep normal renal function.

## SAQ 6 CSIC.

P3

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 3, Q3e

*Taken to mean **CISC – clean intermittent self-catheterisation**.*

### Definition and principle

- Regular, **clean (not sterile)** passage of a catheter by the patient (or carer) to empty the bladder, then removal.
- **Lapides (1971–72)**: infection comes from **stasis and overdistension** (poor bladder-wall blood flow), not from bacteria alone. Regular complete emptying protects the bladder wall and upper tracts, even though bacteriuria develops.
- Gold standard for chronic emptying failure.

### Indications

- **Neurogenic bladder:** spina bifida, spinal cord injury, multiple sclerosis, cauda equina, diabetic cystopathy.
- **Underactive or acontractile detrusor**; chronic retention; post-operative retention.
- **Detrusor–sphincter dyssynergia**: CISC bypasses the obstructed sphincter.
- After **augmentation or continent diversion**; **valve bladder**; high residual after botulinum toxin.
- **Contraindications:** poor hand function or cognition without a carer, urethral false passage or stricture, unwillingness.

### Technique

- Wash hands → clean meatus with water → lubricated catheter (or pre-lubricated **hydrophilic** catheter) → pass until urine flows → drain fully → withdraw slowly.
- Sizes: adults about 12–14 Fr; children by age (8–10 Fr).
- **Frequency:** 4–6 times a day, keeping each volume **below about 400–500 mL** (child: below expected bladder capacity).
- Combine with **antimuscarinics** for overactivity; bladder diary.

### Catheters

| Type | Note |
|---|---|
| Plain PVC (polyvinyl chloride), reusable after washing | Cheap; common in India |
| **Hydrophilic-coated** single-use | Less urethral trauma; longer time to first UTI (urinary tract infection) |
| Compact or closed-system | Convenience, outdoor use |

### Children

- Start **early (infancy)** in spina bifida with hostile bladder; families adapt best; child involved from 3–5 years.
- **Mitrofanoff appendicovesicostomy** (1980) if urethral catheterisation is difficult (wheelchair, sensate urethra, girls).

### Complications

- **Asymptomatic bacteriuria** (common): **do not treat or give prophylaxis**; treat only symptomatic UTI.
- Urethral trauma, bleeding, **false passage**, stricture, epididymitis, bladder stones (hair, mucus), rarely lost catheter.

### Outcome

- Less pyelonephritis, epididymo-orchitis and urosepsis than indwelling catheters; preserves the upper tracts; with drugs, continence in many.

**Diagram to draw:** neurogenic bladder management ladder: CISC + antimuscarinic → botulinum toxin → augmentation + channel.

> **Key points to remember:**
> - Lapides: overdistension, not bacteria, causes infection; clean is enough.
> - 4–6 times daily, under about 500 mL each.
> - Do not treat asymptomatic bacteriuria.
> - Mitrofanoff when urethral access is hard.

## SAQ 7 Discuss diagnostics and operative pediatric laparoscopy in urology

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 3, Q3a

## Evolution

- Diagnostic (non-palpable testis, 1976) → nephrectomy (1994) → pyeloplasty (1995) → robotic (2002).

## Diagnostic laparoscopy

- **Non-palpable testis** (procedure of choice, after examination under anaesthesia): note vessels, vas, testis and internal ring.

| Finding | Meaning |
|---|---|
| Blind-ending vessels and vas, closed ring | Vanishing testis: stop |
| Vessels and vas entering ring | Explore groin; excise remnant |
| Intra-abdominal testis | Orchidopexy (primary or Fowler-Stephens) |
| - **DSD (differences of sex development):** gonadal inspection and biopsy, Müllerian structures. | |

## Operative laparoscopy

- **Testis:** primary orchidopexy (>90% success); Fowler-Stephens one- or two-stage; gonadectomy in DSD.
- **Kidney:** nephrectomy, heminephrectomy, **pyeloplasty** (commonest MIS (minimally invasive surgery) procedure).
- **Ureter and bladder:** extravesical or vesicoscopic (cross-trigonal) reimplantation, Mitrofanoff, augmentation, bladder neck reconstruction, utricle and urogenital sinus.
- **Also know:** RPLND (retroperitoneal lymph node dissection) for paratesticular rhabdomyosarcoma.

## Technical principles

- **Access:** open Hasson or Veress at umbilicus (Palmer's point if scarred); equal safety.
- 3–5 mm ports; robots use 8 mm. Transperitoneal (most space), retroperitoneal (spares peritoneum, useful in peritoneal dialysis), vesicoscopic.
- **Pressures:** $CO_2$ absorption, reduced venous return, raised airway pressure; reversible oliguria (no fluid bolus).

| Age | 0–1 month | 2–12 months | 1–2 years | 2–8 years |
|---|---|---|---|---|
| Max IAP (intra-abdominal pressure) | ≤6 mmHg | ≤8 | ≤10 | ≤12 |
| - **Positioning:** modified flank for kidney; supine steep Trendelenburg for pelvis. Infants: minimise inhalational anaesthetic, rectal tube to reduce bowel distension. | | | | |
| - **Only absolute contraindication:** inability to tolerate pneumoperitoneum. | | | | |

## Complications

- Right common iliac vessel injury (behind umbilicus; commonest major), inferior epigastric (commonest minor), bowel injury, subcutaneous emphysema, **port-site hernia 0.5–3%** (close fascia).

### Pros and cons

- Shorter stay, better cosmesis, deep pelvic access; pain benefit unproven. Robot: cost, learning curve; infant working space only about 1 L.

**Diagram to draw:** laparoscopic view of internal ring with blind-ending vessels.

> **Key points to remember:**
> - Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
> - Pressure ceiling 6–12 mmHg by age.
> - Right common iliac at Veress entry; close port fascia.

## SAQ 8 Vesicostomy.

P4

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 4, Q3f

### Definition

- **Cutaneous vesicostomy**: the bladder dome is brought to the lower abdominal skin as an incontinent stoma draining into the nappy. It "vents" the bladder, protecting the upper tracts; usually a **temporary** bridge to reconstruction.

### Indications

- **PUV (posterior urethral valves)**: infant too small for a resectoscope; upper tract deterioration despite drainage or ablation.
- **Prune-belly syndrome**: acute renal failure, urosepsis, urethral atresia. A larger stoma is needed because stenosis is common.
- **Neurogenic bladder**: high pressure despite CIC (clean intermittent catheterisation) and drugs, or CIC impossible; worsening hydronephrosis; febrile UTI (urinary tract infection) with high-grade reflux.
- **Also know:** urethral atresia; MMIHS (megacystis-microcolon-intestinal hypoperistalsis syndrome); urogenital sinus anomalies.

### Blocksom technique (Duckett modification)

1. Transverse 2 cm incision midway between umbilicus and pubis. (In prune-belly, excise any urachal diverticulum.)
2. Incise the rectus sheath; separate the recti.
3. Fill the bladder; mobilise the **dome** up to the urachus, sweeping peritoneum off.
4. Bring the **dome** (not the anterior wall) to the wound. This reduces prolapse of the posterior wall.
5. Anchor detrusor to rectus fascia; stoma should take about 24 Fr.
6. Open the dome; suture mucosa to skin. No catheter needed.

### Advantages

- Simple; no catheter or appliance.
- The bladder still fills and empties through the stoma at low leak pressure, so compliance is **not** lost; it remains usable for reconstruction or transplantation.

### Complications

- **Stomal stenosis**, mucosal **prolapse**, peristomal dermatitis, stones, UTI; leakage harder to manage as the child grows.

### Closure

- When the cause is treated (valve ablation, urethral growth) or at definitive reconstruction; check bladder function first (cystogram or urodynamics).
- Excise the stoma, free the bladder from fascia, close in two layers, catheter drainage for a few days.

### Alternatives

| Option | Use |
|---|---|
| Cutaneous ureterostomy or pyelostomy | Rare; persistent upper tract obstruction |
| Ileovesicostomy | Unable to catheterise; 10–15 cm ileum |
| Appendicovesicostomy (Mitrofanoff, 1980) | Continent catheterisable channel |

**Diagram to draw:** sagittal view of bladder dome sutured to fascia and skin.

> **Key points to remember:**
> - Blocksom: dome-based stoma midway between umbilicus and pubis.
> - Bladder keeps cycling; compliance preserved.
> - Stenosis and prolapse are the main complications.

## SAQ 9 Management of Defecation Disorders resulting into LUTS

P4

**Type:** SAQ (10 marks)
**Source:** 2025-Winter Paper 4, Q3f

*Assumed meaning: bladder and bowel dysfunction, where constipation or defecation disorders cause LUTS (lower urinary tract symptoms).*

### Why the bowel causes LUTS

- **Shared space:** a loaded rectum compresses the bladder, reducing capacity and obstructing outflow.
- **Shared nerves:** bladder and bowel afferents converge on the same spinal segments; cross-organ sensitisation causes urgency and detrusor overactivity.
- **Shared muscle:** a pelvic floor that contracts during defecation (**dyssynergic defecation**) also contracts during voiding (**dysfunctional voiding**) → high PVR (post-void residual), UTI (urinary tract infection), reflux.
- Result: urgency, daytime incontinence, enuresis, retention, recurrent UTI.

### Evaluation

- Stool frequency, pain, **Bristol chart**; 3-day bladder and bowel diary.
- **Rome IV (child):** at least 2, weekly for 1 month: ≤2 defecations/week; faecal incontinence weekly; retentive posturing; painful or hard stools; large rectal mass; large-diameter stools.
- **Rome IV (adult), for 3 months:** ≥2 of straining, hard stools, incomplete evacuation, blockage, manual manoeuvres (each in >25% of defecations), <3 stools/week.

- Examine abdomen, spine and perineum; exclude organic causes (Hirschsprung disease, hypothyroidism, neurogenic).
- **Also know:** rectal diameter ≥30 mm on ultrasound; no routine X-ray; manometry if refractory.

### Management

**Rule: treat the bowel first (EAU, European Association of Urology/ESPU, European Society for Paediatric Urology).**

1. **Disimpaction:** oral PEG (polyethylene glycol) 3350, 1–1.5 g/kg/day for 3–6 days; enemas if needed.
2. **Maintenance:** PEG 0.4 g/kg/day (range 0.2–0.8), at least 2 months; lactulose alternative; stimulants (senna) for expulsion.
3. **Urotherapy:** toileting after meals (gastrocolic reflex); footstool posture; normal fluid and fibre.
4. **Biofeedback, pelvic floor physiotherapy** for dyssynergia.
5. **Refractory: transanal irrigation** (water via rectal cone; a quarter to a third stop); **ACE (antegrade continence enema, Malone)**; botulinum toxin to the sphincter; sacral or tibial neuromodulation. - **Adults:** laxatives, prucalopride or linaclotide, biofeedback; rectocele repair; irrigation in neurogenic bowel. - **Caution:** anticholinergics worsen constipation; LUTS return.

**Diagram to draw:** pelvic cross-section showing loaded rectum compressing bladder.

> **Key points to remember:**
> - Treat the bowel first.
> - Rome IV: ≥2 criteria for 1 month (child).
> - PEG disimpaction 1–1.5 g/kg/day for 3–6 days.

# Rapid Revision Sheet · Section 5

### Bladder embryology and exstrophy-epispadias complex (LAQ)

- Bud from Wolffian duct; position decides VUR or ectopia.
- Urorectal septum weeks 5–6 (P2: weeks 4–6); mesonephric duct entry is the landmark.
- Vesicourethral canal forms the bladder; urachus from allantois.
- Trigone: common excretory duct absorption; mostly bladder muscle (current view).
- Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.
- Marshall–Muecke: overdeveloped cloacal membrane, failed mesodermal reinforcement; rupture timing decides variant.
- CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis about 4.8 cm; VUR in 100%.
- Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.
- MSRE: closure at birth, epispadias repair 6–12 months, YDL / BNR at 5–7 years at ≥100 mL.
- Osteotomy if closure after 72 h or diastasis >4 cm.
- CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
- Capacity ≥100 mL predicts continence; template size and first closure success predict continence.

### Prune belly syndrome (LAQ)

- Triad; 95% male; dysplasia and lungs decide survival.
- Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
- Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
- Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

### Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency (LAQ)

- S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
- PMC = switch; medial prefrontal cortex holds it off.
- EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
- DLPP ≥40 cmH2O = hostile; treat bowel first.
- Always ask about bowels; treat constipation first.
- Rule out UTI, PUV, meatal stenosis and occult dysraphism.
- Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

### Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies (LAQ)

- UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
- Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
- ANH (antenatal hydronephrosis) = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
- UTD A2–3 = APRPD ≥7 mm or any abnormal feature.
- Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
- First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.

- Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
- Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
- ARPKD large bright kidneys; MCDK non-communicating cysts.
- Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
- PLUTO: survival up, renal function unchanged.

**Embryological development of urinary bladder and management of Hutch Diverticulum? (LAQ)**

- Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
- Hutch: herniation through the ureteric hiatus → VUR.
- Ureter entering the sac: VUR will not resolve.
- Secondary: treat the outlet first; else diverticulectomy + reimplantation.

**Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias) (LAQ)**

- Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
- About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
- Diastasis over 3 cm (>3 cm) → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
- Continence after YDL: males 47–82%, females 70–87%.

**Posterior urethral valves: embryology, management and valve bladder (SAQ)**

- Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
- Urorectal septum divides the cloaca at weeks 5–6.
- COPUM: one membrane torn into leaflets by catheterisation.
- Dysplasia is fixed in utero; ablation cannot reverse it.
- Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
- Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
- Valve bladder: polyuria, poor sensation, residual urine.
- Valve bladder: overnight drainage and CIC; overnight drainage breaks the cycle.
- Ureteric augmentation rather than bowel; optimise before transplant.

**Ultrasonography in neonates (SAQ)**

- First postnatal scan after 48 h, except suspected PUV; normal APD <10 mm.
- Echogenic cortex and prominent pyramids are normal in neonates.
- UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
- Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

**Nocturnal enuresis (SAQ)**

- ≥5 years; secondary = after 6 months dry.
- Alarm (durable) and desmopressin (rule of thirds) are first line.
- Treat daytime symptoms and constipation first.

**Paediatric neurogenic bladder (myelodysplasia): assessment, management and bladder myoplasty (SAQ)**

- Level does not predict bladder: UDS within 3 months.
- DLPP 40 cm $H_2O$ or more = hostile (McGuire 1981); early CIC + antimuscarinic.
- Early CIC plus antimuscarinic protects kidneys.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

**Anterior urethral valve (SAQ)**

- 25–30 times rarer than PUV; diverticulum common.
- Penile bulge on voiding; diagnosis by VCUG.
- Endoscopic ablation usually definitive; 78% keep normal renal function. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**CSIC. (SAQ)**

- Lapides: overdistension, not bacteria, causes infection; clean is enough.
- 4–6 times daily, under about 500 mL each.
- Do not treat asymptomatic bacteriuria.
- Mitrofanoff when urethral access is hard.

**Discuss diagnostics and operative pediatric laparoscopy in urology (SAQ)**

- Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
- Pressure ceiling 6–12 mmHg by age.
- Right common iliac at Veress entry; close port fascia. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Vesicostomy. (SAQ)**

- Blocksom: dome-based stoma midway between umbilicus and pubis.
- Bladder keeps cycling; compliance preserved.
- Stenosis and prolapse are the main complications.

**Management of Defecation Disorders resulting into LUTS (SAQ)**

- Treat the bowel first.
- Rome IV: ≥2 criteria for 1 month (child).
- PEG disimpaction 1–1.5 g/kg/day for 3–6 days. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 6 · Female Urology

## Long answer questions (LAQ)

### LAQ 1 Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair

P1 · P2 **merged from 3**

*Asked as:* P1 LAQ "Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia" · P1 SAQ "Describe clinical presentation of pelic floor prolapse. diagnostic evaluation and brief discussion of various surgical options." · P2 SAQ "Vaginal prolapse repair."

#### Introduction and definitions

- Assumed meanings: "procedentia" is read as procidentia (complete uterine prolapse); "fertility sparing" / "fertility preserving" means uterus-preserving treatment for a woman who wants future pregnancy; "pelic floor prolapse" and "vaginal prolapse repair" mean pelvic organ prolapse (POP) in women and the surgical repair of vaginal wall and vault prolapse.
- **SUI (stress urinary incontinence):** involuntary leakage on effort, coughing or sneezing.
- **POP (pelvic organ prolapse):** descent of the anterior wall, posterior wall, uterus/cervix or vault. Anterior is commoner than posterior, which is commoner than apical. It is a disease only if symptomatic: a quality-of-life condition.
- Pelvic organs are supported by the **levator ani**, the **endopelvic fascia and its ligaments** and the **perineal membrane and perineal body**, anchored to the **bony pelvis**. Failure causes POP and SUI. Every SUI operation uses one of these anchors.

#### Bony pelvis and surgical anchors

- **Bony landmarks** and dangers: **sacrospinous ligament** (pudendal bundle), **Cooper's ligament** (bleeding), **sacral promontory** (left common iliac vein, right ureter).
- **Sacrospinous ligament (SSL)**, 7–8 cm, ischial spine to sacrum: the pudendal nerve and vessels pass behind the spine, the sacral plexus just posterior.
- **Sacral promontory:** left common iliac vein and right ureter each about 2.7 cm away.

#### Pelvic floor muscles: pelvic diaphragm (levator ani + coccygeus)

| Muscle | Attachment | Function |
|---|---|---|
| Puborectalis | Back of pubis, loops (as a sling) behind the anorectal junction | Anorectal angle; anorectal sling |
| Pubococcygeus (pubovaginal part) | Posterior pubis and arcus tendineus to viscera (vagina, perineal body, anus) and anococcygeal raphe | Slings urethra and vagina; closes hiatus, lifts urethra |
| Iliococcygeus | ATLA (arcus tendineus levator ani), ischial spine to raphe, coccyx | Horizontal **levator plate** |
| Coccygeus (not levator; separate) | Ischial spine to sacrum, coccyx | Lies on sacrospinous ligament |

- Levator tone (tonic contraction) closes the **urogenital hiatus**, through which all POP passes; the uterus and upper vagina rest on the levator plate.
- Levator is palpable through the lateral vaginal walls for PFMT (pelvic floor muscle training) assessment.
- **Muscle fails first, fascia second:** once injury (birth injury, avulsion) opens the hiatus, ligaments carry the load and fail; the fascia then carries the whole load and stretches.
- Levator defects in up to **20% of primiparous women**; avulsion predicts advanced prolapse.

### Endopelvic fascia and ligaments

- **Cardinal ligament** (side wall to cervix; carries the uterine artery) and **uterosacral ligament** (sacrum to cervix): the **cardinal–uterosacral complex** suspends the apex.
- The ureter is 4 cm from the uterosacral ligament near the sacrum but 0.9 cm near the cervix.
- **Pubocervical (pubovesical) fascia** attaches laterally to the **ATFP (arcus tendineus fasciae pelvis)**, the "white line" from the lower pubis to just above the ischial spine.
- **Rectovaginal fascia** supports the posterior wall.
- **Pubourethral ligaments** suspend the urethra from the pubis and anchor the mid-urethra; **urethropelvic ligaments** anchor it to the ATFP.

### Perineal membrane, perineal body and urethra

- The perineal membrane spans the pubic arch and is pierced by the urethra and vagina; above it lies the striated urogenital sphincter. The perineal body anchors the distal vagina.
- **Striated urogenital sphincter:** sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**.
- Urethra 4–5 cm; the continence zone is the **proximal and mid-urethra**. The **oestrogen-sensitive vascular submucosa** gives up to about 30% of closure force.

### DeLancey's levels of support (1992)

| Level | Region | Structures | Loss produces |
|---|---|---|---|
| **I – suspension** | Cervix/upper third | Cardinal–uterosacral complex | **Uterine or vault prolapse**, enterocele |
| **II – attachment** | Middle third | Paravaginal attachment to ATFP, levator fascia | Cystocele (central or paravaginal), rectocele; anterior or posterior wall prolapse |
| **III – fusion** | Distal third | Urethra, perineal membrane, perineal body | **Urethral hypermobility**, perineal descent, distal rectocele |

- **Central cystocele:** midline weakness, rugae lost. **Paravaginal cystocele:** ATFP detachment (Richardson 1981), rugae preserved.
- Level I is the keystone: always correct the apex.

**Diagram to draw:** sagittal female pelvis with levator plate and DeLancey levels I–III, with sacrospinous and sacrocolpopexy fixation points; hammock cross-section.

### Innervation

- **Pudendal nerve (S2–S4):** sphincters and perineal muscles; runs through Alcock's canal near the ischial spine; stretched in delivery.

- **Nerve to levator ani (S3–S5):** runs on the muscle's pelvic surface, separate from the pudendal; neuropathy in about a quarter of primiparous women.
- Autonomic pelvic splanchnic and hypogastric nerves: injured in radical pelvic surgery.

### Theories of continence and the operations they produced

| Theory | Mechanism | Surgical answer |
|---|---|---|
| **Pressure transmission (Enhorning 1961)** | Hypermobile urethra drops out of the abdominal pressure zone | Restore position: Burch, MMK (Marshall–Marchetti–Krantz) |
| **Hammock (DeLancey 1994)** | Urethra lies on anterior vaginal wall and endopelvic fascia attached to ATFP and levator; abdominal pressure compresses it against this firm backing; repositioning not essential | Restore backing |
| **Integral theory (Petros & Ulmsten 1990)** | Pubourethral ligament laxity at the **mid-urethra** | Tension-free mid-urethral sling |
| **Intrinsic sphincter deficiency (ISD) (McGuire 1978)** | Urethra fails to close | Sling, bulking, AUS (artificial urinary sphincter) |

- ISD: VLPP (Valsalva leak point pressure) <60 $cmH_2O$ or maximal urethral closure pressure <20 $cmH_2O$. Most SUI includes some ISD. So slings work in hypermobility, but Burch does poorly in severe ISD with a fixed urethra.

### Why prolapse occurs: risk factors (Bump & Norton)

- Muscle fails first (birth trauma, levator avulsion, denervation). The hiatus widens and the ligaments carry the whole load and stretch.
- **Predisposing:** genetics, collagen disorders (Ehlers–Danlos, Marfan).
- **Inciting: vaginal delivery** (strongest; forceps worse), parity, hysterectomy.
- **Promoting:** obesity, constipation, cough.
- **Decompensating:** ageing.

### Clinical presentation of prolapse

- **Vaginal bulge** (most specific), pressure, heaviness, worse by evening.
- **Splinting/digitation** to void or defecate (high-grade POP).
- Urinary: voiding difficulty (urethral kinking), OAB (overactive bladder); SUI in about 40%, which may be **occult**.
- Bowel (rectocele): obstructed defecation, stool trapping.
- Sexual dysfunction, discharge, ulceration; procidentia may cause hydronephrosis.

### Examination

- Lithotomy (supine) and **standing** with Valsalva / strain; Sims or split speculum for each compartment; lateral sulci (paravaginal defect); rectovaginal examination.
- **Cough stress test**, repeated with prolapse reduced (**occult SUI**); **Q-tip** angle change >30° = hypermobility; levator strength and avulsion.

## POP-Q (Pelvic Organ Prolapse Quantification, Bump 1996)

- Nine points in cm relative to the **hymen (0)**: inside (above) negative, outside positive; measured on maximal strain, recorded on a 3 × 3 grid.

| Point | Definition |
|---|---|
| Aa / Ap | Anterior / posterior wall 3 cm inside hymen (−3 to +3) |
| Ba / Bp | Most distal point of anterior / posterior wall (−3 if no prolapse) |
| C | Cervix or vault cuff |
| D | Posterior fornix (omitted after hysterectomy) |
| gh, pb, tvl | Genital hiatus, perineal body, total vaginal length (at rest) |

| Stage | Most distal point (leading edge) |
|---|---|
| 0 | No prolapse |
| I | >1 cm above hymen (< −1 cm) |
| II | Within ±1 cm of hymen (−1 to +1 cm) |
| III | >1 cm below hymen (> +1 cm) but < (tvl − 2) cm |
| IV | Complete eversion (≥ tvl − 2) |

- **Also know:** Baden–Walker grades 0–4; stage I–II often normal (POSST study).

## Investigations

- Urinalysis, PVR (post-void residual), bladder diary.
- **Cough test or urodynamics with prolapse reduced** for occult SUI; urodynamics also if ISD or mixed symptoms.
- Renal ultrasound in procidentia (hydronephrosis).
- Dynamic MRI or translabial ultrasound if complex or recurrent.
- A pessary trial confirms that the POP causes the symptoms.

## Principles of prolapse repair

- Operate for bother, not anatomy. The first operation gives the best chance.
- Identify each compartment by POP-Q. **The apex is the cornerstone:** failing to support it greatly raises recurrence; uncorrected level I loss causes recurrent prolapse after SUI repair.
- **Three categories: restorative** (native tissue), **compensatory** (graft or mesh, for weak tissue or failed repair), **obliterative**.
- Vaginal route: fewer complications, quicker recovery. Abdominal route: more durable.
- Screen for **occult SUI** with prolapse reduced and consider a concomitant sling (OPUS, CARE trials). Finish with **cystoscopy** to confirm ureteric efflux.

## Surgical options by compartment

| Compartment | Operations | Key points |
|---|---|---|
| **Anterior** (cystocele) | **Anterior colporrhaphy** (midline plication of pubocervical fascia); **paravaginal repair** (fascia reattached to the ATFP) | Central versus lateral defect; does not support the apex |

| Compartment | Operations | Key points |
|---|---|---|
| **Apical, vaginal** (uterus, vault) | **Uterosacral suspension** (sutures; ureter injury 1–11%; cystoscopy); **sacrospinous fixation** (suture 1.5–3 cm medial to the ischial spine, avoiding the pudendal bundle; gluteal pain 15%); iliococcygeus suspension; McCall culdoplasty; vaginal hysterectomy; uterus-sparing hysteropexy; Manchester | OPTIMAL trial: uterosacral = sacrospinous at 2 years |
| **Apical, abdominal** (compensatory) | **Sacrocolpopexy** (open, laparoscopic, robotic): macroporous polypropylene mesh from vaginal apex to the anterior longitudinal ligament over S1 / sacral promontory | Most durable (>90%); mesh exposure about 3–10% (P2: about 10% long term, CARE) |
| **Posterior** (rectocele) | **Midline fascial plication** without levatorplasty; site-specific repair; perineorrhaphy (levels II–III) | Levator plication causes dyspareunia; constipation may persist |
| **Obliterative** | **Le Fort** (partial, uterus in situ, lateral channels) or total colpocleisis | Elderly, not sexually active; success >90%; coital regret ≤13% |

- Cochrane 2016: vaginal apical repair has more recurrence and dyspareunia than sacrocolpopexy, but suits many older women.
- **April 2019:** the US FDA (Food and Drug Administration) ordered transvaginal synthetic mesh for (anterior) POP withdrawn; slings unaffected.
- Anterior colporrhaphy is not an SUI operation.

## Importance of anatomy in SUI surgery

| Operation | Anatomy used | Anatomical hazards |
|---|---|---|
| **Burch colposuspension** | Space of Retzius (avascular: symphysis in front, bladder behind, endopelvic fascia below). Paravaginal fascia at mid-urethra and bladder neck sutured to **Cooper's (iliopectineal) ligament** | Bleeding from Cooper's ligament and perivesical veins; ureteric kinking (cystoscopy); needs a mobile vagina |
| Paravaginal repair / VOS (vagino-obturator shelf) | Lateral vaginal sulcus to **ATFP** or obturator fascia (level II) | Obturator vessels and nerve |
| **Retropubic MUS (mid-urethral sling), TVT (tension-free vaginal tape; Ulmsten 1995)** | Tape under mid-urethra (integral theory), through **endopelvic fascia** and Retzius; suprapubic exits about 2 cm from midline | Bladder perforation (4.5%), vessels, bowel |
| **TOT (transobturator tape; Delorme 2001)** | Through obturator foramen / membrane, **obturator internus and externus**; exit at adductor longus tendon notch | Groin pain (6.4%), obturator neurovascular bundle; bladder injury rare (0.6%) |
| **SIMS (single-incision mini-sling)** | Anchors in obturator internus or endopelvic fascia | Works by restricting mobility |
| **Autologous PVS (pubovaginal sling)** | Rectus fascia at the **bladder neck** (proximal urethra), arms to rectus | Voiding dysfunction; best for ISD (compression) |
| Bulking agents | **Submucosa** coaptation | Repeated injections |

- **Why mid-urethra:** it is the zone of maximal closure pressure and pubourethral support. A loose tape lets the posterior urethral wall be compressed and kinked during stress.
- **TOMUS (2010):** retropubic 80.8% versus transobturator 77.7% objective success, not different.
- **SISTEr:** autologous sling more successful than Burch at 2 years, with more voiding dysfunction.
- **Also know:** needle suspensions abandoned (sutures pulled through).

**Diagram to draw:** cross-section of urethra on the vaginal hammock (DeLancey) with Burch, bladder-neck sling and mid-urethral sling sites.

### Procidentia: definition and assessment

- **Procidentia:** complete uterine (or vault) prolapse with the uterus wholly outside the hymen, **POP-Q stage IV**.
- Assess all compartments, occult SUI with prolapse reduced, PVR, hydronephrosis (ureteric kinking at the hiatus), cervical length and ulceration.
- Abnormal bleeding or cervical pathology needs gynaecological evaluation first, because it may argue against keeping the uterus.

### Fertility-sparing treatment of procidentia

**Principle:** restore **level I** support while keeping uterus and cervix. Defer definitive surgery until the family is complete where possible, because pregnancy and delivery cause recurrence.

**1. Conservative (first line)**

- PFMT, weight loss, treat constipation.
- **Pessary:** space-filling type (Gellhorn, cube) for procidentia; removed for conception; review about 6-monthly for discharge and erosion.

**2. Uterus-preserving surgery**

| Operation | Technique | Fertility points |
|---|---|---|
| **Sacrospinous hysteropexy** (vaginal) | Cervix sutured into the SSL **1.5–3 cm medial to the ischial spine** | Non-inferior to vaginal hysterectomy with uterosacral suspension (SAVE U trial); later data suggest superiority. Buttock pain, pudendal injury |
| **Uterosacral hysteropexy** (vaginal or laparoscopic) | Shorten and reattach uterosacral ligaments to cervix, sutures lateral to medial | No mesh; cystoscopy for ureteric kinking |
| **Sacrohysteropexy** (open, laparoscopic, robotic) | Mesh from posterior cervix to anterior longitudinal ligament over the promontory | Most durable apical support; later delivery usually by caesarean |
| **Manchester (Fothergill)** | Cervical amputation, cardinal ligaments sutured in front of the stump, anterior colporrhaphy | Suits elongated cervix, but risks **cervical stenosis, incompetence, miscarriage, preterm labour**: avoid if pregnancy wanted |

- **Also know:** Indian abdominal (cervicopexy) slings for young nulliparous prolapse (Shirodkar, Purandare, Khanna, Virkud). **Le Fort colpocleisis** keeps the uterus but not fertility (elderly only).

**3. Choosing** `Procidentia, wants pregnancy -> PFMT + pessary -> fails -> sacrospinous hysteropexy or laparoscopic sacrohysteropexy (avoid Manchester) -> recurrence after childbearing -> hysterectomy + vault suspension or sacrocolpopexy`

### Complications and follow-up

- Bladder, ureter or rectal injury; retention; de novo SUI; dyspareunia.
- Recurrence (mostly posterior overall; anterior after sacrospinous fixation).
- Mesh exposure: 3.4–5.4% after sacrocolpopexy in meta-analyses (P1 SAQ: about 3–10%; P2: about 10% long term, CARE); ureteric kinking.
- The retained uterus needs cervical screening.

> **Key points to remember:**
>
> - Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
> - DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone: always restore apical support.
> - Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
> - Integral theory: mid-urethral sling; hammock: backing, not repositioning.
> - Burch to Cooper's ligament; TOT through obturator foramen; PVS at bladder neck.
> - Bulge is most specific; splinting means high grade.
> - Central cystocele rugae lost, paravaginal preserved; POP-Q stage II = −1 to +1 cm (±1 cm of hymen).
> - Sacrocolpopexy most durable; colpocleisis for frail elderly.
> - Cystoscopy after repair; screen for occult SUI.
> - Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
> - Manchester amputates the cervix: poor choice for future pregnancy.

## LAQ 2 Vesicovaginal fistula: management, timing of repair, eponymous techniques and recurrent VVF

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 LAQ "Vesicovaginal fistula (VVF): management options and early vs delayed repair" · P2 SAQ "VVF repair: early versus delayed timing, and surgery for recurrent VVF" · P1 SAQ "Eponymous techniques of VVF repair"

### Definition and aetiology

- VVF (vesicovaginal fistula): an epithelialised bladder–vagina tract (communication) causing continuous leakage; the commonest acquired urinary fistula (>75% of genitourinary fistulae).
- **Developing world (low-resource settings):** obstetric (obstructed labour) in >90%; the fetal head impacted for days causes pressure necrosis.
- **Developed world (high-resource settings):** iatrogenic in >75%, mainly after hysterectomy, usually supratrigonal at the vaginal cuff.
- Other causes: radiation, malignancy, mesh, foreign body.
- Common mechanism: **ischaemic necrosis**; continuous urine flow keeps the tract open until it epithelialises with time.
- Time to presentation: cystotomy, days; thermal injury, 7–10 days; radiation, months to years.

### Classification

| Goh (distance from meatus) | Size | Tissue |
|---|---|---|
| 1: >3.5 cm; 2: 2.5–3.5; 3: 1.5–2.5; 4: <1.5 cm | A <1.5 cm; B 1.5–3 cm; C >3 cm | i mild fibrosis; ii moderate/severe; iii radiation, ureteric, circumferential, previous repair |
| - **Also know:** Waaldijk (closing mechanism involvement); WHO (World Health Organization) simple vs complex. | | |

### Evaluation (guides choice)

- Speculum; fluid creatinine.
- **Dye tests:** single (blue backfill, tampon); **double** (oral phenazopyridine plus blue): orange = UVF (ureterovaginal fistula), blue = VVF.
- **Cystoscopy and vaginoscopy:** number, size, relation to ureteric orifices and bladder neck, mesh; **biopsy the tract if any history of cancer** (the fistula may signal recurrence).
- **Upper tract imaging (CTU, computed tomography urography) is mandatory:** to exclude a coexisting ureteric (ureterovaginal) fistula (up to 12–25%).
- Urodynamics: capacity, compliance, SUI (stress urinary incontinence) in 10–15%.
- Prepare: treat UTI (urinary tract infection), optimise nutrition, vaginal oestrogen for atrophy; counsel about residual incontinence, dyspareunia and possible reoperation.

### Principles of management

- Optimise nutrition, treat infection, drain continuously, relieve obstruction, exclude malignancy.
- Surgery: excise scar, wide mobilisation, tension-free, multilayer, **non-overlapping**, watertight closure; remove foreign material; consider interposition; drain afterwards. **The first repair has the best chance.**
- At repair: treat infection; ureteric stents if within 1 cm of a ureter; vaginal or abdominal route (open, laparoscopic, robotic); **interposition flap** (Martius, omentum, peritoneum) if irradiated or recurrent; catheter drainage for about 2 weeks (≥10–14 days).

### Option 1: Conservative and endoscopic (conservative trial while waiting)

- **Catheter drainage** cures about 15%. Best if small (<1 cm, ideally <3 mm), fresh, started (catheterised) within **3 weeks**, and non-irradiated. Once epithelialised (≥6 weeks), closure is unlikely. EAU (European Association of Urology): if tried, at least 12 weeks.
- **Fulguration** plus 2 weeks' drainage for tiny tracts <3.5 mm.
- **Sealants** (fibrin glue, cyanoacrylate, collagen): tiny series; not recommended by EAU.

### Timing: early versus delayed repair (the debate)

| | **Early repair** | **Delayed repair** |
|---|---|---|
| Definition (varies) | Days, <6 weeks or <3 months | Traditionally ≥3 months |
| Rationale | Clean iatrogenic tissue heals well; ends wetness and distress early | Inflammation settles, necrotic tissue sloughs, margins become clear |
| Evidence | **Comparable closure** in most series; Waaldijk: immediate obstetric repair gave very high first-attempt closure | One small series: more recurrence when repaired <6 weeks |
| Drawback | Friable tissue if inflamed | Prolonged distress, no better outcome |

- Some authors avoid the **4–12 week** window (active healing phase).
- **EAU consensus:** no minimum delay if there is **no active infection and no recent irradiation**.
- **Timing by cause:**
- **Clean iatrogenic / post-hysterectomy:** early repair.
- **Obstetric with necrosis (necrotic edges):** catheter trial, then repair at about 3 months.
- **Radiation:** wait **≥12 months** after radiotherapy; biopsy to exclude recurrence; interposition flap; consider diversion if the bladder is poor.
- **Recurrent (redo) fistula:** wait until inflammation settles (commonly about 3 months); radiation ≥12 months after radiotherapy; failed radiation repairs were redone after **≥6 months**.

### Option 2: Surgical route

| **Vaginal** | **Abdominal (open, laparoscopic, robotic)** |
|---|---|
| Low, distal or trigonal, accessible apex | Supratrigonal, narrow vagina, no descent |

| Vaginal | Abdominal (open, laparoscopic, robotic) |
|---|---|
| Ureters distant | Ureter involved, needs reimplantation |
| Less pain, blood loss, shorter stay | Augmentation, **Tanagho bladder flap** or diversion possible |
| Flap-splitting, offset flaps | Large (≥4 cm primary, **≥2 cm recurrent**), radiation |
| Failed **Latzko** can be cured by a second Latzko | **O'Conor** bivalving finds the scarred plane |
| Risk: vaginal shortening | |
| - First-attempt closure about **90% by either route**; choose the route the surgeon knows best. | |

### Eponymous and named techniques: transvaginal

- **Sims:** first successful systematic vaginal repair (silver wire).
- **Flap-splitting (Raz):**
- Ureteric stents if within 1 cm; Foley through the tract for traction.
- Inverted J or U incision with a 1–3 mm cuff; raise anterior and posterior vaginal flaps.
- Layer 1: bladder (3-0 or 4-0), including the tract edge. Layer 2: perpendicular perivesical layer. Fill test 200–300 mL.
- Optional Martius or peritoneal flap; advance the opposite vaginal flap ≥2 cm beyond. Closure ≥90%.
- **Latzko (1942), high partial colpocleisis:**
- Denude vaginal epithelium 1–2 cm around the tract; the bladder is **not opened or dissected**.
- Close the vaginal walls over the tract in layers.
- ≥90% success in non-irradiated post-hysterectomy apical VVF; favoured for radiation VVF and tracts near the ureters. Drawback: **vaginal shortening and dyspareunia**.
- **Also know:** vaginal cuff excision (all apical scar excised as an inverted cone).

### Eponymous and named techniques: abdominal

- **O'Conor and Sokol (1951), suprapubic extraperitoneal:**
- Midline cystotomy at the dome, **bivalved down** to the fistula; ureteric catheters.
- Excise the tract; separate bladder from vagina 2–3 cm beyond.
- Close the vagina; **omental or peritoneal flap** between; close the bladder in layers; suprapubic catheter.
- Modifications: **transvesical** (limited cystotomy) and **extravesical** (no cystotomy; fewer LUTS [lower urinary tract symptoms]).
- **Laparoscopic (1994) and robotic (2005):** equal closure (85–100%) with less blood loss, shorter stay and less morbidity; vaginal stent, backfill, ICG (indocyanine green) perfusion check. Robotic repair of recurrences gives high closure.

### Interposition flaps (radiation, recurrent, ischaemic or large VVF)

| Flap | Blood supply | Route | Use |
|---|---|---|---|
| **Martius (1928)** (labial fibrofatty pad) | Internal pudendal (inferior), external pudendal (superior) | Vaginal | Mid–distal VVF, bladder neck; does not reach the apex |
| Peritoneal (vaginal) | Local; pouch of Douglas fold | Vaginal or abdominal | High apical post-hysterectomy VVF |
| **Omentum (J-flap)** | Gastroepiploic arteries | Abdominal | Radiation, recurrent, large |
| Gracilis, Singapore, rectus | Myocutaneous | Vaginal/perineal | Refractory, large tissue loss |
| - RCT (randomised controlled trial): Martius added nothing in simple VVF, with donor-site pain in nearly a third. | | | |
| - **Interposition is widely used in recurrent fistula** (accepted, though not proven superior). | | | |
| - **Also know:** rectus abdominis flap (last resort). | | | |

### Recurrent VVF

- Persists after a failed repair; redo tissue is scarred and poorly vascularised, so the first repair has the best chance.
- Failure causes: tension, ischaemia, overlapping suture lines, infection, poor drainage, **malignancy, radiation**, missed second tract.
- Before redo: biopsy, cystoscopy and vaginoscopy, CT urography and dye test to exclude a **ureterovaginal fistula**, urodynamics; treat infection, optimise nutrition.
- Route: abdominal for recurrent ≥2 cm; a failed Latzko can be redone vaginally; robotic repair gives high closure.
- Interposition flap; catheter ≥10–14 days, then cystogram.
- More than one repair predicts residual incontinence.

### Postoperative care

- Catheter ≥10–14 days (urethral and/or suprapubic); antimuscarinic or β3-agonist for spasm; cystogram before removal; stents out at 4–6 weeks.

### Option 3: Urinary diversion (when repair is unwise)

- Multiple failures, severe radiation damage, massive tissue loss: conduit or continent reservoir/diversion.
- Unfit: ureteric occlusion with nephrostomies.
- Resource-poor: ureterosigmoidostomy as a last resort.

### Outcomes

- Residual urgency or incontinence despite closure (Goh 3–4 predicts; more than one repair predicts); retention (vertical bladder closure); dyspareunia (narrow vagina, fistula ≥3 cm).

ICS (International Continence Society): pelvic floor physiotherapy.

### Diagrams to draw

- Flap-splitting repair: perpendicular bladder layers, Martius flap, advanced vaginal flap.
- Latzko denuded area versus flap-splitting layers.
- Timing timeline: early window, 4–12 week zone, 3 months (obstetric), 12 months (radiation).
- Martius flap between bladder and vaginal closures.

**Key points to remember:**

- Image upper tracts; exclude ureteric fistula and cancer (biopsy before redo).
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.
- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- First repair has the highest success.
- Interposition flap (Martius, peritoneum, omentum) for complex and recurrent VVF.
- Diversion after multiple failures or severe radiation.
- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

## LAQ 3 Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula

P3 **merged from 2**

*Asked as:* P3 LAQ "Urological complications in OBS-GYN and uretero-vaginal fistula" · P3 LAQ "Urological complications of OBGY surgery and urethrovaginal fistula"

*Assumed meaning: OBS-GYN / OBGY = obstetric and gynaecological practice and surgery.*

### Part A: Urological complications in OBS-GYN

- The bladder, distal ureter and urethra lie against the cervix, uterine vessels and anterior vaginal wall. Risk rises with distorted anatomy: fibroids, endometriosis, cancer, previous caesarean, radiation, heavy bleeding, prolapse or mesh surgery.

#### By setting

| Setting | Complications |
|---|---|
| **Obstetric** | **Obstructed labour** (fetal head impacted for days): pressure necrosis, so large VVF (vesicovaginal fistula), urethral loss, ureteric involvement; over 90% of VVF in the developing world. **Caesarean**: bladder injury, ureteric injury, **vesicouterine fistula** (Youssef syndrome: cyclical haematuria, amenorrhoea, continence preserved). Post-partum or post-epidural **retention** (overdistension) |
| **Gynaecological surgery** | **Bladder injury** (commonest organ). **Ureteric injury** (over 90% distal third; ligation commonest). **Fistulae**: VVF (over 75% follow hysterectomy in the developed world), UVF (ureterovaginal fistula), urethrovaginal fistula |
| **Radical hysterectomy** | Pelvic plexus injury: retention, poor compliance; manage with CIC (clean intermittent catheterisation) |
| **Prolapse and continence surgery** | Ureteric kinking at apical suspension; mesh or sling exposure, perforation, outlet obstruction, pain |
| **Radiotherapy** | Radiation cystitis, VVF, ureteric stricture |

#### By complication: key facts

| Complication | Key facts |
|---|---|
| **Bladder injury** (commonest organ injured) | Caesarean 0.27–0.3%; hysterectomy about 0.3–0.75% (laparoscopic highest); retropubic tape 3–9%. Repair: debride, 2-layer closure, fill test, catheter 7–14 days; check ureters if trigonal |
| **Ureteric injury** | Gynaecology causes over half of iatrogenic injuries; over 90% distal third. Hysterectomy overall 0.2–2.4%: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery 1.7–4.3%. **Ligation** is the commonest gynaecological mechanism |
| **Fistulae** | VVF about 0.8 per 1000 hysterectomies; UVF; vesicouterine fistula after caesarean (Youssef syndrome); urethrovaginal fistula |
| **Voiding dysfunction** | Radical hysterectomy (pelvic plexus injury) 16–80% historically: retention, poor compliance; CIC and wait. Post-partum or post-epidural overdistension |
| **Mesh and sling** | Exposure, erosion into bladder or urethra, outlet obstruction, pain, dyspareunia |
| **Radiation** | Radiation cystitis; VVF in 0.6–2% after cervical cancer radiotherapy; ureteric stricture |

- **Ureteric danger sites (four):** pelvic brim (near the infundibulopelvic ligament); **under the uterine artery** in the cardinal ligament; lateral to (beside) the uterosacral ligament; anterolateral vaginal fornix near the UVJ (ureterovesical junction). Most injuries are in the **distal 6 cm**.
- **Prevention and recognition:** haemostasis and anatomy; know the danger sites; prophylactic stents only selectively, if high risk (they aid recognition but do not prevent injury); cystoscopy with dye (IV indigo carmine or oral phenazopyridine) after hysterectomy where indicated, minimally invasive hysterectomy, and prolapse or continence surgery.
- Only about one-third of ureteric injuries are recognised at open surgery. A missed injury presents with fever, ileus, flank pain, urinoma or leakage; do CTU (computed tomography urography) with delayed films, then retrograde stenting. Distal repair: ureteroneocystostomy ± psoas hitch or Boari flap (see Part B).

## Part B: Ureterovaginal fistula (UVF)

### Definition

- Abnormal communication between the ureter and the vagina, causing continuous leakage.

### Aetiology and pathogenesis

- Usually an **unrecognised injury to the distal third of the ureter** during pelvic surgery. About **three-quarters follow an often uneventful abdominal hysterectomy**.
- **Mechanisms:** ligation, crush, laceration, thermal injury, **devascularisation** or ischaemia. The injured segment necroses and leaks into the vaginal cuff, which is the weakest point.
- Diagnosis is delayed in 18–88%; **delayed recognition raises fistula risk about fourfold**.
- Other causes: obstetric, cancer, radiation, endometriosis, foreign body.

### Clinical features

- **Continuous leakage with normal voiding** (the other kidney fills the bladder), usually appearing **within 4 weeks**, often with fever, flank pain or ileus, as thermal or ischaemic injury necroses.

| | VVF | UVF |
|---|---|---|
| Voiding | Reduced | Normal |
| Upper tract | Normal | Hydronephrosis common |
| First line | Catheter, repair | Stent |

### Investigations (confirm before treatment)

| Test | Method and reading |
|---|---|
| **Double dye test** | Oral phenazopyridine 100–200 mg (orange urine); after 1 hour, blue dye instilled into the bladder; tampon; walk about. **Orange = UVF**; blue = VVF; both = both |
| **Three-swab test** | Blue dye in the bladder; three swabs stacked in the vagina. Blue upper swab = VVF; **wet, unstained swab = UVF** |
| **CT urography, delayed phase** | Hydronephrosis, extravasation; MR (magnetic resonance) urography if contrast is contraindicated |
| **Cystoscopy + RGP (retrograde pyelogram)** | Abrupt cut-off or contrast leak; stent at the same sitting |

| Test | Method and reading |
|---|---|
| - **Exclude a coexisting VVF** (up to 25% of VVF have a UVF) and bilateral injury. | |

**Management**

- **Goals:** dryness and preservation of renal function. Management has moved from nephrectomy, through reimplantation, to endourology.
- **1. Endourological (first line): stent first**
- **Retrograde double-J stent as soon as possible** if a wire passes. Stenting **within 2 weeks** of detection works best, succeeding far more often than at 2–6 weeks, because the ureter heals over the stent before fibrosis.
- Wire will not pass: **PCN (percutaneous nephrostomy)** with antegrade stent, or rendezvous.
- Stent for **4–6 weeks**, then remove and re-image. Most stentable fistulae close; a stricture may follow.
- **2. Surgical repair (stent fails or impossible)**
- **Timing:** no consensus. Early repair equals delayed: early repair took less time and blood loss with equal success; a systematic review found equal success before and after 6 weeks. Delay only if tissue is still inflamed.
- **Ureteroneocystostomy** (distal third, so usually feasible):
  1. Transperitoneal or extraperitoneal approach; identify the ureter above the fistula.
  2. Divide above the diseased segment; dissect as distally as possible for length; **preserve periureteric adventitia**, especially laterally where vessels enter.
  3. Mobilise the bladder; spatulate; tension-free mucosa-to-mucosa anastomosis over a double-J stent; refluxing or tunnelled.
  4. Drain; catheter 7–14 days.
- **Adding length:**

| Technique | Extra reach |
|---|---|
| **Psoas hitch** (sutures parallel to psoas fibres, spare genitofemoral nerve) | Up to about 5 cm |
| **Boari flap** (base ≥4 cm, length:base ≤3:1) | 10–15 cm |
| Downward nephropexy; ileal ureter | Longer defects |
| - Open, **laparoscopic and robotic repair give equal results**; vaginal reimplantation reported. **Success ≥90%.** | |
| - The vaginal opening is left to heal; no separate vaginal repair is needed. | |

**Algorithm** `Leak + normal voiding -> exclude VVF (dye test, cystoscopy) -> CT urography / RGP -> wire/stent passes? Yes -> double-J 4-6 wk -> re-image -> healed / persistent No -> PCN + antegrade stent -> fails -> ureteroneocystostomy + psoas hitch / Boari flap`

**Complications and follow-up**

- Hydronephrosis, infection, ischaemic stricture.

- Ultrasound 1–2 weeks after stent removal; some add a renal scan at 6 weeks. Stricture usually appears within a year but may appear **up to 30 months** later, so follow beyond this.

## Part C: Urethrovaginal fistula

### Aetiology

- Developing world: **obstructed labour**, often with a VVF; also trauma, radiation, catheter injury.
- Developed world (iatrogenic): **urethral diverticulectomy** (fistula in 1–8%; causes 20–25% of urethrovaginal fistulae), slings, bulking agents, anterior repair or mesh, indwelling catheters (pressure necrosis), pessaries, radiation, trauma; also gender-affirming and paediatric genital surgery.

### Clinical features

- Depend on site relative to the sphincter. **Distal** fistula: often asymptomatic, split stream, vaginal voiding, post-void dribble. **Proximal** (bladder neck): continuous leakage or SUI (stress urinary incontinence).
- Dyspareunia, recurrent UTI (urinary tract infection), perineal irritation; partner discomfort with exposed mesh.

### Evaluation

- **Speculum examination** usually suffices; **methylene blue test** if rugae hide it (distal stain on tampon). Assess prolapse, SUI, atrophy.
- **VCUG (voiding cystourethrogram), lateral view;** VUDS (videourodynamics) for associated SUI or detrusor overactivity.
- **Cystourethroscopy** (flexible is better: outflow at the lens distends the urethra): size, site, bladder neck, exposed mesh, other fistulae. **About 20% have a separate VVF.** Image the upper tract.
- Goh location (distance from external meatus): type 1 >3.5 cm, 2: 2.5–3.5, 3: 1.5–2.5, 4: <1.5 cm. Closer means worse closure and continence.
- Optimise: treat UTI, vaginal oestrogen, nutrition; biopsy if previous cancer.

### Management

- Asymptomatic distal fistula: may need no repair. Timing: early if tissue is healthy; otherwise after inflammation settles.
- **Standard: transvaginal rotational-flap repair:**

| Step | Details |
|---|---|
| 1. Exposure | Lithotomy (prone gives better anterior exposure); cystourethroscopy, cannulate tract; suprapubic catheter (Lowsley tractor), urethral catheter |
| 2. Incision | Hydrodissect; circumscribe tract with 1–2 mm cuff, minimal excision; inverted **J- or U-shaped** vaginal flap incision |
| 3. Flaps | Raise vaginal flap; mobilise periurethral tissue widely to bladder neck; open retropubic space if sling planned |
| 4. Closure | **Close urethra** over a urethral sound: **4-0 absorbable** sutures, transverse or longitudinal; **second layer perpendicular**; buccal graft or flap onlay for large defects; leak test |

| Step | Details |
|---|---|
| 5. Interposition | **Martius flap** (labial fat pad, external pudendal pedicle) most often; omentum, gracilis, rectus fascia occasionally |
| 6. Cover | Advance vaginal flap beyond closure (2-0) |

- **Autologous pubovaginal sling** for SUI or bladder-neck involvement; **never synthetic** (erodes the ischaemic urethra). Remove any mesh in the field.
- **Near-total urethral loss:** tubularised anterior bladder flap neourethra. Irreparable cases: diversion.
- **After repair:** urethral catheter out at 2–3 days; suprapubic catheter ≥14 days; lateral VCUG before removal; nothing per vagina for 6 weeks.

**Outcomes and complications**

- Closure **90–95%**, but continence only **50–80%**; postoperative SUI in up to half. Sling then or staged later.
- Urethral stricture, recurrence, dyspareunia.

## Diagrams to draw

- Ureter passing under the uterine artery; double dye test; ureteroneocystostomy with psoas hitch.
- Urethrovaginal repair: inverted-U vaginal flap, two perpendicular urethral layers, Martius flap.

> **Key points to remember:**
>
> - Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
> - Ureteric injury is mostly distal and missed.
> - UVF: distal ureteric injury at hysterectomy; continuous leak within 4 weeks with normal voiding.
> - Orange tampon = UVF on double dye test; exclude coexisting VVF.
> - Stent early (within 2 weeks); remove at 4–6 weeks.
> - Reimplant with psoas hitch or Boari flap if stenting fails; success ≥90%; follow 30 months.
> - Urethrovaginal fistula: diverticulectomy is the commonest Western cause; obstetric in developing countries.
> - Look for a coexisting VVF (about 20%).
> - Layered closure plus Martius flap; autologous sling only.
> - Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

## LAQ 4 Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery

P3 **merged from 2**

*Asked as:* P3 LAQ "Female pelvic floor, continence mechanism, classification and management of SUI" · P3 LAQ "SUI: evaluation in females, role of UDS, and surgical therapy in males"

### Definition and mechanism

- **SUI (stress urinary incontinence)**: involuntary leakage on effort, exertion, coughing or sneezing (ICS, International Continence Society).
- Continence needs **urethral pressure greater than bladder pressure** at rest and during stress.
- Support comes from the **levator ani, endopelvic fascia and ligaments, perineal membrane and perineal body**.
- Two mechanisms: **urethral hypermobility** (loss of the suburethral support, the DeLancey "hammock") and **ISD (intrinsic sphincter deficiency)**. Most women have some ISD.

### Anatomy of the female pelvic floor

#### 1. Pelvic diaphragm

| Muscle | Attachment | Function |
|---|---|---|
| Puborectalis | Back of pubis, loops behind anorectal junction | Anorectal angle |
| Pubococcygeus (pubovaginal) | Pubis, ATFP (arcus tendineus fasciae pelvis) to viscera, raphe | Slings urethra and vagina |
| Iliococcygeus | Arcus tendineus levator ani, ischial spine to coccyx | Horizontal **levator plate** |
| Coccygeus (not levator) | Ischial spine to sacrum, coccyx | Lies on sacrospinous ligament |
| - Nerve to levator ani **(S3–5)** on its pelvic surface. Resting tone closes the **urogenital hiatus**; widening of the hiatus or tilting of the plate causes prolapse. Muscle fails first, then fascia carries the load. | | |

#### 2. Perineal membrane and perineal body

- The perineal membrane spans the pubic arch; urethra and vagina pierce it. Above it lies the **striated urogenital sphincter**: sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**. The perineal body anchors the distal vagina.

#### 3. Endopelvic fascia and ligaments

- **Pubocervical fascia** attaches laterally to the **ATFP** (pubis to ischial spine); **rectovaginal fascia** posteriorly; **cardinal–uterosacral complex** at the apex; **pubourethral ligaments** anchor the mid-urethra.

#### 4. DeLancey levels (1992)

| Level | Structures | Defect |
|---|---|---|
| I suspension | Cardinal–uterosacral | Apical prolapse, enterocele |
| II attachment | ATFP, levator fascia | Cystocele (central or paravaginal), rectocele |
| III fusion | Perineal membrane, perineal body | **Urethral hypermobility**, perineal descent |

## Continence mechanism in the female

**1. Intrinsic urethral factors** (smooth and striated sphincters, vascular submucosa)

- **Mucosal seal**: folded urothelium coapts. The **oestrogen-dependent submucosal vascular plexus** provides up to **30%** of closure force.
- **Smooth muscle** (internal sphincter, bladder neck and proximal urethra): involuntary tone. The female bladder neck is weak and often open in continent women.
- **Striated urogenital sphincter**: the rhabdosphincter, mostly **slow-twitch** fibres, gives tonic mid-urethral closure. Distal parts are the **compressor urethrae** and **urethrovaginal sphincter**.
- The **continence zone is the proximal and mid-urethra.**

**2. Support (extrinsic)**

- The urethra lies on the **anterior vaginal wall and endopelvic fascia**, attached laterally to the **ATFP** and levator ani.
- **Pubourethral ligaments** anchor the mid-urethra; **levator ani** (pubococcygeus) tone keeps the backing firm.

**3. Neural**

- **Pudendal nerve (S2–4)** drives the sphincter (**guarding reflex**). Urethral pressure rises **just before** a cough reaches it, so closure is active, not just transmitted.

**4. Theories of continence and SUI**

| Theory | Mechanism | Surgical implication |
|---|---|---|
| **Pressure transmission (Enhorning 1961)** | Hypermobile urethra drops out of the abdominal pressure zone | Restore position (Burch) |
| **Hammock (DeLancey 1994)** | Abdominal pressure compresses the urethra against a firm suburethral layer; a lax layer fails | Restore backing |
| **Integral theory (Petros & Ulmsten 1990)** | Pubourethral ligament laxity at the mid-urethra | Tension-free MUS (mid-urethral sling) |
| **ISD (McGuire & Lytton 1978)** | Urethra itself fails to close (VLPP [Valsalva leak point pressure] <60, MUCP [maximum urethral closure pressure] <20 $cmH_2O$) | Sling, bulking, AUS (artificial urinary sphincter) |
| - Most SUI has some ISD. **Hypermobility alone never causes SUI.** | | |

## Classification of SUI

**1. Blaivas & Olsson (1988), on video-urodynamics**

| Type | Bladder neck at rest | On stress |
|---|---|---|
| **0** | Closed, above lower border of symphysis | Descends and opens, but no leak seen (history of SUI) |
| **I** | Closed, above symphysis | Descends **<2 cm**; leaks; little or no cystocele |
| **IIA** | Closed, above symphysis | Descends **>2 cm** with cystourethrocele; leaks |
| **IIB** | Closed but **at or below** lower border of symphysis | Proximal urethra opens; leaks with or without further descent |
| **III** | **Open at rest** without detrusor contraction | Leaks with minimal stress = **ISD** |

### 2. Functional: hypermobility versus ISD

| | Urethral hypermobility | ISD |
|---|---|---|
| Q-tip test | Angle change **>30°** | Often fixed urethra |
| VLPP / ALPP | **≥90 cm$H_2O$** | **<60 cm$H_2O$** (60–90 equivocal) |
| MUCP | Normal | **≤20 cm$H_2O$** |
| Cause | Childbirth, ageing, obesity | Previous surgery, radiation, neuropathy, ageing |

### 3. Other

- **Severity (Stamey):** grade 1 leak on cough or sneeze; grade 2 on minimal effort (walking, rising); grade 3 at rest.
- **Index (uncomplicated)** versus **complicated** SUI (recurrent, prolapse, neurogenic, high PVR [post-void residual], radiation); pure versus mixed incontinence.

## Evaluation of SUI in women

### History

- Leakage with cough or activity means SUI; after urgency, UUI (urgency urinary incontinence); both, MUI (mixed). A positive SUI history predicts urodynamic SUI in about three-quarters.
- Severity (pads/day), bother; voiding symptoms, prolapse, obstetric history, prior surgery or radiation, neurological disease, drugs.
- **Questionnaires:** ICIQ-UI SF (International Consultation on Incontinence Questionnaire–UI Short Form); UDI-6 (Urogenital Distress Inventory: bother); IIQ-7 (Incontinence Impact Questionnaire: impact). **Bladder diary** (3 days).

### Examination (comfortably full bladder)

- Neurological (S2–S4 sensation, anal tone, bulbocavernosus reflex); atrophy; prolapse by **POP-Q (Pelvic Organ Prolapse Quantification)**.
- **Cough stress test:** cough supine, then standing if negative. Leakage synchronous with cough confirms SUI. Reduce prolapse to unmask **occult SUI**.
- **Q-tip test:** swab to bladder neck, Valsalva; rotation **>30°** = hypermobility. Visual mobility examination is now routine.

### Tests

| Test | Method and use |
|---|---|
| Urinalysis | Required; exclude infection, haematuria |
| PVR | Required before surgery; ultrasound; repeat if high (no agreed cut-off) |

| Test | Method and use |
|---|---|
| 1-hour pad test (ICS) | 500 mL fluid, standard exercises; >1.4 g positive; 1–10 g mild, 11–50 g moderate, >50 g severe |
| 24-hour pad test | Normal activities; more reproducible |
| Cystoscopy | Haematuria, irritative symptoms, previous mesh |

## Role of urodynamics (UDS) in SUI

- UDS (urodynamic studies) assess storage and voiding. The key question in SUI is: **will the result change the operation?** Pure, demonstrable SUI gains little. **UDS is not routine.**

| Parameter | Method and meaning |
|---|---|
| USI (urodynamic stress incontinence) | Leak with raised abdominal pressure, no detrusor contraction |
| ALPP/VLPP (abdominal/Valsalva leak point pressure) | Standing, from 150 mL, Valsalva or cough; **<60 cm$H_2O$ = ISD**, 60–90 equivocal, ≥90 = little ISD (McGuire 1993) |
| MUCP | Urethral pressure profile; a low MUCP (≤20 cm$H_2O$) also suggests ISD; does not diagnose or grade SUI |
| Filling cystometry | DO (detrusor overactivity), compliance, capacity; cough-induced DO mimics SUI |
| Pressure–flow study | Voiding dysfunction, poor contractility (retention risk after sling) |
| Prolapse reduction | Occult SUI |
| - If no leak with the catheter in place, repeat with it removed. | |

### Evidence

- **VALUE (ValUE) trial (2012), 630 women** with uncomplicated SUI: office evaluation alone was **non-inferior** to office evaluation plus UDS before surgery (success 77.2% vs 76.9%).
- **Dutch VUE study:** immediate midurethral sling was non-inferior to UDS-tailored treatment.
- **SISTEr:** UDS did not predict outcome.

### Guidelines

- **AUA (American Urological Association)/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023:** UDS may be omitted in the index patient with demonstrable SUI.
- **UDS advised in non-index patients:** unclear diagnosis or SUI not demonstrated; mismatch between symptoms and findings; urgency-predominant MUI; neurogenic bladder; raised PVR or voiding dysfunction; prolapse stage ≥3 without demonstrable SUI; failed previous anti-incontinence or prolapse surgery.
- **EAU (European Association of Urology):** no routine UDS for uncomplicated SUI; do not use leak point or urethral pressure to grade severity.
- **Men:** before AUS or sling for post-prostatectomy incontinence, UDS confirms USI and finds DO, poor compliance or weak contractility. The MASTER trial recruited men with USI.

## Conservative management (first line)

- Weight loss, fluid and caffeine adjustment.
- **Supervised PFMT (pelvic floor muscle training)** for at least 3 months (Cochrane: more cure than no treatment).

- **Pessary**; vaginal oestrogen for atrophy.
- **Duloxetine** (serotonin–noradrenaline reuptake inhibitor): increases Onuf's nucleus drive to the sphincter; approved in Europe, not the USA.

### Surgical options for female SUI: principles

| Operation | Principle (how it works) | Best for | Key facts |
|---|---|---|---|
| **MUS**: retropubic (TVT, tension-free vaginal tape, Ulmsten 1995), transobturator (TOT; Delorme 2001; inside-out de Leval), single-incision | Tension-free type I polypropylene tape under the **mid-urethra** acts as a new pubourethral ligament; the urethra kinks against it during stress | First-line index SUI | **First line** (AUA/SUFU strong recommendation). **TOMUS**: retropubic 80.8% vs transobturator 77.7% cure (equivalent); retropubic often preferred in ISD. Retropubic: more bladder perforation; transobturator: more groin pain |
| **SIMS (single-incision mini-sling)** | Short tape, vaginal only | Option | Non-inferior to TOT for subjective cure in most studies; conditional recommendation (2023) |
| **Autologous pubovaginal sling (PVS)** (rectus fascia or fascia lata) | At the **bladder neck**, minimal tension; restores the hammock, giving bladder-neck support and compression | **All types, including ISD**, recurrent SUI, neurogenic SUI, failed MUS, after mesh removal, mesh-averse | Gold standard. **SISTEr**: sling beat Burch (47% vs 38%) but more voiding dysfunction and UTI (urinary tract infection) |
| **Burch colposuspension** (open, laparoscopic or robotic) | Paravaginal fascia beside the mid-urethra and bladder neck sutured to **Cooper's ligament**; lifts the bladder neck back into the pressure zone and gives a stable backing | Hypermobility; women declining mesh; at abdominal surgery | 85–90% at 1 year, about 70% at 5 years. Needs a mobile urethra; poor in ISD |
| Paravaginal repair, VOS (vagino-obturator shelf) | Vagina to ATFP or obturator fascia | | Flatter, less obstructive |
| **MMK (Marshall–Marchetti–Krantz)** | Bladder neck to symphysis periosteum | Historical | Osteitis pubis 0.7–2.5%; abandoned |
| **Urethral bulking agents** | Submucosal injection at bladder neck or proximal urethra improves coaptation | ISD with fixed urethra; frail patients; minimal surgery | Less durable; repeat injections |
| **AUS** | Bladder-neck cuff; constant circumferential closure | Severe refractory ISD, fixed urethra | Second-line; erosion, infection, revision |

| Operation | Principle (how it works) | Best for | Key facts |
|---|---|---|---|
| - **Also know:** needle suspensions abandoned (sutures pull through); anterior colporrhaphy is not an SUI operation; CARE (prophylactic Burch at sacrocolpopexy); Ward–Hilton (TVT = colposuspension). | | | |

**Burch steps (algorithm):** Pfannenstiel → space of Retzius → stay lateral to urethra → 2–4 sutures each side (mid-urethra, then bladder neck) → Cooper's ligament → tie loosely (two fingers behind pubis) → **cystoscopy** for ureteric efflux.

**Retropubic vs transobturator MUS**

| | Retropubic | Transobturator |
|---|---|---|
| Bladder perforation | 4.5% | 0.6% |
| Voiding dysfunction (TOMUS surgery) | More (2.7% vs 0%) | Less |
| Groin pain | 1.3% | 6.4% |
| ISD, long-term durability | Often preferred | Possibly more repeat surgery |

## Complications of female SUI surgery

- **MUS complications:** perforation, retention, de novo urgency, mesh exposure (about 2%), pain.
- Bladder trocar injury: re-pass and continue (cystoscopy after every pass); **urethral injury: abandon mesh**.
- Voiding dysfunction (sling too tight or proximal): urethrolysis or sling incision; de novo urgency; UTI; haematoma.
- **Mesh complications** (IUGA/ICS [International Urogynecological Association/International Continence Society] terms): exposure, extrusion, perforation into bladder or urethra, pain; manage by excision. FDA (Food and Drug Administration) 2019 stopped transvaginal mesh for prolapse, not slings.

## Male SUI: surgical therapy

- **Aetiology:** commonest after **radical prostatectomy (RP)**, which removes the internal sphincter, leaving continence to the rhabdosphincter. Also TURP (transurethral resection of the prostate), radiation, pelvic fracture.
- **Work-up:** 24-h pad weight (mild <150 g, moderate <400 g, severe above); standing cough test; **cystoscopy** (anastomotic stricture, sphincter coaptation); UDS if mixed or poor contractility.
- **Timing:** PFMT first; no surgery for 6–12 months, as continence keeps improving (other P3 answer: continence improves for up to 2 years, but do not delay surgery beyond 6 months for severe leak).

| Device | Mechanism | Use and results |
|---|---|---|
| **Transobturator sling (AdVance XP)** | Repositions the bulb 3–4 cm proximally; needs residual sphincter | Mild–moderate leakage, no radiotherapy; cure about 60% |
| Quadratic sling (Virtue) | Compression plus elevation | Mild; limited data |
| Adjustable sling (ATOMS, adjustable transobturator male system) | Cushion adjusted via scrotal port | Mild–moderate; about two-thirds dry |
| ProACT balloons | Bladder-neck compression | More erosion |
| Bulking agents | Submucosal bulk | Small leaks, unfit men; rarely durable; not advised |
| **AUS** | Circumferential compression | **Gold standard**; all severities, moderate–severe, after radiation; about 80% continent |

**AMS 800 AUS**

- **Components:** urethral **cuff** (bulbar, usually 4.0–4.5 cm); **PRB (pressure-regulating balloon) 61–70 cm$H_2O$** (51–60 high risk; 71–80 bladder neck) in a preperitoneal pocket; scrotal **pump**. Filled with 23 mL.
- **Technique:** perineal incision; cuff on the proximal bulbar urethra; **deactivate for 4–6 weeks**.
- **Needs** dexterity and cognition. Social continence (0–1 pad) 55–77%; satisfaction about 90%; lifespan 7–10 years.
- **Complications:** infection 1–3% (explant; reimplant after 3–6 months); **erosion** 2–15% (explant, catheter about 3 weeks); urethral **atrophy** (late recurrent leak); mechanical failure; retention.
- **Erosion risks:** radiation, prior erosion, urethroplasty; use a **transcorporal cuff** (tunica adds bulk) or lower PRB. Atrophy: smaller or tandem cuff.

**Algorithm (AUA/SUFU 2019)** `Severe SUI or any radiotherapy -> AUS Moderate, no radiotherapy -> AUS or male sling Mild, no radiotherapy -> sling (or PFMT, AUS) Sling failure -> AUS`

- **MASTER RCT (randomised controlled trial):** sling was **non-inferior** to AUS for continence at 12 months; AUS remains preferred for heavy leak (>250 g/day) or radiation.

## Diagrams to draw

- DeLancey levels and hammock cross-section.
- Sites of Burch sutures, bladder-neck sling and mid-urethral tape; sagittal pelvis showing retropubic and transobturator tape paths.
- Cystometrogram trace with abdominal pressure spike and leak (ALPP).
- AMS 800 (cuff, pump, PRB) and male transobturator sling path.

**Key points to remember:**

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD.
- Q-tip >30° = hypermobility; VLPP / ALPP <60 $cmH_2O$ = ISD; MUCP ≤20 $cmH_2O$.
- Burch corrects position; slings and AUS correct function.
- VALUE (ValUE) trial: UDS not needed in uncomplicated (index) SUI.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
- PFMT first; MUS first-line surgery.
- SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
- PVS for ISD; urethral injury → no mesh.
- Men: AUS gold standard; sling for mild, non-irradiated.
- AUS PRB 61–70 $cmH_2O$; deactivate 4–6 weeks.
- Radiation or severe male SUI: AUS first; MASTER: sling non-inferior.

## LAQ 5 Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management

P3

**Asked as:**

- Etio-pathogenesis, diagnosis and management of Recto-urethral fistula? — *Summer 2020 Paper III Q1 (LAQ)*
- Etiopathogenesis, diagnosis and management of Recto-Prostatic fistula — *Winter 2025 Paper III Q1 (LAQ)* (also answered in 2020-Summer Paper 3, Q1, as recto-urethral fistula)

**Type:** LAQ (25 marks)

*Assumed meaning (recto-prostatic): rectourethral fistula at the level of the prostatic urethra or the vesicourethral anastomosis.*

### Definition

- **RUF (rectourethral fistula):** a communication between the posterior urethra / prostatic urethra (or the vesicourethral anastomosis after prostatectomy) and the distal rectum.
- Rare but very morbid. Almost never seen in females, because the vagina lies between urethra and rectum.

### Aetiology

| Group | Causes |
|---|---|
| Congenital | Anorectal malformation in boys (rectobulbar or rectoprostatic fistula) |
| Surgical (commonest acquired) | **RP (radical prostatectomy)**; historically TURP (transurethral resection of the prostate) and open (simple) prostatectomy |
| Radiation and energy | **EBRT (external beam radiotherapy) plus brachytherapy** (commonest), then brachytherapy alone, then pelvic radiotherapy; cryotherapy; HIFU (high-intensity focused ultrasound): low after primary, up to about 9% after salvage HIFU |
| Others | Trauma (penetrating injury, pelvic fracture); Crohn disease; advanced rectal or prostate cancer |

### Pathogenesis

- **Surgical:** the rectum is torn during apical or posterior dissection behind Denonvilliers fascia, especially with small glands and salvage surgery. An unrecognised tear (near the apex) leaks urine (urine leaks into it), an abscess forms and the tract epithelialises.
- Rectal injury at RP: laparoscopic about 1%, open and robotic under 0.5%, salvage RP about 7%. (Another source: rectal injury is commonest with retropubic and laparoscopic RP, lowest with robotic RP, and was over 15% in historical open salvage series.)
- **Radiation or ablation:** obliterative endarteritis makes the tissue hypoxic and fibrotic. Necrosis follows months to years later. These fistulae are large and heal poorly.
- **Congenital:** abnormal cloacal development leaves the rectum ending in the posterior urethra.

### Classification: Martini (after prostate cancer treatment)

| Axis | Category | Definition |
|---|---|---|
| Stage (size on CT (computed tomography) urography or cystography) | I / II / III | ≤1.5 cm / >1.5 cm / any size with sphincter damage |
| Grade (tissue) | 0 | Surgical injury, no prior non-surgical treatment |
| | 1 | Prior radiation, brachytherapy, HIFU or other energy |
| | 2 | Salvage RP or salvage ablation |
| - **Tissue quality (grade), not size, chiefly decides the repair.** | | |

### Clinical features

- **Faecaluria and pneumaturia** (most typical); urine per rectum on voiding.
- Recurrent UTI (urinary tract infection), dysuria, urosepsis; coexisting SUI (stress urinary incontinence).

### Diagnosis

- **DRE (digital rectal examination):** may feel the defect on the anterior rectal wall.
- **Cystoscopy and proctosigmoidoscopy:** show the tract and tissue vitality; **biopsy** to exclude recurrent cancer.
- **VCUG (voiding cystourethrogram) or RUG (retrograde urethrogram):** usually conclusive; gives size and site for planning.
- Upper-tract imaging to exclude ureteric injury; pelvic MRI (magnetic resonance imaging) for complex cases.
- **UDS (urodynamics)** with rectal packing: fistula repair alone will not cure severe SUI; an AUS (artificial urinary sphincter) may be needed later; a later AUS erodes more often.

### Prevention

- At RP: a recognised tear without gross spillage → two-layer (multilayer) closure, air test, **omental or peritoneal interposition**, drain, antibiotics; colostomy if (previously) irradiated.
- Radiotherapy: a **hydrogel rectal spacer** lowers rectal dose and late rectal toxicity (randomised trial: late rectal toxicity 2.0% vs 7.0%).

### Management

- **Principles:** control sepsis; divert urine ± faeces; let inflammation settle (about 12 weeks); exclude cancer; then excise the tract and close urethra and rectum separately in **tension-free, non-overlapping layers with vascularised (interposition) tissue interposed**.
- **Conservative:** no sepsis and no faecaluria → catheter (urethral or suprapubic), low-residue diet ± colostomy. A simple fistula may close in about 12 weeks; irradiated or complex ones rarely do.
- **Surgery:** no single accepted operation; choice depends on site, radiation and prior repair. **A diverting colostomy usually comes first.**

| Approach | Essentials | Strengths | Weaknesses |
|---|---|---|---|
| **Perineal** (Young; Goodwin) | Separate and close urethra and rectum separately; **gracilis flap** between (or dartos; buccal graft) | High closure; most stomas reversed | Urinary incontinence common |
| **Transsphincteric (York-Mason)** | Prone jackknife; sphincter divided in tagged layers; through anterior rectal wall; three-layer closure | Avoids scarred plane; spares sexual function | Poor membranous exposure; no flap; faecal incontinence up to 20% |
| **Abdominal** (transvesical, extravesical, O'Conor-like) / robotic | Omental flap | Large or irradiated fistulae, extensive resection | Morbid; poor deep-pelvic view; small robotic series |
| **Laparoscopic / robotic** | Open pouch of Douglas; close rectum in two layers; omental or peritoneal flap; can combine with perineal gracilis | Deep-pelvic vision | Small series only |
| **Salvage** | Cystoprostatectomy, ileal conduit ± permanent colostomy | Unreconstructable irradiated field | Major surgery |

`RUF → sepsis control, catheter ± colostomy, biopsy small, non-irradiated, no faecaluria → conservative ~12 weeks persistent, grade 0 → perineal (gracilis) or York-Mason irradiated or large → abdominal/robotic (omentum) or perineal (gracilis) unreconstructable → urinary + faecal diversion Congenital → newborn colostomy → posterior sagittal anorectoplasty`

- **PSARP (posterior sagittal anorectoplasty)** separates the rectum from the urethra and places it within (in) the sphincter complex.

### Complications

- **Recurrence** (the greatest challenge), urinary or faecal incontinence, urethral stricture, ED (erectile dysfunction), wound infection, permanent stoma.

**Diagram to draw:** sagittal pelvis showing (with) the fistula and a gracilis flap between urethra and rectum.

> **Key points to remember:**
> - Most acquired RUFs follow prostate cancer treatment.
> - Martini: stage = size; grade = tissue (surgery, radiation, salvage).
> - VCUG or RUG plus endoscopic biopsy to exclude cancer.
> - Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

## Short answer questions (SAQ)

## SAQ 1 Leak point pressures and ALPP

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications" · P2 SAQ "ALPP"

## Definition

- **LPP (leak point pressure):** the pressure at which leakage is first seen, spontaneous or provoked. There are two independent types:

| Feature | ALPP (abdominal LPP), incl. VLPP (Valsalva LPP) and CLPP (cough LPP) | DLPP (detrusor LPP) |
|---|---|---|
| Definition | Total Pves (vesical pressure) at leak during raised abdominal pressure, **no detrusor contraction** | Lowest Pdet (detrusor pressure) at leak, **without contraction or straining** |
| Measures | Sphincter strength against abdominal pressure | Bladder pressure against outlet resistance |
| Bad result | **Low = bad** (weak sphincter) | **High = bad**: >40 $cmH_2O$ means upper-tract risk; hydronephrosis or reflux in 85% (McGuire 1981) |
| Used in | SUI (stress urinary incontinence) | Neurogenic or low-compliance bladder |

- **ALPP (abdominal leak point pressure):** the **intravesical pressure** at which urine leaks because of raised abdominal pressure, **without a detrusor contraction**. It measures the urethra's ability to resist abdominal pressure: **lower ALPP = weaker sphincter**.
- Provoked by Valsalva (**VLPP**, the intravesical pressure at which urine leaks during Valsalva) or cough (**CLPP**).
- Seen only in SUI; there is **no normal value** (continent women never leak).
- Newer term: "urodynamic stress test" (ICS, International Continence Society).

## Technique (VLPP/ALPP)

1. Small dual-lumen urethral catheter plus rectal catheter; zero to atmosphere.
2. Patient **standing or sitting**; fill to **150 mL** (original method; some use 200 mL).
3. Ask for a **graded Valsalva** or cough; record the **lowest total Pves at visible leak** (include baseline, not the change).
4. No leak: repeat every 50 mL to capacity; then, at capacity, **remove the urethral catheter** and read pressure from the rectal line.
5. Report volume, catheter size and provocation (Valsalva or cough).

## Interpretation (McGuire 1993)

| VLPP / ALPP ($cmH_2O$) | Inference |
|---|---|
| **<60** | **ISD (intrinsic sphincter deficiency)** |
| **60–90** | Equivocal; mixed |
| **≥90** (>90) | **Urethral hypermobility**, little ISD |

## Clinical implications and uses of ALPP

- Separates ISD from hypermobility; a low VLPP/ALPP (ISD) traditionally favours a sling or bulking agent over colposuspension.
- Post-prostatectomy incontinence: many men leak only once the urethral catheter is removed.

## Limitations

- Not standardised: falls with **larger catheters** and **higher volumes**.

- **VLPP is lower than CLPP** in the same woman (VLPP < CLPP), so state the method.
- Invalid during DO (detrusor overactivity) or poor compliance; prolapse raises it or masks the leak (reduce it).
- ISD and hypermobility coexist; VLPP/ALPP alone **does not predict surgical outcome** well.
- **VALUE trial (Nager 2012):** for uncomplicated SUI, office evaluation was non-inferior to urodynamics before surgery. AUA (American Urological Association)/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023 SUI guideline: urodynamics optional in the index patient.
- **EAU (European Association of Urology):** do not use leak point pressure to grade severity.

### DLPP: clinical implications

- **McGuire (1981):** DLPP **>40 $cmH_2O$** in myelodysplastic children leads to hydronephrosis or VUR (vesicoureteral reflux) in **85%**.
- A strong outlet keeps the patient dry but transmits pressure to the kidneys; a weak outlet "pops off" safely.
- **Hostile bladder** (≥40 $cmH_2O$): CIC (clean intermittent catheterisation) plus antimuscarinic; botulinum toxin, augmentation or vesicostomy if needed.
- **Caveat:** 40 is not a validated "safe" value; 20 $cmH_2O$ is more sensitive in children.

### Also know

- DO leak point pressure (ICS 2017); MUCP (maximum urethral closure pressure) ≤20 $cmH_2O$ suggests ISD.

### Diagram to draw

- Pves, Pabd and Pdet traces marking ALPP (leak on Valsalva with flat Pdet) and DLPP.

> **Key points to remember:**
> - VLPP/ALPP: Pves at leak with Valsalva or strain, no detrusor contraction; start at 150 mL.
> - ALPP low = weak sphincter; DLPP high = upper-tract danger.
> - VLPP <60 ISD; 60–90 equivocal; ≥90 (>90) hypermobility.
> - VLPP < CLPP; catheter size and volume alter it.
> - DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
> - Lower DLPP is safer; no validated safe cut-off.

## SAQ 2 Mid-urethral slings: TOT and TVT

P1 · P2 **merged from 2**

*Asked as:* P2 SAQ "Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT" · P1 SAQ "TOT in SUI"

### Definition and history

- Both are polypropylene **MUS (mid-urethral slings)** for female SUI (stress urinary incontinence), based on the **integral theory** (Petros and Ulmsten, 1990): a loose mid-urethral tape replaces the pubourethral ligaments as a backboard on straining.
- **TVT (tension-free vaginal tape, retropubic):** retropubic polypropylene MUS; Ulmsten, 1995–1998.

- **TOT (transobturator tape):** a synthetic MUS of macroporous monofilament polypropylene, passed through the obturator foramina instead of the retropubic space. Delorme (2001) designed it to avoid bladder, bowel and vessel injury.

### Mechanism

- **Integral theory:** pubourethral ligaments, the suburethral vaginal hammock and pubococcygeus maintain continence at the mid-urethra. Hypermobility is a sign of their damage, not the cause.
- The loose tape at the mid-urethra supports the **posterior urethral wall**, which is compressed against it on straining (dynamic kinking); it is not tight at rest.
- TOT anchors in the obturator muscles (**obturator internus and externus**).

### Material

- **Type 1 (Amid) macroporous monofilament polypropylene**, pores >75 µm: allows macrophage entry and fibrous ingrowth, so it resists infection.

### Indications and evaluation

- SUI or stress-predominant mixed incontinence after failed PFMT (pelvic floor muscle training); index SUI with urethral hypermobility.
- TVT preferred over TOT in **ISD (intrinsic sphincter deficiency)** or recurrent SUI.
- History, examination, **objective stress test**, urinalysis, PVR (post-void residual); urodynamics only if complicated.
- Counsel about permanent mesh, mesh risks and alternatives (colposuspension, autologous sling, bulking) (AUA/SUFU [American Urological Association/Society of Urodynamics and Female Urology]).

### Technique: differences

| | TVT (retropubic) | TOT (transobturator) |
|---|---|---|
| Path | Vagina to retropubic space to suprapubic skin, hugging the pubis; cystoscopy essential | Labiocrural fold to obturator membrane to vagina (outside-in), or reverse (inside-out) |
| Structures at risk | Bladder, bowel, iliac and epigastric vessels | Obturator nerves and vessels, vaginal fornix |

### TVT technique

1. Local, spinal or general anaesthesia; lithotomy.
2. Two suprapubic stab incisions 2 cm lateral to midline; 1.5 cm midline vaginal incision 1.5 cm from the meatus; minimal dissection, endopelvic fascia not opened.
3. **Bottom-up** 5 mm trocar hugging the back of the pubis, through the retropubic space to the skin (catheter guide deflects the bladder).
4. **Cystoscopy with a 70° lens, full bladder**, after each pass; if perforated, re-pass more laterally.
5. Tension **loosely** with a clamp between tape and urethra; remove sheath.

### TOT technique (outside-in)

1. Lithotomy with hips hyperflexed; 1.5 cm midline vaginal incision 1.5 cm below the meatus; dissect laterally to the ischiopubic ramus.

2. Groin stab incision in the labiocrural fold, at the notch between inferior pubic ramus and **adductor longus tendon**.
3. Trocar perforates the obturator membrane (felt as a give), turns medially onto the finger and exits the vaginal incision.
4. Cystourethroscopy; check the vaginal fornices (sulci).
5. Tension-free: a clamp or scissors pass easily between tape and urethra. - **Inside-out** (de Leval; vagina to thigh): fewer vaginal perforations, more voiding dysfunction.

### Outcomes

- TVT: Ulmsten (1998): 91% cured at 1 year. **Ward–Hilton:** equal to colposuspension at 5 years. Long-term cure varies 48–97% by definition.
- **TOMUS (Richter, 2010):** 597 women; 12-month objective cure **80.8% retropubic vs 77.7% transobturator** (not significantly different, equivalent); voiding dysfunction needing surgery 2.7% vs 0%.
- TOMUS at 5 years: 51.3% vs 43.4%; equivalence not shown; satisfaction similar.
- **Cochrane review (Ford, 2017):** cure similar up to and beyond 5 years; repeat surgery possibly commoner after TOT.
- TOT is poorer with **ISD** or a **fixed urethra**: retropubic sling preferred.

### Complications: comparison (Cochrane, 2017)

| Complication | TVT (retropubic) | TOT |
|---|---|---|
| Bladder perforation | **4.5%** | **0.6%** |
| Groin pain | 1.3% | **6.4%** |
| Voiding dysfunction, bleeding, vascular injury | More | Less |
| Mesh exposure | About 2% | About 2% |
| Long-term repeat surgery | Less | Possibly more |

### TVT complications

| Complication | Notes |
|---|---|
| Bladder perforation | 4.5% (TOT 0.6%); re-pass and continue |
| Urethral injury | Abandon mesh; staged repair |
| Bleeding/haematoma | 2–3%; rarely bowel or vascular injury |
| Voiding dysfunction | Surgery needed 2.7%; incise tape in midline within 3 months |
| Mesh exposure | About 2% |
| Others | De novo urgency, suprapubic pain (groin pain less than TOT) |

### TOT complications

- Groin or thigh pain (usually temporary; NSAIDs [non-steroidal anti-inflammatory drugs]), vaginal mesh exposure, voiding dysfunction (incise sling within 4 weeks), de novo urgency, UTI (urinary tract infection), rarely thigh abscess.
- Urethral injury during either sling: **do not place mesh** (AUA guideline).

### Choosing between TVT and TOT

- **TVT preferred: ISD** (ALPP [abdominal leak point pressure] <60 $cmH_2O$ or MUCP [maximum urethral closure pressure] <20 $cmH_2O$), fixed urethra, recurrent SUI. Failure is 12 times likelier after TOT with low leak point pressure (O'Connor, 2006).
- **TOT preferred:** previous retropubic surgery, obesity, avoiding bladder injury.
- AUA: either may be offered to the index patient.
- **Also know:** SIMS (single-incision mini-slings; non-inferior to TOT); mesh warnings (FDA [Food and Drug Administration] 2010; UK pause 2018): counsel about alternatives (Burch colposuspension, autologous fascial sling, bulking agents).

### Diagram to draw

- Coronal pelvis: U-shaped retropubic TVT at mid-urethra versus flat TOT, the tape under the mid-urethra through both obturator foramina.

> **Key points to remember:**
> - Integral theory; loose tape at mid-urethra.
> - Type 1 macroporous monofilament polypropylene.
> - Delorme 2001; TOT anchors in obturator muscles; fewer bladder injuries than retropubic.
> - TOMUS: cure equivalent at 1 year (80.8% retropubic vs 77.7% transobturator).
> - TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
> - Cystoscopy after each TVT trocar pass; bladder perforation 4.5%.
> - Prefer retropubic TVT in ISD or fixed urethra.

## SAQ 3 Burch colposuspension

P1

**Type:** SAQ (10 marks)
**Source:** 2022-Summer Paper 1, Q3b (related: 2017-Winter Paper 3, Q2)

### Definition

- Retropubic suspension (**Burch, 1961**) of the paraurethral and para-bladder-neck vaginal wall to **Cooper's (iliopectineal) ligament**; Tanagho (1976) modified it.

### Principle

- SUI (stress urinary incontinence) from **urethral hypermobility**: the urethra drops out of the retropubic position. Elevation restores **pressure transmission** and a backstop; scarring holds it.
- It treats hypermobility, not ISD (intrinsic sphincter deficiency).

### Indications

- SUI or stress-predominant MUI (mixed urinary incontinence) with hypermobility, especially with **concomitant abdominal surgery** (sacrocolpopexy), **mesh refusal**, or poor vaginal access.
- **Contraindications:** immobile urethra or vagina, ISD alone, pure urgency incontinence.

### Work-up

- Cough stress test, PVR (post-void residual), urinalysis; urodynamics only if SUI is not demonstrated.

### Steps (open)

1. Low lithotomy, Foley; Pfannenstiel incision; enter the **space of Retzius**.
2. Vaginal finger elevates the vagina; sweep bladder medially. **Avoid midline dissection** over urethra.
3. Two to four 0 or 1-0 delayed-absorbable or non-absorbable sutures each side through vaginal wall (not lumen): distal at mid-urethra, proximal at bladder neck, placed laterally.
4. Pass through Cooper's ligament; tie with vaginal elevation, without tension; leave about **two fingerbreadths** between bladder neck and pubis.
5. **Cystoscopy:** ureteric efflux, no suture in bladder. - Laparoscopic and robotic versions match open short-term results.

### Outcomes

- Cure 85–90% at 1 year, about 70% at 5 years (Cochrane).
- **SISTEr (2007):** autologous fascial sling more effective but more voiding dysfunction and UTI.
- **Ward and Hilton:** TVT (tension-free vaginal tape) equivalent to colposuspension; meta-analyses favour midurethral slings.
- **CARE (2006):** prophylactic Burch at sacrocolpopexy reduced postoperative SUI.

### Complications

- Voiding dysfunction 2–27%, de novo storage symptoms 8–27%, **de novo prolapse** 2.5–27%; bladder or ureteric injury; bleeding.
- **Also know:** MMK (Marshall–Marchetti–Krantz; osteitis pubis 0.7–2.5%), paravaginal repair, VOS (vagino-obturator shelf); failure predictors: BMI (body mass index) >25, age >60, prior surgery.

**Diagram to draw:** sagittal pelvis with sutures from paravaginal fascia to Cooper's ligament.

> **Key points to remember:**
> - Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
> - Cure 85–90% at 1 year, ~70% at 5 years.
> - Niche: concomitant abdominal surgery, mesh avoidance.

## SAQ 4 Management of perforated mid urethral sling

P1

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 1, Q3a

### Terminology (IUGA/ICS, International Urogynecological Association / International Continence Society)

- **Perforation:** mesh in a hollow organ (bladder, urethra). **Exposure:** mesh seen through unhealed vaginal epithelium. **Extrusion:** passage out of a tissue. "Erosion" is no longer used.

- Two settings: **intraoperative trocar injury** and **late mesh perforation**.

### A. Intraoperative trocar injury

- Bladder injury: 4.5% retropubic versus **0.6% transobturator**; registries 2.7–3.8%.
- **Detection:** cystoscopy with a **70° lens and full bladder distension** after every trocar pass.

| Injury | Management |
|---|---|
| **Bladder** | Withdraw trocar, **repass more laterally**, confirm on cystoscopy, complete the sling; catheter overnight to 72 h. Benign; no later mesh perforation. |
| **Urethra** | **Do not place mesh** (AUA/SUFU, American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction). Repair, catheter drainage; **staged** anti-incontinence surgery after healing. |

### B. Late perforation of bladder or urethra

- **Presentation:** recurrent UTI (urinary tract infection), haematuria, urgency, pain, stones on mesh, voiding dysfunction, fistula. Often months later.
- **Evaluation:** examination, **cystourethroscopy**, translabial ultrasound (best imaging for mesh); exclude fistula.

**Management**

- **Bladder:** endoscopic excision (scissors, resection or holmium laser) flush with the wall; or transvesical laparoscopic or robotic excision; open cystotomy if large.
- **Urethra:** transvaginal excision of mesh in the field and layered urethral repair; **Martius flap** if complex or fistula.
- **Continence after excision:** SUI recurs in many (44–100% after synthetic sling revision). Options: concomitant **autologous pubovaginal sling** (Shah 2013: good results) or staged. A damaged urethra predicts poor continence.
- Avoid urethral dilation after mesh slings; it can cause perforation.

### Prevention

- Empty bladder; trocar hugging the pubis; bottom-up retropubic route; transobturator route when suitable.

**Also know:** Amid type I macroporous polypropylene integrates best; chronic pain or infection may need excision from both sides of the bone.

**Diagram to draw:** retropubic and transobturator trocar paths with bladder at risk.

> **Key points to remember:**
> - Bladder trocar injury: repass laterally, continue.
> - Urethral injury: abandon mesh; stage the repair.
> - Late perforation: excise mesh, repair, consider autologous sling.

## SAQ 5 Female urethral diverticula

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 1, Q3e

### Definition

- **UD (urethral diverticulum):** a urine-filled periurethral sac within the periurethral fascia, connected to the urethra by an **ostium**.
- Third to seventh decades; prevalence up to 1–6%.

### Aetiology

- Mostly **acquired**. **Routh theory** (most accepted): infection of **periurethral glands** leads to duct obstruction, suburethral abscess, rupture into the lumen and an epithelialised sac.
- Hence the ostium is **posterolateral at mid or distal urethra** in ≥90%.
- **Also know:** childbirth trauma theory (but 20–30% are nulliparous).

### Clinical features

- "Three Ds" (dysuria, dyspareunia, post-void dribbling) in only about 5%.
- Commonest: frequency, urgency, pain, **recurrent UTI (urinary tract infection)**.
- Tender anterior vaginal wall mass; about one-third incontinent.
- **Complications:** stones (4–10%); 10% show metaplasia, dysplasia or carcinoma (**adenocarcinoma** commonest).

### Evaluation

- Urine culture; **cystourethroscopy** (ostium seen in under half).
- **Imaging sensitivity: MRI (magnetic resonance imaging) 97%**, double-balloon urethrography 95%, ultrasound 82%, VCUG (voiding cystourethrogram) 74%.
- MRI shows saddlebag or circumferential extension.
- **Videourodynamics:** true SUI (stress urinary incontinence) versus dribbling from the sac.
- **Differential diagnosis:** Skene cyst (distal, distorts meatus, no communication), Gartner cyst, leiomyoma, caruncle, bulking agent.

### Classification

- **L/N/S/C3 (Leach):** location, number, size, configuration, communication, continence.
- **Also know:** Leng–McGuire (absent fascia = pseudodiverticulum).

### Management

- **Asymptomatic or unfit:** suppressive antibiotics, post-void stripping; counsel about cancer.
- **Symptomatic: transvaginal diverticulectomy.**
- **Principles:** 1. Sterile urine; topical oestrogen if atrophic. 2. **Inverted-U** anterior vaginal wall flap. 3. **Preserve periurethral fascia** as a separate layer. 4. Excise the **entire sac** and close the ostium. 5. **Watertight** urethral closure over a catheter (as small as 12 Fr). 6. **Multilayered, non-overlapping** closure; obliterate dead space. 7. **Martius flap** if poor tissue.
- **Bothersome SUI:** concomitant **autologous pubovaginal sling**; **never synthetic mesh**.
- **Also know:** Spence–Duckett marsupialisation for distal lesions only.
- Pericatheter VCUG at 14–21 days.

### Complications

- Recurrence 10–22%; de novo SUI 7–16%; urethrovaginal fistula 1–8%; stricture; recurrent UTI.

**Diagram to draw:** sagittal section of a mid-urethral sac and its ostium.

> **Key points to remember:**
> - Periurethral gland infection (Routh); posterolateral ostium.
> - MRI most sensitive (97%).
> - Preserve fascia; non-overlapping layers; Martius flap.
> - No synthetic mesh at diverticulectomy.

## SAQ 6 Endoscopic procedure of SUI

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 1, Q3f

*Read as endoscopic injection of urethral bulking agents.*

### Introduction

- **SUI (stress urinary incontinence)** results from urethral hypermobility and **ISD (intrinsic sphincter deficiency)**.
- **Bulking agents**, injected submucosally under cystoscopic vision, are the least invasive surgical option.

### Mechanism

- Bulk at the proximal urethra improves **mucosal coaptation** and resistance.
- It does not correct hypermobility, but most SUI includes some ISD.

### Indications

- ISD with a **fixed, immobile urethra** (low VLPP [Valsalva leak point pressure] <60 $cmH_2O$, historically).
- Women wanting a **low-risk** procedure; elderly or unfit; future pregnancy.
- Persistent leakage after a sling; neurogenic SUI (safer for bladder compliance than a sling).
- **Avoid in:** active UTI (urinary tract infection), untreated detrusor overactivity.

### Agents

| Agent | Note |
|---|---|
| Polyacrylamide hydrogel (Bulkamid) | Non-particulate |
| Polydimethylsiloxane (Macroplastique) | Silicone particles |
| Calcium hydroxylapatite; carbon-coated zirconium beads | Particulate |
| Bovine collagen (skin test); PTFE (polytetrafluoroethylene); autologous fat | Withdrawn or abandoned: allergy, migration, granuloma, embolism |

### Technique

1. Sterile urine; local anaesthesia; cystoscope (0–30° lens) in the urethra.
2. **Transurethral:** needle into the submucosa about 1–1.5 cm distal to the bladder neck; or **periurethral:** needle alongside the urethra, watched cystoscopically.
3. Inject at 2–4 sites until the mucosal cushions **coapt in the midline**.
4. Avoid an indwelling catheter (it moulds the bulk); use CISC (clean intermittent self-catheterisation) for retention.

### Outcomes

- Early improvement is common, but cure is lower than with slings and **declines with time**; repeat injections are often needed.
- **Cochrane review (Kirchin 2017):** limited evidence of benefit.
- **AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction):** an option, with counselling on durability.
- **Men:** EAU (European Association of Urology): **do not offer** for post-prostatectomy incontinence.

### Complications

- Transient retention, UTI, dysuria, haematuria, urgency.
- Sterile abscess, migration, erosion; may mimic a urethral diverticulum on imaging.

**Diagram to draw:** cystoscopic view of coapted mucosal cushions after injection.

> **Key points to remember:**
> - Bulking improves coaptation; does not correct hypermobility.
> - Best for fixed urethra with ISD, or unfit patients.
> - Less durable than slings; repeat injections common.
> - Not recommended for post-prostatectomy incontinence.

## SAQ 7 Endometriosis in urology

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 22 SAQ Sheet, Q3d

### Definition and epidemiology

- Endometriosis is endometrial glands and stroma outside the uterus. Urinary tract involvement occurs in about 1–5% of women with endometriosis, mostly with deep disease.
- Sites: **bladder** (commonest, about 80%) > ureter (about 15%) > kidney, urethra (rare).

### Pathogenesis

- Retrograde menstruation (Sampson), coelomic metaplasia, lymphovascular spread, and **iatrogenic implantation** at pelvic surgery (e.g. caesarean section).
- Bladder lesions start on the **serosa** (vesicouterine pouch) and invade the detrusor.
- Ureteric disease: **extrinsic** (periureteric fibrosis, commoner) or intrinsic (muscularis or mucosa); usually distal, left more often.

### Clinical features

- **Bladder:** cyclical haematuria or frequency, urgency and dysuria with menses; suprapubic pain. Differential: BPS (bladder pain syndrome), vesicouterine fistula (menouria).
- **Ureter:** often **silent**; hydronephrosis and loss of renal function may go unnoticed.

### Evaluation

- Transvaginal and transabdominal ultrasound gauge size and depth; **renal ultrasound in every woman with deep endometriosis**.

- **MRI (magnetic resonance imaging):** extent; high T1 signal from blood products.
- **Cystoscopy:** limited value (disease is extravesical); may show a bluish submucosal lesion. Biopsy excludes carcinoma.
- CTU (computed tomography urography) and MAG3 (mercaptoacetyltriglycine) renogram for obstruction and split function.
- Staging: rASRM (revised American Society for Reproductive Medicine) I–IV; #Enzian records bladder and ureter.

### Management (with gynaecologist)

- **Medical:** progestins (dienogest), combined pill, GnRH (gonadotrophin-releasing hormone) analogues. Symptomatic relief; not for obstructive ureteric disease.
- **Bladder: partial cystectomy** (full-thickness excision with margin), laparoscopic or robotic; stent ureters if near. TUR (transurethral resection) is incomplete, with recurrence and perforation risk.
- **Ureter:**
- Extrinsic, mild: ureterolysis.
- Intrinsic or stricture: segmental resection with ureteroureterostomy, or **reimplantation with psoas hitch or Boari flap**.
- Non-functioning kidney: nephrectomy.
- Treat all pelvic disease at the same operation to lower recurrence.
- **Also know:** risk factor for VVF (vesicovaginal fistula) and ureteric injury at hysterectomy; rare malignant change (clear-cell, endometrioid).

**Diagram to draw:** endometriotic nodule in the vesicouterine pouch invading the detrusor.

> **Key points to remember:**
> - Bladder commonest; cyclical haematuria.
> - Ureteric disease silent: screen kidneys.
> - Partial cystectomy, not TUR.

## SAQ 8 Urethral caruncle.

P2

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 2, Q3f

### Definition

- A benign, red, fleshy, **inflammatory polypoid lesion of the posterior lip of the distal female urethra**. Not a true tumour, but a benign mimic.
- Common; seen mainly in **postmenopausal women**.

### Aetiology and pathogenesis

- **Oestrogen deficiency (GSM, genitourinary syndrome of menopause)**: atrophy and laxity of distal urethral mucosa.
- Related to **mucosal prolapse**: the everted posterior mucosa is chronically irritated, causing haemorrhage, necrosis and inflammatory growth.

## Pathology

- Classical types: **papillomatous, angiomatous and granulomatous**; inflamed vascular stroma.

## Clinical features

- Often asymptomatic, found on examination.
- **Spotting on underwear**, bleeding, pain, dysuria; may thrombose into a dark, tender mass.
- Examination: small, red, soft mass at the **6 o'clock** meatal margin; the rest of the meatus looks normal.

## Differential diagnosis

| Lesion | Distinguishing feature |
|---|---|
| **Urethral mucosal prolapse** | **Circumferential "doughnut"** around whole meatus; prepubertal girls and postmenopausal women |
| **Urethral carcinoma** | Firm, ulcerated, indurated, fixed; bleeding mass; biopsy |
| **Urethral diverticulum** | Mid or proximal anterior vaginal wall mass; dysuria, dyspareunia, dribbling; MRI (magnetic resonance imaging) |
| Skene gland cyst or abscess | Lateral to meatus; distorts it; does not communicate |
| Others | Melanoma, lymphoma, TB (tuberculosis), condyloma, intestinal metaplasia; postmenopausal bleeding from the uterus |
| - Caruncle is listed as a possible risk factor for female urethral cancer, so atypical lesions must be biopsied. | |

## Evaluation

- Clinical diagnosis; exclude a gynaecological bleeding source.
- **Excision biopsy** if atypical, indurated, ulcerated, enlarging or refractory; cystourethroscopy if in doubt.

## Treatment

- **Conservative (first line)**: **topical vaginal oestrogen** (shrinks most), anti-inflammatory creams, sitz baths. Ask about breast or endometrial cancer before oestrogen.
- **Surgical excision** for large, symptomatic, thrombosed, refractory or suspicious lesions: 1. Grasp the tip and apply traction to expose the base. 2. Excise at the base. 3. Reapproximate mucosa with absorbable sutures if the defect is large. 4. Foley catheter for a few days.
- Complications: bleeding, recurrence, meatal stenosis.

**Diagram to draw:** meatus with a 6 o'clock caruncle compared with circumferential prolapse.

> **Key points to remember:**
> - Postmenopausal posterior-lip lesion; oestrogen deficiency.
> - Prolapse is circumferential; caruncle is focal.
> - Topical oestrogen first; excise if atypical or refractory.

## SAQ 9 Female sexual dysfunction

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 1 Paper 3, Q3f

### Definition and prevalence

- FSD (female sexual dysfunction): persistent desire, arousal, orgasm or pain problems causing **personal distress**.
- About **40%** of women in their lifetime; **HSDD (hypoactive sexual desire disorder) commonest**. Almost half of women with UI (urinary incontinence) have FSD.

### Physiology and response models

- Arousal: pudendal afferents, pelvic parasympathetic efferents → **NO (nitric oxide)–cGMP (cyclic guanosine monophosphate)** → engorgement and lubrication. Oestrogen maintains tissues; testosterone drives desire.
- **Masters and Johnson** (linear): excitement, plateau, orgasm, resolution; Kaplan added desire. **Basson** (circular): desire is often responsive.

### Classification

| DSM-5 (Diagnostic and Statistical Manual, 2013) | ICSM (International Consultation on Sexual Medicine)/ISSWSH (International Society for the Study of Women's Sexual Health) |
|---|---|
| Female sexual interest/arousal disorder (≥3 of 6 criteria) | HSDD and arousal disorder kept separate (cognitive vs genital arousal) |
| Female orgasmic disorder | Orgasm disorders |
| Genito-pelvic pain/penetration disorder | Genital-pelvic pain dysfunction |
| – | PGAD (persistent genital arousal disorder) |
| - ≥6 months with distress; lifelong or acquired; generalised or situational. | |

### Causes

- Hormonal (menopause, GSM [genitourinary syndrome of menopause]), drugs (SSRIs [selective serotonin reuptake inhibitors], antihypertensives), diabetes, depression, relationship factors.
- Urological: incontinence, prolapse, sling dyspareunia, cystectomy nerve or clitoral injury.

### Evaluation

- Clinician must ask. **PLISSIT** (permission, limited information, specific suggestions, intensive therapy); BSSC-W (Brief Sexual Symptom Checklist for Women) screen; **FSFI (Female Sexual Function Index, 19 items, six domains)**.
- Examination: atrophy, lichen sclerosus, cotton-swab test, pelvic floor tone. Laboratory tests rarely help.

### Management

- Treat causes; education; sex therapy (sensate focus, mindfulness).

- **HSDD (premenopausal, FDA [US Food and Drug Administration] approved): flibanserin** 100 mg at bedtime (5-HT1A agonist/5-HT2A antagonist; alcohol caution); **bremelanotide** 1.75 mg subcutaneously on demand (melanocortin agonist). Postmenopausal: transdermal testosterone (off-label).
- **Arousal:** testosterone; PDE5 (phosphodiesterase type 5) inhibitors only with neurovascular disease.
- **Orgasm:** education, directed masturbation.
- **Pain/vestibulodynia:** pelvic floor physiotherapy, dilators, biofeedback, topical lidocaine, vaginal oestrogen; vestibulectomy if refractory.
- **Also know:** ospemifene, intravaginal prasterone (menopausal dyspareunia).

**Diagram to draw:** Basson circular sexual response model.

> **Key points to remember:**
> - About 40% prevalence; HSDD commonest; distress required.
> - DSM-5 merges desire and arousal, dyspareunia and vaginismus.
> - Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

## SAQ 10 Duloxetin

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 3, Q3e

*Read as: duloxetine, mainly its role in stress urinary incontinence.*

### Introduction

- **Duloxetine** is an SNRI (serotonin–noradrenaline reuptake inhibitor), also an antidepressant.
- Licensed in Europe (not USA) for **moderate to severe SUI (stress urinary incontinence) in women**; off-label in men.

### Mechanism

- During storage, bladder afferents drive glutamate excitation of pudendal motor neurons in **Onuf's nucleus (S2–S4)**, which keep the striated urethral sphincter closed.
- Serotonin (5-HT2) and noradrenaline (α1) amplify this glutamate drive. Duloxetine blocks their reuptake, so **sphincter tone and the sneeze-induced continence reflex increase**.
- In voiding, glutamate drive switches off, so retention does not occur.
- Central effects also reduce bladder activity.

### Dose

- **40 mg twice daily**; starting at 20 mg twice daily for 2 weeks reduces nausea. Taper on stopping.

### Evidence

- Phase III RCTs (randomised controlled trials) in women: about 50% median fall in incontinence episodes versus about 30–40% with placebo; better quality of life; **cure rare**.
- Benefit is temporary and lasts only while taken.
- Male PPI (post-prostatectomy incontinence): small, short-term benefit.

### Guideline position

- **EAU (European Association of Urology):** may be offered to women for **temporary** improvement, not cure; warn about side effects. Men: offer for SUI (weak), and **inform of off-label use and adverse effects** (strong).
- **NICE (National Institute for Health and Care Excellence):** not first line; not routinely second line; only if surgery is unwanted or unsuitable.
- Adding PFMT (pelvic floor muscle training): evidence conflicting.

### Adverse effects

- **Nausea** (commonest, early), dry mouth, dizziness, insomnia, constipation; **high discontinuation**.
- Raised blood pressure, hyponatraemia (elderly), suicidal ideation, discontinuation syndrome.

### Contraindications and interactions

- MAOIs (monoamine oxidase inhibitors), including **methylene blue** (serotonin toxicity); CYP1A2 inhibitors (fluvoxamine, ciprofloxacin); hepatic impairment; severe renal impairment; uncontrolled hypertension or narrow-angle glaucoma.

**Diagram to draw:** Onuf's nucleus with 5-HT and noradrenaline terminals driving the pudendal nerve.

> **Key points to remember:**
> - SNRI acting on Onuf's nucleus to increase sphincter tone.
> - 40 mg twice daily; titrate from 20 mg to reduce nausea.
> - Temporary benefit only; approved in Europe, not USA.

## SAQ 11 Acute urinary retention in females.

P4

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 4, Q3f

### Introduction

- AUR (acute urinary retention) is sudden inability to void. It is uncommon in women and is often non-obstructive, so search for the cause.

### Causes

| Group | Examples |
|---|---|
| Obstructive | Large cystocele or prolapse; tight sling after incontinence surgery; urethral diverticulum, stricture; pelvic mass, **retroverted gravid uterus**, haematocolpos; cancer |
| Functional | **Fowler syndrome**; dysfunctional voiding; primary bladder neck obstruction (rare) |
| Neurogenic | MS (multiple sclerosis; may present first with retention), cauda equina, transverse myelitis, diabetes |
| Inflammatory | Cystitis, vulvovaginitis, **genital herpes**, Bartholin abscess |
| Drugs | Anticholinergics, opioids, tricyclics, calcium-channel blockers |

| Group | Examples |
|---|---|
| Postoperative/postpartum; psychogenic | Epidural, pain; diagnosis of exclusion |

### Fowler syndrome

- Young women (20–35) with painless retention and bladder capacity over 1 litre.
- **Urethral sphincter EMG (electromyography):** complex repetitive discharges, so the sphincter fails to relax. Urethral closure pressure >100 cm $H_2O$.

### Evaluation

- **History:** drugs, surgery, delivery, neurological and bowel symptoms, genital lesions.
- **Examination:** pelvic (prolapse, mass); neurological (perineal sensation, anal tone).
- **Tests:** drained volume; urine culture; creatinine; pregnancy test; ultrasound.
- MRI (magnetic resonance imaging) spine if neurological signs; cystoscopy.
- **Later:** video-urodynamics (female obstruction, e.g. Qmax ≤12 mL/s with Pdet ≥20 cm $H_2O$); sphincter EMG.

### Management

- **Immediate:** urethral catheter; suprapubic if urethral access fails.
- **Treat the cause:** stop drugs; treat infection or herpes; pessary or prolapse repair; reposition the uterus; sling division; diverticulectomy.
- **TWOC (trial without catheter)** after correction.
- **Persistent retention:**
- **CISC (clean intermittent self-catheterisation)** rather than an indwelling catheter.
- **SNM (sacral neuromodulation):** about 70% success in Fowler syndrome.
- Bethanechol not recommended. Bladder neck incision only with caution (incontinence).

**Diagram to draw:** flowchart: AUR → catheterise → find cause → treat → TWOC → CISC or SNM.

> **Key points to remember:**
>
> - Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
> - Fowler: abnormal sphincter EMG; SNM about 70%.
> - CISC for persistent retention.

# Rapid Revision Sheet · Section 6

**Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair (LAQ)**

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone.
- Always restore apical support.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Integral theory: mid-urethral sling; hammock: backing, not repositioning.
- Burch: Cooper's ligament; TOT: obturator foramen; PVS: bladder neck.
- Bulge is most specific; splinting means high grade.
- Central cystocele rugae lost, paravaginal preserved; POP-Q stage II = −1 to +1 cm (±1 cm of hymen).
- Sacrocolpopexy most durable; colpocleisis for frail elderly.
- Cystoscopy after repair; screen for occult SUI.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

**Vesicovaginal fistula: management, timing of repair, eponymous techniques and recurrent VVF (LAQ)**

- Image upper tracts; exclude ureteric fistula and cancer (biopsy before redo).
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.
- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- First repair has the highest success.
- Interposition flap (Martius, peritoneum, omentum) for complex and recurrent VVF.
- Diversion after multiple failures or severe radiation.
- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

**Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula (LAQ)**

- Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
- Ureteric injury is mostly distal and missed.
- UVF: distal ureteric injury at hysterectomy; continuous leak within 4 weeks with normal voiding.
- Orange tampon = UVF on double dye test; exclude coexisting VVF.
- Stent early (within 2 weeks); remove at 4–6 weeks.
- Reimplant with psoas hitch or Boari flap if stenting fails; success ≥90%; follow 30 months.
- Urethrovaginal fistula: diverticulectomy is the commonest Western cause; obstetric in developing countries.

- Look for a coexisting VVF (about 20%).
- Layered closure plus Martius flap; autologous sling only.
- Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

**Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery (LAQ)**

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD.
- Q-tip >30° = hypermobility; VLPP / ALPP <60 $cmH_2O$ = ISD; MUCP ≤20 $cmH_2O$.
- Burch corrects position; slings and AUS correct function.
- VALUE (ValUE) trial: UDS not needed in uncomplicated (index) SUI.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
- PFMT first; MUS first-line surgery.
- SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
- PVS for ISD; urethral injury → no mesh.
- Men: AUS gold standard; sling for mild, non-irradiated.
- AUS PRB 61–70 $cmH_2O$; deactivate 4–6 weeks.
- Radiation or severe male SUI: AUS first; MASTER: sling non-inferior.

**Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management (LAQ)**

- Most acquired RUFs follow prostate cancer treatment.
- Martini: stage = size; grade = tissue (surgery, radiation, salvage).
- VCUG or RUG plus endoscopic biopsy to exclude cancer.
- Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

**Leak point pressures and ALPP (SAQ)**

- VLPP/ALPP: Pves at leak with Valsalva or strain, no detrusor contraction; start at 150 mL.
- ALPP low = weak sphincter; DLPP high = upper-tract danger.
- VLPP <60 ISD; 60–90 equivocal; ≥90 (>90) hypermobility.
- VLPP < CLPP; catheter size and volume alter it.
- DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
- Lower DLPP is safer; no validated safe cut-off.

**Mid-urethral slings: TOT and TVT (SAQ)**

- Integral theory; loose tape at mid-urethra.
- Type 1 macroporous monofilament polypropylene.
- Delorme 2001; TOT anchors in obturator muscles; fewer bladder injuries than retropubic.
- TOMUS: cure equivalent at 1 year (80.8% retropubic vs 77.7% transobturator).
- TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
- Cystoscopy after each TVT trocar pass; bladder perforation 4.5%.
- Prefer retropubic TVT in ISD or fixed urethra.

**Burch colposuspension (SAQ)**

- Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
- Cure 85–90% at 1 year, ~70% at 5 years.
- Niche: concomitant abdominal surgery, mesh avoidance.

**Management of perforated mid urethral sling (SAQ)**

- Bladder trocar injury: repass laterally, continue.
- Urethral injury: abandon mesh; stage the repair.
- Late perforation: excise mesh, repair, consider autologous sling.

**Female urethral diverticula (SAQ)**

- Periurethral gland infection (Routh); posterolateral ostium.
- MRI most sensitive (97%).
- Preserve fascia; non-overlapping layers; Martius flap.
- No synthetic mesh at diverticulectomy.

**Endoscopic procedure of SUI (SAQ)**

- Bulking improves coaptation; does not correct hypermobility.
- Best for fixed urethra with ISD, or unfit patients.
- Less durable than slings; repeat injections common.
- Not recommended for post-prostatectomy incontinence.

**Endometriosis in urology (SAQ)**

- Bladder commonest; cyclical haematuria.
- Ureteric disease silent: screen kidneys.
- Partial cystectomy, not TUR. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Urethral caruncle. (SAQ)**

- Postmenopausal posterior-lip lesion; oestrogen deficiency.
- Prolapse is circumferential; caruncle is focal.
- Topical oestrogen first; excise if atypical or refractory.

**Female sexual dysfunction (SAQ)**

- About 40% prevalence; HSDD commonest; distress required.
- DSM-5 merges desire and arousal, dyspareunia and vaginismus.
- Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

**Duloxetin (SAQ)**

- SNRI acting on Onuf's nucleus to increase sphincter tone.
- 40 mg twice daily; titrate from 20 mg to reduce nausea.
- Temporary benefit only; approved in Europe, not USA. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Acute urinary retention in females. (SAQ)**

- Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
- Fowler: abnormal sphincter EMG; SNM about 70%.
- CISC for persistent retention. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 7 · Voiding Dysfunction

## Long answer questions (LAQ)

### LAQ 1 Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management

P1 · P2 · P3 **merged from 4**

*Asked as:* P1 LAQ "Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management" · P2 SAQ "Overactive bladder and its drug therapy" · P3 SAQ "OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron" · P1 SAQ "Compare and contrast various detrusor relaxants in management of OAB"

#### Definition

- **OAB (overactive bladder), ICS (International Continence Society): urgency**, usually with frequency and nocturia, with or without UUI (urgency urinary incontinence), in the absence of UTI (urinary tract infection) or other obvious pathology.
- **OAB dry** versus **OAB wet** (with UUI).
- OAB is a **symptom** diagnosis. DO (detrusor overactivity) is a **urodynamic** finding; only 44–69% of patients with urgency show DO.

#### Pathophysiology

All mechanisms converge on **more afferent signal reaching awareness**: the bladder turns the signal up, or the CNS (central nervous system) fails to turn it down.

| Mechanism | Explanation |
|---|---|
| **Afferent (sensory)** | Aδ fibres normally sense filling. Unmyelinated **C fibres** are "silent", but become spontaneously active and hypersensitive in pathological states |
| **Urothelial** | Urothelium acts as a transducer: mechanical, inflammatory or chemical stimuli release ATP (adenosine triphosphate), acetylcholine and other mediators onto suburothelial afferents |
| **Myogenic** | Patchy denervation causes receptor upregulation and spontaneous contraction; exaggerated **micromotions** cause urgency, and coordinated spread across the wall gives DO |
| **Neurogenic / central** | Forebrain and PAG–PMC (periaqueductal grey–pontine micturition centre) normally inhibit voiding; loss of this inhibition lets "afferent noise" be felt as urgency |
| **Other** | Bladder–bowel crosstalk (constipation), ischaemia–reperfusion, chronic inflammation, altered urinary microbiome |

- **Risk factors and associations:** ageing, female sex, BOO (bladder outlet obstruction: an animal model, but only an indirect cause in humans; DO persists in up to 60% after TURP [transurethral resection of the prostate]), anxiety and depression, metabolic syndrome, poor toileting habits.

#### Evaluation

- **Minimum:** history, examination, **urinalysis**.

- Bladder diary (3–7 days; P2: 3 days), questionnaires, PVR (post-void residual) when indicated.
- Urodynamics and cystoscopy only for complicated or refractory cases, or if urodynamics will change an invasive decision.

### Outline of management (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction]; not a compulsory sequence)

`Education + behavioural therapy (first line) ↓ (4–12 weeks) Antimuscarinic or β3-agonist (mirabegron, vibegron); combinations ↓ PTNS / onabotulinumtoxinA / SNM (third line) ↓ Augmentation cystoplasty or urinary diversion`

| Line | Treatment |
|---|---|
| Behavioural | Education; reduce fluids by about 25%; cut caffeine; weight loss; **bladder training** (timed voids, interval increased by 15–30 minutes); **PFMT (pelvic floor muscle training)** with urge suppression: 50–80% fewer leaks |
| Drugs | **Antimuscarinics** (oxybutynin, solifenacin, extended-release trospium, darifenacin, fesoterodine, tolterodine); **β3-agonists** (mirabegron, vibegron); combinations |
| Third line | **PTNS (percutaneous tibial nerve stimulation)** weekly × 12; **onabotulinumtoxinA 100 U** intradetrusor; **SNM (sacral neuromodulation)** at S3 (success about 70–80%; revision >30% at 5 years) |
| Last resort | Augmentation cystoplasty or urinary diversion |

### Non-pharmacological management

#### 1. Education and lifestyle

- Explain normal bladder function; set realistic goals.
- Reduce fluid intake by about 25% if excessive; cut caffeine; treat constipation.
- **Weight loss:** about 8 kg loss in obese women reduced UUI episodes.

#### 2. Bladder training (BT) (breaks the cycle of urgency, frequent voiding and smaller capacity)

1. Keep a bladder diary.
2. Start a fixed interval (often 1 hour while awake).
3. Void by the clock, not by urge.
4. Increase by 15–30 minutes a week as control improves.
5. Target interval 3–4 hours. - Works by **uncoupling the void from the urge**.

#### 3. PFMT and urge suppression

- Kegel exercises: quick 1–2 s flicks and 5–10 s holds, twice daily.
- A pelvic floor contraction **reflexly inhibits the detrusor**. Urge suppression: stop, stay still, squeeze quickly several times, relax, let the urge pass, then walk calmly to the toilet.
- PFMT and BT reduce incontinence episodes by **50–80%**.
- **Also know:** biofeedback helps patients who cannot find the muscle; electrical stimulation at 5–20 Hz inhibits the detrusor; timed and prompted voiding for cognitively impaired patients.

#### 4. PTNS

- 34-gauge needle 3–4 cm above the medial malleolus; 20 Hz, 200 µs; confirm great-toe flexion or sole tingling. **30 min weekly for 12 weeks**, then maintenance.

- The posterior tibial nerve (L4–S3) shares sacral roots with the bladder. **SUmiT trial:** 55% vs 21% (sham) responders.

**5. SNM**

- Mechanism: stimulates **afferents** that modulate brainstem and forebrain control.
- **Test phase:** PNE (percutaneous nerve evaluation, 5–7 days) or **staged tined lead** at **S3** (anal "bellows" and great-toe flexion). ≥50% improvement on diary leads to implanting the pulse generator in the buttock.
- Success about 70–80%. **InSite trial:** 61% vs 42% (standard medical therapy). Revision >30% at 5 years; MRI (magnetic resonance imaging)-conditional rechargeable devices now available.

**6. Intradetrusor onabotulinumtoxinA**

- Cleaves SNAP-25 (synaptosomal-associated protein 25), blocking acetylcholine release; also reduces afferent signalling.
- **100 U in 10 mL at 20 sites** cystoscopically; effect lasts about 6–9 months.
- About 25% become completely dry. Risks: UTI about 20%, retention needing CISC (clean intermittent self-catheterisation). **ROSETTA:** greater UUI reduction than SNM at 6 months, but more UTI.

**7. Last resort:** augmentation cystoplasty or urinary diversion.

## Drug therapy: place and mechanisms

- Drugs are second line, after behavioural therapy, as first-line pharmacotherapy (antimuscarinics and β3 agonists alongside each other; AUA/SUFU). Two main classes: **antimuscarinics** and **β3-adrenoceptor agonists**.
- **Antimuscarinics:** acetylcholine acts on **M3** receptors (raising intracellular calcium) to contract the detrusor; the commoner M2 enhances it. Blockade reduces involuntary contractions and afferent urgency signalling.
- Muscarinic side effects by subtype: **M1** (brain) cognition; **M2** heart rate; **M3** (glands, gut) dry mouth, constipation.
- Selectivity is by **receptor subtype** (M3) or **tissue** (bladder over salivary gland); tissue selectivity matters more.
- **β3 agonists:** β3 is the commonest (most highly expressed) adrenoceptor in human detrusor. Activation raises **cAMP (cyclic adenosine monophosphate)** and relaxes the detrusor during storage, sparing the voiding contraction.

## Agents

| Drug | Distinguishing feature | Usual dose |
|---|---|---|
| Oxybutynin | Non-selective; direct relaxant; metabolite causes most dry mouth; patch, gel | 5 mg two to three times daily; ER (extended-release) |
| Tolterodine | Bladder (over salivary) selective; QT prolongation at high dose | 2 mg twice daily (IR, immediate-release); ER 4 mg daily |
| Fesoterodine | Prodrug of tolterodine's active metabolite | 4–8 mg daily |
| Solifenacin | 10-fold M3 > M2; long half-life | 5–10 mg daily |
| Darifenacin | About 30-fold M3-selective | 7.5–15 mg daily |

| Drug | Distinguishing feature | Usual dose |
|---|---|---|
| Trospium | Quaternary; does not cross the blood–brain barrier (little brain entry) | 20 mg twice daily |
| Mirabegron (β3) | Inhibits CYP2D6 (a liver enzyme): metoprolol interaction; small blood-pressure rise | 25–50 mg daily |
| Vibegron (β3) | No CYP2D6 interaction | 75 mg daily |

### Darifenacin vs solifenacin vs tolterodine

| Feature | Darifenacin | Solifenacin | Tolterodine |
|---|---|---|---|
| Selectivity | **Most M3-selective (about 30-fold)**; not tissue-selective | M3 about 10-fold over M2; bladder-selective | Non-subtype-selective; **bladder-selective** |
| Dose | 7.5–15 mg daily (ER) | 5–10 mg daily | 2 mg twice daily (IR); 4 mg daily (ER) |
| Half-life | Moderate (ER once daily) | **Long** (about 2–3 days): once daily | Short; active metabolite 5-HMT (5-hydroxymethyl tolterodine) |
| Metabolism | CYP3A4 (a liver enzyme) and CYP2D6 | CYP3A4 | CYP2D6 (and 3A4) |
| Dry mouth | Dose-related | Dose-related | Lower with ER |
| Constipation | **Highest** (gut M3) | Moderate–high | Lower |
| CNS | Spares M1: **preferred in elderly** | Better persistence | High dose: QT prolongation |
| Dose limit | Lower dose with CYP3A4 inhibitors or liver disease | 5 mg in severe renal or moderate liver impairment | Halve with renal or liver impairment |

- **Efficacy is similar**; response is individual, so switch agent or dose.
- **STAR trial:** flexible solifenacin beat tolterodine ER for urgency and incontinence.

### Mirabegron (β3 agonist)

- First **β3-adrenoceptor agonist** for OAB; FDA (Food and Drug Administration) approval 2012.
- **Mechanism:** activation of adenylyl cyclase raises **cAMP**, relaxing the detrusor during **storage**. Reduces non-voiding contractions and afferent firing. The voiding contraction (muscarinic M3) is spared, so PVR barely rises.
- **Dose:** oral, extended release, once daily: start **25 mg**, increase to **50 mg** after 4–8 weeks. **25 mg maximum** in severe renal impairment (eGFR [estimated glomerular filtration rate] 15–29) or moderate hepatic impairment; avoid in end-stage renal disease and severe hepatic impairment.
- **Moderate CYP2D6 inhibitor:** raises levels of metoprolol, desipramine, flecainide; monitor digoxin.
- **Efficacy:** phase III trials (SCORPIO, ARIES, CAPRICORN): fewer incontinence episodes and micturitions than placebo. **TAURUS** (12 months): efficacy similar to tolterodine, with less dry mouth.
- **Adverse effects:** hypertension, tachycardia, headache, nasopharyngitis, UTI. **Dry mouth and constipation near placebo rates**, far below antimuscarinics.

- **Contraindicated:** severe uncontrolled hypertension (≥180/110 mmHg). Caution with arrhythmia, and with BOO when combined with antimuscarinics (retention).
- Other uses: add-on to α-blocker in men with BPH (benign prostatic hyperplasia); neurogenic detrusor overactivity.
- **Also know:** vibegron 75 mg (no CYP2D6 interaction).

### Antimuscarinics versus β3 agonists

| Feature | Antimuscarinics | β3 agonists |
|---|---|---|
| Efficacy | Similar | Similar (TAURUS: mirabegron ≈ tolterodine) |
| Dry mouth, constipation, blurred vision | Common; main reason for stopping | Near placebo |
| Cognition | Central M1 blockade; **dementia risk 10–65% higher with use >3 months** | No anticholinergic burden |
| Retention | Up to 3% in men | Low |
| Contraindications | Urinary or gastric retention, uncontrolled narrow-angle glaucoma | Uncontrolled (severe) hypertension (≥180/110 mmHg) |

### Choice of drug and special groups

- **Elderly and cognitive impairment:** antimuscarinic use >3 months is linked to 10–65% more new-onset dementia (SUFU white paper). Prefer a **β3 agonist** (no anticholinergic burden), or ER trospium, darifenacin or fesoterodine. Beers criteria: avoid antimuscarinics in dementia.
- ER forms cause less dry mouth. Poor persistence is the main problem; switch dose or drug before third-line therapy.
- **Combination:** solifenacin plus mirabegron beats each alone (**SYNERGY, SYNERGY II**), without extra cardiovascular effects; in non-responders it equals solifenacin 10 mg with less dry mouth (**BESIDE**).
- **Men:** α-blocker first; add an antimuscarinic (if PVR is low) or β3 agonist; retention up to 3%, so check PVR.
- **Also know:** vaginal oestrogen in atrophy; desmopressin for nocturnal polyuria; imipramine (children, mixed incontinence); intravesical oxybutynin; refractory OAB: onabotulinumtoxinA 100 U and neuromodulation.

### Diagrams to draw

- Bladder showing urothelial afferents, C fibres and micromotions, with brainstem and forebrain (cortical) inhibitory pathways.
- Detrusor cell: acetylcholine → M3 → contraction; noradrenaline → β3 → cAMP → relaxation.
- Muscarinic subtypes M1 (brain), M2 (heart), M3 (bladder, gland, gut) with each drug's selectivity.

**Key points to remember:**

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- Behavioural therapy first: BT and PFMT with urge suppression give 50–80% fewer leaks; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP for storage-phase relaxation; voiding spared.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- STAR: solifenacin beat tolterodine ER.
- Mirabegron 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Elderly: mirabegron (β3), trospium or darifenacin preferred over other antimuscarinics.
- Solifenacin + mirabegron combination works (SYNERGY).
- Refractory OAB: PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

## LAQ 2 Neuromodulation for voiding dysfunction, including sacral neuromodulation

P2 · P3 · P4 **merged from 3**

*Asked as:* P4 LAQ "Neuromodulation for voiding dysfunction, including sacral neuromodulation" · P2 SAQ "Neuromodulation" · P3 SAQ "Sacral neuromodulation"

*Taken as neuromodulation for lower urinary tract dysfunction.*

### Definition and history

- **Neuromodulation:** electrical stimulation of a peripheral nerve that alters the activity of central (reflex) pathways controlling the bladder. It acts on the nervous system (**afferents**), not directly on muscle.
- Direct muscle stimulation failed (painful, local). Stimulating **afferents** needs less current and affects the whole organ.
- **SNM (sacral neuromodulation)** (Tanagho and Schmidt, about 1985): continuous low-amplitude **S3** stimulation through a tined lead and IPG (implantable pulse generator). PTNS (percutaneous tibial nerve stimulation): FDA (Food and Drug Administration) approval 2000.
- **Rule:** exclude anatomical causes first (obstruction, stones, cancer, fistula). Stimulation cannot correct anatomy or restore muscle.

### Mechanism

- Sacral roots are mixed nerves. Current excites **afferents to the spinal cord and forebrain** (areas of awareness and attention), altering PMC (pontine micturition centre) output. It acts on **afferent** nerves, not motor nerves.
- **OAB (overactive bladder):** heightened central control (afferent input) damps PMC firing, so there are fewer involuntary contractions.
- **Non-obstructive retention:** restores awareness of filling (sensation) and pelvic floor tone, so the PMC switches on and the **EUS (external urethral sphincter) relaxes**. Hence one therapy treats **both** opposite disorders.
- **Tibial:** the nerve carries **S2–S3** fibres; inhibition of OAB-type (C-fibre) reflexes is **opioid-mediated** (naloxone blocks it); long-term cortical plasticity.

### Modalities

| Modality | Target and method | Main use |
|---|---|---|
| **SNM** | Tined lead in **S3** foramen + implanted generator | Refractory OAB, NOUR (non-obstructive urinary retention), faecal incontinence |
| **PTNS** | 34-gauge needle above medial malleolus | Refractory OAB |
| TTNS (transcutaneous tibial) | Surface patch, home use | OAB (less effective) |
| Implantable tibial | Small subcutaneous device over tibial nerve | OAB, avoids clinic visits |
| Pudendal | Tined lead in Alcock's canal | IDO (idiopathic detrusor overactivity), NDO (neurogenic DO), SNM failure (off-label) |

| Modality | Target and method | Main use |
|---|---|---|
| DGNS (dorsal genital nerve stimulation) | Dorsal penile/clitoral nerve | Experimental; NDO |
| IVES (intravesical electrical stimulation) | Intravesical cathode | Underactive bladder in children |
| **SARS (sacral anterior root stimulator)** | Anterior S2–S4 roots + dorsal rhizotomy | Complete SCI (spinal cord injury) |

## A. Sacral neuromodulation: indications

- **Third line** for OAB after behavioural and drug therapy (AUA [American Urological Association] and EAU [European Association of Urology] agree).

| FDA approved | Not approved |
|---|---|
| Refractory UUI (urgency urinary incontinence) 1997 | Bladder pain syndrome |
| Urgency–frequency 1999 | Neurogenic bladder (small studies positive; low-level evidence) |
| **Non-obstructive urinary retention** (e.g. Fowler's syndrome) 1999 | Pelvic pain |
| Faecal incontinence 2011 | |

- **Fowler's syndrome**: young women in retention with abnormal sphincter EMG (electromyography) activity; responds well to SNM.
- Retention needs some detrusor function; success falls with age.

## SNM: technique

**Test phase** (with a bladder diary)

| Test phase | PNE (percutaneous nerve evaluation) | Staged tined lead |
|---|---|---|
| Lead | Temporary, office, local anaesthesia (dislodges) | Permanent tined lead, theatre, fluoroscopy |
| Duration | 5–7 days | Up to 4 weeks |
| Use | OAB | More reliable; higher conversion; preferred for retention (NOUR) |

- **Landmark:** prone; needle into S3 about **9 cm above the coccyx tip, 1–2 cm lateral**.

| Root | Pelvic response | Foot response |
|---|---|---|
| S2 | — | Calf and foot (external) rotation |
| **S3 (target)** | **Bellows**, perineal sensation | **Great-toe flexion** |
| S4 | Bellows/perineal only | None |

- **Implant if ≥50% improvement** in a diary parameter (retention: ≥50% fewer catheterisations); IPG in the buttock.
- **Also know:** 10–16 Hz at sensory threshold; bilateral stimulation no better.

### SNM: outcomes

- **InSite:** success 61% vs 42% with standard medical therapy at 6 months; about 70–80% success at 5 years (another source: about half keep benefit at ≥5 years).
- **ROSETTA:** onabotulinumtoxinA 200 U reduced UUI slightly more (−3.9 vs −3.3 episodes/day) with higher satisfaction, but caused more UTI (urinary tract infection) (24% vs 10%) and CIC (clean intermittent catheterisation, self-catheterisation); SNM removal 9% at 2 years.
- NOUR: most stop catheterising.

### SNM: complications and contraindications

- Pain at the IPG site, lead migration, loss of effect, infection; **revision over 30% at 5 years**; costly (costliest option); battery replacement.
- Contraindications: failed test, inability to use the programmer, sacral anomaly.

### B. Tibial nerve stimulation

- **PTNS technique:** 34-gauge needle **3–4 cm (three fingerbreadths) above the medial malleolus**, behind the tibia; ground pad on the same leg; 20 Hz, 0–10 mA. Confirm by **toe flexion/fanning** or sole tingling. **30 min weekly × 12 weeks**, then maintenance.
- **SUmiT** (n = 220, validated sham): response **55% vs 21%**. Extension: 77% relapse-free at 3 years with tapering maintenance.
- **OrBIT:** PTNS comparable to extended-release tolterodine, with fewer side effects.
- **TTNS:** home self-treatment; skin impedance limits amplitude, so less effective.
- **Implantable devices** remove clinic visits: **BlueWind** (miniature, external control; benefit sustained at 3 years); **eCoin** (coin-sized, battery-powered, office insertion under local anaesthesia).

### C. Other targets

- **Pudendal nerve stimulation:** tined lead via a posterior approach to Alcock's canal. Better than SNM in a blinded crossover; **salvage after failed SNM**. Off-label.
- **DGNS:** purely sensory; acute DO suppression in SCI; **conditional (on-demand)** stimulation possible. Electrode fixation is difficult.
- **IVES (Katona):** intravesical cathode in saline activates Aδ afferents, producing detrusor contraction. Needs some afferent and muscle function; daily sessions; used in children; the sham-controlled trial was negative.
- **Brindley SARS:** stimulates anterior S2–S4 roots for voiding, defecation and erection. **Dorsal rhizotomy** abolishes reflex incontinence and DSD (detrusor–sphincter dyssynergia). The sphincter relaxes faster than the detrusor between bursts, giving **post-stimulus voiding**. Needs intact efferents.
- **Also know:** intravaginal stimulation (SUI, stress urinary incontinence); sacral dermatome stimulation; closed-loop SNM.

### Comparison of third-line OAB options

| | PTNS | OnabotulinumtoxinA 100 U | SNM |
|---|---|---|---|
| Invasiveness | Lowest | Moderate | Implant |
| Drawback | 12 weekly visits | UTI, retention/CIC | Revision, cost |
| Retention | No role | Worsens | **Treats** |

### Recent advances

- Rechargeable stimulators (FDA 2019; ≥15 years); non-rechargeable devices lasting ≥10 years (2021).
- **Full-body (whole-body) MRI (magnetic resonance imaging)-conditional** systems at 1.5 T and 3 T.
- **Implantable tibial nerve stimulators** (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives; PTNS and pudendal stimulation as alternatives to SNM.
- Standardised tined-lead placement improves outcomes and lowers revision.
- Bilateral stimulation no better.

### Diagram to draw

- Lead in the S3 foramen with IPG in the buttock, and PTNS needle above the medial malleolus.

> **Key points to remember:**
>
> - Afferent modulation explains SNM benefit in both OAB and retention.
> - S3: bellows plus great-toe flexion; implant if the test improves ≥50%.
> - SNM revision >30% at 5 years.
> - SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
> - ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.
> - Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
> - Full-body MRI-conditional devices at 1.5 T and 3 T.
> - Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
> - Standardised tined-lead placement improves outcomes and lowers revision.

## LAQ 3 Pathophysiology, classification and management of LUTS.

P2

**Type:** LAQ (25 marks)
**Source:** 2024-Winter Paper 2, Q1 (medical and surgical management of BPH also answered in 2015-Summer Paper 2, Q1)

### Definition

- **LUTS (lower urinary tract symptoms):** symptoms of the storage, voiding and post-micturition phases; not disease-specific; up to half of older men.
- **BPH (benign prostatic hyperplasia)** is histology; BPE is enlargement; BPO is prostatic obstruction; **BOO (bladder outlet obstruction)** has any cause. Not every man with LUTS has BPO.

### Classification

#### 1. By phase (ICS, International Continence Society)

| Storage | Voiding | Post-micturition |
|---|---|---|
| Frequency, urgency, **nocturia** (waking ≥1 time to void), urgency incontinence | Weak or split stream, hesitancy, intermittency, straining, terminal dribble | Feeling of incomplete emptying, post-micturition dribble |

- Storage symptoms form **OAB (overactive bladder)**: urgency ± urgency incontinence, usually with frequency and nocturia; most bothersome.

#### 2. Functional (Wein)

| | Because of the bladder | Because of the outlet |
|---|---|---|
| Failure to store | DO (detrusor overactivity), low compliance, hypersensitivity | Sphincter weakness (SUI, stress urinary incontinence) |
| Failure to empty | DUA (detrusor underactivity) | BPO, stricture, bladder neck contracture, sphincter dyssynergia |

**3. Severity:** IPSS (International Prostate Symptom Score), 7 questions scored 0–5 plus a quality-of-life question: mild 0–7, moderate 8–19, severe 20–35.

**4. By cause:** prostate, bladder, urethra, neurological, polyuria, infection, stone, tumour.

### Pathophysiology

- **Prostate:** BPH is an increase in cell number of stroma and glands in the **transition zone** and periurethral region. **Type 2 5α-reductase** in stromal fibroblasts makes **DHT (dihydrotestosterone)**; androgens are permissive, not mitogenic. Ageing, inflammation and metabolic syndrome contribute.
- **Static component:** gland bulk; the **capsule** transmits expansion to the urethra (TUIP, transurethral incision of the prostate, relieves obstruction without removing tissue).
- **Dynamic component: α1A-adrenergic** smooth-muscle tone; the basis of α-blockers.
- Size does not correlate with symptoms or obstruction.
- **Bladder response:** hypertrophy → compensation → decompensation. Collagen causes trabeculation and low compliance; DO causes storage symptoms; late detrusor failure causes retention. One-third keep storage symptoms after surgery.

- **Also know:** nocturnal polyuria (night output >33% of 24 h over age 65); BPO complications: AUR (acute urinary retention), UTI (urinary tract infection), stones, haematuria, renal failure.

### Evaluation (EAU, European Association of Urology, men ≥40)

- **Basic:** history (comorbidity, drugs, sexual function), IPSS, DRE (digital rectal examination) and neurological check, **urinalysis (mandatory)**, PVR (post-void residual), PSA (prostate-specific antigen) if life expectancy >10 years.
- **Frequency–volume chart, 3 days:** for storage symptoms or nocturia.
- **Before treatment:** uroflowmetry (Qmax, maximum flow rate, <10 mL/s suggests BOO); ultrasound for volume and median lobe.
- **Selected:** creatinine and renal ultrasound if large PVR; cystoscopy for haematuria, stricture or before PUL (prostatic urethral lift).
- **Pressure–flow study** (separates BOO from DUA) before surgery if voided volume <150 mL, PVR >300 mL, age <50 or >80, or failed surgery.

### Management: EAU algorithm

```
Bothersome LUTS? No -> watchful waiting, lifestyle advice Absolute surgical indication? Yes -> surgery Otherwise: Nocturnal polyuria -> lifestyle +/- desmopressin Storage predominant -> antimuscarinic or beta3-agonist Prostate >40 mL -> 5ARI (5-alpha-reductase inhibitor) +/- alpha1-blocker Others -> alpha1-blocker or PDE5 inhibitor Residual storage -> add antimuscarinic or beta3-agonist Failure or complication -> surgery
```

- **Watchful waiting:** education, fluid 1.5–2 L/day (none 2 h before bed), less caffeine and alcohol, double voiding, bladder training; yearly review.

| Drug | Example | Mechanism and notes |
|---|---|---|
| α1-blocker | Tamsulosin 0.4 mg, silodosin 8 mg, alfuzosin 10 mg | Relax prostatic smooth muscle; acts in days; dizziness, ejaculatory dysfunction, IFIS (intraoperative floppy iris syndrome) |
| 5ARI | Finasteride 5 mg, dutasteride 0.5 mg | Lower DHT; volume falls about 20% over 6–12 months; **PSA halves**; reduces AUR and surgery |
| Antimuscarinic | Solifenacin, tolterodine | Block detrusor M3 receptors; for storage symptoms if PVR <200 mL |
| β3-agonist | Mirabegron 50 mg, vibegron | Relax detrusor during storage |
| PDE5I (phosphodiesterase-5 inhibitor) | Tadalafil 5 mg daily | Smooth-muscle relaxation; useful with ED (erectile dysfunction) |

- **MTOPS:** progression cut by doxazosin 39%, finasteride 34%, combination 66%; **CombAT** confirmed. Combine when volume >30–40 mL and PSA >1.5 ng/mL.

### Surgery

- **Indications:** refractory retention, recurrent UTI, bladder stones, recurrent haematuria, BPO-related renal insufficiency; failed medical therapy.

| Prostate volume | Options |
|---|---|
| <30 mL | TUIP, TURP (transurethral resection of the prostate) |
| 30–80 mL | **TURP (reference standard)**, bipolar TURP, PVP (photoselective vaporisation of the prostate), enucleation, PUL, Rezum (water vapour), Aquablation |
| >80 mL | **HoLEP (holmium laser enucleation)** or thulium enucleation, open or robotic simple prostatectomy |

- Ejaculation preserved: PUL, Rezum. Anticoagulated: PVP, HoLEP. PAE (prostatic artery embolisation) in selected men.
- **TURP complications:** bleeding, **TUR syndrome** (hyponatraemia with glycine; avoided by bipolar saline), retrograde ejaculation, stricture, incontinence.
- **Also know:** iTind; female LUTS follow OAB and SUI guidelines.

**Diagram to draw:** BPH → BPE → BPO → BOO → bladder changes → LUTS.

> **Key points to remember:**
> - ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
> - Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
> - 5ARI for >40 mL; MTOPS combination cut progression 66%.
> - TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

## LAQ 4 Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male

P2

**Type:** LAQ (25 marks)
**Source:** Practice Set 1 Paper 2, Q2

*Assumed meaning: cervical spinal cord injury (traumatic, operated, or from the surgery itself), giving a suprasacral lesion above T6.*

### Introduction

- A cervical cord lesion lies **between the PMC (pontine micturition centre) and the sacral cord**. It causes tetraplegia and the classic suprasacral pattern.
- Complete transection is rare; incomplete lesions are commonest. Manage by **urodynamics, not neurology**.

### Pathophysiology

- The PMC normally coordinates detrusor contraction with sphincter relaxation. Cutting the descending pathway leaves a **spinal reflex**.
- **Why NDO (neurogenic detrusor overactivity) appears:** unmyelinated C-fibre afferents become sensitised (NGF (nerve growth factor)), sprout and drive a new sacral reflex. Bulbospinal inhibition is lost.

| Phase | Detrusor | Outlet | Clinical |
|---|---|---|---|
| **Spinal shock** (usually 6–12 weeks; up to 1–2 years) | Areflexic, smooth bladder | Bladder neck closed; reduced closure pressure | **Retention**; overflow only |
| **Recovery (suprasacral pattern)** | **NDO**, no sensation below lesion | **Striated DSD (detrusor–sphincter dyssynergia)**; lesion above T7–T8 adds **smooth sphincter dyssynergia** | Reflex leakage, incomplete emptying, high pressure |

- **Blaivas DSD types:** type 1, sphincter relaxes at peak pressure (incomplete lesions); type 2, sporadic contractions; type 3, crescendo–decrescendo obstruction (types 2–3 with complete lesions).
- **Consequences:** high storage and voiding pressure, VUR (vesicoureteric reflux) in 17–25%, hydronephrosis, stones (about 7% in 10 years), UTI (urinary tract infection). Untreated, about half of men with DSD deteriorate (worse if complete, continuous DSD).
- Upper tract risk also rises with low compliance (<12.5 mL/cm $H_2O$) and NDO for over a third of filling.
- **VUR:** treat the bladder first (lower the pressure); reflux surgery in a thick bladder is difficult.
- **Also know:** cervical myelopathy may instead give an underactive detrusor.

### Autonomic dysreflexia

- Occurs with lesions **above T6** (cervical 60%), after spinal shock. Bladder or bowel distension triggers unopposed spinal sympathetic outflow, as medullary inhibition is lost.
- **Features:** pounding headache, severe hypertension (systolic rise ≥20 mmHg defines it at urodynamics), bradycardia, sweating and flushing above the lesion. It can cause stroke or

seizures.

- **Treatment:** sit up, **drain the bladder** (check catheter), then **nitroglycerin 2% paste** above the lesion; intravenous nitroprusside or labetalol if persistent. Prophylaxis with terazosin or prazosin. Monitor blood pressure during cystoscopy and urodynamics; use spinal or general anaesthesia.

## Evaluation

- History, examination, **hand function** (decides self-catheterisation), transfers, carers.
- Urinalysis, creatinine, **renal ultrasound**. Men with good hand function can self-catheterise; others need carers or a suprapubic catheter.
- **VUDS (videourodynamics) after spinal shock**, with blood pressure monitoring: compliance, NDO, DSD, VUR, **DLPP (detrusor leak point pressure) ≥40 cm $H_2O$** = upper tract risk.
- AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2021: suprasacral lesions are "unknown risk" until tested; DSD or poor compliance = high risk.

## Management

**Goals:** protect kidneys, low-pressure storage, low-pressure emptying, no infection, continence, social acceptability. Start with the simplest reversible option.

- **Spinal shock: CIC (clean intermittent catheterisation)** preferred; indwelling catheter early is acceptable. Defer irreversible procedures.

| Aim | Options |
|---|---|
| **Emptying** | **CIC: gold standard**, volumes <500 mL; carer CIC in tetraplegics. **Suprapubic catheter** better than long-term urethral catheter. Credé/Valsalva not recommended |
| **Storage (lower pressure)** | **Antimuscarinics** ± mirabegron (β3 agonist); **intradetrusor onabotulinumtoxinA 200 U** if refractory (repeat when the effect wears off) |
| **Refractory small bladder** | **Augmentation ileocystoplasty** ± continent catheterisable channel (Mitrofanoff) |
| **Men unable to do CIC** | **External sphincterotomy** or intrasphincteric botulinum toxin → reflex voiding into condom catheter. High failure; DLPP ≥40 cm $H_2O$ afterwards = failure. Urethral stents abandoned |
| **Neuromodulation** | **SARS (sacral anterior root stimulator, Brindley) + posterior sacral rhizotomy**: abolishes NDO and dysreflexia, allows voiding on demand |
| **Last resort** | Ileovesicostomy; bladder neck closure + suprapubic catheter; ileal conduit |

- **Also know:** sacral neuromodulation is used for neurogenic bladder without SCI (spinal cord injury); hydrophilic catheters delay first UTI.

## Complications and follow-up

- **UTI:** treat only symptomatic bacteriuria (5–14 days); no routine prophylaxis.
- Stones, VUR (treat the bladder first), bladder cancer (no screening cystoscopy, AUA/SUFU 2021).
- Lifelong: yearly review and renal function, upper tract imaging every 1–2 years, repeat urodynamics if anything changes.

`Cervical SCI → spinal shock (retention): CIC → VUDS at 3 months: NDO + DSD → antimuscarinic + CIC → Botox → augmentation → cannot CIC: sphincterotomy + condom / SARS / ileovesicostomy`

**Diagram to draw:** neural axis: PMC, lesion at cervical cord, sympathetic T10–L2, sacral S2–S4, with the resulting NDO + DSD.

> **Key points to remember:**
> - Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
> - Autonomic dysreflexia above T6: drain bladder first, nitrates next.
> - CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
> - DLPP ≥40 cm $H_2O$ threatens the kidneys.

## Short answer questions (SAQ)

### SAQ 1 Botulinum toxin in urology

P1 · P2 · P3 · P4 **merged from 4**

*Asked as:* P1 SAQ "Botulinum toxin in urology: mechanism, indications and doses" · P2 SAQ "Botulinum neurotoxin and its role in urology" · P3 SAQ "Botulinum toxin in urology" · P4 SAQ "Botulinum toxin in urology"

*"Botulimum" / "Botulinim" in the questions is read as botulinum toxin.*

#### Introduction

- Neurotoxin of *Clostridium botulinum*, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**, the only licensed formulation; type B is short-acting.
- 150 kDa: a heavy chain (100 kDa; binding and entry) and a zinc-dependent light chain (50 kDa).

#### Mechanism of action

1. The heavy chain binds the cholinergic nerve terminal and the toxin is internalised.
2. The light chain (a zinc protease) enters the cytosol.
3. It **cleaves SNAP-25** (synaptosomal-associated protein 25), a SNARE (vesicle-docking) protein. Vesicles cannot fuse, so **acetylcholine release stops**: chemical denervation. - **Afferent action:** less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide) release; fewer TRPV1 (capsaicin) and P2X3 (purinergic) sensory receptors on C fibres, so urgency and pain fall. - The effect lasts about 6–9 months and wanes as nerve terminals sprout, so injections are repeated; repeat injections stay effective and response is maintained.

#### Technique

- Flexible or rigid cystoscopy under local anaesthesia; diluted toxin injected into the detrusor at **about 20 sites (100 U, in 10 mL) or 30 sites (200 U, in 30 mL)**, usually sparing the trigone (reflux concern).
- Sterile urine, antibiotic cover; check PVR (post-void residual) at 2 weeks.

## Urological uses: indications and doses

| Indication | Dose and evidence |
|---|---|
| **Idiopathic OAB (overactive bladder)**, third line | **100 U** (EMBARK): incontinence down about 59%, about 25% dry. **ABC** trial: equal to antimuscarinic, more complete dryness. **ROSETTA**: comparable to SNM (sacral neuromodulation), more UTI (urinary tract infection) (P2, P3: 200 U used, equal to or ≥ SNM; another source: slightly better than SNM) |
| **NDO (neurogenic detrusor overactivity):** SCI (spinal cord injury), MS (multiple sclerosis) | **200 U** (DIGNITY trials): less incontinence, larger capacity, lower detrusor pressure; works with or without antimuscarinics |
| **Paediatric NDO** (spina bifida) | About **10 U/kg (maximum 360 U)**, about 30 sites, in published (Campbell) series; FDA (Food and Drug Administration) label (2021, age ≥5) 6 U/kg, maximum 200 U; continence 65–87%, most avoid augmentation |
| **DSD (detrusor–sphincter dyssynergia)**, underactive detrusor | Intrasphincteric injection 50–100 U, mainly in men unable to do CISC (clean intermittent self-catheterisation) |
| **IC/BPS (interstitial cystitis/bladder pain syndrome)** | 100 U intradetrusor, often trigonal; AUA (American Urological Association) option, off-label |
| **BPH (benign prostatic hyperplasia)** | 100–300 U intraprostatic: an early small trial was positive, but large placebo-controlled RCTs (randomised controlled trials) showed **no benefit over placebo (saline/sham)**; not recommended |

- **Also know:** pelvic floor spasm; paediatric dysfunctional voiding (sphincter); chronic pelvic pain; chronic orchialgia (RCT negative); ED (erectile dysfunction) trials. Units of different toxin products are not interchangeable.

### Complications

- **Retention needing CISC** (the main risk): 0–12% in OAB trials, but about 20% of women and up to 35% of men in practice; commoner in NDO. Each 10 mL rise in baseline PVR raises the risk about 27%. **The patient must accept CISC beforehand.**
- **UTI** about 20%; haematuria; rarely generalised weakness.

### Contraindications

- Active UTI, myasthenia gravis, aminoglycosides, pregnancy, unwillingness to self-catheterise.

### Recent advances

- Paediatric NDO: FDA approved onabotulinumtoxinA for children aged ≥5 years (2021); used from about 2 years of age.
- Other formulations: abobotulinumtoxinA and incobotulinumtoxinA under study for detrusor overactivity; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

### Diagram to draw

- Cholinergic synapse (nerve terminal) showing SNAP-25 cleavage by the light chain.

**Key points to remember:**

- Cleaves SNAP-25; acts on efferent and afferent nerves.
- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U), approved for age ≥5 years (2021); used from about 2 years of age.
- Main risks: retention needing CISC, UTI.
- Other formulations (abobotulinumtoxinA, incobotulinumtoxinA) under study; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

## SAQ 2 Urodynamic studies, including video urodynamics

P2 · P4 **merged from 3**

*Asked as:* P2 SAQ "Uro Dynamic Studies." · P2 SAQ "Vesicourodynamics" · P4 SAQ "Video Urodynamics"

*"Vesicourodynamics" is taken as videourodynamics (VUDS): urodynamics with simultaneous fluoroscopic imaging of the bladder and urethra.*

### Definition and principles

- UDS (urodynamic studies) test bladder and urethral function in storage and voiding.
- ICS (International Continence Society): **decide the question first** and **reproduce the symptom**; a normal study does not exclude disease.
- Before invasive UDS: symptom score, 3-day **frequency–volume chart**, free uroflow and PVR (post-void residual); treat UTI (urinary tract infection) first.

### Components of conventional UDS

| Test | Method | Main values |
|---|---|---|
| **Uroflowmetry** | Voided volume ≥150 mL | Bell curve normal; Qmax <10 mL/s (men) suggests obstruction or weak detrusor; plateau = stricture |
| **PVR** | Bladder scan or catheter | Emptying |
| **Filling cystometry** | Bladder and rectal lines, fill with saline 30–100 mL/min (adults); **Pdet = Pves − Pabd**; zero at atmosphere, transducer at symphysis | Sensation, capacity, **DO (detrusor overactivity)**, **compliance** = ΔV/ΔPdet (<20 mL/cm$H_2O$ impaired) |
| **Leak point pressures** | ALPP (abdominal LPP): cough or Valsalva at 150 mL | Low ALPP = weak sphincter (ISD, intrinsic sphincter deficiency) |
| | DLPP (detrusor LPP) | **>40 cm$H_2O$** risks upper tracts (McGuire) |
| **Pressure–flow study** | Void with lines in | High pressure–low flow = obstruction; low–low = underactivity |
| **EMG (electromyography)** | Perineal surface electrodes | DESD (detrusor–external sphincter dyssynergia) |

| Test | Method | Main values |
|---|---|---|
| **UPP (urethral pressure profile)** | Catheter withdrawn | Limited value |
| **Videourodynamics** | Above with contrast and fluoroscopy | VUR (vesicoureteric reflux), bladder neck, site of obstruction |

### Indices (men)

- **BOOI (bladder outlet obstruction index)** = PdetQmax − 2Qmax: **>40 obstructed**, 20–40 equivocal, <20 unobstructed.
- **BCI (bladder contractility index)** = PdetQmax + 5Qmax: **>150 strong**, 100–150 normal, **<100 weak**.
- Women: no validated nomogram (Nitti video criteria).

### Indications for UDS

- Male LUTS (lower urinary tract symptoms) before surgery when unclear (young men, equivocal flow, failed TURP [transurethral resection of the prostate]); complex, mixed or recurrent incontinence.
- **Neurogenic bladder** (spinal injury, MS [multiple sclerosis], spina bifida), radiation cystitis, before augmentation.
- Children: neurogenic bladder, PUV (posterior urethral valves); fill at **5–10% of expected capacity per minute**; expected capacity = (age + 2) × 30 mL.

### Video urodynamics (VUDS): definition and set-up

- **VUDS:** multichannel urodynamics (pressure–flow, with or without EMG) recorded at the same time as fluoroscopy, using **contrast as the filling fluid**.
- Vesical and rectal catheters (Pdet [detrusor pressure] = Pves − Pabd), flowmeter, EMG; C-arm (most flexible) or a tilting table.
- Position for the question: **SUI (stress urinary incontinence) views standing**; voiding in the usual position (women sitting, men standing).
- **Screen only at key moments** (provocation, pressure rise, voiding) to limit radiation.

### What fluoroscopy adds

| Situation | VUDS finding |
|---|---|
| NLUTD (neurogenic lower urinary tract dysfunction) | **Gold standard** (EAU); DESD and **DISD** (detrusor–internal sphincter dyssynergia, seen only on VUDS); trabeculation, diverticula |
| **VUR** | Pressure, volume and grade at onset; reflux "pops off" pressure, so compliance looks falsely safe |
| **Bladder neck** | Only way to see and assess it; **PBNO** (primary bladder neck obstruction) in young men and women |
| Female obstruction | **Nitti criteria:** radiographic narrowing between bladder neck and meatus during voiding, with a sustained detrusor contraction |
| Dysfunctional voiding | "Spinning-top" urethra at the external sphincter |
| SUI | Leakage with ALPP; open bladder neck at rest suggests ISD |
| Male obstruction | Localises prostatic obstruction, bladder neck contracture or stricture |

- **PBNO types (young men):** I high pressure, low flow; II normal pressure, low flow with bladder-neck narrowing; III delayed bladder-neck opening.

### Indications for VUDS

- Known or suspected NLUTD; unexplained retention in women; young men with LUTS; complex incontinence; prior radical pelvic surgery or radiation; diversion; renal transplant.
- **Children:** PUV, spinal dysraphism; contrast-enhanced ultrasound VUDS avoids radiation.

### Urodynamic risk factors (upper-tract danger)

- Impaired compliance; DESD; DISD; high-pressure DO throughout filling; DLPP >40 $cmH_2O$; poor emptying with high storage pressure.

### Limitations of VUDS

- Radiation, cost, space, availability, contrast allergy.
- EAU (European Association of Urology): optimum test in NLUTD. AUA (American Urological Association) guideline: lowest radiation dose that answers the question.

### Also know

- Ambulatory UDS; bethanechol test; antibiotic prophylaxis for risk groups.

### Diagrams to draw

- Normal trace: Pves, Pabd, Pdet, flow, EMG.
- VUDS trace beside a voiding cystourethrogram image.

> **Key points to remember:**
> - Pdet = Pves − Pabd.
> - Ask a question first; reproduce the symptom.
> - BOOI >40 obstructed; BCI <100 weak.
> - DLPP >40 $cmH_2O$: upper-tract risk.
> - VUDS = urodynamics + fluoroscopy with contrast.
> - VUDS is the only test for bladder neck dysfunction, PBNO and DISD.
> - Reflux pop-off makes compliance look falsely good.

## SAQ 3 Uroflowmetry

P2 · P3 **merged from 2**

*Asked as:* P2 SAQ "Uroflowmetry." · P3 SAQ "Uroflowmetry"

### Definition

- A non-invasive urodynamic test that records the **volume of urine voided per unit time** (mL/s). It reflects detrusor contraction and outlet resistance together.

### Method

- Patient comes with a **comfortably full bladder** (normal desire) and voids in private, in the usual (preferred) position.
- **Flowmeters:** weight transducer (load cell), rotating disc, capacitance dipstick.
- **Voided volume ≥150 mL** (Qmax is unreliable below 125–150 mL). Repeat 2–3 times; ask whether the void was typical and compare with the frequency–volume chart.

- Measure **PVR** (post-void residual) by ultrasound afterwards.

### Parameters

| Parameter | Meaning |
|---|---|
| **Qmax** (maximum flow rate) | Most useful (key) value; normal men >15 mL/s, women >20 mL/s; read from the smoothed curve, ignoring spikes |
| Qave (average flow rate) | Volume ÷ flow time; less useful and less accurate for obstruction |
| Voided volume (VV) | Must be ≥150 mL |
| Flow time vs voiding time | Voiding time includes interruptions |
| Time to Qmax | Normally early, in the first third of the void |
| Voiding efficiency | VV ÷ (VV + PVR) × 100 |

### Curve patterns (ICCS, International Children's Continence Society)

| Pattern | Suggests |
|---|---|
| **Bell-shaped** | Normal |
| Tower (high, short) | Overactive bladder |
| **Plateau (flattened, box, prolonged)** | Fixed obstruction: **urethral stricture**, PUV (posterior urethral valves) |
| Prolonged, low, skewed | BPO (benign prostatic obstruction) or detrusor underactivity |
| Staccato (never reaches zero) | Sphincter overactivity, dysfunctional voiding |
| **Intermittent / interrupted (reaches zero), straining** | Detrusor underactivity with abdominal straining, obstruction, or DSD (detrusor–sphincter dyssynergia) |

### Interpretation

- **Qmax <10 mL/s:** obstruction likely (high specificity). **Qmax <15 mL/s:** more sensitive, less specific.
- **Nomograms / volume correction: Siroky** (men), **Liverpool** (men and women, centiles). Corrected Qmax = Qmax ÷ √VV.
- **Artefacts:** "wag" (moving stream), knocking the flowmeter, straining spikes. Read Qmax from the smoothed curve, not single spikes.
- Children: valid if voided volume ≥50% of expected capacity; accept Qmax only if the peak lasts ≥2 s.

### Limitations

- **Cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity)**; only a pressure–flow study can.
- Varies with volume, position, anxiety and between days.

### Uses

- BPH (benign prostatic hyperplasia): evaluation before active treatment and follow-up after medical or surgical treatment.
- Follow-up after **urethroplasty** or optical urethrotomy (stricture recurrence).

- Children: dysfunctional voiding, BBD (bladder and bowel dysfunction), after posterior urethral valve ablation; with patch EMG (electromyography).
- Women: voiding dysfunction, including before stress incontinence surgery.
- Pressure–flow study before BPH surgery if Qmax >10 mL/s (P2: EAU [European Association of Urology]; P3: AUA [American Urological Association], "may be done").

### Diagram to draw

- Flow curves: normal bell, tower, plateau (stricture), prolonged low (BPO), interrupted/intermittent.

> **Key points to remember:**
> - Voided volume ≥150 mL; Qmax is the key value.
> - Plateau = fixed obstruction (stricture); staccato = sphincter; interrupted = weak detrusor or straining.
> - Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.
> - Cannot distinguish BOO from DUA.

## SAQ 4 Detrusor underactivity and its management

P2 · P4 **merged from 2**

*Asked as:* P2 SAQ "Management of detrusor underactivity" · P4 SAQ "Detrussor underactivity."

### Definition

- **DUA (detrusor underactivity), ICS (International Continence Society) 2002:** "a contraction of reduced strength and/or duration, resulting in prolonged bladder emptying and/or failure to achieve complete bladder emptying within a normal time span."
- **Acontractile detrusor:** no contraction.
- **UAB (underactive bladder):** the symptom complex of slow stream, hesitancy and straining, with or without incomplete emptying.
- DUA is common in elderly men and women having urodynamics: found in 40–48% of men over 70.

### Pathophysiology and causes

- Failure may be **myogenic** (myocyte loss, fibrosis), **efferent** (poor nerve activation) or **afferent** (reduced sensation cannot sustain the voiding reflex).

| Type | Examples |
|---|---|
| Neurogenic | Cauda equina, disc prolapse, spinal injury, diabetic neuropathy, multiple system atrophy, early stroke, radical pelvic surgery |
| Myogenic | Long-standing BOO (bladder outlet obstruction), diabetes, ischaemia, ageing |
| Drugs | Anticholinergics, opioids |
| Functional | Fowler syndrome, dysfunctional voiding |

### Evaluation

- Bladder diary, flow rate, **PVR (post-void residual)**, sacral neurological examination, spinal MRI if indicated (deficit), drug review.

- **Pressure-flow study is the only accepted (diagnostic) test:** low detrusor pressure with low Qmax.
- **BCI (bladder contractility index) = Pdet@Qmax (detrusor pressure at maximum flow) + 5 × Qmax:** <100 weak, 100–150 normal, >150 strong. Also PIP (projected isovolumetric pressure), Watts factor, stop test.
- Ambulatory urodynamics if no contraction is seen on the standard study (the "bashful bladder").

## Management

- **Goals:** protect kidneys, prevent UTI (urinary tract infection) and stones, improve symptoms. No drug reliably improves contraction.

| Aim / option | Details |
|---|---|
| Conservative drainage | Timed and double voiding; treat constipation; stop causative drugs |
| **CISC (clean intermittent self-catheterisation)** | **Mainstay** for high PVR; fewer infections than an indwelling catheter. Suprapubic catheter if CISC impossible |
| Lower outlet resistance | α-blockers (modest benefit); pelvic-floor biofeedback |
| Outlet surgery (TURP, transurethral resection of the prostate) | In selected obstructed men; little help in low-PVR men; may free catheter-dependent fit men |
| **SNM (sacral neuromodulation)** | For non-obstructive retention, especially Fowler syndrome; needs some detrusor function; intact sensation predicts response |
| Sacral anterior root stimulator (Brindley) | Complete spinal cord injury, with posterior rhizotomy |
| Parasympathomimetics (bethanechol, distigmine) | **Not recommended / not routine**: weak evidence; systemic side effects (bronchospasm, rarely cardiac arrest) |
| Intrasphincteric botulinum toxin | Experimental |
| Diversion | Mitrofanoff when urethral CISC fails; rarely ileal conduit |

- Avoid Credé or Valsalva voiding (reflux risk).
- SNM and parasympathomimetics need some intact detrusor and sensory function; a fibrosed detrusor responds to nothing.

## Investigational

- Intravesical electrotherapy, intravesical prostaglandins, TAC-302, latissimus dorsi detrusor myoplasty, muscle-derived stem cells.

## Diagram to draw

- Pressure-flow plot on the Schäfer nomogram with BCI lines at 100 and 150.

> **Key points to remember:**
> - BCI = Pdet@Qmax + 5Qmax; <100 weak.
> - CISC is the mainstay; no effective drug; parasympathomimetics not recommended.
> - SNM helps non-obstructive retention (Fowler syndrome); intact sensation predicts response.

## SAQ 5 Detrusor–external sphincter dyssynergia (DESD / DSD)

P1

**Asked as:**

- External Sphincter Dyssynergia — *Summer 2016 Paper I Q3a (SAQ)*
- Detrusor sphincter dyssynergia — *Practice Set 10 Paper I Q3f (SAQ)* (also answered in 2016-Summer Paper 1, Q3a)

**Type:** SAQ (10 marks)

### Definition

- **DESD / DSD (detrusor–external sphincter dyssynergia):** the striated sphincter contracts involuntarily (involuntary contraction) during detrusor contraction, causing functional obstruction, high pressure and poor emptying.
- **It requires a lesion between the pons and the sacral cord:** suprasacral spinal cord injury (SCI), MS (multiple sclerosis), transverse myelitis, myelomeningocele. Brain lesions (stroke, Parkinson's) do not cause it (**never with brain lesions**).
- Lesions (at or) above T6 may add bladder-neck (**smooth sphincter**) dyssynergia and autonomic dysreflexia.

### Blaivas classification

| Type | EMG (electromyography) pattern | Lesion |
|---|---|---|
| 1 | Activity rises with detrusor pressure, then relaxes suddenly at peak | Incomplete |
| 2 | Sporadic contractions throughout voiding | Complete |
| 3 | Crescendo–decrescendo contraction; obstruction throughout | Complete |
| - Alternative: intermittent versus continuous (continuous means a complete lesion). Type has no relation to lesion level (type is unrelated to level). | | |

### Diagnosis

- **Pressure–flow study with EMG:** EMG rises during the detrusor contraction, with high pressure and interrupted flow. Needle EMG is more reliable than surface.
- **VUDS (videourodynamics):** narrowed (narrow) membranous urethra with dilated posterior urethra; shows VUR (vesicoureteral reflux).
- **Exclude:** guarding reflex (normal); **pseudodyssynergia** (straining or suppressing a contraction); dysfunctional voiding (learned, no neurological lesion; Hinman syndrome in children).

### Significance

- Worse when complete, continuous and **in men**: about half of untreated men get VUR, hydronephrosis, stones or urosepsis. Other risk factors: high filling pressure, indwelling catheter.
- **Management logic:** DSD is both a storage and an emptying failure. Lower bladder pressure, then empty it.

- DLPP (detrusor leak point pressure) ≥40 $cmH_2O$ endangers the kidneys.

### Management

- **First line:** CIC (clean intermittent catheterisation) plus antimuscarinic or β3-agonist; then intradetrusor botulinum toxin.
- **CIC impossible (men):** sphincterotomy or sphincteric botulinum toxin, with condom drainage. The goal is DLPP <40; DLPP ≥40 afterwards means failure; repeat procedures are common.
- Others: suprapubic catheter, Brindley stimulator with deafferentation, diversion. Urethral stents are abandoned; oral drugs disappoint. Credé and Valsalva voiding contraindicated.
- **Also know:** α-blockers for bladder-neck dyssynergia; lifelong upper-tract surveillance.

**Diagram to draw:** urodynamic trace with rising detrusor pressure (Pdet), EMG bursts and interrupted flow.

> **Key points to remember:**
> - Lesion between pons and sacral cord; never brain lesions.
> - Blaivas 1 incomplete; 2 and 3 complete.
> - Worst in men; DLPP ≥40 $cmH_2O$ hostile.
> - CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

## SAQ 6 EMG

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 9 Paper 1, Q3b

*EMG is taken as urodynamic electromyography of the striated urethral/anal sphincter and pelvic floor.*

### Definition and principle

- EMG (electromyography) records the electrical potentials of depolarising striated muscle membranes, here the **external urethral sphincter and pelvic floor**, during filling and voiding.
- It shows **coordination** between detrusor and sphincter, not sphincter strength. The bladder neck (smooth muscle) is invisible to EMG.
- External anal sphincter activity mirrors the urethral sphincter, except in some neurological diseases.

### Electrodes

| Surface (patch) | Concentric needle |
|---|---|
| Perianal patches at 3 and 9 o'clock (ICS [International Continence Society] standard: two skin electrodes plus reference) | Needle into the sphincter (girls beside the meatus; boys transperineal) |
| Comfortable; poorer signal; movement artefact | Best signal; painful; dislodges |
| Coordination only | Also motor unit analysis: acute versus chronic denervation |

### Normal pattern

- Filling: activity **rises gradually** (recruitment, guarding reflex).
- Voiding: EMG **falls silent before** detrusor pressure rises and stays silent until flow ends.
- **Lag time** (uroflow with patch EMG in children): relaxation to flow 2–6 s is normal; <2 s suggests DO (detrusor overactivity); >6 s suggests primary bladder neck dysfunction.

### Abnormal patterns

| Finding | Meaning |
|---|---|
| **DSD (detrusor–sphincter dyssynergia)** | Sphincter contracts during detrusor contraction; needs a **suprasacral** cord lesion (SCI [spinal cord injury], MS [multiple sclerosis]). Blaivas types 1–3 |
| Dysfunctional voiding | Same tracing without neurological lesion (learned) |
| Pseudodyssynergia | EMG rise from straining or trying to suppress a contraction |
| Denervation potentials | Sacral cord or cauda equina lesion; MSA (multiple system atrophy) |
| Complex repetitive discharges, decelerating bursts | **Fowler syndrome** (retention in young women); predicts SNM (sacral neuromodulation) response |

### Clinical uses

- Neurogenic bladder (spina bifida, SCI); voiding dysfunction in children; female retention; before sphincter surgery.
- **Limitation:** surface EMG misdiagnoses some dysfunctional voiders; **VUDS (videourodynamics)** is more accurate.

**Diagram to draw:** tracing of Pdet, flow and EMG: EMG silent during normal voiding versus active in DSD.

> **Key points to remember:**
> - EMG judges coordination, not strength.
> - True DSD needs a suprasacral lesion.
> - Lag time 2–6 s normal.
> - Fowler syndrome: complex repetitive discharges on needle EMG.

## SAQ 7 Bladder dysfunction non genitourinary systemic disease

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 2, Q3a

*Taken as lower urinary tract dysfunction caused by systemic (non-genitourinary) diseases.*

### Principle

- The **site of the lesion**, not the disease name, predicts the bladder:
- Brain above the pons: **DO (detrusor overactivity), no DSD (detrusor–sphincter dyssynergia)**.
- Spinal cord: DO **with DSD**.
- Peripheral nerves or roots: **areflexia, reduced sensation, high PVR (post-void residual)**.

- **Urodynamics** before treatment, because symptoms do not predict findings.

### Diabetes mellitus (most important)

- **Diabetic cystopathy:** sensory then motor autonomic neuropathy. Impaired sensation leads to large capacity, poor contractility, low flow and later high PVR and UTI (urinary tract infection).
- **DO is in fact the commonest urodynamic finding** (early disease or coexisting cerebrovascular disease); hypocontractility comes late.
- Mechanisms: glycation end products, sorbitol pathway, oxidative stress, polyuria.
- Differentiate from BOO (bladder outlet obstruction) by pressure–flow study. Treatment: **timed voiding**, CIC (clean intermittent catheterisation), cautious antimuscarinics.

### Other systemic diseases

| Group | Disease | Usual finding |
|---|---|---|
| Neurological | Stroke | Acute retention, then DO; no true DSD |
| | Parkinson disease | DO; pseudodyssynergia; TURP (transurethral resection of prostate) possible after excluding MSA (multiple system atrophy) |
| | Multiple sclerosis | DO about 38%, DO with DSD 29%, areflexia 26%; changes over time |
| Infective | HIV/AIDS | DO, DSD; areflexia with CMV (cytomegalovirus) |
| | Herpes zoster/simplex, Guillain–Barré | Retention, areflexia; reversible |
| | Spinal tuberculosis, Lyme, HTLV-1 | DO with DSD or areflexia |
| Autoimmune and connective tissue | SLE (systemic lupus erythematosus) | DO; lupus cystitis |
| | Systemic sclerosis | Areflexia, low compliance |
| Infiltrative | Amyloidosis (transthyretin) | Low sensation, underactivity, SUI (stress urinary incontinence) |
| Nutritional | B12 deficiency, alcohol (thiamine) | Mixed; reversible with replacement |
| Endocrine | Hyperthyroidism | Poor emptying; resolves with treatment |

**Also know:** myasthenia gravis; schizophrenia; gastroparesis.

### Management principles

- Treat the disease; reversible therapy.
- Storage: antimuscarinics (trospium, or mirabegron in elderly brains), botulinum toxin, neuromodulation.
- Emptying: CIC; avoid Credé with DSD.

**Diagram to draw:** neuraxis with lesion levels and the bladder pattern at each.

**Key points to remember:**

- Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
- Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
- Urodynamics before treating; reversible treatment first.

## SAQ 8 Urinary bladder compliance

P3

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 3, Q3e

### Definition

- **Compliance** is the change in volume for a change in detrusor pressure: **C = ΔV / ΔPdet**, in **mL/cm$H_2O$**.
- It reflects **low-pressure storage** during filling.

### Physiological basis

- **Viscoelastic wall**: smooth muscle, collagen (about 50% of the wall) and elastin (about 2%). Coiled collagen straightens as the bladder fills.
- More collagen or less elastin makes the wall stiff.

### Measurement (filling cystometry)

- ICS (International Continence Society) points: Pdet at **start of filling** and at **cystometric capacity** (or before a leak-causing contraction), **excluding detrusor contractions**.
- **No agreed normal value**; compliance depends on capacity. Below **20 mL/cm$H_2O$** (some use 10–20) suggests poor accommodation. In children, normal is often taken as **>15 mL/cm$H_2O$**.

### Pitfalls

| Pitfall | Effect | Fix |
|---|---|---|
| Fast filling | False low compliance | Slow or stop filling; pressure falls |
| Sustained low-amplitude detrusor contraction | Mimics low compliance | Stop filling: pressure returns to baseline |
| **Pop-off**: VUR (vesicoureteric reflux), diverticulum, leaking outlet | Falsely good compliance | Video-urodynamics; occlude outlet |
| - A small bladder can give a low number without danger (**compliance paradox**). **Absolute storage pressure matters more than the ratio.** | | |

### Causes of low compliance

- **Neurogenic**: spina bifida, spinal cord injury, lower motor neuron and decentralisation injury (radical pelvic surgery).
- **Outlet obstruction**: PUV (posterior urethral valves), long-standing BPO (benign prostatic obstruction).
- **Structural**: radiation cystitis, tuberculosis, schistosomiasis, multiple bladder operations.

### Consequences and DLPP

- **DLPP (detrusor leak point pressure)**: lowest Pdet at which leakage occurs without contraction or strain.
- **McGuire (1981)**: DLPP **>40 cm$H_2O$** in myelodysplastic children led to hydronephrosis or VUR in 85%. Not a validated safe limit (20 cm$H_2O$ is more sensitive); aim as low as possible.
- A strong outlet keeps the patient dry but transmits high pressure to the kidneys.

### Management

- **Clean intermittent catheterisation + antimuscarinics** (first line).
- **Intradetrusor botulinum toxin**.
- **Augmentation cystoplasty** if refractory; vesicostomy in infants.
- Repeat urodynamics to confirm safety.

**Diagram to draw:** cystometrogram: normal flat versus steep rising filling curve.

> **Key points to remember:**
> - C = ΔV/ΔPdet; <20 mL/cm$H_2O$ suggests poor compliance.
> - McGuire: DLPP >40 cm$H_2O$, 85% upper-tract damage.
> - Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

# Rapid Revision Sheet · Section 7

**Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management (LAQ)**

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- Behavioural therapy first: BT and PFMT with urge suppression give 50–80% fewer leaks; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP for storage-phase relaxation; voiding spared.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- STAR: solifenacin beat tolterodine ER.
- Mirabegron 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Elderly: mirabegron (β3), trospium or darifenacin preferred over other antimuscarinics.
- Solifenacin + mirabegron combination works (SYNERGY).
- Refractory OAB: PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

**Neuromodulation for voiding dysfunction, including sacral neuromodulation (LAQ)**

- Afferent modulation explains SNM benefit in both OAB and retention.
- S3: bellows plus great-toe flexion; implant if the test improves ≥50%.
- SNM revision >30% at 5 years.
- SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
- ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.
- Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
- Full-body MRI-conditional devices at 1.5 T and 3 T.
- Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
- Standardised tined-lead placement improves outcomes and lowers revision.

**Pathophysiology, classification and management of LUTS. (LAQ)**

- ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
- Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
- 5ARI for >40 mL; MTOPS combination cut progression 66%.
- TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

**Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male (LAQ)**

- Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
- Autonomic dysreflexia above T6: drain bladder first, nitrates next.
- CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
- DLPP ≥40 cm $H_2O$ threatens the kidneys.

**Botulinum toxin in urology (SAQ)**

- Cleaves SNAP-25; acts on efferent and afferent nerves.

- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U), approved for age ≥5 years (2021); used from about 2 years of age.
- Main risks: retention needing CISC, UTI.
- Other formulations (abobotulinumtoxinA, incobotulinumtoxinA) under study; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

**Urodynamic studies, including video urodynamics (SAQ)**

- Pdet = Pves − Pabd.
- Ask a question first; reproduce the symptom.
- BOOI >40 obstructed; BCI <100 weak.
- DLPP >40 $cmH_2O$: upper-tract risk.
- VUDS = urodynamics + fluoroscopy with contrast.
- VUDS is the only test for bladder neck dysfunction, PBNO and DISD.
- Reflux pop-off makes compliance look falsely good.

**Uroflowmetry (SAQ)**

- Voided volume ≥150 mL; Qmax is the key value.
- Plateau = fixed obstruction (stricture); staccato = sphincter; interrupted = weak detrusor or straining.
- Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.
- Cannot distinguish BOO from DUA.

**Detrusor underactivity and its management (SAQ)**

- BCI = Pdet@Qmax + 5Qmax; <100 weak.
- CISC is the mainstay; no effective drug; parasympathomimetics not recommended.
- SNM helps non-obstructive retention (Fowler syndrome); intact sensation predicts response.

**Detrusor–external sphincter dyssynergia (DESD / DSD) (SAQ)**

- Lesion between pons and sacral cord; never brain lesions.
- Blaivas 1 incomplete; 2 and 3 complete.
- Worst in men; DLPP ≥40 $cmH_2O$ hostile.
- CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

**EMG (SAQ)**

- EMG judges coordination, not strength.
- True DSD needs a suprasacral lesion.
- Lag time 2–6 s normal.
- Fowler syndrome: complex repetitive discharges on needle EMG.

**Bladder dysfunction non genitourinary systemic disease (SAQ)**

- Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
- Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
- Urodynamics before treating; reversible treatment first.

**Urinary bladder compliance (SAQ)**

- C = ΔV/ΔPdet; <20 mL/cm$H_2O$ suggests poor compliance.
- McGuire: DLPP >40 cm$H_2O$, 85% upper-tract damage.
- Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

# Section 8 · Nocturia & Male Incontinence

## Short answer questions (SAQ)

### SAQ 1 Post-prostatectomy incontinence

P2

**Asked as:**

- Post prostatectomy incontinence. — *Summer 2021 Paper II Q3a (SAQ)*
- Prostatectomy incontinence — *Practice Set 11 Paper II Q3f (SAQ)*

**Type:** SAQ (10 marks)

(Practice Set 11 Paper II Q3f: also answered in 2021-Summer Paper 2, Q3a)

#### Definition and incidence

- PPI (post-prostatectomy incontinence) is mainly **SUI (stress urinary incontinence)** after RP (radical prostatectomy), the commonest cause of male SUI. Rates vary with definition ("no pad" vs "safety pad") and reporting; expert no-pad continence is in the mid-90s % at 1 year; continence improves for up to 2 years.

#### Pathophysiology

- RP removes the internal sphincter (bladder neck, prostatic urethra), so continence depends (entirely) on the external **rhabdosphincter**.
- Causes: apical injury to rhabdosphincter or pudendal branches (sphincter or nerve injury at the apex), short membranous urethra, afferent denervation; also detrusor overactivity, low compliance, **VUAS (vesicourethral anastomotic stenosis)**.
- Nerve sparing and posterior reconstruction (Rocco stitch) give **earlier, not better final** continence.

#### Evaluation

- History (type, severity, radiotherapy, cancer status), bladder diary, **24-hour pad weight**: mild <200 g, moderate 200–400 g, severe >400 g (another source: mild <150 g, moderate <400 g, severe more; thresholds vary).
- **Standing cough test**, grade (grades) 0 to 4 (0–4: leak reported but not seen; delayed drops; early drops; delayed stream; early persistent stream).
- Urinalysis, creatinine, PVR (post-void residual), PSA (prostate-specific antigen). **Cystoscopy** excludes VUAS and judges (shows) residual sphincter coaptation; urodynamics when needed (if mixed symptoms).

#### Management

1. **Wait 6–12 months** (improves up to 2 years). Supervised **PFMT (pelvic floor muscle training)** with biofeedback, ideally preoperative (started before surgery); pads, condom catheter or (penile) clamp.
2. **Duloxetine** (off-label). **Bulking agents are not advised.**
3. **Surgery** (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2019, EAU [European Association of Urology]):

| Situation | Choice |
|---|---|
| Mild–moderate, no radiotherapy | **Male sling**: AdVance / AdVance XP (repositions the bulbar urethra; needs a coapting sphincter) or adjustable systems (ATOMS [adjustable transobturator male system]) |
| Severe, or any radiotherapy | **AUS (artificial urinary sphincter), AMS 800**: gold standard; cuff pressure 61–70 cm $H_2O$ |
| ProACT balloons | Expert centres; avoid after radiotherapy |

- **MASTER trial:** male sling non-inferior to AUS at 12 months.
- AUS: social continence (0–1 pad) 55–77%; revision 10–29% at 3 years; lasts (lifespan) 8–10 years. Erosion risks: radiotherapy (risk higher after radiotherapy), prior erosion, urethroplasty (use a **transcorporal cuff** for fragile urethra).
- **Also know:** climacturia; treat VUAS first.

**Diagram to draw:** AMS 800 with cuff, scrotal pump and pressure-regulating balloon.

> **Key points to remember:**
> - Wait 6–12 months with PFMT before surgery.
> - Sling for mild–moderate; AUS for severe or irradiated.
> - MASTER: sling non-inferior to AUS at 12 months.

## SAQ 2 Desmopressin (DDAVP)

P3

**Asked as:**

- Desmopressin. — *Winter 2023 Paper III Q3b (SAQ)*
- DDAVP — *Practice Set 3 Paper III Q3c (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2023-Winter Paper 3, Q3b)

### Pharmacology

- DDAVP (**1-deamino-8-D-arginine vasopressin**, desmopressin), a synthetic analogue of AVP (arginine vasopressin/antidiuretic hormone).
- Deamination and D-arginine substitution prolong action (give longer action) and make it a **selective V2 agonist (selective V2 agonism) without pressor (V1) effect**.
- **Mechanism:** V2 receptor on collecting-duct cells raises cAMP (cyclic adenosine monophosphate), inserting **aquaporin-2**; water is reabsorbed, so urine is less and concentrated (V2 receptor → cAMP → aquaporin-2 insertion → water reabsorption → less, concentrated urine).
- Endothelial V2 action releases von Willebrand factor and factor VIII (haemostatic use).

### Formulations

| Form | Dose |
|---|---|
| Tablet | 0.2–0.4 mg at bedtime, about 1 h before bed (enuresis) |
| Oral lyophilisate (melt) | 120–240 µg, about 1 h before bed |
| Low-dose nasal spray (nocturia) | 0.83 or 1.66 µg, 30 min before bed |

| Form | Dose |
|---|---|
| Sublingual ODT (orally disintegrating tablet) | 27.7 µg women, 55.3 µg men |
| IV / subcutaneous | 0.3 µg/kg (bleeding) |

### Uses (in urology)

1. **Monosymptomatic nocturnal enuresis:** first line with the alarm. **Rule of thirds**: one-third full, one-third partial, one-third no response. Acts (works) at once but relapses on stopping. Restrict fluids in the evening (avoid evening fluid); add oxybutynin if partial.
2. **Nocturia from nocturnal polyuria** (NPi [nocturnal polyuria index] >20–33%, age-dependent). EAU (European Association of Urology): offer below (under) 65; low dose above (over) 65 with ≥2 voids. Add to α-blocker for refractory nocturia in BPH (benign prostatic hyperplasia).
3. **Neurogenic nocturia** (multiple sclerosis, MSA [multiple system atrophy]).
4. **Bleeding:** uraemic platelet dysfunction, von Willebrand disease (including acquired, in Wilms tumour). Tachyphylaxis after about 3 (repeated) doses.
5. **Diagnosis:** central versus nephrogenic DI (diabetes insipidus).

### Adverse effects

- **Hyponatraemia** (most important): headache, nausea, confusion, seizures.
- Risk factors: **age ≥65**, baseline sodium <135 mmol/L, CrCl (creatinine clearance) <60 mL/min, higher dose; women are more sensitive.
- Most events occur in the **first 2–3 weeks**.

### Monitoring and contraindications

- Sodium at baseline, then within 1 week and at 1 month (EAU adds day 3), and after dose change.
- **Avoid:** hyponatraemia, polydipsia, heart failure, SIADH (syndrome of inappropriate antidiuretic hormone), significant renal impairment, **loop diuretics**.

**Diagram to draw:** V2 receptor–cAMP–aquaporin-2 pathway in the collecting duct.

> **Key points to remember:**
> - Selective V2 agonist; no pressor effect.
> - Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
> - Hyponatraemia: age ≥65, low sodium; check sodium early.

# Rapid Revision Sheet · Section 8

**Post-prostatectomy incontinence (SAQ)**

- Wait 6–12 months with PFMT before surgery.
- Sling for mild–moderate; AUS for severe or irradiated.
- MASTER: sling non-inferior to AUS at 12 months. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Desmopressin (DDAVP) (SAQ)**

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 9 · Andrology & Sexual Medicine

## Long answer questions (LAQ)

### LAQ 1 Spermatogenesis, azoospermia and non-obstructive azoospermia

P1 · P2 · P3 **merged from 3**

*Asked as:* P1 LAQ "Spermatogenesis, azoospermia, non-obstructive azoospermia and maturation arrest" · P2 SAQ "Azoospermia and non-obstructive azoospermia" · P3 SAQ "Spermatogenesis"

*Maturation arrest is taken as spermatogenic maturation arrest on testicular histology.*

#### Spermatogenesis: definition

- Production of haploid spermatozoa from diploid spermatogonia in the seminiferous tubules: about **64 days** (42–76); about 123 million sperm a day.

#### Hormonal control: HPG (hypothalamic–pituitary–gonadal) axis

- The hypothalamus releases **GnRH (gonadotropin-releasing hormone)** in pulses; pulsatility is essential (constant agonist exposure shuts the axis down). Kisspeptin triggers puberty.
- **LH (luteinising hormone)** acts on Leydig cells to make testosterone (about 5 mg/day). **FSH (follicle-stimulating hormone)** acts on Sertoli cells.
- Feedback: testosterone (and oestradiol) on LH; **inhibin B** from Sertoli cells on FSH. So FSH and inhibin B report the tubules, and LH reports the Leydig cells.
- **Intratesticular testosterone** is about 50-fold higher than serum (held by **ABP, androgen-binding protein**) and is essential for postmeiotic maturation. Exogenous testosterone suppresses LH and so stops spermatogenesis.
- The testis is kept 2–4 °C below core temperature by the pampiniform countercurrent exchange.

#### Sertoli cells and the blood–testis barrier

- Nurse cells: nourish germ cells; secrete ABP, inhibin and fluid; phagocytose residual bodies.
- **Sertoli–Sertoli tight junctions** divide the tubule into a **basal** compartment (spermatogonia) and an **adluminal** compartment (spermatocytes, spermatids). Post-meiotic cells are "foreign" to the immune system; the barrier and local tolerance protect them.

#### Phases of spermatogenesis

| Phase | Events |
|---|---|
| **1. Proliferation (spermatocytogenesis)** | **Dark type A (Ad)** reserve stem cells self-renew (GDNF [glial cell line–derived neurotrophic factor] niche); **pale type A (Ap)** divide every 16 days into type B spermatogonia, then primary spermatocytes. Daughter cells stay joined by cytoplasmic bridges |
| **2. Meiosis** | Primary spermatocytes cross into the adluminal compartment. Meiosis I (reduction division; synapsis and crossing-over) gives secondary spermatocytes; meiosis II gives 4 haploid round spermatids |

| Phase | Events |
|---|---|
| **3. Spermiogenesis** | No division: acrosome from Golgi, flagellum from centriole, nucleus condenses to about 10%, mitochondria form the midpiece, cytoplasm shed |
| **4. Spermiation** | Sertoli cell releases sperm into the lumen |

- **Cycle, stage, wave:** six stages in humans; cycles are arranged as a helical wave, so production is continuous.
- Defective meiotic recombination causes maturation arrest and sperm aneuploidy.

### Epididymal maturation and transport

- Testicular sperm are immotile. During 2–12 days of epididymal transit they gain **progressive motility** and fertilising ability; they are stored in the cauda.
- **Capacitation** and the acrosome reaction occur in the female tract.

### Clinical relevance of spermatogenesis

- A semen sample reflects events about 3 months earlier; reassess 3 months after treatment.
- Disruptors: heat (varicocele, cryptorchidism), exogenous testosterone, chemotherapy, radiation.

### Azoospermia: definition

- No sperm in the **centrifuged** ejaculate on **at least two** semen analyses; 10–15% of infertile men.
- **NOA (non-obstructive azoospermia):** azoospermia due to **failure of spermatogenesis** (testicular failure); about 60% of azoospermic men.

### Classification

| Type | Causes | Clues |
|---|---|---|
| **Pre-testicular** (hypogonadotropic) | Kallmann syndrome, pituitary tumour, hyperprolactinaemia, **exogenous testosterone or anabolic steroids** | Low FSH, LH, testosterone |
| **Testicular (NOA, spermatogenic failure)** | Klinefelter, Y microdeletion, cryptorchidism, mumps, chemotherapy or radiation, idiopathic | Small soft testes, **FSH >7.6 IU/L**, long axis ≤4.6 cm |
| **Post-testicular (OA, obstructive azoospermia)** | CBAVD (congenital bilateral absence of the vas deferens), epididymal obstruction, vasectomy, hernia repair, EDO (ejaculatory duct obstruction) | Normal testes, **FSH ≤7.6**, dilated epididymis |
| Ejaculatory | Retrograde ejaculation, anejaculation | Low volume; sperm in post-ejaculate urine |

### Causes of NOA

- **Genetic: Klinefelter (47,XXY)**, the commonest; **Y microdeletion** (AZF [azoospermia factor] a, b, c); other karyotype anomalies.
- **Acquired:** cryptorchidism, mumps orchitis, torsion, **chemotherapy or radiation**, trauma, varicocele.
- **Idiopathic:** commonest overall.

- *Note:* hypogonadotropic hypogonadism is pre-testicular and treatable medically (hCG [human chorionic gonadotropin] ± FSH).

### Evaluation

- History (mumps, surgery, chemotherapy, drugs, testosterone use); examination (testis size, **vasa**, epididymis, varicocele).
- **Hormones:** FSH (reflects germ-cell mass), LH, testosterone, oestradiol, prolactin.
- NOA: small soft testes, palpable vasa, normal ejaculate volume. **FSH >7.6 IU/L and testis long axis ≤4.6 cm** point to NOA; FSH ≤7.6 with normal testes (>4.6 cm) suggests obstruction.

```
Azoospermia (2 centrifuged samples) ├─ Vasa absent → CBAVD → CFTR testing, both partners; renal scan/US ├─ Volume <1 mL → post-ejaculate urine: sperm = retrograde; none → TRUS for EDO └─ Normal volume: FSH ≤7.6 IU/L + testis long axis >4.6 cm → obstructive FSH >7.6 + testis ≤4.6 cm (small) → spermatogenic failure
```

- **Genetics (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine): karyotype** and **Y microdeletion** in all azoospermic men with NOA, with counselling. **AZFa or AZFb complete deletion: no sperm, do not operate**; AZFc: sperm often found. **CFTR (cystic fibrosis transmembrane conductance regulator)** testing in CBAVD.
- **TRUS (transrectal ultrasound):** seminal vesicle >1.5 cm wide or a midline cyst suggests EDO.
- **Testis biopsy:** combined with sperm extraction and freezing.

#### Histology patterns in NOA

| Pattern | Histology / meaning | Micro-TESE retrieval |
|---|---|---|
| Hypospermatogenesis | All stages present, fewer cells | Best |
| **Maturation arrest** | Halts at one germ-cell stage | About 40% |
| SCO (Sertoli cell–only) | No germ cells | About 30% |

## Maturation arrest

### Definition

- A histological pattern of **spermatogenic failure** in which germ cells develop normally up to one stage and **the sequence halts** at a premeiotic, meiotic or postmeiotic step. No mature sperm, or very few, are made.

### Classification

- **Early:** arrest at spermatogonia or **primary spermatocyte** (meiotic failure).
- **Late:** arrest at the **spermatid** stage.
- **Complete** (all tubules) versus **incomplete** (foci progress to sperm): incomplete gives better retrieval.

### Causes

- **Genetic:** complete **AZFb deletion** causes arrest; meiotic recombination defects in about half of men with maturation arrest; karyotype anomalies.
- **Acquired:** heat (varicocele, cryptorchidism), chemotherapy, radiation, drugs, orchitis; often idiopathic.

### Clinical features and diagnosis

- Azoospermia or severe oligospermia, normal semen volume, palpable vasa.
- **Testis size and FSH are often normal**, so it can mimic obstruction (FSH ≤7.6 IU/L, long axis >4.6 cm).
- **Diagnosis needs testicular biopsy** (Bouin's fixative, never formalin), usually done as sperm extraction.
- Karyotype and Y microdeletion testing in all azoospermic men (AUA/ASRM).

### Management

- Optimise: treat varicocele, stop exogenous testosterone; off-label hormonal stimulation (clomiphene, hCG) unproven.
- **Micro-TESE** with ICSI: retrieval **about 40%** (versus 30% in SCO).
- **Complete AZFb deletion: retrieval highly unlikely; do not operate.**
- No sperm: donor sperm or adoption. Round spermatid injection is experimental.
- Genetic counselling: sons inherit Y deletions.

## Management of azoospermia

### 1. Pre-testicular (hypogonadotropic)

- Stop exogenous testosterone. **hCG 1500–5000 IU 2–3 times weekly ± FSH**; pulsatile GnRH in hypothalamic failure; dopamine agonist for prolactinoma.

### 2. Obstructive

- Microsurgical **vasovasostomy** or **vasoepididymostomy** (clear watery fluid or sperm parts in vasal fluid: vasovasostomy; thick toothpaste-like fluid without sperm: vasoepididymostomy).
- **TURED (transurethral resection of the ejaculatory ducts)** for EDO.
- CBAVD or unreconstructable: sperm retrieval by **MESA (microsurgical epididymal sperm aspiration)**, PESA (percutaneous epididymal sperm aspiration) or TESA + **ICSI (intracytoplasmic sperm injection)**; freeze sperm at reconstruction.

### 3. Non-obstructive

- **Optimisation (before retrieval):** stop exogenous testosterone; repair a clinical varicocele (evidence weak; no definitive evidence in NOA); off-label hormonal therapy (clomiphene, anastrozole, hCG) for low testosterone.
- **Micro-TESE (microdissection testicular sperm extraction) + ICSI:**
- **Principle:** spermatogenesis in NOA is **focal**. Under ×20–25 magnification, **larger, dilated, opaque tubules** are chosen; the least tissue is removed and the blood supply is preserved.
- Sperm found in **about 50%**; better than conventional TESE (Klinefelter 68%, Sertoli cell–only 30%).

| Predictor | Retrieval (%) |
|---|---|
| Post-orchidopexy | 74 |
| AZFc deletion | 70 |
| Klinefelter | 68 |
| Post-chemotherapy | 53 |
| **AZFa or AZFb complete deletion** | **0: do not operate** |

- FSH and testis size **do not** reliably predict retrieval.
- **ICSI:** fresh or frozen testicular sperm. Live birth per cycle is modest; miscarriage is higher (sperm DNA damage). Male offspring of AZF deletion carriers inherit the deletion.

**4. Ejaculatory**

- Retrograde: sympathomimetics, or sperm from alkalinised urine. Anejaculation: vibratory or electroejaculation.

**5. Failure (no sperm):** donor sperm or adoption; genetic counselling.

### Also know

- Spermatogonial stem cell renewal by GDNF; in-vitro spermatogenesis and stem-cell transplantation are experimental.

### Diagrams to draw

- Cross-section of a seminiferous tubule showing basal and adluminal compartments with germ cells maturing towards the lumen.
- Hypothalamic–pituitary–testis axis with the three levels of failure.
- Opened testis under the microscope: plump tubules versus thin Sertoli cell–only tubules.
- Maturation arrest: seminiferous tubule showing germ cells up to spermatocytes with no spermatids or sperm.

**Key points to remember:**

- Spermatogenesis about 64 days; Ap divide every 16 days; six stages.
- Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
- FSH → Sertoli, LH → Leydig; tight junctions between Sertoli cells form the blood–testis barrier.
- Confirm azoospermia on two centrifuged samples.
- FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- AZFa/b complete deletion: do not operate (no retrieval).
- NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.
- Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
- FSH and testis size may be normal in maturation arrest: biopsy diagnoses.
- Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

# LAQ 2 Azoospermia: testicular biopsy and surgical sperm retrieval techniques

P1 · P2 · P3 · P4 **merged from 4**

*Asked as:* P4 LAQ "Define Azoospermia and describe types of testicular biopsies and surgical sperm retrival." · P2 SAQ "Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction)" (asked as "Sperm extraction in Obstructive azoospermia", "Testicular aspiration of sperms and assisted reproduction" and "Sperm Retrival techniques") · P1 SAQ "Testicular biopsy and TESA (testicular sperm aspiration)" (asked as "TESA" and "Testicular biopsy") · P3 SAQ "Sperm retrieval techniques"

## Definitions

- **Azoospermia:** no sperm in the **centrifuged pellet** of the ejaculate on **at least two** semen analyses. Found in about 1% of men and 10–15% of infertile men.
- Not **cryptozoospermia** (rare sperm in pellet) or **aspermia** (no ejaculate).
- **OA (obstructive azoospermia):** no sperm in the centrifuged ejaculate despite **normal spermatogenesis**, from blockage of the ducts.
- **TESA (testicular sperm aspiration):** percutaneous needle aspiration of seminiferous tubules under local anaesthesia to retrieve sperm for **ICSI (intracytoplasmic sperm injection)**, which needs only one sperm per oocyte.

## Classification

| Type | Causes | Clues |
|---|---|---|
| Pre-testicular | Kallmann syndrome, pituitary disease, **exogenous testosterone** | Low FSH (follicle-stimulating hormone), LH (luteinising hormone) |
| Testicular: **NOA (non-obstructive azoospermia)** | Klinefelter, Y microdeletion, cryptorchidism, chemotherapy, mumps | Small testes, high FSH |
| Post-testicular: **OA** | CBAVD (congenital bilateral absence of the vas deferens), vasectomy, epididymal or ejaculatory duct obstruction, hernia-repair injury | Normal testes and FSH |
| Ejaculatory | Retrograde ejaculation, anejaculation | Low volume; sperm in post-ejaculate urine |

## Evaluation (brief)

- Absent vasa (CBAVD): CFTR (cystic fibrosis transmembrane conductance regulator) testing of both partners. Volume <1 mL: post-ejaculate urine, then TRUS (transrectal ultrasound).
- **FSH ≤7.6 IU/L and testis long axis >4.6 cm: obstruction likely;** FSH >7.6 and axis ≤4.6 cm: spermatogenic failure.
- **Karyotype and Y microdeletion** in NOA. **AZF (azoospermia factor) a or b complete deletion: no sperm, do not operate.**

## Testicular biopsy

### Indications

- Azoospermia with **normal testicular size, palpable vasa and normal FSH**: separates OA from NOA (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine).
- Since ICSI needs one sperm, biopsy is now **diagnostic and therapeutic**: plan it as TESE (testicular sperm extraction) and **cryopreserve** sperm.
- **Not needed:** CBAVD with normal FSH (spermatogenesis always present); high antisperm antibodies after iatrogenic vasal injury. Do not operate with complete AZFa or AZFb deletion.

### Types

| Type | Method | Use | Limits |
|---|---|---|---|
| **Open biopsy** | Small incision in the albuginea, pea-sized tissue | **Gold standard** for histology and retrieval | Open procedure |
| **Percutaneous core biopsy** | 14 G biopsy gun under cord block | Fresh retrieval in OA; core of tissue | Blind; may injure epididymis or testicular artery; avoid in scarred scrotum |
| **FNA (fine-needle aspiration)**, i.e. TESA | 23 G needle, suction, several passes | OA retrieval; backup to epididymal methods | Few tubules, poor architecture; low yield in NOA |
| **Microdissection biopsy** (micro-TESE) | Operating microscope, plump tubules | NOA | Long operation |

- **Also know:** multi-site FNA "mapping" in NOA.

### Open biopsy technique

1. Local anaesthesia with **spermatic cord block** (or general/spinal); choose the larger testis.
2. 2 cm transverse scrotal incision; open tunica vaginalis.
3. Stay sutures; 3–4 mm albugineal incision avoiding surface vessels (loupes).
4. Excise extruding tubules with iris scissors, **no forceps**.
5. Fix in **Bouin's, Zenker's or glutaraldehyde, never formalin** (distorts histology).
6. **Touch prep** at once, then wet squash prep for sperm; if negative, contralateral biopsy.
7. Close albuginea and tunica vaginalis separately; no antibiotics.

### Histology patterns

| Pattern | Meaning |
|---|---|
| Normal spermatogenesis | Suggests obstruction |
| Hypospermatogenesis | All stages, fewer cells |
| Maturation arrest | Stops at a germ-cell stage |
| SCO (Sertoli cell–only) | No germ cells |
| Tubular sclerosis/hyalinisation | End-stage (e.g. Klinefelter) |

## Surgical sperm retrieval: indications

- **OA with normal spermatogenesis:** CBAVD, unreconstructable obstruction, failed or declined vasectomy reversal, back-up at reconstruction (vasoepididymostomy; frozen for IVF [in vitro fertilisation]), or failed epididymal aspiration.
- **NOA (testicular failure):** only testicular retrieval works; TESA gives poor yield; **micro-TESE (microdissection testicular sperm extraction)** preferred.

- **Anejaculation** (spinal cord injury): vibratory stimulation or electroejaculation first; retrieval after these fail.
- All retrieved sperm need **ICSI**, because sperm from obstructed systems have poor motility; cryopreserve spare sperm.

### Techniques of sperm retrieval

| Technique | Method / site | Best for | Merits, limits and notes |
|---|---|---|---|
| **PESA** (percutaneous epididymal sperm aspiration) | Fine needle into caput epididymis, local anaesthesia | OA | Simple; motile but few sperm; may obstruct epididymis |
| **MESA** (microsurgical epididymal sperm aspiration) | Single tubule opened under microscope at 16–25×; micropipette by **capillary action**, never suction | **OA (best)** | Many (most) motile sperm, multiple frozen vials; needs microsurgical skill |
| **TESA** | Testis, 23 G needle with suction, many passes | OA | Least invasive; immature, immotile sperm, small yield; poor in NOA |
| Percutaneous core biopsy | 14 G biopsy gun | OA | Fresh retrieval; core of tissue; blind; vessel injury |
| **Conventional TESE** | Open biopsy, one or more sites | OA, NOA | When epididymis is absent or empty; gold standard for tissue diagnosis; blind sampling |
| **Micro-TESE** (Schlegel 1999) | Testis bivalved (opened in an avascular plane); **larger, opaque tubules** picked under the microscope | **NOA (best)** | Most sperm, least tissue; sperm in about half; long operation |
| Vasal aspiration | Vasotomy, testicular end flushed | At reconstruction | |

#### MESA technique

1. Testis delivered; epididymis inspected at 16–25×.
2. Dilated tubule opened with a microknife; fluid checked at once.
3. No motile sperm: close and **move proximally** (start distally), up to the efferent ductules (proximal sperm are more motile; the most motile are often in the efferent ductules).
4. Collect by **capillary action** into micropipettes, never suction; 4–12 pipettes.
5. **Cryopreserve** in several aliquots; close tubule with 10-0 suture.

#### TESA technique

1. Spermatic cord block with local anaesthetic.
2. Testis fixed; **23 G needle** (or angiocatheter) on a high-suction syringe.
3. Multiple passes in different directions draw tubules into the needle (often 10–20 passes needed; P1 table: 3–4 passes).
4. Immediate wet examination for sperm; repeat until sperm are found. **Cryopreserve** spare sperm.

#### TESA: advantages and limitations

- **Advantages:** office procedure, no microsurgery, cheap, repeatable.

- **Limitations:** few tubules with poor architecture (not diagnostic histology); small numbers of immature, immotile sperm; often too few to freeze; blind.

**Micro-TESE logic**

- Spermatogenesis in NOA is **focal**. Plump, larger tubules hold sperm; thin white ones are Sertoli cell–only.
- Embryologist examines tissue in theatre; map positive sites; if negative, explore the other testis.
- It removes the least tissue for the most sperm, and is better than conventional TESE.

**Choice:** epididymal sperm first (MESA) in OA; testicular (TESA/TESE) if epididymal sperm are absent or retrieval fails; micro-TESE in NOA.

## Results

- OA: retrieval almost always succeeds; good pregnancy rates.
- NOA: micro-TESE finds sperm in **about half**; fine-needle aspiration yields far fewer sperm than micro-TESE. Retrieval depends on the cause:

| Condition | Micro-TESE retrieval |
|---|---|
| Post-orchidopexy | 74% |
| AZFc deletion | 70% |
| Klinefelter | 68% |
| Post-chemotherapy | 53% |
| Maturation arrest | 40% |
| SCO | 30% |
| **AZFa/AZFb deletion** | **0%: do not operate** |

- **Before retrieval in NOA:** stop testosterone; repair clinical varicocele; off-label clomiphene, anastrozole or hCG (human chorionic gonadotropin).

## Assisted reproduction and use of retrieved sperm

- **IUI (intrauterine insemination):** washed sperm into the uterus; mild male factor.
- **IVF:** oocytes and sperm combined in the laboratory.
- **ICSI:** one sperm injected into each oocyte; essential for surgically retrieved (testicular) sperm (poor motility).
- Frozen retrieved sperm equal fresh, except possibly in spinal cord injury.
- NOA pregnancies: more miscarriage (sperm aneuploidy, DNA damage); genetic counselling. No sperm: donor sperm or adoption.
- ICSI bypasses natural selection: risk of transmitting genetic defects. Offer karyotype, Y-microdeletion and CFTR testing as indicated before ICSI.
- Female burden: ovarian stimulation, multiple pregnancy.

## Complications

- **Haematoma** (commonest), pain, infection (rare), testicular vascular injury (TESA, core biopsy).
- **Inadvertent epididymal biopsy** (most serious: causes obstruction); epididymal obstruction (PESA).

- Testicular atrophy or transient fall in testosterone (TESE).

### Also know

- Testicular sperm may beat ejaculated sperm when DNA fragmentation is high.
- Post-mortem retrieval needs prior consent.
- Microfluidic sperm sorting.

### Diagrams to draw

- Testis and epididymis showing PESA, MESA and TESA sites; testis with 23 G needle on a suction syringe and multiple passes; flow to ICSI.
- Open biopsy: small albugineal window with extruding tubules snipped off.
- Micro-TESE: bivalved testis with plump (dilated) spermatogenic tubules beside thin Sertoli cell–only tubules.

> **Key points to remember:**
>
> - Azoospermia: no sperm in centrifuged pellet on two samples.
> - Biopsy indication: azoospermia, normal testes, vasa and FSH.
> - Open biopsy is the gold standard; fix in Bouin's, never formalin; no forceps; touch prep.
> - OA: normal FSH and testes; MESA is best (capillary action, never suction; move proximally): many motile sperm, multiple vials.
> - TESA: 23 G needle, suction, local anaesthesia; for OA; TESA/TESE when epididymal sperm absent.
> - NOA: micro-TESE best (sperm in about half); TESA yield poor.
> - AZFa/AZFb deletion: 0% retrieval, do not operate.
> - All surgically retrieved sperm need ICSI; cryopreserve extras.
> - Haematoma commonest; epididymal biopsy most serious.

## LAQ 3 Surgical management of ED and penile prostheses (types, biofilms)

P2 · P3 · P4 **merged from 3**

*Asked as:* P2 LAQ "Surgical management of erectile dysfunction and penile prosthesis" (asked as "Discuss surgical management of erectile dysfunction", "Penile Prosthesis" and "Describe various penile prosthesis in ED") · P3 SAQ "Types of penile prosthesis. What are biofilms in penile prosthesis?" · P4 SAQ "Penile implants."

### Introduction

- Surgery for ED (erectile dysfunction) means **penile prosthesis** (main operation) and, rarely, **penile vascular surgery**.
- Prosthesis: most effective, but last-line, treatment of ED; **satisfaction 80–90%**. First devices: Scott inflatable (1972), Small–Carrion rod (1978).
- Lue's goal-directed approach: surgery is for men who fail, cannot use or decline oral PDE5Is (phosphodiesterase type 5 inhibitors), intracavernosal injection, intraurethral alprostadil or a VED (vacuum erection device).
- Detailed vascular testing is needed only before vascular surgery.

### Indications and contraindications for a prosthesis

- **Indications:** ED not responding to, contraindicated for, or unacceptable with (refusal of) oral PDE5Is, injections or vacuum devices; common after prostatectomy, pelvic radiotherapy, diabetes.
- Corporal (cavernosal) fibrosis: after ischaemic priapism (**>36 h**), infection or implant removal; early implant preserves length.
- **Peyronie disease with ED**; penile or pelvic trauma.
- **Contraindications:** active skin, urinary (UTI [urinary tract infection]) or systemic infection; uncontrolled diabetes; unrealistic expectations; reversible ED.
- In insensate men (spinal cord injury) use an inflatable device, because rods erode.

### A. Penile vascular surgery

| | Arterial revascularisation | Venous surgery |
|---|---|---|
| Defect | Arteriogenic (poor inflow) | Veno-occlusive (venous leak) |
| Work-up | Duplex; selective pudendal angiography (also checks the inferior epigastric donor) | DICC (dynamic infusion cavernosometry and cavernosography) locates the leak |
| Technique | Inferior epigastric artery to dorsal artery (revascularisation) or to deep dorsal vein (arterialisation) | Ligation or embolisation of deep dorsal, superficial dorsal or crural veins; crural plication |
| Selection | Age <55, non-smoker, non-diabetic, no venous leak, focal internal pudendal stenosis; best in young men after pelvic or perineal trauma | Not defined |
| Status | Strictly selected men | Investigational |
| Complications | Glans hyperaemia (commonest), shunt thrombosis, inguinal hernia | Glans numbness, shortening, curvature, skin necrosis |

## B. Penile prosthesis: types

| Type | Devices / examples | Features | Best for |
|---|---|---|---|
| **Malleable (semirigid) rods** | Tactra (nitinol core), Genesis (braided silver), Rigi 10 (stainless steel) | Two bendable rods; simple, cheap; almost never fail (mechanical failure exceedingly rare; P3: no mechanical failure); always firm; erode more in insensate men | Poor manual dexterity; cost |
| Mechanical rods | Spectra (discontinued) | Articulating segments | – |
| 2-piece inflatable | Ambicor (discontinued) | Cylinders + scrotal pump; no reservoir; poor flaccidity | – |
| **3-piece inflatable** | AMS 700 CX/LGX/CXR; Coloplast Titan, Titan Touch, Titan Narrow; Rigicon Infla 10 | Two cylinders, scrotal pump, abdominal reservoir; **best rigidity and true flaccidity**; most popular | Most men |

### Three-piece inflatable details

- **Expansion:** girth only (CX, Titan) or girth and length (LGX, up to 25%; Infla 10 AX). **Narrow** cylinders (CXR, Titan Narrow) for fibrotic corpora.
- **Material:** AMS and Rigicon use silicone–fabric–silicone layers; Titan uses polyurethane (Bioflex).
- **Coatings:** InhibiZone (rifampicin + minocycline) on AMS; hydrophilic coating on Titan, dipped in antibiotic.
- **Reservoir:** space of Retzius, or **ectopic submuscular** after pelvic surgery or cystectomy; lockout valve prevents autoinflation.

### Consent

- The erection may be shorter (pseudocapsule scar); the aim is a firm penis for intercourse; sensation and ejaculation may reduce; infection or mechanical failure may need reoperation.

### Preoperative preparation

- Treat phimosis and skin, fungal and urine infection; sterile urine; stop anticoagulants and aspirin.
- Blood glucose **<200 mg/dL on the day of surgery** (more important than HbA1c [glycated haemoglobin]).
- Shave in theatre; chlorhexidine–alcohol skin preparation (about 40% fewer wound infections than povidone-iodine); MRSA (methicillin-resistant *Staphylococcus aureus*) decolonisation if a carrier.
- IV vancomycin or a cephalosporin plus gentamicin, 1 hour before incision, for up to 24 hours.

### Operative steps (3-piece, penoscrotal approach)

`Catheter → upper scrotal incision → expose corpora lateral to urethra → stay sutures → corporotomy → dilate proximally to ischial tuberosity, distally under glans (instruments dorsolateral; 7→13 mm) → goalpost test for crossover → measure → cylinders equal to or 1 cm shorter → test inflation → close`

`corporotomies → pump in dependent subdartos (dartos) pouch → reservoir in space of Retzius → connect tubing → test inflation → irrigate → close in layers`

- Dilate with Hegar or Brooks dilators, instruments pointing dorsolaterally; check for crossover or perforation; cylinders sized to corporal length (equal to or 1 cm shorter than measured length).
- **Approaches:** penoscrotal (commonest; secures the pump; urethra at risk); infrapubic (reservoir placed under direct vision; dorsal nerve at risk); subcoronal or ventral penile for rods.
- Malleable rods are placed slightly shorter: 0.5–1 cm shorter than the measured length; girth chosen with the two-dilator test.
- **Surgical principles:** antibiotics, glucose control, no-touch technique.
- **Intraoperative problems:** crossover (redilate laterally); crural perforation (suture sling); distal urethral perforation or urethral injury (abandon that side for 5–6 months); fibrotic corpora (cavernotomes, narrow cylinders); curvature (modelling for Peyronie's disease); floppy glans (glans fixation).
- **Postoperative:** mummy wrap with glans visible; cylinders partly inflated; drain out by 24 hours; activation at 4–6 weeks.

### Biofilm

- **Definition:** a structured microbial community in a self-made **polysaccharide matrix**, attached to the implant.
- **Stages:** 1. Conditioning film of host proteins coats the device within minutes. 2. Reversible, then irreversible bacterial adhesion. 3. Microcolonies. 4. Maturation into a 3D structure with water channels and quorum sensing. 5. Dispersal of free bacteria, causing clinical infection.
- **Organisms:** coagulase-negative staphylococci commonest; also *S. aureus*, Gram-negatives, *Candida*.
- **Why antibiotics fail:** the matrix blocks penetration, bacteria are slow-growing (persisters), and they survive up to **1000–1500 times** normal killing levels.
- **Relevance:** organisms from the first operation lie dormant; revising an uninfected device gives about 10% infection, cut to about 3% by **antiseptic washout**.

### Complications

- **Mechanical failure** (commonest; inflatables): repair in about **15% at 5 years, 30% at 10 years**.
- **Infection (1–3%)**: biofilm on the device.
- **Prevention:** glucose <200 mg/dL; treat skin and urine; decolonise MRSA carriers; intravenous vancomycin or cephalosporin plus gentamicin (up to 24 h); chlorhexidine–alcohol preparation; **antibiotic-coated devices halve infection** (InhibiZone rifampin–minocycline; hydrophilic Titan dipped in antibiotic); **no-touch technique** (under 0.5%).
- **Infected implant:** pus, fixed pump or exposed parts. Remove **all** components.
- **Mulcahy salvage:** remove all parts; wash with dilute povidone-iodine (3 minutes) then saline or antibiotic; change gloves, drapes and instruments; insert a new device at once. Success **84%**; fails with early onset, aggressive organisms, cellulitis. Delayed reimplantation loses length.
- **Intraoperative:** crural perforation, crossover, urethral injury.

- **Others:** erosion or extrusion (catheterisation, rods in insensate men); reservoir erosion into bladder or bowel; reservoir herniation; pump migration; autoinflation; penile shortening; **glans necrosis** (diabetes, smoking, radiation, previous implant); glans hypermobility; cylinder aneurysm.

### Special situations (also know)

- Ischaemic priapism >36 hours: early implantation preserves length.
- Transplant recipients: no extra infection.
- After phalloplasty: single cylinder, high erosion.
- Peyronie's disease: modelling.

### Outcomes

- Patient and partner satisfaction **80–90%**, the highest of any ED treatment; the main complaint is shorter length.

### Recent advances

- Minimal-dilation technique; flat ectopic reservoirs; penile transplantation. Regenerative therapies (Li-ESWT [low-intensity shock wave therapy], stem cells) remain experimental.

### Diagrams to draw

- 3-piece inflatable prosthesis: two cylinders, scrotal pump, reservoir in the space of Retzius.
- Biofilm life cycle.

> **Key points to remember:**
> - Prosthesis is the last line but gives 80–90% satisfaction.
> - 3-piece inflatable gives the best rigidity and flaccidity.
> - Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
> - Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
> - Biofilm: microbes in a matrix; antibiotics fail.
> - Infection 1–3%; infected implant: remove all parts; Mulcahy salvage 84%.
> - Repairs 15% at 5 years.
> - Arterial revascularisation only for young men with post-traumatic focal stenosis.

## LAQ 4 Priapism: diagnosis and management

P2 · P3 **merged from 2**

*Asked as:* P3 LAQ "Priapism: Diagnosis and Management" (also asked as P3 SAQ "Priapism") · P2 SAQ "Management of priapism"

### Definition

- Full or partial erection lasting **≥4 hours** beyond sexual stimulation, or unrelated to it (AUA/SMSNA [American Urological Association/Sexual Medicine Society of North America] 2021–2022).

### Classification (types)

| | **Ischaemic (low flow)** | **Stuttering (recurrent ischaemic)** | **Non-ischaemic (high flow)** |
|---|---|---|---|
| Share | **95%** | Subset of ischaemic | <5% |
| Lesion | Veno-occlusion | Repeated (recurrent) ischaemic episodes | Cavernosal artery–sinusoid (arterio-sinusoidal) fistula after trauma |
| Pain, rigidity | Painful, fully rigid | Painful, recurrent, often nocturnal | Painless, partial |
| Typical cause | ICI (intracavernosal injection), drugs, SCD (sickle cell disease) | SCD | Perineal or needle trauma |
| Urgency | **Emergency** | Emergency per episode + prophylaxis | Not an emergency |

### Aetiology

- **Ischaemic:** ICI (papaverine, phentolamine, PGE1 [prostaglandin E1]), including recreational use; α-blockers, trazodone, antipsychotics, cocaine; heparin or warfarin withdrawal; **SCD** (29–42% of affected men in their lifetime; 63% of paediatric ischaemic priapism); leukaemia (1–5%, high white cell count); penile metastases; scorpion and spider venom.
- **Non-ischaemic:** straddle or coital injury, pelvic fracture, needle injury. Onset is often delayed, when the clot over the arterial tear lyses.

### Pathophysiology

- **Ischaemic:** failed outflow acts like a compartment syndrome. Hypoxia, hypercarbia and acidosis cause sinusoidal thrombosis, smooth-muscle necrosis and TGF-β (transforming growth factor beta)-driven fibrosis, so ED (erectile dysfunction) follows. Irreversible damage starts within hours.
- **Stuttering (molecular):** low basal NO (nitric oxide) from eNOS (endothelial NO synthase) dysfunction downregulates PDE5 (phosphodiesterase 5). When an erection starts, cGMP (cyclic guanosine monophosphate) is not broken down, so relaxation is unopposed. Low Rho-kinase and high adenosine add to this. This explains why daily PDE5 inhibitors prevent episodes (they reset PDE5 regulation).
- **Non-ischaemic:** unregulated, well-oxygenated arterial inflow, so fibrosis is unusual.

## Diagnosis

- **History:** duration, pain, drugs, trauma, SCD or blood disease, prior episodes, baseline erectile function.
- **Examination:** rigid tender corpora with a soft glans and spongiosum (ischaemic); partial, non-tender (high flow); perineal bruising; vital signs before phenylephrine.
- **Corporal blood gas** (cornerstone; decides the type); high flow resembles arterial blood:

| | Colour | pO2 (mmHg) | pCO2 (mmHg) | pH |
|---|---|---|---|---|
| Ischaemic | Dark red | **<30** | **>60** | **<7.25** |
| Flaccid (mixed venous) | Medium red | 40 | 50 | 7.35 |
| Non-ischaemic (arterial) | Bright red | >90 | <40 | 7.40 |

- **PDUS (penile duplex ultrasound, colour Doppler)** when the gas is equivocal: ischaemic = absent cavernosal flow or PSV (peak systolic velocity) <50 cm/s with no diastolic flow; high flow = PSV >50 cm/s, may show the fistula. Angiography only for embolisation.
- Blood count, reticulocytes, haemoglobin electrophoresis, coagulation, toxicology: in parallel, never delaying treatment.

## Management of ischaemic priapism (AUA/SMSNA 2021)

``` Counsel on ED risk → dorsal penile block <36 h: aspiration (19 G, penoscrotal junction, 3 or 9 o'clock) ± saline irrigation + phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to 1 h (monitor blood pressure, pulse: reflex bradycardia) → fails → distal shunt (Winter, T-shunt, Al-Ghorab) ± tunnelling → proximal shunt (Quackels) or penoscrotal decompression

> 36 h or refractory → early (immediate) penile prosthesis (or analgesia and counselling) ```

- Aspiration alone succeeds in under 1 in 4. Cold (10 °C) saline irrigation was better than body temperature in an RCT (randomised controlled trial); avoid chilled saline in SCD.
- Ice, oral agents, hydration and IV (intravenous) fluids are adjuncts only; in SCD, intracavernosal treatment still comes first, with concurrent haematology care.

| Shunt | Method |
|---|---|
| **Winter** (distal) | Tru-Cut (biopsy) needle through glans; least effective |
| **Ebbehoj / T-shunt** (distal) | Scalpel through glans; T-shunt rotates blade 90° |
| **Al-Ghorab** (distal, open) | Dorsal glans incision; excise corporal (tunica) tips |
| **Snake tunnelling** | 7/8 Hegar dilator passed proximally through distal shunt |
| **Quackels / Sacher** (proximal) | Cavernosum to spongiosum at bulb; stagger ≥1 cm |
| **Grayhack / Barry** (proximal) | Cavernosum to saphenous / dorsal vein |
| **Penoscrotal decompression** | Proximal corporotomy, clot suction (evacuation); allows prosthesis |

- **ED by duration** (Bennett and Mulhall, SCD series): ≤12 h 0%; 12–24 h 22%; 24–36 h 56%; ≥36 h 100%.
- **Early prosthesis** (Zacharakis 2014): less shortening and fewer revisions than delayed insertion: shortening 3% vs 40%, revision 9% vs 27%, satisfaction 96% vs 60%.

## Stuttering priapism

- An episode lasting ≥4 hours is treated as ischaemic; home self-injection of phenylephrine.
```

- **Prophylaxis:** daily PDE5 inhibitor (Burnett 2014 RCT: primary endpoint missed, but promising), pseudoephedrine, etilefrine; hormonal agents (GnRH [gonadotropin-releasing hormone] agonists, antiandrogens, finasteride) at the cost of libido. **Also know:** digoxin; baclofen.

### Non-ischaemic priapism (AUA/SMSNA 2022)

- Observation, perineal compression and ice; about 60% resolve. Reassess at 4 weeks.
- Persistent: PDUS, then **selective cavernosal artery embolisation** (resolution ≥75%, recurrence 30–40%). For recurrence, repeat embolisation with non-resorbable PVA (polyvinyl alcohol), not surgery. Surgical ligation is a last resort (ED about 50%).

### Complications

- ED, corporal fibrosis, penile shortening; shunt-related oedema, haematoma, urethral injury or fistula, skin necrosis, infection.

**Diagram to draw:** distal (glans) and proximal shunt sites on a penile outline.

> **Key points to remember:**
> - Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
> - Aspiration plus phenylephrine first; distal shunt ± tunnelling before proximal.
> - ED risk 0% ≤12 h, 100% ≥36 h; early prosthesis beyond 36 h or if refractory.
> - High flow: observe (about 60% resolve), then embolise; repeat embolisation rather than surgery.

## LAQ 5 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED

P1

**Asked as:**

- Write in detail about physiology of erection. How will you evaluate a diabetic with ED? — *Practice Set 5 Paper I Q2 (LAQ)*
- Neurotransmitters in erectile function — *Practice Set 12 Paper I Q3e (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)

(erection physiology also in Practice Set 5 Paper 1, Q2)

### Introduction

- Erection is **cavernous smooth-muscle relaxation**: low cytosolic $Ca^{2+}$ dilates helicine arteries and sinusoids; expanding sinusoids compress emissary veins (veno-occlusion).
- Pathways: parasympathetic **S2–S4** cavernous nerves (erection), sympathetic **T11–L2** (detumescence), pudendal (sensation, ischiocavernosus rigidity).

### Functional anatomy

- **Corpora cavernosa:** sinusoids lined by endothelium, separated by **trabecular smooth muscle**. Flaccid tone depends on smooth-muscle contraction.
- **Tunica albuginea:** inner circular and outer longitudinal layers. **Emissary veins** pass obliquely between them and are compressed when the corpora expand: the **veno-occlusive mechanism**.
- **Arteries:** internal pudendal → common penile → **cavernous (helicine branches)**, dorsal and bulbourethral arteries. Accessory pudendal arteries in about one-third.
- Spongiosum and glans have a thin or absent tunica, so no veno-occlusion; they stay low-pressure.

### Haemodynamic phases

| Phase | Events | Intracavernous pressure |
|---|---|---|
| Flaccid | Smooth muscle contracted; low inflow | Low |
| Latent (filling) | Arterial dilation; **inflow peaks** | Barely rises |
| Tumescence | Sinusoids fill; venous outflow compressed | Rising |
| Full erection | Maximal veno-occlusion | About 100 mmHg |
| Rigid | **Ischiocavernosus** contraction (pudendal nerve) | Several hundred mmHg |
| Detumescence | Initial, slow, fast phases | Falls to baseline |

### Molecular mechanism and peripheral neurotransmitters

- **Relaxation (erection):** NO (nitric oxide) from **nNOS (neuronal nitric oxide synthase)** in cavernous nerves initiates erection; endothelial NOS (eNOS) sustains it.
- NO → soluble guanylyl cyclase → **cGMP (cyclic guanosine monophosphate)** → protein kinase G (PKG) → $K^+$ channels open, $Ca^{2+}$ channels close, $Ca^{2+}$ sequestered → **low cytosolic $Ca^{2+}$** → relaxation.

- Second route: PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide) → **cAMP (cyclic adenosine monophosphate)** → protein kinase A.
- **PDE5 (phosphodiesterase type 5)** degrades (breaks down) cGMP and ends relaxation (ends / terminates erection); PDE5 inhibitors block it.

### Peripheral pro-erectile transmitters

| Transmitter | Source / mechanism |
|---|---|
| **NO (nitric oxide)** | Principal. **nNOS (neuronal NOS)** initiates; **eNOS (endothelial)** sustains. NO → soluble guanylyl cyclase → **cGMP** → PKG → $K^+$ channels open, $Ca^{2+}$ channels close → relaxation |
| ACh (acetylcholine) | Indirect: endothelial NO release, inhibits adrenergic nerves; atropine does not block erection |
| PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide), CGRP (calcitonin gene-related peptide) | **cAMP** → PKA (protein kinase A); VIP role minimal in man |
| CO, H2S, EDHF (endothelium-derived hyperpolarising factor) | Gaseous/endothelial relaxants via cGMP or $K^+$ channels |

- O2 is a NOS substrate: low flaccid Po2 (~35 mmHg) limits NO; erect Po2 ~90 mmHg.
- **Also know:** carbon monoxide and hydrogen sulfide also raise cGMP; acetylcholine acts indirectly through endothelial NO (atropine does not block erection).

### Anti-erectile transmitters (flaccidity and detumescence)

- **Contraction (flaccidity): noradrenaline (α1)** (principal), endothelin-1, angiotensin II → raised $Ca^{2+}$ → calmodulin → myosin light-chain kinase → contraction.
- **Noradrenaline (α1)**: principal; α2 prejunctional reduces noradrenaline and NO release.
- **Endothelin-1** (ET-A), angiotensin II (AT1), PGF2α, thromboxane A2: Gq → IP3 → $Ca^{2+}$ rise → $Ca^{2+}$–calmodulin → MLC (myosin light-chain) kinase → contraction.
- **Rho-kinase** inhibits myosin (MLC) phosphatase (calcium sensitisation) and maintains (sustaining) tone.
- **Detumescence:** NO release stops, PDE5 degrades cGMP and sympathetic discharge contracts the smooth muscle, so emissary veins reopen.
- Gap junctions (connexin-43) synchronise the smooth muscle.

## Neural control

| Pathway | Origin | Role |
|---|---|---|
| Parasympathetic | **S2–S4** → pelvic plexus → cavernous nerves | Erection |
| Sympathetic | **T11–L2** → hypogastric nerves | Detumescence, emission |
| Somatic | Pudendal nerve (Onuf nucleus) | Penile sensation; ischiocavernosus rigidity |

- **Three types: psychogenic** (brain, chiefly by reducing sympathetic tone), **reflexogenic** (genital touch, sacral arc; preserved in upper cord injury), **nocturnal** (REM [rapid eye movement] sleep).
- **Central:** MPOA (medial preoptic area) and **PVN (paraventricular nucleus)**. Dopamine and oxytocin promote; serotonin, prolactin, GABA (γ-aminobutyric acid) and opioids inhibit.

**Central neurotransmitters (PVN [paraventricular nucleus], MPOA [medial preoptic area])**

| Promote | Inhibit |
|---|---|
| **Dopamine** (D2 in PVN; apomorphine) | **Serotonin** (5-HT1A, 5-HT2; but 5-HT2C pro-erectile) |
| **Oxytocin** | GABA (γ-aminobutyric acid) |
| Central NO | Opioids (high dose), cannabinoids |
| Central noradrenaline; melanocortins (MC4R) | **Prolactin** |

- Dopamine and oxytocin converge on NO in the PVN. Noradrenaline is pro-sexual centrally but anti-erectile peripherally.

**Clinical relevance**

- PDE5 inhibitors, intracavernosal alprostadil (cAMP), papaverine, phenylephrine for priapism (α1), SSRIs (selective serotonin reuptake inhibitors) cause dysfunction, hyperprolactinaemia lowers libido.

**Diagram to draw:** NO–cGMP pathway in the cavernous smooth-muscle cell.

## Evaluation of a diabetic with ED (erectile dysfunction)

**Why diabetics:** ED is commoner, earlier and more severe in DM (diabetes mellitus). Mechanisms: endothelial NO loss, autonomic neuropathy, arteriopathy, smooth-muscle loss and fibrosis (veno-occlusive failure), AGEs (advanced glycation end products) and low testosterone. ED is a **sentinel of cardiovascular disease**.

**1. History**

- Onset, duration (≥3 months), severity, morning erections, libido, ejaculation; partner.
- **IIEF-5 (International Index of Erectile Function, SHIM):** 5–7 severe, 8–11 moderate, 12–16 mild-to-moderate, 17–21 mild, 22–25 no ED.
- Diabetes: type, duration, **glycaemic control**, complications (retinopathy, nephropathy, neuropathy, foot disease), autonomic symptoms (postural giddiness, gastroparesis, bladder dysfunction).
- Cardiovascular: angina, exercise tolerance, smoking, hypertension, dyslipidaemia, nitrates.
- Drugs: antihypertensives, antidepressants. Depression screen.

**2. Examination**

- BMI, waist, blood pressure, peripheral pulses.
- Genitalia: Peyronie plaques, testicular size, gynaecomastia.
- Neurological: genital sensation, **bulbocavernosus reflex**, foot sensation. Penile neuropathy can exist without foot neuropathy.

**3. Laboratory**

- **HbA1c (glycated haemoglobin)** and fasting glucose, **lipid profile**, creatinine/eGFR (estimated glomerular filtration rate), urine albumin.
- **Morning total testosterone**; if low, repeat with free testosterone, LH (luteinising hormone), prolactin. TSH (thyroid-stimulating hormone) if indicated.

**4. Cardiovascular risk stratification (Princeton Consensus)**

- **High risk** (unstable angina, recent infarction, uncontrolled hypertension, some arrhythmias): defer sexual activity and treatment; cardiology referral. **Indeterminate:**

cardiac assessment (e.g. stress test), then restratify. **Low:** start treatment.

**5. Specialised tests (selected cases)**

- **CIS (combined intracavernous injection and stimulation):** alprostadil 10–20 µg; a rigid erection excludes major vascular disease.
- **Penile duplex Doppler:** PSV (peak systolic velocity) **<25 cm/s = arterial insufficiency** (>35 normal); EDV (end-diastolic velocity) **>5 cm/s** and RI (resistive index) **<0.75 = veno-occlusive dysfunction**.
- **NPTR (nocturnal penile tumescence and rigidity), RigiScan:** separates psychogenic from organic ED.
- **Neurophysiological:** bulbocavernosus reflex latency, dorsal nerve conduction velocity, biothesiometry.
- DICC (dynamic infusion cavernosometry and cavernosography) and angiography: rarely, as diffuse diabetic vascular disease is not suited to revascularisation.

```
Diabetic with ED → history + IIEF-5 + examination → HbA1c, lipids,
testosterone, renal function → Princeton cardiac risk → treat risk factors →
PDE5 inhibitor trial → failure → duplex/CIS → ICI or prosthesis
```

> **Key points to remember:**
> - nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
> - Noradrenaline α1 is the main anti-erectile transmitter.
> - Dopamine/oxytocin promote; serotonin/prolactin inhibit.
> - Rigid phase: ischiocavernosus, suprasystolic pressure.
> - Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
> - Diabetic ED: check HbA1c, testosterone and cardiac risk.

## LAQ 6 Discuss the role of urologist in reproductive endocrinology and male infertility practice.

P2

**Type:** LAQ (25 marks)
**Source:** 2022-Summer Paper 2, Q1

### Introduction

- **Infertility:** no conception after 12 months of regular unprotected intercourse; about **15% of couples** (WHO, World Health Organization). The male contributes about half, yet many men are never examined.
- The urologist (andrologist) has five roles: **diagnose** the male factor, **treat endocrine causes**, **correct surgical causes**, **retrieve sperm for ART (assisted reproductive technology)**, and protect **general health and future fertility**.
- Commonest diagnoses: idiopathic, **varicocele** (commonest correctable), obstruction.

### 1. Diagnostic role

- **History:** duration, coital timing (semen is best after 1–2 days' abstinence), partner's age (fecundity falls after 35), cryptorchidism, mesh herniorrhaphy, RPLND (retroperitoneal lymph node dissection), diabetes, spinal injury, smoking, cannabis, opioids, **exogenous testosterone or anabolic steroids**.
- **Examination:** androgenisation; **testis size** (reflects spermatogenic mass); epididymis; **both vasa present?**; varicocele (standing, Valsalva).
- **Semen analysis**, at least two. WHO 6th edition (2021) 5th centiles: volume 1.4 mL, concentration 16 million/mL, total count 39 million, progressive motility 30%, normal forms 4%. These describe fertile men, not fertility cut-offs.
- **Genetics:** karyotype and **Y-chromosome microdeletion** (AZF, azoospermia factor) in non-obstructive azoospermia or severe oligospermia; **CFTR (cystic fibrosis transmembrane conductance regulator)** testing of both partners in CBAVD (congenital bilateral absence of the vas deferens).
- **Imaging:** scrotal US (ultrasound); **TRUS (transrectal US)** if volume <1 mL (ejaculatory duct obstruction, EDO), after post-ejaculate urine excludes retrograde ejaculation; renal US in vasal agenesis.

### 2. Endocrine evaluation (the urologist's reproductive endocrinology)

- Spermatogenesis needs a **high intratesticular testosterone (T)**. Morning total T + SHBG (sex hormone-binding globulin) + albumin gives bioavailable T; low is below about **280–300 ng/dL**; repeat borderline values (T is pulsatile and circadian).

| Test | What it tells |
|---|---|
| LH (luteinising hormone) | Localises low T: high → testicular (Leydig) failure; low/normal → pituitary or hypothalamus |
| FSH (follicle-stimulating hormone) / inhibin B | Germ-cell mass. Azoospermia with **FSH >7.6 IU/L and testis long axis ≤4.6 cm → spermatogenic failure**; FSH ≤7.6 and >4.6 cm → obstruction |
| Oestradiol (E2) | **T (ng/dL) : E2 (pg/mL) <10** → aromatase excess (obesity) |

| Test | What it tells |
|---|---|
| Prolactin | Only with low T, low libido/ED (erectile dysfunction), headache or field defect |
| TSH (thyroid-stimulating hormone), cortisol, 17-hydroxyprogesterone | Only if signs; CAH (congenital adrenal hyperplasia) |

## 3. Medical (endocrine) treatment

| Condition | Mechanism | Treatment |
|---|---|---|
| **Hypogonadotropic hypogonadism** (Kallmann, pituitary) | Testis normal, signal missing: most treatable cause | **hCG (human chorionic gonadotropin) 1500–5000 IU 2–3 times weekly**, add hMG (human menopausal gonadotropin) 75 IU or recombinant FSH; pulsatile GnRH (gonadotropin-releasing hormone) pump for hypothalamic failure. Sperm may take 1–2 years |
| **Anabolic steroids / exogenous T** | Negative feedback suppresses LH → intratesticular T falls → azoospermia | Stop; hCG ± clomiphene to speed recovery. **Never give T to a man wanting fertility** (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine) |
| **Hyperprolactinaemia** | Prolactin inhibits GnRH | Repeat level; pituitary MRI (magnetic resonance imaging) if high; **cabergoline** (fewer side effects than bromocriptine) |
| Partial hypogonadism, obesity | Aromatisation of T to E2 suppresses LH | Weight loss; **clomiphene 25 mg daily** (off-label), **anastrozole 1 mg daily** or letrozole if T:E2 <10 |
| **CAH** | Adrenal androgens suppress gonadotropins; testicular adrenal rest tumours | Glucocorticoids |
| Primary testicular failure (Klinefelter 47,XXY) | Germ cells absent or few | No drug restores sperm; exogenous T worsens it; **micro-TESE** (microdissection testicular sperm extraction) |

## 4. Surgical role

| Problem | Operation |
|---|---|
| Clinical varicocele + abnormal semen | **Microsurgical subinguinal varicocelectomy** or embolisation: improves semen, DNA fragmentation and pregnancy. Not for US-only varicocele |
| Vasal or epididymal obstruction | **Vasovasostomy / vasoepididymostomy** (microsurgical intussusception) |
| EDO | **TURED** (transurethral resection of ejaculatory ducts) |
| Non-obstructive azoospermia | **Micro-TESE**, best technique; no operation for **AZFa/AZFb** deletions (no sperm found) |
| Unreconstructable obstruction | PESA/MESA (percutaneous/microsurgical epididymal sperm aspiration), TESA (testicular sperm aspiration) |
| Retrograde ejaculation; anejaculation | Sympathomimetics, urine sperm retrieval; **penile vibratory stimulation**, electroejaculation |

### 5. Partner of the ART team, counselling and men's health

- Choose the least invasive route: natural conception, IUI (intrauterine insemination), IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection). **Cryopreserve sperm** at retrieval.
- **Fertility preservation:** sperm banking before chemotherapy, radiotherapy or RPLND; **onco-TESE** if azoospermic.
- **Genetic counselling:** AZFc deletion passes to sons; CFTR risk to offspring; paternal age ≥40 years raises offspring risk.
- **Infertility is a marker of general health:** higher risk of **testis cancer** (scan every azoospermic man), later comorbidity and mortality.

**Recent advances:** sperm DNA fragmentation testing; microfluidic sperm selection; genomic panels; prepubertal testicular tissue cryopreservation (experimental).

**Also know:** antisperm antibodies; globozoospermia and ciliary dyskinesia (ICSI); androgen receptor defects (high T and LH).

**Diagram to draw:** hypothalamic–pituitary–testis axis with feedback loops (T and E2 on LH; inhibin B on FSH).

> **Key points to remember:**
> - FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
> - Exogenous testosterone is a contraceptive: never give it for infertility.
> - Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
> - Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

## Short answer questions (SAQ)

## SAQ 1 Male reproductive immunology and immunobead test

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "Male reproductive immunology" · P3 SAQ "Immunobead test for semen"

### Introduction

- Haploid germ cells appear at puberty, after self-tolerance is established, so their antigens are "foreign". The testis is therefore **immune-privileged**.

### Immune privilege

#### Anatomical barrier: blood–testis barrier (BTB)

- **Tight junctions between Sertoli cells**, formed at puberty, divide a **basal compartment** (spermatogonia, plasma-like fluid) from an **adluminal compartment** (later spermatocytes, spermatids). If the barrier is breached, sperm antigens meet the immune system.

#### Functional barrier

- A small antigen leak at the **rete testis, efferent ductules and epididymis** induces suppressor T cells and tolerance; a large load provokes immunity.
- Sertoli and local mediators: galectin-1, TGF-β (transforming growth factor-β), interferon-γ, soluble Fc receptor, **androgens**.

- Insult before puberty does not induce antibodies; the barrier also limits chemotherapy (isolated testicular relapse).

### Antisperm antibodies (ASA)

- **Causes:** obstruction (**vasectomy** commonest); torsion, trauma, biopsy; infection (orchitis, prostatitis, STI [sexually transmitted infection]); heat (varicocele, cryptorchidism); genetic (human leukocyte antigen HLA-B28).
- **Effects:** agglutination, poor motility, failure to cross (poor penetration of) cervical mucus, failure to bind the zona and impaired fertilisation; head-bound antibody matters most.
- **Indications to test:** sperm agglutination or isolated asthenozoospermia; unexplained infertility; after vasectomy reversal.

### Tests for ASA: immunobead test (IBT) and MAR

- The IBT detects ASA bound to sperm (direct) or free in body fluids (indirect).
- **Principle:** polyacrylamide beads coated with rabbit anti-human **IgG, IgA or IgM**.

| Test | Method | Use |
|---|---|---|
| **Direct IBT** | Patient's motile sperm washed free of seminal plasma; mixed with beads coated with anti-human IgG/IgA; motile sperm with attached beads counted | Sperm-bound antibody: % bound, class and site (head/tail) |
| **Indirect IBT** | Antibody-free donor sperm incubated with heat-inactivated patient serum, seminal plasma or female serum/cervical mucus; then direct IBT done | Free antibody; patient sperm too few or immotile; female partner |
| **MAR (mixed antiglobulin reaction)** | Fresh unwashed semen + immunoglobulin (Ig) G (or IgA)-coated latex beads + antiglobulin (anti-IgG); beads stick to antibody-coated motile sperm; mixed sperm–bead clumps counted | Quick, cheap screen; IgG/IgA only |

- Report: **% of motile sperm with beads**, immunoglobulin class, and **site** (head, midpiece, tail). Tail-tip binding is ignored.
- **WHO (World Health Organization) 2010 (5th edition) threshold: ≥50% of motile sperm bound is clinically significant.**
- **Head binding** and **IgA** matter most (IgA is produced locally and blocks mucus penetration; IgM is too large to enter semen).
- Both direct tests need motile sperm.
- **Also know:** ELISA (enzyme-linked immunosorbent assay) and flow cytometry; serum tests correlate poorly with sperm-bound antibody.

### Management of ASA

| Treatment | Comment |
|---|---|
| Condoms | Lack good evidence |
| Corticosteroids | Two controlled trials conflict; risk of avascular necrosis; not recommended |
| Sperm washing + intrauterine insemination (IUI) | Bypasses cervical mucus; for mild binding; modest benefit, lacks good evidence (washing removes unbound, not bound, antibody) |

| Treatment | Comment |
|---|---|
| **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)** | Preferred, usual treatment; fertilisation largely unaffected, bypassing antibody effects |

### Infection and inflammation

- **Leukocytospermia:** >1 × $10^6$ peroxidase-positive leukocytes/mL. Reactive oxygen species and cytokines damage sperm.
- **Orchitis:** mumps (post-pubertal, atrophy); bacterial (*Chlamydia*/gonococcus if young, *E. coli* if older); tuberculous; **autoimmune**, primary (antibodies with azoospermia) or secondary to vasculitis.
- **Also know:** sperm DNA fragmentation testing; prostatitis harms sperm via oxidative stress and ASA; seminal plasma suppresses female tract immunity.

### Diagrams to draw

- Seminiferous tubule with Sertoli tight junctions and compartments.
- Motile sperm with beads attached at head and tail.

> **Key points to remember:**
> - BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
> - ASA: vasectomy commonest; ≥50% bound significant; head-bound IgA matters most.
> - Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
> - Treatment of choice: IVF/ICSI.
> - Leukocytospermia >1 million/mL.

## SAQ 2 Premature ejaculation

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "Premature Ejaculation" · P3 SAQ "Premature ejaculation."

### Definition

- **ISSM (International Society for Sexual Medicine) unified definition (2014):** all three needed: 1. Ejaculation always or nearly always within **about 1 minute** of vaginal penetration (lifelong), or a clinically significant, bothersome fall in latency, often to **≤3 minutes** (acquired). 2. Inability to delay ejaculation on all or nearly all penetrations. 3. Negative personal consequences: distress, frustration, avoidance of intimacy.
- **AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2020:** lifelong PE (premature ejaculation) within about **2 minutes**.
- **Types:** lifelong, acquired, variable and subjective (normal latency).

### Physiology and aetiology

- Ejaculation = **emission** (sympathetic T10–L2) then **expulsion** (somatic S2–S4, pudendal). **Serotonin brakes and dopamine accelerates**.
- **Lifelong PE:** neurobiological and genetic; 5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity.

- **Acquired:** performance anxiety, relationship problems, **ED (erectile dysfunction)**, chronic prostatitis, hyperthyroidism, drug withdrawal.

### Evaluation

- Diagnosis is by **history**: latency, control, bother; lifelong or acquired.
- **IELT (intravaginal ejaculatory latency time):** self-estimate in clinic; stopwatch in trials. Population median 5.4 minutes.
- **PEDT (Premature Ejaculation Diagnostic Tool):** 5 items; score ≥11 suggests PE, 9–10 probable PE. Also PEP (Premature Ejaculation Profile) and IPE (Index of Premature Ejaculation).
- IIEF-5 (International Index of Erectile Function) for ED; examination; TSH (thyroid-stimulating hormone) only if suggested.

**Office algorithm (diagram to draw):**

1. Confirm PE.
2. Secondary to ED? → treat the cause.
3. Otherwise → classify lifelong or acquired.

### Management

- **Psychosexual education** for all; behavioural therapy (stop–start, squeeze).
- Acquired PE: treat the cause first (ED, prostatitis, thyroid).

| Drug | Dose and timing |
|---|---|
| **Dapoxetine** (short-acting SSRI [selective serotonin reuptake inhibitor]; only licensed oral drug) | 30–60 mg, 1–3 h before intercourse |
| Paroxetine (strongest delay) 10–40 mg, sertraline, fluoxetine, citalopram | Daily, off-label; full effect 2–3 weeks; do not stop abruptly |
| Clomipramine | 12.5–50 mg daily or 3–4 h before |
| **Topical lidocaine–prilocaine** spray (licensed in Europe 2014) | 5–30 min before (P1: at least 5 min before); condom avoids partner numbness |
| PDE5Is (phosphodiesterase type 5 inhibitors) | With ED, alone or with SSRI |
| Tramadol | Low evidence; addiction risk |

- SSRIs raise synaptic serotonin, which delays ejaculation. Avoid in bipolar disorder.
- **Also know:** high dropout from drug therapy.

> **Key points to remember:**
> - ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
> - Lifelong is neurobiological; acquired is usually secondary.
> - Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

## SAQ 3 Oligospermia and oligoasthenozoospermia

P2 · P3 **merged from 2**

*Asked as:* P3 SAQ "Oligospermia and oligoasthenozoospermia" (asked as "Oligoanaesthospermia" and "Oligospermia") · P2 SAQ "Oligospermia."

*Assumed meaning (oligoanaesthospermia): oligoasthenozoospermia (low sperm count with poor motility).*

### Definition (WHO [World Health Organization] 2021, 6th edition, 5th centiles)

- **Oligozoospermia:** sperm concentration below the lower reference limit: **WHO 2021 (6th edition) 16 million/mL** (5th centile; total count 39 million per ejaculate). Earlier thresholds were 20 million/mL (1999, the legacy cut-off) and 15 million/mL (2010).
- **Severity:** mild to moderate 5–16 million/mL; **severe oligozoospermia <5 million/mL**; cryptozoospermia: sperm only in a centrifuged pellet.

| Parameter (WHO 2021 5th centiles) | Lower reference value |
|---|---|
| Concentration | 16 million/mL (total 39 million) |
| Volume | 1.4 mL |
| Progressive motility | 30% |
| Total motility | 42% |
| Normal forms (strict) | 4% |
| Vitality | 54% |

- These centiles come from fertile men; they are a reference, not pass/fail fertility cut-offs. Define oligospermia in context: read a single value with testis size, FSH (follicle-stimulating hormone) and repeated values.

### Causes

- **Varicocele** (commonest correctable; heat and oxidative stress) and **idiopathic** (largest group).
- **Pre-testicular (hormonal):** hypogonadotropic hypogonadism, **exogenous testosterone or anabolic steroids**, obesity (raised oestradiol), hyperprolactinaemia, opioids.
- **Testicular: varicocele**, cryptorchidism, mumps orchitis, torsion, chemotherapy or radiotherapy, gonadotoxins; genetic (Klinefelter, Klinefelter mosaic, **AZFc [azoospermia factor c] deletion**).
- **Post-testicular (motility):** infection or **leucocytospermia** (over 1 million/mL), antisperm antibodies, partial ejaculatory duct obstruction (EDO; low volume), PCD (primary ciliary dyskinesia).
- **Lifestyle:** smoking, alcohol, cannabis, heat, drugs, gonadotoxins.

### Evaluation

- **At least two semen analyses, 2–3 weeks apart**, same abstinence (WHO 2–7 days of abstinence; parameters peak at 1–2 days).
- History and examination: testis size/volume, varicocele, vasa, secondary sex characters.
- Morning **testosterone, LH (luteinising hormone), FSH**; oestradiol and prolactin if testosterone is low.
- **Under 5 million/mL** (especially with raised FSH or small testes): **karyotype and Y-microdeletion** testing (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine] 2020 guideline).
- Selected: vitality test (eosin or hypo-osmotic swelling), antisperm antibodies, semen culture, scrotal Doppler, TRUS (transrectal ultrasonography) if volume is low (<1.0 mL), DNA fragmentation.

### Management

- **General:** stop smoking, weight loss, avoid heat; **stop exogenous testosterone** and gonadotoxins; treat infection.
- **Varicocelectomy** (microsurgical subinguinal) for a **palpable** varicocele with abnormal semen; it improves semen parameters and pregnancy rates. Never treat a subclinical one.
- **Medical:**
- Hypogonadotropic: hCG (human chorionic gonadotropin) ± FSH.
- Idiopathic with low testosterone: off-label clomiphene 25 mg daily, or anastrozole 1 mg daily if the testosterone:oestradiol ratio is low (<10).
- Antioxidants: weak evidence.
- **ART (assisted reproductive technology):** mild: IUI (intrauterine insemination); severe or failed IUI: **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)**. Bank sperm in AZFc carriers because counts may fall.
- Spermatogenesis takes about 64 days, so judge response (reassess) after 3–6 months.

**Diagram to draw:** hypothalamic–pituitary–testis axis with sites of treatment.

> **Key points to remember:**
> - WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
> - Karyotype and Y-microdeletion if under 5 million/mL.
> - Repair palpable varicoceles only; never give testosterone.

## SAQ 4 Management of Peyronie's disease

P1

**Type:** SAQ (10 marks)
**Source:** 2017-Winter Paper 1, Q3a

### Definition

- **PD (Peyronie's disease)** (AUA (American Urological Association) 2015): acquired fibrosis of the tunica albuginea, with pain, deformity, ED (erectile dysfunction) and/or distress. Prevalence 3–20%; up to half worsen untreated.

### Phases

- **Acute (active, first 12–18 months):** pain, worsening deformity; aim to stop progression. **Stable:** pain gone, deformity unchanged for ≥3–6 months; aim to straighten.

### Evaluation

- History: pain, deformity, Dupuytren disease, ability to penetrate. **Key question:** "If your penis were straight, would your current rigidity allow intercourse?" (predicts postoperative function).
- Plaque, SPL (stretched penile length); **office intracavernosal injection ± duplex Doppler ultrasound** before any invasive treatment (curvature by goniometer, calcification, vascular status).
- **Kelami grading:** ≤30° mild, 31–60° moderate, >60° severe; note hourglass or hinge. PDQ (Peyronie's Disease Questionnaire).

### Non-surgical treatment

| Option | Evidence/position |
|---|---|
| NSAIDs (non-steroidal anti-inflammatory drugs) | For pain (only oral agent AUA supports) |
| Oral vitamin E, tamoxifen, potaba, pentoxifylline | No reliable curvature benefit; not recommended |
| **CCH (collagenase *Clostridium histolyticum*)** | Stable disease, curve 30–90°, intact erections. **IMPRESS I/II:** 34% improvement vs 18% placebo. Up to 4 cycles (2 injections of 0.58 mg) with modelling. Risk: corporal fracture |
| Verapamil, interferon α-2b (intralesional) | May offer; stabilise and modestly improve |
| **Traction therapy** | Gains length, reduces curvature; useful in acute phase |
| ESWT (extracorporeal shock wave therapy) | Pain only; no effect on curvature |

### Surgery

- **Indications:** stable disease (≥12 months, stable ≥6 months), deformity preventing intercourse, failed conservative therapy.

```
Erection adequate (± PDE5I (phosphodiesterase-5 inhibitor))? ├─ No → penile prosthesis (inflatable) + manual modelling; │ residual >30° → tunical incision; graft if defect >2 cm └─ Yes → curve <60–70°, no hourglass/hinge → plication (Nesbit, 16-dot, tunica albuginea plication) curve >60–70°, hourglass, hinge, short penis → plaque incision/partial excision + graft (pericardium, small intestinal submucosa)
```

- **Plication** shortens the convex side: simple, low ED risk, but loses length.
- **Grafting** restores length and girth but carries higher ED risk (5–54%).

**Diagram to draw:** Nesbit wedge excised opposite the plaque.

> **Key points to remember:**
> - Ask the "straight penis" question; duplex before surgery.
> - CCH: IMPRESS, stable 30–90° curve, intact erections.
> - Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

## SAQ 5 Haematospermia

P1

**Asked as:**

- Hematospermia — *Winter 2025 Paper I Q3c (SAQ)*
- Hematospermia — *Practice Set 9 Paper I Q3d (SAQ)* (also answered in 2025-Winter Paper 1, Q3c)

**Type:** SAQ (10 marks)

### Definition

- Blood in the ejaculate. It alarms patients but is usually **benign and self-limiting**, most often from non-specific prostatic or seminal vesicle inflammation.

- **Primary (idiopathic):** no cause found, often young men. **Secondary:** a cause is identified.
- **Pseudo-haematospermia:** blood from the partner (partner's blood; condom test) or haematuria.

## Causes

| Group | Examples |
|---|---|
| **Iatrogenic (commonest identifiable)** | Prostate biopsy (about 10–50%, lasts about a month), TURP (transurethral resection of the prostate), radiotherapy, BPH (benign prostatic hyperplasia) procedures |
| Inflammation and infection | Prostatitis, vesiculitis, urethritis, STI (sexually transmitted infection), **genitourinary tuberculosis**, schistosomiasis |
| Obstruction and cysts | Ejaculatory duct obstruction or calculi; seminal vesicle cyst (Zinner syndrome) or utricle cyst |
| Tumours | **Prostate cancer**; rarely urethral, bladder or seminal vesicle |
| Vascular | Prostatic varices, BPH vessels, AV (arteriovenous) malformation |
| Systemic | Hypertension, bleeding disorders, anticoagulants, liver disease; also abstinence, trauma |

## Evaluation

- **History:** episodes, recent biopsy, LUTS (lower urinary tract symptoms), haematuria, TB (tuberculosis) contact, anticoagulants.
- **Examination:** blood pressure, genitalia, **DRE (digital rectal examination)**.
- **Basic tests:** urinalysis and culture, semen culture, STI screen, **PSA (prostate-specific antigen)** in men over 40, coagulation if indicated.

**Who needs full work-up (full work-up if):** age **≥40**, persistent beyond several weeks or recurrent, haematuria, abnormal DRE or PSA, risk factors for TB or cancer, or a distressed patient.

- **TRUS (transrectal ultrasound):** cysts, calculi, dilated seminal vesicles (>1.5 cm) or ejaculatory ducts (>2.3 mm) suggest obstruction.
- **MRI (magnetic resonance imaging) pelvis:** best for seminal vesicle haemorrhage, cysts and prostate cancer.
- **Cystoscopy and urine cytology**; early-morning urine for acid-fast bacilli.

## Management

- **Young, single episode, normal tests: reassure**.
- **Treat the cause:** antibiotics or antitubercular drugs; control BP (blood pressure); correct coagulopathy.
- **Persistent prostatic bleeding:** 5-ARI (5-alpha-reductase inhibitor), e.g. finasteride.
- **Refractory or obstructive:** TURED (transurethral resection of ejaculatory ducts), cyst unroofing or aspiration, **seminal vesiculoscopy** (4–6 Fr small scope through the ejaculatory duct or utricle, irrigation, laser for stones).

**Diagram to draw:** algorithm: <40 and single episode → reassure; ≥40 or persistent → PSA, TRUS/MRI, cystoscopy.

**Key points to remember:**

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
- Exclude prostate cancer and genitourinary tuberculosis.
- TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

## SAQ 6 Avanafil

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 21 SAQ Sheet, Q3e

### Introduction

- Second-generation oral **PDE5I (phosphodiesterase type 5 inhibitor)**; **FDA (US Food and Drug Administration) approved 2012**; Europe 2013. Structure differs most from the other PDE5Is.

### Mechanism

- Sexual stimulation releases NO (nitric oxide), which raises cGMP (cyclic guanosine monophosphate) and relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; avanafil blocks this.
- **Enhances, does not initiate, erection**: stimulation is needed.
- **More selective for PDE5** over PDE6 (retina) and PDE1 than sildenafil, so **no visual disturbance** in the comparison table.

### Pharmacokinetics (Campbell Table 72.7)

| Feature | Avanafil | Sildenafil | Tadalafil |
|---|---|---|---|
| Time to peak | **0.3–0.5 h (fastest)** | 0.8 h | 2 h |
| Half-life | 3–5 h | 3–5 h | 17.5 h |
| Fatty food | Reduces absorption | Reduces | No effect |
| Doses | **50, 100, 200 mg** | 25–100 mg | 5–20 mg |
| Blue vision | No | Yes | Rare |

- Taken about 15–30 minutes before sex; start 100 mg; maximum once daily. Metabolised by CYP3A4 (cytochrome P450 3A4).

### Efficacy

- Effective in general ED (erectile dysfunction), **diabetes** and **after nerve-sparing radical prostatectomy** (REVIVE trial programme): improves IIEF (International Index of Erectile Function) erectile scores and successful-intercourse rates, some within 15 minutes. Efficacy equivalent across PDE5Is; no head-to-head superiority.

### Adverse effects

- Headache, flushing, nasal congestion, dyspepsia; back pain rare.

### Contraindications and interactions

- **Nitrates in any form**: absolute; no nitrate within 12 hours of a dose.
- Avoid strong CYP3A4 inhibitors (ketoconazole, ritonavir); maximum 50 mg with moderate inhibitors.
- **α-blockers:** start 50 mg once stable (additive hypotension).
- Caution: aortic stenosis, recent myocardial infarction or stroke, hypotension, retinitis pigmentosa. Not recommended in severe renal impairment, dialysis or severe hepatic impairment.
- **Also know:** proposed prophylaxis for recurrent ischaemic priapism; can cause priapism rarely.

### Advantages

- **Fastest onset**, fewer visual and muscle effects; suits on-demand spontaneous use.

**Diagram to draw:** NO–cGMP pathway in cavernosal smooth muscle with the PDE5 block.

> **Key points to remember:**
> - Peak 0.3–0.5 h; doses 50, 100, 200 mg.
> - Highly PDE5-selective: no blue vision.
> - Nitrates absolutely contraindicated.

## SAQ 7 CRE & BEP

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 2, Q3f

*"CRE" is read as carbapenem-resistant Enterobacterales, and BEP as bleomycin, etoposide and cisplatin chemotherapy for germ cell tumours; the two are answered separately.*

### Part A: CRE (carbapenem-resistant Enterobacterales)

- **Definition:** Enterobacterales (Klebsiella pneumoniae, Escherichia coli, Enterobacter) resistant to at least one carbapenem, or producing a carbapenemase.

| Mechanism | Example |
|---|---|
| Class A carbapenemase | KPC (Klebsiella pneumoniae carbapenemase) |
| Class B metallo-β-lactamase | **NDM (New Delhi metallo-β-lactamase)**: common in India |
| Class D | OXA-48 |
| Non-enzymatic | ESBL (extended-spectrum β-lactamase) or AmpC plus porin loss |

- **Urological risk:** catheters, stents, prolonged stay, prior carbapenem or fluoroquinolone use, transrectal biopsy, ICU (intensive care unit) care.
- **Treatment** (by culture and carbapenemase type; ID (infectious disease) input):
- KPC, OXA-48: ceftazidime–avibactam; meropenem–vaborbactam or imipenem–relebactam for KPC.
- **NDM:** ceftazidime–avibactam plus aztreonam, or cefiderocol.
- Cystitis: single-dose aminoglycoside, fosfomycin; polymyxins as last resort. Avoid tigecycline for UTI (urinary tract infection) (low urine levels).

- **Prevention:** antibiotic stewardship, contact precautions, rectal screening of high-risk patients, early catheter removal, transperineal rather than transrectal biopsy.

### Part B: BEP (bleomycin, etoposide, cisplatin)

- Standard first-line chemotherapy for metastatic **GCT (germ cell tumour)**; replaced PVB (less neuromuscular toxicity).
- **Usual 21-day cycle:** cisplatin 20 mg/m² days 1–5, etoposide 100 mg/m² days 1–5, bleomycin 30 units days 1, 8, 15.

| IGCCCG (International Germ Cell Cancer Collaborative Group) risk | Regimen |
|---|---|
| Good | BEP ×3 or EP ×4 |
| Intermediate/poor | BEP ×4 (VIP ×4 if lungs poor) |
| Stage I NSGCT (non-seminomatous GCT), high risk | BEP ×1 adjuvant |

- **Toxicity:** bleomycin lung fibrosis (keep $FiO_2$ low at later RPLND (retroperitoneal lymph node dissection)) and Raynaud phenomenon; cisplatin nephrotoxicity, ototoxicity, neuropathy; myelosuppression; infertility (bank sperm); late cardiovascular disease and second cancers.

**Diagram to draw:** BEP 21-day cycle timeline.

> **Key points to remember:**
> - CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
> - BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
> - Bleomycin lung: low $FiO_2$ and restricted fluids at surgery.

## SAQ 8 Genetic causes of male infertility. Enumerate and classify.

P3

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 3, Q3d

### Introduction

- Genetic defects underlie many cases of **azoospermia** and **severe oligozoospermia**. They decide **sperm retrieval odds** and **transmission to offspring** through ICSI (intracytoplasmic sperm injection).

### Classification

| Group | Condition | Key features |
|---|---|---|
| **1. Chromosomal, numerical** | **Klinefelter syndrome (47,XXY)** | **Commonest genetic cause**; 1 in 500–1000; small testes, high FSH (follicle-stimulating hormone), low testosterone, usually azoospermia. Non-mosaic in 80–90%. Testicular sperm in about 50% (micro-TESE, microdissection testicular sperm extraction) |
| | 47,XYY; **46,XX male** (SRY gene translocated to X) | XX male: azoospermia, no sperm retrievable |

| Group | Condition | Key features |
|---|---|---|
| **2. Chromosomal, structural** | **Robertsonian and reciprocal translocations**, inversions, ring Y | Oligozoospermia; recurrent miscarriage |
| **3. Y chromosome microdeletions (Yq, AZF (azoospermia factor))** | **AZFa** (USP9Y, DDX3Y), **AZFb**, **AZFc** (DAZ genes) | Complete AZFa or AZFb deletion: **no sperm at retrieval**. AZFc: sperm often in ejaculate or testis; sons inherit the deletion |
| **4. Single-gene: excurrent ducts** | **CFTR (cystic fibrosis transmembrane conductance regulator) mutations**: CBAVD (congenital bilateral absence of vas deferens); idiopathic ejaculatory duct obstruction | Low-volume azoospermia, normal spermatogenesis; **test both partners** (ΔF508) |
| **5. Single-gene: hypothalamic–pituitary** | **Kallmann syndrome** (KAL1, anosmia); GnRH (gonadotropin-releasing hormone) receptor mutations | Low LH (luteinising hormone) and FSH; treat with hCG ± FSH |
| **6. Androgen action** | **Androgen receptor** mutation (long CAG repeat); 5α-reductase type 2 deficiency | High testosterone and LH, poor sperm |
| **7. Sperm structure and motility** | **Globozoospermia** (SPATA16, DPY19L2: no acrosome); macrocephalic sperm (AURKC); **PCD (primary ciliary dyskinesia)**/Kartagener (DNAH5, DNAI1: immotile sperm) | ICSI; macrocephalic sperm: ICSI not advised (aneuploidy) |

### Genetic testing indications

- **Karyotype and Y microdeletion:** non-obstructive azoospermia and severe oligozoospermia (**<5 million/mL**, ASRM/AUA (American Society for Reproductive Medicine/American Urological Association) 2021; EAU (European Association of Urology) uses <10 million/mL for karyotype).
- **CFTR:** CBAVD, idiopathic obstructive azoospermia; screen the partner; counsel before ICSI.
- **Also know:** Noonan syndrome, myotonic dystrophy.

**Diagram to draw:** Y chromosome long arm showing AZFa, b, c.

> **Key points to remember:**
> - Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
> - AZFa or AZFb complete deletion: do not attempt retrieval.
> - CBAVD = genital form of cystic fibrosis; test both partners.

## SAQ 9 Ejaculatory duct obstruction

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 3, Q3b

*Assumed meaning: ejaculatory duct obstruction (EDO).*

## Anatomy

- Intraprostatic continuations of the seminal vesicles, joined by the vasal ampullae; they lack muscle and open at the **verumontanum**. Seminal vesicles give about 60% of the ejaculate and its fructose.

## Aetiology

- **Congenital (commonest):** midline (Müllerian/utricular) cyst, Wolffian duct cyst, atresia. Idiopathic EDO carries CFTR (cystic fibrosis transmembrane conductance regulator) mutations in about half.
- **Acquired:** infection, instrumentation, duct stones, compression.
- **Zinner syndrome:** seminal vesicle cyst, ipsilateral renal agenesis, EDO.
- Complete bilateral EDO causes under 5% of azoospermia. Partial EDO gives oligoasthenospermia.

## Clinical features

- Infertility, painful ejaculation, haematospermia, perineal pain.

## Diagnosis

### Semen (complete bilateral EDO)

- Volume **<1.0 mL**, **acidic pH**, low or absent **fructose**, azoospermia.
- Palpable vas, normal FSH (follicle-stimulating hormone) and testes.
- Exclude retrograde ejaculation and CBAVD (congenital bilateral absence of the vas deferens).

### Investigations

| Test | Method / finding |
|---|---|
| **TRUS (transrectal ultrasound)** | Seminal vesicle AP diameter **>1.5 cm**, duct >2.3 mm, midline cyst. Low specificity |
| **Seminal vesicle aspiration** | **≥3 sperm per high-power field** confirms obstruction |
| Vesiculography | Contrast into vesicles; invasive |
| Chromotubation | Indigo carmine into vesicles; dye at the veru on cystoscopy |

## Management

### TURED (transurethral resection of the ejaculatory ducts)

- For azoospermic or severely oligoasthenospermic men, after the couple declines IVF (in vitro fertilisation).
- **Steps:** TRUS-guided aspiration (cryopreserve sperm) → instil indigo carmine with contrast → 24-Fr resectoscope, finger in rectum → resect the veru until dye effluxes. Spare bladder neck, sphincter and rectum. Catheter overnight.
- **Results:** volume rises in most; sperm appear in about half. Cold-knife incision alone re-obstructs.

### Complications

- **Urine reflux** into the ducts, **epididymitis**, retrograde ejaculation, rectal injury, re-obstruction.

### Alternatives

- **Seminal vesiculoscopy** (4–6 Fr via utricle): avoids reflux; restenosis unknown. Sperm retrieval with ICSI (intracytoplasmic sperm injection).

**Diagram to draw:** sagittal prostate with midline cyst compressing the ducts at the veru.

> **Key points to remember:**
> - Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
> - TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
> - TURED: sperm return in about half; reflux and epididymitis are its price.

## SAQ 10 Tadalafil

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 3, Q3b

### Pharmacology

- Selective **PDE5 (phosphodiesterase type 5) inhibitor**; FDA (Food and Drug Administration) approved 2003.
- **Mechanism:** sexual stimulation releases NO (nitric oxide), raising cGMP (cyclic guanosine monophosphate), which relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; tadalafil blocks this. It **augments, not initiates,** erection.
- Structurally distinct from sildenafil; cross-reacts with **PDE11**, little with PDE6 (retina).

| Property | Tadalafil (vs sildenafil) |
|---|---|
| Time to peak | 2 h (0.8 h) |
| Half-life | **17.5 h** (3–5 h) |
| Fatty food | No effect (reduces absorption) |
| Window | Widest (up to 36 h) |

### Uses and doses

1. **ED (erectile dysfunction):** 10–20 mg on demand, or 2.5–5 mg once daily. Intercourse success about 70%; 40–50% in diabetes and after prostatectomy.
2. **BPH (benign prostatic hyperplasia) with LUTS (lower urinary tract symptoms), with or without ED: 5 mg daily**, the only PDE5 inhibitor licensed. IPSS (International Prostate Symptom Score) falls about 6 points (placebo 3.6) **without Qmax (peak flow rate) change**; mechanism is NO–cGMP relaxation, perfusion and reduced afferent activity, not de-obstruction. Equal to tamsulosin for IPSS. EAU (European Association of Urology) 2022 and AUA (American Urological Association) 2021: option for moderate–severe LUTS.
3. **Also know:** fixed finasteride–tadalafil (FDA 2021); penile rehabilitation (benefit unproven); prevention of stuttering priapism (PDE5I paradox); pulmonary arterial hypertension.

### Adverse effects

- Headache, dyspepsia, **back pain and myalgia** (characteristic), flushing, nasal congestion. Visual effects rare.
- NAION (non-arteritic anterior ischaemic optic neuropathy): no proven link; stop if sudden visual loss.

### Contraindications and interactions

- **Nitrates in any form: absolute.** After tadalafil, no nitrates for **48 h** (24 h for sildenafil).
- Caution: α-blockers (start lowest dose of the added drug), aortic stenosis, hypotension. CYP3A4 inhibitors (ketoconazole, ritonavir) raise levels.
- Not studied: MI (myocardial infarction) or stroke within 6 months, unstable angina, BP <90/50 or >170/100 mmHg, retinitis pigmentosa.

**Diagram to draw:** NO–cGMP–PDE5 pathway in cavernosal smooth muscle.

> **Key points to remember:**
> - Half-life 17.5 h; food-independent; back pain.
> - 5 mg daily for LUTS: IPSS improves, Qmax does not.
> - No nitrates for 48 h after a dose.

## SAQ 11 Retrograde ejaculation

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 3, Q3f

### Definition

- RE (retrograde ejaculation): semen passes backwards into the bladder at orgasm because the **bladder neck fails to close**; orgasm is preserved, ejaculate absent or low volume.

### Physiology

| Phase | Event | Control |
|---|---|---|
| Emission | Vas, seminal vesicles, prostate deliver semen to posterior urethra | Sympathetic T10–L2 |
| Ejection | **Bladder neck closes**, external sphincter relaxes, bulbospongiosus contracts | Somatic S2–S4 + sympathetic reflex |
| Orgasm | Cerebral perception | Cortical |

### Causes

- **Anatomical: TURP (transurethral resection of prostate)** 42–100%, TUIP (incision) 5–45%; most other resections and enucleations; previous bladder neck reconstruction. Ejaculation-sparing techniques, PUL (prostatic urethral lift) and Aquablation preserve ejaculation.
- **Neurogenic: diabetic autonomic neuropathy** (commonest medical), RPLND (retroperitoneal lymph node dissection; less with nerve sparing), colorectal surgery, sympathectomy, aortic surgery, spinal cord injury, multiple sclerosis.
- **Drugs:** α-blockers (silodosin, tamsulosin), antipsychotics, antidepressants, thiazides.
- **Congenital:** Müllerian duct cyst, Wolffian duct anomaly, prune belly syndrome.

### Evaluation

- Low-volume or absent ejaculate with orgasm; infertility.
- **Post-ejaculatory urinalysis:** void, ejaculate, void again; centrifuge. **>5–10 sperm per HPF (high-power field)** confirms RE; significant if urinary sperm approach or exceed

antegrade sperm.
- Sperm and fructose in urine separate RE from **failure of emission**; TRUS (transrectal ultrasound) if EDO (ejaculatory duct obstruction) suspected.

### Treatment

- Reassure if fertility not desired; stop offending drug.
- **Sympathomimetics** (close bladder neck): **pseudoephedrine** 60–120 mg (120 mg 6-hourly the day before and twice on the day); midodrine, ephedrine, phenylpropanolamine. Side effects: dry mouth, hypertension.
- **Imipramine** 25 mg twice daily; antihistamines (brompheniramine) add to sympathomimetics.
- **Bladder neck reconstruction:** poor results; collagen injection experimental.
- **Sperm retrieval:** alkalinise urine (sodium bicarbonate), void or catheterise after orgasm, wash sperm in buffer → IUI (intrauterine insemination), IVF (in vitro fertilisation) or ICSI (intracytoplasmic sperm injection).
- **Prevention:** nerve-sparing RPLND, ejaculation-preserving procedures for BPH (benign prostatic hyperplasia).

**Diagram to draw:** bladder neck open at ejaculation with semen entering the bladder.

> **Key points to remember:**
> - TURP and diabetic neuropathy are the commonest causes.
> - Post-ejaculatory urine: sperm and fructose confirm RE.
> - Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

## SAQ 12 CBAVD

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 3, Q3f

### Definition

- **CBAVD (congenital bilateral absence of the vas deferens):** both vasa absent, causing **obstructive azoospermia** with normal spermatogenesis.
- Often a **genital form of CF (cystic fibrosis)**: nearly all men with CF lack both vasa.

### Genetics

- **CFTR (cystic fibrosis transmembrane conductance regulator)**: chloride channel gene; more than 1600 mutations.
- Severe mutation on both alleles: CF. **Mild mutation on one or both alleles: CBAVD.** Commonest severe mutation **ΔF508**; the **5T variant** (intron 8) is a common mild allele.
- A minority have other mesonephric duct gene defects instead, with more **renal agenesis**.

### Clinical features

- **Vasa not palpable** (Meacham: a presumed vas slipping away three times is absent).
- Caput epididymis present and distended; body and tail often absent.
- Normal testis size, FSH (follicle-stimulating hormone) and testosterone.

- Seminal vesicles absent or hypoplastic: **semen volume <1.0 mL, acidic, fructose-negative**, azoospermic.
- Sinus or chest symptoms may reveal mild CF.

### Evaluation

- Semen analysis; FSH; **TRUS (transrectal ultrasound)** for seminal vesicles.
- **Renal ultrasound** (renal agenesis in about 1 in 10).
- **CFTR testing of the man and his female partner** (carrier rate about 1 in 20 in northern Europeans). Panels cover 25–40 mutations, so a negative panel leaves residual risk.
- Testis biopsy unnecessary when FSH and testis size are normal.

### Management

- **Genetic counselling**: if the partner carries a mutation, offspring may have CF; PGT (preimplantation genetic testing) or donor sperm can be offered.
- Reconstruction is impossible, so **surgical sperm retrieval + ICSI (intracytoplasmic sperm injection)**:

| Technique | Points |
|---|---|
| **MESA (microsurgical epididymal sperm aspiration)** | Best: many motile sperm; cryopreserve several vials |
| PESA (percutaneous epididymal aspiration) | Local anaesthetic; fewer sperm |
| TESA/TESE (testicular aspiration/extraction) | If epididymal retrieval fails |
| - ICSI results are good; one study reported more miscarriage and stillbirth. | |
| - **Also know:** CUAVD (unilateral absence) is a Wolffian duct defect with ipsilateral renal agenesis. | |

**Diagram to draw:** absent vas and seminal vesicle with intact testis and caput epididymis.

> **Key points to remember:**
> - CBAVD = mild CFTR mutation; test both partners.
> - Low-volume, acidic, fructose-negative azoospermia; normal FSH.
> - Renal ultrasound; MESA + ICSI.

## SAQ 13 Sertoli cell only syndrome

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 3, Q3a (non-obstructive azoospermia also answered in 2025-Winter Paper 2, Q3a)

### Definition

- SCOS (Sertoli cell only syndrome), **germ cell aplasia** (Del Castillo syndrome): seminiferous tubules lined only by Sertoli cells, with **complete loss of spermatogonia**. Leydig cells usually normal.
- The worst histological pattern of NOA (non-obstructive azoospermia).

## Types

| Type | Feature |
|---|---|
| Pure (complete, congenital) | All tubules lack germ cells |
| Mixed (focal, acquired) | Some tubules keep foci of spermatogenesis, the basis of sperm retrieval |

## Aetiology

- **Developmental:** primordial germ cells fail to migrate into the testis.
- **Genetic: complete AZFa (azoospermia factor a) deletion** gives pure SCOS; **AZFc** may give SCOS with foci; Klinefelter (47,XXY).
- **Acquired:** cryptorchidism, chemotherapy or radiation, mumps orchitis, torsion; idiopathic.

## Clinical features

- Normal virilisation; **azoospermia** with normal ejaculate volume; small soft testes.

## Evaluation

- Semen: azoospermia on two centrifuged samples.
- **FSH (follicle-stimulating hormone) high**, **inhibin B low** (Sertoli marker); LH (luteinising hormone) normal or high; testosterone normal or low.
- **FSH >7.6 IU/L and testis long axis ≤4.6 cm** suggest NOA.
- **Karyotype and Y microdeletion** before surgery.
- **Histology** confirms, usually at sperm retrieval.

| Pattern | Meaning | Micro-TESE retrieval |
|---|---|---|
| Hypospermatogenesis | All stages, fewer | Best |
| Maturation arrest | Stops at one stage | About 40% |
| **SCOS** | No germ cells | **About 30%** |

## Management

- **Counsel** on genetic findings.
- **Complete AZFa or AZFb deletion:** retrieval essentially nil; do not operate; donor sperm or adoption.
- **Micro-TESE (microdissection testicular sperm extraction):** at ×20–25, **larger, opaque, yellow tubules** (foci of spermatogenesis) are selected; SCOS tubules are thin, white and stringy. Live births are reported even in SCOS.
- Sperm used for **ICSI (intracytoplasmic sperm injection)**.
- Optimise first: stop exogenous testosterone; hormonal stimulation evidence weak.
- **Also know:** spermatogonial stem cell transplantation and in-vitro spermatogenesis are experimental.

**Diagram to draw:** tubule cross-section: normal layered epithelium versus Sertoli cells only.

> **Key points to remember:**
> - Complete loss of spermatogonia; high FSH, low inhibin B.
> - Complete AZFa deletion: pure SCOS, no retrieval.
> - Micro-TESE retrieval about 30%: worst histology, not hopeless.

## SAQ 14 Oxidative stress and male infertility

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 3, Q3b

### Definition

- **Oxidative stress (OS):** imbalance where ROS (reactive oxygen species) exceed the antioxidant capacity of semen. The term **MOSI (male oxidative stress infertility)** has been proposed.
- Low ROS levels are **physiological**: needed for capacitation, hyperactivation, acrosome reaction and sperm–oocyte fusion.

### Sources of ROS

| Source | Examples |
|---|---|
| **Leucocytes** (main) | Infection, prostatitis, epididymitis, leucocytospermia |
| **Immature sperm** | Excess residual cytoplasm |
| Varicocele | Heat and hypoxia |
| Lifestyle | **Smoking**, alcohol, obesity, heat |
| Others | Ageing, diabetes, pollutants, radiation |

### Why sperm are vulnerable

- Membrane rich in **polyunsaturated fatty acids** → lipid peroxidation.
- Little cytoplasm, so few antioxidant enzymes; limited DNA repair.
- **Seminal plasma antioxidants:** superoxide dismutase, catalase, glutathione peroxidase; taurine, vitamins C and E, zinc.

### Effects

- Lipid peroxidation → loss of **motility** and membrane fluidity.
- **Sperm DNA fragmentation** → poor embryo quality, miscarriage, lower ART (assisted reproductive technology) success.
- Mitochondrial damage and apoptosis.

### Evaluation

- Semen analysis: peroxidase-positive WBC (white blood cells) **>1 million/mL** = leucocytospermia.
- **Direct ROS:** chemiluminescence (luminol).
- **Indirect:** total antioxidant capacity, malondialdehyde (lipid peroxidation), ORP (oxidation-reduction potential).
- **DNA fragmentation tests:** TUNEL (terminal deoxynucleotidyl transferase dUTP nick-end labelling), SCSA (sperm chromatin structure assay), SCD (sperm chromatin dispersion), Comet.
- Use selectively (unexplained infertility, recurrent miscarriage, failed ART); not routine.

### Management

- **Lifestyle:** stop smoking, reduce weight and alcohol, avoid heat.

- **Treat cause:** antibiotics for infection; **varicocelectomy** for clinical varicocele (reduces DNA fragmentation).
- **Antioxidants** (vitamins C and E, CoQ10 [coenzyme Q10], zinc, selenium, carnitine): Cochrane review, low-quality evidence; **MOXI trial (2020)** showed no gain in semen parameters, DNA fragmentation or live birth. Avoid excess: **reductive stress** also harms.
- High DNA fragmentation with failed ICSI (intracytoplasmic sperm injection): **testicular sperm** has less fragmentation.

**Diagram to draw:** balance of ROS sources against seminal antioxidants, with effects on sperm.

> **Key points to remember:**
> - Leucocytes and immature sperm are main ROS sources.
> - Lipid peroxidation and DNA fragmentation are the key injuries.
> - Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

## SAQ 15 Dhat syndrome

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 3, Q3c

### Definition

- **Dhat syndrome:** a **culture-bound syndrome** of South Asia. The man believes he is losing **semen ("dhat")** in urine or nocturnal emissions, and attributes weakness, fatigue and sexual problems to it.
- Term coined by **Wig** (India); listed in ICD-10 (International Classification of Diseases, 10th revision) and in the DSM-5 (Diagnostic and Statistical Manual, 5th edition) glossary of cultural concepts of distress.
- Related syndromes: **Shen-kui** (China), **Koro** (genital retraction).

### Cultural basis

- Traditional belief that semen is a **precious vital fluid** made from many drops of blood; its loss is felt to drain body and mind.

### Epidemiology

- Young men, often unmarried or newly married, rural, lower education; common in urology and sexual medicine clinics.

### Clinical features

- **Somatic:** fatigue, weakness, aches, poor appetite.
- **Psychological:** anxiety, low mood, guilt (about masturbation), poor concentration, sleep loss.
- **Sexual:** fear of ED (erectile dysfunction) or PE (premature ejaculation), worry about penis size and fertility.
- Whitish urine is usually phosphaturia, oxaluria or normal secretions.
- **Comorbidity: depression** and **anxiety** in many; somatisation.

### Evaluation

- Non-judgmental sexual history and belief system.
- Examination of genitalia; urine microscopy (phosphates, oxalates, sperm, pus cells).
- Exclude organic disease: **UTI (urinary tract infection), prostatitis, STI (sexually transmitted infection), diabetes**, true spermatorrhoea.
- Screen for depression and anxiety.

### Management

**1. Counselling (main)**

- **Sex education:** semen is continuously produced; nocturnal emission and masturbation are normal and harmless.
- Reassure without dismissing.

**2. Psychological therapy:** CBT (cognitive behavioural therapy) for misbeliefs and anxiety.

**3. Drugs**

- **SSRIs (selective serotonin reuptake inhibitors)** for depression, anxiety or associated PE; short anxiolytic course if needed.
- Treat genuine ED or PE.

**4. Avoid harm**

- Avoid unnecessary "tonics", antibiotics and repeated tests; they reinforce the belief.
- Refer to psychiatry when depression is significant.

### Prognosis

- Good with counselling; drop-out is common.

**Diagram to draw:** flowchart: complaint of semen loss → exclude organic cause → counsel → treat depression or anxiety.

> **Key points to remember:**
> - Culture-bound semen-loss anxiety of South Asia.
> - Depression and anxiety common; exclude UTI, prostatitis, STI.
> - Counselling and sex education first; SSRIs for comorbidity.

## SAQ 16 Male menopause (andropause, late-onset hypogonadism)

P4

**Asked as:**

- Male menopause. — *Summer 2017 Paper IV Q3e (SAQ)*
- Andropause — *Winter 2017 Paper IV Q3e (SAQ)*
- Male menopause — *Summer 2019 Paper IV Q3f (SAQ)*

**Type:** SAQ (10 marks)

*Read as: late-onset hypogonadism (testosterone deficiency of ageing men, "andropause"). Same topic as 2017-Summer Paper 4, Q3e and 2017-Winter Paper 4, Q3e; full answer given here.*

## Definition

- **Symptoms plus consistently low serum testosterone** in an ageing man.
- "Menopause" (or "Andropause") is a misnomer: the fall is gradual and partial. Preferred terms: PADAM (partial androgen deficiency of the ageing male), LOH (late-onset hypogonadism).

## Pathophysiology

- Testosterone falls slowly from middle age.
- Testosterone in the 70s is about one-third lower than in young men. (Another source: mean testosterone in the seventh decade is about **one-third lower** than in young men.)
- **Testis:** fewer Leydig cells, lower sperm output; FSH (follicle-stimulating hormone) rises about threefold.
- **Centre:** blunted feedback, irregular GnRH (gonadotrophin-releasing hormone) pulses.
- **SHBG (sex hormone-binding globulin) rises with age**, so free testosterone (bioavailable [free + albumin-bound] testosterone) falls more.
- Worse with obesity, diabetes and **opioids**.

## Clinical features

- **Sexual (most specific):** low libido, ED (erectile dysfunction), fewer morning erections.
- Fatigue, low muscle mass, osteoporosis, anaemia, low mood.
- **ADAM (Androgen Deficiency in the Aging Male)** questionnaire: sensitive, not specific.

## Diagnosis

- **Morning (7–11 am) total testosterone, twice (repeated).** AUA (American Urological Association): **<300 ng/dL** (≈10.4 nmol/L). EAU (European Association of Urology): about **12 nmol/L**; calculated free testosterone if borderline. Calculated bioavailable testosterone when SHBG is abnormal.
- **LH (luteinising hormone):** high in primary, low or normal in secondary hypogonadism. Check **prolactin** if LH is low (Low LH → **prolactin**); pituitary MRI (magnetic resonance imaging) if testosterone <150 ng/dL or prolactin high.
- Baseline: PSA (prostate-specific antigen), DRE (digital rectal examination), haematocrit.

## Treatment

- Lifestyle first: weight loss, exercise, treat comorbidities.
- **TRT (testosterone replacement therapy)** for symptoms + confirmed low level: gels, injections (enanthate every 2–3 weeks; undecanoate every 10–14 weeks), oral undecanoate. Stop if no benefit at 3–6 months (another source: at about 3 months).
- **Contraindications:**
- Prostate or breast cancer.
- **Wish for fertility** (TRT stops sperm production [suppresses spermatogenesis]; use hCG [human chorionic gonadotrophin] or clomiphene).
- Haematocrit >50% (raised haematocrit).
- Untreated severe sleep apnoea; recent MI (myocardial infarction) or stroke.

## Monitoring

- At 3–6 months, then yearly: testosterone, **haematocrit** (stop/hold if >54%), PSA.

### Safety evidence

- **TRAVERSE (2023):** 5,246 men aged 45–80 at high cardiovascular risk. TRT was **non-inferior for major cardiac events**; more atrial fibrillation and pulmonary embolism.
- **Testosterone Trials**: modest sexual benefit, **no cognitive benefit**.

**Diagram to draw:** hypothalamic–pituitary–testis axis with ageing defects.

> **Key points to remember:**
> - Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
> - SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
> - TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

## SAQ 17 Klinefelter's syndrome.

P4

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 4, Q3b

### Definition and genetics

- **47,XXY**: commonest sex chromosome aneuploidy and **commonest genetic cause of male infertility**; 1 in 500–1000 male births.
- Non-mosaic in **80–90%**; the rest mosaic **46,XY/47,XXY** (milder, more sperm) or 48,XXXY variants. Extra X from non-disjunction (maternal age).
- Underdiagnosed: most never diagnosed; increasingly detected on NIPT (non-invasive prenatal testing).

### Pathophysiology

- Seminiferous tubules **hyalinise** progressively from puberty; germ cells are lost. Leydig cells fail, so testosterone falls. Puberty starts on time, then stalls.

### Clinical features

| Age | Features |
|---|---|
| Infant | Usually normal; micropenis, cryptorchidism, hypospadias |
| Adolescent/adult | **Small firm testes (<3.5 cm, 1–4 mL)**, gynaecomastia, tall stature with long legs (only ~30%), sparse hair, female fat distribution |
| Neurodevelopment | Language, learning and attention difficulties |

### Hormones and tests

- **High FSH (follicle-stimulating hormone) and LH (luteinising hormone), low or low-normal testosterone, high oestradiol** (raised oestradiol:testosterone ratio). Inhibin B low.
- Semen: **azoospermia** in most (~8% have ejaculated sperm).
- **Karyotype** confirms (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine, guideline: karyotype all men with non-obstructive azoospermia or severe oligozoospermia).

### Management

- **Fertility: micro-TESE** (microdissection testicular sperm extraction) + ICSI (intracytoplasmic sperm injection). Sperm found in **~40–50%** (68% in one series); offspring are usually chromosomally normal.
- Spermatogenesis declines with age, so retrieve early (late adolescence or young adulthood). Some give clomiphene, anastrozole or hCG (human chorionic gonadotropin) beforehand.
- **TRT (testosterone replacement therapy):** for hypogonadism (bone, muscle, mood, libido). It **suppresses spermatogenesis**, so offer sperm retrieval or banking **before** starting.
- Gynaecomastia: reduction mammoplasty if persistent. Speech, educational and psychological support.

### Associated risks

- Osteoporosis, metabolic syndrome, diabetes, VTE (venous thromboembolism), autoimmune disease.
- **Mediastinal (extragonadal) germ cell tumours**; male breast cancer (~1% lifetime).

**Diagram to draw:** hypothalamic–pituitary–testis axis showing high FSH and LH, low testosterone.

> **Key points to remember:**
> - 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
> - Small firm testes, high FSH and LH, azoospermia.
> - Micro-TESE ~50% success; retrieve sperm before testosterone.
> - Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

## SAQ 18 Semen analysis nomenclature.

P4

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 4, Q3e

### Collection (WHO, World Health Organization, 2021, 6th edition)

- Masturbation into a clean wide-mouthed non-toxic container (special condom if needed); complete sample.
- Abstinence **2–7 days** (evidence: 1–2 days best). Keep at 20–37 °C; analyse within **1 hour**, after liquefaction (~30 min).
- At least **two samples, 2–3 weeks apart**, because of variability.

### Lower reference limits (5th centile, fertile men)

| Parameter | Lower limit |
|---|---|
| Volume | **1.4 mL** |
| Concentration | **16 million/mL** |
| Total count | **39 million/ejaculate** |
| Total motility | **42%** |
| Progressive motility | **30%** |

| Parameter | Lower limit |
|---|---|
| Normal forms (strict) | **4%** |
| Vitality | **54%** |
| Peroxidase-positive leucocytes | <1 million/mL |

- These are reference values, not fertility cut-offs. The 6th edition grades motility as progressive, non-progressive or immotile.

## Nomenclature

| Term | Meaning |
|---|---|
| Normozoospermia | All parameters at or above reference |
| Oligozoospermia | Total count (or concentration) below reference |
| Asthenozoospermia | Progressive motility below reference |
| Teratozoospermia | Normal forms below reference |
| OAT (oligoasthenoteratozoospermia) | All three abnormal |
| Azoospermia | No sperm, **including centrifuged pellet** |
| Cryptozoospermia | Sperm absent in fresh smear, present only after centrifugation |
| Aspermia | No ejaculate (dry orgasm) |
| Hypospermia | Volume below reference |
| Necrozoospermia | Low vitality (many dead sperm) |
| Leucocytospermia | Peroxidase-positive white cells above 1 million/mL |
| Haemospermia | Blood in ejaculate |
| Globozoospermia | Round-headed sperm without acrosome |

## Notes

- **Low volume or aspermia:** post-ejaculatory urine for retrograde ejaculation; TRUS (transrectal ultrasound) for EDO (ejaculatory duct obstruction).
- **Vitality:** eosin–nigrosin, or **HOS (hypo-osmotic swelling)** test, which keeps sperm usable for ICSI (intracytoplasmic sperm injection).
- Only azoospermia is definitive; values overlap between fertile and infertile men.
- **Also know:** Kruger strict morphology; DNA fragmentation; antisperm antibodies (MAR, mixed antiglobulin reaction); semen pH no longer recommended.

**Diagram to draw:** normal sperm (head, midpiece, tail) beside a WHO reference box.

> **Key points to remember:**
> - WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
> - Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
> - Two samples 2–3 weeks apart; abstinence 2–7 days.

# Rapid Revision Sheet · Section 9

### Spermatogenesis, azoospermia and non-obstructive azoospermia (LAQ)

- Spermatogenesis about 64 days; Ap divide every 16 days; six stages.
- Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
- FSH → Sertoli, LH → Leydig; tight junctions between Sertoli cells form the blood–testis barrier.
- Confirm azoospermia on two centrifuged samples.
- FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- AZFa/b complete deletion: do not operate (no retrieval).
- NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.
- Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
- FSH and testis size may be normal in maturation arrest: biopsy diagnoses.
- Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

### Azoospermia: testicular biopsy and surgical sperm retrieval techniques (LAQ)

- Azoospermia: no sperm in centrifuged pellet on two samples.
- Biopsy indication: azoospermia, normal testes, vasa and FSH.
- Open biopsy is the gold standard; fix in Bouin's, never formalin; no forceps; touch prep.
- OA: normal FSH and testes; ICSI essential.
- OA: MESA (capillary action); MESA is best: many motile sperm, multiple vials.
- Move proximally; capillary action, never suction.
- TESA: 23 G needle, suction, local anaesthesia; for OA with normal spermatogenesis.
- TESA/TESE when epididymal sperm absent.
- NOA: micro-TESE best; finds sperm in about half of NOA; TESA yield poor.
- AZFa/AZFb deletion: 0% retrieval (no sperm), do not operate.
- All surgically retrieved (testicular) sperm need ICSI; cryopreserve extras.
- Haematoma commonest; epididymal biopsy most serious.

### Surgical management of ED and penile prostheses (types, biofilms) (LAQ)

- Prosthesis is the last line but gives 80–90% satisfaction.
- 3-piece inflatable gives the best rigidity and flaccidity.
- Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
- Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
- Biofilm: microbes in a matrix; antibiotics fail.
- Infection 1–3%; coated devices and no-touch technique; infected implant: remove all parts; Mulcahy salvage 84%.
- Satisfaction 80–90%; repairs 15% at 5 years.
- Arterial revascularisation only for young men with post-traumatic focal stenosis.

### Priapism: diagnosis and management (LAQ)

- Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
- Aspiration plus phenylephrine first; distal shunt ± tunnelling before proximal.
- ED risk 0% ≤12 h, 100% ≥36 h; consider early prosthesis beyond 36 h.

- Beyond 36 h or refractory: early penile prosthesis (or analgesia and counselling).
- High flow: observe (about 60% resolve), then embolise; repeat embolisation rather than surgery.

**Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED (LAQ)**

- nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
- Noradrenaline α1 is the main anti-erectile transmitter.
- Dopamine/oxytocin promote; serotonin/prolactin inhibit.
- Rigid phase: ischiocavernosus, suprasystolic pressure.
- Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
- Diabetic ED: check HbA1c, testosterone and cardiac risk.

**Discuss the role of urologist in reproductive endocrinology and male infertility practice. (LAQ)**

- FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
- Exogenous testosterone is a contraceptive: never give it for infertility.
- Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
- Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

**Male reproductive immunology and immunobead test (SAQ)**

- BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
- ASA: vasectomy commonest; ≥50% motile sperm bound is significant; head-bound IgA matters most.
- Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
- Treatment of choice: IVF/ICSI.
- Leukocytospermia >1 million/mL.

**Premature ejaculation (SAQ)**

- ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

**Oligospermia and oligoasthenozoospermia (SAQ)**

- WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
- Karyotype and Y-microdeletion if under 5 million/mL.
- Repair palpable varicoceles only; never give testosterone.

**Management of Peyronie's disease (SAQ)**

- Ask the "straight penis" question; duplex before surgery.
- CCH: IMPRESS, stable 30–90° curve, intact erections.
- Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

**Haematospermia (SAQ)**

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
- Exclude prostate cancer and genitourinary tuberculosis.

- TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

**Avanafil (SAQ)**

- Peak 0.3–0.5 h; doses 50, 100, 200 mg.
- Highly PDE5-selective: no blue vision.
- Nitrates absolutely contraindicated. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**CRE & BEP (SAQ)**

- CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
- BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
- Bleomycin lung: low $FiO_2$ and restricted fluids at surgery. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Genetic causes of male infertility. Enumerate and classify. (SAQ)**

- Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
- AZFa or AZFb complete deletion: do not attempt retrieval.
- CBAVD = genital form of cystic fibrosis; test both partners.

**Ejacultory duct obstruction (SAQ)**

- Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
- TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
- TURED: sperm return in about half; reflux and epididymitis are its price.

**Tadalafil (SAQ)**

- Half-life 17.5 h; food-independent; back pain.
- 5 mg daily for LUTS: IPSS improves, Qmax does not.
- No nitrates for 48 h after a dose.

**Retrograde ejaculation (SAQ)**

- TURP and diabetic neuropathy are the commonest causes.
- Post-ejaculatory urine: sperm and fructose confirm RE.
- Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

**CBAVD (SAQ)**

- CBAVD = mild CFTR mutation; test both partners.
- Low-volume, acidic, fructose-negative azoospermia; normal FSH.
- Renal ultrasound; MESA + ICSI.

**Sertoli cell only syndrome (SAQ)**

- Complete loss of spermatogonia; high FSH, low inhibin B.
- Complete AZFa deletion: pure SCOS, no retrieval.
- Micro-TESE retrieval about 30%: worst histology, not hopeless.

**Oxidative stress and male infertility (SAQ)**

- Leucocytes and immature sperm are main ROS sources.
- Lipid peroxidation and DNA fragmentation are the key injuries.
- Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

**Dhat syndrome (SAQ)**

- Culture-bound semen-loss anxiety of South Asia.
- Depression and anxiety common; exclude UTI, prostatitis, STI.
- Counselling and sex education first; SSRIs for comorbidity. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Male menopause (andropause, late-onset hypogonadism) (SAQ)**

- Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
- SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
- TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

**Klinefelter's syndrome. (SAQ)**

- 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
- Small firm testes, high FSH and LH, azoospermia.
- Micro-TESE ~50% success; retrieve sperm before testosterone.
- Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

**Semen analysis nomenclature. (SAQ)**

- WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
- Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
- Two samples 2–3 weeks apart; abstinence 2–7 days.

# Section 10 · Testis, Penis & Urethra

## Long answer questions (LAQ)

### LAQ 1 Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma

P1 · P2 **merged from 2**

*Asked as:* P1 LAQ "Lymphatic drainage of testis and scrotum, RPLND templates and penile cancer implications" (asked as "Lymphatic drainage of testis and principles for designing templates for RPLND" and "Anatomy of lymphatic drainage of testis and scrotum. Describe management implications of penile malignancies") · P2 LAQ "Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis."

*The P1 second part (Summer 2021) is taken as the management implications of lymphatic drainage in penile cancer, i.e. management of the inguinal and pelvic nodes.*

#### Embryological basis

- **Why retroperitoneal:** the testis develops at L1–L2 beside the mesonephros (near the kidney) and descends, carrying its vessels and lymphatics; its artery, vein and lymphatics stay retroperitoneal. So its lymph returns to where it began: the **retroperitoneal (lumbar) nodes near the renal hilum**, not the groin. The scrotum is skin and drains to the **groin**.

#### Lymphatic drainage of the testis

- **Route:** lymphatics of the testis and the caput and corpus of the epididymis run up the **spermatic cord** with the testicular (gonadal) vessels, through the internal ring and along the gonadal vessels to the lumbar nodes near the renal hilum (**para-aortic and interaortocaval nodes**). Lymph capillaries do not enter seminiferous tubules.
- **Onward:** lumbar trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta, behind the right crus; hence retrocrural nodes), then the **thoracic duct** to the left innominate vein. This explains **left supraclavicular** node metastases.
- Lymph flows **cephalad and right to left**.
- **Epididymis:** caput and corpus drain with the testis; **cauda epididymis** drains with the vas to the **external iliac** nodes.

**Retroperitoneal nodal regions**

| Region | Extent |
|---|---|
| Right paracaval (with precaval, retrocaval) | IVC (inferior vena cava) midline to right ureter |
| Interaortocaval | IVC midline to aortic midline (between IVC and aorta) |
| Left para-aortic (with pre-aortic, retroaortic) | Aortic midline to left ureter |

- **Also know:** ten mapped regions include right and left common iliac, interiliac and right and left suprahilar.

### Primary landing zones

| | Right testis | Left testis |
|---|---|---|
| Primary (landing zone) | **Interaortocaval** (below renal vessels) | **Left para-aortic** (and left hilar) |
| Secondary / next echelon | Right paracaval, precaval, then para-aortic | Interaortocaval; pre-aortic |
| Crossover / contralateral | Small but real to left para-aortic (another source: common; right to left occurs) | None significant; rare to the right, bulky disease only (another source: rare; implies bulk) |

- **Crossover is only right to left**, matching the overall lymph flow.
- **Donohue mapping (1982):** contralateral and suprahilar spread is rare but rises with bulk; interiliac spread is very rare.
- **Pelvic, iliac and inguinal nodes** are involved only when drainage is altered (or with retrograde spread from bulky disease): previous scrotal or inguinal surgery (orchidopexy, hernia repair), scrotal violation, or tumour invading the tunica vaginalis or scrotum.
- The retroperitoneum is the first metastatic site in 70–80% (choriocarcinoma spreads by blood). CT (computed tomography) understages 25–35% of stage I NSGCT (non-seminomatous germ cell tumour); 5–9 mm landing-zone nodes are suspicious.

### Lymphatic drainage of the scrotum

- Scrotal skin drains to the **ipsilateral superficial inguinal nodes**. Lymphatics, like vessels and nerves, **do not cross the median raphe**.

### Clinical implications: testis and scrotum

- **Trans-scrotal orchidectomy or biopsy is contraindicated.** It leaves cord behind and opens scrotal skin lymphatic routes, so tumour can reach inguinal and pelvic nodes, causing local and inguinal or pelvic recurrence. Use **radical inguinal orchidectomy** with high cord clamping.
- Inguinal or pelvic spread from the testis means altered drainage: scrotal violation, previous orchidopexy or hernia repair, scrotal invasion.
- RPLND (retroperitoneal lymph node dissection) templates and para-aortic or "dog-leg" radiotherapy fields follow the landing zones.
- Scrotal skin cancer is staged at the ipsilateral groin.

### Principles of designing RPLND templates

**1. Map-based design: follow the side-specific landing zones** to remove likely disease and spare the rest.

- Historical dissection: bilateral infrahilar + suprahilar + interiliac: most men anejaculatory.
- **Ray (1974)** first templated landing zones; **Donohue (1982)** mapping confirmed side-specific spread; **Weissbach and Boedefeld (1987)** reduced left template with frozen section.
- So a template must include the primary landing zone, the next echelon and the known crossover route.

**2. Fixed boundaries**

- **Superior:** renal hila (renal vessels). **Lateral:** ureters. **Inferior:** ureter crossing the common iliac artery. Always remove the **gonadal vein and cord stump** to the internal ring.

**3. Omit suprahilar and interiliac regions** in non-bulky disease, as spread there is rare. This reduces chylous ascites, renal, pancreatic and duodenal injury, and protects the hypogastric plexus.

**4. Unilateral modified templates** leave contralateral sympathetic fibres untouched. Because of right-to-left crossover, both templates take the para-aortic nodes **above the IMA (inferior mesenteric artery)**.

| Template | Content |
|---|---|
| **Right modified** | Paracaval, precaval, retrocaval, interaortocaval, right common iliac; para-aortic **above IMA**; omit para-aortic below IMA |
| **Left modified** | Left para-aortic, preaortic, retroaortic, left common iliac; interaortocaval above IMA; omit paracaval/precaval/retrocaval |
| **Full bilateral infrahilar** | All three regions both sides (paracaval, precaval, retrocaval, interaortocaval, pre-aortic, para-aortic, retroaortic, both common iliacs), between renal vessels, ureters and ureteric crossing of iliacs |
| Historical extended | Plus suprahilar and interiliac (thoracoabdominal); most anejaculatory |

- Omitting para-aortic above the IMA (right) or interaortocaval (left) is controversial.

**5. Nerve sparing to preserve ejaculation:** antegrade ejaculation (emission and bladder-neck closure) depends on **L1–L4 postganglionic sympathetic fibres** (about three each side) joining the **superior hypogastric plexus at or below the IMA** and running over the sacral promontory. Identify them prospectively before dividing tissue.

- **Unilateral templates** leave contralateral fibres untouched (antegrade ejaculation 75–97%). **Prospective nerve sparing** (Jewett 1988) identifies fibres: 90–100%.

**6. Completeness ("duplicate open RPLND"): split-and-roll** over the great vessels, **divide lumbar vessels** to clear retrocaval and retroaortic tissue, clip lymphatics at the cephalad limit (left renal vein) to prevent chylous ascites. Incomplete RPLND or mass excision gives worse outcomes (GETUG T93BP; 33% local relapse after post-chemotherapy mass-only resection).

**7. Extend if needed:** enlarged nodes at operation call for the full bilateral template (frozen section, Weissbach 1987). Convert to bilateral if nodes are suspicious at surgery or the primary has somatic-type malignancy.

## Template by clinical setting

| Setting | Template |
|---|---|
| **Primary RPLND** (high-risk stage I or low-volume stage II NSGCT, normal markers) | Default: modified unilateral nerve-sparing **or** full bilateral infrahilar nerve-sparing; no randomised trial. Modified allowed: low-volume, clinically negative nodes. Main risk: out-of-template nodes 3–23% (right > left) |
| **PC-RPLND (post-chemotherapy)**, residual mass >1 cm, normal markers | **Full bilateral infrahilar + residual mass** (standard). Main risk: crossover with bulk; viable tissue unrecognisable |
| Modified PC template allowed only if | Mass ≤5 cm in landing zone, normal markers, IGCCCG (International Germ Cell Cancer Collaborative Group) good or intermediate risk |

- Against unilateral templates: 3–23% of node-positive men have nodes outside the modified template, more with right tumours.
- *Sources differ:* Campbell testis-neoplasm chapter calls full bilateral nerve-sparing the standard; the surgery chapter accepts either.

### RPLND outcomes and complications

- Primary RPLND: pN+ about 20–30%; cancer-specific survival (CSS) about 98.8–100%; retroperitoneal recurrence <2% after full template.
- **Antegrade ejaculation >90%** with templates plus nerve sparing (modified template alone: early 75–87%, refined 97%).
- Complications: ileus (commonest), wound infection, chylous ascites (0.2–2.1% primary; 2–12% post-chemotherapy), pulmonary (bleomycin lung toxicity; low $FiO_2$), ejaculatory loss.
- **Also know:** nephrectomy needed at PC-RPLND (left-sided mass >10 cm); GETUG T93BP showed incomplete RPLND predicts relapse.

### Stage I seminoma: definition, staging and work-up

- Disease confined to the testis: any pT, N0, M0 on imaging. Stage IA = pT1; IB = pT2–4; IS = markers still raised after orchidectomy.
- **Work-up:** AFP (alpha-fetoprotein), β-hCG (human chorionic gonadotropin) and LDH (lactate dehydrogenase) before and after orchidectomy. CT abdomen and pelvis; chest CT or radiograph. Baseline testosterone; **offer sperm banking**. A raised AFP means non-seminoma.
- **Radical inguinal orchidectomy:** early cord control at the internal ring. It cures 80–85% of stage I seminoma.
- **Risk factors for relapse: tumour >4 cm** and **rete testis invasion**. Five-year relapse is 12%, 16% and 32% with 0, 1 and 2 factors. LVI (lymphovascular invasion) predicts relapse in non-seminoma, not seminoma.

### Stage I seminoma: options after orchidectomy (cancer-specific survival about 100% with each)

| | Surveillance | Carboplatin | Radiotherapy |
|---|---|---|---|
| Method | Clinical review, markers, CT | **One cycle, AUC (area under the curve) 7** | **20 Gy** in 10 fractions to the para-aortic strip |
| Relapse | 14–20% | About 5% | 3–5% |
| Relapse site | Retroperitoneum (84–100%) | Retroperitoneum | Pelvis, chest, left supraclavicular |
| Drawbacks | Late relapse, many scans, compliance | Underdosing; late effects unknown | Second cancers, GI (gastrointestinal) toxicity, oligospermia |
| Status | **Preferred** | If unwilling or unsuitable for surveillance; risk factors | If unwilling or unsuitable for surveillance |

- **Carboplatin (Calvert):** 7 × (GFR [glomerular filtration rate] + 25) mg, using isotope GFR.
- **Trials: MRC TE18** (Medical Research Council): 20 Gy equals 30 Gy, with less toxicity. **MRC TE19/EORTC (European Organisation for Research and Treatment of Cancer) 30982:** one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE10: para-aortic field equals dog-leg, with 2% pelvic relapse.

- **Risk-adapted (EAU [European Association of Urology]):** surveillance for low risk; with risk factors discuss surveillance or carboplatin (treating all would overtreat two-thirds).

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Inguinal orchidectomy + markers + CT │ Stage I seminoma │ Surveillance (preferred) ── unwilling / unreliable ──► Carboplatin AUC7 ×1(–2) │ (or radiotherapy 20 Gy) Relapse (mostly retroperitoneal) ├─ small nodes ► dog-leg radiotherapy with boost, or chemotherapy └─ bulky/distant ► BEP ×3 or EP ×4
```

- BEP: bleomycin, etoposide, cisplatin; EP: etoposide, cisplatin.

### Stage I seminoma: relapse, IGCCCG and follow-up

- Relapses are almost always cured; median 14 months, but 10–20% occur after 4 years.
- **IGCCCG:** seminoma is good risk unless there are non-pulmonary visceral metastases (intermediate). **There is no poor-risk seminoma**, and markers do not change the risk group.
- **Surveillance follow-up:** examination and markers every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly; CT every 4–6 months (years 1–3), then 6-monthly to year 7, then yearly. CT finds >85% of relapses.

### Lymphatic drainage of the penis

| Superficial system | Deep system |
|---|---|
| Prepuce and shaft skin | Glans, frenulum, corpora |
| Dorsal network divides at the base and drains to both superficial inguinal groups | Trunks circle the corona and run under Buck's fascia to the presymphyseal plexus, then to femoral triangle nodes |

- **Two rules:** drainage is **bilateral and unpredictable in side** (presymphyseal crossover), but **orderly in sequence**: superficial inguinal, then deep inguinal, then external iliac and obturator. **No skip metastases.**
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**; sentinel nodes are mostly **superomedial**, then central. **Deep nodes** (1–3) lie medial to the femoral vein; **Cloquet's node** is the highest.

### Management implications in penile cancer

#### 1. Prognosis and timing

- Nodal status is the **strongest prognostic factor**. Penile SCC (squamous cell carcinoma) has a long **locoregional phase**, so the groin is a **curable station**.
- **Kroon (2005):** early ILND (inguinal lymph node dissection) for impalpable nodes gave 3-year disease-specific survival about **84% vs 35%** when delayed until palpable.
- Manage both groins **within 3 months**; remove **at least 8 nodes per side**.

#### 2. Bilateral management

- Crossover means **both groins** are staged or treated; with one palpable groin, the other holds micrometastasis in about 30%.

#### 3. Impalpable groin (cN0): risk-based

| Risk | EAU-ASCO (European Association of Urology–American Society of Clinical Oncology) 2023 | Groin |
|---|---|---|
| Low | pTis, pTa, pT1a G1 | Surveillance |

| Risk | EAU-ASCO (European Association of Urology–American Society of Clinical Oncology) 2023 | Groin |
|---|---|---|
| Intermediate | pT1a G2 | Surveillance if compliant, or surgical staging |
| High | **pT1b or higher**, G3 | **DSNB (dynamic sentinel node biopsy)**; ILND if DSNB unavailable |

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk and is staged surgically (ILND first).
- **DSNB:** ultrasound ± FNAC (fine-needle aspiration cytology) first, because a tumour-replaced node diverts flow (false negative). Then technetium-99m nanocolloid, blue dye and gamma probe.
- **No-skip spread and superomedial sentinels** justify **modified ILND (Catalona)**: smaller field, saphenous vein preserved; a positive frozen section leads to complete radical ILND.

**4. Palpable nodes (cN+)**

- About half are reactive, but do not delay with antibiotics. Ultrasound-guided **FNAC**; if positive, **radical ILND** on that side and stage the other.
- **Bulky (≥4 cm), fixed or bilateral nodes:** neoadjuvant **TIP** (paclitaxel, ifosfamide, cisplatin) × 4, then surgery in responders.

**5. Pelvic nodes**

- Stepwise spread means a negative groin needs no pelvic dissection.
- **PLND (pelvic lymph node dissection)**, ipsilateral, if **more than 2 inguinal nodes or ENE (extranodal extension)**. ENE in any node is pN3.

**6. Complications of ILND**

- Wound infection, **flap necrosis**, **lymphocele**, **lymphoedema**, DVT (deep vein thrombosis). Reduced by modified templates, saphenous sparing and **VEIL (video-endoscopic inguinal lymphadenectomy)**.
- **Also know:** adjuvant therapy for pN2–3; Cabanas sentinel biopsy abandoned.

## Recent advances

- **Robotic RPLND** for low-volume disease (same templates, same rules).
- RPLND for small-volume stage II seminoma (**SEMS, PRIMETEST**).
- Serum **miR-371a-3p** microRNA for selection, and it detects relapse better than standard markers.
- MRI (magnetic resonance imaging) instead of CT for seminoma surveillance (TRISST trial) to reduce radiation.

## Diagrams to draw

- Retroperitoneum showing great vessels with the three nodal regions, right (interaortocaval) and left (para-aortic) landing zones and the right-to-left crossover arrow; outline the right and left modified templates, IMA level and hypogastric plexus.
- Beside it, the femoral triangle with Daseler's zones and Cloquet's node.

**Key points to remember:**

- Right testis to interaortocaval; left to left para-aortic; crossover only right to left.
- Boundaries: renal hila, ureters, ureter–iliac crossing; omit suprahilar and interiliac.
- Nerve sparing of L1–L4 fibres gives antegrade ejaculation >90%; the fibres meet at the IMA, so unilateral templates spare ejaculation.
- PC-RPLND: full bilateral template remains standard unless mass ≤5 cm in landing zone.
- Scrotum (ipsilateral superficial inguinal) and penis drain to inguinal nodes; never operate trans-scrotally.
- Seminoma relapse risk factors: >4 cm and rete testis invasion (not LVI).
- Stage I seminoma: surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).
- Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
- pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.

## LAQ 2 Grafts in urology and buccal mucosa

P1 · P3 **merged from 2**

*Asked as:* P1 LAQ "Enlist various grafts used in urology and role of BMG in urology" · P3 SAQ "Uses of buccal mucosa in urology"

### Graft versus flap

| | Graft | Flap |
|---|---|---|
| Blood supply | Separated; gets a new supply from the bed | Moved with its own pedicle |
| Depends on | A well-vascularised host bed | Its pedicle |
| Example | BMG (buccal mucosal graft) | Penile fasciocutaneous island flap |

### Graft take (about 96 hours)

| Phase | Duration | Mechanism |
|---|---|---|
| **Imbibition** | about 48 h | Graft "drinks" nutrients from the bed; graft is cooler than core |
| **Inosculation** | about 48 h | Bed and graft vessels connect; true microcirculation returns |

- A graft lives on its bed, so it does best on thick vascular beds. Take fails with poor beds (scar, radiation), haematoma, seroma, infection or shear. Hence **quilting or pie-crusting** and **immobilisation for 5–10 days** (bolster or vacuum dressing).

### Grafts used in urology

| Group | Graft | Main uses |
|---|---|---|
| **Skin** | **STSG** (split-thickness skin graft), meshed | Staged urethroplasty only; penile and scrotal cover (Fournier gangrene, buried penis) |
| | **FTSG** (full-thickness skin graft) | Fossa navicularis, penile urethra (avoid in lichen sclerosus) |
| | Posterior auricular (Wolfe) FTSG | Staged hypospadias repair |
| **Mucosa** | **Buccal, lingual, labial (oral mucosa)** | Urethra, hypospadias, ureter |
| | Bladder epithelium (mucosa) | Urethra, hypospadias; meatal prolapse |
| | Rectal (colonic) mucosa | Panurethral stricture when oral mucosa is unavailable |
| **Other autologous** | Dermis | Tunica albuginea in Peyronie disease |
| | Tunica vaginalis | Peyronie disease, urethral cover; easy harvest |
| | Saphenous vein | Peyronie defects; vascular repair |
| | Rectus fascia, fascia lata | Pubovaginal sling |
| **Allograft / xenograft** | Pericardium, porcine SIS, porcine dermis, cadaveric fascia | Peyronie grafting ("off the shelf"); slings |
| **Synthetic** | Polypropylene (macroporous type I mesh) | Mid-urethral sling, prolapse |
| | Dacron, PTFE (polytetrafluoroethylene) | Vascular grafts; not advised for Peyronie (infection, fibrosis) |

| Group | Graft | Main uses |
|---|---|---|
| **Newer** | Collagen fleece (TachoSil); acellular matrix, tissue-engineered oral mucosa | Peyronie; experimental urethra |

- Vein is not a true graft: it survives by direct endothelial perfusion and its vasa vasorum.
- **Peyronie grafts:** pericardium (thin, strong, does not contract) and SIS (cut about 25% oversize; contracts) are commonest; AUA gives no preference.
- **STSG versus FTSG:** STSG carries little dermis, takes easily, can be meshed (up to sixfold) but contracts. FTSG carries full dermis, contracts less, looks better but is more "fastidious" in take.

### Buccal mucosa: history and why it works

- **BMG** is the workhorse graft for urethral reconstruction. **Humby 1941** (hypospadias); **Bürger 1992** and **El-Kasaby 1993** (stricture).
- Thick, **non-keratinised**, elastin-rich, hairless epithelium used to a **wet environment**; resists infection.
- Thin, **highly vascular lamina propria** with a panlaminar plexus, so imbibition and inosculation are rapid and take is excellent.
- Little contraction; handles and holds sutures well; low donor morbidity.

### Harvest

1. Nasotracheal intubation; mouth gag; mark the graft (about 4–5 × 2 cm per cheek; P3: 4–6 × 2–2.5 cm).
2. Stay **1 cm from the commissure** and avoid **Stensen's (parotid) duct** opposite the upper second molar.
3. Infiltrate, incise, raise the mucosa off buccinator; haemostasis.
4. Donor site: leave open or close; both acceptable.
5. Back table: **defat** the graft, cut about 10% oversize, fenestrate with a needle; quilt to the bed. - **Lingual** (under the tongue) and **labial (lip)** grafts are alternatives. Buccal and lingual give equal urethroplasty success; lip mucosa is thinner, suits the glans.

### Uses in urethroplasty

| Technique | Principle | Best for |
|---|---|---|
| **Dorsal onlay** (Barbagli, 1995) | Urethra rotated, dorsal urethrotomy; graft spread-fixed on corpora | Bulbar; reliable bed |
| **Ventral onlay** | Ventral urethrotomy; graft covered by spongioplasty | Bulbar with healthy spongiosum |
| **Dorsal inlay** (Asopa) | Ventral approach (urethrotomy); dorsal plate incised; graft inlaid | Penile and bulbar, no mobilisation |
| **Augmented anastomosis** | Excise narrowest part, floor anastomosis, graft roof | Long, dense bulbar stricture |
| **Kulkarni** (2012) | One-sided dissection keeps ventral blood supply; penile invagination; dorsal graft (onlay) | Panurethral, one stage |
| **Staged** (Johanson, Bracka) | Stage 1 lay open and graft; stage 2 tubularise after 6 months or more | **Lichen sclerosus**, failed hypospadias |

| Technique | Principle | Best for |
|---|---|---|
| **BMG perineal urethrostomy** | Graft completes dorsal wall to skin | Panurethral or failed repairs |

- Dorsal and ventral results are similar (dorsal equals ventral); graft equals flap; onlay beats tubes.
- **Skin is avoided in lichen sclerosus** because the disease recurs in skin; use staged BMG.

### Other uses of BMG

- **Ureter:** BMG onlay ureteroplasty (first human report 1999), now robotic with an **omental wrap**; for long proximal or mid strictures in dense scar. CORRUS (Collaborative of Reconstructive Robotic Ureteral Surgery) failure about 15–25% by site.
- Redo and staged hypospadias; urethral closure in rectourethral fistula; bladder neck contracture; female urethral stricture; Peyronie grafting (small defects only).

### Complications

- **Donor:** pain, swelling (first month), numbness (up to 6 months), mouth tightness, salivary duct injury.
- **Recipient:** graft failure, restricture, sacculation, fistula, ED.

### Outcomes

- Bulbar BMG urethroplasty about 85–90% success; most failures within 12 months.
- Results fall with time and in penile, redo and lichen sclerosus cases.

### Recent advances

- Tissue-engineered oral mucosa; acellular matrix; robotic BMG ureteroplasty.

**Diagram to draw:** bulbar cross-section showing dorsal onlay, ventral onlay and dorsal inlay (Asopa) positions.

> **Key points to remember:**
> - Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
> - BMG: non-keratinised, thin vascular lamina propria (excellent take), wet-adapted.
> - Avoid Stensen's duct; buccal equals lingual in success.
> - Staged BMG, not skin, in lichen sclerosus.
> - Bulbar success about 85–90%.

## LAQ 3 Urethral stricture: etiology, classification, DVIU, management and failed VIU

P2 · P3 **merged from 2**

*Asked as:* P2 LAQ "Urethral stricture disease: etiology, presentation and management (including recent advances)" (asked as "Describe etiology, clinical presentation and management of urethral strictures" and "Describe etiology, clinical presentation and recent advances in management of Urethral Structure Disease") · P3 LAQ "Urethral stricture: classification, DVIU and surgery after failed VIU" (asked as "Classify urethral strictures. Discuss in brief, surgical management of stricture after failed VIU" and SAQ "DVIU (Direct Visual Internal Urethrotomy)")

*"Structure" is taken to mean urethral stricture disease.*

### Definition

- A **urethral stricture** is a fixed narrowing of the **anterior** urethra (inside the corpus spongiosum) that will not accept instruments without tearing the mucosa. It is an epithelial scar, often with variable **spongiofibrosis**.
- Posterior narrowing is named differently (WHO [World Health Organization] 2004, ICUD [International Consultation on Urological Diseases] 2010): after pelvic fracture, **PFUI (pelvic fracture urethral injury)** or distraction defect; after prostate surgery or radiation, **stenosis or contracture**.
- Area falls with the square of diameter: halving calibre (30 to 15 Fr) loses about three-quarters of the lumen area, so symptoms appear late.

### Anatomy relevant to surgery

- **Bulbar urethra lies eccentric (dorsal)** in the spongiosum; the penile urethra is central.
- **Dual blood supply** (bulbar arteries plus retrograde flow from the glans) lets the bulb be divided and mobilised.

### Aetiology

| Category | Examples and mechanism |
|---|---|
| **Idiopathic** (commonest, about equal to iatrogenic) | No cause found |
| **Iatrogenic** | Catheters (very common in LMICs [low- and middle-income countries]), TURP (transurethral resection of prostate), TURBT (transurethral resection of bladder tumour), cystoscopy, hypospadias repair, radiotherapy. Mechanism: mucosal injury, pressure ischaemia, infection |
| **Traumatic** | **Straddle injury**: bulb crushed against the pubis; pelvic fracture (PFUI) gives a posterior obliterative defect |
| **Inflammatory** | Gonorrhoea (historically commonest, now rare); **lichen sclerosus (LS)**. Chlamydia: no proven link |
| Congenital | Rarest; infants only (short anterior stricture) |

- **LS**: chronic skin disease starting at the glans and meatus. High-pressure voiding forces urine into Littre glands, causing inflammation and deep spongiofibrosis. It **recurs in any genital skin** used to repair it.

## Classification

### 1. By site (WHO 2004 / ICUD 2010 terminology)

| Segment | Term |
|---|---|
| Anterior: meatus, fossa navicularis, penile (pendulous), bulbar | **Stricture** |
| Membranous urethra after pelvic fracture | **PFUI** / distraction defect |
| Posterior urethra after prostatectomy or radiotherapy; bladder neck | **Stenosis or contracture** |

**2. By aetiology:** idiopathic, iatrogenic, traumatic, inflammatory, congenital (table above).

**3. By length:** short (up to about 2 cm), long, multifocal, **panurethral** (whole anterior urethra).

**4. By depth (Jordan's spectrum):** (A) mucosal fold, (B) iris constriction, (C) full-thickness with minimal spongiofibrosis, (D) full-thickness (deep) spongiofibrosis, (E) fibrosis beyond the spongiosum (periurethral fibrosis), (F) complex stricture with fistula or abscess.

**5. By complexity:** simple (short, bulbar, first treatment) versus complex (long, multiple, LS, radiation, failed previous treatment, fistula).

- **PFUI** (Ch 125): major distraction, minor distraction, partial injury.
- **Also know: LSE system** (Lumen calibre, Segment involved, Etiology) for standard reporting.

## Clinical presentation

- Poor stream, straining, spraying, incomplete emptying, post-void dribble; UTI (urinary tract infection), prostatitis, epididymitis.
- **Retention** after long-tolerated symptoms; periurethral abscess, fistula ("watering-can perineum"), stones, upper-tract damage.

## Evaluation (define the 4 Ls: location, length, depth, density)

| Test | Role |
|---|---|
| **Uroflowmetry and PVR (post-void residual)** | Low, flat plateau-shaped flow curve; screen and follow-up |
| **Dynamic RUG (retrograde urethrogram) + VCUG (voiding cystourethrogram)** | Site and length; steep oblique views; done by the surgeon |
| **Flexible urethroscopy** | Calibre, appearance; paediatric scope sees beyond |
| **Sonourethrography** | Length and depth of spongiofibrosis |
| MRI (magnetic resonance imaging) | PFUI defect alignment only; not routine |

- **Urethral rest** (suprapubic catheter 6–8 weeks) shows the true length. Avoid blind dilation and instrumentation before surgery. In retention, place a **suprapubic catheter** rather than force a dilation.
- Before urethroplasty: treat infection; look for LS at the meatus; examine the mouth; record erectile function.

## Management principles

- First agree the aim: **palliation or cure**.

- The old "reconstructive ladder" (repeat endoscopy before surgery) is abandoned (archaic), because repeated endoscopy makes later urethroplasty harder.

## 1. Dilation

- Soft or balloon dilation over a guidewire. It stretches, not tears, the scar. Cures only epithelial strictures without spongiofibrosis.

## 2. DVIU (direct vision internal urethrotomy, VIU)

### Principle

- Incision of the stricture scar into healthy tissue under vision (Sachse cold-knife urethrotome), so the lumen heals enlarged by **secondary intention**: a **race between epithelialisation and wound contraction**. The stricture recurs when contraction wins, which is usual when spongiofibrosis is deep. Deep spongiofibrosis cannot be cured endoscopically.

### Indications

| Favourable | Unfavourable |
|---|---|
| **Bulbar** | Penile, fossa navicularis, any non-bulbar |
| **Short (<1.5–2 cm)** | Longer |
| Superficial, no dense spongiofibrosis | Dense scar (straddle, LS) |
| **First procedure** | Repeat; luminal narrowing >74% |

### Technique

1. RUG and VCUG define length and site.
2. Saline irrigation; **guidewire** through the stricture.
3. **Cold-knife incision traditionally at 12 o'clock** through scar to healthy spongiosum. Campbell notes the spongiosum is thinnest from 10 to 2 o'clock; deep cuts enter the corpora (ED [erectile dysfunction], fistula).
4. Catheter **3–7 days**; longer catheterisation adds no benefit. - **Laser** (holmium, Nd:YAG [neodymium:yttrium–aluminium–garnet], others): **no advantage over cold knife** (laser = cold knife).

### Adjuncts

- **CISC (clean intermittent self-catheterisation)/self-dilation**, tapering over 3–6 months: delays recurrence, which returns when stopped.
- Mitomycin C injection (reduces recurrence), steroid injection (limited data); removable stents; permanent UroLume stents withdrawn.

### Complications

- **Recurrence** (commonest); bleeding (often with erections); **irrigant extravasation**; UTI, sepsis; ED from deep cuts; spongio-cavernous fistula; incontinence if sphincter incised.

## Why repeat VIU is futile after failure

- VIU succeeds only in **bulbar strictures <1.5 cm without dense spongiofibrosis, at the first attempt**. Recurrence shows deep spongiofibrosis, which an incision cannot remove.
- **Steenkamp 1997, Heyns 1998 (randomised trials):** dilation = urethrotomy; a second urethrotomy helps few, a third none. **A second urethrotomy adds virtually nothing.**

- **Santucci & Eisenberg (2010):** about **8% stricture-free** long term after a first DVIU; repeat procedures almost always fail.
- Each repeat dilation or urethrotomy lengthens the scar and lowers later urethroplasty success; urethroplasty is more **cost-effective** after one failure.
- **AUA (American Urological Association) guideline (2016)** and EAU (European Association of Urology): DVIU or dilation may be offered for **short (<2 cm) bulbar** strictures; offer **early urethroplasty, not repeated endoscopy**, after one failed dilation or DVIU; no permanent stents (no stent if urethroplasty is possible).

### 3. Urethroplasty (including surgery after failed VIU)

| Stricture | Operation | Principle and key points |
|---|---|---|
| **Bulbar, short (1–2 cm, up to 3–5 cm proximally)** | **EPA (excision and primary anastomosis)**: gold standard, about 90–95% success | Excise all scar, spatulate both ends widely, tension-free anastomosis. **Non-transecting** or **vessel-sparing** variants keep bulbar arteries |
| **Bulbar, longer** | **BMG (buccal mucosal graft) onlay**: dorsal (**Barbagli**), ventral, dorsal inlay (**Asopa**) | Graft spread-fixed on a well-vascular bed (corpora or spongiosum) |
| Bulbar with an obliterated segment; long dense | **Augmented anastomosis** | Excise the narrowest part, anastomose the floor, graft the roof |
| **Penile** | One-stage BMG inlay or onlay; **penile skin flap** (Orandi, Quartey, circular) | Skin island carried on dartos; not in LS; avoid EPA here (chordee) |
| **LS, failed hypospadias, infection** | **Staged repair** (Johanson): stage 1 open and graft; tubularise after about 6 months or more | Oral mucosa, not genital skin, in LS |
| **Panurethral** | Kulkarni one-sided, one-stage dorsal graft, or staged | — |
| **Elderly, unfit, complex, multiple failures** | **Perineal urethrostomy** (BMG-augmented) | Simple, durable bypass |
| Meatal stenosis; fossa navicularis | Meatotomy; flap or graft | Topical clobetasol for early LS meatal disease |
| **PFUI** (including after failed VIU) | Suprapubic catheter, then delayed **perineal anastomotic urethroplasty** after 3 months or more | Progressive steps: bulbar mobilisation, intracrural separation, inferior pubectomy, rerouting of the urethra |

- **Onlay is better than tube** repairs. Graft and flap results are equal; grafts are simpler.
- **Avoid genital skin in LS** (use oral mucosa).
- **Not fit or declines surgery: Optilume** paclitaxel-coated balloon; intermittent self-dilation; long-term suprapubic catheter.
- **Urethral stents** (UroLume) are withdrawn.

### Recent advances in management

1. **Ladder abandoned**: repeated endoscopy lowers later urethroplasty success, so **early urethroplasty** is offered after one failed DVIU (AUA and EAU guidelines).
2. **Optilume paclitaxel-coated balloon** (FDA [Food and Drug Administration] 2023): paclitaxel inhibits fibroblasts. **ROBUST III**: stricture-free 74.6% vs 26.8% at 1 year after (repeated) failed endoscopy; IPSS (International Prostate Symptom Score) 9 vs 19.9.
3. **Anti-scarring adjuncts after DVIU**: mitomycin C injection reduces recurrence; intermittent self-dilation; removable stents. Permanent UroLume stents withdrawn.
4. **Vessel-sparing and non-transecting EPA**: preserve bulbar arteries, theoretically less ED.
5. **Augmented anastomosis**: excise the narrowest segment, graft the rest.
6. **Kulkarni one-sided dorsal BMG** through a perineal incision for panurethral strictures.
7. **Staged oral mucosa repair for LS**; avoid skin. **BMG perineal urethrostomy** reduces neomeatal stenosis.
8. **Robotic reconstruction** of bladder neck contracture and posterior stenosis (Y-V plasty, BMG).
9. **Tissue-engineered oral mucosa** (investigational). **Prevention**: silicone and hydrophilic catheters, smaller endoscopes.
10. **PROMs (patient-reported outcome measures)**: USS-PROM (Urethral Stricture Surgery PROM) is increasingly used to judge success. - **Also know:** dartos-island fossa repair; lingual graft; clobetasol for early LS.

### Complications of urethroplasty

- Recurrence, ED (overall similar to circumcision; higher with long repairs), chordee, post-void dribble, graft-site pain, numbness.

### Follow-up

- Voiding cystourethrogram at catheter removal (about 3 weeks); then uroflow, PVR and symptom score; RUG, cystoscopy or urethroscopy if flow falls.
- Most EPA recurrences occur in the first 6 months; graft results decline with time.

### Diagrams to draw

- RUG of a bulbar stricture.
- Cross-section of dorsal onlay versus ventral onlay.
- EPA with spatulated ends.
- Urethrotome cutting at 12 o'clock over a guidewire.

> **Key points to remember:**
>
> - Idiopathic and iatrogenic are commonest; LS recurs in skin.
> - Define location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
> - DVIU only once, for bulbar strictures <1.5–2 cm at first attempt; do not repeat it.
> - Santucci: about 8% long-term success; repeat DVIU futile.
> - AUA: urethroplasty after failure; laser = cold knife; ladder abandoned.
> - EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
> - Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
> - ROBUST III: Optilume 74.6% vs 26.8% stricture-free at 1 year.

## LAQ 4 Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy

P2 · P3 · P4 **merged from 3**

*Asked as:* P2 LAQ "Importance and management of Inguinal lymph nodes in CA penis." · P3 SAQ "Inguinal Block Dissection?" · P4 SAQ "Videoendoscopic inguinal lymphadenectomy"

*"Inguinal block dissection" is taken as ILND (inguinal lymph node dissection), mainly for penile cancer.*

### Importance

- Presence and extent of inguinal metastasis is the **strongest prognostic factor** in penile SCC (squamous cell carcinoma).
- A long **locoregional phase** precedes distant spread, so the groin is a **curable station**; lymphadenectomy alone can cure.
- **Kroon (2005)**: early ILND for impalpable nodes gave 3-year disease-specific survival about 84% vs about 35% when delayed until nodes were palpable (ENE [extranodal extension] about 20% vs 95%).
- Manage both groins **within 3 months** of diagnosis; aim for **at least 8 nodes per side**.
- About 80% 5-year survival with 2 or fewer unilateral nodes, no ENE, no pelvic nodes.

### Lymphatic drainage and surgical anatomy

- **Superficial system** (prepuce, shaft skin) drains to superficial inguinal nodes. **Deep system** (glans, frenulum) drains to the corona, then within Buck's fascia to the presymphyseal plexus, then to femoral triangle nodes.
- Presymphyseal crossover makes drainage **bilateral** and unpredictable in side; with one palpable groin, the other has micrometastasis in about 30%.
- **Stepwise, no skip**: superficial inguinal, then deep inguinal, then external iliac; so superficial status gates deep dissection and inguinal status gates pelvic dissection.
- **Femoral triangle:** superior **inguinal ligament**, lateral **sartorius**, medial **adductor longus**; floor fascia lata; apex where the muscles cross.
- **Superficial nodes** (4–25, above fascia lata) lie in **Daseler's five zones**:

| Zone | Related vessel |
|---|---|
| Central | Saphenofemoral junction |
| Superolateral | Superficial circumflex iliac vein |
| Inferolateral | Lateral femoral cutaneous and superficial circumflex veins |
| **Superomedial** (sentinel nodes mostly here) | Superficial external pudendal and superficial epigastric veins |
| Inferomedial | Great saphenous vein |

- **Deep nodes** (1–3) medial to the femoral vein; **Cloquet's node** is the highest.

### Staging (TNM [tumour–node–metastasis] 8th edition)

| N | Clinical | Pathological |
|---|---|---|
| 1 | One mobile unilateral node | 2 or fewer unilateral, no ENE |

| N | Clinical | Pathological |
|---|---|---|
| 2 | Mobile, 2 or more unilateral or bilateral | 3 or more (unilateral or bilateral) |
| 3 | Fixed mass or pelvic nodes | ENE or pelvic nodes |

- **T1a**: no LVI (lymphovascular invasion), no PNI (perineural invasion), not high grade. **T1b**: LVI or PNI or high grade.

### Evaluation

- **Careful bilateral palpation**: cN0 or cN+ (number, side, mobile or fixed). Imaging (US [ultrasound], CT [computed tomography], MRI) only if examination is unreliable (obesity, previous groin surgery).
- **cN+ or high risk**: FDG-PET/CT (fluorodeoxyglucose positron emission tomography–CT) or CT chest–abdomen–pelvis.
- About half of palpable nodes are reactive, yet an antibiotic trial is **no longer recommended** (delays treatment). US-guided **FNAC (fine-needle aspiration cytology)** is specific; a negative result never ends the pathway.

### Management of the impalpable groin (cN0): risk-based

| Risk (EAU-ASCO [European Association of Urology–American Society of Clinical Oncology] 2023) | Primary | Groin |
|---|---|---|
| Low | pTis, pTa, pT1a G1 | Surveillance |
| Intermediate | pT1a G2 | Surveillance (compliant) or surgical staging |
| High | **pT1b or more**, G3 | **DSNB (dynamic sentinel node biopsy)**; ILND if unavailable or DSNB positive |

- NCCN (National Comprehensive Cancer Network): pT1b G1–2 is intermediate risk, staged surgically (ILND first).
- **DSNB**: US ± FNAC first (a tumour-replaced node diverts flow and causes a false negative); then 99mTc (technetium-99m) nanocolloid, blue dye and gamma probe; serial sectioning. False-negative about 7% at expert centres.
- **Also know:** Cabanas anatomical sentinel biopsy abandoned.

### Management of palpable nodes (cN+)

`Palpable node → US-guided FNAC positive → radical ILND same side + stage other side (superficial ILND, FS-ILND or DSNB) negative → excisional biopsy → negative: observe Bulky (≥4 cm), fixed, bilateral or pelvic nodes → neoadjuvant TIP × 4 → responders: ilioinguinal resection → cisplatin-unfit: chemoradiation`

- **PLND (pelvic lymph node dissection)**: if **more than 2 inguinal nodes or ENE** (pN2–3); ipsilateral; external iliac commonest, then obturator.
- **Neoadjuvant TIP** (paclitaxel, ifosfamide, cisplatin), **Pagliaro (2010)**: objective response 50%; survival mainly in responders. InPACT (International Penile Advanced Cancer Trial) is testing the sequence.

- **Adjuvant**: chemotherapy for pN2–3 (weak evidence); radiotherapy for multiple nodes or ENE (at least 50 Gy; 54 Gy for ENE).

### Inguinal block dissection (ILND): indications

- Penile SCC: **proven cN+**; **cN0 high risk** (pT1b or more, G3) if DSNB is positive or unavailable. Also vulval cancer, melanoma.
- Within 3 months; at least 8 nodes per side. Bulky or fixed nodes: chemotherapy first.

### ILND templates

| Template | Extent |
|---|---|
| **Radical** (cN2 or more) | Superficial + deep; saphenous often ligated; sartorius transposed |
| **Fascia-sparing ILND (FS-ILND)** | May be used for cN1 |
| **Modified (Catalona)** | Smaller field: nothing lateral to femoral artery or below fossa ovalis; **saphenous spared**; no sartorius transposition; positive frozen section: convert to complete (radical) ILND |
| Superficial | Above fascia lata only |

### Technique: open radical ILND

1. Frog-leg position; incision **2 cm below and parallel** to the inguinal ligament.
2. Flaps **deep to Camper's fascia** (preserves the skin capillaries).
3. Boundaries: inguinal ligament, sartorius, adductor longus. Divide saphenous at apex; open fascia lata; clear along the femoral vessels; **never lateral to the femoral artery** (femoral nerve).
4. **Sartorius transposition** over the vessels; suction drains until <30 mL/day.
5. Pelvic dissection if more than 2 inguinal nodes or ENE.

### Videoendoscopic inguinal lymphadenectomy (VEIL)

#### Definition and aim

- **VEIL:** ILND through thigh ports, beneath Scarpa's fascia; laparoscopic (Tobias-Machado) or robotic (**RVEIL**).
- **Aim:** open-surgery template without the groin incision that causes flap necrosis.

#### Indications

- Penile SCC, **impalpable nodes**, intermediate/high risk (pT1b+, G3, LVI).
- **Mobile, non-bulky palpable nodes.**
- **Not for** bulky, fixed (cN3) or skin-involving nodes: these need chemotherapy, then open surgery.
- Also vulval cancer, melanoma.
- DSNB still comes first for impalpable nodes (EAU-ASCO).

#### Technique

- Mark an inverted triangle on the skin **before insufflation**, because gas distorts landmarks.
  `Position: supine frog-leg / low lithotomy → 2 cm camera incision below the femoral triangle apex → finger dissection beneath Scarpa's fascia → two lateral and medial working ports (+ assistant port) → CO2 at 15 mmHg (higher`

`briefly to create space) → develop space up to the inguinal ligament → remove nodal packet off fascia lata, apex to ligament → preserve saphenous vein if possible; clip tributaries → frozen section → deep dissection medial to femoral vein if positive → bag retrieval, suction drain in dependent part`

- **Template:** same as open: superficial ± deep; Daseler's five zones; Cloquet's node is the highest deep node. **Never dissect lateral to the femoral artery** (femoral nerve).

**Outcomes vs open**

| | Open ILND | VEIL/RVEIL |
|---|---|---|
| Wound infection, necrosis, dehiscence | Higher | **Lower** |
| Hospital stay, drain time | Longer | Shorter |
| Operative time | Shorter | About 2 h (slightly longer per side) |
| Nodal yield, recurrence | Reference | **Similar** |
| Evidence | Retrospective | Retrospective, selection-biased |

- Commonest complication of VEIL: **lymphocele**, then lymphoedema and cellulitis.

**Robotic VEIL**

- 3D vision, wristed instruments, ICG (indocyanine green) for lymphatics; adequate dissection on open re-exploration (phase I). No VEIL vs RVEIL comparison.

## Complications of ILND

- Wound infection, **flap necrosis**, dehiscence, seroma, **lymphocele**, **lymphoedema**, DVT/PE (deep vein thrombosis/pulmonary embolism).
- Reduced by transverse incision, correct flap plane, modified template, saphenous sparing, drain removal by volume (<30 mL/day), thromboprophylaxis and minimally invasive access (VEIL); sartorius transposition increases them.

## Follow-up

- Examination 3-monthly for 2 years, then 6-monthly to 5 years (recurrence mostly within 2 years); then self-examination.

## Diagrams to draw

- Femoral triangle with Daseler's zones and Cloquet's node.
- Femoral triangle with VEIL port sites below the apex.

> **Key points to remember:**
> - Nodal status is the strongest prognostic factor; ILND can cure.
> - pT1b or higher: stage groins surgically (DSNB preferred) within 3 months; at least 8 nodes per side.
> - Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
> - PLND if more than 2 nodes or ENE; bulky nodes: neoadjuvant TIP (InPACT testing the sequence).
> - Femoral triangle: ligament, sartorius, adductor longus; flaps deep to Camper's; stay medial to femoral artery.
> - VEIL: same template, fewer wound complications, equal nodal yield; not for bulky, fixed or skin-involving nodes.

## LAQ 5 Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias

P1

**Asked as:**

- Describe the embryology and anatomy of male Urethra. Describe different endoscopic management techniques of treatment of urethral strictures. — *Summer 2019 Paper I Q1 (LAQ)*
- Discuss anatomy of urethra. Classify its congenital anomalies and discuss management of epispadias. — *Practice Set 5 Paper I Q1 (LAQ)* (male urethral anatomy also answered in 2019-Summer Paper 1, Q1; isolated epispadias in 2022-Summer Paper 3, Q3f)

**Type:** LAQ (25 marks)

### Embryology

- **Cloaca** is divided by the **urorectal septum** (weeks 4–7) into the anterior **UGS (urogenital sinus)** and the posterior anorectal canal.
- The UGS has three parts: **vesical** (bladder), **pelvic** (prostatic and membranous urethra) and **phallic** (urethral plate of the penis).
- **Posterior urethra:** from the pelvic UGS. Prostatic buds appear at 10 weeks; the **utricle** (Müllerian remnant) and ejaculatory ducts (Wolffian) open on the verumontanum.
- **Anterior urethra:** Leydig cells make testosterone from about 8 weeks; type 2 5α-reductase converts it to DHT (dihydrotestosterone), which masculinises the genital tubercle. Germ layers: ectoderm gives skin and prepuce, mesoderm the corpora, endoderm the **urethral plate**.
- **Penile urethra: "double zipper"** 1. **Opening zipper:** the urethral plate canalises proximal to distal into a urethral groove between the urethral folds. 2. **Closing zipper:** the folds fuse in the ventral midline, forming the tube and the median raphe. Folds fused by 16 weeks; urethra complete by about 18 weeks.

| Glanular urethra | Ectodermal ingrowth theory (Glenister) | Endodermal theory (current) |
|---|---|---|
| Formed by | Surface ectoderm grows into the glans to meet the endodermal urethra | Direct canalisation of the urethral plate inside the glans |
| Epithelium | Mixed ectoderm and endoderm | Endoderm (UGS) along the whole urethra |

- **Clinical link:** the shaft–glans junction (change of mechanism) is the commonest hypospadias site. **Also know:** urethral duplication; PUV (posterior urethral valves); utricular cyst.

### Anatomy of the male urethra

- About 18–20 cm long in adults; normal diameter 8–9 mm (wide). **Posterior urethra** runs from the bladder neck to the **perineal membrane**; **anterior urethra** from there (perineal membrane) to the meatus, inside the corpus spongiosum.

| Segment | Length / features | Lining |
|---|---|---|
| Prostatic | Length of prostate; urethral crest, **verumontanum** with utricle (Müllerian remnant) and ejaculatory ducts; angle divides preprostatic and prostatic parts | Urothelium |
| Membranous | 2–2.5 cm; shortest and narrowest; striated sphincter | Pseudostratified columnar |
| Bulbar | Widens as the bulb; **Cowper ducts at 3 and 9 o'clock**; lies **eccentric, close to (nearer) the dorsum** of spongiosum | Pseudostratified columnar |
| Penile | About 15 cm; central, uniform calibre; glands of Littré | Pseudostratified columnar |
| Fossa navicularis | Glanular; lacuna magna on the roof | Stratified squamous |

- **Sphincters:** (1) **internal (preprostatic) smooth muscle sphincter** at the bladder neck, autonomic, gives continence at rest; (2) **external striated rhabdosphincter**, horseshoe-shaped around the membranous urethra, supplied by the pudendal nerve (mainly S3, some S2; Onuf nucleus), plus its intrinsic smooth muscle.
- **Blood supply (arteries):** internal pudendal → common penile artery → artery to the bulb (proximal; bulbourethral branches) and **retrograde flow via the dorsal arteries through the glans**. This **dual supply** allows bulbar transection in anastomotic urethroplasty.
- **Venous** (veins) to pudendal and prostatic plexuses; **lymph** to internal and external iliac nodes.
- **Female urethra:** about 4 cm; smooth muscle plus striated sphincter over the distal two-thirds; **Skene glands** (prostate homologue) near the meatus; oestrogen-dependent mucosal seal.

## Endoscopic management of stricture

**Principle:** healing after incision or dilation is by **secondary intention**, a race between epithelialisation and wound contraction. No endoscopic method removes **spongiofibrosis**, so cure is limited to short, superficial bulbar strictures.

| Favourable | Unfavourable |
|---|---|
| Bulbar, **<1.5–2 cm** | Penile, fossa, longer |
| No dense spongiofibrosis | Straddle injury, LS (lichen sclerosus) |
| First procedure | Repeat; luminal narrowing >74% |

### 1. Dilation

- Stretches the scar without tearing it (bleeding means tearing).
- **Safest: balloon dilation over a guidewire** under vision; blind filiforms condemned. Sounds or self-dilation for palliation.

### 2. DVIU (direct vision internal urethrotomy)

1. RUG (retrograde urethrogram) and VCUG (voiding cystourethrogram) define site and length.
2. Guidewire across; **normal saline** irrigation.
3. **Sachse cold knife** at 12 o'clock, through scar to healthy tissue; avoid deep cuts (corpora: ED (erectile dysfunction), fistula).

4. Catheter **3–7 days**; longer adds nothing. - **Laser urethrotomy** (holmium, others): **no advantage over cold knife**. - Results: **Steenkamp and Heyns** (randomised): dilation equals urethrotomy; a second attempt rarely and a third never succeeds. **Santucci (2010):** about 8% stricture-free long term.

**3. Adjuncts against recurrence**

- **CISC (clean intermittent self-catheterisation)** tapering over 3–6 months: delays recurrence, which returns on stopping.
- **Mitomycin C** injection reduces recurrence; steroid injection, colchicine: weak data.

**4. Urethral stents**

- **UroLume** (permanent): **withdrawn** (ingrowth). **Memokath**, removable stents: small series only.

**5. Optilume (paclitaxel-coated balloon)**

- FDA (Food and Drug Administration) 2023. **ROBUST III:** 74.6% vs 26.8% stricture-free at 1 year after failed endoscopy.

**Complications:** recurrence (commonest), bleeding with erections, irrigant extravasation, UTI (urinary tract infection) and sepsis, ED, spongio-cavernous fistula, incontinence.

**Guideline (AUA (American Urological Association) 2016):** urethrotomy or dilation for short (<2 cm) bulbar strictures; **urethroplasty, not repeat endoscopy**, after failure. The reconstructive ladder is abandoned.

## Classification of congenital urethral anomalies

| Group | Anomalies |
|---|---|
| Abnormal meatal position | **Hypospadias** (ventral); **epispadias** (dorsal) |
| Obstructive | **PUV (posterior urethral valves)**, Young types I–III; **anterior urethral valves** (with diverticulum); urethral atresia or hypoplasia (congenital stricture); congenital meatal stenosis |
| Dilatation | **Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); anterior urethral diverticulum; Cowper duct cyst (syringocele); enlarged prostatic utricle |
| Duplication | **Effmann classification**: type I blind incomplete; type II complete (IIA2 "Y-type"); type III with caudal duplication. The ventral channel is the functional one |
| Accessory openings | Congenital urethrocutaneous fistula; parameatal urethral cyst |
| Female | Epispadias; urethral prolapse; persistent urogenital sinus |

## Epispadias: definition and classification

- A **dorsal urethral wall defect**, replaced by a broad dorsal mucosal strip (urethral plate); part of the **exstrophy–epispadias complex**, mostly seen with exstrophy.
- Embryology: an overdeveloped cloacal membrane blocks mesenchymal ingrowth; genital tubercles fuse caudal to it.

| | Male | Female |
|---|---|---|
| Incidence | About 1 in 117,000 | About 1 in 484,000 |

| | Male | Female |
|---|---|---|
| Types | Glanular, penile, **penopubic** (about 70%, incontinent) | **Davis:** vesicular (patulous meatus), subsymphyseal, retrosymphyseal (incontinent) |
| Genitalia | Short, upturned penis with **dorsal chordee** | **Bifid clitoris**, flat glabrous mons |
| VUR (vesicoureteric reflux) | 30–40% | 30–75% |

- The more proximal the meatus, the worse the continence (glanular never incontinent). Pubic diastasis is less than in exstrophy.

## Evaluation

- Renal ultrasound; VCUG (voiding cystourethrogram) for reflux; cystoscopy and **bladder capacity** under anaesthesia.
- **Key decision:** use the exstrophy pathway (osteotomy and closure) if pubic diastasis >3 cm, poor intrasymphyseal bar, absent sphincter tissue or tiny bladder.

## Management

**Aims:** continence with safe upper tracts, and a straight penis of adequate length that works for intercourse.

- **Preoperative testosterone** (2 mg/kg intramuscular, 5 and 2 weeks before).

**Stage 1: urethroplasty and penile reconstruction (6–12 months)**

1. Release dorsal chordee: divide suspensory ligaments; free corpora from the inferior pubic rami.
2. **Modified Cantwell–Ransley:** an 18-mm plate strip is outlined; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; the plate is tubularised over an 8-Fr stent and moved **ventral to the corpora**, which are rotated medially over it (diamond corporotomies if curvature persists).
3. Glans wings closed; skin flaps resurface the dorsum; base Z-plasty; stent 10–12 days. - **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias. - Urethroplasty adds outlet resistance, so the bladder grows before stage 2.

**Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)**

- **Young–Dees–Leadbetter** repair with Marshall–Marchetti–Krantz suspension and **ureteric reimplantation**. Osteotomy is rarely needed.
- **Good capacity at presentation:** single-stage repair is possible.
- **Female:** urethroplasty and genitoplasty (clitoral halves and labia joined, mons reconstructed) at 6–12 months; bladder neck repair at 4–5 years.

## Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula about 1 in 6 (penile base); stricture rare; neourethra catheterisable; erections and paternity preserved.
- Better than exstrophy: the bladder was never exposed or scarred. Failures: injectables, augmentation.

**Diagram to draw:** sagittal male urethra with segments, sphincters and a 12 o'clock urethrotomy; Cantwell–Ransley with plate transposed under the corpora.

**Key points to remember:**

- Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
- Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock; dual retrograde glanular blood supply.
- DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
- Optilume ROBUST III 74.6% vs 26.8%.
- Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
- About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

## LAQ 6 Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy.

P1

**Type:** LAQ (25 marks)
**Source:** 2024-Winter Paper 1, Q1 (related: 2019-Summer Paper 1, Q2; 2020-Summer Paper 1, Q3b)

### Anatomy: definition and boundaries

- The space behind the parietal peritoneum. **Anterior:** posterior peritoneal reflection; **posterior:** posterior abdominal wall; **cranial:** diaphragm; **caudal:** extraperitoneal pelvis.

| Group | Contents |
|---|---|
| Primarily retroperitoneal (formed there) | Kidneys, ureters, adrenals, aorta, IVC (inferior vena cava) |
| Secondarily retroperitoneal (mesentery fused to the wall) | Ascending and descending colon, duodenum D2–D3, pancreas (not the tail) |
| Vessels, lymphatics, nerves | Lumbar nodes, cisterna chyli, sympathetic trunks, autonomic plexuses, lumbosacral plexus |

### Posterior abdominal wall

- **Flank layers:** external oblique, internal oblique, transversus abdominis, transversalis fascia. **Posterior:** psoas, quadratus lumborum, erector spinae.
- **Lumbodorsal (thoracolumbar) fascia:** three layers that fuse into one at the **tip of the 12th rib**, the standard entry point for retroperitoneoscopy.
- **Ribs:** access above the 12th rib risks the pleura; the intercostal bundle runs under the rib above.

### Fasciae and spaces (three embryological strata)

| Space | Boundaries | Contents |
|---|---|---|
| Anterior pararenal | Posterior peritoneum and anterior renal fascia | Colon, duodenum, pancreas |
| Perirenal | Anterior (Toldt) and posterior (Zuckerkandl) laminae of Gerota's fascia | Kidney, adrenal, perirenal fat, hilum, upper ureter, gonadal vessels |
| Posterior pararenal | Posterior lamina and transversalis fascia | Fat only |

- The laminae fuse laterally as the **lateroconal fascia**. The perirenal cone is **open inferiorly**, so urine, blood or pus track into the pelvis.
- Perirenal fat is fine and pale; pararenal fat is coarse and yellow-orange.

### Vessels, lymphatics and nerves

- **Aorta** enters at T12. Unpaired anterior branches supply the gut (coeliac, SMA (superior mesenteric artery), IMA (inferior mesenteric artery)); paired lateral branches supply adrenals, kidneys (L1–L2) and gonads; lumbar arteries go to the wall.

- At the hilum the **renal vein lies anterior to the artery**. The **left renal vein is long** (gonadal, adrenal and lumbar tributaries); the right is short.
- Lymph flows right to left; **cisterna chyli** at L1–L2.
- **Nerves at risk:** sympathetic trunk (medial to psoas), genitofemoral (on psoas), subcostal, iliohypogastric, ilioinguinal (on quadratus); hypogastric plexus (ejaculation).

### Principles of laparoscopic retroperitoneal surgery

1. **No natural cavity:** the working space must be created (finger and balloon dissection) before insufflation.
2. **Psoas is the key landmark:** it forms the floor. Keep it horizontal on screen; the kidney in Gerota's fascia lies anterior, and the peritoneum is pushed medially.
3. **Direct posterior route to the hilum:** the renal artery, posterior to the vein, is met first, allowing early arterial control.
4. **Peritoneum stays intact:** no bowel mobilisation or adhesiolysis; no spill of tumour, urine or pus into the peritoneum.
5. **Small space:** closely spaced ports, limited triangulation, lens smudging.
6. **Gas behaviour:** $CO_2$ (carbon dioxide) absorbs through fat, so hypercarbia and subcutaneous emphysema are commoner; haemodynamic swings are smaller than transperitoneal.

| Preferred for | Limitations |
|---|---|
| Adhesions, peritoneal dialysis, morbid obesity, pregnancy; posterior tumours, pyeloplasty, cyst surgery, renal biopsy | Small space, unfamiliar landmarks, harder suturing, learning curve |

- **Evidence:** randomised trials and meta-analysis: blood loss, complications and oncology similar to transperitoneal; the consistent advantage is **shorter operative time**.

### Salient steps: retroperitoneal laparoscopic nephrectomy

1. **Position:** full flank (90°), table flexed, axillary roll; surgeon behind the patient.
2. **Access (open Hasson):** 1.5–2.5-cm incision just below the 12th rib tip; split muscle, open the lumbodorsal fascia; finger confirms psoas behind and Gerota-covered lower pole in front.
3. **Space creation:** balloon dilator (commercial, or Gaur glove balloon with 600–800 mL saline) anterior to psoas, posterior to Gerota's fascia; blunt-tip balloon-cuff trocar seals the port.
4. **Ports:** two more trocars under vision, one off the 12th rib tip, one posterior and superior to the camera.
5. **Dissection:** along psoas, find the ureter and gonadal vein medially; lift ureter and lower pole to put the hilum on stretch; see arterial pulsation.
6. **Hilum:** clip or staple the **artery first**, then the vein; on the left a **lumbar vein** usually needs division.
7. **Mobilise:** upper pole (adrenal preserved or taken), then lateral and anterior attachments; radical = outside Gerota's fascia.
8. **Extract:** clip ureter, bag the specimen, remove via an extended port; close. - **Problems:** peritoneal tear (space collapses), emphysema, pneumothorax, vascular injury, flank hernia (intramuscular balloon), rhabdomyolysis (about 1%). **Bail-out:** open the peritoneum and convert.

**Diagram to draw:** transverse section showing the three spaces, Gerota's laminae and the balloon anterior to psoas.

> **Key points to remember:**
>
> - Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
> - Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
> - Psoas is the landmark; artery met first.
> - Main advantage: shorter operative time, peritoneum untouched.

## LAQ 7 Classify testicular tumors and describe management of seminoma

P3

**Type:** LAQ (25 marks)
**Source:** Practice Set 1 Paper 3, Q1 (stage I seminoma also answered in 2018-Summer Paper 2, Q2)

### Introduction

- Over 95% of testicular tumours are GCTs (germ cell tumours). Seminoma is 52–56% of GCNIS (germ cell neoplasia in situ)-derived GCTs, peaks in the fourth–fifth decade.
- **Risk factors:** cryptorchidism, previous contralateral GCT (twelvefold), family history, GCNIS, White race, infertility.
- All postpubertal GCTs except spermatocytic tumour carry **isochromosome 12p**.

### WHO (World Health Organization) classification (2016, retained 2022)

| Group | Tumours |
|---|---|
| **1. GCNIS-derived GCTs** | GCNIS; **seminoma**; **NSGCT (non-seminomatous GCT)**: embryonal carcinoma, postpubertal yolk sac tumour, choriocarcinoma and other trophoblastic tumours, postpubertal teratoma, teratoma with somatic-type malignancy; mixed GCT |
| **2. Non-GCNIS-derived GCTs** | **Spermatocytic tumour** (formerly spermatocytic seminoma); prepubertal teratoma, dermoid and epidermoid cyst; prepubertal yolk sac tumour |
| **3. Sex cord–stromal tumours** (<5%) | Leydig cell, Sertoli cell (including large cell calcifying), granulosa cell (adult, juvenile), fibroma–thecoma, mixed |
| **4. Mixed germ cell–sex cord stromal** | Gonadoblastoma (DSD [difference of sex development]) |
| **5. Others** | Lymphoma (commonest testicular tumour in older men), rete testis and collecting duct tumours, adnexal and paratesticular tumours, metastases |

- Any NSGCT element, however small, makes the tumour an NSGCT and is treated as such.
- **Seminoma pathology:** sheets of clear cells with lymphocyte-rich fibrovascular septa. Immunohistochemistry: CD117 and PLAP (placental alkaline phosphatase) positive, CD30 negative. Syncytiotrophoblasts (hCG [human chorionic gonadotropin]) in about 15%. "Anaplastic seminoma" is no longer recognised.
- **Spermatocytic tumour** (<1%): older men, no GCNIS, no i(12p), benign unless sarcomatous.

### Evaluation and staging

- Scrotal ultrasound; AFP (alpha-fetoprotein), β-hCG and LDH (lactate dehydrogenase) before and after orchidectomy. **A raised AFP means NSGCT**, whatever the histology.
- **Radical inguinal orchidectomy** with early cord control at the internal ring; never trans-scrotal. Offer sperm banking and prosthesis.
- CT (computed tomography) of chest, abdomen and pelvis; TNMS (tumour, node, metastasis, serum markers) staging.

- About 80% of seminomas are clinical stage (CS) I; 15–20% CS II (nodes IIA ≤2 cm, IIB 2–5 cm, IIC >5 cm).

### Management of seminoma by stage

**CS I** (cure by orchidectomy alone 80–85%; cancer-specific survival about 100% with any option)

| | **Surveillance (preferred)** | **Carboplatin** | **Radiotherapy** |
|---|---|---|---|
| Method | Examination, markers, CT for at least 5 years | 1 cycle **AUC (area under the curve) 7**, dose = 7 × (GFR [glomerular filtration rate] + 25) mg | **20 Gy** para-aortic strip |
| Relapse | 14–20% | About 5% (1 cycle) | 3–5% |
| Drawback | Late relapses (10–20% after 4 years) | Underdosing; late effects unknown | Second cancers, GI (gastrointestinal) toxicity |

- **Relapse risk factors:** tumour **>4 cm** and **rete testis invasion** (relapse 12%, 16%, 32% with 0, 1, 2 factors). LVI (lymphovascular invasion) is not a factor in seminoma.
- **Trials:** MRC TE19/EORTC (Medical Research Council/European Organisation for Research and Treatment of Cancer) 30982: one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE18: 20 Gy equals 30 Gy.
- Surveillance relapses are 84–100% retroperitoneal and almost all salvaged.

**CS IIA–B**

- **Non-bulky (≤3 cm):** dog-leg radiotherapy with nodal boost (25–30 Gy in the chapter; EAU [European Association of Urology] uses 30 Gy for IIA and 36 Gy for IIB), or chemotherapy, or **primary RPLND (retroperitoneal lymph node dissection)** (SEMS trial).
- **Bulky (>3 cm) or multiple nodes:** BEP (bleomycin, etoposide, cisplatin) ×3 or EP (etoposide, cisplatin) ×4.

**CS IIC–III: chemotherapy by IGCCCG (International Germ Cell Cancer Collaborative Group)**

| **Risk** | **Seminoma definition** | **5-year survival** | **Regimen** |
|---|---|---|---|
| Good (90%) | Any primary, **no NPVM (non-pulmonary visceral metastases)**, normal AFP, any hCG/LDH | 86% | BEP ×3 or EP ×4 |
| Intermediate (10%) | NPVM present | 72% | BEP ×4 |
| Poor | **None for seminoma** | – | – |
| - Carboplatin alone is inferior in advanced disease. | | | |

### Post-chemotherapy residual mass

Residual mass after chemotherapy ├ <3 cm ——————► observe (necrosis about 90%) └ >3 cm ► FDG-PET (fluorodeoxyglucose positron emission tomography) at ≥6 weeks ├ negative ► observe └ positive ► resection (full template PC-RPLND) or salvage

- Surgery is difficult because of desmoplasia; complete resection only 58–74%.

### Relapse and follow-up

- After surveillance or radiotherapy: radiotherapy for small retroperitoneal relapse, otherwise first-line BEP. After chemotherapy: VeIP (vinblastine, ifosfamide, cisplatin) or high-dose chemotherapy.
- Surveillance schedule: review every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly. CT detects >85% of relapses.

### Recent advances

- Serum **miR-371a-3p** microRNA: sensitive marker of residual disease and relapse.
- MRI (magnetic resonance imaging) for surveillance to cut radiation; primary RPLND for low-volume stage II.

**Diagram to draw:** WHO classification tree (GCNIS-derived vs non-GCNIS-derived vs sex cord–stromal) and the residual-mass flowchart.

> **Key points to remember:**
> - WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
> - CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
> - Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
> - No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

## Short answer questions (SAQ)

## SAQ 1 Surgical management of primary scrotal cancer

P1 · P4 **merged from 2**

*Asked as:* P1 SAQ "Surgical management of primary scrotal cancer" · P4 SAQ "Surgical management of primary scrotal cancer."

### Introduction and aetiology

- Rare. **SCC (squamous cell carcinoma)** is commonest. **Percivall Pott (1775)** linked it to chimney soot in chimney sweeps: the first occupational cancer.
- **Risks:** soot, tar, mineral oils, arsenic, PUVA (psoralen + ultraviolet A), HPV (human papillomavirus), chronic inflammation.
- **Other types:** EMPD (extramammary Paget disease), BCC (basal cell carcinoma), melanoma, sarcoma.
- Lymph drains to **superficial inguinal nodes, crossing the midline**: both groins at risk.

### Evaluation

- Painless nodule or ulcer on the anterolateral scrotum, often presenting late; **biopsy** confirms.
- Examine both groins (lymphatics cross the midline); FNAC (fine-needle aspiration cytology) of palpable nodes; CT (computed tomography) of abdomen and pelvis for staging.

### Staging (Lowe modification of Ray and Whitmore)

| Stage | Extent |
|---|---|
| A1 | Confined to scrotum |
| A2 | Invades testis, cord, penis, pubis or perineum |
| B | Inguinal nodes only |
| C | Pelvic nodes, no distant spread |
| D | Distant metastases |

### Management of the primary

1. **Wide local excision with a 2 cm margin**, full thickness including dartos; frozen sections for margins.
2. Testes and cord preserved unless invaded; hemi- or total scrotectomy for large tumours.
3. Stage A2: en bloc removal of involved testis, penis, perineum or pubis.
4. Positive margin: re-excise.

### Reconstruction (Chapter 87)

- Scrotal laxity usually allows primary closure.

| Defect | Management |
|---|---|
| <50%, no tension | Primary closure |
| <50%, tension | Scrotal advancement flap or secondary intention |
| >50% | Testes in thigh pockets (pouches); local or thigh flaps; later STSG (split-thickness skin graft) or flap |

### Management of nodes (extrapolated from penile cancer)

- **Palpable, cytology positive:** ILND (inguinal lymph node dissection) ± pelvic dissection. Nodal status is the main prognostic factor.
- **Impalpable:** surveillance; dynamic sentinel node biopsy or modified ILND for high-grade/invasive tumours.
- **Bulky or fixed:** neoadjuvant cisplatin-based chemotherapy, then surgery; RT (radiotherapy) for palliation.
- Routine prophylactic dissection is not advised.

### Other histologies

- **EMPD:** wide excision (up to 3 cm) guided by **mapping biopsies**; screen for associated GU (genitourinary) and other cancers.
- **Also know:** BCC, excision or Mohs surgery; melanoma, excision by Breslow depth ± sentinel node; sarcoma, wide excision.

**Diagram to draw:** elliptical wide excision with 2 cm margin and bilateral inguinal node drainage.

> **Key points to remember:**
>
> - Pott 1775: chimney sweeps, first occupational cancer.
> - Wide local excision, 2 cm margin; testes spared unless invaded.
> - Nodes: managed as in penile cancer; nodes decide prognosis.

## SAQ 2 Varicocele: pathophysiology, controversies and surgical management

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 SAQ "Varicocele: controversies and present status of treatment" (asked as "Controversies in management of varicocele" and "Present status of varicocele treatment") · P1 SAQ "Varicocele – Pathophysiology" · P2 SAQ "Surgical management of bilateral varicocele"

### Definition and epidemiology

- Abnormal dilatation and tortuosity of the **pampiniform plexus** of the internal spermatic veins.
- About **15% of men, 35% with primary and 75–81% with secondary infertility** (P3: up to 80%). Appears after age 10, peaks at Tanner 3. About 90% left-sided.
- With sensitive tests most are bilateral; the clinical significance of subclinical ones is uncertain.
- **Venous anatomy:** anterior (internal spermatic), middle (deferential) and posterior (cremasteric) groups; the last two give return after repair and cause recurrence.

### Aetiology: why it develops

1. **Left-sided anatomy:** the left vein enters the left renal vein at a right angle, 8–10 cm higher than the right enters the IVC (inferior vena cava). The taller column overloads its valves. The right vein enters obliquely, which acts as a valve.
2. **Valve incompetence** or absence; continuous reflux predicts progression.
3. **Nutcracker:** left renal vein compressed between aorta and SMA (superior mesenteric artery).
4. **Collateral pathways**; **genetic** (4–8 times commoner in first-degree relatives); tall, thin habitus. - **Secondary:** an isolated **right-sided or non-decompressing** varicocele (one that does not decompress lying down) suggests a retroperitoneal or renal mass (tumour), so image the abdomen.

### Pathophysiology: how it damages the testis

| Mechanism | Explanation |
|---|---|
| **Heat (primary)** | Lost countercurrent heat exchange; the testis normally runs 2–4 °C below core |
| Oxidative stress | Raised reactive oxygen species cause sperm DNA fragmentation |
| Hypoperfusion and stasis | Hypoxia, toxin accumulation |
| Apoptosis | Germ cell loss |
| Leydig dysfunction | Lower testosterone than controls |

- The injury is **progressive and duration-dependent** (atrophy), and may affect **both** testes.

## Consequences

- **Testicular hypotrophy** (asymmetry above about 20% guides adolescent repair); poor semen; hypogonadism (lower testosterone); ache on standing (2–11%).
- Repair improves semen, DNA fragmentation, testosterone and pregnancy.

## Grading

| Grade | Finding |
|---|---|
| Subclinical | Doppler only (not treated) |
| I | Palpable with Valsalva only |
| II | Palpable, not visible |
| III | Visible |

## Controversies and present status of treatment (indications)

| Question / situation | Evidence | Current position |
|---|---|---|
| **Does repair improve fertility?** Infertile couple, **palpable** varicocele, abnormal semen | Madgar RCT (randomised controlled trial, 1995): pregnancy 44% versus 10% at 1 year; meta-analyses mixed | **Repair clinical** varicocele with **abnormal semen** in an infertile couple with normal female partner (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine]) |
| **Subclinical varicocele?** | No benefit for semen or pregnancy | **Do not treat** |
| **Normal semen?** | No proven benefit | Not for fertility |
| **Adolescent varicocele** | Testis volume is a surrogate; link to adult fertility unproven | Observe; repair for **testis asymmetry (hypotrophy) above 20%**, pain, or abnormal semen (Tanner 5 or 18 years) |
| **NOA (non-obstructive azoospermia)** | Small series: sperm returns to ejaculate in some | No definitive evidence; may be considered before micro-TESE (microsurgical testicular sperm extraction) |
| **Before IVF/ICSI** (in vitro fertilisation/intracytoplasmic sperm injection) | Repair may improve live-birth rate or allow less invasive ART (assisted reproductive technology) | Discuss, especially with oligozoospermia |
| **High DNA fragmentation** | Repair lowers fragmentation | Reasonable indication |
| **Hypogonadism** (low testosterone) | Repair raises testosterone in hypogonadal men | Emerging indication |
| **Pain** | Dull ache on standing; relief in most after microsurgery | Repair after other causes excluded |

- **Bilateral disease: do not repair subclinical** sides; repair each palpable side.

## Choice of technique

| Technique | Recurrence | Hydrocele | Comment / bilateral use |
|---|---|---|---|
| **Microsurgical subinguinal/inguinal** | Lowest, **0.5–1%** | **About 0** | **Best, preferred**: artery and lymphatics preserved; two small incisions, local or regional anaesthesia |
| Retroperitoneal (Palomo) | High with artery spared (15–25%) | High with mass ligation (about 7%) | Children (artery ligation acceptable) |
| Laparoscopic | 3–15% | Up to 12% | Both sides through same ports; general anaesthesia; serious morbidity possible |
| Embolisation / sclerotherapy | 4–25% (P2: 15–25%) | Nil | Outpatient; femoral (left) or jugular (right) access; technical failure |

## Microsurgical steps (each side)

1. Subinguinal incision over the external ring; deliver the cord.
2. Microscope and **micro-Doppler** to map the testicular artery; papaverine for spasm.
3. Ligate all internal and external spermatic veins (all veins except vasal); preserve **artery, lymphatics, vas and one set of vasal veins**.
4. Deliver the testis to ligate gubernacular and external spermatic veins (lower recurrence).
5. **Impulse test** confirms no missed vein.

## Bilateral varicocele: specific points

- **Never ligate both testicular arteries**: in adults this has caused azoospermia and atrophy. So artery-sparing microsurgery is safest bilaterally.
- Image for a renal or retroperitoneal mass if a varicocele is isolated right-sided or non-decompressing.
- Repair any coexisting hydrocele.

## Complications and outcome

- **Hydrocele** (lymphatic ligation; commonest after non-microsurgical repair); testicular artery injury (atrophy); recurrence (diagnose by upright colour Doppler showing reflux).
- **Predictors of poor result:** recurrence, artery ligation, hydrocele. Earlier repair recovers more.

## Diagrams to draw

- Left and right testicular veins draining into renal vein and IVC.
- Spermatic cord cross-section at microsurgery: artery, lymphatics and vas preserved, veins ligated.

**Key points to remember:**

- Left vein: right-angled entry, 8–10 cm higher column; 90% left.
- Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage; injury is progressive.
- Right or non-decompressing varicocele: image for a renal mass.
- Repair only palpable varicocele (each palpable side) with abnormal semen; never subclinical.
- Madgar 1995: 44% versus 10% pregnancy.
- Adolescents: observe; repair for asymmetry above 20%.
- Microsurgical subinguinal: lowest recurrence (under 1%) and hydrocele (nil).
- Never ligate both testicular arteries.

## SAQ 3 Meatal stenosis and submeatal stricture in males

P1 · P2 **merged from 2**

*Asked as:* P2 SAQ "Management of Urethral meatal stenosis." · P1 SAQ "Treatment of submeatal stricture urethra in males"

### Definition

- **Meatal stenosis:** narrowing of the external urethral meatus below the normal calibre for age.
- **"Submeatal stricture"** is read as a stricture of the meatus and fossa navicularis (glanular urethra). In adults, meatal stenosis is usually the tip of a **fossa navicularis stricture**.

| Age | Normal calibre |
|---|---|
| Under 4 months | 6–10 Fr |
| 1–12 months | 10–12 Fr |
| 1–10 years | 12–14 Fr |
| Over 10 years | 14–18 Fr |

### Causes

- **Children:** almost always after **neonatal circumcision**: wet nappies irritate the exposed meatus (ammoniacal meatitis) and the ventral lips fuse into a thin membrane. Rarely congenital with hypospadias.
- **LS (lichen sclerosus, formerly BXO [balanitis xerotica obliterans]):** main inflammatory cause in adults today; begins at the glans and meatus and progresses proximally.
- **Iatrogenic:** catheter or instrument trauma, TURP (transurethral resection of prostate).
- **Failed hypospadias repair.**

### Clinical features

- Thin, forceful stream deflected **upwards**; spraying; straining, frequency, incomplete emptying; pain at the start of voiding.

## Evaluation

- Child: watch the child void; calibrate with a bougie à boule. Uroflowmetry. Urethrography only for UTI (urinary tract infection) or suspected proximal lesion.
- Adult: uroflowmetry; **RUG (retrograde urethrogram) before any meatal surgery**, as a proximal stricture often coexists, especially in LS.
- Biopsy suspicious or atypical lesions: LS turns malignant (squamous cell carcinoma) in up to 8%.

## Treatment

| Option | Method | Indication / comment |
|---|---|---|
| **Topical steroid** | Clobetasol 0.05% once or twice daily for 1–2 months, then alternate days; gentle dilation and nappy hygiene in children | Meatitis and early narrowing in a child; early LS |
| **Dilation or self-calibration** | Meatal dilator lubricated with clobetasol | Elderly, unfit; seldom durable |
| **Ventral meatotomy (child)** | EMLA (eutectic mixture of local anaesthetics) cream in clinic, or general anaesthesia; parents spread the edges with a dilator tip 3 times daily for 7–10 days | Symptomatic stenosis in a child |
| **Ventral meatotomy (adult)** | Crush and incise ventrally; **three sutures** (apex and each side), mucosa to glans skin; dilate afterwards | Simple stenosis; in LS extend it with an inverted-V relaxing incision (extended meatotomy) |
| **Dorsal meatotomy** (Y-V plasty) | Excise the scar ridge, advance a flap | Scarred or recurrent; effective but cosmesis poorer |
| **Fossa navicularis reconstruction (flaps)** | **Jordan flap:** ventral urethrotomy through the stenosis; **transverse ventral penile skin island on dartos** transposed and inverted into the defect. **De Sy:** ventral longitudinal island advanced | Fossa navicularis involved, non-LS; excellent function and cosmesis; not in LS |
| **Graft** | **BMG (buccal mucosal graft)** inlay or onlay, one or two stages; glans wings developed | **LS: oral mucosa, never genital skin** |
| **Perineal urethrostomy** | Bypass | Pan-anterior LS, failed repairs, elderly |

- LS affects genital skin, so skin flaps recur; it spares the proximal urethra, so perineal urethrostomy works.

## Complications

- Bleeding, restenosis (commoner in LS), spraying, poor cosmesis.

## Diagrams to draw

- Ventral meatotomy: crush, incise ventrally, suture mucosa to glans at apex and both sides.
- Jordan flap: transverse ventral skin island on dartos inverted into the ventral meatotomy.

**Key points to remember:**

- Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
- Adult: RUG first; adult meatal stenosis is usually the tip of a fossa navicularis stricture.
- Non-LS: meatotomy or Jordan/De Sy skin island flap.
- LS: clobetasol, buccal mucosa (never genital skin), perineal urethrostomy if extensive.

## SAQ 4 Perineal urethrostomy and eponymous urethroplasty techniques

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 SAQ "Eponymous urethroplasty techniques and perineal urethrostomy" · P1 SAQ "Perineal urethrostomy." · P2 SAQ "Perineal urethrostomy."

### Introduction: urethroplasty options

- Options: **EPA (excision and primary anastomosis)**, the gold standard for short bulbar strictures; **graft or flap onlay**; **staged repair**.
- **Principles:** grafts need a vascular bed; onlay beats tube; graft equals flap.

### Eponymous urethroplasty techniques

| Eponym | Technique | Use |
|---|---|---|
| **Johanson** | Two-stage: stricture laid open, edges sutured to skin (urethrostomy); tubularised later | LS (lichen sclerosus), failed hypospadias, infection |
| **Blandy** | Inverted-U scrotal or perineal skin flap into a ventral urethrotomy | Bulbar strictures; classic perineal urethrostomy (PU) |
| **Turner-Warwick** | Two-stage scrotal inlay; transpubic approach, omental wrap | Complex, posterior |
| **Webster** | Progressive perineal approach: mobilisation, crural separation, inferior pubectomy, rerouting | PFUI (pelvic fracture urethral injury) |
| **Barbagli** | **Dorsal onlay** BMG (buccal mucosal graft) spread-fixed to corpora | Bulbar |
| **Asopa** | **Dorsal inlay** graft through a ventral urethrotomy, no mobilisation | Penile and bulbar |
| **Kulkarni** | One-sided dorsolateral onlay; panurethral repair via perineum with penile invagination | Panurethral (LS) |
| **Orandi** | Ventral longitudinal penile skin island flap on dartos | Pendulous |
| **Quartey** | Showed skin island is a passenger on a fascial (dartos) flap | Basis of all genital skin flaps |
| **McAninch** | Circular fasciocutaneous penile skin flap | Long anterior strictures (up to 15 cm) |
| **Jordan** | Vessel-sparing EPA; ventral transverse island for fossa | Proximal bulbar; fossa navicularis |

- **Also know:** Monseur dorsal urethrotomy; Andrich–Mundy non-transecting anastomosis; Bracka two-stage BMG.

- **Diagram to draw:** bulbar cross-section showing dorsal onlay (Barbagli), dorsal inlay (Asopa) and ventral onlay.

### Perineal urethrostomy: definition

- PU (perineal urethrostomy): the proximal bulbar urethra is opened onto perineal skin as a permanent or staged new meatus; the patient voids sitting. It **bypasses**, rather than reconstructs, a diseased anterior urethra.

### Indications

- **Complex anterior strictures:** LS panurethral disease, multiple failed urethroplasties, failed hypospadias repair.
- **Elderly or high-risk men** unwilling or unfit for staged reconstruction.
- **First stage** of a staged (Johanson-type) urethroplasty.
- After **penectomy** or distal urethrectomy for penile or urethral cancer.
- Temporary: stricture after phalloplasty; Fournier gangrene; resectoscope access in morbid obesity.
- AUA (American Urological Association) guideline (2016): PU is an acceptable long-term alternative to urethroplasty.
- LS spares the proximal urethra, so PU suits LS and is easy in LS.

### Contraindications and limits

- Disease reaching the proximal bulbar or membranous urethra; prior pelvic radiation and LS raise restricture.

### Techniques

**1. Classical (Blandy) PU:** inverted-U perineal skin flap (a random-pattern advancement flap) advanced into a ventral bulbar urethrotomy (the opened bulbar urethra).

**2. BMG PU, steps:**

1. High lithotomy; inverted-U incision; split (divide) bulbospongiosus in the midline.
2. Free the spongiosum and **transect** it; **dorsal urethrotomy** into healthy mucosa.
3. Bring the healthy ventral urethra to perineal skin; calibrate proximally to **30 Fr**; close the distal urethra.
4. Quilt a BMG on the corpora to complete the dorsal wall up to the skin. - The graft prevents the circumferential scar that narrows a deep PU.

**Diagram to draw:** sagittal perineum with the inverted-U flap sutured into the opened bulbar urethra.

### Complications and outcomes

- **Stenosis of the urethrostomy** (main failure): revision or periodic dilation.
- Spraying, dribbling, perineal wetness, wound infection, body-image concerns.
- Compared with staged reconstruction it is a shorter, single operation with less morbidity.
- High mid-term success and satisfaction in selected patients.

**Key points to remember:**

- Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
- Orandi and McAninch flaps; Quartey flap anatomy.
- Johanson staged; Blandy inverted-U PU.
- PU bypasses, rather than reconstructs, the diseased anterior urethra; patient voids sitting.
- Ideal for LS, failed repairs and elderly patients.
- BMG PU prevents circumferential scarring; calibrate to 30 Fr.

## SAQ 5 Minimally invasive and robotic RPLND

P3 · P4 **merged from 2**

*Asked as:* P3 SAQ "Minimal invasive retroperitoneal lymph node dissection" · P4 SAQ "Robotic RPLND"

### Introduction

- **MI-RPLND (minimally invasive retroperitoneal lymph node dissection)** for testicular GCT (germ cell tumour): laparoscopic (1992) or robotic (2006). **It must duplicate open RPLND exactly.**

### Indications

- **Primary:** marker-negative stage I NSGCT (non-seminomatous GCT), especially high-risk IB; small-volume stage II NSGCT or seminoma (1–2 nodes ≤3 cm in the landing zone).
- **Postchemotherapy:** only a small solitary residual mass with little desmoplasia; **open** for large masses or vessel involvement.
- **Expanding indication:** primary RPLND for small-volume stage II seminoma to avoid chemotherapy and radiotherapy (SEMS, PRIMETEST).
- **Patient selection:** serum miR-371a-3p (microRNA) may select stage I patients with occult nodes for RPLND.

### Templates and nerve sparing

- Boundaries: renal hila, ureters, ureter crossing the common iliacs; plus spermatic cord.
- **Primary:** extended **unilateral modified template** with nerve sparing (right: paracaval, precaval, interaortocaval; left: para-aortic, preaortic; both take nodes above the IMA (inferior mesenteric artery)). If nodes are enlarged, the dissection becomes bilateral.
- **Postchemotherapy:** bilateral full template.
- **Nerve sparing:** identify the sympathetic chains and L1–L4 postganglionic fibres to the hypogastric plexus below the IMA.
- **Diagram to draw:** right and left modified templates.

### Steps (robotic)

1. Positioning: **modified lateral decubitus** (unilateral) or **supine steep Trendelenburg with peritoneal hammock** (any template).
2. Mobilise colon; Kocherise duodenum (right); divide gonadal vein.
3. **Split-and-roll:** split over the great vessel from common iliac to renal vein; roll packets off.
4. **Divide lumbar vessels** to clear retrocaval and retroaortic tissue.

5. Clip lymphatics (left renal vein); bag each packet.

| Feature | Laparoscopic | Robotic |
|---|---|---|
| Retraction | Assistant | Fourth arm |
| Instruments | Straight | Wristed |
| Suturing, vessel control | Difficult | Easier |

### Outcomes

- Stay 1–3 days, blood loss mostly ≤200 mL; conversion about 5%; **antegrade ejaculation ≥90%** (robotic nerve sparing after primary surgery); chylous ascites as open. Node yields approach open (about 30).
- **Major bleeding:** pressure, raise pneumoperitoneum, convert if uncontrolled (open set ready).

### Concerns and guidelines

- Concerns: staging-only dissections, frequent adjuvant chemotherapy, unindicated surgery, catastrophic bleeding, follow-up <5 years; no randomised trials.
- **Guidelines:** AUA (American Urological Association): experienced surgeons may offer it; EAU (European Association of Urology): specialist centres only.

### Recent advances

- Serum miR-371a-3p for selection of stage I patients with occult nodes.
- RPLND for small-volume stage II seminoma (SEMS, PRIMETEST) to avoid chemotherapy and radiotherapy.

> **Key points to remember:**
> - Same template, split-and-roll, lumbar division and nerve sparing as open.
> - Best for chemotherapy-naive low-volume disease.
> - Conversion about 5%; ejaculation preserved ≥90%.
> - Expanding indication: primary RPLND for small-volume stage II seminoma.
> - Serum miR-371a-3p may select stage I patients with occult nodes.

## SAQ 6 Testicular microlithiasis

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 1, Q3a

### Definition

- **TM (testicular microlithiasis):** multiple small (<3 mm) bright echogenic foci **without acoustic shadowing** scattered in the testicular parenchyma on ultrasound.
- **Classic TM:** ≥5 microliths per testis (or per image); **limited TM:** <5, of no clinical significance.

### Pathology

- Laminated calcified bodies **inside seminiferous tubules**, formed on degenerated tubular cells that Sertoli cells fail to clear. Affected tubules are often obliterated.

### Prevalence

| Population | Prevalence |
|---|---|
| General population | 5–10% (population-based 4.2%, rising with age) |
| Multicentre review, >37,000 males | 2.9% |
| Boys with cryptorchidism | 3.5% |
| Previously cryptorchid testis vs contralateral | 14% vs 19% (no difference) |

### Associations

- Cryptorchidism, subfertility, **GCNIS (germ cell neoplasia in situ)** and TGCT (testicular germ cell tumour), Klinefelter syndrome, Down syndrome.

### Cancer significance

- Tumours were found in 4.64% with TM vs 0.33% without, but cryptorchidism was not accounted for.
- **TM is not clearly an independent risk factor** for TGCT. An EAU (European Association of Urology) paediatric panel review of several hundred children found only one cancer through puberty.
- **Exception:** contralateral TM in a man with previous GCT (germ cell tumour) signals a higher GCNIS risk.

### Management

- **Isolated TM, no risk factors:** reassure; **no further evaluation, no routine ultrasound follow-up**.
- **TM with risk factors** (previous GCT, cryptorchidism, infertility, atrophic testis): teach **monthly self-examination from puberty**; ultrasound only for a palpable change.
- **Biopsy:** not for TM alone. Contralateral biopsy at orchidectomy is considered when GCNIS risk is high: atrophic testis (<12 mL), cryptorchidism, age <40.
- Counsel carefully: incidental TM causes anxiety and unnecessary tests.

### Differential diagnosis on ultrasound

- Coarse calcification, "burned-out" tumour scar, stellate scar (these need orchidectomy if intratesticular), scrotal pearls, rete testis ectasia.

**Also know:** routine screening ultrasound for infertility is discouraged because of incidental TM; follow-up schedules remain undefined.

**Diagram to draw:** ultrasound sketch of a testis with scattered non-shadowing echogenic dots.

> **Key points to remember:**
> - Classic TM ≥5 microliths; limited <5.
> - Not an independent cancer risk factor in isolation.
> - Self-examination only with risk factors; no routine imaging.

## SAQ 7 Growing teratoma syndrome

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 20 SAQ Sheet, Q3a

### Definition

- GTS (growing teratoma syndrome; Logothetis, 1982): metastatic masses of an NSGCT (non-seminomatous germ cell tumour) **enlarge during or after chemotherapy while AFP (alpha-fetoprotein) and hCG (human chorionic gonadotrophin) are normal or falling**. Resection shows **mature teratoma only**.
- Uncommon: about 2–8% of metastatic NSGCT.

### Pathogenesis

- Teratoma is **histologically benign but genetically malignant** (i(12p), aneuploidy) and **chemoresistant**.
- Chemotherapy kills the sensitive elements (embryonal, yolk sac), so markers fall; the teratoma keeps growing, mostly by enlarging cysts. Chemotherapy-induced differentiation is a second theory.
- Main risk factor: teratoma in the orchidectomy specimen.

### Clinical features

- Retroperitoneum commonest; also lung, mediastinum, supraclavicular nodes.
- Often silent; or compression: ureteric obstruction, IVC (inferior vena cava) compression or thrombosis, bowel obstruction, back pain.

### Evaluation

- Markers before every cycle; **rising markers mean viable cancer, not GTS**.
- CT (computed tomography): enlarging, well-defined, **cystic** mass, sometimes with fat or calcification.
- **FDG-PET (fluorodeoxyglucose positron emission tomography) has no role**: teratoma is not FDG-avid.
- **Differential:** progressing viable GCT (germ cell tumour; markers rise); malignant transformation (somatic sarcoma or carcinoma); necrosis or fibrosis (mass shrinks).

### Management

- **Stop chemotherapy**; do not escalate to salvage chemotherapy (ineffective).
- **Prompt complete surgical resection**: PC-RPLND (post-chemotherapy retroperitoneal lymph node dissection), with resection of every other site; adjacent vessel, kidney or bowel resection may be needed. Delay lets the mass become unresectable.
- Radiotherapy is ineffective.
- **Outcome:** favourable after complete resection (long-term survival 75–90% after resection of residual teratoma). Incomplete resection leads to regrowth, **malignant transformation** (rhabdomyosarcoma commonest) or **late relapse**.
- **Also know:** CDK4/6 (cyclin-dependent kinase 4/6) inhibitors studied for unresectable teratoma; lifelong CT and marker follow-up.

**Diagram to draw:** graph of mass size rising while marker curve falls during chemotherapy cycles.

> **Key points to remember:**
> - Enlarging mass plus normal or falling markers equals GTS.
> - Teratoma is chemoresistant and not FDG-avid.
> - Treatment is complete surgical resection, not more chemotherapy.

## SAQ 8 Prognostic index score in carcinoma penis

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 3, Q3e

*Assumed meaning: prognostic factors, risk-group systems and nomograms in penile SCC (squamous cell carcinoma). No single universally used "prognostic index score" exists.*

### Introduction

- Prognosis rests on primary histology and stage and, above all, **inguinal node status**.

### 1. Nodal status (most important factor)

| Nodal finding | 5-year survival |
|---|---|
| Node negative | About 73% |
| ≤2 nodes, unilateral, no ENE (extranodal extension) | About 80% |
| >2 nodes | 0–50% |
| **ENE** or **pelvic nodes** | Poor (pelvic about 14%) |

- TNM (tumour-node-metastasis) 8th edition: pN1 (1–2 unilateral nodes) about 90% 3-year DSS (disease-specific survival); pN3 (ENE or pelvic) about one-third.

### 2. Primary tumour factors (predict occult nodes in cN0)

- **Depth:** corporal invasion → about 60% node positive.
- **Grade:** any grade III component = grade III.
- **LVI (lymphovascular invasion)** and **PNI (perineural invasion)**: strongest predictors; define pT1b.
- **Subtype:** verrucous and PeIN (penile intraepithelial neoplasia) do not metastasise; basaloid and sarcomatoid aggressive.

### 3. Risk-group systems

| System | Low | Intermediate | High |
|---|---|---|---|
| **Earlier EAU (European Association of Urology), after Solsona** | Tis, Ta, T1 G1 | T1 G2 | T2 or more, or G3 |
| **Current EAU–ASCO (American Society of Clinical Oncology) / TNM** | PeIN, Ta, **T1a** (nodes 0–18%) | — | **T1b or more** (one-third to over half) |

- Low risk → groin surveillance; high risk → **DSNB (dynamic sentinel node biopsy)** or ILND (inguinal lymph node dissection).

### 4. Nomograms and indices

- **Ficarra nomogram:** predicts nodal involvement (thickness, growth pattern, grade, LVI, corporal invasion, clinical node status).

- **Kattan nomogram:** predicts cancer-specific survival.
- A pathological **prognostic index** (grade + anatomical level of invasion + PNI; Cubilla group) has been proposed.
- **Status:** none externally validated; not used routinely.

### 5. Molecular markers

- **HPV (human papillomavirus)/p16 positive:** better DSS (not node status).
- Adverse: p53, PD-L1 (programmed death-ligand 1), Ki-67, high **NLR** (neutrophil-to-lymphocyte ratio), raised CRP (C-reactive protein).

### 6. Other

- Inguinal recurrence after observation: only about one-third survive 5 years. Size >5 cm worse.

**Diagram to draw:** flowchart: primary pathology → low or high risk → surveillance or DSNB/ILND.

> **Key points to remember:**
> - Node status is the strongest prognostic factor.
> - LVI, PNI, grade III define high-risk T1b.
> - Nomograms (Ficarra, Kattan) unvalidated; p16 favourable.

## SAQ 9 Testis sparing surgeries for CA testis

P4

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 4, Q3b

### Definition and rationale

- **Partial orchidectomy**: inguinal excision of the tumour, preserving the rest of the testis. The standard remains **radical inguinal orchidectomy**; sparing aims to keep androgen production and avoid lifelong testosterone.

### Indications

**Suspected germ cell tumour (GCT), all of:**

- Synchronous bilateral tumours or tumour in a **solitary testis**.
- Tumour **≤2 cm** (up to 2–3 cm, under 30% of testicular volume), preferably polar.
- Adequate preoperative testosterone.
- Not with a normal contralateral testis.

**Suspected benign or indeterminate lesions:**

- Small impalpable lesion (<10 mm) with normal markers: most benign (cyst, infarct, Leydig nodule), but 20–50% are small seminomas. Lesions <5 mm are rarely malignant.
- **Prepubertal**: teratoma and epidermoid cyst with normal AFP (alpha-fetoprotein): partial orchidectomy is the standard.

### Preoperative work-up

- Ultrasound of both testes; AFP, beta-hCG (human chorionic gonadotropin), LDH (lactate dehydrogenase); testosterone, LH (luteinising hormone); **sperm banking**; consent for

radical orchidectomy.

### Technique

1. Inguinal incision; early cord control with a soft clamp.
2. Deliver the testis; isolate with towels.
3. **Intraoperative ultrasound** localises the lesion.
4. Cold ischaemia optional if resection <30 minutes.
5. Incise the tunica parallel to the segmental vessels; excise the mass with a rim of normal tubules.
6. **Frozen section** (benign versus malignant) and biopsies of adjacent parenchyma for GCNIS (germ cell neoplasia in situ).
7. Close the tunica; fix the testis in the scrotum.

### Management of GCNIS

- **Radiotherapy about 18–20 Gy** to the remnant: prevents invasive GCT, keeps Leydig function, but causes sterility; up to 40% later need testosterone.
- German Testicular Cancer Study Group: no local recurrences with radiotherapy; most men who omitted it recurred.
- Radiotherapy may be deferred for paternity under close follow-up. Local recurrence → completion orchidectomy.

### Complications and follow-up

- Haematoma, atrophy, hypogonadism, local recurrence. Follow with self-examination, ultrasound, markers and testosterone.

**Diagram to draw:** testis with polar tumour excised with a rim of normal tissue.

> **Key points to remember:**
> - Solitary or bilateral tumour, ≤2 cm, normal testosterone.
> - Frozen section plus adjacent biopsies for GCNIS.
> - GCNIS: 18–20 Gy keeps Leydig function but sterilises.

## SAQ 10 Muscle assisted urethroplasty

P4

**Asked as:**

- Muscle assisted urethroplasty — *Winter 2023 Paper IV Q3b (SAQ)*
- Muscle assisted urethroplasty — *Practice Set 5 Paper IV Q3a (earlier version) (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3b)

*Read as muscle-flap-assisted graft urethroplasty: a vascularised muscle flap used as bed or cover for a graft. "Muscle-sparing" bulbar urethroplasty is noted at the end.*

### Rationale

- A free graft (e.g. **BMG [buccal mucosal graft]**) has no blood supply. It survives by **imbibition** (about 48 h, nutrients from the bed), then **inosculation** (new microcirculation). So **"a graft lives on its bed"**.

- **Poor beds:** radiation, multiple failed repairs, fistula, pelvic-fracture scar, infection. Here a **muscle flap brings blood supply**, fills dead space and separates suture lines.

### Muscle flaps used

| Flap | Supply | Use |
|---|---|---|
| **Gracilis** (muscle-only, axial, multiple pedicles) | Medial circumflex femoral branch, proximal thigh | Irradiated posterior urethra, rectourethral fistula, complex perineal repair |
| **Bulbospongiosus** | Perineal branches | Covers a ventral graft; support in bulbar repair |
| Rectus abdominis (VRAM, vertical rectus abdominis myocutaneous) | Inferior epigastric | Large pelvic or perineal defects |

### Technique (gracilis-assisted BMG)

1. Lithotomy; perineal exposure; excise scar or fistula.
2. Close rectal side; urethra opened.
3. Gracilis harvested through a medial thigh incision; pedicle preserved; tunnelled to perineum.
4. Muscle interposed between rectum and urethra, or used as the graft bed.
5. **BMG quilted onto the muscle** as onlay; urethra closed over a catheter.
6. Catheter drainage; urethrogram before removal.

### Results (Chapter 86)

- Gracilis with or without BMG for rectourethral fistula: **non-irradiated, all closed**, no later strictures. **Irradiated:** most closed in one stage, but more strictures and permanent bowel diversion.
- Muscle interposition is integral; BMG is invaluable for urethral closure.

### Complications

- Thigh wound problems; graft contraction, restricture, fistula recurrence.

### "Muscle-sparing" bulbar urethroplasty (Barbagli)

- The **bulbospongiosus muscle, central perineal tendon and perineal nerves** are left intact; dorsal BMG onlay through a lateral approach.
- Aim: less post-void dribbling and ejaculatory dysfunction.

**Diagram to draw:** gracilis flap tunnelled to the perineum, lying between rectum and urethra with BMG on it.

> **Key points to remember:**
> - Graft survival: imbibition then inosculation.
> - Muscle flap rescues a poor (irradiated) bed.
> - Gracilis is axial, robust, easy to tunnel.
> - Muscle-sparing: preserves bulbospongiosus and nerves.

## SAQ 11 Gracilis myocutaneous flap in Uro.

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 4, Q3d (gracilis-assisted urethroplasty also in 2023-Winter Paper 4, Q3b)

## Definition

- A pedicled **axial** flap of gracilis muscle, with or without skin, rotated from the medial thigh to bring **vascularised bulk** to the perineum, groin or vagina.

## Anatomy

| Feature | Detail |
|---|---|
| Origin–insertion | Inferior pubic ramus and body of pubis → medial proximal tibia (pes anserinus) |
| Blood supply | **Mathes–Nahai type II**: dominant pedicle from the **medial circumflex femoral artery** (profunda femoris), entering about 8–10 cm below the pubic tubercle; minor distal pedicles from the superficial femoral artery |
| Nerve | Anterior branch of the **obturator nerve** |
| Skin paddle | Reliable over the proximal two-thirds only |
| Function | Expendable (other adductors compensate) |

## Why it is used

- Brings blood supply into **irradiated or scarred** fields, fills dead space and **separates suture lines**. Near the perineum, easy to tunnel.

## Uses in urology

1. **Rectourethral fistula** (perineal): non-irradiated all closed; irradiated mostly closed, but more strictures and permanent colostomy.
2. **Complex or irradiated VVF (vesicovaginal fistula)** with tissue loss.
3. **Urethroplasty** in a poor bed: muscle as bed for BMG (buccal mucosal graft).
4. Extensive perineal or scrotal loss after **Fournier gangrene**.
5. **Groin cover** of femoral vessels after inguinal node dissection with skin excision.
6. Vaginal reconstruction after exenteration.

## Technique

1. Lithotomy; line from pubic tubercle to medial femoral condyle; gracilis lies just behind it.
2. Medial thigh incision; skin paddle over the proximal muscle if needed.
3. Divide the distal tendon; ligate minor pedicles; **preserve the dominant pedicle**.
4. Tunnel subcutaneously to the perineum; rotate without twisting.
5. Interpose and fix; drain the donor site.

## Complications

- Donor seroma, dehiscence, **distal skin-paddle necrosis**, thigh numbness; flap loss rare.
- **Also know:** alternatives: Martius, omentum, VRAM (vertical rectus abdominis myocutaneous), Singapore flap.

**Diagram to draw:** gracilis tunnelled on its proximal pedicle between urethra and rectum.

**Key points to remember:**

- Type II flap: medial circumflex femoral pedicle 8–10 cm below the pubic tubercle.
- Skin paddle unreliable over the distal third.
- Main uses: rectourethral fistula, complex VVF, Fournier, groin cover.

# Rapid Revision Sheet · Section 10

### Lymphatic drainage of testis and scrotum, RPLND templates and stage I seminoma (LAQ)

- Right testis to interaortocaval; left to left para-aortic; crossover only right to left.
- Boundaries: renal hila (renal vessels), ureters, ureter–iliac crossing; omit suprahilar and interiliac.
- Nerve sparing of L1–L4 fibres gives antegrade ejaculation >90%; the fibres meet at the IMA, so unilateral templates spare ejaculation.
- PC-RPLND (post-chemotherapy): full bilateral template remains standard unless mass ≤5 cm in landing zone.
- Scrotum to ipsilateral superficial inguinal nodes; scrotum and penis drain to inguinal nodes; never operate trans-scrotally.
- Penile drainage is bilateral and stepwise without skips: stage both groins, PLND only if inguinal nodes positive.
- pT1b or higher: DSNB or ILND within 3 months; Kroon 84% vs 35%.
- Seminoma risk factors: >4 cm and rete testis invasion (not LVI).
- Stage I seminoma: surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).

### Grafts in urology and buccal mucosa (LAQ)

- Take: imbibition 48 h, then inosculation 48 h; graft lives on its bed.
- BMG: non-keratinised, wet-adapted; thin vascular lamina propria gives excellent take.
- Avoid Stensen's duct; buccal equals lingual in success.
- Staged BMG, not skin, in lichen sclerosus.
- Bulbar success about 85–90%.

### Urethral stricture: etiology, classification, DVIU, management and failed VIU (LAQ)

- Idiopathic and iatrogenic are commonest; LS recurs in skin.
- Define location, length, depth, density (dynamic RUG + VCUG, sonourethrography).
- DVIU (VIU) only once, for short bulbar strictures (<1.5–2 cm) at first attempt; do not repeat it.
- Santucci: about 8% long-term success; repeat DVIU futile.
- AUA: urethroplasty after failure; laser = cold knife; early urethroplasty, ladder abandoned.
- EPA is the gold standard (about 90–95%); BMG onlay for longer bulbar strictures.
- Staged oral mucosa for LS; perineal urethrostomy for the unfit or multiply failed.
- Optilume (ROBUST III): 74.6% vs 26.8% stricture-free at 1 year.

### Inguinal lymph nodes in carcinoma penis: management, block dissection and videoendoscopic lymphadenectomy (LAQ)

- Nodal status is the strongest prognostic factor; ILND can cure.
- pT1b or higher: stage groins surgically (DSNB preferred) within 3 months.
- Kroon 2005: early vs delayed ILND, 84% vs 35% 3-year survival.
- PLND if more than 2 nodes or ENE; bulky nodes: neoadjuvant TIP (InPACT [International Penile Advanced Cancer Trial] testing the sequence).
- Femoral triangle: ligament, sartorius, adductor longus.

- Flaps deep to Camper's; stay medial to femoral artery.
- Within 3 months; at least 8 nodes per side.
- VEIL: same template as open; fewer wound complications; equal nodal yield.
- VEIL not for bulky, fixed or skin-involving nodes.

**Male urethra: embryology and anatomy, endoscopic stricture management, congenital anomalies and epispadias (LAQ)**

- Double zipper: canalisation, then fold fusion; glanular urethra endodermal.
- Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock; dual retrograde glanular blood supply.
- DVIU: bulbar, <1.5–2 cm, first attempt; Santucci about 8% long-term.
- Optilume ROBUST III 74.6% vs 26.8%.
- Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
- About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

**Describe in brief the anatomy of the retroperitoneum. Explain the principles of laparoscopic retroperitoneal surgery while mentioning the salient steps of retroperitoneal laparoscopic nephrectomy. (LAQ)**

- Three spaces: anterior pararenal, perirenal, posterior pararenal; perirenal cone open below.
- Entry at the 12th rib tip, where the lumbodorsal fascia fuses.
- Psoas is the landmark; artery met first.
- Main advantage: shorter operative time, peritoneum untouched.

**Classify testicular tumors and describe management of seminoma (LAQ)**

- WHO 2016: GCNIS-derived vs non-GCNIS-derived; spermatocytic tumour is separate and benign.
- CS I risk factors: >4 cm and rete testis invasion; surveillance preferred.
- Carboplatin AUC 7 ×1 equals radiotherapy (TE19); 20 Gy equals 30 Gy (TE18).
- No poor-risk seminoma; residual >3 cm → PET at ≥6 weeks.

**Surgical management of primary scrotal cancer (SAQ)**

- Pott 1775: chimney sweeps, first occupational cancer.
- Wide local excision, 2 cm margin; testes spared unless invaded.
- Nodes: managed as in penile cancer; nodes decide prognosis.

**Varicocele: pathophysiology, controversies and surgical management (SAQ)**

- Left vein: right-angled entry, 8–10 cm higher column; 90% left.
- Heat (lost countercurrent exchange) is the main injury; also oxidative DNA damage.
- Right or non-decompressing varicocele: image for a renal mass.
- Injury is progressive; subclinical varicocele is not treated.
- Repair only palpable varicocele (each palpable side) with abnormal semen; never subclinical.
- Madgar 1995: 44% versus 10% pregnancy.
- Adolescents: observe; repair for asymmetry above 20%.
- Microsurgical subinguinal: lowest recurrence (under 1%) and hydrocele (nil).
- Never ligate both testicular arteries.

**Meatal stenosis and submeatal stricture in males (SAQ)**

- Child: post-circumcision ventral membrane; ventral meatotomy and parental dilation.
- Adult: RUG first; adult meatal stenosis is usually the tip of a fossa navicularis stricture.
- Non-LS: meatotomy or Jordan/De Sy skin island flap.
- LS: steroid (clobetasol), buccal mucosa (never genital skin), perineal urethrostomy if extensive.

**Perineal urethrostomy and eponymous urethroplasty techniques (SAQ)**

- Barbagli dorsal onlay; Asopa dorsal inlay; Kulkarni one-sided.
- Orandi and McAninch flaps; Quartey flap anatomy.
- Johanson staged; Blandy inverted-U PU.
- PU bypasses, rather than reconstructs, the diseased anterior urethra; patient voids sitting.
- Ideal for LS, failed repairs and elderly patients.
- BMG PU prevents circumferential scarring; calibrate to 30 Fr.

**Minimally invasive and robotic RPLND (SAQ)**

- Same template, split-and-roll, lumbar division and nerve sparing as open.
- Best for chemotherapy-naive low-volume disease.
- Conversion about 5%; ejaculation preserved ≥90%.
- Expanding indication: primary RPLND for small-volume stage II seminoma to avoid chemotherapy and radiotherapy.
- Robotic nerve sparing preserves antegrade ejaculation in ≥90% after primary surgery.
- Serum miR-371a-3p (microRNA) may select stage I patients with occult nodes for RPLND (retroperitoneal lymph node dissection).

**Testicular microlithiasis (SAQ)**

- Classic TM ≥5 microliths; limited <5.
- Not an independent cancer risk factor in isolation.
- Self-examination only with risk factors; no routine imaging.

**Growing teratoma syndrome (SAQ)**

- Enlarging mass plus normal or falling markers equals GTS.
- Teratoma is chemoresistant and not FDG-avid.
- Treatment is complete surgical resection, not more chemotherapy. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Prognostic index score in carcinoma penis (SAQ)**

- Node status is the strongest prognostic factor.
- LVI, PNI, grade III define high-risk T1b.
- Nomograms (Ficarra, Kattan) unvalidated; p16 favourable. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Testis sparing surgeries for CA testis (SAQ)**

- Solitary or bilateral tumour, ≤2 cm, normal testosterone.
- Frozen section plus adjacent biopsies for GCNIS.
- GCNIS: 18–20 Gy keeps Leydig function but sterilises.

**Muscle assisted urethroplasty (SAQ)**

- Graft survival: imbibition then inosculation.
- Muscle flap rescues a poor (irradiated) bed.
- Gracilis is axial, robust, easy to tunnel.
- Muscle-sparing: preserves bulbospongiosus and nerves.

**Gracilis myocutaneous flap in Uro. (SAQ)**

- Type II flap: medial circumflex femoral pedicle 8–10 cm below the pubic tubercle.
- Skin paddle unreliable over the distal third.
- Main uses: rectourethral fistula, complex VVF, Fournier, groin cover. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 11 · Stones, Obstruction & Trauma

## Long answer questions (LAQ)

### LAQ 1 Congenital PUJ obstruction, failed pyeloplasty and salvage procedures

P1 · P2 **merged from 3**

*Asked as:* P1 LAQ "Describe Pathogenesis of Congenital Pelvi Ureteric Junction Obstruction its diagnosis and Management." · P1 SAQ "Management of failed pyeloplasty" · P2 SAQ "Salvage procedures for Complicated PUJO."

#### Definition

- PUJO (pelviureteric junction obstruction) is impaired flow of urine from the pelvis into the ureter. **Koff:** obstruction is "any restriction to urinary outflow that, if left untreated, will lead to progressive kidney damage". **Dilatation is not obstruction.**
- About 1 in 1500 live births; commonest cause of antenatal hydronephrosis. Male and left predominance; **bilateral in 10–40%**.

#### Pathogenesis

| Type | Lesion | Typical patient |
|---|---|---|
| **Intrinsic** (commonest) | Short adynamic segment: interrupted circular muscle, excess collagen, fewer pacemaker (interstitial Cajal-like) cells. The peristaltic wave cannot cross, so the pelvis empties poorly | Neonate or infant, antenatal finding |
| **Extrinsic** | **Lower-pole crossing vessel** anterior to the PUJ; the ureter kinks over it | Older child or adult with intermittent pain |
| Secondary | Kinking by a tortuous ureter in high-grade VUR (vesicoureteral reflux); fibroepithelial polyp; high insertion | Any age |

- **Crossing vessels** are persistent embryonic renal arteries. They occur at a fifth to a quarter of **normal** PUJs, so a vessel alone does not prove cause.
- **Also know:** proposed incomplete ureteric recanalisation; horseshoe kidney and duplex lower moiety associations.

#### Effects of obstruction (pathophysiology)

- Pelvic pressure rises; filtration and perfusion fall.
- **Immature kidney:** apoptosis and impaired nephrogenesis. **Mature kidney:** hypertrophy, then fibrosis (TGF-β [transforming growth factor-beta] signalling). Fibrosis, not dilatation, causes loss of function.
- **Also know:** concentrating defect, salt wasting; rarely renin-mediated hypertension.

#### Clinical features

- **Antenatal ultrasound** (commonest); infants: UTI (urinary tract infection), flank mass, failure to thrive.
- Older children and adults: **intermittent loin pain with vomiting (Dietl crisis)**, haematuria after minor trauma, stones, infection.

- **Associated anomalies (up to 50%):** contralateral PUJO, dysplasia or MCDK (multicystic dysplastic kidney), VUR (up to 40%).

## Diagnosis

### 1. Antenatal grading (APD, anteroposterior renal pelvic diameter)

| | Second trimester | Third trimester |
|---|---|---|
| Mild | 4 to <7 mm | 4 to <9 mm |
| Moderate | 7–10 mm | 9–15 mm |
| Severe | >10 mm | >15 mm |

- **UTD (urinary tract dilation) 2014 consensus:** six parameters (APD, central/peripheral calyces, parenchymal thickness and appearance, ureter, bladder). Prenatal A1 (low) or A2–3; postnatal **P1** (APD 10–<15 mm, central calyces), **P2** (≥15 mm or peripheral calyces or ureter), **P3** (abnormal parenchyma).
- **SFU (Society for Fetal Urology) grades 0–4:** 1 pelvis just seen; 2 pelvis plus major calyces; 3 all calyces, normal parenchyma; **4 parenchymal thinning**.

**2. Postnatal ultrasound** after **48 hours** (neonatal dehydration under-reads dilatation): large pelvis, rounded calyces, ureter not seen.

**3. MAG3 (technetium-99m mercaptoacetyltriglycine) diuretic renogram** ("well-tempered": hydrated, catheterised, furosemide)

- **DRF (differential renal function)** (normal 45–55%); drainage half-time <10 min normal, 10–20 equivocal, **>20 min obstructed** (adult chapter: >40 min).

**4. Others:** VCUG (voiding cystourethrogram) by UTI risk; MRU (magnetic resonance urography) or CT (computed tomography) angiography for vessels; **Whitaker test** (10 mL/min; >15–20 cm $H_2O$ obstructed) if equivocal.

## Primary management

**Conservative:** most resolve, about **one-third need surgery**; serial ultrasound and renogram.

### Indications for pyeloplasty

- **DRF <40%** or a **fall of >10%** on serial scans.
- **Increasing APD** or SFU 4 / P3 with poor drainage.
- **Breakthrough UTI**, symptoms, stones, hypertension.
- Bilateral or solitary kidney.

### Anderson–Hynes dismembered pyeloplasty: gold standard (~95% success)

1. Expose the PUJ; free the lower pole and look for a crossing vessel.
2. Stay sutures; transect the pelvis; reduce a redundant pelvis.
3. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a "no-touch" handle.
4. **Transpose** the PUJ anterior to any vessel (never divide it).
5. Watertight, tension-free, dependent anastomosis (5-0/6-0 absorbable); excise the segment.
6. Double-J or externalised stent (1–4 weeks equivalent); Foley 24–48 h with a double-J.

- **Route:** open preferred in infants; laparoscopic equals open with less pain; **robotic** (easier suturing, shorter learning curve) is now primary at most major centres.

- **Alternatives:** Foley Y-V plasty (high insertion); Culp spiral or vertical flap (long segment); **vascular hitch** (crossing vessel with normal PUJ); **endopyelotomy** (73% overall; **42% with vs 86% without a crossing vessel**), mainly for secondary PUJO.
- **Nephrectomy:** DRF **<15% in adults** (children: no agreed cut-off).

**Diagram to draw:** Anderson–Hynes steps: transection, lateral spatulation, transposition anterior to the crossing vessel, dependent anastomosis.

### Complications after pyeloplasty

- Urine leak, stent problems, UTI.

### Failed pyeloplasty: definition

- Persistent pain, UTI or stones; **obstructive diuretic renogram** (T½ >20 minutes); unchanged or increasing hydronephrosis; falling split function.
- Dilatation settles slowly; judge by renography.
- About 1 in 9 children need a secondary procedure. Most failures occur within 1–2 years (P2: within 2 years), but up to 30% occur later.

### Causes of failure

- **Ischaemia** (over-stripping) and **tension** at the anastomosis.
- **Missed crossing vessel**; inadequate excision of the narrow segment; non-dependent anastomosis; kinking.
- **Urine leak** or infection causing periureteric fibrosis.
- Missed distal stricture or high-grade VUR.

### Early failure (first days or weeks)

- Pain, fever, increasing dilatation: usually **oedema or clot**; temporary nephrostomy or stent, and most settle.

### Evaluation of late failure

- **MAG3 diuretic renogram:** obstruction (drainage) and split function.
- **CT angiography:** crossing vessel.
- Antegrade or retrograde pyelography (stricture length); **Whitaker test** if equivocal.
- Drain first (stent or nephrostomy) if infected or function is uncertain, then reassess.

### Salvage options

| Option | When | Results/notes |
|---|---|---|
| **Endopyelotomy** (antegrade or retrograde ureteroscopic, laser) | Short stricture, no significant crossing vessel, moderate hydronephrosis; first choice after failed open repair | About 70% after failed pyeloplasty; lateral incision; stent 4–6 weeks |
| **Redo pyeloplasty** (open, laparoscopic, robotic) | Most failures; long or dense stricture, crossing vessel; failed endopyelotomy | About 86–89% (P2: about 83–89%); best choice in most children; wide mobilisation of kidney and ureter; transperitoneal (scarred retroperitoneum) |
| **Ureterocalicostomy** | Scarred, small intrarenal pelvis; dilated lower-pole calyx | About 70% in redo series; **resect** lower-pole parenchyma (not nephrotomy) |

| Option | When | Results/notes |
|---|---|---|
| **Buccal mucosa onlay graft** | Long scarred PUJ/UPJ (ureteropelvic junction) or proximal ureter, dense scar | Robotic with omental wrap; failure about 15–25% (CORRUS) |
| **Appendiceal onlay flap** | Right-sided proximal stricture | Pedicled on appendiceal artery |
| Flaps (spiral, vertical), downward nephropexy | Gain length | Combine with redo repair |
| **Ileal ureter** | Long defect / long ureteric loss | Contraindicated if creatinine >2 mg/dL, bladder dysfunction, IBD (inflammatory bowel disease), radiation |
| **Autotransplantation** | Solitary or poor contralateral kidney; other repairs impossible; long ureteric loss | About 90% long term |
| **Nephrectomy** | Split function <15% with normal contralateral kidney; multiple failures | Definitive; laparoscopic |

- Rule of thumb: failed pyeloplasty: try endourology first if anatomy suits; failed endopyelotomy: redo pyeloplasty. Redo pyeloplasty is the most reliable.
- **Also know:** robotic near-infrared fluorescence (indocyanine green, ICG) to identify the ureter and perfusion; Davis intubated ureterotomy (historical).

### Principles of redo surgery

- Transperitoneal route in children (retroperitoneum is scarred).
- Ureteric catheter; wide mobilisation of kidney and ureter; **excise all scar**; spatulate; **tension-free, watertight** anastomosis over a stent; wrap with omentum or perirenal fat.
- **Also know:** near-infrared fluorescence shows perfusion; downward nephropexy.

**Diagram to draw:** ureterocalicostomy: lower-pole parenchyma excised, spatulated ureter sewn to lower calyx.

### Follow-up

- After primary pyeloplasty: renogram and ultrasound after stent removal.
- After redo: stent out at 4–6 weeks; renograms for 2–3 years (late failures).

### Recent advances

- Robotic redo with buccal graft and ICG fluorescence; pyramidal thickness ≤3 mm predicts surgery; urinary biomarkers.

**Key points to remember:**

- Intrinsic adynamic segment (infants) vs crossing vessel (older children, intermittent pain).
- SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
- Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
- Anderson–Hynes ~95%; never divide a crossing vessel.
- Causes of failure: ischaemia, tension, missed crossing vessel, leak.
- Failed open repair: endourology first; redo pyeloplasty most reliable (about 83–89%); endopyelotomy about 70%.
- Ureterocalycostomy for small scarred pelvis: resect lower-pole parenchyma; about 70% in redo series. Buccal onlay emerging.
- Split function <15% with normal other kidney: nephrectomy.

## LAQ 2 Urolithiasis: etiopathogenesis, medical management and prevention of recurrence

P1 · P2 · P4 **merged from 3**

*Asked as:* P4 LAQ "Describe etiopathogenesis of stone formation and medical management of urolithiasis" / "Describe medical management of recurrent stone disease and discuss recent advances in instrumentations of RIRS" · P2 LAQ "Describe etiopathogenesis of stone formation and medical management of urolithiasis" (also P2 SAQ "Describe medical management of renal calculi") · P1 LAQ "Discuss pathophysiology of renal stones and also describe how to prevent its recurrence"

### Introduction

- Urolithiasis is a **chronic metabolic disease** punctuated by stone episodes; recurrence is up to **50% within 5 years** (another source: about half recur within 5–10 years). Removal is surgical; recurrence is metabolic. Recurrent formers need a **metabolic evaluation** and targeted therapy.
- **Composition**: calcium oxalate about 60%, mixed with apatite about 20% (calcium oxalate with or without apatite about 80% in all), uric acid and struvite about 7% each, cystine 1–3%.
- RIRS (retrograde intrarenal surgery) is flexible ureteroscopy inside the kidney.

### Aetiology: risk factors

- **Epidemiological**: male sex (gap narrowing), hot climate and occupation (low urine volume), diet (animal protein, salt), family history.
- **Systemic**: **metabolic syndrome**, obesity, diabetes (insulin resistance lowers urine pH), gout, primary hyperparathyroidism, RTA (renal tubular acidosis), bowel disease or bypass, sarcoidosis.
- **Anatomical (stasis)**: PUJ (pelviureteric junction) obstruction, horseshoe kidney, calyceal diverticulum, medullary sponge kidney (stasis plus metabolic defects).
- **Drugs**: direct (indinavir, triamterene); indirect (acetazolamide, topiramate, vitamin C and D excess).

### Physicochemistry: state of saturation

| State | Concentration product (CP) | What happens |
|---|---|---|
| Undersaturated | Below Ksp (solubility product) | No crystals; stones may dissolve |
| **Metastable** | Between Ksp and Kf (formation product) | No spontaneous nucleation, but growth on existing crystals; inhibitors act here. **Zone of therapy** |
| Unstable | Above Kf | Spontaneous nucleation, growth, aggregation |

- Crystals form in metastable urine by local high CP, stasis, or **heterogeneous nucleation** on cells, debris or other crystals (needs less energy).

### Steps of stone formation

1. **Supersaturation:** low volume; high calcium, oxalate or uric acid; low citrate.

2. **Nucleation:** homogeneous, or **heterogeneous** on debris or other crystals (needs less energy).
3. **Growth and aggregation:** aggregates, not single crystals, can occlude a tubule within the 5–7 minute nephron transit (**free-particle theory**).
4. **Retention (fixed-particle theory): Randall plaque** (calcium apatite) starts in the basement membrane of the thin loop of Henle, spreads through the interstitium, erodes the papillary urothelium and anchors calcium oxalate growth (idiopathic formers). **Ductal plugs** (crystals in collecting ducts) point to a secondary cause: enteric hyperoxaluria, distal RTA, hyperparathyroidism, cystinuria.

### Inhibitors, promoters and matrix

- **Inhibitors**: **citrate** (complexes calcium), **magnesium** (complexes oxalate), pyrophosphate, glycosaminoglycans, nephrocalcin, **Tamm–Horsfall protein**, osteopontin, bikunin. There is **no inhibitor of uric acid**.
- **Matrix** (about 2.5% of stone weight; matrix substance A): role uncertain.

### Pathophysiology by urinary abnormality

| Abnormality (definition) | Mechanism |
|---|---|
| **Hypercalciuria** (commonest; >200 mg/day, or >4 mg/kg/day) | Absorptive (gut), renal leak (secondary high PTH [parathyroid hormone]), resorptive (primary hyperparathyroidism): Pak classification |
| Hyperoxaluria (>40 mg/day) | **Primary:** hepatic enzyme defect (PH1, AGXT gene); **enteric:** malabsorbed fat binds calcium, freeing oxalate; dietary |
| Hyperuricosuria (>600 mg/day) | Urate promotes calcium oxalate nucleation |
| **Hypocitraturia** (<320 mg/day) | Distal RTA, diarrhoea, hypokalaemia, animal protein (acid load, intracellular acidosis) |
| **Uric acid stones** | **Low urine pH (<5.5)** more than excess uric acid; pKa 5.5 |
| **Cystine** | Autosomal recessive defect of tubular cystine reabsorption |
| **Struvite** | **Urease** (Proteus) splits urea → ammonia, pH >7.2 → magnesium ammonium phosphate and carbonate apatite; biofilm |

### Medical management: aims

- Relieve pain, dissolve suitable stones and **prevent recurrence** (up to 50% at 5 years).

### Acute colic and MET (medical expulsive therapy)

- **NSAIDs (non-steroidal anti-inflammatory drugs)** first line; opioids only if refractory. Urgent drainage if infection with obstruction.
- **α-blocker** (tamsulosin; silodosin faster) for **distal ureteric stones 5–10 mm** after migration from the kidney: relaxes ureteric muscle. **SUSPEND (2015)** was negative because it included all sizes and sites. Trial of passage 4–6 weeks.

### Dissolution (chemolysis)

- **Uric acid** stones (radiolucent) dissolve with oral potassium citrate to alkalinise (EAU [European Association of Urology]: pH 7.0–7.2 for dissolution); allopurinol if hyperuricosuric.
- **Cystine:** alkalinise to pH 7.0–7.5 with thiol drugs. **Struvite:** acetohydroxamic acid occasionally dissolves.

## Prevention of recurrence

### 1. Screen every first-time former (AUA [American Urological Association], EAU)

- History, serum electrolytes, creatinine, calcium, uric acid; urine pH and culture; **stone analysis**; imaging. This sorts patients into high and low risk.

### 2. General measures (every former)

- **Fluid for urine ≥2.5 L/day** (drink about 3 L). **Borghi 1996**: urine >2 L cut 5-year recurrence (12.1% vs 27%; recurrence halved).
- **Normal calcium**, low sodium (≤2.3 g/day), less animal protein, more fruit and vegetables (DASH [Dietary Approaches to Stop Hypertension]-type diet). **Borghi 2002**: this diet halved recurrence compared with a low-calcium diet. Restricting calcium raises oxalate absorption.
- Limit high-oxalate foods (spinach, nuts, chocolate); weight loss.
- Citrus juices (lemonade, orange) are adjuncts only.

### 3. Metabolic evaluation that guides therapy (high-risk or recurrent formers, children) (see 2015-Summer Paper 2, Q2)

- **Stone analysis** (infrared spectroscopy or X-ray diffraction); serum chemistry: serum calcium, uric acid, creatinine, electrolytes, PTH if calcium is high-normal.
- One or **two 24-hour urines** on usual diet, at least 4 weeks after an episode: volume, pH, calcium, oxalate, citrate, uric acid, sodium, potassium, creatinine, cystine. Therapy targeted to these results reduces stone events.
- Targets: volume >2.5 L, calcium <250 mg (women) or <300 mg (men), oxalate <40 mg, citrate >320 mg, uric acid <600 mg.

### 4. Selective drug therapy

| Abnormality | Drug and dose | Rationale and notes |
|---|---|---|
| Hypercalciuria | **Thiazide**: hydrochlorothiazide 25 mg twice daily, chlorthalidone 25–50 mg, indapamide 2.5 mg | Distal calcium reabsorption plus volume contraction. Add potassium citrate: thiazide lowers potassium, and hypokalaemia causes hypocitraturia |
| Primary hyperparathyroidism | Parathyroidectomy | Cures resorptive hypercalciuria |
| Hypocitraturia, distal RTA, chronic diarrhoea | **Potassium citrate 30–60 mEq/day** | Alkali load reduces tubular (proximal) citrate reabsorption; avoid urine pH ≥7 |
| Hyperuricosuria / hyperuricosuric calcium oxalate stones | Allopurinol 300 mg/day | Watch hypersensitivity; reduce dose in renal impairment |
| Uric acid stones | Potassium citrate to **pH 6.0–6.5** (avoid >7.0: calcium phosphate); allopurinol only for hyperuricosuria or failed alkalinisation | A pH disease: pKa 5.5; dissolution possible |
| Enteric hyperoxaluria | Low fat and oxalate; **calcium with meals**; liquid potassium citrate; magnesium; cholestyramine | Calcium binds oxalate in the gut |

| Abnormality | Drug and dose | Rationale and notes |
|---|---|---|
| Primary hyperoxaluria type 1 | Pyridoxine; **lumasiran** (RNA [ribonucleic acid] interference against glycolate oxidase) | ILLUMINATE-A: 84% vs 0% reached near-normal urine oxalate |
| Cystinuria | Urine ≥3 L/day (another source: >3 L); low sodium and protein; potassium citrate to pH **7.0–7.5**; then **tiopronin** (or D-penicillamine) | Aim cystine <250 mg/L; thiols form soluble mixed disulfides |
| Struvite | **Complete clearance**; culture-specific antibiotics; AHA (acetohydroxamic acid) 250 mg three times daily (another source: two to three times daily) if surgery impossible | Urease inhibitor; risk of thromboembolism |

- **Also know:** recurrent calcium stones with a normal work-up: thiazide or potassium citrate (AUA: even without abnormality); citrus juices are adjuncts.
- **Trials**: Ettinger 1988 (chlorthalidone) and Barcelo 1993 (potassium citrate) reduced new stones. **NOSTONE (2023)**: hydrochlorothiazide 12.5–50 mg did not reduce recurrence, which questions the older trials.

**5. Monitoring and follow-up**

- Repeat 24-hour urine within **6 months** of starting therapy, then yearly; reanalyse any new stone.
- Serum potassium 1–2 weeks after starting a thiazide or citrate; imaging for new stones.

**Diagram to draw:** saturation zones (Ksp, Kf) and Randall plaque on a renal papilla.

**Diagram to draw:** flowchart from stone analysis and 24-hour urine to abnormality to selective drug therapy.

## Recent advances in prevention

- **ROKS (Recurrence of Kidney Stone) nomogram** predicts individual risk.
- **Also know:** NOSTONE (2023) negative; citrus juices are adjuncts; *Oxalobacter formigenes* loss; lumasiran for PH1.

## Recent advances in RIRS instrumentation

| Advance | Principle | Benefit or limitation |
|---|---|---|
| **Digital flexible ureteroscopes** | CMOS (complementary metal-oxide semiconductor) chip at the tip; LED (light-emitting diode) light | Sharper image that does not degrade; tip larger (~8.5 Fr); no more durable |
| **Single-use scopes** (LithoVue, 2016) | 7.7 Fr tip, ≥270° deflection, new each case | No repair or reprocessing; cheaper below ~99 cases a year; for lower pole and difficult cases |
| **Thulium fibre laser** (about 1940 nm) | Long pulse, very high frequency | Fine dust, minimal retropulsion; faster than Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm) at low settings |
| **Pulse-modulated (Moses) Ho:YAG** | Pulse shaped through a vapour bubble | Less retropulsion, faster dusting |

| Advance | Principle | Benefit or limitation |
|---|---|---|
| Access sheaths (12/14 Fr) | Outflow channel | Lower intrarenal pressure and temperature; severe ureteric injury in ~1 in 8 |
| **Suction sheaths and FANS** (flexible and navigable suction access sheath) | Vacuum at the sheath; flexible tip follows the scope into calyces | Aspirates dust; lower pressure; better stone-free rates in early studies |
| **Intrarenal pressure monitoring** | Sensor in the wire or scope (LithoVue Elite) | High pressure causes backflow, SIRS (systemic inflammatory response syndrome) and sepsis; safe limits not yet defined |
| **Robotic flexible ureteroscopy** (Avicenna Roboflex) | Seated console controls the scope | Ergonomics, less radiation to surgeon |

- **Laser safety**: tissue damage above 43 °C, so keep power ≤10 W in the ureter and ≤25 W in the kidney, with continuous cool irrigation.
- **Also know:** dusting versus fragment-and-basket; nitinol tipless baskets; anti-retropulsion devices; Monarch robotic platform for PCNL (percutaneous nephrolithotomy) access.

**Key points to remember:**

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor for idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi); Borghi 1996: 12% vs 27% recurrence; MET for distal 5–10 mm stones.
- Thiazide (plus potassium citrate) for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

## LAQ 3 Metabolic evaluation in stone disease

P1 · P2 **merged from 2**

*Asked as:* P2 LAQ "Discuss metabolic evaluation in urinary calculus disease." / "Describe metabolic evaluation in stone disease" / "Discuss evaluation for urolithiasis and management of bilateral renal staghorn calculi" · P1 SAQ "Metabolic Evaluation in Stone Disease"

### Introduction and rationale

- Stone disease is a chronic metabolic disorder: first-time formers have about a **50% risk of recurrence within 10 years** (another source: up to 50% recur within 5–10 years).
- Evaluation finds a treatable cause and directs targeted therapy; therapy matched to 24-hour urine results reduces stone events. The **ROKS (Recurrence of Kidney Stone) nomogram** estimates individual risk.

### Physicochemical basis (why we measure these)

- Stones form when urine is **supersaturated**: nucleation, growth, aggregation and retention, often on **Randall plaque**.
- In the **metastable zone** (between solubility and formation products) crystals grow on existing nuclei but do not form spontaneously; above it nucleation is spontaneous.
- **Promoters** (calcium, oxalate, uric acid, low volume) raise supersaturation; **inhibitors** (citrate, magnesium, Tamm–Horsfall protein) raise the supersaturation needed to crystallise. The 24-hour urine measures both.

### Who needs evaluation

- **Screening evaluation** for every first-time former (AUA [American Urological Association], EAU [European Association of Urology]).
- **24-hour urine for high-risk and recurrent formers:** recurrent formers, **children**, solitary kidney, bilateral or large burden, GI (gastrointestinal) disease, bariatric surgery or bypass, obesity, metabolic syndrome, diabetes, primary hyperparathyroidism, RTA (renal tubular acidosis), sarcoidosis, gout, family history; uric acid, cystine or infection stones. Bilateral staghorns are high risk by definition.

### Step 1: Screening evaluation (every first-time former)

| Component | What is assessed |
|---|---|
| History | Colic, UTI (urinary tract infection) and recurrent UTI, previous stones, fluid and diet; GI disease or bariatric surgery, gout, sarcoidosis, family history; lithogenic drugs: calcium, vitamins C and D, acetazolamide, topiramate, steroids, triamterene, indinavir |
| Blood | Creatinine, sodium, potassium (electrolytes), calcium, uric acid, bicarbonate (if RTA suspected); **PTH (parathyroid hormone)** if calcium is high or high-normal |
| Urine | Urinalysis: pH, crystals (hexagonal = cystine), pyuria; culture; cyanide–nitroprusside cystine screen, confirmed by quantitative 24-hour cystine |
| **Stone analysis** | Infrared spectroscopy or X-ray diffraction: uric acid points to metabolic syndrome, cystine to a genetic cause, struvite to infection |
| Imaging | NCCT: burden predicts future events; uric acid usually <500 HU (Hounsfield units) |

- Screening is a **triage**: low risk gets general advice; high risk goes on to 24-hour urine.

### Imaging in evaluation for urolithiasis

| Imaging | Role |
|---|---|
| **NCCT** (non-contrast CT) | Gold standard: sensitivity 97%, specificity 95%. Size, site, burden, density (uric acid 200–400 HU; calcium oxalate 600–1200 HU), skin-to-stone distance |
| Low-dose NCCT (≤3 mSv) | Equal accuracy for stones ≥3 mm |
| Ultrasound | First line in pregnancy and children |
| KUB (plain X-ray) | Follow-up of radio-opaque stones only |
| CT urography | Anatomy before PCNL (percutaneous nephrolithotomy) |
| MAG3 (mercaptoacetyltriglycine) renogram or DMSA (dimercaptosuccinic acid) scan | Split function |

### Step 2: Extended metabolic evaluation (24-hour urine)

- **One or two 24-hour urines** on the usual diet, at least 4 weeks after the episode, stone-free and infection-free. Creatinine checks completeness (men about 20–25, women about 15–20 mg/kg/day).
- Measure volume, pH, calcium, oxalate, uric acid, citrate, sodium, potassium, creatinine (cystine if suspected).

| Parameter | Desirable per day | Abnormality and meaning |
|---|---|---|
| Volume | **>2.5 L** | Low volume: commonest, easiest to fix |
| Calcium | <250 mg women, <300 mg men (or <4 mg/kg) | **Hypercalciuria**: absorptive (PTH suppressed), renal leak (PTH high, serum calcium normal), resorptive (primary hyperparathyroidism) |
| Oxalate | **<40 mg** | Dietary; **enteric** (fat malabsorption, gastric bypass: fat binds calcium, freeing oxalate; often 60–70 mg); **primary** (hepatic, often >100 mg, childhood; type 1 (AGXT gene) ~70%, type 2 (GRHPR), type 3 (HOGA1)) |
| Citrate | **>320 mg** | Hypocitraturia: distal RTA, diarrhoea, hypokalaemia, high animal protein |
| Uric acid | <600 mg (labs: <750 women, <800 men) | Hyperuricosuria promotes calcium oxalate stones |
| pH | 5.8–6.2 | <5.5 uric acid (metabolic syndrome); >6.5–7 calcium phosphate, RTA, infection |
| Sodium | <100 mmol | High sodium raises calcium excretion |
| Cystine | Negative screen | Cystinuria; keep <250 mg/L |

- **Uric acid stones** depend mainly on pH: below the pKa (5.5) uric acid is undissociated and insoluble.
- **Infection stones**: urease-splitting organisms, pH >7, struvite.
- **Also know:** magnesium, potassium, phosphate; hypomagnesiuria; RSS (relative supersaturation) is calculated from these and guides therapy; medullary sponge kidney and calyceal diverticulum (stasis plus metabolic defects); 2,8-dihydroxyadenine stones (APRT [adenine phosphoribosyltransferase] deficiency).

### Interpreting hypercalciuria (Pak classification)

| Type | Defect | Serum calcium | PTH | Fasting urine calcium |
|---|---|---|---|---|
| Absorptive (20–40%) | Increased gut absorption | Normal | Normal or suppressed | Normal |
| Renal leak (5–8%) | Tubular calcium loss | Normal | High (secondary) | High |
| Resorptive (3–5%) | Primary hyperparathyroidism | High | High | High |

- Fast and calcium-load subtyping is no longer advised, because it does not change treatment.

### Special tests

- **Distal RTA:** urine pH >6, low citrate, calcium phosphate stones. Incomplete forms: oral **ammonium chloride load** (0.1 g/kg) or furosemide–fludrocortisone test fails to lower urine pH below 5.5.
- **Cystinuria:** autosomal recessive transport defect (cystine, ornithine, lysine, arginine); quantitative 24-hour cystine; solubility about 300 mg/L at pH 5 and 400 mg/L at pH 7, so aim for <250 mg/L.
- **Primary hyperoxaluria:** genetic testing (type 1, AGXT gene, about 70%).

### Evaluation algorithm

`Stone episode → screening evaluation Low risk → general advice High risk → 24-h urine ×1–2 → targeted therapy → repeat within 6 months → yearly`

**Diagram to draw:** supersaturation zones (undersaturated, metastable, unstable) and the evaluation flowchart (first stone, screening, low risk with general advice or high risk with 24-hour urine and targeted therapy).

### Treatment linked to abnormality

- **General**: urine >2.5 L, sodium ≤2.3 g/day, normal calcium, less animal protein, more fruit and vegetables.

| Abnormality | Treatment |
|---|---|
| Low volume | Fluid for urine >2.5 L/day |
| Hypercalciuria | **Thiazide** (hydrochlorothiazide 25 mg twice daily, chlorthalidone, indapamide) plus potassium citrate; plus sodium restriction |
| Primary hyperparathyroidism | **Parathyroidectomy** |
| Hypocitraturia, distal RTA | **Potassium citrate** 30–60 mEq/day |
| Enteric hyperoxaluria | Low fat and oxalate; **calcium with meals**; magnesium |
| Primary hyperoxaluria type 1 | Pyridoxine, **lumasiran** (RNAi [ribonucleic acid interference]); liver ± kidney transplant |
| Hyperuricosuria | **Allopurinol** 300 mg/day |
| Uric acid stones | **Potassium citrate** to urine pH 6.0–6.5 |
| Cystinuria | Urine ≥3 L/day; pH **7.0–7.5**; **tiopronin** |
| Struvite | Complete clearance; AHA (acetohydroxamic acid) if surgery impossible |

- **Borghi 1996**: urine >2 L/day cut 5-year recurrence (12% vs 27%). **Borghi 2002**: low-salt, low-protein, normal-calcium diet halved recurrence versus low calcium. **NOSTONE (2023)**: hydrochlorothiazide did not reduce recurrence.

### Special groups and follow-up

- **Children:** all evaluated (spot ratios if not toilet-trained; consider Dent disease, monogenic causes). **Pregnancy:** defer 24-hour urine (physiological hypercalciuria).
- **Follow-up (AUA):** repeat 24-hour urine **within 6 months** of starting therapy, then yearly; blood tests for drug side effects (potassium on thiazide or citrate); re-analyse every new stone. Adherence is poor; citrus juices are only adjuncts.

### Recent advances

- **Lumasiran** (RNA interference) for primary hyperoxaluria type 1; **NOSTONE (2023):** hydrochlorothiazide did not reduce recurrence; ROKS nomogram.

### Staghorn calculus (bilateral staghorn question)

- **Definition:** a branched stone filling the renal pelvis and extending into the calyces. **Partial:** pelvis and some calyces; **complete:** pelvis and all major calyces.
- **Composition:** classically struvite (magnesium ammonium phosphate) with carbonate apatite. Bacterial **urease** (Proteus, Klebsiella, Ureaplasma) splits urea into ammonia, so urine pH rises above 7.2 and struvite crystallises inside a bacterial biofilm. Fewer than half of modern staghorns are infection stones; calcium phosphate, cystine and uric acid also occur.
- **Why treat:** untreated staghorns destroy the kidney and cause sepsis, XGP (xanthogranulomatous pyelonephritis) and death.

### Staghorn: work-up before surgery

- NCCT with coronal and 3D views: burden, calyceal anatomy, colon and spleen.
- **Renal function:** eGFR (estimated glomerular filtration rate) and split function, because both kidneys are at risk.
- **Infection:** culture-specific antibiotics before surgery; drain an obstructed or infected kidney (nephrostomy through a calyx usable later for PCNL). Culture the stone.
- **Guy's stone score (2011)** predicts stone-free rate:

| Grade | Stone |
|---|---|
| I | Single stone in mid or lower pole or pelvis, simple anatomy |
| II | Single upper-pole stone; multiple stones in simple anatomy; or single stone in abnormal anatomy |
| III | Multiple stones in abnormal anatomy; calyceal diverticulum; partial staghorn |
| IV | Staghorn; or any stone in spina bifida or spinal injury |

- **Also know:** STONE score, CROES nomogram, S-ReSC score.

### Staghorn: treatment options (AUA, EAU)

- **Goal: complete clearance**, because each fragment is an infected nidus; also clear infection and preserve function.

| Option | Role |
|---|---|
| **PCNL** | **First line.** Multiple tracts if needed; upper-pole access reaches pelvis, lower pole and ureteropelvic junction. Flexible nephroscopy raises clearance; second-look for residue |
| **ECIRS** (endoscopic combined intrarenal surgery) | PCNL (usually supine) plus retrograde flexible ureteroscopy: fewer tracts, less bleeding, higher clearance |
| Staged RIRS (retrograde intrarenal surgery) | Selected partial staghorns; patients who cannot stop anticoagulants |
| SWL (shock wave lithotripsy) | Not as monotherapy; adjunct for residual fragments |
| **Anatrophic nephrolithotomy** (Boyce–Smith) | Incision along Brödel's avascular line, artery clamped, cold ischaemia, infundibula reconstructed. Now rare; when endoscopy cannot clear the stone in a reasonable number of sessions |
| **Nephrectomy** | Poorly functioning kidney (split below about 10–20%) that is infected or painful; XGP |

## Bilateral staghorns: sequence

``` Sepsis or obstruction on either side? Yes -> drain + antibiotics -> treat that side once sterile No -> treat the better-functioning (or symptomatic) kidney first

|

NCCT: stone-free? -> second look if residue

|

Other kidney after recovery (selected: same-session bilateral PCNL, or PCNL + contralateral RIRS) ```

- **Why the better kidney first:** it secures renal function early; some start with the simpler side. No trial defines the order.
- **Same-session bilateral surgery:** feasible at experienced centres if the first side was short and uncomplicated; saves an anaesthetic but risks bilateral bleeding or obstruction.
- Check creatinine after each side. Remove a non-functioning kidney only after the other side is cleared.

## Staghorn: complications and follow-up

- PCNL risks: bleeding (transfusion about 1%; angioembolisation), sepsis, pleural, colon, spleen or liver injury.
- NCCT for stone-free status; treat residual fragments >4 mm. Surveillance cultures.
- 24-hour urine even in struvite formers, then targeted therapy; urine ≥2.5 L/day.
- **AHA** 250 mg three times daily, a urease inhibitor, only when surgery is exhausted. Avoid if creatinine >2.5 mg/dL or in pregnancy; causes DVT (deep vein thrombosis), haemolytic anaemia.

**Diagram to draw:** staghorn with an upper-pole PCNL tract and a retrograde flexible scope (ECIRS).
```

**Key points to remember:**

- Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: ammonium chloride (NH4Cl) fails to lower pH below 5.5; cystine <250 mg/L at pH 7.0–7.5.
- Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months of therapy.
- NCCT gold standard (97%/95%).
- Staghorn: PCNL first line; complete clearance is the goal.
- Bilateral: drain an infected side first, then the better kidney.
- AHA only when surgery is not possible.

## LAQ 4 Retroperitoneal fibrosis: diagnosis and (medical) management

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 LAQ "Discuss the pathology, etiology, Clinical features diagnostic evaluation and treatment of retroperitoneal fibrosis." · P1 SAQ "Retroperitoneal fibrosis" · P2 SAQ "Retro peritoneal fibrosis medical management"

### Introduction

- RPF (retroperitoneal fibrosis) is a rare inflammatory fibrosis (fibrous plaque) encasing the aorta, IVC (inferior vena cava) and ureters, centred on the **distal aorta at L4–L5**.
- Incidence about 1 per 200,000–500,000 a year; age 40–60; male:female 2–3:1.

### Etiology

| Idiopathic (about 70%) | Secondary (about 30%) |
|---|---|
| Part of **chronic periaortitis**, a large-vessel vasculitis | **Drugs** (commonest identifiable cause): methysergide and other ergots, β-blockers, phenacetin |
| Proposed antigen: **ceroid** (oxidised lipid in plaque) | **Malignancy:** lymphoma (commonest), myeloma, carcinoid, pancreas, prostate, sarcoma |
| Probably autoimmune: membranous glomerulonephritis, ankylosing spondylitis, granulomatosis with polyangiitis; ANA (antinuclear antibody) positivity predicts relapse | Radiotherapy; asbestos |
| Part of multifocal fibrosclerosis: sclerosing mediastinitis and cholangitis, orbital pseudotumour, Riedel thyroiditis | Infection: tuberculosis, actinomyces, gonorrhoea, schistosomiasis |

- Many idiopathic cases are now classed as **IgG4 (immunoglobulin G4)-related disease** (raised serum IgG4, IgG4-positive plasma cells, storiform fibrosis).

### Pathology

- **Gross:** flat, tan, dense plaque centred on the **distal aorta at L4–L5**, wrapping the ureters; may invade ureter or psoas.
- **Histology changes with stage:**
- Early (active): collagen with capillary proliferation and lymphocytes, plasma cells and fibroblasts (steroid-responsive).
- Late (burnt-out): acellular, avascular sheets of collagen.
- **Trap:** malignant RPF looks identical except for small islands of tumour cells, so take **deep, multiple biopsies**.
- **Mechanism of obstruction:** extrinsic compression and loss of ureteric peristalsis; the lumen stays patent.

### Clinical features

- **Dull lower back or flank pain** (commonest); weight loss, anorexia, nausea, malaise, fever, hypertension.
- About half present with **uraemia** or oliguria/anuria; symptoms last 4–6 months before diagnosis.

- **Extension:** IVC (leg oedema, DVT [deep venous thrombosis]); renal vein (gross haematuria); spermatic cord and scrotum. Examination is usually unrevealing.

## Diagnostic evaluation

### Laboratory

- Raised ESR (erythrocyte sedimentation rate) and CRP (C-reactive protein) in half to two-thirds; leucocytosis; anaemia; raised creatinine (renal failure); ANA; serum IgG4.

### Imaging

| Test | Findings and role |
|---|---|
| CT (computed tomography) / CT urography | Homogeneous soft-tissue mass enveloping aorta, IVC and ureters; **medial deviation** of proximal and mid-ureters with smooth tapering |
| Retrograde pyelography | When renal function is poor |
| MRI (magnetic resonance imaging) | Low T1; **high T2 signal means active disease**; enhancement falls with treatment |
| **PET-CT (positron emission tomography)** | **Most sensitive test of activity**; guides biopsy and follow-up |
| Renography | Split function (repair or nephrectomy) |

- **Biopsy** (CT-guided or at ureterolysis) if atypical or malignancy is suspected; may be omitted with no malignancy history, no lymphadenopathy and typical imaging.

## Treatment algorithm

```
RPF with hydronephrosis -> Relieve obstruction (stents; nephrostomy if anuric or septic) -> Treat post-obstructive diuresis -> Stop culprit drug; exclude malignancy -> Active disease: steroids (+/- second-line drugs) -> Failure or burnt-out fibrosis: ureterolysis
```

## Initial management

- Emergent **stents or PCN (percutaneous nephrostomy)**; nephrostomy if critically ill, uraemic and anuric with little urine. Replace fluid during **post-obstructive diuresis**.
- **Stop the culprit drug;** treat any secondary cause.

## Medical management

| Agent | Role and regimen |
|---|---|
| **Prednisone** (first line) | About 60 mg daily (about 1 mg/kg), tapered to 5 mg, total course up to 2 years. About 80% respond; relapse on tapering 25–50% |
| Mycophenolate mofetil (with steroids), azathioprine, cyclophosphamide, ciclosporin, colchicine | Second line or steroid-sparing |
| Tamoxifen; medroxyprogesterone | Thought to inhibit fibroblasts |
| **Rituximab** | Refractory or IgG4-related disease |

- Steroids work on the cellular, active phase: pain and symptoms settle in days, ESR falls and the mass shrinks. Active inflammation predicts response; burnt-out fibrosis responds poorly.

- **Vaglio RCT (randomised controlled trial; Lancet 2011):** relapse 17% with prednisone vs 50% with tamoxifen, so steroids remain first line.
- Drainage plus medical therapy clears obstruction in about two-thirds; only about **15% need ureterolysis**.

### Surgical ureterolysis

- For failed medical therapy, burnt-out (fibrotic) disease or tissue diagnosis. **Bilateral**, even if hydronephrosis is unilateral; open, laparoscopic or robotic.
- Steps: 1. Pre-place stents to identify the ureters. 2. Mobilise the colon; take deep biopsies for frozen section. 3. Start at the distal, non-dilated ureter. 4. "Split and roll": right-angle clamp between ureter and plaque; incise the fibrosis over it. 5. Close any ureterotomy with absorbable suture. 6. **Separate the ureter from the fibrotic bed so it cannot be re-encased.**

| Protection method | How |
|---|---|
| Intraperitonealisation | Ureter lateralised; peritoneum closed behind it |
| Lateral retroperitoneal placement (lateralisation) | Ureter displaced laterally (equivalent results) |
| **Omental wrap** | Omentum split on the gastroepiploic arteries, wrapped round each ureter; adds blood supply; best for extensive disease |

- Postoperative steroids may reduce recurrence.
- **Outcomes:** open about 94% stent-free at 1 year; laparoscopic about 90%; robotic with omental wrap about 86%.

### Other options

- Long-term metallic stents (unfit); autotransplantation (rarely needed) if ureterolysis is impossible; nephrectomy for an unrecoverable (dead) kidney.

### Follow-up and monitoring

- Symptoms, ESR/CRP, creatinine, ultrasound; periodic CT/MRI or PET-CT for activity; remove stents once the ureters drain.
- Watch for relapse and steroid side effects (diabetes, osteoporosis).

**Diagram to draw:** periaortic plaque at L4–L5 drawing both ureters medially, with an omental wrap.

> **Key points to remember:**
> - Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
> - Medially deviated ureters; PET-CT best shows activity; biopsy deep.
> - Drain first, then prednisone (about 80% respond).
> - Vaglio 2011: prednisone better than tamoxifen (relapse 17% vs 50%).
> - Rituximab for IgG4-related or refractory disease.
> - About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

## LAQ 5 Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management

P2 · P3 **merged from 2**

*Asked as:* P2 LAQ "Classification and treatment of lower track trauma in # pelvis" (also P2 SAQ "Pelvic # assessment and management" and P2 SAQ "Early surgical intervention in urethral injuries") · P3 LAQ "Describe early and late management of PFUD injuries"

*PFUD is taken as pelvic fracture urethral distraction defect, i.e. PFUI (pelvic fracture urethral injury). Lower urinary tract trauma means bladder and urethral injury in pelvic fracture.*

### Introduction

- Pelvic fracture accompanies **83–95% of bladder ruptures**, yet only **5–10% of pelvic fractures** injure the bladder. Urethral injury (PFUI) occurs in about **10% of male** and up to 6% of female pelvic fractures.
- 80–94% have other major (associated) injuries, which cause the deaths; long-term disability comes from wrong management.

### Mechanism

- **EPR (extraperitoneal rupture):** pelvic disruption (shearing) tears the bladder at its fascial attachments, or bone spicules lacerate it.
- **IPR (intraperitoneal rupture):** a blow on a **full bladder** bursts the dome, the weakest part, covered by peritoneum.
- **PFUI:** the posterior urethra is fixed to the pubis (puboprostatic ligaments, perineal membrane), so displacement shears it, usually at the **bulbomembranous junction**. In prepubertal boys the tear may extend into the prostate and bladder neck.
- Highest risk: straddle fracture of all four pubic rami, vertically unstable fractures, symphyseal diastasis.
- The sphincter is usually avulsed but stays functional, so most men stay continent after repair.
- The gap fills with scar: an obliterative **distraction defect**, not a true stricture.

### Classification

#### Bladder: AAST (American Association for the Surgery of Trauma) grades I–V

| Grade | Injury |
|---|---|
| I | Contusion, intramural haematoma, partial-thickness laceration |
| II | Extraperitoneal wall laceration <2 cm |
| III | Extraperitoneal ≥2 cm, or intraperitoneal <2 cm |
| IV | Intraperitoneal ≥2 cm |
| V | Laceration extending into bladder neck or ureteric orifice (trigone) |

- Practical: contusion, EPR (simple or complex), IPR, combined.

#### Urethra: Goldman I–V

| Type | Injury |
|---|---|
| I | Posterior urethra stretched, intact |

| Type | Injury |
|---|---|
| II | Disruption above the urogenital diaphragm (perineal membrane) |
| III | Disruption of membranous urethra through the urogenital diaphragm, extending below the membrane into the bulb (commonest) |
| IV | Bladder neck injury extending into urethra (IVa: bladder-base injury mimicking IV) |
| V | Pure anterior urethral injury |

- **Campbell-Walsh practical grading: partial injury; minor distraction** (a cystoscope passes retrogradely into the bladder); **major distraction** (a cystoscope cannot pass into the bladder). Practically partial vs complete disruption. Stricture risk and need for urethroplasty rise from partial to major.
- Anterior urethra: contusion, incomplete or complete.

### Assessment and clinical features

- **ATLS (Advanced Trauma Life Support)** resuscitation first; associated injuries take priority; pelvic binder and stabilisation.
- Bladder: **gross haematuria (nearly all)**, suprapubic pain, inability to void, free fluid.
- Urethra: **triad: blood at the meatus, inability to void, palpable bladder**. A "high-riding" prostate and butterfly perineal haematoma are unreliable.
- **Vaginal and rectal examination**; vaginal examination in every female.

### Investigations

- **Blood at the meatus mandates RUG (retrograde urethrogram) before any catheter** (AUA [American Urological Association]). Method: 16 Fr catheter tip in fossa navicularis, 3 mL balloon, penile traction, oblique position, 20 mL contrast, film during injection. Extravasation with bladder filling means partial injury; without filling, complete. Females: urethroscopy.
- **Cystography:** absolute indication: **gross haematuria + pelvic fracture** (rupture in ~29%). Relative: gross haematuria alone; microscopic haematuria with fracture.
- **CT cystography:** retrograde fill with **350 mL** of 2–4% contrast; no drainage film needed.
- Plain cystogram: scout, full-bladder and **drainage films** (posterior leaks otherwise missed).
- Clamping the catheter after IV contrast is inadequate.
- **EPR:** flame-shaped pelvic contrast. **IPR:** contrast outlines bowel loops.

### Bladder injury: management

- **Uncomplicated EPR:** 22 Fr urethral catheter 2–3 weeks; cystogram before removal; repair if leak persists beyond 4 weeks.
- **Immediate repair:** IPR; penetrating injury; bladder neck injury; bone fragments in the bladder; rectal or vaginal injury; poor drainage or clots; open pelvic fixation or laparotomy for other reasons.
- IPR: open or laparoscopic two-layer closure; inspect ureteric orifices; cystogram at 7–10 days. At laparotomy, repair EPR from inside the bladder, leaving the pelvic haematoma undisturbed.
- Suprapubic tube adds nothing over urethral catheter (AUA). Antibiotics only around IPR repair.
- Rectal or vaginal injury: separate the walls, no overlapping suture lines, interpose tissue.

## Posterior urethra (PFUI): early management

- Key decision: **immediate repair versus deliberate delay**. Clean, localised injuries are repaired at once; PFUI is drained and reconstructed later.

```
Blood at meatus → RUG Partial → one gentle catheter attempt (confirm position)
Complete → SPC (suprapubic catheter): percutaneous 16 Fr or open (if bladder
repair needed) ± brief primary realignment if stable (flexible cystoscopy,
antegrade + retrograde) Bladder neck/rectal injury → immediate repair Delayed
(2–3 months) → up-and-down-o-gram → perineal excision and primary anastomosis
(corporal split → pubectomy → rerouting)
```

1. ATLS resuscitation; associated injuries take priority.
2. **SPC (suprapubic cystostomy): the standard of care.** Percutaneous (peel-away sheath, 16 Fr Foley; avoid small "punch" tubes) or open through a lower midline incision, which allows bladder inspection (needed if bladder repair is required). - SPC is safe with pelvic plating (AUA Urotrauma guideline); place and tunnel it high, away from hardware.
3. **Primary realignment (PR)** (AUA Urotrauma option): a brief attempt in a stable patient, acutely or within days, by combined antegrade and retrograde flexible cystoscopy over a guidewire (avoid prolonged attempts). Catheter out at 4–6 weeks; keep the SPC and remove it 7–14 days later if voiding well.
4. Partial tear: one gentle catheter attempt allowed.
5. **Immediate open exploration/repair** only for **bladder neck or rectal injury**, or when laparotomy is needed anyway. - **Also know:** female urethral injuries are usually repaired early; in boys tears may extend into the bladder neck.

## SPC versus primary realignment

| | **SPC + delayed urethroplasty** | **Primary realignment** |
|---|---|---|
| Advantage | Simple; haematoma untouched | Partial tears may heal; aligned ends ease repair |
| Drawback | Complete injuries need urethroplasty | Most still form a 1–2 cm stenosis |
| Evidence | Prospective data: **no significant difference** in need for urethroplasty | Modern PR does not add ED (erectile dysfunction) or incontinence |

## Anterior urethra: early management

| **Injury** | **Early management** |
|---|---|
| Contusion, incomplete straddle tear | Urethral catheter alone |
| Major straddle injury | SPC (standard); primary realignment for lesser injuries; delayed anastomotic urethroplasty |
| **Low-velocity gunshot or stab** | **Immediate primary repair:** debride minimally, spatulate, tension-free anastomosis over a catheter |
| High-velocity or close-range shotgun | SPC, staged repair |
| **Penile fracture with urethral tear** | **Immediate repair:** partial tear oversewn; complete tear debrided and anastomosed; catheter 2–3 weeks |

- Penile fracture: urethral injury in about 20% of Western and 3% of Asian series; **intraoperative flexible cystoscopy** before catheterisation. Limit spongiosal debridement.

## Late management of PFUI

**Timing:** delayed repair at 2–3 months (P3: after **3 months**), once the scar is stable and the patient is ambulant.

### Evaluation

- **"Up-and-down-o-gram":** cystogram plus RUG while the patient tries to void, to measure the gap length; antegrade flexible cystoscopy via the SPC tract if the bladder neck stays closed / does not open. Bladder neck appearance does **not** predict its later function.
- **MRI (magnetic resonance imaging)** for complex cases: gap length, prostatic displacement (cephalad displacement predicts pubectomy).

**Endoscopic:** DVIU (direct vision internal urethrotomy) only for short, incomplete stenosis. "Cut-to-the-light" or repeated DVIU for gaps ≥1 cm fails (false passages).

### Perineal excision and primary anastomosis (EPA; delayed anastomotic urethroplasty): treatment of choice

1. Exaggerated lithotomy (under 5 hours); sounds via meatus and SPC tract.
2. Midline or lambda incision; mobilise the bulbar urethra up to the penoscrotal junction (not beyond: chordee).
3. Divide the urethra at the scar; excise **all** fibrosis until healthy prostatic mucosa appears over the sound (identify the prostatic apex).
4. Spatulate both ends to 30 Fr; 8–10 interrupted 4-0 absorbable sutures; tension-free mucosa-to-mucosa anastomosis over a silicone catheter.

**Progressive perineal approach** (each step shortens the path to bridge a longer gap):

| Step | Manoeuvre |
|---|---|
| 1 | Bulbar urethral mobilisation (enough for most) |
| 2 | Separation of the corporal bodies (corporal splitting, intracrural space) |
| 3 | Inferior wedge pubectomy (Kerrison rongeur) |
| 4 | Supracrural rerouting of the urethra around one corpus |

- **Abdominoperineal or transpubic approach:** urethrorectal fistula, failed repair, bladder neck injury, children.
- **Postoperative:** voiding study at 21–28 days; cystoscopy at about 6 months.

## Outcomes and complications

- Urethroplasty success above 95%. Restenosis 5–15%; often DVIU-treatable because the scar is already excised.
- **ED** up to 82%, mostly from the injury (cavernous nerves, pudendal arteries; nerve, arterial, venous). **Incontinence <4%** (uncommon; bladder neck).
- Missed IPR: ascites, acidosis, azotaemia (peritoneal reabsorption).
- Fistula, abscess, neurogenic voiding dysfunction; follow up ≥1 year.
- **Also know:** chordee, penile shortening, ejaculatory dysfunction, lithotomy neuropathy, penile Doppler before repair in impotent men.

**Diagram to draw:** pelvis with EPR (flame-shaped) vs IPR on cystogram; RUG showing partial vs complete posterior urethral disruption; sagittal pelvis showing the bulbomembranous gap and the four progressive perineal steps.

> **Key points to remember:**
>
> - Blood at meatus: RUG before catheter.
> - Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
> - EPR: catheter 2–3 weeks; IPR: immediate repair.
> - PFUI: SPC is the standard initial step, then delayed perineal anastomotic urethroplasty at 2–3 months (P3: after 3 months).
> - Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
> - Progressive perineal steps: bulbar mobilisation, crural separation, pubectomy, rerouting.
> - Immediate repair: bladder neck or rectal injury, penile fracture urethra, low-velocity penetrating injury.
> - ED mostly from the injury; incontinence <4%.

## LAQ 6 Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management

P2

**Asked as:**

- Traumatic renal vascular parenchymal injuries : diagnosis and management. — *Summer 2017 Paper II Q2 (LAQ)*
- Diagnosis and Treatment of renal trauma — *Practice Set 9 Paper II Q1 (LAQ)*
- Grade 4 Renal trauma. — *Practice Set 3 Paper II Q3c (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)
(Practice Set 9 Paper II Q1: also answered in 2017-Summer Paper 2, Q2; grade IV in Practice Set 3 Paper 2, Q3c) (Practice Set 3 Paper II Q3c: renal trauma also answered in 2017-Summer Paper 2, Q2)

*Assumed meaning: diagnosis and management of traumatic renal parenchymal and renal vascular injuries.*

### Introduction

- Kidney is the **commonest urological organ injured** (up to 5% of all trauma); **blunt injury causes 80–90%**; firearms cause most penetrating injuries.
- **Rapid deceleration** stretches the kidney at its fixed points (hilum and UPJ, ureteropelvic junction): arterial intimal tear and thrombosis, vein disruption, pedicle or UPJ avulsion.
- Main renal artery injury is rare (about 0.05% of blunt trauma).

### Classification: AAST (American Association for the Surgery of Trauma) 2018

| Grade | Injury |
|---|---|
| I | Subcapsular haematoma and/or contusion, no laceration |
| II | Perirenal haematoma within Gerota's fascia; laceration ≤1 cm, no extravasation |
| III | Laceration >1 cm without collecting-system injury; **any vascular injury or active bleeding contained within Gerota's fascia** |
| IV | Laceration into collecting system with extravasation; renal pelvis laceration and/or complete UPJ disruption; **segmental artery or vein injury**; active bleeding beyond Gerota's fascia; segmental or complete infarction (thrombosis) |
| V | **Main renal artery or vein laceration or hilar avulsion**; devascularised kidney with active bleeding; shattered kidney |
| - Advance one grade for multiple injuries, up to grade III. Change from 1989: main vessel injury moved from IV to V; pseudoaneurysm and AVF (arteriovenous fistula) now grade IV. | |

### Definition: AAST 2018, grade IV in detail

- Laceration into the **collecting system with urinary extravasation**.
- **Renal pelvis laceration and/or complete UPJ (ureteropelvic junction) disruption.**
- **Segmental artery or vein injury** (pseudoaneurysm, AVF [arteriovenous fistula]).
- **Active bleeding beyond Gerota's fascia.**
- **Segmental or complete infarction** from thrombosis without active bleeding.
- Change from 1989: main-vessel injury moved to grade V; segmental vascular and UPJ injuries added to grade IV.

## Clinical diagnosis

### History and examination

- Mechanism is the most important (key) history; mechanism matters. Signs: flank bruise or tenderness, **fractured ribs 9–12**, flank or low chest wound.

### Haematuria

- Test the **first urine sample**. Significant: gross haematuria, or microscopic (microhaematuria) (>5 RBC [red blood cells] per high-power field, hpf) with **shock (SBP [systolic blood pressure] <90 mmHg)**.
- Degree does **not** match severity; it may be **absent in pedicle injury** and UPJ avulsion (in about half of UPJ injuries), and in grade IV.

## Indications for imaging (AUA (American Urological Association) and EAU (European Association of Urology))

1. Blunt trauma with gross haematuria.
2. Blunt trauma with microhaematuria and **SBP (systolic blood pressure) <90 mmHg** at any time.
3. Rapid deceleration (high-speed collision, fall), even without haematuria.
4. Penetrating trauma with likely renal injury, any haematuria.
5. Children: image liberally (hypotension appears late). - Microhaematuria without shock or deceleration (microhaematuria alone without shock): observe without imaging.

## Imaging

- **Contrast-enhanced CT (computed tomography)** is the gold standard. Arterial/nephrographic phase (~80 s) shows bleeding; **delayed excretory phase** (5–10 min / minutes) shows urinary extravasation (urine leak).
- **CT red flags:**

| Finding | Suggests |
|---|---|
| Medial haematoma | Pedicle (vascular) injury |
| Medial extravasation, no contrast in distal ureter (without distal ureteric contrast) | Renal pelvis or UPJ avulsion |
| Absent enhancement, cortical rim sign | Main renal artery occlusion |
| Active contrast blush (vascular contrast blush) | Arterial bleeding needing AE (angioembolisation) |

| Finding | Suggests |
|---|---|
| Haematoma rim >4 cm (another source: >3.5–4 cm), medial laceration | High risk of intervention; bleeding likely to need intervention (≥2 of rim, blush, medial laceration predict intervention) |
| - **One-shot IVP (intravenous pyelography)** on table (2 mL/kg, one film at 10 min) in the unstable patient: confirms a **functioning contralateral kidney**. FAST (focused assessment with sonography for trauma) is non-specific. Renal vein injury is poorly seen on CT. | |

## Management

`Unstable despite resuscitation → laparotomy (± damage control) (or angioembolisation / AE if team ready, else laparotomy) Stable → stage with CT Grade I–III → non-operative Grade IV–V → non-operative in selected; AE for active bleeding Active blush / pseudoaneurysm / AVF → superselective AE Extravasation → observe (90% heal); stent ± nephrostomy if urinoma enlarges, fever, obstruction, or UPJ/pelvis injury`

- **Non-operative**: bed rest until gross haematuria clears, serial haemoglobin (**every 4–6 h**); success >90% in blunt injury and selected penetrating ones. **Non-operative success** about 85% for grade IV–V even in penetrating series; most failures need only AE, stent or nephrostomy.
- Failure: instability, or falling haemoglobin despite 2–3 units of blood.
- Urinary extravasation heals spontaneously in **76–90%** (another source: up to 90%).
- **Angioembolisation**: superselective coils for active bleeding, pseudoaneurysm, AVF or delayed bleeding; repeat AE often succeeds.
- **Reimaging** (reimage / repeat CT at ~48 hours) for grade IV–V with deep laceration (deep lacerations), or new fever, pain or falling haematocrit (routine reimaging debated).

## Indications for exploration

| Absolute | Relative |
|---|---|
| Haemodynamic instability, no or transient response | Extravasation with devitalised parenchyma (>25%; lowers the threshold) |
| Expanding or pulsatile retroperitoneal haematoma | Arterial thrombosis |
| Suspected pedicle avulsion | Associated colon or pancreatic injury |
| UPJ avulsion | Incomplete staging |

## Operative technique (operative steps)

- Midline transabdominal (laparotomy); life-threatening injuries first.
- **Vascular control before opening Gerota's fascia**: lift small bowel; incise retroperitoneum over the aorta above the inferior mesenteric artery → find the left renal vein crossing the aorta → loop vein and both arteries; then reflect colon, open Gerota's fascia.
- **Renorrhaphy**: debride → suture-ligate vessels → watertight collecting-system closure → close defect over bolster → omental cover → drain. Polar injury: partial nephrectomy.

- **Nephrectomy** if unreconstructable or unstable. **Damage control**: pack, resuscitate in ICU (intensive care unit), relook (return) at 24–48 h.

### Renovascular injury

- **Main artery thrombosis** (intimal tear): unilateral with normal other kidney → observe (revascularisation rarely succeeds). **Solitary or bilateral → revascularise** (warm ischaemia ideally <120 min); endovascular stent for (or stent an) intimal flap.
- **Vein**: left renal vein can be ligated near the IVC (inferior vena cava) (gonadal, adrenal collaterals); right cannot. Repair with 5-0 vascular suture; segmental veins may be ligated.

### Complications

- **Delayed bleeding** (within 3 weeks; usually within 21 days): AE (embolise).
- Urinoma or abscess: stent, nephrostomy, drain.
- **Hypertension**: Goldblatt (arterial stenosis), **Page kidney** (haematoma compression), segmental ischaemia, AVF.
- Hydronephrosis, loss of function; **also know:** AKI (acute kidney injury).

### Follow-up

- Blood pressure, urinalysis, creatinine; EAU: DMSA (dimercaptosuccinic acid) scan for salvaged grade III–V kidneys; imaging if new symptoms.

**Diagram to draw:** left renal vein crossing the aorta, with vessel loops on the vein and artery. For grade IV: kidney with laceration into a calyx, urinoma and segmental pseudoaneurysm.

> **Key points to remember:**
> - AAST 2018: segmental vessel injury grade IV; main vessel grade V.
> - Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
> - Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
> - Delayed CT phase is essential; blush means AE.
> - Stable patients, even grade IV–V: non-operative ± AE (angioembolisation).
> - Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.
> - Revascularise only a solitary kidney or bilateral injuries.

## LAQ 7 ESWL: mechanism, indications, factors affecting outcome, complications and current status

P2

**Asked as:**

- Describe indication, mechanism of stone fragmentation and current status of ESWL? — *Summer 2019 Paper II Q2 (LAQ)*
- Describe factors affecting outcome of ESWL. Discuss complications of ESWL and strategies to avoid them. — *Summer 2021 Paper II Q1 (LAQ)* (Mechanism, indications and current status are answered in 2019-Summer Paper 2, Q2.)
- Desc. factors influencing outcome of ESWL. Discuss complic. & strategies to avoid them. — *Practice Set 13 Paper II Q2 (LAQ)* (also answered in 2021-Summer Paper 2, Q1; mechanism and indications in 2019-Summer Paper 2, Q2)

**Type:** LAQ (25 marks)

### Introduction

- ESWL (extracorporeal shock wave lithotripsy), now SWL, breaks stones with focused acoustic shock waves generated outside the body (generated at F1 and focused on the stone at F2). First human use 1980 (Dornier HM1).
- A shock wave is a short pulse: peak positive pressure 30–100 MPa, then a negative tail of about −10 MPa.
- Stones break by spallation, shear, squeezing and **cavitation**. Cavitation also causes most tissue injury, so steps that improve outcome usually reduce complications too.
- Outcome = SFR (stone-free rate) and need for retreatment.

### Components of a lithotripter

- Energy source at F1 (first focus); focusing system; coupling (water or gel); fluoroscopy and/or ultrasound to place the stone at F2. A focal zone wider than the stone improves breakage.

| Energy source | Principle | Advantages | Disadvantages |
|---|---|---|---|
| Electrohydraulic (HM3) | Underwater spark; ellipsoid reflector | Most effective | Electrode wear; variable shocks |
| Piezoelectric | Ceramic elements on a spherical dish | Long life; little pain | Low energy; tiny focal zone |
| Electromagnetic (modern machines) | Coil moves a metal membrane; lens focuses | Consistent; high energy at F2 | — |

### Mechanisms of stone fragmentation

- Each shock creates stress inside the stone. Above a critical value, cracks form; repeated shocks make them grow and join (**fatigue**).

| Mechanism | How it works |
|---|---|
| **Spallation** | Wave reflects and inverts at the back surface of the stone; tension breaks off the distal face |
| **Shear and tear (superfocusing)** | Waves inside the stone reflect and focus; cracks form where shear peaks |

| Mechanism | How it works |
|---|---|
| **Quasistatic squeezing** | Wave travels faster in stone than fluid; a ring of pressure compresses the stone |
| **Dynamic squeezing** | Shear waves at the stone edges combine; strongest force in initial breakage |
| **Cavitation** | Negative phase forms bubbles; collapse erodes the surface. Excess bubbles shield the stone and injure tissue |

## Indications

- **Renal (AUA/EAU):** <2 cm outside the lower pole and lower pole <1 cm (SWL or URS [ureteroscopy]); EAU: 10–20 mm equal to endourology; >20 mm only after PCNL (percutaneous nephrolithotomy).
- **Ureteric:** any level, best <10 mm; emergency SWL within 48 hours works well. Patients avoiding anaesthesia and stents; children.
- **Poor choices:** >2 cm or staghorn; lower pole >1 cm (**Lower Pole I**: stone-free 21% SWL vs 91% PCNL); cystine, brushite, calcium oxalate monohydrate; infection stones.

## Contraindications

- **Absolute:** pregnancy; uncorrected coagulopathy or (ongoing) antithrombotics; untreated UTI (urinary tract infection); obstruction distal to the stone; nearby arterial aneurysm (aneurysm near the stone; aortic >4 cm); severe uncontrolled (untreated) hypertension.
- **Relative:** obesity or skeletal deformity preventing targeting; pacemaker.

## Factors affecting (influencing) outcome / predictors of success

### 1. Stone factors

| Factor | Favourable | Unfavourable |
|---|---|---|
| Size (most important) | <10 mm | Renal >20 mm (PCNL [percutaneous nephrolithotomy] first); ureteric ≥10 mm favours URS (ureteroscopy) |
| Density on NCCT (non-contrast computed tomography / non-contrast CT [computed tomography]) | <900 HU (Hounsfield units) clears best | >1000 HU predicts failure |
| Composition | Calcium oxalate dihydrate, uric acid | Cystine, brushite, calcium oxalate monohydrate (dense, homogeneous) |
| Impaction (ureter) | Recent, mobile | Chronic impaction (soft-tissue ring of thick ureteral wall around the stone on CT, unchanged position): may be the strongest predictor |
| Burden and site | Single, renal pelvis or upper calyx | Multiple sites; lower pole >1 cm |

- Infection stones and stones in several sites: prefer a more efficient modality.

### 2. Renal anatomical factors

- **Lower pole (lower pole anatomy):** fragments drain against gravity. **Lower Pole I (Albala, 2001):** stones >1 cm, SFR **21% SWL vs 91% PCNL**. A steep infundibulopelvic

angle and a long (>10 mm), narrow (<5 mm) infundibulum reduce clearance.
- Obstruction distal to the stone (UPJO [ureteropelvic junction obstruction], infundibular stenosis, calyceal diverticulum) traps fragments. Horseshoe and ectopic kidneys drain poorly.

**3. Patient factors**

- **SSD (skin-to-stone distance) >10 cm** (obesity) predicts failure. Skeletal deformity can prevent targeting.
- Respiratory movement of 10–15 mm moves the stone out of F2.
- **Triple D score:** stone density, stone volume and skin-to-stone distance together predict success.

**4. Machine and technique factors (technique to optimise results)**

1. **Energy source:** electrohydraulic (HM3) breaks best; piezoelectric has low energy and a tiny focal zone; modern machines are electromagnetic. A focal zone wider than the stone breaks better.
2. Treat infection; stop antithrombotics; control blood pressure. No routine stent: it does not improve clearance.
3. **Analgesia:** good analgesia, GA (general anaesthesia) or sedation keeps the stone in focus, because stone movement reduces breakage.
4. **Coupling:** gel as a mound, free of air; air pockets in the gel cut fragmentation by up to two-fifths.
5. **Rate 60–90 shocks/min** beats 120/min, because the bubble cloud clears between shocks (bubbles clear between shocks), so less shielding and renal injury.
6. **Ramping:** start at low energy and step up (ramping energy up also helps); this causes protective vasoconstriction.
7. Positioning (prone for mid-ureteric stones over the sacrum), continuous imaging (monitor with imaging), experienced team. Little gain beyond two sessions (more than two sessions add little).
8. **Adjuncts:** α-blockers (faster fragment clearance); potassium citrate; percussion, diuresis and inversion for lower pole fragments.

## Complications

| Group | Complication | Comment |
|---|---|---|
| Fragment-related | Renal colic; **steinstrasse** ("stone street" of fragments in the ureter); residual fragments | Steinstrasse rises several-fold with burden >2 $cm^2$ |
| Infective | UTI (urinary tract infection), pyelonephritis, urosepsis (sepsis) | Infection stones, stents |
| Renal tissue injury | Haematuria (transient, common); **subcapsular or perirenal haematoma** (minority; rarely embolisation) | Risk: age, hypertension, obesity, diabetes, coronary disease, thrombocytopenia, anticoagulation, high energy, many shocks, narrow-focus machines |
| Other organs | Liver, spleen, pancreas, colon / bowel, lung (children), skeletal muscle | Very rare |
| Cardiac | Arrhythmia during treatment | Pause or ECG (electrocardiogram) gating |

| Group | Complication | Comment |
|---|---|---|
| Skin | Petechiae, bruising | Minor |

- **Refuted long-term effects** (**not** proven): new hypertension, diabetes, decline in (loss of) renal function and infertility.

### Strategies to avoid complications

1. **Select.** Absolute contraindications as above. Send large, dense, lower pole or impacted stones to URS or PCNL.
2. **Prepare.** Urine culture; treat bacteriuria. Prophylaxis only with risk factors (stent, nephrostomy, struvite, recent UTI, bowel reconstruction). Stop antithrombotics before and for the day and (about) 48 hours after. Control blood pressure; reprogramme / check pacemakers.
3. **Protect the kidney.** Slow rate; **slow ramping** (ramp over at least 4 minutes) (induces protective renal vasoconstriction and lowers haematoma risk); fewest shocks and lowest effective energy.
4. **Prevent steinstrasse.** PCNL or URS for large burdens. Routine stenting does not improve SFR, though it may reduce steinstrasse with large stones. α-blockers after SWL.
5. **Treat early.** Steinstrasse: MET (medical expulsive therapy), SWL to the lead fragment or URS; **nephrostomy if fever or obstruction**. Haematoma: bed rest, transfusion, embolisation if bleeding continues.
6. **Follow up** with KUB (kidney-ureter-bladder radiograph) and ultrasound.

### Current status

- **Declining use worldwide.** URS has grown with small digital scopes and lasers, giving higher single-session stone-free rates. Newer lithotripters are also less effective than the HM3.
- **SWL vs URS:** SWL has fewer complications, no stent or general anaesthesia, lower cost, but more retreatment. Guidelines still list it first-line for small renal and ureteric stones.

### Recent advances

- **Burst wave lithotripsy:** bursts of 5–100 ultrasound cycles at 4–8 MPa; less cavitation; done awake (no anaesthesia); uniform fragments.
- **Ultrasonic propulsion** moves fragments to help them pass.

**Diagram to draw:** ellipsoid reflector with F1 (spark) and F2 (stone), stone and gel coupling, with spallation and squeezing arrows on the stone; box of failure predictors.

> **Key points to remember:**
> - Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
> - Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
> - Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
> - Slow rate (60–90 shocks/min), ramping and good coupling improve breakage and reduce injury.
> - Steinstrasse risk rises with burden >2 $cm^2$; no routine stent.
> - Use is declining as URS grows; burst wave lithotripsy is coming.

## LAQ 8 Management of upper ureteric stones

P2

**Type:** LAQ (25 marks)
**Source:** Practice Set 2 Paper 2, Q2 (SWL also answered in 2019-Summer Paper 2, Q2 and 2021-Summer Paper 1, Q1; flexible URS in 2019-Summer Paper 2, Q3b)

### Introduction

- **Upper (proximal) ureter:** UPJ (ureteropelvic junction) to the upper border of the sacroiliac joint.
- Proximal stones pass least often, so many need intervention. Dangers: renal damage and sepsis.

### Evaluation

- **NCCT (non-contrast CT)**: gold standard; gives size, density (HU, Hounsfield units), SSD (skin-to-stone distance) and impaction. KUB (kidney–ureter–bladder radiograph) for SWL follow-up; creatinine, urine culture, coagulation.

### 1. Emergency: drain first

- **AUA (American Urological Association) indications for urgent intervention:** infection with obstruction, bilateral obstruction, solitary kidney, immunosuppression, renal deterioration, refractory pain or vomiting.
- **Obstructed infected kidney: drain only** (double-J stent or PCN, percutaneous nephrostomy; equal for sepsis); definitive treatment later. PCN suits large stones (stent fails if transverse diameter ≥9.5 mm).

### 2. Observation and MET (medical expulsive therapy): limits

| Spontaneous passage | Rate |
|---|---|
| By size | 1–4 mm 78%; 5–7 mm 60%; 8–10 mm 39% |
| By site | UVJ (ureterovesical junction) 92%; distal 75%; mid 60%; **proximal 48%** |

- **Observation** suits small (<5–6 mm) uncomplicated stones: NSAIDs (non-steroidal anti-inflammatory drugs) first.
- **MET (alpha-blockers, tamsulosin or silodosin):** reduce spasm so a larger urine bolus pushes the stone. Benefit is clear only for **distal stones 5–10 mm**; meta-analysis shows **no benefit for proximal stones**. SUSPEND (2015) showed no overall benefit.
- Trial of passage is limited to **4–6 weeks**, with imaging.

### 3. Choice of active treatment

- Factors: stone (size, density, impaction), patient (habitus, anticoagulation, preference), surgeon and equipment.

**EAU (European Association of Urology) recommendation for proximal ureteric stones**

| Size | First choice | Alternative |
|---|---|---|
| <10 mm | SWL or URS (ureteroscopy) | – |
| >10 mm | URS (retrograde or antegrade) | SWL |
| Large (>15–20 mm), impacted | PCNL with antegrade URS | Laparoscopic ureterolithotomy |

**SWL versus URS**

| | SWL (shock wave lithotripsy) | URS |
|---|---|---|
| Stone-free rate (AUA/EAU meta-analysis, approximate) | <10 mm about 90%; >10 mm about 70% | <10 mm about 80%; >10 mm about 80% |
| Sessions | More retreatment | Single session; higher SFR at 2 weeks, equal at 3 months |
| Anaesthesia | Sedation or none | Usually general |
| Complications | Fewer | Perforation, stent pain |
| Anticoagulation | Must stop | Laser URS can continue |

- **SWL fails** with size >10–15 mm, impaction (CT "ring" sign), HU >1000, SSD >10 cm, cystine or COM (calcium oxalate monohydrate) stones.
- **SWL technique:** supine; 60–90 shocks/min; ramp energy; no routine pre-stent. Contraindicated in pregnancy, bleeding disorder, untreated UTI (urinary tract infection), distal obstruction, uncontrolled hypertension.

### 4. Ureteroscopy for proximal stones

- **Flexible URS** preferred: it follows a stone that migrates into the kidney.
- Safety wire; access sheath (12/14 Fr); tight ureter: pre-stent and return after 2–4 weeks.
- **Laser:** holmium:YAG (2120 nm) or thulium fibre laser; dusting or fragmenting; keep **≤10 W in the ureter** to avoid thermal injury.
- **Retropulsion** into the kidney: reduced by low-energy settings and antiretropulsion devices (Stone Cone, NTrap); pneumatic lithotripsy works poorly above the brim.
- **Push-back:** stone deliberately pushed into the renal pelvis with a catheter, then treated by SWL, RIRS (retrograde intrarenal surgery) or PCNL; now largely replaced by in-situ flexible URS.
- Stent only for perforation, impaction, infection or a planned second stage.

### 5. Antegrade approach (PCNL with antegrade URS)

- Indications: large (>15–20 mm), hard or **impacted proximal stones**, failed retrograde access, urinary diversion, transplant, concomitant renal stones.
- **Upper or mid-pole posterior calyx** gives a straight line to the UPJ; flexible nephroscope or antegrade URS down the dilated ureter; high SFR.

### 6. Laparoscopic (or robotic) ureterolithotomy

- Indications: large impacted stones, failed SWL/URS, where endourology is unavailable, or with concomitant UPJ obstruction (pyeloplasty).
- Steps: retroperitoneal or transperitoneal access; identify the bulge; **clamp the ureter above the stone** to prevent migration; longitudinal ureterotomy; remove; double-J stent; close with 4-0 absorbable sutures; drain.
- High single-stage SFR; risk of urine leak. **Open ureterolithotomy** is now rare.

### Complications of treatment

| Complication | Points |
|---|---|
| Perforation | 0–4%; stop, stent about 4 weeks |

| Complication | Points |
|---|---|
| **Avulsion** | 0.06–0.5%; **proximal third most at risk**; nephrostomy and delayed reconstruction |
| Stricture | 3–6%; image all after URS |
| Sepsis | About 5% after URS |
| SWL | Steinstrasse, haematoma, colic |

**Also know:** pregnancy: conservative, then stent or URS (no SWL).

### Recent advances

- Single-use digital scopes; pressure sensors; thulium fibre laser; robotic URS; **BWL (burst wave lithotripsy)**.

**Diagram to draw:** algorithm for proximal stone: drain if infected; <10 mm SWL or URS; >10 mm URS; large impacted antegrade or laparoscopic.

> **Key points to remember:**
> - Proximal stones pass in only 48%; MET does not help proximal stones.
> - <10 mm: SWL equals URS; >10 mm: URS first (EAU).
> - Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
> - Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

## LAQ 9 Uric acid calculi: etiopathogenesis, medical management and allopurinol

P2

**Asked as:**

- Discuss etiopathogenesis of uric acid calculi. Describe rationale of medical management, — *Practice Set 9 Paper II Q2 (LAQ)*
- Allopurinol — *Practice Set 10 Paper II Q3a (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)

### Introduction

- Uric acid stones form **8–10%** of stones in the USA and are increasing with obesity and diabetes. They are **radiolucent** and can be **dissolved medically**.

### Etiopathogenesis

#### 1. Why humans form uric acid stones

- Humans and apes lack hepatic **uricase**, which converts uric acid to allantoin (10–100 times more soluble). Blood and urine uric acid are therefore high.

#### 2. Physical chemistry: pH is the key

- Uric acid is a weak acid, **pKa about 5.35–5.5**. At pH 5.5 half is free uric acid and half is urate.
- Free uric acid is poorly soluble (about 96–100 mg/L); sodium urate is about 20 times more soluble. At **pH 6.5, over 90% is ionised**.
- **No natural inhibitor** of uric acid crystallisation is known, so stone formation depends almost entirely on supersaturation.

#### 3. The three determinants

| Determinant | Causes | Importance |
|---|---|---|
| **Low urine pH (<5.5)** | Insulin resistance, obesity, type 2 diabetes, gout, high animal-protein diet, chronic diarrhoea, ileostomy | **Most important**; most formers have normal uric acid excretion |
| **Low urine volume** | Hot climate or occupation, diarrhoea, dehydration | Concentrates uric acid |
| **Hyperuricosuria (>600 mg/day)** | Purine-rich diet, gout, myeloproliferative disorders, tumour lysis, uricosuric drugs, inherited defects | Least often the driver in idiopathic cases |

#### 4. Why urine is too acid: insulin resistance

- Insulin normally stimulates **ammonia production** from glutamine in the proximal tubule and the $Na^+/H^+$ exchanger NHE3.
- Insulin resistance reduces ammonium excretion. Acid is then buffered as titratable acid, so urine pH falls.
- **Renal lipotoxicity:** fat accumulates in the proximal tubule, which then burns fatty acids instead of glutamine, so ammoniagenesis falls further.
- Endogenous acid production is also higher. Uric acid stones are thus a **renal manifestation of the metabolic syndrome**; about a third of stones in diabetics are uric acid.

**5. Aetiological categories**

- **Congenital:** HGPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency (Lesch–Nyhan), PRPP (phosphoribosylpyrophosphate) synthetase overactivity, hereditary renal hypouricaemia (URAT1, the urate transporter).
- **Idiopathic:** low urine pH without gout ("gouty diathesis").

**6. Uric acid and calcium stones**

- In **hyperuricosuric calcium oxalate stone disease**, urate promotes calcium oxalate by removing inhibitors, by "salting out" and by **epitaxy** (urate crystals act as a nidus).
- **Also know:** ammonium acid urate stones (laxative abuse, diarrhoea, ileostomy); theobromine (cocoa) inhibits uric acid crystallisation in vitro.

## Diagnosis in brief

- Radiolucent on X-ray; CT **200–400 HU (Hounsfield units)**; **dual-energy CT** identifies uric acid.
- Urine pH persistently <5.5; serum uric acid; 24-hour urine; screen for diabetes.

## Rationale of medical management

**Principle:** uric acid stones are a **"pH disease" more than a uric acid disease**. Raising pH turns insoluble uric acid into soluble urate. Diluting urine lowers supersaturation. Lowering uric acid production helps only when excretion is high.

**1. Fluids**

- Urine output **at least 2.5 L/day** lowers uric acid concentration.

**2. Diet and lifestyle**

- Limit non-dairy **animal protein** (it gives purines and an acid load; dietary purines provide about 30% of urinary uric acid).
- More fruit and vegetables (alkali load). Weight loss, exercise and glucose control correct the underlying insulin resistance.

**3. Urinary alkalinisation (mainstay)**

- **Potassium citrate 30–60 mEq/day** in two or three doses. Citrate is metabolised to bicarbonate, raising urine pH.
- **Target pH 6.0–6.5** for prevention (EAU [European Association of Urology]: 6.2–6.8); **avoid pH >7.0**, which risks calcium phosphate stones.
- **Why potassium, not sodium:** sodium alkali raises urinary calcium and can cause calcium stones. Sodium bicarbonate or sodium citrate is used if potassium is unsafe (renal impairment, hyperkalaemia).
- Patients **monitor urine pH with dipsticks**.

**4. Dissolution (oral chemolysis)**

- Pure uric acid stones dissolve over weeks to months. **EAU: pH 7.0–7.2**; stent or nephrostomy if obstructed or infected; monitor by imaging.

**5. Xanthine oxidase inhibitors**

- **Allopurinol 300 mg/day** lowers uric acid production. It is **not first-line**, because low pH, not hyperuricosuria, is the usual cause.
- Indications: hyperuricosuria with hyperuricaemia (e.g. gout), recurrence despite alkalinisation (AUA, American Urological Association), hyperuricosuric calcium oxalate

stones.

- Reduce the dose in renal impairment. Risks: rash; **allopurinol hypersensitivity syndrome** (2–8% have reactions; may be fatal); Stevens–Johnson syndrome.
- **Febuxostat** is an alternative. **Rasburicase** (recombinant uricase) prevents uric acid nephropathy in tumour lysis.

**Landmark trial: Ettinger (1986)**, randomised placebo-controlled trial: allopurinol more than halved recurrence in hyperuricosuric, normocalciuric calcium oxalate stone formers.

## Allopurinol (in detail)

### Introduction and mechanism

- A **purine (hypoxanthine) analogue** that inhibits **xanthine oxidase (xanthine dehydrogenase)**. This blocks hypoxanthine → xanthine → uric acid, lowering serum and urinary uric acid.
- Metabolised in the liver to **oxypurinol**, the active metabolite. It has a long half-life and is cleared by the kidney, so the dose is cut in renal impairment.

### Uses in urology

| Indication | Rationale |
|---|---|
| **Recurrent calcium oxalate stones with hyperuricosuria** (>600 mg/day) and normal urinary calcium | Urate "seeds" calcium oxalate (heterogeneous nucleation, salting out, inhibitor adsorption). **Ettinger (1986) RCT (randomised controlled trial): recurrence more than halved**; AUA (American Urological Association) recommends it |
| Uric acid stones | **Not first line:** low urine pH is the main cause, so alkalinise with potassium citrate to pH 6.0–6.5. Allopurinol is added for hyperuricosuria, gout or failure of alkalinisation |
| 2,8-DHA (dihydroxyadenine) stones (APRT (adenine phosphoribosyltransferase) deficiency) | Blocks adenine → DHA; febuxostat more effective |
| Tumour lysis prophylaxis; gout; HPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency, PRPS (phosphoribosylpyrophosphate synthetase) overactivity | Prevents urate nephropathy |

### Dose

- Adult **300 mg/day** (start 100 mg, titrate); child 4–10 mg/kg/day (maximum 300 mg). Reduce for low creatinine clearance.

### Adverse effects

- Rash (commonest), myalgia, gastrointestinal upset; gout flare on starting; interstitial nephritis.
- **Hypersensitivity in 2–8%**, from skin reactions to **allopurinol hypersensitivity syndrome** (fatal in up to 27%) and Stevens-Johnson syndrome. Risk is higher with high dose, renal failure, diuretics and statins. **HLA-B*58:01** carriers (some Asian populations) have higher risk.
- **Xanthine stones** (radiolucent): rare, mainly with high doses in Lesch-Nyhan syndrome.

**Drug interactions**

- **Azathioprine and 6-mercaptopurine:** xanthine oxidase metabolises them, so toxicity (marrow suppression) rises. Reduce their dose sharply (important after transplant).

**Alternatives**

- **Febuxostat:** non-purine xanthine oxidase inhibitor, no dose change in mild to moderate renal impairment. Carries a cardiovascular-death warning.
- **Also know:** rasburicase (urate oxidase) for tumour lysis.

**Diagram to draw:**

- Sigmoid curve of uric acid solubility against urine pH, marking pKa 5.5 and target 6.0–6.5.
- Purine pathway (hypoxanthine → xanthine → uric acid) showing the xanthine oxidase block.

> **Key points to remember:**
>
> - Low urine pH is the main cause; pKa 5.5; no inhibitor exists.
> - Insulin resistance reduces ammoniagenesis, so urine is acid.
> - Potassium citrate to pH 6.0–6.5 (dissolution 7.0–7.2); avoid >7.0.
> - Allopurinol only for hyperuricosuria or failed alkalinisation.
> - Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
> - Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
> - Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

## LAQ 10 Urinary crystals

P2

**Type:** LAQ (25 marks)
**Source:** Practice Set 11 Paper 2, Q2

### Introduction

- Crystals are precipitated urinary solutes seen on microscopy of the sediment. They occur in both normal and diseased states.
- Their value lies in suggesting stone type, metabolic disease, infection or drug toxicity. **Cystine is the one crystal that is never normal.**

### How crystals form (physicochemistry)

- Urine is normally **supersaturated** with calcium oxalate and phosphate. Crystals form only when the concentration product passes a critical level.

| Saturation zone | What happens |
|---|---|
| Undersaturated (below Ksp, solubility product) | No crystals; stones may dissolve |
| Metastable (Ksp to Kf, formation product) | No spontaneous nucleation; growth on existing crystals; inhibitors effective |
| Unstable (above Kf) | Spontaneous nucleation, growth and aggregation |

- **Nucleation** in vivo is mostly heterogeneous (on cell debris or other crystals). Crystals then grow, aggregate and are retained, for example on **Randall plaque**.
- **Inhibitors:** citrate, magnesium, pyrophosphate, Tamm–Horsfall protein, nephrocalcin, osteopontin. No inhibitor exists for uric acid or cystine, so these depend on pH and concentration.
- **Determinants of crystalluria:** urine volume, solute excretion, pH, temperature and inhibitors.

### Examination of urine for crystals

- Fresh midstream specimen, examined within 2 hours at body or room temperature. **Refrigeration precipitates crystals** that may not redissolve.
- Dipstick pH (physiological range 4.5–8.0); centrifuged sediment; polarised light helps identify crystals.
- Phosphaturia (cloudy alkaline urine) **clears with acetic acid**; pyuria does not.
- Candida mimics calcium oxalate and red cells; budding and hyphae separate it.

### Classification by urine pH and shape

| Urine pH | Crystal | Shape | Clinical significance |
|---|---|---|---|
| Acidic | **Uric acid** | Rhomboid or lemon-shaped plates, rosettes; yellow-brown | Uric acid stones (pH <5.5; pKa 5.35), gout, tumour lysis |
| Acidic | Amorphous urates | Pink "brick-dust" granules | Usually normal (concentrated, cooled urine) |

| Urine pH | Crystal | Shape | Clinical significance |
|---|---|---|---|
| Acidic | **Cystine** | **Hexagonal** colourless plates | Always abnormal: cystinuria |
| Acidic or alkaline | **Calcium oxalate dihydrate** | **Envelope** (octahedral) | Common, often normal; hypercalciuria |
| Acidic or alkaline | Calcium oxalate monohydrate | Dumbbell or oval | Hyperoxaluria; ethylene glycol poisoning |
| Alkaline | **Triple phosphate (struvite)** | **Coffin-lid** prisms | Urease-producing infection (Proteus); staghorn stones |
| Alkaline | Amorphous phosphate | White granules | Benign phosphaturia |
| Alkaline | Calcium phosphate (brushite, apatite) | Needles or rosettes; amorphous | RTA (renal tubular acidosis), hyperparathyroidism |
| Alkaline | Ammonium biurate | "Thorny apple" spheres | Old specimens; ammonium urate stones (diarrhoea, laxative abuse, ileostomy) |
| Any | Cholesterol | Notched flat plates | Nephrotic syndrome, lipiduria |

- **Rare metabolic crystals:** 2,8-dihydroxyadenine, brown spiculated spheres (APRT [adenine phosphoribosyltransferase] deficiency, radiolucent stones); xanthine (hereditary xanthinuria, high-dose allopurinol); tyrosine needles and leucine spheres (severe liver disease).

### Drug crystals

- **Two routes:** the drug itself precipitates (direct), or it changes urine chemistry (indirect).

| Drug | Crystal and features | Prevention |
|---|---|---|
| **Indinavir** (protease inhibitor) | Needles or fan-shaped plates; crystalluria in about two-thirds of new users; stones 4–13%; radiolucent, often invisible on CT | Hydration; more soluble below pH 5.5 |
| Atazanavir | Highest stone risk of newer antiretrovirals | Hydration |
| Sulfonamides (sulfadiazine) | "Sheaves of wheat" | Hydration, alkalinise |
| Acyclovir (high-dose intravenous) | Birefringent needles; acute kidney injury | Slow infusion, hydration |
| Amoxicillin, ciprofloxacin (high dose) | Needles; worse in alkaline urine | Hydration |
| Methotrexate (high dose) | Crystal nephropathy | Alkalinise, hydration |
| Triamterene, ephedrine, guaifenesin, silicate antacids | Stones of drug or metabolite | Stop the drug |

- **Indirect:** loop diuretics and steroids (hypercalciuria, calcium stones); acetazolamide and topiramate (alkaline urine, hypocitraturia, calcium phosphate); cytotoxics (uric acid); laxative abuse (ammonium urate).

### Clinical significance

1. **Diagnosis of stone type** before stone analysis is available; crystals in a stone former help direct metabolic work-up.

2. **Recurrence risk:** persistent crystalluria in repeated early-morning samples, and large or aggregated crystals, suggest active stone disease.
3. **Infection:** struvite crystals with urine pH >7.2 signal a urease-producing organism.
4. **Acute kidney injury:** uric acid (tumour lysis), calcium oxalate monohydrate (ethylene glycol) and drug crystals cause intratubular obstruction.
5. **Monitoring therapy:** absence of cystine or uric acid crystals confirms adequate alkalinisation. - Most crystalluria in healthy people is transient and harmless.

### Management principles

- Repeat on a fresh, warm specimen; confirm pH; 24-hour urine metabolic evaluation and stone analysis.
- **Fluids** for urine output above 2–2.5 L/day.
- **pH manipulation:** alkalinise (potassium citrate) for uric acid, cystine, methotrexate and sulfonamides; avoid alkali in calcium phosphate and struvite.
- Treat the cause: antibiotics and stone clearance for struvite; stop or change the drug; specific therapy (thiols for cystine, allopurinol for uric acid).

**Diagram to draw:** sketches of hexagon (cystine), envelope (calcium oxalate), coffin lid (struvite), rhomboid (uric acid), sheaf of wheat (sulfonamide).

> **Key points to remember:**
> - Cystine hexagons are always pathological.
> - Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
> - Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
> - Refrigerated urine precipitates crystals: examine fresh within 2 hours.

## LAQ 11 Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.

P3

**Type:** LAQ (25 marks)
**Source:** 2015-Summer Paper 3, Q1

### Introduction

- A stent keeps the upper tract draining past obstruction, injury or an anastomosis. **Zimskind 1967** placed the first indwelling splint; **Finney 1978** introduced the self-retaining **double-J (double-pigtail)** stent.

### Types and materials

| Feature | Details |
|---|---|
| Size | 3–8.5 Fr (adult usually 6 Fr); adults 20–30 cm by height, children about age + 10 cm |
| Polymers | Polyurethane, silicone (least encrusting, soft); hydrocoated silicone causes less pain |
| **Metallic (Resonance)** | Resists extrinsic compression: malignant obstruction, retroperitoneal fibrosis |
| Others | Hybrid metal-reinforced, thermo-expandable nitinol (Memokath), biodegradable (experimental) |
| Designs | Single-pigtail diversion stent; open-ended catheter to identify the ureter; modified bladder ends (Polaris loop: no benefit) |

**Also know:** anatomy: three narrowings (PUJ (pelviureteric junction), iliac vessels, intramural ureter, the narrowest at 3–4 mm). Placement: retrograde pyelogram first, stent over a wire under fluoroscopy; antegrade route if retrograde fails.

**Diagram to draw:** double-J stent in situ with both coils and side holes.

### Indications

- Obstruction: infected obstructed kidney (stent equals PCN (percutaneous nephrostomy)), malignancy, stricture, pregnancy.
- After URS (ureteroscopy), pyeloplasty, reimplantation, transplant; injury, leak, fistula; pre-stenting a tight ureter.

### Stent symptoms and complications

- About **4 in 5** have urgency, frequency, haematuria or flank pain, measured by the **USSQ (Ureteral Stent Symptom Questionnaire)**. Cause: **intravesical material irritating the bladder and reflux up the stent**. Diameter and length matter little.
- Relief: **alpha-blockers and antimuscarinics**; intraureteral placement; shortest dwell; opioid-free ERAS (enhanced recovery after surgery) analgesia.
- **Misplacement** (submucosal) and **migration** (too short).
- **Stent failure** from compression, debris or encrustation: worst with malignant or irradiated ureter, bilateral obstruction; then PCN.
- **Encrustation** rises with dwell time, infection and stone-forming urine. **Forgotten stent** causes sepsis and renal loss; keep a register.

- **FECal (Forgotten, Encrusted, Calcified) grading**: I minimal linear encrustation; II circular encrustation; III partial encrustation of one or both coils; IV complete encrustation of one coil; V both coils.
- **Encrusted stent**: never pull by force; NCCT (non-contrast computed tomography); clear the **bladder coil**, then **ureter**, then **renal coil** (flexible URS or PCNL (percutaneous nephrolithotomy)).

### Ureteroscopy

| | Semirigid | Flexible |
|---|---|---|
| Tip | 7 Fr or less | 4.5–7.5 Fr; 3.6 Fr channel |
| Use | Distal and mid ureter | Upper ureter, kidney |
| Deflection | None | Over 270° |

- Segura's dictum: always a safety wire. Tight ureter: stent and return after **2–4 weeks**; routine pre-stenting is unnecessary.
- **UAS (ureteral access sheath)**, usually 12/14 Fr: lowers intrarenal pressure and eases repeated passes, but **severe injury in about 1 in 8**; less after pre-stenting.
- **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm)** or **TFL (thulium fibre laser, about 1940 nm)**. **Dusting** (low energy, high frequency) is faster but leaves fines; **fragment and basket** confirms stone clearance and gives stone for analysis. Keep pressure and temperature low with outflow.

**Steps of URS**: (1) sterile urine, antibiotic dose; (2) cystoscopy and **safety guidewire**; (3) semirigid inspection, dilate only if needed; (4) access sheath over a working wire under fluoroscopy (for flexible URS); (5) low-pressure irrigation, laser, basket; (6) inspect the ureter on withdrawal; (7) decide on stent.

- **Also know:** anti-retropulsion devices; digital and single-use scopes; URS suits patients who cannot stop anticoagulants.
- Complications: sepsis about 5% (treat positive urine culture first), stricture about 3%, bleeding under 1%, perforation (stop and stent), avulsion under 1% (PCN, delayed repair).

### Stent after URS?

- **Denstedt 2001** randomised trial and a Cochrane review: URS is safe without a stent.
- **AUA (American Urological Association)**: stent may be omitted when anatomy and renal function are normal, there is no ureteric injury and no second stage is planned. **EAU (European Association of Urology)** agrees.
- **Stent needed**: perforation, stricture or tight ureter, infection, planned second stage, solitary kidney, impacted stone.
- Removal at 3 days gave more unplanned visits than at 7 days; a string allows home removal. Stenting at SWL (shock wave lithotripsy) does not improve stone passage.

```
Uncomplicated URS -> no stent (or short string stent) Complicated -> double-J: injury 2-6 wk; second stage until re-look
```

### Recent advances

- Suction sheaths, intrarenal pressure sensors, biodegradable stents.

**Key points to remember:**

- Finney 1978 double-J; symptoms from intravesical stent and reflux.
- Encrusted stent: bladder coil, ureter, then kidney.
- Denstedt 2001: no routine stent after uncomplicated URS.

## LAQ 12 Laparoscopic pyeloplasty.

P3

**Type:** LAQ (25 marks)
**Source:** 2016-Summer Paper 3, Q2

### Introduction

- **PUJO (pelviureteric junction obstruction)** is impaired flow from pelvis to ureter that, if untreated, causes progressive renal damage (Koff).
- Causes: **intrinsic** adynamic segment (abnormal circular muscle), **extrinsic** crossing lower-pole vessel, high insertion; secondary (failed repair, stone).
- **Anderson–Hynes (1949) dismembered pyeloplasty** is the gold standard by any route. First laparoscopic pyeloplasty **1993 (Schuessler)**; robotic **2002**. Success **over 90–95%**, equal to open, with less morbidity.

### Indications

- Symptoms (pain, Dietl's crisis), infection, stones, hypertension (rare).
- Impaired split function or **fall of 10% or more** on serial renography; rising APD (anteroposterior diameter); obstructed solitary kidney or bilateral disease.
- Children: most antenatal hydronephrosis resolves; about one-third need surgery.
- **Contraindications**: uncorrected coagulopathy, untreated UTI (urinary tract infection), cardiopulmonary disease barring pneumoperitoneum.
- **Nephrectomy instead** if split function under 15% in adults with a normal other kidney.

### Preoperative evaluation

- **USG (ultrasonography)**: large pelvis, small calyces, ureter not seen.
- **MAG3 (mercaptoacetyltriglycine) diuretic renogram**: split function and an obstructive curve that does not wash out after furosemide. Usual teaching: washout half-time under 10 minutes normal, 10–20 minutes equivocal, over 20 minutes obstructed (Ch 93 quotes over 40 minutes). Split function under 40% supports repair.
- **CT (computed tomography) angiography**: anatomy, **crossing vessels**, stones.
- **Retrograde pyelogram** at surgery: confirms the level, excludes ureteric stricture.
- **Whitaker test** if equivocal: perfusion 10 mL/min; pelvic pressure over 15–20 cm H2O means obstruction.
- Sterile urine. Drain beforehand (stent or nephrostomy) only for infection, azotaemia or severe pain.

### Approaches

| | **Transperitoneal** | **Retroperitoneal** |
|---|---|---|
| Space | Large, familiar landmarks | Small, created by balloon |
| Bowel | Colon reflected (left: **transmesocolic** window) | Not entered; leak stays extraperitoneal |
| Use | Commonest; easier suturing | Avoids bowel; harder orientation |

| | Transperitoneal | Retroperitoneal |
|---|---|---|
| - One paediatric randomised trial favoured retroperitoneal (shorter stay); overall outcomes similar. | | |

### Steps (transperitoneal Anderson–Hynes)

1. Cystoscopy, retrograde pyelogram, catheter.
2. 45–60° flank; pneumoperitoneum 12–15 mmHg in adults (under 12 mmHg in children); 3–4 ports, camera at umbilicus.
3. Reflect colon medially, or open the mesocolon over the pelvis.
4. Find the ureter; trace it to the PUJ with its periureteric tissue; minimal diathermy. Look for a **crossing vessel**.
5. Hitch stitch on the pelvis; transect the pelvis above the PUJ; **reduce** a redundant pelvis if needed.
6. **Spatulate the ureter laterally** into healthy ureter; keep the narrow segment as a handle (no-touch).
7. **Transpose** the PUJ in front of any crossing vessel.
8. Anastomose the spatulation apex to the most dependent pelvis with 4-0 absorbable (5-0/6-0 in infants): back wall, stent, front wall, then close the pelvis. Watertight, tension-free, funnel-shaped.
9. **Antegrade double-J** over a wire; clamp the catheter: urine from the stent side holes confirms the bladder coil.
10. Excise the narrow segment; drain near (not on) the suture line.

### Crossing vessel

- It lowers endopyelotomy success (42% vs 86%), so pyeloplasty is preferred. **Never divide** a lower-pole vessel (segmental infarction).
- **Vascular hitch (Hellström)**: vessel moved up onto the pelvis, only if the PUJ is intrinsically normal and peristalsing.

### Stenting and postoperative care

- Double-J for **4–6 weeks** (children 1–4 weeks: no difference). Externalised stent in infants avoids a second anaesthetic.
- Foley for 24–48 hours (bladder pressure refluxes up the stent); drain out when dry. If drain output rises after Foley removal, replace the Foley.
- **MAG3 renogram** after stent removal, then ultrasound; follow for years.

### Robotic pyeloplasty

- 3D vision, wristed suturing, tremor filtration, **shorter learning curve**. Now first-line at major centres.
- Success 94.7–100%, the same as laparoscopy; advantage is ergonomic, at higher cost.

### Outcomes

- Laparoscopic: operating time 2–4 hours; complications 2–16%; conversion 0–5.5%; stay about 2.7 vs 4.2 days (open); success 87–99%.
- **Up to 30% of failures appear after 2 years**, so follow-up must be long.

### Complications

- Urine leak and urinoma, stent symptoms or migration, UTI, haematuria, ileus, bowel or vessel injury, port-site hernia, recurrent obstruction.

### Management of failure

`Early (clot, oedema) -> nephrostomy, wait Persistent obstruction -> CT angiography (missed vessel?) short, no vessel -> endopyelotomy (about 70%) otherwise -> redo laparoscopic/robotic pyeloplasty (about 89% after failed open; buccal graft onlay) small intrarenal pelvis -> ureterocalicostomy poor function -> nephrectomy`

**Also know:** Y-V plasty (high insertion); Culp spiral flap (long narrowing); stones removed through the pyelotomy with a flexible scope; safe in horseshoe and pelvic kidneys.

### Recent advances

- Single-port robot (2018); indocyanine green fluorescence for redo cases; buccal mucosa ureteroplasty.

**Diagram to draw:** Anderson–Hynes steps: transection, lateral spatulation, transposition over the crossing vessel, dependent anastomosis.

> **Key points to remember:**
>
> - Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
> - Transpose over a crossing vessel; never divide it.
> - Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
> - Robotic equals laparoscopic in outcome, with easier suturing.

## LAQ 13 Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury

P3

**Asked as:**

- Iatrogenic ureteric injury — *Practice Set 2 Paper III Q1 (LAQ)* (upper ureteric injury also answered in 2023-Winter Paper 3, Q2)
- Discuss management of ureteric injuries occurring during gynec surgeries — *Practice Set 4 Paper III Q1 (LAQ)* (also answered in Practice Set 2 Paper 3, Q1)
- Discuss Etiology, Diagnosis and Management of Upper ureteric injury. — *Winter 2023 Paper III Q2 (LAQ)*

**Type:** LAQ (25 marks)

*Assumed scope: surgical (iatrogenic) injury at any level of the ureter, in gynaecological, colorectal, vascular and urological surgery. Assumed meaning (upper ureteric injury): injury to the upper (proximal) ureter, from the PUJ (pelviureteric junction) to the pelvic brim, from external trauma or surgery.*

### Introduction

- Ureteric trauma is uncommon: 1–2.5% of genitourinary trauma. **Iatrogenic injury is commoner overall** than external trauma.
- **The proximal ureter is the commonest site of external injury. The distal ureter is the commonest (usual) iatrogenic site** in non-urological surgery.
- Gynaecological surgery has historically caused about half of all iatrogenic ureteric injuries; hysterectomy is the main culprit.
- It is mainly a problem of **recognition** (the core problem): thermal, ligation and devascularisation injuries are invisible at operation and declare themselves days to weeks later. Injury is often **missed** and declares itself later as urinoma, sepsis, stricture or fistula.
- Late diagnosis brings urinoma, sepsis, AKI (acute kidney injury), fistula and renal loss, and needs nephrostomy about ten times more often.

### Aetiology and incidence (EAU (European Association of Urology) trauma guideline data)

#### 1. External trauma

- **Penetrating** (most): gunshot wounds up to about 90%. Blast damages the intramural blood supply up to 2 cm from the wound.
- **Blunt** (up to about 40%): rapid deceleration avulses the ureter at the **PUJ**; linked to lumbar transverse process fractures.
- **Over 90% have associated injuries** (small bowel commonest, colon, vessels, kidney 10–28%). Mortality 6–9% reflects these.

#### 2. Iatrogenic

| Source | Key facts (upper-ureter relevance) |
|---|---|
| **Gynaecological** | Historically about half. Hysterectomy: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery highest (1.7–4.3%). Endometriosis and placenta accreta raise risk |

| Source | Key facts (upper-ureter relevance) |
|---|---|
| Colorectal | 0.15–10%; laparoscopic equals open |
| Colorectal, spinal, retroperitoneal surgery | Ligation, transection, cautery |
| **Urological / Ureteroscopy** | Ureteroscopy is the commonest (iatrogenic) cause in some series: abrasion, perforation 0.2–2%, avulsion under 0.5% (another source: <0.3%) (proximal third most at risk); stricture about 3% (another source: 0.3–3%). Risk: proximal stone, impacted stone, long operation, large sheath, inexperience. Also radical prostatectomy, partial nephrectomy or ablation (late proximal stricture) |
| Partial nephrectomy, thermal ablation | Proximal stricture months later (lower-pole and hilar tumours) |
| Vascular (aortic) surgery | Aorto-iliac bypass: transient hydronephrosis 12–20%; up to 85% unrecognised (missed initially). Devascularisation. Rare ureteroarterial fistula (chronic stent plus pelvic surgery or radiation) |

### Gynaecological incidence (EAU trauma guideline data)

| Procedure | Ureteric injury (%) |
|---|---|
| Caesarean section (emergency) | 0.01–0.06 |
| Vaginal hysterectomy | 0.02–0.5 |
| Abdominal hysterectomy | 0.03–2.0 |
| Minimally invasive hysterectomy | 0.13–6.0 (laparoscopic = robotic, about 0.9) |
| Prolapse surgery | **1.7–4.3 (highest)** |

- **Danger sites in the pelvis (applied anatomy):** pelvic brim near (at) the infundibulopelvic ligament; **under the uterine artery** ("water under the bridge"); beside the uterosacral ligament; vaginal fornix / vaginal cuff near the UVJ (ureterovesical junction).
- Up to a third are recognised during open surgery, fewer during laparoscopy; the uterosacral region is hardest to see.
- **Mechanisms:** ligation, crush, laceration, transection, avulsion, thermal injury, **devascularisation**. About 10% of women get a large share of distal ureteric blood from the uterine artery (the uterine artery supplies much of the distal ureter), so some (ischaemic) injuries are unavoidable.
- **Risk factors:** previous surgery, adhesions, malignancy, radiotherapy, heavy bleeding, long operation, learning curve; endometriosis (encasement, adhesions, medial deviation), placenta accreta.
- **URS (ureteroscopy) risks:** long operation, 12/14-Fr sheath, impacted or proximal stone, inexperience.

## Classification: AAST (American Association for the Surgery of Trauma) ureter scale

| Grade | Injury (description) |
|---|---|
| I | Haematoma or contusion, no devascularisation (without devascularisation) |
| II | Laceration <50% of circumference |
| III | Laceration ≥50% |

| Grade | Injury (description) |
|---|---|
| IV | Complete transection, <2 cm devascularised (with <2 cm devascularisation) |
| V | Avulsion, >2 cm devascularised (with >2 cm devascularisation) |

- Advance one grade for bilateral injury, up to grade III.

## Prevention

- See (identify) the ureter at the **pelvic brim** (crossing the common iliac artery) and trace it before ligating pedicles. Good haemostasis and exposure.
- **Prophylactic stents** (or lighted stents) help recognition in high-risk (complex) cases, but do not clearly reduce (prevent) injury. **ICG (indocyanine green)** instilled into the ureter (intraureteric) makes it fluoresce (fluoresces) for hours.
- **Cystoscopy** after minimally invasive hysterectomy and prolapse surgery, with oral phenazopyridine or IV (intravenous) dye to show efflux.
- URS: safety wire; ≤14-Fr sheath in an unstented ureter; fragment before basketing; "stop, stent, stage".

## Diagnosis and recognition

### Suspicion (trauma)

- Penetrating flank or abdominal wound; high-energy deceleration.
- **Haematuria is absent in 25–45%**, so its absence does not exclude injury.

### Intraoperative (best outcome)

- Inspect the ureter (at laparotomy); IV (intravenous) indigo carmine or methylene blue; cystoscopy for efflux (jets), with retrograde (on-table) pyelogram or stent. Check the other ureter and bladder. About 11% are missed at trauma laparotomy.
- Methylene blue is an MAO (monoamine oxidase) inhibitor: avoid with SSRIs/SNRIs (selective serotonin / serotonin–noradrenaline reuptake inhibitors; serotonin toxicity), G6PD (glucose-6-phosphate dehydrogenase) deficiency and pregnancy.

### Delayed presentation (most cases) / postoperative

- Mean diagnosis about day 10: fever, ileus, flank pain, leucocytosis, rising creatinine, rising drain output, drain or wound urine leak, urinoma, anuria (bilateral), or **ureterovaginal fistula** (UVF): normal voiding plus constant leak, because the other kidney fills the bladder, usually within 4 weeks.

### Imaging and tests

| Test | Role |
|---|---|
| **CT (computed tomography) urography with 10-minute delayed images** (both ureters) | Best test in stable patients. Signs: extravasation, non-opacified distal ureter, delayed nephrogram, urinoma |
| **Retrograde pyelography** | Most sensitive; shows site; allows stenting at once |
| Antegrade nephrostogram | When retrograde fails or gap is wide |
| One-shot IVU (intravenous urography) | Non-diagnostic in 33–100%; never rely on it |

- **Drain-fluid (or vaginal fluid) creatinine** above (far above) serum creatinine confirms urine.
- **Double-dye test** separates VVF (vesicovaginal fistula) from ureterovaginal fistula: oral phenazopyridine plus intravesical methylene blue. Orange pad = UVF; blue = VVF; both =

both.

## Management

### Principles

- **Stability decides timing; level and length decide the operation.**
- Repair: mobilise sparing (preserve) adventitia; debride to bleeding tissue (edges); **spatulated, tension-free, watertight, stented** repair (anastomosis) with fine absorbable suture; drain; retroperitonealise; omentum (omental wrap) for thermal, radiation, blast or vascular injury; non-tunnelled reimplant.
- **No bowel interposition or autotransplantation acutely.**

**By timing** ``` Recognised at surgery -> repair at once (stable patient) Unstable -> ligate + later nephrostomy, or exteriorised stent Recognised late -> try retrograde stent (or nephrostomy + antegrade stent)

|-- succeeds -> stent 4-8 weeks, then image |-- fails -> nephrostomy; early repair if <=7 days, otherwise delayed repair after about 6 weeks Uncontrolled sepsis / collections -> drain; repair regardless of timing Ureteric injury suspected

|-- Unstable -> damage control: ligate + later PCN, | or exteriorised stent / cutaneous ureterostomy |-- Stable, found at surgery -> immediate repair over stent |-- Found late / postoperatively -> Attempt retrograde stent (+/- PCN, drain) -> Fails -> PCN; delayed definitive repair Suspected late injury -> retrograde stent (or nephrostomy + antegrade stent)

```
|-- succeeds -> stent 4-8 weeks, then image
|-- fails -> nephrostomy
     |-- found <=7 days -> early repair
     |-- later -> repair after about 6 weeks
```

Sepsis / uncontrolled collections -> drain; repair regardless of timing ```

### At surgery, by injury

| Injury | Action |
|---|---|
| Suture through wall, minor thermal | Remove suture; stent |
| Partial tear | Close transversely (transverse closure, 4-0/5-0) over stent, 4–8 weeks |
| Ligation | Remove ligature (deligate); stent if viable, excise and repair if not |
| Transection, lower ureter (usual) | **Ureteroneocystostomy ± psoas hitch** (95–100% success) |
| Long distal gap | **Boari flap** + hitch (gaps up to 12 cm) |
| Transection above pelvic brim | Ureteroureterostomy |
| Unstable patient | Ligate + later nephrostomy, or exteriorised stent |

### Repair by grade

- **Grade I–II (contusion, partial):** double-J stent for about 6 weeks; or close a laceration transversely over a stent. Excise non-viable contused ureter.
- **Grade III–V:** formal reconstruction.

### By level (complete injury)

| Level | Repair |
|---|---|
| Upper | Ureteroureterostomy; ureteropyelostomy (PUJ (pelviureteric junction) avulsion); ureterocalicostomy |
| Mid | Ureteroureterostomy ± downward nephropexy; TUU (transureteroureterostomy) when delayed |
| Lower (commonest iatrogenic) | **Ureteroneocystostomy, non-tunnelled ± psoas hitch** (95–100% success); **Boari flap** for gaps up to 12 cm |
| Long loss (delayed only) | Ileal ureter or Yang–Monti; buccal graft ureteroplasty; autotransplantation (last resort) |

### Options for the upper ureter

| Procedure | Key points |
|---|---|
| **Ureteroureterostomy** | Commonest acute repair; short defects (2–3 cm); spatulate 5–6 mm, 180° apart; 5-0 absorbable over double-J; up to 90% success; leak 10–24%, stricture up to 12% |
| **Ureteropyelostomy** (reimplant into pelvis) | PUJ avulsion; spatulated |
| **Ureterocalicostomy** | Pelvis or PUJ destroyed; lower-pole parenchyma resected |
| **Renal descensus** | Mobilise kidney downwards; gains up to 8 cm |
| **TUU (transureteroureterostomy)** | Delayed; long loss when the bladder cannot reach; 90–97% success. Avoid with stones, RPF (retroperitoneal fibrosis), TB (tuberculosis), urothelial cancer or radiation |
| **Ileal ureter / Yang–Monti** | Long or panureteric loss; 81–100% success. Avoid if creatinine >2 mg/dL, bladder dysfunction, inflammatory bowel disease, radiation enteritis. Acidosis about 20% |
| Appendiceal interposition | Right side; about 92% short-term success |
| **Buccal mucosa graft ureteroplasty** | Delayed long strictures; onlay with omental wrap, often robotic; about 87% success |
| **Autotransplantation** | Last resort before nephrectomy; 5–10% renal units lost |
| Nephrectomy | Rarely; unstable with unreconstructable kidney, or poor function |

### Iatrogenic endoscopic injury

- **URS perforation:** stop the procedure and place a **stent** about 4 weeks (± nephrostomy), image later; later reconstruction if stricture develops. **Avulsion:** nephrostomy, delayed reconstruction; avoid nephrectomy.

### Fistulae

- **Ureterovaginal fistula (UVF):** stent first (best / ideally within 2 weeks; most heal); if stenting fails, reimplant with psoas hitch or Boari flap (success ≥90%). Early and delayed repair do equally well.
- **Ureteroarterial fistula:** endovascular covered stent.
- **Also know:** TUU (transureteroureterostomy), ileal ureter, buccal graft ureteroplasty and autotransplantation for long delayed defects; robotic reconstruction (repair) for delayed repair matches open; avoid nephrectomy.

### Complications and follow-up

- Urinoma, abscess, leak, fistula, stricture, loss of renal function.
- Persistent leak: check stent position, take drain off suction, catheterise the bladder, then nephrostomy.
- **Follow-up:** ultrasound or diuretic renogram at 4–12 weeks and (at) 1 year; strictures (late stenosis) can be silent and may appear up to 30 months.

### Medicolegal

- Disclose and document; malpractice claims most often follow gynaecological surgery.

**Diagram to draw:** psoas hitch with non-tunnelled reimplant; ureter's relation to the uterine artery (ureter passing under the uterine artery); spatulated ureteroureterostomy over a double-J stent.

> **Key points to remember:**
>
> - Distal ureter (under the uterine artery is the classic site) commonest iatrogenic site; most injuries present late.
> - Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
> - AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
> - CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
> - Late: stent first; early repair (repair) ≤7 days, otherwise delayed after 6 weeks (or after 6 weeks).
> - Lower ureter: psoas hitch 95–100%; Boari flap (bridges) up to 12 cm.
> - Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

## LAQ 14 Energy sources for breaking stones in urology

P4

**Asked as:**

- Describe energy sources for breaking stones in Urology — *Practice Set 1 Paper IV Q2 (LAQ)*
- Describe energy sources for braking stones in uro — *Practice Set 5 Paper IV Q1 (LAQ)*

**Type:** LAQ (25 marks)
(also covered in 2025-Winter Paper 4, Q2 and Q3b; also answered in Practice Set 1 Paper 4, Q2)

*Read as "energy sources for breaking stones in urology".*

### Introduction

- Stones are **brittle**: they crack under tension and shear. Tissue is **elastic** and survives the same stress. Every lithotripter exploits this difference.
- Energy reaches the stone **extracorporeally** (SWL, shock wave lithotripsy) or **intracorporeally** through an endoscope.

### Classification

| Group | Sources |
|---|---|
| Extracorporeal | SWL: electrohydraulic, electromagnetic, piezoelectric; BWL (burst wave lithotripsy, investigational) |
| Intracorporeal mechanical | Ballistic (pneumatic), ultrasonic, combined ultrasonic–pneumatic |
| Intracorporeal electrical | EHL (electrohydraulic lithotripsy) |
| Intracorporeal light | Lasers: Ho:YAG (holmium:yttrium-aluminium-garnet), TFL (thulium fibre laser), pulsed Tm:YAG (thulium:YAG); historic pulsed dye and FREDDY |

### 1. Shock wave lithotripsy

- **Shock wave:** peak positive pressure 30–100 MPa, then a tensile tail of about −10 MPa. It is generated at F1, focused, and delivered to the stone at F2.

| Generator | Mechanism | Strength | Weakness |
|---|---|---|---|
| Electrohydraulic (original HM3, 1980) | Underwater spark at F1 of an ellipsoid reflector | Very effective | Electrode life <40,000 shocks; focus drifts |
| Electromagnetic (modern standard) | Coil pushes a metal membrane; lens or reflector focuses | Consistent; high energy at F2; long life | — |
| Piezoelectric | Many ceramic elements on a spherical dish | Long life; little pain | Small focal zone, low energy |

- **Fracture mechanisms:** shear and tear forces; **spallation** at the distal face; quasistatic squeezing; **cavitation** (bubble collapse erodes the surface); dynamic squeezing (most powerful). A focal zone wider than the stone works best.
- **Technique:** 60–90 shocks/min beats 120/min, because bubble clouds shield the stone and injure the kidney; voltage ramping; bubble-free gel coupling.

- Poor results: calcium oxalate monohydrate, brushite, cystine, density >1000 HU (Hounsfield units), skin-to-stone distance >10 cm.

### 2. Ballistic (pneumatic) lithotripsy

- Compressed air drives a projectile against a metal probe, a **jackhammer** (about 12 Hz).
- Breaks all stones cheaply with little ureteric injury. Drawbacks: **retropulsion**; rigid probes only, as flexible probes block deflection.

### 3. Ultrasonic lithotripsy

- A piezoceramic transducer vibrates a hollow probe at ultrasonic frequency; contact drills the stone.
- **Suction** through the hollow probe removes fragments at once. Best for soft stones in PCNL (percutaneous nephrolithotomy); weak on hard stones; needs irrigation for cooling.

### 4. Combined devices

- Ultrasonic plus ballistic in one probe (for example Trilogy, ShockPulse-SE): ballistic cracks hard stones while ultrasound and suction clear fragments. Faster PCNL.

### 5. Electrohydraulic lithotripsy

- A spark at the electrode tip forms a **vapour bubble**; expansion and collapse send pressure waves in all directions.
- Cheap; 3–5 Fr flexible probes. Poorly focused, so it injures the ureter and the scope; weak on cystine. AUA (American Urological Association) advises against it as first-line ureteric lithotripsy; still used for bladder stones.

### 6. Lasers

- **Pulse duration decides mechanism.** Short pulses (<10 µs: pulsed dye, FREDDY) are photomechanical, with cavitation shocks. Long pulses (>250 µs: holmium, thulium) are **photothermal**: the stone is heated until it melts, decomposes or flashes its water to steam.

| | **Ho:YAG** | **TFL** |
|---|---|---|
| Wavelength | **2120 nm** (often quoted 2100) | 1940 nm (water absorption peak) |
| Frequency | Up to about 120 Hz | Up to about 2000 Hz |
| Peak power | High | Low → less retropulsion |
| Fibre | ≥200 µm | 50–150 µm (more flow and deflection) |
| Product | Fragments and dust | Finer dust |

- **Holmium:** standard since the 1990s; breaks **every** composition; flexible quartz fibres. Fragmenting 0.6–1.0 J × 6–10 Hz; dusting 0.2–0.3 J × 40–80 Hz; "popcorning" in a calyx.
- **Moses effect:** the start of the pulse boils a vapour channel through which the rest reaches the stone; pulse modulation reduces retropulsion.
- **TFL RCTs (randomised controlled trials):** superior to low-power (30 W) holmium; equal stone-free rate to 120 W pulse-modulated holmium.
- **Retropulsion** rises with pulse energy, not frequency, and falls with longer pulses.

### Safety

- Laser heat follows total power: ≤10 W in the ureter, pauses, high flow, access sheath. Eyewear for all.

- Never fire without seeing the fibre tip on the stone.
- SWL: haematoma, arrhythmia, steinstrasse. Contraindications: pregnancy, coagulopathy or anticoagulants, untreated infection, aneurysm, distal obstruction.

### Choice by site

| Site | Usual energy |
|---|---|
| Ureter | Laser (semirigid or flexible); pneumatic distally; SWL |
| Kidney | SWL; laser via flexible URS (ureteroscopy); PCNL with ultrasonic, combined or laser |
| Bladder | Laser, pneumatic, EHL, mechanical |

### Recent advances

- **BWL:** handheld ultrasound bursts (5–100 cycles) break stones without anaesthesia. **Ultrasonic propulsion** moves fragments.
- Suction sheaths; AI (artificial intelligence)-adjusted laser settings.

**Diagram to draw:** SWL electrohydraulic ellipsoid with F1 and F2, stone at F2.

> **Key points to remember:**
> - SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
> - Holmium 2120 nm, photothermal, breaks every composition.
> - TFL 1940 nm: low peak power, finer dust, less retropulsion.
> - EHL is poorly focused, so it is not first-line in the ureter.

## Short answer questions (SAQ)

## SAQ 1 Flexible ureteroscopy / RIRS and accessories

P1 · P2 · P3 **merged from 3**

*Asked as:* P2 SAQ "RIRS (Retrograde intrarenal surgery)" / "RIRS" / "Flexible Ureteroscopy." / "Flexible URS" · P1 SAQ "Flexible Ureterorenoscopy" · P3 SAQ "Accessories for RIRS"

### Introduction

- RIRS (retrograde intrarenal surgery) is flexible ureterorenoscopy (fURS; flexible ureteroscopy) inside the kidney; it inspects and treats the whole upper tract retrogradely by the natural route.
- Marshall first reported it in 1964; a working channel and active deflection made it therapeutic. Active deflection, miniaturised digital scopes and the holmium laser made it practical.

### Instrument

| Feature | Details |
|---|---|
| Tip | 4.5–7.5 Fr, tapered (shaft larger) (digital tips up to about 8.5 Fr) |
| Working channel | **3.6 Fr** (laser fibre, basket), shared by accessories and irrigation |

| Feature | Details |
|---|---|
| Deflection | **≥270° both ways** (reaches lower pole); lower pole angle averages about 140° (up to 175°) |
| **Fibreoptic** | Smallest tip; image degrades as fibres break (black spots); repair about every 15 cases |
| **Digital** ("chip-on-tip", CCD [charge-coupled device] or CMOS [complementary metal-oxide semiconductor] chip) | Better image; larger tip (about 8.5 Fr); not more durable |
| **Single-use** (LithoVue, 2016) | 7.7 Fr; no repair, reprocessing or sterilisation cost; cheaper below about 99 cases a year |

## Indications

- **Stones:** renal stones **<2 cm**; lower pole <1 cm (or 1–2 cm / larger as an alternative to PCNL [percutaneous nephrolithotomy]); patients who must stay on anticoagulants (first choice when anticoagulants cannot be stopped); failed SWL (shock wave lithotripsy); obesity; bleeding disorders; staged treatment of larger stones.
- **Others:** diagnosis and laser ablation of low-grade UTUC (upper tract urothelial carcinoma); filling defects, positive cytology; short strictures and endopyelotomy; calyceal diverticulum; infundibular stenosis; essential haematuria; foreign bodies.

## Technique

1. Sterile urine; single-dose antibiotic; α-blocker or **pre-stenting** if the ureter is tight.
2. Cystoscopy, **safety guidewire**, retrograde pyelogram.
3. **Ureteral access sheath (UAS)** (usually 12/14 Fr) over a stiff wire under fluoroscopy: lowers intrarenal pressure and temperature and allows repeated passes, but causes ureteric injury in nearly half (severe in about 1 in 8).
4. Low-pressure irrigation. **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm) or thulium fibre laser**: dusting (low energy, high frequency) or fragment-and-basket. Keep power ≤10 W in the ureter and ≤25 W in the kidney (heat above 43 °C damages tissue).
5. Lower pole stones: relocate the stone to an upper calyx with a basket.
6. Inspect the ureter on withdrawal; stent unless uncomplicated. **Omit stent** if the case was uncomplicated with normal anatomy, normal renal function, no ureteric injury and no second stage (AUA [American Urological Association]). - **Difficult (unfavourable) lower pole:** steep infundibulopelvic angle, long (>10 mm) or narrow (<5 mm) infundibulum (EAU [European Association of Urology] factors against SWL; they also hinder fURS).

## Accessories

### Guidewires (0.018–0.038 inch, 145–260 cm)

| Type | Construction | Use | Weakness |
|---|---|---|---|
| Hydrophilic | Nitinol core, slippery when wet | Bypassing impacted stones, tortuous ureter | Displaces easily |
| PTFE (polytetrafluoroethylene)-coated | Steel or nitinol | Safety and working wire | Damages scope channel on backloading |
| **Hybrid** | Floppy hydrophilic tip, stiff PTFE shaft | Both roles | — |

- **Safety wire** keeps access and allows stenting after injury (AUA, EAU).
- **Also know:** open-ended or dual-lumen catheter (pyelogram, second wire); stiff wire for the sheath; balloon dilators.

**UAS (ureteral access sheath)**

- Hydrophilic sheath with tapered obturator; usually **12/14 Fr** (inner/outer); tip below the PUJ (pelviureteric junction).
- **Pros:** lower intrarenal pressure and temperature; repeated passes; less scope wear.
- **Cons:** ureteric injury in nearly half, severe in about **1 in 8** (less if pre-stented); no proven fall in infection.
- **Suction sheaths and FANS (flexible and navigable suction access sheath):** aspirate dust and keep pressure low.

**Retrieval devices**

- **Nitinol tipless baskets** (about 1.5–2.4 Fr): kink-resistant, open flush in calyces; relocate lower-pole stones. Small baskets allow more flow but are fragile. Graspers release easily.

**Laser and fibres**

- Ho:YAG or thulium fibre laser; small-core fibres (about 200 µm) keep deflection and flow; **≤10 W in the ureter**.

**Irrigation and antiretropulsion**

- Gravity, pressure bag, hand or foot pumps (hand pumps: more fever); pressure sensors emerging.
- Stone Cone, NTrap, BackStop gel: mainly ureteric stones.

**Stents**

- **Double-J**; AUA: may omit after uncomplicated URS (ureteroscopy).

## Outcomes

- Stone-free rate is higher than SWL for renal stones <2 cm but falls with size and lower pole location. Larger stones may need staged procedures.
- Reported rates depend on the imaging used (CT strictest).

## Complications

- Stent symptoms (commonest); **sepsis about 5%**; stricture about 3%; perforation (a few per cent); bleeding or subcapsular haematoma <1%; **avulsion <1%** (nephrostomy, delayed reconstruction); scope damage.

## Recent advances

- Single-use digital scopes; **thulium fibre laser** and pulse-modulated (Moses) holmium (less retropulsion, finer dust); **suction access sheaths and FANS**; intrarenal pressure sensors (pressure-sensing scopes); robotic flexible ureteroscopy; AI (artificial intelligence) navigation.

**Diagram to draw:** flexible scope through an access sheath over a wire, deflected into the lower pole calyx, with a tipless basket in the lower calyx.

**Key points to remember:**

- 3.6 Fr channel; ≥270° deflection.
- 12/14 Fr access sheath lowers pressure but severely injures the ureter in about 1 in 8.
- Laser power ≤10 W ureter, ≤25 W kidney.
- RIRS: stones <2 cm; first choice when anticoagulants cannot be stopped.
- Sepsis about 5%, stricture about 3%.
- Hydrophilic wire passes; PTFE wire stays.
- Nitinol tipless basket for calyces.

## SAQ 2 Renal tubular acidosis (distal and type 2)

P1 · P2 · P3 **merged from 3**

*Asked as:* P2 SAQ "Renal tubular acidosis" · P1 SAQ "Distal Renal tubular acidosis." · P3 SAQ "Type 2 Renal tubular acidosis?"

### Definition

- **RTA (renal tubular acidosis)**: hyperchloraemic, normal AG (anion gap) metabolic acidosis from defective tubular $H^+$ secretion or $HCO_3^-$ reabsorption.
- **Distal (type 1) RTA:** the collecting duct cannot secrete $H^+$ despite systemic acidosis. Result: **hypokalaemic, hyperchloraemic, normal anion gap metabolic acidosis** with **urine pH >6**: the only acidosis with alkaline urine.
- **Proximal (type 2) RTA:** failure of the proximal tubule to reclaim bicarbonate, causing hyperchloraemic, normal anion gap metabolic acidosis.

### Comparison of types

| Feature | Type 1 (distal) | Type 2 (proximal) | Type 4 (hyperkalaemic) |
|---|---|---|---|
| Defect | $H^+$ secretion (α-intercalated cell) | Proximal $HCO_3^-$ reabsorption | Aldosterone deficiency or resistance |
| Urine pH in acidosis | **Always >6** | <5.5 at steady state | Low |
| Potassium | Low | Low | **High** |
| Serum $HCO_3^-$ | Low, may be very low | 15–18 mEq/L (P3: 14–18 mEq/L) | Mildly low |
| Urine calcium | High | Normal or mildly high | Low |
| Urine citrate | **Profoundly low** | Near normal | – |
| Stones | **Common: up to 70%; calcium phosphate, nephrocalcinosis** | Uncommon | Uncommon |
| Causes | Sjögren, lupus, inherited | Fanconi | Diabetes, obstruction |
| Treatment | **Potassium citrate** 1–2 mEq/kg/day | Alkali 10–15 mEq/kg/day plus potassium | Fludrocortisone, low-potassium diet, stop offending drugs |

### Distal (type 1) RTA

#### Normal acidification and the defect

- **α-intercalated cells** secrete $H^+$ by apical $H^+$-ATPase; bicarbonate exits by the basolateral $Cl^-/HCO_3^-$ exchanger (AE1). Normally urine pH can fall to 4.5.
- In distal RTA, potassium is secreted in place of $H^+$, causing hypokalaemia.

**Causes**

| Inherited | Acquired (secondary) |
|---|---|
| **SLC4A1** (AE1): mostly dominant | Sjögren syndrome, SLE (systemic lupus erythematosus) |
| **ATP6V1B1, ATP6V0A4** ($H^+$-ATPase): recessive, **sensorineural deafness** | Obstructive uropathy, pyelonephritis, medullary sponge kidney |
| **CA2**: mixed proximal–distal (carbonic anhydrase II deficiency) | Hypercalciuria, hyperparathyroidism, amphotericin B |

**Why stones form (urological importance)**

- **Stones in up to 70%**: **calcium phosphate** stones and **nephrocalcinosis**. Stones lead to diagnosis in up to half.
- **Profound hypocitraturia** (most important): acidosis increases proximal citrate reabsorption.
- **Hypercalciuria:** bone buffering releases calcium. **Fixed alkaline (high) urine pH** precipitates calcium phosphate.

**Clinical features**

- Stones, nephrocalcinosis, weakness (hypokalaemia), osteomalacia or rickets; children fail to thrive (growth failure).
- **Incomplete distal RTA:** normal bicarbonate, but urine pH cannot fall below 5.5 after an acid load.

## Proximal (type 2) RTA

**Pathophysiology**

- The proximal tubule reclaims **85–90% of filtered bicarbonate** (luminal carbonic anhydrase; exit by the basolateral Na–$HCO_3$ cotransporter).
- **Self-limiting:** bicarbonate wasting lowers plasma $HCO_3^-$ until the filtered load drops below the lowered threshold; the intact distal tubule then copes, giving a **steady state at about 14–18 mEq/L (P2: 15–18 mEq/L) with urine pH <5.5**.
- **Hypokalaemia** from distal sodium bicarbonate delivery; worsens on alkali.

**Causes**

| Isolated | Generalised (Fanconi syndrome) |
|---|---|
| Carbonic anhydrase inhibitors (acetazolamide, **topiramate**) | **Cystinosis** (commonest in children), Lowe, Wilson, galactosaemia |
| Carbonic anhydrase II deficiency (mixed) | **Ifosfamide, cisplatin**, aminoglycosides, valproate |
| | Myeloma, amyloidosis, lead |

- **Fanconi:** glycosuria, aminoaciduria, phosphaturia, uricosuria.

**Clinical features**

- Children: growth failure, **rickets**; adults: osteomalacia, weakness.

- **Stones uncommon**: urinary citrate stays near normal and urine is acid at steady state. Topiramate and acetazolamide are exceptions (alkaline urine, calcium phosphate stones).

### Type 4 (hyperkalaemic) RTA

- Hyperkalaemia, low urine pH; diabetes, obstructive uropathy. Amiloride and trimethoprim cause hyperkalaemic distal RTA.

### Diagnosis

- Bloods: low bicarbonate, high chloride, low potassium (types 1 and 2); urine pH >6 on a fasting morning sample in distal RTA; urine can acidify below 5.5 in proximal RTA.
- **UAG (urine anion gap) = ($Na^+$ + $K^+$) − $Cl^-$** reflects urinary ammonium: **positive** (low ammonium excretion) in renal causes; negative in GI (gastrointestinal) loss such as diarrhoea.
- **Ammonium chloride load** (0.1 g/kg orally, hourly urine pH for about 6 h): **failure to fall below 5.5** confirms distal RTA. The **furosemide–fludrocortisone test** is a better-tolerated alternative.
- **Bicarbonate loading:** urine pH rises; **fractional bicarbonate excretion >15%** indicates proximal RTA. Look for Fanconi features; drug history.
- 24-hour urine: low citrate, high calcium (distal); ultrasound for nephrocalcinosis.

### Treatment

- **Distal RTA: potassium citrate** (about 1–2 mEq/kg/day in divided doses): corrects acidosis, hypokalaemia and hypocitraturia, reduces calcium excretion and stone formation. Sodium alkali is less suitable because sodium raises urinary calcium. Treat the cause; manage stones endourologically.
- **Proximal RTA:** treat the cause; **large alkali doses (about 10–15 mEq/kg/day)** as potassium citrate or mixed citrate, replacing potassium; thiazide; phosphate and vitamin D for rickets.
- **Type 4:** fludrocortisone, low-potassium diet, stop offending drugs.

### Urological relevance

- Obstruction causes hyperkalaemic distal RTA, reversible by relief.
- Topiramate and acetazolamide mimic RTA (alkaline urine, low citrate, calcium phosphate stones).
- Potassium citrate beats sodium alkali, which raises calcium excretion.
- **Also know:** CKD (chronic kidney disease) acidosis; amiloride and trimethoprim (hyperkalaemic distal RTA); mixed type (carbonic anhydrase II deficiency); medullary sponge kidney; laxative abuse (ammonium urate stones); inherited forms (recessive with deafness); growth failure in children.

**Diagram to draw:** nephron with defect sites; α-intercalated cell with $H^+$-ATPase, AE1 and the stone mechanism; proximal tubular cell with carbonic anhydrase.

**Key points to remember:**

- Type 1 (distal): urine pH >6 with acidosis, hypokalaemia, hypocitraturia (key), positive UAG; calcium phosphate stones in up to 70%.
- $NH_4Cl$ load: pH fails to fall below 5.5 in distal RTA.
- Potassium citrate is the treatment for distal RTA.
- Type 2 (proximal): bicarbonate wasting; $HCO_3^-$ 14–18 mEq/L (P2: 15–18); urine pH <5.5 at steady state; usually Fanconi (ifosfamide, cisplatin).
- Fractional $HCO_3^-$ excretion >15%; stones uncommon in type 2.
- Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
- Positive UAG = renal cause.

## SAQ 3 PCNL patient positioning: prone vs supine

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "Prone vs Supine PCNL: Indications, positioning, technique, specific complications, advantages and disadvantages of each." · P2 SAQ "Describe various pt positions in PCNL"

### Introduction

- PCNL (percutaneous nephrolithotomy) position decides calyceal access, working space, airway safety and simultaneous retrograde access.

### Indications for PCNL

- Renal stones **>2 cm**, staghorn, lower pole >1 cm, calyceal diverticulum, failed retrograde route or urinary diversion, UPJ (ureteropelvic junction) obstruction.

### Positions

| Position | Set-up | Advantages | Disadvantages |
|---|---|---|---|
| **Prone** (Goodwin, 1955) | Lithotomy for ureteric catheter, then turned prone; chest and pelvic bolsters; horizontal bolsters at nipple line and ASIS (anterior superior iliac spine) bring the kidney down more than vertical ones; arms ≤90°, face and knees padded | **Posterior calyx and upper pole** access; large working area; multiple tracts; familiar | Turning (repositioning); airway hard to reach; pressure and positioning injuries; retrorenal colon commoner prone; needs split-leg table for ECIRS |
| **Prone split-leg / prone-flexed** | Legs apart for flexible cystoscopy; hip flexion opens the costovertebral space | ECIRS (endoscopic combined intrarenal surgery) prone | Needs split-leg table; prone cystoscopy learning curve |
| **Supine** (Valdivia-Uría, 1987) | Ipsilateral flank at table edge on a lumbar bolster; arm across chest | **Airway retained**; no repositioning; single position, shorter time; **gravity drains fragments**, low intrarenal pressure; safer anaesthesia | Smaller working area; collapsed system; anterior calyces; longer tract; mobile kidney; harder upper pole |

| Position | Set-up | Advantages | Disadvantages |
|---|---|---|---|
| **Galdakao-modified supine Valdivia** (Scoffone) | Supine flank tilt with ipsilateral leg extended and other leg in lithotomy | **Simultaneous retrograde access (ECIRS)** in one position | Longer, more horizontal tract |
| **Bart's flank-free modified supine** (Bach) | Bolster under the thorax, flank free | More flank space | Fewer data |
| **Lateral decubitus (flank)** | Kidney-up flank position | Obese, kyphoscoliosis, poor respiratory reserve | Unfamiliar fluoroscopic orientation |

### Technique: prone vs supine

| Feature | Prone | Supine |
|---|---|---|
| Puncture and tract | Posterior calyx puncture under fluoroscopy or ultrasound; tract more vertical; large working area | Puncture in posterior axillary line; tract more horizontal and longer; kidney more mobile; air outlines anterior calyces |
| Drainage | Against gravity; higher pelvic pressure | Gravity drains fragments, low intrarenal pressure |
| Anaesthesia | Airway access difficult; turning needed | Airway retained; no repositioning |

### Positioning safety (prone)

- Bolster shins so toes float; pad knees; arms ≤90° to the torso (brachial plexus); axillary and pectoral rolls; padded face and eyes.

### Specific complications

- **Prone:** ocular injury, neuropraxia (brachial plexus), pressure injury, endotracheal tube dislodgement while turning.
- **Supine: colon injury** (if puncture too anterior), access failure to upper pole, kidney displacement during dilatation.
- **Both:** bleeding (transfusion now about 1%), pleural injury above the 12th rib, sepsis, bowel, spleen or liver injury.

### Choosing a position

- **Prone:** standard anatomy, upper pole or multiple posterior tracts.
- **Supine variants:** obesity, cardiopulmonary disease, kyphosis, ECIRS, ileal conduit or continent channel, combined ureteric stone.
- **Spinal deformity:** individualise; pad every pressure point.
- **Evidence:** stone-free rates and complications similar; supine often shortens operating time; choice is mainly surgeon preference.

**Diagram to draw:** axial CT cross-section showing prone (vertical) and supine (horizontal) tracts.

**Key points to remember:**

- Prone: posterior and upper pole access, large working area.
- Supine: airway safe, ECIRS, gravity drainage, no turning; Galdakao-modified supine gives one position for ECIRS.
- Horizontal bolsters drop the kidney more.
- Similar outcomes; choose by anatomy and experience.

## SAQ 4 Renal autotransplantation

P1 · P3 **merged from 2**

*Asked as:* P3 SAQ "Auto-transplantation of Kidney?" / "Role of auto transplant of kidney in Urology." / "Renal Autotransplantation" · P1 SAQ "Autotransplantation"

### Definition and principle

- Removing a kidney and re-implanting it, usually in the **ipsilateral iliac fossa** on the iliac vessels, with **bench (ex vivo) surgery** if needed. First done in **1963** for proximal ureteric injury.
- **Principle:** bringing the kidney to the bladder replaces the ureter without bowel; the bench gives a cold, bloodless field.

### Role (indications)

| Group | Examples |
|---|---|
| **Ureteric loss (commonest)** | Long-segment loss or stricture; long iatrogenic or traumatic loss; repeated failed repairs (**delayed setting only, never acute**) |
| **RPF (retroperitoneal fibrosis)** | When ureterolysis fails or is impossible and function is satisfactory |
| **Tumour** | Total ureterectomy for UTUC (upper tract urothelial carcinoma) in a solitary kidney; bench partial nephrectomy in a solitary kidney; left RCC (renal cell carcinoma) needing suprarenal caval resection (new drainage for the right kidney) |
| **Vascular** | Complex branch renal artery aneurysm, stenosis or AVM (arteriovenous malformation) unsuitable for endovascular repair; nutcracker syndrome |
| **Pain** | Selected loin pain–haematuria syndrome |

- **Preferred over ileal ureter** when the contralateral kidney is absent or poor, or bowel is unsuitable (creatinine >2 mg/dL, inflammatory bowel disease, radiation enteritis).
- **Contraindications:** poor renal function, severe iliac atherosclerosis, unfit patient.

### Procedure

CT (computed tomography) angiography, renogram, ureteric study → harvest as for live donor nephrectomy (open/laparoscopic/robotic): maximal vessel length, preserve ureteric blood supply → cold perfusion on bench ± ex vivo repair → Gibson incision, extraperitoneal iliac fossa → renal vein to external iliac vein (end-to-side); renal artery to external iliac (end-to-side) or internal iliac (end-to-end) → drainage: pyelovesicostomy (± Boari flap),

ureteroneocystostomy, or native ureter left in continuity → stent, catheter, drain

- Laparoscopic harvest plus Gibson implantation: less analgesia, faster recovery; **robotic autotransplantation** emerging.

### Complications and outcome

- Arterial or venous **thrombosis**, bleeding, urine leak, anastomotic stricture, lymphocele, infection.
- **Renal unit loss 5–10%** (lower in transplant centres); major (Clavien–Dindo ≥III) complications about 15–20%.
- **Long-term success about 90%.**

### Follow-up

- Doppler ultrasound early; renogram and blood pressure later.

### Alternatives

- Boari flap, TUU (transureteroureterostomy), ileal ureter, Yang–Monti, buccal ureteroplasty, renal descensus.

**Diagram to draw:** kidney in the iliac fossa with vascular anastomoses and pyelovesicostomy.

> **Key points to remember:**
> - For long ureteric loss when other repairs fail and the other kidney is poor or bowel unsuitable.
> - Delayed setting only; bench surgery for complex vascular or tumour cases.
> - Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
> - Unit loss 5–10%; success about 90%.

## SAQ 5 Stones in pregnancy

P2 · P3 **merged from 2**

*Asked as:* P2 SAQ "Management of Stones in pregnancy." / "Urolithiasis in pregnancy" · P3 SAQ "Management of lower ureteric stone of size less than 1 cm in pregnancy?"

### Background

- 1 in 200 to 1 in 1500 pregnancies; renal colic is the **commonest non-obstetric cause of admission**; mostly second or third trimester.
- Linked with preterm delivery, pre-eclampsia and premature rupture of membranes.
- **Why calcium phosphate stones:** raised GFR (glomerular filtration rate) and placental calcitriol cause hypercalciuria; urine is alkaline. Raised citrate, magnesium and urine flow protect, so stone rates barely change.
- **Physiological hydronephrosis** (up to 90% by third trimester, right-sided) mimics obstruction.

### Evaluation

- Urinalysis, **urine culture always**, blood count, creatinine; obstetric review.
- **US (ultrasound) first line**; **transvaginal US** for the distal ureter; a rise in Doppler **RI (resistive index)** suggests acute obstruction.

- Second line: **MRI (magnetic resonance imaging)** (stone as a filling defect) or **low-dose CT (computed tomography)**. Keep total fetal dose **≤50 mGy** (under 50 mGy).

### Management algorithm

```
Symptomatic stone in pregnancy (including lower ureteric stone <1 cm) ├─ sepsis, solitary kidney, renal failure, uncontrolled pain │ → urgent stent or nephrostomy (+ antibiotics) ├─ pain controlled, no infection → observe (50–80% pass) └─ failed observation → ureteroscopy + holmium laser or temporary stent/nephrostomy until delivery
```

### Conservative management

- Hydration, paracetamol, small doses of morphine (opioids).
- **NSAIDs (non-steroidal anti-inflammatory drugs) are contraindicated** (ductus arteriosus closure in the third trimester).
- **MET (medical expulsive therapy)** with tamsulosin is off-label.

### Ureteroscopy

- **URS (ureteroscopy) with holmium laser:** first-line intervention and definitive treatment for a non-complex stone in any trimester; ideal for a distal (lower ureteric) stone; second trimester is the safest window.
- Neuraxial anaesthesia, lateral tilt, minimal fluoroscopy (lead shield beneath the patient, US guidance), fetal monitoring.
- Third-trimester URS may raise preterm labour risk.

### Temporising drainage

- **Stent or PCN (percutaneous nephrostomy):** temporising. Encrustation is rapid: change stents about **every 4 weeks** (P3: every 4–6 weeks; 4–6 weeks in series), nephrostomy about every 3 weeks; definitive treatment after delivery.

### Other modalities

- **Asymptomatic renal stones:** observe until after delivery.
- **SWL (shock wave lithotripsy) contraindicated. PCNL (percutaneous nephrolithotomy) deferred** until after delivery except in expert centres.

### After delivery

- Clear residual stones; **metabolic evaluation** postpartum.

**Diagram to draw:** the management flowchart above.

> **Key points to remember:**
> - US first; MRI or low-dose CT second (≤50 mGy).
> - Observation first; no NSAIDs.
> - URS with laser is definitive; SWL contraindicated.
> - Stents encrust: change about every 4 weeks (P3: every 4–6 weeks).

## SAQ 6 Ureterocalycostomy

P2 · P3 **merged from 2**

*Asked as:* P3 SAQ "Indication, procedure and complications of Uretero-Calycostomy?" · P2 SAQ "Uretero-calycostomy."

### Definition

- End-to-side anastomosis of the spatulated proximal ureter to a dilated **lower-pole calyx**, after **resecting lower-pole parenchyma**, bypassing the pelvis and PUJ (pelviureteric junction).

### Indications

- **Failed pyeloplasty** with a scarred pelvis and short ureter (commonest).
- **Small intrarenal pelvis** with PUJ obstruction.
- **Rotation or fusion anomalies** (horseshoe kidney).
- Proximal ureteric or pelvic loss: trauma, iatrogenic, **TB (tuberculosis) PUJ stricture**.
- **Prerequisites:** dilated lower calyx, ureter reaching it without tension, worthwhile function. Cortex <5 mm with poor function predicts failure.

### Procedure (open, laparoscopic or robotic)

`Flank position (45–60°); pre-placed double-J stent → mobilise ureter with generous periureteral tissue → mobilise kidney; partial lower-pole nephrectomy to expose the calyx widely (not simple nephrotomy) → close (ligate) the pelvic outlet; spatulate ureter laterally → apex stitch to lateral calyx wall; second at 180° → interrupted mucosa-to-mucosa sutures, all placed then tied → capsule closed over cut surface, away from anastomosis → cover with perinephric fat, peritoneum or omentum; drain`

- **Why resect:** parenchyma or capsule left round the anastomosis contracts and causes stenosis.

### Postoperative care and follow-up

- Stent out at 4–6 weeks; urogram about 1 month later, then renogram.
- Later endoscopy needs **antegrade upper-pole access**.

### Complications and outcome

- **Anastomotic stricture** (main failure; mostly within the first year).
- Urine leak, urinoma, bleeding from the lower pole, infection, loss of the renal unit.
- Success about **70%** in the largest (mostly redo) series; better in small paediatric (robotic) series. The access route does not affect outcome.

### Alternatives

- Endopyelotomy, redo pyeloplasty, flap repair, Davis intubated ureterotomy, ileal ureter, buccal graft, autotransplantation, nephrectomy (poor kidney); calicovesicostomy for a low-lying kidney.

**Diagram to draw:** kidney with resected lower pole and ureter sewn to the calyx.

> **Key points to remember:**
> - Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
> - Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
> - Stricture is the main failure; success about 70% in redo series.

## SAQ 7 Tubeless PCNL

P2 · P4 **merged from 2**

*Asked as:* P2 SAQ "Tubeless PCNL" · P4 SAQ "Tubeless PCNL"

### Definition

- PCNL (percutaneous nephrolithotomy) that ends **without a nephrostomy tube**.
- **Tubeless**: drainage by a ureteral stent (double-J, or externalised stent or string). **Totally tubeless**: no stent and no tube.
- **Wickham (1984)** did the first tubeless PCNL; **Bellman (1997)** popularised it.

### Rationale

- A large nephrostomy tube causes pain, more analgesia, longer stay and higher cost. Most tracts stop bleeding without tamponade.

### Selection

| Suitable | Keep a nephrostomy |
|---|---|
| Complete clearance, single tract | Pyonephrosis or infected stone |
| Minimal bleeding | Significant bleeding |
| Intact collecting system | Significant perforation of (injury to) the collecting system |
| Supracostal access, bilateral cases (shown safe) | Residual stone needing a second look |

### Technique (exit strategy)

1. Confirm clearance by rigid and **flexible nephroscopy** and fluoroscopy.
2. Antegrade double-J stent, or externalised stent through the urethra, or a string.
3. Withdraw the sheath under vision; the **safety wire is removed last**.
4. Watch the tract; apply pressure (rib as backstop in thin patients). Fibrin glue in the tract did not reduce bleeding in randomised trials; some use haemostatic sealants or tract cautery.
5. Urethral catheter overnight.

### Evidence

- Meta-analyses of randomised trials: shorter stay and operative time, and less pain.
- The pain benefit is over large tubes (**>16 Fr**); there is **no benefit over small 8–9 Fr tubes**; small tubes (≤10 Fr) give similar comfort without more bleeding.
- No difference in fever, infection, transfusion or stone-free rate.
- Stent discomfort: worse WiSQoL (Wisconsin Stone Quality of Life) scores than nephrostomy in two trials. Externalised stents come out before discharge without cystoscopy.
- Children: insufficient evidence.

### Disadvantages and complications

- Loss of tract for a second look.
- Stent symptoms and a further procedure for removal.
- Urine leak or urinoma.
- Bleeding without tamponade: re-insert a balloon or Kaye catheter, then **angioembolisation**.

### Current position

- Choose tubeless, small tube (≤10 Fr) or stent case by case; both stents and small tubes are acceptable (AUA, American Urological Association). Pairs well with mini-PCNL.

**Diagram to draw:** kidney with sealed tract and double-J stent.

> **Key points to remember:**
> - Tubeless = stent only; totally tubeless = neither.
> - Benefit is mainly over large (>16 Fr) tubes.
> - Keep a tube for bleeding, perforation, infection or second look.

## SAQ 8 Mini-PCNL and newer PCNL techniques

P3 · P4 **merged from 2**

*Asked as:* P4 SAQ "Miniperc" / "Newer PCNL techniques and nomenclature of miniaturised PCNL" · P3 SAQ "Mini Perc."

### Definition and why miniaturise

- PCNL (percutaneous nephrolithotomy) through a **smaller tract** than standard, aiming to reduce bleeding and renal trauma. First "mini-perc" by **Jackman (1998)**, 13 Fr, in children.
- In PCNL, bleeding rises with tract size. Smaller tracts mean less bleeding and pain, at the cost of longer operating time and poorer vision.

### Nomenclature (by tract or outer sheath size)

| Name | Tract size | Features |
|---|---|---|
| Standard | 24–34 Fr | Ultrasonic or pneumatic lithotripsy; graspers |
| **Mini** | 15–22 Fr | First "mini-perc": Jackman 1998, 13 Fr; irrigation washes fragments out ("vacuum-cleaner" effect) |
| **Super-mini** | About 10–14 Fr | Sheath with **active suction** of fragments |
| **Ultra-mini** | 11–14 Fr | 3 Fr inline telescope in 11–13 Fr sheath |
| **Micro-perc** | <10 Fr | Single-step "all-seeing needle" with optics; laser only, fragments pass spontaneously |

- **Problem:** names overlap and cut-offs are contested (paediatric chapter: mini <20–24 Fr), so many authors report the **actual sheath size in Fr**.
- The **working channel**, not sheath size, decides which instruments pass: the smallest systems take only a laser fibre.

### Instruments and technique

- Mini nephroscopes (e.g. 12 Fr telescope, 15–18 Fr sheath, 6 Fr channel); laser (holmium/thulium) or small pneumatic probes.
- **Fragment removal**: irrigation outflow around the scope washes fragments out (**"vacuum-cleaner" effect**), or **active suction sheaths** (super-mini PCNL).
- Ultrasound or fluoroscopic access; supine or prone; ECIRS (endoscopic combined intrarenal surgery) possible.
- Exit: often **tubeless** or small nephrostomy (≤10 Fr: less pain).

### Newer PCNL techniques

1. **Supine PCNL and ECIRS:** Galdakao-modified supine; simultaneous retrograde flexible ureteroscopy watches puncture and dilatation; fewer tracts.
2. **Ultrasound-guided access:** less radiation; shows pleura and viscera.
3. **Single-step or balloon dilatation.**
4. **Suction sheaths:** lower pressure, faster clearance.
5. **Lasers:** high-power and pulse-modulated holmium, TFL (thulium fibre laser) for dusting through small tracts.
6. **Flexible nephroscopy** after bulk removal: higher stone-free rate.
7. **Tubeless or small (≤10 Fr) nephrostomy:** less pain.

- **Also know:** robotic or artificial intelligence-assisted puncture under study.

### Indications and contraindications

- Renal stones **1–2 cm**, **lower pole stones** >1 cm (alternative to RIRS [retrograde intrarenal surgery]), failed SWL (shock wave lithotripsy) or RIRS, **paediatric stones**, calyceal diverticular stones; selected larger stones or extra staghorn tracts.
- **Contraindications**: as for PCNL (untreated infection, uncorrected coagulopathy, pregnancy).

### Advantages over standard PCNL

- Less bleeding and haemoglobin drop, **fewer transfusions**, shorter stay; less parenchymal damage; stone-free rates non-inferior to standard PCNL.
- Children: a 24–30 Fr tract in a child equals about a 72 Fr tract in an adult, so miniaturisation matters most here.

### Disadvantages

- **Longer operative time**, poorer vision, slower fragment clearance.
- **Intrarenal pressure**: high-pressure irrigation through a tight sheath, especially when the scope nearly fills it, raises pelvic pressure and causes pyelovenous backflow, **fever and sepsis**. Keep outflow free, use low-pressure irrigation, or suction sheaths.

### Evidence

- Mini vs standard (meta-analysis): **non-inferior stone-free rate**, less bleeding (fewer transfusions), shorter stay, longer operation; same complications.
- Stones >2 cm: equal stone-free rate and fever; mini bleeds less than **30 Fr** tracts, but **every advantage disappears against a 24 Fr tract**.
- Mini-PCNL versus RIRS, large matched study: similar stone-free and retreatment rates; RIRS shorter stay.

**Diagram to draw:** scale bar of tract sizes compared (micro, ultra-mini, mini, standard).

**Key points to remember:**

- Definition by sheath size: standard 24–34 Fr; mini-PCNL 15–22 Fr; ultra-mini 11–14 Fr; micro-perc <10 Fr; Jackman 1998: first mini-perc (13 Fr).
- Main advantage: less bleeding and fewer transfusions, shorter stay, with stone-free rates non-inferior to standard PCNL.
- Main limitation: longer operating time and higher intrarenal pressure (fever, sepsis); the advantage disappears against a 24 Fr tract.
- Working channel decides instruments; suction lowers pressure.

## SAQ 9 Newer methods of stone analysis

P3 · P4 **merged from 2**

*Asked as:* P3 SAQ "Newer methods of stone analysis" / "Newer techniques of stone analysis" · P4 SAQ "Newer methods of stone analysis and their significance in management of calculus disease."

### Why analyse stones

- AUA (American Urological Association) and EAU (European Association of Urology): analyse a stone at least once in every former (every first stone); **repeat if stones recur on treatment** (evolving metabolic condition), or on early recurrence after clearance.
- Composition points to the cause: **uric acid** suggests metabolic disease, **cystine** a genetic one, **struvite** infection. Cystine, struvite, 2,8-DHA (2,8-dihydroxyadenine) and xanthine are diagnostic in themselves.
- Composition can change over time (e.g. calcium oxalate converting to brushite), so re-analyse recurrent stones.

### Why wet chemical analysis is obsolete

- Detects ions, not **crystal phases**, so it cannot separate COM (calcium oxalate monohydrate, whewellite) from COD (dihydrate, weddellite), or apatite from brushite.
- High error rates; misses rare stones (2,8-DHA, drugs); needs a large sample.

### Newer methods

| Method | Principle | Comment |
|---|---|---|
| **FTIR (Fourier-transform infrared) spectroscopy** | Each compound absorbs a characteristic infrared spectrum | Commonest reference method; tiny sample; detects proteins and drugs |
| **XRD (X-ray diffraction)** | Crystal lattice diffracts X-rays in a unique pattern | Gold standard for crystals; misses amorphous material |
| **Polarising microscopy** | Crystal optics (birefringence) | Cheap; needs an experienced observer (observer-dependent) |
| **Morpho-constitutional analysis (Daudon)** | Stereomicroscopy of surface and section, plus FTIR | Subtypes point to cause (e.g. type Ia COM: low urine volume; Ic: primary hyperoxaluria) |
| **SEM (scanning electron microscopy) ± EDX (energy-dispersive X-ray)** | Surface structure and elements | Research |

| Method | Principle | Comment |
|---|---|---|
| **Micro-CT (micro-computed tomography)** | 3-D non-destructive structure | Research |
| **Raman spectroscopy** | Scattered light spectrum | Research |

### In vivo prediction

- **DECT (dual-energy computed tomography):** two tube energies (e.g. 80 and 140 kVp) separate **uric acid from non–uric acid stones** with near-perfect accuracy; it estimates composition in vivo, mainly uric acid versus calcium stones.
- **Hounsfield units:** SWL (shock wave lithotripsy) works best below about 900–1000 HU; >1000 HU predicts SWL failure; cystine and brushite are dense.
- **Endoscopic recognition** of morphology (lost when stones are dusted). Dusting loses the specimen: collect fragments.

### Significance in management

- **Uric acid:** oral dissolution by urinary alkalinisation (potassium citrate).
- **Struvite:** complete clearance, culture-directed antibiotics.
- **Cystine:** hydration, alkalinisation, tiopronin; avoid SWL.
- **COM, brushite, cystine:** hard, SWL-resistant; prefer endoscopy.
- **Brushite:** can convert from calcium oxalate after SWL or infection; avoid over-alkalinisation.
- **2,8-DHA:** APRT (adenine phosphoribosyltransferase) deficiency; treat with allopurinol or febuxostat. **Drug stones** (indinavir, triamterene): stop the drug.

**Diagram to draw:** FTIR spectrum outline and COM (dumbbell) versus COD (envelope) crystal shapes.

> **Key points to remember:**
> - FTIR (Fourier transform infrared spectroscopy) and XRD (X-ray diffraction) are the reference methods; wet chemistry is obsolete.
> - DECT (dual-energy computed tomography) separates uric acid stones in vivo, mainly uric acid versus calcium stones.
> - Composition guides dissolution, SWL choice and prevention.
> - Stone analysis guides prevention: uric acid points to a metabolic cause, cystine to a genetic one, struvite to infection.

## SAQ 10 Radionuclide studies and diuretic renography in PUJ obstruction

P1

**Asked as:**

- Radionuclide studies for PUJ obstruction. — *Summer 2021 Paper I Q3e (SAQ)*
- Radionuclear studies in PUJ obstruction — *Practice Set 4 Paper I Q3e (SAQ)*
- Diureteric renography — *Practice Set 2 Paper I Q3f (SAQ)*

**Type:** SAQ (10 marks)

(Practice Set 4: also answered in 2021-Summer Paper 1, Q3e. Practice Set 2: PUJ obstruction renography also in 2021-Summer Paper 1, Q3e; tracers: see Practice Set 2 Paper 1, Q3e)

*"Diureteric" is read as diuretic renography.*

## Introduction / principle

- **Dilatation is not obstruction.** PUJ (pelviureteric junction) obstruction is impaired drainage that damages the kidney. **Diuretic renography** is the functional standard (functional gold standard for PUJ obstruction) and best follow-up test (best follow-up study).
- A capacious pelvis retains tracer passively and furosemide washes it out; an obstructed system cannot empty even under diuresis.

## Tracers

| Tracer | Handling | Use |
|---|---|---|
| $^{99m}$Tc-**MAG3** (technetium-99m mercaptoacetyltriglycine) | Tubular secretion | **Agent of choice** (tracer of choice): split function and drainage; good images (good curves) even with poor function |
| $^{99m}$Tc-**DTPA** (diethylenetriaminepentaacetic acid) | Glomerular filtration | GFR (glomerular filtration rate); poor in renal impairment; an alternative to MAG3 |
| $^{99m}$Tc-**DMSA** (dimercaptosuccinic acid) | Cortical retention | Static split function and scars; no drainage data |

## "Well-tempered" diuretic renogram (standard protocol)

| Element | How | Why |
|---|---|---|
| **Hydration** | Oral, or IV (intravenous) fluid in children (about 15 mL/kg) | Dehydration mimics obstruction |
| **Bladder catheter** | Urethral catheter | Removes full-bladder (back-pressure) and reflux (VUR, vesicoureteral reflux) effects |
| **Furosemide** | 0.5 mg/kg (adult maximum 40 mg) in adults; 1 mg/kg in children | Maximal diuresis |
| **Timing** | **F+20** (classic; 20 min after tracer, when the pelvis is full; Campbell authors use F+15), **F0** (with tracer) or **F−15** (15 min before, so the system is already in diuresis) | Diuretic before filling (early diuretic into an unfilled system) misleads; always state timing |
| **Regions of interest** | Regions of interest over cortex (for split) and pelvis (dilated pelvis for drainage); then upright images | Wrong placement gives spurious values |

- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- First study after about 4–6 weeks of age (immature function).

## Outputs

- **DRF (differential renal function)** from counts at 1–3 min (1–3 minute counts); normal 45–55%.
- **Drainage half-time (T½, half-clearance time)** after furosemide, from maximal filling, not from injection:

| T½ after furosemide | Interpretation |
|---|---|
| <10 min (minutes) | Not obstructed |
| 10–20 min (minutes) | Equivocal (indeterminate) |

| T½ after furosemide | Interpretation |
|---|---|
| >20 min (minutes) | Obstructed |

- **O'Reilly curves:** I normal; II obstructed (keeps rising); IIIa dilated, non-obstructed (falls after furosemide); IIIb equivocal.
- **Also know:** P40 (activity remaining / left at 40 min) raises sensitivity; output efficiency; NORA (normalised residual activity); a **biphasic curve** and poor upright (post-void) drainage predict progression.

### Indications

- PUJ (pelviureteric junction) obstruction, antenatal hydronephrosis, megaureter, after pyeloplasty, transplant ureter.

### Indications for pyeloplasty from renography

- Consider surgery for:
- **DRF <40%**, **fall in DRF** on serial comparable scans (**>10%** on serial scans; ≥5% in children; 10% in older series), **T½ >20 min** (minutes) or upright retention; also (poor drainage with) symptoms, infection, increasing dilatation.
- Adult DRF <15%: likely non-salvageable; nephrostomy and reassess, or nephrectomy.

### Pitfalls

- Dehydration, early diuretic (mistimed diuretic), full bladder, wrong ROI (region of interest), very poor function, very capacious pelvis.
- One Campbell chapter calls T½ >40 min obstructed; children have no agreed (paediatric T½) cut-off (false positives up to 44%; neonatal false-positive rates reach 44%); one Campbell chapter uses **T½ >40 min** in adults, so **serial trends and curve shape** matter / decide.

### Follow-up and alternatives

- After pyeloplasty: renogram 1 month after stent removal; spontaneous drainage means success.
- Equivocal (cases) → **Whitaker test** (>15–20 cm $H_2O$ obstructed) or MRU (magnetic resonance urography).

**Diagram to draw:** normal, obstructed and dilated non-obstructed renogram curves (O'Reilly curves I, II, IIIa and IIIb) with furosemide arrow.

> **Key points to remember:**
> - MAG3 is the tracer of choice; hydrate and catheterise.
> - Well-tempered: hydrate, catheter, standard diuretic dose and timing.
> - Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
> - DRF normal 45–55%; T½ <10 normal (not obstructed); 10–20 indeterminate; >20 min obstructed.
> - Operate (surgery) for DRF <40%, fall >10% (falling DRF), or poor drainage with symptoms.

## SAQ 11 What are nanobacteria? Describe their role in urolithiasis.

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 1, Q3e

### Definition

- **Nanobacteria**, now called **CNPs (calcifying nanoparticles)**, are spherical particles far smaller than bacteria with a **calcium apatite shell**, described in bovine and human serum by **Kajander and Çiftçioglu (1998)** as *Nanobacterium sanguineum*.
- **Their living nature is disputed.** Proposed as the smallest self-replicating bacteria, but later work showed "replication" is mineral nucleation, and that they are **mineralo-protein complexes** (apatite with serum proteins such as fetuin-A). Their bacterial DNA proved to be contamination. Most now regard them as **non-living**.

### Claimed properties

| Property | Claim |
|---|---|
| Size | Pass through bacteria-retaining filters |
| "Growth" | Very slow, in serum-containing medium |
| Shell | Apatite; resistant to heat and gamma irradiation |
| Detection | Electron microscopy, von Kossa and Hoechst staining, specific antibodies, culture |
| Susceptibility | Tetracycline and EDTA (ethylenediaminetetraacetic acid) inhibit them |

### Proposed role in urolithiasis

- **Nidus theory:** CNPs in blood or urine are filtered or excreted, deposit apatite in renal tubular cells and interstitium, and act as a **nucleating nidus** for crystal growth.
- **Link to Randall plaque:** idiopathic calcium oxalate stones grow on **Randall plaque**, an interstitial **apatite** deposit that starts in the basement membrane of the thin loop of Henle and erodes through the papilla. CNPs are **associated with**, but **not proven to cause**, Randall plaque (Campbell, Chapter 95).
- Early studies found apatite nanoparticles in a large majority of stones; rat studies reported renal calcification after injection.
- Also implicated in atherosclerotic calcification, polycystic kidney cysts and chronic prostatitis.

### Therapeutic implications (unproven)

- Tetracycline-based regimens with EDTA were tried in stones and chronic prostatitis/chronic pelvic pain syndrome; no RCT (randomised controlled trial) evidence supports routine use.

### Current position

- Not in stone guidelines; pathogenesis is explained by supersaturation, inhibitors, Randall plaque and ductal plugs.

**Diagram to draw:** renal papilla with interstitial apatite (Randall plaque) eroding through urothelium, CNPs as nidus, and calcium oxalate stone attached.

**Key points to remember:**

- Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
- Proposed nidus for apatite and Randall plaque; association, not causation.
- Tetracycline/EDTA therapy unproven.

## SAQ 12 Tear drop bladder

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 1, Q3a

### Definition

- A **tear-drop (pear-shaped or inverted-pear) bladder** is a bladder that is symmetrically **compressed from both sides** and elongated upwards on cystogram, IVU (intravenous urogram) or CT (computed tomography).
- It is a radiological sign of a **bilateral perivesical mass or collection**, not a disease itself.

### Causes

| Group | Causes |
|---|---|
| **Collections** | **Pelvic haematoma** (pelvic fracture: commonest in trauma); urinoma (extraperitoneal leak); bilateral lymphocele (after lymphadenectomy or transplant); pelvic abscess |
| **Fat** | **Pelvic lipomatosis** |
| **Muscle** | **Iliopsoas hypertrophy** (muscular young men) |
| **Vascular** | Dilated pelvic venous collaterals (IVC [inferior vena cava] occlusion); bilateral iliac artery aneurysms |
| **Nodes and tumour** | Pelvic lymphadenopathy (lymphoma, metastases); pelvic retroperitoneal fibrosis; pelvic tumours |

### Pelvic lipomatosis (the classic cause)

- Overgrowth of benign, non-encapsulated fat in the perivesical and perirectal spaces; mainly in men, more in Black men.
- **Presentation:** LUTS (lower urinary tract symptoms), suprapubic pain, constipation; often incidental.
- **Imaging:**
- Plain film: pelvic lucency. IVU: elevated, elongated pear-shaped bladder; **medial deviation of both ureters**; hydronephrosis.
- Barium enema: straightened, elongated rectosigmoid.
- **CT or MRI (magnetic resonance imaging):** fat density is diagnostic.
- **Cystoscopy:** high bladder neck, difficult passage; often **cystitis glandularis**, which can progress to adenocarcinoma. Biopsy and surveillance are needed.
- **Management:** observe if no obstruction. Upper-tract obstruction → stent or nephrostomy, ureteric reimplantation or urinary diversion. Weight loss rarely helps; surgical excision of fat is difficult.

### Evaluation of a tear-drop bladder

- History: trauma, pelvic surgery or transplant, malignancy.
- **CT** shows the cause (blood, fat, fluid, muscle, nodes, vessels). Ultrasound for collections; aspiration (creatinine for urinoma, lymph for lymphocele).

### Management

- Treat the cause: pelvic haematoma usually observed; lymphocele drained or marsupialised (laparoscopic fenestration); urinoma drained and leak repaired.

**Diagram to draw:** cystogram outline of a bladder squeezed into a tear drop by bilateral pelvic haematoma.

> **Key points to remember:**
> - Tear-drop bladder = bilateral extrinsic compression.
> - Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
> - Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

## SAQ 13 Retrograde endopyelotomy

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 9 Paper 1, Q3f

### Definition and principle

- Endoscopic **full-thickness incision of the UPJ (ureteropelvic junction)** from inside the ureter, through to peripelvic fat, followed by stenting so the ureter heals with a wider lumen (Davis "intubated ureterotomy" concept, 1943).
- **Retrograde ureteroscopic holmium laser endopyelotomy** is now the standard endourological route: direct vision without a percutaneous tract.

### Indications and contraindications

| Favourable | Unfavourable / contraindicated |
|---|---|
| Functionally significant UPJO (UPJ obstruction); short segment | **Long stenotic segment** |
| **Secondary UPJO** (failed pyeloplasty) | **Upper-tract stones** (treat percutaneously instead) |
| Moderate hydronephrosis, fair function | **Crossing vessel**, **gross hydronephrosis**, poor function |
| - Pre-operative CT (computed tomography) angiography or Doppler to look for crossing vessels. | |

### Technique

1. General anaesthesia; retrograde pyelogram; hydrophilic safety wire into the pelvis.
2. Access sheath; flexible ureteroscope to the UPJ (semirigid in some women). Dilate a narrow intramural ureter or pre-stent.
3. Holmium laser, 200-µm fibre, 0.8–1.2 J, 10–15 Hz.
4. **Lateral incision** (no crossing vessels laterally), deepened into fat and extended into normal ureter and pelvis.

5. Optional balloon dilation to 24 Fr; DJ (double-J) stent and Foley catheter.
6. Stent out after some weeks; **diuretic renogram 4 weeks later**; follow up for 24–36 months.

### Results

- Holmium series 70–80% at 5 years (range 44–89%). Excluding crossing vessels >4 mm raises success to about 90%.
- Crossing vessel present versus absent: 42% versus 86% (antegrade series).
- Still below pyeloplasty (about 95%), but most cost-effective endourological option.

### Complications

- Urine leak, stent migration, infection, bleeding (lateral cut spares vessels), stricture (rare with small scopes); ureteroarterial fistula reported.
- **Also know:** antegrade (percutaneous) endopyelotomy for UPJO with stones; cautery wire balloon (more haemorrhage); balloon dilation alone (about 42% success); failure then needs pyeloplasty.

**Diagram to draw:** UPJ with lateral full-thickness incision and crossing vessel lying anterior/posterior.

> **Key points to remember:**
> - Lateral full-thickness incision over a stent.
> - Crossing vessel and gross hydronephrosis predict failure.
> - Success 70–80%, below pyeloplasty (about 95%).

## SAQ 14 Post-obstructive diuresis

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 19 SAQ Sheet, Q3b

### Definition

- POD (post-obstructive diuresis): marked polyuria after relief of obstruction, as the kidney sheds retained salt, water and urea.
- No agreed rate; a common working definition is **>200 mL/h for 2 consecutive hours or >3 L in 24 hours**. Up to 300 mL/h is common.
- **Physiological** (appropriate, self-limited) versus **pathological** (continues after euvolaemia, causing hypovolaemia and electrolyte loss).

### Incidence and setting

- Unilateral obstruction: **about 2%** (the other kidney compensates). Bilateral or solitary kidney: **about 50%**.
- Commonest after catheterising **chronic retention with renal impairment**; also bilateral ureteric obstruction and PUV (posterior urethral valves).

### Mechanisms

1. **Transporter dysregulation:** down-regulation of $Na^+/K^+$-ATPase and aquaporin-2 → concentrating defect (nephrogenic DI (diabetes insipidus)); loss of medullary tonicity.

2. **Volume expansion:** on relief, raised ANP (atrial natriuretic peptide), low aldosterone and sympathetic tone → salt and water loss.
3. **Osmotic diuresis** from retained urea.

### Risk factors

- Bilateral obstruction, raised creatinine, oedema, heart failure, hypertension, confusion.
- After valve ablation: oligohydramnios, preterm birth, raised creatinine.

### Clinical features

- Starts within hours of relief; most settle within 24–48 hours.
- Danger signs: postural hypotension, tachycardia, weight loss, hypernatraemia (free-water loss), low potassium, magnesium, phosphate.

### Management

``` Relieve obstruction (catheter / nephrostomy)

|

Monitor hourly urine output, vitals, weight

|

Alert, can drink -> free oral fluids (most resolve)

|

Output persists / cannot drink / unstable -> admit; electrolytes every 4-6 h -> intravenous 0.45% saline: about 75% of urine output -> replace K, Mg, phosphate; adjust to urine sodium/osmolality ```

- **Do not overcorrect:** replacing every millilitre sustains the diuresis.
- **Gradual bladder decompression has no role**: it does not prevent haematuria or diuresis.
- PCN (percutaneous nephrostomy) for upper-tract obstruction; hydration alone for physiological POD.
- **Also know:** decompression haematuria; neonatal PUV concentrating defect; no approved drug therapy.

**Diagram to draw:** graph of urine output against time after catheterisation.

> **Key points to remember:**
> - Unilateral about 2%, bilateral about 50%.
> - Mechanisms: aquaporin-2 loss, volume expansion, urea osmotic diuresis.
> - Replace about 75% of output with 0.45% saline; do not overcorrect.
> - Gradual decompression has no role.

## SAQ 15 Stone complexity scores

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 20 SAQ Sheet, Q3b
```

### Purpose

- Grade stone and kidney complexity, chiefly to **predict stone-free rate (SFR) after PCNL (percutaneous nephrolithotomy)**. Used for counselling, choosing the surgeon, auditing and comparing series.

### Main scores

| Score | Variables and scoring | Predicts |
|---|---|---|
| **Guy's Stone Score** (Thomas, 2011) | **Grade I:** single stone in mid or lower pole, or pelvis, simple anatomy. **II:** single upper-pole stone, or multiple stones with simple anatomy, or single stone with abnormal anatomy. **III:** multiple stones with abnormal anatomy, calyceal diverticulum stone, or partial staghorn. **IV:** complete staghorn, or any stone in spina bifida or spinal injury | SFR on KUB (kidney, ureter, bladder radiograph); not complications |
| **S.T.O.N.E. nephrolithometry** | **S**ize (area), **T**ract length (≤100 or >100 mm), **O**bstruction (hydronephrosis), **N**umber of involved calyces, **E**ssence (density, ≤950 or >950 HU [Hounsfield units]); total 5–13 | SFR, blood loss, operative time, stay; needs CT (computed tomography) |
| **CROES nomogram** (Clinical Research Office of the Endourological Society) | Stone burden, location, staghorn, number, prior treatment, **annual centre volume** | SFR; only multicentre derivation, but SFR on KUB |
| **S-ReSC** (Seoul National University Renal Stone Complexity) | Number of **9 sites** involved on CT: pelvis, upper and lower major calyces, anterior and posterior minor calyces of the three groups; score 1–9 | SFR; not complications |
| **RUSS** (Resorlu–Unsal Stone Score) | 1 point each: stone >20 mm; lower-pole stone with infundibulopelvic angle <45°; stones in more than one calyx; abnormal anatomy; score 0–4 | SFR after **RIRS** (retrograde intrarenal surgery) |

### Limitations

- Each score added what the previous lacked: anatomy, then CT depth and density, then centre factors, then calyceal distribution.
- Mostly predict clearance, **not complications**. Meta-analysis: heterogeneous; no single generalisable score.
- **Also know:** Triple D score predicts SWL (shock wave lithotripsy) success, not PCNL.

**Diagram to draw:** kidney outline showing the 9 S-ReSC sites.

> **Key points to remember:**
>
> - Guy's: grades I–IV; complete staghorn or spina bifida is grade IV.
> - S.T.O.N.E.: five variables, score 5–13.
> - CROES includes centre volume; RUSS is for RIRS.

## SAQ 16 Calyceal diverticulum – classification and management

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 20 SAQ Sheet, Q3c

## Definition

- Cystic cavity in the renal cortex, **lined by transitional epithelium**, connected to a calyx (or pelvis) by a **narrow neck**. Fills passively with urine.
- About 4.5 per 1000; equal in children and adults; no side or sex preference.

## Classification

| | Type I | Type II |
|---|---|---|
| Frequency | Commoner | Less common |
| Connects to | Minor calyx (usually upper pole) | Renal pelvis or major calyx |
| Size, symptoms | Small, usually silent | Larger, often symptomatic |

## Aetiology

- **Congenital** (favoured): persistent third- and fourth-generation ureteric bud branches.
- **Acquired:** cortical abscess draining into a calyx, infundibular stenosis, trauma, reflux.

## Clinical features

- Stasis: **stones in up to 40%**, milk of calcium, infection, pain, haematuria.

## Evaluation

- Ultrasound mimics a simple cyst. **CT (computed tomography) urography with delayed images shows contrast pooling**; milk of calcium layers and shifts with position.
- Retrograde pyelography maps the neck before intervention.
- **Differential:** hydrocalycosis (dilated major calyx from infundibular obstruction), megacalycosis, simple cyst.

## Management

- **Asymptomatic:** observe with ultrasound.
- **Treat** pain, recurrent infection, haematuria, abscess, symptomatic stones.

| Approach | Technique | Best for |
|---|---|---|
| **Ureteroscopy** | Find the ostium, laser-incise or balloon-dilate the neck, remove stones | Upper or mid-pole, small stones |
| **PCNL** (percutaneous nephrolithotomy) | Direct puncture onto stone, clear stones, fulgurate lining, dilate or create neck | Posterior, lower-pole, larger stones; long-term results good |
| **Laparoscopic or robotic** | Unroof, remove stones, close neck, obliterate cavity, pack with fat | Anterior, thin-parenchyma, failed endoscopy |
| Partial nephrectomy | Excision | Rare; destroyed segment |

- Endoscopy keeps the cavity draining by widening the neck; laparoscopy abolishes it by closing the neck.
- **Also know:** SWL (shock wave lithotripsy) gives symptom relief but low stone clearance; fulguration of lining is debated.

**Diagram to draw:** kidney with upper-pole diverticulum, narrow neck to calyx, stone inside.

**Key points to remember:**

- Transitional lined, narrow neck; type I calyceal, type II pelvic.
- Stones in up to 40%; delayed CT shows pooling.
- Ureteroscopy or PCNL first; laparoscopy for anterior or failed cases.

## SAQ 17 Pelvic lipomatosis

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 20 SAQ Sheet, Q3e

### Definition

- Rare benign **overgrowth of non-encapsulated mature fat** in the perivesical and perirectal spaces, compressing the bladder, ureters, rectum and pelvic veins.
- Mostly men aged 20–60; associated with obesity and hypertension. Cause unknown; obesity alone does not explain it (weight loss rarely shrinks the fat). Chronic inflammation and venous obstruction are proposed.

### Clinical features

- LUTS (lower urinary tract symptoms), haematuria, recurrent UTI (urinary tract infection), suprapubic or perineal pain, constipation.
- Leg oedema or venous thrombosis from venous compression.
- **Hydronephrosis and renal failure** from ureteric compression in a substantial minority.
- **Proliferative cystitis (cystitis glandularis)** is common; it can transform to **adenocarcinoma**.

### Evaluation

- **IVU (intravenous urography) or cystogram:** elevated, elongated **"pear-" or "teardrop-shaped" bladder**; medially displaced pelvic ureters; high prostate.
- **Barium enema:** elongated, narrowed rectosigmoid.
- **Ultrasound:** echogenic pelvic fat; hydronephrosis.
- Medial deviation of **both** pelvic ureters also occurs in retroperitoneal fibrosis and after abdominoperineal surgery.
- **CT (computed tomography) or MRI (magnetic resonance imaging): diagnostic**; homogeneous fat density surrounding pelvic organs, separates it from tumour.
- **Cystoscopy:** elevated trigone, long prostatic urethra (rigid scope may be difficult); biopsy for cystitis glandularis.
- **Differential of pear-shaped bladder:** pelvic haematoma, lymphadenopathy, liposarcoma, iliopsoas hypertrophy, venous collaterals, pelvic fibrosis.

### Management

- **Surveillance** (often progressive): ultrasound for hydronephrosis, renal function, periodic cystoscopy for cystitis glandularis.
- Weight loss, antibiotics and steroids give limited or no benefit; COX-2 (cyclo-oxygenase-2) inhibitors for cystitis glandularis are case reports only.
- **Ureteric obstruction:** stents or nephrostomy; ureteric reimplantation (difficult, dense fat).

- **Advanced or failed cases:** urinary diversion; cystectomy if adenocarcinoma or unmanageable bladder.
- Surgical excision of fat is difficult: no tissue planes, heavy bleeding.
- Resect cystitis glandularis transurethrally and treat obstruction or inflammation.

**Diagram to draw:** cystogram outline of pear-shaped bladder with medially deviated ureters.

> **Key points to remember:**
> - Pear-shaped bladder with medial ureters; CT shows fat.
> - Cystitis glandularis common; adenocarcinoma risk.
> - Watch for hydronephrosis; diversion if obstruction progresses.

## SAQ 18 Cystinuria

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 21 SAQ Sheet, Q3a

### Definition and genetics

- **Autosomal recessive** defect of proximal tubular (and jejunal) reabsorption of the dibasic amino acids **COLA: cystine, ornithine, lysine, arginine**. Only cystine is insoluble, so only cystine forms stones.
- Transporter subunits and ICC (International Cystinuria Consortium) types:

| Gene | Subunit | Chromosome | Type |
|---|---|---|---|
| SLC3A1 | rBAT (heavy) | 2p | A |
| SLC7A9 | $b^{0,+}$AT (light) | 19q | B |
| Both | — | 2 and 19 | AB |

- Fractional excretion of cystine rises from about 0.4% to up to 100%. Incidence 1 in 1000 to 1 in 17,000; **1–3% of adult stones, up to 10% of paediatric stones**; males more severe.

### Pathophysiology

- **Pure supersaturation disease**: no specific inhibitor. Solubility is pH dependent: about 300 mg/L at pH 5, 400 mg/L at pH 7, 1000 mg/L at pH 9.
- Coexisting hypocitraturia (commonest), hyperuricosuria or hypercalciuria cause mixed calcium–cystine stones.

### Clinical features

- First stone at mean about 12 years; recurrent, bilateral, staghorn stones; CKD (chronic kidney disease) from repeated surgery (about a quarter of adults have eGFR [estimated glomerular filtration rate] <60).

### Diagnosis

- **Hexagonal crystals** in urine (always abnormal); cyanide–nitroprusside screen; 24-hour urinary cystine (normal <60 mg/day/1.73 $m^2$; homozygotes often >400).
- Stone: yellow, waxy, faintly radiopaque (sulphur); **resists SWL (shock wave lithotripsy)**.

### Management

- **Fluids:** urine output **≥3 L/day**, including at night.
- **Diet:** sodium ≤2300 mg/day; limit animal protein (not in children).
- **Alkalinise to pH 7.0–7.5** with potassium citrate 30–60 mEq/day; avoid >7.5 (calcium phosphate stones) and sodium alkali (raises cystine excretion).
- **Thiols** if cystine stays >250 mg/L: thiol–cystine mixed disulfide is soluble.
- **Tiopronin** (choice): 300 mg twice daily, up to 1200 mg/day.
- **D-penicillamine:** 250 mg/day, titrated; rash, proteinuria, nephrotic syndrome, pancytopenia.
- Captopril: weak evidence.
- Monitor blood count, liver tests, urine protein.
- **Surgery:** ureteroscopy or PCNL (percutaneous nephrolithotomy) with laser; nephron-sparing approach.

**Diagram to draw:** proximal tubule cell with rBAT/$b^{0,+}$AT transporter and hexagonal crystal.

> **Key points to remember:**
> - Autosomal recessive; SLC3A1 (type A), SLC7A9 (type B).
> - Hexagonal crystals; target cystine <250 mg/L.
> - Urine ≥3 L, pH 7.0–7.5, then tiopronin.

## SAQ 19 Trans Uretero Ureteric anastomoses

P2

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 2, Q3a

### Definition

- **TUU (transureteroureterostomy)**: the injured (donor) ureter is brought across the midline and joined **end-to-side** to the opposite (recipient) ureter. First clinical use in 1934.

### Indications

- Lower ureteric loss that reimplantation, psoas hitch or Boari flap cannot bridge: small, scarred or irradiated bladder; long ureteric loss; pelvic phlegmon; associated rectal, vascular or extensive bladder injury.
- One ureter short while the other is reimplanted with a psoas hitch (hitch only one side).
- Undiversion and reconstruction in children (Hendren).
- Usually a **delayed** repair, not acute.

### Contraindications

| Absolute | Relative (disease that may involve both ureters) |
|---|---|
| Donor ureter too short to reach without tension | Stone disease, **urothelial carcinoma** |
| Diseased recipient ureter | Retroperitoneal fibrosis, pelvic radiation |

| Absolute | Relative (disease that may involve both ureters) |
|---|---|
| | Chronic pyelonephritis, genitourinary tuberculosis |
| - Recipient reflux must be corrected at the same time, so do a preoperative VCUG (voiding cystourethrogram). Less favoured in children (long life, both kidneys at risk). | |

### Technique (steps)

1. Midline transperitoneal incision; reflect the colon.
2. Mobilise the **donor ureter with its adventitia** (blood supply); divide it just above the diseased segment.
3. Expose about 5 cm of recipient ureter with minimal mobilisation.
4. Tunnel the donor ureter **behind the sigmoid mesentery, above the IMA (inferior mesenteric artery)** to avoid kinking.
5. **Anteromedial ureterotomy** of the recipient (at least 2 cm); **spatulated, tension-free end-to-side** anastomosis with fine absorbable interrupted sutures.
6. **Double-J stent** from donor kidney to bladder; drain. (Laparoscopic and robotic routes are feasible.)

### Complications

- Urine leak (commonest), urinoma, stricture or obstruction, kinking under the IMA.
- **Main risk: one unilateral problem becomes bilateral obstruction.**
- Later endoscopic access is difficult; use the antegrade route.

### Results

- Success 90–97% in adults; complications (mostly leak) in about one-quarter of complex cases.

**Diagram to draw:** donor ureter crossing above the IMA to an end-to-side anastomosis, with stent.

> **Key points to remember:**
> - End-to-side, spatulated, tension-free; cross above the IMA.
> - Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
> - Risk: unilateral injury becomes bilateral.

## SAQ 20 Medical expulsive therapy (MET) in urinary calculi

P2

**Asked as:**

- Medical expulsive therapy in urinary calculi. — *Summer 2021 Paper II Q3d (SAQ)*
- Medical expulsive Rx for urinary calculi. — *Practice Set 11 Paper II Q3e (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2021-Summer Paper 2, Q3d)

## Definition

- MET (medical expulsive therapy): drugs that help a ureteric stone pass into the bladder. **α-blockers** are the main agents.

## Rationale: natural history

- Spontaneous passage depends on **width**, size and site:

| Width | Spontaneous passage | Location | Spontaneous passage |
|---|---|---|---|
| 0–3 mm | 98% | Vesicoureteric junction (small stones) | 92% |
| 5 mm | 65% | Distal ureter | 75% |
| >6.5 mm | 9% | Proximal ureter | 48% |

## Mechanism

- **α1 receptors** (mainly **α1A and α1D**) are abundant in (dominate in) the distal ureter and VUJ (vesicoureteric junction). Blockade reduces spasm and the force and frequency of peristalsis (peristaltic force and frequency), so a larger urine bolus pushes the stone / builds higher pressure above the stone.
- Others: calcium-channel blockers (nifedipine); PDE5 (phosphodiesterase-5) inhibitors under study. **NSAIDs (non-steroidal anti-inflammatory drugs) relieve pain but do not aid passage.**

## Drugs

- **Tamsulosin 0.4 mg daily** (most studied); **silodosin 8 mg daily** (better expulsion in meta-analyses but twice the retrograde ejaculation); alfuzosin, doxazosin.

## Evidence

| Study | Result |
|---|---|
| **SUSPEND (Pickard, 2015):** 1136 patients; tamsulosin, nifedipine or placebo | No difference in need for intervention at 4 weeks (19% vs 20%) |
| **STONE (Meltzer, 2018):** stones <9 mm (up to 9 mm) | Tamsulosin no better than placebo |
| Meta-analyses and a large distal-stone RCT (randomised controlled trial) | Benefit for **distal stones >5 mm** |

- Stratified, the trials agree: benefit lies in **distal stones of 5–10 mm** (smaller ones pass anyway; larger rarely pass).

## Indications and guideline position (AUA [American Urological Association] and EAU [European Association of Urology])

- AUA and EAU: offer MET for **ureteric stones <10 mm**, especially distal, when immediate intervention is not needed (EAU: distal >5 mm).
- **After SWL (shock wave lithotripsy)** for faster fragment clearance (to clear fragments). Children: off-label but effective.

## Contraindications (need urgent drainage instead; drain urgently)

- Infection with obstruction or sepsis, solitary or bilateral obstruction, renal deterioration (impairment), refractory pain or vomiting.

### Protocol

- NSAID analgesia; counsel on off-label use, dizziness, hypotension, retrograde ejaculation.
- Periodic KUB (kidney–ureter–bladder radiograph) or ultrasound follow-up; **intervene after 4–6 weeks** if not passed.

**Diagram to draw:** ureter with α1 receptors concentrated distally (at the distal ureter and VUJ [vesicoureteric junction]) and a stone above it (above the vesicoureteric junction).

> **Key points to remember:**
> - Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
> - SUSPEND 2015: no benefit in an unselected population.
> - Trial of passage 4–6 weeks, then intervene.

## SAQ 21 Physiology of UPJ in relation to urine transport

P2

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 2, Q3f

*UPJ = ureteropelvic (pelviureteric) junction.*

### Anatomy relevant to function

- The UPJ is the apex of a funnel-shaped pelvis; its spiral smooth muscle is continuous with calyces and ureter.

### Pacemaker and electrical activity

- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves carry pain and only modulate.
- **Primary pacemaker: atypical smooth muscle cells** at the pelvicalyceal border; they fire spontaneously (HCN, hyperpolarisation-activated cyclic nucleotide-gated, channels).
- **ICC-like (interstitial cells of Cajal-like) cells** (c-Kit positive) in the pelvis and UPJ relay the signal to typical smooth muscle and act as back-up pacemakers.
- Signal spreads through **gap junctions**. Calcium entry gives a plateau action potential; calcium–calmodulin activates myosin light-chain kinase, causing contraction.

### Bolus formation and transport

1. Urine fills the pelvis at low pressure; the UPJ is closed.
2. Pelvic contraction pushes urine through the opening UPJ into the upper ureter.
3. The UPJ closes behind it, forming a **bolus**.
4. Ureteric walls **coapt** behind the bolus; the wave (2–6 per minute, 20–80 cm $H_2O$) carries it to the bladder. - At **low flow**, the UPJ acts as a filter: several pelvic contractions occur for each ureteric contraction. - As **flow rises**, ureteric frequency reaches 1:1, then bolus volume rises. - At **very high flow**, boluses merge into a continuous column and the ureter dilates **without obstruction**.

### Effect of obstruction

- **Acute:** stronger, more frequent contractions.
- **Chronic:** dilatation, walls fail to coapt, peristalsis becomes ineffective; raised pelvic pressure lowers GFR (glomerular filtration rate).

- **Intrinsic UPJO (UPJ obstruction):** abnormal muscle, excess collagen, fewer ICC-like cells, so an **aperistaltic segment** cannot transmit the wave.
- **Also know:** endotoxin and progesterone reduce peristalsis. Diuretic renogram and Whitaker test (chapter: pressure difference >12 cm $H_2O$ suggests obstruction; classic teaching: <15 unobstructed, >22 obstructed) assess transport.

**Diagram to draw:** pelvis, UPJ and ureter showing bolus formation and coaptation.

> **Key points to remember:**
> - Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
> - Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
> - Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.

## SAQ 22 Complications of PCNL, including bleeding and persistent haematuria after PCNL

P2

**Asked as:**

- Complications of PCNL — *Practice Set 2 Paper II Q3e (SAQ)*
- What are the causes of bleeding in PCNL? How will you manage persistent hematuria 6 days after PCNL!? — *Practice Set 6 Paper II Q3b (SAQ)*

**Type:** SAQ (10 marks)
(mini-PCNL complications in 2022-Summer Paper 1, Q1) (PCNL complications also in Practice Set 2 Paper 2, Q3e)

### Introduction

- PCNL (percutaneous nephrolithotomy) is the standard for renal stones >2 cm. Most complications arise from **access and tract dilatation**.
- Graded by the **Clavien–Dindo** system (I: no intervention; II: drugs or transfusion; III: intervention; IV: ICU [intensive care unit]; V: death).
- Bleeding is the commonest serious complication of PCNL. Transfusion was historically ~7%, now nearer 1% with smaller tracts. Angioembolisation is needed in <1%.

### Incidence (systematic review, Seitz 2012, quoted by EAU [European Association of Urology])

| Complication | Rate |
|---|---|
| Fever | 10.8% |
| Transfusion | 7% (nearer 1% with modern and mini tracts) |
| Thoracic | 1.5% |
| Sepsis | 0.5% |
| Organ injury | 0.4% |
| Angioembolisation | 0.4% |
| Death | 0.05% |

## Main complications

| Complication | Cause | Prevention and management |
|---|---|---|
| **Acute bleeding** | Infundibular or segmental vessel injury | Fornix puncture; place sheath, tamponade with balloon or **Kaye catheter**, clamp nephrostomy; transfuse |
| **Delayed bleeding** | Pseudoaneurysm, AV (arteriovenous) fistula | **Selective angioembolisation** |
| **Infection, SIRS (systemic inflammatory response syndrome), sepsis** | Infected stone, high intrarenal pressure | Sterile urine, culture-specific antibiotics, low-pressure irrigation; stop and drain if pus |
| **Pleural** (pneumothorax, hydrothorax) | Supracostal access, highest above 11th rib | Subcostal where possible, puncture in expiration; chest drain |
| **Colon** | Retrorenal colon, left lower pole | CT review; extraperitoneal: double-J stent and withdraw nephrostomy into colon (colostomy tube) |
| **Spleen, liver** | Upper-pole access, organomegaly | Ultrasound guidance; conservative or surgery |
| **Collecting system** | Perforation, extravasation | Stop, drain; later infundibular stenosis |
| **Fluid absorption** | Long operation, perforation | **Normal saline**; hyponatraemia, overload |

**Also know:** venous air embolism, DVT (deep vein thrombosis) and pulmonary embolism, neuropraxia and eye injury (prone), residual fragments, loss of renal function (multiple tracts), urinoma, nephrocutaneous fistula, retained hardware.

## Risk factors

- Staghorn, multiple tracts, supracostal access, large tract, long operation, CKD (chronic kidney disease), diabetes, positive urine culture, inexperience.
- **Bleeding risk factors:** infundibular or upper-pole puncture, multiple or large tracts, staghorn stone, long operation, antithrombotic drugs, coagulopathy, diabetes.

## Reducing risk

- **Tranexamic acid** reduces transfusion (Cochrane).
- Ultrasound-guided access, mini-PCNL, ECIRS (endoscopic combined intrarenal surgery).
- **Bleeding prevention:** puncture through the **calyceal fornix** along a posterior calyx (Sampaio: no arterial injury); correct clotting; mini-tracts.

## Bleeding in PCNL

### Causes

| Timing | Cause |
|---|---|
| **Intraoperative** | Parenchymal or venous tract bleeding; infundibular artery injury; through-and-through puncture; posterior segmental artery injury (upper-pole access); torn infundibulum or pelvis |
| **Early postoperative** | Tract bleeding after sheath removal; clot retention |

| Timing | Cause |
|---|---|
| **Delayed (about 1 week)** | **Pseudoaneurysm**, **AV (arteriovenous) fistula** |

**Mechanism of delayed bleeding**

- **Pseudoaneurysm:** clot seals an arterial tear, then ruptures intermittently.
- **AV fistula:** artery and vein heal together; the thin vein cannot withstand arterial pressure.

**Management at day 6 (persistent haematuria)**

- At day 6, assume an **arterial lesion** until proved otherwise. `Resuscitate: IV access, crossmatch, transfuse; Hb, clotting → Stop anticoagulants; catheter, irrigate clots → Stable, mild: bed rest, clamp nephrostomy, serial Hb → Persistent or recurrent bleeding or Hb fall → CT angiography (arterial blush) or direct angiography → SUPERSELECTIVE ANGIOEMBOLISATION (coils) → Fails or unstable after embolisation → open exploration, partial or total nephrectomy`
- **Angiography** diagnoses >90% and treats in the same sitting. It may miss intermittent bleeds; non-selective embolisation loses renal tissue.
- Clot colic or obstruction: double-J stent or nephrostomy.
- Follow-up: Hb and creatinine; check for infection and abscess.

**Intraoperative ladder (for completeness)**

- Pressure on tract → nephrostomy clamped (tamponade) → balloon reinflation or **Kaye tamponade catheter** → angioembolisation.

**Diagram to draw:** kidney cross-section with posterior calyx, Brödel line, pleura and retrorenal colon; angiogram sketch: pseudoaneurysm before and after coiling.

> **Key points to remember:**
> - Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
> - Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
> - Early bleeding venous: tamponade.
> - Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
> - Treatment: superselective angioembolisation.
> - Supracostal access increases pleural injury; colon injury at left lower pole.

## SAQ 23 Steinstresse.

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 7 Paper 2, Q3e

*Assumed meaning: steinstrasse.*

### Definition

- **Steinstrasse** (German, "stone street"): a column of stone fragments in the ureter that fails to pass and may obstruct. Typically after **SWL (shock wave lithotripsy)**; occasionally after PCNL (percutaneous nephrolithotomy) or URS (ureteroscopy).
- Incidence is low overall (about 4–7% after SWL, EAU [European Association of Urology]) but rises several-fold with stone burden **>2 cm²**.

### Risk factors

- Large burden (>2 cm), renal pelvic stones, hard stones that break into large fragments, high energy, impaired ureteric drainage.

### Classification (Coptcoat)

| Type | Composition |
|---|---|
| I | Small fragments only |
| II | Larger lead fragment (4–5 mm) followed by small fragments |
| III | Large fragments throughout |

### Clinical features

- Asymptomatic (silent obstruction), colic, nausea; fever and sepsis if infected; anuria and AKI (acute kidney injury) in a solitary kidney. Most lie in the distal ureter.

### Evaluation

- KUB (kidney, ureter, bladder) X-ray, ultrasound, or non-contrast CT (computed tomography); creatinine, urine culture, blood counts.

### Management

```
Steinstrasse ├─ infection / sepsis → urgent PCN (percutaneous nephrostomy) + antibiotics ├─ asymptomatic, no obstruction → MET + analgesia, imaging every 1–2 weeks └─ failure, pain, obstruction: ├─ lead fragment (type II/III) → SWL to lead fragment └─ URS with laser lithotripsy (most definitive)
```

- **MET (medical expulsive therapy):** α-blockers (tamsulosin) improve fragment clearance and reduce pain.
- **PCN** is preferred in infected obstruction: drains without manipulating the ureter, and fragments often pass once the kidney is decompressed. A stent is an alternative.
- Retained fragments with poor progress: URS clears the column.

### Prevention

- Choose PCNL or RIRS (retrograde intrarenal surgery) for large burdens.
- Energy ramping, slower shock rate; MET after SWL.
- **Stent before SWL:** may reduce steinstrasse with large stones but does not improve stone-free rate and adds stent symptoms. The chapters disagree: Chapter 97 says it may reduce steinstrasse; Chapter 98 advises against stenting at SWL.

**Diagram to draw:** ureter with lead fragment and trailing column (Coptcoat types I–III).

> **Key points to remember:**
> - Risk rises several-fold with burden >2 cm².
> - Coptcoat I–III by fragment size.
> - Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

## SAQ 24 Treatment of inferior calyceal calculus

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 2, Q3b

## Introduction

- Lower pole is the **commonest site** of renal stones. Gravity-dependent position makes fragments hard to clear, so **SWL (shock wave lithotripsy) works poorly** here.

## Factors in decision

- **Stone:** size, density (<900 HU [Hounsfield units] favours SWL), composition.
- **Anatomy unfavourable for SWL clearance:** acute **infundibulopelvic angle**, long and narrow infundibulum, skin-to-stone distance >10 cm.
- **Patient:** obesity, anticoagulants (favour URS [ureteroscopy]), pregnancy, preference.
- Evaluation: NCCT (non-contrast CT), urine culture, renal function.

## Treatment by size

| Size | Options |
|---|---|
| **Asymptomatic, small** | Observation acceptable; about half have an event in 5 years. Treat if growing, symptomatic, infected or occupational |
| **<10 mm** | SWL or **RIRS** (retrograde intrarenal surgery). AUA (American Urological Association): either |
| **10–20 mm** | EAU (European Association of Urology): SWL only if favourable anatomy; otherwise **RIRS or PCNL** (percutaneous nephrolithotomy) |
| **>10 mm (AUA)** | **PCNL** (highest SFR [stone-free rate]) or URS; SWL not first-line |
| **>20 mm** | PCNL first; staged RIRS an alternative |

## Evidence

- **Lower Pole I (Albala 2001):** stones >1 cm: SFR 91% PCNL vs 21% SWL.
- Lower Pole II: stones ≤1 cm, URS vs SWL: no significant SFR difference.

## Technique points

- **RIRS:** flexible ureteroscope (deflection up to 270°); relocate the stone to an upper calyx with a basket before laser (holmium or thulium fibre) lithotripsy; access sheath.
- **PCNL:** direct lower pole posterior calyx puncture; mini-PCNL reduces bleeding (transfusion near 1%).
- **SWL adjuncts:** percussion, diuresis, inversion therapy and alpha-blockers improve fragment clearance.
- **Also know:** lower pole calyceal diverticulum or infundibular stenosis favours PCNL; residual fragments >4 mm after URS need treatment.

## Follow-up

- Imaging for residual fragments; metabolic evaluation if recurrent.

**Diagram to draw:** infundibulopelvic angle and infundibular length/width on an IVU (intravenous urogram) tracing.

> **Key points to remember:**
> - Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
> - <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.
> - Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
> - Reposition stone during RIRS.

## SAQ 25 Primary Hyperoxaluria

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 10 Paper 2, Q3f

### Definition

- Rare **autosomal recessive** errors of **hepatic glyoxylate metabolism**: glyoxylate is not converted to glycine and is oxidised to oxalate, an end product cleared only by the kidney.
- Hyperoxaluria = urinary oxalate >40 mg/day; in PH (primary hyperoxaluria) often **>100 mg/day**.

### Types

| | PH1 (~70%) | PH2 (~10%) | PH3 (~10%) |
|---|---|---|---|
| Enzyme (gene) | **AGT** (alanine-glyoxylate aminotransferase; **AGXT**), liver peroxisome, **pyridoxine (B6) cofactor** | GRHPR (glyoxylate reductase/hydroxypyruvate reductase) | HOGA1 (4-hydroxy-2-oxoglutarate aldolase), mitochondria |
| Urine marker | **Glycolate** | **L-glycerate** | 4-hydroxy-2-oxoglutarate |
| Course | **Most severe**; ESKD (end-stage kidney disease) in 2nd–3rd decade | Milder, can reach ESKD | Childhood stones; rarely ESKD |

### Pathophysiology

- High urinary oxalate → calcium oxalate crystals in tubules → **nephrocalcinosis** and recurrent stones.
- **Vicious circle:** falling GFR (glomerular filtration rate) reduces oxalate clearance → plasma oxalate rises → **systemic oxalosis** (bone, retina, heart, vessels, skin).

### Clinical features

- Spectrum from **infantile oxalosis with renal failure** to occasional adult stones; some diagnosed only after a transplant fails.
- Clues: childhood stones, nephrocalcinosis, family history, **calcium oxalate monohydrate** stones.

### Diagnosis

`24-h urinary oxalate ↑ (child: spot oxalate:creatinine ratio) → urinary glycolate / L-glycerate / 4-hydroxy-2-oxoglutarate → genetic testing (AGXT, GRHPR, HOGA1) all normal → secondary (enteric, dietary) hyperoxaluria`

- Plasma oxalate when GFR is low; screen siblings and related living donors.

### Management

- **Hyperhydration** (about 3 L/m²/day), **potassium citrate**, normal calcium intake.
- **Pyridoxine** trial in all PH1: responsive mainly with the Gly170Arg mutation; it restores AGT activity and peroxisomal targeting.
- **Lumasiran** (RNA interference against glycolate oxidase, which cuts glyoxylate supply): **ILLUMINATE-A**: 84% vs 0% reached urinary oxalate ≤1.5 × upper limit of normal. **Also know:** nedosiran.
- Stones: ureteroscopy or PCNL (percutaneous nephrolithotomy); limit SWL (shock wave lithotripsy).
- **ESKD:** intensive dialysis, then **combined liver–kidney transplant for PH1** (only the liver replaces the enzyme; a lone kidney would be re-damaged). Kidney alone usually suffices for PH2.

**Diagram to draw:** hepatic glyoxylate pathway showing the AGT block (PH1) and the lumasiran target (glycolate oxidase).

> **Key points to remember:**
> - PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.
> - Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
> - Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.

## SAQ 26 Fracture penis

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 15 Paper 2, Q3c

### Definition and mechanism

- **Penile fracture** is a tear of the **tunica albuginea** with rupture of the corpus cavernosum, in the **erect** penis.
- Abnormal bending of the erect penis causes an abrupt rise in intracavernosal pressure beyond tunical strength → **transverse tear**.
- The tunica (inner circular, outer longitudinal layers) is **thinnest ventrolaterally**, so most tears are ventral or lateral, proximal to mid shaft.
- **Causes:** intercourse (commonest; rear-entry), masturbation, rolling over; self-inflicted "taqaandan" (Middle East); collagenase injection for Peyronie disease.
- **Bilateral tears in about 10%**; urethral injury about 20% in the West vs about 3% in Asia and the Middle East.

### Clinical features

- **Cracking or popping sound → pain → rapid detumescence → swelling and discoloration.**
- **"Eggplant deformity"** when Buck's fascia is intact; if torn, haematoma spreads to scrotum, perineum and suprapubic area.
- Penis deviates **away from the tear**; tunical defect may be palpable.
- **Urethral injury:** blood at meatus, haematuria, retention.

### Investigations

- **Diagnosis is clinical** (AUA [American Urological Association] Urotrauma Guideline).
- **Ultrasound** for equivocal cases (localises the tear); **MRI** (magnetic resonance imaging) if still equivocal (100% negative predictive value). Cavernosography is discouraged.
- Urethra: **intraoperative flexible cystoscopy** (AUA and EAU [European Association of Urology] preference) or RUG (retrograde urethrogram).
- **"False fracture":** dorsal vein or artery rupture.

### Management

- **Immediate surgical repair** (less curvature, ED [erectile dysfunction] and nodules than conservative care). 1. Incision: ventral penoscrotal or lateral over the defect; subcoronal degloving if site unclear (preputial ischaemia risk). 2. Evacuate haematoma; find the tear (artificial erection helps). 3. Close the tunica with **interrupted 2-0/3-0 absorbable sutures**; avoid debriding erectile tissue. 4. Urethra: partial → close over catheter; complete → tension-free repair; catheter 2–3 weeks.
- Abstinence about 1 month. Delay up to 7 days does not worsen outcome. After collagenase injection without urethral injury, conservative care is acceptable.

### Complications

- About 20%: nodule, curvature, ED, painful erection, infection, stricture or fistula.

**Diagram to draw:** cross-section of penis showing tunical tear, Buck's fascia and haematoma spread.

> **Key points to remember:**
> - Snap, detumescence, eggplant deformity.
> - Urethral injury about 20% (West); suspect if bilateral.
> - Immediate exploration and interrupted absorbable tunical closure.

## SAQ 27 Ovarian vein syndromes

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 15 Paper 2, Q3e

### Definition

- **Ovarian vein syndrome** (Clark, 1964) is extrinsic compression of the **right ureter** by a dilated or thrombosed **right ovarian vein**, causing intermittent hydroureteronephrosis. It is a **controversial** entity; many cases are now attributed to other causes.
- The plural also covers ovarian vein thrombosis and pelvic congestion.

### Anatomy (basis)

- Ovarian veins run in the infundibulopelvic ligaments. The **right** enters the IVC (inferior vena cava) obliquely; the **left** enters the left renal vein at a right angle.
- Gonadal vessels **cross anterior to the ureter**; ovarian veins dilate greatly in pregnancy.

### Types

| Syndrome | Mechanism | Features |
|---|---|---|
| **Ovarian vein (ureteric) syndrome** | Dilated right ovarian vein compresses the ureter at the crossing | Multiparous women; right flank pain (premenstrual or in pregnancy); recurrent UTI (urinary tract infection) |
| **Ovarian vein thrombosis** | Postpartum hypercoagulability and venous stasis; mostly right-sided | Fever and right lower-quadrant or flank pain after delivery; may obstruct ureter; risk of pulmonary embolism |
| **Pelvic congestion syndrome** | Incompetent, refluxing ovarian veins with pelvic varices; sometimes secondary to nutcracker (left renal vein compression) | Chronic pelvic pain worse on standing, dyspareunia, post-coital ache |

### Controversy

- Right-sided dilatation in pregnancy is mostly **physiological hydronephrosis** (progesterone plus uterine compression; stops at the pelvic brim). Venous compression is rarely proven, so diagnosis is by **exclusion**.

### Investigations

- Ultrasound; **CT urography** or IVU (intravenous urography): right hydroureter to a smooth oblique extrinsic indentation, normal ureter below.
- **Diuretic renogram** to prove obstruction; retrograde pyelography.
- CT or MR (magnetic resonance) venography, Doppler; **gonadal venography** (gold standard for reflux).
- **Exclude:** stone, retrocaval ureter, iliac vessel crossing, retroperitoneal fibrosis, endometriosis.

### Management

- **Asymptomatic:** observe. **Obstruction:** stent or nephrostomy if infected or pregnant; definitive **ureterolysis with ligation and division of the ovarian vein** (open or laparoscopic).
- **Thrombosis:** anticoagulation and antibiotics.
- **Pelvic congestion:** progestogens or GnRH (gonadotropin-releasing hormone) analogues; **coil or sclerotherapy embolisation**; ligation.

**Diagram to draw:** right ovarian vein crossing anterior to the right ureter, with proximal hydroureteronephrosis.

> **Key points to remember:**
> - Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
> - Postpartum ovarian vein thrombosis: right side, anticoagulate.
> - Treatment: ureterolysis with vein division, or embolisation.

## SAQ 28 Post traumatic bladder neck injury: acute and delayed management?

P3

**Type:** SAQ (10 marks)
**Source:** 2021-Summer Paper 3, Q3c

### Introduction

- The bladder neck is the **internal sphincter**. It is torn mainly in **pelvic fracture**, often with prostatic urethral, rectal or vaginal injury; **commoner in children**.
- Missed injury causes **incontinence, fistula and stricture**.

### Diagnosis

- Pelvic fracture with gross haematuria, retention or blood at the meatus; **vaginal examination** in females.
- **RUG (retrograde urethrogram)**, then **retrograde CT or plain cystography** (350 mL).
- Grading: AAST (American Association for the Surgery of Trauma) bladder **grade V** = laceration into the bladder neck or ureteric orifice; Goldman type IV = bladder rupture extending into the proximal urethra.

### Acute management

- Resuscitate; if unstable, damage control (drainage), repair later.
- **Bladder neck injury needs immediate repair** (AUA (American Urological Association) and EAU (European Association of Urology)), unlike isolated posterior urethral distraction (suprapubic tube, delayed repair). `Lower midline incision → anterior cystotomy → identify ureteric orifices (stent if near) → minimal debridement; re-approximate neck to prostatic urethra, layered absorbable sutures over a 22 Fr catheter → rectal or vaginal tear: separate layers, no overlapping suture lines, interpose omentum (colostomy for rectum) → remove bone spicules; drain ± suprapubic tube`
- Catheter 2–3 weeks; cystogram before removal.

### Delayed management (3–6 months)

| Problem | Management |
|---|---|
| Associated urethral distraction | Cystogram + RUG + endoscopy; **anastomotic urethroplasty first**, not combined with bladder neck surgery |
| Bladder neck contracture | Endoscopic incision or resection; open Y-V plasty if refractory |
| **Incontinence** (both sphincters damaged) | Urodynamics; **AUS (artificial urinary sphincter)**; bladder neck reconstruction (Young–Dees–Leadbetter) or fascial sling; last resort **bladder neck closure + Mitrofanoff** ± augmentation |
| Fistula | Layered repair with tissue interposition |
| - Bladder neck appearance on imaging **does not predict** continence; restore continuity first, then treat incontinence. | |
| - Continence is poor whenever the bladder neck is involved (paediatric series: incontinence up to 20% overall). | |

**Diagram to draw:** pelvic fracture with bladder neck tear and repair over a catheter.

**Key points to remember:**

- Bladder neck injury = immediate repair.
- Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
- Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

## SAQ 29 Boari flap

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 3, Q3d

### Definition

- A **tubularised flap of anterior bladder wall**, based posterolaterally on the side of the lesion, swung up to bridge a distal or mid-ureteric gap. Described in dogs 1894, humans 1947; laparoscopic 2001, robotic 2008.

### Reconstructive ladder (distal ureter)

| Option | Gap bridged |
|---|---|
| Ureteroneocystostomy | Distal 4–5 cm |
| + Psoas hitch | Adds up to 5 cm (total 6–10 cm) |
| **Boari flap (+ hitch)** | **10–15 cm**; spiral flap may reach pelvis (right) |
| Downward nephropexy | Adds up to 5–8 cm |
| - Beyond: TUU (transureteroureterostomy), ileal ureter, autotransplantation. | |

### Indications

- Long distal or mid-ureteric loss: gynaecological injury (commonest), stricture, failed reimplantation, distal TCC (transitional cell carcinoma) ureter, tuberculosis.

### Contraindications

- **Small, contracted or poorly compliant bladder** (radiation, tuberculosis); uncorrected outlet obstruction or neurogenic bladder. Not usually for acute unstable trauma.
- Preoperative: urodynamics or cystogram for capacity; antegrade and retrograde studies for length.

### Technique

1. Mobilise bladder; divide contralateral superior vesical pedicle if needed.
2. Mark flap on the **non-distended** bladder: **base ≥4 cm, apex ≥3 cm, length:base ≤3:1**; length = gap + 3 cm if tunnelled.
3. Raise flap; **psoas hitch** its base (sutures superficial, parallel to fibres: spare genitofemoral and femoral nerves).
4. Anastomose spatulated ureter: **non-refluxing** 3 cm submucosal tunnel, or **refluxing** direct (apex about 3 times ureteric diameter).
5. Stent (double-J); tubularise in two layers 2-0 absorbable; drain, catheter 7–14 days. - Laparoscopic and robotic: transperitoneal, refluxing anastomosis, results excellent.

### Complications

- **Recurrent stricture** (commonest; ischaemia or tension), reduced capacity with frequency and urgency, leak, nerve palsy, iliac vessel injury, pseudodiverticulum.
- eGFR (estimated glomerular filtration rate) fell in 44% within 2 years after reimplantation: counsel.
- **Alternative:** bladder advancement flap, similar reach and simpler.

**Diagram to draw:** bladder outline with oblique flap (4 cm base, 3 cm apex), hitched and tubularised to ureter.

> **Key points to remember:**
> - Bridges 10–15 cm; combine with psoas hitch.
> - Base ≥4 cm, length:base ≤3:1, measure bladder empty.
> - Contraindicated in small contracted bladder.

## SAQ 30 Struvite stones.

P4

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 4, Q3a

### Definition and epidemiology

- **Infection stones:** magnesium ammonium phosphate hexahydrate ($MgNH_4PO_4 \cdot 6H_2O$), usually mixed with **carbonate apatite**.
- 5–15% of stones (falling); female:male ~2:1; highest in India and Pakistan. Risk: neurogenic bladder, spinal injury, diversion, catheters, obstruction, diabetes, elderly.

### Pathogenesis

```
Urease-producing bacteria Urea + H2O -> 2NH3 + CO2 NH3 -> NH4+ (urine pH > 7.2) CO2 -> carbonate; phosphate -> PO4(3-) + Mg -> struvite; + Ca -> carbonate apatite
```

- Sterile urine has no urease, so infection is essential. Ammonia damages the GAG (glycosaminoglycan) layer; bacteria live inside the stone in a **biofilm**, so antibiotics cannot sterilise a retained stone.
- **Organisms: Proteus mirabilis** (commonest), Klebsiella, Pseudomonas, Providencia, Staphylococcus, **Ureaplasma** (easily missed). E. coli is rarely urease-positive.

### Clinical features

- Recurrent UTI (urinary tract infection), fever, flank pain, haematuria; often few symptoms. Grow fast into **staghorn** calculi. Untreated: pyonephrosis, XGP (xanthogranulomatous pyelonephritis), sepsis, renal loss.

### Evaluation

- Urine pH >7; culture (urine and stone cultures often differ). NCCT (non-contrast computed tomography): burden and anatomy; DMSA (dimercaptosuccinic acid) or renogram for split function. Stone analysis; 24-h urine (many have hypercalciuria or hypocitraturia).

### Management

- **Complete stone clearance** is the goal: each fragment is an infected nidus.
- Culture-specific antibiotics before surgery; drain an obstructed infected kidney first.
- **PCNL (percutaneous nephrolithotomy): first line** for staghorns (AUA, American Urological Association). Add flexible nephroscopy, ECIRS (endoscopic combined intrarenal surgery) or second look. SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery) for small stones only.
- Anatrophic nephrolithotomy rarely; nephrectomy for a poorly functioning kidney.
- **Prevention:** surveillance cultures, suppressive antibiotics, fluids, correct stasis or metabolic defects.
- **AHA (acetohydroxamic acid)** 250 mg three times daily (urease inhibitor), only when surgery is exhausted. Side effects in up to 30% (haemolytic anaemia, DVT (deep vein thrombosis)); avoid if creatinine >2.5 mg/dL or pregnancy.
- **Also know:** hemiacidrin irrigation (historical); matrix stones (radiolucent, Proteus).

**Diagram to draw:** urease cascade; staghorn with PCNL tract.

> **Key points to remember:**
> - Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
> - Proteus mirabilis commonest; Ureaplasma easily missed.
> - Complete clearance with PCNL; AHA only when surgery is not possible.

## SAQ 31 Role of open and laparoscopic stone surgeries in modern era of endourology.

P4

**Type:** SAQ (10 marks)
**Source:** 2021-Summer Paper 4, Q3f

### Introduction

- SWL (shock wave lithotripsy), URS (ureteroscopy) and PCNL (percutaneous nephrolithotomy) treat almost all stones. Open surgery is now a small minority in centres with full endourology, but remains in resource-limited settings.
- **AUA (American Urological Association)/Endourological Society 2016:** consider laparoscopic, robotic or open surgery **only when SWL, URS and PCNL fail or are expected to fail**; not first line.

### Indications

| Category | Examples |
|---|---|
| Endourology failed or unlikely | Very complex staghorn needing many PCNL sessions; failed multiple procedures |
| Anatomical problem treated together | **PUJ (pelviureteric junction) obstruction + stone:** pyeloplasty with pyelolithotomy; anterior calyceal diverticulum; infundibular stenosis |
| Difficult access | Pelvic or ectopic kidney (laparoscopy-assisted PCNL); morbid obesity; skeletal deformity |

| Category | Examples |
|---|---|
| Poor renal function | **Nephrectomy** for a non-functioning kidney; partial nephrectomy for a non-functioning pole |
| Large impacted ureteric stone | Laparoscopic ureterolithotomy after failed URS/SWL |
| Concomitant surgery | Other open or laparoscopic procedure |
| Bladder | Very large stone; with open prostatectomy, diverticulectomy or augmentation; small bladder; urethral stricture |

### Procedures

- **Pyelolithotomy:** simple, extended (Gil-Vernet, sinus dissection) or coagulum.
- **Anatrophic nephrolithotomy (Smith and Boyce):** for complex staghorn. Clamp renal artery; cool with ice slush; give mannitol; incise along **Brödel's avascular line**; remove stone; repair infundibula; close.
- Radial nephrotomy, ureterolithotomy, cystolithotomy.

### Laparoscopic and robotic surgery

- Transperitoneal or retroperitoneal pyelolithotomy and ureterolithotomy; robotic anatrophic nephrolithotomy reported.
- Best for large pelvic stone in an extrarenal pelvis: high single-stage clearance, less bleeding than PCNL. Drawbacks: longer operation, calyceal stones need a flexible scope.

| Feature | Open | Laparoscopic/robotic |
|---|---|---|
| Morbidity | Large incision, pain, long stay | Less pain, shorter stay |
| Kidney | Ischaemic loss of function (anatrophic) | Similar repair, less wound morbidity |
| Later surgery | Adhesions make redo difficult | Fewer |

### Complications

- Bleeding, urinary leak, infection, loss of renal function, residual fragments.

**Diagram to draw:** Brödel's line incision for anatrophic nephrolithotomy.

> **Key points to remember:**
> - Open/laparoscopic only when endourology fails or is unsuitable.
> - Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
> - Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.

## SAQ 32 Diet in stone disease

P4

**Asked as:**

- Diet in stone disease. — *Winter 2023 Paper IV Q3f (SAQ)*
- Diet in stone disease. — *Practice Set 5 Paper IV Q3f (earlier version) (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3f)

## Principle

- Diet lowers urinary **supersaturation** by **dilution**, by reducing promoters (calcium, oxalate, uric acid, sodium) and by raising inhibitors (citrate).
- General advice for all; recurrent formers tailored to **24-hour urine**.

## General measures

| Measure | Target | Mechanism and evidence |
|---|---|---|
| **Fluid** | Urine **≥2.5 L/day** (about 3 L intake), spread through the day | Most important single factor. **Borghi 1996 RCT (randomised controlled trial):** recurrence 12.1% vs 27% at 5 years |
| **Normal calcium** | 1000–1200 mg/day, with meals | Binds oxalate in the gut. **Restriction increases stones**. **Borghi 2002:** normal-calcium, low-salt, low-protein diet halved stone events vs low-calcium diet |
| **Low sodium** | Salt about 4–5 g/day | Sodium increases urinary calcium, lowers citrate |
| **Limit animal protein** | 0.8–1.0 g/kg/day | Acid and purine load: hypercalciuria, hypocitraturia, hyperuricosuria |
| **Fruit and vegetables** (DASH [Dietary Approaches to Stop Hypertension] diet) | Daily | Alkali and citrate; DASH linked to fewer stones |
| **Weight** | BMI (body mass index) 18–25 | Obesity: low urine pH, more oxalate and uric acid |

- **Citrus:** lemonade (most evidence) and orange juice raise citrate; orange juice adds alkali. **Volume matters more than drink type.**
- Avoid sugar-sweetened drinks and **high-dose vitamin C** (converted to oxalate).

## Stone-specific diet

| Stone or abnormality | Diet |
|---|---|
| **Hyperoxaluria** | Avoid large portions of **spinach, beets, chocolate, nuts, tea**; calcium with meals |
| **Enteric hyperoxaluria** | Strict low oxalate and **low fat**; **calcium up to 1 g/day with meals**; fluid; magnesium |
| **Uric acid** | Less animal protein (purines give about 30% of urinary uric acid); fruit and vegetables; weight loss |
| **Cystine** | Fluid **3–5 L/day** including night; sodium **≤2300 mg/day**; less animal protein (not in children) |
| **Struvite** | Infection control; diet has little role |

## Adherence

- Poor: only half reach 2.5 L at 6 months; apps and smart bottles help.

**Diagram to draw:** "stone-prevention plate": water, fruit and vegetables, normal dairy, little meat and salt.

**Key points to remember:**

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

## SAQ 33 ECIRS (endoscopic combined intrarenal surgery)

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 4, Q3b

### Definition

- **ECIRS:** PCNL (percutaneous nephrolithotomy) and retrograde flexible ureteroscopy performed **at the same time, in one position**, by two surgeons.
- Term and technique popularised by **Scoffone (2008)** in the **GMSV (Galdakao-modified supine Valdivia) position**. Combined access dates back to the 1980s (Lawson's retrograde needle placement, 1983).

### Rationale

- Each route covers the other's blind spots: the retrograde scope guides and watches the puncture, and reaches calyces the rigid nephroscope cannot.
- Result: one tract, better clearance.

### Positioning

| Position | Features |
|---|---|
| **GMSV** (preferred) | Supine; flank on a bolster at the table edge; ipsilateral leg extended, contralateral leg in lithotomy |
| Bart's modified supine | Variant of supine |
| Prone split-leg | Easier posterior calyces; needs a split-leg table |

- **Supine advantages:** airway access, gravity drainage of fragments, no turning, simultaneous retrograde access.

### Technique

1. Cystoscopy; UAS (ureteral access sheath) placed; flexible ureteroscope maps the stones and chooses the target calyx.
2. Ultrasound or fluoroscopic puncture **watched endoscopically** (double bull's-eye technique).
3. Wire grasped and brought out: through-and-through access.
4. Tract dilated under vision (balloon) and sheath placed.
5. Antegrade lithotripsy and suction; the retrograde scope **relocates stones** from inaccessible calyces into the nephroscope's view, or lasers them.
6. Final check with both scopes; stent or tubeless exit. - Mini-ECIRS uses a mini-PCNL tract.

### Indications

- Staghorn and complex multi-calyceal stones, combined renal and ureteric stones, anomalous kidneys, failed previous treatment, urinary diversion, obese or cardiopulmonary patients (supine).

### Advantages (meta-analysis)

- Single position, **fewer tracts, less bleeding**, fewer ancillary procedures, **higher stone-free rate**; less fluoroscopy; the UAS keeps intrarenal pressure low.

### Disadvantages

- Two surgeons and two sets of equipment; cost; longer set-up; ureteric injury from the sheath; less working space in supine.

**Diagram to draw:** GMSV position with nephroscope from the flank and flexible ureteroscope from below in the same calyx.

> **Key points to remember:**
> - Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
> - Retrograde scope guides the puncture and relocates stones.
> - Fewer tracts, less bleeding, higher stone-free rate.

# Rapid Revision Sheet · Section 11

### Congenital PUJ obstruction, failed pyeloplasty and salvage procedures (LAQ)

- Intrinsic adynamic segment (infants) vs crossing vessel (older children, intermittent pain).
- SFU 4 = parenchymal thinning; UTD P3 = abnormal parenchyma.
- Operate for DRF <40%, fall >10%, rising APD, UTI or symptoms.
- Anderson–Hynes ~95%; never divide a crossing vessel.
- Causes of failure: ischaemia, tension, missed crossing vessel, leak.
- Failed open repair: endourology first; redo pyeloplasty most reliable (about 83–89%); endopyelotomy about 70%.
- Ureterocalycostomy for small scarred pelvis: resect lower-pole parenchyma; about 70% in redo series. Buccal onlay emerging.
- Split function <15% with normal other kidney: nephrectomy.

### Urolithiasis: etiopathogenesis, medical management and prevention of recurrence (LAQ)

- Metastable zone between Ksp and Kf is where inhibitors and therapy act.
- Randall plaque = interstitial apatite anchor for idiopathic calcium oxalate stones; ductal plugs suggest a secondary cause.
- Urine ≥2.5 L/day; normal calcium, low salt and animal protein (Borghi); Borghi 1996: 12% vs 27% recurrence; MET for distal 5–10 mm stones.
- Thiazide (plus potassium citrate) for hypercalciuria; potassium citrate 30–60 mEq/day for hypocitraturia; NOSTONE 2023 negative.
- Uric acid pH 6.0–6.5; cystine pH 7.0–7.5 and <250 mg/L.
- RIRS: single-use digital scopes, thulium laser, suction sheaths, pressure sensing.

### Metabolic evaluation in stone disease (LAQ)

- Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: ammonium chloride ($NH_4Cl$) fails to lower pH below 5.5; cystine <250 mg/L at pH 7.0–7.5.
- Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months of therapy.
- NCCT gold standard (97%/95%).
- Staghorn: PCNL first line; complete clearance is the goal.
- Bilateral: drain an infected side first, then the better kidney.
- AHA only when surgery is not possible.

### Retroperitoneal fibrosis: diagnosis and (medical) management (LAQ)

- Idiopathic 70% (periaortitis); methysergide and lymphoma classic secondary causes.
- Medially deviated ureters; PET-CT best shows activity; biopsy deep.
- Drain first, then prednisone (about 80% respond).
- Vaglio 2011: prednisone better than tamoxifen (relapse 17% vs 50%).
- Rituximab for IgG4-related or refractory disease.
- About 15% need bilateral ureterolysis with intraperitonealisation or omental wrap.

**Lower urinary tract trauma in pelvic fracture and PFUD injuries: early and late management (LAQ)**

- Blood at meatus: RUG before catheter.
- Gross haematuria + pelvic fracture = retrograde CT cystogram (350 mL).
- EPR: catheter 2–3 weeks; IPR: immediate repair.
- PFUI: SPC is the standard initial step, then delayed perineal anastomotic urethroplasty at 2–3 months (P3: after 3 months).
- Primary realignment: brief, endoscopic; no difference in final urethroplasty rates.
- Progressive perineal steps: bulbar mobilisation, crural separation, pubectomy, rerouting.
- Immediate repair: bladder neck or rectal injury, penile fracture urethra, low-velocity penetrating injury.
- ED mostly from the injury; incontinence <4%.

**Renal trauma (renal parenchymal and vascular injuries, including grade IV): diagnosis and management (LAQ)**

- AAST 2018: segmental vessel injury grade IV; main vessel grade V.
- Grade IV = collecting system, UPJ, segmental vessel, bleeding beyond Gerota's, infarction.
- Image if gross haematuria, microhaematuria with SBP <90, deceleration or penetrating injury.
- Delayed CT phase is essential; blush means AE.
- Stable patients, even grade IV–V: non-operative ± AE (angioembolisation).
- Most stable grade IV heal non-operatively; extravasation resolves in up to 90%.
- Revascularise only a solitary kidney or bilateral injuries.

**ESWL: mechanism, indications, factors affecting outcome, complications and current status (LAQ)**

- Spallation, squeezing, shear, cavitation; dynamic squeezing strongest.
- Size is the main predictor; HU >1000 and SSD >10 cm predict failure.
- Lower Pole I: SFR 21% SWL vs 91% PCNL for lower pole stones >1 cm.
- Slow rate (60–90 shocks/min), ramping and good coupling improve breakage and reduce injury.
- Steinstrasse risk rises with burden >2 $cm^2$; no routine stent.
- Use is declining as URS grows; burst wave lithotripsy is coming.

**Management of upper ureteric stones (LAQ)**

- Proximal stones pass in only 48%; MET does not help proximal stones.
- <10 mm: SWL equals URS; >10 mm: URS first (EAU).
- Flexible URS chases migrated fragments; laser ≤10 W in the ureter.
- Large impacted: antegrade PCNL or laparoscopic ureterolithotomy; avulsion risk highest in proximal third.

**Uric acid calculi: etiopathogenesis, medical management and allopurinol (LAQ)**

- Low urine pH is the main cause; pKa 5.5; no inhibitor exists.
- Insulin resistance reduces ammoniagenesis, so urine is acid.
- Potassium citrate to pH 6.0–6.5 (dissolution 7.0–7.2); avoid >7.0.
- Allopurinol only for hyperuricosuria or failed alkalinisation.

- Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
- Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
- Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

**Urinary crystals (LAQ)**

- Cystine hexagons are always pathological.
- Acidic: uric acid, cystine; alkaline: struvite, phosphates; calcium oxalate at any pH.
- Indinavir: crystalluria in two-thirds, stones 4–13%, radiolucent.
- Refrigerated urine precipitates crystals: examine fresh within 2 hours.

**Discuss various ureteric stents. Discuss ureteroscopy with or without stenting. (LAQ)**

- Finney 1978 double-J; symptoms from intravesical stent and reflux.
- Encrusted stent: bladder coil, ureter, then kidney.
- Denstedt 2001: no routine stent after uncomplicated URS.

**Laparoscopic pyeloplasty. (LAQ)**

- Anderson–Hynes dismembered pyeloplasty is gold standard by any route; laparoscopic success over 90%.
- Transpose over a crossing vessel; never divide it.
- Double-J 4–6 weeks; MAG3 follow-up; up to 30% of failures are late.
- Robotic equals laparoscopic in outcome, with easier suturing.

**Ureteric injury: iatrogenic (gynaecological) and upper ureteric injury (LAQ)**

- Distal ureter (under the uterine artery is the classic site) commonest iatrogenic site; most injuries present late.
- Proximal ureter commonest site of external injury; haematuria absent in 25–45%.
- AAST: 50% circumference (II vs III) and 2 cm devascularisation (IV vs V).
- CT with 10-minute delays; retrograde pyelogram is most sensitive and allows stenting.
- Late: stent first; early repair (repair) ≤7 days, otherwise delayed after 6 weeks (or after 6 weeks).
- Lower ureter: psoas hitch 95–100%; Boari flap (bridges) up to 12 cm.
- Stable: immediate stented ureteroureterostomy; unstable: damage control; bowel and autotransplant only delayed.

**Energy sources for breaking stones in urology (LAQ)**

- SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
- Holmium 2120 nm, photothermal, breaks every composition.
- TFL 1940 nm: low peak power, finer dust, less retropulsion.
- EHL is poorly focused, so it is not first-line in the ureter.

**Flexible ureteroscopy / RIRS and accessories (SAQ)**

- 3.6 Fr channel; ≥270° deflection.
- 12/14 Fr access sheath lowers pressure but severely injures the ureter in about 1 in 8.
- Laser power ≤10 W ureter, ≤25 W kidney.
- RIRS: stones <2 cm; first choice when anticoagulants cannot be stopped.
- Sepsis about 5%, stricture about 3%.

- Hydrophilic wire passes; PTFE wire stays.
- Nitinol tipless basket for calyces.

**Renal tubular acidosis (distal and type 2) (SAQ)**

- Type 1 (distal): urine pH >6 with acidosis, hypokalaemia, hypocitraturia (key), positive UAG; calcium phosphate stones in up to 70%.
- $NH_4Cl$ load: pH fails to fall below 5.5 in distal RTA.
- Potassium citrate is the treatment for distal RTA.
- Type 2 (proximal): bicarbonate wasting; $HCO_3^-$ 14–18 mEq/L (P2: 15–18); urine pH <5.5 at steady state; usually Fanconi (ifosfamide, cisplatin).
- Fractional $HCO_3^-$ excretion >15%; stones uncommon in type 2.
- Type 4: hyperkalaemia, low urine pH; obstructive uropathy.
- Positive UAG = renal cause.

**PCNL patient positioning: prone vs supine (SAQ)**

- Prone: posterior and upper pole access, large working area.
- Supine: airway safe, ECIRS, gravity drainage, no turning; Galdakao-modified supine gives one position for ECIRS.
- Horizontal bolsters drop the kidney more.
- Similar outcomes; choose by anatomy and experience.

**Renal autotransplantation (SAQ)**

- For long ureteric loss when other repairs fail and the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for complex vascular or tumour cases.
- Iliac vessels; pyelovesicostomy or ureteroneocystostomy.
- Unit loss 5–10%; success about 90%.

**Stones in pregnancy (SAQ)**

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change about every 4 weeks (P3: every 4–6 weeks).

**Ureterocalycostomy (SAQ)**

- Salvage for failed pyeloplasty, intrarenal pelvis, horseshoe kidney.
- Requires a dilated lower calyx and lower-pole resection, not simple nephrotomy.
- Stricture is the main failure; success about 70% in redo series.

**Tubeless PCNL (SAQ)**

- Tubeless = stent only; totally tubeless = neither.
- Benefit is mainly over large (>16 Fr) tubes.
- Keep a tube for bleeding, perforation, infection or second look.

**Mini-PCNL and newer PCNL techniques (SAQ)**

- Definition by sheath size: standard 24–34 Fr; mini-PCNL 15–22 Fr; ultra-mini 11–14 Fr; micro-perc <10 Fr; Jackman 1998: first mini-perc (13 Fr).

- Main advantage: less bleeding and fewer transfusions, shorter stay, with stone-free rates non-inferior to standard PCNL.
- Main limitation: longer operating time and higher intrarenal pressure (fever, sepsis); the advantage disappears against a 24 Fr tract.
- Working channel decides instruments; suction lowers pressure.

**Newer methods of stone analysis (SAQ)**

- FTIR (Fourier transform infrared spectroscopy) and XRD (X-ray diffraction) are the reference methods; wet chemistry is obsolete.
- DECT (dual-energy computed tomography) separates uric acid stones in vivo, mainly uric acid versus calcium stones.
- Composition guides dissolution, SWL choice and prevention.
- Stone analysis guides prevention: uric acid points to a metabolic cause, cystine to a genetic one, struvite to infection.

**Radionuclide studies and diuretic renography in PUJ obstruction (SAQ)**

- MAG3 is the tracer of choice; hydrate and catheterise.
- Well-tempered: hydrate, catheter, standard diuretic dose and timing.
- Furosemide 0.5 mg/kg (adult maximum 40 mg); 1 mg/kg in children.
- DRF normal 45–55%; T½ <10 normal (not obstructed); 10–20 indeterminate; >20 min obstructed.
- Operate (surgery) for DRF <40%, fall >10% (falling DRF), or poor drainage with symptoms.

**What are nanobacteria? Describe their role in urolithiasis. (SAQ)**

- Nanobacteria = calcifying nanoparticles; apatite shell; living nature disputed.
- Proposed nidus for apatite and Randall plaque; association, not causation.
- Tetracycline/EDTA therapy unproven.

**Tear drop bladder (SAQ)**

- Tear-drop bladder = bilateral extrinsic compression.
- Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
- Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

**Retrograde endopyelotomy (SAQ)**

- Lateral full-thickness incision over a stent.
- Crossing vessel and gross hydronephrosis predict failure.
- Success 70–80%, below pyeloplasty (about 95%).

**Post-obstructive diuresis (SAQ)**

- Unilateral about 2%, bilateral about 50%.
- Mechanisms: aquaporin-2 loss, volume expansion, urea osmotic diuresis.
- Replace about 75% of output with 0.45% saline; do not overcorrect.
- Gradual decompression has no role.

**Stone complexity scores (SAQ)**

- Guy's: grades I–IV; complete staghorn or spina bifida is grade IV.
- S.T.O.N.E.: five variables, score 5–13.
- CROES includes centre volume; RUSS is for RIRS.

**Calyceal diverticulum – classification and management (SAQ)**

- Transitional lined, narrow neck; type I calyceal, type II pelvic.
- Stones in up to 40%; delayed CT shows pooling.
- Ureteroscopy or PCNL first; laparoscopy for anterior or failed cases.

**Pelvic lipomatosis (SAQ)**

- Pear-shaped bladder with medial ureters; CT shows fat.
- Cystitis glandularis common; adenocarcinoma risk.
- Watch for hydronephrosis; diversion if obstruction progresses.

**Cystinuria (SAQ)**

- Autosomal recessive; SLC3A1 (type A), SLC7A9 (type B).
- Hexagonal crystals; target cystine <250 mg/L.
- Urine ≥3 L, pH 7.0–7.5, then tiopronin. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Trans Uretero Ureteric anastomoses (SAQ)**

- End-to-side, spatulated, tension-free; cross above the IMA.
- Avoid with stones, urothelial cancer, retroperitoneal fibrosis, radiation, tuberculosis.
- Risk: unilateral injury becomes bilateral.

**Medical expulsive therapy (MET) in urinary calculi (SAQ)**

- Tamsulosin 0.4 mg; best for distal stones 5–10 mm.
- SUSPEND 2015: no benefit in an unselected population.
- Trial of passage 4–6 weeks, then intervene.

**Physiology of UPJ in relation to urine transport (SAQ)**

- Myogenic peristalsis from atypical smooth muscle pacemakers; ICC-like cells back up.
- Low flow: several pelvic beats per ureteric beat; very high flow: dilatation without obstruction.
- Intrinsic UPJO: aperistaltic segment with fewer ICC-like cells.

**Complications of PCNL, including bleeding and persistent haematuria after PCNL (SAQ)**

- Fever 10.8%, transfusion 7%, thoracic 1.5%, death 0.05%.
- Fornix puncture avoids arterial injury; delayed bleeding = pseudoaneurysm, embolise.
- Early bleeding venous: tamponade.
- Bleeding at about 1 week arterial: pseudoaneurysm or AV fistula.
- Treatment: superselective angioembolisation.
- Supracostal access increases pleural injury; colon injury at left lower pole.

**Steinstresse. (SAQ)**

- Risk rises several-fold with burden >2 $cm^2$.
- Coptcoat I–III by fragment size.
- Infected: PCN; uncomplicated: MET; failed: SWL to lead fragment or URS.

**Treatment of inferior calyceal calculus (SAQ)**

- Lower Pole I: PCNL 91% vs SWL 21% for >1 cm.
- <1 cm: SWL or RIRS; >1 cm: PCNL or RIRS.

- Unfavourable: acute angle, long narrow infundibulum, >900 HU, SSD >10 cm.
- Reposition stone during RIRS.

**Primary Hyperoxaluria (SAQ)**

- PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.
- Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
- Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.

**Fracture penis (SAQ)**

- Snap, detumescence, eggplant deformity.
- Urethral injury about 20% (West); suspect if bilateral.
- Immediate exploration and interrupted absorbable tunical closure.

**Ovarian vein syndromes (SAQ)**

- Right ovarian vein compresses right ureter; controversial, diagnosis of exclusion.
- Postpartum ovarian vein thrombosis: right side, anticoagulate.
- Treatment: ureterolysis with vein division, or embolisation.

**Post traumatic bladder neck injury: acute and delayed management? (SAQ)**

- Bladder neck injury = immediate repair.
- Posterior urethral distraction alone = suprapubic tube and delayed urethroplasty.
- Late incontinence: AUS, bladder neck reconstruction, or closure + Mitrofanoff.

**Boari flap (SAQ)**

- Bridges 10–15 cm; combine with psoas hitch.
- Base ≥4 cm, length:base ≤3:1, measure bladder empty.
- Contraindicated in small contracted bladder.

**Struvite stones. (SAQ)**

- Urease → ammonia → pH >7.2 → struvite + carbonate apatite.
- Proteus mirabilis commonest; Ureaplasma easily missed.
- Complete clearance with PCNL; AHA only when surgery is not possible.

**Role of open and laparoscopic stone surgeries in modern era of endourology. (SAQ)**

- Open/laparoscopic only when endourology fails or is unsuitable.
- Laparoscopic pyeloplasty + pyelolithotomy for PUJ obstruction with stone.
- Anatrophic nephrolithotomy: Brödel's line, cold ischaemia.

**Diet in stone disease (SAQ)**

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

**ECIRS (endoscopic combined intrarenal surgery) (SAQ)**

- Simultaneous PCNL plus flexible ureteroscopy in one position (Scoffone, GMSV).
- Retrograde scope guides the puncture and relocates stones.
- Fewer tracts, less bleeding, higher stone-free rate. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 12 · Bladder Cancer & Diversion

## Long answer questions (LAQ)

### LAQ 1 Intestinal segments in urology and orthotopic neobladder

P1 · P2 **merged from 2**

*Asked as:* P2 LAQ "Use of intestinal segments in urology" / "Various uses of intestines in Uro" / "Principles of uses of bowel segments in urology. Describe and compare techniques of orthotopic neobladder." (also P2 SAQ "Neobladder") · P1 SAQ "Instestinal segments in urology"

#### Introduction

- Bowel is used to divert urine (conduit, continent pouch, neobladder), store urine, carry it to the skin, enlarge the bladder (augmentation), replace the ureter, make catheterisable channels and line a neovagina.
- Choice depends on renal function, previous surgery or irradiation, bowel disease, mesenteric length and the reconstruction needed.

#### Uses at a glance

| Use | Segment and operation |
|---|---|
| **Incontinent diversion** | **Ileal conduit** (10–15 cm ileum, 10–15 cm from the ileocaecal valve); colon conduit (transverse) after pelvic radiation; jejunal conduit only if nothing else (avoid) |
| **Continent cutaneous pouch** | Indiana (right colon, plicated ileum as channel), Mainz I, Kock (ileal nipple valves); with appendix (Mitrofanoff) or tapered ileum channel |
| **Orthotopic neobladder** | Studer (40–44 cm ileum plus afferent limb), Hautmann (~70 cm ileum, W shape); about 50–70 cm detubularised ileum joined to the urethra |
| **Rectal reservoir** | Ureterosigmoidostomy, Mainz II: now rare (acidosis, cancer) |
| **Augmentation cystoplasty** | "Clam" ileocystoplasty (20–40 cm ileum; ~25 cm ileum), sigmoid, caecum; gastrocystoplasty in renal failure. For refractory neurogenic or small contracted bladder; needs CIC (clean intermittent catheterisation) |
| **Ureteric replacement** | **Ileal ureter**, isoperistaltic ileum, for long defects; Yang–Monti tubes |
| **Catheterisable channels** | **Mitrofanoff** appendicovesicostomy; **Yang–Monti** ileal tube (1–2 cm ileum retubularised transversely) when the appendix is absent; MACE (Malone antegrade continence enema) for bowel |
| **Neovagina** | **Sigmoid** preferred (wide lumen, mobile mesentery, less stenosis); ileum if pelvis irradiated; MRKH (Mayer–Rokitansky–Küster–Hauser), cloaca, gender-affirming vaginoplasty |

- Ileal ureter contraindications: creatinine >2 mg/dL, bladder outlet obstruction, inflammatory bowel disease or radiation enteritis.

## Choice of segment

| Segment | Electrolyte disturbance | Advantages / use | Drawbacks |
|---|---|---|---|
| **Stomach** | Hypochloraemic, hypokalaemic metabolic alkalosis | Secretes $H^+$ and $Cl^-$ (acid); less mucus; usable for renal failure, short gut, dense adhesions | **Haematuria–dysuria syndrome**, ulcers |
| **Jejunum** | Hyponatraemic, hypochloraemic, **hyperkalaemic acidosis** | Outside radiation field | Severe salt loss; avoid |
| **Ileum** | **Hyperchloraemic metabolic acidosis** | Default: mobile, compliant, constant (reliable) blood supply; default conduit, ureteric replacement | B12 and bile-salt loss; obstruction ~10% |
| **Colon** | Hyperchloraemic metabolic acidosis, hypokalaemia | Easy antireflux tunnel; transverse colon outside (spared by) the pelvic radiation field | Diarrhoea if ileocaecal valve used; obstruction ~5% |
| Appendix | — | Mitrofanoff catheterisable channel | — |

## Technical principles of using bowel

- **Preparation:** ERAS (enhanced recovery after surgery); routine mechanical bowel preparation not needed. Assess renal function. Mark the stoma site before surgery when a stoma is possible, through the rectus.
- **Mesentery:** transilluminate, preserve a major mesenteric arcade, wide base; clear ≤8 cm of mesentery from a segment end; avoid watershed areas (Griffiths point at splenic flexure, Sudeck point at rectosigmoid).
- **Bowel anastomosis:** well-vascularised, no tension, no spillage, close the mesenteric window.
- **Detubularisation** (Kock): opening the bowel on its antimesenteric border and refolding it **abolishes coordinated (peristaltic) contractions** and gives more volume for the same surface area; a **sphere** gives the largest volume for its surface area; by Laplace's law (pressure ∝ wall tension/radius) a larger radius means lower pressure, so pressure stays low.

## Ureterointestinal anastomoses

| Technique | Type | Stricture |
|---|---|---|
| **Bricker** | Refluxing, end to side | 4–22% (mean 6%) |
| **Wallace** | Refluxing, ureters joined end to end | ~3% (lowest) |
| Leadbetter–Clarke, Goodwin | Nonrefluxing (antireflux) tunnel in colon | 8–14% |

- Principles: spatulate, preserve periureteric tissue, tension-free, mucosa to mucosa, soft stents (leak 3–5% without). Bring the bowel to the ureter, not the ureter to the bowel.
- Refluxing (Bricker, Wallace; stricture about 3–6%) versus antireflux tunnels (more strictures).
- Fix and retroperitonealise the anastomosis. The left ureter passes behind the sigmoid mesentery; avoid angulation under the IMA (inferior mesenteric artery), a common stricture site.
- Reflux with normal unobstructed ureters has not been shown to harm the adult kidney; antireflux tunnels stricture more. Obstruction, not low-pressure reflux, damages kidneys.

### Stomas

- **Protruding (nipple, "rosebud") stoma** for appliance wearers: less stenosis, better appliance fit. **Flush stoma** for continent pouches that are catheterised.
- **Loop-end (Turnbull) stoma** for obese patients with a short mesentery. Site is marked before surgery, through the rectus muscle.

### Selecting conduit versus continent diversion

- **Renal function needed for continent diversion:** creatinine <2 mg/dL or GFR (glomerular filtration rate) >35 mL/min; urine pH ≤5.8 after ammonium chloride load; urine osmolality ≥600 mOsm/kg after water deprivation; minimal proteinuria; normal liver function; ability to self-catheterise.
- **Contraindications to continent diversion:** poor renal or hepatic function, inability or unwillingness to self-catheterise, inflammatory bowel disease or short gut. A neobladder also needs a tumour-free urethra and a competent sphincter.
- A conduit is simplest and fastest; continent options improve body image but carry more reoperations and metabolic problems.

### Complications of bowel in the urinary tract

**Metabolic** (worse with longer segment, longer contact time, poorer renal function)

- **Hyperchloraemic acidosis (ileum, colon):** $NH_4^+$ (ammonium) replaces $Na^+$ on the $Na^+/H^+$ exchanger, coupled with $Cl^-/HCO_3^-$ exchange, so ammonium chloride is absorbed. Treat with sodium bicarbonate or citrate (potassium citrate if hypokalaemic); chlorpromazine or nicotinic acid limit chloride transport.
- **Hypokalaemia:** worse with colon; replace potassium before correcting acidosis (paralysis risk).
- **Jejunal syndrome:** treat with sodium chloride and thiazide. **Gastric alkalosis:** H2 blockers or PPI (proton-pump inhibitor).
- **Altered sensorium:** hyperammonaemia (drain the pouch/diversion, lactulose), hypomagnesaemia.
- **Drug reabsorption:** phenytoin, methotrexate; drain pouches during chemotherapy.
- **Osteomalacia:** acidosis, vitamin D resistance, calcium loss; growth retardation in children.
- **Nutrition:** B12 deficiency (give parenteral B12 after 5–7 years of ileal diversion), bile-salt diarrhoea, short-gut effects.

**Other**

- Infection (bacteriuria in about three-quarters of conduits; treat only symptoms or urea-splitting organisms, or pure Proteus or Pseudomonas).
- Mucus.
- **Stones:** colon conduit 3–4%, ileal conduit 10–12%, continent (caecal) reservoirs ~20% (mucus, staples).
- **Augment perforation (pouch rupture):** surgical emergency; regular CIC prevents it.
- Ureterointestinal stricture (lifelong risk), renal deterioration, bowel obstruction, stomal stenosis and parastomal hernia.
- **Secondary malignancy:** ureterosigmoidostomy 6–29% after 10–20 years (endoscopy from 5 years); augments (endoscopy from 10 years). Neovagina: stenosis, mucus discharge, adenocarcinoma reported.

## Orthotopic neobladder: definition and principles

- An **orthotopic** detubularised bowel (usually ileal) reservoir joined to the native urethra after radical cystectomy, when the urethra is tumour-free, to allow **volitional voiding** per urethra without a stoma. 80–90% of men and about 75% of women are medically eligible.
- **Principles:** 1. **Reliable outlet:** intact rhabdosphincter (pudendal nerve), healthy, tumour-free urethra. 2. **Low pressure:** complete **detubularisation** abolishes coordinated contractions; folding into a **sphere** gives most volume for surface area. 3. **Adequate capacity:** ≥500 mL when mature (44 cm double-folded ileum: <200 mL early, **500–600 mL** by 1 year). 4. Unobstructed upper tracts.

## Neobladder: selection and contraindications

- **Absolute contraindications:** tumour at the **urethral margin** (positive urethral margin on frozen section), urethral stricture, poor renal function (eGFR [estimated glomerular filtration rate] <35–40 mL/min), severe liver disease, inability to self-catheterise.
- **Relative:** prostatic stromal invasion (men); bladder-neck or anterior vaginal tumour (women); poor sphincter; prior radiation (irradiated pelvis); frailty. Age, obesity and CIS (carcinoma in situ) alone are not bars.

## Neobladder techniques compared

| Design | Segment and shape | Ureters | Advantage | Drawback |
|---|---|---|---|---|
| **Studer** | 40–44 cm ileum, double-folded U + afferent limb | Refluxing, isoperistaltic limb | Simple, no staples, short ureters; stricture <3% | Reflux on cystogram (benign) |
| **Hautmann W** | About 70 cm ileum, W | Refluxing (since 1997) | Largest early capacity, early continence | More late retention, more absorption |
| T pouch | 44 cm + 15 cm T limb | Antireflux flap valve | Accepts dilated or short ureters | More reoperations than Studer |
| Abol-Enein | 40 cm ileum, W | Serous-lined tunnel | Durable antireflux | Needs long, non-dilated ureters |
| Camey II | 65 cm ileum, U | Le Duc antireflux tunnel | Early detubularised design | Le Duc strictures 7–29% |
| Hemi-Kock | 61 cm ileum | Intussuscepted nipple | Good early results | Nipple stenosis, staple stones |
| Padua (VIP) | About 42 cm ileum, funnel | Le Duc | Anatomical funnel; robotic | — |
| Mainz III | 10–15 cm caecum + 20–30 cm ileum (ileocaecal) | Submucosal tunnel | Good compliance | Loses ileocaecal valve |
| Sigmoid (Reddy) | 35 cm sigmoid colon, U | Tunnelled | Spares ileum (short bowel) | Poor compliance, urgency |

- **Hautmann vs Studer:** larger initial capacity gives earlier continence but more late retention and absorption.
- **Studer steps (brief):** isolate ileum 15–20 cm proximal to the ileocaecal valve; open distal 40–44 cm antimesenterically; sew into a U and fold into a sphere; stented ureters end-to-side to the afferent limb; dependent point to urethra.

- **USC (University of Southern California) RCT (randomised controlled trial):** Studer vs T pouch, no difference in renal function at 3 years; T pouch needed more secondary procedures.

### Neobladder outcomes

| Outcome | Result |
|---|---|
| Day continence | 80–90% (3–6 months) |
| Night incontinence | 20–50% (no sphincter reflex, mucosal water secretion) |
| Retention needing CIC | Men 4–10%; women 20–30%: men leak, women fail to empty |
| Urethral recurrence (men) | About 7% at 5 years |

### Neobladder complications

- **Early:** urine leak, ileus, infection, mucus plugging.
- **Late: hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, ureteroileal stricture (<3%), stones, pouch rupture, incisional hernia, urethral recurrence.

### Care and follow-up

- Neobladder: mucus irrigation, bicarbonate; voiding training (sit, relax, strain), timed voiding, night alarm. Annual metabolic panel, B12, upper tracts; urethral cytology if CIS.
- Lifelong for all bowel reconstructions: electrolytes, bicarbonate, renal function, B12, upper-tract imaging (ultrasound or CT) for stricture and stones; endoscopy of colonic or augmented segments as above.

### Recent advances

- Robotic intracorporeal diversion (complications similar to open in experienced hands); robotic intracorporeal neobladder comparable to open; tissue-engineered bladder (experimental) and alternatives to bowel (ureterocystoplasty, autoaugmentation) remain limited.
- **Also know:** neobladder uptake is low despite 80–90% of men being eligible.

**Diagram to draw:** ileal conduit (loop) with mesenteric arcade and Wallace anastomosis.

**Diagram to draw:** Studer U-pouch with afferent limb, ureters and urethral anastomosis.

> **Key points to remember:**
> - Ileum and colon: hyperchloraemic acidosis ($NH_4Cl$ absorption).
> - Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
> - Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
> - Wallace has the lowest stricture rate; detubularisation gives low pressure.
> - Sigmoid is the default neovagina; Yang–Monti when the appendix is absent.
> - Detubularise and make a sphere; ≥500 mL; 500–600 mL by 1 year.
> - Neobladder absolute bar: positive urethral margin, stricture, poor renal function.
> - Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
> - Day continence 80–90%, night 20–50%; women retain (20–30%).

## LAQ 2 Bladder-preserving treatment of TCC

P2 · P4 **merged from 2**

*Asked as:* P2 LAQ "Classification of bladder cancer and bladder-preserving treatment of TCC" · P4 LAQ "Discuss bladder preservation surgery for TCC of urinary bladder."

*Read as: classification of bladder cancer, and bladder-preserving treatment of TCC (transitional cell carcinoma, urothelial carcinoma), including TURBT (transurethral resection of bladder tumour), radical TURBT, partial cystectomy and trimodality therapy.*

### Classification of bladder cancer

#### 1. Histological

| Type | Share | Note |
|---|---|---|
| **Urothelial carcinoma** | ~90% | Variants: micropapillary, plasmacytoid, sarcomatoid, nested; squamous or glandular differentiation |
| Squamous cell carcinoma | ~5% | Schistosomiasis, catheter, stones |
| Adenocarcinoma | ~1–2% | Urachal (dome), exstrophy |
| Small cell / others | <1% | Neuroendocrine; sarcoma |

#### 2. Grade (WHO 2004/2016, World Health Organization)

- PUNLMP (papillary urothelial neoplasm of low malignant potential), low-grade and high-grade carcinoma. WHO 1973 (G1–G3) is still reported by some.
- Low-grade (FGFR3 pathway) recurs but rarely progresses. High-grade and CIS (carcinoma in situ; TP53/RB pathway) progress.

#### 3. Stage: TNM 2017

| T | Definition |
|---|---|
| Ta / Tis | Non-invasive papillary / flat CIS |
| T1 | Lamina propria |
| T2a / T2b | Inner / outer half of muscle |
| T3a / T3b | Perivesical fat, microscopic / macroscopic |
| T4a / T4b | Prostatic stroma, uterus, vagina / pelvic or abdominal wall |

- N1 one true-pelvic node; N2 ≥2 pelvic nodes; N3 common iliac; M1a distant nodes; M1b organs.

#### 4. Clinical groups

- **NMIBC (non-muscle-invasive, ~70%)**: Ta, Tis, T1. Risk groups: low (solitary primary low-grade Ta <3 cm), intermediate, high (high-grade, T1 or CIS).
- **MIBC (muscle-invasive)**: T2–T4a. Metastatic.
- **Molecular:** luminal papillary versus basal/squamous subtypes.

### Why bladder preservation (introduction)

- Standard for MIBC (muscle-invasive bladder cancer) is cisplatin NAC (neoadjuvant chemotherapy) plus RC (radical cystectomy) with PLND (pelvic lymph node dissection).

- RC is morbid and patients are often elderly. Bladder preservation suits **(1) unfit patients** and **(2) fit, motivated patients with limited disease**.
- **Why single modalities fail:** they leave either the remaining urothelium or occult nodes untreated (cT2 understaged to pT3 in up to about 40%; occult nodes about 25%).

### Options

| Option | Use / status |
|---|---|
| Radical TURBT alone | Not a primary treatment |
| **Partial cystectomy + PLND** | Highly selected solitary tumour **<3 cm** in mobile bladder (dome), **no CIS**, 1–2 cm margin, frozen sections; **urachal adenocarcinoma** |
| RT (radiotherapy) alone | Inferior; quasi-palliative |
| Chemotherapy alone | Inferior; not recommended |
| **TMT (trimodality therapy)** | **Best-studied, guideline-accepted** |

### TURBT for NMIBC: the standard

- Complete resection with **detrusor muscle in the specimen**; separate base biopsies; cold-cup biopsy of suspicious areas.
- **En bloc resection** (loop or laser) keeps architecture, gives more detrusor and fewer perforations.
- **Repeat TURBT within 6 weeks** for all T1, incomplete resection or no muscle. Residual tumour in ≥40%.
- Then intravesical chemotherapy or BCG (bacillus Calmette–Guérin) by risk group.

### Radical TURBT for MIBC

- Rationale: about 10% are pT0 at cystectomy. Only for tumour limited to superficial muscle, with a negative re-resection.
- About a third later progress to MIBC on long follow-up; selected series report 5-year CSS (cancer-specific survival) about 82%. Not a primary treatment.

### Partial cystectomy

| Ideal candidate | Contraindication |
|---|---|
| Small (**<3 cm**), **solitary** tumour in a mobile part of bladder (dome) | **CIS**, multifocal tumour |
| No CIS, no hydronephrosis | Trigone or bladder neck tumour |
| 1–2 cm margin achievable away from the ureteric orifice and trigone; adequate residual capacity | Small bladder; prostatic urethral involvement |
| **Urachal adenocarcinoma** (treatment of choice: en bloc dome, urachus and umbilicus) | |

#### Technique (steps):

1. NAC (neoadjuvant cisplatin chemotherapy) if cisplatin-eligible.
2. PLND (always).
3. Cystotomy away from tumour.
4. Full-thickness excision with perivesical fat, **frozen-section margins**.

5. Ureteric reimplantation if needed.
6. Watertight closure; irrigation (prevents seeding); drain.
7. Cystogram before catheter removal.

**Outcomes:** 5-year OS (overall survival) 67–70%; intact bladder 65–74%; salvage RC up to a quarter. Late recurrence, so lifelong follow-up.

### Trimodality therapy (TMT)

**Components:** (1) **maximal, visibly complete TURBT**; (2) **RT** to bladder ± pelvic nodes; (3) concurrent **radiosensitising chemotherapy**.

**Selection criteria:**

- cT2 (selected cT3–4a), small solitary tumour.
- **No hydronephrosis, no extensive CIS (no CIS)**, visibly complete TURBT.
- Good bladder function; motivated, compliant patient.
- Only 6–19% of MIBC meet strict criteria.

**Radiosensitising regimens:**

| Regimen | Evidence |
|---|---|
| **Mitomycin C + 5-FU (fluorouracil)** | **BC2001**: 2-year locoregional DFS (disease-free survival) **67% vs 48%** vs RT alone, no excess late toxicity |
| Cisplatin (± 5-FU or paclitaxel) | MGH (Massachusetts General Hospital)/RTOG (Radiation Therapy Oncology Group) protocols |
| Gemcitabine | Cisplatin-ineligible |
| Carbogen + nicotinamide (BCON) | Hypoxia modification |

**Radiation dose and schedules:** 64–66 Gy in 32 fractions or **55 Gy in 20** (non-inferior).
`Split course (MGH/RTOG): TURBT → chemoRT ~40 Gy → cystoscopy + biopsy CR → consolidation to 60–66 Gy Residual tumour → immediate cystectomy Continuous (current): TURBT → chemoRT 64 Gy/32 fr or 55 Gy/20 fr → cystoscopy + biopsy and imaging at 3–4 months ├─ Complete response → surveillance (NMIBC recurrence: TURBT + BCG) └─ Persistent / recurrent MIBC (invasive) → salvage cystectomy (10–30%)`

**Response assessment:** cystoscopy with biopsy of tumour site, cytology, CT (computed tomography)/MRI (magnetic resonance imaging). **CR (complete response)** leads to surveillance. Non-response leads to salvage RC.

**Salvage cystectomy:**

- Needed in **10–30%** for persistent or recurrent muscle-invasive disease; it is the designed safety net.
- NMIBC (non-muscle-invasive) recurrence: TURBT and intravesical BCG.

**Outcomes:**

- 5-year CSS 50–84%, OS 36–74%.
- MGH/RTOG: CR 69%; 5-year DSS (disease-specific survival) 71%; 80% of 5-year survivors keep the bladder.
- Propensity-matched RC vs TMT: equal 5-year metastasis-free survival (about 75%) and similar CSS (81% vs 84%).

- Low late toxicity; good quality of life.

### Comparison with radical cystectomy

| | Radical cystectomy | Bladder preservation (TMT) |
|---|---|---|
| Staging | Pathological | Clinical only |
| Nodes | Removed | Irradiated or not treated |
| Function | Diversion; sexual dysfunction | Native bladder; RT (bowel/bladder) toxicity |
| Follow-up | Upper tract, urethra | **Intensive cystoscopy: 3-monthly for 2 years, to 10 years**, plus CT |
| Evidence | Standard | No completed RCT (SPARE closed for poor accrual) |

### Follow-up after preservation

- Cystoscopy every 3–4 months for 2 years, 6-monthly in years 3–4, then yearly to 10 years.
- CT urogram and chest CT every 3–6 months for 2 years, then yearly to 5 years; cytology.

### Recent advances

- **Immunotherapy + chemoradiation:** PD-1/PD-L1 (programmed death-1 / ligand-1) inhibitors with TMT under trial (e.g. SWOG/NRG S1806 with atezolizumab; KEYNOTE-992 with pembrolizumab); results awaited.
- Hypofractionation 55 Gy/20 non-inferior (BC2001–BCON meta-analysis: same toxicity).
- Bladder sparing after complete response to NAC (± checkpoint inhibitor) with biomarker selection (DNA repair gene mutations, circulating tumour DNA); RETAIN: about half recurred.
- Robotic partial cystectomy; molecular subtypes.
- **Also know:** RT alone is quasi-palliative; chemotherapy alone not recommended.

**Diagram to draw:** TNM T-stage diagram of the bladder wall, with the TMT flowchart from TURBT to chemoradiation, response check, surveillance or salvage cystectomy.

> **Key points to remember:**
> - WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
> - Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
> - Partial cystectomy: solitary <3 cm (dome tumour), no CIS, frozen-section margins, always PLND.
> - TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
> - BC2001: MMC/5-FU (mitomycin C/5-FU) + RT, 2-year locoregional DFS 67% vs 48%.
> - Salvage cystectomy 10–30%; follow up the bladder for 10 years.

## LAQ 3 Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic

P2 · P4 **merged from 2**

*Asked as:* P4 LAQ "Continent urinary diversion: orthotopic neobladder (Studer) versus non-orthotopic diversion" · P2 LAQ "Discuss principles of continence in continent urinary diversion. Application of Mitrofanoff principle in continent urinary and GI reconstruction."

*Read as: principles of continence in continent diversion, orthotopic neobladder versus non-orthotopic (heterotopic) urinary diversion after radical cystectomy, and the Mitrofanoff principle in urinary and GI (gastrointestinal, bowel) reconstruction, i.e. antegrade continence enema channels.*

### Introduction, definition and types

- A continent diversion stores urine in a bowel reservoir and empties it **at will**: by voiding through the native urethra (**orthotopic neobladder**) or by CIC (clean intermittent catheterisation) of a stoma (**continent cutaneous pouch**).
- **Neobladder definition**: an **orthotopic** reservoir of detubularised bowel (usually ileum) joined to the native urethra after radical cystectomy, allowing **volitional voiding** without a stoma.
- Evolution: ileal conduit (Bricker, 1950), then continent cutaneous (Kock, 1982), then orthotopic (Camey, 1979; Hautmann, 1988; Studer, 1989).
- **Ileal conduit** (Bricker, 1950) remains the commonest diversion worldwide; **ureterosigmoidostomy** is the oldest continent diversion.
- AUA (American Urological Association)/SUO (Society of Urologic Oncology) and EAU (European Association of Urology): counsel every patient on all options. **80–90% of men and ~75% of women** are medically eligible for a neobladder, but far fewer receive one.

| Group | Examples | How urine is held and emptied |
|---|---|---|
| Orthotopic | Studer, Hautmann W, T pouch, Padua, Mainz III | Native rhabdosphincter; voids by relaxing and straining |
| Incontinent cutaneous | Ileal conduit; transverse colon, sigmoid or jejunal conduit | Stoma and appliance |
| Continent cutaneous | Indiana pouch; right colon pouch with appendix (Mitrofanoff) or Yang–Monti channel; Kock (historical) | Self-catheterisation through a valved channel |
| Rectal | Ureterosigmoidostomy, Mainz II (sigma-rectum pouch) | Anal sphincter; urine mixed with stool |

### Principles of continence

- **Continence is a pressure balance:** low reservoir pressure must stay below adequate outlet resistance. Failure of either side causes leakage.

#### A. Low-pressure, adequate-capacity reservoir

- **Detubularisation** (opening bowel on its antimesenteric border) and cross-folding abolish coordinated peristaltic contractions.
- **Spherical shape:** greatest volume for a given surface area. By **Laplace's law** ($P = 2T/R$; wall tension = pressure × radius), a larger radius gives lower pressure for the same wall

tension. Result: a low-pressure reservoir.

- **Capacity ≥500 mL** when mature. Bowel stretches with time (ileum more than colon); 44 cm of double-folded ileum holds <200 mL at first and 500–600 mL within a year.
- Ileum is most compliant; colon has pressure waves.

**B. Reliable outlet**

- **Orthotopic:** intact **rhabdosphincter** (pudendal nerve), healthy urethra, careful apical dissection, nerve sparing. Voiding is by Valsalva and pelvic floor relaxation. Night leakage (20–50%) reflects loss of the guarding reflex.
- **Continent cutaneous:** a stable efferent channel that is easy to catheterise and has a durable, dependable continence mechanism (see continent cutaneous mechanisms below).

**C. Protect the upper tracts and metabolism**

- **Unobstructed ureterointestinal anastomosis** to protect the kidneys.
- **Enough renal function** (eGFR, estimated glomerular filtration rate, about ≥35–40 mL/min) to clear the acid load reabsorbed by bowel mucosa.
- Ability to self-catheterise.

**Studer as an example of the principles: orthotopic low-pressure ileal reservoir** (**Studer, 1989**): detubularised ileal reservoir with a long **isoperistaltic afferent limb**, joined to the urethra. **Detubularisation** stops coordinated contractions, so pressure stays low; a **sphere** (Laplace) gives most volume for surface area. The **isoperistaltic afferent limb** limits reflux (damps reflux during Valsalva voiding) and accepts **short ureters**. No staples in the reservoir. Continence depends on the **rhabdosphincter**.

## Choice of bowel segment

| Segment | Points |
|---|---|
| **Ileum** (default) | Most compliant, lowest pressure; mesentery reaches the urethra; mucosa atrophies, so absorption falls |
| Ileum + caecum (Mainz) | Near-ileal compliance; loses ileocaecal valve |
| Colon (right, sigmoid) | Less distensible, pressure waves, so more night leakage; used when ileum is short or diseased |
| Stomach | Secretes acid: helps in renal impairment; haematuria–dysuria; rarely used |

## Reflux versus obstruction

- Voiding by Valsalva raises pressure equally throughout, so reflux into a low-pressure pouch is benign, while **every antireflux valve can obstruct** (Le Duc tunnel stricture 7–29%; late Kock nipple stenosis).
- Direct refluxing anastomosis (Wallace, Leadbetter): stricture about 3–6%.
- **USC (University of Southern California) RCT (randomised controlled trial)**, 484 patients: Studer vs T pouch, no difference in renal function or infection at 3 years; the antireflux T pouch gave no renal benefit and needed more secondary procedures (reoperations).
- Studer: reflux on cystogram, yet kidneys preserved in about 95%.

## Continent cutaneous mechanisms

| Mechanism | Example | How it works | Weakness |
|---|---|---|---|
| **Nipple valve** (intussuscepted) | Kock (historical), Mainz I | Reservoir pressure compresses the intussuscepted segment | Same pressure effaces it; failure ≥15%, reoperation about one-third; late nipple stenosis |
| **Flap valve**: narrow tube (appendix) tunnelled submucosally or in taenia | **Mitrofanoff** (Penn, right colon pouch), Yang–Monti ileal tube if no appendix, Kropp | Narrow tube in a submucosal tunnel is coapted as pressure rises | No pop-off; stomal stenosis |
| **Imbricated ileocaecal valve + tapered ileum** | **Indiana** pouch | Continence zone lies at the plicated valve | Bowel frequency, B12 loss |
| - Used when continence is wanted but the urethra is unusable; umbilical stoma preferred. | | | |
| - **Also know** (native outlet): Kropp, Pippi Salle, slings, AUS (artificial urinary sphincter), bulking, bladder neck closure. | | | |

## The Mitrofanoff principle

- **Mitrofanoff (1980):** any supple, narrow tube implanted in a **submucosal tunnel** of a low-pressure reservoir, with its other end at skin, gives a **continent catheterisable channel**.
- As the reservoir fills, pressure transmitted through the mucosa flattens the tunnelled segment against the muscle backing: a **flap valve**. Continence lies in the tunnel, not in the conduit.
- Appendix: only a **2 cm tunnel** suffices because of its small calibre.

### Conduits

| Conduit | Points |
|---|---|
| **Appendix** (first choice) | Natural tube, minimal harvest morbidity; needs length ≥5–6 cm and ≥12 Fr; caecal cuff or **Mitchell caecal extension** adds length |
| **Yang–Monti** | 2–3 cm ileum opened on the antimesenteric border and retubularised transversely over 12 Fr; double or spiral Monti for obese; twice the revisions |
| Tapered ileum | Long enough; kinks if redundant |
| Ureter, fallopian tube | More stenosis |
| **Casale** bladder flap | No bowel; consumes bladder capacity |

### Technique (appendicovesicostomy)

1. Mobilise caecum; detach appendix with a caecal cuff, preserving mesoappendix.
2. Spatulate tip; tunnel submucosally (posterolateral or anterior bladder wall), intra- or extravesically.

3. Bring the base to the umbilicus or right iliac fossa with skin flaps, without tension or twist; keep it short and straight; fix the bladder to the abdominal wall.
4. Catheterise after each step. Channel catheter 2–4 weeks; then CIC every 4 hours by day.

### Mitrofanoff applications in urinary reconstruction

- **Neurogenic bladder** (spina bifida, spinal injury), often with augmentation, when urethral CIC is difficult (wheelchair-bound, sensate urethra).
- After bladder neck reconstruction or closure; exstrophy; valve bladder.
- **Continent cutaneous diversion** after cystectomy: Penn and right colon pouch (appendix in taenia; up to 100% continence); Yang–Monti if no appendix.
- Unusable urethra: complex stricture, urethrectomy.

### Mitrofanoff applications in GI reconstruction: MACE

- **MACE (Malone antegrade continence enema; Malone, 1990):** appendix brought to skin as an appendicocaecostomy with an antireflux tunnel (flap valve) so stool does not leak.
- Antegrade washouts empty the colon, giving **faecal continence** in neurogenic bowel (spina bifida), anorectal malformations, cloaca and refractory constipation.
- Options: Monti-MACE; **split appendix** for both MACE and urinary channel; caecostomy button. Yang–Monti for the urinary channel when the appendix is absent or used for MACE.

### Results and complications of continent catheterisable channels

- **Continence >95%** for flap valves; nipple valves about 85%.
- **Stomal stenosis is the commonest problem**; also false passage, channel polyps or granulation, pouch stones, revision (suprafascial or subfascial).
- Missed CIC: no pop-off, so **reservoir rupture** or upper tract damage.

### Patient selection

| | Orthotopic only | All continent diversions |
|---|---|---|
| Absolute contraindications | Tumour at the urethral margin (frozen section); significant urethral stricture | Severe renal (eGFR <40 mL/min) or hepatic dysfunction (liver failure); physical or mental inability to self-catheterise |
| Relative | Poor external sphincter; bladder-neck or anterior vaginal wall tumour (women); prostatic stromal invasion (men); irradiated sphincter (irradiated pelvis) | Frailty, poor cognition, social problems, prior radiation or bowel resection |

- **Renal function**: creatinine <1.7–2.2 mg/dL or eGFR **>35–40 mL/min** (below about 35–40 mL/min = severe renal impairment; another source: <40 mL/min), because the kidney must excrete acid reabsorbed from stored urine. Formal tests: urine pH ≤5.8 after ammonium chloride, osmolality ≥600 mOsm/kg after water deprivation.
- **Urethral recurrence**: in men the strongest predictor is prostatic stromal invasion (prostatic urethral tumour raises urethral recurrence risk); in women, bladder-neck or anterior vaginal wall tumour. **Intraoperative frozen section** of the urethral margin decides. CIS (carcinoma in situ) or multifocality alone do not exclude a neobladder.
- **Age and obesity alone are not contraindications**; comorbidity, cognition and support decide. A conduit suits the frail or unsupported patient.

- Continent cutaneous pouch: continence wanted but urethra unusable.
- Mark a conduit stoma site in everyone: up to ~10% change plan intraoperatively.

### Orthotopic versus non-orthotopic: comparison

| Feature | Orthotopic neobladder | Ileal conduit | Continent cutaneous | Ureterosigmoidostomy |
|---|---|---|---|---|
| Appliance | None | Bag | None; small stoma | None |
| Body image | Best | Worst | Good | Good |
| Complexity | High | Lowest | Highest | Moderate |
| Continence | Day 80–90%; night incontinence 20–50% | Not applicable | High (Indiana, Mitrofanoff) | Depends on anal sphincter |
| Catheterisation | 10–20% of men, ~30% of women | None | Always | None |
| Bowel used | Long ileal segment (Studer 54–56 cm) | 10–15 cm ileum | Right colon + terminal ileum | None isolated |
| Typical late problems | Night leakage, retention, acidosis | Stomal stenosis (20–24%), parastomal hernia, renal deterioration | Stomal stenosis, stones, difficult catheterisation | Severe acidosis, pyelonephritis, cancer at the anastomosis |

### Neobladder types

| Design | Segment | Ureters |
|---|---|---|
| **Studer** | 40–44 cm ileum + afferent limb | Refluxing, isoperistaltic limb |
| **Hautmann W** | About 70 cm ileum | Refluxing; largest capacity |
| T pouch | 44 cm + T limb | Antireflux flap valve |
| Camey II; Padua | Ileum | Antireflux (Le Duc) |
| Mainz III; sigmoid | Ileocaecal; colon | Tunnelled |

| | Studer | Hautmann |
|---|---|---|
| Ileum | 54–56 cm, U + afferent limb | ~70 cm folded in a W |
| Initial capacity | Smaller early capacity | Larger: earlier continence |
| Late | Less late retention | More retention, reabsorption |

### Studer technique (operative)

- Brief: isolate ileum 15–25 cm from the ileocaecal valve; open the distal part on the antimesenteric border, sew into a U, fold into a sphere; end-to-side ureteric anastomoses to the afferent limb; anastomose the dependent point to the urethra. 1. **Isolate 54–56 cm of terminal ileum**, ending **15–25 cm** above the ICV (ileocaecal valve) (another source: **15–20 cm proximal to the ICV**); restore bowel continuity. 2. **Reservoir: distal 40–44 cm** (two 20–22-cm limbs) opened on the antimesenteric border; sew the two limbs (medial edges)

into a **U** (back wall). 3. **Afferent limb:** leave the proximal **10–15 cm** tubular as the afferent limb (another source: **about 15 cm**; original 20 cm; 10–12 cm in later figures); close its end. 4. **Fold the U in the opposite direction** (across) and close the anterior wall into a **sphere**. 5. **Ureters:** spatulated **end-to-side, refluxing** ureteroileal anastomoses to the afferent limb, interrupted 4-0, **stented**. 6. **Urethral anastomosis** at the most dependent point with pre-placed sutures; catheter and suprapubic tube.

- Capacity rises from <200 to **500–600 mL** in a year.
- **Intracorporeal robotic Studer** follows the same steps.

### Postoperative care, voiding training and follow-up

- Mucus irrigation; oral bicarbonate; cystogram before catheter removal.
- **Voiding training**: void **sitting**, relax the pelvic floor, strain gently; **timed voiding** every 2 hours, extended to 3–4 hours as capacity grows (under 200 mL to 500–600 mL in a year). **Night alarm.**
- Lifelong follow-up: annual metabolic panel, B12, upper-tract imaging; urethral wash cytology if CIS (carcinoma in situ).

### Current status of orthotopic neobladder

**1. Preferred where suitable:** closest to a native bladder. AUA/SUO and EAU: counsel every cystectomy patient on all options and refer to an experienced surgeon.

**2. Under-used.** 80–90% of men and about 75% of women are eligible, yet only about 9% of US patients receive any continent diversion, and this is falling; continent diversion in RAZOR 21%, iROC 11%. Causes: comorbid patients, surgeon training, intracorporeal robotics.

**3. Designs in use. Studer** (40–44 cm U plus isoperistaltic afferent limb) and **Hautmann W** (about 70 cm; refluxing since 1997) dominate; T pouch or serous-lined tunnel when antireflux is wanted.

**4. Oncological safety.**

- Men: urethral recurrence about 7% at 5 years; main risk is prostatic stromal involvement. The neobladder seems protective (meta-analysis HR (hazard ratio) 0.44).
- Women: risk lies with bladder-neck tumour; no skip lesions, so preserve the urethra. **Frozen section** of the urethral margin decides in both sexes.
- Local recurrence <10%, rarely disabling the pouch; nodal disease is not a bar. Selected **post-radiation** patients do well.

**5. Function (outcomes)**

| Outcome | Result |
|---|---|
| Day continence | 80–90% (most by 3–6 months); another source (Studer): 90–98%; about 90% |
| Night incontinence | 20–50% (up to half leak): no sphincter (guarding) reflex, mucosal water secretion, nocturnal diuresis |
| Retention needing CIC | Men 4–10%; **women 20–30%** (pouch angulation): men leak, women retain |
| Self-catheterisation | **10–20% of men, ~30% of women** |

**6. Complications.** Early morbidity equals conduit. Late: hyperchloraemic metabolic acidosis, vitamin B12 deficiency, mucus, ureteroileal stricture, stones, pouch rupture, pouch–vaginal fistula (5–10%), hernia.

7. **Robotic intracorporeal neobladder:** comparable to open; long-term data lacking.
8. **Quality of life:** no diversion proven superior; body image favours neobladder.

- **Also know:** nerve sparing and uterine preservation aid continence.

### Metabolic complications (bowel in the urinary tract)

| Segment | Disturbance |
|---|---|
| Ileum, colon | **Hyperchloraemic metabolic acidosis** (ammonium chloride absorbed); hypokalaemia, worse with colon |
| Jejunum | Hyponatraemic, hypochloraemic, hyperkalaemic acidosis |
| Stomach | Hypochloraemic, hypokalaemic alkalosis |

- Acidosis is severe in most ureterosigmoidostomies; most colonic (Mainz) pouches need alkali. Treat with sodium bicarbonate or citrate, and replace potassium as well.
- **Also know:** osteomalacia; vitamin B12 deficiency and bile-salt diarrhoea after ileal loss (liver B12 stores last 3–5 years); stones (ileal conduit 10–12%, continent caecal reservoirs ~20%); bacteriuria; drug reabsorption (methotrexate, phenytoin); ammoniagenic encephalopathy; growth delay in children.
- **Secondary cancer**: ureterosigmoidostomy 6–29% after 10–20 years (yearly endoscopy from 5 years); ileal conduit and ileal neobladder lowest.

### Surgical complications

- **Early (≤90 days)**: ileus, urine leak (long pouch suture lines; drain percutaneously), infection, mucus retention; not more than with a conduit.
- **Late**: ureteroileal stricture ~3–6% (Wallace or direct end-to-side) (Studer/neobladder: **<3%**); neobladder retention, urethral stricture, rare pouch rupture; ventral and parastomal hernia; **hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, stones, pyelonephritis (commonest late, Studer), urethral recurrence.
- Continent cutaneous channels: stomal stenosis commonest (see results above).

### Quality of life

- No randomised trial. Body image favours the neobladder, urinary function the conduit; no diversion is superior on HRQoL (health-related quality of life).

### Recent advances

- RARC (robot-assisted radical cystectomy): **RAZOR (2018)** non-inferior 2-year progression-free survival (open diversion in both arms); **iROC**: intracorporeal diversion gave more days alive and out of hospital at 90 days.
- Intracorporeal neobladders and nerve- or organ-sparing cystectomy are increasing.

**Diagrams to draw:**

- Studer neobladder and ileal conduit side by side; Studer U-pouch with afferent limb, ureters and urethral anastomosis; ileal segment marked (afferent limb plus 40–44-cm U), then the folded spherical pouch with afferent limb.
- Flap valve (tunnelled appendix collapsing as pressure rises) beside a nipple valve.

**Key points to remember:**

- Continence = low reservoir pressure (detubularise, sphere, ≥500 mL; 500–600 mL by 1 year) against a reliable outlet.
- Detubularised ileal sphere: low pressure, large capacity; continence depends on the rhabdosphincter.
- Neobladder needs eGFR >35–40 mL/min, a tumour-free urethral margin and ability to self-catheterise. Absolute bar: positive urethral margin, stricture, eGFR <40.
- 54–56 cm ileum, 15–20 cm from ICV: 40–44-cm U (sphere) plus 10–15 cm afferent limb.
- Refluxing end-to-side ureters; stricture <3%; kidneys preserved ~95%; obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Day continence 80–90% (about 90%); night incontinence 20–50% (up to half); women retain (20–30%).
- Sitting Valsalva voiding, timed voiding, night alarm.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- Mitrofanoff 1980: submucosal flap valve; appendix needs only a 2 cm tunnel; Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Flap-valve continence >95%; stomal stenosis is the commonest complication.
- No diversion is proven best for quality of life; counsel all options.

## LAQ 4 Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy

P2 · P4 **merged from 2**

*Asked as:* P2 LAQ "Muscle-invasive bladder cancer (TCC): management, neoadjuvant and adjuvant chemotherapy, neobladder types" · P4 LAQ "Role of neoadjuvant therapy and Lymphadenectomy for muscle invasive bladder cancer and upper tract urothelial cell cancer?"

### Introduction

- MIBC (muscle-invasive bladder cancer; TCC, transitional cell carcinoma, now called urothelial carcinoma) is cT2–T4a disease.
- Nearly half of patients treated by RC (radical cystectomy) alone develop metastases from occult micrometastases, so cure needs local plus systemic therapy, planned by an MDT (multidisciplinary team).
- Cure improves with **systemic therapy before surgery** and a **meticulous lymph node dissection** (accurate staging, possible therapeutic value).
- UTUC (upper tract urothelial carcinoma) behaves alike, but RNU (radical nephroureterectomy) removes a kidney, changing chemotherapy timing.

### Staging

- Complete TURBT / Re-TURBT (transurethral resection of bladder tumour) with detrusor muscle; **MRI (magnetic resonance imaging) with VI-RADS** (Vesical Imaging-Reporting and Data System; ≥3 suggests invasion); CT (computed tomography) urogram and chest CT; bone scan if symptomatic.
- Assess fitness and cisplatin eligibility.

### Management algorithm

```
cT2–T4a N0 M0 ├─ Cisplatin-eligible → cisplatin NAC (neoadjuvant chemotherapy) │ → RC + bilateral PLND (pelvic lymph node dissection) ├─ Cisplatin-ineligible → RC + PLND directly │ (pT3–4 or N+ after either → adjuvant nivolumab or chemotherapy) ├─ Selected, motivated → trimodality therapy (TMT) └─ Unfit → chemoradiation / palliative radiotherapy cN+ → induction chemotherapy → consolidation RC in responders M1 → systemic therapy
```

**1. Cystectomy pathway: radical cystectomy (standard of care for cT2–T4a N0 M0)**

- **Cisplatin-based NAC:** ddMVAC (dose-dense methotrexate, vinblastine, doxorubicin, cisplatin) or GC (gemcitabine–cisplatin) ×4 (details below).
- **Cisplatin-ineligible:** direct cystectomy.
- **RC + standard bilateral PLND:** obturator, external and internal iliac nodes. **SWOG S1011:** extended PLND gave no survival benefit (details below).
- Men: cystoprostatectomy. Women: anterior exenteration; organ-sparing in selected cases.
- Do not delay beyond **12 weeks** from diagnosis (worse survival). Diversion: ileal conduit, orthotopic neobladder or continent pouch.
- **Adjuvant:** cisplatin chemotherapy if no NAC; **nivolumab** for pT3–4 or N+ (details below).

**2. Bladder preservation: TMT (trimodality therapy)**

- Maximal (visibly complete) TURBT + RT (radiotherapy) (64 Gy, or 55 Gy in 20 fractions) + radiosensitiser. **BC2001:** mitomycin C + 5-FU (fluorouracil) improved 2-year locoregional disease-free survival (67% vs 48%).
- Best candidates: cT2, solitary tumour, **no CIS (carcinoma in situ), no hydronephrosis**, good bladder function.
- 5-year cancer-specific survival 50–84%; salvage cystectomy 10–30%.
- **Partial cystectomy:** solitary dome or urachal tumour, no CIS.
- **Also know:** chemotherapy alone or radical TURBT alone is not recommended.

**3. Metastatic disease**

- First line: **EV (enfortumab vedotin) + pembrolizumab** (EV-302: about doubled overall survival (OS) vs platinum). Alternatives: platinum chemotherapy then **avelumab maintenance** (JAVELIN Bladder 100); erdafitinib for FGFR3 (fibroblast growth factor receptor 3) alterations.

## Neoadjuvant chemotherapy (NAC) in MIBC

### Rationale

- Treats micrometastases early, while tumour burden is low; better tolerated before surgery, so more patients complete it.
- Downstages; **pT0 at cystectomy** gives about 90% 5-year recurrence-free survival.
- Drawback: delays surgery in non-responders.

**Eligibility:** cT2–T4a N0 M0 and cisplatin-fit. **Cisplatin-ineligible** if any one: GFR (glomerular filtration rate) <60 mL/min, ECOG (Eastern Cooperative Oncology Group) performance status ≥2, grade ≥2 hearing loss, grade ≥2 neuropathy, NYHA (New York Heart Association) class III–IV failure.

| Trial | Regimen | Result |
|---|---|---|
| **SWOG 8710** (Southwest Oncology Group) | MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) ×3 then RC vs RC | pT0 38% vs 15%; median survival 77 vs 46 months (P4: 5-year OS 57% vs 43%); **P = 0.06** |
| **BA06 30894** (EORTC/MRC, European Organisation for Research and Treatment of Cancer/Medical Research Council) | CMV (cisplatin, methotrexate, vinblastine) ×3 | 16% relative reduction in mortality; 10-year OS 36% vs 30% |
| **ABC meta-analysis** (Advanced Bladder Cancer) | Cisplatin combinations | OS HR (hazard ratio) 0.86: **5% absolute 5-year survival gain** |
| **VESPER** | ddMVAC ×6 vs GC ×4 | ddMVAC better than GC: organ-confined response 77% vs 63%; 3-year PFS (progression-free survival) 66% vs 56%; more toxicity |
| **NIAGARA** | Perioperative durvalumab + GC, then adjuvant durvalumab, vs GC | Better event-free survival (HR 0.68) and OS (HR 0.75) |

- **Guidelines (EAU, European Association of Urology; AUA, American Urological Association; NCCN, National Comprehensive Cancer Network):** cisplatin-based NAC for all eligible patients; ddMVAC 3–6 cycles or GC 4 cycles.

- **No survival benefit:** single-agent cisplatin, carboplatin regimens, neoadjuvant radiotherapy.
- **Cisplatin-ineligible:** proceed straight to RC; neoadjuvant pembrolizumab (PURE-01) and atezolizumab (ABACUS) are investigational.
- The chapter calls neoadjuvant immunotherapy unproven; NIAGARA has since made perioperative durvalumab + GC an option.

## Adjuvant therapy in MIBC

| | Neoadjuvant | Adjuvant |
|---|---|---|
| Selection | Clinical stage (understaging risk) | True pathological stage, but still overtreats |
| Delivery | Better tolerated | Post-RC morbidity; a quarter never receive it |
| Evidence | Level 1 | Weaker: small, early-closed trials |

### Adjuvant chemotherapy

- Trials were small or closed early. **ABC 2022 update:** cisplatin-based adjuvant chemotherapy improves 5-year OS by about 6%. Benefit greatest in node-positive and extravesical disease.
- Use: cisplatin-naive patients (no NAC) with **pT3–4 or N+**: ddMVAC or GC.

### Adjuvant immunotherapy

- **CheckMate 274:** nivolumab for 1 year after RC (pT3–4a or N+, with or without prior NAC): median disease-free survival **20.8 vs 10.8 months**. It also included UTUC.
- **IMvigor010** (atezolizumab): negative overall (included pT2); benefit in ctDNA (circulating tumour DNA)-positive patients.
- Use nivolumab if cisplatin-ineligible or declining chemotherapy, or with ypT3 or worse after NAC.
- **Also know:** adjuvant pelvic radiotherapy (Cairo trials) improves pelvic control in pT3–4; ctDNA-guided therapy is emerging.

## Lymphadenectomy in MIBC

- **Primary drainage:** obturator, internal and external iliac, presacral. **Secondary:** common iliac, para-aortic, paracaval.
- Node positivity: pT2 8–26%, pT3 33–50%, pT4 36–80%.

| Template | Extent |
|---|---|
| **Standard** | Bilateral external and internal iliac and obturator nodes; genitofemoral nerve laterally, node of Cloquet caudally, up to the common iliac bifurcation |
| **Extended** | Adds common iliac and presacral nodes, up to the aortic bifurcation |
| Super-extended | Up to the inferior mesenteric artery |

- **Evidence:** German phase III (limited vs extended): no survival benefit. **SWOG S1011** (most had NAC): median nodes 24 vs 39, **no difference** in disease-free survival (HR 1.10) or OS; more grade ≥3 events. So **standard bilateral PLND** is the standard of care.
- **Node count:** fewer than 10 nodes doubled the risk of death (SWOG 8710 reanalysis). **Also know:** lymph node density, extranodal extension (prognostic).

- Unexpected nodes at RC: frozen section, proceed if resectable within the pelvis. Clinically node-positive: chemotherapy first, then RC in responders.

### Neoadjuvant and adjuvant therapy in UTUC

- **Why neoadjuvant:** CKD (chronic kidney disease) is common and GFR falls after RNU, losing cisplatin eligibility. Give platinum **while both kidneys work**.
- Biopsy under-stages, so select by high-risk features (high grade, invasion on imaging, hydronephrosis).
- **EA8141** (phase II): accelerated MVAC, or gemcitabine–carboplatin if CrCl (creatinine clearance) 30–49; complete response 14%, below ypT2N0 60%.
- **AUA/SUO (Society of Urologic Oncology):** neoadjuvant platinum is a standard option for high-risk UTUC, especially if postoperative CrCl will fall below 50.
- **Adjuvant: POUT** (phase III, pT2–T4 or N+ after RNU): gemcitabine–platinum vs surveillance; disease-free survival HR 0.45, 3-year 71% vs 46%. Weaker with carboplatin, favouring neoadjuvant cisplatin. Adjuvant nivolumab (CheckMate 274) also included UTUC.

### Lymphadenectomy in UTUC

- Node positivity 12–25%. Lymph drains **right to left and top to bottom**, so the template depends on side and level.

| Primary site | Right | Left |
|---|---|---|
| Renal pelvis, upper two-thirds ureter | Hilar, paracaval, retrocaval, **interaortocaval** | Hilar, para-aortic |
| Lower third ureter | Ipsilateral common, external, internal iliac, obturator, presacral | Same |

- Value: pN0 better than pNx; ≥8 nodes improved disease-specific survival above pT1. Indicated for high-grade or invasive disease; robotic yield equals open. Risk: chylous leak.

### Urinary diversion options

- Ileal conduit (simplest, commonest), continent cutaneous pouch, or **orthotopic neobladder**. No diversion is proven superior for QoL (quality of life).

### Neobladder: principles

1. **Reliable outlet:** intact rhabdosphincter, tumour-free urethra (frozen section).
2. **Low pressure:** complete **detubularisation** and cross-folding abolish peristalsis; a **sphere** gives most volume (Laplace).
3. **Capacity:** 44 cm of folded ileum holds <200 mL at first, **500–600 mL by 1 year**. - **Contraindications:** positive urethral margin, stricture, eGFR (estimated GFR) <35–40 mL/min, liver failure, inability to self-catheterise. Age, obesity and nodes are not bars.

### Critical evaluation of neobladder types

| Type | Bowel; ureteric technique | Strengths | Weaknesses |
|---|---|---|---|
| **Studer** | 40–44 cm ileum + isoperistaltic afferent limb; refluxing | Simple, no staples, takes short ureters; stricture <3%; good long-term function | Night incontinence; reflux on cystogram (harmless) |

| Type | Bowel; ureteric technique | Strengths | Weaknesses |
|---|---|---|---|
| **Hautmann W** | About 70 cm ileum; refluxing since 1997 | Largest capacity, early continence; stricture fell 9.5% → 1% | More retention, acidosis, B12 loss |
| **T pouch** (Stein–Skinner) | 44 cm + tunnelled T-limb flap valve | Antireflux; accepts dilated ureters | Complex; more secondary procedures (USC [University of Southern California] RCT) |
| **Abol-Enein serous tunnel** | 40 cm W, serosal troughs | Antireflux; stricture about 3.8% | Long ureters needed; no dilated ureters |
| **Hemi-Kock** | Intussuscepted nipple | Good early results | Late nipple stenosis, stones; abandoned |
| **Camey II; Padua (VIP)** | 65 cm or 42 cm ileum; Le Duc tunnel | Early designs | Le Duc stricture 7–29% |
| **Mainz III; right colon; sigmoid (Reddy)** | Colon ± ileum | When ileum short or diseased | High-pressure contractions, poor compliance, more leakage |

**Critical points**

- **Reflux vs obstruction:** Valsalva voiding pressurises the whole system equally, so reflux is benign; every antireflux valve can obstruct. **USC RCT** (484 patients): Studer = T pouch for renal function and infection; T pouch needed more secondary procedures. Hence **Studer and Hautmann** are the most popular.
- **Segment:** ileum is most compliant; colon and stomach give high pressures.
- **Capacity trade-off:** a larger pouch gives earlier continence but more retention and metabolic load.
- **Robotic intracorporeal neobladder:** comparable outcomes; long-term data awaited.

## Neobladder outcomes, complications and follow-up

- **Day continence 80–90%; night 20–50%.** Retention: men 4–10%, **women 20–30%**.
- **Late:** hyperchloraemic metabolic acidosis, B12 deficiency, mucus, stones, ureteroileal stricture, pouch rupture, urethral recurrence (men about 7% at 5 years).
- **Follow-up:** blood gas, B12, upper-tract imaging, urethral cytology if CIS.

**Diagrams to draw:** bladder wall T-staging (T2a, T2b, T3, T4) beside the standard PLND template; standard vs extended pelvic templates, and right vs left UTUC templates; Studer pouch with afferent limb and urethral anastomosis.

**Key points to remember:**

- NAC: cisplatin combinations only; ABC 5% absolute 5-year survival gain; SWOG 8710 pT0 38% vs 15%, P = 0.06.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab improved event-free and overall survival.
- CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
- SWOG S1011: extended PLND gives no survival benefit over standard.
- UTUC: neoadjuvant before kidney loss; POUT disease-free survival HR 0.45; right-sided needs interaortocaval nodes.
- TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
- USC RCT: Studer = T pouch; T pouch more reoperations.
- Day continence 80–90%, night 20–50%; women retain 20–30%.

## LAQ 5 Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation

P2 · P3 · P4 **merged from 3**

*Asked as:* P4 LAQ "Intravesical therapy for urothelial (transitional cell) carcinoma of bladder, including combination intravesical therapy" · P2 SAQ "Intravesical therapy for TCC of bladder" · P3 SAQ "Immediate post TURBT intravesical Therapy for TCC"

### Introduction

- NMIBC (non-muscle-invasive bladder cancer: Ta, T1, CIS [carcinoma in situ]) is 70–80% of bladder cancers. "Urothelial carcinoma" now replaces "transitional cell carcinoma" (TCC).
- Two problems: **recurrence** (40–60%) and **progression** (about 6% in low-grade Ta to 17% in high-grade T1). Intravesical therapy follows complete TURBT (transurethral resection of bladder tumour).
- **Rule:** chemotherapy reduces recurrence only. BCG (bacille Calmette-Guérin) is the only agent that also reduces progression.

### Risk-adapted use (AUA [American Urological Association]/SUO)

| Risk | Criteria | Therapy |
|---|---|---|
| Low | LG (low-grade) solitary (primary) Ta ≤3 cm; PUNLMP | Single post-op instillation only |
| Intermediate | LG Ta recurrent, >3 cm or multifocal; LG T1; HG (high-grade) Ta ≤3 cm | 6-week chemotherapy or BCG induction; maintenance 1 year |
| High | HG T1; HG Ta large, multifocal or recurrent; any CIS; LVI (lymphovascular invasion); variant histology | BCG induction + maintenance 3 years |

- EAU (European Association of Urology) adds a "very high" group; EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables.

### 1. Single immediate instillation after TURBT

#### Rationale

- After TURBT, NMIBC recurs in 40–60%. Many early recurrences come from **floating cells implanting** on the raw resection bed. A single chemotherapy instillation (SI) kills them before implantation.

#### Timing

- Implantation occurs within hours, so instil **as early as possible: within 6 hours** (P3: ideally within 2–6 h, in theatre or recovery). Guidelines allow up to 24 h, but MMC given at 24 h lost its benefit (no benefit at 24 hours).

#### Agents

| Agent | Dose | Note |
|---|---|---|
| **MMC (mitomycin C)** | 40 mg in 20–40 mL | Alkylating; skin rash |
| **Gemcitabine** | 2 g in 100 mL, 1 h | Standard in USA; mild toxicity |
| Epirubicin | 50 mg | Chemical cystitis |
| - **Also know:** pirarubicin works; thiotepa does not. | | |

**Technique:** exclude perforation; instil via catheter; retain 1–2 h; drain.

**Evidence**

- **Sylvester 2016 individual-patient meta-analysis:** about 35% relative reduction in recurrence; **no benefit** with >1 recurrence/year or EORTC recurrence score ≥5.
- **SWOG (Southwest Oncology Group) S0337:** gemcitabine 2 g versus saline in suspected low-grade NMIBC: 4-year recurrence 35% vs 47%; no excess toxicity.
- No effect on progression.

**Who benefits**

- **Low risk** (primary solitary low-grade Ta ≤3 cm): SI alone; greatest benefit.
- Intermediate risk: SI then adjuvant course. High risk: BCG follows.

**Contraindications**

- **Suspected perforation**, extensive resection, bleeding needing irrigation: withhold.
- **Never BCG immediately**: intravasation of live bacteria causes sepsis and death.

**Complications**

- Irritative symptoms (commonest), chemical cystitis, skin desquamation (MMC), contracted bladder; serious harm mainly after perforation.
- **Also know:** continuous saline irrigation may also reduce recurrence.

## 2. Adjuvant chemotherapy

- **MMC** (alkylating), 40 mg in 20 mL weekly ×6. Optimise it (Au, 2001): empty bladder, fluid restriction, sodium bicarbonate (alkalinised urine), concentrated dose. Skin rash.
- **Gemcitabine** (nucleoside analogue): minimal absorption, well tolerated. **Anthracyclines** (epirubicin, doxorubicin) cause chemical cystitis.
- Intermediate risk: 6-week induction, then monthly maintenance for 1 year.
- **Gem/Doce:** gemcitabine 1 g, drain, then docetaxel 37.5 mg; weekly ×6, then monthly for 2 years. Used in BCG-unresponsive disease and during BCG shortage (details under combination therapy).

## 3. BCG immunotherapy

- **Mechanism:** BCG binds **fibronectin**. Cytokines follow (IFN-γ [interferon-gamma], IL [interleukin]-2, IL-12, TNF-α [tumour necrosis factor-alpha]). A **Th1 (T-helper 1)** response gives cell-mediated tumour kill; TRAIL release is important.
- **Technique:** 1 vial in 50 mL saline. Start **2–4 weeks** after TURBT (re-epithelialisation). Urinalysis first; retain 1–2 hours.
- **SWOG schedule (Lamm, 2000):** 6 weekly induction doses, then 3 weekly doses at 3 and 6 months, then every 6 months to 3 years. Median recurrence-free survival 76.8 vs 35.7 months.
- **EORTC 30962:** full dose; 1 year for intermediate, 3 years for high risk. **NIMBUS:** reduced frequency was inferior.
- CIS complete response up to 84%; BCG beats MMC only with maintenance. Only maintenance BCG reduces progression.
- **Contraindications:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk intravasation); previous BCG sepsis; total incontinence; immunosuppression (now relative). Relative: active UTI (urinary tract infection), liver disease, previous TB (tuberculosis).

## BCG toxicity

| Grade | Features | Management |
|---|---|---|
| 1 | Cystitis, fever <38.5 °C, <48 h | Anticholinergics, analgesics, NSAIDs (non-steroidal anti-inflammatory drugs) |
| 2 | Severe or >48 h; persistent fever | Culture; isoniazid + rifampicin; resume at ½–⅓ dose |
| 3 | Sepsis, organ involvement | Stop BCG; isoniazid + rifampicin + ethambutol (triple therapy); steroids only with cover |

- BCG is resistant to **pyrazinamide**. BCG sepsis 0.5%.

## BCG failure

- **Adequate BCG:** ≥5 of 6 induction + ≥2 of 3 maintenance doses.
- **BCG-unresponsive:** persistent or recurrent CIS or HG Ta/T1 despite adequate BCG, or HG relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.
- **Also know:** refractory, intolerant, exposed, delayed relapse.

## BCG-unresponsive options

- **Timely radical cystectomy** is the standard (EAU, AUA).
- If unfit or refusing:
- **Pembrolizumab** (KEYNOTE-057): IV every 3 weeks; CIS complete response 41%.
- **Nadofaragene firadenovec:** adenovirus with interferon-α2b gene, every 3 months; CIS complete response 53%.
- **N-803:** IL-15 superagonist with BCG; CIS complete response 71%.
- Gem/Doce; valrubicin; **TAR-200** (gemcitabine-releasing device); cretostimogene (trials).

## Device-assisted therapy

- **EMDA (electromotive drug administration):** current increases MMC tissue delivery 4–7-fold. Sequential BCG + EMDA-MMC improved outcomes in T1.
- **Chemohyperthermia:** RITE (radiofrequency, 42 °C); HIVEC (heated recirculating MMC).
- **Also know:** UGN-102 MMC gel; interferon with low-dose BCG.

## Combination intravesical therapy

### Rationale

- Single agents fail: chemotherapy reduces **recurrence** only; BCG reduces **progression** but fails in about a third.
- Combinations add **different mechanisms**, overcome resistance, and are useful during **BCG shortage** and in **BCG-unresponsive** NMIBC.

### 1. Sequential chemotherapy (doublets)

| Regimen | Method | Results |
|---|---|---|
| **Gem/Doce** (gemcitabine–docetaxel) | Gemcitabine 1 g in 50 mL (60–90 min), drain, then docetaxel 37.5 mg (60–120 min); weekly ×6, then monthly for 2 years | High-grade recurrence-free survival about 65% at 1 year, 52% at 2 years; CIS complete response 60% at 1 year; **84% avoid cystectomy** |
| Gem/MMC | Gemcitabine 1 g, then MMC 40 mg | Durable recurrence-free about 38% |

- **Why it works:** gemcitabine (antimetabolite) and docetaxel (microtubule stabiliser) act on different cell-cycle phases.
- Oral sodium bicarbonate and antiemetics aid tolerance. No level 1 evidence; phase III BRIDGE compares it with BCG in BCG-naive patients.

**2. Chemotherapy with BCG**

- Sequential MMC–BCG: **no benefit** over BCG alone.
- **Exception: BCG alternating with EMDA-MMC** in T1: lower recurrence and progression than BCG alone. A later trial was non-significant.
- Chemohyperthermia (RITE, radiofrequency-induced thermochemotherapeutic effect) potentiates BCG.

**3. BCG with immunomodulators**

- **BCG + interferon-α (α2b):** no gain over BCG; allows BCG dose reduction to one-third.
- **N-803 (nogapendekin alfa)** = an IL-15 (interleukin-15) superagonist activating CD8 T and NK (natural killer) cells. With BCG in BCG-unresponsive CIS: complete response 71%, 45% at 12 months. FDA (Food and Drug Administration)-approved in 2024 (chapter: "under review").

**4. BCG with checkpoint inhibitors (BCG-naive high risk)**

- **CREST** (subcutaneous sasanlimab + BCG) and **POTOMAC** (durvalumab + BCG): improved event-free or disease-free survival over BCG alone.
- **ALBAN** (atezolizumab + BCG): negative. KEYNOTE-676 (pembrolizumab) ongoing.

**5. Device-assisted and novel combinations**

- **Chemohyperthermia** (MMC at 42 °C: RITE; HIVEC, hyperthermic intravesical chemotherapy) increases uptake.
- **TAR-200:** gemcitabine-releasing intravesical device (± cetrelimab); very high early complete response in BCG-unresponsive CIS (SunRISe-1).
- **Cretostimogene** (oncolytic adenovirus) + pembrolizumab: about two-thirds complete response at 12 months.

**Toxicity of combinations**

- Additive cystitis; immune-related events with checkpoint inhibitors.

**Diagrams to draw:** BCG mechanism (fibronectin, cytokines, Th1, kill) with SWOG schedule timeline; Gem/Doce sequence timeline (instil, dwell, drain, instil); floating cells implanting on the resection bed and killed by instillation.

**Key points to remember:**

- Single instillation within 6 h (at most 24 h); never BCG immediately; withhold if perforation; best for low-risk solitary tumours.
- SWOG S0337: 35% vs 47% recurrence with gemcitabine 2 g.
- Only maintenance BCG reduces progression; SWOG 3-year schedule; NIMBUS reduced frequency fails.
- BCG-unresponsive: cystectomy; pembrolizumab (41% CR), nadofaragene, N-803.
- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

## LAQ 6 CA bladder, etiology, newer investigations and treatment.

P2

**Type:** LAQ (25 marks)
**Source:** 2016-Summer Paper 2, Q1

### Introduction

- Commonest urinary tract cancer; over 90% **UC (urothelial carcinoma)**. Men 3 times commoner; women present later. About 70% are **NMIBC (non-muscle-invasive bladder cancer)**; it recurs often.

### Aetiology (risk factors)

| Factor | Key facts |
|---|---|
| **Smoking** | 30–40% of UC; risk 2–3 times. Aromatic amines form DNA adducts; **NAT2 (N-acetyltransferase 2) slow acetylators** at higher risk. Risk nears never-smokers >15 years after quitting |
| **Occupation** (5–10%) | Aromatic amines (dye, rubber, leather, painters, hairdressers); polycyclic hydrocarbons (aluminium, chimney sweeps) |
| Environment | **Arsenic** in water (West Bengal, Bangladesh, Taiwan) |
| Radiation | Pelvic radiotherapy 1.5–4 times; latency 15–30 years |
| Drugs | **Cyclophosphamide** (dose-dependent, above 20 g); pioglitazone (conflicting) |
| Chronic inflammation | *Schistosoma haematobium*, long-term catheter, stones → **SCC (squamous cell carcinoma)**; exstrophy → adenocarcinoma |
| Genetic | First-degree relatives 2 times; **Lynch syndrome** (MSH2); GSTM1 deletion |
| - **Also know:** obesity raises risk; diet has no proven role. | |

### Pathology and staging

- **WHO (World Health Organization) 2004 grading**: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade. Grade predicts progression.
- **Papillary pathway** (FGFR3; recurs, rarely progresses) versus **flat CIS (carcinoma in situ)** (TP53, RB; progresses).

| TNM (tumour, node, metastasis), 2017 | Extent |
|---|---|
| Ta / Tis | Papillary non-invasive / flat CIS |
| T1 | Lamina propria |
| T2a / T2b | Inner / outer half of muscle |
| T3a / T3b | Perivesical fat, microscopic / macroscopic |
| T4a / T4b | Prostatic stroma, uterus, vagina / pelvic or abdominal wall |
| N1 / N2 / N3 | One pelvic node / several / common iliac |

### Clinical features and standard work-up

- **Painless gross haematuria (85%)**; irritative symptoms suggest CIS.

- **Cystoscopy and biopsy (gold standard)**; **CT (computed tomography) urography**; **urine cytology** (sensitivity 31–62%, specificity 94–100%, detects high grade; **Paris System** reporting).

### Newer investigations

| Test | Principle | Value |
|---|---|---|
| **Blue-light cystoscopy (PDD, photodynamic diagnosis)** with HAL (hexaminolevulinate) | Porphyrins in tumour fluoresce red under blue light | About one-third more CIS; recurrence benefit inconsistent |
| **NBI (narrow-band imaging)** | 415 and 540 nm light absorbed by haemoglobin highlights vessels | More tumours; many false positives |
| **Urine biomarkers** | NMP22 (nuclear matrix protein 22), BTA (bladder tumour antigen), **UroVysion FISH (fluorescence in situ hybridisation)** (chromosomes 3, 7, 17, 9p21), ImmunoCyt, Cxbladder | More sensitive, less specific than cytology; **none replaces cystoscopy** |
| **mpMRI (multiparametric MRI) with VI-RADS (Vesical Imaging-Reporting and Data System)** | Muscle-line integrity scored 1–5 | **≥3 suspicious for muscle invasion**; about 80–90% accurate; before TURBT |
| **En bloc TURBT (transurethral resection of bladder tumour)** | Tumour removed in one piece with detrusor | Better staging and muscle sampling, fewer perforations; recurrence similar |
| - **Also know:** optical coherence tomography; urinary tumour DNA; PET-CT (positron emission tomography) for doubtful metastases. | | |

### Treatment

**1. TURBT**: complete, with muscle in the specimen. **Re-TURBT within 6 weeks** for T1, incomplete resection or no muscle. **Single chemotherapy instillation within 6 hours** (mitomycin C, epirubicin or gemcitabine) for low-risk tumours; never BCG (bacillus Calmette–Guérin) at this time.

**2. NMIBC by risk (AUA (American Urological Association))**

| Risk | Definition | Treatment |
|---|---|---|
| Low | Solitary low-grade Ta ≤3 cm; PUNLMP | Single instillation; surveillance |
| Intermediate | Low-grade Ta recurrent, >3 cm or multifocal; low-grade T1; high-grade Ta ≤3 cm | Chemotherapy or BCG; maintenance 1 year |
| High | High-grade T1; large, multifocal or recurrent high-grade Ta; any CIS; variant histology; lymphovascular invasion | **BCG induction + 3-year maintenance**; early cystectomy if very high risk |

| Risk | Definition | Treatment |
|---|---|---|
| - **BCG** induces a Th1 immune response; six weekly doses from 2–4 weeks after TURBT, then **SWOG (Southwest Oncology Group) maintenance** (recurrence-free survival 76.8 vs 35.7 months). Only agent that reduces progression. | | |
| - **BCG-unresponsive**: radical cystectomy; if unfit, **pembrolizumab** (KEYNOTE-057), **nadofaragene firadenovec**, gemcitabine–docetaxel, TAR-200. | | |

**3. MIBC (muscle-invasive bladder cancer), cT2–T4a N0M0**

- **Cisplatin-based NAC (neoadjuvant chemotherapy)** (dose-dense MVAC (methotrexate, vinblastine, doxorubicin, cisplatin) or gemcitabine–cisplatin) → **radical cystectomy + pelvic lymphadenectomy**. NAC adds 5–8% absolute survival (SWOG 8710: pT0 38% vs 15%). Cisplatin-ineligible: GFR (glomerular filtration rate) <60 mL/min, poor performance status, hearing loss, neuropathy, heart failure.
- **Diversion**: ileal conduit or orthotopic neobladder. Robotic equals open (**RAZOR**).
- **Trimodality therapy**: complete TURBT + chemoradiation (5-FU (fluorouracil) + mitomycin C, **BC2001**). For solitary T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- **Adjuvant nivolumab** for ≥pT3 or N+ (**CheckMate 274**: disease-free survival 20.8 vs 10.8 months). Partial cystectomy for selected solitary dome tumours.

**4. Metastatic**: **enfortumab vedotin + pembrolizumab** (**EV-302**: nearly doubled survival versus platinum). Otherwise platinum then **avelumab maintenance**; **erdafitinib** for FGFR3 alterations.

## Follow-up and recent advances

- Cystoscopy at 3 months, then by risk; upper-tract imaging in high risk.
- Molecular subtypes (basal tumours respond to cisplatin); **NIAGARA** (2024): perioperative durvalumab with NAC improved survival.

**Diagram to draw:** bladder wall layers with Ta, Tis, T1–T4 marked.

> **Key points to remember:**
>
> - Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
> - Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
> - High-risk NMIBC: BCG with 3-year maintenance.
> - MIBC: cisplatin NAC + cystectomy; trimodality in selected.

## LAQ 7 Management of T1G3 (high-grade T1) TCC of urinary bladder, including T1 with CIS in a young woman

P2

**Asked as:**

- Describe in detail the management of T1G3 TCC of urinary bladder? — *Summer 2019 Paper II Q1 (LAQ)*
- Discuss management of T1G3 bladder tumor — *Practice Set 7 Paper II Q3a (SAQ)*
- Discuss treatment of 25 yr female with bladder TCC T1NoMo and C13 — *Practice Set 4 Paper II Q1 (LAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)
(Practice Set 7 Paper II Q3a: also answered in 2019-Summer Paper 2, Q1) (Practice Set 4 Paper II Q1: T1G3 management also answered in 2019-Summer Paper 2, Q1)

*Practice Set 4 Paper II Q1: "C13" is read as CIS (carcinoma in situ): T1 N0 M0 urothelial carcinoma with concurrent CIS in a 25-year-old woman.*

### Introduction and risk category

- T1G3 is high-grade (HG) urothelial carcinoma (the current name for TCC, transitional cell carcinoma) that invades (invading) the lamina propria but not the muscularis propria (not muscle). TCC is now urothelial carcinoma. CIS is flat and always high grade.
- T1 is 20–30% of NMIBC (non-muscle-invasive bladder cancer) and its most dangerous form: recurrence >80%, progression to muscle invasion 30–50%.
- Grade, not stage, drives progression. So every HG T1 is **high risk** (AUA [American Urological Association]/SUO [Society of Urologic Oncology] and EAU [European Association of Urology]).
- **T1 with CIS is high-risk NMIBC** (AUA and EAU). CIS is the second strongest progression factor after grade. Untreated CIS progresses to muscle invasion in 40–83%.
- **EAU 2021 very high risk:** T1 high grade (HG) with CIS plus at least one of age >70, multiple papillary tumours, or diameter ≥3 cm. In this woman (25-year-old), age does not count, so number and size decide. LVI (lymphovascular invasion) and variant histology also make it very high risk.
- **Young woman:** long life expectancy, fertility, pregnancy and sexual function shape every choice.

### Risk factors for progression

- **Adverse factors:** concurrent CIS (carcinoma in situ), LVI (lymphovascular invasion), variant histology, size >3 cm, multifocality, prostatic urethral involvement, residual T1 at re-resection.

| Factor | Significance |
|---|---|
| Concurrent CIS (carcinoma in situ) | Second most important factor after grade |
| Depth of invasion (substaging) | Invasion deep to the muscularis mucosae is worse; T1a/T1b proposed, no agreed method |
| LVI (lymphovascular invasion) | Suggests occult spread |

| Factor | Significance |
|---|---|
| Variant histology | Micropapillary (any amount), plasmacytoid, sarcomatoid, nested |
| Size >3 cm, multifocal, recurrent | More residual tumour and progression |
| Prostatic urethra, diverticulum | Poor BCG contact; diverticulum lacks muscle |
| Residual T1 at re-TURBT | Progression up to 80% |
| HG T1 at 3 months after BCG | Progression about 82% |
| - **Also know:** EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables; hydronephrosis suggests muscle invasion. | |

### Evaluation

- Cystoscopy, urine cytology (usually positive in CIS), CT (computed tomography) urography (upper tract tumour <5%, more with CIS), bimanual examination under anaesthesia (including vagina).
- Smoking and occupational history.
- **Pregnancy test** before CT and BCG (bacille Calmette-Guérin); expert pathology review.

### Step 1: Complete, good-quality TURBT (transurethral resection of bladder tumour)

- Resect all visible tumour **with detrusor muscle** in the specimen; send the base separately.
- Blue-light cystoscopy or NBI (narrow-band imaging) finds more CIS (**blue-light cystoscopy** finds about one-third more CIS). **En bloc resection** (where possible) keeps orientation for substaging.
- Biopsy the prostatic urethra if tumour is at the bladder neck or trigone, or with CIS.
- Biopsy red velvety areas and the **bladder neck** (relevant to a later neobladder).

### Step 2: Re-TURBT (restaging) in all T1

- Within **2–6 weeks** (within 6 weeks), before BCG (bacille Calmette-Guérin).
- **Why:** residual tumour in ≥40%; upstaging to muscle invasion in 25–30% (about half if no muscle in the first specimen). **Residual T1: progression up to 80%.** Re-TURBT improves recurrence- and progression-free survival: Divrik's RCT (randomised controlled trial) showed better recurrence-free and progression-free survival.

```
HG T1 on TURBT → re-TURBT at 2–6 weeks ├─ ≥T2 → neoadjuvant chemotherapy + cystectomy ├─ residual T1 / very high risk → offer early radical cystectomy └─ T0, Ta or CIS → BCG induction + 3-year maintenance
```

- Practice Set 7 version: HG T1 → re-TURBT; T0, Ta, CIS, T1 → BCG (bacille Calmette-Guérin); very high risk → offer early radical cystectomy.
- Practice Set 4 version: T1 + CIS → re-TURBT within 6 weeks; T0/Ta/CIS → BCG induction + 3-year maintenance.

### Step 3a: BCG – standard for most patients (bladder preservation)

- **Mechanism:** BCG attaches to (binds) **fibronectin**, triggers cytokines (IFN-γ [interferon-gamma], IL [interleukin]-2) and a **Th1 (T-helper 1)** immune (cytokine) response that kills tumour. It is the **only intravesical agent that reduces progression**.

- **Schedule:** start 2–4 weeks after TURBT, never immediately (intravasation and sepsis); induction weekly ×6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** 3 weekly doses at 3 and 6 months, then every 6 months (6-monthly) to **3 years**. Median recurrence-free survival 76.8 vs 35.7 months. **EORTC 30962:** full dose for 3 years in high risk.
- **CIS:** tumour-free up to 84%, about half durable; persistent CIS at 3 months: second course gives a further 30–50% complete response.
- BCG is live: avoid pregnancy and breastfeeding during treatment.
- **Contraindications (avoid with):** gross haematuria, traumatic catheterisation, symptomatic UTI (urinary tract infection), previous BCG sepsis.
- **Toxicity:** cystitis common; BCG sepsis needs isoniazid, rifampicin, ethambutol and steroids (antituberculous drugs and steroids).

### Step 3b: Early (immediate) radical cystectomy

- **Rationale:** clinical T1 is understaged in one-third to one-half. Patients who progress under surveillance do worse than those presenting with muscle invasion (progressors do worse than primary muscle-invasive disease). Immediate cystectomy gives about **92% 10-year disease-free survival**.
- **AUA:** offer to fit patients with persistent HG T1 on re-resection, or T1 with CIS, LVI or variant histology.
- **EAU very-high-risk group (2021 model; risk factors: age >70, multiple papillary tumours, diameter ≥3 cm):** T1 HG with CIS plus ≥1 risk factor; T1 HG without CIS but with all three; also LVI, variant histology, CIS of the prostatic urethra (prostatic urethral CIS), persistent T1 at re-TURBT. Offer immediate cystectomy; BCG if declined or unfit.
- **Offer for (Practice Set 4):** very high risk, residual HG T1, LVI, variants, diffuse CIS, or patient preference.
- Also: diverticular tumour, tumour not controllable endoscopically.

### Step 3c: Cystectomy tailored to a young woman

- Standard female operation is anterior exenteration (bladder, urethra, anterior vagina, uterus, cervix, ovaries).
- **Organ-sparing** (≤cT2, bladder neck uninvolved): keep the **uterus and ovaries** (fertility, hormones, avoids surgical menopause; ovaries also make testosterone). Keep the **vagina and neurovascular tissue** (vaginal dissection no more dorsal than 2 and 10 o'clock) for sexual function and continence. Oncological safety of sparing is not fully known.
- **Orthotopic neobladder** suits her: needs a negative bladder-neck biopsy and **urethral frozen section** (negative bladder neck means negative urethra; CIS does not predict urethral involvement). Day continence 80–90%; **retention 20–30% in women**, so she must accept CIC (clean intermittent self-catheterisation). See Practice Set 4 Paper 2, Q3e.
- **Fertility:** if uterus or ovaries must go, offer oocyte or embryo cryopreservation first. Pregnancy after a neobladder is possible; caesarean uses a high midline incision to avoid the pouch.

### Step 4: BCG failure (BCG-unresponsive disease)

- **Adequate BCG for T1:** ≥5 of 6 induction doses.
- **BCG-unresponsive:** HG T1 at the 3-month check after induction; persistent or recurrent CIS or HG Ta/T1 despite adequate BCG; or HG relapse within 6 months (Ta/T1) or 12 months

(CIS) of the last dose.

- **Standard:** timely radical cystectomy (EAU, AUA).
- **Unfit or refusing (declined):** pembrolizumab (KEYNOTE-057: CIS complete response 41%); nadofaragene firadenovec (53%); N-803 with BCG; gemcitabine–docetaxel; TAR-200; trials.
- **Also know:** BCG-intolerant; delayed relapse may respond to BCG rechallenge.

### Follow-up (AUA/SUO high risk)

- Bladder preserved: cystoscopy and cytology every 3–4 months (3-monthly) for 2 years, 6-monthly for 2 years, then yearly for life.
- Upper-tract imaging yearly for 2 years (regular upper-tract imaging).
- Positive cytology with normal cystoscopy: check prostatic urethra and upper tracts.
- After neobladder: B12, metabolic panel, upper tracts.

### Recent advances

- Checkpoint inhibitors with BCG in BCG-naive disease (trials); urine markers predicting BCG response; TAR-200.

**Diagram to draw:** bladder wall layers showing CIS, Ta, T1a/T1b (above and below the muscularis mucosae) and T2.

> **Key points to remember:**
> - Re-TURBT in all T1 at 2–6 weeks (within 6 weeks); residual T1 means progression up to 80%.
> - BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
> - T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
> - Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy (92% 10-year disease-free survival).
> - BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.
> - Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

## Short answer questions (SAQ)

### SAQ 1 Haemorrhagic cystitis: diagnosis and management

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "Diagnosis and management of hemorrhagic cystitis" · P3 SAQ "Management of Haemorrhagic cystitis."

### Definition

- Diffuse inflammation and bleeding of the bladder mucosa, after other sources are excluded (cystoscopy, cytology, upper tract imaging).

### Causes and mechanism

| Cause | Mechanism |
|---|---|
| **Cyclophosphamide, ifosfamide** | Liver forms **acrolein**, excreted in urine, which sloughs urothelium; onset within 48 h |

| Cause | Mechanism |
|---|---|
| **Pelvic radiation** (about 5%; 6 months to 10 years later) | **Obliterative endarteritis**, ischaemia, fibrosis, fragile neovessels |
| **Viral** | **BK polyomavirus**; adenovirus 11 and 35; after bone marrow or renal transplant |
| Others | Bacterial infection; busulfan, thiotepa; penicillins; amyloidosis |

### Diagnosis

- History: drugs, radiotherapy, transplant, anticoagulants. Severity: microscopic, gross, clots, clot retention.
- Blood count, coagulation, renal function, urine culture, viral PCR (polymerase chain reaction) if transplanted.
- **Exclude bladder cancer first:** cytology, cystoscopy with biopsy, upper tract imaging (CT urography). Cyclophosphamide and radiation are also risk factors for bladder cancer.

### Prevention

- **Mesna** binds and inactivates acrolein; hyperhydration with forced diuresis, CBI (continuous bladder irrigation).

### Management (escalate by severity; stability decides the branch)

`Resuscitate, transfuse, correct coagulopathy, treat infection → large-bore 3-way catheter + CBI → cystoscopy under anaesthesia: clot evacuation, fulguration, biopsy → intravesical alum 1% → bleeding after 48 h: hyperbaric oxygen (+ nephrostomies if bleeding continues) → formalin 1%, then 4% (cystogram first) → cystectomy + conduit diversion (Unstable at any stage: internal iliac embolisation)`

### Intravesical agents

| Agent | Mechanism and method | Caution / key point |
|---|---|---|
| **Alum 1%** (first line) | Protein precipitation, vasoconstriction; 200–300 mL/h; no anaesthesia | Avoid in renal failure (aluminium toxicity) |
| Aminocaproic acid | Antifibrinolytic CBI, 200 mg/L | Only in a clot-free bladder; thromboembolism |
| Prostaglandins (carboprost), silver nitrate, hyaluronate (second-line options) | Vasoconstriction, coagulation, GAG (glycosaminoglycan) repair | Silver nitrate: cystogram |
| **Formalin 1–4%** | Protein fixation; **80–90% success**; under anaesthesia, instilled by gravity for 10–15 min | **Cystogram to exclude reflux** (occlude ureters if present); fibrosis, contracted (small) bladder, ureteric stricture |

### Other measures

- **HBOT** (hyperbaric oxygen therapy): 100% oxygen at 2–3 atmospheres, 30–40 sessions; raises tissue oxygen and drives neovascularisation; 80–90% response, but many relapse.

**RICH-ART** trial: improved urinary symptom scores in radiation cystitis.

- **Nephrostomies:** divert urokinase-rich urine if bleeding continues.
- **Angioembolisation:** selective anterior division of internal iliac; spare the posterior division (gluteal pain); initial control about 90%. Used if unstable.
- **Cystectomy with conduit diversion:** last resort; remove the bladder (pyocystis risk).
- **Viral:** supportive (hydration, irrigation); cidofovir (intravenous or intravesical; evidence anecdotal).
- **Also know:** tranexamic acid and laser ablation are unproven; oral pentosan polysulfate in children.

**Diagram to draw:** stepwise escalation ladder.

> **Key points to remember:**
> - Acrolein (prevent with mesna); radiation endarteritis; BK virus.
> - Exclude bladder cancer.
> - Alum first; formalin only after cystogram.
> - Hyperbaric oxygen for radiation cystitis (RICH-ART); embolise if unstable.

## SAQ 2 BCG therapy and BCG failure

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "BCG therapy in Bladder Cancer" · P2 SAQ "BCG failure"

### Introduction

- BCG (bacille Calmette-Guérin), a live attenuated mycobacterium (Morales, 1976), is standard for high-risk NMIBC (non-muscle-invasive bladder cancer) and the only intravesical agent that reduces both **recurrence and progression**.

### Mechanism

- BCG binds **fibronectin** on the urothelium; cytokines (IFN-γ [interferon-gamma], IL-2 [interleukin-2] and TNF-α [tumour necrosis factor-alpha]) and immune cells follow.
- A **Th1 (T-helper 1)** response gives cell-mediated tumour kill. It needs repeated stimulation, which is why **maintenance** is essential.

### Indications

- **High-risk NMIBC:** HG (high-grade) T1, large/multifocal/recurrent HG Ta, **CIS (carcinoma in situ)** (initial complete response up to 84%).
- **Intermediate risk:** option. Not for low risk.

### Technique and schedule

- Start **2–4 weeks after TURBT (transurethral resection of bladder tumour)**; urinalysis first; atraumatic catheter; retain **1–2 hours**.
- **Induction:** weekly × 6.
- **Maintenance (SWOG [Southwest Oncology Group], Lamm 2000):** weekly × 3 at 3 and 6 months, then every 6 months to **3 years** (recurrence-free survival 76.8 vs 35.7 months).
- **EORTC (European Organisation for Research and Treatment of Cancer) 30962:** intermediate risk, full dose for 1 year; high risk, full dose for 3 years. **NIMBUS:** reduced

frequency was inferior.

### Contraindications

- **Absolute:** immediately after TURBT, gross haematuria, traumatic catheterisation (all risk **intravasation**); previous BCG sepsis; total incontinence; immunosuppression (now debated).
- **Relative:** UTI (urinary tract infection), liver disease, prior TB (tuberculosis).

### Toxicity

| Grade | Features | Management |
|---|---|---|
| 1 | Cystitis, fever <38.5 °C, <48 h | Symptomatic |
| 2 | Severe or >48 h | Isoniazid + rifampicin; resume at ½–⅓ dose |
| 3 | Sepsis, organ involvement | Stop BCG; add ethambutol; steroids only with cover |
| - BCG is resistant to **pyrazinamide**. | | |

### BCG failure: definitions

- "Failure" was poorly defined; consensus categories now guide treatment.

| Category | Definition | Implication |
|---|---|---|
| **Adequate BCG** | ≥5 of 6 induction + ≥2 of 3 maintenance, or ≥2 doses of a second induction; for T1, ≥5 of 6 induction alone | Threshold for "unresponsive" |
| **BCG-unresponsive** | Persistent or recurrent CIS or HG Ta/T1 despite adequate BCG; or HG relapse within **6 months (Ta/T1) or 12 months (CIS)** of last dose; **HG T1 at first 3-month check** | More BCG futile; cystectomy |
| **BCG-resistant** | CIS or HG Ta at 3 months after induction | Reinduction: 30–50% respond |
| Delayed relapse | HG relapse beyond these windows | Rechallenge may work |
| **BCG-intolerant** | Toxicity prevented adequate BCG | Salvage chemotherapy |

- **Why BCG fails:** inadequate dosing; PD-L1 (programmed death-ligand 1) upregulation on tumour cells.

### Evaluation of BCG failure

- Cystoscopy, cytology, **re-TURBT** and biopsies to exclude invasion.
- **Extravesical disease:** prostatic urethral biopsy (24–30%) and CT urography (upper tract 0.8–25%).

### Management of BCG failure

1. **Timely radical cystectomy** is standard (EAU [European Association of Urology], AUA [American Urological Association]): HG T1 at 3 months progresses in 82%; CIS failure in about 50%.
2. **Bladder sparing** (unfit or unwilling):

| Option | Key result (CIS complete response, CR) |
|---|---|
| **Pembrolizumab** IV (KEYNOTE-057) | 41% at 3 months, 19% at 12 months; grade 3–4 toxicity 13% |
| **Nadofaragene firadenovec** (adenovirus carrying IFN-α2b [interferon alpha-2b] gene) every 3 months | 53% at 3 months, 24% at 12 months |
| **Gemcitabine–docetaxel** sequential | 60% at 1 year; 84% avoid cystectomy (retrospective) |
| **N-803 + BCG** (IL-15 [interleukin-15] superagonist) | 45% at 12 months |

- Salvage CR rates halve by 12 months.
- **Also know:** BCG-refractory (older term); gemcitabine, gemcitabine–mitomycin, valrubicin, chemohyperthermia, EMDA (electromotive drug administration), chemoradiation; TAR-200; cretostimogene.

### Recent advances

- PD-1/PD-L1 (programmed death-1/ligand-1) inhibitor + BCG in BCG-naive high risk improved event-free survival (CREST, POTOMAC).
- TAR-200 gemcitabine-releasing device.

**Diagrams to draw:** SWOG schedule timeline; flowchart: HG recurrence after BCG, was BCG adequate?, unresponsive, then cystectomy or bladder sparing.

> **Key points to remember:**
> - Only BCG reduces progression; maintenance is essential.
> - Never immediately after TURBT; start at 2–4 weeks.
> - Adequate BCG = 5/6 induction + 2/3 maintenance.
> - Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
> - Timely cystectomy is standard; salvage CR rates halve by 12 months.

## SAQ 3 Malignancy in augmented bladder and after urinary diversion

P1 · P4 **merged from 2**

*Asked as:* P4 SAQ "Secondary tumours after urinary diversion" · P1 SAQ "Cancer in augmented bladder"

### Definition

- New cancers arising in or next to bowel used for urinary diversion or bladder augmentation, usually after a **long latency** (10–20 years or more; usually decades). Distinct from recurrent urothelial cancer.
- Bladder augmentation (enterocystoplasty) is mainly for neurogenic bladder. **Earliest reported tumour 4 years after surgery**; tumours after augmentation are often **aggressive and advanced at diagnosis**.

### Risk by diversion (German registry, Kälble 2011)

| Diversion | Risk | Surveillance |
|---|---|---|
| **Ureterosigmoidostomy** | 2.6% (6–29% in long-term series); mean latency about 26 years | Regular colonoscopy **from 5 years** |
| Augmentation cystoplasty | 1.6% | Endoscopy, often from 10 years (see surveillance below) |
| Colonic neobladder | 1.3% | — |
| Ileocaecal pouch | 0.14% | If symptomatic |
| Ileal neobladder | 0.05% | No routine |
| Ileal conduit | 0.02% | No routine |

- **Pattern:** risk is highest where urothelium meets colon and is bathed in faeces; purely ileal diversions are lowest.

### Risk factors

- **Gastrocystoplasty** (highest among augments), **exstrophy**, immunosuppression after transplant, smoking, chronic infection, long latency.
- The underlying disease may itself carry risk: bladder cancer in spina bifida is about 2–4%, mostly stage III or IV, and non-gastric augmentation did not appear to increase that risk.

### Pathogenesis

- **Urothelium–colonic junction** exposed to urine and faeces; tumours arise at the ureterocolic anastomosis. Ureterosigmoidostomy is the extreme model: urothelium beside colonic mucosa bathed in faeces.
- **N-nitrosamines (N-nitroso compounds)** from bacterial action on urinary nitrates in stagnant, infected urine (probably enhancers, not sole cause).
- Chronic inflammation at the anastomosis releases growth factors (EGF, epidermal growth factor) that drive proliferation; ornithine decarboxylase; abnormal mucin; perianastomotic urothelium is genetically unstable.
- Removing the urothelium prevents tumours experimentally.
- **Defunctioned anastomoses still form cancer:** when converting a ureterosigmoidostomy, excise the old anastomosis.

### Pathology

- Mostly **adenocarcinoma**, usually at the anastomosis (ureterocolic or **enterovesical**); also adenomatous polyps, urothelial carcinoma (aggressive, often metastatic), signet-ring and other types, sarcoma.
- **Nephrogenic adenoma:** benign; probably the commonest tumour after cystoplasty.

### Clinical features

- **Gross haematuria**, rectal bleeding, change in bowel habit, ureteric obstruction (hydronephrosis), recurrent UTI (urinary tract infection), pelvic pain, new incontinence, mass.
- Neurogenic patients may notice symptoms late.

### Surveillance and prevention

- **Ureterosigmoidostomy:** regular colonoscopy from year 5.

- **Augmentation:** chapters differ: Campbell bladder-surgery chapter advises **endoscopy yearly from 10 years**; paediatric chapter: no consensus, not shown cost-effective (Mayo group: unnecessary). Higher-risk groups benefit most.
- Investigate any haematuria promptly: cystoscopy with biopsy, cytology, CT (computed tomography) urography; biopsy and staging.
- **Prevention:** prefer ileal segments; counsel patients; regular clean intermittent catheterisation, daily bladder irrigation (less stasis, mucus and stones), treat urea-splitting infection, stop smoking, lifelong (transitional) follow-up.

### Management

- **Ureterosigmoidostomy:** resection of the segment with the anastomosis; conversion to conduit or continent diversion.
- **Augmentation:** partial or total (radical) cystectomy with **excision of the augment**, lymphadenectomy and new urinary diversion (conduit or continent pouch).
- Chemotherapy as for histology; prognosis poor because of late stage.
- **Also know:** after cystectomy, recurrent urothelial cancer may arise in the upper tracts or a retained urethra (urethral recurrence about 7–9%); needs separate surveillance. Stones, perforation and metabolic acidosis also occur after augmentation.

**Diagrams to draw:** ureterosigmoid anastomosis with tumour at the urothelial–colonic junction; clam ileocystoplasty with tumour at the enterovesical suture line.

> **Key points to remember:**
> - Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
> - Mostly adenocarcinoma at the anastomosis; earliest at 4 years after augmentation.
> - Excise defunctioned anastomoses.
> - Gastric segment, exstrophy, immunosuppression: highest risk in augments.
> - Haematuria needs cystoscopy; treatment is cystectomy.

## SAQ 4 Optical coherence tomography in carcinoma bladder

P2 · P4 **merged from 2**

*Asked as:* P4 SAQ "Optical coherence tomography in carcinoma bladder" · P2 SAQ "What is optical coherence tomography. What is its potential role in management of CA bladder?"

### Introduction

- **WLC (white-light cystoscopy)** is the gold standard but subjective: many biopsies are negative and small lesions are missed. WLC misses flat lesions and cannot show depth.
- **OCT (optical coherence tomography)** is an **"optical biopsy"** technique listed among enhanced cystoscopic methods for NMIBC (non-muscle-invasive bladder cancer): real-time cross-sectional imaging of the bladder wall. It works like ultrasound but uses light.

### What is OCT: principle

- OCT is the **optical analogue of B-mode ultrasound**. **Near-infrared light** (about 1300 nm) is shone on tissue, and the echo delay of back-scattered light is measured instead of sound, using **low-coherence interferometry** (back-scattered light measured against a reference beam).
- The delay of reflected light gives depth. Scanning across builds a cross-sectional image.

- **Resolution about 10–20 µm** (near histology). **Depth about 2–3 mm** (P2: only **1–2 mm**): urothelium, lamina propria and superficial muscle.
- First described in 1991 (ophthalmology).

### Technique

- A fibre-optic probe passes through the cystoscope's working channel.
- WLC or blue light finds the lesion; the probe is placed **in contact** with it and read at once (real time).

### Image interpretation

| Tissue | OCT appearance |
|---|---|
| Normal | Three layers: thin, dark, low-scatter urothelium; bright lamina propria with clear boundary; darker muscularis visible below |
| Inflammation | Layers preserved but thickened |
| Papillary tumour (Ta) | Thickened urothelium, papillary pattern; lamina propria boundary intact |
| Invasive (T1) | Layers lost; **loss (break) of the urothelium–lamina propria boundary** = invasion |

### Potential role in carcinoma bladder

1. **Diagnosis:** tell tumour from inflammation at suspicious areas; separate benign from malignant flat lesions in real time; fewer unnecessary biopsies.
2. **Staging aid: Ta vs T1** (lamina propria invasion) during TURBT (transurethral resection of bladder tumour); helps in diverticula and thin walls.
3. **Resection margins:** checks margins and completeness of TURBT.
4. **Surveillance** of NMIBC, including after BCG (bacille Calmette-Guérin).
5. **Upper tract:** through the ureteroscope, OCT correlates well with final grade and stage of UTUC (upper tract urothelial carcinoma).

### Evidence and limitations

- Small studies show **high sensitivity but lower (moderate) specificity**. Inflammation, scar, recent TURBT and post-BCG change mimic tumour (false positives). Adding OCT to fluorescence (blue-light) cystoscopy improves specificity.
- Tiny field of view: needs WLC or PDD (photodynamic diagnosis) to guide it.
- Too shallow to judge (deep) muscle invasion; cannot grade (no cellular detail); cost; learning curve.
- Still **investigational**; not recommended routinely by guidelines.

### Other enhanced cystoscopy techniques

| Technique | Principle |
|---|---|
| **PDD (blue light, HAL, hexaminolevulinate)** | Tumour porphyrins fluoresce red; detects about one-third more CIS (carcinoma in situ) |
| **NBI (narrow-band imaging)** | 415/540 nm light absorbed by haemoglobin; shows vessels; many false positives |
| **CLE (confocal laser endomicroscopy)** | Cell-level surface images after fluorescein |

| Technique | Principle |
|---|---|
| - **Also know:** Raman spectroscopy. | |

**Diagram to draw:** OCT image showing the three normal bladder layers beside a tumour with loss of layering.

> **Key points to remember:**
> - Near-infrared interferometry ("ultrasound with light"); 10–20 µm resolution; 2–3 mm depth (P2: 1–2 mm).
> - Loss of layering suggests tumour; boundary loss suggests T1.
> - Role: Ta vs T1; cannot assess deep muscle.
> - High sensitivity, moderate specificity, false positives from inflammation; investigational.

## SAQ 5 Bladder diverticulum.

P2

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 2, Q3c

### Definition

- Herniation of bladder mucosa through the **muscularis propria**, forming a sac with a neck. With no functioning muscle it **cannot empty** (stasis).
- Male:female about 9:1.

### Classification

| Feature | Congenital (primary) | Acquired (secondary) |
|---|---|---|
| Age | Children (<10 years) | Men >60 years |
| Cause | Weak detrusor at the ureteric hiatus | **BOO (bladder outlet obstruction)**: BPH (benign prostatic hyperplasia) in about 70%, stricture; neurogenic bladder |
| Number, site | Solitary; **paraureteral (Hutch)** | Often multiple, anywhere |
| Bladder wall | Smooth | Trabeculated |
| Associations | VUR (vesicoureteral reflux); Ehlers–Danlos, Williams, Menkes syndromes | Stones, cancer |
| - **Also know:** iatrogenic (after cystotomy or reimplantation); cellule, saccule and diverticulum form a continuum. | | |

### Clinical features

- Usually incidental while investigating LUTS (lower urinary tract symptoms), haematuria or UTI (urinary tract infection).
- **Double voiding**, recurrent UTI, stones, retention, ureteric obstruction.

- **Malignancy** in 0.8–10% (urothelial 70–80%, squamous 20–25%). With no muscle layer, tumours invade the fat early (beyond lamina propria = **T3a**).

### Investigations

- **VCUG (voiding cystourethrogram)**: size, VUR, emptying, **paradoxical filling** during voiding.
- **Cystoscopy**: enter and inspect the sac (flexible scope); cytology; biopsy carefully (perforation risk).
- **CT (computed tomography) urography**: tumour, stones, ureteric deviation, hydronephrosis.
- **Urodynamics**: BOO or neurogenic dysfunction (urine "sinks" into the sac, so contractility looks falsely low).

### Management

- **Treat the cause first** (BOO); a small diverticulum may then empty and need only surveillance.
- **Observation** (cystoscopy, cytology) if asymptomatic; **CIC (clean intermittent catheterisation)** if unfit with poor emptying.
- **Endoscopic**: neck incision or resection ± fulguration in frail patients.
- **Diverticulectomy indications**: refractory symptoms, recurrent UTI, stones, upper-tract damage or VUR, premalignant change.
- **Tumour inside**: partial or radical cystectomy.

### Diverticulectomy technique

- Open, laparoscopic or robotic; endoscopic prostate surgery first, same session.
- Steps: ureteric catheter if near the ureter → cystotomy → evert or dissect the sac from its pseudocapsule → close in two layers → catheter about 1 week, cystogram before removal.
- Complications: ureteric injury, leak.

**Diagram to draw:** paraureteral (Hutch) diverticulum with narrow neck and ureter.

> **Key points to remember:**
> - No muscle → stasis, UTI, stones, cancer.
> - Acquired type: treat the BOO first.
> - Tumour invading beyond lamina propria is T3a.

## SAQ 6 Intravesical gene therapy.

P2

**Type:** SAQ (10 marks)
**Source:** 2022-Summer Paper 2, Q3a

### Concept

- A gene is instilled into the bladder so that urothelial and tumour cells make an anticancer product, or the vector itself kills them.
- **Why the bladder suits it:** direct contact, little systemic exposure, response checked by cystoscopy and cytology.

- **Barrier:** the glycosaminoglycan layer resists vectors → **transduction enhancers** (Syn3; DDM, n-dodecyl-β-D-maltoside).
- **Indication:** high-risk **BCG (bacille Calmette–Guérin)-unresponsive NMIBC (non-muscle-invasive bladder cancer)** in patients unfit for or declining radical cystectomy. BCG-unresponsive: persistent or recurrent CIS (carcinoma in situ) or high-grade Ta/T1 after adequate BCG, or relapse within 6 months (Ta/T1) or 12 months (CIS) of the last dose.

## Agents

| Agent | Vector, mechanism | Key data |
|---|---|---|
| **Nadofaragene firadenovec** | **Replication-incompetent adenovirus type 5** carrying the **interferon-α2b gene**, with **Syn3** surfactant. Urothelium keeps secreting interferon → apoptosis (TRAIL, TNF-related apoptosis-inducing ligand) and immunity, overcoming the short dwell of interferon protein | 75 mL every **3 months**, 1-hour dwell. CIS: **complete response 53% at 3 months, 24% at 12 months**; papillary high-grade RFS (recurrence-free survival) 44% at 12 months; serious events 2%. **FDA (Food and Drug Administration) approved December 2022**, first bladder gene therapy |
| **Cretostimogene grenadenorepvec (CG0070)** | **Replication-competent oncolytic adenovirus**: E2F promoter limits replication to RB (retinoblastoma) pathway-defective tumour cells; **GM-CSF (granulocyte-macrophage colony-stimulating factor)** transgene adds immunity | Two 6-week inductions + maintenance; single-arm phase III BOND-003 in BCG-unresponsive CIS: high CR (complete response) rate; combination with pembrolizumab under study |
| **Inodiftagene vixteplasmid (BC-819)** | Plasmid: **diphtheria toxin A** gene under tumour-active **H19** promoter | Investigational |

## Administration and toxicity

- Catheter instillation, patient turned during dwell; biosafety precautions.
- Mostly grade 1–2: **bladder spasm, urgency, discharge around catheter, fatigue**.

## Place in practice

- Alternatives: radical cystectomy (standard), **pembrolizumab** (KEYNOTE-057), gemcitabine–docetaxel, **N-803** (interleukin-15 superagonist) + BCG.
- Limits: responses wane (median CR duration about 10 months); single-arm trials; cost.

**Also know:** non-viral IL-12 (interleukin-12) plasmid (EG-70); early adenoviral attempts failed through poor transduction.

**Diagram to draw:** adenovirus transducing a urothelial cell → interferon secretion → tumour apoptosis.

> **Key points to remember:**
> - Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
> - CIS CR 53% at 3 months, 24% at 12 months.
> - Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

## SAQ 7 Von Bruns Nest

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 10 Paper 2, Q3e

*Read as: von Brunn nests of the urothelium.*

### Definition

- **Von Brunn nests** are solid nests of benign urothelial cells lying in the **lamina propria**. They form when the basal layer of the surface urothelium buds downward and the buds pinch off.
- A common normal or reactive finding in most adult bladders, especially at the **trigone and bladder neck**; also in the ureter and renal pelvis.

### Pathogenesis: the proliferative cystitis spectrum

`Chronic irritation (infection, stones, catheter, outlet obstruction) → urothelial buds → von Brunn nests (solid) → central degeneration and fluid → cystitis cystica (cysts) → glandular (columnar / goblet cell) metaplasia → cystitis glandularis`

| Lesion | Histology | Malignant potential |
|---|---|---|
| **Von Brunn nests** | Solid urothelial nests, lamina propria | None |
| **Cystitis cystica** | Nests with central cyst, urothelial lining | None |
| **Cystitis glandularis** (typical) | Cuboidal or columnar lining | Low |
| Cystitis glandularis, **intestinal type** | Goblet cells, mucin | Possible link to **adenocarcinoma** |

### Clinical features

- Usually incidental. When florid: irritative symptoms, haematuria, infection; cystoscopy shows submucosal nodules or cysts that can mimic tumour.
- **Associations:** chronic inflammation, outlet obstruction, **pelvic lipomatosis** (florid cystitis glandularis), bladder exstrophy.

### Differential diagnosis (pathologist's trap)

- **Nested variant of urothelial carcinoma:** deceptively bland but aggressive (male:female 6:1). Clues: deep infiltrative nests reaching muscle, nuclear atypia.
- **Inverted papilloma:** endophytic nests with trabeculae, usually at the trigone.
- Nephrogenic adenoma; urothelial carcinoma with glandular differentiation.

### Management

- None for isolated nests; treat the cause (infection, stone, catheter, obstruction).
- **TURBT (transurethral resection) or biopsy** if a mass is present, to exclude carcinoma.
- **Cystitis glandularis:** remove the irritant and arrange **endoscopic surveillance**, because rare change to adenocarcinoma is reported (especially the intestinal type and in exstrophy).
- Upper tract: benign, but coexisting upper-tract urothelial cancer cannot be excluded, so follow up.

**Diagram to draw:** urothelium budding into lamina propria, forming a solid nest, then a cyst, then glandular metaplasia.

> **Key points to remember:**
> - Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
> - Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
> - The nested variant of urothelial carcinoma mimics von Brunn nests.

## SAQ 8 Mitomycin

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 11 Paper 2, Q3b

### Pharmacology

- MMC (mitomycin C) is an antitumour antibiotic from *Streptomyces caespitosus*.
- **Mechanism:** after reductive activation in the cell it acts as an **alkylating agent**, cross-linking DNA and stopping DNA synthesis; cell-cycle non-specific.
- Molecular weight 334 Da: **little systemic absorption** from the bladder.

### Uses in NMIBC (non-muscle-invasive bladder cancer)

1. **Single immediate post-TURBT (transurethral resection of bladder tumour) instillation:** 40 mg. Given **within 6 hours** it reduces recurrence; at 24 hours it does not, because it works by preventing implantation of floating cells. Best for a solitary low-grade papillary tumour. Avoid if perforation or extensive resection is suspected.
2. **Adjuvant induction:** 40 mg weekly for 6 weeks (± maintenance) in low- and intermediate-risk disease. Reduces recurrence but **not progression**; BCG (bacille Calmette–Guérin) with maintenance is superior in high risk.
3. **Optimised delivery (Au, 2001):** empty bladder, overnight fluid restriction, sodium bicarbonate (less drug breakdown), 40 mg in 20 mL; retain 1–2 hours.
4. **Device-assisted:** EMDA (electromotive drug administration) increases tissue uptake; chemohyperthermia at 42 °C (RITE [radiofrequency-induced thermochemotherapeutic effect], HIVEC [hyperthermic intravesical chemotherapy]).
5. **Sequential gemcitabine then MMC** for BCG-exposed disease.

### Other urological uses

- **Upper tract:** single bladder dose after nephroureterectomy (ODMIT-C: bladder recurrence 28% to 16% at 1 year). **UGN-101 (mitomycin gel)**, FDA (Food and Drug Administration) approved 2020 for low-grade UTUC (upper tract urothelial carcinoma): complete response 59%, ureteric stricture about 44%. UGN-102 for low-grade intermediate-risk bladder tumours.
- **Bladder-sparing chemoradiation:** 5-FU (5-fluorouracil) + MMC (BC2001: better locoregional control than radiation alone).
- **Antifibrotic:** injected after incision of recurrent bladder neck contracture or urethral stricture.

### Side effects

- Chemical cystitis (30–40%); **skin rash, palmar or genital dermatitis** (signature toxicity; wash after voiding); contracted bladder (about 5%); encrustation cystitis; myelosuppression rare.
- Extravasation through a perforation can cause pelvic necrosis.

**Diagram to draw:** timeline: TURBT, single MMC within 6 hours, then weekly induction ×6.

> **Key points to remember:**
> - Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.
> - Reduces recurrence, not progression.
> - Signature toxicity: skin rash; contracted bladder about 5%.

## SAQ 9 Mullerionosis of UB

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 2, Q3b

*"Mullerionosis of UB" is read as müllerianosis of the urinary bladder.*

### Definition

- Benign lesions of **müllerian-type epithelium** in the bladder wall, in three forms:

| Lesion | Tissue |
|---|---|
| **Endometriosis** (commonest) | Endometrial glands **and stroma** |
| **Endocervicosis** | Endocervical mucinous glands |
| **Endosalpingiosis** | Ciliated tubal epithelium |

- **Müllerianosis** strictly means two or more of these together.
- Women of reproductive age, often after **caesarean section**. Usually **posterior wall or dome** (vesicouterine pouch).
- Bladder is the commonest urinary tract site of endometriosis, ahead of the ureter.

### Pathogenesis

- **Implantation:** retrograde menstruation (Sampson) into the vesicouterine pouch.
- **Iatrogenic implantation** at caesarean or pelvic surgery (primary vs iatrogenic forms).
- **Metaplasia** of coelomic or subperitoneal müllerian remnants (the "secondary müllerian system").
- Lymphatic or vascular spread; direct extension of anterior uterine adenomyosis.

### Clinical features

- **Cyclical haematuria** or irritative LUTS (lower urinary tract symptoms) with menses; suprapubic pain, dysmenorrhoea, dyspareunia.
- Endocervicosis: non-cyclical mass or LUTS.
- **Also know:** menouria differential includes vesicouterine fistula (Youssef syndrome).

### Evaluation

- **Ultrasound** (transabdominal plus transvaginal): size and depth of a hypoechoic nodule.

- **MRI (magnetic resonance imaging):** extent; haemorrhagic foci **bright on T1**, fibrotic thickening dark on T2.
- **Cystoscopy:** bluish submucosal nodule with neovascularity, best seen during menses. Of **limited value**, because disease is mainly **extravesical** (from serosa inwards).
- **Biopsy:** excludes adenocarcinoma, especially with endocervicosis. Immunohistochemistry: ER/PR (oestrogen/progesterone receptor) positive; CD10 marks endometrial stroma.
- Ultrasound of upper tracts for ureteric involvement.
- **Differential:** adenocarcinoma, cystitis glandularis, nephrogenic adenoma, urachal remnant.

### Management

- **Hormonal suppression** (combined pill, progestins, GnRH [gonadotropin-releasing hormone] agonists): symptom relief; recurrence after stopping.
- **TUR (transurethral resection):** incomplete for transmural disease; perforation risk; recurs.
- **Partial cystectomy** (laparoscopic or robotic) with a margin: **primary treatment**. Ureteric stent or reimplantation if the orifice is involved.
- Excise other pelvic endometriosis with the gynaecologist.
- Malignant change (endometrioid or clear cell adenocarcinoma) is rare.

**Diagram to draw:** sagittal pelvis with a dome nodule growing from the vesicouterine pouch into the detrusor.

> **Key points to remember:**
> - Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
> - Cyclical haematuria; MRI bright on T1.
> - Disease is extravesical: cystoscopy limited, partial cystectomy preferred.

## SAQ 10 Methods of preoperative bowel preparation.

P3

**Type:** SAQ (10 marks)
**Source:** 2022-Summer Paper 3, Q3d

### Rationale

- Bowel used in urinary diversion is opened into a sterile urinary tract. Faecal spillage causes wound infection, abscess, leak and sepsis.
- Flora: aerobes (*E. coli*, *Enterococcus faecalis*); anaerobes (*Bacteroides*, *Clostridium*).

| Segment | Organisms per gram |
|---|---|
| Jejunum | $10–10^5$ |
| Distal ileum | $10^5–10^7$ |
| Ascending colon | $10^6–10^8$ |
| Descending colon | $10^{10}–10^{12}$ |

### Methods

**1. Dietary:** clear liquids the day before; clear drinks until 2 hours before surgery.

**2. Mechanical bowel preparation (MBP)** reduces faecal **amount**, so total bacteria, not concentration.

- **PEG (polyethylene glycol)–electrolyte** lavage: non-absorbable; preferred in elderly, renal or heart failure, cirrhosis.
- Sodium phosphate, magnesium citrate; enemas.
- Historical: 3-day purgation, whole-gut irrigation by nasogastric tube (9–12 L saline), mannitol (feeds bacteria).
- Harms: dehydration, electrolyte loss; liquid stool may spill more.

**3. Oral antibiotic preparation** reduces bacterial **concentration**.

- **Neomycin + erythromycin base** (Nichols–Condon, 1977: septic complications 43% with MBP alone vs 9% with added antibiotics); neomycin + metronidazole; kanamycin.
- Harms: diarrhoea, *C. difficile* colitis, candidiasis; useless in obstruction.

**4. Intravenous prophylaxis (most important):** anaerobic plus aerobic cover (e.g. cephalosporin + metronidazole) within 1–2 hours before incision; it must be in tissue before contamination.

**5. ERAS (enhanced recovery after surgery): no routine MBP**; prehabilitation, multimodal opioid-sparing analgesia, chewing gum, **alvimopan** (peripheral μ-opioid antagonist, 12 mg before surgery), early feeding and mobilisation.

### Evidence

- Colorectal meta-analyses and Cochrane: MBP does not reduce leak or abscess; may raise SSI (surgical site infection).
- Oral antibiotics added to MBP reduce SSI and leaks in large colectomy series.
- Small cystectomy trials: no difference without preparation.

### Who still needs full preparation

- Salvage cystectomy after radiation, or cystectomy for benign disease (radiation cystitis, fistula, neurogenic bladder); immunosuppression.

**Also know:** wound protectors reduce SSI; enema before transrectal biopsy.

**Diagram to draw:** bacterial load along the gut.

> **Key points to remember:**
> - Mechanical lowers amount; antibiotics lower concentration.
> - Nichols–Condon: neomycin + erythromycin.
> - Intravenous antibiotics before incision are essential; ERAS omits routine MBP.

## SAQ 11 Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder.

P3

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 3, Q3e

### Definition

- **CIS (carcinoma in situ, Tis):** flat, non-invasive, **high-grade** urothelial carcinoma. It is not "premalignant".
- Histology: severe atypia, loss of polarity, non-cohesive pleomorphic cells, **loss of umbrella cells** (dysplasia keeps them).
- **Types:** primary (isolated, about 10%), secondary or concurrent (with papillary tumour, about 90%).

### Pathogenesis

- **Flat pathway:** dysplasia → CIS → high-grade invasive cancer.
- Driven by **TP53, RB and PTEN** loss; **no FGFR3** mutation (unlike low-grade papillary tumours).
- **Field change:** multifocal; spreads to prostatic urethra, distal ureters and upper tract.

### Clinical importance

- Untreated, **40–83% progress** to muscle invasion. Primary CIS progresses in >50%.
- "CIS alone" hides invasion at cystectomy in up to 20%.
- **Second strongest prognostic factor after grade**; any CIS makes NMIBC (non-muscle-invasive bladder cancer) **high risk**.
- About 80% have **irritative symptoms**; it mimics cystitis.

### Diagnosis

- **Urine cytology:** most useful in CIS, because the cells are poorly cohesive. Positive cytology with normal cystoscopy needs mapping.
- Cystoscopy: red velvety patch or normal-looking mucosa; **blue-light cystoscopy** (hexaminolevulinate) detects about one-third more CIS; NBI (narrow-band imaging).
- **Biopsies:** cold-cup from suspicious and random sites; prostatic urethra; upper tract imaging.

### Management

- **BCG (bacillus Calmette–Guérin) is first line:**
- Induction: 6 weekly instillations.
- **Maintenance** (SWOG [Southwest Oncology Group] 2000: 3 weekly doses at 3 and 6 months, then 6-monthly for 3 years): median recurrence-free survival 76.8 vs 35.7 months.
- Complete response up to 84%; durable in about half.
- **Persistent CIS at 3 months:** second BCG course; 30–50% more respond.
- **BCG-unresponsive** (persistent or recurrent CIS within 12 months of adequate BCG): **radical cystectomy**.
- Unfit or unwilling: **pembrolizumab** (KEYNOTE-057: complete response 41%), nadofaragene firadenovec, gemcitabine–docetaxel.
- Prostatic urethral CIS: TURP plus BCG; ductal involvement needs cystoprostatectomy.

### Follow-up

- Cystoscopy and cytology 3-monthly for 2 years, then 6-monthly; upper-tract imaging.

**Diagram to draw:** flat versus papillary pathways of bladder cancer.

> **Key points to remember:**
>
> - Flat, high-grade; TP53/RB; no FGFR3.
> - Untreated 40–83% progress; always high risk.
> - BCG with maintenance; cystectomy if BCG-unresponsive.

## SAQ 12 Re-look (second-look) TUR bladder tumour

P4

**Asked as:**

- Re-look TUR bladder tumour. — *Summer 2017 Paper IV Q3d (SAQ)*
- Re-look TUR bladder tumour. — *Practice Set 5 Paper IV Q3d (earlier version) (SAQ)*

**Type:** SAQ (10 marks)

*Read as second-look (repeat or restaging) TURBT.*

### Definition

- A **planned second (restaging) TURBT (transurethral resection of bladder tumour)** of the primary site, including detrusor muscle, a few weeks after the first. It completes resection and corrects staging. It is a quality indicator of NMIBC (non-muscle-invasive bladder cancer) care.

### Rationale

- **TURBT is a staging operation first.** Completeness and muscle in the specimen are the quality markers.
- **Incomplete resection** — residual tumour is common:
- At the resection site in **≥40%** when re-resected within weeks; tumour at first check cystoscopy in up to 45%.
- **HG (high-grade) Ta:** residual up to 50%, upstaging up to 15%.
- **Understaging:** in **HG T1**, re-resection shows worse disease or upstaging to muscle invasion in **25–30%**. This rises to about half if the first specimen had **no muscle**.
- Understaging is about twice as likely without muscle.

### Indications (AUA [American Urological Association] / EAU [European Association of Urology])

| Recommended | Consider | Not needed |
|---|---|---|
| Incomplete first (initial) resection | HG Ta, large or multifocal (AUA) (residual up to 50%, upstaging up to 15%) | Low-grade Ta |
| **All T1 tumours** | Outside resection with no slides for review | Primary CIS (carcinoma in situ) |
| **No detrusor muscle** in specimen (except low-grade Ta and primary CIS) | | Clear muscle invasion |
| | | Cystectomy already planned |

### Timing

- **2–6 weeks** after the first TURBT (EAU); **within 6 weeks** (AUA), before BCG (bacille Calmette–Guérin). This allows healing yet avoids delaying intravesical BCG.

### Technique

- Resect the **primary-site scar and base deep into detrusor**, plus any residual tumour; send the base / specimens separately. Use blue light or NBI (narrow-band imaging) if available; it helps.

### Benefits / evidence

- Corrects stage and **changes treatment in about one-third**.
- **Divrik RCT (randomised controlled trial, 2010)** in T1: re-TUR improved recurrence-free and progression-free survival (lower recurrence and progression).
- Better BCG response with minimal residual tumour / when BCG follows re-TUR.

### Result-based plan (decision after re-TUR)

- **Residual T1 at re-resection** means (predicts) progression in up to **80%**: consider early cystectomy. `No tumour (T0) / Ta → BCG induction + maintenance Residual HG T1 → early cystectomy (or BCG with close review) Muscle invasion (≥T2) → treat as muscle-invasive: neoadjuvant chemotherapy + cystectomy`

### Recent advances

- **En bloc resection** gives muscle (more detrusor) more often, less residual tumour, and may reduce re-TUR. It is not yet a proven substitute.
- Expert pathology review changes management in about one-third.
- **Also know:** maximal TURBT before trimodality therapy.

**Diagram to draw:** flowchart: T1 or no muscle → re-TUR 2–6 weeks → T2: cystectomy pathway / residual T1: consider cystectomy / T0–Ta: BCG; and bladder wall showing first resection scar and deep second resection into detrusor.

> **Key points to remember:**
> - Re-TUR indications: all T1, incomplete resection, no muscle; consider HG Ta.
> - Timing 2–6 weeks; residual ≥40%; HG T1 upstaged (upstaging) in 25–30% (about half if no muscle / without muscle).
> - Residual T1 → progression (progresses) in up to 80%.

## SAQ 13 VI-RADS

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 4, Q3b

### Definition

- **VI-RADS (Vesical Imaging–Reporting and Data System):** a 5-point score for the likelihood of **muscle (detrusor) invasion** by bladder cancer on mpMRI (multiparametric magnetic resonance imaging). Panebianco et al., Eur Urol 2018.

- Standardises bladder MRI reporting, as PI-RADS (Prostate Imaging Reporting and Data System) does for the prostate.

### Technique

- Multiplanar **T2W (T2-weighted)**, **DWI (diffusion-weighted imaging)** with ADC (apparent diffusion coefficient) map, and **DCE (dynamic contrast-enhanced)** T1.
- Done **before TURBT (transurethral resection of bladder tumour)**: resection oedema mimics invasion. Distend the bladder adequately.

### Principle

- On T2W the detrusor is a **low-signal (dark) line**; urine, mucosa and submucosa are bright. The question is whether tumour **interrupts that dark line**.

### Scoring

| Score | Invasion | T2W | DCE | DWI |
|---|---|---|---|---|
| 1 | Highly unlikely | <1 cm, ± stalk; dark line intact | No early muscle enhancement | Muscle continuous |
| 2 | Unlikely | >1 cm with stalk and thick inner layer, or sessile; line intact | Inner layer enhances only | As 1 |
| 3 | Equivocal | No category 2 features; line intact | No muscle interruption | As 1 |
| 4 | Likely | **Line interrupted** | Enhancement extends into muscle | High signal into muscle |
| 5 | Highly likely | **Into perivesical fat** | Through wall into fat | High signal through wall and fat |

- **Cut-off ≥3 = suspicious for MIBC (muscle-invasive bladder cancer)**; the "equivocal" category already counts as positive.
- Sensitivity and specificity roughly 80–90%; DWI is the main functional sequence.

### Clinical uses

- Local staging before TURBT; earlier referral for neoadjuvant chemotherapy and cystectomy.
- Patients who cannot have iodinated contrast (renal impairment, allergy).
- **MRI-first pathway** (UK BladderPath trial) shortened time to correct treatment for MIBC.
- Response after neoadjuvant chemotherapy or bladder preservation: under study.

### Limitations

- Needs an experienced radiologist; post-TURBT or BCG (bacille Calmette–Guérin) inflammation; small, bladder-neck or diverticular tumours; poor for CIS (carcinoma in situ) and nodes; does not replace histology.

**Diagram to draw:** bladder wall with tumour breaching the dark detrusor line (VI-RADS 4).

> **Key points to remember:**
> - Five points; tests the dark muscle line on T2W.
> - ≥3 suspicious for MIBC; 4 line broken, 5 fat.
> - Scan before TURBT.
> - Panebianco 2018; accuracy about 80–90%.

## SAQ 14 Virtual cystoscopy

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 4, Q3a

### Definition and principle

- **Virtual cystoscopy (VC)** is a computer-generated, three-dimensional (3D) endoluminal view of the bladder. It is reconstructed from thin-section **CT (computed tomography)** or **MRI (magnetic resonance imaging)** data of a distended bladder.
- Software "flies through" the lumen using **surface or volume rendering**; multiplanar images also show wall thickness and extravesical spread.

### Technique

| Step | CT virtual cystoscopy | MR virtual cystoscopy |
|---|---|---|
| Distension | Air or CO2 through a catheter (negative contrast), or excretory-phase contrast (positive contrast, no catheter) | Urine-filled bladder; urine bright on T2 |
| Acquisition | Thin slices, supine and prone (moves dependent urine off the posterior wall) | 3D T2 or T1 contrast sequences |
| Drawback | Radiation; catheter for gas | Cost, time, motion artefact |

### Advantages

- Non-invasive; no anaesthesia.
- Sees where a scope cannot: through a **tight urethral stricture**, inside **diverticula**, beyond a large median lobe.
- With **CTU (CT urography)**, upper tracts and staging are imaged in one study (Chapter 5: high accuracy for bladder tumours if well distended).

### Limitations

- Detects polypoid tumours ≥5 mm well; misses **small and flat lesions, including CIS (carcinoma in situ)**.
- No colour or mucosal detail, **no biopsy and no treatment**.
- False positives: trabeculation, clot, debris, folds of an underdistended bladder, prostatic impression.

### Indications (selective)

- Unfit for, or refusing, cystoscopy; urethral stricture; diverticulum; bladder evaluation during CTU for haematuria.
- **Guideline position:** EAU (European Association of Urology) and AUA (American Urological Association) do not recommend VC in place of cystoscopy. Flexible cystoscopy remains the standard.

### Related advances

- **Also know:** MRI **VI-RADS (Vesical Imaging–Reporting and Data System)** scores muscle invasion (Chapter 6); blue-light and narrow-band cystoscopy improve CIS detection (Chapter 8); virtual ureteroscopy uses the same principle.

**Diagram to draw:** thin-slice CT of a gas-distended bladder, then a 3D endoluminal fly-through showing a polypoid tumour.

> **Key points to remember:**
> - 3D endoluminal reconstruction from thin-section CT or MRI.
> - Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
> - No biopsy; does not replace cystoscopy.
> - Useful beyond strictures and inside diverticula.

## SAQ 15 Bladder transplantation

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 4, Q3f

### Rationale

- After cystectomy, or when the bladder cannot be reconstructed, the usual substitute is **bowel**. Bowel in the urinary tract causes mucus, metabolic acidosis, stones, infection and malignancy risk.
- Chapter 135: two frontiers aim to **remove bowel from urinary reconstruction**: tissue engineering and bladder transplantation.

### Tissue engineering (why it has not succeeded)

- **Atala et al. (2006):** first seeded biodegradable construct for neurogenic augmentation. A later multi-institution trial showed **no significant gain in compliance or capacity** (Chapter 48).
- Engineered conduit trial: **half developed stenosis or stricture**, because of stiff material, poor cell coverage and **no blood supply** (Chapter 135).
- Engineered bladder: storage only (Chapter 111). See 2022-Summer Paper 4, Q2.

### Bladder transplantation: development (Chapter 135)

- A **VCA (vascularised composite allograft)**: the donor bladder is transplanted with its own blood vessels.
- Stepwise preclinical work at USC (University of Southern California): pigs, perfused cadavers, then brain-dead human donors.

### Technique (robotic)

1. **Robotic retrieval** of the bladder on its vesical vascular pedicles from the donor.
2. **Back-table vascular reconstruction**.
3. **Robotic implantation**: vessels to recipient iliac vessels; ureters joined to the graft.
4. **ICG (indocyanine green) fluorescence** confirms perfusion. Vascularity and mucosal integrity held for over 12 hours.

### First clinical case

- Reported in **2025** from Los Angeles: a combined kidney and bladder transplant from a deceased donor, in a dialysis patient who had lost most of his bladder.

### Problems

- **Lifelong immunosuppression**: infection, malignancy, drug nephrotoxicity. This is the major limitation (Chapter 135), especially after cancer.
- **Denervated graft**: stores urine; voiding may need straining or CIC (clean intermittent catheterisation).
- Rejection monitoring, donor supply, unknown long-term function.

### Likely candidates

- Already immunosuppressed patients (kidney transplant recipients) with an absent or non-functioning bladder.

**Diagram to draw:** donor bladder on its vascular pedicles anastomosed to recipient iliac vessels, ureters entering the graft.

> **Key points to remember:**
> - Goal: avoid bowel in the urinary tract.
> - Tissue-engineered bladders failed on blood supply and function.
> - Robotic VCA; first human case 2025.
> - Limits: immunosuppression and denervation.

## SAQ 16 Artificial urinary bladder

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 4, Q3d

*Read as a man-made bladder substitute (prosthetic, engineered or transplanted). Bowel neobladder: see Practice Set 4 Paper 2, Q3e.*

### Need

- Bowel substitutes absorb urine and secrete mucus, causing acidosis, mucus, stones, infection, perforation and cancer. The aim is a substitute that **is bladder-like, not bowel**.

### Requirements of an ideal substitute

- Low-pressure storage of adequate volume; continence.
- Complete, voluntary emptying; no reflux or upper-tract damage.
- Biocompatible: no encrustation, infection, rejection or cancer risk; durable.

### Approaches

| Approach | Principle | Status |
|---|---|---|
| **Alloplastic (prosthetic) bladder** | Synthetic reservoir (silicone, polymers) joined to ureters and urethra | Failed: encrustation, infection, leak at the tissue–prosthesis junction, migration, fibrosis |
| **Tissue-engineered bladder** | Autologous urothelial and muscle cells seeded on a biodegradable scaffold, omental wrap | Atala (2006): 7 patients improved; phase II trial showed no significant gain in capacity or compliance; experimental |
| **Engineered conduit** | Seeded scaffold as urinary conduit | Neo-Urinary Conduit trial: half developed stenosis or stricture; stiff, no blood supply |

| Approach | Principle | Status |
|---|---|---|
| **Bladder transplantation** | Robotic retrieval and transplant of a vascularised composite bladder allograft | Feasible in models and brain-dead donors; first human case reported in 2025; needs immunosuppression |

### Why progress is slow

- **Vascularity:** thick constructs die centrally.
- **Innervation and contraction:** an engineered bladder stores urine but cannot yet empty voluntarily.
- **Urine–material interface:** foreign material encrusts and becomes infected.
- Fibrosis and contracture.

### Alternatives in clinical use

- Augmentation (bowel, ureterocystoplasty, autoaugmentation); continent or orthotopic diversion; ileal conduit.
- **Also know:** neuroprosthetic emptying (sacral anterior root stimulator), artificial urinary sphincter, intraurethral valve-pump devices for underactive bladders.

### Future

- Bioprinted vascularised constructs, stem cells (urine-derived, iPSCs, induced pluripotent stem cells), smart scaffolds.

**Diagram to draw:** biopsy → cell culture → seeding on bladder-shaped scaffold → omental wrap → anastomosis to bladder base.

> **Key points to remember:**
> - Prosthetic bladders failed (encrustation, infection, leak).
> - Atala 2006 promising; later trial no gain; still experimental.
> - Barriers: vascularity, innervation, urine–material interface.

# Rapid Revision Sheet · Section 12

### Intestinal segments in urology and orthotopic neobladder (LAQ)

- Ileum and colon: hyperchloraemic acidosis ($NH_4Cl$ absorption).
- Jejunum: hyperkalaemia; stomach: hypochloraemic alkalosis.
- Continent diversion: GFR >35, pH ≤5.8, osmolality ≥600.
- Wallace has the lowest stricture rate; detubularisation gives low pressure.
- Sigmoid is the default neovagina; Yang–Monti when the appendix is absent.
- Detubularise and make a sphere; ≥500 mL; 500–600 mL by 1 year.
- Neobladder absolute bar: positive urethral margin, stricture, poor renal function.
- Studer and Hautmann (refluxing) are commonest; USC RCT: Studer = T pouch.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

### Bladder-preserving treatment of TCC (LAQ)

- WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
- Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
- Partial cystectomy: solitary <3 cm (dome tumour), no CIS, frozen-section margins, always PLND.
- TMT = complete TURBT + RT + radiosensitiser; select T2, no hydronephrosis, no CIS.
- BC2001: MMC/5-FU (mitomycin C/5-FU) + RT, 2-year locoregional DFS 67% vs 48%.
- Salvage cystectomy 10–30%; follow up the bladder for 10 years.

### Continent urinary diversion: continence principles, Mitrofanoff, orthotopic vs non-orthotopic (LAQ)

- Continence = low reservoir pressure (detubularise, sphere, ≥500 mL) against a reliable outlet; 500–600 mL by 1 year.
- Detubularised ileal sphere: low pressure, large capacity; continence depends on the rhabdosphincter.
- Neobladder needs eGFR (estimated glomerular filtration rate) >35–40 mL/min, a tumour-free urethral margin and ability to self-catheterise. Absolute bar: positive urethral margin, stricture, eGFR <40.
- 54–56 cm ileum, 15–20 cm from ICV: 40–44-cm U (sphere) plus 10–15 cm afferent limb.
- Refluxing end-to-side ureters; stricture <3%; kidneys preserved ~95%; refluxing ureteric anastomosis is acceptable (USC (University of Southern California) trial).
- Obstruction, not reflux, destroys kidneys; USC RCT: Studer = T pouch.
- Day continence 80–90% (about 90%); night incontinence 20–50% (up to half); women retain (20–30%).
- Sitting Valsalva voiding, timed voiding, night alarm.
- Eligible 80–90% men, 75% women; actual uptake about 9% in the US.
- Ileum/colon: hyperchloraemic acidosis; jejunum: hyperkalaemic; stomach: alkalosis.
- Mitrofanoff 1980: submucosal flap valve; appendix needs only a 2 cm tunnel.
- Yang–Monti when appendix absent or used for MACE (Malone 1990).
- Flap-valve continence >95%; stomal stenosis is the commonest complication.
- No diversion is proven best for quality of life; counsel all options.

**Muscle-invasive bladder cancer: management, neoadjuvant therapy and lymphadenectomy (LAQ)**

- NAC: cisplatin combinations only; ABC 5% absolute 5-year survival gain; SWOG 8710 pT0 38% vs 15%, P = 0.06.
- VESPER: ddMVAC beat GC; NIAGARA: perioperative durvalumab (adds durvalumab) improved event-free and overall survival.
- CheckMate 274: adjuvant nivolumab disease-free survival 20.8 vs 10.8 months.
- SWOG S1011: extended PLND gives no survival benefit over standard.
- UTUC: neoadjuvant before kidney loss; POUT disease-free survival HR 0.45; right-sided needs interaortocaval nodes.
- TMT: T2, no CIS, no hydronephrosis; salvage cystectomy 10–30%.
- Studer and Hautmann (refluxing) dominate: obstruction, not reflux, destroys kidneys.
- USC RCT: Studer = T pouch; T pouch more reoperations.
- Day continence 80–90%, night 20–50%; women retain 20–30%.

**Intravesical therapy for urothelial carcinoma, including combination and immediate post-TURBT instillation (LAQ)**

- Single instillation within 6 h (at most 24 h); never BCG immediately (SWOG S0337 gemcitabine).
- SWOG S0337: 35% vs 47% recurrence with gemcitabine 2 g.
- Withhold if perforation; best for low-risk solitary tumours.
- Only maintenance BCG reduces progression; SWOG 3-year schedule.
- NIMBUS reduced frequency fails.
- BCG-unresponsive: cystectomy; pembrolizumab (41% CR), nadofaragene, N-803.
- Gem/Doce: gemcitabine 1 g then docetaxel 37.5 mg; 84% cystectomy-free.
- Only EMDA-MMC improves BCG sequentially.
- N-803 + BCG: CIS complete response 71%.
- BCG + checkpoint: CREST, POTOMAC positive; ALBAN negative.

**CA bladder, etiology, newer investigations and treatment. (LAQ)**

- Smoking causes 30–40%; aromatic amines; NAT2 slow acetylators.
- Blue light finds more CIS; VI-RADS ≥3 suggests invasion; no marker replaces cystoscopy.
- High-risk NMIBC: BCG with 3-year maintenance.
- MIBC: cisplatin NAC + cystectomy; trimodality in selected.

**Management of T1G3 (high-grade T1) TCC of urinary bladder, including T1 with CIS in a young woman (LAQ)**

- Re-TURBT in all T1 at 2–6 weeks (within 6 weeks); residual T1 means progression up to 80%.
- BCG induction + 3-year maintenance (SWOG); only agent that reduces progression.
- T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
- Very high risk (CIS, LVI, variants, prostatic urethra): offer early cystectomy (92% 10-year disease-free survival).
- BCG-unresponsive: cystectomy; pembrolizumab or nadofaragene if unfit.
- Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

**Haemorrhagic cystitis: diagnosis and management (SAQ)**

- Acrolein (prevent with mesna); radiation endarteritis; BK virus.
- Exclude bladder cancer.
- Alum first; formalin only after cystogram.
- Hyperbaric oxygen for radiation cystitis (RICH-ART); embolise if unstable.

**BCG therapy and BCG failure (SAQ)**

- Only BCG reduces progression; maintenance is essential.
- Never immediately after TURBT; start at 2–4 weeks.
- Adequate BCG = 5/6 induction + 2/3 maintenance.
- Unresponsive: HG within 6 months (Ta/T1) or 12 months (CIS); HG T1 at 3 months.
- Timely cystectomy is standard; salvage CR rates halve by 12 months.

**Malignancy in augmented bladder and after urinary diversion (SAQ)**

- Ureterosigmoidostomy: highest risk; colonoscopy from year 5.
- Mostly adenocarcinoma at the anastomosis; earliest at 4 years after augmentation.
- Excise defunctioned anastomoses.
- Gastric segment, exstrophy, immunosuppression: highest risk.
- Haematuria needs cystoscopy; treatment is cystectomy.

**Optical coherence tomography in carcinoma bladder (SAQ)**

- Near-infrared interferometry ("ultrasound with light"); 10–20 μm resolution; 2–3 mm depth (P2: 1–2 mm).
- Loss of layering suggests tumour; boundary loss suggests T1.
- Role: Ta vs T1; cannot assess deep muscle.
- High sensitivity, moderate specificity, false positives from inflammation; investigational.

**Bladder diverticulum. (SAQ)**

- No muscle → stasis, UTI, stones, cancer.
- Acquired type: treat the BOO first.
- Tumour invading beyond lamina propria is T3a.

**Intravesical gene therapy. (SAQ)**

- Nadofaragene: non-replicating adenovirus + interferon-α2b + Syn3, every 3 months.
- CIS CR 53% at 3 months, 24% at 12 months.
- Cretostimogene: replication-competent oncolytic adenovirus + GM-CSF.

**Von Bruns Nest (SAQ)**

- Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
- Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
- The nested variant of urothelial carcinoma mimics von Brunn nests.

**Mitomycin (SAQ)**

- Alkylating antibiotic; 40 mg; single dose within 6 hours of TURBT.
- Reduces recurrence, not progression.
- Signature toxicity: skin rash; contracted bladder about 5%.

**Mullerionosis of UB (SAQ)**

- Endometriosis, endocervicosis, endosalpingiosis; ≥2 = müllerianosis.
- Cyclical haematuria; MRI bright on T1.
- Disease is extravesical: cystoscopy limited, partial cystectomy preferred. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Methods of preoperative bowel preparation. (SAQ)**

- Mechanical lowers amount; antibiotics lower concentration.
- Nichols–Condon: neomycin + erythromycin.
- Intravenous antibiotics before incision are essential; ERAS omits routine MBP.

**Pathogenesis, Clinical importance and Management of Carcinoma in situ urinary bladder. (SAQ)**

- Flat, high-grade; TP53/RB; no FGFR3.
- Untreated 40–83% progress; always high risk.
- BCG with maintenance; cystectomy if BCG-unresponsive. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Re-look (second-look) TUR bladder tumour (SAQ)**

- Re-TUR indications: all T1, incomplete resection, no muscle; consider HG Ta.
- Timing 2–6 weeks; residual ≥40%; HG T1 upstaged (upstaging) in 25–30% (about half if no muscle / without muscle).
- Residual T1 → progression (progresses) in up to 80%.

**VI-RADS (SAQ)**

- Five points; tests the dark muscle line on T2W.
- ≥3 suspicious for MIBC; 4 line broken, 5 fat.
- Scan before TURBT.
- Panebianco 2018; accuracy about 80–90%.

**Virtual cystoscopy (SAQ)**

- 3D endoluminal reconstruction from thin-section CT or MRI.
- Good for polypoid tumours ≥5 mm; misses flat lesions and CIS.
- No biopsy; does not replace cystoscopy.
- Useful beyond strictures and inside diverticula.

**Bladder transplantation (SAQ)**

- Goal: avoid bowel in the urinary tract.
- Tissue-engineered bladders failed on blood supply and function.
- Robotic VCA; first human case 2025.
- Limits: immunosuppression and denervation.

**Artificial urinary bladder (SAQ)**

- Prosthetic bladders failed (encrustation, infection, leak).
- Atala 2006 promising; later trial no gain; still experimental.
- Barriers: vascularity, innervation, urine–material interface. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 13 · Kidney & UTUC

## Long answer questions (LAQ)

### LAQ 1 Nephron-sparing surgery for renal masses: indications, approaches and recent advances

P1 · P2 · P3 · P4 **merged from 4**

*Asked as:* P1 LAQ "Embryology and anatomy of kidney; nephron-sparing surgery in small renal mass and renal malignancy" · P2 LAQ "Nephron-sparing surgery / partial nephrectomy in RCC: role, indications and approaches" · P3 LAQ "Nephron-sparing surgery for renal masses: methods, recent advances and neoadjuvant therapy" · P4 SAQ "Nephron Saving Surgery (NSS)"

*"Nephron sparing nephrectomy" and "nephron saving surgery" are read as nephron-sparing surgery (NSS): partial nephrectomy and ablation. "Nepron" is read as nephron.*

#### Embryology of the kidney

- Three successive kidneys form from **intermediate mesoderm**, in cranial-to-caudal order:

| Kidney | Time and site | Fate |
|---|---|---|
| Pronephros | Week 3–4, cervical | Non-functional; regresses |
| Mesonephros | Week 4, thoracolumbar | Functions briefly; tubules become efferent ductules; **mesonephric (Wolffian) duct** persists |
| **Metanephros** | Week 5, sacral | **Definitive kidney** |

- **Reciprocal induction:** the **ureteric bud (UB)** grows from the caudal Wolffian duct (about week 5) into the **metanephric blastema**.
- UB branches into ureter, pelvis, calyces and collecting ducts.
- Blastema, induced by UB tips, forms nephrons (glomerulus to distal tubule).
- **Molecular control:** GDNF (glial cell line-derived neurotrophic factor) from the blastema acts on RET on the UB; SLIT2/ROBO2 restricts GDNF to one bud. Also WT1, PAX2.
- Urine production is established by 14 weeks; nephrogenesis ends before birth.
- **Ascent and rotation (weeks 6–9):** the kidney rises from the pelvis to the lumbar region and rotates 90° so the hilum faces medially. It takes successively higher arteries; persistent lower ones become accessory arteries.
- **Clinical correlation:** no UB gives agenesis; abnormal induction gives MCDK (multicystic dysplastic kidney); failed ascent gives ectopia; fused lower poles trapped by the IMA (inferior mesenteric artery) give horseshoe kidney; also malrotation, duplication.

#### Surgical anatomy of the kidney

- **Size and position:** about 11 cm, 125–170 g; T12–L3, lies on the cone-shaped psoas, so the upper pole is more medial and posterior, and the kidney is angled 30–50° behind the coronal plane. The right kidney is 1–2 cm lower (liver).
- **Relations:**

| Aspect | Right | Left |
|---|---|---|
| Superior | Liver, adrenal, 12th rib | Spleen, stomach, adrenal, 11th–12th ribs |

| Aspect | Right | Left |
|---|---|---|
| Medial/anterior | Duodenum (2nd part), hepatic flexure | Pancreatic tail with splenic vessels, splenic flexure, jejunum |
| Posterior (both) | Diaphragm, psoas, quadratus lumborum, subcostal nerve | Same |

- **Coverings (inside out):** true capsule, perirenal fat, **Gerota fascia**, pararenal fat. Gerota fascia is closed above and laterally but open below. So it contains tumour and abscess (collections), yet fluid tracks into the pelvis.
- **Hilum (front to back):** renal vein, artery, pelvis, then the posterior segmental artery.
- **Vascular anatomy (the basis of partial nephrectomy)**
- **Arteries:** renal artery from the aorta at L1–L2; the right passes behind the IVC (inferior vena cava).
- **Five segmental end arteries**: posterior (first branch, behind the pelvis) plus four from the anterior division (apical, upper, middle, lower). With no anastomoses, occlusion causes segmental infarction, but each segment can be clamped or resected separately. Then interlobar (columns of Bertin), arcuate, interlobular arteries.
- **Accessory arteries** in 25–28%, end arteries too.
- **Brödel line:** an avascular plane between anterior (two-thirds) and posterior (one-third) territories, just behind the lateral border. Incision here bleeds least (anatrophic nephrolithotomy).
- **Veins anastomose freely**, so a segmental vein can be ligated safely. The long left renal vein receives adrenal, gonadal and lumbar veins.
- **Parenchyma / internal structure:** cortex and 8–18 pyramids; 0.4–1.2 million nephrons per kidney.
- **Collecting system:** papilla, minor calyces (7–9), major calyces, pelvis. Puncture through the **calyceal fornix**, not the infundibulum (Sampaio).
- **Also know:** lymph drains to para-aortic (left) or paracaval/interaortocaval nodes (right); sympathetic T8–L1.

## Small renal mass (SRM)

- **Definition:** an enhancing renal mass **≤4 cm (cT1a)**, mostly incidental.
- **Biology:** 20–30% of masses ≤4 cm and about 40% ≤2 cm are benign (oncocytoma, fat-poor angiomyolipoma) (P2: about **20%** of enhancing cT1 masses are benign); most cancers are indolent (<2% of the cancers metastasise). About 20% benign, 60% indolent, 20% potentially aggressive.

## Definition and rationale of NSS

- NSS (nephron-sparing surgery) is complete local removal of the tumour with **maximal preservation of functioning parenchyma**. PN (partial nephrectomy) is the standard form; thermal ablation is the non-excisional form.
- **Why spare nephrons:** RN (radical nephrectomy) causes CKD (chronic kidney disease), which raises cardiovascular events and death. PN gives equal cancer control for small masses (cT1a) with better function and keeps about **90%** of global function (RN 65–70%).
- **CKD from surgery alone** is stable, with near-normal survival if the new GFR (glomerular filtration rate) stays >45; **CKD from medical disease** (diabetes, hypertension) progresses.

So NSS matters most when medical renal disease coexists.

- **Concept:** function after PN depends mainly on the **parenchyma preserved** (preserved parenchymal volume), less on warm ischaemia time.

### Indications for nephron-sparing surgery

| Category | Situations |
|---|---|
| **Imperative (absolute)** | Solitary (functioning) kidney; bilateral tumours (bilateral RCC, renal cell carcinoma); RN would mean dialysis |
| **Relative** | Contralateral kidney threatened: stones, renal artery stenosis, hydronephrosis, diabetes, (severe) hypertension, proteinuria; hereditary RCC |
| **Elective** | Localised tumour with a normal contralateral kidney; **cT1a (≤4 cm) with normal contralateral kidney: standard of care**; cT1b–T2 when feasible |
| **Benign** | Symptomatic AML (angiomyolipoma); focal atrophy in duplex kidney; infected calyceal diverticulum |

- Now extended (increasingly used) for selected cT1b–T2, complex or hilar tumours when function matters (better function).
- **Relative contraindications: <20% of nephron mass retained**; pedicle encasement; collecting-system invasion; **venous tumour thrombus**; cT4; cN1 (nodes).

### Role of NSS by clinical setting

| Setting | Role of NSS |
|---|---|
| **cT1a (≤4 cm)** | PN is the **standard of care** when intervention is needed; RN is overtreatment |
| cT1b–T2 | PN when feasible and function matters; otherwise debatable |
| Hereditary RCC | Observe until the largest tumour is **3 cm**, then enucleate (except HLRCC [hereditary leiomyomatosis and RCC]: wide excision) |
| Bilateral tumours, solitary kidney | Imperative PN, staged if bilateral; avoids dialysis |
| Frail, comorbid, cT1a <3 cm | Thermal ablation or active surveillance |
| Locally advanced, thrombus, cN1, metastatic | Generally RN or cytoreductive surgery |

### Choosing the option for a small renal mass (AUA, American Urological Association, 2021)

| Option | Role |
|---|---|
| **Active surveillance** | <2 cm or predominantly cystic; elderly, comorbid (frail patients), life expectancy <5 years. Imaging every 3–6 months. Intervene if >3 cm or growth >5 mm a year (DISSRM registry data: almost no metastases) |
| **Thermal ablation** (cryoablation, radiofrequency) | cT1a **<3 cm**, percutaneous, biopsy first. Higher local recurrence than PN, salvaged by repeat ablation |
| **Partial nephrectomy** | **Prioritise PN for cT1a** needing treatment. Imperative: solitary kidney, bilateral tumours, CKD or proteinuria; also familial RCC, young patients |

| Option | Role |
|---|---|
| **Radical nephrectomy** | Preferred only if **all three** apply: high complexity; no CKD or proteinuria; normal contralateral kidney with new (post-RN) eGFR likely **>45**. Otherwise PN |

- **PN principles:** negative margin (width does not matter); maximise preserved parenchyma; avoid long warm ischaemia; enucleation for familial or multifocal tumours.

## Preoperative evaluation and planning

- Multiphase CT (computed tomography) or MRI (magnetic resonance imaging) with 3D (three-dimensional) reconstruction; chest imaging; eGFR (estimated glomerular filtration rate) and proteinuria (CKD stage).
- **Renal mass biopsy (RMB)** when it may change management, and before (preferred) or at ablation (non-diagnostic about 14%). Genetic counselling if age ≤46, multifocal or bilateral.
- **Nephrometry scores** predict complexity, complications, ischaemia time and conversion:

| R.E.N.A.L. component | Scored as |
|---|---|
| **R**adius | ≤4, 4–7, ≥7 cm |
| **E**xophytic/endophytic | ≥50% exophytic, <50%, entirely endophytic |
| **N**earness to collecting system | ≥7, 4–7, ≤4 mm |
| **A**nterior/posterior | Descriptor |
| **L**ocation to polar lines | Polar, crossing, central |
| - Total: **low 4–6, moderate 7–9, high 10–12** complexity. **PADUA** (Preoperative Aspects and Dimensions Used for an Anatomical) and C-index are alternatives. 3D models help (shorten ischaemia in a randomised trial). | |

## Methods of NSS

| Group | Methods |
|---|---|
| Surgical PN | Open, laparoscopic, robot-assisted (RAPN) |
| Excision technique | Standard (wedge) resection with a rim; enucleation; polar resection or heminephrectomy (≥30% of parenchyma; large polar tumours) |
| Thermal ablation | Cryoablation, RFA (radiofrequency ablation), MWA (microwave ablation) |
| Developmental | SBRT (stereotactic body radiotherapy), HIFU (high-intensity focused ultrasound), irreversible electroporation |

## Surgical approaches for PN

| Approach | Features |
|---|---|
| **Open, extraperitoneal flank** (supracostal 11th/12th rib, 8 cm mini-flank) | Posterior exposure; solitary kidney, complex or central tumours, cold ischaemia |
| **Open transperitoneal** (subcostal, chevron, midline) | Bilateral or large tumours, vascular reconstruction |
| **Laparoscopic transperitoneal** | Large space (more space), easier suturing; anterior tumours |

| Approach | Features |
|---|---|
| **Laparoscopic retroperitoneal** | No bowel mobilisation, direct hilum; posterior tumours |
| **Robot-assisted (RAPN)** | Easier suturing, shorter warm ischaemia, fewer conversions; transperitoneal for anterior and retroperitoneal for posterior tumours; equal oncology |

- Open PN suits complex tumours in a solitary kidney.
- First laparoscopic PN 1993 (Winfield).
- **Robotic vs laparoscopic:** shorter warm ischaemia, fewer conversions. **Robotic vs open:** less blood loss, shorter stay, fewer complications; same margins and eGFR.
- Haemostasis: sliding-clip renorrhaphy, barbed sutures, sealants.

## Operative steps of PN

1. Flank extraperitoneal or transperitoneal (open), laparoscopic or robotic approach.
2. **Early hilar control** of artery, vein and accessory vessels, even if off-clamp is planned.
3. Expose and defat the kidney; **intraoperative ultrasound** maps the tumour borders and depth.
4. Vascular control: clamp the artery (± vein); selective or off-clamp for small tumours. Use **ice slush** (cold ischaemia) if ischaemia is expected to be **≥25–30 minutes** (solitary kidney, CKD).
5. Excise with a thin rim of normal parenchyma, or **enucleate** along the pseudocapsule (hereditary, familial, multifocal). **A negative margin is the priority; its width does not matter.**
6. **Two-layer renorrhaphy** (sliding clip): inner layer closes vessels and collecting system; outer capsular layer over a haemostatic bolster.
7. Unclamp; check haemostasis; close Gerota fascia; drain.

## Enucleation vs standard resection

| | Standard (wedge) resection | Enucleation |
|---|---|---|
| Plane | Through normal parenchyma, thin rim | Along the tumour pseudocapsule |
| Advantage | Fewer positive margins | Less bleeding, maximal parenchyma, eases zero ischaemia |
| Use | Default | Hereditary RCC (e.g. VHL [von Hippel–Lindau]), familial, multifocal, solitary kidney, severe CKD |

## Vascular control and ischaemia

| Strategy | Verdict |
|---|---|
| En bloc or artery-only clamp | Equivalent; traditional warm ischaemia <25–30 minutes; ice slush if longer is expected |
| Early unclamping (after inner layer) | May reduce pseudoaneurysm |
| Selective or super-selective segmental clamping | Avoids global ischaemia; no functional gain in a randomised trial |
| Off-clamp (zero ischaemia) | Small exophytic tumours; more blood loss; no long-term eGFR benefit in randomised trials |

| Strategy | Verdict |
|---|---|
| Parenchymal compression | Peripheral tumours |
| Hypothermia, mannitol | No benefit with a normal contralateral kidney |

- Function depends mainly on the **quantity of parenchyma preserved**; ischaemia is secondary within accepted limits.
- **Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes. RAPN: shorter ischaemia than laparoscopy, less blood loss than open.

### Thermal ablation (ablative NSS)

- **Cryoablation** (double freeze–thaw), **RFA**, **MWA**; mostly **percutaneous** (CT or ultrasound guided).
- AUA 2021: alternative for **cT1a <3 cm**, especially elderly, comorbid, solitary kidney or VHL; biopsy before (preferred) or at ablation.
- **Cryoablation:** cell death at ≤−40 °C; iceball 5–10 mm beyond the tumour. **RFA/MWA:** heat causes coagulative necrosis.
- Success = loss of enhancement on CT/MRI. Local recurrence is higher than PN but treatable by repeat ablation (equal after repeat ablation).
- **Also know:** SBRT for non-surgical candidates; active surveillance for frail patients.

### Special situations

- **Hereditary RCC (VHL):** observe until the largest tumour is **3 cm**, then enucleate (except HLRCC: wide excision).
- **Bilateral tumours:** bilateral PN, usually staged; if one side needs RN, do the PN first.
- **Solitary kidney:** temporary dialysis about 3.5%; local recurrence 3–5%.
- **Also know:** bilateral Wilms tumour; kidney-sparing upper-tract urothelial cancer surgery.

### Role of neoadjuvant therapy

- **Aim:** downsize a tumour so that PN becomes possible, mainly an **imperative indication** (solitary kidney, bilateral tumours) where radical nephrectomy means dialysis.
- **TKIs (tyrosine kinase inhibitors):** pazopanib, sunitinib, axitinib. Tumour volume falls **10–30%**; resection becomes possible in **40–45%**, including PN candidates. Best in clear cell RCC.
- **Toxicity:** serious adverse events about 20%, mainly **wound healing**. Stop TKIs several half-lives before and after surgery.
- **Checkpoint inhibitors:** nivolumab is safe but shrinks little; trials ongoing.
- **Status:** not standard (trials only); **upfront surgery whenever feasible**. Adjuvant pembrolizumab (KEYNOTE-564) suits high-risk resected disease.

### Complications of PN

- **Urine leak** (drain, then stent); **haemorrhage, pseudoaneurysm or AV (arteriovenous) fistula** (**delayed haematuria**: angioembolisation); AKI (acute kidney injury), renal insufficiency; temporary dialysis in a solitary kidney.
- **Positive margin 0–7%:** surveillance, not reoperation.
- **Hyperfiltration injury** when >50% of a solitary kidney is lost: proteinuria, FSGS (focal segmental glomerulosclerosis); give an ACE (angiotensin-converting enzyme) inhibitor.

## Follow-up

- AUA risk-based imaging; eGFR and urine protein at every visit.

## Outcomes and evidence

- Elective PN: **local recurrence 1–2%**, cancer-free survival >90%; imperative 3–5%. Margin width does not matter if negative.
- **EORTC (European Organisation for Research and Treatment of Cancer) 30904 (2011):** RN vs PN for ≤5 cm tumours: PN better function, similar oncology (cancer control); overall survival slightly favoured RN (10-year overall survival 81% RN vs 76% PN), but the trial is flawed.
- **DISSRM (Delayed Intervention and Surveillance for Small Renal Masses) registry:** surveillance almost never led to metastasis; about 24% had delayed intervention.

## Recent advances

| Area | Advance and evidence |
|---|---|
| **Planning** | R.E.N.A.L. and PADUA scores; **3D models**: randomised trial showed shorter operation and ischaemia, less blood loss |
| **Platform** | **RAPN:** shorter warm ischaemia than laparoscopic; less blood loss than open. **Single-port robot:** shorter stay, less opioid |
| **Vascular control** | Off-clamp, segmental, selective or super-selective clamping (guided by **NIRF (near-infrared fluorescence)** with ICG, indocyanine green), early unclamping. **Randomised trials: no long-term eGFR gain** |
| **Excision** | **Enucleation** along the pseudocapsule: less blood loss; hereditary, multifocal, solitary kidney |
| **Reconstruction** | Sliding-clip and **barbed suture** renorrhaphy; single-layer gives better function, fewer pseudoaneurysms; sutureless with sealants in selected |
| **Neoadjuvant** | TKI downsizing in imperative cases (see above) |
| Outcome reporting | **Trifecta:** negative margin + no major complication + warm ischaemia ≤25 minutes |

**Diagrams to draw:** segmental arteries with the Brödel line on a transverse kidney section; clamped renal artery, tumour excised with a rim, two-layer renorrhaphy; RAPN with selective segmental clamp and renorrhaphy.

**Key points to remember:**

- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; veins anastomose; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard for cT1a, ablation <3 cm (after biopsy), surveillance <2 cm or frail.
- Imperative, relative, elective; elective PN is standard for cT1a.
- AUA: RN only if complex + no CKD + new GFR likely >45.
- R.E.N.A.L. 4–6 / 7–9 / 10–12.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Preserved parenchyma, not clamp time, governs function; trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Enucleation for hereditary/multifocal.
- Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials; 3D models: randomised benefit.
- EORTC 30904: PN better function, similar cancer control.
- Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.

## LAQ 2 Metastatic RCC: assessment and management

P2 · P3 **merged from 2**

*Asked as:* P2 LAQ "Metastatic RCC (mRCC): assessment and management, including recent concepts" · P3 SAQ "Discuss the management of Metastatic Renal Cell Carcinoma."

### Introduction

- About one-third of RCC (renal cell carcinoma) patients have metastases at diagnosis; 20–40% of localised tumours metastasise later.
- Five-year survival is about 12%, but median survival now approaches 3–5 years.
- Treatment moved from cytokines to VEGF (vascular endothelial growth factor)-targeted TKIs (tyrosine kinase inhibitors), and now to ICI (immune checkpoint inhibitor) combinations. Surgery has become selective.

### Assessment (before treatment)

- **Clinical:** symptoms of metastases (cough, bone pain, neurological deficit), weight loss; paraneoplastic syndromes (hypercalcaemia, polycythaemia, hypertension, Stauffer syndrome); performance status (Karnofsky or ECOG [Eastern Cooperative Oncology Group]); comorbidity.
- **Laboratory:** haemoglobin, corrected calcium, neutrophils, platelets, LDH (lactate dehydrogenase), ALP (alkaline phosphatase), creatinine, liver tests.
- **Imaging:** contrast CT (computed tomography) of chest, abdomen and pelvis (or MRI, also for IVC [inferior vena cava] thrombus). Bone scan only with bone pain, raised ALP or poor performance status; brain MRI (magnetic resonance imaging) with neurological symptoms; that is, bone scan or brain MRI only when symptoms or tests suggest spread; PET not routine.
- **Histology:** nephrectomy specimen or **percutaneous biopsy** before systemic therapy. It separates clear cell from non-clear cell RCC and shows sarcomatoid change, which alters drug choice.

### Prognostic risk groups (decide first-line therapy)

| Adverse factor | MSKCC (Memorial Sloan Kettering), cytokine era | IMDC (International Metastatic RCC Database Consortium), TKI era |
|---|---|---|
| Karnofsky <80% | Yes | Yes |
| Haemoglobin below lower limit of normal | Yes | Yes |
| Corrected calcium raised | >10 mg/dL | > upper limit of normal (ULN) |
| LDH >1.5 × ULN | Yes | No |
| Time factor | No prior nephrectomy | <1 year from diagnosis to systemic therapy |
| Neutrophils, platelets > ULN | No | Yes (two factors) |

- IMDC 6 factors; MSKCC model: 5 factors including LDH.
- Favourable 0 factors, intermediate 1–2, poor ≥3. IMDC median survival: 43.2, 22.5 and 7.8 months.

### Management: principles

- Multidisciplinary care: systemic therapy is the backbone; surgery and radiotherapy are added selectively.

### 1. First-line systemic therapy, clear cell RCC (NCCN [National Comprehensive Cancer Network] and EAU [European Association of Urology])

| IMDC risk | Preferred | Alternatives (if IO, immuno-oncology, not possible) |
|---|---|---|
| Favourable | Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib | Sunitinib or pazopanib; active surveillance for slow, low-volume (indolent) disease |
| Intermediate/poor | The same three, **or nivolumab + ipilimumab** | Cabozantinib; sunitinib; pazopanib |

- **Rationale:** VHL (von Hippel–Lindau) loss lets HIF (hypoxia-inducible factor) accumulate and drive VEGF (the TKI target). ICIs block PD-1 (programmed death-1) or CTLA-4 (cytotoxic T-lymphocyte antigen-4), releasing T-cell brakes and giving durable responses.
- **Trials (all versus sunitinib):** CheckMate 214 (nivolumab + ipilimumab: OS [overall survival] HR [hazard ratio] 0.63; overall survival benefit in intermediate/poor risk); KEYNOTE-426 (pembrolizumab + axitinib); CheckMate 9ER (nivolumab + cabozantinib); CLEAR (pembrolizumab + lenvatinib: PFS [progression-free survival] 23.9 vs 9.2 months). Benefit is independent of PD-L1 (programmed death ligand-1) status, so PD-L1 does not select patients.

### 2. Later lines: switch mechanism

- After an IO combination: a VEGF TKI not used before (cabozantinib, axitinib, tivozanib) or **belzutifan** (HIF-2α inhibitor). IO rechallenge is not advised (CONTACT-03 negative).
- After both IO and TKI: **belzutifan**; LITESPARK-005 showed better PFS than everolimus.
- After TKI alone: nivolumab (CheckMate 025) or cabozantinib (METEOR), both with OS gain.

### 3. Cytoreductive nephrectomy (CN): from routine to selective

- Cytokine era: SWOG 8949 and EORTC (European Organisation for Research and Treatment of Cancer) trials showed longer survival with CN + interferon-α.
- **CARMENA (2018):** sunitinib alone was non-inferior to CN + sunitinib in MSKCC intermediate/poor risk (median OS 18.4 vs 13.9 months).
- **SURTIME:** deferred CN after sunitinib was safe; OS favoured deferral.
- **Now:** immediate (upfront) CN only for fit patients with low metastatic burden who do not need immediate systemic therapy; **deferred CN** in responders. Avoid in poor risk, poor performance status or brain metastases.
- Palliative nephrectomy or angioembolisation for haematuria (bleeding) or pain.

### 4. Metastasectomy and local therapy (radiotherapy)

- Complete resection of solitary or few metastases gives 5-year survival of 35–50%. Better with complete resection, solitary lung lesion, disease-free interval >1 year, metachronous disease.
- **SBRT (stereotactic body radiotherapy)** or ablation for oligometastases; radiosurgery for brain metastases; palliative radiotherapy or surgery for cord compression and weight-

bearing bones; palliative radiotherapy for bone pain.

### 5. Adjuvant therapy after complete resection

- **KEYNOTE-564:** pembrolizumab for 1 year after surgery in high-risk clear cell RCC, including **M1 NED** (no evidence of disease after complete metastasectomy): disease-free survival HR 0.63; FDA (Food and Drug Administration) approval 2021; a later report showed OS benefit.

### 6. Supportive care and special groups (non-clear cell RCC)

- Zoledronic acid or denosumab for bone metastases. Immune-related adverse events (colitis, hepatitis, hypophysitis, thyroiditis) need steroids and drug interruption.
- Non-clear cell RCC: clinical trials preferred; cabozantinib or sunitinib; gemcitabine + carboplatin (platinum) for collecting duct carcinoma.
- **Also know:** high-dose IL-2 (interleukin-2) gives durable complete response in 7–9% (another source: in a few percent); temsirolimus for poor risk; chemotherapy and hormones are ineffective.

### Follow-up

- CT every 8–12 weeks on therapy (response by RECIST [Response Evaluation Criteria in Solid Tumours]); monitor thyroid, liver, blood pressure and cortisol.

### Recent advances

- Belzutifan, also approved for VHL-associated tumours; neoadjuvant ICI trials; biomarkers to choose IO/IO or IO/TKI.

### Algorithm

```
Metastatic RCC -> biopsy + IMDC risk -> IO/TKI (all risks) or nivo + ipi (intermediate/poor) -> deferred CN / metastasectomy / SBRT in responders -> progression: switch class (TKI, belzutifan)
```

**Diagram to draw:** the flowchart above (metastatic RCC, biopsy and IMDC risk, IO combination ± deferred CN in responders, metastasectomy or SBRT, switch drug class on progression).

> **Key points to remember:**
>
> - IMDC (6 factors): Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
> - IO/TKI combinations for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
> - CARMENA and SURTIME: no routine upfront CN.
> - Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.
> - Belzutifan after IO and TKI.

## LAQ 3 Upper tract urothelial carcinoma

P2 · P3 **merged from 2**

*Asked as:* P3 LAQ "Upper Tract Urothelial Carcinoma (including TCC of Ureter): Diagnosis, Staging, Minimally Invasive, Endoscopic and Radical Management" · P2 LAQ "Etiopathogenesis and management of Upper tract TCC"

*TCC (transitional cell carcinoma), including TCC of the ureter, is taken as UTUC (upper tract urothelial carcinoma).*

### Introduction and epidemiology

- **UTUC** is 5–10% of urothelial cancers. Pelvicalyceal tumours are about twice as common as ureteric ones. Incidence is rising, **driven by ureteric tumours**.
- Peak age 70–90; male:female 2:1.
- About 60% are invasive at diagnosis, because the upper tract has a thin muscle wall. Concurrent bladder cancer in 17%.
- Bladder recurrence follows in 22–47%, mostly from downstream seeding.
- **Site (ureter): distal ureter 70%, mid 25%, proximal 5%**, explained by downstream implantation.
- Standard is **RNU (radical nephroureterectomy) with bladder cuff**; kidney-sparing versus radical surgery is chosen by **risk**.
- URS (ureteroscopy) biopsy is small and shallow, so it **cannot stage**. Stage is inferred from grade, endoscopic appearance, imaging and cytology together.

### Aetiology (etiopathogenesis)

| Factor | Key facts and mechanism |
|---|---|
| **Tobacco** | Commonest and most important modifiable factor; risk 2.5–7-fold. Aromatic amines are activated to mutagenic N-hydroxylamines |
| **Aristolochic acid** | Balkan endemic nephropathy (UTUC 40% of renal tumours there) and Chinese herb nephropathy. Aristolactam–DNA adducts cause A:T to T:A transversions in TP53. Familial but **not inherited**; often bilateral |
| **Lynch syndrome** (HNPCC, hereditary nonpolyposis colorectal cancer) | Germline MMR (mismatch repair) mutation: MLH1, MSH2, MSH6, PMS2. Risk ~14-fold, highest with MSH2; found in 5–9% of screened UTUC |
| **Occupation** | "Amino tumours": aniline dyes, benzidine, β-naphthylamine (dyes, rubber, petrochemicals); latency up to 20 years |
| **Analgesic abuse** | Phenacetin: papillary necrosis; capillarosclerosis is pathognomonic; history in 11% of ureteric tumours |
| **Arsenic** | Well water in south-west Taiwan (blackfoot disease) |

- **Also know:** cyclophosphamide (acrolein); chronic infection and stones (squamous carcinoma); alcohol.
- **Lynch screening:** suspect if young (age <60), or a personal or family history of Lynch-spectrum cancer (colon, endometrium). Test the tumour for MSI (microsatellite instability) or MMR immunohistochemistry, then germline testing.

## Pathogenesis

- Two theories both operate. **Intraluminal seeding** (one clone spreading downstream) explains distal predominance and frequent bladder recurrence (22–47%). **Field effect / field change** (carcinogen in urine, independent clones) explains multifocality.
- Sequence: hyperplasia, dysplasia, CIS (carcinoma in situ). Variant histology in ~25%. Molecular subtypes include FGFR3-mutant (low grade), TP53/MDM2-mutant and hypermutated (Lynch).

## Pathology and spread

- 85% papillary, 15% sessile; WHO (World Health Organization) 2004/2016 low or high grade. Ureteric tumours are low grade and low stage in 55–75%.
- **Why ureteric tumours do worse:** the renal parenchyma is a barrier for pelvic tumours, but periureteric fat holds a rich lymphovascular plexus. At equal stage, ureteric tumours have more local and distant failure.
- Spread: direct, lymphatic (**mid and distal ureter to pelvic nodes**; upper ureter to hilar and retroperitoneal), blood (lung, liver, bone).

## Clinical features

- **Haematuria** (56–98%), **flank pain** (20–32%, dull from hydronephrosis/obstruction or colic from clots); lumbar mass about 10%; incidental about 15%.
- Weight loss, bone pain or a mass suggest advanced disease.

## Diagnosis (investigations)

**1. Cystoscopy:** always, to exclude coexisting bladder cancer (concurrent in 17%). Positive cytology with a normal bladder and urethra means the upper tracts must be examined.

### 2. Imaging

| Test | Role and key points |
|---|---|
| **CTU (computed tomography urography)** | First line; best test; highest accuracy (sensitivity ~92%, specificity ~95%). Radiolucent **filling defect**, hydronephrosis, ureteric wall thickening, non-visualisation. Tumour ~46 HU (Hounsfield units) versus uric acid stone ~100 HU. Misses flat lesions (CIS) |
| MRU (magnetic resonance urography) | If contrast or radiation is contraindicated; sensitivity ~75% for tumours <2 cm |
| Retrograde pyelography | Poorly filled calyces; accuracy ~75%; done with ureteroscopy. **Goblet (champagne-glass) sign**, the ureter dilated just below the tumour; **Bergman's sign**, the catheter coiling below it |
| FDG-PET/CT (fluorodeoxyglucose positron emission tomography) | Nodes and metastases when contrast is contraindicated |
| Renogram | Split function; contralateral kidney must be assessed |
| - **Hydronephrosis** predicts invasive disease (invasive ureteric tumour in up to 80%). | |

### 3. Urine cytology

- Sensitivity ~65%, grade-dependent (grade 1 ~20%, grade 3 ~75%). **Selective** ureteric washings are more accurate than voided urine. Take it before contrast. FISH (fluorescence in

situ hybridisation) is optional for surveillance.

**4. Flexible URS with biopsy: the key test**

- Adds accuracy (75–85% to ~90%). Biopsy gives **grade correctly in ~90%**, but **stage cannot be assessed** (shallow samples; a Ta biopsy is often T3 at surgery).
- Indications: diagnostic doubt, planned kidney-sparing treatment, solitary kidney.
- Diagnostic URS before RNU raises bladder recurrence, without changing survival.

## Staging (TNM [tumour, node, metastasis] 8th edition, 2017)

| T | Renal pelvis | Ureter |
|---|---|---|
| Ta / Tis | Non-invasive papillary / carcinoma in situ | Papillary non-invasive / carcinoma in situ |
| T1 | Subepithelial connective tissue | Subepithelial connective tissue |
| T2 | Muscularis | Muscularis |
| T3 | **Peripelvic fat or renal parenchyma** | Periureteric fat |
| T4 | Adjacent organs, or **through the kidney into perinephric fat** | Adjacent organs |
| - **N1** single node ≤2 cm; **N2** node >2 cm or multiple. **M1** distant metastasis. | | |
| - **AJCC (American Joint Committee on Cancer) grouping:** stage I = T1, II = T2, III = T3; **any N+, T4 or M1 = stage IV**. | | |
| - **Grade:** WHO 2004/2016: PUNLMP (papillary urothelial neoplasm of low malignant potential), low grade, high grade. | | |
| - Regional nodes: pelvis and upper ureter to hilar and retroperitoneal; lower ureter to pelvic. | | |

## Risk stratification

### EAU (European Association of Urology)

| Low risk (all needed) | High risk (any one) |
|---|---|
| Unifocal | Multifocal |
| Under 2 cm (<2 cm) | 2 cm or more (P2: >2 cm) |
| Low-grade cytology and biopsy | High-grade cytology or biopsy |
| No invasion on CTU | Hydronephrosis; invasion on imaging |
| — | Previous radical cystectomy; variant histology |

- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023** splits each group into favourable and unfavourable (obstruction, multifocality, lower-tract involvement). Ablation is **preferred** in low-risk favourable, **offered** in low-risk unfavourable, **rare** in high-risk favourable, **palliative** in high-risk unfavourable. Endoscopic therapy suits tumours **under 1.5 cm**.

- **Prognostic factors:** stage (strongest), grade, LVI (lymphovascular invasion), nodes, sessile architecture, necrosis >10%, ureteric site, surgical delay >12 weeks.

## Management A: kidney-sparing (minimally invasive and endoscopic) treatment

### Indications

- **Elective**: low-risk disease, whatever the other kidney.
- **Imperative**: solitary kidney, bilateral tumours, CKD (chronic kidney disease), unfit for RNU, after shared decision.
- The patient must accept frequent recurrence and **lifelong endoscopic surveillance**.

### Ureteroscopic (retrograde) management

- Closed system, lowest morbidity; for small ureteric and renal tumours.
- **Steps**: cystoscopy, then selective cytology and saline barbotage, then no-touch semirigid then flexible URS, map all lesions, biopsy (3 Fr forceps or basket) into fixative, ablate the base with laser (holmium or thulium, after biopsy), stent optional.

| Laser | Penetration | Note |
|---|---|---|
| **Ho:YAG (holmium:yttrium-aluminium-garnet)** | Under 0.5 mm | Safe in thin ureter |
| **Nd:YAG (neodymium:YAG)** | Up to 5–6 mm | Coagulative necrosis; transmural injury risk |
| **Thulium** | Under 0.2 mm | Strong haemostasis |

- Avoid circumferential fulguration or ablation (stricture); stage large tumours.
- Results: upper-tract recurrence 23–90% (P2: about half), bladder recurrence 15–45%, progression 0–20%. **Grade is the strongest predictor.**

### Percutaneous (antegrade) management

- Percutaneous resection for large pelvic tumours, lower-pole or inaccessible calyces and urinary diversions.
- **Steps**: retrograde catheter, puncture a calyx away from the tumour (upper or middle calyx for pelvic tumours), dilate to 30 Fr, cold-cup, bipolar resection or laser with the base sent separately, nephrostomy left, **second-look nephroscopy**.
- Removes more tumour and can **stage**; the tract allows topical therapy. But it is an **open system**: tract seeding about 0.3%, bleeding.

### Segmental ureterectomy

- **Distal ureterectomy with bladder cuff and reimplantation** (psoas hitch or Boari flap) for distal tumours, including high-grade ones when the kidney must be saved. It gives a staging specimen; add pelvic lymphadenectomy if high grade.
- First examine the whole tract above for multifocality. Lower-ureteric tumours with unfavourable features need resection, not endoscopy.

| Site | Operation |
|---|---|
| **Distal ureter** | Distal ureterectomy with bladder cuff and reimplantation; psoas hitch (~5 cm extra) or Boari flap (10–15 cm) |

| Site | Operation |
|---|---|
| Mid or proximal | Excision with 1–2 cm margins (frozen section) and ureteroureterostomy (rare) |
| Whole ureter | Ileal ureter or autotransplantation |
| - Refluxing reimplant eases surveillance; non-refluxing limits seeding (debated). | |
| - Laparoscopic or robotic routes are feasible; outcomes equal RNU in selected patients. | |
| - Adjuvant upper-tract BCG (bacillus Calmette–Guérin) or MMC (mitomycin C); UGN-101 mitomycin gel for low grade (see topical therapy). Lifelong endoscopic surveillance. | |

### Management B: RNU with bladder cuff (standard for high risk)

- **Principles**: kidney outside Gerota's fascia; clip the ureter early; never open the collecting system; remove the **whole intramural ureter and cuff** (stump recurrence 30–75% if left); bag extraction intact; watertight bladder closure; template lymphadenectomy for high risk.
- **Laparoscopic and robotic RNU steps**: modified flank, 3–4 ports; reflect colon; clip ureter; artery then vein; kidney outside Gerota's fascia (adrenal spared); ureter traced under the superior vesical artery; cuff excised (extravesical or open); two-layer closure.
- Abandoned (higher recurrence): transurethral (endoscopic) "pluck", stripping, laparoscopic stapling.
- Open, laparoscopic and robotic routes are oncologically equivalent: laparoscopic **equivalent to open** (AUA/SUO), with less blood loss and shorter stay; less favoured in bulky T3/T4. Robotic yields more nodes.
- **Template lymphadenectomy** for high risk: hilar plus paracaval (right) or para-aortic (left); pelvic nodes for lower ureteric tumours.

### Management C: adjuvant topical therapy

- **Single postoperative intravesical chemotherapy** (AUA/SUO strong): **ODMIT-C (2011)** MMC 40 mg cut bladder recurrence 28% to 16% at 1 year; **Ito (2013)** pirarubicin, 42% to 17% at 2 years.
- **Upper-tract instillation** via nephrostomy (low pressure, 25 cm H2O), ureteric catheter or reflux up a double-J. **BCG** (or MMC): 6-week course for upper-tract CIS, and after complete ablation of high-risk favourable Ta/T1. Risks: sepsis, stricture.
- **UGN-101 (Jelmyto)**: mitomycin reverse-thermal gel, **FDA (Food and Drug Administration) 2020** for low-grade UTUC not manageable endoscopically. **OLYMPUS**: complete response 59%, 56% disease-free at 12 months; ureteric stricture about 44%.

### Management D: perioperative (systemic) chemotherapy

- **POUT (2020)**: adjuvant platinum after RNU for pT2–T4 or N+: 3-year DFS (disease-free survival) **71% vs 46%**. Gemcitabine–cisplatin; carboplatin if creatinine clearance 30–49.
- **Neoadjuvant** platinum is preferred when renal function is borderline, i.e. when post-RNU renal function would rule out cisplatin (RNU removes cisplatin eligibility).
- **Also know:** adjuvant nivolumab (CheckMate 274) if platinum-ineligible, but no clear benefit in the UTUC subgroup.

### Management E: metastatic disease

- First line: **enfortumab vedotin + pembrolizumab** (EV-302, 2023): overall survival ~31 vs ~16 months versus platinum chemotherapy. Alternative: gemcitabine–platinum, then avelumab maintenance.
- Later: pembrolizumab (KEYNOTE-045), erdafitinib for FGFR3 alterations. Node-positive: chemotherapy first, then consolidative surgery.

### Prognosis

- 5-year disease-specific survival: localised ~89%, regional ~63%, distant ~17%. Stage is the strongest factor.

### Surveillance and follow-up (AUA/SUO)

- **Kidney-sparing**: cystoscopy and URS within 3 months, URS again at 6 and 12 months; imaging 3–6-monthly (high risk) or 6–9-monthly (low risk), then yearly to 5 years; URS and CTU at closer intervals, for life.
- **After RNU**: cystoscopy and cytology from 3 months; CTU yearly (more often if pT2 or more).

### Recent advances

- Lynch syndrome screening; thulium fibre laser; single-use digital ureteroscopes.

**Diagrams to draw:** ureteroscopic laser ablation of a pelvic tumour beside RNU with bladder cuff; RNU specimen with bladder cuff beside lymph node templates; TNM levels of a renal pelvic tumour (T3 parenchyma versus T4 perinephric fat), beside the diagnostic flowchart (CTU, cystoscopy, URS biopsy, risk group); distal ureterectomy with psoas-hitch reimplantation, beside the goblet sign.

> **Key points to remember:**
>
> - Tobacco, aristolochic acid (A:T to T:A in TP53) and Lynch (MSH2) are the key causes.
> - CTU is the best test; URS biopsy gives grade, not stage.
> - Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
> - Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
> - EAU low risk needs all favourable features (unifocal, under 2 cm, low grade, no invasion); any single adverse feature makes it high risk.
> - Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
> - Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
> - Distal ureterectomy for selected distal tumours.
> - RNU with complete bladder cuff + single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C).
> - POUT: adjuvant platinum, 3-year DFS 71% vs 46%; EV-302 for metastatic disease.

## LAQ 4 Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy.

P3

**Type:** LAQ (25 marks)
**Source:** 2019-Summer Paper 3, Q1

### Definition and biology

- **SRM (small renal mass):** an enhancing solid or complex cystic renal mass **≤4 cm (cT1a)**, usually found incidentally.
- **Benign share falls with size:** 20–30% of masses ≤4 cm and about 40% of masses ≤2 cm are benign. At ≥7 cm few are benign and about half are high grade.
- Male sex triples the risk of malignancy. Under 2% of small cancers metastasise.

### Differential diagnosis

| Group | Entities | Pointers |
|---|---|---|
| RCC (renal cell carcinoma) | **Clear cell** (commonest), papillary (10–15%), chromophobe (3–5%), cystic RCC, rare molecular subtypes | Clear cell enhances strongly; papillary enhances weakly |
| Other malignant | Urothelial carcinoma, lymphoma, metastasis, sarcoma, epithelioid AML (angiomyolipoma) | Infiltrative growth |
| Benign solid | Oncocytoma, fat-poor AML, papillary adenoma, metanephric adenoma | See below |
| Cystic | Haemorrhagic or complex cyst, cystic nephroma | Bosniak category |
| Inflammatory, vascular | Abscess, focal pyelonephritis, tuberculosis; aneurysm, AVM (arteriovenous malformation), infarct, pseudotumour | Clinical picture |

- **Oncocytoma:** commonest benign enhancing SRM (up to half of benign SRMs). Central stellate scar and segmental enhancement inversion suggest it but are not specific. A core cannot separate it from chromophobe RCC, which has oncocytic areas. Sestamibi SPECT (single-photon emission computed tomography) is under evaluation.
- **Fat-poor AML:** no visible fat (fat below −20 HU [Hounsfield units] diagnoses classic AML); homogeneous, T2-dark, mimics RCC; biopsy HMB-45 and Melan-A positive.
- **Also know:** metanephric adenoma (hypovascular, polycythaemia, WT1 positive); papillary adenoma (≤15 mm).

### Narrowing the differential

- **Multiphase CT (computed tomography):** enhancement over 15–20 HU means solid tissue. MRI (magnetic resonance imaging) helps below 2 cm.
- **RMB (renal mass biopsy)** if it will change management (AUA [American Urological Association]); seeding almost unreported; non-diagnostic about 14%.
- Then surveillance, ablation or **PN (partial nephrectomy)**, preferred for cT1a.

## Haemostasis in LPN (laparoscopic partial nephrectomy)

**Principle:** a clear field for excision, then secure closure of the bed. Plan with R.E.N.A.L. score and 3D (three-dimensional) CT; intraoperative ultrasound.

**1. Vascular control**

| Technique | Method | Comment |
|---|---|---|
| Main artery ± vein clamp | Bulldog clamps, or laparoscopic Satinsky en bloc | Artery-only equals artery–vein. Aim warm ischaemia ≤25 min (trifecta) |
| Selective or super-selective | Clamp the tumour's segmental branch; 3D CT, NIRF (near-infrared fluorescence) | Avoids global ischaemia; no functional gain in an RCT (randomised controlled trial) |
| Off-clamp ("zero ischaemia") | No clamping; small exophytic tumours | More blood loss; same long-term eGFR (estimated glomerular filtration rate) |
| Early unclamping | Release after the inner suture layer | Shorter ischaemia; possibly fewer pseudoaneurysms |
| Cold ischaemia | Ice slush, cold saline via ureteric catheter, cold arterial perfusion | Solitary kidney, CKD (chronic kidney disease), long repair; benefit unproven |

- Function depends mainly on parenchyma preserved. Mannitol: no benefit in an RCT. **Also know:** parenchymal clamp for peripheral tumours.

**2. Excision:** score the capsule, cut with cold scissors, clip visible vessels. Enucleation along the pseudocapsule bleeds less.

**3. Renorrhaphy (most reliable step)**

- **Inner layer:** running 3-0 or 2-0 absorbable suture closes cut vessels and collecting system (opened in about 60%).
- **Outer layer, sliding-clip technique:** mattress sutures anchored by a locking clip; clips slid down to the capsule give tension without knots. Pledgets prevent capsular tears; oxidised cellulose bolster fills the defect.
- **Barbed suture:** holds tension without knots, so ischaemia is shorter.
- Single layer gave better function and fewer pseudoaneurysms than two layers.

**4. Energy:** argon beam or bipolar coagulation of the bed. Argon can destroy collecting-system sutures and adds gas, so vent the ports.

**5. Haemostatic agents (adjuncts, no proven benefit in RCTs):** gelatin–thrombin matrix; fibrin sealant; oxidised cellulose; PEG (polyethylene glycol) hydrogel; cyanoacrylate. Sutureless closure only for selected small tumours.

**6. Final check:** unclamp, inspect, **lower pneumoperitoneum** to unmask venous bleeding; drain.

## Postoperative bleeding

- Early (hypotension, falling haemoglobin): resuscitate, CT, angioembolisation.
- **Delayed haematuria (about day 12):** pseudoaneurysm or arteriovenous fistula; selective angioembolisation preserves function. Endophytic and posterior tumours carry higher risk.

**Diagram to draw:** sliding-clip renorrhaphy over a bolster, with bulldog clamp on the artery.

**Key points to remember:**

- SRM ≤4 cm: 20–30% benign; ≤2 cm about 40%.
- Oncocytoma is the commonest benign mimic; biopsy cannot exclude chromophobe RCC.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Delayed haematuria after PN means pseudoaneurysm: angioembolise.

## LAQ 5 Radiology for solid renal masses.

P4

**Type:** LAQ (25 marks)
**Source:** 2018-Summer Paper 4, Q1

### Introduction

- Most renal masses are now found incidentally. Imaging is the **strongest predictor of malignancy**; size and **enhancement** are the two key variables.
- About 20% of solid enhancing cT1 masses are benign (oncocytoma, fat-poor AML (angiomyolipoma)).
- **Goals:** solid or cystic? enhancing? fat? subtype? stage and surgical anatomy.

### A. CT (computed tomography): the standard test

**Renal mass protocol** (dedicated, thin slices):

| Phase | Timing | Use |
|---|---|---|
| Non-contrast | – | Baseline HU (Hounsfield units), fat, calcification, haemorrhage |
| Corticomedullary | 30–70 s | Hypervascular masses; arteries; clear cell vs other subtypes |
| Nephrographic | 90–120 s | **Most sensitive** for detection (tumour enhances less than parenchyma); renal vein and IVC (inferior vena cava) thrombus |
| Excretory | 7–10 min | Collecting system relation |

**Enhancement thresholds** (rise from unenhanced value):

| Rise | Meaning |
|---|---|
| ≥20 HU | Enhancing: solid tumour |
| 10–20 HU | Equivocal (MRI, magnetic resonance imaging, or CEUS) |
| <10 HU | Non-enhancing: cyst |

- **Pitfall:** pseudoenhancement of small (<10–15 mm) cysts; dual-energy CT iodine maps help.
- **Macroscopic fat = AML:** fat ≤ −10 to −20 HU on unenhanced CT is diagnostic. Fat **with calcification** suggests RCC (renal cell carcinoma).
- **Fat-poor AML:** hyperdense on unenhanced CT, homogeneous, avid early enhancement and washout.
- **Subtype clues:** clear cell RCC enhances early, avidly and heterogeneously (necrosis); papillary RCC is homogeneous and **hypoenhancing**; chromophobe and oncocytoma intermediate, with a central scar. Overlap is large.

### B. MRI

- **Indications:** contrast allergy, CKD (chronic kidney disease), equivocal CT enhancement, young patients, thrombus. Enhancement = signal rise ≥15–20%. Macrocyclic gadolinium carries very low NSF (nephrogenic systemic fibrosis) risk.

| Feature | Clear cell | Papillary | Chromophobe / oncocytoma | AML |
|---|---|---|---|---|
| T2 signal | High, heterogeneous | **Low**, homogeneous (haemosiderin) | Central T2-bright scar | Fat signal |
| Enhancement | Avid (>200%) | Weak | Intermediate / avid | Avid |
| Diffusion (ADC, apparent diffusion coefficient) | High | Lowest | Intermediate | – |
| Chemical shift | **Signal drop** (intracellular lipid, ~59%) | No drop | – | **India-ink** artefact; loss on fat saturation |

- **ccLS (clear cell likelihood score, 1–5):** structured MRI algorithm for solid cT1a masses using T2 signal, corticomedullary enhancement and microscopic fat, plus diffusion and **segmental enhancement inversion** (favours oncocytoma). Score 4–5 suggests clear cell RCC; 1–2 a non-clear-cell tumour. It guides biopsy versus surveillance.

### C. Ultrasound and CEUS (contrast-enhanced ultrasound)

- Grey-scale: cyst versus solid; poor below 2 cm. AML is hyperechoic, but so are some small RCCs.
- **CEUS** (microbubbles, not nephrotoxic): shows enhancement; used in CKD or allergy, indeterminate lesions, ablation and surveillance.

### D. Cystic masses: Bosniak 2019 (CT and MRI)

| Class | Key feature | Malignancy / action |
|---|---|---|
| I | Simple, thin wall ≤2 mm, no septa | Benign; none |
| II | 1–3 thin septa; homogeneous hyperdense (≥70 HU unenhanced) | Benign; none |
| IIF | Wall or septa 3 mm, or ≥4 thin septa | <10%; follow 6, 12 months, then yearly for 5 years |
| III | Wall or septa ≥4 mm, or irregular | ~50%; surgery or surveillance |
| IV | **Enhancing nodule** | ~90%; treat |

### E. Functional and molecular imaging

- **99mTc-sestamibi SPECT/CT** (single-photon emission CT): mitochondria-rich **oncocytoma** and hybrid oncocytic or chromophobe tumours take it up; clear cell RCC does not.
- **Girentuximab PET (positron emission tomography)** targets CA-IX (carbonic anhydrase IX) on clear cell RCC. **ZIRCON** ($^{89}$Zr-girentuximab, indeterminate cT1 masses): about 85% sensitivity and 87% specificity for clear cell RCC.
- **FDG-PET** (fluorodeoxyglucose PET): not recommended (low sensitivity). PSMA (prostate-specific membrane antigen) PET: experimental.

### F. Renal mass biopsy

- **Indications:** result may change management; suspected lymphoma, metastasis or infection; before ablation or surveillance; metastatic disease. Not needed if surgery is planned regardless.

- Cores: PPV (positive predictive value) 98.8%; subtype accuracy ≥90%, grade 60–80%; non-diagnostic ~14%. Seeding almost nil. Oncocytoma cannot be confirmed on cores.

### G. Staging and surgical planning

- **Venous thrombus:** CT sensitivity 78% (renal vein), 96% (IVC). MRI or multiplanar CT shows the cephalad tip; enhancing thrombus is tumour, non-enhancing is bland.

| Mayo level | Extent | T stage |
|---|---|---|
| 0 | Renal vein | T3a |
| I | IVC ≤2 cm above renal vein | T3b |
| II | IVC below hepatic veins | T3b |
| III | Intrahepatic IVC, below diaphragm | T3b |
| IV | Above diaphragm or atrium | T3c |

- Nodes ≥2 cm usually malignant. Chest CT; bone scan only if symptomatic.
- **Nephrometry:** R.E.N.A.L. (Radius, Exophytic/endophytic, Nearness to collecting system, Anterior/posterior, Location), PADUA, C-index: predict complexity and complications; guide partial versus radical nephrectomy.
- **After ablation:** persistent or new enhancement = residual or recurrence.
- **Recent:** radiomics and AI (artificial intelligence).

**Diagram to draw:** algorithm: incidental mass → renal-protocol CT/MRI → fat (AML) / cyst (Bosniak) / enhancing solid → biopsy or treat.

> **Key points to remember:**
> - Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
> - MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
> - Sestamibi for oncocytoma; $^{89}$Zr-girentuximab PET (ZIRCON) for clear cell.
> - Biopsy PPV ~99%, non-diagnostic ~14%.

## Short answer questions (SAQ)

### SAQ 1 Immunotherapy / immunomodulation in RCC

P2 · P4 **merged from 2**

*Asked as:* P2 SAQ "Immunotherapy in (metastatic) RCC" (2024-Winter Paper II Q3e; Practice Set 4 Paper II Q3f) · P4 SAQ "Immuno modulation in Renal, cell carcinoma" (2025-Winter Paper 4 Q3a)

*Assumed meaning: immunomodulation (immunotherapy) in renal cell carcinoma. Overall management of metastatic RCC: see 2024-Winter Paper 3, Q3b.*

#### Rationale

- RCC (renal cell carcinoma) is immunogenic and chemoresistant: rare (<1%) spontaneous regression after nephrectomy, and tumour-infiltrating T cells.
- **Checkpoints:** PD-1 (programmed death-1) on CD8 T cells binds tumour PD-L1 and switches the T cell off; CTLA-4 (cytotoxic T-lymphocyte antigen-4) on regulatory T cells damps

priming. Blockade **releases the brake** on existing anti-tumour immunity.
- VEGF (vascular endothelial growth factor) is immunosuppressive, so TKIs (tyrosine kinase inhibitors) synergise with checkpoint inhibitors.

### Cytokine era (non-specific stimulation)

| | IFN-α (interferon-alpha) | High-dose IL-2 (interleukin-2) |
|---|---|---|
| Response | 10–15% | 15–20% |
| Durable complete response | <2% | **7–9%** |
| Toxicity | Flu-like | **Capillary leak**; needs intensive care |
| Role | Obsolete | Selected fit patients only |

- High-dose IL-2 dose: 600,000–720,000 IU/kg intravenously every 8 hours, maximum 15 doses; over 60% of complete responders stay disease-free.

### Checkpoint inhibitors: trials (all clear cell)

| Trial | Regimen vs comparator | Result |
|---|---|---|
| CheckMate 025 | Nivolumab vs everolimus, second line after TKI | OS (overall survival) 25.0 vs 19.6 months |
| **CheckMate 214** | Nivolumab + ipilimumab vs sunitinib | Intermediate/poor risk: OS HR (hazard ratio) 0.63; complete response 9% |
| KEYNOTE-426 | Pembrolizumab + axitinib vs sunitinib | Better PFS (progression-free survival) and OS |
| CheckMate 9ER | Nivolumab + cabozantinib vs sunitinib | Better PFS and OS; PFS 16.6 vs 8.3 months |
| **CLEAR** | Pembrolizumab + lenvatinib vs sunitinib | PFS 23.9 vs 9.2 months; response 71% |
| JAVELIN | Avelumab + axitinib | No OS benefit |

### Treatment sequencing

- **First line:** an IO/TKI (immuno-oncology plus TKI) combination for all IMDC (International Metastatic RCC Database Consortium) risk groups; nivolumab + ipilimumab for intermediate/poor risk only.
- **After IO:** switch to a VEGF-targeted TKI. IO rechallenge is not recommended: it failed in CONTACT-03 and TiNivo-2. Belzutifan after IO and TKI.

### Adjuvant

- **KEYNOTE-564:** pembrolizumab for 1 year after nephrectomy in high-risk clear cell RCC, including resected metastases (M1 NED, no evidence of disease). DFS (disease-free survival) HR 0.68 (P4 and another source: HR 0.63); later OS benefit (HR 0.62).
- Atezolizumab and nivolumab-based adjuvant trials were negative.

### Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis, rash.

| Grade | Action |
|---|---|
| 1 | Continue, monitor |

| Grade | Action |
|---|---|
| 2 | Hold; prednisolone 0.5–1 mg/kg |
| 3–4 | Stop; high-dose steroids (methylprednisolone 1–2 mg/kg); infliximab or mycophenolate if refractory |

- Endocrinopathies: hormone replacement.

### Biomarkers and also know

- **PD-L1 does not select patients.** Sarcomatoid tumours respond well. Gene signatures are investigational; antibiotics may reduce benefit.
- Non-clear cell data are scant.
- Vaccines and allogeneic transplant failed.

### Diagram to draw

- T cell–tumour synapse with PD-1/PD-L1 and CTLA-4/B7 blocked.

> **Key points to remember:**
> - IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
> - High-dose IL-2: durable complete response 7–9%.
> - CheckMate 214: nivolumab + ipilimumab, OS HR 0.63, complete response 9%.
> - KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
> - irAE grade ≥2: hold drug and give steroids.

## SAQ 2 Renal angiomyolipoma and its management

P2 · P3 **merged from 2**

*Asked as:* P2 SAQ "Renal aangiomyolipoma" (Practice Set 7 Paper 2 Q3f) · P3 SAQ "Management of AML." (2021-Summer Paper 3 Q3d)

*Assumed meaning: AML = renal angiomyolipoma (not acute myeloid leukaemia).*

### Introduction and pathogenesis

- Benign **PEComa (perivascular epithelioid cell tumour)** of dysmorphic **vessels, smooth muscle and fat**. Female predominance; fourth–fifth decades.
- **Sporadic** (solitary, small) or with **TSC (tuberous sclerosis complex**; AML in 55–90%: younger, bilateral, multifocal, larger) or LAM (lymphangioleiomyomatosis).
- **Mechanism:** loss of **TSC2** (tuberin, 16p13) or TSC1 (hamartin, 9q34) leads to unchecked **mTOR (mammalian target of rapamycin)** signalling, which drives growth and angiogenesis.
- **Epithelioid AML**: little fat; aggressive (venous extension, metastasis); treat as RCC (renal cell carcinoma).

### Clinical features

- Mostly incidental. Large tumours bleed: **Wunderlich syndrome** (spontaneous retroperitoneal haemorrhage; Lenk triad: flank pain, mass, shock).
- CKD (chronic kidney disease) in TSC.

### Diagnosis

- **CT (computed tomography): macroscopic fat (below about −15 to −20 HU [Hounsfield units]) on non-contrast images is diagnostic.**
- US (ultrasound): hyperechoic (so are some RCCs).
- MRI (magnetic resonance imaging): signal loss on fat suppression, India-ink artefact.
- Fat-poor lesion: MRI ± biopsy (HMB-45 [human melanoma black-45], Melan-A, smooth muscle actin positive).
- **Aneurysm ≥5 mm** on CT angiography predicts bleeding.

### Management algorithm (Restrepo 2022)

| Situation | Management |
|---|---|
| Acute haemorrhage | Resuscitate, then **SAE (selective arterial embolisation)**; nephrectomy if SAE fails or is unavailable |
| Symptomatic (pain, haematuria) or poor access to care | SAE, NSS (nephron-sparing surgery), ablation in selected cases |
| Asymptomatic sporadic with **>6 cm, aneurysm >5 mm, growth >0.25 cm/year, childbearing age or pregnancy** | Active treatment |
| None of these | **Surveillance**: imaging (US or CT) at 6 and 12 months, then yearly |
| TSC, asymptomatic, **>3 cm** | **Everolimus**; MRI every 1–3 years |

- The old **4 cm** cut-off overtreats and should not be a general trigger; EAU (European Association of Urology) 2023 favours surveillance for most.

### Treatments

- **SAE:** first line; **post-embolisation syndrome** (fever, pain, leucocytosis) is common; abscess, infarction.
- **Partial nephrectomy:** preferred surgery; essential in TSC (multifocal, recurrent).
- **Thermal ablation** (radiofrequency, cryo): limited data.
- **Everolimus (EXIST-2):** ≥50% shrinkage in **42% vs 0%** (placebo); side effects stomatitis, acne, hyperlipidaemia.

### Diagram to draw

- Management algorithm by bleeding, symptoms, size and TSC.

> **Key points to remember:**
> - Macroscopic fat on non-contrast CT is diagnostic.
> - Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
> - SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

## SAQ 3 Role of renal nephrometry in management of RCC.

P1

**Type:** SAQ (10 marks)
**Source:** 2020-Summer Paper 1, Q3f

## Definition

- **Nephrometry scores** grade the anatomical complexity of a renal mass on CT (computed tomography) or MRI (magnetic resonance imaging), to standardise RCC (renal cell carcinoma) reporting and plan treatment.

## R.E.N.A.L. score (Kutikov and Uzzo, 2009)

| Component | 1 point | 2 points | 3 points |
|---|---|---|---|
| **R**adius (maximum diameter) | ≤4 cm | >4 to <7 cm | ≥7 cm |
| **E**xophytic/endophytic | ≥50% exophytic | <50% exophytic | Entirely endophytic |
| **N**earness to collecting system or sinus | ≥7 mm | >4 to <7 mm | ≤4 mm |
| **A**nterior/posterior | Suffix a, p or x (no points) | | |
| **L**ocation to polar lines | Entirely above or below | Crosses a polar line | >50% across, crosses midline, or entirely between |
| - Suffix "h": touches main renal vessels. **Total 4–12:** low 4–6, moderate 7–9, high 10–12. | | | |

## PADUA (Preoperative Aspects and Dimensions Used for an Anatomical classification) score (Ficarra, 2009)

- Polar location, exophytic rate, renal rim (lateral or medial), renal sinus, collecting system and size, plus anterior/posterior face. **Total 6–14:** 6–7 low, 8–9 intermediate, ≥10 high risk.

## C-index (centrality index)

- Distance from tumour centre to kidney centre ÷ tumour radius. **Lower value = more central, more complex.**

## Role in management

| Decision | Use |
|---|---|
| **Partial vs radical nephrectomy** | Feasibility and risk of PN (partial nephrectomy) vs RN (radical nephrectomy) |
| **Approach** | Open vs minimally invasive; hilar clamping |
| **Predict outcomes** | Warm ischaemia, blood loss, complications, conversion, collecting-system entry; urine leak (R and N components) |
| **Function** | Parenchymal loss, eGFR (estimated glomerular filtration rate) decline |
| **Ablation** | Cryoablation complications rise with complexity |
| **Pathology** | Higher scores: malignancy, higher grade |

- **AUA (American Urological Association) 2021:** with high oncological potential, prefer RN only if PN would be challenging even for experts, there is no CKD (chronic kidney disease) or proteinuria, and new baseline eGFR will likely exceed 45.

## Limitations

- Interobserver variation; ignores patient factors and surgeon experience.

**Also know:** ABC (arterial based complexity), DAP (diameter–axial–polar) scores; **MAP (Mayo Adhesive Probability) score** for adherent perinephric fat.

**Diagram to draw:** kidney with polar lines and the five R.E.N.A.L. components.

> **Key points to remember:**
> - RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
> - PADUA 6–14; C-index lower = more central.
> - Predicts PN feasibility, ischaemia time and complications.
> - MAP score: adherent perinephric fat.

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## SAQ 4 Acquired renal cystic disease

P2

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 2, Q3b

### Definition

- **ARCD (acquired renal cystic disease)**: bilateral cysts in kidneys of patients with **ESRD (end-stage renal disease)** not due to inherited cystic disease. Diagnosis: **≥3 cysts in each kidney**.
- Described in 1977; occurs with haemodialysis and peritoneal dialysis.

### Epidemiology

| Duration of renal failure | Prevalence |
|---|---|
| Before dialysis | About 10% |
| 3 years of dialysis | 44% |
| 5 years | 60% |
| ≥10 years | >90% |
| - Male:female 3:1; independent of the cause of renal failure. | |

### Aetiology

- **Uraemia, not dialysis**, is the cause; dialysis only gives cysts time to form.
- Tubular obstruction (fibrosis, oxalate crystals); uraemic toxins; growth factors (EGF (epidermal growth factor)); compensatory hyperplasia.
- Cysts may regress after transplant (data conflict); cancer risk persists.

### Clinical features

- Mostly asymptomatic.
- **Bleeding into cysts (up to 50%)**: loin pain, haematuria, retroperitoneal haematoma; worsened by uraemic coagulopathy and heparin.
- Cyst infection, stones, polycythaemia (erythropoietin).
- **Neoplasia**: adenomas in 20–25%. **RCC (renal cell carcinoma)** risk about 3 times that without ARCD. **Acquired cystic disease-associated RCC** is a distinct subtype (oxalate crystals, good prognosis); papillary RCC is also common.

### Evaluation

- **Ultrasound**: small echogenic kidneys, many small cysts.
- **CT (computed tomography) with and without contrast**: best for tumours. MRI (magnetic resonance imaging) for small lesions; gadolinium risks NSF (nephrogenic systemic fibrosis).

| Feature | ARCD | ADPKD (autosomal dominant polycystic kidney disease) |
|---|---|---|
| Kidney size | Small (<300 g) | Large (>800 g) |
| Cysts | Small | Large |
| Extrarenal | None | Liver cysts, aneurysms |

### Management

- **Bleeding**: conservative; persistent → **embolisation or nephrectomy**; peritoneal dialysis avoids heparin.
- Painful cyst: aspiration; infected cyst: drainage, nephrectomy if it fails.
- **Tumour >3 cm: nephrectomy**. <3 cm: nephrectomy in fit patients, or yearly CT and excise if growing.
- **Screening** (debated): younger fit patients after 3 years of dialysis, and before transplant.
- After transplant, native-kidney RCC is more aggressive; >70% of transplant RCCs arise there.

**Diagram to draw:** small kidney with multiple cysts and a small solid tumour.

> **Key points to remember:**
> - ≥3 cysts per kidney in ESRD; >90% after 10 years.
> - Uraemia causes it; transplant may regress cysts, not risk.
> - RCC risk 3 times; image dialysis patients with haematuria.

## SAQ 5 ADPKD (Adult polycystic kidney disease)

P2

**Type:** SAQ (10 marks)
**Source:** 2017-Winter Paper 2, Q3d

### Genetics

- ADPKD (autosomal dominant polycystic kidney disease): commonest inherited cystic kidney disease (about 1 in 400–1000); 7–15% of patients on haemodialysis.

| Gene | Chromosome | Share | ESRD (end-stage renal disease) |
|---|---|---|---|
| **PKD1** (polycystin-1) | 16 | ~85% | 50s |
| **PKD2** (polycystin-2) | 4 | ~15% | 70s |

- Polycystins (primary cilium) restrain proliferation.

### Clinical features (30–50 years)

- **Pain** (commonest): mass, cyst haemorrhage, infection, stones. **Haematuria** about 50%; **stones** 20–30%.

- **Hypertension:** about 50% at 20–35 years with normal function; stretched vessels release renin.
- **Renal failure:** tracks kidney volume; worse with PKD1, male sex.
- **Extrarenal:** liver cysts (commonest; up to 90% by 50); **intracranial aneurysm 10–30%**; seminal vesicle (40%), pancreatic, arachnoid cysts; mitral valve prolapse; diverticulosis.
- RCC (renal cell carcinoma) is not commoner, but is more often bilateral and sarcomatoid.

### Diagnosis (ultrasound in at-risk relatives)

| Age | Ravine (PKD1 families) | Pei unified (2009) |
|---|---|---|
| 15–29 | ≥2 cysts, one or both kidneys | ≥3 cysts in total (15–39) |
| 30–59 | ≥2 cysts in each kidney | ≥2 in each kidney (40–59) |
| ≥60 | ≥4 cysts in each kidney | ≥4 in each kidney |

- CT or MRI shows TKV (total kidney volume), which predicts progression. Genetic testing if equivocal or for a living donor.

### Management

- **BP <130/80 mmHg** with ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker); HALT-PKD: tighter control slowed TKV growth.
- **Tolvaptan** (V2-receptor antagonist; lowers cyclic adenosine monophosphate): **TEMPO 3:4 (2012)** slowed TKV growth and function loss; **REPRISE (2017)** slowed eGFR (estimated glomerular filtration rate) decline in later disease. Causes polyuria and liver toxicity.
- **Pain:** exclude infection, stone, tumour; avoid NSAIDs (non-steroidal anti-inflammatory drugs); aspiration, then laparoscopic unroofing.
- **Infected cyst:** lipid-soluble antibiotics (trimethoprim–sulfamethoxazole, fluoroquinolones); drain if no response.
- **Haemorrhage:** usually settles with rest. **Aneurysm screening** (MR angiography) if family history.
- **ESRD:** dialysis or transplant (no recurrence). **Native nephrectomy** before or at transplant for recurrent infection, bleeding, pain, suspected tumour or no room for the graft.

**Diagram to draw:** large cystic kidneys, liver cysts and berry aneurysm.

> **Key points to remember:**
> - PKD1: chromosome 16, 85%, earlier ESRD.
> - Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
> - Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

## SAQ 6 Biomarkers for patients with advanced RCC

P2

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 2, Q3d

### Introduction

- A **prognostic** biomarker predicts outcome regardless of treatment; a **predictive** biomarker predicts benefit from a specific drug.

- In advanced RCC (renal cell carcinoma), only clinical risk models are used routinely. No molecular biomarker is yet validated to select treatment (EAU, European Association of Urology; NCCN, National Comprehensive Cancer Network).

### Clinical and laboratory: risk models

| Factor | MSKCC (Memorial Sloan-Kettering Cancer Center) | IMDC (International Metastatic RCC Database Consortium) |
|---|---|---|
| Karnofsky performance status <80% | Yes | Yes |
| Haemoglobin < lower limit of normal | Yes | Yes |
| Corrected calcium high | >10 mg/dL | > upper limit of normal |
| LDH (lactate dehydrogenase) >1.5 × normal | Yes | No |
| Time factor | No prior nephrectomy | <1 year from diagnosis to treatment |
| Neutrophils, platelets > normal | No | Yes |

- IMDC: favourable 0 factors (median survival 43 months), intermediate 1–2 (22.5 months), poor ≥3 (7.8 months).
- IMDC group guides first-line choice (nivolumab–ipilimumab not for favourable risk).
- **Also know:** neutrophil-to-lymphocyte ratio, CRP (C-reactive protein), liver or bone metastases (poor prognosis).

### Tissue biomarkers

- **PD-L1 (programmed death-ligand 1):** benefit of checkpoint combinations is **independent of PD-L1**, so it is not used to select patients.
- **Sarcomatoid features:** respond well to checkpoint inhibitors.
- **Gene expression signatures:** IMmotion150 showed high T-effector signature tumours do better with atezolizumab–bevacizumab; high angiogenic signature tumours do well with sunitinib. Investigational.
- **Genomic:** BAP1 loss and CDKN2A loss indicate poor prognosis; PBRM1 data conflicting. MET alterations in papillary RCC (cabozantinib); FH (fumarate hydratase) deficiency. TMB (tumour mutational burden) is low and not predictive.

### Circulating and other biomarkers

- **KIM-1 (kidney injury molecule-1):** high serum levels predict recurrence and benefit from adjuvant immunotherapy (investigational).
- **ctDNA (circulating tumour DNA):** RCC sheds little DNA; methylation assays are promising.
- **Also know:** antibiotics during checkpoint therapy reduced survival (microbiome); sunitinib-induced hypertension is linked to better outcome.

**Diagram to draw:** IMDC risk groups with survival curves.

> **Key points to remember:**
> - IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
> - PD-L1 does not select patients in RCC.
> - Sarcomatoid tumours respond well to immunotherapy.

## SAQ 7 Tivozanib

P2

**Type:** SAQ (10 marks)
**Source:** 2024-Winter Paper 2, Q3b

### Introduction

- Oral **VEGFR TKI (vascular endothelial growth factor receptor tyrosine kinase inhibitor)** used in advanced RCC (renal cell carcinoma).
- Targets **VEGFR-1, -2 and -3** (also c-KIT and PDGFR-β, platelet-derived growth factor receptor-β); highly potent and selective.

### Mechanism and rationale

- In clear cell RCC, **VHL (von Hippel–Lindau) loss** lets HIF (hypoxia-inducible factor) accumulate, driving VEGF and angiogenesis.
- Tivozanib blocks VEGFR signalling, starving the tumour of new vessels.
- High selectivity means fewer off-target effects (less hand-foot skin reaction and diarrhoea than sorafenib).

### Dose and pharmacology

- **1.34 mg orally once daily, 21 days of a 28-day cycle.**
- Long half-life (about 4–5 days); dose reduction for toxicity or hepatic impairment.

### Key trials

| Trial | Setting | Result |
|---|---|---|
| **TIVO-1** (phase III) | 0–1 prior therapy, versus sorafenib | PFS (progression-free survival) 11.9 vs 9.1 months; response 33% vs 23%; OS (overall survival) not better (28.8 vs 29.3 months) |
| **TIVO-3** (phase III) | After 2–3 prior systemic therapies including a VEGFR TKI, versus sorafenib | PFS 5.6 vs 3.9 months (HR, hazard ratio, 0.73); response 18% vs 8%; OS not different (HR 0.97); benefit also after prior checkpoint inhibitor |
| **TiNivo-2** (phase III) | After checkpoint inhibitor; tivozanib ± nivolumab | **Negative:** adding nivolumab did not improve PFS (no immunotherapy re-challenge) |

### Approval and place

- FDA (Food and Drug Administration) declined in 2013 because of the OS concern in TIVO-1; EMA (European Medicines Agency) approved first-line use in 2017.
- **FDA March 2021:** relapsed or refractory advanced RCC **after ≥2 prior systemic therapies**.
- NCCN (National Comprehensive Cancer Network): later-line option, mainly after immunotherapy combinations.

### Adverse effects

- **Hypertension** (commonest; control before starting), dysphonia, fatigue, diarrhoea, proteinuria, hypothyroidism.
- Bleeding, arterial thrombosis, cardiac failure, **impaired wound healing** (stop before surgery), RPLS (reversible posterior leukoencephalopathy syndrome).
- **Monitoring:** blood pressure, thyroid tests, urine protein, liver tests.

**Diagram to draw:** VHL loss → HIF → VEGF → VEGFR, blocked by tivozanib.

> **Key points to remember:**
> - Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
> - TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
> - FDA 2021: after ≥2 prior systemic therapies.
> - TiNivo-2 negative; main toxicity hypertension.

## SAQ 8 VHL gene and von Hippel–Lindau disease

P2

**Asked as:**

- VHL gene. — *Practice Set 2 Paper II Q3f (SAQ)*
- Von Hippel Lindau disease — *Practice Set 8 Paper II Q3c (SAQ)*

**Type:** SAQ (10 marks)

(Practice Set 8 Paper II Q3c: VHL gene also answered in Practice Set 2 Paper 2, Q3f)

### Introduction

- **VHL (von Hippel–Lindau)** is a **tumour suppressor gene on chromosome 3p25**, identified in 1993; three exons, 213-amino-acid protein.
- Germline mutation causes **VHL disease**: an **autosomal dominant** familial cancer syndrome (about 1 in 36,000), penetrance >95% by 65 years.
- It is the familial form of **clear cell RCC** (renal cell carcinoma). Somatic VHL loss occurs in most sporadic clear cell RCC; chromosome 3p is altered in over 90%.

### Function / genetics and pathogenesis: the oxygen-sensing pathway

- VHL protein binds elongin B, elongin C, cullin-2 and RBX1 to form (VHL protein forms) an **E3 ubiquitin ligase** that degrades **HIF (hypoxia-inducible factor)** in normoxia.
- **Normoxia:** hydroxylated **HIF -1α and -2α** are tagged for proteasomal destruction.
- **VHL loss (or hypoxia):** loss of VHL stabilises HIF, which (chiefly HIF-2α) accumulates, joins ARNT (aryl hydrocarbon receptor nuclear translocator) and switches on **VEGF (vascular endothelial growth factor)**, PDGF (platelet-derived growth factor), TGF-α (transforming growth factor-α), GLUT1, erythropoietin and CA-IX (carbonic anhydrase IX).
- Hence the **hypervascularity** (**highly vascular** tumours) of clear cell RCC.
- **Knudson two-hit:** one germline plus one somatic hit, so VHL tumours are **early, bilateral and multifocal**.

## VHL disease manifestations

| Lesion | Frequency |
|---|---|
| Craniospinal haemangioblastoma (cerebellum commonest) | 60–80% |
| Retinal angioma (earliest) | 25–60% |
| **Clear cell RCC** (mean age 39; commonest cause of death) | 25–75% |
| Renal cysts (commonest renal lesion) | Up to 76% |
| **Phaeochromocytoma** | 10–25% |
| Pancreatic cysts / neuroendocrine tumours | 45–75% / 11–17% |
| Epididymal cystadenoma | 25–60% of men |
| - Also endolymphatic sac tumour. | |
| - **RCC is now the commonest cause of death.** | |

## Genotype–phenotype

| Type | Mutation | Phaeochromocytoma | RCC |
|---|---|---|---|
| 1 | Deletions, truncations | Low | High |
| 2A | Missense | High | Low |
| 2B | Missense | High | High |
| 2C | Missense | Phaeochromocytoma only | Absent |

## Diagnosis

- Family history plus one lesion; otherwise ≥2 haemangioblastomas, or 1 plus a visceral lesion. Genetic testing of relatives.

## Surveillance (VHL Alliance 2020)

- Eye check (examination) yearly from infancy; metanephrines yearly from 5 years; CNS (brain and spine) MRI every 2 years from 11; **abdominal MRI every 2 years from 15**.

## Clinical implications / urological management

- **3 cm rule:** observe solid renal tumours <3 cm; intervene when largest reaches 3 cm; then nephron-sparing surgery (enucleation) or ablation, after excluding phaeochromocytoma.
- **Nephron-sparing surgery** (enucleation) is standard; not curative: local recurrence 75–85%, yet 10-year disease-specific survival 81–94%. Ablation for repeated small tumours.
- Exclude phaeochromocytoma before any surgery (normetanephrine raised; α-blockade). **Partial (cortical-sparing) adrenalectomy.**
- **Targeted therapy:** VEGF TKIs (tyrosine kinase inhibitors), bevacizumab, mTOR (mammalian target of rapamycin) inhibitors.
- **Belzutifan** (HIF-2α inhibitor): FDA (US Food and Drug Administration) approved 2021 for VHL disease; renal response 30–50% (another source: responses in about half of renal tumours); anaemia.
- **Also know:** PBRM1, BAP1, SETD2 on 3p.

**Diagram to draw:** VHL–HIF pathway in normoxia versus (vs) VHL loss, with drug targets.

**Key points to remember:**

- AD; 3p25 tumour suppressor; VHL protein degrades HIF.
- VHL loss: HIF-2α drives (→) VEGF, giving vascular clear cell RCC.
- Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing (nephron-sparing surgery).
- Belzutifan (HIF-2α inhibitor) approved 2021.

## SAQ 9 Zero-ischaemia partial nephrectomy (laparoscopic and robotic)

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 4, Q3e (clamping strategies and laparoscopic partial nephrectomy haemostasis also in 2019-Summer Paper 3, Q1)

### Definition and rationale

- **Zero-ischaemia PN (partial nephrectomy):** excision and renorrhaphy **without clamping the main renal artery**, so the kidney stays perfused (Gill and colleagues, 2011).
- **Why:** warm ischaemia injures nephrons (historical limit 30 minutes; trifecta ≤25). Most relevant in a **solitary kidney or CKD (chronic kidney disease)**.

### Techniques

| Technique | Method |
|---|---|
| **Off-clamp** | Hilum looped for rescue but not clamped; cold-scissor excision with strong suction |
| **Anatomical (super-selective)** | Microdissect the higher-order arterial branches entering the tumour; occlude only these with small bulldogs |
| **Selective segmental clamping** | Only the tumour's segment is ischaemic ("minimal ischaemia") |
| **Parenchymal compression** | Parenchymal clamp across a polar tumour |
| **Enucleation** | Pseudocapsule plane bleeds less, so suits off-clamp work |

### Aids

- **3D (three-dimensional) CT (computed tomography) reconstruction** maps segmental arteries; intraoperative Doppler ultrasound.
- **NIRF (near-infrared fluorescence) with ICG (indocyanine green)** on robotic consoles: after selective clamping, perfused kidney glows and the ischaemic tumour zone stays dark.
- **Robot:** wristed instruments and fast suturing ease off-clamp work.
- **Haemostasis:** clip or suture bed vessels as met; sliding-clip or barbed renorrhaphy; clamp if bleeding hides the margin.

### Selection

- **Ideal:** small (cT1a), exophytic, peripheral tumour; experienced surgeon.
- **Caution:** endophytic, hilar or central tumours.

### Evidence

- Off- versus on-clamp: **better early eGFR (estimated glomerular filtration rate), more blood loss**; **no difference beyond the early period**.
- **Robotic RCTs (randomised controlled trials, e.g. CLOCK):** no difference in eGFR or split function up to 2 years.
- RCT of super-selective clamping with NIRF: no 6-month gain; stopped for futility.
- **Concept:** parenchymal volume preserved, not ischaemia time within accepted limits, governs function.
- **Also know:** early unclamping; hypothermia for long ischaemia; mannitol no benefit.

**Diagram to draw:** hilum with a micro-bulldog on a tertiary branch; tumour zone dark on NIRF.

> **Key points to remember:**
> - Zero ischaemia: no main-artery clamp (Gill, 2011); off-clamp or super-selective.
> - NIRF with ICG confirms the right branch.
> - RCTs: no long-term eGFR gain; more blood loss.
> - Parenchyma preserved matters more than clamp time.

# Rapid Revision Sheet · Section 13

**Nephron-sparing surgery for renal masses: indications, approaches and recent advances (LAQ)**

- Ureteric bud gives collecting system; blastema gives nephrons (GDNF–RET).
- Five segmental end arteries; veins anastomose; Brödel line avascular; puncture at the fornix.
- SRM ≤4 cm: 20–30% benign; PN is standard for cT1a, ablation <3 cm, surveillance <2 cm.
- Imperative, relative, elective; elective PN is standard for cT1a; ablation or surveillance for frail patients with tumours <3 cm.
- R.E.N.A.L. 4–6 / 7–9 / 10–12; trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- AUA: RN only if complex + no CKD + new eGFR/GFR likely >45.
- Retroperitoneal for posterior, transperitoneal for anterior tumours.
- Preserved parenchyma, not clamp time, governs function.
- Enucleation for hereditary/multifocal; ablation for cT1a <3 cm after biopsy.
- Off-clamp and super-selective clamping: no long-term eGFR gain in randomised trials.
- 3D models: randomised benefit.
- EORTC 30904: PN better function, similar cancer control.
- Neoadjuvant TKI shrinks tumours 10–30%; for imperative PN only.

**Metastatic RCC: assessment and management (LAQ)**

- IMDC (6 factors): Karnofsky, haemoglobin, calcium, neutrophils, platelets, <1 year to therapy; poor = ≥3.
- IO/TKI (IO combinations) for all risk groups; nivolumab + ipilimumab for intermediate/poor only.
- CARMENA and SURTIME: no routine upfront CN.
- Adjuvant pembrolizumab (KEYNOTE-564), including M1 NED.
- Belzutifan after IO and TKI.

**Upper tract urothelial carcinoma (LAQ)**

- Tobacco, aristolochic acid (A:T → T:A in TP53) and Lynch (MSH2) are the key causes.
- CTU is the best test; URS biopsy gives grade, not stage.
- Ureteroscopy is closed but cannot stage; percutaneous stages but can seed the tract.
- Pelvic tumour into parenchyma is T3; through kidney to perinephric fat is T4; any N+ is stage IV.
- EAU low risk needs all favourable features (unifocal, under 2 cm, low grade, no invasion); any single adverse feature makes it high risk.
- Ureteric tumours: 70% distal; hydronephrosis predicts invasion (up to 80%).
- Ureteric T3 does worse than pelvic T3 (no parenchymal barrier).
- Distal ureterectomy for selected distal tumours.
- RNU with complete bladder cuff + single postoperative intravesical MMC: bladder recurrence 28% to 16% (ODMIT-C).
- POUT: adjuvant platinum, 3-year DFS 71% vs 46%; EV-302 for metastatic disease.

**Discuss differential diagnosis of small renal tumours. Describe techniques to achieve hemostasis during laparoscopic partial nephrectomy. (LAQ)**

- SRM ≤4 cm: 20–30% benign; ≤2 cm about 40%.
- Oncocytoma is the commonest benign mimic; biopsy cannot exclude chromophobe RCC.
- Trifecta: negative margin, no major complication, warm ischaemia ≤25 min.
- Delayed haematuria after PN means pseudoaneurysm: angioembolise.

**Radiology for solid renal masses. (LAQ)**

- Enhancement ≥20 HU = solid tumour; macroscopic fat = AML.
- MRI: clear cell T2-bright with chemical-shift drop; papillary T2-dark and hypoenhancing; ccLS 1–5.
- Sestamibi for oncocytoma; $^{89}$Zr-girentuximab PET (ZIRCON) for clear cell.
- Biopsy PPV ~99%, non-diagnostic ~14%.

**Immunotherapy / immunomodulation in RCC (SAQ)**

- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
- High-dose IL-2: durable complete response 7–9%.
- CheckMate 214: nivolumab + ipilimumab, OS HR 0.63, complete response 9%.
- KEYNOTE-564: adjuvant pembrolizumab, DFS and OS benefit.
- irAE grade ≥2: hold drug and give steroids.

**Renal angiomyolipoma and its management (SAQ)**

- Macroscopic fat on non-contrast CT is diagnostic.
- Treat if >6 cm, aneurysm >5 mm, growth, pregnancy or symptoms.
- SAE first for bleeding; partial nephrectomy preferred; everolimus for TSC >3 cm.

**Role of renal nephrometry in management of RCC. (SAQ)**

- RENAL 4–12: low 4–6, moderate 7–9, high 10–12.
- PADUA 6–14; C-index lower = more central.
- Predicts PN feasibility, ischaemia time and complications.
- MAP score: adherent perinephric fat. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Acquired renal cystic disease (SAQ)**

- ≥3 cysts per kidney in ESRD; >90% after 10 years.
- Uraemia causes it; transplant may regress cysts, not risk.
- RCC risk 3 times; image dialysis patients with haematuria.

**ADPKD (Adult polycystic kidney disease) (SAQ)**

- PKD1: chromosome 16, 85%, earlier ESRD.
- Ravine: <30 years ≥2 cysts; 30–59 ≥2 each kidney; ≥60 ≥4 each.
- Tolvaptan (TEMPO 3:4, REPRISE); ACE inhibitor or ARB.

**Biomarkers for patients with advanced RCC (SAQ)**

- IMDC: 6 factors; poor risk ≥3; LDH only in MSKCC.
- PD-L1 does not select patients in RCC.
- Sarcomatoid tumours respond well to immunotherapy.

**Tivozanib (SAQ)**

- Selective VEGFR-1–3 TKI; 1.34 mg daily, 3 weeks on, 1 off.
- TIVO-3: PFS 5.6 vs 3.9 months; OS not different.
- FDA 2021: after ≥2 prior systemic therapies.
- TiNivo-2 negative; main toxicity hypertension.

**VHL gene and von Hippel–Lindau disease (SAQ)**

- AD; 3p25 tumour suppressor; VHL protein degrades HIF.
- VHL loss: HIF-2α drives (→) VEGF, giving vascular clear cell RCC.
- Type 2 = phaeochromocytoma; 3 cm rule for nephron sparing (nephron-sparing surgery).
- Belzutifan (HIF-2α inhibitor) approved 2021.

**Zero-ischaemia partial nephrectomy (laparoscopic and robotic) (SAQ)**

- Zero ischaemia: no main-artery clamp (Gill, 2011); off-clamp or super-selective.
- NIRF with ICG confirms the right branch.
- RCTs: no long-term eGFR gain; more blood loss.
- Parenchyma preserved matters more than clamp time. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 14 · Adrenal & Retroperitoneum

## Long answer questions (LAQ)

### LAQ 1 Phaeochromocytoma: adrenal anatomy, hormones and perioperative management

P1 · P2 **merged from 2**

*Asked as:* P1 LAQ "Describe anatomy of arenal glands AND steps in production of adrenal hormones & perioperative management of a patient of pheochromocytoma" (Practice Set 6 Paper 1 Q1) · P2 SAQ "Phaeochromocytoma: overview, pre-operative and intra-operative management" (asked as "Pre operative management of pheochromocytoma", Summer 2020 Paper II Q3c; "Intra op mgt of pheo", Practice Set 7 Paper II Q3c; "Pheochromocytoma", Practice Set 1 Paper II Q3d)

*Assumed meaning: "arenal" = adrenal glands.*

#### Adrenal anatomy

**Gross**

- Paired retroperitoneal glands on the upper renal poles (11th–12th ribs), inside Gerota fascia, separated from the kidney by thin connective tissue. Yellow-orange, 4–6 × 2–3 cm, about 5 g each.

| Feature | Right | Left |
|---|---|---|
| Shape, level | Triangular; higher | Crescentic; lower (to L1) |
| Relations | Liver anteriorly, duodenum anteromedially, IVC (inferior vena cava) medially, diaphragmatic crus behind | Pancreas body and splenic vessels anteriorly, aorta medially, crus behind |
| Vein | **Short (<1 cm), into posterior IVC**: tears easily | **Longer, into left renal vein**, often joined by inferior phrenic vein |

**Blood supply**

- **"Many arteries in, one vein out."** Superior adrenal (from inferior phrenic), middle (aorta) and inferior (renal artery) arteries break into 10–50 small branches.
- Cortical sinusoids drain into the medulla, so cortisol-rich blood reaches the chromaffin cells. **Cortisol induces PNMT (phenylethanolamine-N-methyltransferase)**, which is why the medulla makes adrenaline.

**Nerves:** preganglionic sympathetic fibres (T11–L2, via splanchnic nerves, no synapse in the coeliac ganglion) end directly on chromaffin cells. The medulla is a modified sympathetic ganglion.

**Embryology:** cortex from coelomic mesoderm (adrenogonadal ridge, weeks 3–4); medulla from neural crest. Hence adrenals stay orthotopic in renal agenesis, and adrenal rests occur along the path of gonadal descent.

**Histology (zones from outside in: glomerulosa, fasciculata, reticularis = "salt, sugar, sex")**

| Zone | Share | Product | Main control |
|---|---|---|---|
| Glomerulosa | ~15% of cortex | Aldosterone | Angiotensin II, potassium |
| Fasciculata | ~80% | Cortisol (up to 20 mg/day) | ACTH (adrenocorticotropic hormone) |
| Reticularis | 5–7% | DHEA (dehydroepiandrosterone), DHEA-S, androstenedione | ACTH |
| Medulla | ~10% of gland | Adrenaline 80%, noradrenaline 19%, dopamine 1% | Sympathetic nerves |

## Steps in adrenal hormone production

### 1. Steroids (cortex)

- Cholesterol (mainly from LDL [low-density lipoprotein]) is carried into mitochondria by **StAR (steroidogenic acute regulatory protein)**: the rate-limiting step, driven by ACTH.
- **Side-chain cleavage enzyme (CYP11A1)** converts it to **pregnenolone**. The enzymes present in each zone then decide the product: `Pregnenolone -3βHSD-> Progesterone -21OH-> 11-deoxycorticosterone -11βOH-> Corticosterone -aldosterone synthase (CYP11B2)-> ALDOSTERONE (glomerulosa only) Pregnenolone -17αOH-> 17-OH pregnenolone/progesterone -21OH-> 11-deoxycortisol -11βOH-> CORTISOL (fasciculata) 17-OH pregnenolone -17,20-lyase-> DHEA -> androstenedione (reticularis)`
- 3βHSD (3β-hydroxysteroid dehydrogenase). Steroids are not stored; they diffuse out and act on intracellular receptors.
- **Regulation:** CRH (corticotropin-releasing hormone) drives ACTH, which drives cortisol, with negative feedback and a morning peak; RAAS (renin–angiotensin–aldosterone system) and potassium control aldosterone.
- **Clinical link:** 21-hydroxylase deficiency is the commonest cause of CAH (congenital adrenal hyperplasia).

**2. Catecholamines (medulla)** `Tyrosine -tyrosine hydroxylase (rate-limiting)-> L-DOPA -decarboxylase-> Dopamine -dopamine β-hydroxylase-> Noradrenaline -PNMT (cortisol-induced)-> Adrenaline`

- Stored in vesicles, released by exocytosis. **COMT (catechol-O-methyltransferase)** inside chromaffin cells forms **metanephrine and normetanephrine** continuously, which is why metanephrines are the best test. MAO (monoamine oxidase) and the liver form VMA (vanillylmandelic acid).

## Phaeochromocytoma: definition and epidemiology

- Catecholamine-secreting tumour of adrenal medullary chromaffin cells; **paraganglioma** is the extra-adrenal form (organ of Zuckerkandl, bladder).
- 1–2 per 100,000 a year; about 5% of adrenal incidentalomas.
- **"Rule of 10" is outdated:** extra-adrenal up to 25%, **hereditary about one-third**, bilateral commoner in hereditary disease.

### Genetics

- **MEN2 (multiple endocrine neoplasia type 2)**, RET: about 50% risk, adrenal, low malignancy.
- **VHL (von Hippel–Lindau)**: normetanephrine only.
- **NF1 (neurofibromatosis type 1)**.
- **SDHB (succinate dehydrogenase subunit B):** extra-adrenal, **30–50% metastatic**. (SDHB/SDHD = succinate dehydrogenase subunits.)
- Offer genetic testing to all: about one-third hereditary (MEN2, VHL, NF1, SDHB/SDHD).

### Clinical features

- Triad: **headache, sweating, palpitations**. Sustained hypertension is commoner than paroxysmal; pallor, hyperglycaemia; >20% asymptomatic. Epinephrine-dominant tumours can cause hypotensive episodes.

### Why preparation is needed

- Handling the tumour releases catecholamines: hypertensive crisis, arrhythmia, MI (myocardial infarction), stroke. After vein ligation, sudden withdrawal causes hypotension and hypoglycaemia.
- Phaeochromocytoma surgery is a **two-phase haemodynamic problem**: catecholamine **surge** while the tumour is handled, then catecholamine **withdrawal** once the vein is tied.
- Surges occur at intubation, positioning, pneumoperitoneum and tumour manipulation.
- Mortality fell from up to 50% to <3% with blockade, now advised for **all** proven tumours, even normotensive.

### Pre-operative step 1: confirm, localise, test genes

- **Biochemistry first:** plasma free or 24-hour urinary fractionated **metanephrines**. Tumours convert catecholamines continuously, so metanephrines are more sensitive than episodic catecholamines. Chromogranin A; clonidine suppression for borderline results.
- **Imaging:** CT (computed tomography): >10 HU (Hounsfield units) unenhanced, no rapid washout; MRI (magnetic resonance imaging).
- **Functional imaging** if extra-adrenal or hereditary: $^{68}$Ga-DOTATATE PET (positron emission tomography) or MIBG (metaiodobenzylguanidine); FDG (fluorodeoxyglucose) PET (best for SDHB); $^{123}$I-MIBG (best in MEN2).
- **Never biopsy** (hypertensive crisis). Iodinated contrast is safe.
- Genetic testing (about one-third hereditary).

### Pre-operative step 2: cardiac assessment

- ECG (electrocardiogram) and echocardiography for catecholamine cardiomyopathy; joint plan with the anaesthetist.

### Pre-operative step 3: drug blockade (7–14 days)

| Drug | Action | Points |
|---|---|---|
| **Phenoxybenzamine** (first line) | Irreversible non-selective α-blocker | 10 mg twice daily, increased every 2–3 days (titrated); about 1 mg/kg total |
| Doxazosin, prazosin | Selective α1-blockers | Fewer side effects |

| Drug | Action | Points |
|---|---|---|
| Calcium channel blocker | Smooth muscle relaxation | Alternative strategy or add-on |
| β1-blocker (atenolol, metoprolol) | Controls tachycardia | **Only after adequate α-blockade** |
| Metyrosine | Blocks tyrosine hydroxylase | Add-on; sedation, extrapyramidal effects |

- **α-blockade for all**, 7–14 days.
- **Never β before α:** β-blockade removes β2 vasodilatation, leaving α-vasoconstriction unopposed, which causes crisis.
- **Target:** seated BP (blood pressure) 120–130/80 mmHg; a mild postural drop (systolic >90 mmHg) shows adequate blockade.

### Pre-operative step 4: volume expansion

- High-salt diet and fluids once blockade starts; 1–2 L IV (intravenous) fluid the night before. This prevents post-excision hypotension.

### Pre-operative step 5: day of surgery

- Last phenoxybenzamine dose the night before surgery; omit the morning dose.

### Intra-operative management

#### Prerequisites

- α-blockade 7–14 days (phenoxybenzamine or doxazosin), β-blocker only after α, volume expansion; ECG and echocardiography; joint plan with the anaesthetist.

#### Monitoring

- **Arterial line before induction**, central venous line, ECG, urine output, frequent glucose; cardiac output or transoesophageal echo if cardiomyopathy.

#### Anaesthesia

- Deep, smooth induction (propofol, opioid such as fentanyl or remifentanil); attenuate the intubation response.
- **Avoid triggers:** histamine-releasing drugs (morphine, atracurium), ketamine, ephedrine, metoclopramide, droperidol, desflurane.
- Sevoflurane or isoflurane; vecuronium or rocuronium; magnesium sulphate as an adjunct (reduces catecholamine release and arrhythmia).

#### Haemodynamic control

| Event | Action |
|---|---|
| **Hypertensive surge** | **Pause manipulation**; nitroprusside, nicardipine or clevidipine, nitroglycerin, phentolamine |
| **Tachyarrhythmia** | **Esmolol**, lidocaine; amiodarone if needed |
| **Hypotension after vein ligation** | **Aggressive volume**, then α-agonist (phenylephrine, noradrenaline); vasopressin if refractory |
| **Hypoglycaemia** (insulin rebound) | Dextrose infusion, glucose checks |

### Surgical principles

- **Laparoscopic adrenalectomy** (transperitoneal or posterior retroperitoneoscopic) is standard; open for large or invasive tumours.
- **Minimal handling**; "no-touch" dissection away from the tumour.
- **Early adrenal vein ligation** where possible, after warning the anaesthetist.
- Remove intact in a bag (avoid capsule rupture and seeding).
- **Cortical-sparing** (partial) adrenalectomy in MEN2, VHL and bilateral tumours.
- Steroid cover if bilateral adrenalectomy.

## Post-operative management

- HDU (high dependency unit) overnight: blood pressure, heart rate, glucose.
- Fluids ± vasopressors for **hypotension**.
- Glucose monitoring for **rebound hypoglycaemia** (insulin no longer suppressed).
- Steroid cover after bilateral adrenalectomy.

## Follow-up and metastatic disease

- Metanephrines after surgery, then **yearly for life** (recurrence up to 16%).
- Malignancy is defined only by **metastases**. Treat with metastasectomy, $^{131}$I-MIBG, CVD (cyclophosphamide, vincristine, dacarbazine).

## Diagrams to draw

- Posterior view of both adrenals with the three arteries and the two veins.
- Flowchart: metanephrines, imaging, α-blockade, β-blockade, salt and fluids, surgery.
- Intra-operative BP curve: surges during handling, fall after vein ligation, with drugs marked.
- Catecholamine pathway to metanephrines (COMT).

> **Key points to remember:**
>
> - Right vein short into IVC; left vein into left renal vein.
> - StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
> - Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
> - Metanephrines first; never biopsy.
> - α-blockade 7–14 days for all; never β before α (α before β), plus volume expansion.
> - Volume expansion prevents hypotension; watch glucose.
> - Arterial line before induction; avoid histamine releasers and ephedrine.
> - Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
> - After vein ligation: fluids, vasopressors, glucose.
> - One-third hereditary; SDHB most often metastatic.

## LAQ 2 Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma

P2 · P4 **merged from 2**

*Asked as:* P4 LAQ "Adrenal mass evaluation and adrenocortical carcinoma: diagnosis and management" (asked as "Diagnosis and management for Adrenal Carcinoma", Summer 2018 Paper IV Q2 LAQ; "Approach to evaluation of adrenal mass/s", Summer 2022 Paper IV Q3f SAQ) · P2 SAQ "Algorithm for management of an incidentaloma" (Practice Set 3 Paper 2 Q3d)

*Assumed meaning (P2): adrenal incidentaloma.*

### Definition and epidemiology

- **Adrenal incidentaloma:** mass **≥1 cm** found on imaging done for another reason.
- Two questions: **Is it malignant? (imaging)** and **Is it functioning? (biochemistry).**
- ACC (adrenocortical carcinoma): rare (0.7–2 per million per year), aggressive cortical tumour. **Bimodal** (children <10 years; adults 40–50 years); slight female excess.
- **ACC syndromes:** Li–Fraumeni (TP53), Beckwith–Wiedemann (IGF-2, insulin-like growth factor 2), Lynch, MEN-1 (multiple endocrine neoplasia 1), Carney complex, FAP (familial adenomatous polyposis). Surgery is the only cure.

### Differential diagnosis of an adrenal mass

- Non-functioning adenoma (commonest); cortisol-secreting adenoma 5.3%; phaeochromocytoma 5.1%; aldosteronoma 1%; ACC 4.7%; metastasis 2.5%; myelolipoma, cyst, haemorrhage.
- **Bilateral:** metastases, lymphoma, **tuberculosis** and histoplasmosis, haemorrhage, CAH (congenital adrenal hyperplasia), macronodular hyperplasia, bilateral phaeochromocytoma (MEN-2 [multiple endocrine neoplasia type 2], VHL [von Hippel–Lindau]).

### Clinical features of ACC

- **Functional in 50–79% of adults** (90% of children):
- Cortisol (commonest): rapidly progressive Cushing syndrome, often with virilisation.
- Androgens: virilisation in women. **Oestrogen in a man (feminisation) suggests carcinoma.**
- Aldosterone: rare.
- Non-functional: pain, fullness, weight loss, or incidental.

### Incidentaloma algorithm

Adrenal mass ≥1 cm ├─ IMAGING: unenhanced CT (computed tomography) │ ≤10 HU (Hounsfield units), homogeneous → benign (lipid-rich) adenoma │ (or MRI chemical-shift signal drop-out → lipid-rich adenoma) │ >10 HU → washout CT (15 min) or chemical-shift MRI │ APW >60% or RPW >40% → adenoma │ indeterminate → MDT (multidisciplinary team): repeat imaging 6-12 months, FDG-PET or surgery │ Macroscopic fat → myelolipoma ├─ HORMONES (all patients) │ 1-mg overnight DST (dexamethasone suppression test) │ plasma or urine metanephrines │ ARR (aldosterone-renin ratio) if hypertension or low potassium │ sex steroids only if ACC suspected └─ DECISION functioning, suspicious, >4 cm, or growing → adrenalectomy benign, non-functioning → discharge

### Hormonal work-up (all lesions; ESE–ENSAT, European Society of Endocrinology–European Network for the Study of Adrenal Tumours)

| Axis / hormone | Tests |
|---|---|
| Glucocorticoid (cortisol) | **1-mg overnight DST** (or 24-h urine or late-night salivary cortisol): ≤1.8 µg/dL (50 nmol/L) excludes; 1.8–5.0 µg/dL = possible autonomous secretion (**MACS**, mild autonomous cortisol secretion); >5.0 µg/dL autonomous; ACTH (adrenocorticotropic hormone) |
| Sex steroids and precursors | DHEA-S (dehydroepiandrosterone sulphate), androstenedione, testosterone, 17-OH-progesterone, 11-deoxycortisol; oestradiol in men and postmenopausal women (only if ACC suspected) |
| Mineralocorticoid (aldosterone) | **ARR** (aldosterone-to-renin ratio) if hypertension or hypokalaemia |
| Catecholamines | **Plasma free or 24-h urinary fractionated metanephrines**: must exclude phaeochromocytoma |
| Bilateral | 17-hydroxyprogesterone (CAH); exclude adrenal insufficiency |

- Why: confirms cortical origin, plans steroid cover, gives **tumour markers** for follow-up.
- **Never biopsy or operate before excluding phaeochromocytoma.** Biopsy only for suspected metastasis in known cancer.

### Imaging

- Abbreviations: CT, HU, MRI (chemical-shift magnetic resonance imaging), MDT, FDG-PET (fluorodeoxyglucose positron emission tomography).
- **APW** (absolute percentage washout) = (enhanced − delayed) ÷ (enhanced − unenhanced) × 100; **RPW** (relative) = (enhanced − delayed) ÷ enhanced × 100.
- Malignant features: heterogeneous, irregular, invasion, >4 cm. Macroscopic fat = myelolipoma.
- **Size and malignancy:** <4 cm 0.1%; 4–6 cm 2.4%; >6 cm 19.5% (another source: carcinoma in ~2% <4 cm, ~6% at 4–6 cm, ~25% >6 cm). Benign imaging overrides size.
- **ACC size:** mean 10–12 cm; >90% exceed 5 cm.
- **CT in ACC:** irregular, heterogeneous, necrosis, calcification; **unenhanced >10 HU** (mean ~39 HU vs ~8 HU in adenoma); **slow washout** (absolute <60%, relative <40% at 15 min). Growth ~1.9 cm/year.
- **MRI in ACC:** no chemical-shift signal loss; best for IVC (inferior vena cava) thrombus and invasion.
- **FDG-PET/CT:** ACC is avid, but PET cannot separate ACC from metastasis.
- **Staging:** CT chest, abdomen, pelvis; bone or brain only if symptomatic.
- **No biopsy** if ACC is suspected (seeding, poor accuracy); only for unresectable or metastatic disease, after excluding phaeochromocytoma.

### Management of incidentaloma (ESE–ENSAT 2023)

- **Surgery:** cortisol excess, phaeochromocytoma (after alpha-blockade), aldosteronoma (after AVS, adrenal vein sampling), malignant features or suspicious imaging, >4 cm in fit patients, significant growth (≥5 mm and >20%). Laparoscopic for benign; open for suspected ACC.
- **MACS:** screen for hypertension, diabetes and osteoporosis; surgery individualised.

- **Follow-up:** homogeneous **≤10 HU, non-functioning: benign, no further imaging whatever the size**; metanephrines may be omitted. Indeterminate non-resected: re-image at 6–12 months; growth >20% and ≥5 mm is concerning.
- **Bilateral:** metastases, lymphoma, tuberculosis, haemorrhage, CAH (17-hydroxyprogesterone); exclude adrenal insufficiency.

### ACC pathology: Weiss score (≥3 of 9 = malignant)

| Domain | Criteria |
|---|---|
| Nuclear / cytology | High nuclear grade; >5 mitoses per 50 high-power fields; atypical mitoses |
| Architecture | ≤25% clear cells; diffuse architecture; necrosis |
| Invasion | Venous; sinusoidal; capsular |

- SF-1 (steroidogenic factor 1) confirms cortical origin. **Ki-67** and mitotic rate grade the tumour (low vs high grade).

### ENSAT staging

| Stage | Definition | 5-year disease-specific survival (ENSAT 2009) |
|---|---|---|
| I | T1 (≤5 cm), N0 M0 | ~80% |
| II | T2 (>5 cm), N0 M0 | ~60% |
| III | T3 (periadrenal fat) or T4 (adjacent organ, venous thrombus) or N1 | ~50% |
| IV | Distant metastasis only | ~13% |

- ENSAT differs from older UICC (Union for International Cancer Control) staging by moving T4 and N1 without metastasis into stage III.

### Management of ACC

#### 1. Surgery (stages I–III)

- **Open transperitoneal adrenalectomy** for suspected ACC, especially >6 cm or invasive. Laparoscopy risks capsule rupture and peritoneal carcinomatosis; only for small organ-confined tumours by experts.
- **R0, intact capsule:** rupture gives dismal results. **En bloc** removal of involved kidney, liver, spleen, pancreas or IVC; IVC thrombectomy (bypass if needed).
- **Locoregional lymphadenectomy** (periadrenal fat, renal hilum, para-aortic or paracaval) improves staging and may reduce recurrence.
- High-volume surgeon (ESE–ENSAT: >6, preferably ≥20 adrenalectomies a year). Hydrocortisone cover if cortisol-secreting.

#### 2. Adjuvant therapy

- **Mitotane** (adrenolytic DDT [dichlorodiphenyltrichloroethane] derivative) for **high risk**: stage III, R1/Rx, or Ki-67 >10% (ESE–ENSAT 2018), for at least 2 years.
- Target blood level **14–20 mg/L**; needs **glucocorticoid replacement** (it destroys the cortex and raises cortisol-binding globulin).
- Toxicity: gastrointestinal, neurological, hepatic, hypothyroidism, gynaecomastia.

- **ADIUVO** (RCT, randomised controlled trial; R0, stage I–III, Ki-67 ≤10%): no recurrence-free survival benefit, so observe low-risk patients.
- **Adjuvant radiotherapy** (tumour bed): R1 or Rx, or stage III. It cuts local recurrence but not overall survival.

**3. Advanced or metastatic (stage IV)**

- **EDP-M** (etoposide, doxorubicin, cisplatin + mitotane) is first line. **FIRM-ACT** (2012): versus streptozotocin–mitotane, response 23% vs 9% and PFS (progression-free survival) 5.0 vs 2.1 months; overall survival not significantly better.
- Mitotane alone for low-volume, slow disease.
- **Surgery:** resect oligometastases or late recurrence; debulk only if >90% removable.
- **Hormone control:** metyrapone, ketoconazole, mifepristone, etomidate. Radiotherapy for bone and brain metastases.

### Follow-up of ACC

- CT chest, abdomen, pelvis and baseline-raised hormones **every 3 months for 2 years**, then 6-monthly to 5 years.

### Prognosis of ACC

- Overall 5-year survival **20–47%**. Bad factors: stage, R1, high Ki-67 or mitoses, cortisol excess, age, size >12 cm.

### Recent advances

- Molecular classes (CIMP, CpG island methylator phenotype) predict prognosis.
- Checkpoint inhibitors give modest responses; IGF-1R (IGF-1 receptor) inhibitor linsitinib failed (GALACCTIC).
- ADIUVO-2 tests mitotane ± platinum–etoposide in high risk.

### Diagrams to draw

- The incidentaloma imaging and biochemistry algorithm above.
- ACC algorithm: 1. Adrenal mass: hormones + CT (HU, washout). 2. Suspected ACC: open R0 en bloc resection + LND (lymph node dissection). 3. Risk-based adjuvant mitotane or radiotherapy.

> **Key points to remember:**
>
> - ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
> - 1-mg DST: >1.8 µg/dL = MACS.
> - Exclude phaeochromocytoma before biopsy or surgery.
> - Surgery for incidentaloma: functioning, suspicious, >4 cm or growing.
> - Feminisation in a man, >6 cm, >10 HU, poor washout: think ACC. Never biopsy before resection.
> - Weiss ≥3/9; ENSAT stage IV = distant metastasis only.
> - Open, R0, intact capsule, en bloc + lymphadenectomy.
> - Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

## LAQ 3 How will you evaluate and manage 38 yr male with uncontrolled hypertension

P2

**Type:** LAQ (25 marks)
**Source:** Practice Set 7 Paper 2, Q1 (renovascular hypertension also answered in Practice Set 2 Paper 2, Q1; phaeochromocytoma in Practice Set 6 Paper 1, Q1)

*Assumed focus: secondary (urological, renal and adrenal) causes of hypertension in a young adult.*

### Introduction

- **Resistant hypertension:** BP (blood pressure) above goal despite 3 drugs of different classes, including a diuretic, at optimal doses; or needing 4 or more drugs.
- Poor control below 40 years is a classic trigger to search for a **secondary cause**; many are curable.

### Causes

| Group | Causes | Clues |
|---|---|---|
| Renovascular | FMD (fibromuscular dysplasia), atherosclerotic RAS (renal artery stenosis), Takayasu arteritis, reninoma | Bruit; creatinine rise >30% on ACE (angiotensin-converting enzyme) inhibitor; flash pulmonary oedema; kidneys differing >1.5 cm |
| Renal parenchymal | CKD (chronic kidney disease), ADPKD (autosomal dominant polycystic kidney disease), reflux nephropathy, glomerulonephritis, Page kidney | Raised creatinine, proteinuria, haematuria |
| Obstructive uropathy | Bilateral obstruction (about 77% hypertensive), unilateral (about 20%), chronic retention | LUTS (lower urinary tract symptoms), palpable bladder, hydronephrosis |
| Primary aldosteronism (PA) | Adenoma (Conn), bilateral hyperplasia | Hypokalaemia (present in <40% only) |
| Phaeochromocytoma, paraganglioma | Sporadic or hereditary | Paroxysmal headache, sweating, palpitations |
| Cushing syndrome | Exogenous steroid, adrenal adenoma, pituitary | Central obesity, striae, diabetes |
| - **Also know:** sleep apnoea; drugs (NSAIDs [non-steroidal anti-inflammatory drugs], steroids, decongestants, liquorice); alcohol; thyroid and parathyroid disease; coarctation. | | |

### Evaluation

#### Step 1: Exclude pseudo-resistance

- Correct cuff technique; home or ambulatory monitoring to exclude white-coat effect; check adherence, doses, interfering drugs, salt and alcohol.

**Step 2: History and examination**

- Paroxysms, weakness or polyuria (hypokalaemia), stones, UTI (urinary tract infection) or childhood reflux, snoring, family history.
- BP both arms and radio-femoral delay; abdominal bruit; palpable kidneys or bladder; cushingoid habitus; café-au-lait spots (NF1 [neurofibromatosis type 1]); fundus.

**Step 3: Baseline tests and target-organ damage**

- Creatinine, eGFR (estimated glomerular filtration rate), electrolytes, urinalysis with albumin–creatinine ratio, glucose, lipids, ECG (electrocardiogram), echocardiography.
- **Ultrasound KUB (kidneys, ureters, bladder):** renal size and asymmetry, scars, cysts, masses, hydronephrosis, residual urine.

**Step 4: Targeted screening (method and interpretation)**

| Cause | Screen | Confirm or localise |
|---|---|---|
| PA | Morning **ARR (aldosterone-to-renin ratio)** >20–30 with aldosterone ≥15–20 ng/dL; stop spironolactone first; correct potassium | **Intravenous saline 2 L over 4 h** (aldosterone stays >5 ng/dL) or oral salt loading → adrenal CT (computed tomography) → **AVS (adrenal vein sampling)**; skip AVS if age <40 with a clear unilateral adenoma >1 cm |
| Phaeochromocytoma | **Plasma free or 24-h urinary fractionated metanephrines** | CT or MRI (magnetic resonance imaging); $^{68}$Ga-DOTATATE PET (positron emission tomography) or MIBG (metaiodobenzylguanidine); genetic testing |
| Cushing | **1-mg overnight dexamethasone suppression** (cortisol ≤1.8 µg/dL excludes); late-night salivary cortisol; 24-h urinary free cortisol | ACTH (adrenocorticotropic hormone): low = adrenal, high = pituitary or ectopic |
| Renovascular | **Duplex Doppler**: peak systolic velocity >180–200 cm/s, renal/aortic ratio >3.5 | CT or MR angiography; catheter angiography with pressure gradient if intervention planned |
| Parenchymal, obstructive | Ultrasound, urine tests | Renogram (DTPA [diethylenetriaminepentaacetic acid] or MAG3 [mercaptoacetyltriglycine]) for split function and drainage; CT urography |

```
Uncontrolled BP, age 38 → exclude pseudo-resistance → baseline tests + ultrasound KUB → ARR, metanephrines, dexamethasone test, renal Doppler ├─ positive → confirm → localise → treat cause └─ all negative → essential: optimise drugs
```

## Management

### 1. General (all patients)

- Lifestyle: salt restriction, weight loss, exercise, limit alcohol.
- Optimise **ACE inhibitor or ARB (angiotensin receptor blocker) + calcium channel blocker + thiazide-like diuretic**. Fourth drug: **spironolactone** (PATHWAY-2, 2015: best add-on in resistant hypertension).

**2. Cause-specific**

| Cause | Treatment |
|---|---|
| PA, lateralised | **Laparoscopic adrenalectomy**: biochemical cure about 94%, but only 37–52% need no antihypertensives |
| PA, bilateral or unfit | Spironolactone 25–50 mg/day (up to 400 mg) or eplerenone |
| Phaeochromocytoma | **α-blockade 7–14 days**, β-blocker only after α, volume expansion → laparoscopic adrenalectomy; metanephrines yearly for life |
| Adrenal Cushing | Adrenalectomy with steroid cover; pituitary disease → trans-sphenoidal surgery |
| FMD | **Angioplasty alone** (cure common in the young) |
| Atherosclerotic RAS | Medical therapy (**ASTRAL 2009, CORAL 2014**: stenting no benefit); stent for flash pulmonary oedema or falling function |
| Branch disease, aneurysm | Surgical bypass or ex vivo repair with autotransplantation |
| Reninoma | Partial nephrectomy (cures most) |
| Small non-functioning kidney (reflux, segmental hypoplasia) | **Nephrectomy** if renal vein renin lateralises (ratio >1.5) |
| Page kidney | Drain or decapsulate |
| Obstruction | **Relieve it** (catheter, nephrostomy, pyeloplasty, prostate surgery); watch post-obstructive diuresis |
| CKD | ACE inhibitor or ARB for proteinuria; scheduled diuretics |

**3. Follow-up:** home BP, creatinine, potassium; duplex after angioplasty.

### Recent advances

- **Catheter renal denervation** (radiofrequency or ultrasound), approved in the USA in 2023 for selected resistant hypertension; SPARTACUS trial questioned routine AVS.

**Diagram to draw:** stepwise algorithm from pseudo-resistance to cause-specific treatment.

> **Key points to remember:**
> - Hypertension below 40 years: screen for secondary causes.
> - ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
> - Metanephrines for phaeochromocytoma; α before β.
> - FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

## Short answer questions (SAQ)

### SAQ 1 Cushing syndrome and the dexamethasone suppression test

P2

**Asked as:**

- Cushing Syndrome — *Practice Set 2 Paper II Q3c (SAQ)*
- Dexamethasone suppression test — *Practice Set 5 Paper II Q3c (SAQ)*

**Type:** SAQ (10 marks)

## Definitions

- **Cushing syndrome:** symptomatic hypercortisolism. **Cushing disease:** due to a pituitary ACTH (adrenocorticotropic hormone)-secreting adenoma.
- **MACS (mild autonomous cortisol secretion):** autonomous cortisol without clinical signs (replaces "subclinical Cushing"); raises cardiovascular, bone and diabetes risk.

## Causes

| Group | Cause | Share |
|---|---|---|
| Exogenous | Glucocorticoids (oral, inhaled, topical) | Commonest overall |
| ACTH-dependent (80–85% of endogenous) | Pituitary adenoma (Cushing disease) | About 70% |
| | Ectopic ACTH (bronchial carcinoid, small cell lung cancer, phaeochromocytoma) | 10–15% |
| ACTH-independent | Adrenal adenoma, ACC (adrenocortical carcinoma), AIMAH (macronodular hyperplasia), PPNAD (pigmented nodular disease; Carney complex) | About 10% |

## Clinical features

- Central obesity, moon face, plethora (90–100%); purple striae, hypertension, hirsutism, ED (70–90%).
- **Discriminating:** proximal weakness, bruising.
- **Urological:** nephrolithiasis in up to 50%; hypogonadotropic hypogonadism (low libido, ED).

## Diagnosis

**Step 1: exclude exogenous steroids (including creams and sprays) first. Step 2: confirm cortisol excess (two tests)**

| Test | Method | Abnormal |
|---|---|---|
| **Overnight 1-mg DST (dexamethasone suppression test)** | 1 mg at 11 pm; 8 am cortisol | >1.8 µg/dL (50 nmol/L) |
| Late-night salivary cortisol | 11 pm, twice | Raised |
| 24-h UFC (urinary free cortisol) | Two collections | Raised; used in pregnancy |

**Step 3: locate the source**

- **Low ACTH:** adrenal; adrenal CT or MRI.
- **High ACTH:** pituitary MRI; **IPSS (inferior petrosal sinus sampling) with CRH (corticotropin-releasing hormone)**: central gradient = pituitary; none = ectopic (chest CT).
- Pitfalls: pseudo-Cushing (depression, alcohol), oestrogens, enzyme inducers.

## Dexamethasone suppression test

### Principle

- **Dexamethasone** is a potent synthetic glucocorticoid that is **not detected by the cortisol assay**. In a normal axis it suppresses pituitary ACTH (adrenocorticotropic hormone) by negative feedback, so morning cortisol falls.

- **Autonomous** cortisol (adrenal adenoma) or excess ACTH drive (pituitary, ectopic) fails to suppress.
- "Low-dose" 1 mg is still 3–4 times physiological glucocorticoid.

### Indications

- Suspected Cushing syndrome; **every adrenal incidentaloma** (5–8% secrete cortisol); suspected ACC (adrenocortical carcinoma).

### Types

| Test | Method | Abnormal |
|---|---|---|
| **Overnight 1-mg LD-DST** (low-dose DST; preferred screen) | Dexamethasone 1 mg at 11 pm–midnight; serum cortisol 8–9 am | See below |
| 48-hour 2-mg LD-DST | 0.5 mg 6-hourly for 48 hours; cortisol at 48 hours | >1.8 µg/dL; improves accuracy |
| High-dose DST | 8 mg overnight or 2 mg 6-hourly for 48 hours | Over 50% suppression suggests pituitary source; **no longer routine** (some ectopic tumours also suppress) |

### Interpreting the 1-mg test

| Morning cortisol | Meaning |
|---|---|
| **≤1.8 µg/dL (50 nmol/L)** | Normal; >90% sensitive, rules out |
| 1.8–5.0 µg/dL | Possible autonomous cortisol secretion |
| **>5.0 µg/dL (140 nmol/L)** | Autonomous secretion; >95% specific |
| - ESE (European Society of Endocrinology) 2023 incidentaloma guideline: cortisol >1.8 µg/dL without overt signs = **MACS (mild autonomous cortisol secretion)**. | |
| - If abnormal: repeat testing and measure **morning ACTH** (low = adrenal; high = pituitary or ectopic, IPSS [inferior petrosal sinus sampling]). | |

### Pitfalls

- **False positive:** oral contraceptives (raise cortisol-binding globulin; up to 50%); CYP3A4 inducers (phenytoin, carbamazepine, rifampicin); missed dose; pseudo-Cushing states (depression, alcohol, obesity, stress).
- **False negative:** CYP3A4 inhibitors (itraconazole, ritonavir, diltiazem).
- **Choice of test:** pregnancy → UFC (urinary free cortisol); epilepsy → LNSC (late-night salivary cortisol); renal failure → DST (not affected by GFR [glomerular filtration rate]).

## Management

| Cause | Treatment |
|---|---|
| Cushing disease | Trans-sphenoidal surgery (cure 60–80%); radiotherapy |
| Adrenal adenoma | **Laparoscopic adrenalectomy** |
| ACC | Open radical adrenalectomy, mitotane |

| Cause | Treatment |
|---|---|
| Ectopic | Resect source; otherwise bilateral adrenalectomy |
| AIMAH, PPNAD | Bilateral adrenalectomy |

- **Bilateral adrenalectomy:** for failed pituitary surgery or life-threatening hypercortisolism. Cost: lifelong replacement, **Nelson syndrome** (8–29%).
- **Medical (bridge):** metyrapone, ketoconazole, etomidate, mitotane, mifepristone.

### Perioperative care (urological relevance)

- **Contralateral adrenal is suppressed:** autonomous cortisol suppresses the opposite adrenal; adrenal insufficiency in about 99.7% after surgery for overt Cushing.
- Give **perioperative steroid cover** and taper after adrenalectomy: **Hydrocortisone 100 mg IV** on the day of surgery, then slow oral taper; recovery takes a median 11 months.

**Diagram to draw:** HPA (hypothalamic–pituitary–adrenal) axis with dexamethasone feedback and with ACTH-dependent and ACTH-independent sites.

> **Key points to remember:**
> - Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
> - 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
> - Oral contraceptives and enzyme inducers cause false positives.
> - ACTH low = adrenal; high = pituitary/ectopic (IPSS); ACTH separates adrenal from pituitary/ectopic.
> - High-dose DST no longer routine.
> - Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

## SAQ 2 MIBG scan

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 2, Q3c

### Principle

- MIBG (meta-iodobenzylguanidine) is a guanethidine analogue that resembles **noradrenaline**.
- It enters chromaffin and sympathetic cells through the **noradrenaline transporter** and is stored in **adrenergic secretory vesicles**. So it maps catecholamine-handling tissue: function, not just anatomy.

### Isotopes

| | $^{123}$I-MIBG | $^{131}$I-MIBG |
|---|---|---|
| Use | **Diagnostic** (preferred) | Older diagnostic; mainly **therapeutic** |
| Image quality | Better; allows SPECT-CT (single-photon emission computed tomography) | Poorer |
| Emission | Gamma | Beta (therapy) plus gamma |

### Technique

- **Thyroid blockade** with potassium iodide before and after injection (free iodine).
- **Stop interfering drugs:** labetalol, tricyclic antidepressants, sympathomimetics, reserpine, some calcium channel blockers.
- Intravenous injection; planar and SPECT-CT images at about 24 hours.
- **Normal uptake:** salivary glands, liver, heart, bladder, thyroid if unblocked; normal adrenals faint.

### Uses

**1. Phaeochromocytoma and paraganglioma**

- **Extra-adrenal, multiple, metastatic or recurrent disease**; biochemistry positive but imaging negative.
- **Isolated normetanephrine excess** may indicate an extra-adrenal tumour; MIBG finds it.
- Tumour >5 cm: MIBG or FDG (fluorodeoxyglucose) PET (positron emission tomography) to exclude metastases before surgery.
- **Best in MEN-2 (multiple endocrine neoplasia type 2).** Raised metanephrines after surgery: MIBG can unmask metastases.

**2. Neuroblastoma**

- $^{123}$I-MIBG is **required by COG (Children's Oncology Group) for staging**, replacing skeletal survey and bone scan. More sensitive for relapse. Non-avid tumours: PET.

**3. Therapy ($^{131}$I-MIBG)**

- Confirm uptake first. Metastatic phaeochromocytoma: symptom response in up to two-thirds; complete response <5%. High-specific-activity form FDA (Food and Drug Administration)-approved 2018.
- Relapsed or refractory neuroblastoma: response about 28–39%.

### Limitations

- **Cold scans in SDHB (succinate dehydrogenase subunit B) and VHL (von Hippel–Lindau)**; aggressive metastases lose uptake.
- **FDG PET is now gold standard** for staging, and **$^{68}$Ga-DOTATATE PET/CT** is replacing it at some centres.
- **Also know:** carcinoid and medullary thyroid cancer imaging; cardiac sympathetic imaging.

**Diagram to draw:** whole-body MIBG showing an adrenal hot spot with normal liver, heart and bladder uptake.

> **Key points to remember:**
> - Noradrenaline analogue; block thyroid; stop labetalol.
> - $^{123}$I diagnostic, $^{131}$I therapeutic.
> - Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

## SAQ 3 Adrenal Sx.

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 2, Q3d (preoperative phaeochromocytoma management also in

2020-Summer Paper 2, Q3c)

*Assumed meaning: surgery of the adrenal gland (adrenalectomy).*

## Surgical anatomy

- Right: pyramidal; **short vein into posterior IVC (inferior vena cava)**, so it is the riskier side. Left: crescentic; long vein into left renal vein. Three arteries (inferior phrenic, aorta, renal), one vein.

## Indications (AAES [American Association of Endocrine Surgeons] 2022)

- **Functional, any size:** aldosteronoma (Conn; after AVS [adrenal venous sampling]), cortisol-secreting adenoma, phaeochromocytoma, sex-hormone tumour, failed pituitary surgery for Cushing disease.
- **Malignant potential:** ≥4 cm; growth; >10 HU (Hounsfield units) unenhanced; washout <50%; suspected ACC (adrenocortical carcinoma); solitary metastasis.
- Myelolipoma >7 cm or symptomatic; cysts.
- **Contraindications (correct first):** uncontrolled hypertension, coagulopathy, peritonitis (transperitoneal); pregnancy relative.

## Preoperative preparation

- Biochemistry: metanephrines in **every** case before surgery.
- **Phaeochromocytoma:** α-blockade 2 weeks (phenoxybenzamine or doxazosin), β-blocker only after α, salt and fluids.
- Cushing: hydrocortisone 100 mg IV on the day of surgery. Conn: correct potassium.

## Approaches

| Approach | Use |
|---|---|
| **Laparoscopic transperitoneal** (gold standard since 1992) | Familiar view, larger tumours |
| **Retroperitoneoscopic** (lateral or prone) | Prior abdominal surgery, small tumours |
| Robotic | Either route |
| **Open** (subcostal, chevron, thoracoabdominal, flank) | **ACC**, large or invasive tumours |

## Steps (laparoscopic transperitoneal)

- **Right:** elevate liver → Kocher manoeuvre → follow IVC → **control adrenal vein early** (warn anaesthetist) → free gland to psoas → bag intact.
- **Left:** reflect colon and spleen → follow left renal vein to adrenal vein → clip → free from pancreas and splenic vessels.
- Kidney attachment last (handle). ACC: open en bloc, no rupture.
- **Partial adrenalectomy:** bilateral or hereditary phaeochromocytoma, solitary gland.

## Complications

- Bleeding (adrenal vein avulsion, IVC); spleen, pancreatic tail, liver, bowel injury; pneumothorax.
- **Phaeochromocytoma:** surges during handling, **hypotension after vein ligation**, rebound hypoglycaemia.
- **Adrenal insufficiency** after Cushing surgery (~99.7%): steroid taper.
- Conn: **hyperkalaemia** for weeks (high-salt diet).

**Diagram to draw:** right and left adrenal veins and relations.

> **Key points to remember:**
> - Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
> - Control the vein early; right vein short into IVC.
> - Open en bloc surgery for ACC.

# Rapid Revision Sheet · Section 14

### Phaeochromocytoma: adrenal anatomy, hormones and perioperative management (LAQ)

- Right vein short into IVC; left vein into left renal vein.
- StAR is rate-limiting for steroids; tyrosine hydroxylase for catecholamines.
- Cortisol induces PNMT; COMT forms metanephrines inside the tumour.
- Metanephrines first; never biopsy.
- α-blockade 7–14 days for all; never β before α (α before β), plus volume expansion.
- Volume expansion prevents hypotension; watch glucose.
- Arterial line before induction; avoid histamine releasers and ephedrine.
- Surge: pause handling, nitroprusside or phentolamine; esmolol for tachycardia.
- After vein ligation: fluids, vasopressors, glucose.
- One-third hereditary; SDHB most often metastatic.

### Adrenal mass evaluation, incidentaloma and adrenocortical carcinoma (LAQ)

- ≤10 HU = lipid-rich adenoma; APW >60%, RPW >40%.
- 1-mg DST: >1.8 µg/dL = MACS.
- Exclude phaeochromocytoma before biopsy or surgery.
- Surgery for incidentaloma: functioning, suspicious, >4 cm or growing.
- Feminisation in a man, >6 cm, >10 HU, poor washout: think ACC. Never biopsy before resection.
- Weiss ≥3/9; ENSAT stage IV = distant metastasis only.
- Open, R0, intact capsule, en bloc + lymphadenectomy.
- Mitotane level 14–20 mg/L; ADIUVO (low risk: no benefit); FIRM-ACT: EDP-M first line.

### How will you evaluate and manage 38 yr male with uncontrolled hypertension (LAQ)

- Hypertension below 40 years: screen for secondary causes.
- ARR → saline suppression → CT → AVS (skip if <40 with adenoma >1 cm).
- Metanephrines for phaeochromocytoma; α before β.
- FMD: angioplasty; atherosclerotic RAS: medical (ASTRAL, CORAL).

### Cushing syndrome and the dexamethasone suppression test (SAQ)

- Exclude exogenous steroids; 1-mg DST cortisol >1.8 µg/dL abnormal.
- 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
- Oral contraceptives and enzyme inducers cause false positives.
- ACTH low = adrenal; high = pituitary/ectopic (IPSS); ACTH separates adrenal from pituitary/ectopic.
- High-dose DST no longer routine.
- Steroid cover after adrenalectomy; Nelson syndrome after bilateral.

### MIBG scan (SAQ)

- Noradrenaline analogue; block thyroid; stop labetalol.
- $^{123}$I diagnostic, $^{131}$I therapeutic.
- Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

**Adrenal Sx. (SAQ)**

- Function (any size) or malignancy risk (≥4 cm, >10 HU) indicates surgery.
- Control the vein early; right vein short into IVC.
- Open en bloc surgery for ACC. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 15 · Renal, Nephrology & Transplant

## Long answer questions (LAQ)

### LAQ 1 Renovascular hypertension: pathophysiology, evaluation and treatment

P1 · P2 **merged from 2**

*Asked as:* P2 LAQ "Renovascular hypertension and ischaemic nephropathy: renal vascular anatomy, pathophysiology, evaluation and treatment" (asked as "Pathophysiology of renovascular hypertension (RVH) and iscemic nephropathy. Describe treatment of RVH in detail", Summer 2022 Paper II Q2; "Discuss evaluation of & management of reno-vascular hypertension", Practice Set 2 Paper II Q1; "Anat. of Renal Vasculature + Dx modalities & Rx optns for Renovasc. HTN.", Practice Set 4 Paper II Q2) · P1 LAQ "Pathophysiology of Renovascular hypertension" (Practice Set 8 Paper 1 Q2)

*"Iscemic" is read as ischaemic nephropathy.*

#### Anatomy of the renal vasculature

##### Renal arteries

- Arise from the aorta at the **L1–L2 disc**. The longer **right artery passes behind the IVC (inferior vena cava)**. Branches supply the adrenal, renal pelvis and proximal ureter.
- **Hilum, anterior to posterior:** renal vein, renal artery, pelvis; the posterior segmental artery lies behind the pelvis.
- **Anterior division** gives four segmental branches (apical, upper, middle, lower). **Posterior division** gives the posterior segmental artery: the first, most constant branch, passing behind the pelvis. If it crosses in front of the ureter, it can cause UPJO (ureteropelvic junction obstruction).
- **Five segmental arteries are end arteries** with no significant anastomosis. Occlusion causes segmental infarction; each segment can be resected separately.
- **Sequence of branching:** 1. Segmental 2. Lobar 3. Interlobar (columns of Bertin) 4. Arcuate (corticomedullary junction) 5. Interlobular 6. Afferent arteriole, glomerulus, efferent arteriole 7. Peritubular capillaries or vasa recta
- **Brödel line:** avascular plane between anterior and posterior territories, just behind the lateral border. Used for anatrophic nephrolithotomy and posterior calyceal access.
- **Variations (25–40%):** accessory arteries in 25–28% (T11–L4, rarely iliac or SMA [superior mesenteric artery]). They are end arteries, usually to the lower pole. Relevant to donor nephrectomy and endopyelotomy.

##### Renal veins

- Veins **anastomose freely** (venous collars, subcapsular plexus), so segmental vein ligation is safe. 3–5 segmental veins form the renal vein, anterior to the artery.

| | Right renal vein | Left renal vein |
|---|---|---|
| Length | 2–4 cm | 6–10 cm |
| Tributaries | None | Left adrenal, left gonadal, lumbar |
| Course | Direct to IVC | Between SMA and aorta (nutcracker syndrome) |

| | Right renal vein | Left renal vein |
|---|---|---|
| Anomalies | Accessory veins commoner | Circumaortic (2–16%); retroaortic |

**Relevance to RVH (renovascular hypertension):** atheroma affects the **ostium and proximal third**; **FMD (fibromuscular dysplasia)** affects the **mid-distal artery and branches**; a stenosed accessory artery can cause segmental RVH.

## Definitions

- **RVH:** hypertension caused by RAS (renal artery stenosis) that is relieved by revascularisation or nephrectomy. RAS is common in older (vascular) adults; RVH is proved only by cure.
- **Ischaemic nephropathy:** loss of GFR (glomerular filtration rate) from haemodynamically significant RAS affecting the **whole renal mass** (bilateral RAS or stenosis to a solitary kidney).

## Causes

| | Atherosclerotic RAS | FMD |
|---|---|---|
| Share | Most (about 90%) | Most of the rest |
| Patient | Older, smokers, diabetic, vascular disease | Young women |
| Site | **Ostium and proximal third** (aortic plaque) | **Mid and distal artery, branches**; "string of beads" (medial fibroplasia, commonest type) |
| Course | Progresses to occlusion; renal function lost | Rarely occludes |
| Treatment | Medical first | **Balloon angioplasty without stent**; good BP (blood pressure) response |

**Also know:** Takayasu arteritis, neurofibromatosis, midaortic syndrome (children), renal artery aneurysm, dissection, embolism, **Page kidney** (perinephric compression, e.g. subcapsular haematoma), transplant RAS.

## The RAAS (renin–angiotensin–aldosterone system)

- **Renin** is released by juxtaglomerular cells of the afferent arteriole in response to: 1. low afferent arteriolar pressure (intrarenal baroreceptor); 2. low sodium chloride delivery to the **macula densa**; 3. sympathetic **β1** stimulation.
- Renin cleaves hepatic **angiotensinogen** to angiotensin I. **ACE (angiotensin-converting enzyme)**, mainly in lung endothelium, converts it to **angiotensin II**.
- **Angiotensin II (AT1 receptor) actions:** systemic arteriolar vasoconstriction; **aldosterone** release (sodium retention, potassium loss); proximal tubular sodium reabsorption; thirst and ADH (antidiuretic hormone); sympathetic facilitation; **efferent arteriolar constriction**, which maintains GFR; cardiac and vascular hypertrophy and fibrosis.
- Normally, rising BP and sodium suppress renin (negative feedback). In RVH the stenosed kidney "sees" low pressure, so this feedback fails.

### Pathophysiology of RVH: Goldblatt models (basis of the tests)

- **Goldblatt (1934)** clipped the canine renal artery and produced sustained hypertension, reversed by early clip removal. Loop: low perfusion pressure releases juxtaglomerular renin, which produces angiotensin II (vasoconstriction) and aldosterone (sodium retention).

| | **Two-kidney, one-clip (2K1C)** | **One-kidney, one-clip (1K1C)** |
|---|---|---|
| Human equivalent | Unilateral RAS, normal opposite kidney | Bilateral RAS or stenosed solitary kidney |
| Sodium | Normal contralateral kidney excretes the extra sodium (**pressure natriuresis**) | No normal kidney: sodium and water retained |
| Renin | **High** from stenotic side; contralateral suppressed | Rises, then returns to normal as volume expands |
| Type | **Renin (angiotensin)-dependent** | **Volume-dependent** |
| ACE inhibitor effect | BP falls; captopril test positive | Less BP effect; may cause **acute renal failure / acute kidney injury** |

### Phases of 2K1C hypertension

1. **Acute phase:** renin-dependent; ACE inhibitor or clip removal reverses it.
2. **Transition:** sodium retention and secondary hyperaldosteronism; renin falls towards normal.
3. **Chronic phase:** hypertension damages the contralateral kidney (nephrosclerosis) and resistance vessels. BP no longer depends on the stenosis, so **revascularisation cures less** (cure falls with duration). This explains poorer results in long-standing hypertension.

### Other contributing mechanisms

- Renal afferent nerve signals raise **sympathetic activity**.
- Angiotensin II generates **oxidative stress** (reactive oxygen species), lowers **nitric oxide** and raises **endothelin**, worsening vasoconstriction.
- Inflammation and fibrosis; left ventricular hypertrophy and diastolic dysfunction.

### Pathophysiology of ischaemic nephropathy

- The kidney receives about **25% of cardiac output** (P1: 20–25%), far more than its metabolic need. **Autoregulation** holds GFR until stenosis is severe (about **70% or more**, or a significant translesional pressure gradient).
- Beyond this, the stenotic kidney keeps its GFR by angiotensin II-mediated **efferent constriction**. Blocking it (ACE inhibitor or ARB [angiotensin receptor blocker]) drops GFR: hence the **creatinine rise >30%** after these drugs.
- Chronic hypoperfusion causes hypoxia, inflammation, microvascular rarefaction, **tubulointerstitial fibrosis** and atrophy. Atheroemboli and hypertensive injury add damage.
- Early disease is reversible ("hibernating" kidney); established fibrosis is not.
- **Flash pulmonary oedema:** bilateral RAS causes volume retention plus angiotensin II-driven diastolic dysfunction.

### Evaluation: clinical clues (who to test)

- Onset <30 years (FMD) or severe hypertension after 55 years; **resistant** (≥3 drugs) or accelerated hypertension; sudden worsening of controlled BP.

- **Creatinine rise >30% after ACE inhibitor/ARB**; **flash pulmonary oedema**; kidneys differing in size by >1.5 cm; abdominal bruit; **hypokalaemia** (secondary hyperaldosteronism).

### Evaluation: basic tests

- Creatinine and eGFR (estimated GFR), potassium, urinalysis and proteinuria, ECG; exclude primary aldosteronism and phaeochromocytoma.

### Evaluation: imaging and functional tests

| Test | Method and finding | Comment |
|---|---|---|
| **Duplex Doppler US (ultrasound)** | High PSV (peak systolic velocity), above about 180–200 cm/s; renal/aortic velocity ratio >3.5; **RI (resistive index) = (PSV − EDV)/PSV** | First line, no contrast; operator dependent. **RI ≥0.8 (>0.8) means parenchymal disease and predicts poor response** |
| **CT angiography** | Stenosis, ostial plaque, beads | Best non-invasive anatomy; contrast nephropathy |
| **MR (magnetic resonance) angiography** | Gadolinium-enhanced | Caution if eGFR <30 mL/min |
| **Captopril renography (DTPA, diethylenetriaminepentaacetic acid)** | Day 1 after oral captopril: delayed uptake, low peak; repeat off ACE inhibitor and calcium channel blocker. Improvement of ≥10% = high probability RVH | Removing angiotensin II (efferent constriction) drops GFR in the stenosed kidney, proving renin-dependence; DTPA is cleared purely by filtration; use declining |
| **Catheter angiography** | Gold standard; **translesional (pressure) gradient** (systolic ≥20 or mean ≥10 mmHg = significant) | Done when intervention is planned |
| Renal vein renin | Ratio >1.5 lateralises | Before nephrectomy |

**Algorithm** `Clinical clue -> Duplex US (or CTA/MRA if eGFR allows) -> no RAS: treat as essential hypertension -> RAS: FMD -> angioplasty Atherosclerotic -> optimal medical therapy -> high-risk features + viable kidney -> angiography + gradient -> stent`

### Treatment 1: medical (first line for atherosclerotic RAS)

- **ACE inhibitor or ARB** controls renin-dependent BP; check creatinine and potassium at 1–2 weeks. Calcium channel blockers, diuretics.
- **Statin, antiplatelet, smoking cessation, diabetes control**: most deaths in these patients are cardiovascular.

### Treatment 2: landmark trials (atherosclerotic RAS)

- **ASTRAL (2009):** stenting plus medical therapy versus medical therapy alone gave **no benefit** in renal function, BP, events or mortality; procedural complications.
- **CORAL (2014):** stenting versus optimal medical therapy gave **no reduction** in cardiovascular or renal events; only a small BP fall.

- Hence **no routine revascularisation** of atherosclerotic RAS (ESC [European Society of Cardiology] 2017).

### Treatment 3: indications for revascularisation

- **FMD with hypertension** (first choice).
- Atherosclerotic RAS with **flash pulmonary oedema** or recurrent unexplained heart failure, **rapidly declining function** with bilateral or solitary-kidney RAS, or truly resistant hypertension.
- Kidney must be viable (not small and atrophic; RI <0.8).

### Treatment 4: endovascular

- **PTRA (percutaneous transluminal renal angioplasty) alone for FMD**; stent only for dissection or failure.
- **Ostial atherosclerotic lesion: angioplasty plus stent**, because a balloon alone fails from elastic recoil of aortic plaque. Restenosis is the main late problem.
- Complications: contrast nephropathy, **atheroembolism**, dissection, rupture, thrombosis, access haematoma.

### Treatment 5: surgical revascularisation

- Indications: failed endovascular therapy, **branch disease or aneurysm**, children, coexisting aortic surgery.

| Technique | Use |
|---|---|
| **Aortorenal bypass** (saphenous vein; internal iliac artery in children; PTFE [polytetrafluoroethylene]) | Standard |
| **Transaortic endarterectomy** | Bilateral ostial plaque |
| **Splenorenal (left) / hepatorenal (right) bypass** | Hostile (diseased) aorta |
| **Ex vivo bench repair + autotransplantation** | Branch FMD, aneurysms |
| Nephrectomy (partial or total) | Small non-functioning kidney with lateralising renin |

### Transplant RAS

- Raised velocity on Doppler; contrast imaging confirms; angioplasty or stent first, surgery if this fails.

### Follow-up

- BP, creatinine, duplex at intervals for restenosis; cure is commoner in FMD than atherosclerosis.

### Diagrams to draw

- 2K1C versus 1K1C with the renin–angiotensin loop (low pressure, renin, angiotensin II, aldosterone, hypertension).
- RAAS cascade from renin release to angiotensin II actions.
- Renal artery dividing into anterior (four segments) and posterior divisions, with the Brödel line.

**Key points to remember:**

- Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
- Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
- Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
- Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
- Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
- RVH is proved only by cure; duplex first, angiography with gradient before intervention.
- ASTRAL 2009 and CORAL 2014: stenting adds nothing to medical therapy in atherosclerotic RAS; no routine stenting.
- FMD gets balloon angioplasty without stent; ostial atheroma: stent.
- Revascularise for flash pulmonary oedema, declining GFR in a viable kidney (RI <0.8), resistant hypertension.

# LAQ 2 Allograft rejection and immunobiology of renal transplantation

P1 · P4 **merged from 2**

*Asked as:* P4 LAQ "Allograft rejection, immunobiology of renal transplant, marginal kidney donor and newer immunosuppressant drugs" (asked as "Discuss pathophysiology of allograft rejection. Discuss newer immunosuppressive drugs in renal transplant", Winter 2017 Paper IV Q2; "Describe immunobiology of renal transplant and newer immunosuppressant drugs", Summer 2020 Paper IV Q1; "Immunobiology of renal transplant, describe. Also, describe newer immunosuppressant drugs", Practice Set 3 Paper IV Q1; "Discuss marginal kidney donor and describe newer immunosuppressant drugs in renal transplantation", Summer 2015 Paper IV Q2) · P1 SAQ "Renal allograft rejection." (2020-Summer Paper 1 Q3d)

## Definition and introduction

- Rejection is the recipient's immune attack on donor (allo) antigens, which the recipient's immune system recognises as foreign: **HLA (human leucocyte antigen)** class I (A, B, C) and II (DR, DQ, DP), ABO and non-HLA antigens.
- Immunosuppression has cut acute rejection from over 60% (pre-ciclosporin era) to **below 10%**. Slow, subclinical **chronic antibody-mediated injury** is now the main problem and the main cause of late graft loss.
- Organ shortage drives use of **marginal donors**. Their kidneys carry more DGF (delayed graft function) and shorter graft survival, but transplantation still beats dialysis.

## Transplant (target) antigens

| Antigen | Features |
|---|---|
| **HLA = MHC (major histocompatibility complex)** | Class I (A, B, C) on all nucleated cells; class II (DR, DQ, DP) on antigen-presenting and endothelial cells. Class II mismatch is most immunogenic |
| **ABO** blood group | On endothelium; incompatibility causes hyperacute rejection unless desensitised |
| Minor and non-HLA antigens | Rejection despite HLA identity |

- **Sensitisation:** transfusion, pregnancy or previous transplant creates preformed anti-HLA antibodies.
- **Testing:** CDC (complement-dependent cytotoxicity) and flow crossmatch (donor lymphocytes with recipient serum); **Luminex single-antigen beads** define each recipient's antibodies and allow a **virtual crossmatch**; cPRA (calculated panel-reactive antibody) expresses sensitisation.

## Innate immunity: the first trigger

- **Ischaemia–reperfusion injury** releases danger signals (DAMPs, damage-associated molecular patterns). These activate **Toll-like receptors**, complement and macrophages, and up-regulate MHC on the graft. Hence DGF raises rejection risk.
- **Dendritic cells** bridge innate and adaptive immunity: activated, they up-regulate MHC and costimulatory B7 and migrate to lymph nodes.

## Allorecognition (how T cells see the graft)

| Pathway | Mechanism | Main role |
|---|---|---|
| **Direct** | Recipient T cells recognise **intact donor MHC** on donor "passenger" dendritic cells (donor antigen-presenting cells) | Very strong (high T-cell frequency); early acute TCMR (T-cell-mediated rejection) |
| **Indirect** | Recipient APCs (antigen-presenting cells) present **processed donor peptides on self-MHC** | Lifelong; helps B cells make DSA (donor-specific antibody), so alloantibody and chronic rejection |
| **Semi-direct** | Recipient APCs acquire **intact donor MHC** (membrane transfer) and present it | Links both pathways |

## Three-signal model of T-cell activation

1. **Signal 1 (antigen):** TCR (T-cell receptor) binds MHC–peptide; **calcineurin** dephosphorylates NFAT (nuclear factor of activated T cells), which drives IL-2 (interleukin-2) transcription.
2. **Signal 2 (costimulation): CD28 binds CD80/86 (B7)**. Signal 1 without signal 2 causes **anergy**. CTLA-4 (cytotoxic T-lymphocyte antigen-4) is the natural brake.
3. **Signal 3:** IL-2 binds **CD25** (IL-2 receptor) and signals through **mTOR (mammalian target of rapamycin)**, moving the cell cycle from G1 to S; clonal proliferation follows, which needs de novo purine synthesis.

## Effector mechanisms

- **Cellular:** CD4 Th1 cells (IL-2, IFN-γ [interferon-γ]) recruit macrophages; CD8 cytotoxic T cells kill by perforin and granzyme, causing **tubulitis and endarteritis (arteritis)**.
- **Humoral:** B cells with T-cell help become plasma cells that make **DSA**. DSA binds endothelium and fixes complement, leaving **C4d** on peritubular capillaries, with **microvascular inflammation** (glomerulitis, capillaritis). NK (natural killer) cell injury is complement-independent, so AMR (antibody-mediated rejection) may be C4d-negative.
- **Regulation: Tregs** (regulatory T cells, FoxP3+) suppress responses; the basis of tolerance research.

## Types of rejection

| Type | Timing | Mechanism and features |
|---|---|---|
| **Hyperacute** | Minutes–hours | Preformed ABO/HLA antibody; complement, thrombosis; graft turns blue; **prevented by crossmatch** |
| Accelerated acute | 2–5 days | Memory (anamnestic) response in sensitised recipients |
| **Acute TCMR** | Days–months (mostly first 6 months) | Interstitial infiltrate, tubulitis, arteritis |
| **Acute AMR** | Any time | DSA, C4d, microvascular inflammation (capillaritis) |
| **Chronic active AMR** | Months–years | **Transplant glomerulopathy** (GBM [glomerular basement membrane] double contours), arterial intimal fibrosis; commonest (main) cause of late graft loss |
| Subclinical | — | Histological rejection, stable function |

- **Risk factors:** HLA mismatch, DSA, ABO incompatibility, long ischaemia and DGF, **non-adherence**, retransplant, previous rejection.

### Diagnosis and monitoring

- **Clinical:** rising creatinine, oliguria; fever and graft tenderness are now rare.
- **Exclude:** obstruction, urine leak, lymphocele, renal artery stenosis or thrombosis (ultrasound with Doppler), CNI (calcineurin inhibitor) toxicity, BK virus nephropathy, recurrent disease.
- **Monitoring: de novo DSA** predicts graft loss; **dd-cfDNA (donor-derived cell-free DNA)** has negative predictive value >90% and avoids unnecessary biopsy; gene-expression panels.
- **Biopsy is the gold standard;** Banff grades the findings and classifies TCMR and AMR.

### Banff classification

| Category | Key criteria |
|---|---|
| Borderline | Foci of tubulitis with mild interstitial inflammation |
| TCMR IA / IB | Interstitial inflammation >25% with moderate (IA) or severe (IB) tubulitis |
| TCMR IIA / IIB | Mild-moderate (IIA) or severe (IIB) intimal arteritis |
| TCMR III | Transmural arteritis or fibrinoid necrosis |
| Active AMR | All three: (1) tissue injury (microvascular inflammation, arteritis, TMA [thrombotic microangiopathy]); (2) antibody–endothelium interaction (C4d, microvascular inflammation or gene transcripts); (3) **circulating DSA** |

### Treatment of rejection

- **TCMR:** IV (intravenous) pulse methylprednisolone; **rATG (rabbit antithymocyte globulin) / ATG** for Banff II–III or steroid-resistant rejection.
- **AMR** (not standardised): steroids, **plasmapheresis, IVIG (intravenous immunoglobulin)**, rituximab; eculizumab in selected patients; anti-IL-6 (interleukin-6) for chronic active AMR.
- Optimise maintenance (tacrolimus, mycophenolate, steroids); address adherence.

### Marginal kidney donor: categories

**1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002:** age **≥60**, or **50–59** with **≥2** of **hypertension**, **creatinine >1.5 mg/dL**, **death from CVA (cerebrovascular accident)**. About 70% higher risk of graft loss.

**2. KDPI (Kidney Donor Profile Index):** percentile of a 10-factor risk index. **KDPI >85%** replaced ECD in the US (December 2014). Low-KDPI kidneys go to recipients with the longest EPTS (Estimated Post-Transplant Survival): **longevity matching**.

**3. DCD (donation after circulatory death): Maastricht**

| Category | Description | Type |
|---|---|---|
| I / II | Dead on arrival / failed resuscitation | Uncontrolled |
| III | Arrest after withdrawal of therapy | **Controlled (commonest)** |
| IV | Arrest in a brain-dead donor | Either |

| Category | Description | Type |
|---|---|---|
| V | Euthanasia | Controlled |

- Warm ischaemia needs rapid cannulation and cooling, and DCD kidneys usually go on machine perfusion. DGF is **higher (roughly 40–50%)**, but long-term survival is similar to DBD (donation after brain death).

**4. Others: AKI (acute kidney injury)** donors; **HCV (hepatitis C virus)-viraemic** donors into negative recipients with DAAs (direct-acting antivirals); **paediatric donors <15 kg** (en bloc); older or hypertensive living donors (KDIGO [Kidney Disease: Improving Global Outcomes] 2017 risk-based approach).

- **Absolute exclusions:** metastatic or haematological cancer, melanoma, untreatable viral infection. Treated bacterial infection is acceptable; screen by serology and nucleic acid testing.

### Marginal kidney donor: assessment and optimisation

- **Remuzzi biopsy score** (glomeruli, tubules, interstitium, arteries, each 0–3): **0–3 single**, **4–6 dual kidney transplant**, **≥7 discard**.
- **Cold storage** at 4 °C in UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution slows metabolism.
- **HMP (hypothermic machine perfusion)** reduces DGF (**Moers, NEJM 2009**: about 21% vs 27%; greatest benefit in ECD kidneys); Campbell notes no long-term survival difference. Normothermic perfusion allows viability testing (limited experience).
- **ESP (Eurotransplant Senior Program):** donors **≥65** to recipients **≥65**, prioritising short cold ischaemia over HLA matching. Informed consent is essential.
- **Ojo (JASN 2001):** ECD transplantation improves survival over dialysis, although graft survival is lower than with standard kidneys.
- With DGF, give antibody induction and delay nephrotoxic CNI therapy.

### Standard immunosuppression (current practice)

- **Induction** (over 90% receive it) plus **tacrolimus + mycophenolate** ± steroids (mycophenolate: GI [gastrointestinal] upset, teratogenic).
- Acute rejection is below 10%; chronic AMR is now the main problem. Newer drugs aim to avoid CNI nephrotoxicity, improve adherence and treat AMR.

### Newer immunosuppressive drugs

| Drug | Mechanism (site) | Key points |
|---|---|---|
| **Belatacept** | **CTLA4-Ig**: binds CD80/86, blocks **CD28 costimulation (signal 2)** | FDA (Food and Drug Administration) 2011; IV 10 mg/kg induction, then **5 mg/kg 4-weekly** (monthly); CNI-free. **BENEFIT**: better GFR (glomerular filtration rate), more early acute rejection; 7-year better patient and graft survival. **Contraindicated if EBV (Epstein–Barr virus)-seronegative** (PTLD [post-transplant lymphoproliferative disorder]) |
| **Extended-release tacrolimus** | CNI: binds FKBP12 and inhibits calcineurin (**signal 1**), so less IL-2 transcription | Once daily, better adherence, steadier levels; nephrotoxicity, NODAT (new-onset diabetes after transplant), tremor |

| Drug | Mechanism (site) | Key points |
|---|---|---|
| Mycophenolate | Inhibits IMPDH (inosine monophosphate dehydrogenase), blocking lymphocyte purine synthesis | Standard partner of tacrolimus |
| **mTOR inhibitors** (sirolimus, everolimus) | Bind FKBP12 and block mTOR (**signal 3**), arresting G1 to S | Not nephrotoxic; fewer CMV (cytomegalovirus)/BK infections; antitumour effect. **TRANSFORM (2018)**: everolimus + reduced (low-dose) CNI non-inferior. Poor wound healing, **lymphocele**, proteinuria, dyslipidaemia |
| **Basiliximab** | Anti-CD25 (IL-2 receptor), non-depleting (**signal 3**) | Low-risk induction, **20 mg IV days 0 and 4** |
| **rATG**; **alemtuzumab** | T-cell depletion; anti-CD52 | High-risk induction, DGF; infection (CMV), PTLD |
| **Rituximab; bortezomib** | Anti-CD20 (B cells); proteasome inhibitor (plasma cells) | AMR, desensitisation |
| **Eculizumab** | Anti-C5 (complement) | Selected AMR; aHUS (atypical haemolytic uraemic syndrome) recurrence |
| **Imlifidase** | Enzyme that **cleaves IgG** | **EMA (European Medicines Agency) 2020**: crossmatch-positive, highly sensitised patients |
| **Anti-IL-6 therapy** (tocilizumab, clazakizumab) | Blocks IL-6 pathway | Chronic active AMR |

- **Also know:** anti-CD40 costimulation blockade is under trial. **Voclosporin** is approved for lupus nephritis, **not transplantation**.

## Recent advances

- **Tolerance induction** by mixed chimerism (combined kidney and donor stem-cell transplant) and Treg therapy; normothermic repair; **pig-kidney xenotransplant** into a living human (March 2024).

## Diagrams to draw

- Three-signal T-cell activation (APC–T-cell synapse showing signals 1–3) with drug sites: CNI at signal 1, belatacept at 2, basiliximab and mTOR inhibitors at 3, mycophenolate at DNA synthesis.
- Banff biopsy features: tubulitis, arteritis, peritubular capillary C4d.

**Key points to remember:**

- Direct pathway drives early TCMR; indirect pathway drives antibody (DSA) and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Hyperacute rejection prevented by crossmatch.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept blocks signal 2; better GFR, contraindicated if EBV-negative; imlifidase (IgG cleavage) for highly sensitised patients; dd-cfDNA NPV >90%.
- ECD: ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5, CVA death; KDPI >85% since 2014.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP reduces DGF; ESP ≥65 to ≥65.

# LAQ 3 Deceased donor kidney: selection, maintenance and retrieval techniques

P1 · P2 **merged from 2**

*Asked as:* P2 LAQ "Deceased (cadaver) kidney donor: selection, identification, maintenance, distribution and retrieval" (asked as "Describe selection criteria for cadaver renal donor. Describe in details kidney retrieval surgery", Summer 2021 Paper II Q2; "Describe procedure of identification, maintenance, distribution and retrieval of kidney in a 25 yr old brain dead pt", Practice Set 3 Paper II Q1) · P1 SAQ "Kidney retrieval techniques for organ donation" (2023-Winter Paper 1 Q3c)

*Assumed meaning: the Indian deceased-donor pathway: identification and brain-stem death certification, donor maintenance, allocation and retrieval. Marginal donor: see 2016-Summer Paper 3, Q3f; brain death certification: 2018-Summer Paper 4, Q3c; also answered in 2022-Summer Paper 4, Q3b.*

## Types of donor

- **Deceased (cadaver): DBD** (donation after brain death: circulation maintained until cold perfusion) or **DCD** (donation after circulatory death: warm ischaemia).
- **Living:** related or unrelated (THOA, Transplantation of Human Organs Act, in India).
- A 25-year-old brain-dead patient is an ideal **standard-criteria DBD** donor: circulation continues until cold perfusion, so warm ischaemia is minimal.
- India: **THOA 1994**, amended 2011, **Rules 2014**, makes brain-stem death legal death; **NOTTO** (National Organ and Tissue Transplant Organisation) allocates organs.

## Identification and selection criteria

**Potential donor:** ventilated patient with catastrophic brain injury (head injury, intracranial bleed, hypoxia). ICU (intensive care unit) informs the **transplant coordinator**; **mandatory request** to the family.

### Donor evaluation

- Cause of death, medical and social (high-risk behaviour) history; kidney: terminal creatinine, urine output, hypertension, diabetes, ultrasound.
- Laboratory: ABO and HLA (human leucocyte antigen) typing; serology and NAT (nucleic acid testing) for HIV (human immunodeficiency virus), hepatitis B and C, CMV (cytomegalovirus), EBV (Epstein–Barr virus), syphilis, toxoplasma.

### Absolute contraindications

- Active or metastatic cancer, haematological malignancy, past melanoma, untreatable viral infection.
- Treated bacterial infection (even bacterial meningitis) is acceptable; viral encephalitis needs care.

### Standard versus marginal donor

| Category | Definition |
|---|---|
| **ECD (expanded criteria donor), UNOS (United Network for Organ Sharing)** | Age ≥60, or 50–59 with 2 of: hypertension, creatinine >1.5 mg/dL, death from CVA (cerebrovascular accident) |

| Category | Definition |
|---|---|
| **KDPI (Kidney Donor Profile Index)** | Replaced ECD in the US (2014); >85% = marginal; longevity matching with EPTS (Estimated Post-Transplant Survival) |
| Paediatric <15 kg | En bloc transplant |

- **Remuzzi biopsy score** for marginal kidneys: 0–3 single, 4–6 dual, ≥7 discard.
- **Also know:** HCV (hepatitis C virus)-viraemic donors with antivirals.

### Legal death and consent: certification of brain-stem death (THOA)

- **DBD preconditions:** apnoeic coma with a known irreversible cause; exclude drugs (sedatives, relaxants), hypothermia, metabolic causes and shock.
- **Board of four (4-member Board):** doctor in charge of the hospital; independent specialist from an approved panel; neurologist or neurosurgeon (else a nominated surgeon, physician, anaesthetist or intensivist); treating doctor. **None from the transplant team.**

| Brainstem reflex (all absent) | Method | Cranial nerves |
|---|---|---|
| Pupillary | Bright light: pupils fixed, no reaction | II, III |
| Corneal | Touch cornea: no blink | V, VII |
| Oculocephalic (doll's eye) | Turn head: eyes move with the head | VIII, III, VI |
| Oculovestibular (caloric) | Ice-cold water into each ear (drum seen): no eye movement | VIII, III, VI |
| Cranial motor response | Supraorbital pressure: no grimace | V, VII |
| Gag and cough | Touch pharynx; suction catheter in trachea: no response | IX, X |

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect the ventilator with oxygen insufflation; **no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg**. Abort for hypotension, desaturation or arrhythmia.
- **Two sets of tests at least 6 hours apart**; **Form 10**. Ancillary tests (angiography, perfusion scan) only if examination is impossible.
- **DCD (Maastricht):** I dead on arrival; II failed resuscitation (both uncontrolled); **III arrest after withdrawal of support (controlled, commonest)**; IV arrest in a brain-dead donor; V euthanasia.
- **Consent:** next-of-kin (near relative) consent after a **mandatory request** by ICU staff; in medico-legal cases inform police.

### Donor maintenance

- **Why:** herniation causes a **catecholamine storm**, then loss of sympathetic tone (hypotension), **diabetes insipidus (DI)**, hypothermia, DIC (disseminated intravascular coagulation), low thyroid hormone and cortisol, and pulmonary oedema.
- **"Rule of 100s":** systolic pressure >100 mmHg, urine >100 mL/h, $PaO_2$ (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Measures:** fluids guided by CVP (central venous pressure), no starch colloids (nephrotoxic); **vasopressin** first (pressor and treats DI), then noradrenaline; methylprednisolone 15 mg/kg, thyroid hormone, insulin; lung-protective ventilation; warming; correct sodium, potassium and clotting.

### Distribution (allocation)

- **NOTTO**, New Delhi, with **ROTTOs** (regional) and **SOTTOs** (state organisations), keeps registries and waiting lists.
- **Sequence:** hospital and state list, then region, then national.
- **Kidney criteria:** ABO compatibility, waiting time on dialysis, HLA match, priority for children and previous living donors, **negative crossmatch** (donor spleen or nodes). Many states let the retrieving hospital keep one kidney.
- 2023 policy removed the domicile requirement and upper age limit for registration.

### Deceased multi-organ kidney retrieval (DBD)

1. **Checks:** death certificate, consent, blood group; coordinate thoracic, liver, pancreas and kidney teams.
2. **Incision:** midline xiphoid-to-pubis laparotomy with median sternotomy.
3. **Exposure:** mobilise right colon and small-bowel mesentery off the retroperitoneum (Cattell–Braasch manoeuvre); expose aorta and IVC (inferior vena cava).
4. **Prepare cannulation:** encircle the distal aorta above the bifurcation; ligate the inferior mesenteric artery; prepare the supracoeliac aorta for clamping. **Look for and preserve aberrant (accessory) renal arteries**; keep accessory arteries from the iliacs.
5. **Cold perfusion:** systemic heparin; cannulate the distal aorta; clamp the supracoeliac aorta; **vent** the IVC (or right atrium); flush cold UW (University of Wisconsin) or HTK (histidine–tryptophan–ketoglutarate) solution; crushed (surface) ice; open Gerota's fascia laterally to cool the kidneys.
6. **Remove heart, lungs, liver and pancreas first.**
7. **Kidneys en bloc (last):** lift both with aorta, IVC, perinephric fat and ureters off the posterior wall; keep periureteric tissue; divide ureters low. Divide aorta and IVC (upper limit set by the liver team, lower at the bifurcation); lift everything anteriorly and divide prevertebral tissue.
8. **Extras:** iliac vessels for reconstruction; spleen and lymph nodes for crossmatch.

**DCD:** after death and a no-touch period, **super-rapid** laparotomy, direct aortic cannulation and IVC venting limit warm ischaemia; kidneys usually go on machine perfusion.

### Back-bench (back-table) preparation

- Separate the kidneys: open the aorta posteriorly to see the renal ostia and divide it between them, keeping an aortic **Carrel patch** for each.
- Keep the IVC with the right kidney to lengthen the short right renal vein.
- Ligate tributaries and lymphatics (prevents lymphocele); trim fat without decapsulation; spare the lower pole–ureter triangle.
- Document anatomy and injuries; biopsy marginal kidneys; triple-bag in cold solution on ice at 4 °C; label.

### Living donor nephrectomy

- **Principles:** donor safety first; maximal vessel length; keep periureteric tissue and the lower pole–ureter triangle; minimal handling; short warm ischaemia.
- **Left kidney preferred** (longer vein).

| Technique | Features |
|---|---|
| Open (flank or mini-incision) | Complex anatomy, more often right side |
| **Laparoscopic** (Ratner, 1995) | Preferred; less pain, faster recovery |
| **Hand-assisted** | About two-thirds of US cases; tactile control |
| Retroperitoneoscopic, LESS (laparoendoscopic single-site), robotic | Selected centres |

- **Sequence:** 1. Hilar dissection. 2. Ureter divided. 3. Artery then vein stapled or clipped. 4. Rapid extraction. 5. Cold flush.
- **Risks:** complications about 16%; mortality 3 per 10,000; conversion 0.26–3.34%.

### Preservation

- Cooling to 4 °C slows metabolism. **Hypothermic machine perfusion** reduces DGF (delayed graft function) versus static cold storage (Moers, 2009), but not long-term graft loss.
- Keep cold ischaemia short: cold ischaemia >24 hours raises thrombosis risk.
- Normothermic perfusion is experimental.

### Pitfalls

- Polar artery or intimal injury, short right vein, ureteric devascularisation, capsular tears.

### Diagram to draw

- Aorta and IVC with distal aortic cannula, supracoeliac clamp, IVC vent and en bloc kidneys.

> **Key points to remember:**
> - THOA: 4-member Board (Board of four), none from the transplant team; two tests ≥6 h apart; Form 10.
> - Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
> - ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death; KDPI >85%.
> - Rule of 100s; vasopressin for hypotension and DI.
> - Cannulate, clamp, vent, cool: distal aortic cannula, supracoeliac clamp, IVC vent, cold UW/HTK and ice.
> - Kidneys removed last, en bloc; Carrel patch, IVC for right vein.
> - Left kidney preferred for living donation.
> - Machine perfusion reduces DGF.

## LAQ 4 Vascular grafts and vascular surgery in urology

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 LAQ "Vascular grafts and vascular surgery in urology" (asked as "Vascular Grafts in Urology", Summer 2021 Paper III Q2 LAQ; "Vascular Surgery in Urology", Winter 2024 Paper III Q3a SAQ) · P1 SAQ "Vascular grafts in urology" (Practice Set 5 Paper 1 Q3e) · P2 SAQ "Role of vascular grafts in urology" (2015-Summer Paper 2 Q3e)

*Assumed meaning: blood-vessel grafts and patches (conduits) used in urological surgery: transplantation, renal and IVC (inferior vena cava) tumour surgery, RPLND (retroperitoneal lymph node dissection), sarcoma, renovascular repair and autotransplantation, dialysis access and penile surgery. Tissue grafts (buccal mucosa) are not included.*

*Assumed meaning (vascular surgery): vascular procedures performed or met by the urologist (control and repair of vessels, caval surgery, transplant and renovascular surgery, penile vascular surgery and access), not vascular surgery as a specialty.*

### Introduction

- A vascular graft **replaces, bypasses, extends or widens** a vessel when primary repair would fail or leave tension or narrowing.
- **Patch:** widens a partly resected wall. **Interposition graft:** replaces a segment. **Bypass:** routes flow around a lesion.

### Classification of graft materials

| Type | Examples | Advantages | Disadvantages |
|---|---|---|---|
| **Autologous vein** | Great saphenous, gonadal, superficial femoral, donor external iliac vein | Resists infection; suits small vessels and contaminated fields | Limited length; valves; harvest wound |
| **Autologous artery** | Internal iliac (hypogastric), **Carrel aortic patch** | Best for renal artery repair | Limited supply |
| **Allograft (homograft)** | Cadaveric vein or artery | Off the shelf; avoids xenograft | Availability; degeneration |
| **Xenograft (biological)** | **Bovine or porcine pericardium** | Pliable, strong patch, low infection | Patch only; cost |
| **Synthetic** | **PTFE (polytetrafluoroethylene, Gore-Tex)**, ringed PTFE, Dacron (polyethylene terephthalate), collagen-impregnated Dacron | Off-the-shelf, large calibre, any length | Infection needs removal; venous grafts thrombose |

### Principles

- **Control first:** proximal and distal control with vessel loops, Rummel tourniquets and atraumatic clamps (bulldog, Satinsky, DeBakey). Artery before vein.
- **Choice of graft: autologous** vein or artery for small vessels and contaminated fields; **synthetic** (PTFE or Dacron) for large, high-flow vessels with no bowel contamination.
- **Venous (low-pressure) grafts thrombose** more than arterial ones, so use the **widest graft** possible.
- Patch: **smooth side luminal**, rough thrombogenic side outward; edges must not invert.

- Systemic heparin before clamping; flush air and debris before tying; tension-free, everting, spatulated anastomosis with fine monofilament polypropylene (Prolene).
- **Sutures:** polypropylene (Prolene), double-armed. IVC or aorta 3-0/4-0; renal vessels 4-0/5-0; transplant anastomoses 5-0/6-0.
- Antiplatelet or anticoagulant afterwards as indicated.
- Wrap synthetic grafts in **omentum** or retroperitoneal fat, away from bowel and urine.

### Application 1: renal transplantation

- Renal vein and artery end-to-side to external iliac vessels; Carrel patch.
- **Short right renal vein:** extend with donor IVC, free donor external iliac vein, or **spiral gonadal vein graft**.
- **Multiple arteries (18–43%):** Carrel patch; on the back table, conjoined side-to-side or end-to-side (bench) reconstruction. **Preserve the lower-pole artery** (it supplies the ureter).
- **Atherosclerotic recipient iliac artery:** mount the graft artery on a **Gore-Tex or bovine pericardial patch**.
- **Children under 15 kg:** aorta and IVC anastomosis, transperitoneal.
- **Transplant renal artery stenosis (1–23%):** angioplasty or stent first; surgical interposition graft or bypass if it fails, which risks graft loss.

### Application 2: RCC (renal cell carcinoma) with IVC tumour thrombus

- **Mayo levels 0–IV:** level I milking and partial caval clamp; II full caval control; III liver mobilisation ± venovenous bypass; IV cardiopulmonary bypass ± deep hypothermic arrest.

| Situation | Repair |
|---|---|
| Residual lumen ≥50% | Primary closure |
| Residual lumen <50% | **Patch cavoplasty**: bovine or autologous pericardium, PTFE, Dacron or saphenous vein |
| Circumferential wall invasion | **Caval replacement: PTFE 16–20 mm**; contralateral renal vein reimplanted into the graft |
| Infrarenal bland thrombus | Filter or caval interruption, sparing lumbar collaterals |

- **Leibovich–Bower patch width** = normal caval diameter × % diameter resected × 3.
- Left tumour with suprarenal caval resection: the right kidney has no collaterals, so give it new drainage (autotransplantation, or **saphenous vein graft** to the splenic, portal or inferior mesenteric vein).
- After caval replacement: low-dose heparin, then **lifelong aspirin 81 mg**.

### Application 3: PC-RPLND (post-chemotherapy RPLND)

- Vascular procedures in **9–15%**; aorta most often.
- **IVC:** narrowing <25%: primary repair; >25%: patch or interposition graft. **Ligation is tolerated** (slow occlusion builds collaterals).
- **Aorta:** never ligated; must be repaired or grafted. Small injury: repair or pericardial patch; resection: **PTFE or Dacron** (about 1% need replacement). Graft also if the duodenum is injured, to prevent aortoduodenal fistula.

### Application 4: retroperitoneal sarcoma

- Caval resection ± reconstruction (Dacron, PTFE, homograft); grafts **thrombose in 10–40%**; anticoagulate at least 3 months.

### Application 5: renovascular repair and autotransplantation

- Aortorenal bypass (saphenous vein, internal iliac artery, PTFE); splenorenal or hepatorenal bypass.
- Ex vivo bench repair of branch aneurysm or stenosis (or ureteric loss) with internal iliac artery or saphenous vein, then **autotransplantation** (see Q3e). Angioplasty or stent first for fibromuscular dysplasia.
- Traumatic renal artery injury: endovascular stent for the main renal artery; attempt open repair only for solitary kidney or bilateral injury (warm ischaemia <120 minutes).

### Application 6: dialysis access

- Arteriovenous fistula first; **AV (arteriovenous) graft (PTFE)** when no native fistula is possible (poor veins): usable in 2–3 weeks; more stenosis, thrombosis and infection than AVF (arteriovenous fistula).

### Application 7: iatrogenic or traumatic vascular injury

- Interposition graft.
- SMA (superior mesenteric artery): repair or graft bypass. SMV (superior mesenteric vein): autologous vein graft; Gore-Tex only if no vein.

### Application 8: penile surgery

- Priapism: **Grayhack** saphenous vein–cavernosal shunt.
- Vein grafts for Peyronie tunical defects (survive by endothelial perfusion); synthetic grafts not recommended.
- **Arterial revascularisation** (inferior epigastric artery to dorsal artery) for young men with post-traumatic pudendal stenosis; venous ligation investigational.

### Application 9: endovascular

- Angioembolisation for trauma, AML (angiomyolipoma) bleeding, pseudoaneurysm and AVF after partial nephrectomy.

### Complications

- **Thrombosis** (caval grafts 10–40%), anastomotic stenosis (intimal hyperplasia), bleeding, pseudoaneurysm.
- **Graft infection** (synthetic grafts usually need removal); graft–enteric or urinary fistula.
- **Ureteric:** transient hydronephrosis after aorto-iliac grafts (12–20%), symptomatic stricture 1–2%, **ureteroarterial fistula** (treat with endovascular covered stent).

### Recent advances

- Endovascular covered stent grafts; bioengineered acellular vessels (under study); robotic autotransplantation.

### Diagrams to draw

- IVC patch cavoplasty and PTFE caval replacement with left renal vein reimplanted.
- Mayo IVC thrombus levels with the matching vascular control.

- Bench arterial reconstruction.

> **Key points to remember:**
> - Proximal and distal control, heparin, flush, everting Prolene anastomosis.
> - Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
> - Mayo levels decide caval control and bypass.
> - IVC: patch if lumen <50%; PTFE 16–20 mm replacement if wall invaded.
> - PC-RPLND: IVC may be ligated; aorta must be grafted.
> - Transplant: Carrel patch, bench reconstruction; preserve the lower-pole artery.
> - Bench surgery plus autotransplantation for complex renovascular lesions.

## LAQ 5 Vascular anastomosis and vascular access for dialysis

P2 · P4 **merged from 2**

*Asked as:* P2 LAQ "Principles of vascular anastomosis and vascular access surgery for HD" (asked as "Describe principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD", Practice Set 5 Paper II Q2 LAQ and Practice Set 12 Paper II Q3a SAQ) · P4 SAQ "Vascular access for dialysis." (2020-Summer Paper 4 Q3b)

*Assumed meaning: HD = haemodialysis. Vascular access also answered in 2025-Winter Paper 2, Q3c.*

### Introduction

- A vascular anastomosis joins two vessels so that flow is smooth, leak-free and lasting (transplantation, autotransplantation, caval repair, dialysis access).
- **Types:** end-to-end, **end-to-side** (commonest: transplant, fistula), side-to-side, and patch (Carrel aortic patch).
- Haemodialysis needs a blood flow of about **250–350 mL/min**; access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

### Principles of vascular anastomosis

| Principle | How and why |
|---|---|
| Exposure and control | Proximal and distal control (vessel loops, Rummel tourniquets, bulldog or Satinsky clamps) |
| Anticoagulation | Systemic heparin before clamping; heparinised saline flush; prevents thrombosis in the static column |
| Atraumatic handling | Hold the adventitia only, never the intima; clamp clear of atheroma, because clamp injury causes intimal dissection and stenosis |
| Healthy ends | Trim diseased ends; resect venous valve cusps at the suture line; prevents stenosis and leak |
| No tension, twist or kink | Adequate length; trial the final position before suturing |
| **Intima-to-intima eversion** | Smooth endothelium faces the lumen; adventitia and rough surfaces stay outside, as they are thrombogenic |
| Wide opening | Spatulate or bevel small vessels, or use end-to-side with an elliptical arteriotomy (aortic punch), to avoid stenosis (stricture) |
| Suture | **Non-absorbable monofilament polypropylene**, double-armed: smooth, low drag, low thrombogenicity. IVC (inferior vena cava) or aorta 3-0/4-0; transplant vessels 5-0/6-0; wrist fistula about 6-0/7-0 |
| Needle direction | Equal, small, close bites; pass **inside-out on the artery** so plaque and intima are tacked down, not lifted into a flap |
| Continuous or interrupted | Continuous is quick and haemostatic; **interrupted** for small vessels and children, allowing growth |
| Technique | **Carrel triangulation** (three stay sutures keep the back wall clear); **parachute** (sew loosely, then draw up; one knot) or tack both apices and run to the midpoint |

| Principle | How and why |
|---|---|
| Flush and release | Before the last knot, briefly release clamps to flush air, clot and debris; leave slack ("growth factor") to avoid purse-stringing; release venous/distal clamp first, then arterial |
| Confirm | Haemostasis, pulse, thrill, Doppler |

- **Failure:** early from technical error (thrombosis, leak, stenosis); late from **intimal hyperplasia**.
- **Also know:** vascular staplers; grafts (autologous vein for small or contaminated fields; PTFE [polytetrafluoroethylene] for large vessels).

## Options for vascular access surgery for HD

| Option | Construction / description | Ready | Comment |
|---|---|---|---|
| **1. Non-tunnelled CVC** (central venous catheter) | Dual-lumen 10–12 Fr in internal jugular or femoral vein | At once | AKI (acute kidney injury) and emergencies; acute dialysis, days only |
| **2. Tunnelled cuffed CVC** | Right internal jugular; cuff blocks bacterial migration | At once | Bridge or last resort; infection, central stenosis |
| **3. Native AVF** (arteriovenous fistula) | Artery joined to a superficial vein, usually vein end to artery side | 6–8 weeks (matures) | **First choice:** best patency, fewest infections |
| a. Snuffbox or **radiocephalic (Brescia–Cimino, 1966)** | Radial artery to cephalic vein at wrist | 6–8 weeks | Most distal; preserves proximal sites; higher failure to mature (most fail to mature) |
| b. Brachiocephalic | Brachial artery to cephalic vein at elbow | 6–8 weeks | Higher flow; more steal |
| c. Brachiobasilic transposition | Deep basilic vein mobilised and moved superficially | Longer, often two stages | When the cephalic vein is unusable |
| **4. AVG** (arteriovenous graft) | PTFE loop or straight: forearm loop, upper-arm or thigh | 2–3 weeks (usable) | Poor veins; more stenosis (venous anastomosis), thrombosis, infection |
| **5. HeRO (Haemodialysis Reliable Outflow) graft** | Graft outflow bypasses occluded central veins into the right atrium | | Central vein occlusion |
| **6. Endovascular AVF** (WavelinQ, Ellipsys) | Percutaneous radiofrequency or thermal anastomosis between adjacent forearm vessels | | No incision; selected anatomy |

- **Also know:** lower-limb femoral vein transposition when arms are exhausted; peritoneal dialysis needs no vascular access.

## Planning

- **Hierarchy:** AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first**; individualised (KDOQI [Kidney Disease Outcomes Quality Initiative] 2019 "ESKD Life-Plan").
- **Timing:** refer at eGFR (estimated glomerular filtration rate) <30; create the AVF about **6 months** before dialysis.

- **Vein preservation:** preserve arm veins; no venepuncture, cannula or PICC (peripherally inserted central catheter) in that arm; avoid subclavian lines.
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm).
- **Maturity "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, at 6 weeks.

### Complications of access

| Complication | Treatment |
|---|---|
| Failure to mature (juxta-anastomotic stenosis) | Angioplasty (PTA), ligate side branches |
| Stenosis (neointimal hyperplasia) | PTA (percutaneous transluminal angioplasty), stent graft |
| **Thrombosis** (usually on a stenosis) | Thrombectomy, then treat the stenosis |
| **Steal syndrome** (hand ischaemia) | Banding, DRIL (distal revascularisation–interval ligation) |
| Aneurysm; infection (CVC > AVG > AVF) | Repair; antibiotics, excise graft, remove catheter |

- **Also know:** venous hypertension; high-output cardiac failure.

### Monitoring

- Thrill and bruit, arm-elevation test, prolonged bleeding after needling, rising venous pressure; duplex when abnormal.

### Recent advances

- **Endovascular AVF** (WavelinQ, Ellipsys); drug-coated balloons; bioengineered vessels.

### Diagram to draw

- Radiocephalic end-to-side fistula at the wrist with Carrel triangulation stay sutures.

> **Key points to remember:**
> - Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
> - AVF first, distal, non-dominant arm, about 6 months ahead.
> - Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.
> - Thrombosis sits on a stenosis; steal: banding or DRIL.

## LAQ 6 Describe in details basic patho-physiology of urine production and transport

P1

**Type:** LAQ (25 marks)
**Source:** 2015-Summer Paper 1, Q1

### Introduction

- Urine is formed by filtration, reabsorption, secretion and concentration, then propelled to a low-pressure bladder.

### A. Functional anatomy

- Glomerular barrier (fenestrated endothelium, basement membrane, podocytes) restricts by size and negative charge, so albumin is excluded.
- **Juxtamedullary nephrons** (long loops) build the medullary gradient.

### B. Renal blood flow (RBF) and autoregulation

- RBF ~25% of cardiac output. The medulla is hypoxic (10–20 mmHg), endangering the thick ascending limb and S3 segment.
- **Autoregulation** keeps RBF and GFR (glomerular filtration rate) constant at mean arterial pressure 80–180 mmHg:
- **Myogenic:** the afferent arteriole constricts when stretched and dilates (prostaglandins) when pressure falls; NSAIDs (non-steroidal anti-inflammatory drugs) abolish this.
- **Tubuloglomerular feedback:** low NaCl at the macula densa releases renin; angiotensin II constricts the efferent arteriole. Angiotensin blockade abolishes this.

### C. Glomerular filtration

- GFR ~125 mL/min (180 L/day); filtration fraction ~20%.
- **Net filtration pressure** ≈ glomerular hydrostatic 60 – Bowman's space 18 – oncotic 32 = **~10 mmHg**.
- Efferent constriction lowers RBF but **raises** GFR; afferent constriction lowers both. Obstruction raises Bowman's space pressure, so GFR falls.
- **Also know:** GFR estimated from creatinine or cystatin C (race-free 2021 equations).

### D. Tubular handling

| Segment | Key transporter | Handling | Drug/disease link |
|---|---|---|---|
| Proximal tubule | $Na^+/H^+$ exchanger, SGLT2 (sodium–glucose cotransporter 2) | ~**65%** $Na^+$ and water (iso-osmotic); all glucose; most bicarbonate and phosphate; secretes drugs | SGLT2 inhibitors |
| Thin descending limb | Aquaporin-1 | Water out, solute stays: fluid concentrates | – |
| Thick ascending limb | NKCC2 ($Na^+$-$K^+$-$2Cl^-$) | ~**25%** NaCl; water-impermeable "diluting segment" | Loop diuretics |

| Segment | Key transporter | Handling | Drug/disease link |
|---|---|---|---|
| Distal convoluted tubule | NCC ($Na^+$-$Cl^-$ cotransporter) | 5–7% $Na^+$; $Ca^{2+}$ (parathyroid hormone) | Thiazides lower urinary $Ca^{2+}$ |
| Collecting duct | ENaC (epithelial $Na^+$ channel), AQP2 | Principal cells: $Na^+/K^+$ (aldosterone), water (ADH); intercalated cells: acid | Amiloride, spironolactone; distal RTA (renal tubular acidosis) |

## E. Concentration

- **Countercurrent multiplication:** ascending limbs absorb NaCl without water, raising medullary osmolality from 300 to ~**1200 mOsm/kg**; urea recycling adds about half. Vasa recta exchange without washing out the gradient.
- **Antidiuretic hormone (ADH)** is released by high osmolality or low volume. Signal chain: V2 receptor → cAMP (cyclic adenosine monophosphate) → protein kinase A → aquaporin-2 (AQP2) inserted into the apical membrane → water reabsorbed. Concentration needs both a hypertonic medulla and AQP2; urine osmolality ranges ~50–1200 mOsm/kg.

| Hormone | Site | Effect |
|---|---|---|
| Angiotensin II, aldosterone | Efferent arteriole, proximal tubule; collecting duct | $Na^+$ retention, $K^+/H^+$ loss, GFR maintained |
| Atrial natriuretic peptide | Afferent arteriole, collecting duct | Natriuresis, lowers renin |
| Parathyroid hormone | Proximal and distal tubule | Phosphaturia, $Ca^{2+}$ retention |

## F. Ureteric transport

- **Pacemaker:** atypical smooth muscle cells at the minor calyces/pelvicalyceal border.
- Peristalsis is **myogenic**: the excised or denervated ureter keeps contracting. Nerves (T11–L1, S2–4) carry only pain.
- Calcium entry triggers action potentials spread by gap junctions; calcium–calmodulin then activates myosin light-chain kinase.
- Walls coapt behind each bolus: 2–6 waves/min at 20–80 $cmH_2O$. Very high flow makes boluses coalesce, dilating the ureter **without obstruction**.
- Ureterovesical tunnel length:diameter 5:1 prevents reflux; bladder pressure >40 $cmH_2O$ stops efflux.

| Factor | Effect on transport |
|---|---|
| Obstruction | Acute: stronger contractions; chronic: dilatation, lost coaptation |
| Infection | Gram-negative endotoxin inhibits peristalsis (dilatation without obstruction) |
| Pregnancy | Right > left dilatation: compression at pelvic brim, progesterone |
| Drugs | α1-blockers and calcium-channel blockers relax (medical expulsive therapy); NSAIDs lower pelvic pressure in colic |

## G. Bladder storage and voiding

- **Storage** below ~10 $cmH_2O$: compliance, sympathetic β3 detrusor relaxation, α1 bladder-neck and pudendal sphincter tone.
- **Voiding:** pelvic parasympathetic (S2–4) acetylcholine on M3 receptors contracts the detrusor; nitric oxide relaxes the urethra.

### H. Pathophysiology of obstruction (unilateral: triphasic)

`Phase 1 (0–1.5 h): RBF up, pressure up (afferent dilatation: prostaglandins, nitric oxide) Phase 2 (1.5–5 h): RBF down, pressure up (efferent constriction) Phase 3 (>5 h): RBF down, pressure down (afferent constriction: angiotensin II, thromboxane A2, endothelin)`

- Ischaemic injury starts **within 2 hours** of complete block.
- **Tubules:** down-regulated aquaporins and $Na^+$ transporters cause concentrating defect, salt wasting and distal RTA.
- **Fibrosis** (final path to failure): stretch, inflammation, TGF-β (transforming growth factor-β), angiotensin II, microRNAs.
- **Bilateral obstruction or solitary kidney:** pressure stays high longer and natriuretic peptides and urea accumulate, so **post-obstructive diuresis** follows relief in ~50% (vs ~2% unilateral).
- **Clinical:** hypertension (~77% bilateral, ~20% unilateral); contralateral hypertrophy; pyelorenal backflow acts as a pop-off; forniceal rupture causes urinoma.
- **Also know:** immature kidneys fail to develop; outlet obstruction causes detrusor hypertrophy then decompensation; diuretic MAG3 (mercaptoacetyltriglycine) renogram (half-time >40 min = obstructed); Whitaker test (>15–20 $cmH_2O$ = obstructed); tolvaptan in polycystic disease.

**Diagram to draw:** Nephron with transporters; triphasic graph.

> **Key points to remember:**
> - RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
> - Proximal 65%, loop 25%, distal 5–7% of $Na^+$; ADH → V2 → AQP2.
> - Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

## LAQ 7 Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL

P1

**Asked as:**

- Describe anatomy of pelvicalyceal system in detail and its application in endourological surgeries — *Winter 2017 Paper I Q1 (LAQ)*
- Anatomy of renal collecting system and its relevance to PCNL — *Practice Set 1 Paper I Q3d (SAQ)* (also answered in 2022-Summer Paper 1, Q1 and 2017-Winter Paper 1, Q1)
- Describe anatomy of kidney in relation to PCNL and describe indication, contraindication, procedure and complications of mini-PCNL. — *Summer 2022 Paper I Q1 (LAQ)* (also answered in 2017-Winter Paper 1, Q1 (pelvicalyceal anatomy) and 2017-Winter Paper 4, Q3d (Miniperc))

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)

### Introduction

- Papilla → minor calyx → infundibulum → major calyx → renal pelvis → UPJ (ureteropelvic junction); lined by urothelium. Anatomy is very variable, so map it (CT (computed tomography) urogram or pyelogram) before every endourological procedure.
- Safe PCNL (percutaneous nephrolithotomy) access depends on the kidney's position, neighbours, vessels and calyces.

### Surgical anatomy relevant to PCNL: position and orientation

- Lies T12–L3 (right 1–2 cm lower); descends about 3 cm on inspiration.
- It lies on the cone-shaped psoas, so the upper pole is medial and posterior and the lower pole lateral and anterior.
- Tract: skin, flank muscles, thoracolumbar fascia, paranephric fat, Gerota fascia, perirenal fat, cortex, then **pyramid and papilla** into the calyx.

### Anatomy of the calyces (collecting system)

- **Minor calyx:** a cup around one papilla; its two side projections are the **fornices**; it drains by an infundibulum to a major calyx and pelvis. **Compound calyces** (2–3 papillae) are commonest at (cluster at) the poles. Average 7–9 calyces per kidney (range 4–19 or more).
- **Distribution:** upper pole usually 3 calyces, interpolar 3–4, lower pole 2–3.
- **Drainage pattern:** upper pole by a single midline infundibulum; lower pole by a single midline infundibulum or paired calyces; hilar region (hilar ones) by **anterior and posterior rows** of paired calyces.
- **Orientation:** the kidney lies on psoas, rotated 30–50° behind the coronal plane (posteriorly), so **posterior calyces point posterolaterally** (towards the prone surgeon). On a pyelogram anterior and posterior calyces (rows) overlap.
- **Brödel vs Hodson pattern (Kaye and Reinke):** in a Brödel-type kidney the posterior calyces lie lateral and the anterior calyces medial; in a Hodson-type kidney the reverse. Brödel type is commoner on the right, Hodson type on the left, so confirm calyx orientation with contrast/air or ultrasound before puncture.

### Renal pelvis

- Pyramidal, apex to the ureter; capacity 3–10 mL. **Intrarenal** (in the sinus), **extrarenal** (larger, longer infundibula, mimics dilatation) or mixed.
- **Duplex (divided) pelvis:** upper and lower parts join at the hilum, with no direct connection between upper and lower calyces (does not link upper and lower calyces). Suspect it when one pole cannot be reached from a puncture.

### Sampaio classification (Sampaio types; 140 resin endocasts)

| Type | Frequency | Midzone (hilar) drainage |
|---|---|---|
| A-I | 45% | Midzone calyces drain into the superior or inferior calyceal group (via a polar group) |
| A-II | 17.2% | **Crossed calyces**: one to the superior and one to the inferior group (to both groups) |
| B-I | 21.4% | Independent major calyceal group (major calyx) from the midzone |
| B-II | 16.4% | 1–4 minor calyces enter the pelvis directly (straight into the pelvis) |

- **Perpendicular minor calyx (11.4%):** drains straight into the pelvis, hidden on a pyelogram (pyelography); narrow neck (≤4 mm) hinders clearance.
- **Crossed calyces:** the limb to the inferior group is **ventral in 87.5%**.

### Vascular relations (the basis of safe access)

- Hilum, anterior to posterior: vein, artery, pelvis, **posterior segmental artery** (the only branch passing behind the pelvis).
- Five segmental **end arteries**; interlobar arteries run in the **columns of Bertin**, hugging the anterior and posterior faces of the infundibula (beside the infundibula).
- **Brödel line:** avascular plane between anterior and posterior arterial territories, just posterior to (behind) the lateral border, along the posterior calyces. Posterior calyces along it are the target.

| Puncture route / site (Sampaio) | Vascular injury |
|---|---|
| Upper-pole infundibulum | 67% (arterial 26%); posterior segmental artery |
| Mid-pole infundibulum | Arterial 23% |
| Lower-pole infundibulum | Infundibular artery (present) in 38% |
| **Calyceal fornix** | Venous <8%, **no arterial injury** |

- So the needle passes **through the papilla along the calyx axis**, never into an infundibulum or the pelvis.

### Relations (the hazards of access)

- **Relations:** pleura reaches the 12th rib posteriorly, lung the 10th intercostal space; liver and spleen descend in inspiration; **retrorenal colon** is commoner at the lower pole and when prone.

| Structure | Relation | Practical point |
|---|---|---|
| Pleura, lung | Pleura reaches the 12th rib; lung the 10th intercostal space | Prefer subcostal or 11th-space access; risk rises above the 12th rib, highest above the 11th; puncture in expiration |
| Liver, spleen | Over the upper poles; descend in inspiration | Risk with supracostal access, inspiration, organomegaly |

| Structure | Relation | Practical point |
|---|---|---|
| Colon | Retrorenal colon, mainly at the lower pole, commoner prone and on the left | Review CT; ultrasound shows bowel |

## Applications in endourology

### 1. PCNL (percutaneous nephrolithotomy) access

- Ideal target: a **posterior calyx with a short, wide infundibulum**, punctured (entered) **through the papilla and fornix** along the calyx axis, never into (through) the infundibulum or pelvis; identified by injected air, C-arm rotation or ultrasound. This passes through a pyramid and avoids interlobar arteries.
- **Upper-pole posterior access** gives the straightest route (best reach) to the pelvis, UPJ and most calyces (staghorn; staghorn branches). Chest risk (pneumothorax, hydrothorax) rises above the 12th rib, highest above the 11th. Puncture in expiration.
- **Lower-pole access** is chest-safe but risks (retrorenal) colon (left) and reaches the UPJ poorly (poor UPJ reach).
- Identifying (identify) posterior calyces: rainbowing the C-arm towards the surgeon shortens them (they shorten when the C-arm rotates towards the surgeon); injected air fills non-dependent (posterior, when prone) calyces (rises posteriorly when prone); ultrasound also shows bowel and pleura.
- **Also know:** traps: perpendicular or crossed calyces, duplex pelvis.

### 2. RIRS (retrograde intrarenal surgery)

- Entering the lower-pole infundibulum needs about **140° deflection** (104–175°); ≥270° scopes see it, but working there is harder.
- **Unfavourable lower-pole anatomy:** steep infundibulopelvic angle, long calyx (>10 mm), narrow infundibulum (<5 mm). Relocate stones to an upper calyx with a basket.
- Lower-pole stone >1 cm: PCNL preferred (**Lower Pole I, 2001:** stone-free 91% PCNL vs 21% SWL (shock wave lithotripsy)).

### 3. Endopyelotomy

- Incise the UPJ **laterally**, where crossing vessels are absent. A crossing vessel lowers success (42% vs 86%).
- Antegrade route: mid-posterior or superolateral calyx; not lower pole (acute angle to the UPJ).

### 4. Infundibulotomy and calyceal diverticulum

- Incise infundibular stenosis **superiorly or inferiorly**; never anteriorly.
- **Calyceal diverticulum:** cortical cavity joined to the fornix by a narrow neck; treat if symptomatic. Upper or interpolar: ureteroscopy, incise or dilate the neck. Lower pole: direct puncture onto the stone, mini-PCNL; fulguration optional.

### 5. Open/anatrophic nephrolithotomy: incision along the Brödel line.

- **Also know:** horseshoe kidney (medial access, upper pole usually subcostal); pelvic kidney (ultrasound or laparoscopic guidance); transplant kidneys; megacalycosis mimics obstruction; AV (arteriovenous) fistula or pseudoaneurysm → angioembolisation.

## Mini-PCNL: definition

| Term | Tract size |
|---|---|
| Standard | 24–34 Fr |
| **Mini** | 15–22 Fr |
| Ultra-mini | 11–14 Fr |
| Micro-perc | <10 Fr |

- First mini-perc: Jackman, 1998 (13 Fr).

## Indications

- Renal stones 1–2 cm; lower-pole stones >1 cm; failed SWL (shock wave lithotripsy) or RIRS (retrograde intrarenal surgery).
- Calyceal diverticulum (lower pole), children, poor retrograde access (urinary diversion, ureteric stricture), selected stones >2 cm or extra tracts for staghorns.

## Contraindications

- Untreated UTI (urinary tract infection), uncorrected coagulopathy, pregnancy (EAU, European Association of Urology).
- Tumour in the access tract or suspected malignant renal tumour; no safe window; unfit for anaesthesia.

## Procedure

1. Urine culture, blood count, coagulation, creatinine; **non-contrast CT (computed tomography)**; antibiotic prophylaxis.
2. Lithotomy: ureteric catheter and retrograde pyelogram.
3. Prone, or supine (Galdakao-modified) for single-position retrograde access.
4. 18 G puncture of a posterior calyx under fluoroscopy (bull's-eye or triangulation) or ultrasound; urine return; guidewire and safety wire.
5. One-step, serial or balloon dilatation to 15–22 Fr; place the sheath.
6. Mini-nephroscope; holmium/thulium laser or pneumatic lithotripsy. Fragments wash out by the irrigation **"vacuum-cleaner" effect** or a **suction sheath**.
7. Flexible nephroscopy to clear calyces.
8. Exit: tubeless, small nephrostomy (≤10 Fr: less pain) or double-J stent.

## Advantages and limitations

- Meta-analysis: **non-inferior stone-free rate**, smaller haemoglobin drop, **fewer transfusions**, shorter stay.
- Longer operation, poorer vision. A tight sheath raises **intrarenal pressure**, causing pyelovenous backflow and **fever or sepsis**. Advantages disappear against 24 Fr tracts.

## Complications (grade by Clavien–Dindo)

| Type | Points |
|---|---|
| Bleeding | Acute tract bleeding: tamponade sheath, balloon or Kaye catheter. Delayed: pseudoaneurysm or AV (arteriovenous) fistula, treated by angioembolisation |
| Infection | Fever, SIRS (systemic inflammatory response syndrome), urosepsis |
| Thoracic | Pneumothorax, hydrothorax: chest drain |

| Type | Points |
|---|---|
| Visceral | Colon (stent plus nephrostomy withdrawn into the colon), spleen, liver, duodenum |
| Collecting system | Perforation, extravasation, later infundibular stenosis; residual stones |
| Others | Irrigation absorption, venous gas embolism, DVT (deep vein thrombosis), neuropraxia, renal function loss, death |

**Diagram to draw:** transverse section of kidney showing anterior and posterior calyces (posterior calyx), Brödel line and the needle through the posterior calyceal fornix.

> **Key points to remember:**
>
> - Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
> - Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
> - Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.
> - Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
> - Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.
> - Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
> - Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
> - High intrarenal pressure in mini-PCNL causes fever and sepsis.

## LAQ 8 Lymphatic drainage of the genitourinary system and chyluria (etiopathogenesis and management)

P1

**Asked as:**

- Describe anatomy of lymphatic drainage of genitourinary system, etiopathogenesis and management of Chyluria — *Summer 2018 Paper I Q1 (LAQ)*
- Describe lymphatic system of urinary tract with reference to etiopathogenesis of chyluria. — *Practice Set 8 Paper I Q1 (LAQ)*
- Management of chyluria — *Practice Set 10 Paper I Q1 (LAQ)*

**Type:** LAQ (25 marks)
(also answered in 2018-Summer Paper 1, Q1 and 2024-Winter Paper 3, Q2)

### General plan of lymphatic drainage

- Lymph flows **cephalad and right → left** and returns to the left innominate (brachiocephalic) vein.
- Pelvic nodes (external iliac, internal iliac, obturator, presacral) → common iliac → lumbar (para-aortic, interaortocaval, paracaval) nodes → **lumbar trunks**. These join the intestinal trunk at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). The **thoracic duct** rises behind the aorta to the left innominate vein.
- Pelvic lymph crosses freely between right and left (the two sides). Organs supplied by lateral aortic branches (kidney, adrenal, ureter, gonad) drain to lumbar nodes.
- **Lumbar groups:** left lumbar (preaortic, left para-aortic, retroaortic), interaortocaval, and right lumbar (precaval, paracaval, retrocaval). They drain organs supplied by lateral aortic branches: kidneys, adrenals, ureters and gonads.
- Preaortic nodes (coeliac, superior and inferior mesenteric) form the **intestinal trunk**, which carries **chyle** from the intestinal lacteals.
- Lumbar and intestinal trunks join at the **cisterna chyli** (in front of L1–L2, behind the right side of the aorta). It continues as the **thoracic duct**, which rises behind the aorta to the left innominate vein.
- **Lymphatic anatomy (summary):** intestinal lacteals → intestinal trunk → **cisterna chyli** (L1–L2) → **thoracic duct** → left subclavian–jugular junction. Renal lumbar lymphatics join the cisterna; valves normally prevent backflow.

### Drainage by organ

| Organ | Primary nodes | Notes |
|---|---|---|
| **Left kidney** | Left lateral para-aortic | Capsular and deep hilar networks drain to renal vein nodes; may go retrocrural |
| **Right kidney** | Right paracaval, interaortocaval | May cross to left para-aortic |
| **Ureter** | Abdominal: para-aortic (left), paracaval/interaortocaval (right); middle: common iliac; pelvic: common, external, internal iliac | Plexus in muscle and adventitia |

| Organ | Primary nodes | Notes |
|---|---|---|
| **Bladder** | Mainly **external iliac**; also internal iliac, obturator | Begins in lamina propria; bilateral cross-drainage |
| **Prostate** | **Obturator and internal iliac**; presacral | Three routes: superolateral (side wall), posterior (presacral), apical (internal pudendal) |
| **Testis** | Left: left para-aortic; right: interaortocaval, then right paracaval | Follows gonadal vessels; crossover only right → left. Epididymal cauda and vas → external iliac |
| **Penis** | Skin → superficial inguinal (both groins); glans and corpora → superficial and deep inguinal, then pelvic | Shaft lymphatics converge dorsally |
| **Scrotum** | Ipsilateral superficial inguinal | Does not cross the raphe |
| **Urethra (male)** | Penile → inguinal; bulbomembranous and prostatic → internal/external iliac, obturator | Female: proximal → pelvic; distal → inguinal |

**Lymphatic drainage of the urinary tract (intrinsic lymphatics and regional nodes)**

| Organ | Intrinsic lymphatics | Regional nodes |
|---|---|---|
| **Kidney** | Superficial capsular and deep hilar networks along the interlobular, arcuate and interlobar arteries, to nodes at the renal vein | **Left:** left lateral para-aortic (sometimes retrocrural, or thoracic duct above the diaphragm). **Right:** interaortocaval and right paracaval (may cross to left para-aortic) |
| **Ureter** | Plexuses in the muscle and adventitia | Abdominal: left para-aortic (left); paracaval and interaortocaval (right). Middle: common iliac. Pelvic: common, external and internal iliac |
| **Bladder** | Begins in the lamina propria; a few perivesical nodes | Mainly **external iliac**; also internal iliac and obturator; cross-drainage to both sides |
| **Prostate** | Periprostatic plexus | Obturator and internal iliac; presacral; external iliac |
| **Urethra** | Submucosal plexus | Male prostatic and membranous: internal iliac, obturator; penile: inguinal, then external iliac. Female: proximal to pelvic, distal to inguinal nodes |

- **Applied:** upper tract cancers spread to lumbar nodes; bladder and prostate cancers to pelvic nodes. Renal drainage is asymmetric: right renal lymph can cross left; left stays left.
- Valves in the renal hilar lymphatics normally stop backflow from the cisterna chyli. **Failure of these valves is the basis of chyluria.**

## Chyluria: definition

- Passage of **chyle** (intestinal lymph rich in chylomicrons, triglyceride and protein) in urine, giving milky urine (making it milky; urine looks milky). Caused by an abnormal **lymphourinary (chylourinary) fistula**, usually at the renal fornices / a renal calyceal fornix (**pyelolymphatic fistula**).

## Etiology

| Parasitic (commonest; India, China) | Non-parasitic |
|---|---|
| ***Wuchereria bancrofti*** (lymphatic filariasis; *Culex* mosquito) | Congenital lymphangiectasia, renal lymphangioma |

| Parasitic (commonest; India, China) | Non-parasitic |
|---|---|
| *Brugia* (lymphatic filariasis that spares the genitals; rarely urological) | Trauma, surgery (e.g. partial nephrectomy, RPLND (retroperitoneal lymph node dissection)) |
| | Retroperitoneal tumour or fibrosis, tuberculosis, pregnancy |

## Pathogenesis (obstructive theory)

- **Life cycle:** mosquito-borne larvae mature into adult worms that live for years in retroperitoneal and genital lymphatics. Females release microfilariae with **nocturnal periodicity**.
- Adult worms live in retroperitoneal lymphatics and cause lymphangitis and fibrosis. This **obstructs** flow between the intestinal lymph trunks and the thoracic duct.
- Proximal lymphatics dilate and their valves fail, so chyle **refluxes** into renal hilar lymphatics.
- Raised pressure ruptures them into a calyceal fornix → **pyelolymphatic fistula**. Rarely the fistula is in the ureter or bladder.
- **Obstructive (reflux) theory:** `Adult worms + host immune reaction -> lymphangitis and fibrosis -> obstruction between intestinal trunks/cisterna chyli and thoracic duct -> dilatation of retroperitoneal lymphatics, valve incompetence -> reflux of chyle into renal hilar and peripelvic lymphatics -> raised pressure ruptures thin-walled lymphatics into a calyceal fornix -> pyelolymphatic fistula -> chyluria`
- **Pathogenesis (filarial):** adult *Wuchereria bancrofti* worms and the host reaction cause **lymphangitis and fibrosis** between the intestinal trunks, cisterna chyli and thoracic duct. Lymphatics below the block dilate and their valves fail. Chyle refluxes into renal hilar lymphatics, which rupture into a calyx.
- The fistula is usually renal (hilar lymphatics lie against thin fornices); rarely ureteric or vesical.
- Chyluria appears **earlier than genital elephantiasis**, is usually **intermittent** and may remit spontaneously. It worsens after fatty meals, when lacteal flow is highest. So every treatment is judged against natural remission.
- **Non-parasitic causes** work the same way: lymphatic obstruction (tumour, fibrosis), direct disruption (surgery, trauma) or congenital lymphangiectasia.
- **Effects:** loss of fat, protein and lymphocytes causes weight loss, hypoproteinaemia and lowered immunity. Fibrin in chyle forms clots.

## Clinical features

- Milky urine, worse after fatty meals; **chylous clots** (retention, colic); **haematochyluria**.
- Loss of fat and protein → weight loss, **hypoproteinaemia**, oedema, anaemia, fatigue, lowered immunity.
- **Severity:** mild (intermittent, no systemic effect); moderate (persistent, clots); severe (haematochyluria, clot retention, weight loss, hypoalbuminaemia).

| Grade | Features |
|---|---|
| Mild | Intermittent; no systemic effect |
| Moderate | Persistent; clots |

| Grade | Features |
|---|---|
| Severe | Haematochyluria, clot retention, weight loss, hypoalbuminaemia |

## Evaluation (before treatment)

**1. Confirm chyle**

- **Ether (chloroform) test**: milky urine shaken with ether clears because fat dissolves. **Urine triglycerides** (and protein) positive/raised; Sudan III stains fat globules.
- Distinguish from / exclude **phosphaturia** (clears with acetic acid) and **pyuria** (leucocytes, no clearing).

**2. Filarial status (filarial tests)**

- Night blood smear (often negative at this stage / late), **circulating filarial antigen** (card) test, eosinophilia; PCR (polymerase chain reaction).

**3. Lateralise the side and localise the fistula**

- **Cystoscopy after a fatty meal** shows milky efflux from the ureteric orifice (one, sometimes both).
- **Retrograde pyelography** shows **pyelolymphatic backflow**.
- **Define fistula: lymphoscintigraphy** (non-invasive, shows tracer in kidney and bladder), pedal lymphangiography (historical gold standard), MR (magnetic resonance) lymphangiography or CT.

**4. Baseline:** serum albumin, haemoglobin, renal function; urine culture before instillation.

## Management (outline / algorithm)

`Mild → diet + DEC Persistent / moderate → add sclerotherapy (can repeat) Severe or failed sclerotherapy → renal hilar lymphatic disconnection Mild -> low-fat, high-protein diet + MCT oil + DEC Moderate/persistent -> endoscopic sclerotherapy (silver nitrate 0.5-1%, povidone-iodine) Severe or failed -> renal hilar lymphatic disconnection (open, laparoscopic, robotic) Confirm chyle -> grade severity Mild -> diet + antifilarial drugs; observe Moderate/persist -> add endoscopic sclerotherapy (repeatable) Severe or failed -> renal hilar lymphatic disconnection`

**1. Conservative (all patients)**

- **Low-fat, high-protein diet with MCT (medium-chain triglyceride) oil**, plenty of fluids.
- **Why MCT:** MCTs (medium-chain fats) are absorbed straight/directly into the portal blood, not into lacteals, so chyle formation falls (less chyle is formed).
- **DEC (diethylcarbamazine)** 6 mg/kg/day for 12 days kills microfilariae (another source: 2 mg/kg orally three times daily, traditionally 12 days; a 1-day course is as effective). It has only modest effect on adults.
- **Doxycycline** 200 mg daily for 4–8 weeks kills *Wolbachia* (endosymbiont; anti-*Wolbachia*) and so sterilises or shortens the life of adult worms. Albendazole and ivermectin are alternatives.
- Correct anaemia and protein deficit. Most mild cases settle on this, and many remit spontaneously.

**2. Endoscopic sclerotherapy (renal pelvic instillation sclerotherapy)**

- Ureteric catheter on the affected (leaking) side, head-down position. Instil **silver nitrate** (0.5–1%) or **povidone-iodine**, repeated over several sessions.
- **Mechanism:** chemical lymphangitis and fibrosis seal the fistula.
- **Contraindications:** active UTI (urinary tract infection), bilateral simultaneous instillation (risk of anuria).
- Good initial success/response; recurrence is common, but it is repeatable and avoids surgery.
- **Complications:** flank pain, haematuria, papillary necrosis, ureteric stricture, anuria; rarely renal failure or death (silver nitrate).

**3. Surgery: renal hilar lymphatic disconnection (chylolymphatic disconnection)**

- **Indications:** severe disease, failed sclerotherapy, haematochyluria with clot retention, marked weight loss.
- **Steps:** 1. Full renal mobilisation (nephrolysis). 2. **Skeletonise the renal artery and vein**, ligating, clipping or sealing every/all hilar lymphatics. 3. Strip the upper ureter; remove perinephric fat. 4. Nephropexy ± optional **omental wrap** around the hilum as a barrier. 5. **On-table cystoscopy** confirms clear urine/efflux.
- Reflux occurs only through **hilar lymphatics**, so upper-pole dissection can be limited.
- **Approach:** open, laparoscopic (transperitoneal or retroperitoneal) or robotic. **Retroperitoneoscopic** ligation beats open in meta-analysis: shorter operation and stay, less blood loss, faster bowel recovery, fewer complications.
- **Complications:** lymphocele, renal vein injury, recurrence.
- **Also know:** lymphovenous anastomosis; autotransplantation (rare); nephrectomy (now) obsolete; prevent by clipping lymphatics during renal and nodal surgery.

**4. Follow-up and prevention**

- Urine triglycerides and weight; recurrence needs repeat sclerotherapy or surgery.
- Mass drug administration (albendazole with DEC or ivermectin), bednets; clip lymphatics during renal and nodal surgery.

**Diagram to draw:** Retroperitoneal lymphatics with cisterna chyli, obstruction (a block above it / obstructed thoracic duct), dilated hilar lymphatics, and reflux to a pyelolymphatic fistula at a calyceal fornix.

> **Key points to remember:**
> - Lymph flows right → left; cisterna chyli at L1–L2, behind the right side of the aorta.
> - Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
> - Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually filarial (*W. bancrofti*); often intermittent.
> - Ether test and (urine) triglycerides confirm (ether test clears milky urine); cystoscopy after (a) fatty meal lateralises.
> - Diet (MCT) + DEC (diet with MCT plus DEC) → (silver nitrate) sclerotherapy → laparoscopic hilar lymphatic disconnection; disconnection surgery for severe or refractory cases.
> - Retroperitoneoscopic disconnection beats open.

## LAQ 9 Haemodialysis principles, dialysis techniques and dialysis access (vascular and PD)

P2

**Asked as:**

- Discuss principles of hemodialysis and vasular access in details. — *Winter 2017 Paper II Q1 (LAQ)*
- Vascular access for hemodialysis — *Winter 2025 Paper II Q3c (SAQ)*
- Hemodialysis access. — *Practice Set 10 Paper II Q3c (SAQ)*
- Access techniques in dialysis — *Practice Set 12 Paper II Q3f (SAQ)*
- Discuss techniques of dialysis — *Practice Set 5 Paper II Q3e (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)

(Winter 2025 Paper II Q3c: also answered in 2020-Summer Paper 4, Q3b) (Practice Set 10 Paper II Q3c: also answered in 2020-Summer Paper 4, Q3b and 2025-Winter Paper 2, Q3c) (Practice Set 12 Paper II Q3f: CAPD also answered in Practice Set 1 Paper 2, Q3a) (Practice Set 5 Paper II Q3e: HD principles also answered in 2017-Winter Paper 2, Q1; CAPD in 2018-Summer Paper 2, Q3e)

*Assumed meaning (Winter 2017): "vasular" is a typo for "vascular" access for haemodialysis. Assumed meaning (Practice Set 12): access for PD (peritoneal dialysis) and HD (haemodialysis); emphasis on PD catheter insertion, since HD vascular access is answered in full in Practice Set 12 Paper 2, Q3a. Assumed meaning (Practice Set 5): principles and techniques of HD (haemodialysis) and PD (peritoneal dialysis), with continuous therapies.*

### Introduction

- Haemodialysis (HD) removes solutes and water from blood across a semipermeable membrane in an extracorporeal circuit. It is the commonest KRT (kidney replacement therapy) for ESKD (end-stage kidney disease) and severe AKI (acute kidney injury). It needs an access giving about **250–350 mL/min** of blood flow.
- Access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

### Principles of haemodialysis (physical principles)

| Process | Driving force | What it removes |
|---|---|---|
| **Diffusion** | Concentration gradient across the membrane | Small solutes (urea, creatinine, potassium); bicarbonate enters blood |
| **Convection** | Transmembrane pressure; solute is dragged with water | Middle molecules (e.g. β2-microglobulin) |
| **Ultrafiltration** | Transmembrane (hydrostatic) pressure (HD) or osmotic (PD) gradient | Excess water, down to "dry weight" |

- **Countercurrent flow:** dialysate runs opposite to blood (**countercurrent**), keeping (to keep) a gradient along the whole dialyser. Clearance rises with flows, membrane area and time.
- **Dialyser:** blood (250–350 mL/min) passes through hollow fibres (hollow-fibre) of **synthetic membrane**; high-flux membranes clear more middle molecules.

- **Dialysate:** ultrapure (reverse-osmosis) water with sodium about 140 mmol/L, potassium about 2 mmol/L, bicarbonate and calcium, adjusted to the patient.
- **Anticoagulation:** heparin; heparin-free or regional citrate if bleeding risk is high (e.g. after surgery).
- **Schedule:** about 4 h (hours) three times (3 times) a week. **Also know:** haemofiltration, haemodiafiltration (HDF), CRRT (continuous renal replacement therapy).
- **Access:** AVF (arteriovenous fistula), AVG (arteriovenous graft) or CVC (central venous catheter).

### Adequacy

- **Kt/V** (dialyser urea clearance × time ÷ urea distribution volume): target 1.4, minimum 1.2 (Kt/V ≥1.2) per thrice-weekly session (KDOQI [Kidney Disease Outcomes Quality Initiative] 2015). **URR (urea reduction ratio) ≥65%.**

### Indications

- **Acute ("AEIOU"), refractory to medical treatment:** Acidosis; Electrolytes (hyperkalaemia); Ingestions (lithium, salicylates, alcohols); Overload with respiratory distress; Uraemia (encephalopathy, pericarditis, bleeding).
- **Chronic:** symptomatic ESKD, usually at eGFR (estimated glomerular filtration rate) 5–10; IDEAL (2010): early start gave no benefit.

### Complications of haemodialysis

- **During dialysis:** hypotension (commonest), cramps, arrhythmia, air embolism, bleeding. **Disequilibrium syndrome:** rapid urea removal in early sessions shifts water into the brain (cerebral oedema); prevent with short, gentle first sessions.
- **Long term:** cardiovascular disease (main cause of death), mineral–bone disorder, calciphylaxis, β2-microglobulin amyloidosis (carpal tunnel), infection, hepatitis B and C; acquired cystic kidney disease (about 80% in ESKD; RCC [renal cell carcinoma] in 1–2% of these).
- **Access-related:** access infection, thrombosis, steal.

### Peritoneal dialysis

- The **peritoneum** is the membrane: diffusion for solutes; **glucose or icodextrin** draws water osmotically.
- **Tenckhoff catheter** (two Dacron cuffs), placed open, laparoscopically (omentopexy) or percutaneously (see PD access below).
- **CAPD (continuous ambulatory PD):** about 4 exchanges of 2 L daily. **APD (automated PD):** night cycler.
- **Adequacy:** weekly Kt/V ≥1.7; PET (peritoneal equilibration test) classifies transport.
- **Complications:** peritonitis, exit-site infection, catheter malfunction, hernia, hyperglycaemia, peritoneal fibrosis.

### Continuous and hybrid therapies

- **CRRT (continuous renal replacement therapy):** CVVH (continuous veno-venous haemofiltration; convection), CVVHD (haemodialysis; diffusion), CVVHDF (both). For haemodynamically unstable AKI (acute kidney injury).

- **Also know:** SLED (sustained low-efficiency dialysis); HDF (haemodiafiltration: CONVINCE 2023 lowered mortality versus high-flux HD); home and nocturnal HD.

### HD versus PD

| | HD | PD |
|---|---|---|
| Setting | Centre or home | Home, continuous |
| Haemodynamics | Rapid shifts | Gentle; preserves residual function |
| Favoured in | Abdominal surgery, failed PD | Children, cardiac instability, no vascular access |

### Types of vascular access

| Access | Construction | Ready | Comment |
|---|---|---|---|
| **Radiocephalic AVF** (arteriovenous fistula; Brescia–Cimino, 1966) | Radial artery to cephalic vein at the wrist, end-to-side (now usually vein end to artery side) | ≥6–8 weeks | **First choice**: best patency, fewest infections; highest failure to mature |
| Brachiocephalic AVF | Brachial artery to cephalic vein at the elbow | ≥6–8 weeks | Higher flow; more steal |
| Brachiobasilic (transposition) AVF | Deep basilic vein transposed superficially | Longer | When the cephalic vein is unusable |
| **AVG** (arteriovenous graft) | PTFE (polytetrafluoroethylene), forearm loop or upper-arm straight | 2–3 weeks | More stenosis, thrombosis, infection |
| **Tunnelled cuffed CVC** (central venous catheter) | Right internal jugular; cuff blocks bacterial migration | At once | Bridge or last resort; highest infection and central stenosis |

- Non-tunnelled 10–12 Fr jugular or femoral catheters are for acute dialysis only.

### Planning the access

- **Hierarchy:** AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first** (US **Fistula First**, 2003). **KDOQI 2019** asks for (favours) an individual "ESKD Life-Plan": the right access for the right patient at the right time.
- **Timing:** refer at eGFR <30; create an AVF at least (about) **6 months** before dialysis is expected.
- **Vein preservation:** no venepuncture, cannula or PICC (peripherally inserted central catheter) in the non-dominant arm (that arm); avoid subclavian catheters (lines).
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm), venography if central stenosis is suspected.
- **Maturity, "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, by (at) 6 weeks.

### Access complications

| Complication | Key point | Treatment |
|---|---|---|
| Failure to mature | Juxta-anastomotic stenosis; side veins | Angioplasty; ligate (side) branches |
| Stenosis | Neointimal hyperplasia; AVG at venous anastomosis | PTA (percutaneous transluminal angioplasty), stent graft, revision |

| Complication | Key point | Treatment |
|---|---|---|
| **Thrombosis** | Usually on an underlying stenosis | Thrombectomy, then treat (fix) the stenosis |
| **Steal syndrome** | Hand ischaemia; diabetes, brachial-based access | Banding, DRIL (distal revascularisation–interval ligation), ligation |
| Aneurysm | Repeated needling at one site | Repair if skin is thin |
| Infection | CVC > AVG > AVF | Antibiotics; excise graft; remove catheter |

- **Also know:** brachiobasilic transposition AVF; venous hypertension (central stenosis); high-output cardiac failure (banding); ischaemic monomelic neuropathy; graft seroma.

## Monitoring

- Each session: thrill and bruit, arm-elevation test (a healthy fistula collapses), prolonged bleeding after needling, rising venous pressure, falling Kt/V. Duplex scan when abnormal; treat a stenosis causing clinical change. Rotate needle sites ("rope-ladder").

## Peritoneal dialysis access

**Catheter: Tenckhoff** silicone catheter with **two Dacron cuffs**. The deep cuff sits in the rectus muscle; the superficial cuff about 2–4 cm from the exit. The tip lies in the **pelvis (rectovesical or rectouterine pouch)**; the exit site points **downwards** to reduce infection.

### Insertion techniques

| Technique | Steps | Comment |
|---|---|---|
| **Open (mini-laparotomy)** | Paramedian incision → split rectus → open posterior sheath and peritoneum → catheter to pelvis on a stylet → purse-string round deep cuff → tunnel → exit site | Simple; general or local anaesthesia |
| **Laparoscopic (advanced)** | Rectus sheath tunnel; **omentopexy**, adhesiolysis, hernia repair as needed | Best function; fixes causes of outflow failure |
| **Percutaneous (Seldinger)** | Needle, guidewire, dilator and peel-away sheath under local anaesthesia, ± ultrasound or fluoroscopy | Bedside; avoid after major abdominal surgery |
| Peritoneoscopic | Small-scope guided placement | Variant of percutaneous |
| - **Break-in:** delay full-volume exchanges about 2 weeks to prevent leak; urgent start uses small supine volumes. | | |
| - **Complications:** outflow failure (constipation, omental wrap, migration), leak, hernia, exit-site or tunnel infection, peritonitis, bowel perforation. | | |
| - **Contraindications:** peritoneal scarring, uncorrectable hernia, inability to perform exchanges. | | |

### Haemodialysis catheter access (brief, insertion)

- **Non-tunnelled CVC (central venous catheter):** dual-lumen 10–12 Fr, internal jugular or femoral vein by **ultrasound-guided Seldinger**; for AKI (acute kidney injury); avoid subclavian (central stenosis).
- **Tunnelled cuffed CVC:** right internal jugular, tip at the cavoatrial junction under fluoroscopy; bridge to AVF.
- **AVF (arteriovenous fistula)** first choice, then **AVG (arteriovenous graft)**; HeRO and endovascular AVF in selected patients (see Practice Set 12 Paper 2, Q3a).

### Recent advances

- **Endovascular AVF** (percutaneous creation: WavelinQ, Ellipsys); drug-coated balloons; bioengineered vessels; home and nocturnal HD.
- **CONVINCE (2023):** high-dose HDF lowered all-cause mortality compared with high-flux HD.

**Diagram to draw:** dialyser with countercurrent blood and dialysate (beside a PD catheter in the pelvis), and a radiocephalic end-to-side fistula at the wrist; Tenckhoff catheter with two cuffs, tunnel, downward exit and pelvic tip.

> **Key points to remember:**
>
> - Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
> - HD Kt/V minimum 1.2 (target 1.4); URR ≥65%; PD weekly Kt/V ≥1.7.
> - AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
> - Thrombosis usually sits on a stenosis; steal is treated by (→) banding or DRIL.
> - Tenckhoff: two cuffs, pelvic tip, downward exit.
> - Laparoscopic placement with omentopexy gives best function.
> - Avoid subclavian lines; AVF first for long-term HD.
> - CRRT for unstable AKI; PD preferred in children.

## LAQ 10 Acute renal failure (AKI) and acute tubular necrosis

P2

**Asked as:**

- Etiology, pathogenesis and management of acute renal failure. — *Winter 2024 Paper II Q2 (LAQ)* (acute tubular necrosis also answered in 2022-Summer Paper 2, Q3c)
- Acute tubular necrosis. — *Summer 2022 Paper II Q3c (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)

*Acute renal failure is now called AKI (acute kidney injury).*

### Definition and staging: KDIGO (Kidney Disease: Improving Global Outcomes)

- AKI (per KDIGO) = rise in serum creatinine **or** fall in urine output; either alone is enough.

| Stage | Serum creatinine | Urine output |
|---|---|---|
| 1 | Rise ≥0.3 mg/dL in 48 h, or ×1.5 baseline in 7 days | <0.5 mL/kg/h for 6 h |
| 2 | ×2 baseline | <0.5 mL/kg/h for 12 h |
| 3 | ×3, or ≥4 mg/dL, or RRT (renal replacement therapy) | <0.3 mL/kg/h for 24 h, or anuria 12 h |

- One-fifth to one-third of inpatients; mortality about 25%, about 50% (about half) if RRT is needed.
- **ATN definition:** ATN is the commonest **intrinsic** cause of AKI (acute kidney injury): tubular injury from **ischaemia or toxins**. A misnomer: cell death is mainly **apoptosis**.

### Etiology

| Type | Causes |
|---|---|
| **Prerenal** | Hypovolaemia (bleeding, vomiting, burns), heart failure, sepsis, hepatorenal syndrome; worsened by NSAIDs (non-steroidal anti-inflammatory drugs) and ACE (angiotensin-converting enzyme) inhibitors |
| **Intrinsic** | **ATN (acute tubular necrosis):** ischaemia, contrast, aminoglycosides, cisplatin, myoglobin. **AIN (acute interstitial nephritis):** β-lactams, NSAIDs, PPIs (proton pump inhibitors), checkpoint inhibitors. Glomerulonephritis; vascular (renal artery or vein thrombosis, atheroemboli) |
| **Postrenal** | BOO (bladder outlet obstruction), bilateral ureteric obstruction or obstructed single kidney: stones, tumours, retroperitoneal fibrosis, ureteric injury at surgery |

- Prerenal AKI and ATN cause about three-quarters of hospital AKI.

#### Causes of ATN

| Ischaemic | Toxic |
|---|---|
| Prolonged prerenal state: shock, sepsis, haemorrhage, burns, cardiac surgery, intraoperative hypotension, renal artery clamping, **transplant (delayed graft function)** | **Exogenous:** aminoglycosides, amphotericin, cisplatin, iodinated contrast. **Endogenous:** myoglobin (rhabdomyolysis, e.g. prolonged flank position), haemoglobin, light chains |

### Pathogenesis (pathophysiology)

- **Autoregulation** holds GFR (glomerular filtration rate) by afferent dilatation (prostaglandins, myogenic reflex) and efferent constriction (angiotensin II, tubuloglomerular feedback).

- **Prerenal:** low perfusion lowers glomerular pressure. Tubules are intact, so they **conserve sodium and water**. NSAIDs block afferent dilatation; ACE inhibitors block efferent constriction. Prolonged prerenal state becomes ATN.
- **Ischaemic ATN:**
- **Why tubules:** the medulla works near hypoxia (medullary oxygen tension is only 10–20 mmHg), while the **S3 proximal tubule (segment)** and **medullary thick ascending limb** consume much oxygen and are most vulnerable.
- **Loss of polarity:** $Na^+/K^+$-ATPase loses its basolateral position (moves off the basolateral membrane) → sodium reabsorption fails → high sodium at macula densa → **tubuloglomerular feedback** constricts afferent arteriole (afferent constriction) → GFR falls (protective).
- Cells die (mostly apoptosis) and slough (dead cells slough) → **muddy-brown casts** block (obstruct) tubules; filtrate leaks back; endothelial injury and inflammation worsen hypoxia; autoregulation is lost.
- Phases (clinical course): initiation → **maintenance** (oliguria, rising creatinine) → **recovery** (polyuria: tubules regenerate before concentrating ability returns).
- **Toxic ATN:** vasoconstriction plus direct tubular injury. **Contrast:** risk with age, diabetes, CKD (chronic kidney disease). **Rhabdomyolysis** (long lateral positioning): myoglobin causes toxicity, cast obstruction and hypoperfusion.
- **Postrenal:** raised tubular pressure, then afferent vasoconstriction; recovery depends on duration (dog studies of complete obstruction: full at 1 week, 70% at 2 weeks, none at 6 weeks).

## Evaluation (diagnosis: ATN versus prerenal)

- History (fluid loss, hypotension, drugs, contrast), volume status, bladder scan.
- **Ultrasound early** in unexplained AKI to exclude obstruction.

| Index | Prerenal | ATN |
|---|---|---|
| Urine sodium | <20 mmol/L | >20 mmol/L |
| FeNa (fractional excretion of sodium) | <1% | **>1%** |
| FeUrea (on diuretics) | <35% | >35% |
| Specific gravity | >1.015 | <1.010 |
| Sediment | Bland | **Muddy-brown granular casts**, tubular cells |
| Fluid challenge (response to fluids) | Quick (rapid) recovery | Lag of days to weeks |

- Creatine kinase; white cells (AIN); red cell casts (glomerulonephritis); biopsy only if recovery lags (fails).

## Management

### General

- Treat (remove) the cause; **avoid hypotension** (MAP, mean arterial pressure, >65 mmHg or near baseline) and nephrotoxins; stop NSAIDs and ACE inhibitors or ARBs (angiotensin receptor blockers) (RAAS (renin–angiotensin–aldosterone system) blockers); adjust drug doses.

- Isotonic crystalloid (fluids) for hypovolaemia; aggressive fluids in rhabdomyolysis; intravenous fluid before contrast (pre-contrast hydration).
- **Postrenal:** urethral or suprapubic catheter; **double-J stent or PCN (percutaneous nephrostomy)**; infected obstructed kidney is an emergency.
- AIN: stop drug ± steroids; glomerulonephritis: nephrology and biopsy.

**Complications**

- **Hyperkalaemia:** calcium gluconate 10 mL of 10% (protects heart) → insulin 10 U with 50 mL 50% dextrose, salbutamol nebuliser (shift) → binders, diuretics or dialysis (remove).
- **Volume overload:** loop diuretics in escalating doses. **Diuretics (only for volume overload) do not treat ATN.**
- Acidosis: bicarbonate. Uraemic bleeding: desmopressin 0.3 µg/kg.

**Dialysis indications (AEIOU), refractory to medical therapy**

| | **Indication** |
|---|---|
| A | Acidosis |
| E | Electrolytes, mainly hyperkalaemia |
| I | Intoxications / ingestions (lithium, salicylate, alcohols) |
| O | Overload with respiratory distress |
| U | Uraemia: encephalopathy, pericarditis, bleeding |

- Intermittent haemodialysis via 10–12 Fr jugular or femoral catheter; **CRRT (continuous RRT)** if haemodynamically unstable. **STARRT-AKI (2020):** early start gave no survival benefit.

**Post-obstructive diuresis**

- Causes: retained urea (osmotic diuresis), volume expansion with ANP (atrial natriuretic peptide), and down-regulated aquaporin-2 and sodium transport.
- Up to 300 mL/h for about 2 days. Free oral fluids; if unable to drink, replace about 75% of output with 0.45% saline; electrolytes every 4–6 hours. Gradual bladder decompression has no role.

## Outcome

- Full or partial recovery, or dialysis dependence; persistence ≥90 days = CKD. ATN recovery takes days to weeks.
- **Also know:** (early) biomarkers NGAL (neutrophil gelatinase-associated lipocalin), KIM-1 (kidney injury molecule-1); AKD (acute kidney disease, under 3 months); gadolinium (groups I and III) avoided in AKI.

**Diagram to draw:** tubular cell losing polarity → sloughing → casts and back-leak.

**Key points to remember:**

- KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
- FeNa <1% prerenal, >1% ATN; specific gravity <1.010, muddy-brown casts in ATN; ultrasound to exclude obstruction.
- S3 segment and thick ascending limb are most vulnerable.
- Diuretics only for overload (do not treat ATN); dialysis for refractory AEIOU.
- Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

## LAQ 11 Discuss the role of tissue typing and cross match in live and cadaveric

P3

**Type:** LAQ (25 marks)
**Source:** 2018-Summer Paper 3, Q2

*The question is incomplete; it is taken to mean "... live and cadaveric (deceased-donor) renal transplantation".*

### Introduction

- Two immune barriers decide compatibility: **preformed antibodies** (anti-A/B isoagglutinins and anti-HLA (human leukocyte antigen) antibodies) and **T-cell responses** to mismatched HLA.
- **Tissue typing** measures how well donor and recipient HLA match. **Antibody screening and crossmatch** detect preformed antibodies before the kidney meets them. Together they prevent **hyperacute rejection**, now practically eliminated.

### The HLA system

- HLA is the human MHC (major histocompatibility complex), a gene cluster on **chromosome 6 (short arm)**. Genes are inherited as two **haplotypes** (one from each parent) and expressed codominantly.

| | Class I | Class II |
|---|---|---|
| Loci | HLA-A, -B, -C | HLA-DR, -DQ, -DP |
| Expression | All nucleated cells | Antigen-presenting cells, endothelium |
| Presents to | CD8 (cytotoxic) T cells | CD4 (helper) T cells |

- **Sibling** chance: 25% two-haplotype identical, 50% one-haplotype, 25% none. A parent or child always shares one haplotype.

### Tissue typing methods

| Method | Principle | Note |
|---|---|---|
| **Serological** (microlymphocytotoxicity) | Lymphocytes + typed antisera + complement; dead cells take up dye | Low resolution; largely replaced |
| **Molecular** (DNA-based) | PCR (polymerase chain reaction) with SSP (sequence-specific primers), PCR-SSO (sequence-specific oligonucleotide probes), sequence-based typing, **NGS** (next-generation sequencing) | High (allele-level) resolution; now standard |

- **Matching:** usually reported as mismatches at **A, B and DR** (0–6). **DR matters most**, then B, then A; DQ increasingly typed. Fewer mismatches mean better graft survival and less sensitisation for a future transplant (Collaborative Transplant Study, Opelz). The effect is larger in deceased-donor grafts.

### ABO compatibility

- Same rules as transfusion: **O universal donor, AB universal recipient**.
- **A2 (non-A1) donors** can go to O or B recipients with low anti-A titre.

- **ABO-incompatible** living transplant is possible: lower titres (plasmapheresis or immunoadsorption, rituximab, IVIG (intravenous immunoglobulin)) to a centre target before surgery.

### Sensitisation and antibody screening

- Anti-HLA antibodies come from **pregnancy, blood transfusion, previous transplant** (and infection).
- **PRA (panel reactive antibody):** recipient serum tested against a panel of lymphocytes; result as % of panel killed.
- **Solid-phase (Luminex) single-antigen beads:** identify each antibody and its strength (MFI, mean fluorescence intensity). Antibodies against a donor are **DSA (donor-specific antibodies)**.
- **cPRA (calculated PRA):** % of donors the patient would be incompatible with, calculated from listed "unacceptable antigens". It replaced cell-panel PRA for allocation; very high cPRA gets allocation priority.

### Crossmatch

| Type | Method | Meaning if positive |
|---|---|---|
| **CDC** (complement-dependent cytotoxicity) | Donor T and B lymphocytes + recipient serum + complement; cell death scored. **DTT** (dithiothreitol) removes IgM autoantibody; AHG (antihuman globulin) increases sensitivity | **Positive T-cell CDC with IgG DSA = absolute contraindication** (hyperacute rejection; Patel and Terasaki, 1969) |
| **Flow cytometry crossmatch** | Antibody binding on donor cells detected by fluorescent anti-IgG | More sensitive; CDC-negative/flow-positive = higher risk of AMR (antibody-mediated rejection); desensitise or exchange |
| **Virtual crossmatch** | Compare recipient's bead-defined antibodies with donor HLA type | Speeds deceased-donor allocation; confirm by cell-based crossmatch |

- **B-cell positive** alone: class II antibody or non-specific binding; interpret with bead results.
- Crossmatch uses a **recent serum** plus historical peak sera.

### Living versus deceased donor

| | Living donor | Deceased (cadaveric) donor |
|---|---|---|
| Typing | ABO and HLA early in donor work-up; choose best of several donors | ABO and HLA at donor referral |
| Crossmatch | At work-up and a **final crossmatch shortly before surgery** | At the time of offer, often virtual first; done against the list rapidly (cold ischaemia) |
| Role of matching | Less critical: even unrelated living donors do better than well-matched deceased donors (Terasaki, 1995) | Allocation uses HLA match, sensitisation (cPRA), waiting time and donor quality (KDPI, Kidney Donor Profile Index) |
| If incompatible | **Paired (swap) exchange**, desensitisation, ABO-incompatible protocol | Patient waits; acceptable-mismatch programmes |

- **India:** swap (paired) donation is legal under the Transplantation of Human Organs (THOA) amendment, 2011.

### Managing the incompatible or sensitised recipient

- **Kidney paired donation**, including chains with altruistic donors.
- **Desensitisation:** plasmapheresis + low-dose IVIG, or high-dose IVIG, ± rituximab; **imlifidase** (cleaves IgG) for highly sensitised deceased-donor recipients (conditional EU approval). Desensitised HLA-incompatible living transplant gave better survival than waiting on dialysis (Montgomery, 2011).

### After transplant

- Monitor for **de novo DSA** (especially class II) with dd-cfDNA (donor-derived cell-free DNA) and biopsy; de novo DSA predicts graft loss.

### Recent advances

- **Eplet (epitope) matching**, high-resolution NGS typing, virtual crossmatch, imlifidase.

**Diagram to draw:** CDC crossmatch: donor lymphocytes + recipient serum + complement, dead cells take up dye (positive).

> **Key points to remember:**
> - HLA on chromosome 6; match A, B, DR; DR matters most.
> - Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
> - Flow and bead assays are more sensitive; cPRA guides allocation.
> - Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

## Short answer questions (SAQ)

## SAQ 1 Brain-stem death: pathophysiology, guidelines and declaration

P1 · P2 · P4 **merged from 3**

*Asked as:* P2 SAQ "Brainstem death: biology and clinical implications" (asked as "Brain stem Death: Biology and clinical implications", Winter 2024 Paper II Q3d; "Brainstem death", Practice Set 13 Paper II Q3c) · P1 SAQ "Patho-physiology of Brain stem death" (2015-Summer Paper 1 Q3d) · P4 SAQ "Brain-stem death: guidelines for declaration and the Board of Medical Experts (brain dead committee)" (asked as "Brain dead committee", Summer 2018 Paper IV Q3c; "Guidelines for declaration of brain stem death", Summer 2022 Paper IV Q3b)

*Assumed meaning (P4): the Board of Medical Experts that certifies brain-stem death under Indian law.*

### Definition and legal basis

- **Brain-stem death:** irreversible loss of the capacity for consciousness and of the capacity to breathe, from permanent brainstem failure; equivalent to death of the person (legally death).
- **India: THOA (Transplantation of Human Organs Act) 1994** legalised brain-stem death; **amended 2011** (Transplantation of Human Organs and Tissues Act); **Rules 2014**.
- Internationally: Harvard criteria (1968), AAN (American Academy of Neurology) criteria (updated 2023).

- Donation after brain death (DBD) keeps the circulation until procurement, so warm ischaemia is minimal.

## Biology (pathophysiology)

### Sequence

1. Brain injury causes oedema and raised ICP (intracranial pressure).
2. Cerebral perfusion falls, leading to herniation ("coning").
3. Pons: Cushing reflex (hypertension, bradycardia).
4. Medulla: catecholamine storm (hypertension, tachyarrhythmia).
5. Loss of sympathetic tone: vasodilatation, hypotension.

| System | Consequence |
|---|---|
| Heart and vessels | Storm causes myocardial injury and arrhythmia, then **hypotension** |
| Lungs | Neurogenic pulmonary oedema |
| Pituitary | **Diabetes insipidus (DI)** (no ADH, antidiuretic hormone): polyuria, hypernatraemia; low thyroid hormone and cortisol |
| Hypothalamus | **Hypothermia** (poikilothermia) |
| Blood | DIC (disseminated intravascular coagulation) from brain thromboplastin |
| Whole body | Cytokine injury to organs; hyperglycaemia |

## Composition of the Board of Medical Experts (India; Rules 2014)

| Member | Role |
|---|---|
| 1. Registered medical practitioner **in charge of the hospital** | Hospital authority |
| 2. **Independent specialist**, nominated by the in-charge from a panel approved by the Appropriate Authority | Independent check |
| 3. **Neurologist or neurosurgeon** from the approved panel (if unavailable: surgeon, physician, anaesthetist or intensivist) | Expert |
| 4. Medical practitioner **treating the patient** | Clinical history |

- **No member may belong to the transplant team.**

## Certification: preconditions

- Apnoeic coma on a ventilator with a **known irreversible cause**.
- Exclude drugs (**sedatives, relaxants**), hypothermia (35–36 °C by protocol), metabolic or endocrine causes, shock.

## Certification: brainstem reflexes (all must be absent)

| Brainstem reflex | Method | Cranial nerves |
|---|---|---|
| Pupillary light reflex | Bright light: pupils fixed, no reaction | II, III |
| Corneal | Touch cornea: no blink | V, VII |
| Oculocephalic (doll's eye) | Turn head: eyes move with the head | VIII, III, VI |

| Brainstem reflex | Method | Cranial nerves |
|---|---|---|
| Oculovestibular (caloric) | ~50 mL ice-cold water into each ear (drum visible): no eye movement | VIII, III, VI |
| Motor response in cranial nerve area | Central painful stimulus (supraorbital pressure): no facial grimace | V, VII |
| Gag and cough | Touch pharynx; pass suction catheter down trachea: no response | IX, X |

### Certification: apnoea test and timing

- **Apnoea test (last):** pre-oxygenate with 100% oxygen; disconnect the ventilator with oxygen insufflation; watch for breathing effort; **no breathing effort at $PaCO_2$ (arterial carbon dioxide tension) ≥60 mmHg** (or ≥20 mmHg above baseline) is a positive test. Abort for hypotension, desaturation or arrhythmia.
- **Two sets of tests at least 6 hours apart**, recorded on **Form 10**.
- **Ancillary tests** are not mandatory; used when clinical examination cannot be completed (e.g. facial trauma): EEG (electroencephalogram), cerebral angiography, perfusion scan, transcranial Doppler.

### Clinical implications: linked duties after declaration

- **Mandatory request:** ICU (intensive care unit) staff must inform relatives of the option to donate; a **transplant coordinator** counsels; next of kin give consent.
- **NOTTO** (National Organ and Tissue Transplant Organisation), with regional and state bodies (ROTTO, SOTTO: regional and state organ and tissue transplant organisations), allocates organs and keeps the registry.

### Clinical implications: donor management

- **Rule of 100s:** systolic pressure >100 mmHg, urine >100 mL/h, $PaO_2$ (arterial oxygen tension) >100 mmHg, haemoglobin >100 g/L.
- **Vasopressin** preferred (pressor, also treats DI); thyroid hormone, methylprednisolone, insulin; warming; protective ventilation.
- **Also know:** cold storage at 4 °C (University of Wisconsin solution); donor exclusions (metastatic cancer, melanoma, untreatable viral infection).

### Recent advances

- Hypothermic machine perfusion reduces delayed graft function.

### Diagrams to draw

- Herniation sequence leading to storm, DI, hypothermia, DIC.
- Certification flowchart: 1. Coma. 2. Preconditions. 3. Test 1. 4. 6 hours. 5. Test 2. 6. Form 10. 7. Consent. 8. NOTTO.

**Key points to remember:**

- Raised ICP, Cushing reflex, catecholamine storm, then hypotension, DI, hypothermia, DIC.
- Six brainstem reflexes absent; apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- THOA: four-member Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10; Rule of 100s.

## SAQ 2 GFR estimation in prospective renal donor

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "GFR estimation in a prospective renal donor and creatinine clearance" (asked as "Describe methods to estimate GFR in a prospective renal donor", Summer 2019 Paper I Q3a; "Creatinine clearance", Practice Set 4 Paper I Q3d) · P2 SAQ "GFR estimation: clinical assessment and in the prospective transplant donor" (asked as "Methods for GFR estimation in prospective tx donor", Practice Set 9 Paper II Q3a; "GFR clinical assessment", Winter 2023 Paper II Q3a)

*Assumed meaning: tx = transplant (prospective living kidney donor).*

### Definition

- GFR (glomerular filtration rate) is the plasma volume filtered by both kidneys per minute; normal about 120 mL/min/1.73 $m^2$. It is the standard index of kidney function.
- The ideal marker is freely filtered and neither secreted nor reabsorbed.

### Why it matters in the donor

- The donor loses about half the nephron mass; function settles at 60–80% of the pre-donation level.
- **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** sets GFR thresholds and asks for **measurement, not just estimation**, at a decision point: GFR is assessed by at least two methods, **including a measured one**.

### Method 1: serum creatinine and eGFR (estimated GFR)

- Creatinine comes from muscle and is freely filtered (not reabsorbed), but generation depends on **muscle mass**, diet (cooked meat, creatine supplements) and age. It is also secreted, so it **overestimates GFR**, increasingly as function falls. Trimethoprim blocks secretion.
- **CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration) 2021 creatinine equation:** age and sex, **no race term**; currently recommended; used for staging. MDRD (Modification of Diet in Renal Disease) and Cockcroft–Gault are older.
- Less accurate above 60 mL/min, where donors lie: a screening test only.

### Method 2: cystatin C

- 13-kDa protein made by all nucleated cells; freely filtered, then fully reabsorbed and broken down. Independent of muscle mass; affected by thyroid state, steroids, inflammation.
- The **combined creatinine–cystatin C equation** is more accurate than either alone; it is the most accurate estimate.

## Method 3: measured 24-hour creatinine clearance

- **Definition:** creatinine clearance (CrCl) is the volume of plasma cleared of creatinine per minute. It estimates GFR because creatinine is freely filtered and not reabsorbed.
- Normal about 120 mL/min/1.73 m² in young men, slightly lower in women; falls with age.
- **CrCl = (U × V) / P**: U = urine creatinine, V = urine flow (mL/min), P = plasma (serum) creatinine.
- **Method:** discard the first morning void, collect all urine for 24 hours including the next morning's first void; take serum creatinine during the collection.
- **Completeness check:** creatinine excretion should be **20–25 mg/kg/day in men, 15–20 mg/kg/day in women**.

### Limitations of creatinine clearance

- **Tubular secretion** of creatinine is small in health but rises as function falls, so CrCl **overestimates GFR** in CKD (chronic kidney disease).
- Collection errors are common. Creatinine generation depends on muscle mass, diet and age.
- Drugs blocking secretion (**trimethoprim, cimetidine**) raise serum creatinine without real injury.
- Invalid in AKI (acute kidney injury).
- **Also know:** cimetidine pretreatment blocks secretion to improve accuracy; averaging urea and creatinine clearance helps in advanced CKD.

### Estimated clearance: Cockcroft–Gault

- **CrCl = (140 − age) × weight (kg) / (72 × serum creatinine mg/dL)**, × 0.85 in women.
- Used for **drug dosing**; not normalised to body surface area; inaccurate in obesity.

## Method 4: mGFR (measured GFR), the reference standard

| Marker | Method / comment |
|---|---|
| **Inulin** | Urinary clearance; gold standard, research only |
| **Iohexol** | Plasma clearance; safe and accurate (another source: plasma or urinary clearance) |
| **Iothalamate** | Urinary or plasma clearance |
| **$^{51}$Cr-EDTA (ethylenediaminetetraacetic acid)** | Plasma clearance |
| **$^{99m}$Tc-DTPA (diethylenetriaminepentaacetic acid)** | Plasma sampling, or gamma camera (Gates method); purely filtered; also gives **split function** |

## Method 5: split function

- DTPA renogram gives each kidney's share (differential function); MAG3 or DMSA (dimercaptosuccinic acid) are alternatives.
- A difference >10% (beyond 45:55) matters: **leave the better kidney with the donor**.

## KDIGO 2017 donor decision

- Initial **eGFRcr**; confirm with **mGFR, measured creatinine clearance or eGFRcr-cys** (creatinine–cystatin C eGFR).

| GFR (mL/min/1.73 m²) | Decision |
|---|---|
| **≥90** | Accept |

| GFR (mL/min/1.73 m²) | Decision |
|---|---|
| **60–89** | Individualise (age, lifetime risk) |
| **<60** | Decline |

- Indian practice: **DTPA GFR** with split function plus CT (computed tomography) angiography is standard.

### Clinical uses of GFR and creatinine clearance

- **CKD staging (KDIGO):** G1 ≥90, G2 60–89, G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m², always with albuminuria (ACR, albumin-to-creatinine ratio).
- **Living donor evaluation:** KDIGO 2017 table above; confirm with measured GFR.
- **Urology:** drug and contrast dosing; **cisplatin eligibility** (CrCl <60 mL/min usually unfit); continent diversion (>35 mL/min); partial versus radical nephrectomy.
- Extremes of body size, amputees and pregnancy, where eGFR is unreliable: measure creatinine clearance.
- **Acute change:** eGFR is invalid when function is unstable; AKI uses creatinine change and urine output. NGAL (neutrophil gelatinase-associated lipocalin) is emerging.

### Diagrams to draw

- Plasma tracer clearance curve after a single injection.
- Nephron showing filtration plus a small secreted fraction of creatinine.

> **Key points to remember:**
> - Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
> - Cystatin C adds accuracy; creatinine–cystatin C combined equation is most accurate.
> - Inulin is the gold standard; iohexol or DTPA in practice; DTPA gives GFR and split function.
> - KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
> - Better kidney stays with the donor.
> - CrCl = UV/P; check completeness by creatinine excreted per kg.
> - Secretion makes CrCl overestimate GFR as function falls.
> - Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

## SAQ 3 Immunosuppression in renal transplantation: protocols, induction and steroid-free regimens

P1 · P2 · P4 **merged from 3**

*Asked as:* P1 SAQ "Immunosuppressants in renal transplantation (classification, protocols, ciclosporin)" (asked as "Immunosuppressants", Summer 2021 Paper I Q3f; "Classify immunosuppresants. Describe few common protocols", Practice Set 3 Paper I Q3b; "Cyclosporins", Practice Set 4 Paper I Q3f) · P2 SAQ "Pre transplant Immunosupression treatment options" (2023-Winter Paper 2 Q3e) · P4 SAQ "Steroid-free immunosuppression in renal transplantation" (Practice Set 3 Paper 4 Q3f)

*Taken as immunosuppressant drugs in renal transplantation; "Cyclosporins" taken as ciclosporin (cyclosporin A); "pre-transplant" taken as induction immunosuppression and*

*desensitisation given before or at transplantation. Also answered in 2020-Summer Paper 4, Q1; tacrolimus in 2017-Winter Paper 3, Q3c.*

## Principle

- T-cell activation needs three signals: **signal 1** (TCR [T-cell receptor] binds donor antigen; calcineurin then drives IL-2 [interleukin-2] transcription), **signal 2** (costimulation, CD28–CD80/86) and **signal 3** (IL-2 receptor signalling through mTOR [mammalian target of rapamycin], driving proliferation). Each drug blocks a step.
- Rejection risk is highest in the first weeks.

## Phases

- **Induction** (peri-transplant intense immunosuppression), **maintenance** (lifelong), **rescue** (treatment of rejection).
- **Desensitisation** removes preformed antibody in HLA (human leucocyte antigen)- or ABO-incompatible recipients.

## Classification

| Class | Drug(s) | Mechanism / action | Main toxicity |
|---|---|---|---|
| **CNI** (calcineurin inhibitors) | **Tacrolimus**, ciclosporin | Block calcineurin (signal 1) | **Nephrotoxicity**, diabetes, tremor, hypertension; ciclosporin: gum hypertrophy, hirsutism |
| **Antimetabolites** (antiproliferatives) | **Mycophenolate**, azathioprine | Block purine synthesis (IMPDH [inosine monophosphate dehydrogenase]) | Diarrhoea, leucopenia |
| **mTOR inhibitors** | Sirolimus, everolimus | Block signal 3 | **Lymphocele, poor wound healing**, proteinuria, dyslipidaemia; not nephrotoxic, antitumour |
| **Corticosteroids** | Prednisolone, methylprednisolone | Block cytokine transcription (IL-1, IL-2) and redistribute lymphocytes | Diabetes, bone loss, cataract |
| **Costimulation blocker** | **Belatacept** | CTLA4-Ig binds CD80/86 (signal 2) | **PTLD** (post-transplant lymphoproliferative disorder); avoid if EBV (Epstein–Barr virus)-negative |
| **Depleting antibodies** | rATG (rabbit antithymocyte globulin), alemtuzumab | T-cell depletion | Infection, PTLD |
| **Non-depleting antibody** | **Basiliximab** | Anti-CD25 (IL-2 receptor) | Few |

| Class | Drug(s) | Mechanism / action | Main toxicity |
|---|---|---|---|
| **Antibody / B-cell agents** | Rituximab, IVIG (intravenous immunoglobulin), plasmapheresis, bortezomib, eculizumab, imlifidase | Antibody removal or blockade; antibody-mediated rejection, desensitisation | — |

### Pre-transplant assessment of immunological risk

- cPRA (calculated panel-reactive antibody), DSA (donor-specific antibody) on Luminex beads, crossmatch (CDC [complement-dependent cytotoxicity] and flow).

### Induction agents

| Agent | Mechanism | Use and drawbacks |
|---|---|---|
| **Basiliximab** | Anti-CD25 (IL-2 receptor); non-depleting | Low-risk recipients; 20 mg IV (intravenous) on days 0 and 4; few side effects |
| **rATG** | Polyclonal T-cell depletion | High-risk (sensitised, retransplant, delayed graft function); less rejection than basiliximab; cytokine release, leucopenia, infection, PTLD |
| **Alemtuzumab** | Anti-CD52; profound depletion | Allows steroid avoidance; off-label |
| IV methylprednisolone | Cytokine suppression | Given intraoperatively with all regimens |

- **Over 90%** receive induction, then **tacrolimus + mycophenolate** ± steroids; steroids in more than two-thirds. Acute rejection has fallen from ≥60% to <10%.
- In living-donor transplants, tacrolimus and mycophenolate often start days before surgery.

### Common protocols

1. **Standard-risk (most recipients):** basiliximab 20 mg IV on days 0 and 4 + intraoperative methylprednisolone; then **tacrolimus + mycophenolate + tapering prednisolone**.
2. **High-immunological-risk** (sensitised, retransplant, DGF [delayed graft function] expected): **rATG** induction, then triple therapy.
3. **Steroid-avoidance or early withdrawal:** low-risk recipients, usually after depleting induction (see below).
4. **CNI-sparing:** belatacept- or mTOR-based regimens to spare nephrotoxicity; TRANSFORM: everolimus with reduced CNI non-inferior.
5. **ABO-incompatible or HLA-sensitised:** desensitisation with rituximab, plasmapheresis and IVIG; imlifidase for crossmatch-positive (see below).
6. **Rejection:** T-cell-mediated: pulse IV methylprednisolone, then rATG (for Banff grade II–III); antibody-mediated: steroids, plasmapheresis, IVIG, rituximab.

- **Monitoring: tacrolimus trough levels** (higher early, lower later); CYP3A4 (cytochrome P450 3A4) interactions (azoles, macrolides raise levels; rifampicin lowers).

### Desensitisation: HLA-incompatible

- **Plasmapheresis** (removes antibody) plus **low-dose IVIG** after each session; or **high-dose IVIG** (immunomodulation).

- **Rituximab** (anti-CD20) reduces B cells that renew antibody. Bortezomib (plasma cells) and eculizumab (anti-C5) in selected cases.
- **Imlifidase:** bacterial enzyme that **cleaves all IgG** within hours, turning a positive crossmatch negative; EMA (European Medicines Agency) conditional approval 2020 for highly sensitised patients.
- Desensitised live-donor transplant gives better survival than staying on dialysis (Montgomery, 2011).

## Desensitisation: ABO-incompatible

- **Rituximab** 2–4 weeks before; **plasmapheresis or immunoadsorption** to lower anti-A/B titres to a centre target (commonly ≤1:8); IVIG; tacrolimus–mycophenolate started before surgery. Splenectomy is now obsolete.
- Post-transplant titre monitoring; A2 donors carry lower risk.
- **Alternatives: kidney paired exchange** avoids desensitisation; acceptable-mismatch programmes.

## Steroid-free immunosuppression

### Why avoid steroids

- **Metabolic:** PTDM (post-transplant diabetes mellitus), hypertension, dyslipidaemia, weight gain, hence **cardiovascular death** (leading cause of death with a functioning graft).
- **Bone:** osteoporosis, avascular necrosis; **growth failure** in children.
- Cataract, infection, poor healing.

### Strategies

| Strategy | Timing |
|---|---|
| **Steroid avoidance** | No steroids, or perioperative doses only |
| **Early (rapid) withdrawal** | Stopped within the first week; most studied |
| **Late withdrawal** | Months later; more rejection, less favoured |

### Regimen

- **Induction is essential:** depleting rATG or alemtuzumab; basiliximab in low risk.
- **Maintenance:** tacrolimus at target levels plus MMF (mycophenolate mofetil).

### Selection

- **Suitable:** low immunological risk: first transplant, non-sensitised, no DSA, prompt graft function, good adherence.
- **Unsuitable:** highly sensitised, DSA, retransplant after immunological loss, DGF, early rejection, **recurring steroid-responsive disease** (IgA nephropathy, lupus), steroids needed for other illness.

### Evidence

- **Woodle et al. (2008):** withdrawal after day 7 versus low-dose maintenance: similar 5-year outcomes; **more mild, steroid-sensitive rejection**.
- **HARMONY (Lancet 2016):** steroids stopped on day 8: similar rejection and graft survival, **less PTDM**.
- **Cochrane review (2016):** more acute rejection, but no difference in patient or graft survival up to 5 years.

- **KDIGO (Kidney Disease: Improving Global Outcomes) 2009:** low-risk recipients with induction may stop steroids within the first week; if continued beyond it, continue rather than withdraw.
- **Practice:** over two-thirds still take maintenance steroids (Chapter 91); steroid-free maintenance is increasing in low-risk children.

#### Risks and safeguards

- Early rejection; possibly more chronic injury and recurrence. Safeguards: tacrolimus levels, adherence, DSA monitoring, biopsy for rising creatinine.

### Ciclosporin (cyclosporin A)

#### Introduction

- Cyclic peptide of 11 amino acids from the fungus *Tolypocladium inflatum*; the first **CNI**.
- Its introduction in the early 1980s cut acute rejection from at least 60% (pre-ciclosporin era) towards today's <10%. Tacrolimus has now largely replaced it.

#### Mechanism

1. Ciclosporin binds cyclophilin in the cytoplasm.
2. The complex inhibits calcineurin.
3. NFAT (nuclear factor of activated T cells) stays phosphorylated.
4. No IL-2 transcription, so T-cell activation is blocked (signal 1).

#### Pharmacokinetics and monitoring

- Variable oral absorption; the **microemulsion form** absorbs more predictably.
- Metabolised by **CYP3A4** and P-glycoprotein; narrow therapeutic index, so whole-blood **trough (C0) or 2-hour (C2) levels** guide dosing.

#### Uses

- **Transplantation:** maintenance with an antimetabolite and steroids, when tacrolimus is not tolerated.
- **IC/BPS (interstitial cystitis/bladder pain syndrome):** refractory **Hunner lesions**, 1–3 mg/kg/day; better than pentosan polysulfate.
- **Others:** retroperitoneal fibrosis after steroid failure; frequently relapsing nephrotic syndrome; severe psoriasis and other dermatoses.

#### Toxicity

| System | Effect |
|---|---|
| Kidney | **Nephrotoxicity**: acute afferent arteriolar vasoconstriction (reversible); chronic striped fibrosis, arteriolar hyalinosis; TMA (thrombotic microangiopathy) |
| Metabolic | Hypertension, hyperkalaemia, hypomagnesaemia, **hyperuricaemia** (gout), dyslipidaemia |
| Cosmetic | **Gingival hyperplasia, hirsutism** |
| Neurological | Tremor, PRES (posterior reversible encephalopathy syndrome) |
| Immune | Infection, skin cancer, PTLD |

#### Interactions

- **Levels rise:** azoles, erythromycin and clarithromycin, diltiazem, verapamil, grapefruit.
- **Levels fall:** rifampicin, phenytoin, carbamazepine, St John's wort.

- More nephrotoxicity with NSAIDs (non-steroidal anti-inflammatory drugs) and aminoglycosides.

**Versus tacrolimus**

- Ciclosporin: more hirsutism, gum hypertrophy, hypertension and hyperlipidaemia; less diabetes and tremor.
- **ELITE-Symphony (2007):** low-dose tacrolimus gave less rejection and better GFR (glomerular filtration rate) than ciclosporin regimens.

### Urological relevance

- **Infection:** UTI (urinary tract infection), **BK virus** (ureteric stricture, nephropathy), CMV (cytomegalovirus).
- **Malignancy:** skin cancer, PTLD, RCC (renal cell carcinoma, mostly native kidneys), **bladder cancer about 3-fold** and more aggressive.
- **Surgery:** lymphocele and wound problems with mTOR inhibitors; stones (CNI, steroids): CNIs promote transplant stones (hyperuricaemia, hyperoxaluria, hypocitraturia).
- **Also know:** belatacept (BENEFIT trial: better GFR), extended-release tacrolimus, TRANSFORM (everolimus with reduced CNI non-inferior).

### Diagrams to draw

- APC (antigen-presenting cell)–T cell synapse with signals 1–3 and the drug acting at each.
- T cell with the cyclophilin–calcineurin–NFAT–IL-2 pathway.
- Antibody titre falling with plasmapheresis sessions to target before transplant.
- Timeline: induction, steroids stopped by day 7, tacrolimus–MMF maintenance.

> **Key points to remember:**
>
> - Tacrolimus + mycophenolate ± steroids (maintenance) after induction (basiliximab or rATG).
> - Basiliximab for low risk; rATG for high immunological risk.
> - CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
> - Belatacept blocks signal 2; avoid if EBV-negative.
> - Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.
> - HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.
> - ABO-incompatible: rituximab + plasmapheresis to target titre.
> - Steroid-free: avoidance, early (first week) or late withdrawal; needs induction plus tacrolimus–MMF; low-risk recipients only.
> - More mild rejection, same survival; less PTDM (HARMONY). KDIGO 2009: stop within week 1, or continue.
> - Ciclosporin: binds cyclophilin, inhibits calcineurin, NFAT stays phosphorylated, IL-2 blocked.
> - Ciclosporin: nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
> - Ciclosporin urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.

## SAQ 4 Evaluation and selection criteria of the living renal donor

P1 · P2 · P3 **merged from 3**

*Asked as:* P2 SAQ "Pre-operative evaluation of the living (related and unrelated) renal donor" (asked as "Pre operative investigations for renal donor", Summer 2017 Paper II Q3b; "Pre-op assessment of unrelated donor", Practice Set 8 Paper II Q3a) · P3 SAQ "Assessment criteria for kidney donor." (2022-Summer Paper 3 Q3a) · P1 SAQ "Extended Criteria for Selection of Live Renal Donor" (2025-Winter Paper 1 Q3a)

*Assumed meaning: evaluation and acceptance criteria for a living kidney donor; for the unrelated question, a living unrelated kidney donor (Indian context); "extended criteria" read as the expanded (marginal, "medically complex") living donor. Related: 2023-Winter Paper 4, Q3e; marginal donor in 2024-Winter Paper 4, Q3e.*

## Principles

- **Donor safety first. KDIGO (Kidney Disease: Improving Global Outcomes) 2017** replaced fixed cut-offs with a **risk-based approach**: the donor's estimated lifetime ESKD (end-stage kidney disease) risk is compared with the centre's acceptable threshold.
- Multidisciplinary team with an **independent donor advocate**.
- For an unrelated donor the medical work-up equals that of a related donor; the extra step is **legal and ethical**: proof of genuine, unpaid motive.
- India: **THOA (Transplantation of Human Organs Act) 1994, amended 2011** (Rules 2014): near relatives may donate; others need authorisation committee approval; swap donation allowed.

## Legal assessment of the unrelated donor (THOA)

- A donor "other than near relative" donates for **affection or attachment**: **Authorisation Committee approval is mandatory**.
- Committee: hospital-based if >25 transplants a year, otherwise district or state; members are not from the transplant team.
- Process: joint application; **video-recorded interview** of donor and recipient; verify identity, documents, financial status and long association; exclude middlemen and payment; confirm free, informed consent.
- Foreign recipient with unrelated Indian donor not approved; appeal within 30 days.

## Medical assessment sequence

### 1. Compatibility (immunological)

- ABO group, HLA (human leukocyte antigen) typing, **crossmatch** (CDC [complement-dependent cytotoxicity] and flow), DSA (donor-specific antibody). Incompatible pairs: paired exchange.
- Unrelated donor: more HLA mismatch is expected, yet living unrelated grafts outperform deceased-donor grafts.

### 2. Kidney function and urine

- GFR (glomerular filtration rate) by two methods, one measured (e.g. DTPA [diethylenetriamine pentaacetic acid] GFR or creatinine clearance).
- Urinalysis, culture; persistent microscopic haematuria: cystoscopy, imaging, ± kidney biopsy.

### 3. Medical and metabolic

- Blood pressure (ambulatory if borderline), fasting glucose, **HbA1c (glycated haemoglobin)**, OGTT (oral glucose tolerance test) if at risk; lipids, liver tests, coagulation,

BMI (body mass index).

- Active cancer, chronic infection and cardiovascular disease exclude.

**4. Infection screen**

- HIV (human immunodeficiency virus), HBsAg (hepatitis B surface antigen), anti-HBc, HCV (hepatitis C virus), CMV (cytomegalovirus), syphilis; **tuberculosis** screen.

**5. Cardiopulmonary:** ECG (electrocardiogram), chest X-ray, echo; stress test if indicated.

**6. Age-appropriate cancer screening:** Pap smear, mammogram, PSA (prostate-specific antigen), stool test or colonoscopy.

**7. Urological:** stones (metabolic work-up), haematuria, malignancy, recurrent UTI (urinary tract infection).

**8. Anatomy**

- **CT (computed tomography) angiography:** number of vessels, early branching, stones, masses.
- **Split function** (DTPA or DMSA [dimercaptosuccinic acid]) if asymmetric: the **better kidney stays with the donor**; left preferred (longer vein).

**9. Psychosocial:** voluntary, no coercion or payment; independent donor advocate.

### Acceptance criteria: standard, extended and exclusion

| Factor | Standard (accept) | Extended (individualise) | Exclude (decline) |
|---|---|---|---|
| **Age** | Adult | Older donors (often >60) acceptable; no upper limit | Under legal age |
| **GFR**, mL/min/1.73 m² | ≥90 | **60–89** | <60 |
| **Albuminuria** | <30 mg/g ACR (P1: <30 mg/day) | **30–100** mg/g ACR (P1: mg/day) | >100 mg/g ACR (P1: mg/day) |
| **Hypertension** | None | Controlled on 1–2 drugs, no target-organ damage | Uncontrolled, organ damage |
| **Obesity** | BMI <30 | **30–35**; individualised risk | Many centres: >35 |
| **Glucose** | Normal | Prediabetes; selected type 2 diabetes | Type 1 diabetes |
| **Stones** | None | Single past stone, low recurrence risk; small stone removed ex vivo | Bilateral or recurrent; cystinuria, primary hyperoxaluria |
| **Anatomy** | Single vessels | Multiple arteries, simple cyst, mild FMD (fibromuscular dysplasia) | Major anomaly |

- ACR = albumin–creatinine ratio.
- P2 (unrelated-donor answer) states that diabetes, significant obesity and uncontrolled hypertension exclude.

### Extended criteria (marginal, medically complex living donor)

- A living donor with a risk factor that would once have excluded donation; used because of organ shortage.

- **Why older donors are acceptable:** their remaining lifespan is short, so lifetime ESKD risk is low. Young donors carry the highest lifetime risk. Healthy older donors are acceptable.
- **Also know:** isolated microscopic haematuria after negative work-up; treated low-risk cancer; APOL1 (apolipoprotein L1) high-risk genotype (test in African ancestry); future pregnancy (more pre-eclampsia in later pregnancy).
- **Safeguards:** measured GFR and split function (better kidney stays with the donor); CT angiography; laparoscopic retrieval; THOA authorisation; independent donor advocate.
- **Outcome:** grafts from older living donors do about as well as standard deceased-donor kidneys.

### Counselling figures (Muzaale 2014)

- Function settles at 60–80% of baseline.
- Lifetime ESKD (ESRD, end-stage renal disease) 90 per 10,000 donors vs 14 in matched healthy non-donors; mortality 3 per 10,000.

### Diagrams to draw

- Donor work-up flowchart: 1. Legal approval. 2. Compatibility. 3. Medical. 4. Anatomy. 5. Surgery.
- Risk-threshold scale (accept, individualise, decline).

> **Key points to remember:**
> - KDIGO 2017: risk-based lifetime ESKD threshold; GFR ≥90 accept, <60 decline; ACR >100 decline.
> - GFR 60–89 and albuminuria 30–100 are individualised.
> - Controlled hypertension on 1–2 drugs may donate; type 1 diabetes and bilateral stones exclude.
> - ABO and crossmatch first; CT angiography for anatomy.
> - Better kidney stays with the donor.
> - THOA: near relatives, else authorisation committee; unrelated donor needs Authorisation Committee approval.
> - Interview video-recorded; no commerce.

## SAQ 5 CAPD and CAPD peritonitis

P1 · P2 **merged from 2**

*Asked as:* P2 SAQ "CAPD (Continuous Ambulatory Peritoneal Dialysis)" (Summer 2018 Paper II Q3e; Practice Set 1 Paper II Q3a) · P1 SAQ "Management of CAPD peritonitis" (Practice Set 21 SAQ Sheet Q3d)

### Definition and principle

- CAPD (continuous ambulatory peritoneal dialysis): home dialysis using the **peritoneum** (about 1–2 $m^2$) as the membrane; dialysate dwells in the abdomen.
- **Diffusion:** urea, creatinine and potassium move into dialysate down a concentration gradient.
- **Ultrafiltration:** an osmotic agent draws water by osmosis: **glucose** (1.5%, 2.5%, 4.25%) or **icodextrin** (for the long dwell).

- **CAPD regimen:** about **4 exchanges of 2 L daily**, 4–6-hour dwells, overnight long dwell. **APD** (automated peritoneal dialysis) uses a night cycler.

### Catheter

- **Tenckhoff catheter:** silicone with **two Dacron cuffs** (deep cuff in rectus muscle; superficial cuff about 2 cm from the exit), exit site pointing downward, tip in the pelvis.
- **Insertion:** open; **laparoscopic** (rectus sheath tunnelling, omentopexy, adhesiolysis); percutaneous Seldinger. Delay full volumes about 2 weeks to prevent leaks.

### Adequacy

- Weekly **Kt/V urea ≥1.7** (peritoneal plus residual renal urea clearance × time ÷ urea distribution volume), with clinical well-being.
- **PET** (peritoneal equilibration test) identifies high transporters, who do better on short dwells (APD).

### Advantages over haemodialysis

- Home-based, no vascular access, **haemodynamically gentle**, **preserves residual renal function**. Favoured in children, poor vascular access and unstable heart disease.

### Contraindications

- **Absolute:** peritoneal adhesions or fibrosis (scarring), loss of membrane function (membrane failure), uncorrectable hernia, nobody able to do exchanges.
- **Relative:** recent abdominal surgery, stoma, abdominal wall infection, massive polycystic kidneys, morbid obesity, severe malnutrition.

### Complications (overview)

- **Peritonitis** (see below).
- **Exit-site and tunnel infection:** prevent with topical mupirocin or gentamicin.
- **Mechanical:** outflow failure (constipation, omental wrap, migration), pericatheter **leak**, **hernias** (raised abdominal pressure), hydrothorax, scrotal oedema (patent processus vaginalis).
- **Metabolic:** hyperglycaemia, weight gain, hypokalaemia, protein loss.
- **Long term:** ultrafiltration failure; **EPS (encapsulating peritoneal sclerosis)**: rare, presents as bowel obstruction.
- **Also know:** the PD (peritoneal dialysis) catheter is removed after a successful transplant; retroperitoneal laparoscopy protects the peritoneum for PD patients.

### CAPD peritonitis: introduction

- The leading cause of technique failure and transfer to haemodialysis. Commonest organism: **coagulase-negative *Staphylococcus*** (touch contamination).
- ISPD (International Society for Peritoneal Dialysis) 2022 target: centre rate **≤0.40 episodes per patient-year**.

### CAPD peritonitis: diagnosis (any 2 of 3)

1. Abdominal pain and/or **cloudy effluent**.
2. Effluent **WCC (white cell count) >100/µL with >50% neutrophils** (dwell ≥2 hours).
3. Positive effluent culture.

### CAPD peritonitis: initial steps

- Effluent cell count, Gram stain, **culture in blood-culture bottles** (culture-negative rate should be <15%).
- Examine exit site and tunnel; analgesia; heparin in bags if fibrin.

### CAPD peritonitis: empirical antibiotics

- **Intraperitoneal route preferred** (intravenous if septic); start immediately after samples.
- Cover **Gram-positives** (vancomycin or first-generation cephalosporin, e.g. cefazolin) **and Gram-negatives** (third-generation cephalosporin, e.g. ceftazidime, or aminoglycoside). Choice follows centre sensitivities.
- Intermittent (once daily in the long dwell) or continuous dosing; avoid prolonged aminoglycoside (residual function, ototoxicity).
- **Antifungal prophylaxis** (nystatin or fluconazole) during antibiotics.

### CAPD peritonitis: organism-specific duration

| Organism | Duration |
|---|---|
| Coagulase-negative *Staphylococcus*, streptococci, culture-negative | 2 weeks |
| *S. aureus*, enterococcus, Gram-negatives | 3 weeks |
| *Pseudomonas* | 3 weeks, two antipseudomonal drugs |
| Polymicrobial enteric | Add metronidazole; surgical review (bowel perforation) |
| **Fungal** | **Immediate catheter removal** + antifungal |

### CAPD peritonitis: catheter removal

- **Refractory** (effluent not clear after 5 days of the right antibiotics), relapsing, fungal, refractory exit-site or tunnel infection.
- Haemodialysis meanwhile; new catheter after at least 2 weeks.

### CAPD peritonitis: definitions

- **Relapse:** same organism (or sterile) within 4 weeks of finishing therapy.
- **Recurrent:** different organism within 4 weeks.
- **Repeat:** same organism after 4 weeks.

### CAPD peritonitis: prevention

- Training, exit-site mupirocin or gentamicin, prophylaxis before colonoscopy or gynaecological procedures.

### Diagrams to draw

- Tenckhoff catheter: two cuffs (deep and superficial), tunnel, downward exit site, pelvic tip.
- Peritonitis management flowchart from cloudy bag to catheter removal.

**Key points to remember:**

- Peritoneum is the membrane; glucose or icodextrin drives ultrafiltration.
- Tenckhoff catheter with two cuffs; Kt/V ≥1.7 per week.
- Peritonitis: two of cloudy fluid or pain, WCC >100/μL with >50% neutrophils, positive culture.
- Intraperitoneal Gram-positive plus Gram-negative cover.
- Remove catheter for fungal or refractory (5 days) peritonitis.

## SAQ 6 THOA/THOTA legislation and hospital authorization committee

P2 · P4 **merged from 2**

*Asked as:* P2 SAQ "THOA 1994 and its 2011 amendment (THOTA) and 2014 Rules" (Amendments to THOA: 2011; HOTA and its modification) · P4 SAQ "Hospital authorization committee for live related kidney donation."

*Indian law; not covered in the chapter notes. Assumed meaning (Practice Set 11): India's Transplantation of Human Organs Act (THOA) 1994, its 2011 amendment and 2014 Rules.*

### Legal framework and background: THOA 1994 aims

- **THOA (Transplantation of Human Organs Act) 1994**; amended 2011 as **THOTA (Transplantation of Human Organs and Tissues Act)**; **Rules 2014**.
- THOA 1994 regulates removal, storage and transplantation of organs, **prevents commercial dealing** (banned commercial dealing) and **recognised brain-stem death as legal death**, allowing deceased donation.
- Purpose: **prevent organ trade** while allowing genuine donation.
- Organ trade continued and deceased donation stayed low. The **Transplantation of Human Organs (Amendment) Act 2011** responded; **Rules followed in 2014**.

### Main provisions of THOA 1994

- **Living donors: near relatives** (spouse, parents, children, siblings) after verification. Others, donating "by reason of affection or attachment", need **Authorisation Committee** approval.
- **Appropriate Authority** registers and inspects hospitals; only registered centres may transplant.
- **Brain-stem death certification by a Board of four:** doctor in charge of the hospital; independent specialist from an approved panel; neurologist or neurosurgeon; treating doctor. **None from the transplant team.** Two sets of tests ≥6 hours apart.
- **Organ sale is a punishable offence.**

### 2011 amendment (THOTA)

| Area | Change | Purpose |
|---|---|---|
| **Scope** | **Tissues** added; renamed the **Transplantation of Human Organs and Tissues Act (THOTA)**; **tissue banks** registered | Regulates tissue banks |
| **Near relatives** | **Grandparents and grandchildren** added to spouse, children, parents and siblings | Wider related donor pool |

| Area | Change | Purpose |
|---|---|---|
| **Swap donation** | **Paired exchange** (swap donation) between near-relative pairs allowed, with Authorisation Committee approval | Overcomes ABO or crossmatch incompatibility |
| **Retrieval centres** | Hospitals with ICU (intensive care unit) facilities can register as **non-transplant organ retrieval centres** | |
| **Transplant coordinator** | Mandatory in every registered transplant hospital | Organises donation |
| **Mandatory ("required") request** | ICU doctor or staff must ask families of a potential brain-dead donor about donation and inform the retrieval centre | More deceased donors |
| **Brain-death board** | If a neurologist or neurosurgeon is unavailable, a surgeon, physician, anaesthetist or intensivist nominated by the hospital can serve | Smaller hospitals can certify |
| **Eye donation** | Trained technicians may remove corneas | |
| **Registry and network** | National human organ and tissue removal and storage network and **national registry** of donors and recipients (now run by NOTTO, the National Organ and Tissue Transplant Organisation) | Fair allocation |
| **Penalties** | **Higher penalties** for organ trade (up to 10 years' imprisonment; fine up to ₹1 crore) (see below) | Deterrence |

### Penalties under the amendment

- **Removal of organs without authority:** imprisonment up to **10 years** and fine up to ₹20 lakh.
- **Commercial dealing in organs:** imprisonment **5–10 years** and fine **₹20 lakh to ₹1 crore** (P4: imprisonment up to 10 years and fine up to ₹1 crore).
- Registered medical practitioner: name removed from the register (doctors' registration removed) for **3 years** for a first offence and **permanently** for a later offence.

### Rules 2014 (THOT Rules)

- Standard forms for consent and certification (**Form 10** for brain-stem death); brainstem reflexes and apnoea test set out.
- Authorisation Committee interviews donor and recipient to exclude coercion or payment.
- **Also know:** regional and state networks (ROTTO, SOTTO); 2023 guidelines removed the domicile rule and age limit for deceased-donor listing.

### Living donor categories and approval

| Donor | Approval |
|---|---|
| **Near relative:** spouse, son, daughter, father, mother, brother, sister, grandparents, grandchildren | Verified by the hospital's competent authority |
| **Other than near relative** (affection or attachment) | **Authorisation Committee approval mandatory** |
| **Swap (paired) donation** between near-relative pairs (2011) | Authorisation Committee |
| **Foreign donor or recipient** | Authorisation Committee; embassy certifies relationship |

### Hospital authorisation committee: constitution

- **Hospital-based** if a centre does more than **25 transplants a year**; otherwise **district or state committee**.
- **Members:** medical director or superintendent (chair); two senior doctors of the hospital **not in the transplant team**; two persons of high integrity and social standing (preferably one woman), nominated by the State; state committees add health-department nominees.

### Hospital authorisation committee: functions and procedure

1. Receive the **joint donor–recipient application** with consent forms, fitness certificates and identity and relationship documents.
2. **Personal interview** of donor, recipient and families; proceedings **video-recorded**.
3. Verify **relationship** (documents; **HLA [human leukocyte antigen] or DNA testing** if doubtful).
4. Exclude **commercial dealing**: financial status, reasons, long association, middlemen.
5. Confirm donor understands risks and consents freely, without coercion.
6. Check **domicile** (other-state donors).
7. Decide promptly; decision recorded and displayed; reasons given if refused. **Appeal** to the State Government within **30 days**.

### Safeguards and restrictions

- Unrelated donors and foreigners need **Authorisation Committee** approval.
- **Minors and mentally ill persons** cannot be living donors except in exceptional, approved cases.
- **Foreign recipient with unrelated Indian donor** not approved.

### Diagrams to draw

- Deceased donation flowchart: potential donor (brain-dead patient) in ICU → required request → transplant coordinator → brain-stem death board (Board of four) → family consent → retrieval → national registry and allocation (NOTTO allocation).
- Living donor approval flowchart: application → near relative (competent authority) or other (committee) → interview, verify → decision → appeal.

> **Key points to remember:**
>
> - THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors; amended 2011; Rules 2014.
> - 2011: tissues added; grandparents and grandchildren became near relatives.
> - Swap donation, retrieval centres, transplant coordinators, required request, higher penalties.
> - Commercial dealing: 5–10 years and ₹20 lakh to ₹1 crore.
> - Rules 2014: Form 10; Board of four; tests ≥6 hours apart.
> - Hospital-based committee if >25 transplants a year; no transplant team member.
> - Non-relatives, swap and foreigners need committee approval.
> - Interview video-recorded; appeal within 30 days.

## SAQ 7 Marginal kidney donor and Remuzzi score

P2 · P3 · P4 **merged from 3**

*Asked as:* P4 SAQ "Marginal donor and Remuzzi score for pre-implantation biopsy" (Define Marginal donor. Mention components of Remuzzi scoring system for preemptive renal biopsy of marginal donor kidneys.) · P2 SAQ "Marginal kidney donor" · P3 SAQ "Marginal Donor"

*"Preemptive" is read as pre-implantation (procurement) biopsy.*

### Definition

- A **marginal (extended) donor** is one whose kidney carries a **higher risk of DGF (delayed graft function) and shorter graft survival** than a standard donor. Such kidneys are used because of organ shortage.
- **Ojo (2001):** ECD transplantation still gives better survival than staying on dialysis.

### Deceased marginal donors

**1. ECD (expanded criteria donor), UNOS (United Network for Organ Sharing) 2002**

- Age **60 or over**, or
- Age **50–59 with 2 or more of:** hypertension, **creatinine >1.5 mg/dL**, death from **CVA (cerebrovascular accident)**.
- Graft-loss risk about 1.7 times that of a standard donor (about 70% higher).

**2. KDPI (Kidney Donor Profile Index) >85%**

- Replaced ECD in the US (2014); 10 donor factors (age, height, weight, ethnicity, hypertension, diabetes, cause of death, creatinine, HCV [hepatitis C virus], DCD).
- **Longevity matching:** lowest-KDPI kidneys go to recipients with the best EPTS (Estimated Post-Transplant Survival).

**3. DCD (donation after circulatory death), Maastricht**

| Category | Description | Type |
|---|---|---|
| I / II | Dead on arrival / failed resuscitation | Uncontrolled |
| III | Arrest after withdrawal of support | **Controlled (commonest)** |
| IV | Arrest in a brain-dead donor | Either |
| V | Euthanasia | Controlled |
| - Warm ischaemia raises DGF (roughly 40–50%); long-term survival is similar to DBD (donation after brain death). | | |

**4. Others**

- Donor AKI (acute kidney injury).
- **HCV-positive (HCV-viraemic)** donors with antiviral treatment.
- **Paediatric donors under 15 kg (en bloc).**
- **Excluded:** metastatic or haematological cancer, melanoma, untreatable viral infection.

### Living marginal donors

- Older donors, controlled hypertension (older or hypertensive living donors), obesity, small stones, multiple renal arteries. Acceptable after careful evaluation and consent.

### Remuzzi score (Karpinski/Remuzzi)

- Pre-implantation biopsy (ideally at least 25 glomeruli), read on rapid sections.
- **Four components (glomeruli, tubules, interstitium, vessels), each 0–3: total 0–12.**

| Component | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| **Glomerular global sclerosis** | None | <20% | 20–50% | >50% |
| **Tubular atrophy** | None | <20% | 20–50% | >50% |
| **Interstitial fibrosis** | None | <20% | 20–50% | >50% |
| **Arterial and arteriolar narrowing** | None | Wall thickness < lumen diameter | Wall = or slightly > lumen | Wall far > lumen, severe narrowing or occlusion |

### Allocation by score

| Total | Grade | Use |
|---|---|---|
| **0–3** | Mild | **Single** kidney transplant |
| **4–6** | Moderate | **Dual** kidney transplant |
| **7–12** | Severe | **Discard** |

- Remuzzi (NEJM 2006): biopsy-guided older-donor kidneys, single or dual, had graft survival similar to young-donor kidneys.
- **Limitations:** sampling error; wedge biopsy overestimates sclerosis; interobserver variation.

### Optimising use of marginal kidneys

- Short cold ischaemia.
- **HMP (hypothermic machine perfusion)** reduces DGF versus cold storage (Moers, 2009), without changing long-term survival; normothermic perfusion tests viability.
- **Old-for-old** matching: **ESP (Eurotransplant Senior Program)**, donors 65 or over to recipients 65 or over, with short cold ischaemia.
- Recipient: informed consent; induction; delayed CNI (calcineurin inhibitor) if DGF.

### Urological relevance

- Older and ECD kidneys have more **urine leak and ureteric stricture** (ischaemic distal ureter), as do multiple arteries and DGF.

### Diagrams to draw

- Remuzzi grid with four compartments scored 0–3.
- Maastricht categories on a timeline of circulatory arrest.

**Key points to remember:**

- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
- KDPI >85% replaced ECD (2014).
- Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
- 0–3 single, 4–6 dual, ≥7 discard.
- HMP reduces DGF; Maastricht III is the commonest DCD.

## SAQ 8 Renin-angiotensin system

P1

**Type:** SAQ (10 marks)
**Source:** 2015-Summer Paper 1, Q3c

### Definition

- Hormonal cascade controlling **blood pressure, extracellular volume, $Na^+/K^+$ balance and GFR (glomerular filtration rate)**.

### Cascade

```
Angiotensinogen (liver) ↓ renin (juxtaglomerular cells) — rate-limiting
Angiotensin I ↓ ACE (angiotensin-converting enzyme), mainly lung Angiotensin II
→ AT1 / AT2 receptors ↓ ACE2 Angiotensin (1-7) → Mas receptor (protective)
```

### Juxtaglomerular apparatus and renin release

- Granular cells in the afferent arteriole secrete renin and act as baroreceptors; the **macula densa** senses tubular NaCl.

| Renin stimulated by | Renin inhibited by |
|---|---|
| Low renal perfusion pressure (baroreceptor) | High perfusion pressure |
| Low NaCl at macula densa | High NaCl at macula densa |
| Sympathetic β1 activity | Angiotensin II (negative feedback) |
| Prostaglandins, hypovolaemia | Atrial natriuretic peptide, β-blockers, NSAIDs (non-steroidal anti-inflammatory drugs) |

### Actions of angiotensin II

- **AT1 receptor:** vasoconstriction; **aldosterone** (distal $Na^+$ retention, $K^+$ loss); direct proximal $Na^+$ reabsorption; thirst and ADH (antidiuretic hormone); **fibrosis**.
- **AT2 receptor:** vasodilatation; fetal urinary tract development.
- **Intrarenal:** efferent constriction maintains GFR when perfusion falls, so ACE inhibitors and ARBs (angiotensin receptor blockers) drop GFR in bilateral or solitary-kidney renal artery stenosis.

### Urological relevance

- **Renovascular hypertension (Goldblatt kidney):** atherosclerosis (older men, ostial) or fibromuscular dysplasia (young women). Diagnose by duplex ultrasound, computed tomography or magnetic resonance angiography, captopril renography. **Renal vein renin**

**ratio >1.5** with contralateral suppression predicts cure by revascularisation or nephrectomy.
- **Page kidney:** subcapsular haematoma compresses parenchyma; ACE inhibitor/ARB, then drainage.
- **Reninoma (juxtaglomerular cell tumour):** young women, ≤3 cm mass, refractory hypertension, **hypokalaemia**; excision cures most.
- **Obstruction:** hypertension (~77% bilateral, ~20% unilateral); angiotensin II drives fibrosis.
- **Also know:** post-traumatic hypertension (arteriovenous fistula, scarring); reflux nephropathy (renin-guided segmental resection); hyperkalaemia with angiotensin blockade in CKD (chronic kidney disease).
- **Drugs:** ACE inhibitors (cough, angio-oedema from bradykinin), ARBs, aliskiren (direct renin inhibitor), aldosterone antagonists; all contraindicated in pregnancy.

**Diagram to draw:** Juxtaglomerular apparatus and cascade.

### Recent advances

- Renal denervation (US approval 2023) for resistant hypertension.

> **Key points to remember:**
> - Renin is rate-limiting; released by low pressure, low NaCl, β1.
> - Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
> - Renin ratio >1.5 lateralises; reninoma excision cures most.

## SAQ 9 Drugs associated with Acute Renal Failure.

P1

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 1, Q3c

### Introduction

- AKI (acute kidney injury), KDIGO (Kidney Disease: Improving Global Outcomes): creatinine rise ≥0.3 mg/dL in 48 hours or ≥1.5 × baseline in 7 days, or urine <0.5 mL/kg/h for 6 hours.

### Classification by mechanism

| Mechanism | Drugs | How |
|---|---|---|
| **Haemodynamic (prerenal)** | **NSAIDs** (non-steroidal anti-inflammatory drugs) | Block prostaglandin afferent dilatation |
| | **ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers)** | Remove efferent constriction that holds GFR (glomerular filtration rate) |
| | Ciclosporin, tacrolimus, contrast; diuretics | Afferent vasoconstriction; volume loss |
| **ATN (acute tubular necrosis)** | **Aminoglycosides**, amphotericin B, vancomycin, cisplatin, ifosfamide, contrast | Direct tubular toxicity |

| Mechanism | Drugs | How |
|---|---|---|
| **AIN (acute interstitial nephritis)** | **β-lactams**, sulfonamides, quinolones, rifampicin, NSAIDs, **PPIs (proton pump inhibitors)**, allopurinol, **checkpoint inhibitors** | Cell-mediated hypersensitivity |
| **Crystals** | Aciclovir, methotrexate, indinavir, sulfadiazine; tumour lysis | Intratubular obstruction |
| **Microangiopathy** | Gemcitabine, VEGF (vascular endothelial growth factor) inhibitors | Endothelial injury |
| **Rhabdomyolysis** | Statins (with fibrates) | Myoglobin toxicity, precipitation |

- **"Triple whammy":** NSAID + ACE inhibitor/ARB + diuretic in a dehydrated patient.
- **False creatinine rise:** trimethoprim and cimetidine block creatinine secretion.

### Recognition

| Type | Urine |
|---|---|
| Prerenal | FeNa (fractional excretion of sodium) <1%, bland |
| ATN | FeNa >1%, muddy-brown casts |
| AIN | Sterile pyuria, white cell casts, ± rash, fever, eosinophilia |
| - AIN appears days to weeks after the drug starts; it is the commonest biopsy lesion with checkpoint inhibitors. Urine eosinophils are unreliable. | |

### Prevention and management

- Identify risk (CKD (chronic kidney disease), age, diabetes). Dose by eGFR (estimated GFR); monitor drug levels; once-daily aminoglycoside.
- Hydrate before contrast or cisplatin; alkalinise for methotrexate.
- **Stop the culprit**, withhold NSAIDs and ACE inhibitors, restore volume. **Early steroids for AIN** (only about half of antibiotic AIN recovers fully).
- Dialysis for refractory AEIOU: acidosis, electrolytes, intoxication, overload, uraemia.

**Diagram to draw:** glomerulus with afferent (prostaglandin) and efferent (angiotensin II) control.

> **Key points to remember:**
> - NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
> - ATN: aminoglycosides, amphotericin, cisplatin, contrast.
> - AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
> - Crystals: aciclovir, methotrexate, indinavir.

## SAQ 10 Berger's Disease.

P1

**Type:** SAQ (10 marks)
**Source:** 2020-Summer Paper 1, Q3a

*Berger's disease is IgA (immunoglobulin A) nephropathy (IgAN).*

## Definition

- **Commonest primary glomerulonephritis (GN) worldwide**, in children too. Dominant **mesangial IgA deposits** on immunofluorescence.
- Commonest cause of **glomerular haematuria** (about one-third). Second–third decades; male predominance.
- **IgA vasculitis** (Henoch–Schönlein purpura) is the systemic form.

## Pathogenesis

- "Multi-hit" model: abnormal (galactose-deficient) IgA1 → autoantibodies → immune complexes → mesangial deposition, complement activation, proliferation. Mucosal (respiratory) infection triggers flares.

## Clinical features

- **Recurrent painless macroscopic haematuria during or within days of an upper respiratory infection** ("synpharyngitic"), or after exercise. Post-streptococcal GN follows infection by weeks.
- Persistent microscopic haematuria, proteinuria, hypertension; occasional loin pain; rarely nephrotic syndrome or RPGN (rapidly progressive GN).

## Evaluation (urologist's role)

- **Dysmorphic RBCs (red blood cells), RBC casts, proteinuria** → glomerular source. **Normal C3 and C4.**
- Without casts, complete the urological evaluation (imaging ± cystoscopy) to exclude a urinary tract cause.
- **Renal biopsy** confirms. Once diagnosed, repeat haematuria work-ups are not needed.

**Oxford MEST-C score (biopsy)**

| Item | Grades |
|---|---|
| M: mesangial hypercellularity | M0, M1 |
| E: endocapillary hypercellularity | E0, E1 |
| S: segmental sclerosis | S0, S1 |
| T: tubular atrophy/interstitial fibrosis | T0 ≤25%, T1 26–50%, T2 >50% |
| C: crescents | C0 none, C1 <25%, C2 ≥25% of glomeruli |

## Prognosis

- Often benign, but **20–30% reach ESKD (end-stage kidney disease)**.
- Poor: persistent proteinuria, hypertension, raised creatinine, older onset, high MEST-C.

## Management

- No cure. **Supportive:** ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker), BP (blood pressure) control, low salt, stop smoking.
- **Newer drugs:** SGLT2 (sodium–glucose cotransporter 2) inhibitors; sparsentan (dual endothelin–angiotensin receptor blocker); targeted-release budesonide acting on gut

mucosal lymphoid tissue.

- Persistent proteinuria above about 0.75–1 g/day despite supportive care: consider a 6-month steroid course (KDIGO (Kidney Disease: Improving Global Outcomes) 2021); weigh infection risk.
- Crescentic RPGN: steroids plus cyclophosphamide.
- **Transplant:** recurs histologically often, clinically rarely.

**Diagram to draw:** glomerulus with mesangial IgA deposits.

> **Key points to remember:**
> - Commonest primary GN; haematuria with upper respiratory infection.
> - Normal complement; biopsy shows mesangial IgA.
> - Oxford MEST-C; 20–30% ESKD.
> - ACE inhibitor or ARB is the mainstay.

## SAQ 11 Vascular complications after renal transplantation, including transplant renal artery stenosis

P1

**Asked as:**

- Vascular Complications after Renal Transplantation in Recipient — *Winter 2025 Paper I Q3d (SAQ)*
- Transplant renal artery stenosis — *Practice Set 20 SAQ Q3d (SAQ)*

**Type:** SAQ (10 marks)

(vascular complications of transplantation also answered in 2025-Winter Paper 1, Q3d)

### Introduction

- Uncommon but a major cause of **early graft loss**. Doppler ultrasound is the first test.
- **Concept:** no inflow gives anuria; blocked outflow gives a swollen, tender graft with haematuria; reduced inflow gives hypertension and fluid retention.

### Main complications

| | Arterial thrombosis | Venous thrombosis | TRAS (transplant renal artery stenosis) |
|---|---|---|---|
| Incidence | 0.2–5% | 0.1–8.2% | 1–23% (commonest) |
| Timing | First days | First week | Months (usually 3 months–2 years) |
| Presentation | Sudden anuria | Painful swollen graft, haematuria, oliguria | Refractory hypertension, rising creatinine, flash pulmonary oedema |
| Doppler | No arterial flow | **Reversed diastolic flow**, clot in vein | High velocity at stenosis; **parvus-tardus** waveform distally |

| | Arterial thrombosis | Venous thrombosis | TRAS (transplant renal artery stenosis) |
|---|---|---|---|
| Treatment | Immediate exploration; thrombectomy (catheter thrombolysis described); nephrectomy if non-viable | Immediate exploration, thrombectomy; nephrectomy if necrotic | **Angioplasty or stenting** (first choice); surgical revision if that fails |

- **Thrombosis risks:** extreme donor or recipient age, **cold ischaemia >24 hours**, multiple arteries, technical error (kink, intimal flap), hypovolaemia, rejection, thrombophilia (factor V Leiden, antiphospholipid antibodies).
- **TRAS causes:** anastomotic technique, clamp injury, recipient atherosclerosis (iliac inflow stenosis mimics it). Contrast angiography confirms.

## Transplant renal artery stenosis (TRAS) in detail

### Definition and incidence

- TRAS (transplant renal artery stenosis): narrowing of the graft artery or its inflow; **commonest vascular complication (1–23%)**, usually 3 months to 2 years after transplant.

### Types and causes

- **Anastomotic:** suture technique, end-to-end anastomosis to the internal iliac artery, no aortic (Carrel) patch.
- **Post-anastomotic (distal):** clamp or perfusion-cannula intimal injury, kinking, rejection.
- **Pre-anastomotic ("pseudo-TRAS"):** recipient iliac atherosclerosis.
- **Also know:** CMV (cytomegalovirus) infection, delayed graft function, deceased donor.

### Pathophysiology and clinical features

- Reduced graft perfusion activates the **renin–angiotensin–aldosterone system**.
- **Refractory hypertension**, rising creatinine, fluid retention, **flash pulmonary oedema**; graft bruit.
- Clue: creatinine rises sharply after an ACE (angiotensin-converting enzyme) inhibitor.

### Evaluation

- **Doppler ultrasound first:** high peak velocity at stenosis (about >200–250 cm/s), and distal **parvus-tardus** waveform with low resistive index.
- MRA (magnetic resonance angiography) or $CO_2$ angiography to spare contrast.
- **Contrast angiography is the gold standard** and allows treatment at once. Iodinated contrast risks nephrotoxicity; gadolinium is avoided at low GFR (glomerular filtration rate).
- **Differential of graft dysfunction with hypertension:** rejection, calcineurin-inhibitor toxicity, ureteric obstruction.

### Management

- **Medical:** for mild stenosis with stable function: blood pressure control (calcium-channel blockers), statin, antiplatelet; avoid or monitor ACE inhibitors.
- **Percutaneous angioplasty with or without stent: treatment of choice** for significant stenosis (usually >50–60%). High technical success; reverses hypertension and graft dysfunction. Stents reduce restenosis.

- **Surgical revision** (resection and re-anastomosis, interposition graft, bypass) if angioplasty fails or anatomy unsuitable; risks graft loss.
- Follow-up Doppler after angioplasty detects restenosis; untreated TRAS ends in graft loss.

### Other vascular complications

- **Haemorrhage:** hilar vessels or anastomosis; re-explore if unstable.
- **Pseudoaneurysm (<1%):** technical or **mycotic**; rupture risk; endovascular or open repair, often graft nephrectomy, prolonged antibiotics.
- **Post-biopsy AV (arteriovenous) fistula:** persistent haematuria; usually closes; large or with steal: **coil embolisation**. Subcapsular haematoma can cause a **Page kidney**.
- **Graft torsion:** intraperitoneal grafts; prevent with nephropexy.
- **DVT (deep vein thrombosis)** of the iliac vein; **lymphocele** compressing the iliac or renal vein.
- **Limb ischaemia** from iliac dissection; internal iliac use: gluteal claudication, erectile dysfunction.

### Prevention

- Anastomose away from atheroma; check the lie for kinking; avoid clamp injury and kinking; use a Carrel patch where possible.
- Extend a short right renal vein with donor IVC (inferior vena cava).
- Short cold ischaemia; anticoagulate thrombophilic patients; avoid hypovolaemia.
- At reperfusion, a soft kidney suggests poor inflow; a tense, engorged kidney suggests outflow block.

**Diagram to draw:** Doppler waveforms: normal, parvus-tardus (TRAS; distal to stenosis) and reversed diastolic flow (venous thrombosis).

> **Key points to remember:**
> - TRAS is the commonest vascular complication; presents months later; angioplasty or stent first (first-line); angiography is gold standard.
> - Refractory hypertension plus rising creatinine: Doppler first.
> - Venous thrombosis: reversed diastolic flow, swollen tender graft.
> - Arterial thrombosis: sudden anuria; explore immediately.

## SAQ 12 Management of acute hyperkalaemia

P1

**Asked as:**

- Management of Acute Hyperkalemia — *Winter 2025 Paper I Q3e (SAQ)*
- Mgt. of acute hyperkalemia. — *Practice Set 6 Paper I Q3e (SAQ)*

**Type:** SAQ (10 marks)

(also answered in 2025-Winter Paper 1, Q3e)

### Definition and physiology

- Serum potassium **>5.5 mmol/L**; mild 5.5–5.9, moderate 6.0–6.4, **severe ≥6.5** (UK Kidney Association). Above 6.0 is generally urgent.

- About 98% of potassium is intracellular; the ratio sets the resting membrane potential, so a high level (high levels) causes **arrhythmia and cardiac arrest**.
- Defence: rapid shift into cells (insulin, $\beta_2$-agonists via $Na^+/K^+$-ATPase), then renal excretion (needs distal sodium delivery, urine flow, aldosterone).

### Causes

| Mechanism | Examples |
|---|---|
| Pseudohyperkalaemia | Haemolysis, tourniquet, fist clenching, thrombocytosis |
| Reduced excretion | AKI (acute kidney injury), CKD (chronic kidney disease), **obstructive uropathy** (type 4 RTA [renal tubular acidosis]), adrenal insufficiency; drugs: ACE (angiotensin-converting enzyme) inhibitors, ARBs (angiotensin receptor blockers), spironolactone, NSAIDs (non-steroidal anti-inflammatory drugs), tacrolimus |
| Shift out of cells | Acidosis, insulin deficiency, β-blockers, suxamethonium |
| Cell release | **Rhabdomyolysis** (prolonged lithotomy or flank position), tumour lysis, crush |
| Urological | **Jejunal conduit** (reabsorbs urinary potassium) |

### Assessment

- Repeat the sample to **exclude pseudohyperkalaemia** if the ECG (electrocardiogram) is normal.
- **ECG changes:** peaked T (earliest) → prolonged PR, wide QRS → loss of P → sine wave → ventricular fibrillation (VF). Ascending weakness.
- Creatinine, glucose, blood gas, CK (creatine kinase); stop potassium sources.

### Emergency treatment (in order)

| Step | Drug and dose | Onset and effect |
|---|---|---|
| **1. Protect the heart** | **Calcium gluconate 10%, 10 mL IV (intravenous)** over 5–10 min; repeat if ECG unchanged | 1–3 min; lasts 30–60 min; does not lower potassium |
| **2. Shift into cells** | **Insulin 10 units regular IV with 50 mL of 50% dextrose** | 15–30 min; lowers about 0.6–1 mmol/L; check (monitor) glucose (hypoglycaemia) |
| | **Salbutamol 20 mg nebulised** | Additive to insulin |
| | Sodium bicarbonate | Only with metabolic acidosis |
| **3. Remove** | Loop diuretic if passing urine; **binders**: sodium zirconium cyclosilicate (about 1 h), patiromer (about 7 h), sodium polystyrene sulfonate (slow; colonic necrosis) | Hours |
| | **Haemodialysis** | Refractory or oliguric, ESKD (end-stage kidney disease) |

- **Relieve obstruction** (catheter, nephrostomy); watch for post-obstructive diuresis.
- Monitor ECG, potassium and glucose; treat the cause.

**Diagram to draw:** flowchart: ECG change or $K^+ \geq 6.5$ → calcium → shift → remove.

**Key points to remember:**

- Calcium stabilises the membrane but does not lower potassium.
- Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
- Binders, diuretics and dialysis remove potassium; relieve obstruction.

## SAQ 13 Hypokalemia

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 1, Q3f

### Definition and physiology

- **Hypokalaemia:** serum potassium **<3.5 mEq/L**; severe <2.5 mEq/L.
- About **98% of body potassium is intracellular**; the ratio sets the resting membrane potential.
- Two defences: **shift into cells** (insulin, β2-adrenergic stimulation, alkalosis) and **renal excretion** (distal nephron). Distal secretion rises with high urine flow, distal sodium delivery, **aldosterone** and alkalosis.

### Causes

| Mechanism | Causes |
|---|---|
| Low intake | Malnutrition |
| **Shift into cells** | Insulin, catecholamines (stress, β-agonists), alkalosis, refeeding, periodic paralysis |
| **GI (gastrointestinal) loss** | Diarrhoea, villous adenoma, **ureterosigmoidostomy and colonic diversions** (colon secretes potassium, ileum absorbs it) |
| **Renal loss, high BP (blood pressure)** | **Primary hyperaldosteronism (Conn)**, renin-secreting juxtaglomerular tumour, Cushing, Liddle, **abiraterone** (mineralocorticoid excess) |
| **Renal loss, normal BP** | Loop and thiazide diuretics, **postobstructive diuresis**, osmotic diuresis, **RTA (renal tubular acidosis) types 1 and 2**, Bartter, Gitelman, hypomagnesaemia, amphotericin, cisplatin |

- **Vomiting:** potassium is lost mainly in urine (alkalosis → kaliuresis), not in gastric juice.

### Clinical features

- Muscle weakness, cramps, ileus, **flaccid paralysis**, rhabdomyolysis.
- Arrhythmias; potentiates digoxin toxicity.
- Polyuria (nephrogenic diabetes insipidus); **hypocitraturia** → calcium stones.
- **ECG (electrocardiogram):** flat T wave, **prominent U wave**, ST depression, prolonged QU; ventricular arrhythmias.

### Evaluation

- Repeat potassium; magnesium, bicarbonate, glucose.
- **24-h urine potassium:** <20 mmol/day suggests extrarenal loss; higher suggests renal loss.

- Renal loss: check blood pressure and acid–base; **aldosterone–renin ratio** if hypertensive (only a minority of Conn patients are hypokalaemic).

### Management

- Treat the cause; stop offending drugs.
- **Potassium chloride** preferred (potassium citrate if acidotic, phosphate if hypophosphataemic). **Oral** 20–40 mEq repeated when possible.
- **Intravenous** for severe or symptomatic cases: usually ≤10 mEq/h peripherally; faster only centrally with ECG monitoring. Never as a bolus.
- **Correct magnesium**, or potassium will not correct.
- **Urinary diversion:** replace potassium **before or with bicarbonate**; correcting acidosis alone drives potassium into cells → paralysis.
- Correct before abiraterone and adrenalectomy for Conn syndrome (spironolactone preoperatively).

**Diagram to draw:** ECG strip showing flat T and U wave.

> **Key points to remember:**
> - Shift or loss; urine potassium separates renal from extrarenal.
> - ECG: flat T, U wave.
> - Diversion acidosis: give potassium with or before bicarbonate.
> - Correct magnesium first.

## SAQ 14 Lymphocele and postoperative lymphatic leak (chylous ascites)

P1

**Asked as:**

- Lymphocele — *Practice Set 8 Paper I Q3a (SAQ)*
- Postoperative lymphatic leak — *Practice Set 14 Paper I Q3b (SAQ)*

**Type:** SAQ (10 marks)

*Postoperative lymphatic leak taken as lymphocele and chylous ascites after RPLND (retroperitoneal lymph node dissection), nephrectomy, pelvic node dissection and transplantation.*

### Definition and types

- **Lymphocele:** a collection of lymph **without an epithelial lining** (a pseudocyst with a fibrous wall), caused by divided lymphatics that keep leaking. It is the **commonest complication of PLND (pelvic lymph node dissection)**.
- **Chylous ascites:** lipid-rich lymph in the peritoneum from the cisterna chyli or major trunks.

### Settings and incidence

- **PLND** (radical prostatectomy, cystectomy): radiological in up to 60%, clinically significant 2–5%. An extended template roughly doubles the rate.
- **Renal transplantation:** from lymphatics around the recipient iliac vessels or donor hilum. Radiological 0.6–33.9% (symptomatic 0.03–26%); most appear within 6 months, peaking at 6 weeks.

- Also after RPLND (retroperitoneal lymph node dissection; lymphocele ≤1.7%), inguinal dissection, nephrectomy.
- **Chylous ascites:** RPLND: 0.2–2.1% primary, **2–12% post-chemotherapy**; left nephrectomy with hilar dissection.

| Type | Definition | Setting and frequency |
|---|---|---|
| **Lymphocele** | Lymph collection **without epithelial lining** | **PLND (pelvic lymph node dissection):** radiological up to 60%, significant 2–5%. **Transplant:** symptomatic 0.03–26%, peak 6 weeks. **RPLND:** ≤1.7% |
| **Chylous ascites** | Lipid-rich lymph in the peritoneum from the cisterna chyli or major trunks | RPLND: 0.2–2.1% primary, **2–12% post-chemotherapy**; left nephrectomy with hilar dissection |

- **Risk factors:** extensive dissection, extended templates, suprahilar dissection near the cisterna chyli (L1–L2), post-chemotherapy surgery, obesity, heparin prophylaxis, mTOR (mammalian target of rapamycin) inhibitors, acute rejection, retransplantation (re-transplant).

## Clinical features

- **Lymphocele:** most are silent (usually silent). Large ones compress:
- iliac vein: leg oedema, **DVT (deep vein thrombosis)**;
- bladder: frequency, urgency, pelvic pressure;
- ureter or graft: hydronephrosis, rising creatinine (graft dysfunction).
- Infection (may become infected): fever, rigors, sepsis.
- **Chylous ascites:** rising girth, early satiety, dyspnoea; milky drain fluid.

## Diagnosis

- **Ultrasound:** anechoic, often septated, thin-walled collection. **CT (computed tomography)** maps it before intervention. Air or rim enhancement means infection.
- **Aspirate:** lymph has creatinine equal to (equals) serum (urinoma: much higher), lymphocytes; culture excludes abscess. Chyle: **triglyceride 2–8 × serum**, protein >3 g/dL.

## Management

**Lymphocele** (treat by symptoms, not size alone)

| Situation | Treatment |
|---|---|
| Asymptomatic, <5 cm | Observe; most resolve |
| Symptomatic | **Percutaneous drain** ± sclerotherapy (povidone-iodine, doxycycline) |
| Recurrent or thick-walled | **Laparoscopic fenestration (marsupialisation)** into a peritoneal window (best) |
| Infected | Antibiotics + percutaneous drainage; exclude urine leak |
| Laparoscopy unsafe | Open drainage |

```
Asymptomatic, <5 cm → observe Symptomatic → percutaneous drain (± sclerotherapy) Recurrent / thick-walled → laparoscopic peritoneal fenestration (best) Infected → drainage + antibiotics; exclude urine leak
```

- **Success:** laparoscopic fenestration best, open next, drain or sclerotherapy intermediate, **aspiration alone worst** (reaccumulates, recurs, risks infection).

- Sclerosants (povidone-iodine, doxycycline) may hinder later fenestration.
- Fenestration lets the peritoneum absorb lymph (internal drainage): the definitive cure.
- **Lymphangiography with embolisation:** high success; long-term data awaited.

**Chylous ascites** (stepwise)

1. Paracentesis; drain if it reaccumulates.
2. **Very low-fat, high-protein diet with MCT (medium-chain triglycerides)**, which are absorbed directly into portal blood, bypassing lacteals; plus **octreotide**.
3. Nil by mouth with **TPN (total parenteral nutrition)**.
4. Refractory: lymphangiography with embolisation, peritoneovenous shunt or surgical ligation. - Continue a low-fat diet for **≥4 weeks** after resolution.

### Prevention

- **Clip or ligate visible lymphatics** (clips often preferred; no robust data) (cephalad at the renal hilum in RPLND; perivascular at transplant). Spare lymphatics over the external iliac artery (leg lymph). Avoid unneeded suprahilar dissection.
- **Peritoneal interposition flap** fixed to the bladder (peritoneal flap after robotic PLND) reduces clinically significant lymphocele. Transperitoneal access is **not** protective (fluid loculates within 48–72 h).
- Drain after transplantation and major nodal dissections.

**Diagram to draw:** pelvic lymphocele compressing the external iliac vein and bladder, with a laparoscopic peritoneal window.

> **Key points to remember:**
> - Lymphocele: commonest PLND complication; radiological up to 60%, significant 2–5%.
> - Lymph creatinine = serum; urinoma creatinine higher.
> - Laparoscopic fenestration best (is best); aspiration alone worst.
> - Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
> - Ladder: diet with MCT and octreotide → TPN → embolisation or shunt.

## SAQ 15 Hyperfiltration injury in AKI and CKD

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 1, Q3c

### Definition

- **Hyperfiltration:** a rise in **SNGFR (single-nephron glomerular filtration rate)** in surviving nephrons when nephron mass falls; whole-kidney GFR (glomerular filtration rate) may look normal.
- **Brenner hypothesis (1982):** the adaptation itself damages glomeruli.

### Physiology

- Net filtration pressure ≈ 10 mmHg (hydrostatic 60 − Bowman's 18 − oncotic 32).
- Afferent dilatation raises flow and filtration; **efferent constriction (angiotensin II) raises filtration pressure**.

### Mechanism

`Nephron loss → afferent dilatation + efferent constriction → raised glomerular capillary pressure and flow (hyperfiltration) → glomerular hypertrophy; podocytes cannot divide → stretch, detachment → proteinuria → FSGS (focal segmental glomerulosclerosis) → tubulointerstitial inflammation, fibrosis → more nephron loss (vicious cycle)`

### In AKI (acute kidney injury)

- Incomplete repair leaves fewer nephrons, which hyperfilter; creatinine may normalise while **renal reserve is lost**.
- This drives the **AKI-to-CKD (chronic kidney disease) transition**; AKI persisting ≥90 days becomes CKD.
- Each perioperative AKI episode is a **modifiable risk factor** for CKD progression.

### In CKD

- Final common pathway of progression; marked by **albuminuria (KDIGO (Kidney Disease: Improving Global Outcomes) A stages)**, hypertension, falling eGFR (estimated GFR).

### Urological settings

| Setting | Point |
|---|---|
| Nephrectomy or donation | Compensatory hypertrophy 20–30%; donor function 60–80% of pre-donation |
| Partial nephrectomy, solitary kidney | >50% parenchymal loss → proteinuria, FSGS; marked when total nephron mass falls >80%; ESRD (end-stage renal disease) if <20–30% of one kidney remains |
| Congenital solitary kidney, MCDK (multicystic dysplastic kidney) | Monitor BP (blood pressure), proteinuria, yearly ultrasound |

- **Also know:** reflux nephropathy, infundibulopelvic stenosis, diabetes, obesity, pregnancy, low birth weight (low nephron endowment).

### Prevention and treatment

- Nephron sparing; avoid NSAIDs (non-steroidal anti-inflammatory drugs) and nephrotoxins.
- **BP <130/80 mmHg**; **ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)** in proteinuric disease: efferent dilatation lowers glomerular pressure.
- **SGLT2 (sodium–glucose cotransporter-2) inhibitors:** restore tubuloglomerular feedback, causing afferent constriction. **DAPA-CKD** and **EMPA-KIDNEY** slowed progression with or without diabetes.
- Finerenone (type 2 diabetic CKD); dietary protein restriction.

**Diagram to draw:** glomerulus with dilated afferent and constricted efferent arterioles.

> **Key points to remember:**
> - Hyperfiltration = raised SNGFR; Brenner 1982.
> - Leads to FSGS, proteinuria and the AKI-to-CKD transition.
> - ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

## SAQ 16 UW solution and organ preservation solutions

P2

**Asked as:**

- UW solution. — *Summer 2017 Paper II Q3c (SAQ)*
- Organ preservation solution — *Practice Set 7 Paper II Q3b (SAQ)*

**Type:** SAQ (10 marks)

*Assumed context: preservation of the kidney for transplantation.*

### Introduction

- **University of Wisconsin solution** (Belzer, late 1980s): the **gold standard** cold-storage solution for kidney, liver and pancreas.

### Principles of cold preservation

- Cooling / **Hypothermia** to **4 °C** cuts (slows) metabolism to roughly a tenth, but does not stop it, so injury accumulates with time.
- Ischaemia causes ATP (adenosine triphosphate) loss (depletion) → **$Na^+/K^+$-ATPase pump failure** → **cell swelling**; anaerobic glycolysis → **acidosis**; calcium influx. On reperfusion, hypoxanthine is converted by xanthine oxidase to **oxygen free radicals** → ischaemia–reperfusion injury, seen clinically as (clinically this appears as) **DGF (delayed graft function)**.
- A solution must therefore prevent swelling and oedema, buffer acid, scavenge radicals and supply ATP precursors. It therefore needs:
- **Impermeants** (lactobionate, raffinose, mannitol, glucose) to prevent cell swelling.
- **Buffers** (phosphate, histidine) against acidosis.
- **Colloid** (hydroxyethyl starch, polyethylene glycol) against interstitial oedema.
- **Antioxidants and energy precursors** (glutathione, allopurinol, adenosine).
- Electrolytes: "intracellular" (high $K^+$, low $Na^+$) or "extracellular" (low $K^+$, high $Na^+$) type.

### UW composition (per litre)

| Component | Amount | Purpose |
|---|---|---|
| **Lactobionate** ($K^+$ salt) | 100 mmol | Impermeant anion; prevents cell swelling; chelates calcium |
| **Raffinose** | 30 mmol | Impermeant trisaccharide |
| **Hydroxyethyl starch** | 50 g | Colloid; prevents interstitial oedema |
| $KH_2PO_4$ | 25 mmol | Buffer |
| $MgSO_4$ | 5 mmol | Membrane stability |
| **Adenosine** | 5 mmol | ATP precursor |
| **Glutathione** | 3 mmol | Antioxidant |
| **Allopurinol** | 1 mmol | Xanthine oxidase inhibitor; fewer free radicals |
| Insulin, dexamethasone, antibiotic | Small | Additives |

| Component | Amount | Purpose |
|---|---|---|
| - **Intracellular-type**: $K^+$ about 125 mmol/L, $Na^+$ about 30 mmol/L; osmolality ~320 mOsm/kg; pH 7.4. | | |

## UW drawbacks

- **Viscous** (starch): slow flush. **High $K^+$**: flush before reperfusion (hyperkalaemia, vasospasm). May crystallise (filter).

## Comparison of solutions

| Feature | UW | HTK (Custodiol) | Euro-Collins |
|---|---|---|---|
| Type | Intracellular (high $K^+$) | Low $K^+$, low $Na^+$ | Intracellular |
| Impermeant | Lactobionate, raffinose | Mannitol | Glucose (enters cells; weaker) |
| Buffer | Phosphate | **Histidine** (strong) | Phosphate |
| Colloid | HES (hydroxyethyl starch) | None (low viscosity) | None |
| Others | Antioxidants, adenosine | Tryptophan, ketoglutarate | — |

| Solution | Type and key components | Points |
|---|---|---|
| **UW (University of Wisconsin, Belzer)** | Intracellular; lactobionate, raffinose, hydroxyethyl starch, glutathione, adenosine, allopurinol | Standard for static storage; viscous; flush out high $K^+$ before reperfusion |
| **HTK (histidine–tryptophan–ketoglutarate, Custodiol)** | Low $K^+$; histidine buffer, tryptophan (membrane stabiliser), ketoglutarate (substrate), mannitol | Low viscosity, fast cooling; large volume; outcomes similar to UW |
| Euro-Collins | Intracellular; glucose impermeant, phosphate | First generation; largely abandoned |
| Celsior | Extracellular; lactobionate, mannitol, glutathione, histidine | Designed for heart; also abdominal organs |
| IGL-1 (Institut Georges Lopez) | UW-like, but polyethylene glycol and reversed $Na^+/K^+$ | Extracellular; less viscous |
| - Kidney outcomes with UW and HTK (histidine–tryptophan–ketoglutarate) are similar; Euro-Collins largely abandoned. **Also know:** Celsior, IGL-1, Marshall's (hypertonic) citrate. | | |

## Practical use and methods

- Living donor, very short cold ischaemia: cold lactated Ringer's suffices; deceased donor or exchange: UW or HTK.
- **Static cold storage:** in situ aortic flush, IVC (inferior vena cava) venting, ice slush; storage at 4 °C. Keep (aim) cold ischaemia short (ideally <24 h / hours): DGF and thrombosis rise with time (beyond).
- **Hypothermic machine perfusion** (Belzer MPS (machine perfusion solution), gluconate-based): pulsatile cold perfusion; flow and resistance help assess viability. Reduces DGF vs static storage; **Moers (2009)**: DGF 20.8% vs 26.5%, 1-year graft survival 94% vs 90%,

without long-term survival gain (the chapter notes no difference in long-term survival). Used especially for DCD (donation after circulatory death) kidneys.
- **Normothermic perfusion:** experimental; allows assessment and repair; limited experience.

**Diagram to draw:** ischaemia(–reperfusion) cascade with the UW / solution component acting at (that blocks) each step.

> **Key points to remember:**
> - Hypothermia slows, not stops, metabolism.
> - Lactobionate and raffinose (impermeants), HES / starch (colloid).
> - Allopurinol, glutathione (antioxidants); adenosine (ATP).
> - High $K^+$: flush before reperfusion.
> - HTK: histidine buffer, tryptophan, ketoglutarate, low viscosity, low $K^+$.
> - Machine perfusion reduces DGF.

## SAQ 17 Pre transplant nephrectomy

P2

**Type:** SAQ (10 marks)
**Source:** 2022-Summer Paper 2, Q3e

*Assumed meaning: removal of the native kidney(s) before or at kidney transplantation.*

### Principle

- **Default: leave native kidneys.** They give residual urine (less fluid restriction), erythropoietin and vitamin D activation; nephrectomy adds morbidity.
- Remove them only when they will threaten the patient or graft under **immunosuppression**.

### Indications

| Group | Indication | Reason |
|---|---|---|
| **Infection** | Chronic pyelonephritis, infected stones, recurrent upper UTI (urinary tract infection) with **high-grade VUR (vesicoureteral reflux)** or obstruction, XGP (xanthogranulomatous pyelonephritis) | Sepsis under immunosuppression |
| **ADPKD (autosomal dominant polycystic kidney disease)** | No room for graft (kidney below iliac crest), recurrent cyst infection or bleeding, pain, early satiety | Space and symptoms |
| **Malignancy** | Suspicious mass or complex cyst (acquired cystic disease), high tumour risk (e.g. Denys–Drash) | Most RCC (renal cell carcinoma) in recipients arises in native kidneys |
| **Hypertension** | Uncontrolled despite multiple drugs | Renin-driven |
| **Heavy proteinuria** | Nephrotic syndrome with malnutrition or thrombosis (e.g. congenital nephrotic syndrome) | Hides graft proteinuria, hypercoagulability |
| Others | Massive polyuria, large stone burden | Graft fluid balance, infection |

### Timing

- **Before transplantation:** infection or tumour; reconstruction 6–12 weeks earlier.
- **Simultaneous:** ADPKD needing space, ipsilateral through the transplant incision; avoided in small children.
- **After transplantation:** symptomatic ADPKD once the graft works.

### Technique

- **Laparoscopic** (transperitoneal or retroperitoneal), open, or **embolisation** in high-risk patients.
- **PD (peritoneal dialysis) patients:** retroperitoneal route keeps PD going; transperitoneal surgery needs temporary haemodialysis.
- Bilateral: prone approach for small kidneys; morcellation or open for large ADPKD kidneys.
- **Ureter:** transect high and leave a normal ureter (for later ureteroureterostomy); **remove a dilated refluxing ureter completely**; keep it as a possible continent stoma if catheterisation may be needed.

### Complications

- Bleeding, transfusion (sensitisation), infection, bowel injury, hypotension after bilateral removal, loss of residual function.

**Also know:** pre-transplant urodynamics; bladder augmentation if capacity <75% expected or pressure >40 $cmH_2O$.

**Diagram to draw:** enlarged ADPKD kidney reaching the iliac fossa beside the planned graft site.

> **Key points to remember:**
> - Default: leave native kidneys.
> - Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
> - Retroperitoneal route for PD; leave a normal ureter.

## SAQ 18 Post transplant malignancy

P2

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 2, Q3c

### Introduction

- Cancer risk after kidney transplantation is about 2–3 times the general population; a leading cause of late death with a functioning graft.

### Causes

- **Lost immune surveillance** from immunosuppression (dose and duration matter; depleting antibodies add risk).
- **Oncogenic viruses:** EBV (Epstein–Barr virus) → PTLD (post-transplant lymphoproliferative disorder); HHV-8 (human herpesvirus-8) → Kaposi sarcoma; HPV (human papillomavirus) → skin and anogenital cancer; hepatitis B/C → liver cancer.
- **Direct drug effects:** azathioprine sensitises skin to UV (ultraviolet) light; calcineurin inhibitors promote tumour growth. mTOR (mammalian target of rapamycin) inhibitors are

**antitumour**.

- **Pre-existing risk:** ESRD (end-stage renal disease), acquired cystic disease, analgesic nephropathy; rarely donor-transmitted.

### Common cancers

| Cancer | Key facts |
|---|---|
| **Non-melanoma skin** | Commonest; squamous exceeds basal cell (reverse of general population); multiple, aggressive |
| **PTLD** | Mostly B-cell, EBV-driven; highest in first year, children and **EBV-negative recipient of EBV-positive donor**; may present as graft mass |
| **Kaposi sarcoma** | HHV-8; regresses with less immunosuppression and switch to sirolimus |
| **RCC (renal cell carcinoma)** | 4.6% of post-transplant cancers; **90% in native kidneys**; graft tumours mostly papillary, low grade |
| **Bladder cancer** | About **3-fold** risk; more aggressive |
| Prostate | Same incidence as general population |

### Management

- **Reduce immunosuppression** and switch to an mTOR inhibitor; sirolimus conversion reduced new skin squamous cancers (Euvrard, 2012).
- **PTLD:** reduce immunosuppression, rituximab, then chemotherapy (R-CHOP).
- RCC, prostate and bladder: treat as in the general population; protect graft vessels and ureter at surgery. BCG (bacillus Calmette–Guérin) is usable in selected bladder cancer.

### Prevention and screening

- Pre-transplant cancer screening and waiting periods (none after curative treatment of small RCC <3 cm, Gleason 6 prostate cancer, superficial bladder cancer).
- Sun protection, yearly skin examination, age-appropriate screening.
- KDIGO (Kidney Disease: Improving Global Outcomes) does **not** endorse routine native-kidney screening; screen only high-risk recipients.

**Diagram to draw:** pie chart of post-transplant cancers (skin largest).

> **Key points to remember:**
> - Skin cancer commonest; squamous more than basal cell.
> - PTLD: EBV-negative recipient of EBV-positive donor, first year.
> - RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
> - Reduce immunosuppression and switch to mTOR inhibitor.

## SAQ 19 Anomalies of renal artery.

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 1 Paper 2, Q3c

### Normal pattern

- One renal artery from the aorta at L1–L2; anterior and posterior divisions; **five segmental end arteries** (apical, upper, middle, lower, posterior). The right artery passes behind the

IVC (inferior vena cava).

## Terminology

| Term | Definition |
|---|---|
| **Multiple renal arteries** (preferred) | More than one vessel from the aorta to one kidney |
| **Aberrant** | Arises from a vessel other than aorta or main renal artery (iliac, SMA (superior mesenteric artery), gonadal) |
| **Accessory** | Two or more branches supplying the same segment |

## Incidence

- Single artery in **71–85%**; variants in 25–40%. Accessory arteries 25–28% (aorta T11–L4), usually to the **lower pole**. True aberrant vessels are rare except in **ectopic and horseshoe kidneys**.
- **Also know:** early prehilar branching; right artery anterior to IVC in about 5%.

## Embryology

- The ascending kidney takes successive aortic branches; the lower ones normally degenerate. **Failure of degeneration** gives polar or multiple arteries.

## Clinical significance

- **PUJ (pelviureteric junction) obstruction** from a lower-pole crossing vessel; but crossing vessels occur in a fifth to a quarter of normal PUJs.
- Compression of an infundibulum: hydrocalyx, pain, stones, UTI (urinary tract infection).
- **Surgery:** bleeding at endopyelotomy (incise posterolaterally); polar arteries are end arteries, so ligation infarcts that segment; **donor nephrectomy** needs CT (computed tomography) angiography; aortic surgery.
- Not linked to hypertension.

## Renal artery aneurysm

- About **0.1%**. **Abeshouse types:** saccular (about 93%), fusiform, dissecting, arteriovenous.
- Mostly asymptomatic; hypertension, haematuria, pain; wreath-like calcification.
- **Treat** (endovascular or open) if rupture, **>2 cm**, expanding, **pregnancy or planned pregnancy**, symptoms or emboli.

## Renal arteriovenous fistula

| | Congenital (cirsoid AVM (arteriovenous malformation)) | Acquired |
|---|---|---|
| Share | <25% | Majority (biopsy, PCNL (percutaneous nephrolithotomy), trauma) |
| Features | Bruit, haematuria, high-output failure | Often resolves |
| Treatment | Selective embolisation | Observe; embolise if bleeding |

**Diagram to draw:** kidney with main artery, lower-pole accessory artery crossing the PUJ.

**Key points to remember:**

- Single artery in 71–85%; accessory arteries usually lower pole.
- Crossing vessel can cause PUJ obstruction.
- Treat aneurysm >2 cm or in pregnancy.

## SAQ 20 Mycophenolate Mofetil.

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 2, Q3b (immunosuppressants also answered in 2021-Summer Paper 1, Q3f and 2020-Summer Paper 4, Q1)

### Introduction

- **MMF (mycophenolate mofetil):** an antimetabolite immunosuppressant; morpholinoethyl ester **prodrug** of MPA (mycophenolic acid). With tacrolimus it forms maintenance therapy in **over 90%** of kidney recipients.

### Mechanism

- MPA is a reversible, non-competitive inhibitor of **IMPDH (inosine monophosphate dehydrogenase)**, the key enzyme of **de novo guanosine nucleotide synthesis**.
- **Why lymphocyte-selective:** T and B lymphocytes depend on the de novo pathway; other cells use the salvage pathway. MPA also inhibits the type II isoform found in activated lymphocytes.
- Result: arrest of T- and B-cell proliferation and reduced antibody formation.

### Pharmacology

| Feature | Detail |
|---|---|
| Dose (kidney) | **1 g twice daily** oral or IV; EC-MPS (enteric-coated mycophenolate sodium) 720 mg twice daily is equivalent |
| Metabolism | Hydrolysed to MPA; liver glucuronidation to inactive MPAG (MPA glucuronide); **enterohepatic recirculation** |
| Excretion | MPAG in urine; accumulates in renal failure |
| Interactions | **Ciclosporin lowers MPA** (blocks enterohepatic recirculation), tacrolimus does not; antacids, cholestyramine reduce absorption; never with azathioprine |

### Uses

- Kidney transplant maintenance (tacrolimus + MMF ± steroids); other solid organs.
- Lupus nephritis, vasculitis; steroid-refractory immune-related adverse events of checkpoint inhibitors.

### Adverse effects

- **GI (gastrointestinal):** diarrhoea, nausea, colitis (commonest; dose split or EC-MPS).
- **Marrow:** leucopenia, anaemia, thrombocytopenia: blood count weekly in month 1.
- **Infection:** CMV, BK virus (ureteric stricture), rarely PML (progressive multifocal leucoencephalopathy).
- **Malignancy:** skin cancer, PTLD (post-transplant lymphoproliferative disorder).

- **Teratogenic:** miscarriage, ear and facial malformations; contraception mandatory; switch to azathioprine before conception.
- Advantage: **not nephrotoxic**; no hypertension, diabetes or dyslipidaemia.

### Evidence

- Three mid-1990s randomised trials (US, European, Tricontinental) showed fewer early acute rejections than azathioprine or placebo, so MMF replaced azathioprine.
- **ELITE-Symphony (2007):** low-dose tacrolimus + MMF + daclizumab + steroids gave the lowest rejection and best GFR.

**Diagram to draw:** purine synthesis pathway showing IMPDH block (de novo) and the salvage pathway spared.

> **Key points to remember:**
> - Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
> - 1 g twice daily; ciclosporin lowers MPA levels.
> - Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

## SAQ 21 Metabolic abnormalities in CKD

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 2, Q3e

### Definition

- **CKD (chronic kidney disease), KDIGO (Kidney Disease: Improving Global Outcomes):** abnormal kidney structure or function for **≥3 months**. Staged by GFR (glomerular filtration rate) G1–G5 (G3a 45–59, G3b 30–44, G4 15–29, G5 <15 mL/min/1.73 m²) and albuminuria A1–A3.
- Each abnormality is a lost kidney function: excretion, acid–base balance, hormone production.

### Abnormalities

| Abnormality | Mechanism | Management |
|---|---|---|
| **Sodium and water** | Fixed urine concentration; sodium retention → overload, hypertension | Salt restriction; diuretics (chlorthalidone works even in advanced CKD) |
| **Hyperkalaemia** | Less distal sodium delivery; RAAS (renin–angiotensin–aldosterone system) blockers lower aldosterone | Potassium binders and diuretics, so RAAS blockers can continue |
| **Metabolic acidosis** | Less ammonium excretion; retained sulphate and phosphate (anion gap rises late); bicarbonate loss | Sodium bicarbonate or citrate to keep bicarbonate **≥22 mEq/L**; slows progression, protects bone and muscle |
| **CKD-MBD** (mineral and bone disorder) | Phosphate retention → FGF-23 (fibroblast growth factor 23) rises → less **calcitriol** → hypocalcaemia → **secondary hyperparathyroidism** | Phosphate restriction, binders with meals, active vitamin D; calcimimetic; parathyroidectomy if tertiary |

| Abnormality | Mechanism | Management |
|---|---|---|
| **Anaemia** | Erythropoietin falls, mainly below GFR 30 | Iron first, then erythropoietin; Hb (haemoglobin) target 10–11.5 g/dL (harm above 13) |
| **Hyperuricaemia** | Reduced excretion | Treat gout; asymptomatic lowering unproven (CKD-FIX) |
| **Lipids** | High triglycerides, low HDL (high-density lipoprotein) | Statin ± ezetimibe if ≥50 years, not on dialysis (SHARP) |
| **Glucose** | Insulin resistance, but slower insulin clearance → hypoglycaemia | Lower insulin doses |

### Bone disease (renal osteodystrophy)

- **High turnover:** osteitis fibrosa (PTH [parathyroid hormone] excess).
- **Low turnover:** adynamic bone (oversuppressed PTH), osteomalacia.
- **Tertiary hyperparathyroidism:** autonomous gland, hypercalcaemia; persists after transplant.
- **Vascular calcification** adds cardiovascular death.

### Other endocrine and uraemic effects

- **Also know:** hypogonadism, hyperprolactinaemia, erectile dysfunction (better after transplant); growth failure in children; protein-energy wasting; dialysis amyloid; uraemic platelet dysfunction (desmopressin).
- **Urological relevance:** dose drugs by GFR; avoid gadolinium if eGFR (estimated GFR) <30; each AKI (acute kidney injury) speeds progression.

**Diagram to draw:** CKD-MBD cascade: phosphate ↑ → FGF-23 ↑ → calcitriol ↓ → calcium ↓ → PTH ↑ → bone disease.

> **Key points to remember:**
> - FGF-23 rises first; secondary then tertiary hyperparathyroidism.
> - Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
> - Hyperkalaemia: binders rather than stopping RAAS blockade.

## SAQ 22 Renal transplant in neurogenic bladder

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 2, Q3a

### Why the bladder matters

- A neurogenic bladder with **poor compliance, low capacity or detrusor overactivity** transmits high pressure to the graft even without reflux. Poor emptying adds infection. Recurrent pyelonephritis, not augmentation, causes graft loss.
- **Five goals:** unimpeded drainage, low-pressure capacious storage, effective emptying, no infection, fewest procedures.

### Pretransplant evaluation

- Renal ultrasound, VCUG (voiding cystourethrogram) and **urodynamics** (capacity, compliance, overactivity, emptying, sphincter). Repeat them; the bladder changes over time.
- **Target:** storage to expected capacity at detrusor pressure **<40 cm $H_2O$**.
- **Defunctionalised bladder** (anuria or diversion): **bladder cycling** (saline fills of increasing volume via catheter), then urodynamics.

### Pretransplant optimisation (stepwise)

`Antimuscarinic (oxybutynin), mirabegron, botulinum toxin → CIC (clean intermittent catheterisation) ± overnight drainage → bladder cycling if defunctionalised → still >40 cm H₂O, <75% expected capacity or CIC ≤3-hourly → augmentation`

- **Augmentation:** ileum or sigmoid most used; dilated ureter (ureterocystoplasty) preferred when available (no mucus or metabolic effects). **Stomach avoided** (haematuria–dysuria syndrome in anuric patients). Augmentation does not clearly raise transplant complications.
- **Mitrofanoff** channel if urethral CIC is difficult; site it **medial** to the planned transplant incision.
- Do reconstruction **before** transplant; allow **6–12 weeks** to heal. Neuropathic bladders may deteriorate under high graft output, so earlier reconstruction is justified.
- **Native nephrectomy** only for named reasons: recurrent upper-tract infection, massive reflux, stones, uncontrolled hypertension.

### At transplant

- Extravesical non-refluxing ureteroneocystostomy into bladder or augment (know the augment pedicle). Ureteroureterostomy to a native ureter is an option.
- Ileal conduit drainage if the bladder is unusable.

### Post-transplant care

- Continue CIC; colonisation is expected, so treat only symptomatic UTI (urinary tract infection).
- **Creatinine rise with hydronephrosis:** catheter drainage for 1–2 weeks first. If it improves, the bladder is the cause; otherwise stent or nephrostomy and consider rejection.
- Febrile UTI with graft reflux: reimplant after optimising the bladder.
- Lifelong surveillance of augments (malignancy, stones).

**Diagram to draw:** stepwise bladder optimisation ladder.

> **Key points to remember:**
> - Target <40 cm $H_2O$ storage before transplant.
> - Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
> - Pyelonephritis, not augmentation, loses grafts.

## SAQ 23 Pre-emptive renal transplant

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 2, Q3a

### Definition

- Kidney transplantation **before starting maintenance dialysis** in a patient with advanced CKD (chronic kidney disease).

### Rationale

- Every year on dialysis adds cardiovascular disease, vascular calcification and bone–mineral damage that the graft later inherits. Pre-emptive transplantation **avoids this exposure**.
- Transplantation beats dialysis: Wolfe (1999) showed early risk is higher, equal at 106 days, then 68% lower long term.

### Advantages

- Better patient and graft survival than transplantation after dialysis; longer dialysis time worsens outcome (registry data).
- No dialysis access surgery, access infection or catheter complications.
- Fewer transfusions, so less HLA (human leucocyte antigen) sensitisation.
- Better quality of life, employment; lower cost.
- Children: better growth and development.

### Frequency

- 28.7% of US transplants in 2021 (living donor 51%, deceased donor 21%, children 56%). Commonest with a **living donor**.

### Timing and evaluation

- Evaluate early, ideally before dialysis. KDIGO (Kidney Disease: Improving Global Outcomes) 2020: refer when eGFR (estimated glomerular filtration rate) is <30 mL/min/1.73 m², 6–12 months before expected dialysis. Living-donor transplant is usually done at eGFR <20 with progressive decline.
- Deceased-donor listing (US) at eGFR ≤20.
- Full recipient work-up: cardiac, vascular (iliac calcification), infection (TB [tuberculosis], hepatitis B and C, HIV [human immunodeficiency virus]), malignancy, psychosocial; recurrence risk (primary FSGS [focal segmental glomerulosclerosis]).
- **Urological:** bladder function and lower tract (valves, neurogenic bladder); native nephrectomy only for specific indications.
- India: mainly living related donors under THOA (Transplantation of Human Organs Act).

### Limitations

- Needs early referral and a ready donor.
- Transplant may come earlier than needed, using up dialysis-free native function; adherence untested.
- In advanced CKD, bowel augmentation may speed decline.

**Also know:** ABO-incompatible or paired exchange programmes widen pre-emptive options.

**Diagram to draw:** Wolfe curve: relative risk of death after transplant vs dialysis over time.

> **Key points to remember:**
> - Transplant before dialysis = best outcome.
> - Refer at eGFR <30; transplant usually at eGFR <20.
> - Commonest with living donors and children.

## SAQ 24 Tacrolimus.

P3

**Type:** SAQ (10 marks)
**Source:** 2017-Winter Paper 3, Q3c

### Introduction

- **Tacrolimus (FK506)**: macrolide CNI (calcineurin inhibitor) from *Streptomyces tsukubaensis*; about 10–100 times more potent than ciclosporin. With mycophenolate it is maintenance in **over 90%** of kidney recipients.

### Mechanism

`Tacrolimus + FKBP12 (FK506-binding protein 12) → complex inhibits calcineurin (Ca²⁺/calmodulin phosphatase) → NFAT (nuclear factor of activated T cells) stays phosphorylated, cannot enter nucleus → no IL-2 (interleukin-2) transcription → T-cell activation blocked (signal 1)`

- Ciclosporin binds **cyclophilin** instead; same end result.

### Pharmacokinetics

- Poor, variable oral bioavailability; reduced by food. Binds red cells, so **whole-blood trough (C0)** levels are monitored (narrow therapeutic index).
- Metabolised by **CYP3A4/3A5**; effluxed by **P-gp (P-glycoprotein)**. CYP3A5 expressers need higher doses.

### Formulations

- Immediate-release (twice daily); prolonged-release (Advagraf, once daily); LCP-tacrolimus (Envarsus, once daily, higher bioavailability); intravenous; topical. Once-daily forms aid adherence.

### Dosing and monitoring

- Started about 0.1–0.2 mg/kg/day in two doses, then adjusted to trough. Targets vary by centre and risk; **ELITE-Symphony low-dose arm: 3–7 ng/mL**.

### Toxicity

| System | Effect |
|---|---|
| Kidney | **Nephrotoxicity**: acute afferent vasoconstriction (reversible); chronic arteriolar hyalinosis, striped fibrosis; TMA (thrombotic microangiopathy) |
| Metabolic | **NODAT/PTDM (new-onset/post-transplant diabetes)**, more than ciclosporin; **hyperkalaemia**, **hypomagnesaemia**, hyperuricaemia |
| Neurological | **Tremor**, headache, paraesthesia, seizures, PRES (posterior reversible encephalopathy syndrome) |

| System | Effect |
|---|---|
| Others | Hypertension, GI (gastrointestinal) upset, alopecia, infection (CMV (cytomegalovirus), BK polyomavirus), malignancy (skin, PTLD (post-transplant lymphoproliferative disorder)) |

### Drug interactions

- **Raise levels (CYP3A4 inhibitors):** azoles, clarithromycin/erythromycin, diltiazem, verapamil, protease inhibitors, grapefruit; diarrhoea.
- **Lower levels (inducers):** rifampicin, phenytoin, carbamazepine, St John's wort.
- Additive nephrotoxicity: NSAIDs (non-steroidal anti-inflammatory drugs), aminoglycosides; ACE (angiotensin-converting enzyme) inhibitors add to hyperkalaemia.

### Versus ciclosporin

| | Tacrolimus | Ciclosporin |
|---|---|---|
| Acute rejection | Lower | Higher |
| Diabetes, tremor, alopecia | More | Less |
| Hirsutism, gum hypertrophy, hyperlipidaemia, hypertension | Less | More |
| - **ELITE-Symphony (2007):** low-dose tacrolimus + mycophenolate + daclizumab + steroids gave the **lowest rejection, best GFR (glomerular filtration rate) and graft survival** at 1 year versus ciclosporin or sirolimus regimens. | | |
| - **Also know:** CNI minimisation with belatacept or everolimus; topical tacrolimus in genital dermatoses (lichen planus, Zoon balanitis). | | |

**Diagram to draw:** T-cell showing FKBP12–calcineurin–NFAT–IL-2 pathway.

> **Key points to remember:**
> - FKBP12 → calcineurin → NFAT → IL-2 blocked.
> - CYP3A4/P-gp; trough-guided; many interactions.
> - Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.
> - ELITE-Symphony: low-dose tacrolimus best.

## SAQ 25 Urological Complications post Kidney Transplantation.

P3

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 3, Q3d

### Introduction

- Urological complications are the **commonest surgical complications** (8–9.5%). Most arise from the **ureter and ureterovesical anastomosis**.
- The distal donor ureter depends on a single renal-artery blood supply, so ischaemia is the root cause. **Denervation** masks pain.
- A **routine stent** reduces leak and stricture; remove at **14–21 days** (earlier increases leak; later increases UTI [urinary tract infection]).

### Complications

| Complication | Key facts | Management |
|---|---|---|
| **Urine leak** (1.2–8.9%; commonest early) | Distal ureteric necrosis; lost lower-pole artery; technical | Fluid creatinine far above serum. Foley + PCN (percutaneous nephrostomy) + antegrade stent; early reimplant or native ureteroureterostomy if large |
| **Ureteric stricture** (0.5–9%; commonest beyond 3 months) | Ischaemia; long cold ischaemia, rejection, BK (BK polyomavirus) and CMV (cytomegalovirus) | Nephrostomy first; balloon or laser endoureterotomy; **open reimplant, native ureteroureterostomy or Boari flap best long-term** |
| **Lymphocele** (symptomatic 0.03–26%; peak 6 weeks) | Divided lymphatics; mTOR (mammalian target of rapamycin) inhibitors | Treat only if symptomatic; **laparoscopic peritoneal fenestration** best |
| VUR (vesicoureteric reflux) | 50–86% radiological; <1% significant | Treat cause; endoscopic injection low grade; reimplant |
| Haematuria | Anastomotic; persistent after biopsy = AV fistula | Self-limiting; selective embolisation |
| Stones (about 1%) | Hyperparathyroidism; calcineurin inhibitors | As native kidney; watch silent obstruction |
| UTI (commonest infection) | Stent, reflux, steroids | Treat 14–21 days if pyelonephritis |

- **Obstruction:** early is technical, oedema or compression (haematoma, lymphocele); late is stricture, stone or lymphocele.
- **Encrusted stent:** CT, nephrostomy, then PCNL (percutaneous nephrolithotomy), RIRS (retrograde intrarenal surgery) or ESWL (shock-wave lithotripsy).
- **Lower tract:** bladder capacity recovers; BPH (benign prostatic hyperplasia) treated by TURP (transurethral resection of prostate) after 3 weeks, never with a stent in place.
- **Also know:** ED (erectile dysfunction) persists in 20–50%; bladder cancer risk about threefold; RCC (renal cell carcinoma) mostly in native kidneys; renal artery stenosis and thrombosis.

**Diagram to draw:** transplant kidney in iliac fossa showing leak, stricture and lymphocele sites.

> **Key points to remember:**
> - Ureter is the weak point; ischaemia is the cause.
> - Leak early, stricture late; open repair best.
> - Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele.

## SAQ 26 Female kidney donors: special precautions

P4

**Asked as:**

- Female kidney donors: enumerate special precautions — *Winter 2023 Paper IV Q3d (SAQ)*
- Female kidney donors: enumerate special precautions — *Practice Set 4 Paper IV Q3a (earlier version) (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2023-Winter Paper 4, Q3d)

### Introduction

- Most living donors, especially in India, are **women** (wives, mothers). Guidance: **KDIGO (Kidney Disease: Improving Global Outcomes) 2017** living donor guideline.

### Before donation

1. **Free, uncoerced consent:** interview the donor alone; independent donor advocate; family pressure and gender inequity screened. In India the THOA (Transplantation of Human Organs Act) authorisation committee verifies this.
2. **Pregnancy:** pregnancy test before contrast CT (computed tomography) and surgery; **reliable contraception** during work-up. Pregnant women do not donate.
3. **Obstetric history:** previous **pre-eclampsia, gestational hypertension or gestational diabetes** signals higher future hypertension, diabetes and kidney risk; evaluate and counsel.
4. **Future pregnancy counselling (KDIGO):** donation **increases gestational hypertension and pre-eclampsia** (Garg et al., NEJM 2015: about 11% vs 5% in non-donors). Maternal and fetal outcomes are otherwise similar (Chapter 91).
5. **Oestrogen therapy:** combined OCPs (oral contraceptive pills) and HRT (hormone replacement therapy) raise **VTE (venous thromboembolism)** risk; stop them some weeks before surgery and give barrier or progestogen-only contraception; VTE prophylaxis.
6. **Gynaecological and cancer screening:** cervical cytology, breast examination or mammography (age-appropriate); pelvic ultrasound if indicated.
7. **Urine tests:** repeat urinalysis **outside menstruation** to avoid false haematuria; exclude recurrent UTI (urinary tract infection).
8. **GFR (glomerular filtration rate):** low muscle mass makes creatinine-based estimates unreliable; **measured GFR** preferred. Note nephron mismatch with a large recipient.
9. Correct **iron-deficiency anaemia**.

### Operative

- Laparoscopic donor nephrectomy; **Pfannenstiel** extraction for cosmesis; transvaginal extraction possible.

### After donation

- **Defer pregnancy** until recovery and stable function; many programmes advise at least 6–12 months (another source: often 6–12 months).
- In pregnancy: inform obstetrician; **high-risk antenatal care**; monitor BP (blood pressure), proteinuria and creatinine; low-dose aspirin may be considered.
- **Lifelong annual BP, creatinine and proteinuria**; avoid NSAIDs (non-steroidal anti-inflammatory drugs).

**Diagram to draw:** flowchart: consent → pregnancy test → hormone review → GFR → surgery → pregnancy follow-up.

> **Key points to remember:**
>
> - Ensure free consent (Indian gender gap).
> - Pregnancy test; contraception; stop oestrogens.
> - Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
> - Defer pregnancy; high-risk antenatal follow-up.

## SAQ 27 Laparoscopic donor nephrectomy (including pros and cons)

P4

**Asked as:**

- Laparoscopic donor nephrectomy — *Practice Set 2 Paper IV Q3a (SAQ)*
- Pros and cons of lap donor nephrectomy — *Practice Set 2 Paper IV Q3d (SAQ)*

**Type:** SAQ (10 marks)
(living donation also in 2023-Winter Paper 1, Q3c; the pros and cons are asked again in Practice Set 2 Paper 4, Q3d) **See Practice Set 2 Paper 4, Q3a (full answer).**

### Introduction

- **Ratner (1995)**; now the preferred approach (open under 2% in the United States). **Donor safety comes first.**
- Work-up includes measured GFR (glomerular filtration rate) and CT (computed tomography) angiography.

### Side selection

- **Left preferred:** longer renal vein, simpler dissection.
- **Right** if the left has multiple arteries or complex veins, or the right is smaller or abnormal: the donor keeps the better kidney. The right vein is short.

### Techniques

| Technique | Features |
|---|---|
| Pure transperitoneal laparoscopic | 3–4 ports; colon reflected; Pfannenstiel extraction |
| **Hand-assisted** (about two-thirds of US cases) | Hand port is also the extraction site; tactile feedback, quick haemostasis |
| Retroperitoneoscopic | Balloon-made space; no bowel mobilisation; direct hilum; less room |
| Robotic; LESS (laparoendoscopic single-site surgery) | Wristed dissection; single umbilical incision; cost |

### Key steps

1. Reflect colon; divide gonadal, adrenal and lumbar veins (left).
2. Keep periureteric tissue and the gonadal–ureter triangle (ureteric blood supply).
3. Hydration and diuretic before vessel division.
4. Divide ureter, artery, then vein (stapler; see Q3f on Hem-o-lok).
5. Extract quickly; cold flush.

### Warm ischaemia

- Arterial occlusion to cold flush: a few minutes, slightly longer than open, without effect on long-term graft function.

### Pros and cons (versus open)

| Pros | Cons |
|---|---|
| Less pain, shorter stay, faster (earlier) return to work | Longer operation; learning curve |
| Better cosmesis; **more people volunteer to donate** (more willing donors) | Slightly longer warm ischaemia |
| Magnified view of hilum | Shorter vessels, especially right vein (on the right) |
| Equal graft function and survival (graft function equals open) | Pneumoperitoneum lowers renal blood flow; cost |

- **Risks (safety):** complications about 16% (mostly gastrointestinal); mortality **3 per 10,000**; conversion 0.26–3.34%. Lifetime ESRD (end-stage renal disease) 90 per 10,000 versus 14 in matched non-donors.

**Diagram to draw:** left hilum with gonadal, adrenal and lumbar veins and the stapled artery and vein.

> **Key points to remember (including pros and cons angle):**
> - Ratner 1995; left kidney preferred (longer vein).
> - Hand-assisted in about two-thirds of US cases.
> - Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.
> - Cons: longer operation, learning curve, slightly longer warm ischaemia, shorter vessels (especially on the right), cost.
> - Safety: mortality 3 per 10,000; conversion 0.26–3.34%.
> - Never rely on Hem-o-lok alone on the donor renal artery (do not use Hem-o-lok clips alone).

## SAQ 28 Artificial kidney

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 4, Q3e

*Read as devices that replace kidney function: the dialyser (classic "artificial kidney") and newer wearable, implantable and bioartificial kidneys.*

### Definition

- An **artificial kidney** removes solutes and water from blood across a semipermeable membrane. The haemodialysis dialyser is the standard example.
- Pioneered by **Kolff (1943)**; long-term use followed the Scribner shunt (1960) and Brescia–Cimino AV (arteriovenous) fistula (1966).

## Principles

| Process | Mechanism | Removes |
|---|---|---|
| **Diffusion** | Concentration gradient across membrane | Small solutes (urea, potassium) |
| **Convection** (haemofiltration) | Solute dragged with water | Middle molecules |
| **Ultrafiltration** | Hydrostatic pressure gradient | Water |
| Adsorption | Binding to membrane or sorbent | Some proteins, toxins |

- **Hollow-fibre dialyser**; blood and dialysate flow **countercurrent** to keep the gradient.
- **Schedule:** blood flow 250–350 mL/min, 3–5 hours, 3–5 times a week; adequacy by Kt/V (urea clearance × time ÷ volume), at least 1.2 per session.
- **Peritoneal dialysis** uses the peritoneum as membrane (glucose or icodextrin draws water).

## Limitations of current dialysis

- Intermittent: fluid and solute swings cause cramps.
- No endocrine (erythropoietin, vitamin D activation) or tubular metabolic function.
- Access thrombosis, infection, steal; cardiovascular disease, vascular calcification.
- **Transplantation remains the best renal replacement therapy.**

## Newer artificial kidneys

| Device | Principle | Status |
|---|---|---|
| **WAK (wearable artificial kidney)** | Belt-worn; sorbent regenerates a small volume of dialysate | Early human trials |
| Wearable or portable peritoneal dialysis | Continuous sorbent-regenerated peritoneal fluid | Development |
| **Implantable bioartificial kidney** | Silicon nanopore haemofilter plus bioreactor of renal tubule cells; driven by blood pressure, no pump | Preclinical |
| **RAD (renal assist device)** | Human proximal tubule cells in a hollow-fibre cartridge with continuous dialysis | Phase II in acute kidney injury; not adopted |
| Decellularised or organoid kidneys; pig kidney xenotransplant | Regenerative and xenograft routes | Experimental (see Practice Set 6 Paper 4, Q2) |

## Urologist's role

- AV fistula and graft work-up, laparoscopic PD (peritoneal dialysis) catheter placement with omental fixation, transplant.

**Diagram to draw:** hollow-fibre dialyser with countercurrent blood and dialysate flow.

> **Key points to remember:**
> - Diffusion, convection, ultrafiltration across a semipermeable membrane.
> - Dialysis lacks endocrine and tubular function; transplant is best.
> - Wearable, implantable bioartificial and RAD devices are investigational.

## SAQ 29 Robotic kidney transplantation

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 6 Paper 4, Q3f

### Introduction

- **RAKT (robot-assisted kidney transplantation)**: the graft is implanted intracorporeally on the iliac vessels using the robot.
- Milestones: Hoznek (2002, robot-assisted anastomosis); Giulianotti (2010, obese recipient); **Menon and Ahlawat (Vattikuti–Medanta, 2014)** standardised the transperitoneal technique with **regional hypothermia**, assessed by the IDEAL (idea, development, exploration, assessment, long-term study) framework.

### Indications and selection

- Living-donor grafts mainly; increasingly deceased donors.
- **Obese recipients** gain most, as open transplant carries more wound complications.
- **Contraindications (relative):** heavy iliac calcification on non-contrast CT (computed tomography), multiple graft vessels early in experience, previous transplant on that side, dense adhesions.

### Technique (Vattikuti–Medanta)

1. Lithotomy, steep Trendelenburg; transperitoneal ports.
2. Periumbilical **GelPOINT** incision for graft and ice insertion.
3. Peritoneal flap; expose the **external iliac vessels**.
4. Graft wrapped in gauze with ice slush; **ice slush in the iliac fossa** for regional hypothermia.
5. **Vein first:** renal vein end-to-side to external iliac vein, to shorten arterial ischaemia.
6. Renal artery end-to-side to external iliac artery; unclamp and reperfuse.
7. Extravesical **Lich–Gregoir** ureteroneocystostomy over a double-J stent.
8. Graft retroperitonealised with the peritoneal flap to prevent torsion.

### Advantages

- Small incision: **fewer wound infections, less lymphocele**, less pain, faster recovery, better cosmesis.
- Magnified 3D (three-dimensional) view and precise suturing in deep pelvis or obese patients.

### Disadvantages

- **Longer rewarming (anastomosis) time**; cost; learning curve; pneumoperitoneum and Trendelenburg effects.
- No tactile feedback for calcified vessels.

### Outcomes

- Multicentre and ERUS (European Association of Urology Robotic Urology Section) series: graft function and survival comparable with open transplant, with lower surgical-site complications.

### Complications

- Graft thrombosis, haemorrhage, urine leak, ureteric stricture, lymphocele (less), port-site hernia; DGF (delayed graft function).

**Diagram to draw:** pelvis with graft on external iliac vessels and ureter to bladder dome; ice slush via GelPOINT.

> **Key points to remember:**
> - Vattikuti–Medanta technique with regional hypothermia (2014).
> - Vein first, then artery, then Lich–Gregoir.
> - Best for obese recipients; graft function equals open.

# Rapid Revision Sheet · Section 15

### Renovascular hypertension: pathophysiology, evaluation and treatment (LAQ)

- Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
- Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
- Goldblatt 1934: 2K1C = renin-dependent; 1K1C = volume-dependent.
- Chronic phase: contralateral nephrosclerosis makes RVH stenosis-independent.
- Angiotensin II keeps GFR by efferent constriction; ACE inhibitor causes acute kidney injury in bilateral RAS.
- RVH is proved only by cure; duplex first, angiography with gradient before intervention.
- ASTRAL 2009 and CORAL 2014: stenting adds nothing to medical therapy in atherosclerotic RAS; no routine stenting.
- FMD gets balloon angioplasty without stent; ostial atheroma: stent.
- Revascularise for flash pulmonary oedema, declining GFR in a viable kidney (RI <0.8), resistant hypertension.

### Allograft rejection and immunobiology of renal transplantation (LAQ)

- Direct pathway drives early TCMR; indirect pathway drives antibody (DSA) and chronic rejection.
- Signal 1 calcineurin, signal 2 CD28–B7, signal 3 IL-2R–mTOR.
- Hyperacute rejection prevented by crossmatch.
- Banff AMR = tissue injury + antibody–endothelium interaction (C4d) + DSA.
- TCMR: pulse steroids, ATG; AMR: plasmapheresis, IVIG.
- Chronic active AMR (transplant glomerulopathy) is the main cause of late graft loss.
- Belatacept blocks signal 2; better GFR, contraindicated if EBV-negative; imlifidase (IgG cleavage) for highly sensitised patients; dd-cfDNA NPV >90%.
- ECD: ≥60, or 50–59 with ≥2 of hypertension, creatinine >1.5, CVA death; KDPI >85% since 2014.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard; HMP reduces DGF; ESP ≥65 to ≥65.

### Deceased donor kidney: selection, maintenance and retrieval techniques (LAQ)

- THOA: 4-member Board (Board of four), none from the transplant team; two tests ≥6 h apart; Form 10.
- Apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death; KDPI >85%.
- Rule of 100s; vasopressin for hypotension and DI.
- Cannulate, clamp, vent, cool: distal aortic cannula, supracoeliac clamp, IVC vent, cold UW/HTK and ice.
- Kidneys removed last, en bloc; Carrel patch, IVC for right vein.
- Left kidney preferred for living donation.
- Machine perfusion reduces DGF.

### Vascular grafts and vascular surgery in urology (LAQ)

- Proximal and distal control, heparin, flush, everting Prolene anastomosis.

- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- Mayo levels decide caval control and bypass.
- IVC: patch if lumen <50%; PTFE 16–20 mm replacement if wall invaded.
- PC-RPLND: IVC may be ligated; aorta must be grafted.
- Transplant: Carrel patch, bench reconstruction; preserve the lower-pole artery.
- Bench surgery plus autotransplantation for complex renovascular lesions.

**Vascular anastomosis and vascular access for dialysis (LAQ)**

- Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
- AVF first, distal, non-dominant arm, about 6 months ahead.
- Rule of 6s: flow >600 mL/min, >6 mm wide, <6 mm deep, 6 weeks.
- Thrombosis sits on a stenosis; steal: banding or DRIL.

**Describe in details basic patho-physiology of urine production and transport (LAQ)**

- RBF 25% of cardiac output; GFR 125 mL/min; net filtration pressure ~10 mmHg.
- Proximal 65%, loop 25%, distal 5–7% of $Na^+$; ADH → V2 → AQP2.
- Myogenic peristalsis from calyceal pacemaker; post-obstructive diuresis 2% vs 50%.

**Pelvicalyceal (renal collecting system) anatomy, its relevance to PCNL and endourology, and mini-PCNL (LAQ)**

- Sampaio: A-I 45%, A-II (crossed) 17.2%, B-I 21.4%, B-II 16.4%.
- Fornix puncture: no arterial injury; upper-pole infundibulum 67% vascular injury.
- Target a posterior calyx with a short, wide infundibulum; upper pole reaches most.
- Lower pole: steep angle, long calyx >10 mm, infundibulum <5 mm unfavourable.
- Endopyelotomy lateral; infundibulotomy superior or inferior, never anterior.
- Pleura to 12th rib; retrorenal colon at the lower pole, commoner prone.
- Mini 15–22 Fr (Jackman 1998): fewer transfusions, longer operation.
- High intrarenal pressure in mini-PCNL causes fever and sepsis.

**Lymphatic drainage of the genitourinary system and chyluria (etiopathogenesis and management) (LAQ)**

- Lymph flows right → left; cisterna chyli at L1–L2, behind the right side of the aorta.
- Bladder drains mainly to external iliac nodes; left kidney to left para-aortic nodes.
- Chyluria = pyelolymphatic fistula from lymphatic obstruction and valve failure, usually filarial (*W. bancrofti*); often intermittent.
- Ether test and (urine) triglycerides confirm (ether test clears milky urine); cystoscopy after (a) fatty meal lateralises.
- Diet (MCT) + DEC (diet with MCT plus DEC) → (silver nitrate) sclerotherapy → laparoscopic hilar lymphatic disconnection; disconnection surgery for severe or refractory cases.
- Retroperitoneoscopic disconnection beats open.

**Haemodialysis principles, dialysis techniques and dialysis access (vascular and PD) (LAQ)**

- Diffusion clears small solutes, convection middle molecules, ultrafiltration water.
- HD Kt/V minimum 1.2 (target 1.4); URR ≥65%; PD weekly Kt/V ≥1.7.

- AVF first, distal, non-dominant arm, ≥6 months ahead; rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis usually sits on a stenosis; steal is treated by (→) banding or DRIL.
- Tenckhoff: two cuffs, pelvic tip, downward exit.
- Laparoscopic placement with omentopexy gives best function.
- Avoid subclavian lines; AVF first for long-term HD.
- CRRT for unstable AKI; PD preferred in children.

**Acute renal failure (AKI) and acute tubular necrosis (LAQ)**

- KDIGO: creatinine ≥0.3 mg/dL or ×1.5; urine <0.5 mL/kg/h for 6 h.
- FeNa <1% prerenal, >1% ATN; specific gravity <1.010, muddy-brown casts in ATN; ultrasound to exclude obstruction.
- S3 segment and thick ascending limb are most vulnerable.
- Diuretics only for overload (do not treat ATN); dialysis for refractory AEIOU.
- Post-obstructive diuresis: oral fluids; replace 75% with 0.45% saline if needed.

**Discuss the role of tissue typing and cross match in live and cadaveric (LAQ)**

- HLA on chromosome 6; match A, B, DR; DR matters most.
- Positive T-cell CDC crossmatch (IgG DSA) = no transplant: hyperacute rejection.
- Flow and bead assays are more sensitive; cPRA guides allocation.
- Incompatible pair: paired exchange (legal in India since 2011) or desensitisation.

**Brain-stem death: pathophysiology, guidelines and declaration (SAQ)**

- Raised ICP, Cushing reflex, catecholamine storm, then hypotension, DI, hypothermia, DIC.
- Six brainstem reflexes absent; apnoea test: no breathing at $PaCO_2$ ≥60 mmHg.
- THOA: four-member Board; no transplant-team member.
- Two tests ≥6 h apart; Form 10; Rule of 100s.

**GFR estimation in prospective renal donor (SAQ)**

- Creatinine overestimates GFR (secretion); CKD-EPI 2021 is race-free.
- Cystatin C adds accuracy; creatinine–cystatin C combined equation is most accurate.
- Inulin is the gold standard; iohexol or DTPA in practice; DTPA gives GFR and split function.
- KDIGO 2017: ≥90 accept, 60–89 individualise, <60 decline.
- Better kidney stays with the donor.
- CrCl = UV/P; check completeness by creatinine excreted per kg.
- Secretion makes CrCl overestimate GFR as function falls.
- Cockcroft–Gault: (140 − age) × weight / (72 × SCr), × 0.85 women.

**Immunosuppression in renal transplantation: protocols, induction and steroid-free regimens (SAQ)**

- Tacrolimus + mycophenolate ± steroids (maintenance) after induction (basiliximab or rATG).
- Basiliximab for low risk; rATG for high immunological risk.
- CNI nephrotoxic; mTOR inhibitors cause lymphocele and poor healing.
- Belatacept blocks signal 2; avoid if EBV-negative.
- Rejection: steroids for T-cell-mediated; plasmapheresis, IVIG, rituximab for antibody-mediated.
- HLA-incompatible: plasmapheresis + IVIG ± rituximab; imlifidase cleaves IgG.

- ABO-incompatible: rituximab + plasmapheresis to target titre.
- Steroid-free: avoidance, early (first week) or late withdrawal; needs induction plus tacrolimus–MMF; low-risk recipients only.
- More mild rejection, same survival; less PTDM (HARMONY). KDIGO 2009: stop within week 1, or continue.
- Ciclosporin: binds cyclophilin, inhibits calcineurin, NFAT stays phosphorylated, IL-2 blocked.
- Ciclosporin: nephrotoxic; gum hyperplasia and hirsutism; CYP3A4 interactions.
- Ciclosporin urology uses: transplant, refractory Hunner lesions, retroperitoneal fibrosis.

**Evaluation and selection criteria of the living renal donor (SAQ)**

- KDIGO 2017: risk-based lifetime ESKD threshold; GFR ≥90 accept, <60 decline; ACR >100 decline.
- GFR 60–89 and albuminuria 30–100 are individualised.
- Controlled hypertension on 1–2 drugs may donate; type 1 diabetes and bilateral stones exclude.
- ABO and crossmatch first; CT angiography for anatomy.
- Better kidney stays with the donor.
- THOA: near relatives, else authorisation committee; unrelated donor needs Authorisation Committee approval.
- Interview video-recorded; no commerce.

**CAPD and CAPD peritonitis (SAQ)**

- Peritoneum is the membrane; glucose or icodextrin drives ultrafiltration.
- Tenckhoff catheter with two cuffs; Kt/V ≥1.7 per week.
- Peritonitis: two of cloudy fluid or pain, WCC >100/µL with >50% neutrophils, positive culture.
- Intraperitoneal Gram-positive plus Gram-negative cover.
- Remove catheter for fungal or refractory (5 days) peritonitis.

**THOA/THOTA legislation and hospital authorization committee (SAQ)**

- THOA 1994: brain-stem death legal; anti-commercial; Authorisation Committee for unrelated donors; amended 2011; Rules 2014.
- 2011: tissues added; grandparents and grandchildren became near relatives.
- Swap donation, retrieval centres, transplant coordinators, required request, higher penalties.
- Commercial dealing: 5–10 years and ₹20 lakh to ₹1 crore.
- Rules 2014: Form 10; Board of four; tests ≥6 hours apart.
- Hospital-based committee if >25 transplants a year; no transplant team member.
- Non-relatives, swap and foreigners need committee approval.
- Interview video-recorded; appeal within 30 days.

**Marginal kidney donor and Remuzzi score (SAQ)**

- ECD: ≥60, or 50–59 with 2 of hypertension, creatinine >1.5, CVA death.
- KDPI >85% replaced ECD (2014).
- Remuzzi: glomeruli, tubules, interstitium, vessels; 0–3 each.
- Remuzzi 0–3 single, 4–6 dual, ≥7 discard.
- HMP reduces DGF; Maastricht III is the commonest DCD.

**Renin-angiotensin system (SAQ)**

- Renin is rate-limiting; released by low pressure, low NaCl, β1.
- Efferent constriction maintains GFR, so ACE inhibitors risk renal failure in bilateral stenosis.
- Renin ratio >1.5 lateralises; reninoma excision cures most.

**Drugs associated with Acute Renal Failure. (SAQ)**

- NSAIDs block afferent dilatation; ACE inhibitors/ARBs block efferent constriction.
- ATN: aminoglycosides, amphotericin, cisplatin, contrast.
- AIN: β-lactams, PPIs, NSAIDs, checkpoint inhibitors; stop and give steroids.
- Crystals: aciclovir, methotrexate, indinavir.

**Berger's Disease. (SAQ)**

- Commonest primary GN; haematuria with upper respiratory infection.
- Normal complement; biopsy shows mesangial IgA.
- Oxford MEST-C; 20–30% ESKD.
- ACE inhibitor or ARB is the mainstay.

**Vascular complications after renal transplantation, including transplant renal artery stenosis (SAQ)**

- TRAS is the commonest vascular complication; presents months later; angioplasty or stent first (first-line); angiography is gold standard.
- Refractory hypertension plus rising creatinine: Doppler first.
- Venous thrombosis: reversed diastolic flow, swollen tender graft.
- Arterial thrombosis: sudden anuria; explore immediately.

**Management of acute hyperkalaemia (SAQ)**

- Calcium stabilises the membrane but does not lower potassium.
- Insulin 10 units + 50 mL 50% dextrose; salbutamol 20 mg nebulised.
- Binders, diuretics and dialysis remove potassium; relieve obstruction.

**Hypokalemia (SAQ)**

- Shift or loss; urine potassium separates renal from extrarenal.
- ECG: flat T, U wave.
- Diversion acidosis: give potassium with or before bicarbonate.
- Correct magnesium first.

**Lymphocele and postoperative lymphatic leak (chylous ascites) (SAQ)**

- Lymphocele: commonest PLND complication; radiological up to 60%, significant 2–5%.
- Lymph creatinine = serum; urinoma creatinine higher.
- Laparoscopic fenestration best (is best); aspiration alone worst.
- Chylous ascites: 2–12% after post-chemotherapy RPLND; triglyceride 2–8 × serum.
- Ladder: diet with MCT and octreotide → TPN → embolisation or shunt.

**Hyperfiltration injury in AKI and CKD (SAQ)**

- Hyperfiltration = raised SNGFR; Brenner 1982.
- Leads to FSGS, proteinuria and the AKI-to-CKD transition.
- ACE inhibitors and SGLT2 inhibitors lower glomerular pressure.

**UW solution and organ preservation solutions (SAQ)**

- Hypothermia slows, not stops, metabolism.
- Lactobionate and raffinose (impermeants), HES / starch (colloid).
- Allopurinol, glutathione (antioxidants); adenosine (ATP).
- High $K^+$: flush before reperfusion.
- HTK: histidine buffer, tryptophan, ketoglutarate, low viscosity, low $K^+$.
- Machine perfusion reduces DGF.

**Pre transplant nephrectomy (SAQ)**

- Default: leave native kidneys.
- Indications: infection/VUR, ADPKD space, tumour, hypertension, proteinuria.
- Retroperitoneal route for PD; leave a normal ureter.

**Post transplant malignancy (SAQ)**

- Skin cancer commonest; squamous more than basal cell.
- PTLD: EBV-negative recipient of EBV-positive donor, first year.
- RCC mostly in native kidneys; bladder cancer 3-fold and aggressive.
- Reduce immunosuppression and switch to mTOR inhibitor.

**Anomalies of renal artery. (SAQ)**

- Single artery in 71–85%; accessory arteries usually lower pole.
- Crossing vessel can cause PUJ obstruction.
- Treat aneurysm >2 cm or in pregnancy.

**Mycophenolate Mofetil. (SAQ)**

- Prodrug of MPA; IMPDH inhibitor; lymphocytes need de novo purine synthesis.
- 1 g twice daily; ciclosporin lowers MPA levels.
- Toxicity: diarrhoea, leucopenia, CMV/BK, teratogenic; not nephrotoxic.

**Metabolic abnormalities in CKD (SAQ)**

- FGF-23 rises first; secondary then tertiary hyperparathyroidism.
- Bicarbonate ≥22 mEq/L; Hb 10–11.5 g/dL.
- Hyperkalaemia: binders rather than stopping RAAS blockade.

**Renal transplant in neurogenic bladder (SAQ)**

- Target <40 cm $H_2O$ storage before transplant.
- Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
- Pyelonephritis, not augmentation, loses grafts.

**Pre-emptive renal transplant (SAQ)**

- Transplant before dialysis = best outcome.
- Refer at eGFR <30; transplant usually at eGFR <20.
- Commonest with living donors and children.

**Tacrolimus. (SAQ)**

- FKBP12 → calcineurin → NFAT → IL-2 blocked.
- CYP3A4/P-gp; trough-guided; many interactions.
- Nephrotoxicity, NODAT, tremor, hyperkalaemia, hypomagnesaemia.

- ELITE-Symphony: low-dose tacrolimus best.

**Urological Complications post Kidney Transplantation. (SAQ)**

- Ureter is the weak point; ischaemia is the cause.
- Leak early, stricture late; open repair best.
- Stent out at 2–3 weeks; laparoscopic fenestration for lymphocele. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Female kidney donors: special precautions (SAQ)**

- Ensure free consent (Indian gender gap).
- Pregnancy test; contraception; stop oestrogens.
- Donors: gestational hypertension or pre-eclampsia about 11% vs 5%.
- Defer pregnancy; high-risk antenatal follow-up.

**Laparoscopic donor nephrectomy (including pros and cons) (SAQ)**

- Ratner 1995; left kidney preferred (longer vein).
- Hand-assisted in about two-thirds of US cases.
- Pros: less pain, shorter stay, earlier return to work, better cosmesis, and more willing donors; graft function equals open.
- Cons: longer operation, learning curve, slightly longer warm ischaemia, shorter vessels (especially on the right), cost.
- Safety: mortality 3 per 10,000; conversion 0.26–3.34%.
- Never rely on Hem-o-lok alone on the donor renal artery (do not use Hem-o-lok clips alone).

**Artificial kidney (SAQ)**

- Diffusion, convection, ultrafiltration across a semipermeable membrane.
- Dialysis lacks endocrine and tubular function; transplant is best.
- Wearable, implantable bioartificial and RAD devices are investigational.

**Robotic kidney transplantation (SAQ)**

- Vattikuti–Medanta technique with regional hypothermia (2014).
- Vein first, then artery, then Lich–Gregoir.
- Best for obese recipients; graft function equals open. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 16 · BPH

## Long answer questions (LAQ)

### LAQ 1 Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate

P1 · P3 · P4 **merged from 3**

*Asked as:* P1 LAQ "Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer" (Anatomy of prostate and its neurovascular bundles with its clinical details; Describe in detail, anatomy of prostate with special reference to concepts in surgical management if localised CA prostate) · P4 LAQ "Prostate anatomy, newer surgical concepts in CA prostate and neurovascular bundle preservation in radical prostatectomy" (Discuss in detail anatomy of prostate with special reference to newer concepts in surgical management of CA prostate; also SAQ: Neurovascular bundle preservation in Radical prostatectomy: recent techniques) · P3 SAQ "Describe in details anatomy of prostate with special reference to newer concept in surgical management of localized CA prostate"

*"if localised" is read as "of localised" prostate cancer.*

#### Introduction

- A fibromuscular, glandular organ of about 20 g around the proximal urethra, from bladder neck (**base**) to urogenital diaphragm (**apex**), in front of the rectum.
- Buds from the urogenital sinus at week 10, driven by androgen acting on the **mesenchymal** AR (androgen receptor).
- RP (radical prostatectomy) aims at the "trifecta" of cancer control, continence and potency. Modern RP is anatomical: each anatomical step or new technique either secures a clear margin or saves a named structure (nerves, sphincter, anterior support).

#### Relations

- **Anterior:** retropubic space (of Retzius), puboprostatic ligaments, DVC (dorsal venous complex).
- **Posterior:** Denonvilliers fascia, then rectal ampulla, hence DRE (digital rectal examination) and TRUS (transrectal ultrasound).
- **Lateral:** levator ani with its fascia; NVB (neurovascular bundle) posterolaterally.
- **Base:** bladder neck; seminal vesicles behind. **Inside:** the urethra bends about 35° at the **verumontanum**, where the ejaculatory ducts open beside the utricle.

#### Zonal anatomy (McNeal 1981)

| Zone | Share of glands | Position | Clinical / surgical relevance |
|---|---|---|---|
| Peripheral (PZ) | About 70% | Posterior and lateral; longest ducts | **70% of cancers**; felt on DRE; biopsy target; posterolateral, so ECE (extracapsular extension) runs along the nerves |

| Zone | Share of glands | Position | Clinical / surgical relevance |
|---|---|---|---|
| Central (CZ) | About 25% | Cone around the ejaculatory ducts at the base | Few cancers; mimics HGPIN (high-grade prostatic intraepithelial neoplasia) |
| Transition (TZ) | About 5% | Two lobes beside the preprostatic sphincter; shortest ducts | **BPH (benign prostatic hyperplasia)**; anterior cancers (about 15%) missed by DRE |
| Periurethral glands | Small | Within the preprostatic sphincter | Middle-lobe BPH |
| Anterior fibromuscular stroma (AFMS) | No glands | Anterior | Anterior tumours reach it; anterior margin risk in posterior approaches |

- **Capsule:** there is **no true capsule**, so cancer breaching the gland shows "extraprostatic extension". In BPH the compressed PZ forms the **surgical capsule**, the plane of enucleation. The capsule transmits growth pressure to the urethra, so TUIP (transurethral incision) relieves obstruction.

### Fascial investments and dissection planes

| Layer | Position | Surgical rule |
|---|---|---|
| **Prostatic fascia** | On the gland; carries DVC tributaries anteriorly | Stays on the prostate in nerve sparing |
| **Levator fascia** | Covers levator; fuses with prostatic fascia as the lateral pelvic fascia | Incised to reach the NVB |
| **Denonvilliers fascia** | Between prostate and rectum; thickest at base and vesicles | Excised completely for the posterior margin |

- **Denonvilliers fascia** stays on the rectum when ECE risk is low; it is excised when posterolateral (posterior) ECE is suspected.

| Nerve-sparing plane | Lies | Trade-off |
|---|---|---|
| **Intrafascial** | Under the prostatic fascia (between capsule and prostatic fascia) | Most nerves kept; more positive margins |
| **Interfascial** | Between prostatic and levator fascia | **Standard nerve-sparing plane** for organ-confined disease |
| **Extrafascial** | Outside levator and Denonvilliers fascia | Widest margin; ED (erectile dysfunction) likely |

- Chapter 160 (Walsh) calls intrafascial "never used"; Chapter 161 accepts it in selected men.

### Blood supply and lymphatics

- **Inferior vesical artery** ends as: **urethral group** (enters at the vesicoprostatic junction, 5 and 7 o'clock; supplies the adenoma) and **capsular group** (posterolateral, in the lateral pelvic fascia, marking the cavernous nerves).
- **Accessory pudendal arteries** run over the anterolateral prostate; dividing them reduces penile inflow. Preserve them; they may be the main penile supply.

- **Veins:** the deep dorsal vein passes under the pubis into a superficial branch and lateral (**Santorini**) plexuses, draining to the internal iliac vein. They anastomose freely, so tears bleed heavily.
- DVC: $CO_2$ pneumoperitoneum tamponades it, so robotic blood loss is low.
- **Lymph:** obturator, internal and external iliac, common iliac, presacral nodes.

### Neurovascular bundle

- **Pelvic plexus:** parasympathetic S2–S4 plus sympathetic (T10–L2) via the hypogastric nerves; beside the rectum, centred on the seminal vesicle tip. It gives the **cavernous nerves** (parasympathetic).
- **NVB of Walsh (Walsh and Donker, 1982):** capsular vessels plus **cavernous nerves**. The nerves are invisible; the capsular vessels mark them.
- It runs posterolaterally in the lateral pelvic fascia **between prostatic and levator fascia**, **3–5 mm from the capsule**, outside Denonvilliers fascia.
- At the apex they lie at **5 and 7 o'clock**, posterolateral to the sphincter. The nerves then swing to about **3 and 9 o'clock** at the mid-urethra and enter the corpora behind the dorsal penile artery. The apex is where most nerve injuries occur.
- Distribution varies (is diffuse): main posterolateral bundle, posterior "hammock", anterolateral fibres (the basis of high anterior release).
- **Injury mechanisms:** incision, entrapment (clip or suture), **heat** (thermal), **traction**; so dissect athermally without traction.
- **Striated sphincter:** a vertical tube over the apex (Oelrich), pudendal nerve; passive continence.

### Continence anatomy

| Passive | Active |
|---|---|
| Smooth muscle of bladder neck and urethra; anterior support / suspensory system (endopelvic fascia, detrusor apron, puboprostatic ligaments, DVC) | Horseshoe striated sphincter at the apex, supplied by pudendal nerve branches that enter it at the apex |

### Clinical details

- **BPH surgery:** TURP (transurethral resection of the prostate) stays proximal to the verumontanum (sphincter lies just distal); urethral-group vessels at 5 and 7 o'clock bleed.
- **Cancer:** PZ cancer exits posterolaterally along nerves; anterior TZ tumours are missed by DRE and biopsy, so MRI (magnetic resonance imaging) helps.
- **Radical prostatectomy (RP) steps:** 1. Incise endopelvic fascia; spare the pubourethral ligament. 2. Ligate the DVC over the urethra before division. 3. Preserve accessory pudendal arteries. 4. Excise Denonvilliers fascia; the apex is the commonest positive-margin site.
- Excise an NVB for palpable extracapsular disease; margins are equal with or without nerve sparing.
- **Also know:** rectal injury at the ampulla; PDE5 (phosphodiesterase type 5) inhibitor rehabilitation.

### Concepts and newer concepts in surgical management of localised cancer

1. **Selection:** RP suits localised disease (mainly intermediate and selected high risk) with a life expectancy of more than 10 years; low risk is usually offered active surveillance.

2. **RARP (robot-assisted RP):** 3D, 10× magnified view and wristed instruments removed the suturing barrier. **Yaxley 2016** RCT (randomised controlled trial): equal early function and early oncology versus open; less blood loss (300 versus 1400 mL) and shorter stay. Surgeon matters more than platform.
3. **Risk-graded (risk-stratified) nerve sparing:** each side is graded from full (intrafascial) through partial to none (for example the Tewari four-grade scale), using grade, positive cores, mpMRI (multiparametric magnetic resonance imaging) capsular contact and nomograms, side by side.
4. **NeuroSAFE:** intraoperative frozen section of each (the whole) posterolateral surface; if positive, that NVB is resected. It aims to allow more nerve sparing without more positive margins; the UK NeuroSAFE PROOF RCT tested its effect on erectile function.
5. **Anterior-support preservation:** - **Retzius-sparing** (Bocciardi, 2010): fully posterior through the pouch of Douglas. Earlier continence at 3 months, equal at 12; more positive margins (anteriorly), so avoid for anterior tumours. - **Hood technique** (Tewari, 2021): anterior approach keeping detrusor apron, puboprostatic ligaments and DVC; early data. - Retzius-sparing and Hood are aimed at continence; potency equal at 12 months.
6. **Apical dissection:** the apex is the commonest positive-margin site. Keep maximal urethral length, watch for an apical lip, divide sharply without cautery (cold division); control the DVC distally or late. Posterior (Rocco) reconstruction may aid early continence.
7. **PLND (pelvic lymph node dissection) templates:**

| Template | Nodes | Drainage captured |
|---|---|---|
| Limited | Obturator | Least |
| Standard | + External iliac | About 38% |
| **Extended** | + Internal iliac | About 63% |
| + Common iliac | + Common iliac | About 75% |

- Extended PLND when nodal risk exceeds 5% (EAU [European Association of Urology]). It stages but does not improve survival (staging, not therapeutic); lymphocele is the commonest complication. 8. **Single-port robot (da Vinci SP, 2018):** one 2.5-cm cannula; extraperitoneal or transvesical; same-day discharge. 9. **Focal therapy anatomy:** ablate the **index lesion**. Transrectal HIFU (high-intensity focused ultrasound) suits posterior PZ lesions; anterior lesions need transperineal cryotherapy or TULSA (transurethral ultrasound ablation). Spare sphincter, NVB and rectum.
- **Also know:** PSMA PET (prostate-specific membrane antigen positron emission tomography) staging; salvage robotic RP; new platforms; ERAS (enhanced recovery after surgery).

### Nerve-sparing techniques (NVB preservation): recent techniques

- **Walsh anatomic (open):** release apex to base from the rectal surface; small clips; **no thermal energy**.
- **High anterior release** (open) and **Veil of Aphrodite** (robotic): lateral prostatic fascia lifted high anteriorly as a "veil". Select by **Tsuzuki criteria**: ≥2 of PSA (prostate-specific antigen) >10 ng/mL, Gleason >6, abnormal DRE, perineural invasion, high core involvement → >10% risk, so avoid.
- **Antegrade release (robotic):** incise levator fascia, sweep NVB posterolaterally, clip pedicle at the "groove".

- **Athermal, traction-free:** clips or bulldogs, cold scissors.
- Graded nerve sparing, NeuroSAFE, Retzius-sparing and Hood: see above.

### Adjuncts

- **Penile rehabilitation:** PDE5i (phosphodiesterase type 5 inhibitor), then injections.
- **Also know:** sural nerve grafts and nerve stimulation tests (abandoned); amniotic membrane wraps, fluorescence nerve imaging (experimental).

### Diagram to draw

- Axial prostate showing the three fasciae, the NVBs at 5 and 7 o'clock and the intra-, inter- and extrafascial planes.

> **Key points to remember:**
> - PZ gives 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts; no true capsule.
> - NVB between prostatic and levator fascia, 5 and 7 o'clock; apex is the commonest margin and nerve-injury site.
> - Interfascial plane is the standard nerve-sparing plane; avoid heat and traction; clips, cold scissors.
> - Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.
> - Veil and high release for low-risk sides (Tsuzuki).
> - NeuroSAFE frozen section: more sparing, safe margins.
> - Retzius-sparing: earlier continence, more positive margins.
> - Extended PLND captures about 63% of drainage.

## LAQ 2 HoLEP

P1 · P4 **merged from 2**

*Asked as:* P4 LAQ "HoLEP (holmium laser enucleation of the prostate)" (Holep; also SAQ: HOLEP.) · P1 SAQ "HoLEP in BPH"

*Read as HoLEP (holmium laser enucleation of the prostate) for BPH (benign prostatic hyperplasia).*

### Introduction and principle

- **HoLEP** removes the whole transition-zone adenoma endoscopically **along the surgical capsule**. It is the endoscopic equivalent of open simple prostatectomy.
- **Evolution:** HoLAP (holmium laser ablation, 1994, slow) → HoLRP (resection, strips removed with a grasper) → **HoLEP (1998)**, made practical by the **tissue morcellator**.
- "Enucleation is enucleation": any energy giving a true **AEEP (anatomic endoscopic enucleation of the prostate)** gives durable results.

### Laser physics

- **LASER:** light amplification by stimulated emission of radiation. Ho:YAG (holmium:yttrium–aluminium–garnet) = holmium atoms in a YAG (yttrium–aluminium–garnet) crystal.
- **Wavelength about 2120 nm**; chromophore **water** (absorbed by water); **pulsed**; optical penetration **0.4 mm**. It cuts by vaporisation with a thin coagulation zone; heat spreads 2–3 mm and coagulates small and medium vessels.
- Flexible quartz fibres; **normal saline** irrigation, so no TUR (transurethral resection) syndrome.

| Laser | Wavelength | Chromophore | Penetration |
|---|---|---|---|
| **Holmium:YAG** | 2120 nm | Water | 0.4 mm |
| Thulium fibre | 1940 nm | Water | 0.2 mm |
| KTP (potassium-titanyl-phosphate, green light) | 532 nm | Haemoglobin | 0.8 mm |
| Nd:YAG | 1064 nm | – | About 1 cm (sloughing) |

### MOSES (pulse modulation)

- **Moses effect:** the first part of each pulse boils a **vapour channel** in the water. Vapour absorbs the light far less, so the rest of the pulse passes through to the tissue ("parting the water").
- **Moses technology** splits each pulse to exploit this: more energy reaches tissue, and it **may shorten operating time**.

### Indications

- **Size-independent** (<30, 30–80, >80 mL), with or without a median lobe (AUA [American Urological Association] Statement 38). Only enucleation spans every size band.
- **High bleeding risk or anticoagulated men** (AUA recommendation, Statement 43).
- Large glands otherwise needing open prostatectomy (alternative to open prostatectomy for large glands); retention; failed medical therapy.

## Technique (steps)

- **Equipment:** 60–150 W Ho:YAG laser; **550 µm end-firing fibre**; 24–28 Fr continuous-flow laser resectoscope with 30° lens; saline.
- **Settings:** about 2.0 J × 40–50 Hz to enucleate; 1.0–2.0 J × 20 Hz to coagulate.
- **Three-lobe (bottom-up, early apical release):** 1. Median lobe: **bladder-neck incisions at 5 and 7 o'clock** to the capsule, joined proximal to the verumontanum; median lobe enucleated into the bladder. No median lobe: one 6 o'clock incision. 2. **Perimontanal inverted-U incision**; scope pushes laterally to open the lobe–capsule plane at the apex; **early apical release** to protect (spare) the sphincter. 3. Capsular plane developed anteriorly to 11–1 o'clock (round to 12 o'clock); **apical mucosal strip incised early** to avoid sphincter traction. 4. Anterior commissure divided; lateral lobes enucleated (detached) into the bladder (**en bloc, 2-lobe or 3-lobe**). 5. **Haemostasis** with a defocused beam.
- **Two-lobe:** 6 and 12 o'clock grooves; each half enucleated with its share of median lobe.
- **En bloc:** whole adenoma in one piece after circumferential apical release; usually paired with early apical release.
- **Morcellation:** nephroscope, two open inflow lines, **full bladder** and good haemostasis. Morcellators: Piranha (rotating), Versacut (oscillating). Fragments sent for histology.
- **Catheter:** 22 Fr three-way; same-day or next-day removal.

## Outcomes and trials

- **Versus TURP (transurethral resection of the prostate):** RCT (randomised controlled trial) in glands <100 mL: better symptom score and PVR (post-void residual), less haemoglobin drop (less bleeding), shorter catheter time and stay, longer operation.
- **Meta-analyses:** better **IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate)**, durable at **3–8 years**. **HoLEP is the only endoscopic procedure with efficacy superior to TURP.**
- **Versus open prostatectomy** for large glands (Naspro 2006; Kuntz 2008, 5 years): equal function, less morbidity, less blood loss, shorter catheter time and stay.
- **Durable; retreatment rare**: whole transition zone removed.
- **Anticoagulated men:** feasible; slightly longer stay; warfarin worse than DOACs (direct oral anticoagulants).
- **Guidelines:** EAU (European Association of Urology) and AUA list HoLEP (enucleation) as a first-line alternative to TURP and to open surgery for large (>80 mL) glands.

## Complications

- **Intraoperative:** capsular perforation; ureteric orifice injury (glands >80 mL); **morcellator bladder injury** (stop and drain; intraperitoneal → repair); retained adenoma.
- **Early:** transient storage symptoms and dysuria; **transient (early) SUI (stress urinary incontinence)** from adenoma volume removed, falling over 6 months; persistent SUI rare.
- **Late: retrograde ejaculation about 75%**; BNC (bladder neck contracture) **0–3.2%** (small glands <40 mL, consider bladder-neck incision); meatal stricture; urethral stricture.
- Erectile function unchanged. Transfusion rare; conversion to TURP almost never.

### Limitation: learning curve

- The main barrier is the **steep learning curve**: from 20 supervised cases; **efficiency plateaus after about 50 cases**. Mentoring and simulators help.

### Recent advances

- High-power **Moses pulse modulation**; **ThuLEP** (thulium laser enucleation) and bipolar enucleation.
- **Ejaculation-preserving (ejaculation-sparing) and median-lobe-only HoLEP** (antegrade ejaculation about 90%).
- **Same-day discharge** and catheter removal.

### Diagram to draw

- Sagittal prostate showing the enucleation plane along the surgical capsule, with 5 and 7 o'clock incisions.

> **Key points to remember:**
>
> - Ho:YAG 2120 nm, water chromophore, 0.4 mm; saline, no TUR syndrome.
> - Size-independent; best (preferred) in anticoagulated men.
> - Only endoscopic method superior to TURP; better IPSS and Qmax; durable; retreatment rare.
> - Retrograde ejaculation about 75%; learning curve about 50 cases.

## LAQ 3 BPH: medical management and surgical modalities

P2

**Asked as:**

- Describe medical management of BPH and role of various surgical modalities in its management. — *Summer 2015 Paper II Q1 (LAQ)*
- Medical treatment of BPH — *Practice Set 6 Paper II Q3f (SAQ)* (also answered in 2015-Summer Paper 2, Q1)

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)

### Introduction, pathophysiology and rationale

- **BPH (benign prostatic hyperplasia)** is stromal and epithelial hyperplasia of the transition zone; it may cause BPO (benign prostatic obstruction) and **LUTS (lower urinary tract symptoms)**.
- Obstruction has a **static** component (gland bulk, DHT (dihydrotestosterone)-dependent / DHT-driven gland bulk) and a **dynamic** component (dynamic part: α1A-mediated smooth-muscle tone; static part: DHT-driven gland bulk). Drugs target each. The bladder responds with detrusor overactivity or later underactivity.

### Evaluation (before treatment)

- **IPSS (International Prostate Symptom Score)** 0–35: mild 0–7, moderate 8–19, severe 20–35.
- DRE (digital rectal examination), urinalysis, **PSA (prostate-specific antigen)**, frequency-volume chart, Qmax (maximum flow), PVR (post-void residual); PVR >300 mL is managed as chronic retention (PVR >300 mL = chronic retention).
- Measure prostate volume before surgery (AUA (American Urological Association)); pressure-flow study if unclear.

### Medical management

| Option / class | Doses | Key facts |
|---|---|---|
| **Watchful waiting** | — | Mild or non-bothersome LUTS; lifestyle advice |
| **α1-blockers** (first line) | Tamsulosin 0.4 mg; alfuzosin 10 mg; silodosin 8 mg; doxazosin 4–8 mg (titrate) | First line; act within days; no effect on gland size, AUR (acute urinary retention) or surgery (do not prevent AUR or surgery). Hypotension, ejaculatory dysfunction; silodosin causes most anejaculation. **IFIS (intraoperative floppy iris syndrome)** |
| **5-ARIs (5α-reductase inhibitors)** | Finasteride 5 mg; dutasteride 0.5 mg | For glands (prostate) >30–40 mL or PSA >1.5 ng/mL. Shrink the gland about 18–25% (another source: ~20%) (maximal at 6 months; act after 6 months), halve PSA. PLESS (Proscar Long-Term Efficacy and Safety Study): AUR or surgery down (↓) 51%. Low libido, ED (erectile dysfunction). Also BPH haematuria |
| **Combination** | — | For high-risk men. The α-blocker gives early relief; the 5-ARI prevents hard events |

| Option / class | Doses | Key facts |
|---|---|---|
| **Antimuscarinic or β3-agonist** (mirabegron, vibegron) | Add-on | Add for persisting storage symptoms; trials included only men with low PVR (PVR <150 mL) |
| **Tadalafil 5 mg daily** | PDE5 (phosphodiesterase-5) inhibitor | LUTS with or without (±) ED; not with nitrates |
| **Desmopressin** | — | Nocturnal polyuria (night urine >33% of 24-h output); check (monitor) sodium, especially ≥65 years |

### Combination therapy

- **MTOPS (Medical Therapy of Prostatic Symptoms)**: progression fell (↓) 39% (doxazosin), 34% (finasteride) and **66% (combination)**; only finasteride or combination (finasteride-containing arms) reduced AUR and surgery.
- **CombAT (Combination of Avodart and Tamsulosin)**: combination cut AUR or surgery by about 66% versus tamsulosin.
- Combine when **PV (prostate volume) >30–40 mL and PSA >1.5 ng/mL**; α-blocker may stop (withdrawable) after 6 months of combination (SMART (Symptoms Management After Reducing Therapy) trial).
- Tadalafil + finasteride fixed dose approved (FDA [Food and Drug Administration] 2021).

### Side effects and important cautions

- **Side effects**: α-blockers cause dizziness, postural hypotension, ejaculatory dysfunction and **IFIS (intraoperative floppy iris syndrome)**, which persists after stopping, so avoid starting them before planned cataract surgery. 5-ARIs: low libido, ED, gynaecomastia.
- **IFIS:** billowing iris, prolapse, miosis; stopping the drug does not prevent it.
- 5-ARIs: double the PSA; not for cancer prevention (more high-grade tumours in PCPT [Prostate Cancer Prevention Trial] and REDUCE).
- **AUR**: catheterise (catheter), then TWOC (trial without catheter) on an α-blocker (ALFAUR (Alfuzosin in Acute Urinary Retention): success 62% vs 48%).
- **Also know:** phytotherapy (saw palmetto) not recommended by guidelines.

## Surgical management

**Indications**: refractory retention, recurrent UTI (urinary tract infection) or haematuria, bladder stones, renal insufficiency from BPO, failed medical therapy (surgery if refractory).

| Modality | Size (AUA 2023) | Key feature |
|---|---|---|
| **TURP (transurethral resection)**, monopolar or bipolar | 30–80 mL | Reference standard; retrograde ejaculation ~two-thirds |
| **TUIP (transurethral incision)** | ≤30 mL | Less retrograde ejaculation; more reoperation |
| **PVP (photoselective vaporization)** | ≤80 mL | Safe when anticoagulation cannot stop; no histology |
| **HoLEP/ThuLEP (laser enucleation)** | Any size | Lowest retreatment, least bleeding |

| Modality | Size (AUA 2023) | Key feature |
|---|---|---|
| **Simple prostatectomy** (open or robotic) | >80 mL | Complete removal; diverticula or stones |
| **Aquablation** | 30–80 mL (consider 80–150) | Heat-free; preserves ejaculation |
| **PUL (prostatic urethral lift), Rezūm (water vapour)** | 30–80 mL (PUL: no median lobe) | Preserve sexual function; higher retreatment |
| **iTind (temporary implant)** | 25–75 mL, no median lobe | Removed after a few days; modest efficacy |
| **PAE (prostate artery embolisation)** | Larger glands (AUA 2023: may be offered) | Local anaesthesia; less effective than TURP; radiation, non-target embolisation |

- **TUR syndrome** (monopolar TURP): hypo-osmolar glycine or water is absorbed through open sinuses, causing dilutional hyponatraemia, fluid overload and confusion. Risk rises with gland >45 mL and resection >90 minutes. Treat with diuretics and, if severe, 3% saline, correcting slowly. **Bipolar TURP** uses saline and eliminates it.
- **Complications**: bleeding, stricture, bladder neck contracture, retreatment.
- **GOLIATH**: 180 W PVP non-inferior to TURP. **WATER (Waterjet Ablation Therapy for Endoscopic Resection)**: Aquablation non-inferior, with less retrograde ejaculation (10% vs 36%).
- **Also know:** TUVP (transurethral vaporization); TUMT (transurethral microwave therapy) dropped as legacy; Optilume paclitaxel-coated balloon (2023).

### Follow-up

- Reassess at 4–12 weeks (α-blocker), 3–6 months (5-ARI) (another source: 6 months for 5-ARIs), then yearly; surgery if refractory.

**Diagram to draw:** Size-based surgical algorithm (≤30, 30–80, >80 mL); algorithm: LUTS → size, PSA, storage symptoms → drug choice.

> **Key points to remember:**
> - α-blockers: rapid relief, no prevention.
> - 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
> - MTOPS: combination cut progression 66% (↓66%); only 5-ARI-containing arms cut AUR and surgery.
> - AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
> - TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

## Short answer questions (SAQ)

### SAQ 1 Millin's prostatectomy

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "Millin's prostatectomy" · P2 SAQ "Millin's prostatectomy"

## Definition

- **Retropubic transcapsular (extravesical) simple prostatectomy** (Millin, 1945): adenoma enucleated through a transverse anterior capsulotomy **without opening the bladder**. Plane: the surgical capsule (compressed peripheral zone around the transition-zone adenoma).

## Indications

- BPH (benign prostatic hyperplasia), gland **>80–100 g**, with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or upper-tract damage (obstructive complications), or when lithotomy is impossible.
- Prefer suprapubic (Freyer) for a large median lobe, diverticulum or bladder stones.
- EAU (European Association of Urology) and AUA (American Urological Association): simple prostatectomy or HoLEP (holmium laser enucleation) for large glands.

## Work-up

- Exclude cancer: PSA (prostate-specific antigen), MRI (magnetic resonance imaging) ± biopsy; TRUS (transrectal ultrasound) volume; cystoscopy; urodynamics if bladder function is doubtful.

## Steps

1. Lower midline or Pfannenstiel; extraperitoneal approach to the **space of Retzius**.
2. Control the dorsal venous complex: optional prophylactic ligation of DVC (dorsal venous complex) and pedicles (Walsh–Oesterling).
3. **Transverse capsulotomy ~2 cm distal to (below) the bladder neck** (a longitudinal cut may extend into the sphincter).
4. Find the adenoma–capsule plane with scissors; **finger enucleation**.
5. **Divide the apical urethra under vision** proximal to the sphincter, sparing it.
6. Trim the bladder neck lip; advance bladder neck mucosa into the fossa.
7. **Figure-of-eight sutures at 5 and 7 o'clock**, where prostatic arteries enter.
8. 22 Fr three-way catheter, **watertight capsule closure**, drain.

## Millin versus Freyer

| Aspect | Millin (retropubic) | Freyer (suprapubic) |
|---|---|---|
| Bladder | Not opened | Opened |
| Apex | Divided under vision | Pinched by finger |
| Fossa haemostasis | Directly visible | Less visible |
| Stones, diverticulum, median lobe | Difficult | Ideal |
| Specific risk | (Dorsal) venous bleeding | Cystotomy leak |

## Complications

- Bleeding (transfusion <5%): control by catheter traction, Malament purse-string, O'Conor capsular plication.
- Urine leak, UTI (infection), ureteric injury, bladder neck contracture 2–6%, stricture, DVT (deep vein thrombosis) and thromboembolism.
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence very rare.

### Versus TURP (transurethral resection of prostate)

- Complete removal, low (lower) retreatment, no TUR syndrome; but incision, longer stay.
- **Also know:** vesicocapsular variant.

### Recent advances and modern alternatives

- HoLEP; laparoscopic, robotic multiport and single-port transvesical simple prostatectomy: less bleeding, shorter stay.

### Diagram to draw

- Sagittal view of transverse capsulotomy and enucleation plane.

> **Key points to remember:**
> - Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
> - Apex divided under vision; 5 and 7 o'clock sutures.
> - Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

## SAQ 2 TUR syndrome

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "TUR Syndrome" · P3 SAQ "TUR Syndrome."

### Definition

- **TUR (transurethral resection) syndrome:** neurological and cardiovascular disturbance from **absorption of hypo-osmolar, non-conducting irrigant** during monopolar TURP (transurethral resection of the prostate). Incidence low and falling; about 1–2% in large series.

### Pathophysiology

- Prostatic venous pressure is about **10 mmHg**; irrigant at higher pressure enters open sinuses, or the periprostatic space after perforation.
- Even routine resection absorbs 800–1000 mL.
- Monopolar needs ion-free fluid: **glycine 1.5% (~200 mOsm/kg)**, sorbitol, mannitol or water (all hypo-osmolar; plasma ~290).
- **Fluid overload** → hypertension, pulmonary oedema.
- **Dilutional hyponatraemia and hypo-osmolality** → water moves into brain → **cerebral oedema**.
- **Glycine toxicity:** ammonia → encephalopathy; inhibitory retinal neurotransmitter → **transient blindness**.
- Sorbitol → hyperglycaemia; mannitol → osmotic diuresis.

### Risk factors

- Gland **>45 mL**, resection **>90 minutes** (Veterans Affairs Cooperative Study), high irrigation pressure, capsular perforation, sterile water.

### Clinical features

- Awake (spinal) patient: restlessness, confusion, **nausea, vomiting, visual disturbance**.
- Cardiovascular: **hypertension with bradycardia**, tachypnoea, later hypotension, ECG (electrocardiogram) changes.

- Severe: seizures, coma, herniation, death. **Na <120 mEq/L** is dangerous.

### Prevention

- **Bipolar TURP or laser with saline** (eliminates it; NNT [number needed to treat] about 50).
- Bag **~60 cm** above patient (each extra 10 cm more than doubles absorption); continuous-flow resectoscope; low-pressure irrigation.
- Limit resection time; stage huge glands; check sodium in long cases.
- Spinal anaesthesia for early detection; **1% ethanol-tagged irrigant** (breath test) monitors absorption.

### Treatment

1. Stop surgery, secure haemostasis, insert catheter.
2. ABC (airway, breathing, circulation), oxygen, monitoring; send **sodium, osmolality**, glucose.
3. Fluid overload: **furosemide**; restrict fluids.
4. Severe neurological signs or Na <120: **3% saline 100 mL bolus** (raises Na 2–3 mEq/L), repeated at 10 minutes, maximum 300 mL. **Target rise 4–6 mEq/L.**
5. Correction limit: **≤12 mEq/L in 24 h** for acute post-TURP hyponatraemia (8–10 mEq/L if symptoms start >48 h after surgery), avoiding **osmotic demyelination (central pontine myelinolysis)**.
6. Seizures: benzodiazepines; ICU (intensive care unit) care.

### Other settings

- **Also know:** occurs in TURBT (transurethral resection of bladder tumour), PCNL (percutaneous nephrolithotomy) and hysteroscopy.

### Diagram to draw

- Flowchart: absorption → overload + hyponatraemia → cerebral oedema.

> **Key points to remember:**
>
> - Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
> - Glycine: ammonia and transient blindness.
> - 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
> - Bipolar saline TURP prevents it.

## SAQ 3 TURP instruments, techniques and monopolar vs bipolar

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "TURP instruments and techniques" (TURP instruments; TURP techniques) · P2 SAQ "Monopolar v/s Bipolar TURP."

### Introduction

- **TURP (transurethral resection of the prostate):** an electrified (wire) loop removes transition-zone adenoma **from bladder neck to verumontanum, down to the surgical capsule**. The benchmark BPH (benign prostatic hyperplasia) operation; gold standard for

glands up to 80 mL (AUA [American Urological Association] 2023); larger glands need enucleation or simple prostatectomy.

### Resectoscope

| Part | Description |
|---|---|
| **Outer sheath** | 24–28 Fr; **insulated beak** prevents arcing to the urethra |
| **Inner sheath** | Creates **continuous flow**: inflow through the inner, outflow between sheaths (passive or suction). Lower bladder pressure, clear vision, no repeated emptying |
| **Obturator** | Blunt or visual; atraumatic passage |
| **Working element** | Moves the loop. **Passive:** spring holds loop inside sheath; cutting stroke by spring recoil. **Active:** surgeon's thumb pulls the extended loop in |
| **Telescope** | **Hopkins rod-lens**, usually **30°** (12° alternative) |

### Electrodes

- **Cutting loop:** tungsten (monopolar); **platinum–iridium (bipolar plasma)**. The plasma loop cuts at lower temperature and seals vessels.
- Roller ball or barrel (coagulation), vaporisation button, **Collins knife** (TUIP [transurethral incision of the prostate]).

### Energy: generator and waveforms

- **Generator:** about 500 kHz, too fast to stimulate nerves.
- **Cut:** continuous waveform. **Coagulation:** interrupted, higher voltage. **Blend:** better haemostasis, cuts less well.

### Monopolar versus bipolar: principles

- **Monopolar (m-TURP):** current runs from the loop through the patient to a grounding (dispersive) pad. Saline would disperse the current, so a **non-conducting, hypo-osmolar irrigant** (glycine 1.5%, sorbitol–mannitol, water) is needed.
- **Bipolar (b-TURP):** active and return poles are on the same instrument (the loop), so current stays local and **normal saline** can be used. The Plasma system turns saline into a plasma layer (orange glow) that cuts at lower temperature and seals vessels.
- **AUA:** no preference between monopolar and bipolar; choose by expertise.

### Monopolar versus bipolar: comparison

| Feature | Monopolar | Bipolar |
|---|---|---|
| Current path | Loop → patient → **dispersive pad** | Active and return poles on the loop; stays within the loop |
| Irrigant | **Non-conducting, hypo-osmolar:** glycine 1.5%, sorbitol, water | **Normal saline** |
| TUR (transurethral resection) syndrome | Yes; keep resection <90 min | Practically (virtually) eliminated |
| Resection time | Limited (speed over haemostasis) | No time pressure; large glands |
| Loop | Tungsten | Platinum–iridium (Plasma) |

| Feature | Monopolar | Bipolar |
|---|---|---|
| Bleeding, transfusion, clot retention | Higher | Lower |
| Catheter time, hospital stay | Longer | Shorter |
| Obturator reflex | Risk | Reduced |
| Efficacy: IPSS (International Prostate Symptom Score), QoL, PVR | Equivalent (Qmax marginally better with bipolar) | Equivalent |
| Stricture, BNC (bladder neck contracture), sexual function, retreatment | Similar | Similar |

### Imaging and irrigation

- **Light source:** xenon or LED (light-emitting diode) with fibreoptic cable; camera and monitor.
- **Irrigant** at body temperature; bag about **60 cm** high (each extra 10 cm more than doubles absorption); lowest height giving vision.

### Accessories

- **Ellik evacuator or Toomey syringe** for chips.
- Van Buren sounds or Otis urethrotome: dilate meatus 2 Fr above sheath size (prevents stricture).
- **Three-way Foley catheter** (22–24 Fr) with catheter guide; continuous irrigation set.

### Preparation

- Sterile urine; spinal or general anaesthesia; lithotomy with peroneal padding.
- **Inspect bladder first** (tumour, diverticula); identify ureteric orifices, verumontanum, external sphincter.

### Resection techniques

- **Median lobe first**, then lateral lobes.
- **Modified Nesbit "encirclement":** start at 11–9 and 1–3 o'clock; expose but do not resect bladder neck fibres.
- **Floor-first:** channel at 5 or 7 o'clock to the capsule (sets depth early); widen laterally; lateral lobes "fall" into the fossa.
- **Anterior start:** 12 o'clock bladder neck, then quadrants down to the apex.
- **Chips:** long "canoe-like" chips with synchronised rocking.
- **Anterior tissue and apex last** in a bloodless field; leaving a little tissue is better than incontinence.
- Haemostasis: arterial bleeders must be fulgurated; venous ooze settles with catheter traction.
- Evacuate chips (Ellik evacuator); three-way catheter.
- **Also know:** TUVP (transurethral vaporisation, button electrode; ≤80 mL); TUIP (transurethral incision, <30 mL, no median lobe); bipolar enucleation; high bleeding risk: AUA prefers HoLEP (holmium laser enucleation) or PVP (photoselective vaporisation).

### Troubleshooting a loop that will not cut

`Scope–generator connection → foot pedal → irrigant matches generator (glycine for monopolar, saline for bipolar) → patient grounding (pad contact)`

### TUR syndrome (monopolar)

- Irrigant absorbed through open sinuses (venous pressure about 10 mmHg) → **dilutional hyponatraemia**: confusion, visual disturbance, nausea, bradycardia, hypertension, seizures; glycine also yields ammonia.
- **Risk factors:** gland >45 mL, resection >90 minutes, high irrigation bag.
- **Treatment:** stop surgery, check sodium; 3% saline 100 mL boluses for neurological symptoms; avoid over-rapid correction (central pontine myelinolysis).

### Outcomes and evidence

- About 80% symptom improvement; low retreatment; retrograde ejaculation about two-thirds.
- Meta-analyses: monopolar and bipolar have equal efficacy up to 5 years; b-TURP safer (NNT about 50 to prevent one TUR syndrome, about 20 to prevent one clot retention).

### Recent advances

- Bipolar plasma resection; chip-on-tip digital optics; laser enucleation scopes; single-use resectoscopes.
- **Also know:** TURis (sheath as return) caused strictures; bipolar enucleation and vaporisation (PlasmaButton).

### Diagrams to draw

- Assembled resectoscope labelling sheaths, working element, loop and flow channels.
- Resection field with clock positions, bladder neck, verumontanum and sphincter.
- Monopolar circuit (loop → patient → pad) versus bipolar (loop to return electrode on the same resectoscope).

> **Key points to remember:**
>
> - Continuous-flow 24–28 Fr sheath, 30° telescope.
> - Tungsten loop (monopolar) vs platinum–iridium (bipolar).
> - Glycine 1.5% for monopolar, saline for bipolar; bag 60 cm.
> - Bladder neck to verumontanum, depth to capsule.
> - Median lobe first; apex last.
> - Bipolar allows saline; TUR syndrome practically eliminated.
> - Equal efficacy; bipolar has less bleeding and shorter catheter time.
> - TUR syndrome risk: >45 mL gland, >90 minutes.

## SAQ 4 Lasers in BPH, including green light

P2 · P3 · P4 **merged from 3**

*Asked as:* P3 SAQ "Lasers in BPH, including HoLEP" (Various lasers in treatment of BPH; HOLEP) · P2 SAQ "LASERS in BPH." · P4 SAQ "Green light lasers"

*HOLEP read as: holmium laser enucleation of the prostate for BPH (benign prostatic hyperplasia).*

## Principle

- LASER (light amplification by stimulated emission of radiation) turns light into local heat. The tissue effect depends on wavelength and **chromophore** (water or haemoglobin), which set **penetration depth**.
- About 60 °C coagulates tissue (delayed sloughing); ≥100 °C vaporises it (immediate removal). Coagulation lasers are obsolete.

## Lasers used in BPH

| Laser | Wavelength | Chromophore | Penetration | Procedure |
|---|---|---|---|---|
| **Ho:YAG** (holmium:yttrium-aluminium-garnet), pulsed | ~2120 nm | Water | 0.4 mm | **HoLEP** (holmium laser enucleation of the prostate) |
| **Thulium:YAG** (continuous) / **TFL** (thulium fibre laser) | 2013 / 1940 nm | Water (TFL at absorption peak, about 4× holmium) | 0.2 mm | ThuLEP (enucleation), ThuVEP (vapoenucleation), ThuVARP (vaporesection), vaporisation |
| **KTP (potassium titanyl phosphate) / LBO (lithium triborate)** | 532 nm | Haemoglobin | 0.8 mm | **PVP** (photoselective vaporisation), 120–180 W |
| Diode | 940–1470 nm (various) | Water and haemoglobin (mixed) | 0.5–5 mm | Vaporisation, enucleation |
| Nd:YAG (neodymium) | 1064 nm | Weak | About 1 cm | Coagulation; abandoned (sloughing) |

## Enucleation versus vaporisation

| Feature | Enucleation (HoLEP, ThuLEP) | Vaporisation (PVP) |
|---|---|---|
| Tissue | Whole transition zone to the capsule, then morcellated | Removed layer by layer |
| Histology | Yes | No |
| Gland size | **Any (size-independent)** | Up to 80 cc |
| Retreatment | Rare | Higher, especially >80 cc |

## HoLEP

- **HoLEP** removes the whole adenoma along the **surgical capsule**: the endoscopic equivalent of open simple prostatectomy.
- **Ho:YAG, 2120 nm**: water absorbs it; pulsed; **0.4 mm penetration**; coagulates vessels. Saline irrigation, so no TUR (transurethral resection) syndrome.
- **Indications: size-independent** (<30, 30–80, >80 mL), with or without median lobe; **anticoagulated or high bleeding risk** (AUA [American Urological Association]); alternative to open prostatectomy for large glands.

- **Steps:** 60–150 W laser, 550 µm end-firing fibre, 24–28 Fr scope; **about 2.0 J × 40–50 Hz** for enucleation. 1. Bladder-neck incisions at 5 and 7 o'clock (median lobe) or one at 6 o'clock. 2. Perimontanal incision; **early apical release** protects the sphincter. 3. Develop the capsular plane to 12 o'clock; push lobes into the bladder (**en bloc, two- or three-lobe**). 4. Haemostasis with a defocused beam. 5. **Morcellation** with a full bladder; three-way catheter.
- **Limitation: steep learning curve** (about 50 cases).

### Green light (KTP/LBO) lasers: physics

- A 1064 nm Nd:YAG (neodymium:yttrium-aluminium-garnet) beam is passed through a **KTP or LBO crystal**. This doubles its frequency and halves the wavelength to **532 nm** (green).
- Green light is absorbed by **haemoglobin**, not water, so vascular prostate absorbs it avidly and vaporisation is bloodless.
- Optical penetration **0.8 mm**, tissue effect 1–3 mm, thin (0.2 mm) coagulation rim. The result is bloodless **photoselective vaporisation**.

### Green light platforms

| System | Power | Note |
|---|---|---|
| KTP | 80 W (later 100 W) | Popularised PVP (photoselective vaporisation of the prostate) |
| LBO HPS (High Performance System) | 120 W | Faster vaporisation |
| LBO XPS | **180 W** | 750 µm MoXy fibre: metal tip, fluid cooling |

### PVP technique

- 23–24 Fr cystoscope; side-firing fibre emitting at **70°**; saline irrigation.
- Open the bladder neck first, away from the orifices; sweep 15–30° in 1–2 seconds at ≤2 mm.
- Large bubbles mean efficient vaporisation. Pulsatile bleeding: coagulation mode (about 30 W).
- More than 5 kJ/cc predicts complete vaporisation.

### Indications by guideline (AUA)

- HoLEP or ThuLEP for any size.
- PVP for prostates <30 cc and 30–80 cc (under 80 cc); not >80 cc (high retreatment).
- HoLEP, ThuLEP or PVP for men at **higher bleeding risk** (anticoagulated men, who can continue anticoagulants with PVP).

### Evidence and outcomes

- **HoLEP: size-independent**; the only endoscopic procedure **superior to TURP** (transurethral resection of the prostate) for IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate), durable. Less bleeding, shorter catheter time and stay; longer operating time. Retreatment rare (whole transition zone removed).
- **ThuLEP:** outcomes equal to HoLEP.
- **GOLIATH RCT** (randomised controlled trial; 29 European centres, about 270 men, prostate ≤100 cc; 180 W PVP vs TURP): PVP **non-inferior** for IPSS, Qmax and complications at 2

years; shorter catheter time and stay; TURP faster. Retrograde ejaculation (about two-thirds) and reoperation equal. Retreatment high above 80 cc.

### Complications

- Dysuria and storage symptoms (dysuria worst after PVP, the highest of transurethral techniques); UTI (urinary tract infection) and epididymitis from coagulum after PVP.
- Transient storage symptoms and early / transient stress incontinence (SUI, stress urinary incontinence).
- Retrograde ejaculation in most (**about 75%**, three-quarters, after HoLEP).
- Bladder neck contracture (BNC) 0–3.2% (P2: 0–3%); stricture; capsular perforation (0.2–1% with PVP); ureteric orifice injury; **morcellator bladder injury**.
- No TUR syndrome (saline); transfusion very rare; erectile function unchanged; PVP leaves **no tissue for histology**.
- **Safety:** wavelength-specific eyewear for all; laser safety officer.

### Recent advances

- **Moses pulse modulation**; ThuLEP (thulium enucleation); median-lobe-only HoLEP (antegrade ejaculation about 90%); same-day discharge.
- GreenLight enucleation; office and same-day catheter-free pathways.

### Diagrams to draw

- HoLEP: sagittal prostate with enucleation plane along the capsule; adenoma enucleated to the capsule, pushed into the bladder, then morcellated.
- PVP: side-firing fibre at 70° vaporising a lateral lobe.

> **Key points to remember:**
> - Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
> - Ho:YAG 2120 nm, water chromophore, 0.4 mm.
> - HoLEP is size-independent and the only endoscopic method superior to TURP; suits anticoagulated men.
> - Retrograde ejaculation about 75% after HoLEP.
> - Green light: 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration; 180 W XPS with MoXy fibre.
> - GOLIATH: PVP non-inferior to TURP; best for anticoagulated men; avoid PVP above 80 cc.

## SAQ 5 Silodosin

P2 · P4 **merged from 2**

*Asked as:* P4 SAQ "Silodosin" (Silodosin.) · P2 SAQ "Silodosin."

### Definition and pharmacology

- **Uroselective α1-blocker**, the **most α1A-selective** agent (well over 100-fold vs α1B, or by another source several hundredfold over α1B; about 50-fold vs α1D), for LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia).
- Sparing vascular α1B gives **minimal hypotension** (few blood-pressure effects). Blocking α1A in the vas and seminal vesicles causes **failure of emission**.
- Onset: flow improves within 2–6 hours, symptoms within 3–4 days. Metabolised by CYP3A4.

## Mechanism of action

- **α1A receptors** (about 70% of prostatic α1 receptors) dominate in the prostate, bladder neck and prostatic urethra. Noradrenaline on them causes the **dynamic component** of BOO (bladder outlet obstruction).
- Blockade relaxes this smooth muscle.

## Dose and contraindications

| Item | Detail |
|---|---|
| Dose | **8 mg once daily with a meal**; no titration |
| Renal: CrCl (creatinine clearance) 30–50 mL/min | **4 mg daily** |
| Contraindicated | CrCl <30; severe liver (hepatic) impairment; strong CYP3A4 inhibitors (e.g. ketoconazole) |
| Caution | Other α-blockers (avoid), PDE5 (phosphodiesterase 5) inhibitors, planned cataract surgery |

## Indications and efficacy

| Indication | Evidence |
|---|---|
| **LUTS due to BPE (benign prostatic enlargement)** | Japanese, American and European 12-week RCTs (randomised controlled trials): IPSS (International Prostate Symptom Score) and flow better than placebo; **non-inferior to tamsulosin 0.4 mg** (**Chapple 2011**, European trial), not better; effective in severe LUTS |
| **MET (medical expulsive therapy)** | Distal ureteric stones 5–10 mm; α-blockers reduce ureteric spasm; meta-analyses show higher, faster expulsion than tamsulosin; **SUSPEND (2015)** found no overall tamsulosin benefit |
| CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome) | α-blocker-naive men with voiding symptoms; of recent onset |

- **Greatest fall in BOOI (bladder outlet obstruction index)** of all α1-blockers (subgroup data).
- Does not shrink the prostate; add a 5-ARI (5α-reductase inhibitor) for large glands.
- **Also know:** before ureteroscopy (easier access); with propiverine for storage symptoms; nocturia (modest).

## Adverse effects

- **Ejaculatory dysfunction / anejaculation (signature effect):** over a quarter of men in registration trials (about 14% vs 2% with tamsulosin head-to-head), yet few stop the drug; men with anejaculation had better responses. Mechanism: **emission failure** (vas and seminal vesicle α1A blockade) and bladder-neck relaxation. Reversible. Counsel young men.
- **Cardiovascular:** dizziness; orthostatic hypotension rare; no QT change; no excess hypotension in the elderly or with antihypertensives.
- Falls in older men: falls and fractures rise in men ≥66 on any α-blocker.
- **IFIS (intraoperative floppy iris syndrome):** not specifically studied but reported; class effect; warn the ophthalmologist before cataract surgery (stopping the drug does not remove risk).

### Comparison with other α-blockers

| Drug | Selectivity | Dose | Hypotension | Ejaculatory dysfunction | Efficacy in LUTS |
|---|---|---|---|---|---|
| **Silodosin** | α1A (highest); α1A ≫ α1D, α1B | 8 mg OD | Lowest | **Highest** | Equal |
| Tamsulosin | α1A, α1D > α1B | 0.4 mg OD | Low | Moderate | Equal |
| Alfuzosin | Clinically uroselective | 10 mg OD | Low | Lowest | |
| Naftopidil | α1D | 25–75 mg | Some | Less | |
| Doxazosin, terazosin | Non-selective; titrate | 1–8 mg, 1–10 mg | Highest | Low | |

### Diagram to draw

- α1A in prostate and vas, α1B in vessels, α1D in bladder; prostate and bladder neck showing α1A receptor sites blocked by silodosin.

> **Key points to remember:**
> - Most α1A-selective α-blocker; 8 mg with a meal; no titration.
> - 4 mg if CrCl 30–50; avoid if <30.
> - Equal efficacy to tamsulosin 0.4 mg; greatest BOOI fall; better for MET.
> - Anejaculation is the signature effect; least hypotension.

## SAQ 6 Methods to estimate prostate size

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 1, Q3f (TRUS also answered in 2017-Winter Paper 1, Q3e)

### Why measure

- Chooses treatment: 5ARI (5α-reductase inhibitor) if >30 mL (AUA (American Urological Association)) or >40 mL (EAU (European Association of Urology)); type of BPH (benign prostatic hyperplasia) surgery.
- **PSAD (PSA density)** = PSA (prostate-specific antigen) ÷ volume; >0.15 suggests cancer. Also AUR (acute urinary retention) risk, brachytherapy planning.
- Normal adult gland 20–25 g; about 48 g at 60–70 years. **1 cm³ ≈ 1 g**.

### Methods

| Method | Technique | Accuracy and use |
|---|---|---|
| **DRE (digital rectal examination)** | Estimate size, consistency | **Underestimates** above 30 mL (lateral lobes impalpable); flat gland predicts <30 mL |
| **PSA as surrogate** | Age-specific cut-offs for **>40 mL:** >1.6 ng/mL (50s), >2.0 (60s), >2.3 (70s) | AUC (area under curve) 0.76–0.78; within ±20% of TRUS in most |
| **TAUS (transabdominal ultrasound)** | Full bladder; transverse width and height, sagittal length | Less precise than TRUS; also PVR (post-void residual) and **IPP (intravesical prostatic protrusion)** |

| Method | Technique | Accuracy and use |
|---|---|---|
| **TRUS (transrectal ultrasound)** | Ellipsoid formula; planimetry | **Method of choice** |
| **MRI (magnetic resonance imaging)** | T2-weighted ellipsoid or segmentation (manual or automated) | Most accurate and reproducible |
| **CT (computed tomography)** | Axial images | Overestimates; poor capsule definition |
| **Cystoscopy** | Urethral length, lobes, median lobe | Rough; guides procedure choice |

### TRUS formulas

| Shape assumed | Formula |
|---|---|
| **Ellipsoid (standard)** | **π/6 (0.52) × transverse × AP (anteroposterior) × length** |
| Sphere | π/6 × transverse³ |
| Prolate spheroid | π/6 × transverse² × AP |
| **Planimetry** | Sum of serial transverse areas |

- Transverse and AP at the widest axial section; length sagittally. All formulas correlate >0.90 with specimen weight.
- **Planimetry:** probe on a stepper, images every 3–5 mm, areas summed; most precise; used for brachytherapy.

### IPP grading (suprapubic, sagittal, bladder 100–200 mL)

- Grade I <5 mm, II 5–10 mm, III >10 mm; ≥10 mm predicts BOO (bladder outlet obstruction) and failed trial without catheter.

**Diagram to draw:** TRUS transverse and sagittal views with three measured diameters.

> **Key points to remember:**
> - Ellipsoid volume = 0.52 × T × AP × L; 1 mL ≈ 1 g.
> - DRE underestimates above 30 mL; PSA >1.6/2.0/2.3 by decade predicts >40 mL.
> - Planimetry most precise; MRI most reproducible.

## SAQ 7 Catheter free trial

P2

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 2, Q3e

*Assumed meaning: trial without catheter (TWOC) after acute urinary retention.*

### Definition

- **TWOC (trial without catheter)**: catheter removal after decompression to test whether the patient voids adequately.
- **AUR (acute urinary retention)**: sudden, painful inability to void; commonest in older men with LUTS (lower urinary tract symptoms).

### Types of AUR

| Spontaneous | Precipitated |
|---|---|
| BPH (benign prostatic hyperplasia) progression | Anaesthesia, surgery, UTI (urinary tract infection), overdistension, anticholinergic or sympathomimetic drugs |
| TWOC fails more often; about 75% later need surgery | Better TWOC outcome; about a quarter need surgery |

### Initial management

- Catheterise; record drained volume; check creatinine; treat the trigger.
- Start an **α1-blocker** (alfuzosin or tamsulosin).

### Rationale for α-blockers

- Sympathetic bladder-neck tone (dynamic component) adds to retention; α1-blockers relax it.
- **ALFAUR (Alfuzosin in Acute Urinary Retention)**: alfuzosin 10 mg for 3 days → TWOC success **62% vs 48%**.
- Meta-analysis: success relative risk about 1.5; fewer recurrences.

### Procedure

- Remove the catheter after **2–3 days of α-blocker**, usually in the morning (no evidence favours short or long catheterisation).
- Encourage fluids; record voided volumes and **PVR (post-void residual)** by bladder scan.
- **Success**: comfortable voiding, low residual (unit threshold). **Failure**: recatheterise or start CISC (clean intermittent self-catheterisation).

### Predictors of failure

- Advanced age; **drained volume >1 L**; spontaneous AUR; severe LUTS.
- Ultrasound: **IPP (intravesical prostatic protrusion) ≥10 mm**; thick bladder wall.

### Results

- Success about **60%**; a second or third TWOC still succeeds in over a quarter.
- Recurrence after success is common, mostly within **6 months**; then offer surgery.

### After TWOC

- **Success**: continue α-blocker; add a **5-ARI (5α-reductase inhibitor)** for large glands, because 5-ARIs, not α-blockers, prevent further AUR.
- **Failure**: TURP (transurethral resection of prostate) or other surgery; CISC or long-term catheter if unfit.
- **Also know:** immediate surgery has higher morbidity; TWOC also follows prostate surgery.

**Diagram to draw:** algorithm: AUR → catheter + α-blocker → TWOC at 2–3 days → success / failure.

> **Key points to remember:**
> - TWOC is first choice after AUR; success about 60%.
> - ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
> - Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

## SAQ 8 Irrigation fluids for TURP.

P2

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 2, Q3a

### Ideal irrigant

- Transparent, **non-conducting** (monopolar), isotonic, non-toxic, non-haemolytic, cheap.
- **Why non-conducting?** In monopolar TURP (transurethral resection of the prostate), current flows loop → patient → pad; ionic saline would disperse it. **Bipolar** TURP completes the circuit in the instrument, so it uses **normal saline**.

### Fluids

| Fluid | Osmolality (approx.) | Points |
|---|---|---|
| Sterile water | 0 | Best vision, cheap; haemolysis, hyponatraemia; now avoided for TURP |
| **Glycine 1.5%** | ~200 mOsm/kg | Commonest (UK, North America); glycine → ammonia, serine; transient blindness (retinal inhibitory neurotransmitter), encephalopathy |
| Sorbitol 3% | ~165 | Metabolised to fructose and glucose → hyperglycaemia; lactic acidosis |
| Mannitol 5% | ~275 | Near iso-osmolar; not metabolised; osmotic diuresis; volume overload in renal failure |
| Sorbitol 2.7% + mannitol 0.54% | ~180 | Continental Europe |
| **Normal saline 0.9%** | ~308 | Bipolar TURP, lasers (HoLEP, PVP); no TUR syndrome |
| - Plasma is about 290 mOsm/kg, so monopolar fluids are **hypo-osmolar**. | | |

### TUR (transurethral resection) syndrome

- **Mechanism**: prostatic venous pressure is ~10 mmHg; irrigant at higher pressure enters open sinuses → volume overload, **dilutional hyponatraemia** and hypo-osmolality → cerebral oedema; plus glycine toxicity.
- **Features**: confusion, nausea, vomiting, visual disturbance, hypertension then hypotension, bradycardia, seizures, coma.
- **Risk factors**: gland >45 mL, resection >90 minutes (VA (Veterans Affairs) study), high irrigation pressure, open sinuses or capsular perforation.
- **Prevention**: bag height ~60 cm (each extra 10 cm more than doubles absorption), continuous-flow resectoscope, time limit, spinal anaesthesia (awake patient), bipolar TURP. Ethanol 1% tagging can monitor absorption.
- **Treatment**: stop surgery, check sodium; Na <120 mEq/L serious. Furosemide for overload; **3% saline** 100 mL boluses (max 300 mL) for neurological signs; raise sodium by 4–6 mEq/L initially, not over 12 mEq/L in 24 h (risk of central pontine myelinolysis).

### Bipolar advantage

- Meta-analyses: no TUR syndrome, less transfusion and clot retention; NNT (number needed to treat) about 50 to prevent one TUR syndrome.

**Diagram to draw:** monopolar vs bipolar current pathways.

> **Key points to remember:**
> - Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
> - Glycine 1.5%: ~200 mOsm/kg; transient blindness.
> - Risk: gland >45 mL, time >90 min; bag height 60 cm.
> - 3% saline; ≤12 mEq/L rise in 24 h.

## SAQ 9 Dutasteride.

P2

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 2, Q3c

### Pharmacology

- A **dual 5ARI (5α-reductase inhibitor)**: it blocks **type 1** (liver, skin) and **type 2** (prostate stroma, genital tissue) isoenzymes. Finasteride blocks type 2 only.
- **Mechanism:** blocks conversion of testosterone to DHT (dihydrotestosterone), the main prostatic androgen; serum DHT falls about **90%**. Epithelial atrophy shrinks the gland and relieves the **static** component of obstruction.
- **Dose: 0.5 mg once daily**; hepatic metabolism; **half-life about 5 weeks**.

### Effects

- **Prostate volume falls about 25%** (maximum by 6 months).
- Symptoms and Qmax (maximum flow rate) improve modestly and slowly (**at least 6 months**).
- **Changes natural history:** pivotal trials: AUR (acute urinary retention) down 57%, BPH (benign prostatic hyperplasia) surgery down 48%.
- **PSA (prostate-specific antigen) halves** by 6–12 months: **double the PSA** to interpret it. A rise from the nadir on treatment needs evaluation.

### Indications

- Moderate-to-severe LUTS (lower urinary tract symptoms) with an enlarged prostate: **>40 mL (EAU [European Association of Urology])** or PSA about >1.5 ng/mL; AUA (American Urological Association): >30 mL.
- **Combination with tamsulosin** for men at risk of progression (fixed-dose capsule available).
- Off-label: BPH haematuria (lowers prostatic microvessel density); before TURP (transurethral resection of the prostate) to reduce bleeding (inconsistent evidence).

### Landmark trials

| Trial | Result |
|---|---|
| **CombAT** (Combination of Avodart and Tamsulosin, 4 years; PV [prostate volume] ≥30 mL, PSA ≥1.5) | Combination best for symptoms and progression; dutasteride (and combination) beat tamsulosin for AUR and surgery |
| **REDUCE** (Reduction by Dutasteride of Prostate Cancer Events, 4 years) | Prostate cancer down about 23%, but slightly more Gleason 8–10; **not licensed for prevention** |

### Side effects

- **Sexual:** ED (erectile dysfunction), low libido, ejaculatory disorders; lessen with time.
- Gynaecomastia and breast tenderness; depression (FDA [Food and Drug Administration] warning).
- **No IFIS** (intraoperative floppy iris syndrome) and no hypotension, unlike α-blockers.

### Precautions

- **Teratogenic:** pregnant women must not handle leaking capsules. **No blood donation for 6 months** after the last dose.

**Diagram to draw:** testosterone to DHT pathway with 5α-reductase types 1 and 2 blocked by dutasteride.

> **Key points to remember:**
> - Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
> - Volume down 25%; AUR and surgery down about half; needs 6 months.
> - Double the PSA; CombAT and REDUCE.

## SAQ 10 Measures to decrease complications of TURP

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 9 Paper 2, Q3b (TUR syndrome also answered in 2017-Winter Paper 3, Q3f)

### Introduction

- TURP (transurethral resection of the prostate) is the benchmark; overall morbidity approaches one-fifth, mostly preventable.

### Preoperative measures

- **Selection:** AUA (American Urological Association) 2023: TURP for glands up to 80 mL; larger glands need HoLEP (holmium laser enucleation) or simple prostatectomy.
- **Sterile urine:** treat UTI (urinary tract infection) first; single-dose prophylaxis covering gram-negatives and enterococci.
- **Anticoagulants:** stop after advice from the prescriber; if they cannot be stopped, choose HoLEP, ThuLEP or PVP (photoselective vaporisation).
- **Finasteride** for recurrent BPH (benign prostatic hyperplasia) bleeding.
- Spinal anaesthesia shows early TUR syndrome; pad the peroneal nerve.

### Intraoperative measures

| Complication | Prevention |
|---|---|
| **TUR syndrome** (hypo-osmolar fluid absorbed, dilutional hyponatraemia) | **Bipolar TURP with saline**; else irrigant ≤60 cm high, resection <90 min, avoid capsular perforation, check sodium in long resections |
| **Bleeding** | Systematic resection to capsule; control arterial bleeders as you go; bipolar "cut and seal"; catheter traction for venous ooze |
| **Ureteric orifice injury** | Identify orifices first; take a large median lobe down first |
| **Perforation** | Respect the capsule; visible fat means perforation |
| **Incontinence** | **Stop at the verumontanum**; resect apex last in a bloodless field |
| **Stricture** | Calibrate or dilate meatus, lubricate generously, correct sheath size |
| **Bladder neck contracture** | Avoid over-resection and excess fulguration; TUIP (transurethral incision of the prostate) for small glands |

### Postoperative measures

- Large-bore three-way catheter; careful **continuous bladder irrigation** (blocked outflow distends the bladder and worsens bleeding).
- Even routine cases absorb 800–1000 mL of irrigant; check sodium if confused.
- TUR syndrome treatment: **3% saline 100 mL** boluses; raise sodium by no more than 12 mEq/L in 24 h (pontine myelinolysis).
- Stool softeners; no cycling for 4–6 weeks; counsel about **retrograde ejaculation** (about two-thirds).

**Diagram to draw:** resection field from bladder neck to verumontanum with the external sphincter marked.

> **Key points to remember:**
> - Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
> - Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
> - Verumontanum is the distal limit.

## SAQ 11 Tamsulosin

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 12 Paper 2, Q3d

### Introduction

- A **uroselective α1-adrenergic antagonist**, first-line drug for bothersome male LUTS (lower urinary tract symptoms) due to BPH (benign prostatic hyperplasia) (AUA [American Urological Association], EAU [European Association of Urology]).

## Receptor basis

| Subtype | Site | Effect of blockade |
|---|---|---|
| **α1a** (about 70% in prostate) | Prostate, bladder neck, prostatic urethra, iris dilator | Relieves dynamic obstruction; ejaculatory dysfunction; IFIS (intraoperative floppy iris syndrome) |
| α1b | Blood vessels | Hypotension (tamsulosin has 10-fold less affinity) |
| α1d | Detrusor, spinal afferents | Storage symptom relief |

## Mechanism

- Blocks noradrenaline at α1a receptors → relaxes prostatic and bladder-neck smooth muscle → lowers the **dynamic component** of BOO (bladder outlet obstruction). It does not shrink the prostate.

## Pharmacology and dose

- **0.4 mg once daily** after the same meal (up to 0.8 mg); no titration.
- Modified release; hepatic metabolism (CYP3A4, CYP2D6); avoid strong CYP3A4 inhibitors (ketoconazole).
- Flow rises within 4–8 hours, symptoms by 1 week; 0.2 mg is not better than placebo.

## Indications

1. **BPH/LUTS:** IPSS (International Prostate Symptom Score) falls about 30–40%; Qmax rises; storage and nocturia improve.
2. **Combination with 5ARI** (5α-reductase inhibitor) for prostate >30–40 mL: **CombAT** (dutasteride + tamsulosin) and MTOPS (combination cut progression by 66%).
3. **MET (medical expulsive therapy):** distal ureteric stones 5–10 mm; SUSPEND (2015) negative overall but included all sizes.
4. **AUR (acute urinary retention):** before trial without catheter.
5. **Also know:** ureteric stent symptoms; female functional outlet obstruction.

## Adverse effects

- **Ejaculatory dysfunction** (dose-dependent; less than silodosin): retrograde ejaculation from bladder-neck relaxation, or anejaculation (emission failure).
- **IFIS:** floppy, billowing iris, iris prolapse, progressive miosis during cataract surgery; **highest risk with tamsulosin**. Stopping the drug does not remove the risk, so warn the ophthalmologist.
- Dizziness, rhinitis, headache; less orthostatic hypotension than non-selective agents. Caution: sulfonamide allergy.

## Comparison

| | Tamsulosin | Silodosin | Alfuzosin |
|---|---|---|---|
| Selectivity | α1a/α1d | Most α1a | Clinical only |
| Ejaculatory effect | Moderate | Highest | Lowest |

**Diagram to draw:** prostate cross-section showing α1a receptors in stroma and bladder neck.

> **Key points to remember:**
> - 0.4 mg once daily; α1a/α1d selective; no titration.
> - IFIS highest with tamsulosin; persists after stopping.
> - MET benefit in distal stones 5–10 mm.

## SAQ 12 Prostatic stents

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 2, Q3f (urethral stents also answered in Practice Set 4 Paper 4, Q3f and 2015-Summer Paper 4, Q3d)

### Definition

- A device placed in the **prostatic urethra**, between bladder neck and verumontanum, to relieve BOO (bladder outlet obstruction) without resection. Mainly for men **unfit for surgery**.

### Classification

| Type | Examples | Features |
|---|---|---|
| **Temporary, first generation** | Urospiral, Prostakath (spirals) | Steel or gold-plated coil; not epithelialised |
| **Temporary, second generation** | **Memokath** (thermo-expandable nitinol); Allium (covered nitinol) | Expands with warm water, softens with cold water for removal |
| Temporary polyurethane | Intraurethral catheter, Spanner | Bulbar anchor; sphincter left free |
| Biodegradable | Polylactic acid stents | Dissolve; trial use |
| **Permanent, epithelialising** | **UroLume** (woven mesh) | Urothelium covers the mesh; withdrawn |

### Indications

- Retention or severe BOO from BPH (benign prostatic hyperplasia) in men unfit for surgery: avoids a long-term catheter.
- Drainage after thermal treatments; trial of de-obstruction.
- **Also know:** DSD (detrusor–sphincter dyssynergia), now abandoned; bulbar stricture.

### Contraindications

- Poor detrusor contractility (do urodynamics if in doubt); untreated UTI (urinary tract infection); large median lobe; bladder stones; stricture.

### Technique

- Measure prostatic urethral length cystoscopically. Place under local anaesthesia with endoscopic or fluoroscopic guidance. The stent must **end above the external sphincter**, or incontinence follows.

### Complications

- **Migration** (commonest with temporary stents); **encrustation**; UTI; irritative symptoms, perineal pain, haematuria.

- **Hyperplastic ingrowth** and restenosis with permanent stents (difficult removal); incontinence if the sphincter is crossed.

### Current status

- Most unfit men void after placement, but many stents are later removed.
- **EAU (European Association of Urology):** alternative to catheterisation in men unfit for surgery (weak). Not in the **AUA (American Urological Association) 2023** BPH list.
- **Recent: iTind**: temporary nitinol device left 5–7 days; it remodels the urethra by ischaemic pressure necrosis. AUA 2023 option for 25–75 mL without median lobe. **Optilume BPH** (paclitaxel balloon).

**Diagram to draw:** sagittal prostate with a Memokath from bladder neck to verumontanum, above the external sphincter.

> **Key points to remember:**
> - Temporary (Memokath) vs permanent (UroLume, withdrawn).
> - Need detrusor contractility; never cross the sphincter.
> - Migration, encrustation, ingrowth; iTind is the modern temporary device.

## SAQ 13 Role of UDS before prostate surgery

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 2, Q3a

### Why consider UDS (urodynamic study)

- Symptoms, flow rate, PVR (post-void residual), prostate volume and cystoscopy **cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity)**: low Qmax (maximum flow rate) occurs in both. DUA is found in 9–48% of men with LUTS (lower urinary tract symptoms) studied.
- Surgery helps obstruction, not a weak bladder. **PFS (pressure–flow study) is the gold standard** for BOO.

### Components

- **Filling cystometry:** sensation, compliance, **DO (detrusor overactivity)**.
- **PFS:** detrusor pressure at Qmax plotted against flow (ICS (International Continence Society) nomogram).

| Index | Formula | Interpretation |
|---|---|---|
| BOOI (bladder outlet obstruction index) | PdetQmax − 2Qmax | >40 obstructed; 20–40 equivocal; <20 unobstructed |
| BCI (bladder contractility index) | PdetQmax + 5Qmax | >150 strong; 100–150 normal; <100 weak |

- High pressure, low flow = BOO; low pressure, low flow = DUA.
- **Videourodynamics** adds anatomy (diverticula, reflux).

### Indications before BPH (benign prostatic hyperplasia) surgery (EAU (European Association of Urology))

- Previous failed invasive treatment.
- Cannot void >150 mL (unreliable flow).
- PVR >300 mL.
- Age >80 years with mainly voiding LUTS.
- Age <50 years with mainly voiding LUTS.
- **Also:** suspected neurogenic bladder, prior pelvic surgery or radiation, Qmax >10 mL/s (AUA (American Urological Association): diagnostic uncertainty).

### How results change management

- **Unobstructed or DUA:** poorer surgical results; counsel, consider CIC (clean intermittent catheterisation).
- **DO with BOO:** storage symptoms persist in about one-third after TURP (transurethral resection of the prostate); warn the patient.
- **Low compliance:** the only "absolute" urodynamic indication to relieve BOO (protects upper tracts).

### Evidence and limitations

- Not routine. **UPSTREAM RCT (randomised controlled trial):** routine UDS did not change IPSS (International Prostate Symptom Score) or surgery rates.
- Invasive: UTI (urinary tract infection), discomfort, cost. Before **radical prostatectomy**, not routine.

**Diagram to draw:** ICS nomogram with obstructed, equivocal and unobstructed zones.

> **Key points to remember:**
> - PFS is the only test that separates BOO from DUA.
> - BOOI >40 obstructed; BCI <100 weak.
> - EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
> - UPSTREAM: routine UDS did not change outcomes.

## SAQ 14 IPSS score

P3

**Type:** SAQ (10 marks)
**Source:** 2022-Summer Paper 3, Q3e

### Definition

- **IPSS (International Prostate Symptom Score):** validated, self-administered questionnaire grading male LUTS (lower urinary tract symptoms) over the past month.
- Based on the **AUA-SI (American Urological Association Symptom Index, 1992)**; adopted by WHO (World Health Organization) in 1993 by adding one quality-of-life question.

### Items

| No. | Symptom | Phase |
|---|---|---|
| 1 | Incomplete emptying | Voiding |

| No. | Symptom | Phase |
|---|---|---|
| 2 | Frequency (again within 2 hours) | Storage |
| 3 | Intermittency | Voiding |
| 4 | Urgency | Storage |
| 5 | Weak stream | Voiding |
| 6 | Straining | Voiding |
| 7 | Nocturia | Storage |

- **Scoring:** items 0–5. 0 not at all; 1 less than 1 time in 5; 2 less than half the time; 3 about half; 4 more than half; 5 almost always. **Nocturia scored by number of times** (0 to ≥5).
- **Total 0–35:** voiding subscore 4 items, storage 3 (not validated).

| Severity | Score |
|---|---|
| Mild | 0–7 |
| Moderate | 8–19 |
| Severe | 20–35 |

- **QoL (quality of life)/bother question:** "If you were to spend the rest of your life with your urinary condition as it is now, how would you feel?" Scored 0 (delighted) to 6 (terrible); reported separately, outside the 35.

### Uses

- First step in LUTS work-up (AUA and EAU [European Association of Urology]); grades severity and bother.
- Guides treatment: mild, non-bothersome → watchful waiting; bothersome → medical therapy or surgery.
- Monitors response: **a 3-point fall is the minimum perceptible change**.
- Standard end point in trials.

### Limitations

- **Not disease-specific:** women score similarly; does not diagnose BOO (bladder outlet obstruction) or BPH (benign prostatic hyperplasia); correlates poorly with urodynamics, prostate size and flow.
- No incontinence, pain or post-micturition items.
- Needs literacy.

### Alternatives

- **VPSS (Visual Prostate Symptom Score):** pictures, for low literacy.
- BPH Impact Index; ICIQ-MLUTS (International Consultation on Incontinence Questionnaire–Male LUTS); DAN-PSS (Danish Prostate Symptom Score); OAB-q (overactive bladder questionnaire); LURN (Lower Urinary Tract Research Network) SI-29 adds incontinence and pain.

**Diagram to draw:** IPSS chart with scoring columns.

**Key points to remember:**

- 7 items × 0–5 = 0–35; bands 0–7, 8–19, 20–35.
- QoL 0–6 is separate; IPSS = AUA-SI + QoL.
- 3-point change is noticeable; IPSS does not diagnose BOO.

## SAQ 15 Laparoscopic Trans vesical Prostatectomy

P3

**Type:** SAQ (10 marks)
**Source:** 2025-Winter Paper 3, Q3d

*Assumed meaning: laparoscopic transvesical (suprapubic) simple prostatectomy for large BPH (benign prostatic hyperplasia).*

### Introduction

- Laparoscopic version of the Freyer suprapubic operation: the adenoma is enucleated through a **cystotomy** in the **surgical capsule plane** (compressed peripheral zone). First LSP (laparoscopic simple prostatectomy): Mariano, 2002.

### Indications

- Gland **>80–100 g** with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or obstructive nephropathy.
- **Transvesical route preferred** for a large intravesical median lobe, bladder stones or a diverticulum needing surgery.
- Work-up: PSA (prostate-specific antigen) ± MRI (magnetic resonance imaging) to exclude cancer; TRUS (transrectal ultrasound) volume; cystoscopy.

### Steps

1. General anaesthesia, supine, Trendelenburg. **Ports:** 10-mm umbilical camera, two 10-mm working ports, 5-mm and 12-mm assistant ports. Transperitoneal or extraperitoneal (Retzius).
2. **Cystotomy** with stay sutures; identify ureteric orifices.
3. **Circumferential bladder-neck incision** (start 3 to 9 o'clock posteriorly), away from orifices.
4. Develop the **posterior plane first** to the apex, then the lateral lobes.
5. Divide anterior commissure; **transect urethra just proximal to the verumontanum** (protects sphincter).
6. **Figure-of-eight sutures at 5 and 7 o'clock** (prostatic arteries enter posterolaterally).
7. **Bladder-neck mucosal advancement** to urethral mucosa covers the fossa and reduces bleeding.
8. 22 Fr three-way catheter; two-layer cystotomy closure.

### Outcomes versus open

- Less blood loss, shorter catheter time and stay, less analgesia; **same functional results** (IPSS (International Prostate Symptom Score), flow rate, residual).
- Limitations: steep learning curve, poor ergonomics; **robotic simple prostatectomy** has largely replaced it. **Single-port transvesical robotic enucleation** (2.7 cm suprapubic

incision, pneumovesicum) allows same-day discharge.

### Complications

- Bleeding (transfusion rare), urine leak, ureteric orifice injury, bladder neck contracture (<1%; open 2–6%).
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence rare.

**Diagram to draw:** sagittal view of cystotomy, bladder-neck incision and enucleation plane.

> **Key points to remember:**
> - Indication: gland >80 g, especially with median lobe, stones or diverticulum.
> - Posterior plane first; divide urethra proximal to verumontanum.
> - 5 and 7 o'clock sutures plus mucosal advancement.
> - Same function as open, less morbidity.

## SAQ 16 UroLift (prostatic urethral lift)

P4

**Asked as:**

- Urolift. — *Winter 2024 Paper IV Q3d (SAQ)*
- Urolift. — *Practice Set 6 Paper IV Q3b (earlier version) (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2024-Winter Paper 4, Q3d)

*Read as PUL (prostatic urethral lift, UroLift) for BPH (benign prostatic hyperplasia).*

### Introduction

- A **non-ablative MIST (minimally invasive surgical therapy)**. Permanent implants pull (retract) the lateral lobes apart and open the prostatic urethra.
- **FDA (Food and Drug Administration) approval 2013**; extended to ≤100 mL in 2020.
- **Main advantage:** preserves erectile and **ejaculatory** function.

### Implant and mechanism

- Spring-loaded implant with T-shaped ends:
- **Capsular tab:** nitinol.
- **Urethral end:** stainless steel (resists encrustation).
- **Connector:** polyethylene terephthalate suture under tension.
- Delivered through a **19-gauge needle** from a cystoscopic device.
- The gland is more compliant than the capsule, so tension compresses the lobe outward; later local atrophy and fibrosis. Implants are MRI (magnetic resonance imaging)-conditional (≤3 T).

### Indications

- **AUA (American Urological Association): prostate 30–80 mL, no obstructing median lobe** (confirm by cystoscopy or ultrasound). EAU (European Association of Urology): <70 mL, no middle lobe, men wanting ejaculation preserved.
- Men wishing to stop medication.

- **Avoid:** obstructing median lobe (MedLift data non-randomised), large gland, active infection.

### Technique

1. Local anaesthesia; rigid cystoscope.
2. Compress lateral lobe, fire needle, deploy implant; repeat on the other side.
3. **Usually 4–6 implants**, anterolateral, from bladder neck to apex, avoiding the neurovascular bundles.
4. **Keep implants >1.5 cm distal to the bladder neck** (encrustation).
5. Most go home without a catheter.

### Evidence

| | **L.I.F.T. (2013)** | **BPH6 (2015)** |
|---|---|---|
| Design | PUL vs sham, 2:1, 19 centres | PUL vs TURP (transurethral resection of prostate), 10 European centres |
| Result | IPSS (International Prostate Symptom Score) about halved, twice the sham effect; Qmax (maximum flow) +4 mL/s; no new sexual dysfunction | Composite end point met 52.3% vs 20.0%; antegrade ejaculation 100% vs 60.6% |
| Durability / retreatment | 13.6% surgical retreatment at 5 years | 13.6% vs 5.7% at 2 years; TURP better IPSS, Qmax, PVR (post-void residual) |

- TURP wins on IPSS, Qmax and PVR (post-void residual).

### Complications

- Transient dysuria, haematuria, urgency (<2 weeks); encrustation; rare pelvic haematoma; device malfunction (misfire).
- Lower major adverse events but **higher retreatment than TURP**.

**Diagram to draw:** axial prostate with implants pulling lateral lobes to the capsule.

> **Key points to remember:**
> - 30–80 mL, no median lobe; local anaesthesia.
> - Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
> - TURP wins (better) on IPSS, flow and durability.

# Rapid Revision Sheet · Section 16

### Prostate anatomy, neurovascular bundles and surgical concepts in localised CA prostate (LAQ)

- PZ gives 70% of cancers; TZ gives BPH; CZ around ejaculatory ducts; no true capsule.
- NVB between prostatic and levator fascia, 5 and 7 o'clock; apex is the commonest margin and nerve-injury site.
- Interfascial plane is the standard nerve-sparing plane; avoid heat and traction; clips, cold scissors.
- Ligate DVC; excise Denonvilliers fascia; spare accessory pudendals.
- Veil and high release for low-risk sides (Tsuzuki).
- NeuroSAFE frozen section: more sparing, safe margins.
- Retzius-sparing: earlier continence, more positive margins.
- Extended PLND captures about 63% of drainage.

### HoLEP (LAQ)

- Ho:YAG 2120 nm, water chromophore, 0.4 mm; saline, no TUR syndrome.
- Size-independent; best (preferred) in anticoagulated men.
- Only endoscopic method superior to TURP; better IPSS and Qmax; durable; retreatment rare.
- Retrograde ejaculation about 75%; learning curve about 50 cases.

### BPH: medical management and surgical modalities (LAQ)

- α-blockers: rapid relief, no prevention.
- 5-ARIs: prevent AUR and surgery in glands >30–40 mL.
- MTOPS: combination cut progression 66% (↓66%); only 5-ARI-containing arms cut AUR and surgery.
- AUA sizes: TUIP ≤30; TURP/PVP/MISTs (minimally invasive surgical therapies) 30–80; HoLEP any size; simple prostatectomy >80 mL.
- TUR syndrome: gland >45 mL, resection >90 min; bipolar TURP avoids it.

### Millin's prostatectomy (SAQ)

- Millin 1945: transverse capsulotomy ~2 cm below bladder neck; bladder not opened.
- Apex divided under vision; 5 and 7 o'clock sutures.
- Retrograde ejaculation 80–90%; transfusion <5%; bladder neck contracture 2–6%.

### TUR syndrome (SAQ)

- Venous pressure 10 mmHg; risk gland >45 mL, time >90 min.
- Glycine: ammonia and transient blindness.
- 3% saline 100 mL boluses; ≤12 mEq/L in 24 h.
- Bipolar saline TURP prevents it.

### TURP instruments, techniques and monopolar vs bipolar (SAQ)

- Continuous-flow 24–28 Fr sheath, 30° telescope.
- Tungsten loop (monopolar) vs platinum–iridium (bipolar).
- Glycine 1.5% for monopolar, saline for bipolar; bag 60 cm.
- Bladder neck to verumontanum, depth to capsule.

- Median lobe first; apex last.
- Bipolar allows saline; TUR syndrome practically eliminated.
- Equal efficacy; bipolar has less bleeding and shorter catheter time.
- TUR syndrome risk: >45 mL gland, >90 minutes.

**Lasers in BPH, including green light (SAQ)**

- Chromophore sets penetration: thulium 0.2, holmium 0.4, KTP 0.8 mm, Nd:YAG 1 cm.
- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- HoLEP is size-independent and superior to TURP (only endoscopic method superior to TURP); suits anticoagulated men.
- Retrograde ejaculation about 75%.
- 532 nm, frequency-doubled Nd:YAG; haemoglobin chromophore; 0.8 mm penetration.
- 180 W XPS with MoXy fibre.
- GOLIATH: PVP non-inferior to TURP; best for anticoagulated men; avoid PVP above 80 cc.

**Silodosin (SAQ)**

- Most α1A-selective α-blocker; 8 mg with a meal; no titration.
- 4 mg if CrCl 30–50; avoid if <30.
- Equal efficacy to tamsulosin 0.4 mg; greatest BOOI fall; better for MET.
- Anejaculation is the signature effect; least hypotension.

**Methods to estimate prostate size (SAQ)**

- Ellipsoid volume = 0.52 × T × AP × L; 1 mL ≈ 1 g.
- DRE underestimates above 30 mL; PSA >1.6/2.0/2.3 by decade predicts >40 mL.
- Planimetry most precise; MRI most reproducible. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Catheter free trial (SAQ)**

- TWOC is first choice after AUR; success about 60%.
- ALFAUR: alfuzosin 10 mg raises success to 62% vs 48%.
- Fails more with age, >1 L drained, spontaneous AUR; 5-ARI prevents recurrence.

**Irrigation fluids for TURP. (SAQ)**

- Monopolar: non-ionic hypo-osmolar fluid; bipolar: saline.
- Glycine 1.5%: ~200 mOsm/kg; transient blindness.
- Risk: gland >45 mL, time >90 min; bag height 60 cm.
- 3% saline; ≤12 mEq/L rise in 24 h.

**Dutasteride. (SAQ)**

- Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
- Volume down 25%; AUR and surgery down about half; needs 6 months.
- Double the PSA; CombAT and REDUCE.

**Measures to decrease complications of TURP (SAQ)**

- Bipolar saline TURP removes TUR syndrome risk; fewer transfusions.
- Monopolar: irrigant ≤60 cm, resection <90 min, gland <45 mL safer.
- Verumontanum is the distal limit.

**Tamsulosin (SAQ)**

- 0.4 mg once daily; α1a/α1d selective; no titration.
- IFIS highest with tamsulosin; persists after stopping.
- MET benefit in distal stones 5–10 mm.

**Prostatic stents (SAQ)**

- Temporary (Memokath) vs permanent (UroLume, withdrawn).
- Need detrusor contractility; never cross the sphincter.
- Migration, encrustation, ingrowth; iTind is the modern temporary device.

**Role of UDS before prostate surgery (SAQ)**

- PFS is the only test that separates BOO from DUA.
- BOOI >40 obstructed; BCI <100 weak.
- EAU indications: failed surgery, <150 mL voided, PVR >300 mL, age <50 or >80.
- UPSTREAM: routine UDS did not change outcomes.

**IPSS score (SAQ)**

- 7 items × 0–5 = 0–35; bands 0–7, 8–19, 20–35.
- QoL 0–6 is separate; IPSS = AUA-SI + QoL.
- 3-point change is noticeable; IPSS does not diagnose BOO.

**Laparoscopic Trans vesical Prostatectomy (SAQ)**

- Indication: gland >80 g, especially with median lobe, stones or diverticulum.
- Posterior plane first; divide urethra proximal to verumontanum.
- 5 and 7 o'clock sutures plus mucosal advancement.
- Same function as open, less morbidity.

**UroLift (prostatic urethral lift) (SAQ)**

- 30–80 mL, no median lobe; local anaesthesia.
- Ejaculation preserved; L.I.F.T. retreatment 13.6% at 5 years.
- TURP wins (better) on IPSS, flow and durability.

# Section 17 · Prostate Cancer

## Long answer questions (LAQ)

### LAQ 1 Staging and management of locally advanced (non-organ-confined) CA prostate

P1 · P2 **merged from 2**

*Asked as:* P2 LAQ "Staging of prostate cancer and management of locally advanced (non-organ-confined) prostate cancer; complications of bilateral orchidectomy" (Describe staging of CA prostate and management of non-organ confirmed CA prostate; Desc. mgt optns for locally adv. CaP. Discuss complic. of b/l orchidectomy.; Describe management of LAPC) · P1 LAQ "Describe staging of CA prostate and management of nonorgan confined CA prostate"

*Read as: staging of prostate cancer, and management of non-organ-confined (locally advanced and node-positive, non-metastatic) disease; describe the management options for locally advanced carcinoma prostate, and discuss the complications of bilateral orchidectomy. LAPC is read as locally advanced prostate cancer.*

#### Introduction and definition

- **Locally advanced CaP (carcinoma prostate) / LAPC: cT3–4, N0 or N1 (N+), M0:** extracapsular extension (cT3a), seminal vesicle invasion (cT3b), fixed or invading adjacent organs (cT4), ± pelvic nodes.
- At least 10% of new prostate cancers are locally advanced; these men contribute a disproportionate share of cancer deaths.
- Non-organ-confined disease has a high chance of micrometastases. Any single treatment carries a high recurrence risk (single-modality treatment usually fails), so **multimodal (combined) therapy**, local therapy plus systemic ADT (androgen deprivation therapy), is the rule.
- In the PSA (prostate-specific antigen) era, most "high-risk" disease is defined by grade and PSA rather than palpation.
- Untreated, 15-year disease-specific survival is 57% for T3–T4 vs 81% for T1–T2 (Swedish cohort). Untreated, Gleason 8–10 disease carries high cancer mortality, so fit men need treatment.

#### Staging: TNM (tumour, node, metastasis) 2017 (AJCC [American Joint Committee on Cancer] 8th edition)

| Stage | Definition |
|---|---|
| T1a / T1b | Not palpable; incidental at TURP (transurethral resection of prostate): ≤5% / >5% of resected tissue |
| T1c | Not palpable; found on needle biopsy (for raised PSA) |
| T2a / T2b / T2c | Palpable, confined: ≤ half of one lobe / > half of one lobe / both lobes |
| **T3a** | Extraprostatic extension (unilateral or bilateral), including microscopic bladder neck invasion |
| **T3b** | **Seminal vesicle invasion** (not T4) |
| **T4** | Fixed or invading external sphincter, rectum, bladder, levators, pelvic wall |

| Stage | Definition |
|---|---|
| N0 / N1 | No nodes / regional (pelvic) nodes |
| M1a / M1b / M1c | Non-regional nodes / bone / other sites ± bone |

- **Traps:** seminal vesicle invasion is T3b, not T4; non-regional nodes are M1a.
- **Prognostic stage groups** combine T, N, M, PSA and grade group: stage III = T3–4, PSA ≥20 ng/mL or grade group 5; IVA = N1; IVB = M1. pT2 is not subdivided.
- **Grade group (ISUP, International Society of Urological Pathology):** GG1 ≤6; GG2 3+4; GG3 4+3; GG4 8; GG5 9–10.

## Risk groups

| System | High risk | Very high / locally advanced |
|---|---|---|
| D'Amico | T2c, PSA >20 or Gleason 8–10 (any one) | – |
| NCCN (National Comprehensive Cancer Network) | cT3a, GG4–5 or PSA >20 | cT3b–T4, primary Gleason pattern 5, ≥2 high-risk features, or >4 cores GG4–5 |
| EAU (European Association of Urology) | PSA >20 ng/mL, ISUP 4–5 or cT2c | **Locally advanced: cT3–4 or cN+ (cN1), any PSA, any grade** |

- Nomograms (CAPRA [Cancer of the Prostate Risk Assessment] score, Partin tables, Kattan nomograms) and genomic classifiers (Decipher) refine risk.

## Staging work-up (evaluation)

- **DRE (digital rectal examination)** sets clinical T stage but under- and over-stages. Biopsy ISUP grade group.
- PSA predicts extent: organ-confined about 80% if PSA <4, under 50% if >10; nodes in about 20% if PSA >20 ng/mL and about 75% if >50 ng/mL.
- **mpMRI (multiparametric magnetic resonance imaging):** local extent; separates T2 from T3 (extraprostatic extension and seminal vesicle invasion; PI-RADS v2 reporting); best specificity in higher-risk men.
- **Bone scan + CT (computed tomography) or MRI** for high risk, PSA >20 or GG4–5.
- **PSMA (prostate-specific membrane antigen) PET-CT (positron emission tomography)** is more accurate; **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Assess life expectancy: a fit man in his 70s with high-grade disease will live long enough to die of it.

## Management of locally advanced N0 (cT3–4 N0 M0)

### A. Radiotherapy (EBRT, external-beam RT) + long-term ADT: the standard

- Dose-escalated IMRT (intensity-modulated RT) with image guidance, ≥78–81 Gy (P1: about 76–78 Gy), with or without pelvic node RT, with or without brachytherapy boost (**ASCENDE-RT:** LDR [low-dose-rate] boost halved biochemical failure, improving biochemical control; overall survival (OS) equal).
- **Why add ADT:** it shrinks the target and is synergistic with radiation: neoadjuvant, concurrent and adjuvant, **18–36 months** for high risk (NCCN; EAU 2–3 years).

| Trial | Finding |
|---|---|
| **EORTC 22863** | 3 years of goserelin with RT improved 5-year OS, 78% vs 62% (other sources: 79% vs 62%) |
| **RTOG 85-31 / 92-02** | Long adjuvant ADT better; RTOG 92-02: 28 vs 4 months ADT better on all endpoints except overall OS; OS gain in Gleason 8–10 |
| **TROG 96.01** | 6 months better than 3 months |
| **SPCG-7; MRC PR07/NCIC PR.3** | **RT is still needed with ADT:** adding RT to ADT reduced cancer death and gave an overall survival gain: SPCG-7 cancer death RR (relative risk) 0.44; MRC PR07 OS HR (hazard ratio) 0.77 (P1: HR 0.46). ADT alone is inferior |
| **STAMPEDE** (high-risk M0) | Adding 2 years of abiraterone: metastasis-free survival HR 0.53 |
| **RTOG 0521; GETUG 12** | Docetaxel added modest benefit |

- **Duration:** NCCN: ADT 18–36 months with EBRT (12–36 with brachytherapy boost); 18 vs 36 months gave similar OS.
- Pelvic nodal RT: RTOG 94-13 inconclusive; POP-RT (Tata Memorial) improved biochemical failure-free survival in high-risk N0.

**B. Systemic intensification**

- **STAMPEDE (abiraterone, M0):** for N1, or two of cT3–4, Gleason ≥8 or PSA ≥40 ng/mL, **abiraterone + prednisolone for 2 years** with ADT + RT gave metastasis-free survival HR 0.53 and OS HR 0.60. NCCN now lists it for very high risk.
- Docetaxel with RT and ADT (RTOG 0521, OS HR 0.69): not routine.

**C. Radical prostatectomy (RP) + extended PLND (pelvic lymph node dissection, ePLND), within multimodal therapy: selected men**

- For fit men, life expectancy >10 years, with a completely resectable tumour (mainly cT3a; selected cT3b, limited nodes). Clinical overstaging of cT3 occurs in 7–27%, so cT3 alone is no bar.
- **Advantages:** accurate pathological staging; removes the primary, the source of local morbidity (an untreated prostate causes retention, haematuria and ureteric obstruction that ADT does not prevent); RT kept for salvage.
- RP alone gives disease-free survival in at least half at 8–10 years; disease-specific survival 85–92% at 5 years and 79–82% at 10 years (10-year disease-specific survival about 80–90%), but biochemical recurrence is common, so multimodal care is planned.
- **Neoadjuvant ADT before RP:** clinical downstaging and fewer positive margins, but **no survival or PSA benefit**.
- **Adjuvant vs early salvage RT after RP:** EORTC 22911, SWOG 8794 (metastasis-free survival and OS improved) and ARO 96-02 favour adjuvant RT over observation. RADICALS-RT, RAVES and GETUG-AFU 17 favour **observation with early salvage RT** (at PSA rise) for most men with pT3 or a positive margin. Adjuvant RT is still considered for very high-risk pathology (pT3b, GG4–5 / Gleason 8–10, positive margins). Dose 64–66 Gy; adding short-term ADT to salvage RT improved progression-free survival (GETUG-AFU 16).

**D. ADT alone**

- Only for men unfit for or declining local therapy. Consider immediate ADT if symptomatic, PSA doubling time is short (<12 months) or PSA is very high (>50 ng/mL); otherwise

deferred ADT is acceptable. Early ADT delays progression; survival gain mainly in aggressive disease. In fit men it is inferior to ADT + RT.

- Options: bilateral orchidectomy, LHRH (luteinising hormone-releasing hormone) agonist or antagonist; bicalutamide 150 mg monotherapy spares sexual interest (harmful in localised disease); intermittent ADT.

**E. Others**

- Watchful waiting with deferred ADT in elderly comorbid men.
- Cryotherapy and HIFU (high-intensity focused ultrasound): not standard, investigational in high risk.
- **Also know:** TURP or stents for local obstruction; trial enrolment.

## Node-positive disease (N1 M0)

| Setting | Management |
|---|---|
| **cN1 M0** | RT to prostate and pelvic nodes + long-term ADT 2–3 years + abiraterone 2 years (STAMPEDE). RP + ePLND in selected men |
| **pN1 (pN+) after RP** | **Immediate ADT** (ECOG 7887: improved overall survival) **or ADT + adjuvant pelvic RT** (beats ADT alone); observation with undetectable PSA and ≤2 nodes |

## Metastatic (M1), in brief

- ADT plus an **ARPI (androgen receptor pathway inhibitor)** (abiraterone, enzalutamide, apalutamide) or a triplet with docetaxel; prostate RT for low-volume disease (see 2025-Winter Paper 3, Q3b).

## Algorithm

```
Non-organ-confined, cT3–4 / cN1, M0 (PSMA PET or CT + bone scan) ├─ Fit, cT3a, resectable → RP + ePLND → early salvage RT ± ADT ├─ Most (cT3–4 or cN1) → EBRT + ADT 18–36 months │ very high risk / N1 → + abiraterone 2 years └─ Unfit / refuses → ADT alone (immediate if symptomatic, or deferred)
```

## Complications and local sequelae

- ADT: hot flushes, osteoporosis, metabolic syndrome, cardiovascular risk, loss of libido; give bone protection.
- RT: proctitis, cystitis. RP: incontinence, erectile dysfunction.
- A prostate left in place can cause bladder outlet obstruction (channel TURP), ureteric obstruction (stent or nephrostomy) and haematuria (palliative RT).

## Bilateral orchidectomy

- Removes about 95% of testosterone: **castrate level (<20 ng/dL) within 12 hours**. Cheap, no compliance issues, no flare, so useful in impending cord compression. Simple or subcapsular (all Leydig tissue must be removed).
- Versus LHRH agonists: fewer cardiac events, fractures and peripheral arterial disease reported.

## Complications of bilateral orchidectomy

**A. Surgical complications**

- Scrotal haematoma (commonest), wound infection, chronic scrotal or groin pain.

- Psychological: empty scrotum and altered self-image. Prevent with subcapsular or epididymis-sparing technique, or a testicular prosthesis.
- Irreversible: intermittent ADT is impossible.

**B. Systemic effects of castration**

| Effect | Details and management |
|---|---|
| Hot flushes | Commonest (50–80%); loss of hypothalamic oestrogen signalling; venlafaxine, gabapentin, megestrol or cyproterone if bothersome |
| Sexual dysfunction | Loss of libido, ED (erectile dysfunction), penile shortening; ED aids help selected men |
| Osteoporosis, fractures | Fracture risk up 20–45%; baseline DEXA (dual-energy X-ray absorptiometry), FRAX; calcium, vitamin D, exercise; **denosumab 60 mg 6-monthly** or zoledronic acid |
| Metabolic | Weight gain, fat gain, **sarcopenia**, insulin resistance, diabetes (HR about 1.34), metabolic syndrome |
| Cardiovascular | Coronary events, sudden cardiac death, thromboembolism |
| Anaemia | Normocytic; reversible after stopping |
| Others | Gynaecomastia (≤5%), fatigue, depression, cognitive change |

- **Also know:** risk of castration resistance over time; check testosterone with rising PSA.

## Follow-up

- PSA every 3–6 months; bone health, metabolic and cardiovascular monitoring on ADT.

## Recent advances

- PSMA PET staging; trials adding apalutamide or enzalutamide (ATLAS, ENZARAD); SPCG-15 (RP vs RT + ADT).

## Diagrams to draw

- TNM T-stage diagram / sagittal pelvis: T2 confined, T3a extracapsular, T3b seminal vesicle, T4 adjacent organs.
- Subcapsular orchidectomy.

> **Key points to remember:**
> - Seminal vesicle invasion is T3b; non-regional nodes are M1a.
> - Locally advanced = T3–4, N0/N+, M0; treat with combined therapy.
> - RT + long-term ADT (18–36 months; EAU 2–3 years) is the standard; ADT alone is inferior (SPCG-7).
> - EORTC 22863 overall survival 79% vs 62%.
> - STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
> - Neoadjuvant ADT before RP: no survival gain.
> - Early salvage RT preferred over routine adjuvant RT for most men after RP.
> - Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

## LAQ 2 Metastatic prostate cancer, including newly diagnosed and bone metastases

P2 · P3 **merged from 2**

*Asked as:* P2 LAQ "Metastatic prostate cancer with bone metastases (retention and bone pain; bone mets; zoledronic acid)" (Treatment of metastatic CA prostate in 60 yr male with retention of urine and bone pain; Management of bone mets in prostatic CA; also SAQ: Zolendronic acid) · P3 SAQ "Management of newly Metastatic prostate cancer"

*Zolendronic acid: read as zoledronic acid.*

### Introduction

- Bone is the commonest metastatic site; visceral metastasis occurs in <20%.
- Aims: control the cancer, relieve pain, **prevent SREs (skeletal-related events)** and treat emergencies (cord compression, fracture).
- **Retention + bone pain case:** this is newly diagnosed (synchronous) **mHSPC (metastatic hormone-sensitive prostate cancer)** with two urgent problems: an obstructed outlet and painful bone metastases. Treat the emergencies first, start castration safely (no flare), then intensify systemic therapy.

### Definitions

- **mHSPC:** no prior hormones. **Synchronous** (at diagnosis) or **metachronous** (after local therapy).
- **CHAARTED high volume:** ≥4 bone metastases with ≥1 outside the axial skeleton, or any visceral metastasis.
- **LATITUDE high risk:** ≥2 of Gleason ≥8, ≥3 bone lesions, visceral metastasis.

### Pathophysiology of bone metastases

- Tumour cells lodge in axial marrow, adhering through integrins and collagen receptors.
- Metastases are mainly **osteoblastic (sclerotic)**, but resorption also rises. Tumour cytokines induce **RANKL (receptor activator of nuclear factor-κB ligand)**, which activates osteoclasts; resorbed bone releases growth factors that feed the tumour (**vicious cycle**). ADT (androgen deprivation therapy) adds osteoporosis through loss of oestrogen.
- **SREs:** pathological fracture, spinal cord compression, need for radiotherapy or surgery to bone. Marrow replacement causes anaemia. Hypercalcaemia is rare and suggests neuroendocrine disease.

### Evaluation (immediate assessment and work-up)

- **Clinical:** pain, neurological signs, performance status. **Ask and examine for cord compression:** back pain, leg weakness, sensory level, bowel symptoms. DRE (digital rectal examination).
- **Bloods:** PSA (prostate-specific antigen), ALP (alkaline phosphatase), calcium, creatinine, haemoglobin, testosterone.
- **Imaging:** ultrasound for hydronephrosis and residual; **bone scan and CT (computed tomography)** define volume; PSMA (prostate-specific membrane antigen) PET (positron emission tomography) finds more and is more sensitive, but trials used conventional imaging. **Plain X-ray or CT of painful weight-bearing sites** (lytic lesion, impending

fracture). **Whole-spine MRI (magnetic resonance imaging)** if any back pain or neurological sign.
- **Histology:** prostate biopsy (or TURP [transurethral resection of the prostate] chips).
- **Germline (and somatic) genetic testing** for all men with metastatic disease (>10% have DNA-repair mutations).
- **Volume (CHAARTED):** see definitions above.

### Step 1: Relieve the retention

- **Urethral catheter** (suprapubic if urethral fails). Watch for post-obstructive diuresis.
- Persisting renal impairment with hydronephrosis means ureteric obstruction by tumour at the trigone: **nephrostomy or stent**.
- ADT shrinks the prostate, so try **TWOC (trial without catheter)** after a few months of ADT.
- If TWOC fails or symptoms persist: **channel (palliative) TURP**. ADT alone does not prevent local progression (older series: about a third needed TURP within 60 days of orchidectomy).

### Step 2: Systemic treatment, hormone-sensitive (mHSPC): start ADT at once, without flare

- **ADT is the backbone, never alone** (orchidectomy, LHRH [luteinising hormone-releasing hormone] agonist or antagonist):

| Option | Point |
|---|---|
| **Bilateral orchidectomy** | Castrate within hours; best for impending cord compression |
| **GnRH (gonadotropin-releasing hormone) antagonist**: degarelix 240 mg subcutaneously, then 80 mg monthly; oral relugolix | **No flare**; castrate by day 3 in >95% (degarelix); for impending cord compression |
| **LHRH agonist + antiandrogen** | Agonist raises LH (luteinising hormone) up to 10-fold, raising testosterone, for 10–20 days (**flare**). Give bicalutamide 50 mg daily with or before the agonist, for 21–28 days |

- **Why avoid flare:** the testosterone surge can worsen bone pain, retention and cord compression in spinal metastases. Cover it with an antiandrogen for 21–28 days, or use an antagonist (degarelix) or orchidectomy.
- Target castrate testosterone **<50 ng/dL**. Antiandrogen alone is not recommended.

### Intensified systemic therapy in mHSPC (doublets and triplets)

- ADT alone is no longer standard. Classify volume (CHAARTED) and risk (LATITUDE).

| Add to ADT | Trial | Overall survival HR (hazard ratio) |
|---|---|---|
| Docetaxel 75 mg/m², 6 cycles | **CHAARTED** (benefit in high volume only); STAMPEDE | 0.61; 0.78 |
| Abiraterone 1000 mg + prednisone 5 mg | **LATITUDE**; STAMPEDE arm G | 0.62; 0.63 |
| Enzalutamide (ARPI, androgen receptor pathway inhibitor) | ARCHES (radiographic PFS [progression-free survival]); ENZAMET | 0.67 (ENZAMET) |

| Add to ADT | Trial | Overall survival HR (hazard ratio) |
|---|---|---|
| Apalutamide (ARPI) | **TITAN** | 0.67 |
| **Triplet:** docetaxel 75 mg/m² × 6 + abiraterone | **PEACE-1** (synchronous) | 0.75 (with docetaxel) |
| **Triplet:** docetaxel + darolutamide | **ARASENS** | 0.68 |

- Hormonal agents work **regardless of volume**; docetaxel helps only high volume (GETUG-AFU 15 negative), so it is now given within a triplet. Triplets suit chemotherapy-fit men, especially high volume.
- *Guideline note:* the chapter lists ADT plus docetaxel as an option; current EAU (European Association of Urology) guidance gives docetaxel only within a triplet.

### Local and metastasis-directed therapy in mHSPC

- **RT (radiotherapy) to the prostate in low-volume synchronous disease only:** STAMPEDE arm H, survival HR 0.68 (3-year 81% vs 73%); it also helps local control. HORRAD negative overall.
- Cytoreductive prostatectomy: trials only.
- **MDT (metastasis-directed therapy)** / **SBRT (stereotactic body RT)** for oligometastases (STOMP, ORIOLE) delays ADT; SBRT also for spinal re-irradiation.

### Castration-resistant disease (mCRPC)

- ARPIs, docetaxel then cabazitaxel, PARP (poly-ADP-ribose polymerase) inhibitors for HRR (homologous recombination repair) mutations.
- **Radium-223** (alpha emitter, path <100 µm, spares marrow): for **symptomatic bone-only** disease (licensed only in castration-resistant disease); 50 kBq/kg every 4 weeks × 6. **ALSYMPCA:** overall survival 14.9 vs 11.3 months; delayed SREs. **Never with abiraterone** (more fractures and deaths).
- **$^{177}$Lu-PSMA-617** (beta emitter, PSMA-avid disease at any site): after ARPI and taxane; **VISION:** survival 15.3 vs 11.3 months.

### Step 3: Bone pain

- **Analgesia** by the WHO (World Health Organization) analgesic ladder; neuropathic pain needs gabapentinoids or tricyclics; steroids for diffuse pain.
- **Focal pain: EBRT (external beam radiotherapy) single 8 Gy fraction.** Pain relief equals multiple fractions (30 Gy in 10 fractions, RTOG 9714); retreatment is commoner.
- X-ray or CT of painful weight-bearing sites to exclude lytic lesions and impending fracture; **orthopaedic fixation** then RT.
- **Diffuse pain:** steroids, wide-field RT; radiopharmaceuticals (strontium-89, samarium-153 relieve pain only; radium-223 adds survival, licensed only in castration-resistant disease).

### Bone-targeted agents

| Feature | Zoledronic acid | Denosumab |
|---|---|---|
| Mechanism / target | Bisphosphonate: inhibits osteoclast enzyme | Antibody against RANKL |
| SRE dose (mCRPC) / route | 4 mg IV every 4 weeks, adjusted for creatinine clearance | 120 mg SC (subcutaneous) every 4 weeks |

| Feature | Zoledronic acid | Denosumab |
|---|---|---|
| Renal adjustment | Yes | No |
| Evidence | Fewer SREs than placebo | Time to first SRE 20.7 vs 17.1 months vs zoledronate |
| Toxicity | Fever, myalgia, renal impairment, hypocalcaemia, ONJ | Hypocalcaemia, hypophosphataemia, ONJ 2–4% |

- **Neither improves survival**; they protect the skeleton only.
- **In mHSPC, SRE doses are not indicated** (no benefit: CALGB 90202, STAMPEDE). SRE doses (zoledronic acid 4 mg or denosumab 120 mg 4-weekly) are for CRPC (castration-resistant prostate cancer). Use osteoporosis doses for ADT bone loss: denosumab 60 mg 6-monthly or zoledronic acid yearly, by DEXA (dual-energy X-ray absorptiometry) and FRAX.
- **ONJ (osteonecrosis of the jaw) prevention:** dental check before starting; calcium 1000 mg and vitamin D 800 IU daily.
- Give a bone agent with radium-223 or any ARPI.

### Zoledronic acid (in detail)

- **Introduction:** a potent IV (intravenous) **nitrogen-containing bisphosphonate**.
- **Mechanism:**
- Binds hydroxyapatite at sites of active bone turnover and is taken up by **osteoclasts**.
- Inhibits **farnesyl pyrophosphate synthase** (mevalonate pathway), causing osteoclast dysfunction and apoptosis, so resorption falls.
- Breaks the "vicious cycle".
- Excreted unchanged by the kidneys.

| Indication | Dose |
|---|---|
| **SRE prevention, mCRPC (metastatic castration-resistant prostate cancer)** and other bone metastases (RCC [renal cell carcinoma], bladder) | **4 mg IV over ≥15 min every 4 weeks**, adjusted for creatinine clearance |
| ADT bone loss (by DEXA and FRAX) | 4–5 mg IV yearly |
| Hypercalcaemia of malignancy | 4 mg IV with saline hydration |

- 12-weekly dosing after the first doses is non-inferior to 4-weekly (CALGB 70604).
- **Not indicated at SRE doses in mHSPC:** no benefit (CALGB 90202, STAMPEDE).
- **Evidence: Zometa 039 trial (mCRPC):** fewer SREs than placebo (about 33% vs 44%). **Denosumab vs zoledronate (1904 men, mCRPC):** time to first SRE 20.7 vs 17.1 months. **Neither improves survival.**
- **Precautions and monitoring:** creatinine before every dose; avoid if creatinine clearance <30 mL/min. **Dental review before starting**; avoid invasive dental work during therapy. Calcium and vitamin D supplements.
- **Adverse effects: acute-phase reaction** (fever, myalgia) after the first infusion; nephrotoxicity, hypocalcaemia, hypophosphataemia; **ONJ**, atypical femoral fracture; ocular inflammation (rare).

### Emergencies

**Spinal cord compression: watch constantly** (every focal back pain is cord compression until proven otherwise): `Back pain ± weakness, sensory level, sphincter change →`

dexamethasone 10 mg IV load, then 4–10 mg 6–12-hourly → urgent whole-spine MRI → decompressive surgery + RT (unstable spine, progression on RT, prior RT) or RT alone → orchidectomy or antagonist if hormone-naive (never agonist alone)

- Surgery plus RT beat RT alone for walking ability (**Patchell, 2005**).

**Pathological or impending fracture:** orthopaedic fixation then RT. **Mirels score** ≥9 (site, pain, lytic type, size) favours prophylactic fixation.

### Algorithms

Retention + bone pain + metastases ├─ Catheter ± nephrostomy; cord signs? → dexamethasone → MRI → RT ± surgery ├─ Orchidectomy / antagonist / agonist + antiandrogen ├─ Analgesia + 8 Gy single-fraction RT; bone protection ├─ Add ARPI or triplet; prostate RT if low volume └─ TWOC after ADT → channel TURP if it fails

- Newly diagnosed mHSPC: split by volume and fitness to doublet or triplet ± prostate RT.

### Follow-up

- PSA, ALP, liver tests, calcium, renal function and blood count every 3–6 months; testosterone if PSA rises; new scan if symptoms or PSA rise. Rising PSA with castrate testosterone = CRPC.
- PSA nadir ≤0.2 ng/mL at 7 months predicts long survival (SWOG 9346).
- PCWG3 (Prostate Cancer Working Group 3): bone scan progression needs ≥2 new lesions, confirmed later (early "flare" misleads).
- Manage ADT effects: bone loss (DEXA), diabetes, cardiovascular risk, hot flushes.
- **Also know:** stent or nephrostomy for obstruction; marrow-failure anaemia; hospice care.

### Diagrams to draw

- The retention + bone pain flowchart above.
- Vicious cycle: tumour → RANKL → osteoclast → growth factors → tumour, with the sites blocked by denosumab and zoledronate.
- Algorithm splitting newly metastatic disease by volume and fitness to doublet or triplet ± prostate RT.

**Key points to remember:**

- ADT alone is no longer standard: ADT plus abiraterone or an ARPI, or a triplet (PEACE-1, ARASENS), for every fit man.
- CHAARTED volume vs LATITUDE risk definitions.
- Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).
- Never start an LHRH agonist alone with spinal metastases: antiandrogen cover, antagonist or orchidectomy.
- Single 8 Gy fraction gives the same pain relief for focal pain; focal back pain = cord compression until MRI proves otherwise; dexamethasone and MRI.
- Denosumab 120 mg or zoledronate 4 mg IV 4-weekly prevent SREs in mCRPC, not in mHSPC; adjust zoledronate for creatinine clearance; no survival benefit.
- Dental check first (ONJ); calcium and vitamin D.
- Denosumab delays SREs longer (20.7 vs 17.1 months).
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

## LAQ 3 Radiotherapy and brachytherapy for CA prostate

P2 · P4 **merged from 2**

*Asked as:* P4 LAQ "Role and impact of Radiotherapy methods for management of focal and advanced carcinoma prostate?" · P2 SAQ "Radiotherapy and brachytherapy for CA prostate" (Radiotherapy for CA prostate.; Brachytherapy for CA prostate)

*"Focal" is taken to mean localised (organ-confined and locally advanced) disease.*

### Radiobiology

- RT (radiotherapy) causes DNA double-strand breaks. Prostate cancer has a **low α/β ratio (~1.5 Gy)**, lower than the rectum (~3 Gy). So larger fractions kill more tumour for the same late toxicity: the basis of **hypofractionation**. The low α/β ratio also favours high dose per fraction (HDR brachytherapy).

### EBRT (external beam radiotherapy) techniques

| Step | Advance |
|---|---|
| 3D-CRT (three-dimensional conformal RT) | CT (computed tomography)-planned beams shaped to the prostate |
| **IMRT (intensity-modulated RT)/VMAT (volumetric modulated arc therapy)** | Inverse planning, steep dose gradients; less rectal toxicity; now standard; allows dose escalation |
| **IGRT (image-guided RT)** | Daily fiducials or cone-beam CT; margins 3–7 mm |
| MRI fusion, PSMA (prostate-specific membrane antigen) PET (positron emission tomography) | Better targeting; PSMA PET changes plans in up to half of high-risk men |

### EBRT fractionation schedules

| Schedule | Dose per fraction | Example | Evidence |
|---|---|---|---|
| Conventional | 1.8–2 Gy | 78 Gy in 39 | Dose escalation (RTOG 0126): better PSA control, no survival gain |
| **Moderate hypofractionation** (standard) | 2.25–4 Gy | **60 Gy in 20** | **CHHiP** (non-inferior to 74 Gy in 37), **PROFIT**: non-inferior |
| **SBRT (stereotactic body RT)** | ≥5 Gy | 36.25 Gy in 5 | **HYPO-RT-PC** non-inferior; **PACE-B** equal toxicity |

### Brachytherapy: definition and principle

- Brachytherapy (BT; Greek *brakhus*, short): radioactive sources placed **within the prostate**.
- **Steep dose fall-off** gives an intraprostatic dose far above what EBRT can safely give.

### Brachytherapy: types

| Feature | LDR (low dose rate), permanent seeds | HDR (high dose rate), temporary |
|---|---|---|
| Source | $^{125}$I (half-life 59.6 days), $^{103}$Pd (17 days), $^{131}$Cs (9.7 days) | **$^{192}$Ir** via remote afterloader |
| Monotherapy dose | $^{125}$I 145 Gy (P2: 140–160 Gy); $^{103}$Pd 110–125 Gy | 27 Gy in 2, 31.5 Gy in 3, 34–38 Gy in 4 fractions (single 19 Gy fraction fails) |

| Feature | LDR (low dose rate), permanent seeds | HDR (high dose rate), temporary |
|---|---|---|
| Monotherapy use | Low and favourable intermediate risk | |
| Dosimetry / quality | Post-implant CT at 14–30 days: **D90** (dose covering 90% of gland) **>90%** of prescription | Real-time planning; none afterwards |

### Brachytherapy: technique

- **TRUS (transrectal ultrasound)-guided transperineal** implant under anaesthesia: high lithotomy, stepper, template grid, needles, seeds deployed on withdrawal.
- **Peripheral loading** spares the urethra. Intraoperative planning gives best coverage.
- Quality: **D90 >90%** of prescription and **V100 >85%** of gland; urethral and rectal (V100 <1 $cm^3$) limits. Low D90 predicts failure.

### Brachytherapy: indications

- **Monotherapy:** low and favourable intermediate risk (**RTOG 0232**: adding EBRT to implant gave no benefit in intermediate risk, more toxicity).
- **Boost** (EBRT 45 Gy + implant ± ADT [androgen deprivation therapy]) for unfavourable intermediate and high risk: **ASCENDE-RT:** LDR boost halved biochemical failure vs EBRT boost, with more grade 3 urinary toxicity (18.4% vs 5.2%).
- Also salvage and focal BT.

### Brachytherapy: contraindications (relative)

- **IPSS (International Prostate Symptom Score) >20, Qmax <10 mL/s**, large median lobe; recent or large TURP (transurethral resection of the prostate) defect; gland >60 $cm^3$ (cytoreduce with ADT); pubic arch interference; prior pelvic RT; inflammatory bowel disease; cannot lie in lithotomy.

### Brachytherapy: complications and follow-up

- Acute obstructive and irritative symptoms; retention (29% if IPSS >20, 2% if <10); α-blockers help.
- Chronic grade ≥2 urinary toxicity about 20%; urethral stricture; rectal complications 1–2%; rare rectourethral fistula; ED (erectile dysfunction).
- Nadir is deeper and later (about 2.5 years). **Benign PSA bounce** in about one-third: watch for 30 months before biopsy, do not salvage.

### ADT with RT

- **Why:** ADT radiosensitises (less DNA repair) and causes apoptosis; RT up-regulates androgen-receptor signalling, which ADT blocks.

| Risk group | ADT with EBRT |
|---|---|
| Low, favourable intermediate | None |
| Unfavourable intermediate | **4–6 months** |
| High / locally advanced | **18–36 months** (P2: 24–36 months) (EORTC 22863, PCS IV); add **abiraterone** if very high risk (STAMPEDE) |

- **RT added to ADT** improves survival in locally advanced disease: **SPCG-7, NCIC PR.3/MRC PR07** (about 8–10% absolute).

### After radical prostatectomy

- **Adjuvant RT** (pT3/positive margin): EORTC 22911, ARO 96-02 (biochemical control), **SWOG 8794** (metastasis-free and overall survival).
- **RADICALS-RT, RAVES, GETUG-AFU 17** (ARTISTIC meta-analysis): **early salvage at PSA (prostate-specific antigen) 0.1–0.2 ng/mL** equals adjuvant RT and spares over 60% of men RT.
- Salvage dose ≥64–66 Gy. **RTOG 9601:** 2 years bicalutamide improved overall survival. **SPPORT:** pelvic nodes + short ADT improve freedom from progression.

### Advanced and metastatic disease

| Setting | RT role | Evidence |
|---|---|---|
| Low-volume de novo mHSPC (metastatic hormone-sensitive) | **RT to prostate** + ADT | **STAMPEDE arm H**: overall survival HR (hazard ratio) 0.68 (3-year 73% → 81%); none in high volume |
| Oligometastases (≤3–5) | SBRT to all lesions (MDT, metastasis-directed therapy) | STOMP, **ORIOLE** (less progression), SABR-COMET |
| Painful bone metastases | Palliative EBRT, e.g. **8 Gy single fraction**; cord compression | Equal pain relief to multiple fractions |
| mCRPC (metastatic castration-resistant), bone-only symptomatic | **Radium-223** (α-emitter, calcium mimic, range <100 µm) 50 kBq/kg 4-weekly × 6 | **ALSYMPCA**: overall survival 14.9 vs 11.3 months; fewer skeletal events |
| mCRPC, PSMA-positive, after taxane and ARPI (androgen-receptor pathway inhibitor) | **$^{177}$Lu-PSMA-617** (β-emitter radioligand) | **VISION**: survival 15.3 vs 11.3 months (HR 0.62); **TheraP**: better PSA response than cabazitaxel |

- **Never combine radium-223 with abiraterone** (more deaths and fractures).
- **Also know:** protons (no proven gain over IMRT); strontium-89, samarium-153 (pain palliation); elective pelvic nodal RT (RTOG 94-13, GETUG-01: no survival gain).

### Outcomes

- **ProtecT** (2016; 15-year 2023): equally low prostate-cancer mortality with RT, surgery or active monitoring; RT more bowel effects, surgery more incontinence and ED.

### Response assessment

- **Phoenix failure:** PSA nadir + 2 ng/mL.
- Biopsy-proven local failure: salvage prostatectomy, brachytherapy, HIFU (high-intensity focused ultrasound) or cryotherapy.

### Toxicity of RT

| Type | Features and management |
|---|---|
| Acute (weeks) | Frequency, dysuria, diarrhoea; α-blockers |
| Late GI (gastrointestinal) | **Radiation proctitis** with bleeding at 6–18 months: sucralfate/steroid enemas, argon plasma coagulation, hyperbaric oxygen |

| Type | Features and management |
|---|---|
| Late GU (genitourinary) | Cystitis, haematuria, stricture |
| Sexual | ED, vascular; worse with ADT; PDE5 (phosphodiesterase type 5) inhibitors work in most |
| Rare | Second cancers, fistula |

### Recent advances

- **Hydrogel rectal spacer** (late rectal toxicity 2% vs 7%); MR-linac adaptive SBRT; **FLAME** focal boost to the MRI lesion; protons; PSMA radioligands moving earlier (PSMAfore).

### Diagram to draw

- TRUS-guided perineal template brachytherapy (transperineal implant set-up).

> **Key points to remember:**
>
> - α/β ~1.5 Gy: basis of hypofractionation, 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
> - ADT 4–6 months unfavourable intermediate; 18–36 months (P2: 24–36 months) high risk.
> - Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
> - STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.
> - Phoenix: nadir + 2 ng/mL.
> - LDR $^{125}$I 145 Gy; HDR $^{192}$Ir; D90 >90%, V100 >85%.
> - IPSS >20 or Qmax <10 = relative contraindication.
> - ASCENDE-RT: boost halves biochemical failure.

## LAQ 4 Castration-resistant prostate cancer (CRPC): definition and treatment

P2

**Asked as:**

- CRPC in Ca prostate. — *Summer 2020 Paper II Q3a (SAQ)*
- Discuss treatment of CRPC — *Practice Set 6 Paper II Q1 (LAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)
(also answered in 2020-Summer Paper 2, Q3a)

### Definition (EAU [European Association of Urology])

- CRPC (castration-resistant prostate cancer): castrate testosterone **<50 ng/dL (1.7 nmol/L)** plus either:
- **Biochemical progression:** three consecutive PSA (prostate-specific antigen) rises 1 week apart, two of them 50% above nadir, with PSA >2 ng/mL; or
- **Radiological progression:** ≥2 new bone lesions, or soft-tissue progression by RECIST (Response Evaluation Criteria in Solid Tumours).

### Why it happens: the AR still drives growth

- Castration-resistant is **not** androgen-independent, so treatment first targets the AR (androgen receptor) more deeply.
- Mechanisms: AR overexpression and amplification; AR mutation (activated by other steroids); **splice variants lacking the ligand-binding domain**; adrenal and intratumoural androgen synthesis (CYP17 overexpressed); bypass pathways (PI3K/AKT, PTEN loss in ~40%).
- **Lineage plasticity:** neuroendocrine change with low PSA, visceral and lytic disease (visceral disease).

### Evaluation

- Testosterone (exclude non-compliance), PSA, **PSADT (PSA doubling time)**; CT and bone scan; PSMA (prostate-specific membrane antigen) PET increasingly.
- Prognosis: performance status, pain, haemoglobin, ALP (alkaline phosphatase), LDH (lactate dehydrogenase), visceral disease.
- **Germline testing** in all metastatic disease; somatic sequencing of tumour tissue (germline and tumour testing) for HRR (homologous recombination repair) genes such as BRCA2, and MSI (microsatellite instability).
- Biopsy a metastasis if disease progresses fast with a low PSA (neuroendocrine change).
- PCWG3 (Prostate Cancer Clinical Trials Working Group 3): early bone-scan lesions may be **flare**; re-baseline at 12–16 weeks, then progression = ≥2 new lesions.

**Clinical states model (PCWG3):** `Localised → rising PSA → mCSPC (castration-sensitive) ↓ (on ADT) nmCRPC → mCRPC 1st line → 2nd line → 3rd line`

### General principles of management

1. **Continue ADT (androgen deprivation therapy) lifelong**, including during chemotherapy; try antiandrogen withdrawal.

2. **Antiandrogen withdrawal** (e.g. bicalutamide): watch PSA 4–8 weeks; brief responses in a minority.
3. Choose by prior therapy, symptoms, visceral disease, genomics and comorbidity.
4. **Sequencing: switch mechanism**; a second ARPI rarely works (cross-resistance). Cross-resistance means a second ARPI (AR-pathway inhibitor) helps only 15–30%.

### nmCRPC (non-metastatic)

- **PSADT ≤10 months:** add an ARPI. Each improved MFS (metastasis-free survival) and overall survival (improve metastasis-free and overall survival):
- Apalutamide 240 mg (**SPARTAN**: MFS HR 0.28); enzalutamide 160 mg (**PROSPER**); darolutamide 300 mg twice daily (**ARAMIS**: MFS 40.4 vs 18.4 months).
- Slower PSADT: continue ADT and monitor.

### mCRPC: life-prolonging options

| Class | Agent and dose | Key trial (overall survival) |
|---|---|---|
| ARPI (AR-pathway inhibitor) | **Abiraterone** 1000 mg fasting + prednisone | COU-AA-301: 14.8 vs 10.9 months; COU-AA-302 (pre-chemotherapy) |
| | **Enzalutamide** 160 mg | AFFIRM: 18.4 vs 13.6 months; PREVAIL |
| Taxane | **Docetaxel** 75 mg/m² 3-weekly + prednisone | TAX 327: 18.9 vs 16.4 months (mitoxantrone) |
| | **Cabazitaxel** 20–25 mg/m² after docetaxel | TROPIC: 15.1 vs 12.7; CARD: beats a second ARPI |
| Radioligand (mCRPC) | **Radium-223** 50 kBq/kg 4-weekly ×6 (symptomatic bone-only, no visceral) | ALSYMPCA: 14.9 vs 11.3 months |
| | **$^{177}$Lu-PSMA-617** (PSMA-positive, after ARPI + taxane) | VISION: 15.3 vs 11.3 months |
| PARP (poly-ADP-ribose polymerase) inhibitor | Olaparib (HRR genes); rucaparib (BRCA) | PROfound; TRITON3 |
| | Olaparib + abiraterone; niraparib + abiraterone; talazoparib + enzalutamide (first line) | PROpel; MAGNITUDE; TALAPRO-2 |
| Immunotherapy | **Sipuleucel-T** (asymptomatic, no visceral disease) | IMPACT: 25.8 vs 21.7 months |
| | Pembrolizumab for MSI-high/dMMR (mismatch repair-deficient), 2–5% | Tumour-agnostic approval |

- **Abiraterone** blocks CYP17, which also lowers cortisol. ACTH (adrenocorticotropic hormone) then rises and causes mineralocorticoid excess (hypokalaemia, hypertension, oedema), so prednisone is co-prescribed. Never combine with radium-223 (more deaths and fractures).
- **Enzalutamide:** seizures (~1%), fatigue, falls. Apalutamide: rash, hypothyroidism.
- **PARP inhibitors** cause synthetic lethality in HRR-deficient cells; benefit is greatest with BRCA2 and weak with ATM.

### Sequencing by prior mCSPC therapy

| Prior therapy | Next option |
|---|---|
| Docetaxel only | ARPI |
| ARPI only | Docetaxel, radium-223, sipuleucel-T |
| Both | Cabazitaxel, $^{177}$Lu-PSMA, PARP inhibitor if HRR mutation |

### Neuroendocrine/aggressive variant

- Neuroendocrine variant: low PSA, liver or brain metastases, lytic bone lesions with hypercalcaemia. Treat with **platinum + etoposide** (or carboplatin + taxane) ± radiotherapy; survival <12 months.

### Supportive and palliative care

- **Bone health:** denosumab 120 mg SC or zoledronic acid 4 mg IV (renal dose adjustment) every 4 weeks against skeletal events, with calcium and vitamin D. Denosumab delayed the first SRE (skeletal-related event) longer (20.7 vs 17.1 months); neither prolongs life. Risk: osteonecrosis of the jaw.
- Focal bone pain: external-beam radiotherapy (radiotherapy for bone pain).
- **Cord compression** needs high-dose dexamethasone → urgent MRI → radiotherapy ± decompression (Patchell trial: surgery + radiotherapy better).
- Ureteric obstruction: stent or nephrostomy. Anaemia; hospice care.

### Recent advances

- PSMA-ligand therapy earlier (PSMAfore); AR degraders (ARV-110); AKT inhibition in PTEN loss; bipolar androgen therapy; antibody–drug conjugates.

**Diagram to draw:** clinical states: rising PSA → nmCRPC → mCRPC first, second, third line.

**Diagram to draw:** clinical states model with drugs placed at each state.

> **Key points to remember:**
> - Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
> - AR still drives CRPC.
> - nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
> - mCRPC: ARPI, taxanes (docetaxel, cabazitaxel), radium-223, $^{177}$Lu-PSMA, PARP inhibitors, sipuleucel-T.
> - Switch mechanism; never radium-223 with abiraterone.

## LAQ 5 PSA screening for prostate cancer: evidence, problems and solutions

P2

**Asked as:**

- What are the various problems in PSA screening — *Practice Set 8 Paper II Q1 (LAQ)*
- Screening of CA prostate. — *Practice Set 13 Paper II Q3d (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)

### Introduction (concept)

- Screening tests healthy, symptom-free (asymptomatic) men; its aim is lower prostate cancer mortality through earlier treatment (cure). The price is **overdiagnosis** (cancers that would never cause harm) and **overtreatment**.
- PSA (prostate-specific antigen) is a kallikrein (hK3) made by prostatic epithelium. It reaches blood when glandular architecture is disrupted. It is **organ-specific, not cancer-specific**.
- Since PSA testing began, (US) mortality has fallen by >40% (over 40%; SEER data), but at the price of overdiagnosis.

### Tools

- **PSA:** biopsy threshold 4 ng/mL (many use 2.5–3; ERSPC used 3). Repeat a raised PSA before biopsy.
- **DRE (digital rectal examination):** little value alone; abnormal DRE raises risk.
- **Refinements:** PSA density ≥0.15; free PSA ≤25%; PHI (Prostate Health Index), 4Kscore, Stockholm3; urine tests (PCA3 [prostate cancer antigen 3], ExoDx, SelectMDx).
- **MRI (magnetic resonance imaging) before biopsy:** fewer biopsies and less low-grade overdiagnosis.

### Evidence from screening trials

| Trial | Design | Result |
|---|---|---|
| **ERSPC** (European Randomised Study of Screening for Prostate Cancer) | >182,000 men; PSA every 2–4 years vs none | **20% relative fall** in (prostate) cancer death at 16 years; 570 invited and 18 extra diagnoses per death prevented |
| **PLCO** (Prostate, Lung, Colorectal and Ovarian), USA | Annual PSA vs usual care | **No difference** at 17 years: control arm heavily PSA-tested (contamination), lower power |
| **Göteborg-1** (Sweden) | Biennial PSA, men 50–64 | 44% fall in cancer death at 14 years (larger mortality reduction with biennial screening) |
| **CAP** (Cluster randomised trial of PSA testing for prostate cancer), UK | Single PSA invitation | No mortality difference at 10 years; only a small fall at 15 years |
| **Göteborg-2 (2022)** | MRI-targeted biopsy only | Halved overdiagnosis |
| - Corrected for non-attendance and contamination, ERSPC benefit is 27–31%; it appears only after about 10 years. | | |

## Problems in PSA screening

### 1. Poor specificity (false positives)

- BPH (benign prostatic hyperplasia), prostatitis, retention, catheterisation, biopsy and TURP (transurethral resection of prostate) all raise PSA.
- At PSA 4–10 ng/mL with a normal DRE (digital rectal examination), only 17–32% of biopsies find cancer. Most men biopsied therefore have no cancer.

### 2. Poor sensitivity: no safe cut-off

- PCPT (Prostate Cancer Prevention Trial): risk is continuous; about 1 in 11 men biopsied at PSA 0–1 ng/mL has cancer.
- A high threshold misses curable cancer; a low threshold adds biopsies and overdiagnosis. Thresholds used: 4, 3 or 2.5 ng/mL.

### 3. Biological variability of PSA

- Rises with age, prostate volume (about 4% per mL) and Black race; physiological fluctuation; assay differences.
- Lowered by 5α-reductase inhibitors (about 50% at 12 months, so double the value), obesity, hypogonadism and statins.

### 4. Overdiagnosis (the central problem)

- Detection of cancers that would never have caused symptoms or death. Autopsy prevalence is high: 29% at 60–69 years and 59% after 79.
- Mostly Grade Group 1 cancers.

### 5. Overtreatment

- Overdiagnosed men treated radically suffer ED (erectile dysfunction), incontinence and bowel toxicity for no gain, especially above 65 years.

### 6. Harms of biopsy

- Pain, haematuria, haematospermia, retention and sepsis (transrectal route); anxiety from repeat biopsies.

### 7. Modest and delayed benefit

- Screening prevents only 20–30% of cancer deaths; no overall survival benefit has been shown. Men with life expectancy <10 years cannot benefit.

### 8. Biases in interpreting screening data

- **Lead-time bias**: earlier diagnosis lengthens apparent survival without delaying death.
- **Length-time bias**: screening preferentially finds slow-growing tumours.
- **Healthy-user bias**: health-conscious men seek testing, confounding risk-factor studies.

### 9. Unsettled policy

- Start age, stop age, interval and threshold are all debated; further large randomised trials are unlikely.
- Opportunistic screening is common and inequitable.

## Guideline positions (guidelines)

| Body | Recommendation |
|---|---|
| **USPSTF (US Preventive Services Task Force) 2012** | Grade D: against screening |

| Body | Recommendation |
|---|---|
| **USPSTF 2018 (current)** | **Grade C, 55–69 years: individual shared decision; Grade D ≥70** (not ≥70; was Grade D in 2012) |
| AUA (American Urological Association) | Shared decision 55–69; every 2 years (2-yearly interval) |
| NCCN (National Comprehensive Cancer Network) | Baseline PSA and discussion at 45 |
| EAU (European Association of Urology) | Risk-adapted early detection: informed men from 50; earlier start (45) with family history or African descent, 40 for BRCA2 carriers |
| - **No body recommends (advises) unselected population screening.** | |
| - **High risk:** Black men, father or brother diagnosed <65, BRCA2 and other DNA-repair mutations: start earlier. | |
| - Stop when life expectancy is under 10–15 years. | |

### Solutions: risk-adapted, MRI-first pathway (reducing harm)

`Informed man, life expectancy >10–15 y → PSA (repeat if raised) → risk stratify: PSA density, free/total PSA, risk calculator, PHI / 4Kscore / Stockholm3 → MRI (magnetic resonance imaging) PI-RADS 1–2, low density → no biopsy, keep in follow-up PI-RADS ≥3 → targeted ± systematic biopsy (transperineal) → Grade Group 1 → active surveillance`

- **Baseline PSA** predicts future risk (Malmö): low PSA at 40–60 allows longer intervals (intervals by baseline PSA).
- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.
- **PRECISION (2018):** MRI-targeted pathway found more significant cancer (38% vs 26%) and less insignificant cancer; 28% avoided biopsy.
- **Göteborg-2 (2022):** MRI with targeted biopsy in screening roughly halved overdiagnosis.
- **Active surveillance** of low-risk disease (Grade Group 1) uncouples overdiagnosis from overtreatment; MRI and biomarkers before biopsy.
- **Also know:** germline testing (BRCA2, ATM, HOXB13); polygenic risk scores; transperineal biopsy lowers sepsis.

**Diagram to draw:** iceberg of latent vs clinically significant cancer, plus the MRI-first flowchart (pathway: PSA → repeat → risk calculator or MRI → targeted biopsy → active surveillance or treatment).

> **Key points to remember:**
> - ERSPC: (about) 20% fall in prostate cancer death; PLCO null from contamination.
> - Core problems: low specificity, no safe cut-off, overdiagnosis, overtreatment.
> - USPSTF 2018: Grade C at 55–69, Grade D ≥70.
> - Fix: risk-adapted PSA, MRI before biopsy, active surveillance for GG1; MRI-first and active surveillance reduce overdiagnosis and overtreatment.

## LAQ 6 65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis

P2

**Type:** LAQ (25 marks)
**Source:** Practice Set 12 Paper 2, Q1 (biochemical recurrence also discussed in 2019-Summer Paper 4, Q2)

*Assumed meaning: 6 months after RP (radical prostatectomy) the PSA (prostate-specific antigen) never became undetectable and is now ≥2 ng/mL, i.e. PSA persistence with early rise, suggesting residual or systemic disease.*

### Introduction

- After RP, PSA (half-life about 3 days) should become undetectable (<0.1 ng/mL) within about 6 weeks.
- **PSA persistence:** failure to become undetectable after RP. **BCR (biochemical recurrence):** a rise after an undetectable nadir.
- **Gleason 4+3 = grade group 3** (unfavourable intermediate risk).

### Definitions of BCR after RP

| Body | Definition |
|---|---|
| AUA (American Urological Association) | PSA ≥0.2 ng/mL |
| NCCN (National Comprehensive Cancer Network) | Failure to reach undetectable, two consecutive rises, or PSA ≥0.1 ng/mL |
| **EAU (European Association of Urology) risk groups** | **Low risk:** PSADT (PSA doubling time) >1 year **and** grade group <4. **High risk:** PSADT ≤1 year **or** grade group 4–5 |

### Why is the PSA high? (sources)

1. **Residual local disease:** positive margin (the only modifiable factor), residual benign tissue (low, stable PSA).
2. **Occult nodal metastases** (limited or no lymph node dissection).
3. **Distant micrometastases:** likeliest with persistent PSA and a fast rise.

### Evaluation

- **Repeat PSA** to confirm; calculate PSADT.
- **Review pathology:** pT stage, margin (length >1 mm, Gleason 4 at margin), seminal vesicles, nodes. **Also know:** CAPRA-S (Cancer of the Prostate Risk Assessment–Surgery) nomogram; Decipher genomic classifier.
- **PSMA (prostate-specific membrane antigen) PET/CT (positron emission tomography/computed tomography):** best test; detection about half below 0.5 ng/mL and nearly 100% above 2 ng/mL. Choline or fluciclovine if PSMA is unavailable.
- **mpMRI (multiparametric MRI) pelvis:** best for local recurrence in the prostatic fossa.
- CT and bone scan are rarely positive below PSA 10 ng/mL.

## Treatment

```
Confirm PSA, PSADT, pathology → PSMA PET ± mpMRI ├ Negative or fossa only →
salvage RT to bed + ADT ├ Pelvic nodes only → bed + pelvic node RT + ADT ├
Oligometastatic (≤3) → ADT ± metastasis-directed therapy └ Polymetastatic →
treat as mHSPC: ADT + ARPI ± docetaxel
```

**1. Salvage RT (radiotherapy): the only curative option**

- Dose **≥64–66 Gy** to the bed (SAKK 09/10: 70 Gy gave no gain over 64 Gy); 40–45 Gy to pelvic nodes.
- Best at the lowest PSA: each 0.1 ng/mL costs about 2.6% recurrence-free survival.
- **RADICALS-RT, RAVES, GETUG-AFU 17 (ARTISTIC meta-analysis):** early salvage at PSA 0.1–0.2 equals adjuvant RT and spares ≥60% from RT. These apply to men who become undetectable, not to this patient.

**2. Add ADT (androgen deprivation therapy) to salvage RT: essential here**

| Trial | Arms | Result |
|---|---|---|
| **RTOG 9601** (Radiation Therapy Oncology Group) | RT ± 2 years bicalutamide 150 mg; PSA 0.2–4.0 | 12-year OS (overall survival) 76.3% vs 71.3% |
| **GETUG-AFU 16** | RT ± 6 months goserelin; PSA 0.2–2.0 | 10-year BCR-free 64% vs 49%; MFS (metastasis-free survival) better; OS unchanged |
| **SPPORT (RTOG 0534)** | Bed / bed + 4–6 months ADT / pelvis + ADT | 5-year freedom from progression 70.9% / 81.3% / 87.4% |
| **RADICALS-HD** | None vs 6 vs 24 months ADT | 24 months better MFS than 6 months |
| - At PSA ≥2 ng/mL with grade group 3, give whole-pelvis RT with **long-course (about 2 years) ADT**. | | |

**3. Systemic therapy**

- **PSMA-positive metastases:** ADT + ARPI (androgen receptor pathway inhibitor: abiraterone, enzalutamide, apalutamide) ± docetaxel (LATITUDE, ARCHES, PEACE-1, ARASENS).
- **Oligometastases (≤3):** MDT (metastasis-directed therapy) by SBRT (stereotactic body RT) delays ADT (**STOMP:** ADT-free survival 21 vs 13 months; **ORIOLE:** progression 19% vs 61%).
- **High-risk BCR not fit for salvage: EMBARK** (PSADT ≤9 months, PSA ≥1 ng/mL): enzalutamide + leuprolide, MFS HR (hazard ratio) 0.42.
- **Also know:** castrate testosterone <50 ng/dL; intermittent ADT acceptable (PR.7); observation only for EAU low-risk BCR or short life expectancy; salvage lymph node dissection is investigational.

## Prognosis

- **Untreated BCR (Pound):** metastasis in about one-third, median 8 years; death a median 5 years later. Predictors: **BCR <2 years, PSADT <10 months, Gleason 8–10.**
- **This patient is poor-risk:** persistent PSA, high level within 6 months, likely short PSADT. Persistent PSA carries a higher risk of metastasis than late BCR.

- **Salvage RT:** one-third to one-half BCR-free at 4–5 years; poorer above PSA 0.5–1 ng/mL. Cancer-specific survival gain mainly with PSADT <6 months treated within 2 years.
- Lifelong PSA follow-up.

**Diagram to draw:** the treatment flowchart above, with PSA kinetics curve after RP.

> **Key points to remember:**
> - EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
> - PSMA PET first: almost 100% detection above PSA 2 ng/mL.
> - Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
> - Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

## LAQ 7 48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief.

P2

**Type:** LAQ (25 marks)
**Source:** Practice Set 14 Paper 2, Q1 (related: prognosis in 2019-Summer Paper 4, Q3e; staging of non-organ-confined disease in Practice Set 10 Paper 2, Q1)

*"Mixed voiding symptoms" is read as mixed storage and voiding LUTS (lower urinary tract symptoms).*

### Introduction

- In a 48-year-old, a PSA (prostate-specific antigen) of 18 ng/mL is far above what BPH (benign prostatic hyperplasia) usually produces. **Prostate cancer must be excluded first**; the LUTS are then treated on their own merits.
- PSA is organ-specific, not cancer-specific: BPH, prostatitis, UTI (urinary tract infection), retention and instrumentation also raise it. At PSA >10 ng/mL, biopsy yield is 43–65% with a normal DRE (digital rectal examination) and 70–90% with a suspicious one.

### History

- **LUTS:** IPSS (International Prostate Symptom Score) with bother score; storage versus voiding; retention, haematuria, recurrent UTI.
- **Red flags:** bone pain, weight loss, leg oedema, cord compression.
- Family history (prostate cancer <65 years; BRCA (breast cancer gene)-related cancers), recent catheter, fever or perineal pain (prostatitis), 5ARI (5α-reductase inhibitor) use, neurological disease, erectile function.

### Examination

- **DRE:** size, nodule, induration, extension, fixity (clinical T stage). Palpable bladder, neurological and spinal examination.

### Investigations

| Test | Purpose |
|---|---|
| Urinalysis, culture | Exclude UTI or prostatitis; treat and repeat PSA later (no empirical antibiotics without symptoms) |
| Repeat PSA, free PSA | Confirm the rise; low free PSA favours cancer |
| Creatinine, ultrasound KUB (kidney, ureter, bladder) | Upper tracts, prostate volume (PSA density) |
| Uroflowmetry, PVR (post-void residual) | Baseline LUTS assessment |
| **mpMRI (multiparametric MRI) before biopsy** | PI-RADS (Prostate Imaging Reporting and Data System) score; guides targeted cores; local staging |
| **Biopsy** | Targeted (PI-RADS ≥3) **plus systematic** 12 cores; **transperineal** preferred (lower sepsis) |

- At PSA 18, biopsy is needed **even if MRI is negative**. Blood or urine biomarkers (PHI (Prostate Health Index), 4Kscore, PCA3) matter mainly in the 2–10 ng/mL grey zone.

### Staging if cancer is found

- Grade group, cores involved, PSA, T stage → risk group. PSA 10–20 makes him at least **intermediate risk** (D'Amico, NCCN (National Comprehensive Cancer Network)).
- **High risk:** bone scan with CT (computed tomography), or **PSMA PET (prostate-specific membrane antigen positron emission tomography)**. Organ-confined disease is <50% at PSA >10; nodes about 20% at PSA >20.
- **Germline testing** (BRCA2, mismatch repair genes) for high-risk or metastatic disease or strong family history.

### Management of the cancer (life expectancy >25 years)

| Situation | Management |
|---|---|
| Low or favourable intermediate risk | AS (active surveillance) is possible, but uncommon at PSA 18 |
| Localised intermediate or high risk | **RP (radical prostatectomy)**, usually robotic, with extended PLND (pelvic lymph node dissection) when node risk exceeds the nomogram threshold; or **EBRT (external beam radiotherapy) + ADT (androgen deprivation therapy)**: 4–6 months intermediate, 18–36 months high risk |
| Locally advanced (T3–T4, N1) | Multimodal: RP ± adjuvant or early salvage RT, or EBRT + long-term ADT; abiraterone added for high-risk node-positive or high-risk N0 disease (STAMPEDE) |
| Metastatic | ADT + ARPI (androgen receptor pathway inhibitor) ± docetaxel; RT to the prostate for low volume |

- **Why surgery suits him:** young, gives pathological staging, and **removes the obstructing gland**, curing the LUTS. SPCG-4: surgical benefit is concentrated in men <65.
- Brachytherapy is relatively contraindicated with severe LUTS, a large gland or prior TURP (transurethral resection of the prostate) (retention, incontinence).

### If biopsy is negative

- PSA 18 still needs explaining (prostatitis, large gland, missed anterior tumour): repeat MRI-targeted transperineal biopsy; %free PSA, PCA3, ConfirmMDx; close PSA follow-up.

### Management of LUTS

- **Behavioural measures** (fluid timing, less caffeine) and an **α-blocker** (e.g. tamsulosin) at once; it does not alter PSA.
- **Avoid starting a 5ARI until cancer is excluded:** it halves PSA and masks cancer; if used later, record a baseline PSA and double readings after 12 months.
- Persistent storage symptoms with low PVR: add an antimuscarinic or β3-agonist (mirabegron).
- **Surgery** (TURP, HoLEP (holmium laser enucleation of the prostate)) for retention, refractory symptoms or complications, only once cancer is excluded. In locally advanced cancer with retention, a channel TURP relieves obstruction.
- **Also know:** pressure-flow study before surgery at age <50 with mainly voiding LUTS (EAU (European Association of Urology)).

`PSA 18 + LUTS → history, DRE, urine culture → repeat PSA → mpMRI → targeted + systematic biopsy (transperineal) Cancer → risk group + staging → RP / RT + ADT / systemic Benign → explain PSA, re-biopsy if needed → BPH pathway LUTS throughout: α-blocker; no 5ARI until cancer excluded`

**Diagram to draw:** the flowchart above.

> **Key points to remember:**
>
> - PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
> - PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
> - Young fit man: RP treats both cancer and obstruction.
> - α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

## LAQ 8 Discuss management of organ confined CA prostate

P2

**Type:** LAQ (25 marks)
**Source:** Practice Set 15 Paper 2, Q2

### Introduction

- Organ-confined prostate cancer is **cT1–T2 N0 M0**. Treatment must balance cancer mortality against **competing mortality**: life expectancy, comorbidity, baseline urinary, sexual and bowel function, and patient preference (shared decision-making).

### Risk stratification (NCCN, National Comprehensive Cancer Network)

| Risk group | Criteria |
|---|---|
| Very low | All of: T1c, GG (grade group) 1, PSA (prostate-specific antigen) <10 ng/mL, <3 positive cores with ≤50% cancer in each, PSA density <0.15 |
| Low | All of: T1–T2a, GG1, PSA <10 |
| Favourable intermediate | 1 intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive |
| Unfavourable intermediate | 2–3 intermediate factors, GG3, or ≥50% cores positive |
| High | Any of: T3a, GG4–5, PSA >20 |

- **Grade groups:** GG1 = 3+3; GG2 = 3+4; GG3 = 4+3; GG4 = 8; GG5 = 9–10.

### Evaluation

- PSA, DRE (digital rectal examination), mpMRI (multiparametric MRI) with targeted and systematic biopsy.
- **Staging imaging** from unfavourable intermediate risk: bone scan + CT/MRI, or PSMA (prostate-specific membrane antigen) PET (positron emission tomography). **proPSMA:** accuracy 92% vs 65% for conventional imaging.
- Life expectancy (life tables, Charlson index); IPSS (International Prostate Symptom Score) and erectile function.
- **Also know:** Partin tables, Kattan nomogram, CAPRA (Cancer of the Prostate Risk Assessment) score; genomic tests (Decipher, Prolaris, Oncotype DX) for borderline surveillance decisions.

### Options by risk

| Risk | Preferred options |
|---|---|
| Very low / low | **Active surveillance (AS)**; watchful waiting if life expectancy short |
| Favourable intermediate | AS in selected men; RP (radical prostatectomy); EBRT (external-beam radiotherapy) or brachytherapy alone |
| Unfavourable intermediate | RP + PLND (pelvic lymph node dissection); EBRT + **4–6 months ADT** (androgen deprivation therapy) |
| High (organ-confined, e.g. GG4–5 or PSA >20) | RP + extended PLND (multimodal); EBRT + **18–36 months ADT** |

## 1. Active surveillance vs watchful waiting

- **AS:** deferred treatment with **curative intent**. Monitor by PSA every 3–6 months, DRE, **confirmatory biopsy or MRI within 6–12 months**, then periodic MRI and biopsy. Trigger for treatment: grade progression (GG ≥2–3), increasing volume or patient choice.
- Entry criteria: Johns Hopkins (strict, Epstein), PRIAS (Prostate Cancer Research International Active Surveillance), Toronto (inclusive). About half are treated by 10–15 years.
- **Watchful waiting:** no curative intent; ADT for symptomatic progression. For limited life expectancy.

## 2. Radical prostatectomy

- Open retropubic or **robot-assisted** (similar outcomes in the randomised trial). Nerve sparing when there is no same-side extracapsular risk.
- **PLND** when nodal risk is significant on nomograms (extended template).
- **Complications:** incontinence, ED (erectile dysfunction), anastomotic stricture, rectal injury.
- **Advantages:** pathological staging; relieves obstruction; RT kept for salvage.

## 3. Radiotherapy

- **EBRT:** IMRT (intensity-modulated RT) with image guidance. Fractionation: conventional 78 Gy in 39; **moderate hypofractionation 60 Gy in 20** (CHHiP: non-inferior); **SBRT (stereotactic body RT) 36.25 Gy in 5** (PACE-B: similar toxicity).
- **Why add ADT:** synergistic cell kill and target shrinkage. **RTOG 94-08:** 4 months ADT improved OS in intermediate risk; **DFCI 95-096:** 6 months ADT improved 8-year OS (61% → 74%).
- **Brachytherapy:** LDR (low-dose-rate) seeds for low and favourable intermediate risk. Relative contraindications: gland >60 mL, high IPSS, recent TURP. HDR (high-dose-rate) or LDR **boost** with EBRT for higher risk (ASCENDE-RT: half the biochemical failure).
- **Toxicity:** proctitis, irritative urinary symptoms, late ED, second cancers (controversial).

## 4. Focal and whole-gland ablation

- HIFU (high-intensity focused ultrasound), cryotherapy, focal laser, IRE (irreversible electroporation), VTP (vascular-targeted photodynamic therapy).
- Based on the **index lesion hypothesis**. Best for MRI-visible, biopsy-confirmed, unifocal **intermediate-risk (GG2)** disease. Investigational; ideally in trials. VTP is the only modality with a randomised trial (versus AS in low risk).

## Landmark trials

| Trial | Comparison | Result |
|---|---|---|
| **SPCG-4** (pre-PSA era) | RP vs watchful waiting | Cancer death RR (relative risk) 0.56 at 23 years; NNT 8; benefit <65 years |
| **PIVOT** | RP vs observation (PSA era) | No mortality difference; benefit only in intermediate risk |
| **ProtecT** | Monitoring vs RP vs RT + ADT | 15-year cancer mortality 2.7%, 2.2%, 2.9% (no difference); more metastases with monitoring |

- **ProtecT function:** RP worst for continence and sexual function; RT transient bowel effects.

### Algorithm

```
Organ-confined CaP → risk group + life expectancy ├─ short life expectancy → watchful waiting ├─ very low / low → active surveillance ├─ favourable intermediate → AS (selected), RP, EBRT, brachytherapy └─ unfavourable intermediate / high → RP + PLND or EBRT + ADT (4–6 or 18–36 months)
```

### Follow-up

- PSA every 6–12 months. **Biochemical recurrence:** after RP, PSA ≥0.2 ng/mL confirmed; after RT, **nadir + 2 ng/mL (Phoenix)**.

### Recent advances

- PSMA PET staging; MRI-led surveillance; SBRT; genomic classifiers.

**Diagram to draw:** stepwise decision tree (treat or not → partial or whole gland → surgery or radiation).

> **Key points to remember:**
> - Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
> - AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
> - EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
> - Focal therapy rests on the index lesion hypothesis; still investigational.

## LAQ 9 Discuss advances in management of CA prostate

P4

**Type:** LAQ (25 marks)
**Source:** Practice Set 5 Paper 4, Q2 (related: 2025-Winter Paper 3, Q3b; 2024-Winter Paper 4, Q3f)

### Introduction

- Three aims drive recent change: **find significant cancer and avoid diagnosing insignificant cancer**; **match local treatment to risk** with less morbidity; give **earlier, intensified and targeted** systemic therapy.

### 1. Screening and diagnosis

- **ERSPC (European Randomized Study of Screening for Prostate Cancer)** and Göteborg-1: PSA (prostate-specific antigen) screening lowers cancer-specific mortality, but PSA alone causes negative biopsies and overdiagnosis of GG (Grade Group) 1. Hence risk-stratified pathways.
- **Biomarkers** (reflex tests before biopsy):

| Test | Sample | Content |
|---|---|---|
| PHI (Prostate Health Index) | Blood | (p2PSA ÷ free PSA) × √total PSA |
| 4Kscore | Blood | Four kallikreins + clinical data; % risk of GG ≥2 |
| STHLM3 | Blood | 4Kscore kallikreins + 232 risk SNPs (single-nucleotide polymorphisms) + clinical variables |
| PCA3 (prostate cancer antigen 3), SelectMDx | Urine after DRE (digital rectal examination) | RNA ratio (PCA3); DLX1 and HOXC6 (SelectMDx) |
| ExoDx IntelliScore | Urine, no DRE | Exosomal RNA of PCA3, ERG, SPDEF |

- Tissue genomic classifiers (Decipher, Prolaris, Oncotype DX) refine risk for surveillance and adjuvant decisions.
- **MRI-first pathway:** mpMRI (multiparametric magnetic resonance imaging), scored by **PI-RADS (Prostate Imaging–Reporting and Data System)** 1–5, before biopsy.
- **PROMIS (2017):** mpMRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; NPV (negative predictive value) 89%; 27% could avoid biopsy.
- **PRECISION (2018):** MRI with targeted biopsy of PI-RADS ≥3 found more significant cancer (38% vs 26%); 28% avoided biopsy.
- So PI-RADS ≥3 lesions are biopsied, while PI-RADS 1–2 with low PSA density may safely avoid biopsy.
- **Transperineal biopsy** under local anaesthesia: near-zero sepsis; MRI–ultrasound fusion targeting. **Micro-ultrasound** (PRI-MUS scoring) matches mpMRI in registry data.
- **PSMA (prostate-specific membrane antigen) PET (positron emission tomography): proPSMA:** accuracy 92% vs 65% for CT (computed tomography) plus bone scan (sensitivity 85%, specificity 98%). Used for high-risk staging, biochemical recurrence and selecting radioligand therapy.

## 2. Localised disease

- **AS (active surveillance):** standard for GG1; extended to low-volume favourable intermediate risk (GG2), with MRI and PSA density. **ProtecT (15 years):** prostate cancer mortality about 2–3% in monitoring, surgery and radiotherapy arms; more metastasis on monitoring, no survival difference.
- **Robotic RP (radical prostatectomy): Retzius-sparing** (Bocciardi) gives earlier continence (1 week to 3 months; equal by 12 months); single-port robot; frozen-section-guided nerve sparing (NeuroSAFE).
- **Radiotherapy:**
- Moderate hypofractionation **60 Gy in 20** (CHHiP) is standard.
- **SBRT (stereotactic body radiotherapy) 36.25 Gy in 5:** PACE-B showed similar late toxicity at 2 years (chapter); its 5-year report (2024) showed non-inferior biochemical or clinical control. HYPO-RT-PC (42.7 Gy in 7) was also non-inferior.
- **FLAME:** MRI-lesion boost improved biochemical PFS (progression-free survival) (HR [hazard ratio] 0.45). **Hydrogel rectal spacer** reduces rectal toxicity.
- High-risk M0: **abiraterone added to ADT (androgen deprivation therapy) and RT (STAMPEDE):** MFS (metastasis-free survival) HR 0.53.
- **Focal therapy:** based on the **index lesion hypothesis** (the largest or highest-grade focus drives progression). Energies: HIFU (high-intensity focused ultrasound), cryotherapy, IRE (irreversible electroporation), focal laser, VTP (vascular-targeted photodynamic therapy, padeliporfin). **HEAT registry (1379 men):** failure-free survival 68% (intermediate risk); continence 90–95%. Still within trials or registries.

## 3. Recurrent and advanced disease

| Setting | Advance | Evidence |
|---|---|---|
| High-risk BCR (biochemical recurrence) | PSMA-guided salvage; enzalutamide + leuprolide | **EMBARK:** MFS HR 0.42 |
| mHSPC (metastatic hormone-sensitive) | ADT + ARPI (androgen receptor pathway inhibitor) | LATITUDE, TITAN, ENZAMET, ARCHES |
| | Triplet (ADT + docetaxel + ARPI) | **PEACE-1, ARASENS** |
| | Prostate RT, low volume | STAMPEDE arm H |
| | MDT (metastasis-directed therapy) | STOMP, ORIOLE |
| nmCRPC (non-metastatic castration-resistant) | Apalutamide, enzalutamide, darolutamide | SPARTAN, PROSPER, ARAMIS |
| HRR (homologous recombination repair)-mutated mCRPC | **PARP (poly-ADP-ribose polymerase) inhibitors** | **PROfound:** olaparib imaging PFS 7.4 vs 3.6 months; PROpel, MAGNITUDE, TALAPRO-2 |
| PSMA-positive mCRPC | **$^{177}$Lu-PSMA-617** | **VISION:** OS (overall survival) 15.3 vs 11.3 months (HR 0.62); TheraP: PSA response 65% vs 37% vs cabazitaxel |

- **ADT alone is no longer standard** for mHSPC; every fit man gets intensification.

- **Germline and somatic testing** for all metastatic disease; **pembrolizumab** for MSI-high (microsatellite instability-high) tumours.
- **Relugolix:** oral GnRH (gonadotropin-releasing hormone) antagonist; HERO showed better sustained castration than leuprolide.
- **Also know:** radium-223 (ALSYMPCA, OS HR 0.70); PSMAfore (lutetium before chemotherapy); AI in MRI reading and pathology.

**Diagram to draw:** MRI-first pathway: raised PSA → mpMRI → PI-RADS ≥3 targeted transperineal biopsy; PI-RADS 1–2 with low PSA density → surveillance.

> **Key points to remember:**
> - PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
> - proPSMA: PSMA PET accuracy 92% vs 65%.
> - ProtecT: no mortality difference at 15 years; AS is standard for GG1.
> - VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

## Short answer questions (SAQ)

### SAQ 1 TRUS and prostate biopsy

P1 · P2 **merged from 3**

*Asked as:* P2 SAQ "Transrectal ultrasonography (TRUS)" (Transrectal ultrasonography; TRUS) · P2 SAQ "TRUS / prostate biopsy" (TRUS biopsy.; Prostate biopsy) · P1 SAQ "Transrectal ultrasound study of prostate"

*Read as: transrectal ultrasonography (TRUS) of the prostate, and TRUS-guided prostate biopsy.*

#### Physics, principle and technique of TRUS

- Piezoelectric crystals; higher frequency gives better resolution but less penetration (depth).
- The prostate lies directly in front of the rectum. Probes: **6–10 MHz** end-fire, side-fire or biplane.
- Left lateral position (or lithotomy for transperineal work), after DRE (digital rectal examination); transverse and sagittal planes, base to apex.

#### Normal sonographic anatomy

| Zone (McNeal) | Glandular share | TRUS appearance | Cancer share |
|---|---|---|---|
| Peripheral | ~70% | Homogeneous, echogenic (posterior) | ~70% |
| Transition | 5–10% (BPH [benign prostatic hyperplasia]) | Heterogeneous, anterior | ~20–25% |
| Central | ~25% | Homogeneous, echogenic, at base | ~5–10% |
| Anterior fibromuscular stroma | Non-glandular | Hypoechoic | – |

- Zones are not seen as such; TRUS separates echogenic posterior PZ (peripheral zone) from heterogeneous TZ (transition zone).
- **Corpora amylacea** mark the surgical capsule.
- Seminal vesicles about 4.5–5.5 × 2 cm; absent seminal vesicle → ipsilateral renal agenesis in 79%.

### Measurements

- **Volume (ellipsoid) = 0.52 (π/6) × transverse × AP (anteroposterior) × length**; 1 mL ≈ 1 g; normal 20–25 g. **Planimetry** (serial sections) for brachytherapy.
- **PSAD (PSA [prostate-specific antigen] density)** = PSA/volume; **>0.15 ng/mL/cc** is suspicious (suggests cancer).

### Findings

- **Cancer:** classically a hypoechoic peripheral-zone lesion, but **39% are isoechoic**; false positives are BPH nodules, prostatitis and infarcts. Capsular bulge suggests extracapsular (extraprostatic) extension.
- **Cysts:** Müllerian (midline); ejaculatory duct cyst (just off midline); seminal vesicle cyst (lateral; renal agenesis).
- Abscess, calculi.
- After ADT (androgen deprivation therapy) or radiotherapy the gland shrinks and becomes diffusely hypoechoic.

### Indications for TRUS

- **Guided biopsy:** 12-core systematic ± MRI (magnetic resonance imaging)-targeted fusion.
- Volume before BPH surgery, brachytherapy or cryotherapy; fiducial markers.
- **Azoospermia:** ejaculatory duct obstruction (seminal vesicles >1.5 cm wide, dilated ducts, midline cyst).
- Drainage of prostatic abscess or cysts; haematospermia.

### Prostate biopsy: introduction

- TRUS-guided systematic 12-core biopsy has been the standard for diagnosing prostate cancer; about 30% are positive at first biopsy.

### Prostate biopsy: indications and contraindications

- Raised PSA with risk factors; suspicious DRE; MRI PIRADS (Prostate Imaging Reporting and Data System) ≥3; suspected metastasis.
- **Repeat:** rising PSA; ASAP (atypical small acinar proliferation; 50% cancer risk); positive biomarkers (PHI [Prostate Health Index], 4Kscore, PCA3); active surveillance. Saturation biopsy only after negative biopsies.
- **Contraindications:** coagulopathy, acute UTI (urinary tract infection) or prostatitis, severe immunosuppression; no anus (use transperineal route).

### MRI-first pathway

- **PROMIS (2017):** mpMRI (multiparametric MRI) sensitivity 93% vs 48% for TRUS biopsy.
- **PRECISION (2018):** MRI-targeted biopsy found more significant cancer (38% vs 26%) and avoided 28% of biopsies.
- Targeting: cognitive, MRI/US fusion or in-bore; 2–3 cores per target ± systematic cores.

## Prostate biopsy: preparation

- Continue low-dose aspirin; stop warfarin and clopidogrel 5–7 days, DOACs (direct oral anticoagulants) 2–5 days before.
- **Transrectal: antibiotics for all** (contaminated route), ≤24 hours: fluoroquinolone, or cephalosporin ± aminoglycoside. Augmented or swab-directed if risk factors (antibiotics in last 6 months, diabetes, previous infection). Povidone-iodine rectal cleansing.
- Enema optional. **Periprostatic block:** 1–2% lidocaine, 5 mL each side at the junction of prostate base and seminal vesicle.

## Prostate biopsy: technique

| | Transrectal | Transperineal |
|---|---|---|
| Position | Left lateral | Lithotomy |
| Needle route | Rectal wall (contaminated) | Sterile perineal skin |
| Antibiotics | Mandatory | Skin-flora only or none |
| Anterior/apex | Poor | Good |
| Anaesthesia | Periprostatic block | Perineal and periapical triangle block |

- Left lateral position; DRE; 6–10 MHz probe; measure volume (PSA density).
- 18-gauge spring-loaded needle; indent the capsule, withdraw slightly, then fire, so the core samples peripheral zone.
- **12 cores:** base, mid-gland and apex on each side, medial and lateral (sextant alone is inadequate; it missed lateral cancers). Add MRI-targeted cores (cognitive or fusion). TZ (transition zone) cores if gland >50 mL.
- **EAU (European Association of Urology) 2023 prefers transperineal biopsy.**

## Complications (transrectal)

| Complication | Rate |
|---|---|
| Haematuria | 23–63% |
| Haematospermia | 10–50% |
| Rectal bleeding | 2–22% |
| Infection | 0.1–7%; sepsis 0.3–3% |
| Retention | 0.2–0.4% |
| Vasovagal reaction | 1–5% |

- Bleeding is commonest; sepsis is most serious.
- **Sepsis** from fluoroquinolone-resistant *E. coli*: carbapenem until cultures return; prevent with povidone-iodine rectal cleansing plus targeted or augmented prophylaxis.
- Rectal bleeding: probe pressure, balloon tamponade, endoscopy, embolisation.

## Recent advances

- **Transperineal biopsy** under local anaesthesia: fewer infections, better apex and anterior sampling; preferred by EAU 2023.
- **MRI-first pathway** (PROMIS, PRECISION, above).
- Colour/power Doppler, contrast-enhanced TRUS, **elastography**, multiparametric ultrasound.

- **Micro-ultrasound (29 MHz; 70 µm resolution)** with PRI-MUS scoring, detection comparable to mpMRI.
- **Also know:** PSMA (prostate-specific membrane antigen) PET.

### Diagrams to draw

- 12-core biopsy template on a transverse prostate outline.
- Transverse and sagittal TRUS views labelling PZ, TZ, seminal vesicles and urethra.

> **Key points to remember:**
> - Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
> - 39% of cancers isoechoic; hypoechoic PZ lesion is classic.
> - Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.
> - MRI before biopsy: PROMIS and PRECISION established the MRI-first pathway.
> - 12 cores; antibiotics ≤24 hours for all (transrectal); sepsis 0.3–3%.
> - Bleeding commonest; sepsis most serious.
> - Transperineal now preferred.

## SAQ 2 Multiparametric MRI, PI-RADS and fusion biopsy

P1 · P4 **merged from 2**

*Asked as:* P1 SAQ "Prostate multiparametric MRI and endorectal coil MRI" (Multiparametric MRI : its components and interpretation for prostate cancer.; Endorectal coil MRI) · P4 SAQ "PI-RADS and MRI–ultrasound fusion biopsy"

### Introduction

- **mpMRI (multiparametric magnetic resonance imaging):** anatomical plus functional sequences, usually at 3 T without endorectal coil. Reported by **PI-RADS v2.1 (Prostate Imaging Reporting and Data System, 2019)**.
- PI-RADS scores the likelihood of **clinically significant cancer** (Gleason ≥7, volume >0.5 cm³ or EPE [extraprostatic extension]).

### Components

| Sequence | Principle | Cancer appearance |
|---|---|---|
| **T2W (T2-weighted)**, triplanar | Anatomy; normal PZ (peripheral zone) bright | Low-signal lesion; "erased charcoal" in TZ (transition zone); shows EPE and SVI (seminal vesicle invasion) |
| **DWI (diffusion-weighted imaging)** with **ADC (apparent diffusion coefficient)** map; high b-value (≥1400) | Cellularity: cellular tumour restricts water movement | Bright on high-b DWI, **dark on ADC**; ADC falls as Gleason rises |
| **DCE (dynamic contrast-enhanced)** T1 with gadolinium | Vascularity (angiogenesis): early enhancement and washout | Early focal enhancement (type 3 curve) |

- **Biparametric MRI (T2W + DWI):** no contrast, cheaper, similar detection; poorer staging.
- **PI-QUAL (Prostate Imaging Quality):** image quality score (1–5) before reporting.

## Interpretation: PI-RADS v2.1

| Score | Clinically significant cancer |
|---|---|
| 1 | Highly unlikely |
| 2 | Unlikely |
| 3 | Equivocal |
| 4 | Likely; lesion <1.5 cm |
| 5 | Highly likely; ≥1.5 cm or definite EPE |

| Zone | Dominant sequence | Tie-breaker at score 3 |
|---|---|---|
| Peripheral | **DWI** | DWI 3 + DCE positive → 4 |
| Transition | **T2W** | T2W 3 + DWI 5 → 4; (v2.1) T2W 2 + DWI ≥4 → 3 |

- **DCE only upgrades PZ 3 to 4**; it never changes 4–5 or TZ scores.
- Cancer in about one-third of PI-RADS 3, over half of 4, 90% of 5.
- **PRECISE** score (1–5) tracks change during active surveillance.

## Clinical use and evidence for MRI

- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; 27% could avoid biopsy.
- **PRECISION (2018):** MRI pathway with targeted biopsy of PI-RADS ≥3 detected more significant cancer (38% vs 26%); 28% avoided biopsy.
- EAU (European Association of Urology) and AUA (American Urological Association): MRI **before biopsy**; also staging, active surveillance, focal therapy planning.
- Limitations: low specificity (Cochrane 0.37); reader-dependent.

## MRI–ultrasound fusion biopsy: principle

- Lesion contoured on MRI → software **co-registers it with live TRUS** in three dimensions; **elastic deformation** corrects probe distortion → needle aimed at the displayed target.
- Freehand electromagnetic or mechanical-arm tracking; transrectal or **transperineal** (fewer infections).

| | Cognitive | Software fusion | In-bore MRI |
|---|---|---|---|
| Method | Aims from memory of MRI | Software overlay | Needle under MRI |
| Drawback | Operator-dependent; small lesion in large gland | Cost; registration error | Time, cost |

- Meta-analysis (2022): no significant difference among the three; one RCT (randomised controlled trial) favoured fusion over cognitive.

## Fusion biopsy: indications and evidence

- PI-RADS ≥3 lesion with raised PSA (prostate-specific antigen), first or repeat biopsy; active surveillance; focal therapy.
- **AUA/SUO (American Urological Association/Society of Urologic Oncology) 2023:** 2–3 cores per target ± systematic cores.
- Targeting reduces overdiagnosis; MRI-invisible cancers keep systematic cores relevant.

### Endorectal coil MRI: principle

- **ERC (endorectal coil):** a receive-only surface coil inside an inflatable balloon placed in the rectum against the prostate, used with an external pelvic phased-array coil.
- **"The currency of MRI is signal."** A coil close to the prostate gives higher **SNR (signal-to-noise ratio)**, allowing **higher spatial resolution**.

### Endorectal coil MRI: technique

- Antispasmodic to reduce rectal motion artefact; balloon filled with air or perfluorocarbon (fluid reduces susceptibility artefact).
- Sequences as for mpMRI: **T2W** triplanar for anatomy, **DWI with ADC** map, **DCE**; ERC was traditionally needed for **MRSI (MR spectroscopic imaging)**.

### Endorectal coil MRI: uses

- **Local staging:** extraprostatic extension and seminal vesicle invasion (both low T2 signal) before radical prostatectomy.
- **Local recurrence** after prostatectomy: recurrence isointense on T1, slightly hyperintense on T2; DCE raises accuracy at low PSA.
- **Also know:** focal therapy planning (ERC recommended at 1.5 T); seminal vesicles; female urethral diverticulum.

### ERC versus no ERC

| | **With ERC** | **Without ERC (3 T, pelvic coil)** |
|---|---|---|
| Signal and resolution | Higher | Adequate at 3 T |
| csPCa (clinically significant cancer) detection | GG (Grade Group) ≥3 equal; GG2 slightly better | GG ≥3 equal |
| Comfort, cost, time | Worse | Better |
| Gland | Deformed (fusion and radiotherapy planning harder) | Natural shape |

### ERC: disadvantages and contraindications

- Artefacts: **near-field signal flare** next to the coil (needs intensity correction), air susceptibility at the rectal wall, malposition.
- **Contraindications:** absent rectum (abdominoperineal resection), recent rectal surgery, anorectal fistula or inflammatory bowel disease, severe haemorrhoids or stricture, latex allergy.

### ERC: current position

- At **3 T the ERC is largely historical**; PI-RADS v2.1 does not require it. **Consider it at 1.5 T** or with older scanners, or when body-coil quality is poor.

### Diagrams to draw

- Prostate zones with a PZ lesion dark on ADC.
- MRI target on live TRUS with needle path.
- Sagittal pelvis with inflated endorectal coil against the posterior prostate.

**Key points to remember:**

- T2W anatomy, DWI/ADC cellularity, DCE vascularity.
- PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
- PRECISION and PROMIS: MRI before biopsy.
- Fusion: co-registration with elastic deformation; cognitive, fusion, in-bore: similar yield.
- Endorectal coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.

## SAQ 3 Maximum / combined androgen blockade

P2 · P4 **merged from 2**

*Asked as:* P2 SAQ "Maximum (combined) androgen blockade and bicalutamide" (Maximum androgen blockade; Bicalutamide) · P4 SAQ "Combined Androgen Blockade."

### Definition and rationale

- **MAB (maximum androgen blockade)**, also **CAB (combined androgen blockade)** or total androgen blockade: castration (orchidectomy or LHRH [luteinising hormone-releasing hormone] agonist) **plus a non-steroidal antiandrogen**.
- **Rationale (Labrie):** castration removes about 90–95% of testosterone, but **adrenal androgens** (DHEA, dehydroepiandrosterone; androstenedione) are converted to DHT (dihydrotestosterone) inside the prostate. The antiandrogen blocks these at the AR (androgen receptor).

### Antiandrogens

| Drug | Dose | Signature toxicity |
|---|---|---|
| Flutamide | 250 mg three times daily | Diarrhoea, hepatotoxicity |
| Nilutamide | 300 mg, then 150 mg daily | Delayed dark adaptation, pneumonitis |
| **Bicalutamide** | **50 mg daily** (with LHRH agent) | Best tolerated; gynaecomastia |
| Cyproterone (steroidal) | 100 mg two to three times daily | Cardiovascular, hepatic; no CAB benefit, not used for CAB now |

- Class effects: gynaecomastia, hepatotoxicity, pneumonitis.

### Evidence for MAB

- **Crawford (1989):** leuprolide + flutamide improved survival (partly flare control); **Eisenberger (1998):** orchidectomy + flutamide, **no** benefit.
- **PCTCG (Prostate Cancer Trialists' Collaborative Group) meta-analysis, 2000** (27 trials, about 8,000 men): with non-steroidal antiandrogens, 5-year survival improved by only **about 2–3%** (borderline significance); cyproterone gave no benefit (slightly worse).
- Small gain, more cost and toxicity: **AUA (American Urological Association) advises against long-term CAB**.

### Current uses of MAB

1. **Flare prevention:** an LHRH agonist raises LH (luteinising hormone) up to 10-fold, and testosterone, for 10–20 days, risking cord compression, pain or obstruction with bone metastases. Start antiandrogen **with or before** the agonist for **21–28 days**. Antagonists (degarelix, relugolix) or orchidectomy cause no flare.
2. **Antiandrogen withdrawal:** on progression, stopping bicalutamide gives a PSA (prostate-specific antigen) fall >50% in **15–30%** for 3.5–5 months (mutant AR turns the drug into an agonist).

### Bicalutamide: introduction and mechanism

- Non-steroidal **AR antagonist**, called first- or second-generation. It competes with testosterone and DHT at the ligand-binding domain.
- Given alone, it blocks central feedback, so **LH and testosterone rise** (about 1.5 times); aromatisation to oestradiol causes gynaecomastia; libido is partly preserved.
- **Half-life about 6 days** (once daily); unaffected by age or renal impairment. Most potent and best tolerated of its class.

### Uses of bicalutamide

| Use | Dose | Evidence |
|---|---|---|
| **CAB** with LHRH agonist, metastatic | 50 mg daily | Small survival gain (2–3%, PCTCG) |
| **Flare prevention** | 50 mg, with or before the agonist, 2–4 weeks | Main use today |
| Monotherapy, locally advanced | 150 mg daily | **EPC (Early Prostate Cancer) programme:** better progression-free survival; **harm in localised disease** (death HR [hazard ratio] 1.23) |
| With salvage radiotherapy | 150 mg for 2 years | **RTOG 9601:** 12-year overall survival 76.3% vs 71.3% |
| CRPC (castration-resistant prostate cancer) | 50 mg | Second-line step; largely replaced by newer agents |

- 150 mg monotherapy roughly equals castration in locally advanced disease, with better sexual interest; not FDA (US Food and Drug Administration) approved alone, but used in some countries.
- **Also know:** stuttering priapism; feminising hormone therapy; studied in BPH (benign prostatic hyperplasia).

### Adverse effects of bicalutamide

- **Gynaecomastia and breast pain** (two-thirds at 150 mg): prevent with breast radiotherapy (10–15 Gy single fraction) or tamoxifen 10 mg daily.
- **Hepatotoxicity** (monitor liver tests); rare **interstitial pneumonitis**; erectile dysfunction; fewer hot flushes than castration.

### Modern intensification (replaces CAB in mHSPC, metastatic hormone-sensitive prostate cancer)

| Trial | Addition to ADT (androgen deprivation therapy) | Result |
|---|---|---|
| LATITUDE, STAMPEDE (arm G) | Abiraterone + prednisone | OS (overall survival) HR 0.62 / 0.63 |
| ENZAMET | Enzalutamide (vs ADT + older antiandrogen, a CAB control arm) | OS HR 0.67; enzalutamide beat CAB |
| ARCHES | Enzalutamide | Radiographic PFS (progression-free survival) HR 0.39 |
| TITAN | Apalutamide | OS HR 0.67 |
| PEACE-1, ARASENS | **Triplet**: + docetaxel + abiraterone or darolutamide | OS benefit (ARASENS HR 0.68) |

- Docetaxel is also an option with ADT.
- **Comparison:** enzalutamide beat bicalutamide in CRPC (TERRAIN, STRIVE) and in **ENZAMET**.

### Diagrams to draw

- Testicular and adrenal androgen sources with LHRH agonist and antiandrogen sites of action.
- Hypothalamus–pituitary–testis axis with bicalutamide blocking the AR.

> **Key points to remember:**
> - MAB/CAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
> - PCTCG 2000: only 2–3% 5-year survival gain.
> - Main use now: flare cover for 21–28 days.
> - ARPIs (androgen receptor pathway inhibitors) or triplets now standard in mHSPC; newer AR pathway inhibitors beat CAB (ENZAMET).
> - Bicalutamide half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
> - Monotherapy raises testosterone; gynaecomastia two-thirds.
> - EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
> - Withdrawal response 15–30%.

## SAQ 4 Saturation biopsy of prostate

P2 · P4 **merged from 2**

*Asked as:* P2 SAQ "Saturation biopsy in CA prostate" · P4 SAQ "Saturation biopsies of prostate"

### Definition

- Systematic prostate biopsy with **more cores than the extended 12-core scheme**, usually **18–21 or more** (often about 20–30), sampling the peripheral zone, transition zone, anterior gland and apex.
- Route: transrectal, or **transperineal template mapping biopsy** through a brachytherapy grid with a core every **5 mm**.

## Rationale

- Sextant biopsy (Hodge, 1989) under-sampled the lateral peripheral zone; the 12-core scheme (adds apical and far-lateral cores) is today's standard.
- Saturation reduces sampling error, mainly for **anterior and apical cancers** missed transrectally (anterior-only cancer 6–20%).

## Indications

- **Repeat biopsy** after a negative biopsy with persistent suspicion: rising PSA (prostate-specific antigen), ASAP (atypical small acinar proliferation), negative or unavailable MRI (magnetic resonance imaging).
- **Active surveillance** confirmation.
- **Focal therapy** mapping (shows multifocality; TRUS [transrectal ultrasound] biopsy misses almost half of disease relevant to hemiablation).

## Templates and technique

| Template | Division | Cores |
|---|---|---|
| Original Barzell | Octants (anterior/posterior × apex/base, each side) | 4–8 per octant |
| Modified Barzell | Apex and base of each side × 5 sectors | 20 |
| Ginsburg | Anterior, mid, posterior sectors per side | 4 per sector |

- Lithotomy, stepper and grid; general or spinal anaesthesia, or periprostatic block (up to 22 mL of 1% lidocaine transrectally).

## Results

- **Initial biopsy:** 12 to 18–21 cores did not raise detection.
- **Repeat setting:** early series gained; a later large prospective study did not. Extra cancers are often insignificant (overdiagnosis).
- **PROMIS (2017)** used 5-mm template mapping as the reference: mpMRI (multiparametric MRI) sensitivity 93% vs TRUS biopsy 48%.

## Complications

- More haematuria, haematospermia, pain and **urinary retention** than standard biopsy; anaesthesia and theatre time. Transperineal route has less infection.

## Current position

- Largely replaced by **mpMRI with targeted biopsy**: **PRECISION (2018)** detected more significant cancer (38% vs 26%) and 28% avoided biopsy.
- **EAU (European Association of Urology) 2023** prefers the transperineal route.
- Still used for MRI-negative high suspicion and focal therapy planning (mapping).

## Diagram to draw

- Transperineal grid over the prostate with sampling sectors.

> **Key points to remember:**
> - Saturation = >12 cores (18–21 or more); 5-mm template mapping.
> - No gain at initial biopsy; used after negatives and for focal therapy.
> - PROMIS: template mapping as reference; mpMRI sensitivity 93%.

## SAQ 5 Cryotherapy for CA prostate

P2 · P4 **merged from 2**

*Asked as:* P4 SAQ "Cryotherapy for carcinoma prostate" · P2 SAQ "Biological effects of cryosurgery"

### Principle

- Tissue destruction by extreme cold. **Argon** expanding in the probe tip cools by the **Joule–Thomson effect**; **helium** expanding warms (active thaw).
- Used for small renal masses and prostate cancer.
- Prostate cryotherapy is **transperineal, image-guided** freezing; suits **anterior** tumours. HIFU (high-intensity focused ultrasound) is the main alternative.
- Thermocouples guard sphincter and neurovascular bundles; a **urethral warmer** prevents sloughing.
- Generations: liquid nitrogen without imaging; TRUS (transrectal ultrasound) plus warmer; argon/helium 17-gauge probes; focal with MRI (magnetic resonance imaging) fusion.

### Biological effects: mechanisms of cell death

Tissue dies during **both freezing and thawing**, by four mechanisms:

| Mechanism | What happens |
|---|---|
| **1. Direct cell injury** | *Freeze:* extracellular water freezes first and becomes hyperosmolar; cells dehydrate and membranes distort. Rapid cooling forms **intracellular ice**, which ruptures organelles. *Thaw:* water rushes back, cells swell and burst; ice recrystallises into larger crystals |
| **2. Vascular injury** | Endothelial damage, platelet activation and **thrombosis** cause **vascular stasis** and ischaemic necrosis; the main delayed killing effect |
| **3. Apoptosis** | Sublethal cold at the periphery triggers programmed cell death |
| **4. Immunomodulation** | Necrotic cells release antigens and DAMPs (damage-associated molecular patterns); dendritic cells activate T cells against tumour |

- **End result:** coagulative necrosis, then fibrous scar.

### Factors deciding lethality

- **Temperature:** normal renal tissue dies at about −19.4 °C; tumour may need −40 to −50 °C. **Target ≤−40 °C.**
- **Fast freeze, slow thaw** maximise injury.
- **Double freeze–thaw cycle** enlarges the necrotic zone (standard of care); each freeze about 8–10 minutes.
- **Heat sink:** nearby large vessels warm tissue; more probes or longer freezing needed.

### Zones of the ice ball

- Centre: complete necrosis. Edge: 0 °C and sublethal; below −20 °C only about 3 mm inside the edge. The lethal −40 °C isotherm lies well inside the visible ice ball.
- So the ice ball must extend **5–10 mm beyond the tumour**.

### Indications in prostate cancer

1. **Salvage after radiotherapy** (main role): biopsy-proven local recurrence, no metastases, **PSA (prostate-specific antigen) <10 (ideally <4)**, doubling time ≥16 months.

2. **Primary whole-gland:** men unfit for surgery or radiotherapy.
3. **Focal or hemi-gland:** MRI-visible grade group 2–3 lesion; rests on the index-lesion hypothesis.

### Technique

- TRUS-guided transperineal template: place cryoprobes, thermocouples and urethral warmer; real-time imaging shows the ice ball; freeze anterior to posterior until −40 °C is reached; helium thaw; repeat the cycle.
- **Also know:** passive versus active thaw (both used); insulated probe shafts reduce skin injury.

### Outcomes

- Salvage: biochemical control **55–71% at 3–5 years** (Phoenix: nadir + 2).
- **Donnelly, Cancer 2010** (RCT, randomised controlled trial): **non-inferior** to external beam radiotherapy; worse sexual function.

### Complications

| | Primary / focal | Salvage |
|---|---|---|
| **ED (erectile dysfunction)** | **Almost always** (whole gland) | Very common |
| Incontinence | Rare | Single digit |
| **Rectourethral fistula** | ≤0.8% (focal) | 1–3.4% |
| Retention | Commonest after focal | ~4% |

- **Also know:** urethral sloughing, stricture; focal ablation preserves erections better (COLD [Cryo On-Line Data] registry).

### Comparison with other ablative energies

| | Cryotherapy | HIFU | IRE (irreversible electroporation) |
|---|---|---|---|
| Energy | Freezing | Focused ultrasound heat (≥55 °C) | Electric pulses, non-thermal |
| Route | Transperineal | Transrectal | Transperineal |
| Best for | Anterior lesions | Posterior lesions | Lesions near nerves, vessels |

### Guidelines and advances

- **EAU (European Association of Urology):** primary or focal only in trials or registries. PSMA (prostate-specific membrane antigen) PET selects salvage patients.

### Diagram to draw

- Prostate (or tumour) with probe and ice ball: −40 °C isotherm inside the 0 °C edge, and tumour margin.

> **Key points to remember:**
> - Argon freezes, helium thaws (Joule–Thomson); ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm, urethral warmer.
> - Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
> - Main role is salvage: PSA <10, doubling time ≥16 months.
> - Whole-gland ED near universal; fistula ≤0.8% focal, up to 3.4% salvage.

## SAQ 6 GnRH (LHRH) analogues: agonists and antagonists

P2 · P4 **merged from 2**

*Asked as:* P4 SAQ "GnRH (LHRH) analogues: agonists and antagonists" (GNRH Analogue; LHRH agonist) · P2 SAQ "LHRH analogue"

*Read as: GnRH (gonadotrophin-releasing hormone, also LHRH, luteinising hormone-releasing hormone) agonists and antagonists, chiefly in prostate cancer.*

### Physiology

- **LHRH (GnRH)** is a hypothalamic decapeptide released in **pulses**. Pulsatile GnRH drives pituitary LH (luteinising hormone) and FSH (follicle-stimulating hormone). LH → Leydig cells → testosterone (about 95% testicular; about 95% of total).
- **Continuous** exposure **desensitises and down-regulates** the pituitary receptor. This is the basis of agonist therapy.

### GnRH (LHRH) agonists: drugs

| Drug | Formulation |
|---|---|
| Leuprolide | Depot 1, 3, 6 monthly |
| Goserelin | SC (subcutaneous) implant 1 or 3 monthly |
| Triptorelin | IM (intramuscular) depot |
| Histrelin | Implant, 1 year |
| Buserelin | Nasal or SC |

- Substitution at **position 6** of the decapeptide raises potency and half-life; depots last 1 month to 1 year.

### Agonists: mechanism

```
Agonist → initial LH surge (flare, testosterone up) → receptor down-regulation (1–2 weeks) → LH falls → castrate testosterone in 2–4 weeks
```

- Survival **equals orchidectomy**.

### Flare phenomenon

- LH rises up to **10-fold for 10–20 days** → bone pain, **spinal cord compression**, urinary retention.
- Dangerous with spinal or weight-bearing metastases.
- **Prevent with an antiandrogen** (e.g. bicalutamide, flutamide) started with or before the agonist, for **21–28 days**.
- Impending cord compression → **antagonist** (degarelix) or orchidectomy instead.

### GnRH antagonists

- Immediate competitive receptor block: LH falls 84% in 24 hours, **no flare**, FSH falls.
- **Degarelix** (SC monthly): castrate by day 3 in >95%; injection-site reactions.
- **Relugolix** (oral daily, FDA 2020). **HERO** vs leuprolide: sustained castration 97% vs 89%; faster recovery after stopping; **fewer MACE (major adverse cardiovascular events)** (about 3% vs 6%).
- Preferred for **impending cord compression** or severe bone pain when orchidectomy is unsuitable.

- **Also know:** abarelix restricted (severe allergy); PRONOUNCE (degarelix vs leuprolide, stopped early) showed no MACE difference.

### Agonist vs antagonist

| Agonist | Antagonist |
|---|---|
| Flare; needs antiandrogen | No flare; castrate by day 3 (degarelix) |
| FSH rises | FSH falls |
| Depot to yearly | Monthly SC or oral (relugolix; HERO) |

### Castrate level

- **<50 ng/dL** (traditional); **<20 ng/dL** is achievable (orchidectomy median about 15) and may be better (aim **<20 ng/dL**). Check testosterone when PSA (prostate-specific antigen) rises.

### Indications in urology

- Metastatic hormone-sensitive prostate cancer (with ARPI [androgen receptor pathway inhibitor] or docetaxel).
- With radiotherapy for intermediate/high-risk and locally advanced disease.
- Recurrent prostate cancer: BCR (biochemical recurrence) with short doubling time; continued in CRPC (castration-resistant prostate cancer).
- **Also know:** intermittent ADT (androgen deprivation therapy) acceptable (PR.7); BPH (benign prostatic hyperplasia) volume reduction (not routine); central precocious puberty; cryptorchidism (GnRH or hCG [human chorionic gonadotrophin]: low success, not first-line); endometriosis.

### Adverse effects

- Hot flushes (50–80%), loss of libido, ED (erectile dysfunction).
- **Osteoporosis:** DEXA (dual-energy X-ray absorptiometry); calcium, vitamin D, denosumab or zoledronic acid.
- Diabetes, cardiovascular events, sarcopenia, anaemia, gynaecomastia (≤5%), cognitive change.
- Slow testosterone recovery after stopping.

### Diagram to draw

- Hypothalamic–pituitary–testis axis showing agonist and antagonist sites, and the flare.

> **Key points to remember:**
> - Agonist: surge, then continuous stimulation → down-regulation → castration in 2–4 weeks.
> - Flare: LH 10-fold for 10–20 days; cover with antiandrogen 21–28 days.
> - Antagonist: no flare; degarelix castrate by day 3; relugolix oral (HERO, fewer MACE); preferred in cord compression.
> - Castrate <50 ng/dL; aim <20 ng/dL.

## SAQ 7 Robotic radical prostatectomy

P1

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 1, Q3b

### Introduction

- RARP (robot-assisted radical prostatectomy): Walsh's anatomical nerve-sparing principles done with **3D, 10× magnified vision and wristed instruments**. The robot removed the laparoscopic suturing barrier; da Vinci from 2000 (S, Si, Xi, SP).

### Indications and contraindications

- Localised cancer (cT1–T2, selected cT3); life expectancy over 10 years.
- **Absolute:** uncorrectable bleeding diathesis; unfit for general anaesthesia.
- **Relative:** morbid obesity, prostate >80 g, median lobe, prior TURP (transurethral resection of the prostate), salvage surgery.

### Set-up

- Low lithotomy, arms tucked, **steep Trendelenburg (22–25°)**, padding; pressure 12–15 mmHg; camera port 15–17 cm above pubis, three robotic and two assistant ports.

### Steps (transperitoneal anterior approach)

1. Drop bladder; enter space of Retzius.
2. Incise endopelvic fascia; ligate DVC (dorsal venous complex).
3. Divide bladder neck (fat line, balloon traction); check for median lobe.
4. Seminal vesicles and vasa; incise Denonvilliers fascia.
5. Clip pedicles athermally; **interfascial nerve sparing** (extrafascial if high risk).
6. Apical dissection preserving urethral length.
7. Extended PLND (pelvic lymph node dissection) if indicated.
8. Posterior (Rocco) reconstruction; **Van Velthoven running anastomosis**.

### Postoperative care

- ERAS (enhanced recovery after surgery); discharge within 24 h; catheter about 1 week (not before day 5: retention); pelvic floor training; penile rehabilitation with PDE5 (phosphodiesterase type 5) inhibitors.

### Outcomes

- **Yaxley 2016** randomised trial: equal function and early margins versus open; less blood loss (300 versus 1400 mL) and shorter stay.
- Margins about 15% (9% if pT2); 1-year continence mid-90s% in expert hands.

### Complications

- Transfusion 1–2%; bladder neck stricture <2% (open up to 16%); rectal injury; ureteric injury; lymphocele; positioning neuropathy, corneal abrasion; robot failure <0.5%.

### Recent advances

- **Retzius-sparing** (earlier continence, more positive margins); Hood technique; single-port robot; same-day discharge.

**Diagram to draw:** port layout and fascial planes (intra-, inter-, extrafascial).

**Key points to remember:**

- Surgeon, not platform, decides outcome (Yaxley 2016).
- Interfascial plane is the standard nerve-sparing plane.
- Bladder neck stricture <2%; transfusion 1–2%.

## SAQ 8 Docetaxel

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 1, Q3b

### Introduction

- Semisynthetic **taxane** cytotoxic. Since **TAX 327 (2004)** it is the standard first-line chemotherapy for mCRPC (metastatic castration-resistant prostate cancer).

### Mechanism of action

- Binds **β-tubulin** and **stabilises microtubules** (GTP [guanosine triphosphate]-independent polymerisation). Cells arrest in **G2/M**.
- Induces apoptosis independent of TP53, through **BCL2 phosphorylation** and p27 induction.
- Prostate-specific: blocks microtubule transport of ligand-bound **AR (androgen receptor)** into the nucleus.

### Dose and administration

- **75 mg/m² IV (intravenous) every 3 weeks with prednisone 5 mg twice daily**; 6 cycles in mHSPC (metastatic hormone-sensitive prostate cancer). Steroid premedication limits fluid retention and hypersensitivity. Hepatic metabolism: caution in liver dysfunction.

### Uses in urology

| Setting | Evidence |
|---|---|
| **mCRPC** | **TAX 327:** overall survival (OS) 18.9 vs 16.4 months versus mitoxantrone (HR [hazard ratio] 0.76); better pain and PSA (prostate-specific antigen) response. SWOG 9916 (with estramustine) also positive |
| **mHSPC** | **CHAARTED:** OS 57.6 vs 44.0 months (HR 0.61), benefit in **high volume**. STAMPEDE: OS HR 0.78. GETUG-AFU 15 not significant |
| **Triplet** | ADT (androgen deprivation therapy) + docetaxel + abiraterone (**PEACE-1**) or darolutamide (**ARASENS**) improves OS |
| High-risk localised | RTOG 0521: added to RT (radiotherapy) + long-term ADT, OS HR 0.69 |
| Bladder | Second-line single agent (response about 13%); **intravesical gemcitabine–docetaxel** for BCG (bacillus Calmette–Guérin)-unresponsive or BCG-naive high-grade NMIBC (non-muscle-invasive bladder cancer) |
| Penis | **TPF** (docetaxel, cisplatin, 5-fluorouracil) |

**CHAARTED high volume:** ≥4 bone metastases with ≥1 outside the axial skeleton, or visceral metastasis.

### Toxicity

- **Neutropenia** and febrile neutropenia (dose-limiting), fatigue, **fluid retention** (peripheral oedema), **peripheral neuropathy**, hyperlacrimation, **nail dystrophy**, alopecia, liver dysfunction, hypersensitivity.
- 3-weekly: more neutropenia; weekly: more fatigue, diarrhoea, tearing.

### Resistance and sequencing

- P-glycoprotein efflux pump mediates resistance. **Cabazitaxel** has poor P-glycoprotein affinity, so it works after docetaxel (**TROPIC**; **CARD** beat a second ARPI [androgen receptor pathway inhibitor]).
- **Also know:** failed combinations (bevacizumab, atrasentan).

**Diagram to draw:** microtubule stabilised by docetaxel causing G2/M arrest and blocking AR entry into the nucleus.

> **Key points to remember:**
> - 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
> - CHAARTED: benefit in high-volume mHSPC (HR 0.61).
> - Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure.

## SAQ 9 PIN

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 1 Paper 2, Q3f

*Assumed meaning: prostatic intraepithelial neoplasia.*

### Definition

- PIN (prostatic intraepithelial neoplasia): **architecturally benign** acini or ducts lined by **cytologically atypical** cells, with an intact or patchy **basal cell layer**.
- Graded **LGPIN (low-grade PIN)** and **HGPIN (high-grade PIN)**. LGPIN is not reported: it is irreproducible and carries no excess risk.

### Histology

- HGPIN: nuclear enlargement, hyperchromasia, **prominent nucleoli**.
- Patterns: tufting, micropapillary, flat. **Cribriform lesions are now classed as IDC-P (intraductal carcinoma of the prostate) or AIP (atypical intraductal proliferation).**
- Immunohistochemistry: basal cells stain with **HMWCK (high-molecular-weight cytokeratin) and p63**; AMACR (α-methylacyl-CoA racemase) positive. Mainly in the peripheral zone.

### Evidence that HGPIN is a precursor

- More and larger HGPIN foci in cancerous prostates; carcinoma seen budding from HGPIN.
- About 20% carry **TMPRSS2:ERG fusion** (about 50% of cancers). Extra genomic changes are needed for invasion.

### Incidence

- About **4–5%** of biopsies (range 0–25%, due to observer variation).

### Related lesions

| Lesion | Meaning | Action on isolated finding |
|---|---|---|
| HGPIN | Weak risk marker | **No repeat biopsy** |
| PINATYP (HGPIN with adjacent atypical glands) | As "atypical" | Manage as ASAP (atypical small acinar proliferation) |
| ASAP | Cancer on re-biopsy about 40% | Repeat biopsy ± MRI (magnetic resonance imaging) |
| AIP | May herald IDC-P | Consider re-biopsy |
| IDC-P | Usually high-grade cancer present | Definitive treatment; excludes active surveillance |

### Management

- Isolated HGPIN, unifocal or multifocal: **no immediate repeat biopsy**. Decide further testing on PSA (prostate-specific antigen), DRE (digital rectal examination) and **mpMRI (multiparametric MRI)**.
- HGPIN on TURP (transurethral resection of the prostate) chips: no further sampling on current data.
- **Chemoprevention:** none proven; green tea showed reduced progression in small trials.
- *Guideline note:* EAU (European Association of Urology) guidelines still list extensive HGPIN (≥3 biopsy sites) as a reason for repeat biopsy; the chapter does not.

**Diagram to draw:** benign gland, HGPIN gland (atypical cells, intact basal layer) and cancer (no basal layer).

> **Key points to remember:**
> - Atypical cells, benign architecture, basal layer retained.
> - Isolated HGPIN needs no repeat biopsy; ASAP does.
> - Cribriform "PIN" is now IDC-P or AIP.

## SAQ 10 Gleason's score and its significance

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 2, Q3f

### Introduction

- **Gleason grading** (Donald Gleason, late 1960s) grades prostate adenocarcinoma by **glandular architecture at low magnification**, not cytology. Patterns run from 1 (most differentiated) to 5. Modified by ISUP (International Society of Urological Pathology) in **2005** and **2014**.

### Patterns (modern)

| Pattern | Architecture |
|---|---|
| 1–2 | Not used on biopsy now |
| **3** | Discrete, well-formed glands of any size |
| **4** | **Cribriform** (including glomeruloid), poorly formed or fused glands |

| Pattern | Architecture |
|---|---|
| **5** | Sheets, cords, single cells, solid nests, **comedonecrosis** |

### Scoring rules

- **Radical prostatectomy:** primary (commonest) + secondary pattern.
- **Biopsy (2005):** commonest + **highest** pattern, however small. Single pattern is doubled (3+3).
- A tertiary pattern 5 under 5% at prostatectomy is reported as a comment; it adds risk but does not change the grade group.
- Report **% pattern 4** in score 7; report cribriform and intraductal carcinoma (both adverse).

### ISUP Grade Groups (2014; WHO [World Health Organization] 2016)

| Grade group | Gleason score | 5-year BCR (biochemical recurrence)-free after prostatectomy |
|---|---|---|
| 1 | 3+3=6 | 96% |
| 2 | 3+4=7 | 88% |
| 3 | 4+3=7 | 63% |
| 4 | 8 (4+4, 3+5, 5+3) | 48% |
| 5 | 9–10 | 26% |

- Grade groups separate 3+4 from 4+3 and 8 from 9–10. Starting at "1" avoids patients misreading the lowest score, 6, as moderate cancer.

### Significance

- **Most powerful biopsy predictor** of pathological stage, EPE (extraprostatic extension), nodal spread (0% in GG1, 4.3% in GG3), recurrence and death.
- **Risk groups** (D'Amico, NCCN [National Comprehensive Cancer Network]) combine grade, PSA (prostate-specific antigen) and stage.
- **Treatment choice:** GG1 → active surveillance; selected GG2 with little pattern 4 → surveillance; GG ≥4 → high risk, longer ADT (androgen deprivation therapy) with radiotherapy; also selects for brachytherapy and focal therapy.
- **Limitations:** biopsy undergrades about one-third of GG1 (sampling error); inter-observer variation; after ADT or radiotherapy, grading is unreliable.
- **Also know:** pure GG1 does not metastasise, so "is it cancer?" is debated.

**Diagram to draw:** Gleason's pattern diagram 3 → 4 → 5.

> **Key points to remember:**
> - Biopsy score = commonest + highest pattern.
> - 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
> - GG4 = score 8; GG5 = 9–10.

## SAQ 11 HIFU

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 5 Paper 2, Q3d

## Definition and principle

- HIFU (high-intensity focused ultrasound) focuses ultrasound from a bowl-shaped transducer or lens onto a point. Energy is absorbed only where the beams converge, so tissue there is heated while the path is spared.
- **Intensity 1500–10,000 W/cm²** (diagnostic ultrasound causes negligible heating).
- **Mechanisms: thermal coagulative necrosis** (≥55 °C; prostate exceeds 100 °C within seconds) plus **cavitation** (bubbles collapse and damage cells).
- Each lesion is a **rice grain**; the target is "painted" with overlapping lesions. Ablated and viable tissue look alike on ultrasound, so gaps occur.

## Devices and routes

- **Transrectal** probe that both monitors and treats (posterior zone).
- **TULSA (MRI [magnetic resonance imaging]-guided transurethral ultrasound ablation): MR thermometry** confirms lethal temperature; reaches the anterior gland.

## Indications

| Setting | Use |
|---|---|
| **Focal therapy, prostate cancer** | MRI-visible, unilateral, grade group 2–3 index lesion; hemiablation; based on the index-lesion hypothesis |
| Whole-gland ablation | Localised cancer; retreatment 29–38% |
| **Salvage after EBRT** (external beam radiotherapy) | Biopsy-proven local recurrence, **≥2 years after EBRT** |
| BPH (benign prostatic hyperplasia) | Approved |
| Kidney | Investigational (incomplete ablation) |

## Limitations

- **Anterior lesions** (energy lost, oedema displaces target), large gland, gland height and **calcification** (some advise TURP [transurethral resection of the prostate] first).

## Outcomes

- **Focal HIFU:** continence 90–95%, potency 50–78%; infield positive biopsy 16.7–54% (csPCa [clinically significant prostate cancer] 0–21.8%).
- **HEAT registry** (1379 men): 7-year failure-free survival 69%.
- **Salvage HIFU:** biochemical progression-free survival about 48% at 3 years; continence 64–87.5%.

## Complications

- Retention (about 8%), UTI (urinary tract infection) 6–18%, stricture (4%), ED (erectile dysfunction); rectourethral fistula rare (about 2% after salvage); osteitis pubis.

## Follow-up

- PSA (prostate-specific antigen) falls only 50–80%; **mpMRI (multiparametric MRI) + targeted biopsy at 3–6 months**, then at 1–2 and 5 years. EAU (European Association of Urology): focal therapy within trials or registries.

**Also know: histotripsy** (>10,000 W/cm², mechanical liquefaction, visible bubble cloud).

**Diagram to draw:** transrectal transducer focusing beams on a rice-grain lesion in the posterior prostate.

> **Key points to remember:**
> - Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
> - Transrectal HIFU suits posterior lesions; TULSA for anterior.
> - Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

## SAQ 12 Imaging measures in CA prostate

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 2, Q3e (mpMRI detail in 2024-Winter Paper 1, Q3d; PSMA PET in 2019-Summer Paper 4, Q3a)

### 1. Ultrasound

- **TRUS (transrectal ultrasound):** guides biopsy, measures volume (PSA [prostate-specific antigen] density). Cancer is a hypoechoic PZ (peripheral zone) lesion, but **39% are isoechoic**.
- **Also know:** Doppler, contrast and elastography (adjuncts); **micro-ultrasound** (29 MHz; PRI-MUS [Prostate Risk Identification using Micro-Ultrasound] score; close to MRI accuracy).

### 2. MRI: detection and local staging

- **mpMRI (multiparametric magnetic resonance imaging):**

| Sequence | Shows | Cancer |
|---|---|---|
| T2-weighted | Anatomy; EPE (extraprostatic extension), SVI (seminal vesicle invasion) | Low signal |
| DWI (diffusion-weighted imaging) with ADC (apparent diffusion coefficient) | Cellularity | Bright on DWI, dark on ADC |
| DCE (dynamic contrast-enhanced) | Vascularity | Early enhancement, washout |

- **PI-RADS v2.1 (Prostate Imaging Reporting and Data System)** 1–5: DWI dominant in PZ, T2 in TZ (transition zone); DCE only upgrades PZ 3 to 4.
- **PROMIS** (sensitivity 93% vs 48% for TRUS biopsy) and **PRECISION** (more significant, less insignificant cancer): **MRI before biopsy**. Cochrane: sensitivity 0.91, specificity 0.37.
- **Also know:** biparametric MRI; **PI-QUAL** (Prostate Imaging Quality); **PRECISE** (scores change on surveillance); fusion biopsy.

### 3. Staging by risk group

- **Low or intermediate risk:** no routine CT (computed tomography) or bone scan.
- **High risk:** CT plus **$^{99m}$Tc-MDP (methylene diphosphonate) bone scan**, or **PSMA (prostate-specific membrane antigen) PET (positron emission tomography)**.
- **proPSMA:** accuracy 92% vs 65% for conventional imaging. Nodal sensitivity only 30–40%, specificity >90%: a negative scan does not exclude micrometastases.

### 4. Biochemical recurrence

- **PSMA PET** works at low PSA. **Calais:** 56% vs 26% for fluciclovine. **CONDOR:** management changed in 63.9%.
- Choline and fluciclovine PET are older; FDG (fluorodeoxyglucose) PET is poor except in neuroendocrine change.

### 5. Advanced disease

- PSMA PET selects men for **$^{177}$Lu-PSMA-617** (VISION); whole-body MRI. No proof yet that PSMA-guided treatment improves survival.

**Diagram to draw:** prostate zones with a PZ lesion dark on ADC; PSMA PET nodal uptake.

> **Key points to remember:**
> - MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
> - High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
> - PSMA nodal sensitivity 30–40%.

## SAQ 13 Expectant treatment of CA prostate

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 2, Q3d

### Rationale

- Overdiagnosis: autopsy prevalence of microfocal cancer roughly equals age as a percentage; lifetime risk of death is about 3%.
- **Grade group 1 (Gleason 6) has virtually no metastatic potential.** Deaths on observation come from higher grade missed at biopsy (about 30% upgraded at prostatectomy).

### Two forms of expectant treatment

| | **AS (active surveillance)** | **WW (watchful waiting)** |
|---|---|---|
| Patient | Fit for radical therapy, life expectancy ≥10 years | Limited life expectancy (age, comorbidity) |
| Intent | **Curative, deferred** | **Palliative** |
| Monitoring | PSA (prostate-specific antigen), DRE (digital rectal examination), MRI, biopsy | Infrequent PSA and review; no biopsy |
| Action | Radical treatment on reclassification | ADT (androgen deprivation therapy) or palliation for symptoms |

### Evidence

| **Trial** | **Result** |
|---|---|
| **SPCG-4** (pre-PSA) | Prostatectomy cut cancer death (19.6% vs 31.3%), mainly <65 years and higher grade |
| **PIVOT** | No significant mortality benefit; low-risk cancer death 2.7% |

| Trial | Result |
|---|---|
| **ProtecT** (15 years) | Cancer death <3% in all arms; metastasis doubled on active monitoring (9.4% vs about 5%) |

### Selection for AS

- **Very low risk (Epstein):** grade group 1, ≤2 positive cores, no core >50%, PSA density <0.15.
- **Low risk:** grade group 1, PSA <10 ng/mL; all guidelines prefer AS.
- **Favourable intermediate risk** (small-volume grade group 2): selected cases.
- Caution: **BRCA2** carriers, high-volume disease.

### AS protocol

- MRI and **confirmatory biopsy** within 1–3 years; PSA 6-monthly; DRE every 1–2 years; MRI every 18–24 months; biopsy every 3–5 years.
- **Triggers:** grade group ≥3 → treat; volume or PSA rise → MRI or biopsy first.

### Outcomes

- Cancer mortality **0.5–5% at 15 years** by entry criteria (Johns Hopkins vs Toronto); about **25% treated by 5 years**.

### Watchful waiting

- Switch AS to WW when life expectancy falls below about 5–10 years.
- **Also know:** genomic tests refine selection; 5ARI (5α-reductase inhibitor) has no proven role.

**Diagram to draw:** AS flowchart: diagnosis → MRI + confirmatory biopsy → surveillance → trigger → treatment.

> **Key points to remember:**
> - AS is deferred cure; WW is palliation.
> - Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
> - ProtecT: no mortality difference at 15 years; more metastases on monitoring.
> - About 25% leave AS by 5 years.

## SAQ 14 Molecular staging of CA prostate

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 2, Q3e (related: prognosis in 2019-Summer Paper 4, Q3e)

### Definition and rationale

- Clinical staging (DRE (digital rectal examination), PSA (prostate-specific antigen), grade, imaging) misses occult spread and cannot fully predict behaviour.
- **Molecular staging** uses DNA, RNA or protein markers in tissue, blood or nodes to find hidden disease and predict outcome.

### 1. Detection of occult spread

- **RT-PCR (reverse transcription polymerase chain reaction)** for PSA or PSMA (prostate-specific membrane antigen) mRNA in blood, marrow or "negative" nodes: detects micrometastases, but inconsistent, so not used clinically.
- **CTCs (circulating tumour cells):** counts are prognostic in metastatic disease; **AR-V7** (splice variant lacking the ligand-binding domain) predicts resistance to ARPIs (androgen receptor pathway inhibitors).
- **ctDNA (circulating tumour DNA)** liquid biopsy: DNA-repair mutations; clonal haematopoiesis gives false calls.

### 2. Tissue genomic classifiers

| Test | Basis | Predicts |
|---|---|---|
| **Decipher** | 22-RNA classifier | Metastasis, cancer death; post-prostatectomy radiotherapy decision |
| **Oncotype DX GPS (Genomic Prostate Score)** | 17 genes (12 cancer, 5 reference) by RT-PCR | Adverse pathology; AS (active surveillance) choice |
| **Prolaris** | 31 cell-cycle progression genes | Progression, cancer death |

- **Also know:** PORTOS (24-gene radiotherapy response score); ConfirmMDx (methylation, after negative biopsy).

### 3. Genetic markers

- **Germline:** BRCA2 (aggressive disease), BRCA1, ATM, CHEK2, Lynch genes, HOXB13 (risk, not aggressiveness); test in metastatic or high-risk disease.
- **Somatic:** PTEN loss, TMPRSS2:ERG fusion, homologous recombination repair defects (guide PARP (poly ADP-ribose polymerase) inhibitors).

### 4. Molecular imaging

- **PSMA PET (positron emission tomography):** more sensitive and specific than CT and bone scan for nodes and metastases (proPSMA), but cannot exclude micrometastases.

### Clinical use

- Refines **AS vs treatment** and post-operative radiotherapy decisions; guides targeted therapy.

### Limitations

- Mostly retrospective; **multifocality and heterogeneity** (the sampled focus may not be the worst); cost; added value still being tested (G-MAJOR trial).

**Diagram to draw:** clinical stage vs molecular stage (tissue, blood, nodes, imaging).

> **Key points to remember:**
> - Molecular staging finds occult disease and predicts behaviour beyond TNM.
> - Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
> - AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
> - Limitation: heterogeneity and mostly retrospective data.

## SAQ 15 Prostate cancer prevention trials

P4

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 4, Q3a

### Rationale

- Prostate cancer is common, slow and androgen-dependent. Men with 5α-reductase deficiency have rudimentary prostates. This supported **5ARI (5α-reductase inhibitor)** trials.

### Landmark trials

| | PCPT (2003) | REDUCE (2010) | SELECT (2009) |
|---|---|---|---|
| Agent | Finasteride 5 mg | Dutasteride 0.5 mg | Selenium 200 µg ± vitamin E 400 IU |
| Men | 18,882; PSA (prostate-specific antigen) ≤3 | PSA 2.5–10, prior negative biopsy | 35,533 |
| Design | 7 years; end-of-study biopsy | Biopsy at 2 and 4 years | Stopped early for futility |
| All cancers | ↓24.8% | ↓22.8% | No benefit |
| High grade | Gleason 7–10 ↑ (6.4% vs 5.1%) | Gleason 8–10 ↑ (0.9% vs 0.6%) | Vitamin E alone ↑ risk |

### The high-grade controversy

- Proposed artefacts: better PSA performance and a smaller gland make cancer easier to find.
- The FDA (Food and Drug Administration) did **not accept** these. In 2011 it refused a prevention licence and added a **high-grade warning** to 5ARI labels.
- PCPT at 18.4 years: non-significant 25% fewer prostate cancer deaths. Real-world data show no mortality effect.
- **Conclusion:** 5ARIs reduce detection of low-grade cancer, not death.

### Vitamins and other agents

- **ATBC** trial: low-dose vitamin E cut prostate cancer 32% in smokers (secondary end point). SELECT then failed.
- **VITAL** (vitamin D): no prostate effect. High-dose zinc, calcium and antioxidants may harm.
- **Aspirin/NSAIDs** (non-steroidal anti-inflammatory drugs): small effect. **Statins:** observational benefit only. **Metformin:** none.
- Diet: whole foods; avoid obesity.

### Current position

- No agent is approved for prevention. 5ARIs may be discussed with men taking them for BPH (benign prostatic hyperplasia).

**Diagram to draw:** bar chart: all cancers ↓ vs high-grade ↑ in PCPT and REDUCE.

> **Key points to remember:**
> - PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
> - FDA warning; no approved preventive agent.
> - SELECT negative; vitamin E alone harmful.

## SAQ 16 PSMA PET Ct scan

P4

**Type:** SAQ (10 marks)
**Source:** 2019-Summer Paper 4, Q3a

### Principle

- **PSMA (prostate-specific membrane antigen):** transmembrane glycoprotein over-expressed by prostate cancer, rising with grade.
- **PET/CT (positron emission tomography/computed tomography):** positron-labelled ligand binds PSMA; uptake as SUV (standardised uptake value).
- **Tracers:** $^{68}$Ga-PSMA-11 (half-life 68 min); $^{18}$F-DCFPyL, $^{18}$F-rhPSMA-7.3, $^{18}$F-PSMA-1007 (110 min, better images; PSMA-1007 little urinary excretion).
- **Specificity >90%**; beats choline and fluciclovine.

### Indications

**1. Primary staging (high risk):**

- **proPSMA (2020):** accuracy **92% vs 65%** vs CT + bone scan.
- Nodal sensitivity only 30–40% against PLND (pelvic lymph node dissection): a negative scan does not exclude micrometastases.
- NCCN (National Comprehensive Cancer Network): may **replace** conventional imaging; AUA (American Urological Association): add-on.

**2. BCR (biochemical recurrence):**

- Detection rises with PSA (prostate-specific antigen): about **half positive below 0.5 ng/mL**, nearly all above 2.
- **Calais trial:** PSMA 56% vs fluciclovine 26%. **CONDOR:** management changed in 63.9%.
- Guides salvage radiotherapy (disease outside field in about 1 in 5).

**3. Intraprostatic detection: PRIMARY** trial: adding PSMA PET to MRI (magnetic resonance imaging) improved negative predictive value; 5-point PRIMARY score.

**4. Theranostics:** selects patients for **$^{177}$Lu-PSMA-617**. **VISION (2021):** mCRPC (metastatic castration-resistant prostate cancer) after ARPI (androgen receptor pathway inhibitor) and taxane: OS 15.3 vs 11.3 months.

### Reporting

- **PROMISE miTNM** (molecular imaging TNM): miT, miN, miM staging plus expression score (vs blood pool, liver, parotid). PSMA-RADS also used.

### Pitfalls

| False positive | False negative |
|---|---|
| Coeliac/stellate ganglia, fractures, Paget's disease, fibrous dysplasia | Small nodes (<5 mm) |
| Other tumours (RCC (renal cell carcinoma), HCC (hepatocellular carcinoma)) | PSMA-negative tumours (about 5–10%) |
| $^{18}$F-PSMA-1007 benign bone uptake | Neuroendocrine transformation; masking by bladder activity |

- **Also know:** no proof yet that PSMA-guided treatment improves survival (stage migration).

**Diagram to draw:** PSMA ligand binding the cell-surface antigen with fused PET/CT image of a nodal lesion.

> **Key points to remember:**
> - proPSMA: accuracy 92% vs 65%.
> - BCR: about half positive below PSA 0.5 ng/mL.
> - Nodal sensitivity 30–40%: negative scan does not exclude nodes.
> - VISION: $^{177}$Lu-PSMA-617 improves OS in mCRPC.

## SAQ 17 Methods to predict prognosis of patient of Cancer prostate

P4

**Type:** SAQ (10 marks)
**Source:** 2019-Summer Paper 4, Q3e

### 1. Clinical and pathological factors

- **PSA (prostate-specific antigen):** organ-confined about 80% at <4 but <50% at >10 ng/mL; density and velocity add information.
- **Clinical stage (DRE (digital rectal examination), TNM).**
- **Grade group (ISUP (International Society of Urological Pathology)) – strongest single factor:**

| Grade group | Gleason | 5-year BCR (biochemical recurrence)-free |
|---|---|---|
| 1 | ≤6 | 96% |
| 2 | 3+4 | 88% |
| 3 | 4+3 | 63% |
| 4 | 8 | 48% |
| 5 | 9–10 | 26% |

- Biopsy: % positive cores, **cribriform and intraductal** patterns, perineural invasion.
- Prostatectomy: margin (extent, grade), EPE (extraprostatic extension), seminal vesicle and node status.

### 2. Risk groups

| D'Amico | Low | Intermediate | High |
|---|---|---|---|
| PSA (ng/mL) | ≤10 | 10–20 | >20 |

| D'Amico | Low | Intermediate | High |
|---|---|---|---|
| Gleason | ≤6 | 7 | 8–10 |
| Stage | T1–T2a | T2b | T2c |

- **NCCN (National Comprehensive Cancer Network):** very low, low, favourable/unfavourable intermediate, high, very high.
- **CAPRA (Cancer of the Prostate Risk Assessment)** score 0–10 (age, PSA, Gleason, stage, % cores); **CAPRA-S** after surgery.

### 3. Nomograms

- **Partin tables:** organ-confined disease, EPE, seminal vesicle, node involvement.
- **MSKCC (Memorial Sloan Kettering) nomograms:** pre- and post-treatment BCR and 15-year cancer-specific survival.
- **Briganti nomogram:** risk of nodal invasion → decides PLND (pelvic lymph node dissection).

### 4. Imaging

- mpMRI (multiparametric MRI): PI-RADS (Prostate Imaging Reporting and Data System) score, tumour contact length, EPE.
- PSMA PET (prostate-specific membrane antigen PET) for metastatic burden.

### 5. Tissue biomarkers

| Test | Basis | Use |
|---|---|---|
| **Decipher** | 22-RNA classifier | Metastasis risk; post-op radiotherapy |
| **Oncotype DX GPS** | 17 genes | Adverse pathology; active surveillance |
| **Prolaris** | Cell-cycle progression genes | Progression, cancer death |

- Germline BRCA2 (BReast CAncer gene 2): aggressive disease.

### 6. After treatment

- **PSA doubling time (PSADT):** <3 months highest risk; Pound: time to BCR <2 years, PSADT <10 months, Gleason 8–10 predict metastasis.
- **EAU (European Association of Urology) BCR risk groups:** low risk = PSADT >1 year and grade group <4.
- PSA nadir on ADT (androgen deprivation therapy); **Phoenix** failure (nadir + 2) after radiotherapy.

**Diagram to draw:** flowchart from risk group to nomogram, biomarker and treatment decision.

> **Key points to remember:**
> - Grade group is the strongest predictor.
> - D'Amico, NCCN and CAPRA combine PSA, grade and stage.
> - Decipher, Oncotype, Prolaris add genomic risk.

## SAQ 18 PARP inhibitors in Prostate cancer.

P4

**Type:** SAQ (10 marks)
**Source:** 2024-Winter Paper 4, Q3f

## Introduction

- **PARP (poly-ADP-ribose polymerase)** repairs single-strand DNA breaks. **PARPIs (PARP inhibitors):** olaparib, rucaparib, niraparib, talazoparib.

## Mechanism: synthetic lethality

- **Catalytic inhibition** and **PARP trapping** on DNA leave single-strand breaks unrepaired.
- At replication these become **double-strand breaks**, normally repaired by **HRR (homologous recombination repair)**.
- Tumours lacking HRR (e.g. **BRCA2** loss) die; normal cells survive.
- **AR (androgen receptor)–PARP synergy:** AR maintains HRR genes, so AR blockade adds repair deficiency: basis for PARPI + ARPI (androgen receptor pathway inhibitor).

## Who benefits: testing

- **HRR alterations in mCRPC (metastatic castration-resistant prostate cancer): up to 20–25%.** Genes: **BRCA2 (largest benefit)**, BRCA1, ATM (weak), PALB2, CHEK2, CDK12.
- **Germline testing** in all metastatic disease, strong family history or Ashkenazi ancestry; **somatic testing** on tissue or ctDNA (circulating tumour DNA), where clonal haematopoiesis can mislead.

## Key trials

| Trial | Setting | Result |
|---|---|---|
| **PROfound** | Olaparib vs enzalutamide/abiraterone after ARPI | BRCA/ATM cohort: imaging PFS (progression-free survival) **7.4 vs 3.6 months (HR [hazard ratio] 0.34)**; OS (overall survival) 19.1 vs 14.7 months in the final analysis (chapter quotes the interim 18.5 vs 15.1) |
| **TRITON3** | Rucaparib vs physician's choice | BRCA: PFS 11.2 vs 6.4 months (HR 0.50); no ATM benefit |
| **PROpel** | Olaparib + abiraterone, first-line mCRPC | OS 42.1 vs 34.7 months (P = 0.054, not significant); greatest in BRCA |
| **MAGNITUDE** | Niraparib + abiraterone, HRR-altered | BRCA rPFS (radiographic PFS) 19.5 vs 10.9 months (HR 0.55); no benefit without HRR alteration |
| **TALAPRO-2** | Talazoparib + enzalutamide, first line | All-comers rPFS not reached vs 21.9 months (HR 0.63); largest in BRCA |

- **Also know:** TOPARP-A/B, TRITON2, GALAHAD (phase II).

## Approvals (FDA [Food and Drug Administration])

- **Olaparib monotherapy:** HRR-mutated mCRPC after ARPI (2020).
- **Rucaparib:** BRCA-mutated mCRPC.
- **Olaparib or niraparib + abiraterone:** BRCA-mutated; **talazoparib + enzalutamide:** HRR-mutated (both 2023).

## Toxicity

- **Anaemia** (commonest grade 3–4; monitor counts), fatigue, nausea; rare **MDS/AML (myelodysplastic syndrome/acute myeloid leukaemia)**, pulmonary embolism.

### Recent advances

- **mCSPC (metastatic castration-sensitive prostate cancer)** trials: AMPLITUDE, TALAPRO-3; ATR inhibitors for ATM loss.

**Diagram to draw:** synthetic lethality: PARP block + BRCA loss → cell death.

> **Key points to remember:**
> - Synthetic lethality; BRCA2 gains most, ATM least.
> - PROfound: olaparib, PFS 7.4 vs 3.6 months.
> - Combinations: PROpel, MAGNITUDE, TALAPRO-2.

## SAQ 19 Indications of genomic testing in carcinoma prostate

P4

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 4, Q3e (PARP [poly-ADP-ribose polymerase] inhibitors also answered in 2024-Winter Paper 4, Q3f)

### Types of test

| Type | Sample | What it finds | Use |
|---|---|---|---|
| **Germline** | Blood or saliva | Inherited variants: BRCA2, BRCA1, ATM, CHEK2, PALB2, HOXB13, MMR (mismatch repair: MLH1, MSH2, MSH6, PMS2) | Therapy, family testing, screening |
| **Somatic** | Tumour tissue (usually primary) or ctDNA (circulating tumour DNA) | HRR (homologous recombination repair) genes, CDK12, MSI-H (microsatellite instability-high)/dMMR (MMR deficiency) | Targeted therapy |
| **Gene-expression classifiers** | Biopsy or prostatectomy tissue | Decipher (22 RNA markers); Oncotype DX GPS (Genomic Prostate Score, 17 genes); Prolaris (cell-cycle genes) | Prognosis |

### Indications: germline testing

- **All metastatic** disease (AUA [American Urological Association], NCCN [National Comprehensive Cancer Network]): over 10% carry actionable DNA-repair mutations.
- **High- or very-high-risk** localised disease.
- **Intermediate risk with intraductal or cribriform histology** (consider).
- **Family history:** early-onset or metastatic prostate cancer (relative <60 years); BRCA-related (breast, ovarian, pancreatic) or Lynch-related (colorectal, endometrial, upper tract) cancers; known familial mutation.
- **Ashkenazi Jewish ancestry.**
- **Why:** PARP inhibitor eligibility; caution with active surveillance in BRCA2; carrier screening; cascade testing of relatives with counselling.

### Indications: somatic testing

- **Metastatic disease, especially mCRPC (metastatic castration-resistant prostate cancer):** HRR alterations in up to 20–25%.
- **HRR (BRCA2 most)** → PARP inhibitors (olaparib, rucaparib, niraparib, talazoparib).

- **MSI-H/dMMR** (2–5%) → pembrolizumab (tumour-agnostic FDA [Food and Drug Administration] approval, 2017); then germline Lynch testing.
- Tissue preferred; ctDNA alternative, but **CHIP (clonal haematopoiesis of indeterminate potential)** gives false calls.

### Indications: tissue classifiers

- **Low and favourable-intermediate risk:** active surveillance versus treatment; surveillance intensity. Never the sole basis.
- **Intermediate and high risk:** metastasis risk guides treatment intensity.
- **After prostatectomy:** high Decipher score may favour adjuvant or early salvage radiotherapy (retrospective evidence).
- **Limits:** tumour heterogeneity, retrospective data, cost.
- **Also know:** ConfirmMDx methylation before repeat biopsy; polygenic risk scores.

**Diagram to draw:** flowchart: metastatic → germline + somatic; localised high-risk or family history → germline; localised low or intermediate → tissue classifier.

> **Key points to remember:**
> - Germline: all metastatic, high-risk, family history, Ashkenazi.
> - Somatic: HRR → PARP inhibitors; MSI-H → pembrolizumab.
> - BRCA2: aggressive; avoid casual active surveillance.
> - Decipher, Oncotype GPS, Prolaris: prognosis only.

# Rapid Revision Sheet · Section 17

### Staging and management of locally advanced (non-organ-confined) CA prostate (LAQ)

- Seminal vesicle invasion is T3b; non-regional nodes are M1a.
- Locally advanced = T3–4, N0/N+, M0; treat with combined therapy.
- RT + long-term ADT (18–36 months; P1: 2–3 years) is the standard; ADT alone is inferior (SPCG-7).
- RT + ADT: EORTC 22863 overall survival 79% vs 62%.
- STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
- Neoadjuvant ADT before RP: no survival gain.
- After RP: early salvage RT preferred over routine adjuvant RT for most men.
- Orchidectomy: castrate in 12 hours; flushes, osteoporosis (denosumab), metabolic, sexual effects.

### Metastatic prostate cancer, including newly diagnosed and bone metastases (LAQ)

- ADT alone is no longer standard: ADT plus abiraterone or an ARPI, or a triplet (PEACE-1, ARASENS), for every fit man.
- CHAARTED volume vs LATITUDE risk definitions.
- Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).
- Never start an LHRH agonist alone with spinal metastases: cover the flare with antiandrogen, or use an antagonist or orchidectomy.
- Single 8 Gy fraction gives the same pain relief as longer courses for focal pain; dexamethasone and MRI for cord compression.
- Focal back pain = cord compression until MRI proves otherwise.
- Denosumab 120 mg or zoledronate 4 mg IV 4-weekly prevent SREs in mCRPC, not in mHSPC; adjust zoledronate for creatinine clearance.
- No survival benefit from bone agents.
- Dental check first (ONJ); calcium and vitamin D.
- Denosumab delays SREs longer (20.7 vs 17.1 months).
- Radium-223 (ALSYMPCA) for symptomatic bone-only mCRPC; never with abiraterone.

### Radiotherapy and brachytherapy for CA prostate (LAQ)

- α/β ~1.5 Gy: basis of hypofractionation, 60 Gy/20 (CHHiP) and SBRT 36.25 Gy/5.
- ADT 4–6 months unfavourable intermediate; 18–36 months (P2: 24–36 months) high risk.
- Early salvage at PSA 0.1–0.2 = adjuvant (RADICALS, RAVES, GETUG-AFU 17).
- STAMPEDE H: prostate RT for low-volume M1; ALSYMPCA, VISION for mCRPC.
- Phoenix: nadir + 2 ng/mL.
- LDR $^{125}$I 145 Gy; HDR $^{192}$Ir.
- D90 >90%, V100 >85%.
- IPSS >20 or Qmax <10 = relative contraindication.
- ASCENDE-RT: boost halves biochemical failure.

**Castration-resistant prostate cancer (CRPC): definition and treatment (LAQ)**

- Testosterone <50 ng/dL with PSA or radiological progression; keep ADT going.
- AR still drives CRPC.
- nmCRPC with PSADT ≤10 months: apalutamide, enzalutamide or darolutamide.
- mCRPC: ARPI, taxanes (docetaxel, cabazitaxel), radium-223, $^{177}$Lu-PSMA, PARP inhibitors, sipuleucel-T.
- Switch mechanism; never radium-223 with abiraterone.

**PSA screening for prostate cancer: evidence, problems and solutions (LAQ)**

- ERSPC: (about) 20% fall in prostate cancer death; PLCO null from contamination.
- Core problems: low specificity, no safe cut-off, overdiagnosis, overtreatment.
- USPSTF 2018: Grade C at 55–69, Grade D ≥70.
- Fix: risk-adapted PSA, MRI before biopsy, active surveillance for GG1; MRI-first and active surveillance reduce overdiagnosis and overtreatment.

**65 yr male underwent RP for stage T1c Gleason's score (4+3). At 6 months post surgery, his S. PSA has risen to 2 ng/ml above his preop PSA value. Discuss treatment and prognosis (LAQ)**

- EAU high-risk BCR: PSADT ≤1 year or grade group 4–5.
- PSMA PET first: almost 100% detection above PSA 2 ng/mL.
- Salvage RT ≥64 Gy + ADT: RTOG 9601 (OS), GETUG-AFU 16, SPPORT (add pelvis).
- Pound: BCR <2 years, PSADT <10 months, Gleason 8–10 predict death.

**48 yr male presented with mixed voiding symptoms with PSA 18. Discuss management option in brief. (LAQ)**

- PSA 18 at 48: exclude cancer first; MRI before biopsy, transperineal targeted + systematic cores.
- PSA >10: organ-confined <50%; PSA >20: nodes about 20%.
- Young fit man: RP treats both cancer and obstruction.
- α-blocker for LUTS; no 5ARI until cancer excluded (halves PSA).

**Discuss management of organ confined CA prostate (LAQ)**

- Organ-confined = cT1–T2 N0 M0; NCCN risk groups drive choice.
- AS is preferred for low risk; ProtecT: no mortality difference at 15 years.
- EBRT + ADT: 4–6 months (intermediate), 18–36 months (high).
- Focal therapy rests on the index lesion hypothesis; still investigational.

**Discuss advances in management of CA prostate (LAQ)**

- PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
- proPSMA: PSMA PET accuracy 92% vs 65%.
- ProtecT: no mortality difference at 15 years; AS is standard for GG1.
- VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

**TRUS and prostate biopsy (SAQ)**

- Probe 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15 suspicious.
- 39% of cancers isoechoic; hypoechoic PZ lesion is classic.
- Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.

- MRI before biopsy: PROMIS and PRECISION established the MRI-first pathway.
- 12 cores; antibiotics ≤24 hours for all (transrectal); sepsis 0.3–3%.
- Bleeding commonest; sepsis most serious.
- Transperineal now preferred.

**Multiparametric MRI, PI-RADS and fusion biopsy (SAQ)**

- T2W anatomy, DWI/ADC cellularity, DCE vascularity.
- PZ ruled by DWI, TZ by T2W; DCE only lifts PZ 3 to 4.
- PRECISION and PROMIS: MRI before biopsy.
- Fusion: co-registration with elastic deformation.
- Cognitive, fusion, in-bore: similar yield.
- Endorectal coil close to prostate: higher SNR and resolution.
- Useful at 1.5 T; largely unnecessary at 3 T (GG ≥3 detection equal).
- Drawbacks: discomfort, gland deformation, near-field flare.

**Maximum / combined androgen blockade (SAQ)**

- MAB/CAB = castration + non-steroidal antiandrogen (blocks adrenal androgens).
- PCTCG 2000: only 2–3% 5-year survival gain.
- Main use now: flare cover for 21–28 days.
- ARPIs (androgen receptor pathway inhibitors) or triplets now standard in mHSPC; newer AR pathway inhibitors beat CAB (ENZAMET).
- Bicalutamide half-life about 6 days; 50 mg with LHRH agonist, 150 mg monotherapy.
- Monotherapy raises testosterone; gynaecomastia two-thirds.
- EPC: harm in localised disease; RTOG 9601: survival gain with salvage radiotherapy.
- Withdrawal response 15–30%.

**Saturation biopsy of prostate (SAQ)**

- Saturation = >12 cores (18–21 or more); 5-mm template mapping.
- No gain at initial biopsy; used after negatives and for focal therapy.
- PROMIS: template mapping as reference; mpMRI sensitivity 93%.

**Cryotherapy for CA prostate (SAQ)**

- Argon freezes, helium thaws (Joule–Thomson); −40 °C, two cycles, urethral warmer.
- Four mechanisms: direct injury, vascular thrombosis, apoptosis, immune effect.
- ≤−40 °C, two freeze–thaw cycles, margin 5–10 mm.
- Main role is salvage: PSA <10, doubling time ≥16 months.
- Whole-gland ED near universal; fistula ≤0.8% focal, up to 3.4% salvage.

**GnRH (LHRH) analogues: agonists and antagonists (SAQ)**

- Agonist: surge, then continuous stimulation → down-regulation → castration in 2–4 weeks.
- Flare: LH 10-fold for 10–20 days; cover with antiandrogen 21–28 days.
- Antagonist: no flare; degarelix castrate by day 3; relugolix oral (HERO, fewer MACE); preferred in cord compression.
- Castrate <50 ng/dL; aim <20 ng/dL.

**Robotic radical prostatectomy (SAQ)**

- Surgeon, not platform, decides outcome (Yaxley 2016).

- Interfascial plane is the standard nerve-sparing plane.
- Bladder neck stricture <2%; transfusion 1–2%.

**Docetaxel (SAQ)**

- 75 mg/m² 3-weekly + prednisone; TAX 327 HR 0.76.
- CHAARTED: benefit in high-volume mHSPC (HR 0.61).
- Neutropenia, oedema, neuropathy, nail change; cabazitaxel after failure. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**PIN (SAQ)**

- Atypical cells, benign architecture, basal layer retained.
- Isolated HGPIN needs no repeat biopsy; ASAP does.
- Cribriform "PIN" is now IDC-P or AIP.

**Gleason's score and its significance (SAQ)**

- Biopsy score = commonest + highest pattern.
- 3+4 (GG2) and 4+3 (GG3) differ: BCR-free 88% vs 63%.
- GG4 = score 8; GG5 = 9–10.

**HIFU (SAQ)**

- Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
- Transrectal HIFU suits posterior lesions; TULSA for anterior.
- Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

**Imaging measures in CA prostate (SAQ)**

- MRI before biopsy (PROMIS, PRECISION); PI-RADS v2.1.
- High risk: bone scan + CT or PSMA PET (proPSMA 92% vs 65%).
- PSMA nodal sensitivity 30–40%.

**Expectant treatment of CA prostate (SAQ)**

- AS is deferred cure; WW is palliation.
- Grade group 1 hardly ever metastasises; risk lies in missed higher grade.
- ProtecT: no mortality difference at 15 years; more metastases on monitoring.
- About 25% leave AS by 5 years.

**Molecular staging of CA prostate (SAQ)**

- Molecular staging finds occult disease and predicts behaviour beyond TNM.
- Decipher 22 genes; Oncotype 17 genes; Prolaris cell-cycle genes.
- AR-V7 on CTCs predicts ARPI resistance; BRCA2 means aggressive disease.
- Limitation: heterogeneity and mostly retrospective data. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Prostate cancer prevention trials (SAQ)**

- PCPT ↓24.8%; REDUCE ↓22.8%; both ↑ high grade.
- FDA warning; no approved preventive agent.
- SELECT negative; vitamin E alone harmful.

**PSMA PET Ct scan (SAQ)**

- proPSMA: accuracy 92% vs 65%.

- BCR: about half positive below PSA 0.5 ng/mL.
- Nodal sensitivity 30–40%: negative scan does not exclude nodes.
- VISION: $^{177}$Lu-PSMA-617 improves OS in mCRPC.

**Methods to predict prognosis of patient of Cancer prostate (SAQ)**

- Grade group is the strongest predictor.
- D'Amico, NCCN and CAPRA combine PSA, grade and stage.
- Decipher, Oncotype, Prolaris add genomic risk.

**PARP inhibitors in Prostate cancer. (SAQ)**

- Synthetic lethality; BRCA2 gains most, ATM least.
- PROfound: olaparib, PFS 7.4 vs 3.6 months.
- Combinations: PROpel, MAGNITUDE, TALAPRO-2.

**Indications of genomic testing in carcinoma prostate (SAQ)**

- Germline: all metastatic, high-risk, family history, Ashkenazi.
- Somatic: HRR → PARP inhibitors; MSI-H → pembrolizumab.
- BRCA2: aggressive; avoid casual active surveillance.
- Decipher, Oncotype GPS, Prolaris: prognosis only.